GRAND TETON NATONAL PARK
“4 LAZY F RANCH” CABINS FOR HOUSING

REHABILITATE “4 LAZY F RANCH” CABINS FOR HOUSING
GRTE - 237343

SPECIFICATIONS

(CONSTRUCTION DOCUMENTS)

NATIONAL PARK SERVICE (NPS)
DENVER SERVICE CENTER (DSC)

July 21, 2026

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SECTION 00 31 32 - GEOTECHNICAL DATA

PART 1 - GENERAL

1.1 SUB-SURFACE SOIL INVESTIGATION REPORT

A. The following report, prepared for this Project in 2025, along with any subsequent memoranda and/or addenda, is available upon request from the Contracting Officer:

1. Report of Jorgensen Geotechnical, LLC,,Geotechnical-Engineering Report GRTE – 4 Lazy F Rehab | Teton County, Wyoming | Grand Teton National Park , dated August 19, 2025

B. Information contained in this report was prepared by Jorgensen Geotechnical, LLC, Hennebery Eddy Architects Inc. and their consultants do not assume any responsibility for the accuracy of this information or any liability associated with this work.

1.2 RELATIONSHIP WITH CONTRACT DOCUMENTS

A. The Subsurface Exploration Services (Soils Report) and the Environmental reports are made available for review so that the Contractor will be able to anticipate the potential conditions that will be encountered in constructing the Project. While the Report in its entirety is not part of the Contract Documents, specific information or recommendations contained in the Report are made part of the requirements of the Project when they are incorporated into the Contract Documents or specifically referred to by them.

1.3 DISCLAIMER

A. The Government does not guarantee the accuracy of the Reports at locations and times other than those given. Ground water elevations, in particular, are subject to change.

1.4 DIFFERENT CONDITIONS

A. If, during construction, conditions are encountered which differ substantially from those indicated in the Report, promptly so notify the Contracting Officer in writing, and do not further disturb such conditions until directed.

B. The Contracting Officer will instruct the Geotechnical Engineer to investigate such conditions.

C. If the Contracting Officer, upon recommendation of the Geotechnical Engineer, determines that conditions do differ substantially from those that reasonably could be anticipated from examination of the site and the Report, and that such conditions will necessitate a change in the Work and/or an adjustment of the Contract Sum and/or the Contract Time, the Contracting Officer may authorize a Change Order, as stipulated in the General Conditions, to enable the necessary changes and adjustments.

PART 2 PRODUCTS

2.1 NOT USED

PART 3 PRODUCTS

3.1 NOT USED

END OF SECTION

GRTE 237343 Rehabilitate "4 Lazy F Ranch" Cabins for Housing

SECTION 01 11 00 - SUMMARY OF WORK

GENERAL

1.1 SUMMARY

A. Section includes the following:

Work Covered by Contract Documents

2. Work Phases
3. Government-Furnished Materials
4. Construction Contractor Use of Site
5. Public Use of Site
6. Occupancy Requirements for Buildings
7. Conduct of Operations
8. Work Restrictions
9. Special Construction Requirements
10. Protection of Parks and Public Lands.
11. Soils Investigation Report
12. Additional Reports

1.2 WORK COVERED BY CONTRACT DOCUMENTS

A. Project Location: The "4 Lazy F Ranch" (4LF) is located on Four Lazy F Road, approximately 1 mile north from the park headquarters building at Moose Wyoming**.**

B. The Work consists of:

1. The GRTE 237343 Rehabilitate "4 Lazy F Ranch" Cabins for Housing project (the Project) will rehabilitate a lodge, eight (8) cabins, a garage, two (2) outbuildings, and a water well structure at 4 Lazy F Ranch, a historic guest ranch which has been unoccupied since 2006. The rehabilitated structures will be utilized as seasonal park employee housing .
2. The Work includes the design, permitting, and construction of a new water distribution system, new sewer collection system including lift station, force main and gravity main, along with new buried telecommunications and underground power. Utility work will include connections to proximate electrical, water, sanitary sewer, and telecommunication services at each lodging building.
3. The work also includes rehabilitation of all structures listed above including roof replacement, foundation repair/replacement, other structural upgrades, exterior log repairs and refinishing, pest exclusion, historic window and door repairs, updating electrical, telecommunications, and plumbing systems to modern codes, installation of new fire suppression, and other preservation work. The sleeping cabins will retain their historic function as bedrooms with attached bathrooms and the Main Lodge will have two bedrooms and bathrooms which will be updated to comply with accessibility

requirements. The Main Lodge will also have a kitchen, dining room, shared laundry room, and living space to be shared by all tenants of the ranch.

4. Associated Site work will include providing a tenant parking area, (1) accessible parking space east of the Main Lodge and (1) accessible parking spot east of the garage, maintained in-kind access road, and pathways connecting buildings.

Project will be constructed under a single prime contract.

1.3 WORK PHASES

A. Phasing will not be required for this project; however, the Contractor is required to schedule Work in accordance with Section 01 32 16 – Construction Schedule, comply with Paragraph 1.11, "Work Restrictions" and the following requirements:

1. Mark construction boundaries and obtain approval of the boundary locations by the Contracting Officer (CO) prior to commencing with any construction.
2. Contractor shall maintain access and wastewater disposal, to the maximum extent as possible, to the Caretaker's Cabin.

GOVERNMENT-FURNISHED MATERIALS

In case of Government -furnished materials:

1. Government will arrange and pay for delivery of Government-furnished items according to Construction Contractor's Construction Schedule.
2. Construction Contractor is responsible for initial inspection, receiving, unloading, and handling Government-furnished items at Project site.
3. Government will inspect delivered items for damage after delivery. Construction Contractor shall be present for and assist in Government's inspection.
4. Government will arrange for replacement if Government-furnished items are damaged, defective, or missing.
5. Construction Contractor is responsible for protecting Government-furnished items from damage during storage and handling, including damage from exposure to the elements.
6. If Government-furnished items are damaged as a result of Construction Contractor's operations, Construction Contractor shall repair or replace them.
7. Construction Contractor shall install and otherwise incorporate Government-furnished items into Work.

B. Government-Furnished Materials:

1. All window sashes and wood screens will be removed and restored by the Government. Completely restored window sashes and wood screens including wood repairs, glazing and glazing putty repairs, and all finishing will be provided to the Construction Contractor for installation. Construction Contractor shall arrange for transportation of the restored window sashes and wood screens from Park Headquarters in Moose to the project site. However, window frame and casing restoration will be performed by the Construction Contractor.

2. Government will provide and install all native seed. Construction Contractor shall prepare site and install seed.

3. 3. Government will provide site conduits and handholes for the General Contractor to install.

4. Government will provide and install communications service and fiber cabling, inside conduit installed by General Contractor.

5. Government will provide and install fiber panels at Main Lodge and cabins.

1.5 CONSTRUCTION CONTRACTOR USE OF SITE

General: Construction Contractor shall have limited use of site for construction operations. Limit use of premises to work in areas indicated on drawings. Do not disturb portions of Project site beyond areas in which the Work is indicated.

1. Limits: Confine constructions operations to the areas within the "limits of disturbance" boundary as shown on the drawings.

2. Vegetation beyond the estimated limits of disturbance shall not be damaged or removed without prior approval via the project documents or by park vegetation management staff.

3. With-in the limits of disturbance boundary, limit ground disturbance to the smallest area possible to reduce disturbance to soil and native plants and reduce the potential for the introduction and/or spread of non-native, invasive plant species.

B. Storage of Materials: Confine storage of materials to the staging areas shown on Landscape drawings.

C. Parking: Confine parking to the staging areas shown on Landscape drawings.

D. Stockpiling: Confine stockpiling to the staging areas shown on Landscape drawings.

E. Preservation of Natural Features:

1. Prevent damage to natural surroundings. Restore damaged areas, repairing or replacing damaged trees and plants, at no additional expense to the Government.
2. Provide temporary barriers to protect existing trees and plants and root zones per Landscape drawings.
3. Do not remove, damage, or destroy trees or other plants without prior approval. Consult with Contracting Officer (CO) and remove agreed on roots and branches that interfere with construction.
4. Do not fasten ropes, cables, or guys to existing trees.
5. Carefully supervise excavating, grading, filling, and other construction operations near trees to prevent damage.
6. Comply with requirements of Section 01 12 00 - Natural Features Protection.

Driveways and Entrances: Keep driveways and entrances serving premises clear and available to Government employees, and emergency vehicles at all times. Do not use for parking or storage of materials.

1. Schedule deliveries to minimize use of driveways and entrances.
2. Schedule deliveries to minimize space and time requirements for storage of materials and equipment on-site.

Construction Camp: Establishment of a camp within park will not be permitted.

Hauling Restrictions: Comply with legal load restrictions in hauling of materials. Load restrictions on park roads are identical to state load restrictions with such additional regulations as may be imposed by the Park Superintendent. Information regarding rules and regulations for vehicular traffic on park roads may be obtained from the Office of the Park Superintendent. A special permit will not relieve Construction Contractor of liability for damage which may result from moving of equipment.

1.6 PUBLIC USE OF SITE

Construction Contractor shall conduct his operations to ensure the least inconvenience to public. Road closures may be permitted, when required, upon specific approval of Contracting Officer for a maximum of 2 days per closure.

B. All traffic on Park Roads must be permitted to pass. If excavation, backfill, or paving operations require a temporary closure of a street, the Contractor shall route traffic on a clearly signed and marked detour. The Contractor will be required to place and maintain signs for all detours.

C. Contractor shall limit the number of temporary closures to Menor's Ferry. Closures will be limited to a maximum of 2 days per closure as approved by the CO no less than 7 days in advance.

D. Comply with requirements specified in Section 01 55 26 - Traffic Control.

E. Implement measures to reduce adverse effects of construction on visitors' health and safety.

OCCUPANCY REQUIRMENTS FOR BUILDINGS

A. Existing Buildings

Partial Government Occupancy: Government will occupy structures located adjacent to the Project site including the 4 Lazy F Caretaker Cabin, Barn and Barn Cluster buildings, during entire construction period, with exception of areas under construction. Cooperate with Government during construction operations to minimize conflicts and facilitate Government usage. Perform Work so as not to interfere with Government's operations. Maintain existing exits, unless otherwise indicated.

a. Maintain access to existing walkways, corridors, and other adjacent occupied or used facilities. Do not close or obstruct walkways, corridors, or other occupied or used facilities without written permission from Contracting Officer.

B. New Buildings

Government Occupancy of Completed Areas of Construction: Government reserves right to occupy and to place and install equipment in completed areas of building, provided such occupancy does not interfere with completion of the Work. Such placement of equipment and partial occupancy shall not constitute acceptance of the entire project.

Contracting Officer will prepare a Beneficial Occupancy letter for each specific portion of the Work to be occupied before Government occupancy.

b. Obtain a Certificate of Occupancy from National Park Service (NPS) before Government occupancy.
c. Before partial Government occupancy, mechanical and electrical systems shall be fully operational, and required tests and inspections shall be successfully completed. On occupancy, Government will operate and maintain mechanical and electrical systems serving occupied portions of building.
d. On occupancy, Government will assume responsibility for maintenance and custodial service for occupied portions of building.

1.8 CONDUCT OF OPERATIONS

A. Construction Contractor shall conduct his operations in conformance with rules and regulations promulgated by the Secretary of the Interior for the National Park Service, and applicable park rules and regulations prescribed by Park Superintendent.

Work on Saturdays, Sundays, Federal holidays or at night may not be performed unless stated in the Work Restrictions below or without prior consent from the Contracting Officer. Submit requests 2 business/calendar days in advance of the work to the Contracting Officer for approval.

C. No signs or advertisements (except those specified herein) shall be displayed on the construction site or within the park unless approved by the Contracting Officer.

1.9 WORK RESTRICTIONS

On-Site Work Hours: On-Site Work Hours are restricted per section 01 35 13.05 Environmental Protection . The following table summarizes those restrictions:

WILDLIFE-RELATED WORK SCHEDULE

	MAY	JUNE	JULY	AUGUST	SEPTEMBER	OCTOBER	NOVEMBER
HOURS OF OPERATION	8AM-6pm *(sage grouse)*		30 minutes after sunrise to 30 minutes prior to sunset		8AM-6pm *(elk)*		
NOTE:	May 15 - June 30: avoid construction / disturbance activities *(elk parturition)*. Consult with Ungulate Biologist if work is desired in this timeframe.						
NOTE:	May 15 - June 30: avoid construction / disturbance activities in the designated elk parturition range, which includes the utility corridor, the 3-Bay Garage, the Nurse's Cabin, the River Cabin, and the staging area by the barn. Consult with Contracting Officer's if work is desired in these areas during this timeframe.".						
NOTE:	February 1 - August 15: potential seasonal closures *(bald eagle nesting)*						
NOTE:	Park will conduct on-going surveys of bat habitation. If roosting bats, especially maternal colonies, are found in buildings, work on those structures could be delayed until after the bats have been effectively excluded (as for non-maternal roosts), and/or the roosting period is over (as for maternal colonies).						

B. Operation of Existing Facilities: The Contractor is not authorized, except under emergency conditions, to operate any Park-owned water valves or equipment.

C. Existing Utilities

1. Existing Utilities: Notify Contracting Officer and utility companies of proposed locations and times for excavation.
2. Construction Contractor shall be responsible for locating and preventing damage to known utilities. If damage occurs, repair utility at no additional expense to the Government.
3. If damage occurs to an unknown utility, repair utility. An equitable adjustment will be made in accordance with the Changes clause of the contract.

Construction Limits: “Limit of Disturbance” lines are provided on the drawings to describe construction boundaries. Do not operate equipment outside the construction boundaries as indicated on the drawings.

Removal of specific trees and bushes are noted on the drawings. Any unauthorized disturbance may result in Grand Teton National Park issuing the appropriate level of law enforcement to the responsible person, or Contractor.

1. All staging and stockpiling areas to be returned to pre-construction conditions following construction as described in drawings and technical specifications.

E. GRTE Mitigation Restrictions

1. Mitigations and best management practices are described in section 01 35 13.05 Environmental Protection and elsewhere in the project manual. Where conflicts between the standard mitigations and this contract occur, this contract supersedes requirements of the standard mitigations.

F. Existing Utility Interruptions: Do not interrupt utilities serving facilities occupied by Government or others unless permitted under the following conditions and then only after arranging to provide temporary utility services according to requirements indicated:

Notify Contracting Officer not less than two business days in advance of proposed utility interruptions.

2. Do not proceed with utility interruptions without Contracting Officer's written permission.
3. Hours and length of Utility Shutdowns: As required by Contracting Officer in advance of proposed utility interruption.
4. Do not conduct planned utility interruptions outside the days of Monday through Wednesday.

G. Nonsmoking Building/Tobacco Use/Vaping: Smoking is not permitted within building or within 25 feet of entrances, operable windows, or outdoor air intakes. Tobacco use and vaping is restricted to inside personal vehicles only.

1.10 SPECIAL CONSTRUCTION REQUIREMENTS

Project Management and Communication Software: Software administered by NPS shall be used to manage communication and document sharing during construction.

See Section 01 31 00 "Project Management and Coordination" for requirements on using NPS project management and communication software.

B. Access to the site is prohibited from December 15-April 1 in any year. Costs associated with plowing snow to gain access to the project site outside of typical park plowing times shall be borne by the Contractor.

C. Pre-construction Video: Prior to execution of Work, Contractor shall submit a video recording of the work site in accordance with Section 01 32 33 - Photographic and Video Documentation.

D. Contractor shall be responsible for repairing utilities damaged during construction. Repairs shall be made within 24 hours of damage.

1.11 PROTECTION OF PARKS AND PUBLIC LANDS

A. Refer to Section 01 35 13.05 – Environmental Protection for additional requirements.

B. Equipment and Vehicle Cleaning and inspection is required for any vehicle and/or equipment that enters the Park and/or is staged in association with this project. Refer to Section 01 35 13.05 – Environmental Protection for specific requirements.

C. Rock, sand, aggregate, gravel, soil or other natural material brought to the project site must be from a Park approved or county weed district approved source or treated to eradicate exotic species seed. Refer to Section 01 35 13.05 – Environmental Protection for specific requirements.

D. Do not operate or park any equipment off the paved or established roadways except in approved areas. Do not operate or store vehicles and equipment leaking oil, gas, or anti- freeze within the Park boundaries. Do not drain oil, hydraulic fluids, anti-freeze, or other chemicals onto the ground within Park boundaries. Completely contain diesel fuel at the work site. Notify CO of any spill and clean up spill immediately. Contractor is required to have a Hazardous Spill Plan in accordance with Section 01 50 00 - Temporary Facilities and Controls.

E. Overnight Camping: Sleeping in vehicles or camping within the Park Boundaries will not be allowed.

F. Explosives: Do not use explosive materials within the Park boundaries.

G. Wildlife Protection: Project requirements related to wildlife protection are described in Section 01 35 13.05 – Environmental Protection. Required protective measures safeguard all animal species living in the park including, migratory birds, special status species, sage grouse, wolves, grizzly bears, bats, and elk. Wildlife protection measures will effect many aspects of the project including construction scheduling, waste management, safety, and temporary facilities and controls. Please refer to Section 01 35 13.05 – Environmental Protection for specific requirements.

1. All on-site construction personnel will be required to attend and complete Wildlife and Bear briefing sessions provided by Park personnel, entitled “Working Bear Country”, reference 36 CFR., The Human-Bear Management Plan for GRTE; and the JDR (1989), Interagency Grizzly Bear Committee; 1986 Interagency Grizzly Bear Guidelines, Missoula, Montana. Construction personnel will receive training with regards to the NPS Bear Management Plan safety protocols, sanitation standards, and food storage regulations. Notify CO two (2) weeks before starting Work so the briefing session can be coordinated

1.12 SOILS INVESTIGATION REPORT

The report Geotechnical-Engineering Report GRTE – 4 Lazy F Rehab | Teton County, Wyoming | Grand Teton National Park , August 19, 2025, prepared by Jorgensen Geotechnical, LLC is available, with this package.

B. In case of conflict between report and drawings or specifications, the drawings and specifications govern.

1.13 ADDITIONAL REPORTS

The Contracting Officer will supply the Asbestos Survey Report, July 8, 2025, completed by Northern Industrial Hygiene, Inc. of Billings, Montana, to the Contractor.

B. The Contracting Officer will supply Indoor Air Sampling for *Histoplasma Capulatum* Report, dated July 8, 2025, completed by Northern Industrial Hygiene, Inc. of Billings, Montana, to the Contractor.

C. The Contracting Officer will supply the Lead-Based Paint Survey Report, dated July 8, 2025. completed by Northern Industrial Hygiene, Inc. of Billings, Montana, to the Contractor.

D. In case of conflict between report and drawings or specifications, drawings and specifications govern.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 11 00

SECTION 01 21 00 - ALLOWANCES

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Cash allowances.

B. Contingency allowance.

1.2 SELECTION AND PURCHASE

A. At the earliest practical date after award of the Contract, advise Architect of the date when final selection and purchase of each product or system described by an allowance must be completed to avoid delaying the Work.

B. At Architect's request, obtain proposals for each allowance for use in making final selections. Include recommendations that are relevant to performing the Work.

C. Purchase products and systems selected by Architect from the designated supplier.

1.3 ACTION SUBMITTALS

A. Submit proposals for purchase of products or systems included in allowances, in the form specified for Change Orders.

1.4 INFORMATIONAL SUBMITTALS

A. Submit invoices or delivery slips to show actual quantities of materials delivered to the site for use in fulfillment of each allowance.

B. Submit time sheets and other documentation to show labor time and cost for installation of allowance items that include installation as part of the allowance.

C. Coordinate and process submittals for allowance items in same manner as for other portions of the Work.

1.5 COORDINATION

A. Coordinate allowance items with other portions of the Work. Furnish templates as required to coordinate installation.

1.6 CASH ALLOWANCES

A. Costs Included in Cash Allowances: Cost of product to Contractor or subcontractor, less applicable trade discounts, less cost of delivery to site , less applicable taxes .

B. Costs Not Included in Cash Allowances: Product delivery to site and handling at the site, including unloading, uncrating, and storage; protection of products from elements and from damage; and labor for installation and finishing.

C. Contracting Officer Responsibilities:

1. Consult with Contractor for consideration and selection of products, suppliers , and installers.

2. Select products in consultation with Contracting Officer and transmit decision to Contractor.

3. Prepare Change Order.

D. Contractor Responsibilities:

1. Assist Contracting Officer in selection of products, suppliers , and installers.

2. Obtain proposals from suppliers and installers and offer recommendations.

3. On notification of which products have been selected, execute purchase agreement with designated supplier and installer.

4. Arrange for and process shop drawings, product data, and samples. Arrange for delivery.

5. Promptly inspect products upon delivery for completeness, damage, and defects. Submit claims for transportation damage.

E. Differences in costs will be adjusted by Change Order.

1.7 CONTINGENCY ALLOWANCE

A. Contractor's costs for products, delivery, installation, labor, insurance, payroll, taxes, bonding, equipment rental, overhead and profit will be included in Change Orders authorizing expenditure of funds from this Contingency Allowance.

B. Funds will be drawn from the Contingency Allowance only by Change Order.

C. At closeout of Contract, funds remaining in Contingency Allowance will be credited to Owner by Change Order.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION

3.1 EXAMINATION

A. Examine products covered by an allowance promptly on delivery for damage or defects. Return damaged or defective products to manufacturer for replacement.

3.2 PREPARATION

A. Coordinate materials and their installation for each allowance with related materials and installations to ensure that each allowance item is completely integrated and interfaced with related work.

3.3 SCHEDULE OF ALLOWANCES

A. Allowance No. 1: Quantity Allowance: Include 2000 cu. yd. (1529 cu. m) of unsatisfactory soil excavation and disposal off-site and replacement with satisfactory soil material from off-site, as specified in Section 31 20 00 - Site Earthwork.

1. Coordinate quantity allowance adjustment with unit-price requirements in Section 01 22 00 - Unit Prices.

B. Allowance No. 2: Quantity Allowance: Include 1000 cu. yd. (765 cu. m) of rock removal and replacement with satisfactory soil material, as specified in Section 31 20 00 - Site Earthwork"

1. Coordinate quantity allowance adjustment with unit-price requirements in Section 01 22 00 - Unit Prices.

C. Allowance No. 3: Lump-Sum Allowance: Include the sum of $10,000 for existing Materials Testing.

1. This allowance includes material testing cost, documentation, submital, and Contractor overhead and profit.

a. Log Wall Assembly Testing:

1) Wood Species

2) Chinking Type

3) Daubing Mixture

b. Chimney Mortar Type Testing.

D. Allowance No. 4: Contingency Allowance for existing structures project. Include a contingency allowance of $100,000.00 for use according to Owner's written instructions.

END OF SECTION

SECTION 01 22 00 - UNIT PRICES

PART 1 GENERAL

1.1 SECTION INCLUDES

A. List of unit prices, for use in preparing Bids.

B. Measurement and payment criteria applicable to Work performed under a unit price payment method.

1.2 COSTS INCLUDED

A. Unit Prices included on the Bid Form shall include full compensation for all required labor, products, tools, equipment, plant, transportation, services and incidentals; erection, application or installation of an item of the Work; overhead and profit.

1.3 PROCEDURES

A. Unit prices include all necessary material, plus cost for delivery, installation, insurance, [applicable taxes,]overhead, and profit.

B. Measurement and Payment: See individual Specification Sections for work that requires establishment of unit prices. Methods of measurement and payment for unit prices are specified in those Sections.

C. Owner reserves the right to reject Contractor's measurement of work-in-place that involves use of established unit prices and to have this work measured, at Owner's expense, by an independent surveyor acceptable to Contractor.

D. List of Unit Prices: A schedule of unit prices is included in Part 3. Specification Sections referenced in the schedule contain requirements for materials described under each unit price.

1.4 MEASUREMENT OF QUANTITIES

A. Measurement methods delineated in the individual specification sections complement the criteria of this section. In the event of conflict, the requirements of the individual specification section govern.

B. Take all measurements and compute quantities. Measurements and quantities will be verified by Contracting Officer.

C. Assist by providing necessary equipment, workers, and survey personnel as required.

D. Measurement Devices:

E. Measurement by Area: Measured by square dimension using mean length and width or radius.

F. Linear Measurement: Measured by linear dimension, at the item centerline or mean chord.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION

4.1 SCHEDULE OF UNIT PRICES

A. Unit Price No. 1 - Unsatisfactory Soil Disposal and Replacement:

1. Description: unsatisfactory soil excavation and disposal off-site and replacement with satisfactory soil material from off-site, as specified in Section 31 20 00 - Site Earthwork.

2. Unit of Measurement: cu. yd.

B. Unit Price No. 2 - Rock Removal and Replacement with Satisfactory Soil:

1. Description: Rock removal and disposal off-site and replacement with satisfactory soil material from off-site, as specified in Section 31 20 00 - Site Earthwork.

2. Unit of Measurement: cu. yd.

C. Unit Price No. 3 - Glazing Pane Replacement:

1. Description: Replace broken glazing panes according to Section 08 80 00 - Glazing

2. Unit of Measurement: Unit.

3. Quantity Allowance: Coordinate unit price with allowance adjustment requirements in Section 01 21 00 - Allowances.

D. Unit Price No. 4 - Roof Plywood Sheathing:

1. Description: Replace roof plywood sheathing according to Section 06 10 00 - Rough Carpentry.

2. Unit of Measurement: Square Foot.

3. Quantity Allowance: Coordinate unit price with allowance adjustment requirements in Section 01 21 00 - Allowances.

E. Unit Price No. 5 - Wood Log Dutchman Repair:

1. Description: Repair logs with a dutchman repair according to Section 06 03 12 - Historic Wood Repair.

2. Unit of Measurement: Linear Foot.

3. Quantity Allowance: Coordinate unit price with allowance adjustment requirements in Section 01 21 00 - Allowances.

F. Unit Price No. 6 - Full Wood Log Replacement:

1. Description: Repair logs with a dutchman repair according to Section 06 13 13 - Historic Log Treatment/Log Repair.

2. Unit of Measurement: Linear Foot.

3. Quantity Allowance: Coordinate unit price with allowance adjustment requirements in Section 01 21 00 - Allowances.

END OF SECTION

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SECTION 01 26 01 – CONTRACT MODIFICATION PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

A. This section consists of administrative and procedural requirements for contract modifications.

1.2 DEFINITIONS AND ALLOWANCES

A. Home Office Overhead: Costs incurred in support of all a contractor's projects and not attributable to a specific job. The cost for home office overhead is only allowed as a percentage of all direct work excluding profit. The following items represent allowable home office overhead costs identified in Part 31 of the Federal Acquisition Regulation (FAR):

1. Rent
2. Utilities
3. Furnishings
4. Office equipment
5. Executive and management staff not exclusively assigned to the project
6. Support, accounting, and administrative staff
7. Preparation of cost proposals, estimating, and schedule analyses connected with Modifications
8. Estimating and preconstruction services
9. Mortgage costs
10. Real estate and corporate taxes
11. Automobile maintenance and travel costs for home office personnel
12. Home office insurances i.e. structure, automotive, umbrella, flood, etc.
13. Depreciation of equipment and other assets
14. Home office supplies (paper, staples, etc.)
15. Legal services
16. Accounting and data processing
17. Professional fees/registration

B. General Conditions (Field Office Overhead): Management and administrative costs incurred on site for the designated project. Costs associated with preparation of modifications will not be allowed. Costs for these items are to be included only in the general conditions of the modification estimate. Only in the case of a contract time extension are additional general conditions included in modifications. The following items, if applicable, are considered allowable costs for calculating General Conditions:

1. Project Manager (PM), Assistant Project Manager
2. Superintendent, Assistant Superintendent
3. Quality Control, Safety Officer, Environmental Manager, etc.
4. Engineers

5. Travel, lodging, and per diem (as established by Federal Travel Regulations)
6. Scheduling
7. Field Office Trailers and associated temporary utilities
8. Field office supplies

    a. Mailing and couriers
    b. Reproduction costs
    c. Storage
    d. Phones
    e. Computers
    f. Copiers

9. Personal vehicles i.e. Superintendent Pickup trucks

C. General Requirements: Costs directly associated with the project and are necessary to perform the actual work of the modification. These costs shall be shown as direct costs in the estimate. The following items, if applicable, are considered allowable costs for calculating General Requirements:

1. Hoisting
2. Material handling
3. Temporary fencing
4. Port-a-lets
5. Trash removal, dumpsters
6. Barricades
7. Small tools
8. Safety supplies
9. Scaffolding
10. Daily cleaning
11. Traffic control
12. Temporary signage
13. Temporary heating and power

D. Personnel Costs: Costs included in the modification must only be for General Conditions staff and workers present and working on the project site. Modification costs for salaried workers are only allowed within the structure of a 40-hour week, and no overtime or holiday pay will be allowed.

1. Worker Hourly Rates are costs directly associated with the individual worker and consist of the following:

    a. Base Rate: The hourly rate paid directly to the worker
    b. Labor Burden: Employer payments of all applicable burdens include insurance and taxes the business must pay on behalf of the worker to government entities and educational forums, such as:

        1) Social Security
        2) Medicare
        3) Workers Compensation – Policy and company calculation to be made available.
        4) Federal Unemployment Tax Act (FUTA) - Cap Rate and percentage to be proportionally allocated over one year.

5) State Unemployment Tax Act (SUTA) - Cap Rate and percentage to be proportionally allocated over one year.
6) Union agreement costs - Other costs required under an enforceable collective bargaining agreement.

c. Fringe Benefits: Various non-wage compensations provided to employees such as:

1) Health Care Insurance Premiums
2) Cell Phone
3) Clothing
4) 401K and Pensions
5) Vehicle allowances
6) Gas allowance
7) Life insurance premiums
8) Disability insurance
9) Other Fringe Benefits required under an enforceable collective bargaining agreement

E. Bonuses or Deferred Compensation: No Bonus or Deferred Compensation will be allowed within any components of pricing including Home Office Overhead, General Conditions, General Requirements, Hourly Worker Rates, or the direct costs of work.

F. General Liability Insurance: An insurance policy that protects Contractor from claims resulting from bodily injury or property damage to a third party. Include as a separate line item within all modification proposals and provide a current insurance quote upon request.

G. Consent of Surety:
1. When any contract is modified, the Contractor shall provide the consent of surety if-

a. An additional bond is obtained from other than the original surety;
b. No additional bond is required and

1) The modification is for new work beyond the scope of the original contract; or
2) The modification does not change the contract scope but changes the contract price (upward or downward) by more than 25 percent or $50,000; or

c. Consent of surety is required for a novation agreement.

2. When a contract for which performance or payment is secured by any of the types of security listed in FAR 28.204 is modified as described in paragraph (a) of this subsection, no consent of surety is required.
3. Agencies shall use Standard Form 1414, Consent of Surety, for all types of contracts.

H. Builder's Risk Insurance: Covers the contractor's loss due to fire, high winds, or other natural forces. Not reimbursed by the National Park Service (NPS) and shall not be included in modification proposals.

1.3 MODIFICATION PROPOSAL PRICING REQUIREMENTS

A. General:

1. Proposal be received in the format and within the time frame specified in the Request for Proposal (RFP) letter. Costs or delays resulting from failure of contractor to submit within the time frame specified will not be compensable.
2. The proposal shall be detailed with itemized lists of equipment, materials, labor, production rates, overhead, profit, and bond markup for each item. Labor costs must be itemized by craft and hourly rate, including Fringe Benefits and Labor Burden. If the costs of Fringe Benefits and Labor Burden are not itemized, it is assumed they are included in the hourly rate shown, or the contractor is not requesting reimbursement. Contractor may utilize the government provided Contractor Estimate Form, or their own form, provided that it contains the same information and level of detail as the Government's form.
3. Requests for extensions of contract time as a result of change must be justified with a Time Impact Analysis (TIA). Refer to Section 01 32 16 "Construction Schedule", for time impact analysis requirements. TIA and associated costs shall be received with the proposal by the date shown within the Request for Proposal letter. Contractor's failure to submit within the specified time frame will be construed as the Contractor waiving right for additional time and no time extension will be allowed.
4. All supporting documentation used to justify the proposed modification will be made available to the Contracting Officer (CO) upon request.
5. Contractor shall review and approve all subcontractor/supplier pricing in detail for proper format, scope, production rates, and pricing prior to submission to NPS. All delay costs associated with not reviewing and approving subcontractor/supplier pricing will be borne by the Contractor.
6. All pricing and production rates within the estimate must be based on fair and reasonable pricing and cannot include built-in contingency.

B. Labor:

1. Contractor shall estimate the cost of labor by itemizing each craft involved, indicating worker hourly rate (base rate + labor burden + fringe benefits) for each and itemizing hours required for each craft directly engaged in modification work. Any work proposed requiring overtime work or premium pay shall be itemized separately. Rates shall be in accordance with the Davis-Bacon Act as incorporated herein. Labor Burden may include payroll taxes, Social Security, unemployment insurance, workers compensation insurance, Federal Insurance Contributions Act (FICA), FUTA, and other direct costs resulting from Federal, State or local laws.
2. Itemize labor costs for equipment operators separate from equipment costs.
3. Labor cost for foremen shall only be costs for related work required for the modification.

C. Materials:

1. Estimated cost for materials shall include quotes from multiple sources. Material prices shall include applicable fees and credits, including but not limited to, sales tax, freight and delivery charges, and tax rebates.
2. No markup shall be applied to any material provided by NPS.

D. Equipment:

1. The equipment used for the project must be appropriately sized for work being performed.
2. Do not include costs for "miscellaneous tools and equipment", in your proposal for a replacement value of $500 or less. Costs shown in excess of $500 shall be broken out separately.
3. Regardless of ownership, rates to be used in determining equipment rental costs shall be the lowest cost from one of the following sources:

    a. United States (U.S.) Army Corps of Engineers, Ownership and Operating Expense Schedule (use latest edition and applicable region)
    b. Construction Blue Book
    c. Local equipment rental rates, documented by actual invoice charges, or itemized vendor quotes.

4. Estimated equipment rates shall include operating costs of all fuel, oil, lubrication, supplies, small tools, necessary attachments, ground engaging components, tires and tracks, routine repairs and maintenance (cost of major repair and overhaul is not allowed per Federal Acquisition Regulation (FAR) 31.105(d)(2)), depreciation, storage, insurance, and all incidentals. Mobilization, if applicable, may be included for equipment solely used on the modification work but must be listed separately.
5. Estimate full rate for equipment only for duration that equipment will be utilized to accomplish work of the modification.
6. Standby unit rates used in accordance with paragraph 1.3, D, 2, above. If the U.S. Army Corp of Engineers is utilized, then their standby rates prevail. If Bluebook or local equipment pricing is accepted, then 1/2 of equipment costs minus any operating costs, major repair and overhaul will be accepted.
7. If equipment is in standby mode due solely to a documented NPS delay, established standby rate shall apply from the first day of the delay.
8. Equipment not used and on the job site for up to five consecutive days may be classified at standby rates, provided the equipment is or has been used solely to perform work on the modification and will be necessary to complete additional modification work. Equipment still on the jobsite but not in use after five consecutive days will not be considered in the modification pricing.
9. Requests for compensation for equipment stand by time must be justified, documented and itemized separately.
10. The estimated timeframe (daily, weekly, monthly) for use of the equipment must reflect the lowest cost to the Government.

E. Establishment and Application of Overhead and Profit Percentages:

Home Office Overhead and Profit (OH&P) shall be applied to direct costs only. Profit shall not be applied to overhead amounts; and overhead shall not be applied to profit. Home office overhead shall contain only allowable, allocable, and reasonable costs per the contract documents and FAR Part 31. Profit percentages are based on risk factors found in FAR Part 31which have been applied to the specific type of work included in this project. Negotiated rates shall not exceed the following percentages for OH&P for contractor self-performed work:

Overhead……10%
Profit……**6.10**%

2. Total aggregate limit of markup (OH&P) for Contractor and Subcontractors on modification work shall not exceed 25%. The NPS will not be responsible for allocation of percentages between contractor and subcontractors at any tier.
3. If Contractors form a partnership, partnership may only receive home office overhead and profit in same amount as an individual Contractor (refer to paragraph 1.3,E,1 above). It is the responsibility of the partners to decide on division of revenue.
4. Combined Increases and Decreases: On proposals involving both increases and decreases in the Contract Price, overhead and profit mark-ups are required on net increases and deducted on net decreases.
5. At no time can profit be calculated on Overhead or itself, it must be calculated on direct costs of work only.

PART 2 - PRODUCTS

PART 3 - EXECUTION

END OF SECTION 012601

SECTION 01 27 00 – DEFINITION OF CONTRACT LINE ITEMS

GENERAL

1.1 SUMMARY

A. Section explains in general, what is and is not included in a contract line item, and limits or cut-off points where one item ends and another begins.

B. If no contract line item exists for a portion of work, include costs in a related item.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 LIST OF CONTRACT LINE ITEMS

A. Base Contract Line Item No. 01A: Sitework - Sewer System
   1. This bid item consists of all Work related to construction of the sewer collection system from within 5 feet of each structure to be served, to connection with the existing Moose Sewer System, as noted in the drawings and technical specifications.
   2. Measurement for payment will be completed installed sewer system for 4LF complete.
   3. Payment will be made at the contract lump sum price bid for the project complete.

B. Base Contract Line Item No. 01B: Sitework -Water System
   1. This bid item consists of all Work related to construction of the water system from the connection of the water main at 4LF, to within 5 feet of each structure to be served, as noted on the drawings and echnical specifications.
   2. Measurement for payment will be completed installed water system for 4LF complete.
   3. Payment will be made at the contract lump sum price bid for the project complete.

C. Base Contract Line Item No. 01C: Sitework -Buried Power
   1. This bid item consists of all Work related to construction of buried power to the new lift station from a supply to be coordinated with local electrical utility company (LVE), direct bury conductor an existing meter and disconnect, to within 5 feet of each structure fo future connection, as noted on the drawings andtechnical specifications.
   2. Measurement for payment will be completed installed buried power for 4LF complete.
   3. Payment will be made at the contract lump sum price bid for the project complete.

D. Base Contract Line Item No. 01D: Sitework -Fiber Optic Communications
   1. This bid item consists of all Work related to construction of buried fiber optic communications from a vault near the Caretaker's Cabin to the new lift station, and

installing conduit for future use to within 5 feet of each structure to be served, as noted on the drawings and technical specifications.

2. Measurement for payment will be completed installed buried fiber optic communications for 4LF complete.

E. Option Contract Line Item Number 002: (B-(HS) 399 4 Lazy F Dude Ranch Main Lodge.

1. This item consists of Rehabilitation of the Main Lodge and associated site work. Building rehabilitation includes foundation and other structural improvements, exterior wall rehabilitation, roof replacement, interior rehabilitation and reconfiguration, electrical and mechanical system replacement, and fire sprinkler installation. Sitework includes excavation, connection to utilities within 5’-0” of the building, and regrading and landscaping associated with the building.
2. Measurement for payment will be completed all rehabilitation work at the building.
3. Payment will be made at the contract lump sum price for the project complete.

F. Option Contract Line Item Number 03: B-(HS) 404 4 Lazy F Dude Ranch George's Cabin

1. This item consists of Rehabilitation of the George's Cabin and associated site work. Building rehabilitation includes foundation and other structural improvements, exterior wall rehabilitation, roof replacement, interior rehabilitation and reconfiguration, electrical and mechanical system replacement, and fire sprinkler installation. Sitework includes excavation, connection to utilities within 5’-0” of the building, and regrading and landscaping associated with the building.
2. Measurement for payment will be completed all rehabilitation work at the building.
3. Payment will be made at the contract lump sum price for the project complete.

G. Option Contract Line Item Number 04: B-(HS) 409 4 Lazy F Dude Ranch Mrs Oliver’s Cabin

1. This item consists of Rehabilitation of the Mrs Oliver’s Cabin and associated site work. Building rehabilitation includes foundation and other structural improvements, exterior wall rehabilitation, roof replacement, interior rehabilitation and reconfiguration, electrical and mechanical system replacement, and fire sprinkler installation. Sitework includes excavation, connection to utilities within 5’-0” of the building, and regrading and landscaping associated with the building.
2. Measurement for payment will be completed all rehabilitation work at the building.
3. Payment will be made at the contract lump sum price for the project complete.

H. Option Contract Line Item Number 05: B-(HS) 408 4 Lazy F Dude Ranch Middle Cabin

1. This item consists of Rehabilitation of the Middle Cabin and associated site work. Building rehabilitation includes foundation and other structural improvements, exterior wall rehabilitation, roof replacement, interior rehabilitation and reconfiguration, electrical and mechanical system replacement, and fire sprinkler installation. Sitework includes excavation, connection to utilities within 5’-0” of the building, and regrading and landscaping or associated with the building.
2. Measurement for payment will be completed all rehabilitation work at the building.
3. Payment will be made at the contract lump sum price for the project complete.

I. Option Contract Line Item Number 06: B-(HS) 407 4 Lazy F Dude Ranch End Cabin

1. This item consists of Rehabilitation of the End Cabin and associated site work. Building rehabilitation includes foundation and other structural improvements, exterior wall rehabilitation, roof replacement, interior rehabilitation and reconfiguration, electrical and mechanical system replacement, and fire sprinkler installation. Sitework includes excavation, connection to utilities within 5’-0” of the building, and regrading and landscaping associated with the building.
2. Measurement for payment will be completed all rehabilitation work at the building.
3. Payment will be made at the contract lump sum price for the project complete.

J. Option Contract Line Item Number 07: B-(HS) 401-402 4 Lazy F Dude Ranch Nurse's Cabin

1. This item consists of Rehabilitation of the Nurse's Cabin and associated site work. Building rehabilitation includes foundation and other structural improvements, exterior wall construction and rehabilitation, roof replacement, interior rehabilitation and reconfiguration, electrical and mechanical system replacement, and fire sprinkler installation. Sitework includes excavation, connection to utilities within 5’-0” of the building, and regrading and landscaping associated with the building.
2. Measurement for payment will be completed all rehabilitation work at the building.
3. Payment will be made at the contract lump sum price for the project complete.

K. Option Contract Line Item Number 08: B-(HS) 400 4 Lazy F Dude Ranch (River) Cabin

1. This item consists of Rehabilitation of the River Cabin and associated site work. Building rehabilitation includes structural improvements, exterior wall rehabilitation, roof replacement, interior rehabilitation and reconfiguration, electrical and mechanical system replacement, and fire sprinkler installation. Sitework includes, connection to utilities within 5’-0” of the building, and regrading and landscaping associated with the building.
2. Measurement for payment will be completed all rehabilitation work at the building.
3. Payment will be made at the contract lump sum price for the project complete.

L. Option Contract Line Item Number 09: B-(HS) 410 4 Lazy F Dude Ranch Harry’s Cabin

1. This item consists of Rehabilitation of the Harry’s Cabin and associated site work. Building rehabilitation includes foundation and other structural improvements, exterior wall rehabilitation, roof replacement, interior rehabilitation and reconfiguration, electrical and mechanical system replacement, and fire sprinkler installation. Sitework includes excavation, connection to utilities within 5’-0” of the building, and regrading and landscaping associated with the building.
2. Measurement for payment will be completed all rehabilitation work at the building.
3. Payment will be made at the contract lump sum price for the project complete.

M. Option Contract Line Item Number 10: B-(HS) 411 4 Lazy F Dude Ranch New Cabin

1. This item consists of Rehabilitation of the New Cabin and associated site work. Building rehabilitation includes structural improvements, exterior wall rehabilitation, roof replacement, interior rehabilitation and reconfiguration, electrical and mechanical system replacement, and fire sprinkler installation. Sitework includes connection to utilities within 5’-0” of the building, and regrading and landscaping associated with the building.
2. Measurement for payment will be no separate measurement will be made.
3. Payment will be made at the contract lump sum price for the project complete.

N. Option Contract Line Item Number 11: Sitework - Parking Lot

1. This item consists of construction of the new parking lot and associated site work .
2. Measurement for payment will be completed all rehabilitation work at the building.
3. Payment will be made at the contract lump sum price for the project complete.

O. Option Contract Line Item Number 12: Sitework - Roadway and Staging Area Restoration

1. This item consists of restoration of the existing roadway and staging area.
2. Measurement for payment will be completed all rehabilitation work at the building.
3. Payment will be made at the contract lump sum price for the project complete.

P. Option Contract Line Item Number 13: B-(HS) 406 4 Lazy F Dude Ranch Ice House

1. This item consists of Rehabilitation of the Ice House and associated site work. Building rehabilitation includes structural improvements, exterior wall rehabilitation, roof replacement, electrical and interior rehabilitation. Sitework includes regrading and landscaping associated with the building.
2. Measurement for payment will be completed all rehabilitation work at the building.
3. Payment will be made at the contract lump sum price for the project complete.

Q. Option Contract Line Item Number 14: B-(HS) 405 4 Lazy F Dude Ranch Mouse House

1. This item consists of Rehabilitation of the Mouse House and associated site work. Building rehabilitation includes structural improvements, exterior wall rehabilitation, roof replacement, electrical and interior rehabilitation. Sitework includes regrading and landscaping associated with the building.
2. Measurement for payment will be completed all rehabilitation work at the building.
3. Payment will be made at the contract lump sum price for the project complete.

R. Option Contract Line Item Number 15: B-(HS) 403 3-Bay Garage 4 Lazy F

1. This item consists of Rehabilitation of the 3-bay Garage and associated site work. Building rehabilitation includes foundation and other structural improvements, exterior wall rehabilitation, roof and roof structure replacement, and interior rehabilitation and reconfiguration. Sitework includes excavation and regrading and landscaping associated with the building.
2. Measurement for payment will be completed all rehabilitation work at the building.
3. Payment will be made at the contract lump sum price for the project complete.

S. Option Contract Line Item Number 16: B-(HS) 416 Well House 4 Lazy F

1. This item consists of Reconstruction of the Well House with new materials to match original materials.
2. Measurement for payment will be completed all rehabilitation work at the building.
3. Payment will be made at the contract lump sum price for the project complete.

END OF SECTION 01 27 00

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SECTION 01 31 00 - PROJECT MANAGEMENT AND COORDINATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative provisions for coordinating construction operations on Project including, but not limited to, the following:

Definitions

2. Construction Coordination
3. Submittals
4. Coordination Drawings
5. Requests for Information (RFIs)
6. NPS Project Management and Communication Software
7. Project Meetings
8. Environmental Coordination
9. Permits

B. Related Requirements:

Section 01 32 16 "Construction Schedule" for preparing and submitting Construction Contractor's construction schedule.

2. Section 01 73 40 "Execution" for procedures for coordinating general installation and field-engineering services, including establishment of benchmarks and control points.
3. Section 01 77 00 "Closeout Procedures" for coordinating closeout of the Contract.

1.2 DEFINITIONS

Agency with Jurisdiction

B. Construction Permits – Contractor Provided

C. Government Furnished Permits

1.3 CONSTRUCTION COORDINATION

Coordination: Coordinate construction operations included in different Sections of Specifications to ensure efficient and orderly installation of each part of the Work. Coordinate construction operations, included in different Sections, which depend on each other for proper installation, connection, and operation.

1. Schedule construction operations in sequence required to obtain best results where installation of one part of the Work depends on installation of other components, before or after its own installation.

2. Coordinate installation of different components with other Construction Contractors to ensure maximum accessibility for required maintenance, service, and repair.
3. Make provisions to accommodate items scheduled for later installation.
4. Where availability of space is limited, coordinate installation of different components to ensure maximum performance and accessibility for required maintenance, service, and repair of components, including mechanical and electrical.
5. Properly plan construction operations to include permit requirements. Allow enough time to execute permit provisions to maintain work schedule, site visits, inspections, and reporting deadlines.

B. Prepare memoranda for distribution to each party involved, outlining special procedures required for coordination. Include such items as required notices, reports, and list of attendees at meetings.

C. Coordinate activities of subcontractors and Park Staff to minimize disruption of normal Park activities. Inform construction workers and supervisors about the special sensitivity of Park values, regulations, and appropriate housekeeping.
1. If the project or activity occurs within or immediately adjacent to park housing areas, coordinate with the park's Housing Program Manager (307-739-3577) at least one week before the project or activity is scheduled to begin and immediately after it is completed.
2. Notify Park Facilities Management Utility Systems staff (307-739-3366) prior to a contractor connecting to or repairing a water or sewer line in the park. Any repairs to potable water lines must be coordinated with the park, and any potable water line that loses pressure during construction will be required to be placed into Boil Order status until a sample can be collected and tested.

D. Administrative Procedures: Coordinate scheduling and timing of required administrative procedures with other construction activities and activities of other Construction Contractors to avoid conflicts and to ensure orderly progress of the Work. Such administrative activities include, but are not limited to:

1. Preparation of Construction Contractor Construction Schedule
2. Preparation of the Schedule of Values
3. Installation and removal of temporary facilities and controls
4. Delivery and processing of submittals
5. Progress meetings
6. Permit requirements
7. Pre-installation conferences
8. Project closeout activities
9. Commissioning activities

1.4 SUBMITTALS

In accordance with paragraph 1.3/A above, submit to the CO a memoranda outlining special procedures required for coordination. Include such items as required notices, reports, and list of attendees at meetings.

B. Coordination Drawings: Prepare Coordination Drawings if limited space availability necessitates maximum utilization of space for efficient installation of different components or if coordination is required for installation of products and materials fabricated by separate entities.

1. Content: Project-specific information, drawn accurately to scale. Do not base Coordination Drawings on reproductions of Contract Documents or standard printed data. Include following information, as applicable:

    a. Indicate functional and spatial relationships of components of architectural, structural, civil, mechanical, and electrical systems.
    b. Indicate dimensions shown on Contract Drawings and make specific note of dimensions that appear to be in conflict with submitted equipment and minimum clearance requirements. Provide alternate sketches to Contracting Officer (CO) for resolution of such conflicts.
    c. Indicate space requirements for routine maintenance and anticipated replacement of components during the life of the installation.
    d. Show location and size of access doors required for access to concealed dampers, valves, and controls.
    e. Indicate required installation sequences.

2. Refer to individual Sections for Coordination Drawing requirements for Work in those Sections.
3. Follow NPS DSC Electronic Drawing Standards. See page DSC Requirements section.

C. Coordination Drawing Organization: Organize coordination drawings as follows:

1. Verification Site Plans: Show all existing and new site elements including utilities and foundations. Include surveyed existing grade and floor elevations. Indicated proposed finish grade elevations at building corners, pathways, and site elements located above grade. Indicate proposed floor, top of footing, and top of foundation wall elevations.
2. Floor Plans and Reflected Ceiling Plans: Show architectural, structural, mechanical, plumbing, fire-protection, fire-alarm, and electrical elements. Show locations of visible ceiling-mounted devices relative to acoustical ceiling grid. Supplement plan drawings with section drawings where required to adequately represent the Work.

3. Review: Contracting Officer will review coordination drawings to confirm Work is being coordinated; details of coordination are Construction Contractor's responsibility. If the Contracting Officer determines coordination drawings are not prepared in scope or detail, or are otherwise deficient, the Contracting Officer will inform Construction Contractor, who shall make changes and resubmit.

Coordination Digital Data Files: Prepare coordination digital data files according to:
File Submittal Format: Submit or post coordination drawing files using Portable Document Format (PDF) file format.
Contracting Officer will furnish Construction Contractor one set of digital data files (AutoCad.dwg) of Drawings for use in preparing coordination digital data files.

    a. Contracting Officer makes no representations as to accuracy or completeness of digital data files as they relate to Drawings.
    b. Digital Data Software Program: Drawings are available in AutoCad.dwg.

E. Division 1 submittals required prior to mobilization:

1. Letter designating Project Superintendent
2. Construction Baseline Schedule

3. A Comprehensive Schedule of Values
4. Accident Prevention Plan
5. A List of Subcontractors for this project
6. Written statements from Subcontractors certifying compliance with applicable labor standard clauses.
7. Certificates of Insurance and Standard Form SF1413 for Contactor and all Subcontractors
8. Waste Management Plan
9. Hazardous Spill Plan
10. Dust Control Plan
11. Emissions Reduction Plan
12. Quality Control Plan
13. Temporary Storm Water Pollution Prevention Plan (SWPP or UPPP)
14. Contractors Commissioning Plan
15. Historic Preservation Treatment Plan (HPTP) (when applicable)
16. List of Required Construction Permits (when applicable). Include the following information for each permit:
    a. Name of Permit
    b. Agency(ies) with Jurisdiction issuing the permit
    c. Information required from the Government to complete permit application

F. All items listed must be provided to the Contracting Officer before the Pre-Construction Conference is held. If all of these documents have not been received one week prior to the scheduled Pre-Construction Conference date, the conference will be cancelled, Notice to Proceed will not be issued, and the Contracting Officer will consider other contractual remedies. Work shall not commence until written Notice to Proceed has been issued.

1.5 REQUESTS FOR INFORMATION (RFIs)

General: Immediately on discovery of the need for additional information or interpretation of Contract Documents, Construction Contractor shall prepare and submit an RFI utilizing form created on NPS project management and communication software.

1. Contracting Officer will not respond to RFIs submitted by other entities controlled by Construction Contractor.
2. Coordinate and submit RFIs in a prompt manner to avoid delays in the work.

Content of RFI: Include detailed, legible description of item needing information or interpretation and the following:

1. RFI number, numbered sequentially
2. Date
3. RFI subject
4. Specification Section number and title and related paragraphs, as appropriate.
5. Drawing number and detail references, as appropriate.
6. Field dimensions and conditions, as appropriate.
7. Construction Contractor's suggested resolution: If suggested resolution impacts Contract Time or Contract Sum, Construction Contractor shall state impact in RFI.
8. Construction Contractor's signature
9. Requested date for response

10. Attachments: Include sketches, descriptions, measurements, photos, Product Data, Shop Drawings, coordination drawings, and other information necessary to fully describe items needing interpretation.

    a. Include dimensions, thicknesses, structural grid references, and details of affected materials, assemblies, and attachments on attached sketches.

RFI Form: Complete and submit on NPS project management and communication software.

D. Contracting Officer's Action: Contracting Officer will review each RFI, determine action required, and respond. The Contracting Officer will determine the critical nature of each RFI and issue response accordingly.

1. The following are not considered to be RFIs and will receive no action:

    a. Requests for approval of submittals.
    b. Requests for approval of substitutions.
    c. Requests for approval of Construction Contractor's means and methods.
    d. Requests for coordination information already indicated in Contract Documents.
    e. Requests for adjustments in Contract Time or Contract Sum.
    f. Requests for interpretation of Architect's actions on submittals.
    g. Incomplete RFIs or inaccurately prepared RFIs.

2. Contracting Officer's action may include a request for additional information; time for response will date from time of receipt of additional information.
3. Contracting Officer's action on RFIs may result in need for a change to Contract Time or Contract Sum. All contract changes will be processed according to the terms and conditions of the contract.

1.6 PROJECT MANAGEMENT AND COMMUNICATION SOFTWARE

Use to manage project communication and document sharing during contract period on:

1. Project directory
2. Project correspondence
3. Meeting agendas and minutes
4. Contract modifications forms and logs
5. RFI form and processing
6. Task and issue management
7. Photo documentation
8. Baseline schedule, schedule updates and calendar management
9. Submittal form and processing
10. Payment coordination documentation
11. Reminder and tracking functions
12. Archiving functions
13. Notification of submittal and RFI statuses and current responsible party
14. Permits and addendums

B. Some documents are not suitable to be shared using NPS project management and communication software. Documents containing Personal Identifying Information (PII) (i.e. certified payrolls) shall not be shared and shall be coordinated with the project team as appropriate.

PROJECT MEETINGS

A. Preconstruction Conference: Before the start of construction, the Contracting Officer will arrange an on-site meeting with Construction Contractor. Meeting agenda will include the following as a minimum:

1. Roles & Responsibilities / Lines of Authority
2. Park rules and regulations
3. Jobsite Safety
4. Resolution of comments on required Division 1 documents
5. Coordination of Subcontractors
6. Labor law application
7. Modifications
8. Payments to Construction Contractor
9. Payroll reports
10. Contract time
11. Liquidated damages
12. Construction Contractor Performance Evaluation
13. Display of Hotline posters
14. Notice to proceed
15. Correspondence procedures
16. NPS project management and communication software
17. Acceptance/rejection of work
18. Progress meetings
19. Submittal procedures
20. NPS Final Accessibility Inspection
21. Environmental requirements
22. Permit requirements
23. As-constructed drawings/operation and maintenance (O&M) manuals.
24. Saturday, Sunday, holiday and night work.
25. Reference materials
26. Value engineering
27. Schedule of Values

B. Progress Meetings: Contracting Officer will schedule weekly meetings with Construction Contractor.

1. Attendees: In addition to Government Representatives, each Construction Contractor, Subcontractor, supplier, and other entity concerned with current progress or involved in planning, coordination, or performance of future activities shall be represented. Participants at meetings shall be familiar with the Project and authorized to conclude matters relating to the Work.
2. Agenda: Review and correct or approve minutes of previous progress meeting. Review other items of significance that could affect progress. Meeting agenda will include:

    a. Approval of minutes of previous meetings

b. Submittal status
c. Review of off-site fabrication and delivery schedules.
d. Requests for information (RFI) and other issues.
e. Modifications
f. Work in progress and projected.
    1) Status of required inspections (Special Inspections, Accessibility, etc.)
g. Inspections of work in progress and projected (Special inspections, Accessibility, etc.)
h. Construction Schedule update (provide updated Critical Path Method (CPM)).
i. Status of Project Record Drawings and O&M manuals.
j. Other business relating to work.
k. Permit requirements

C. Preinstallation Conferences: Conduct at Project site before each construction activity that requires coordination with other construction.

1. Attendees: Installer and representatives of manufacturers and fabricators involved in or affected by installation and its coordination or integration with other materials and installations that have preceded or will follow shall attend meetings. Advise the Contracting Officer of scheduled meeting dates.
2. Agenda: Review progress of other construction activities and preparations for particular activity under consideration, including requirements for:

    a. Contract Documents
    b. Options
    c. Related RFIs
    d. Related Change Orders
    e. Purchases
    f. Deliveries
    g. Submittals
    h. Review of mockups
    i. Possible conflicts
    j. Compatibility requirements
    k. Time schedules
    l. Weather limitations
    m. Manufacturer's written instructions
    n. Warranty requirements
    o. Compatibility of materials
    p. Acceptability of substrates
    q. Temporary facilities and controls
    r. Space and access limitations
    s. Regulations of agency(ies) with jurisdiction
    t. Testing and inspecting requirements
    u. Installation procedures
    v. Coordination with other work
    w. Required performance results
    x. Protection of adjacent work
    y. Protection of construction and personnel

3. Record significant conference discussions, agreements, and disagreements, including required corrective measures and actions.

4. Reporting: Distribute minutes of meetings to each party present and to other parties requiring information.
5. Do not proceed with installation if the conference cannot be successfully concluded. Initiate whatever actions are necessary to resolve impediments to the performance of Work and reconvene the conference at the earliest feasible date.

1.8 ENVIRONMENTAL COORDINATION

A. Construction Contractor's Environmental Manager: Designate on-site party responsible for overseeing Construction Contractor's conformance to environmental goals for project and implementing procedures for environmental protection.

1. Qualifications: Minimum 3 years Construction experience on projects of similar size and scope; with environmental procedures similar to this project; familiarity with Environmental Management Systems (EMSs) such as International Organization for Standardization (ISO) 14001; familiar with environmental regulations applicable to construction operations.
2. Responsibilities: Responsibilities shall include:

Compliance with applicable Federal, State, and local environmental regulations, including maintaining required documentation.

b. Implementation of Waste Management Plan (WMP).
c. Implementation of Indoor Air Quality (IAQ) Management Plan.
d. Implementation of Storm Water Pollution Prevention Plan (SWPPP).
e. Implementation of the Hazardous Spill Plan.
f. Implementation of the Dust Control Plan.
g. Implementation of the Emissions Reduction Plan.
h. Present overview of environmental issues and summarize site specific procedures relating to management plans at Preconstruction conference.
i. Training for Construction Contractor personnel in accordance with position requirements.
j. Monitoring and documentation of environmental procedures.

B. Perform project quality control in accordance with requirements specified in Related Sections, including:

1. Quality Requirements
2. Regulatory Requirements
3. Natural Features Protection
4. Indoor Air Quality (IAQ) Management
5. Noise and Acoustics Management
6. Temporary Storm Water Pollution Prevention Environmental Management
7. Construction Waste Management
8. Hazardous Spill Prevention
9. Dust Control
10. Emission Reduction

C. Construction Contractor's Environmental Training Program: Construction Contractor shall provide environmental training for workers performing work on project site. Training shall include:

1. Overview of environmental issues related to building industry.
2. Overview of environmental issues related to Project.
3. Review of site-specific procedures and management plans:

    a. Construction Waste Management
    b. Indoor Air Quality (IAQ) Management
    c. Noise and Acoustics Management
    d. Temporary Storm Water Pollution Prevention
    e. Hazardous Spill Plan
    f. Dust Control Plan
    g. Emission Reduction Plan
4. NPS Bear Management Plan: All contractors and their employees shall be trained regarding the NPS Beam Management Plan included but not limited to safety protocols and food storage regulations in accordance with Section 01 1100 "Summary of Work".

Pollution Prevention (P2) practices: Submit evidence of P2 training..

5. Compliance with environmental regulations: As specified in Regulatory Requirements. Submit Construction Contractor 40 CFR (Code of Federal Regulations) employee training records upon request of the Contracting Officer.

Provide documentation for environmental procedures as specified herein and in accordance with the approved Waste Management Plan, IAQ Management Plan, and Storm Water Pollution Prevention Plan.

1.9 PERMITS

A. General:

1. Permits and Responsibilities: Construction Contractor shall, without additional expense to the Government, be responsible for obtaining necessary licenses and permits, and for complying with Federal, State and municipal laws, codes, and regulations applicable to the performance of the work. Construction Contractor shall also be responsible for damages to persons or property that occur as a result of Construction Contractor's fault or negligence; and for materials delivered and work performed until completion and acceptance of the work.
2. For the purpose of this contract, Construction Contractor will not be considered an agent of the Government. Construction Contractor shall comply with appropriate Federal, State and local laws.

Government Furnished Permits: During development of the project's design, permits listed below were negotiated and agreed to by the Government. Terms and provisions of these permits shall be adhered to for the duration specified in each permit.

1. A permit was acquired for **WYDEQ Permit to Construct required for modifications to the public water and wastewater systems**. The Agency(ies) with Jurisdiction for this permit is **Wyoming DEQ.**

Potential Permits: Permits listed below were identified during the design process as likely to be required based on typical means and methods of construction. The list is provided to assist Construction Contractor in determining which permits will be required for contract's chosen means and methods. The list shall not

be considered complete; it is the Construction Contractors' responsibility to determine means and methods and obtain required permits. Construction Contractor shall obtain all permits required to legally conduct work.

General Permit to Discharge Storm Water Associated with Large Construction Activity under WYPDES: Prepare a formal permit including the Storm Water Pollution Prevention Plan (SWPPP). Contractor shall obtain written approval of the SWPPP from the CO prior to submission to WYDEQ. At a minimum, SWPPP shall include all erosion control measures described in the project specifications and drawings. The AHJ for this permit is WY DEQ. **Asbestos Abatement Notification Form** The AHJ for this permit is WY DEQ.

2. **General Permit for Temporary Discharge Involving Construction Activities:** This permit is required if the contractor anticipates that water discharge associated with trench dewatering, waterline disinfection, waterline pressure testing, water tank leak testing, and/or water tank disinfection will enter "surface waters of the State" as defined by the WSEO. the AHJ for this permit is WY DEQ.
3. **Hot Work Permit:** Permit required for heat generating work. Permit can be provided by the NPS. Blank hot work permit is attached to this specification.

D. Coordination with Agency(ies) with Jurisdiction Issuing Permits

1. Coordination: Contact the Agency(ies) with Jurisdiction as needed and sufficiently in advance to avoid delaying work: Coordinate meetings, reporting requirements, inspections, and other requirements.

E. Administrative Procedures:

1. Coordinate scheduling and timing of required administrative provisions of project permits with Agency(ies) with Jurisdiction, Construction Manager, and Park to avoid conflicts.
2. Supply needed information to Agency(ies) with Jurisdiction issuing permits, pay fees required and provide material needed to comply with permit's conditions and provisions.
3. Upload permits to NPS project management and communication software when permits are obtained.

1.10 PRE-CONSTRUCTION VIDEO

A. Prior to execution of Work, Contractor shall submit a video recording of the work site in accordance with Section 01 32 33 "Photographic and Video Documentation."

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 31 00

GRTE 237343 Rehabilitate "4 Lazy F Ranch" Cabins for Housing

SECTION 01 32 16 – CONSTRUCTION SCHEDULE

PART 1 - GENERAL

1.1 SUMMARY

A. Section consists of Construction Schedule requirements including:

Schedule of Values

2. Construction Schedule Requirements.
3. Construction Schedule Updates.
4. Time Impact Analysis.

B. Purpose: The Construction Schedule ensures adequate planning, coordination, scheduling, and reporting during execution of the work by the Construction Contractor. It shall assist the Construction Contractor and Contracting Officer (CO) in monitoring the progress of the work, evaluating proposed changes, and processing Construction Contractor's monthly progress payments. It shall include the dates in the contract, phases, milestones, occupancies, holidays, weather consideration, a critical path, and the requirements of this section.

C. Wildlife Protection: Project requirements related to wildlife protection are described in Section 01 35 13.05 – Environmental Protection. Wildlife protection measures will effect construction scheduling including work hours, scheduling of specific activities, and seasonal work stoppages, Please refer to Section 01 35 13.05 – Environmental Protection for specific requirements.

1.2 DEFINITIONS

A. Activity: A discrete part of a project identified for planning, scheduling, monitoring, and controlling the construction project. Activities included in a construction schedule consume time and resources.

1. Critical activities are activities on the critical path. They start and finish on the planned early start and finish times.
2. Predecessor Activity: An activity that precedes another activity in the network.
3. Successor Activity: An activity that follows another activity in the network.

B. Cost Loading: Allocation of the Schedule of Values for completion of an activity as scheduled. The sum of costs for all activities must equal the total Contract Sum, unless otherwise approved by the Contracting Officer.

C. Critical Path Method (CPM): Method of planning and scheduling a construction project where activities are arranged based on activity relationships. Network calculations determine when activities can be performed and the critical path of Project.

D. Critical Path: Longest connected chain of interdependent activities through the network schedule that establishes minimum overall Project duration and contains no float.

E. Float: Measure of leeway in starting and completing an activity.

1. Float: Not for the exclusive use or benefit of the Government or Construction Contractor but is jointly owned.
2. Free Float: Amount of time an activity can be delayed without adversely affecting the early start of the successor activity.
3. Total Float: Measure of leeway in starting or completing an activity without adversely affecting planned Project completion date.

F. Resource Loading: Allocation of manpower and equipment necessary for completion of an activity as scheduled.

G. Fragnet: A partial or fragmentary network that breaks down activities into smaller activities for greater detail.

1.3 SUBMITTALS

A. Electronic Copies: Schedules and reports submitted shall be posted on the NPS project management and communication software as PDF (portable document format) files. The intent of the Government is to limit the number of printed reports to those determined by the project team as essential.

B. Schedule of Values: After contract award and before Pre-Construction conference, submit schedule of dollar values based on Contract Price Schedule.

C. Construction Baseline Schedule: After contract award and before Pre-Construction conference, submit **two** paper copies of baseline schedule, large enough to show entire schedule for entire construction period. Utilize Schedule of Values in preparation of Construction Baseline Schedule.

D. Critical Path Method (CPM) Reports: Concurrent with CPM schedule, submit **three** paper copies of the following computer-generated reports. For each activity, include activity number, activity description, cost and resource loading, original duration, remaining duration, early start date, early finish date, late start date, late finish date, and total float in calendar days.

1. Activity Report: List of activities sorted by activity number and then early start date, or actual start date if known.
2. Logic Report: List of predecessor and successor tasks for activities sorted in ascending order by activity number and then early start date, or actual start date if known.
3. Total Float Report: List of activities sorted in ascending order of total float.

E. Construction Schedule Updates: On or before 7th day preceding progress payment request date, submit estimates of percent completion of each schedule activity and necessary supporting data. Provide two paper copies.

F. Construction Schedule Revisions and Time Impact Analysis: For each Construction Schedule revision, submit **two**paper copies of a Time Impact Analysis. Incorporate a Fragmentary Network

(Fragnet) into currently accepted Construction Schedule demonstrating proposal to incorporate a modification, change, delay, or Construction Contractor request.

1.4 QUALITY ASSURANCE

A. Construction Contractor shall meet with Contracting Officer on day of the preconstruction conference to go over:

Review software limitations, content and format for reports.

2. Verify availability of qualified personnel needed to develop and update schedule.
3. Discuss constraints, including work stages, area separations, and interim milestones
4. Review delivery dates for Government-furnished products.
5. Review schedule for work of separate Government contracts.
6. Review time required for review of submittals and re-submittals.
7. Review requirements for tests and inspections by independent testing and inspecting agencies.
8. Review time required for completion and startup procedures.
9. Review time required for obtaining and activating permits.
10. Review and finalize list of construction activities to be included in schedule.
11. Review baseline schedule comments, resolve issues and progress on incorporating them
12. Review procedures for updating schedule.
13. Discuss reporting requirements and establish protocol for naming and transmitting electronic schedules.

B. Construction Contractor's Schedule Representative: Before the preconstruction conference, designate an authorized representative to be responsible for preparing and maintaining the Construction Schedule. Submit resume outlining qualifications of Scheduler to Contracting Officer for acceptance. Scheduler shall have prepared and maintained at least 5 previous schedules of similar size and complexity similar to this Contract, demonstrating proficiency of using scheduling software. Authorized representative will be responsible for preparing the Baseline Schedule, required updates, revisions, Time Impact Analyses, and reports.

1.5 COORDINATION

A. Coordinate preparation and processing of schedules and reports with performance of construction activities and with scheduling and reporting of separate Construction Contractors.

B. Coordinate Construction Baseline Schedule with Schedule of Values, list of subcontracts, Submittals Schedule, progress reports, payment requests, and other required schedules and reports.

1. In developing Construction Baseline Schedule, ensure Subcontractor’s work at all tiers, and prime Construction Contractor’s work, is included and coordinated.
2. Secure time commitments for performing critical elements of work from parties involved.
3. Coordinate each construction activity in network with other activities and schedule in proper sequence.

PART 2 - PRODUCTS

2.1 SCHEDULE OF VALUES

A. Breakdown each lump-sum item into component work activities used in the schedule for which progress payments may be requested. Work activities broken out within schedule of values shall be integrated into and made a logical part of the construction baseline schedule. Total costs for the component work activities shall equal contract price for that lump-sum item. Contracting Officer may request data to verify accuracy of dollar values. Include mobilization, general condition costs, overhead and profit in the total dollar value of unit price items and in the component work activities for each lump-sum item. Do not include mobilization, general condition costs, overhead or profit as a separate item.

B. Do not break down unit price items. Use only the contract price for unit price items.

C. Total cost of all items shall equal the contract price. The Schedule of Values will form the basis for progress payments and the Construction Schedule.

2.2 CONSTRUCTION SCHEDULE REQUIREMENTS

A. Construction Baseline Schedule: Prepare Construction Baseline Schedule using a computerized, cost and resource-based**,** time-scaled Critical Path Method network analysis diagram for the Work.

Develop and finalize Construction Baseline Schedule so it can be accepted for use no later than [30] Thirty days after date established for the Notice of Award.

a. Failure to include any work item required for performance of this Contract shall not excuse Construction Contractor from completing work within applicable completion dates, regardless of Governments acceptance of schedule.

2. Establish procedures for monitoring and updating Construction Baseline Schedule and for reporting progress. Coordinate procedures with progress meeting and payment request dates.

B. Construction Baseline Schedule Preparation: Prepare a list of all activities required to complete the Work. Using preliminary Critical Path Method network diagram, prepare a skeleton network to identify probable critical paths.

1. Activities: Indicate estimated duration, sequence requirements, and relationship of each activity in relation to other activities.
2. Critical Path Activities: Identify critical path activities, including those for interim completion dates. Scheduled start and completion dates shall be consistent with Contract milestone dates.
3. Processing: Process data to produce output data on a computer-drawn, time-scaled network. Revise data, reorganize activity sequences, and reproduce as often as necessary to produce the Critical Path Method schedule within the limitations of the Contract Time.

4. Show sequence and interdependence of activities required for completion of work. Ensure work sequences are logical and Construction Baseline Schedule shows a coordinated plan of the work.
5. Resource loading of each activity shall include personnel by labor category and equipment type and capacity proposed to complete the activity in duration shown.
6. Consider seasonal weather conditions in planning and scheduling work influenced by high and low ambient temperatures, wind, or precipitation to ensure completion of work within contract time.
7. Time Frame: Proposed duration assigned to each activity shall be Construction Contractor's best estimate of time required to complete activity considering the scope and resources planned for activity.

    a. An early finish date may be shown but the late finish date shall be same date as last day of contract period. An early completion schedule shall contain:

        1) Insert an activity titled "Project Float" as a successor to last activity in early project completion schedule network.
        2) Add a milestone titled "Contract End Date" as a successor to the activity "Project Float".
        3) Add duration to the activity "Project Float" as required so the milestone "Contract End Date" equals the last day of Contract Period.

    b. Contract completion date shall not be changed by submission of a schedule that shows an early completion date.
    c. Construction Contractor shall limit use of lead or lag duration’s between schedule activities.
    d. Project Calendars: Develop and incorporate the following calendars:

        1) Administrative Calendar: Include calendar based on a 7-day week to be used on activities based on calendar days. Apply this calendar to administrative tasks or other tasks not affected by non-working days (Federal Holidays, weather, etc.).
        2) Project Calendar: Include calendar based on planned work week for the project. Include Federal Holidays, weekends, and non-workdays indicated in contract documents. Apply this calendar to activities not anticipated to be affected by weather. Be clear when identifying number of work days in work week.
        3) Weather Calendar: Utilize Project Calendar and show anticipated normal downtime related to weather as non-working time. Weather days shall be based on data for local area from a reliable source like the National Oceanic and Atmospheric Administration (NOAA), National Park Service records, or source acceptable to Contracting Officer. Apply this calendar to activities anticipated to be affected by weather.

Activity Duration: Define so no activity is longer than **[14] fourteen** days, except for non-construction activities including mobilization, shop drawings and submittals, fabrication and delivery of materials and equipment.

    f. Procurement Activities: Include procurement process activities for long lead items and major items, requiring a cycle of more than 60 calendar days, as separate activities in the schedule. Procurement cycle activities can include submittals, approvals, purchasing, fabrication, and delivery.

1) Lodgepole pine logs.

2) Lift Station Pumps and Controls

g. Submittal Review Time: Include review and re-submittal times indicated. Coordinate submittal review times in Construction Baseline Schedule.
h. Startup and Testing Time: Include not less than 21 days for startup, testing activities.
i. Substantial Completion: Allow time for Government administrative procedures necessary for certification of Substantial Completion. (For more information, refer to Specification 01 77 00 "Closeout Procedures.")

8. Constraints: Include constraints and work restrictions indicated in Contract Documents and as follows in schedule and show how the sequence of Work is affected.

Work Restrictions: Show effect of the following on the schedule:
Work hour restrictions related to wildlife protections required by Section 01 10 00 Summary of Work and Section 01 35 13.05 – Environmental Protection. See the referenced sections for specific requirements.

2) Coordination with existing construction
3) Uninterruptible services
4) Use of premises restrictions
5) Seasonal variations

b. Environmental controlPermit provisionsOther Constraints:
1) Uninterruptible water supply and transmission.
2) Work access restrictions due to construction activities.
3) Seasonal weather and temperatures.
c. Work Stages: Indicate important stages of construction for each major portion of the Work.

1) Subcontract awards
2) Submittals
3) Purchases
4) Mockups
5) Fabrication
6) Sample testing
7) Deliveries
8) Installation
9) Tests and inspections
10) Adjusting
11) Curing

Building flush-out.

13) Building commissioning activities.

9. Milestones: Include milestones indicated in Contract Documents in schedule, including, but not limited to, Notice to Proceed, Substantial Completion**,** and the following interim milestones:

a. Winter work stoppage

b. Spring work start and mobilization

c. Completion of each individual construction line item.

C. Joint Review, Revision, and Acceptance:

1. Within seven calendar days of receiving Construction Contractor's proposed Construction Baseline Schedule, Contracting Officer shall review initial Construction Baseline Schedule.
2. Within seven calendar days after review, Construction Contractor shall revise and resubmit Construction Baseline Schedule in accordance with comments presented from review.
3. In the event the Construction Contractor fails to define any element of work, activity, or logic and the Contracting Officer review does not detect this omission or error, such omission or error, when discovered by Construction Contractor or Contracting Officer, shall be corrected by Construction Contractor within seven calendar days and shall not affect contract period.
4. Upon acceptance of the Construction Baseline Schedule, Contracting Officer saves schedule as a baseline and updates on a monthly basis. Construction schedule update will be used to evaluate Construction Contractor's monthly applications for payment based upon information developed at monthly Construction Schedule update meeting.

D. Cost Correlation: In the heading of the schedule, provide a cost correlation line, indicating planned and actual costs. On the line, show dollar volume of Work performed as of dates used to prepare payment requests.

1. Construction Contractor shall assign cost to construction activities on Construction Baseline Schedule. Costs shall not be assigned to submittal activities unless specified otherwise but may, with Contracting Officer’s approval, be assigned to fabrication and delivery activities. Costs shall be included for testing and commissioning activities, operation and maintenance manuals, punch list activities, Project Record Documents, and demonstration and training (if applicable).
2. Each activity cost shall reflect an accurate value based on the Contract Price Schedule.
3. Total cost assigned to activities shall equal total Contract Price.

E. Recovery Schedule: When periodic schedule update indicates Work is 14 or more calendar days behind current accepted schedule, a separate recovery schedule indicating means by which Construction Contractor intends to regain compliance with the schedule shall also be submitted. Indicate changes to working hours, working days, crew sizes, and equipment required to achieve compliance, and date by which recovery shall be accomplished.

F. Computer Software: Prepare schedules using a program developed specifically to manage construction schedules.

1. Use Microsoft Project or Primavera, for Windows 11.

PART 3 - EXECUTION

3.1 CONSTRUCTION SCHEDULE UPDATES

A. Progress Meeting Updates: Provide a [2] two week look-ahead schedule, derived from the currently accepted schedule, before each weekly progress meeting. Utilize look-ahead schedule to facilitate and take notes on discussions held during progress meeting.

B. Monthly Schedule Updates:

1. General: Update Construction Schedule on monthly basis to reflect construction progress and activities throughout entire contract period and until project substantial completion. The status date of each schedule update shall be the 7th day preceding the progress payment request date.
2. Procedure: Construction Contractor shall meet with Contracting Officer each month at Construction Schedule update meeting to review progress made through the status date of the Construction Schedule update, including dates activities were started or completed and percentage of work completed on each activity started or completed.
3. Reports: Concurrent schedule revisions, prepare tabulated reports showing:

Identification of activities that have changed

b. Changes in early and late start dates
c. Changes in early and late finish dates
d. Changes in activity durations in workdays
e. Changes in the critical path
f. Changes in total float or slack time
g. Changes in the Contract Time

4. Narrative: Report shall include a brief description of actual progress made during update period; actual and potential delaying activities; impediments to progress; issues related to inclement weather; progress toward established milestones and project float. Report shall include a brief description of work anticipated to be performed in the next month. Minor revisions to the schedule should be identified for evaluation and acceptance or rejection.
5. As Work progresses, indicate Actual Completion percentage for each activity.
6. If schedule update shows a late finish date after contract completion date, include:

a. Known delays
b. Actions to get back on schedule
c. Pending modifications
d. Impediments or constraints affecting progress

7. Progress Payments: Monthly updating of the currently accepted Construction Schedule shall be an integral part of the process upon which progress payments will be made. If Construction Contractor fails to provide schedule updates or revisions, a portion of the monthly payment may be retained until corrections have been made.

C. Distribution: Distribute copies of accepted schedule to Contracting Officer, Contracting Officers Representative, Construction Management Representative, Subcontractors, testing and inspecting agencies, and other parties identified by Construction Contractor with a need-to-know schedule responsibility.

1. Post copies in Project meeting rooms and temporary field offices.
2. When revisions are made, distribute updated schedules to same parties and post in same locations. Delete parties from distribution when they have completed their assigned portion of the Work.

D. Construction Schedule Revisions:

1. Required Revisions: If, as a result of the monthly schedule update, it appears the currently accepted Construction Schedule no longer represents actual prosecution and progress of the work, Contracting Officer will request, and Construction Contractor shall submit, a revision to the Construction Schedule. Construction Contractor may also request reasonable revisions to currently accepted Construction Schedule in event the Construction Contractor's planning for the work is revised. If Construction Contractor desires to make changes, Construction Contractor shall notify Contracting Officer in writing, stating reason for proposed revision. Accepted revisions shall be incorporated into currently accepted Construction Schedule for next monthly schedule update.
2. Procedure: If revision to currently accepted Construction Schedule is contemplated, Construction Contractor or Contracting Officer shall advise the other in writing at least seven calendar days prior to next monthly schedule update meeting, describing revision and reasons for the revision. Government-requested revisions will be presented in writing to the Construction Contractor, who shall respond in writing within seven calendar days.
3. Reports: Concurrent with making revisions to schedule, prepare tabulated reports showing:

    a. Identification of activities changed
    b. Changes in early and late start dates
    c. Changes in early and late finish dates
    d. Changes in activity durations in workdays
    e. Changes in critical path
    f. Changes in total float or slack time

3.2 TIME IMPACT ANALYSIS FOR CONTRACT MODIFICATIONS CHANGES DELAYS AND CONSTRUCTION CONTRACTOR REQUESTS:

1. Requirements: When contract modifications or changes are initiated, delays experienced, or Construction Contractor desires to revise currently accepted Construction Schedule, Construction Contractor shall submit to Contracting Officer a written time impact analysis illustrating the influence of modification, change, delay, or Construction Contractor request on contract time.
2. Time Extensions: Activity delays, resulting in a late completion date projection, shall not automatically mean an extension of contract time is warranted or due to Construction Contractor. It is possible a modification, change, or delay will not affect existing critical path activities or cause non-critical activities to become critical. A modification, change, or delay may result in absorbing a part of available total float that may exist within an activity chain of the Schedule, not causing any effect on contract time. Time extensions will be granted in accordance with terms of contract.
3. Extension of contract time will be granted only to the extent the equitable time adjustments to activity or activities affected by modification, change, or delay exceeds total (positive or zero) float available on a particular activity.
4. Procedure: Each time impact analysis shall be submitted within time period stated in a request for proposal, or time period designated under the clauses entitled Changes or

Default. In cases where Construction Contractor does not submit a written request for extension of time and a time impact analysis within the designated time, it is mutually agreed that the particular modification, change, delay, or Construction Contractor request does not require an extension of the contract time. Upon acceptance, time impact analysis shall be incorporated into currently accepted Construction Schedule at next monthly schedule update.

5. Contract Modifications: Prepare time-impact analysis using fragnets to demonstrate effect of proposed change on overall Construction Schedule for each proposed contract modification concurrent with submission.

END OF SECTION 01 32 16

GRTE 237343 Rehabilitate "4 Lazy F Ranch" Cabins for Housing

SECTION 01 32 33 – PHOTOGRAPHIC AND VIDEO DOCUMENTATION

GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for:

Existing condition images and video

2. Periodic construction image and video.

B. See Section 01 77 00 "Closeout Procedures" for a complete listing of closeout documents.

C. See Section 01 79 00 "Demonstration and Training" for submitting videotapes of demonstration of equipment and training of National Park Service (NPS) personnel.

1.2 SUBMITTALS

A. Construction Images and Videos: Submit images and videos electronically within seven days of taking the image. Include:

1. Date, time and number (sequentially number all images) in filename.
2. Description of vantage point, indicating location, direction (by compass point), and elevation or story of construction.
3. Submit digital images exactly as originally recorded in digital camera, without alteration, or modifications using image-editing software.

B. Closeout: Submit complete set of digital image and video electronic files as a Project Record Document**.** Submit on Digital Video Disc (DVD).

1. Provide index as separate file on Disc. List each image as a file name with number, date, and time. Include description and or vantage point image was taken.
2. Submit images that have the same aspect ratio as the sensor, un-cropped.

PART 2 - PRODUCTS

2.1 FORMAT REQUIREMENTS

Media: DVD-R Archival Gold

B. Media Labels: Archival DVD labeling markers, archival labels, or direct print.

C. Images: Provide sRGB (standard Red Green Blue) color images in JPEG (Joint Photographic Experts Group) format. Minimum sensor size of 8 megapixels, and at image resolution of not less than 3200 by 2400and 300 dpi (dots per inch).

D. Video: Provide video in WMV or MOV format, produced by a digital video recorder with a 1920 x 1080 resolution. Videos shall be submitted digitally.

PART 3 - EXECUTION

3.1 CONSTRUCTION IMAGES AND VIDEO

A. General: Take digital images and video using the maximum range of depth of field, in-focus, to clearly show the Work. No blurry or out-of-focus areas accepted.

1. Maintain index with each set of Construction images and videos and identify the number, date, time, and description for each.
2. Maintain one set of images and videos accessible in field office at Project site available for reference.
3. Video recordings shall include a complete, clearly spoken narration of the events being photographed. Also, video recordings shall include an unobtrusive time and date indicator on the film, accurately depicting the time and date when the photography was performed.

B. Existing Condition Images: Before starting construction, take color digital images and videos of Project site and surrounding properties, including existing items to remain during construction, from different vantage points.

1. Flag construction limits before recording construction images.
2. Take eight separate images at each structure to show existing conditions adjacent to the structure before starting Work.
3. Take eight separate images of existing buildings, both interior and exterior, to accurately record physical conditions at start of construction.

C. PRE-CONSTRUCTION VIDEO

1. Recorded video information will be utilized to ascertain post construction conditions following contract execution. Video record project site with high resolution equipment.
2. Subsequent to execution of Work, the Contractor shall:
   a. Walk the project site and pipeline alignments in the presence of the CO, or representative thereof, and video record all areas affected by new construction.
   b. Video record the Pump House site including existing building, and all areas affected by new construction.

D. Periodic Construction Images and Video: Take 12 color, digital images and 2 digital videos weekly, with timing each month adjusted to coincide with cutoff date associated with each Application for Payment. Select vantage points to show status of construction and progress since last images were taken.

E. Additional Images or videos: Contracting Officer may issue requests for additional images.

1. Three days advance, where feasible.

2. In emergency situations, take additional images within 24 hours of request.
3. Additional images or videos include, but are not limited to:

Immediate follow-up when on-site events result in construction damage or losses.

b. Fabrication locations away from Project site.
c. Substantial Completion of a major phase or component of Work.
d. Extra record images at time of final acceptance.

END OF SECTION 01 32 33

GRTE 237343 Rehabilitate "4 Lazy F Ranch" Cabins for Housing

SECTION 01 33 23 - SUBMITTAL PROCEDURES

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for submitting Shop Drawings, Product Data, Samples, and other submittals.

1.2 DEFINITIONS

A. Action Submittals: Written, graphic information, and physical samples that require Government's responsive action.

B. Informational Submittals: Written information that does not require Government's responsive action. Submittals may be rejected for not complying with requirements.

C. File Transfer Protocol (FTP): Communications protocol that enables transfer of files to and from another computer over a network and that serves as the basis for standard Internet protocols. An FTP site is a portion of a network located outside of network firewalls within which internal and external users are able to access files.

D. Portable Document Format (PDF): An open standard file format licensed by Adobe Systems used for representing documents in a device-independent and display resolution-independent fixed-layout document format.

1.3 GENERAL SUBMITTAL PROCEDURES

A. General: Prepare and submit submittals required by individual Specification Sections and in some cases as requested in drawings. Types of submittals are indicated in individual specific sections.

1. Contracting Officer (CO) reserves right to require submittals in addition to those called for in individual sections.

2. Provide the Specification Section number for each product on each submittal.

B. Coordination: Coordinate preparation and processing of submittals with performance of construction activities. Review for legibility, accuracy, completeness, and compliance with Contract Documents.

1. Coordinate submittal with fabrication, purchasing, testing, delivery, other submittals, and related activities that require sequential activity.

Coordinate transmittal of different types of submittals for related parts of Work so processing will not be delayed because of need for concurrent review coordination.

a. Contracting Officer reserves right to withhold action on a submittal requiring coordination with other submittals until related submittals are received.

Submittal List: Submittal list is attached to the end of this Specification Section. The intent is to provide an overall summary of submittal requirements. The requirements of individual Specification Sections and terms and conditions of the Contract still apply regardless of what is shown on submittal list.

D. Processing Time: Allow enough time for submittal review and re-submittals. Contract time extensions will not be authorized if submittals are not transmitted with enough time to permit processing, including re-submittals.

Action Submittals

a. Initial Review: Allow 30 days for initial review of each submittal. Allow additional time if coordination with subsequent submittals is required.
b. Re-submittal Review: Allow 30days for review of each re-submittal.

2. Informational submittals

a. Review: Allow 10 days for review of each submittal.

E. Approved Equal Submittal Requirements:

1. For each item proposed, submit supporting data, including:

a. Drawings and samples as appropriate.
b. Comparison of the salient characteristics of the proposed item with that specified.
c. Changes required in other elements of the work because of the substitution.
d. Name, address, and telephone number of vendor.
e. Manufacturer's literature regarding installation, operation, and maintenance, including schematics for electrical and hydraulic systems, lubrication requirements, and parts lists. Describe availability of maintenance service, and state source of replacement materials.

2. A request for approval constitutes a representation that Construction Contractor:

a. Has investigated the proposed item and determined that it is equal or superior in all respects to that specified.
b. Will provide the same warranties for the proposed item as for the item specified.
c. Has determined that the proposed item is compatible with interfacing items.
d. Will coordinate installation of an approved item and make changes required in other elements of the work because of the substitution.
e. Waives claims for additional expenses that may be incurred as a result of the substitution.

F. Submittals: Identify and incorporate information in each submittal file as follows:

1. Transmittal Form (CM-16): Utilize NPS Transmittal Form for all submittals. No action will be taken on a submittal item unless accompanied by this Transmittal Form.

2. Physical samples: Complete Transmittal Form and deliver physical sample to the Contracting Officer (or designee) on site for processing. All comments and actions will be documented on the Transmittal Form.

G. Identification: Submittal number or other unique identifier, including revision identifier.

1. Submittal number shall use a sequential number (e.g. .001). Re-submittals shall include alphabetic suffix after another decimal point (e.g. .001.A).

H. Re-submittals: Make re-submittals using same process used with initial submittal.

1. Note date and content of previous submittal.
2. Note date and content of revision in the title block on the Transmittal Form (CM-16) and clearly indicate extent of revision.
3. Re-submit submittals until they are marked "Approved" or "Approved with notations".
4. Excessive Re-submittals: The contractor is allowed two (2) re-submittals per specification section. If the contractor exceeds this limit, NPS reserves the right to seek additional compensation for review efforts. Additional compensation will be at current NPS labor rates and will be processed through the CO as a contract modification.

I. Distribution: Furnish copies of final submittals to manufacturers, subcontractors, suppliers, fabricators, installers, and others as necessary for performance of construction activities.

J. Use for Construction: Use only final submittals with mark indicating "Approved" or "Approved with notations". Ensure notations have been incorporated and, at a minimum, keep one copy of final approved submittal on site for use during construction.

1.4 CONSTRUCTION CONTRACTOR'S USE OF CAD/BIM FILES

A. General: At Construction Contractor's written request, copies of CAD/BIM (Computer Aided Design/Building Information Modeling) files will be provided to Construction Contractor for use in connection with Project, subject to:

Files provided as is; no format or other changes to files or changes to objects in the drawing will be done by the Government

2. Construction Contractor will be required to sign the Architect's and/or Architect's consultant's electronic file release forms as requested.

3. Any CAD/BIM files will be provided on an information-only basis and shall not to be considered a contract document. Construction contractor is required to verify and coordinate all conditions.

4. Information shown on hard copies of the construction drawings shall take precedence over information provided in electronic CAD files

PART 2 - PRODUCTS

2.1 ACTION SUBMITTALS

A. Product Data: Collect information into a single submittal for each element of construction and type of product or equipment.

1. If information must be specially prepared for submittal because standard printed data are not suitable for use, submit as Shop Drawings, not as Product Data.
2. Mark each submittal to show which products and options are applicable.
3. As applicable, include:

a. Manufacturer's product specifications.
b. Manufacturer's installation instructions: When Contract Documents require compliance with manufacturer's printed instructions, provide one complete set of instructions to Contracting Officer and keep another complete set of instructions at the project site until substantial completion.
c. Manufacturer's catalog cuts: Submit only pertinent pages; mark each page of standard printed data to identify specific products proposed for use.
d. Wiring diagrams showing factory-installed wiring.
e. Printed performance curves.
f. Operational range diagrams.
g. Compliance with specified referenced standards.
h. Testing by recognized testing agency.

4. Submit product data in PDF (portable document format) file format before or concurrent with samples.

B. Shop Drawings: Prepare Project-specific information, drawn accurately to scale. Do not base Shop Drawings on reproductions of the Contract Documents or standard printed data.

1. Preparation: Fully illustrate requirements in Contract Documents. As applicable, include:

Dimensions

b. Identification of products
c. Fabrication and installation drawings
d. Roughing-in and setting diagrams
e. Wiring diagrams showing field-installed wiring, including power, signal, and control wiring
f. Shopwork manufacturing instructions
g. Templates and patterns
h. Schedules
i. Notation of coordination requirements
j. Notation of dimensions established by field measurement
k. Relationship to adjoining construction clearly indicated
l. Seal and signature of professional engineer if specified
m. Wiring Diagrams: Differentiate between manufacturer-installed and field-installed wiring

2. Submit shop drawings as PDF electronic file

C. Samples: Submit Samples for review of kind, color, pattern, and texture for a check of these characteristics with other elements and for a comparison of these characteristics between submittal and actual component as delivered and installed.

1. Transmit Samples that contain multiple, related components such as accessories together in one submittal package.
2. Complete and post the Transmittal Form (CM-16) on the NPS project management and communication software website for processing and documentation of action on submitted samples.
3. Identification: Attach label on unexposed side of Samples that includes:

    a. Generic description of Sample
    b. Product name and name of manufacturer
    c. Sample source
    d. Submittal Number and title of appropriate Specification Section

4. Disposition: Maintain sets of approved Samples at Project site, available for quality-control comparisons throughout course of construction activity. Sample sets may be used to determine final acceptance of construction associated with each set.
5. Samples for Initial Selection: Submit manufacturer's color charts consisting of units or sections of units showing full range of colors, textures, and patterns available.

    a. Number of Samples: Submit two full set(s) of available choices where color, pattern, texture, or similar characteristics are required to be selected from manufacturer's product line. Contracting Officer will return with options selected.

6. Samples for Verification: Submit full-size units or Samples of size indicated, prepared from same material to be used for the Work, cured and finished in manner specified, and physically identical with material or product proposed for use, and that show full range of color and texture variations expected. Samples include: partial sections of manufactured or fabricated components; small cuts or containers of materials; complete units of repetitively used materials; swatches showing color, texture, and pattern; color range sets; and components used for independent testing and inspection.

    a. Number of Samples: Submit four sets of Samples. Contracting Officer will retain three Sample sets; remainder will be returned. Retain Sample set as a Project Record Sample.

D. Construction Materials: Construction Contractor is encouraged to submit products made out of recycled or environmentally responsible material. Every effort will be made by NPS to approve these materials.

2.2 INFORMATIONAL SUBMITTALS

A. General: Prepare and submit Informational Submittals required by individual Specification Sections.

1. Post informational submittals as PDF electronic files directly to the NPS project management and communication software.

2. Certificates and Certifications: Provide a notarized statement with signature of entity responsible for preparing certification. Certificates and certifications shall be signed by officer or other individual authorized to sign documents on behalf of that entity.
3. Informational submittals that do not comply with requirements specified in Contract Documents will be rejected and one copy will be returned.

B. Coordination Drawings: Comply with requirements specified in Section 01 31 00 "Project Management and Coordination."

C. Construction Contractors Construction Schedule: Comply with requirements specified in Section 01 32 16 "Construction Schedule."

D. Accident Prevention Plan: Comply with requirements specified in Section 01 35 23 "Safety Requirements."

E. Schedule of Values: Comply with requirements specified in Section 01 32 16 "Construction Schedule."

F. Waste Recycling Plan: Comply with requirements specified in Section 01 74 19 "Construction Waste Management and Disposal."

G. Quality Control Plan: Comply with requirements specified in Section 01 40 00 "Quality Requirements."

H. Traffic Control Plan: Comply with the requirements specified in Section 01 55 26 “Traffic Control.”

I. Storm Water Pollution Prevention Plan: Comply with requirements specified in Section 01 57 23 "Temporary Storm Water Pollution Prevention" and storm water permit requirements identified in Section 01 31 00 "Project Management and Coordination."

J. Indoor Air Quality Management Plan: Comply with requirements specified in Section 01 57 19.11 "Indoor Air Quality Management."

K. Qualification Data: Prepare written information demonstrating capabilities and experience of firm or person. Include lists of completed projects with project names and addresses, names and addresses of architects and owners, and other information specified.

L. Welding Certificates: Prepare written certification that welding procedures and personnel comply with Contract Documents. Submit record of Welding Procedure Specification (WPS) and Procedure Qualification Record (PQR) on American Welding Society (AWS) forms. Include names of firms and personnel certified.

M. Installer Certificates: Prepare written statements on manufacturer’s letterhead certifying Installer complies with Contract Documents and, where required, is authorized by manufacturer for this specific Project.

N. Manufacturer Certificates: Prepare written statements on manufacturer’s letterhead certifying manufacturer complies with Contract Documents. Include evidence of manufacturing experience where required.

O. Product Certificates: Prepare written statements on manufacturer's letterhead certifying product complies with Contract Documents.

P. Material Certificates: Prepare written statements on manufacturer's letterhead certifying material complies with Contract Documents.

Q. Material Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting test results of material for compliance with requirements in Contract Documents.

R. Product Test Reports: Prepare written reports indicating current product produced by manufacturer complies with requirements in Contract Documents. Base reports on evaluation of tests performed by manufacturer and witnessed by a qualified testing agency, or on comprehensive tests performed by a qualified testing agency.

Research/Evaluation Reports: Prepare written evidence, from a model code organization acceptable to authorities having jurisdiction, that product complies with building code in effect for Project.

T. Preconstruction Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of tests performed before installation of product, for compliance with performance requirements in Contract Documents.

U. Compatibility Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of compatibility tests performed before installation of product. Include written recommendations for primers and substrate preparation needed for adhesion.

V. Field Test Reports: Prepare reports written by a qualified testing agency, on testing agency's standard form, indicating and interpreting results of field tests performed either during installation of product or after product is installed in its final location, for compliance with requirements in Contract Documents.

W. Maintenance Data: Prepare written and graphic instructions and procedures for operation and normal maintenance of products and equipment. Comply with requirements specified in Section 01 78 23 "Operation and Maintenance Data."

X. Design Data: Prepare written and graphic information, including: performance and design criteria, list of applicable codes and regulations, and calculations. Include list of assumptions and other performance and design criteria and a summary of loads. Include load diagrams if applicable. Provide name and version of software, if any, used for calculations. Include page numbers.

Y. Manufacturer's Instructions: Prepare written or published information documenting manufacturer's recommendations, guidelines, and procedures for installing or operating a product or equipment. Include name of product and name, address, and telephone number of manufacturer.

Z. Manufacturer's Field Reports: Prepare written information documenting factory-authorized service representative's tests and inspections. As applicable, include:

1. Statement on condition of substrates and their acceptability for installation of product.

2. Summary of installation procedures being followed, compliance with requirements and, if not, what corrective action was taken.
3. Results of operational and other tests and a statement of whether observed performance complies with the requirements.

AA. Permit Compliance Products: Prepare required information for compliance with permit provisions. Products include written notification of project startup, suspension, and completion of work; photo documentation of site conditions; reports; and drawings.

PART 3 - EXECUTION

3.1 CONSTRUCTION CONTRACTOR'S REVIEW

A. Review each submittal and check for coordination with other Work of Contract and for compliance with Contract Documents. Note corrections and field dimensions.

B. Contractor is responsible for providing complete submittals from their subcontractors in compliance with contract documents. Time expended on excessive review and resubmittals will be the responsibility of Contractor.

3.2 CONTRACTING OFFICER'S ACTION

A. General: Submittals will be disapproved without technical review if identification information is missing, not filled in, or if placed on back of submittal; an incorrect format of submittals is provided; transmittal form is incorrectly filled out; submittals are not coordinated; or submittals do not show evidence of Construction Contractor's approval.

1. Any work done or orders for materials or services placed before approval shall be at Construction Contractor's own risk.

B. Action Submittals: Contracting Officer will review each submittal, generate comments on corrections or modifications required, and indicate appropriate action on the Transmittal Form (CM-16). Submittal will be marked as defined below:

1. APPROVED: Acceptable with no corrections.
2. APPROVED WITH NOTATIONS: Minor corrections or clarifications required. Comments are clear and no further review is required. Construction Contractor shall address review comments when proceeding with the work.
3. DISAPPROVED - RESUBMIT: Rejected as not in accordance with the contract or as requiring major corrections or clarifications. Contracting Officer will identify reasons for disapproval. Construction Contractor shall revise and resubmit with changes clearly identified.

C. Informational Submittals: Contracting Officer will review each submittal and will either accept or reject it.

D. Partial submittals are not acceptable, will be considered non-responsive, and will be returned without review.

END OF SECTION 01 33 23

SUBMITTAL			REQUIREMENTS (Indicate with **X**)							
			INFORMATIONAL				ACTION			
SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
013100	1.4C	Coordination Drawings - Verification Site Plan with existing conditions survey information								X
	1.4D	Division 1 submittals required prior to mobilization		X						
013216	1.3B	Schedule of Values				X				
	1.3C	Construction Baseline Schedule				X				
	1.3D	Critical Path Method (CPM) Reports		X						
	1.3E	Construction Schedule Updates				X				
	1.3F	Construction Schedule Revisions and Time Impact Analysis		X						
013233	3.1B	Existing Conditions Images				X				
	3.1C	Preconstruction Video				X				
	3.1D	Periodic Construction Images and Video				X				
013513.05	1.3A	Daily Work Schedule		X						
013523	1.3A	Accident Prevention Plan (APP)		X						
013591	1.3A	Historic Preservation Treatment Plan		X						
	1.3B	Alternative Methods and Materials		X						
014000	1.4A	Quality Control Plan		X						
	1.4B	Qualification Data			X					
	1.4C	Contractor Quality Control Daily Reports								
	1.4D	Test Reports	X							
	1.4E	Accessibility Inspection Report	X							
	1.4F	Off-Site Inspection Reports	X							
	1.4H	Permits, Licenses, and Certificates	X							
015000	1.4A	Spill Prevention and Response Plan		X						
	1.4B	Emissions Reduction Plan		X						
	1.4C	Dust Control Plan		X						
	2.1C	Safety Barrier Fence							X	
015719_11	1.4A	Indoor Air Quality (IAQ) Management Plan		X						
	1.4B	Product Data - MSDS			X					
	1.4C	Inspection and Test Reports	X							
015723	1.3A	SWPPP showing SWPPP satisfies Federal and State NPDES permit requirements.		X						
	1.3C	Inspection Schedule (SWPPP Measures)				X				
	1.3D	Erosion Control Products			X					
016700	1.3A-1	Affirmative Procurement Reporting Form				X				
	1.3A-2	Environmental Data				X				
017340	1.2A	Survey Certificates	X							
	1.2A	Landfill Receipts								
	1.2C	Certified Surveys		X						
	3.8F	Starting and Adjusting Report		X						
	3.9C	Utility Location Information including GIS Data								
017419	1.4A	Waste Management Plan		X						
	1.4B	Progress Documentation				X				

SUBMITTAL			REQUIREMENTS (Indicate with **X**)							
			INFORMATIONAL				ACTION			
SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
	1.4C	Landfill and Incinerator Disposal Records				X				
	1.4E	Closeout Submittals - Summary of Project Waste Management P		X						
	1.3A	Closeout Submittals								
	1.3A-1	Warranties, Guarantees, Workmanship Bonds, Final Certification	X							
017700	1.3A-2	NPS required Occupancy Forms, Other NPS Forms	X							
	1.3A-3	Project Record Documents				X				
	1.3A-4	Environmental Record Documents				X				
017823	1.2A	Operation and Maintenance Manual				X				
017900	1.3A	Instruction Program				X				
	1.3B	Demonstration and Training Video				X				
	1.5A	Site Plan		X						
	1.6B	Schedule of Selective Demolition Activities				X				
02 41 00	1.6C	Project Record Documents				X				
	1.6E	Statement of Refrigerant Recovery				X				
	1.6F	Inventory: Submit a list of items that have been removed and salvaged.				X				
	1.5B	Proposed Protection Measures		X		X				
02 41 01	1.5C	Schedule of Selective Demolition Activities				X				
	1.5D	Inventory: Submit a list of items to be removed and salvaged				X				
02 41 13	1.4B	Proposed Protection Measures		X		X				
	1.4C	Inventory: Submit a list of trees to be removed				X				
02 43 00	1.4A	Photographs and Videotape				X				
	1.4B	Project Record Documents				X				
	2.4A	Pre-Construction Meeting Submittals				X			X	
02 82 00	2.4B	Submittals During Abatement				X				
	2.5C	Submittals at Completion of Abatement				X				
	2.4A	Pre-Construction Meeting Submittals				X			X	
02 83 00	2.4B	Submittals During Lead Work Activities				X				
	2.4B	Submittals at Completion of Lead Work Activities				X				
	2.5A	Pre-Construction Meeting Submittals				X			X	
02 85 00	2.5B	Submittals During Removal				X				
	2.5C	Submittals at Completion of Removal				X				
03 01 00	1.2A	Product Data							X	
	1.2B	Manufacturer's instruction							X	
	1.2A	Product Data							X	
	1.2B	Design Mixtures							X	
03 30 00	1.2C	Steel Reinforcement Shop Drawings						X		
	1.3A	Material Certificates			X					
	1.4C	Material Test Reports			X					
	1.6A	Product Data							X	
	1.6B	Shop Drawings						X		
	1.6C	Samples for Initial Selection					X			

SUBMITTAL			REQUIREMENTS (Indicate with **X**)							
			INFORMATIONAL				ACTION			
SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
04 03 42	1.6D	Samples for Verification					X			
	1.7A	Qualification Data	X							
	1.7B	Preconstruction Test Reports	X							
	1.7C	Quality Control program		X						
	1.7D	Historic masonry treatment program		X						
05 12 00	1.4A	Product Data							X	
	1.4B	Shop Drawings						X		
	1.5A	Qualification Data	X							
	1.5B	Welding certificates.	X							
	1.5C	Mill test reports for structural steel, including chemical and physical properties.	X							
	1.5D	Source quality-control reports.				X				
	1.5E	Field quality-control and special inspection reports.				X				
05 50 00	1.4A	Shop Drawings						X		
	1.4B	Welder's Qualification Statement				X				
06 03 12	1.5A	Product Data							X	
	1.5B	Shop Drawings						X		
	1.5C	Samples for Initial Selection					X			
	1.5D	Samples for Verification					X			
	1.6A	Preconstruction Test Reports	X							
	1.6B	Wood Species analysis	X							
	1.6C	Qualification Statement				X				
	1.6D	Wood Historic Treatment Program		X						
06 10 00	1.2A	Product Data	X							
	1.3A	Material Certificates	X							
06 13 13	1.6A	Product Data							X	
	1.6B	Shop Drawings						X		
	1.6C	Samples					X			
	1.7A	Qualification Data				X				
	1.7B	Preconstruction Test Reports	X							
	1.7C	Quality Control Program		X						
	1.7D	Log Historic Treatment Program		X						
06 16 00	1.2A	Product Data	X							
	1.3A	Material Certificates	X							
06 18 00	1.2A	Product Data	X							
	1.3A	Certificates of Conformance	X							
06 20 00	1.4A	Product Data							X	
	1.4B	Shop Drawings						X		
	1.4C	Samples					X			
	1.4D	Certificate	X							
	1.3A	Shop Drawings						X		
	1.3B	Product Data							X	

SUBMITTAL			REQUIREMENTS (Indicate with **X**)							
			INFORMATIONAL				ACTION			
SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
06 41 00	1.3C	Samples for selection, countertop material					X			
	1.3D	Samples of cabinet construction					X			
	1.3E	Samples of accessories					X			
	1.3F	Centerline of Blocking Diagram				X				
06 42 00	1.3A	Shop Drawings						X	X	
	1.3B	Samples of finished plywood					X			
	1.3C	Samples of solid wood tongue and groove material					X			
07 01 50.19	1.5A	Product Data							X	
	1.5B	Shop Drawings						X		
	1.5C	Materials Removal Company Qualification Statement				X				
07 11 13	1.3A	Product Data							X	
07 21 00	1.4A	Product Data							X	
	1.4B	Manufacturer's Certificate	X							
07 25 00	1.5A	Product Data							X	
	1.5B	shop Drawings						X		
07 31 13	1.4A	Product Data							X	
	1.4B	Shop Drawings						X		
	1.4C	Samples					X			
	1.4D	Manufacturer's Installation Instructions			X					
	1.4E	Manufacturer's Certificate	X							
	1.4F	Product Test Report	X							
	1.4G	Report of condition of deck following demolition of existing roofing and related substrates	X							
	1.4H	Warranties: Sample of special warranties.				X				
07 46 23	1.4A	Product Data							X	
	1..4B	Samples					X			
	1..4C	Installer's Qualification statement	X							
07 62 00	1.4A	Product Data							X	
	1.4B	Shop Drawings indicating proposed profiles, finish, & fastening.						X		
	1.4C	Samples for selection.					X			
	1.4D	Samples for verification.					X			
07 90 05	1.5A	Preinstallation Field Test Reports	X							
	1.5B	Compatibility and Adhesion Test Reports	X							
	1.5C	Product Data							X	
	1..5D	Samples					X			
08 03 14	1.6A	Product Data							X	
	1.6B	Shop Drawings						X		
	1.6C	Samples for initial Selection					X			
	1.6D	Samples for Verification					X			
	1.7A	Preconstruction Test Reports	X							
	1.7B	Qualification Statements				X				
	1.7C	Wood Door Historic Treatment Program		X						
08 11 33	1.4A	Product Data							X	

SUBMITTAL			REQUIREMENTS (Indicate with **X**)							
			INFORMATIONAL				ACTION			
SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
08 14 33	1.4A	Product Data							X	
	1.4B	Shop Drawings						X		
	1.4C	Samples					X			
08 31 00	1.2A	Product Data							X	
	1.2B	Shop Drawings						X		
	1.2C	Manufacturer's Installation Instructions			X					
08 36 13	1.3A	Product Data							X	
	1.3B	Shop Drawings						X		
	1.3C	Samples for verification					X			
	1.3D	Manufacturer's Installation Instructions			X					
08 52 00	1.4A	Product Data							X	
	1.4B	Shop Drawings						X		
	1.4C	Samples f					X			
	1.4D	Operational Hardware					X			
	1.4E	Grade Substantiation				X				
	1.4F	Field Quality Control Submittals				X				
08 71 00	1.5A	Product Data							X	
	1.5B	Samples					X			
	1.3C	Hardware Schedule								X
	1.3D	Manufacturer's Installation Instructions			X					
	1.3D	Maintenance Data			X					
08 80 00	1.3A	Product Data on Glass Types							X	
	1.3B	Product Data on Glazing Compounds							X	
	1.3C	Samples					X			
08 95 16	1.3A	Product Data							X	
	1.3B	Samples					X			
	1.4A	Evaluation Reports	X							
	1.4B	Sample Warranties				X				
09 21 16	1.4A	Product Data							X	
09 30 00	1.3A	Product Data							X	
	1.3B	Shop Drawings						X		
	1.3C	Samples					X			
	1.3D	Installers Qualification Statement				X				
09 64 29	1.3A	Product Data							X	
	1.3B	Shop Drawings						X		
	1.3C	Samples					X			
	1.3D	Installation Instructions				X				
09 65 00	1.3A	Product Data							X	
	1.3B	Shop Drawings						X		
	1.3C	Samples					X			
	1.5A	Product Data							X	

SUBMITTAL			REQUIREMENTS (Indicate with **X**)							
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SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
09 91 13	1.5B	Samples					X			
	1.5C	Manufacturer's Instructions:			X					
	1.5D	Maintenance Data:			X					
09 91 23	1.4A	Product Data							X	
	1.4B	Samples					X			
	1.4C	Manufacturer's Instructions:			X					
	1.4D	Maintenance Data:			X					
10 14 00	1.2A	Schedule								X
	1.2B	Shop Drawings						X	X	
	1.2C	Material and Finish Samples					X			
	1.2D	Sign Samples					X			
	1.2E	Written Warranty				X				
10 28 00	1.4A	Product Data							X	
10 44 00	1.3A	Product Data							X	
	1.3B	Copies of Warrantees				X				
11 30 13	1.3A	Product Data							X	
12 36 00	1.4A	Product Data							X	
	1.4B	Shop Drawings						X		
	1.4C	Samples					X			
	1.4D	Test Reports	X							
	1.4E	Certificate	X							
	1.4D	Installer's qualification statement.				X				
220500	2.1	Dielectric Fittings			X					
220500	2.2	Sleeves			X					
220500	2.3	Sleeve-Seal Systems			X					
220500	2.6	Escutcheons			X					
220519	2.1	Thermometers			X					
220519	2.2	Thermowells			X					
220519	2.3	Pressure Gages			X					
220519	2.4	Gage Attachments			X					
220523	2.2	Ball Valves			X					
220523	2.3	Swing Check Valves			X					
220529	2.1	Metal Pipe Hangers & Supports			X					
220529	2.2	Trapeze Pipe Hangers			X					
220529	2.3	Thermal-Hanger Shield Inserts			X					
220529	2.4	Fastener Systems			X					
220529	2.5	Pipe Positioning Systems			X					
220529	2.6	Equipment Supports			X					
220529	2.7	Miscellaneous Materials			X					
220548	2.1	Shop Drawings		X				X		
220553	2.1	Equipment Labels			X					

SUBMITTAL			REQUIREMENTS (Indicate with **X**)							
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SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
220553	2.2	Warning Signs & Labels			X					
220553	2.3	Pipe Labels			X					
220553	2.4	Valve Tags			X					
220716	2.1	Insulation Materials			X					
220716	2.2	Insulating Cements			X					
220716	2.3	Adhesives			X					
220716	2.4	Mastics			X					
220716	2.5	Sealants			X					
220716	2.6	Factory-Applied Jackets			X					
220716	2.7	Field-Applied Jackets			X					
220716	2.8	Tapes			X					
220716	2.9	Securements			X					
220716	2.10	Corner Angles			X					
220716	2.11	Protective Shielding Guards			X					
221116	2.1	Piping Material & Fittings			X					
221119	2.3	Vacuum Breakers			X					
221119	2.4	Backflow Preventers			X					
221119	2.5	Water Pressure-Reducing Valves			X					
221119	2.6	Temperature-Actuated, Water Mixing Valves			X					
221119	2.7	Strainers for Domestic Water Piping			X					
221119	2.8	Drain Valves			X					
221119	2.9	Water-Hammer Arresters			X					
221316	2.3	Piping Material & Fittings			X					
221316	2.4	Piping Material & Fittings			X					
221316	2.5	Piping Material & Fittings			X					
221319	2.2	Cleanouts			X					
221319	2.3	Misc. Sanitary Drainage Piping Specialties			X					
221429	2.1	Submersible Sump Pump			X					
223300	2.1	Water Heater			X					
223300	2.2	Water Heater Accessories			X					
224100	2.1	Plumbing Fixtures			X					
230548	2.1	Shop Drawings		X				X		
230553	2.1	Equipment Labels			X					
230553	2.2	Warning Signs & Labels			X					
230713	2.1	Insulation Materials			X					
230713	2.2	Adhesives			X					
230713	2.3	Mastics			X					
230713	2.4	Sealants			X					
230713	2.7	Tapes			X					
233113	2.1	Metal Ducts			X					
233113	2.4	Sealant and Gaskets			X					

SUBMITTAL			REQUIREMENTS (Indicate with **X**)							
			INFORMATIONAL				ACTION			
SPECIFICATION SECTION	PARAGRAPH NUMBER	DESCRIPTION	CERTIFICATIONS OR LABORATORY TESTS	REPORTS OR CALCULATIONS OR PLAN	MANUFACTURER DATA AND INSTRUCTIONS	OTHER	SAMPLES	SHOP DRAWINGS	MANUFACTURER DATA AND INSTRUCTIONS	OTHER
233113	2.5	Acoustic Liner			X					
233113	2.6	Hangers and Supports			X					
233300	2.3	Backdraft Dampers			X					
233300	2.4	Manual Volume Dampers			X					
233300	2.5	Flange Connectors			X					
233300	2.6	Turning Vanes			X					
233300	2.7	Duct-Mounted Access Doors			X					
233300	2.8	Flexible Connectors			X					
233423	2.1	Ceiling-Mounted Ventilators			X					
238231	2.1	Heaters			X					
260519	1.2	Electrical Power Conductors and Cables			X					
260526	1.3	Grounding and Bonding			X					
260529	1.2	Electrical Hangers and Supports			X					
260533	1.2	Raceways and Boxes			X					
260544	1.2	Sleeves and Sleeve Seals			X					
262416	1.4	Panelboards			X			X		
262726	1.4	Wiring Devices			X					
262816	1.3	Enclosed Switches			X					
264313	1.2	Surge Protection			X					
265100	1.4	Lighting			X					
270528	1.2	Communications Pathways			X					
284621	1.5	Fire Alarm System			X			X		
310500	1.3	Geotextile Fabric			X			X		
311000	1.1	Clearing and Grubbing Plan		X						
312000	1.1	Excavation and Shoring Plan		X						
312113	2.3	PVC Piping			X					
312113	2.4	Perforated Piping			X					
312320	1.3	Backfill Materials and Corresponding Proctors							X	
312500	1.5	SWPPP, NOI		X	X					
321243	1.4a	Samples					X			
	1.4b	Product Data		X						
	1.4c	Manufacturer's Installation Instructions			X					
	1.4d	Equipment List				X				
	1.4e	Field quality control compaction reports	X							
321540	1.4b	Material Analysis						X		
	1.4c	Samples					X			
	1.4d	Soil Stabilizer product data		X						
	1.4e	Compaction Testing	X							
	1.4f	Installer Qualification Data	X							
321540	2.2	Landscape Edgings						X		
321723	2.1	Pavement Marking Paint						X		

SUBMITTAL			REQUIREMENTS (Indicate with **X**)							
			INFORMATIONAL				ACTION			
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322000	1.4	Tree Protection Plan		X				X		
329300	1.4a	Product Data		X						
	1.4b	Samples					X			
	1.5a	Product certificates	X							
	1.5b	Sample warranty				X				
	1.6a	Maintenance Data			X					
329113	1.4b	Field quality-control reports		X						
	1.4c	Product Data			X					
	1.4d	Weekly Inspection and Maintenance Reports		X						
	1.4e	Obstruction Below Ground Reporting		X						
	1.4f	Soil Preparation Damage Reporting		X						
	1.4g	Materials List				X				
	1.4h	Construction Equipment Clearing Records		X						
329119	1.5b	Field quality-control reports		X						
	1.5c	Topsoil Schedule			X					
	1.5d	Topsoil Stockpile and Maintenance Plan		X						
	1.5e	Obstruction Below Ground Reporting		X						
	1.5f	Topsoils Damage Reporting		X						
	1.5g	Construction Equipment Cleaning Records				X				
331415	2.2	Gravity sewer pipe and fittings						X		
331415	2.2	Force main piping and fittings						X		
331415	2.2	Lift Station enclosure and piping						X		
331415	2.2	Small packaged grinder/lift station						X		
331415	2.2	Concrete manholes						X		
331415	2.1	Water main and fittings						X		
331415	2.1	Water valves						X		
331415	2.1	Water meter and vault						X		
332000	1.1	Wet Well						X		
333000	1.1	Lift Station Duplex Pump			X			X		

GRTE 237343 Rehabilitate "4 Lazy F Ranch" Cabins for Housing

SECTION 01 35 13.05 – ENVIRONMENTAL PROTECTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section consists of protecting natural resources in and around the construction site.

B. Work includes requirements for wildlife protection, inspection and maintenance of equipment, vegetation/site protection, and pollution prevention.

C. The Government will monitor compliance with this Section throughout construction.

1.2 DEFINITIONS

A. Environmentally Sensitive Area (ESA): Areas designated by the Government requiring special protection, including wetlands, waterways, cultural resources, rare plant habitats, sage-grouse core areas, ungulate ranges, and raptor nesting areas.

B. Daily Work Schedule (DWS): Contractor-prepared document detailing construction activities, methods, and equipment within ESAs.

C. Bear-Resistant Container (BRC): Container certified by the Interagency Grizzly Bear Committee (IGBC) for secure food and garbage storage.

D. Wildlife-Friendly Fencing: Fencing meeting Wyoming Game and Fish Department/NPS standards for ungulate passage, including height and wire spacing requirements.

E. Spill Kit: A readily accessible collection of materials (absorbents, pads, containment berms, PPE) for immediate containment and cleanup of hazardous material spills.

F. Aquatic Invasive Species (AIS): Non-native aquatic organisms identified by the State of Wyoming and NPS as requiring inspection and decontamination prior to water body entry.

1.3 SUBMITTALS

A. Daily Work Schedule (DWS): Submit one week prior to work in ESAs; include start/end dates, temporary facilities, types of construction, equipment, methods, and contingency plans.

B. Spill Prevention and Response Plan: Submit prior to mobilization; identify prevention measures, fueling/containment areas, response procedures, and reporting requirements.

C. Wildlife and Vegetation Protection Plan: Submit locations of protective fencing, staging areas, and sensitive resource avoidance measures.

D. Material Source Certification: Submit documentation verifying sand, gravel, rock, and fill are from approved weed-free sources.

1.4 QUALITY ASSURANCE

A. At least one week before on-site work begins, Contractor shall meet with Contracting Officer and Contracting Officer to discuss Daily Work Schedule, equipment, and special methods used in environmentally sensitive areas. Contractor shall ensure approved Daily Work Schedule is followed throughout construction.

PART 2 - PRODUCTS

2.1 WILDLIFE FRIENDLY FENCING

A. As Specified in Section 01 50 00 Temporary Facilities and Controls

2.2 BEAR-RESISTANT CONTAINERS

2.3 DAILY WORK SCHEDULE

A. Daily Work Schedule is required for work occurring within environmentally sensitive areas. Include all work that is to occur within the area and key the schedule to the drawings to include:

1. Starting and ending dates of ground-disturbing construction.
2. Locations of temporary facilities, such as barriers, field offices, staging areas, sanitary facilities, borrow pits, and haul and access roads.
3. Types of construction, such as clearing, topsoil stripping, structure or trench excavation, landscaping, and post construction clean-up.
4. Methods and equipment used for each type of construction.
5. Plan for relocating work in the event of temporary work stoppages at each environmentally sensitive area

PART 3 - EXECUTION

3.1 BARRICADES

A. Comply with requirements specified in Section 01 50 00 "Temporary Facilities and Controls."

3.2 WILDLIFE PROTECTION

A. Do not disturb any wildlife species (amphibians, reptiles, migratory birds, mammals, raptors, or bats) found nesting, hibernating, estivating (in an inactive dormant state during hot, dry periods), or otherwise living in, or immediately nearby, worksites. Immediately report any observations of wildlife in proximity to work areas. This ensures compliance with wildlife protection measures and reduces risk of unintentional disturbance to sensitive animals.

1. Projects should aim to remove hazardous trees in the Fall (October for this project) to avoid greater impacts to wildlife.
2. Prior to anticipated construction work, buildings scheduled for work shall be effectively bat-proofed between April 1 and April 15 of each year prior to anticipated construction. All exclusion materials shall remain in place for as long as practicable.

B. The potential for special status species in or near the area of the proposed activity exists. Work will cease if a special status species is discovered in the project area, until park staff re-evaluate the project. Protective measures, including potential modification of the work or the work schedule, could be determined necessary.

1. Ensure that all mitigations/conservation measures determined through Endangered Species Act Section 7 consultation with the U.S. Fish and Wildlife Service (USFWS) are followed.
2. In circumstances when it is deemed necessary to conduct activities near sites known to support threatened or endangered species, such work will be performed in a way (specified by the park biologist) to minimize impacts to the listed species (e.g., working quietly on-site, and minimizing time in or near habitats enroute to their work sites).

C. The Contractor shall comply with closures for raptors, swans, sage-grouse, wolves, and other species as specified.

D. The Contractor shall restrict work hours as follows – 30 minutes after sunrise to 30 minutes before sunset. During the Elk rut (Sep 1–Dec 1): Limit work hours to 8 a.m.–6 p.m.

E. The Contractor shall take precautions related to construction in and adjacent to elk parturition (calving) range (May 15–Jun 30) to minimize disturbance during sensitive wildlife periods. Minimize and consolidate vehicle trips on the access road during the calving season (May 15–Jun 30) to reduce noise, visual disturbance, and the risk of vehicle-wildlife interactions in areas used by calving elk.

1. Schedule the loudest and most disruptive construction or rehabilitation activities outside of May 15–June 30. When such activities must occur during the calving season, they will be scheduled for mid-day to avoid peak elk activity periods, minimizing disturbance during sensitive times such as dawn and dusk when cows are more likely to move with calves.
2. If active calving or concentrated elk use is observed near work areas, additional buffers, temporary work pauses, or modifications to activities will be required as directed by the CO and in coordination with wildlife managers.
3. At the direction of the CO and as required by wildlife managers employ temporary visual screening to reduce noise and visual disturbance to adjacent calving areas. Limiting visual cues of human presence helps cows and calves remain undisturbed in nearby habitat.
4. All personnel shall not approach elk or calves and shall immediately report any observations of wildlife in proximity to work areas.
5. No staging areas, equipment laydown, fueling, or material stockpiling shall occur within elk parturition range during the calving period (May 15–June 30).

F. Seasonal closures in the Superintendent's Compendium shall be observed.

G. Eagles are specifically protected under the Bald and Golden Eagle Protection Act of 1940 (16 U.S.C. 668-668c) and the MBTA. Project activities must not lead to the take of bald or golden eagles. The Bald and Golden Eagle Protection Act defines "take" to include disturbing birds.
   1. A ½ mile buffer of no work shall be maintained around occupied bald eagle nests between February 1 and August 15.
   2. It is the responsibility of staff or the contractor to report any eagle activity in the vicinity of proposed activity to park biologists in a timely way so that they may assess whether additional mitigation measures are needed to comply with the BGEPA and MBTA.

H. Contractor shall comply with the Migratory Bird Treaty Act (16 U.S.C. 703–712) and Executive Order 13186. Vegetation clearing, tree felling, fill placement, and excavation shall be scheduled outside of nesting season (May 1–August 1). If work must occur during nesting season, NPS biologists shall conduct surveys prior to work. Provide one week notice prior to vegetation removal to ensure timing of the survey can be met. If active nests are discovered, work shall cease until fledging occurs or NPS biologists approve mitigation. Surveys by NPS biologists are valid for 14 calendar days. Surveys shall be recompleted if vegetation removal is not completed within this time frame.
   1. All project activities must also comply with GRTE's Superintendent's Compendium seasonal closure regulations for raptors, trumpeter swans, great blue herons, and other sensitive bird species.
   2. In general, park biologists recommend that to prevent impacts to nesting migratory birds and to avoid project delays, schedule work involving vegetation clearing, tree felling, fill placement, excavation, or other construction activities for outside of the nesting season. The breeding season is generally as follows for migratory songbirds (May 1 to August 1), or as dictated by nesting chronology; therefore, the above work activities are best outside this time period.
   3. Before commencement of any activities that involve removal or manipulation of vegetation including large trees, grasses, and shrubs during the breeding season (see above) contact park biologists to schedule a survey for nesting birds. Surveys must be conducted by qualified personnel before vegetation removal and/or ground disturbing activities begin. Prior to March 1 (the year of the proposed work), schedule NPS surveys to occur within 2 weeks prior to disturbance.
   4. Inspect hazard trees for potential nest sites. Fell hazard trees that may contain nests outside the breeding season, if possible, unless there is an imminent threat to human health and safety.
   5. Work must be completed within two weeks of the nesting bird survey. If this is not possible, another survey must be scheduled with park biologists.
   6. Active bird nests located during surveys will be protected until nestlings fledge or the nest fails. Park biologists will monitor nests, determine mitigations, and provide updates to the project leader on nesting status.
   7. It is the responsibility of staff or the contractor to report any nesting bird activity in the vicinity of proposed activity to park biologists in a timely way so that they may assess whether additional mitigation measures are needed to comply with the MBTA.

I. The contractor shall adhere to the State of Wyoming Executive Order (Order 2019-3); Greater Sage-grouse Core Area Protection. This project DOES fall within sage-grouse core area.
   1. No shrub-steppe habitat shall be removed within 4 miles of occupied leks between March 15 and June 30.

2. Construction noise shall not exceed 10 decibels above ambient levels at the perimeter of leks between 6:00 p.m. and 8:00 a.m. from March 1 to May 15.
3. Seasonal lek closures shall be enforced as designated in the Superintendent's Compendium.
4. The Contractor shall bury or raptor-proof new power lines within 0.6 miles of leks.
5. The Contractor shall provide 0.5-mile buffers around eagle nests (Feb–Aug).

J. For building work, the NPS will conduct preconstruction bat surveys and may adjust work schedules if roosts are found throughout the duration of the project

K. The Contractor, subcontractor and all employees will be required to attend Park and bear orientation meetings presented by the National Park Service prior to beginning work. Notify the CO 14 days prior to beginning work so orientations can be scheduled.

L. Do not feed or disturb wildlife. To prevent unnatural attractants to wildlife, keep all foods, pet foods, garbage, drinks, trash, or food and drink containers inside vehicles, or enclosed construction equipment, except during actual use. Supply garbage bins, within the construction site that are animal proof. Remove all garbage and food trash from the project site daily.

M. Any mishandling of garbage, trash, food, and other potential bear attractants described above will result in the responsible person, and/or Contractor, receiving a citation which may include a fine. Remove from the project any employee who receives more than one citation for a food storage violation.

N. Immediately report all bear and wolf sightings to the Teton Interagency Dispatch Center (TIDC) at (307)739-3301. Temporarily suspend construction activities according when a migratory bird nest is discovered or identified or when a grizzly bear or wolf comes near an active construction area and creates a potential animal/human conflict. Immediately notify the CO and TIDC for resolution of any potential bear/human conflicts or of actual bear/human confrontations.

O. All project activities must comply with
1. Conservation recommendations (CR) to protect grizzly bears provided in the USFWS 2016 biological opinion on the Historic Properties Management Plan and,
2. the GRTE Superintendent's Compendium regulations related to food storage and park recommended best management practices for living and working in bear country.

P. Provide for proper storage and disposal of materials that may be toxic to bears. All potentially toxic attractants, including petroleum products, must be stored or disposed of in such a way that they are not available to bears.

Q. The Contractor shall fence open excavations and provide escape ramps for wildlife as requested by CO.

R. Bat Exclusion:
1. General requirements:
Contractor shall implement proactive measures to ensure that bats are not actively roosting within buildings or building sites prior to commencement of and during rehabilitation activities.
2. Seasonal Bat-Proofing Requirements:
Prior to anticipated construction work, Buildings shall be effectively bat-proofed between

April 1 and April 15 of each year prior to anticipated construction. All exclusion materials shall remain in place for as long as practicable.

a. Contractor is responsible for providing their own access across existing roadways and shall perform plowing or otherwise traverse snowy or adverse conditions as necessary.
b. Bat exclusion work may also take place after October 1, if an NPS biologist confirms there is no active bat activity through requested monitoring.

3. Protection of Historic Fabric
Materials used for bat exclusion shall not damage or adversely affect historic fabric that is designated to remain. Any proposed adhesives, sealants, or other attachment methods intended for use on historic materials must receive prior approval from the NPS Project Manager.
4. Exclusion Materials and Installation
Buildings may be wrapped in bat-exclusion netting or breathable, lightweight canvas.
   a. Bat-exclusion netting shall have maximum hole apertures of 1/6 inch by 1/6 inch.
   b. Exclusion materials shall be securely affixed to roofs, fascia, and all protruding elements as necessary to prevent damage or displacement caused by wind or other weather conditions.
   c. Exclusion materials extending to the ground plane shall be secured with sandbags or other materials that ensure bats do not enter from the ground.
   d. All exclusion material sections shall be joined to ensure no gaps larger than 1/6 inch by 1/6 inch and shall be secured to both the structure and adjacent ground.
5. Main Lodge – Known Maternal Bat Roost
The Main Lodge is known to contain a large maternal bat roost.
   a. At minimum, exclusion materials shall cover all building facades from the roof eaves to the ground and must be adequately secured at all building edges and along the ground plane.
   b. All roof penetrations, including stone chimneys, shall be fully enclosed using appropriate exclusion materials.
   c. An entry point shall be constructed within the exclusion material that allows controlled access into the building.

3.3 INSPECTION AND MAINTENANCE OF EQUIPMENT

A. All vehicles and equipment including office trailers that enter the park and are staged in association with this project, will be inspected by the CO prior to their entry into the Park for mud, weeds, leaks, and other unwanted substances. Steam clean or power wash all earth-moving equipment (including hauling vehicles), inside of cabs and pile driver equipment so that it is clean of all mud and weed seed before their initial entry into the Park. Subsequent entries of hauling vehicles will not require cleaning unless requested by the CO. Notify the CO a minimum of 48 hours prior to the entry of vehicles and equipment.
   1. The spread of exotic invasive plant species in the Park is a serious concern, and no equipment will be allowed to offload, nor remain within the Park, if dirt or other contaminants with the potential to harbor seeds are apparent. Any equipment which departs the project site (excluding hauling vehicles) must repeat an identical cleaning and inspection process, including formal acceptance by the CO prior to being remobilized and offloaded at the project site.
   2. Notify the CO a minimum of 48 hours prior to the entry of vehicles and equipment. The CO will make arrangements for the inspections.
   3. Prior to offloading of any equipment, obtain verbal approval from the CO.

4. Clean hauling vehicles before their initial entrance into the Park; subsequent entries will not require cleaning or inspection unless equipment is used at another job site or as requested by CO.
5. Any equipment that is unacceptable to the CO shall be immediately removed from the Park.

B. Any equipment entering water bodies shall be inspected for Aquatic Invasive Species by NPS. Notify the CO a minimum of 48 hours prior to water body inspection.

C. Do not operate or park any equipment off the paved roadway except in designated areas. Do not operate or store vehicles and equipment leaking oil, gas, or anti-freeze within the Park or National Forest boundaries. Do not drain oil, hydraulic fluids, anti-freeze, or other chemicals onto the ground within the Park or National Forest boundaries.

D. The Contractor shall inspect all equipment for fluid leaks prior to use. The contractor shall not use equipment that is leaking fluids or is broken. Please see section 01 50 00

E. The Contractor shall fuel only in bermed and lined containment areas.

F. The Contractor shall provide and maintain spill kits at fueling locations.

G. The Contractor shall clean up equipment spills immediately.

3.4 NOISE CONTROL

A. The Contractor shall not idle unused equipment for a duration longer than 15 minutes when weather conditions are above 50 degrees F.

B. The Contractor shall maintain functioning mufflers on all equipment.

C. The Contractor shall prohibit radios and loudspeakers on the construction site.

D. The Contractor shall use broadband or directional backup alarms.

E. The Contractor shall precut and prefabricate materials off-site to minimize chainsaw or loud equipment use in sensitive areas.

3.5 VEGETATION / SITE PROTECTION

A. The Contractor shall use wooden mats or track mats for small entry areas to protect vegetation. No entry areas shall be permitted outside of the designated disturbance area(s).

B. Except in areas indicated on the drawings or specified to be cleared, the Contractor shall not remove, cut, deface, injure, or destroy resources including trees, shrubs, vines, grasses, topsoil, and landforms without approval. No ropes, cables, or guys shall be fastened to or attached to any trees for anchorage unless specifically authorized.

C. The CO may require installation of protective barriers around areas that require special attention, such as staging areas, protected trees and plants, edges of waterways, aquatic habitats, wetlands, sensitive wildlife habitats, or cultural resources.

D. If fencing (snow or silt), barriers, staking, or marking which designates clearing limits or vegetation is damaged, knocked down or destroyed, suspend work in the area until the fencing or other protection is repaired to the satisfaction of the CO at the Contractors expense. If any of the landscape, outside of the construction limits or within the construction limits, that is designated to be protected or to remain is disturbed, the Contractor shall restore, mitigate, or compensate the area as determined by the CO, at the expense of the Contractor.

E. Restore all areas used as staging and/or stockpiling areas to their pre-construction condition at no cost to the Government.

F. All vegetation modification, including tree removal, at National Register of Historic Places -listed or -eligible properties shall be reviewed and approved by the Section 106 specialists (as well as by wildlife and vegetation specialists).. Any vegetation modification or tree removal that varies from the approved plans and specifications will require further review.

3.6 SUPPLY OF MATERIALS

A. Before delivery of rock, sand, aggregate, gravel, soil or other natural material to the project from a material source outside the project area:

B. Obtain clearance or eradicate exotic plants by:
   1. Providing Park approved or county weed district approved source. Teton County Weed and Pest has currently approved as weed-free the various Evans Construction pits in Teton County. (Approval of sources even with a certification by a county extension agent, agronomist, or other qualified invasive species expert that the source is free from exotic plants is NOT guaranteed); or
   2. Notifying the CO at least 1 month before beginning operations or starting crushing to allow for investigation for exotic plants. If exotics are found, the CO will determine if the upper portion of the source will need to be stripped or sprayed with an herbicide approved by the CO. If spraying is required, provide a licensed operator to spray according to applicable state regulations. Do not spray any herbicides until approved in writing by the CO. Spraying or stripping of material does not, necessarily, constitute approval.
   3. If exotic plant clearance is not obtained, heat all material to 150 degrees Centigrade to ensure sterilization of any exotic plants before delivery of uncertified material. Do not perform hearing operations within park boundaries. Do not stockpile the heated material outside the park boundaries prior to delivery. These requirements are waived if the contractor uses a source that meets the requirements stated in items (1) and (2) above.

3.7 WORK STOPPAGE

A. Contractor shall plan, schedule, and execute work to prevent stoppages at one area from stopping all work at construction site.

END OF SECTION 01 35 13.05

GRTE 237343 Rehabilitate "4 Lazy F Ranch" Cabins for Housing

SECTION 01 35 13.22 – ARCHEOLOGICAL PROTECTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section consists of protecting archeological resources contained in soil deposits.

1.2 DEFINITIONS

A. Archeological Resources: Archeological resources are physical evidences of past human activity, including evidences of effects of that activity on the environment. Archeological resources represent both prehistoric and historic time periods, found above and below ground and under water.

B. Archeologically Sensitive Areas: Areas having potential to contain significant (National Register eligible) archeological resources. If National Register eligible or listed archeological resources could not be avoided, an appropriate mitigation strategy would be developed in consultation with state historic preservation officer and, if necessary, associated American Indian tribes.

C. Non-sensitive Areas: Areas with little potential of containing significant (National Register eligible) archeological resources.

D. Archeological Monitor: Representative of Government designated to oversee construction activities that could disturb archeological resources.

E. Archeological Resources Protection Act (ARPA) of 1979 (Public Law (P.L.) 96-95; 93 United States Statues at Large (Stat.) 712): defines archeological resources as any material remains of past human life or activities that are of archeological interest and at least 100 years old; Section 4 of the statute describes the requirements that must be met before Federal authorities can issue a permit to excavate or remove any archeological resource on Federal or Indian lands; the curatorial requirements of artifacts, and other materials excavated or removed.

1.3 SUBMITTALS

A. Daily Work Schedule: Detail construction work in archeologically sensitive areas. Submit to Contracting Officer (CO) 30 days before start of ground disturbing site work.

1.4 QUALITY ASSURANCE

A. At least one week before on-site work begins, Contractor shall meet with Contracting Officer and Archeological Monitor to discuss Daily Work Schedule, equipment, and special methods used in

archeologically sensitive areas. Contractor shall ensure approved Daily Work Schedule is followed throughout construction.

PART 2 - PRODUCTS

2.1 DAILY WORK SCHEDULE

A. Daily Work Schedule is required for work occurring within archeologically sensitive areas. Include all work that is to occur within the area and key the schedule to the drawings to include:

1. Starting and ending dates of ground-disturbing construction.
2. Locations of temporary facilities, such as barriers, field offices, staging areas, sanitary facilities, borrow pits, and haul and access roads.
3. Types of construction, such as clearing, topsoil stripping, structure or trench excavation, landscaping, and post construction clean-up.
4. Methods and equipment used for each type of construction.
5. Plan for relocating work in the event of temporary work stoppages at each archeologically sensitive area

PART 3 - EXECUTION

3.1 BARRICADES

A. Comply with requirements specified in Section 01 50 00 "Temporary Facilities and Controls."

3.2 ARCHEOLOGICAL INVESTIGATION BY NON-NPS PERSONNEL

A. A permit is required for archeological investigations (e.g. excavation, shovel testing, coring, pedestrian survey, underwater archeology, rock art documentation, or other types of reconnaissance including archaeological monitoring of construction) carried out on parklands by non-National Park Service (NPS) personnel, unless carried out under a contract or a cooperative agreement specifically written for archeological investigations. Permits are issued under the Archaeological Resources Protection Act of 1979 (ARPA). The NPS does not issue a permit for archeological investigations carried out by NPS archeologists, or to archeologists working on NPS archeological projects under a contract or cooperative agreement.

B. Applicants should submit a Permit Application (DI Form 1926 (Revision September 2004) Office of Management and Budget (OMB) Number (No.) 1024-0037, approved through 1/31/2008. Permit Application form is available, in PDF (portable document file) format, to the manager of the park in which they propose to work; or to the regional director, with a copy to the park manager.

3.3 OBSERVATION

A. No archeological monitoring is assumed to be needed for the project. If mitigation is needed, work shall stop and the NPS will address unanticipated discoveries within the Park capacity.

3.4 DISCOVERY OF RESOURCES

A. If Archeological Monitor discovers resources, immediate relocation of work to a non-sensitive area may be required for Monitor to identify and document resources and, if necessary, develop appropriate mitigation plan. While Archeological Monitor is documenting resources in sensitive areas, Contractor shall relocate work to non-sensitive areas where monitoring is not normally required.

B. If previously unknown archeological resources and/or human remains are discovered during project work, all work in the immediate vicinity (600 feet) of the discovery shall be halted until the resources are identified and documented and an appropriate mitigation strategy developed. Contact the park's Branch Chief of Cultural Resources (307-739-3671).

C. Federal law prohibits collection of artifacts or damage to archeological sites or historic properties in National Parks. The Government will impose penalties on individuals who break the law. Inform all contractors and subcontractors of the penalties for illegally collecting artifacts or intentionally damaging archeological sites or historic properties.

3.5 WORK STOPPAGE

A. Contractor shall plan, schedule, and execute work to prevent stoppages at one area from stopping all work at construction site.

END OF SECTION 01 35 13.22

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GRTE 237343 Rehabilitate "4 Lazy F Ranch" Cabins for Housing

SECTION 01 35 13.22 – ARCHEOLOGICAL PROTECTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes establishing an effective accident prevention program and providing a safe working environment for personnel and visitors.

1.2 CONDITIONS PRESENT FOR PROJECT

A. In addition to typical hazards associated with construction activities the following site specific hazards are present for the Project:
   1. Severe weather conditions including freezing temperatures and snow, lightning, hail, and high wind events
   2. Large animals including grizzly bears and moose
   3. Hazardous materials including asbestos and lead based paint
   4. Hantavirus
   5. Flooding

1.3 SUBMITTALS

A. Accident Prevention Plan (APP): Submit APP after contract award and before Pre-Construction conference. Contracting Officer (CO) will review proposed APP. If APP requires any revisions or corrections, Contractor shall resubmit Plan within 10 days. No progress payments will be made until the APP is accepted.

1.4 QUALITY ASSURANCE

A. Comply with contract clauses "Accident Prevention" and "Permits and Responsibilities." In case of conflicts between Federal, State, and local safety and health requirements, the most stringent shall apply. Onsite equipment shall meet 29 CFR 1926 (Code of Federal Regulations) (Occupational Safety and Health Administration (OSHA)) requirements. Failure to comply with requirements of this section and related sections may result in suspension of work.

B. Site Safety Supervisor:

   1. Designate authorized onsite representative for preparation and maintenance of the APP.
   2. Shall be responsible for:

      a. Implementation and enforcement of the APP
      b. Daily safety inspections
      c. Conducting and documenting weekly and monthly safety meetings

d. Review of safety requirements at progress meetings
e. Compilation and maintenance of Safety Data Sheets (SDS) and safety reference materials
f. Tracking and resolution of safety violations
g. Site personnel and visitor compliance with site safety and health requirements and APP
h. Investigation and reporting of accidents and injuries

C. Qualifications of Employees:

1. Physically and able to perform their assigned duties in a safe manner.
2. Do not allow employees whose ability or alertness is impaired because of prescription or illegal drug use, fatigue, illness, intoxication, or other conditions that may expose themselves or others to injury to perform work.
3. Provide operating instructions for equipment. Operators of vehicles, hoisting equipment, and hazardous plant equipment shall be able to understand signs, signals, operating instructions, and be fully capable of operating such equipment. Retain copies of operator licenses and certifications onsite.

1.5 ACCIDENT REPORTING

A. Reportable Accidents: Defined as: death, occupational disease, and/or traumatic injury to employees or the public; fires; and/or property damage by accident in excess of $100.

1. Notify Contracting Officer immediately in the event of a reportable accident.
2. Fill out and forward an Accident/Property Damage Report Form (CM-22) to Contracting Officer within 7 days of a reportable accident. Obtain form from Contracting Officer.

PART 2 - PRODUCTS

2.1 ACCIDENT PREVENTION PLAN (APP)

A. APP shall be written to comply with OSHA and project requirements (generic plan is not acceptable) including but not limited to:

1. Name and qualifications of supervisor responsible to carry out program.
2. Weekly and monthly safety meetings shall be documented with topics and attendees.
3. First aid and rescue procedures.
4. Job Hazard Analysis (JHA) for each major phase. List of hazards associated and methods proposed to provide for property protection and safety of the public, National Park Service personnel, and Contractor's employees. Include initial and continuing training.
5. Planning for possible emergency situations, as detailed in Article 1.2. Such planning shall take nature of construction, site conditions, and degree of exposure of persons and property into consideration.

2.2 FIRST AID FACILITIES

A. Provide adequate facilities for number of employees and appropriate to construction hazards.

2.3 PERSONNEL PROTECTIVE EQUIPMENT (PPE)

A. Selection shall conform to OSHA Subpart E.

PART 3 - EXECUTION

3.1 DAILY SAFETY INSPECTIONS

A. Conduct daily safety inspections and maintain daily safety reports which include:
   1. Area/operation inspected
   2. Date of inspection
   3. Identified hazards
   4. Corrective actions taken

3.2 EMERGENCY INSTRUCTIONS

A. Post telephone numbers and reporting instructions for ambulance, physician, hospital, fire department, and police in conspicuous locations at work site.

3.3 FIRE AND LIFE SAFETY

A. Comply with requirements of the 2024 International Fire Code Chapter 33 Fire Safety During Construction and Demolition.

3.4 HAZARDOUS MATERIALS

A. Hazardous materials: Explosive, flammable, poisonous, corrosive, oxidizing, irritating, or otherwise harmful substances that could cause death or injury.

B. Store hazardous materials in accordance with manufacturer's and OSHA Subpart D requirements. Maintain Safety Data Sheets (SDS) for each chemical readily available on site.

   1. Immediately report spills of hazardous materials to the Park.
   2. Maintain a spill emergency response kit.
   3. Train employees how to respond to a spill and use emergency response kit.

3.5 PROTECTIVE EQUIPMENT

A. Inspect personal protective equipment daily and maintain in a serviceable condition. Clean, sanitize, and repair personal items as appropriate before issuing to another individual.

B. Project requirements related to wildlife protection are described in Section 01 35 13.05 – Environmental Protection. Please refer to this section for required NPS briefings and safety considerations related to wildlife.
   1. It is recommended that staff and contractors carry bear spray when away from vehicles and/or buildings.

3.6 SAFETY MEETINGS

A. As a minimum, conduct one weekly 15-minute "toolbox" safety meeting conducted by a foreman or supervisor and attended by construction personnel at worksite. Topics shall coincide with work scheduled for following week. Document and submit meeting minutes to Contracting Officer within one day after meeting.

B. Conduct monthly safety meetings for personnel, contractors, and subcontractors performing work on the site. Notify Contracting Officer of meeting dates and times. Meetings shall be used to: review effectiveness of Contractor's safety effort; resolve current health and safety problems; provide a forum for planning safe construction activities, and for updating Accident Prevention Plan. Contracting Officers Representative will attend meetings and enter results of meetings into the daily log.

3.7 HARD HATS AND PROTECTIVE EQUIPMENT AREAS

A. A hard hat use area shall be designated by Contractor. Hard hat area shall be posted by Contractor in a manner satisfactory to Contracting Officer.

B. It is Contractor's responsibility to require persons working on or visiting site to wear hard hats and PPE in good repair at all times. As a minimum, maintain **six**hard hats and other APP required equipment.

3.8 TRAINING

A. First Aid: Provide training to personnel to ensure prompt and efficient first aid.

B. Hazardous Material: Train and instruct each employee exposed to hazardous material in safe and approved methods of handling and storage.

END OF SECTION 01 35 23

GRTE 237343 Rehabilitate “4 Lazy F Ranch” Cabins for Housing

SECTION 01 35 91 - HISTORIC PRESERVATION TREATMENT PROCEDURES

GENERAL

1.1 SUMMARY

A. Section includes special procedures for historic treatment on Project including, but not limited to:

Definitions

2. Submittals
3. Quality Assurance
4. Storage and protection of existing historic materials
5. Project site conditions
6. Historic Preservation Treatment Plan
7. Protection, General
8. Protection during application of chemicals
9. Protection during use of heat-generating equipment
10. Historic preservation treatment procedures

1.2 DEFINITIONS

"Preservation" - To apply measures necessary to sustain existing form, integrity, and materials of historic property. Work may include preliminary measures to protect and stabilize the property.

"Rehabilitation" - To make possible a compatible use for property through repair, alterations, and additions while preserving those portions or features that convey its historical, cultural, or architectural values.

C. "Restoration" - To accurately return form, features, and character of a property to its appearance at a particular period of time by means of removal of features from other periods in its history and repair and reconstruction of missing and deteriorated features from the restoration period.

D. "Reconstruction" - To reproduce in exact form and detail, a building, structure, or artifact as it appeared at a specific period in time. Reconstructed elements do not possess historic integrity in their own right since they are-not original fabric.

"Stabilize" - To apply measures designed to reestablish a weather-resistant enclosure and structural reinforcement of an item or portion of the building while maintaining essential form as it exists at present. This level of intervention is aimed at retarding or arresting adverse impacts to structures.

F. "Protect and Maintain" - To remove deteriorating corrosion, reapply protective coatings, and install protective measures such as temporary guards; to provide the least degree of intervention.

G. "Repair" - To stabilize, consolidate, or conserve; to retain existing materials and features while employing as little new material as possible. Repair includes patching, piecing-in, splicing, consolidating, or otherwise reinforcing or upgrading materials. Within restoration, repair also

includes limited replacement in kind, rehabilitation, and reconstruction, with compatible substitute materials for deteriorated or missing parts of features when there are surviving prototypes.

H. "Replace" - To duplicate in its entirety, a historic element or feature by matching its historic pattern, detail and appearance. Replacement is justified when original or historic elements are damaged beyond repair or are missing. Replacement conditions and methods include:

1. Replacement with Original or Historic Fabric: Includes fabric salvaged from other locations or projects having identical architectural qualities. Duplication of appearance using identical material possessing historical significance.
2. Replacement with New Materials: Includes replacement with new material of like kind (custom fabricated of manufactured). Duplication of appearance using like material.
3. Replacement with Substitute Materials: Includes replacement with a compatible substitute that is frequently contemporary and unlike the historic fabric. Duplication of appearance using modern (non-traditional) material Use of substitute materials is not approved unless matching materials are not available.

I. "Remove" - To demolish or detach items from existing construction and legally dispose of them off-site unless indicated to be removed and salvaged or removed and reinstalled.

J. "Remove and Salvage" - To detach items from existing construction and deliver them **ready for reuse** to Contracting Officer (CO) or designee.

K. "Remove and Reinstall" - To detach items from existing construction, repair and prepare for reuse, and reinstall where indicated.

L. "Existing to Remain" or "Retain" - Existing items of construction not to be removed and not otherwise indicated to be removed and salvaged or removed and reinstalled.

M. "Material in Kind" - Material that closely matches existing materials through comparison of architectural qualities and salient characteristic such as species, cut, color, grain, dimension, profile, thickness, and finish.

1.3 SUBMITTALS

A. Historic Preservation Treatment Plan:

1. After contract award and before Pre-Construction conference, submit for approval a written Historic Preservation Treatment Plan (HPTP). Approved HPTP required before work begins.
2. If the plan requires revisions or corrections, Contractor shall resubmit plan within 10 days.
3. No change in approved plan may be made without written concurrence by Contracting Officer.

B. Alternative Methods and Materials: If alternative methods and materials to those indicated are proposed for any phase of work, provide a written description including evidence of successful use on other, comparable projects, and program of testing to demonstrate effectiveness for use on this Project.

C. Photographs or Video: In accordance with Section 01 32 33 "Photographic Documentation," show existing conditions of adjoining construction and site improvements including finish surfaces that might be misconstrued as damage caused by historic treatment operations. Submit before work begins.

1.4 QUALITY ASSURANCE

A. Historic Preservation Treatment Specialist Qualifications: 5 year minimum experienced firm with required certifications and training able to demonstrate through past performance they are qualified to perform this work.

B. All work on or near historic buildings, structures, sites, and landscapes shall be conducted in accordance with the Secretary of the Interior's Standards for the Treatment of Historic Properties in consultation with the Branch Chief of Cultural Resources (307-739-3671). Any deviation from the approved drawings and specifications must be submitted for further review.

1.5 PRECONSTRUCTION MEETING

A. PRELIMINARY HISTORIC TREATMENT CONFERENCE: Preliminary Historic Treatment Conference: Before starting historic treatment work, conduct conference at Project site. a. Attendees: In addition to representatives of Owner, Architect, and Contractor, testing service representative, historic treatment specialists, chemical-cleaner manufacturer(s), and installers whose work interfaces with or affects historic treatment shall be represented at the meeting. Agenda: Discuss items of significance that could affect progress of historic treatment work, including review of the following:

1. Historic Treatment Subschedule: Discuss and finalize; verify availability of materials, historic treatment specialists' personnel, equipment, and facilities needed to make progress and avoid delays. Fire-prevention plan. Governing regulations. Areas where existing construction is to remain and the required protection. Hauling routes. Sequence of historic treatment work operations. Storage, protection, and accounting for salvaged and specially fabricated items. Existing conditions, staging, and structural loading limitations of areas where materials are stored. Qualifications of personnel assigned to historic treatment work and assigned duties.
2. Requirements for extent and quality of work, tolerances, and required clearances.
3. Methods and procedures related to historic treatments, including product manufacturers' written instructions and precautions regarding historic treatment procedures and their effects on materials, components, and vegetation.
4. Embedded work such as flashings and lintels, special details, collection of wastes, protection of occupants and the public, and condition of other construction that affect the Work or will affect the work. c. Reporting: Record conference results and distribute copies to everyone in attendance and to others affected by decisions or actions resulting from conference.

1.6 STORAGE AND PROTECTION OF HISTORIC MATERIALS

A. Removed and Salvaged Historic Materials:

1. Clean salvaged historic items.

2. Pack or crate items after cleaning. Identify contents of containers.
3. Store items in secure area until delivery to Contracting Officer.
4. Transport items to storage area at Park Headquarters Complex
5. Protect items from damage during transport and storage.
6. Do not dispose of items removed from existing construction without prior written consent of Contracting Officer.

B. Removed and Reinstalled Historic Materials:

1. Clean and repair historic items to functional condition adequate for intended reuse.
2. Pack or crate items after cleaning and repairing. Identify contents of containers.
3. Protect items from damage during transport and storage.
4. Reinstall items in locations indicated. Comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make item functional for use as designed.

C. Existing Historic Materials to Remain: Protect construction indicated to remain against damage and soiling during historic treatment. When permitted by Contracting Officer, items may be removed to suitable, protected storage location during historic treatment and cleaned and reinstalled in their original locations after historic treatment operations are complete.

D. Storage and Protection: When removed from existing location, store historic materials within weather-tight enclosure protected from wetting by rain, snow, or ground water, and temperature variations. Secure stored materials to protect from theft.

1. Identify removed items with an inconspicuous mark indicating original location.
2. Develop key plan when many similar items are scheduled for removal and reinstallation.

1.7 PROJECT-SITE CONDITIONS

Exterior Cleaning and Repairing:

1. Proceed with work only when forecasted weather conditions are favorable.

a. Wet Weather: Do not attempt repairs during rainy or foggy weather. Do not apply primer, paint, putty, or epoxy when relative humidity is above 80 percent. Do not remove exterior elements of structures when rain is forecast or in progress.
b. Do not perform exterior wet work when air temperature is below 40 degrees Fahrenheit (5 degrees Celsius).
c. Do not begin cleaning, patching, or repairing given likelihood of frost or freezing.
d. Do not begin cleaning when either air or surface temperature is below 45 degrees Fahrenheit (7 degrees Celsius) unless approved means are provided for maintaining 45 degrees Fahrenheit (7 degrees Celsius) temperature of air and materials during, and for 48 hours subsequent to, cleaning.

Perform cleaning and rinsing of the exterior only during daylight hours.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 HISTORIC PRESERVATION TREATMENT PLAN

A. Prepare written plan for preservation work covering preservation components of project. Plan shall verify construction strategy and intent is compatible with Department of the Interior's (DOI) standards for Treatment of Historic Properties, guidelines for Treatment of Cultural Landscapes, and National Park Service management policies for cultural resources. Plan shall satisfy both project scope and resource protection requirements. Plan shall include:

1. Organized list of preservation components of project, systems, and tasks
2. Staging and sequence of work
3. Structures stabilizing and lifting/moving approach
4. Disassembly and reassembly techniques and steps
5. Equipment and tools required
6. Supplies and materials with manufacturer or supplier identified including specific clean up/storage procedure including frequency and documentation of such.
7. Skilled trades and crafts required
8. Anticipated testing and analysis of fabric
9. Additional investigations for extents or magnitude of treatments needed
10. Protective measures
11. Seasonal limitations on work
12. Alternative means if primary treatment method is unfeasible
13. Work conducted off-site (Approval from Contracting Officer required prior to taking resources off-site).

3.2 PROTECTION, GENERAL

A. Comply with manufacturer's written instructions for precautions and effects of products and procedures on adjacent building materials, components, and vegetation.

B. Ensure supervisory personnel are present when work begins and during progress.

C. Temporary Protection of Historic Materials during Construction:

1. Protect existing materials during installation of temporary protections and construction. Do not deface or remove existing materials.
2. Attachments of temporary protection to existing construction shall be approved by Contracting Officer prior to installation.

D. Protect landscape work adjacent to or within work areas as follows:

1. Provide barriers to protect tree trunks.
2. Bind spreading shrubs.
3. Coverings shall allow plants to breathe. Remove coverings at end of day. Do not cover plant material with waterproof membrane more than 8 hours at a time.

4. Set scaffolding and ladder legs away from plants.

E. Existing Drains: Prior to start of work or cleaning operations, test drains and other water removal systems to ensure drains and systems function properly. Notify Contracting Officer immediately of stopped or blocked drains or systems. Do not begin Work of this Section until drains are in working order.

1. Provide method to prevent solids including stone or mortar residue from entering drains or drain lines. Clean out drains and drain lines blocked or filled because of work performed under this Contract.
2. Protect storm drains from pollutants. Block drains or filter out sediments, allowing only clean water to pass.

3.3 PROTECTION DURING APPLICATION OF CHEMICALS

A. Protect persons, motor vehicles, surrounding surfaces of building being restored, building site, plants, and surrounding buildings from harm or damage resulting from applications of chemical cleaners and paint removers.

B. Comply with requirements in Section 01 50 00 "Temporary Facilities and Controls."

C. Cover adjacent surfaces with materials proven to resist chemical cleaners selected for Project unless chemicals being used will not damage adjacent surfaces. Use covering materials containing only waterproof, UV (ultraviolet)-resistant adhesives. Apply masking agents to comply with manufacturer's written instructions. Do not apply liquid masking agent to painted or porous surfaces. Promptly remove masking to prevent adhesive staining on completion.

D. Do not clean surfaces during winds of sufficient force to spread cleaning solutions to unprotected surfaces.

E. Neutralize and collect alkaline and acid wastes and dispose of outside park boundaries.

F. Dispose of runoff from chemical operations by legal means and in a manner preventing soil erosion, undermining of paving and foundations, damage to landscaping, and water penetration into building interiors.

G. Document daily clean-up, proper storage, and disposition of any flammable/combustible materials with daily reports.

3.4 PROTECTION DURING USE OF HEAT-GENERATING EQUIPMENT

Comply with following procedures while performing work with heat-generating equipment, including welding, cutting, soldering, brazing, paint removal with heat, and other operations where open flames or implements utilizing heat are used:

1. Obtain Contracting Officer’s approval for operations involving use of open-flame or welding equipment.

    a. Notification shall be given for each occurrence and location of work with heat-generating equipment.

b. Obtain appropriate permit from the park as required.

2. As far as practical, use heat-generating equipment in shop areas or outside building.
3. Before work with heat-generating equipment commences, furnish fire watch (or watches) for location(s) where work is to be performed.
4. Do not perform work with heat-generating equipment in or near rooms or in areas where flammable liquids or explosive vapors are present or thought to be present. Use combustible gas indicator test to ensure area is safe.
5. Remove and keep area free of combustibles, including, rubbish, paper, waste, etc., within area of operations.

    a. If combustible material cannot be removed, provide fireproof blankets to cover such materials.

6. Where possible, furnish and use baffles of metal or gypsum board to prevent spraying of sparks or hot slag into surrounding combustible material.
7. Prevent extension of sparks and particles of hot metal through open windows, doors, holes, and cracks in floors, walls, ceilings, roofs, and other openings.
8. Inspect each location of day's work not sooner than 30 minutes after completion of operations to detect hidden or smoldering fires; ensure proper housekeeping is maintained.

B. Where sprinkler protection exists and is functional, maintain without interruption while operations are performed. If operations are performed near automatic sprinkler heads, shield individual heads temporarily with guards.

3.5 HISTORIC PRESERVATION TREATMENT PROCEDURES

The principal aim of preservation work is to halt the process of deterioration and stabilize the item's condition to sustain the integrity of the historic element, feature or structure being preserved. Cyclic maintenance is often required as well as repair work. Repair is required where specifically indicated. The following procedures shall be followed:

1. Retain as much existing material as possible; repair and consolidate rather than replace.
2. Use additional material or structure to reinforce, strengthen, prop, tie, and support existing material or structure.
3. Use reversible processes wherever possible.
4. Use traditional replacement materials and techniques if possible. New work shall be distinguishable from old work and original materials and techniques.
5. Record repair work during construction with periodic construction photos and daily inspection reporting. Photo documentation is specified in Section 013233 "Photographic Documentation."

B. Prohibit smoking by personnel performing work on or near historic structures.

C. Notify Contracting Officer of visible changes in integrity of material or components due to environmental causes including biological attack, UV degradation, freezing, or thawing, or due to structural defects including cracks, movement, or distortion.

1. Do not proceed with work in question until directed by Contracting Officer.

D. Where Work requires existing features to be removed, cleaned, and reinstalled, perform operations without damage to material itself, to adjacent materials, or to substrate.

E. Identify new or replacement materials and features with inconspicuous, permanent marks to distinguish from original materials. Record legend of identification marks and locations of these marks on Record Drawings.

F. When cleaning, match samples of existing materials that have been cleaned and identified for acceptable cleaning levels. Avoid over-cleaning to prevent damage. Use gentlest methods available. Initiate cleaning using hand cleaning methods before introducing power cleaning methods and equipment.

END OF SECTION 01 35 91

GRTE 237343 Rehabilitate “4 Lazy F Ranch” Cabins for Housing

SECTION 01 40 00 - QUALITY REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for quality assurance and quality control.

B. Testing and inspecting services are required to verify compliance with requirements specified or indicated. These services do not relieve Contractor of responsibility for compliance with Contract Document requirements. Quality of work shall be responsibility of the Contractor.

1. Specified tests, inspections, and related actions do not limit Contractor's other quality assurance and control procedures that facilitate compliance with Contract Document requirements.

C. See Divisions 2 through 49 Sections for specific test and inspection requirements.

1.2 DEFINITIONS

A. Quality Assurance Services: Activities, actions, and procedures performed before and during execution of the work to guard against defects and deficiencies and substantiate that proposed construction will comply with requirements.

B. Quality Control Services: Tests, inspections, procedures, and related actions during and after execution of the work to evaluate actual products incorporated into the work and completed construction comply with requirements.

Mockups: Full-size physical assemblies constructed on-site. Mockups are constructed from selections made under Sample submittals; to demonstrate aesthetic effects and, where indicated, qualities of materials and execution; to review coordination, testing, or operation; to show interface between dissimilar materials; and to demonstrate compliance with specified installation tolerances. Mockups are not Samples. Approved mockups establish the standard by which the Work will be judged, unless otherwise indicated.

Preconstruction Testing: Tests and inspections performed specifically for project before products and materials are incorporated into work to verify performance or compliance with specified criteria.

E. Product Testing: Tests and inspections performed by a Nationally Recognized Testing Laboratory (NRTL), a National Voluntary Laboratory Accreditation Program (NVLAP), or a testing agency qualified to conduct product testing, to establish product performance and compliance with industry standards.

F. Source Quality Control Testing: Tests and inspections performed at the source, i.e., plant, mill, factory, or shop.

G. Field Quality Control Testing: Tests and inspections performed on-site for installation of work and for completed work.

H. Testing Agency or Laboratory: Entity engaged to perform specific tests, inspections, or both. Testing laboratory shall mean the same as testing agency.

I. Installer/Applicator/Erector: Contractor or another entity engaged by Contractor as an employee, Subcontractor, or Sub-subcontractor, to perform a particular construction operation, including installation, erection, application, and similar operations.

1. Using a term such as "carpentry" does not imply certain construction activities must be performed by accredited or unionized individuals of a corresponding generic name, such as "carpenter." It also does not imply that requirements specified apply exclusively to trades people of corresponding generic name.

1.3 CONFLICTING REQUIREMENTS

Reference Standards: If compliance with two or more standards is specified and standards establish different or conflicting requirements for minimum quality levels, comply with most stringent requirement. Refer uncertainties and requirements that are different, but apparently equal, to Contracting Officer before proceeding.

B. Minimum Quality Levels: Quality level shown or specified shall be minimum provided or performed. Actual installation may comply exactly with minimum quality specified, or it may exceed minimum within reasonable limits. To comply with requirements, indicated numeric values are minimum or maximum, as appropriate, for the context of requirements. Refer uncertainties to Contracting Officer before proceeding.

1.4 SUBMITTALS

A. Quality Control Plan:

1. After contract award and before Pre-Construction conference, submit a written Contractor Quality Control (CQC) plan.
2. If plan requires revisions or corrections, Contractor shall resubmit plan within 10 days.
3. Government reserves the right to require changes in plan during contract period as necessary to obtain the quality specified.
4. No change in the approved plan may be made without written concurrence by Contracting Officer.

B. Qualification Data: For testing agencies specified in "Quality Assurance" Article to demonstrate their capabilities and experience. Include proof of qualifications in form of a recent report on inspection of testing agency by a recognized authority.

C. Contractor Quality Control (CQC) Daily Reports: Submit showing inspections and tests on first workday following date covered by report. Quality Control Supervisor shall utilize DSC Forms.

1. Review Construction Management Representative (CMR) Daily report if applicable and reconcile any differences prior to posting.

D. Test Reports

1. Test reports shall be completed by person performing test.
2. Submit Daily Test Information Sheets with Quality Control Daily Reports.
3. Submit failing test results and proposed remedial actions within four hours of noted deficiency.
4. Submit three copies of complete test results no later than one calendar day after test was performed.

E. Accessibility Inspection Report:

1. Fill out applicable sections of the Accessibility Inspection Report and attach to Contractor Quality Control Daily Report.
2. Utilize attached Accessibility Inspection form to document compliance with Architectural Barriers Act Accessibility Standards (ABAAS).
3. Inspect at various stages of construction as needed to ensure finished product meets standards.
4. Submit report no later than one calendar day after inspection was performed.

F. Off-Site Inspection Reports: Submit prior to shipment.

G. If Contractor Quality Control plan and Quality Control Daily Reports are not submitted as specified, Contracting Officer may retain payments until such time plan(s) is/are accepted and implemented, or may retain payments for work completed on days with no Quality Control Daily Reports.

H. Permits, Licenses, and Certificates: For National Park Service (NPS) records, submit copies of permits, licenses, certifications, inspection reports, releases, jurisdictional settlements, notices, receipts for fee payments, judgments, correspondence, records, and similar documents, established for compliance with standards and regulations bearing on performance of work.

1.5 QUALITY ASSURANCE

A. General: Qualifications paragraphs in this Article establish minimum qualification levels required; individual Specification Sections specify additional requirements.

Contractors Quality Control Staff:
Contractor's Quality Control Supervisor shall be assigned no other duties.

2. Contractor's designated Quality Control Supervisor shall be on the project site whenever contract work is in progress.
3. Contractor's job supervisory staff may be used to assist Quality Control Supervisor supplemented, as necessary, by additional certified testing technicians.

C. Installer Qualifications: Firm or individual experienced in installing, erecting, or assembling work similar in material, design, and extent indicated for Project, whose work has resulted in construction with a record of successful in-service performance.

D. Manufacturer Qualifications: Firm experienced in manufacturing products or systems similar to those indicated for Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

E. Fabricator Qualifications: Firm experienced in producing products similar to those indicated for Project and with a record of successful in-service performance, as well as sufficient production capacity to produce required units.

F. Professional Engineer Qualifications: Professional engineer legally qualified to practice in jurisdiction where Project is located and is experienced in providing engineering services of kind indicated (including Structural Tests and Special Inspections (STSI)). Engineering services are defined as those performed for installations of system, assembly, or products similar to those indicated for Project in material, design, and extent.

Specialists: Certain sections of the Specifications require that specific construction activities shall be performed by entities who are recognized experts in those operations. Specialists shall satisfy qualification requirements indicated and shall be engaged for activities indicated.

1. Requirement for specialists shall not supersede building codes and regulations governing Work.

H. Testing Agency Qualifications: A Nationally Recognized Testing Laboratory (NRTL), a National Voluntary Laboratory Accreditation Program (NVLAP), or an independent agency with experience and capability to conduct testing and inspecting indicated, according to ASTM E 329; and with additional qualifications specified in individual Sections; and where required by Contract, is acceptable to Contracting Officer.

1. Nationally Recognized Testing Laboratory (NRTL): Nationally recognized testing laboratory according to 29 CFR 1910.7 (Code of Federal Regulations).
2. National Voluntary Laboratory Accreditation Program (NVLAP): Testing agency accredited according to National Institute of Standards and Technology's (NIST) National Voluntary Laboratory Accreditation Program.
3. Measuring devices, laboratory equipment, and instruments shall be calibrated at established intervals against certified standards in accordance with NIST requirements. Measuring and testing devices shall be made available for use by Government for verification tests.

I. Factory-Authorized Service Representative Qualifications: Authorized representative of manufacturer who is trained and approved by manufacturer to inspect installation of manufacturer's products similar in material, design, and extent to those indicated for Project.

Mockups: Before installing portions of work requiring mockups, build mockups for each form of construction and finish required to comply with following requirements, using materials indicated for completed work:

1. Build mockups in location and of size indicated; if not indicated, as directed by Contracting Officer.
2. Notify Contracting Officer **fourteen** days in advance of dates and times when mockups will be constructed.
3. Demonstrate proposed range of aesthetic effects and workmanship.
4. Obtain Contracting Officer's approval of mockups before starting work, fabrication, or construction.
5. Maintain mockups in an undisturbed condition as a standard for judging the completed work.

Demolish and remove mockups when directed, unless otherwise indicated.

QUALITY CONTROL

A. Contractor is responsible for testing and inspections, including Structural Tests and Special Inspections (STSI), as identified in attached STSI. Inspect and test work as needed to ensure quality of materials, workmanship, construction, finish, and functional performance are in compliance with applicable specifications, drawings, and those required by the Building Code.

1. Engage qualified testing agency to perform quality-control services.
2. Submit appropriate report for each quality-control service.
3. Testing and inspecting requested by Contractor and not required by Contract Documents are Contractor's responsibility.
4. Contracting Officer may designate test locations.

B. Manufacturer's Field Services: Where indicated, engage factory-authorized service representative to inspect field-assembled components and equipment installation, including service connections. Report results in writing.

C. Re-testing/Re-inspecting: Regardless of whether original tests or inspections were Contractor's responsibility, provide quality-control services, including retesting and re-inspecting, for construction of replaced work that failed to comply with Contract Documents.

D. Testing Agency Responsibilities: Cooperate with NPS and Contractor in performance of duties. Provide qualified personnel to perform required tests and inspections.

1. Notify Contracting Officer and Contractor promptly of irregularities or deficiencies observed in work during performance of services.
2. Determine location from which test samples will be taken and in which in-situ tests are conducted.
3. Conduct and interpret tests and inspections, State in each report whether tested and inspected work complies with or deviates from requirements.
4. Submit 3 copies of certified written report of each test, inspection, and similar quality-control service through Contractor.
5. Do not release, revoke, alter, or increase Contract Document requirements or approve or accept any portion of Work.

E. Associated Services: Cooperate with agencies performing required tests, inspections, and similar quality-control services, and provide reasonable auxiliary services as requested. Notify agency sufficiently in advance of operations to permit assignment of personnel. Provide:

1. Access to Work.
2. Incidental labor and facilities necessary to facilitate tests and inspections.
3. Adequate quantities of representative samples of materials that require testing and inspecting. Assist agency in obtaining samples.
4. Facilities for storage and field curing of test samples.
5. Delivery of samples to testing agencies.
6. Preliminary design mix proposed for material mixes that require control by testing agency.
7. Security and protection for samples and testing and inspecting equipment at Project site.

F. Coordination: Coordinate sequence of activities to accommodate required quality assurance and control services with minimum delay and to avoid removing and replacing construction to accommodate testing and inspecting.

1. Schedule times for tests, inspections, obtaining samples, and similar activities.

PART 2 - PRODUCTS

2.1 QUALITY CONTROL PLAN

A. Quality Control Plan shall include:

1. List of personnel responsible for quality control and assigned duties. Include each person's qualifications. Include alternate(s) and qualifications.
2. Copy of letter of direction to Contractor's Quality Control Supervisor(s) outlining assigned duties and authorities designated by principal or owner.
3. Names, qualifications / accreditations, and descriptions of laboratories to perform sampling and testing, and samples of proposed report forms from laboratories.
4. Methods of performing, documenting, and enforcing quality control of work including Contractor report forms and acknowledgment of NPS forms.
5. Methods of monitoring and controlling environmental pollution and contamination as required by regulations and laws.
6. Specific discussion regarding mockups, off-site visits, receiving inspections, manufacturers representation, startup requirements, and other aspects of performance specific to Project.
7. Provisions for substantial completion(s) and final inspection(s) per Contract.

PART 3 - EXECUTION

3.1 OFF-SITE CONTROL

A. Items fabricated or assembled off-site shall be inspected for quality control at place of fabrication.

3.2 ON-SITE CONTROL

A. Notification:

1. Notify Contracting Officer at least 48 hours in advance of preparatory phase meeting.
2. Notify Contracting Officer at least 24 hours in advance of initial and follow-up phases.

B. Preparatory Phase: Perform before beginning each feature of work.

1. Review control submittal requirements with personnel directly responsible for quality assurance and quantity control of the work. As a minimum, Contractor's Quality Control Supervisor and foreman responsible for feature of work shall be in attendance.
2. Review applicable specifications sections and drawings related to feature of work.
3. Ensure copies of referenced standards related to sampling, testing, and execution for feature of work are available on site.
4. Ensure provisions have been made for field control testing.
5. Examine work area to ensure preliminary work has been completed.

6. Verify field dimensions and advise Contracting Officer of discrepancies with contract documents.
7. Ensure necessary equipment and materials are at project site and they comply with approved shop drawings and submittals.
8. Document preparatory phase activities and discussions on Contractor's Quality Control Daily Report.

C. Initial Phase:

1. As soon as work begins, inspect and test representative portion of particular feature of work for quality of workmanship.
2. Review control testing procedures to ensure compliance with contract requirements.
3. Document initial phase activities and discussions on Contractor's Quality Control Daily Report. Exact location of initial phase shall be indicated for future reference and comparison with follow-up phases.

D. Follow-Up Phase: Inspect and test as work progresses to ensure compliance with contract requirements until completion of work.

E. Additional Preparatory and Initial Phases: Additional preparatory and initial phases may be required on same feature of work for following reasons:

1. Quality of on-going work is unacceptable.
2. Changes in quality control staff, on-site production supervision, or work crew.
3. Work on particular feature of work is resumed after substantial period of inactivity.

3.3 DOCUMENTATION

A. Maintain Quality Control Daily Reports, Daily Test Report Information Sheets, and Accessibility Inspection Reports of quality control activities and tests. (Download from DSC Workflows website > Forms/Templates/Samples/Guidelines page > Construction Forms section.)

B. Quality Control Daily Reports shall not be substituted for other written reports required under clauses of contract, such as Disputes, Differing Site Conditions, or Changes.

C. Quality Control Daily Reports shall accurately portray all work and materials. The materials shall match the materials as approved from the Construction Documents (CD) requirements. Combustible/flammable materials shall be documented as well.

3.4 ENFORCEMENT

A. Contractor shall stop work on any item or feature pending satisfactory correction of deficiency noted by quality control staff or Contracting Officer.

3.5 REPAIR AND PROTECTION

A. General: On completion of testing, inspecting, sample taking, and similar services, repair damaged construction and restore substrates and finishes.

1. Provide materials and comply with installation requirements specified in other Specification Sections. Restore patched areas and extend restoration into adjoining areas with durable seams as invisible as possible.

Protect construction exposed by or for quality-control service activities.

C. Repair and protection are Contractor's responsibility, regardless of the assignment of responsibility for quality-control services.

END OF SECTION 01 40 00

GRTE 237343 Rehabilitate “4 Lazy F Ranch” Cabins for Housing

SECTION 01 42 00 – REFERENCE STANDARDS

GENERAL

1.1 ENVIRONMENTAL DEFINITIONS

A. Definitions pertaining to sustainable development: As defined in ASTM E2114 and as specified herein.

B. Biobased Materials: As defined in the Farm Security and Rural Investment Act, for purposes of Federal procurement of biobased products, "biobased" means a "commercial or industrial product (other than food or feed) that is composed, in whole or in significant part, of biological products or renewable domestic agricultural materials (including plant, animal, and marine materials) or forestry materials." Biobased materials also include fuels, chemicals, building materials, or electric power or heat produced from biomass as defined by The Biomass Research and Development Act of 2000.

1. Biobased content: Amount of biobased carbon in the material or product as a percentage of weight (mass) of total organic carbon in the material or product.

C. Chain-of-Custody: Process whereby a product or material is maintained under physical possession or control during its entire life cycle.

D. Deconstruction: Disassembly of buildings for purpose of recovering materials.

DFE (Design for the Environment): A technique that includes elements of resource conservation and pollution prevention as applied in various product sectors. A technique that incorporates approaches which are part of product (or assembly) concept, need and design. Considerations involve material selection, material and energy efficiency, reuse, maintainability and design for disassembly and recyclability. Refer to International Organization for Standardization (ISO) Guide 64 for additional clarification.

F. Environmentally preferable products: Products and services that have a lesser or reduced effect on the environment in comparison to conventional products and services. Refer to EPA’s Final Guidance on Environmentally Preferable Purchasing Program.

G. Non-Renewable Resource: A resource that exists in a fixed amount that cannot be replenished on a human time scale. Non-renewable resources have potential for renewal only by geological, physical, and chemical processes taking place over of millions of years. Examples include iron ore, coal, and oil.

H. Perpetual Resource: A resource that is virtually inexhaustible on a human time scale. Examples include solar energy, tidal energy, and wind energy.

I. Recycled Content Materials: Products that contain pre-consumer or post-consumer materials as all or part of their feedstock. Recycled content claim shall be consistent Federal Trade Commission (FTC) Guide for Use of Environmental Marketing Claims.

J. Renewable Resource: A resource that is grown, naturally replenished, or cleansed, at a rate which exceeds depletion of the usable supply of that resource. A renewable resource can be exhausted if improperly managed. However, a renewable resource can last indefinitely with proper stewardship. Examples include trees in forests, grasses in grasslands, and fertile soil.

1.2 QUALITY ASSURANCE

A. Applicability of Standards: Unless the Contract Documents include more stringent requirements, applicable construction industry standards have the same force and effect as if bound or copied directly into Contract Documents to the extent referenced. Such standards are made a part of Contract Documents by reference.

B. Publication Dates: Comply with standards in effect as of date of Contract Documents, unless otherwise indicated.

C. Conflicting Requirements: Where compliance with two or more standards is specified, and standards may establish different or conflicting requirements for minimum quantities or quality levels, comply with most stringent requirement. Refer uncertainties and requirements that are different, but apparently equal, to Contracting Officer (CO) for decision before proceeding.

1.3 INDUSTRY STANDARDS

A. Applicability of Standards: Unless Contract Documents include more stringent requirements, applicable construction industry standards have same force and effect as if bound or copied directly into Contract Documents to the extent referenced. Such standards are made a part of Contract Documents by reference.

Publication Dates: Comply with standards in effect as of date of Contract Documents unless otherwise indicated.

Copies of Standards: Each entity engaged in construction on Project should be familiar with industry standards applicable to its construction activity. Copies of applicable standards are not bound with Contract Documents.

1. Where copies of standards are needed to perform a required construction activity, obtain copies directly from publication source.

ABBREVIATIONS AND ACRONYMS

A. Industry Organizations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities found in Section 01 42 00 Sources for Reference Publications, Unified Facilities Guide Specifications (UFGS) (accessible via Masters website > Downloads section > click on UFGS Master (WBDG Website).

Names, telephone numbers, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

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Code Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in following list. Names, telephone numbers, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

DIN	Deutsches Institut fur Normung e.V. www.din.de	49 30 2601-3003
IAPMO	International Association of Plumbing and Mechanical Officials www.iapmo.org	(909) 472-4100
ICC	International Code Council www.iccsafe.org	(888) 422-7233
ICC-ES	ICC Evaluation Service, Inc. icc-es.org	(800) 423-6587 (562) 699-0543

Federal Government Agencies: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the entities in following list. Names, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

ABA & ABAAS United States Access Board	Architectural Barriers Act (ABA) Architectural Barriers Act Accessibility Standards (ABAAS) www.access-board.gov
CoE	Army Corps of Engineers www.usace.army.mil
CPSC	Consumer Product Safety Commission www.cpsc.gov
DOC	Department of Commerce www.commerce.gov
DOD	Department of Defense www.defense.gov
DOJ	Department of Justice www.justice.gov

DOE	Department of Energy www.energy.gov
EPA	Environmental Protection Agency www.epa.gov
FAA	Federal Aviation Administration www.faa.gov
FCC	Federal Communications Commission www.fcc.gov
FDA	Food and Drug Administration www.fda.gov
GSA	General Services Administration www.gsa.gov
HUD	Department of Housing and Urban Development www.hud.gov
LBL	Lawrence Berkeley National Laboratory www.lbl.gov
NCHRP	National Cooperative Highway Research Program (See TRB (Transportation Resource Board))
NIST	National Institute of Standards and Technology www.nist.gov
OSHA	Occupational Safety & Health Administration www.osha.gov
PHS	U.S. Department of Health and Human Services www.hhs.gov
RUS	Rural Utilities Service (See USDA (Department of Agriculture))
SD	State Department www.state.gov
TRB	Transportation Research Board www.nationalacademies.org/trb/transportation-research-board
USDA	Department of Agriculture www.usda.gov
USP	U.S. Pharmacopeia www.usp.org

USPS	Postal Service www.usps.com

Standards and Regulations: Where abbreviations and acronyms are used in Specifications or other Contract Documents, they shall mean the recognized name of the standards and regulations in following list. Names, telephone numbers, and websites are subject to change and are believed to be accurate and up-to-date as of date of Contract Documents.

ABAAS	Architectural Barriers Act Accessibility Standards www.access-board.gov
CFR	Code of Federal Regulations Available from Government Printing Office www.govinfo.gov/app/collection/cfr
DOD	Department of Defense Military Specifications and Standards Available from Department of Defense Single Stock Point www.dsp.dla.mil/Specs-Standards/
DSCC	Defense Supply Center Columbus (See FS (Federal Specification))
FED-STD	Federal Standard (See FS (Federal Specification))
FS	Federal Specification Available from Department of Defense Single Stock Point www.dsp.dla.mil/Specs-Standards/ Available from General Services Administration www.gsa.gov Available from National Institute of Building Sciences www.nibs.org
FTMS	Federal Test Method Standard (See FS (Federal Specification))
MIL	(See MILSPEC (Military Specification and Standards))
MIL-STD	(See MILSPEC (Military Specification and Standards))
MILSPEC	Military Specification and Standards Available from Department of Defense Single Stock Point www.dsp.dla.mil/Specs-Standards/
UFAS	Uniform Federal Accessibility Standards Available from Access Board www.access-board.gov/guidelines-and-standards/buildings-and-sites/about-the-aba-standards/ufas *(UFAS is **only** for housing projects per Fair Housing Act. See also the Fair Housing Act Design Manual, www.huduser.gov/portal/publications/destech/fairhousing)*

1.5 ENVIRONMENTAL REFERENCE STANDARDS

American Forest and Paper Association:

1. Sustainable Forestry Initiative

B. American Society of Heating Refrigerating and Air Conditioning Engineers (ASHRAE):
- **ASHRAE 52.2,** *Method of Testing General Ventilation Air Cleaning Devices for Removal Efficiency by Particle Size*
- **ASHRAE 55,** *Thermal Environmental Conditions for Human Occupancy*
- **ASHRAE 62.1,** *Ventilation for Acceptable Indoor Air Quality*
- **ASHRAE 62.2,** *Ventilation and Acceptable Indoor Air Quality in Low-Rise Residential Buildings*
- **ASHRAE/IESNA 90.1,** *Energy Standard for Buildings, Except Low-Rise Residential Buildings*
- **ASHRAE 90.2,** *Energy Efficient Design of Low-Rise Residential Buildings*

C. American Association of State Highway and Transportation Officials (AASHTO):
D. M288 Geotextile Specification for Highway Applications
E. **MP009-06 Standard Specification for Compost for Erosion/Sediment Control (Filter Berms and Filter Socks)**
F. **MP010-03 Standard Specification for Compost for Erosion/Sediment Control (Compost Blankets)**

G. American Society for Testing and Materials International (ASTM):
- A478 Standard Specification for Chromium-Nickel Stainless Steel Weaving and Knitting Wire
- A580/A580M Standard Specification for Stainless Steel Wire
- A653/A653M Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
- B813 Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube
- C1240 Standard Specification for Silica Fume Used in Cementitious Mixtures
- C128 Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Fine Aggregate
- C131 Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
- C1338 Standard Test Method for Determining Fungi Resistance of Insulation Materials and Facings
- C136 Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
- C1371 Standard Test Method for Determination of Emittance of Materials Near Room Temperature Using Portable Emissometers
- C1549 Standard Test Method for Determination of Solar Reflectance Near Ambient Temperature Using a Portable Solar Reflectometer
- C1601 Standard Test Method for Field Determination of Water Penetration of Masonry Wall Surfaces
- C289 Standard Test Method for Potential Alkali-Silica Reactivity of Aggregates (Chemical Method)
- C311 Test Methods for Sampling and Testing Fly Ash or Natural Pozzolans for Use as a Mineral Admixture in Portland-Cement Concrete

- C33 Standard Specification for Concrete Aggregates
- C593 Standard Specification for Fly Ash and Other Pozzolans for Use With Lime
- C595 Standard Specification for Blended Hydraulic Cements
- C618 Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use as a Mineral Admixture in Concrete
- C67 Standard Test Methods for Sampling and Testing Brick and Structural Clay Tile
- C989 Standard Specification for Ground Granulated Blast-Furnace Slag for Use in Concrete and Mortars
- D1557 Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 foot pound force per cubic foot (ft-lbf/ft3) (2,700 kilonewton meter per cubic meter (kN-m/m3))
- D1972 Standard Practice for Generic Marking of Plastic Products
- D198 Standard Test Methods of Static Tests of Lumber in Structural Sizes
- D2103 Standard Specification for Polyethylene Film and Sheeting
- D217 Standard Test Methods for Cone Penetration of Lubricating Grease
- D2369 Standard Test Method for Volatile Content of Coatings
- D3273 Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber
- D3786 Standard Test Method for Hydraulic Bursting Strength of Textile Fabrics-Diaphragm Bursting Strength Tester Method
- D3792 Standard Test Method for Water Content of Coatings by Direct Injection Into a Gas Chromatograph
- D3864 Standard Guide for Continual On-Line Monitoring Systems for Water Analysis
- D3960 Standard Practice for Determining Volatile Organic Compound (VOC) Content of Paints and Related Coatings
- D4017 Standard Test Method for Water in Paints and Paint Materials by Karl Fischer Method
- D4263 Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method
- D4444 Standard Test Methods for Use and Calibration of Hand-Held Moisture Meters
- D4491 Standard Test Methods for Water Permeability of Geotextiles by Permittivity
- D4552 Standard Practice for Classifying Hot-Mix Recycling Agents
- D4632 Standard Test Method for Grab Breaking Load and Elongation of Geotextiles
- D4716 Test Method for Determining the (In-plane) Flow Rate per Unit Width and Hydraulic Transmissivity of a Geosynthetic Using a Constant Head
- D4833 Standard Test Method for Index Puncture Resistance of Geotextiles, Geomembranes, and Related Product
- D4840 Standard Guide for Sampling Chain-of-Custody Procedures
- D4887 Standard Test Method for Preparation of Viscosity Blends for Hot Recycled Bituminous Materials
- D5106 Standard Specification for Steel Slag Aggregates for Bituminous Paving Mixtures
- D5116 Standard Guide for Small-Scale Environmental Chamber Determinations of Organic Emissions from Indoor Materials/Products
- D5199 Standard Test Method for Measuring the Nominal Thickness of Geosynthetics
- D5261 Standard Test Method for Measuring Mass per Unit Area of Geotextiles
- D5268 Standard Specification for Topsoil Used for Landscaping Purposes
- D5359 Standard Specification for Glass Cullet Recovered from Waste for Use in Manufacture of Glass Fiber
- D5505 Standard Practice for Classifying Emulsified Recycling Agents

- D5509 Standard Practice for Exposing Plastics to a Simulated Compost Environment
- D5512 Standard Practice for Exposing Plastics to a Simulated Compost Environment Using an Externally Heated Reactor
- D5539 Standard Specification for Seed Starter Mix
- D5957 Standard Guide for Flood Testing Horizontal Waterproofing Installations
- D5603 Standard Classification for Rubber Compounding Materials—Recycled Vulcanizate Particulate Rubber
- D5663 Standard Guide for Validating Recycled Content in Packaging Paper and Paperboard
- D5759 Standard Guide for Characterization of Coal Fly Ash and Clean Coal Combustion Fly Ash for Potential Uses
- D5792 Standard Practice for Generation of Environmental Data Related to Waste Management Activities: Development of Data Quality Objectives
- D5834 Standard Guide for Source Reduction Reuse, Recycling, and Disposal of Solid and Corrugated Fiberboard (Cardboard)
- D5851 Standard Guide for Planning and Implementing a Water Monitoring Program
- D5852 Standard Test Method for Erodibility Determination of Soil in the Field or in the Laboratory by the Jet Index Method
- D6002 Standard Guide for Assessing the Compostability of Environmentally Degradable Plastics
- D6006 Standard Guide for Assessing Biodegradability of Hydraulic Fluid
- D6007 Standard Test Method for Determining Formaldehyde Concentration in Air from Wood Products Using a Small Scale Chamber
- D6046 Standard Classification of Hydraulic Fluids for Environmental Impact
- D6081 Standard Practice for Aquatic Toxicity Testing of Lubricants: Sample Preparation and Results Interpretation
- D6108 Standard Test Method for Compressive Properties of Plastic Lumber and Shapes
- D6155 Standard Specification for Nontraditional Coarse Aggregates for Bituminous Paving Mixtures
- D6245 Standard Guide for Using Indoor Carbon Dioxide Concentrations to Evaluate Indoor Air Quality and Ventilation
- D6329 Standard Guide for Developing Methodology for Evaluating the Ability of Indoor Materials to Support Microbial Growth Using Static Environmental Chambers
- D6330 Standard Practice for Determination of Volatile Organic Compounds (Excluding Formaldehyde) Emissions from Wood-Based Panels Using Small Environmental Chambers Under Defined Test Conditions
- D6345 Standard Guide for Selection of Methods for Active, Integrative Sampling of Volatile Organic Compounds in Air
- D6629 Standard Guide for Selection of Methods for Estimating Soil Loss by Erosion
- D6886 Standard Test Method for Speciation of the Volatile Organic Compounds (VOCs) in Low VOC Content Waterborne Air-Dry Coatings by Gas Chromatography
- D692 Standard Specification for Coarse Aggregate for Bituminous Paving Mixtures
- D696 Standard Test Method for Coefficient of Linear Thermal Expansion of Plastics Between -30°C and 30°C With a Vitreous Silica Dilatometer
- D698 Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Standard Effort (12,400 ft-lbf/ft3 (600 kN-m/m3))
- D7186 Standard Practice for Quality Assurance Observation of Roof Construction and Repair
- E1021 Standard Test Methods for Measuring Spectral Response of Photovoltaic Cells

- E1105 Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform or Cyclic Static Air Pressure Difference
- E1171 Standard Test Method for Photovoltaic Modules in Cyclic Temperature and Humidity Environments
- E1333 Standard Test Method for Determining Formaldehyde Concentrations in Air and Emission Rates from Wood Products Under Defined Test Conditions Using a Large Chamber
- E1433 Standard Guide for Selection of Standards on Environmental Acoustics
- E1609 Standard Guide for Development and Implementation of a Pollution Prevention Program
- E1686 Standard Guide for Selection of Environmental Noise Measurements and Criteria
- E1780 Standard Guide for Measuring Outdoor Sound Received from a Nearby Fixed Source
- E1827 Standard Test Methods for Determining Airtightness of Buildings Using an Orifice Blower Door
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- E1861 Standard Guide for Use of Coal Combustion By-Products in Structural Fills
- E1918 Standard Test Method for Measuring Solar Reflectance of Horizontal and Low-Sloped Surfaces in the Field
- E1971 Standard Guide for Stewardship for the Cleaning of Commercial and Institutional Buildings
- E1980 Standard Practice for Calculating Solar Reflectance Index of Horizontal and Low-Sloped Opaque Surfaces
- E1991 Standard Guide for Environmental Life Cycle Assessment of Building Materials/Products
- E2114 Standard Terminology for Sustainability Relative to the Performance of Buildings
- E2128 Standard Guide for Evaluating Water Leakage of Building Walls
- E2129 Standard Practice for Data Collection for Sustainability Assessment of Building Products
- E241 Standard Guide for Limiting Water-Induced Damage to Buildings
- E2432 Standard Guide for General Principles of Sustainability Relative to Buildings
- E408 Standard Test Methods for Total Normal Emittance of Surfaces Using Inspection-Meter Techniques
- E413 Standard Classification for Rating Sound Insulation
- E477 Standard Test Method for Measuring Acoustical and Airflow Performance of Duct Liner Materials and Prefabricated Silencers
- E683 Standard Practice for Installation and Service of Solar Space Heating Systems for One- and Two-Family Dwellings
- E779 Standard Test Method for Determining Air Leakage Rate by Fan Pressurization
- E90 Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements
- E903 Standard Test Method for Solar Absorptance, Reflectance, and Transmittance of Materials Using Integrating Spheres
- F1869 Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride
- F2034 Standard Specification for Sheet Linoleum Floor Covering
- F2170 Standard Test Method for Determining Relative Humidity in Concrete Floor Slabs Using in situ Probes

H. Bat Conservation International:
- Bat Approved Bat Houses

I. Carpet and Rug Institute
- Green Label & Green Label Plus Testing Programs, carpet-rug.org/testing/green-label-plus

J. Center for Resource Solutions
- Green-e program

K. Environmental Protection Agency (EPA):
- **Comprehensive Procurement Guidelines**
- **ENERGY STAR**
- **Environmentally Preferable Purchasing Program Final Guidance**
- **GreenScapes program**
- **Heat Island Initiative**
- **Indoor Air Quality Building Education and Assessment Model (I-BEAM)**
- **National Environmental Performance Track**
- **Pollution Prevention (P2)**
- **Product Stewardship Program**
- **Significant New Alternatives Policy (SNAP) Program**

L. Federal Trade Commission:
- **Guide for the Use of Environmental Marketing Claims**

M. Forest Stewardship Council:
- **Chain-Of-Custody**
- **Forest Management**

N. Green Building Initiative (GBI):

O. **Green Globes - US**

P. Green Seal:
- **GC-03 Anti-Corrosive Paints**
- **GC-12 Occupancy Sensors**
- **GC-13 Split-Ductless Air-Source Heat Pumps**
- **GS-05 Compact Fluorescent Lamps**
- **GS-11 Paints**
- **GS-13 Windows**
- **GS-14 Window Films**
- **GS-31 Electric Chillers**
- **GS-32 Photovoltaic Modules**
- **GS-36 Commercial Adhesives**
- **GS-37 Industrial & Institutional Cleaners**

Q. International Iron and Steel Institute:
- **CO2 Breakthrough Program**

R. International Organization of Standardization:

- **Guide 64; Guide for Inclusion of Environmental Aspects in Product Standards**
- **9660 Information processing -- Volume and file structure of CD-ROM for information interchange**
- **14001 Environmental management systems – Specification with guidance for use**
- **14004 Environmental Management Systems – General Guidelines on Principles, Systems and Supporting Techniques**
- **14020 Environmental labels and declarations – General principles**
- **14024 Environmental labels and declarations – Type I environmental labelling - Principles and procedures**
- **14040 Environmental management – Life cycle assessment – Principles and framework**

S. National Association of Home Builders:
- **Advanced Framing Techniques: Optimum Value Engineering**

T. National Institute of Building Sciences:
- **MOIST program for transfer of heat and moisture**
- **Whole Building Design Guide**

U. National Institute of Standards and Technology:
- **BEES (Building for Environmental and Economic Sustainability) Lifecycle Decision Support Tool**

V. Sheet Metal and Air Conditioning Contractors' National Association (SMACNA):
W. **IAQ Guidelines for Occupied Buildings Under Construction**

X. Southcoast Air Quality Management District:
- **1168 Adhesive And Sealant Applications**

Y. US Composting Council:
A. Seal of Testing Assurance Program

Z. US Department of Agriculture:
1. **Biobased Products – Definitions and Descriptions**

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION (Not Used)

END OF SECTION 01 42 00

SECTION 01 42 16 - DEFINITIONS

PART 1 GENERAL

1.1 SUMMARY

A. This section supplements the definitions contained in the General Conditions.

1.2 DEFINITIONS

A. Typical definitions include - but are not limited to - the following:

1. AHJ: Authorities Having Jurisdiction.

2. Anchorage: A secure connecting point or a terminating component of a fall protection system or rescue system capable of safely supporting the impact forces applied by a fall protection system or anchorage subsystem.

3. Anchorage Connector: A component or subsystem that functions as an interface between the anchorage and a fall protection, work positioning, rope access, or rescue system for the purpose of coupling the system to the anchorage.

4. Carbon Fiber Reinforced Polymer (CFRP): Composite material comprising a polymer matrix reinforced with carbon fiber cloth, mat, or strands.

5. Chinking: The Chinking consists of two parts: first, a dry, bulky, rigid blocking, such as wood slabs or stones is inserted into the joint, followed by a soft packing filler such as oakum, moss, clay, or dried animal dung

6. Daubing: Daubing, which completes the Chinking and Daubing system, is the outer wet-troweled finish layer of varying composition, but often consisting of a mixture of clay and lime or other locally available materials. Instead of daubing, carefully fitted quarter poles or narrow wood strips were sometimes nailed lengthwise across the log joints

7. Delegated Design: Where indicated within specification sections, Contractor shall provide materials or products indicated that will require analysis by a professional engineering service, engage licensed engineering services either directly, through a subcontractor, or through the building component manufacturer’s engineer. Qualified professional engineers must be licensed in the state in which the project is located.

   a. Not all delegated design indicated requires the use of the services of a professional engineer, such materials or products may require the use of manufacturer’s design tables for pre-engineered building components, reference standards or industry accepted reference manuals to fulfill the Delegated Design responsibilities of the Contractor.

   b. Note that Delegated Design may be referred to as Bidder Design.

8. Demolish: See “Remove”.

9. Deferred Submittal: Those portions of the design that are not submitted at the time of building permit application and are being submitted separately to the AHJ.

    a. Refer to OSSC 107.3.4.2 for additional requirements.

10. Dismantle: To disassemble or detach an existing element using gentle methods and equipment to prevent damage to surrounding elements. Such action includes the disposal of elements unless indicated to be salvaged or reinstalled.

11. Disposal: To collect, handle, transport, and deposit materials to a designated collection site for final disposition including – but not limited to - recycling, treatment, incineration, or landfilling. Such action includes compliance with Project requirements and local, state, and federal environmental regulations.

12. Element: A building, structure, assembly, component, item, artifact, feature, material, surface, or part thereof.

13. Existing: An element that is already built and in place at the start of the Project.

14. Existing to Remain or Retain: See Section 01 35 91 - Historic Preservation Treatment Procedures.

15. Face Bedding: Setting of stone with the rift or natural bedding planes (strata) vertical and parallel to the wall plane rather than horizontal or "naturally bedded," which holds bedding planes together by gravity.

16. Fall Arrest System: A system designed to stop you in the process of a fall, typically including an anchor point or series of anchor points, a safety lanyard or self-retracting lifeline, and a harness.

17. Fall Restraint System: A system designed to keep you from getting close enough to the fall hazard to fall, typically including an anchor point or series of anchor points, a safety lanyard or self-retracting lifeline, and a harness.

18. Fall Protection System: System can be either a fall arrest or a fall restraint system.

19. Furnish: To supply, deliver, unload, and inspect for damage.

20. Historic: Existing elements that are determined to have historical significance related to their age, events, persons, characteristics, or potential for cultural value. Such elements are to be protected and handled in accordance with the Project's historic treatment procedures.

21. In kind: Replacement procedure in which new elements are produced to match existing elements as closely as possible, including material, construction, finish, color, and appearance.

22. In situ: Repair procedure in which existing elements remain in-place and are restored without removal from the system.

23. Install: To unpack, assemble, erect, apply, place, finish, cure, protect, clean, start up, and make ready for use.

24. Material in Kind: See Section 01 35 91 - Historic Preservation Treatment Procedures.

25. New: Non-existing elements that have been recently manufactured or procured for the Project.

26. New Elements: New, nonhistoric materials or components added to roofing systems to aid in their ability to resist water infiltration.

27. Patch: Use of substitute repair materials to treat damaged or deteriorated roofing system components in situ

28. Preservation: See Section 01 35 91 - Historic Preservation Treatment Procedures.

29. Product: Material, machinery, components, equipment, fixtures, and systems forming the work result. Not materials or equipment used for preparation, fabrication, conveying, or erection and not incorporated into the work result. Products may be new, never before used, or re-used materials or equipment.

30. Project Manual: The book-sized volume that includes the procurement requirements (if any), the contracting requirements, and the specifications.

31. Protect: To install protective measures such as temporary guards or coverings to prevent damage to an area or element due to construction activities, weather, or human tampering.

32. Protect and Maintain: See Section 01 35 91 - Historic Preservation Treatment Procedures.

33. Provide: To furnish and install.

34. Rebuilding (Setting) Mortar: Mortar used to set and anchor masonry in a structure, distinct from pointing mortar installed after masonry is set in place.

35. Rehabilitation: See Section 01 35 91 - Historic Preservation Treatment Procedures.

36. Remove: See Section 01 35 91 - Historic Preservation Treatment Procedures.

37. Repair: See Section 01 35 91 - Historic Preservation Treatment Procedures.

38. Replace: To duplicate and replace entire features with new material in kind. Replacement includes the following conditions:

    a. Duplication: Includes replacing elements damaged beyond repair or missing. Original material is indicated as the pattern from creating new duplicated elements.

b. Replacement with New Materials: Includes replacement with new material when original material is not available as patterns for creating new duplicated elements.

c. Replacement with Substitute Materials: Includes replacement with compatible substitute materials. Substitute materials are not allowed, unless otherwise indicated.

39. Restoration: See Section 01 35 91 - Historic Preservation Treatment Procedures.

40. Reconstruction: See Section 01 35 91 - Historic Preservation Treatment Procedures.

41. Remove and Salvage: See Section 01 35 91 - Historic Preservation Treatment Procedures.

42. Remove and Reinstall: See Section 01 35 91 - Historic Preservation Treatment Procedures.

43. Rift: The most pronounced direction of splitting or cleavage of a stone. Rift may be obscure in igneous rocks such as granite. Often it is obvious, as with bedding planes in many sedimentary stones.

44. Salvage: See Section 01 35 91 - Historic Preservation Treatment Procedures.

45. Stabilize: See Section 01 35 91 - Historic Preservation Treatment Procedures.

46. Stone Terminology: ASTM C119.

47. Structural Steel: Elements of the structural frame indicated on Drawings and as described in AISC 303, "Code of Standard Practice for Steel Buildings and Bridges.

48. Supply: Same as Furnish.

1.3 PROCESS TERMINOLOGY

A. Typical definitions include - but are not limited to - the following:

1. Very Low-Pressure Spray: Less than [100 (690)psi (kPa).

2. Low-Pressure Spray:

a. Pressure: 100 to 400 (690 to 2750) psi (kPa).

b. Flow Rate: 4 to 6 (0.25 to 0.4) gpm (L/s).

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION - NOT USED

END OF SECTION

GRTE 237343 Rehabilitate "4 Lazy F Ranch" Cabins for Housing

SECTION 01 50 00 - TEMPORARY FACILITIES AND CONTROLS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes requirements for temporary utilities, support facilities, and security and protection facilities.

1.2 DEFINITIONS

A. Permanent Enclosure: As determined by Contracting Officer (CO), permanent or temporary roofing is complete, insulated, and weathertight; exterior walls are insulated and weathertight; and openings are closed with permanent construction or substantial temporary closures.

1.3 USE CHARGES

A. General: Cost or use charges for temporary facilities shall be included in Contract Sum as required.

Water Service: Potable and Nonpotable Water from the Park Headquarters facility in Moose Wyoming, as directed by the Contracting Officer, is available for use without metering and without payment of use charges. The Construction Contractor shall provide a means of transporting, storing, and distributing water at the project site.

C. Electric Power Service: Electric power from existing system is available for use with metering and with payment of use charges. Provide connections and extensions of services as required for construction operations without any additional permit costs. Construction Contractor will be responsible for coordinating temporary power service with the supplier.

1.4 SUBMITTALS

A. Spill Prevention Plan: After contract award and before the Pre-Construction conference, submit for review, a Spill Prevention Plan. The CO will review the proposed Plan. If the plan requires any revisions or corrections, the Contractor shall resubmit the Plan within 10 days. Do not commence with Work until a Plan has been accepted.

B. Emissions Reduction Plan: After contract award and before the Pre-Construction conference, submit for review, an Emissions Reduction Plan. The CO will review the proposed Plan. If the plan requires any revisions or corrections, the Contractor shall resubmit the Plan within 10 days. Do not commence with Work until a Plan has been accepted.

C. Dust Control Plan: After contract award and before the Pre-Construction conference, submit for review, an Dust Control Plan. The CO will review the proposed Plan. If the plan requires any revisions or corrections, the Contractor shall resubmit the Plan within 10 days. Do not commence with Work until a Plan has been accepted.

D. Temporary Safety Fence Plan: After contract award and before the Pre-Construction conference, submit for review, an Temporary Safety Fence Plan. The CO will review the proposed Plan. If the plan requires any revisions or corrections, the Contractor shall resubmit the Plan within 10 days. Do not commence with Work until a Plan has been accepted.

E. Product Data: For safety barrier fence.
   1. Include recommendations for product application and use. Include test data substantiating that products comply with requirements.

1.5 QUALITY ASSURANCE

A. Electric Service: Comply with National Electrical Contractors Association (NECA), National Electrical Manufacturers Association (NEMA), and Underwriter Laboratories (UL) standards and regulations for temporary electric service. Install service to comply with National Fire Protection Association (NFPA) 70.

B. Environmental Protection: Provide environmental protection as required by agency(ies) with jurisdiction and as indicated in Contract Documents. Coordinate with requirements of the following:

   1. Regulatory Requirements
   2. Indoor Air Quality (IAQ) Management
   3. Noise and Acoustics Management
   4. Environmental Management
   5. Construction Waste Management

PROJECT CONDITIONS

A. Temporary Use of Permanent Facilities: Installer of each permanent service shall assume responsibility for operation, maintenance, and protection of each permanent service during its use as a construction facility before NPS acceptance, regardless of previously assigned responsibilities.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Temporary materials may be new or used, but must be adequate in capacity for required usage, must not create unsafe conditions, and must not violate requirements of applicable codes and standards.

B. Pavement: no temporary pavement is anticipated. Coordinate with Contracting Officer if the need for temporary pavement arises.

Safety Barrier Fence: Orange or green plastic fence, minimum height, 4 feet. Coordinate color with the CO.

D. Barrier Tape: Construction zone warning tape, imprinted with “CAUTION: CONSTRUCTION AREA.” Tape shall be 3” thick, yellow, with black lettering. Tape shall have a minimum tensile strength of 400 lbs and meet OSHA CFR Part 1926.502.

TEMPORARY FACILITIES

A. Field Offices, General: Prefabricated or mobile units with serviceable finishes, temperature controls, and foundations adequate for normal loading.

Storage and Fabrication Sheds: Temporary weather tight sheds or other covered facilities for storage of materials subject to weather damage. Number and size of structures shall be subject to Contracting Officer's approval.

C. Toilets: Sufficiently lighted and ventilated toilet facilities in weatherproof, sight proof, handicap accessible, sturdy enclosures with privacy locks.

2.3 EQUIPMENT

A. Fire Extinguishers: Portable, UL rated; with class and extinguishing agent as required by locations and classes of fire exposures.

Heating, Ventilation, and Air Conditioning (HVAC) Equipment: Unless Contracting Officer authorizes use of permanent HVAC system, provide vented, self-contained, liquid-propane-gas or fuel-oil heaters with individual space thermostatic control.

Use of gasoline-burning space heaters, open-flame heaters, or salamander-type heating units is prohibited.

2. Heating Units: Listed and labeled for type of fuel being consumed, by a testing agency acceptable to agency(ies) with jurisdiction, and marked for intended use.

C. Direct Digital Control (DDC) Internet Connection: Furnish, install and maintain a high-speed connection (Digital Subscriber Line (DSL) or similar) between the project’s DDC system and the internet through a Contractor furnished internet service provider. Contractor is responsible for maintenance of this connection and costs associated with internet service provider through warranty period of this contract. Upon completion of warranty period, service shall be transferable to the Government at which time future costs for connection will no longer be the responsibility of Contractor.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Locate facilities Within the staging areas designated on the drawings and where they will serve Project adequately and result in minimum interference with performance of the Work. Relocate and modify facilities as required by progress of the Work. .

B. Provide each facility ready for use when needed to avoid delay. Do not remove until facilities are no longer needed or are replaced by authorized use of completed permanent facilities.

3.2 TEMPORARY UTILITY INSTALLATION

A. General: Install temporary service or connect to existing service.

1. Arrange with utility company, NPS, and existing users for time when service can be interrupted, if necessary, to make connections for temporary services. Acquire necessary permits.

B. Storm Sewers and Drainage: Provide temporary utilities to remove effluent lawfully.

Non-potable water for construction is available from the existing on site hydrant to the west of the main lodge. Install means of water transportation and distribution as required.

Potable water is not available on site. Potable water is available from the existing hydrant located west of the main lodge as directed by the Contracting Officer. Construction contractor shall provide the means of water transportation, storage, and distribution as required. Furnish cool, potable water for construction personnel in locations convenient to workstations. Do not use potable water for anything other than:

1. Drinking,
2. Leak/water tightness testing of new water and sewer facilities.
3. Mortar and daubing preparation as required by the specifications.

E. Sanitary Facilities: Provide temporary toilets, and wash facilities for use by construction personnel.

1. Place in approved locations secluded from public observation and convenient to workstations. Relocate as work progress requires.
2. Maintain and clean toilet facilities at least weekly.
3. Completely remove sanitary facilities on completion of work.

Heating Provide temporary heating required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of low temperatures or high humidity. Select equipment that will not have a harmful effect on completed installations or elements being installed.

1. Use of permanent heating system will not be allowed without written authorization from Contracting Officer. When the permanent heating system is approved for use as temporary heating , pay costs until final acceptance. Permanent heating system shall be sufficiently complete, including controls, to permit safe operation
2. Provide and maintain adequate approved facilities, as required for safety and construction requirements, during the work. Provide ample clearance around stoves, heaters, chimney and vent connections to prevent ignition of combustible material
3. Install and maintain temporary filters when air handing equipment is used for temporary heating Install new filters before final acceptance in addition to any extra sets of filters required. Clean coils as determined by Contracting Officer.
4. Warranties for equipment used for temporary heating shall start on date of Final Acceptance.

G. Ventilation and Humidity Control: Provide temporary ventilation required by construction activities for curing or drying of completed installations or for protecting installed construction from adverse effects of high humidity. Select equipment that will not have a harmful effect on completed installations or elements being installed. Coordinate ventilation requirements to produce ambient conditions required and minimize energy consumption.

Electric Power Service: Use of existing electric power service will be permitted, as long as equipment is maintained in a condition acceptable to NPS.

1. When temporary connections are removed, restore existing utility services to original condition.

Lighting: Provide temporary lighting with local switching that provides adequate illumination for construction operations, observations, inspections, and traffic conditions.

1. Install and operate temporary lighting that fulfills security and protection requirements without operating entire system.

Telephone Service: No telephone service is available on site for Contractor's use. Make arrangements with Telephone Company and pay costs.

3.3 SUPPORT FACILITIES INSTALLATION

A. General: Comply with the following:

1. Provide incombustible construction for offices, shops, and sheds located within construction area or within 50 feet of building lines. Comply with NFPA 241.
2. Maintain support facilities until near Substantial Completion. Remove structures, equipment, and furnishings, and terminate services after punch list is 100 percent completed or when directed by Contracting Officer. Personnel remaining after Substantial Completion will be permitted to use permanent facilities, under conditions acceptable to Contracting Officer.

Temporary Roads and Paved Areas: Construct and maintain temporary roads and paved areas adequate for construction operations. Locate temporary roads and paved areas in same location as permanent roads and paved areas. Extend temporary roads and paved areas, within construction limits indicated, as necessary for construction operations.

1. Coordinate elevations of temporary roads and paved areas with permanent roads and paved areas.

Traffic Controls: Erect and maintain barricades, lights, danger signals, and warning signs in accordance with Manual on Uniform Traffic Control Devices (MUTCD), Part 6, latest edition.

1. Implement a traffic control plan in accordance with Section 01 55 26. Include strategies to maintain safe and efficient traffic flow.
2. Protect existing site improvements to remain including curbs, pavement, and utilities.
3. Maintain access for fire-fighting equipment and access to fire hydrants.
4. Adequately barricade and post open cuts in or adjacent to thoroughfares.
5. Protect pedestrian traffic by guardrails or fences.

6. When pedestrian traffic is detoured onto a roadway, provide temporary walkways with protection as required at ends and overhead. For walkways, use lumber running parallel to direction of traffic movement and provide ramps at changes of elevation.
7. Cover pipes, hoses, and power lines crossing sidewalks and walkways with troughs using beveled edge boards.
8. Install Barrier Tape where directed by Contracting Officer. Keep a minimum of two rolls on site.

D. Parking: Staging areas as indicated on the drawings shall be used for vehicle parking for construction personnel. To the greatest extent possible, Contractor shall limit vehicular traffic to and from work sites within the project area by requiring workers to carpool.

Dewatering Facilities and Drains: Comply with requirements of the agency(ies) with jurisdiction. Maintain Project site, excavations, and construction free of water.

1. Dispose of rainwater in lawful manner that will not result in flooding Project or adjoining properties nor endanger permanent Work or temporary facilities.
2. Remove snow and ice as required to minimize accumulations.

F. Project Identification and Temporary Signs: Provide Project identification and other signs . Fence, barricade, or otherwise block off the immediate work area to prevent unauthorized entry.

1. Communicated to the public and residents (i.e., via signage, media alerts, etc.) any potential closures, detours, and safety messaging. Coordinate messaging with the CO in advance of closure, detaour, or safety concern.
2. Implement measures to close and/or redirect trails in areas that will be affected by construction to ensure visitor health and safety. Provide information on alternatives that will help hikers achieve their goal while staying away from the work area.
3. Provide temporary, directional signs for construction personnel and visitors.
4. Maintain and touchup signs so they are legible at all times.
5. Erect and maintain sufficient detour signs at road closures and along detour routes.

G. Waste Disposal Facilities: Provide waste-collection containers in sizes adequate to handle waste from construction operations. Comply with requirements of agency(ies) with jurisdiction.

H. Concrete Washout Facility: Contractor shall provide a full containment system for washout and cleaning of concrete and flowable fill delivery trucks.
1. Submit plans detailing the size, materials, and equipment to construct the facility.
2. Design the washout facility with sufficient holding capacity for containment of all liquid and concrete generated by washout operations. Design the facility with sufficient freeboard to prevent overflow.
3. Washout facility shall be constructed in the staging areas indicated on the drawings in a location approved by the CO.
    a. Unless directed otherwise, locate facility a minimum of 50 feet from drain inlets, open drainage facilities and watercourses.
4. Collect and dispose of wash water and concrete debris within 24 hours of utilizing the wash water facility.
5. Inspect facility for leaks and damage before and after use and on a daily basis when in use. Repair facility damage immediately upon discovery.
6. Remove washout facility when concrete operations are discontinued.
7. Reclaim site in accordance with the technical specifications.

Lifts and Hoists: Provide facilities necessary for hoisting materials and personnel.

1. Truck cranes and similar devices used for hoisting materials are considered "tools and equipment" and not temporary facilities.

Temporary Stairs: Until permanent stairs are available, provide temporary stairs where ladders are not adequate.

Temporary Use of Permanent Stairs: Cover finished, permanent stairs with protective covering of plywood or similar material so finishes will be undamaged at time of acceptance.

SECURITY AND PROTECTION FACILITIES INSTALLATION

A. Environmental Protection: Provide protection, operate temporary facilities, and conduct construction in ways and by methods that comply with environmental regulations and minimize possible air, waterway, and subsoil contamination or pollution or other undesirable effects. Take every precaution and make every effort to protect the delicate environment of Grand Teton National Park.
   1. Refer to Section 01 35 13.05 – Environmental Protection for additional requirements.

B. Equipment and Vehicle Cleaning and inspection is required for any vehicle and/or equipment that enters the Park and/or is staged in association with this project. Refer to Section 01 35 13.05 – Environmental Protection for specific requirements.

C. Equipment: Provide construction equipment with the best available technology for sound dampening muffler and exhaust systems.

D. Equipment Emissions: To save fuel and reduce noise and emissions, Contractor shall develop and implement Emissions Reduction Plan to prevent excessive idling of all vehicles used in construction.

E. Temporary Erosion and Sedimentation Control: Refer to Section 01 57 23 "Temporary Storm Water Pollution Prevention".

F. Tree and Plant Protection: Refer to Section 01 11 00 "Summary of Work".

G. Wetland and Waterway Protection: Comply with the Federal Water Pollution Control Act (33 USC § 1251 et seq).
   1. Except as authorized by this contract, do not operate mechanized equipment, discharge, or place material within the boundaries of any U.S. waters as identified by the ordinary high-water mark or edge of wetland. This includes wetlands, unless authorized by a permit issued by the U.S. Army Corps of Engineers according to 33 USC 1344, and if required by the state agency having jurisdiction over the discharge of materials into the waters of the U.S. In the event of an unauthorized discharge:
      a. immediately prevent further contamination;
      b. immediately notify appropriate authorities and the CO; and
      c. mitigate damages as required.
   2. Separate work areas, including material sources by the use of a suitable barrier that prevents sediment, petroleum products, chemicals, other liquids, or solid materials from entering the waters of the U.S. Construct and remove barriers to avoid discharge of material into the

waters of the U.S. Remove and properly dispose of sediment or other material collected by the barrier

H. Oil and Hazardous Substances: Submit and follow a Hazardous Spill Plan. Submit the plan at least 15 days before beginning work. Develop a plan describing what actions will be taken in case of a spill and incorporate preventative measures to be implemented (such as the placement of refueling facilities, storage, and handling of hazardous materials, etc.). At a minimum, incorporate the following requirements into the plan.
   1. Comply with federal and state regulations for the storage, handling, and disposal of all hazardous material and waste. If hazardous materials will be used on site, make provisions for storage, containment, and disposal.
   2. Inspect equipment for leaks of oil, fuels, antifreeze, or hydraulic fluids before and during use to prevent soil and water contamination.
   3. Do not use equipment that is leaking fluids. Repair leaks on equipment immediately. Keep a supply of absorbent materials at the job site in the event of spills. Acceptable absorbent materials are those manufactured specifically for the containment and clean-up of hazardous materials.
   4. Store hazardous spill containment devices at the work site when working within 5 meters of a body of water
   5. Immediately notify the CO of all hazardous spills.

I. Grizzly Bears and Wolves”
   1. Contractor supplied garbage bins, within the construction site, must be bear resistant and meet Park requirements. Comply with wildlife protection requirements in Section 01 35 13.05 – Environmental Protection.

J. Pest Control: Follow NPS requirements to minimize attraction and harboring of rodents, roaches, and other pests and perform extermination and control procedures at regular intervals so Project will be free of pests and residues at Substantial Completion. Perform control operations lawfully, using environmentally safe materials.

K. Site Enclosure Fence: Before construction operations begin, furnish and install orange, or green plastic construction fencing to prevent people and animals from easily entering site except by entrance gates. Install fencing within the “limit of disturbance” lines as shown in the drawings.
   1. Install plastic construction fencing around open trenches and excavations to exclude wildlife and people.
   2. Fencing shall be used to confine activity to the minimum area required for construction. Construction activities, including material staging and storage, cannot occur beyond the construction zone fencing.
   3. Fencing shall also be required in sensitive areas or in locations where excavations are to remain open overnight.
   4. Extent of Fence: As required to enclose portions determined sufficient, by the CO, to accommodate construction operations.
   5. Locate vehicular gates to avoid interference with traffic on public thoroughfares.
   6. Locate pedestrian entrance gates as required to provide controlled personnel entry.
   7. Wildlife protection measures may require additional temporary screening. Refer to Section 01 35 13.05 – Environmental Protection and coordination with the CO.

Maintain security by limiting number of keys and restricting distribution to authorized personnel. Provide Contracting Officer with one set of keys.Staking of Construction Corridors: Stake construction boundaries

(limits of disturbance) along water and sewer systems with 3-ft lath and orange flagging at 25 lineal feet intervals.

1. Staking shall be used to confine activity to the minimum area required for construction. Construction activities, including material staging and storage, cannot occur beyond the construction zone staking

M. Security Enclosure and Lockup: Install substantial temporary enclosure around partially completed areas of construction. Provide lockable entrances to prevent unauthorized entrance, vandalism, theft, and similar violations of security. Security of the project site and all associated costs resulting from the lack of security shall be the responsibility of the Contractor.

N. Barricades, Warning Signs, and Lights: Comply with requirements of Manual on Uniform Traffic Control Devices (MUTCD), Part 6, latest edition for erecting structurally adequate barricades, including warning signs and lighting.

O. Temporary Enclosures: Provide temporary enclosures for protection of construction, in progress and completed, from exposure, foul weather, other construction operations, and similar activities. Provide temporary weathertight enclosure for building exterior.

1. Where heating or cooling is needed and permanent enclosure is not complete, insulate temporary enclosures.

Dust Control: Submit a Dust Control Plan prior to construction. Include in the plan the positive methods and dust control materials to minimize raising dust from construction operations, and the means to prevent air-borne dust from dispersing into the atmosphere. Incorporate methods and requirements provided in EPA 832-F-99-003, Storm Water Management Fact Sheet-Dust Control. Incorporate the following measures into the dust control plan:

1. Utilize dust control methods such as watering, covering haul roads and controlling speeds in disturbed areas.
2. Apply soil treatment in the form of a tackifier as directed in Section 01 57 23, “Temporary Storm Water Pollution Prevention.”

Q. Temporary Fire Protection: Install and maintain temporary fire-protection facilities of types needed to protect against reasonably predictable and controllable fire losses. Comply with NFPA 241.

1. Responsible Person: Capable and qualified person shall be placed in charge of fire protection. Responsibilities shall include locating and maintaining fire protective equipment and establishing and maintaining safe torch cutting and welding procedures.

Tobacco Use, Smoking, and Vaping: Smoking within buildings or temporary storage sheds is prohibited. Smoking and vaping is only allowed inside personal vehicles.

3. Supervise welding operations, combustion-type temporary heating units, and similar sources of fire ignition according to requirements of NPS. Check with Park; many require "burn permits" for welding.
4. Develop and supervise overall fire-prevention and -protection program for personnel at Project Site. Review needs with local fire department and establish procedures to be followed. Instruct personnel in methods and procedures. Post warnings and information.
5. Provide temporary standpipes and hoses for fire protection. Hang hoses with warning sign stating hoses are for fire-protection purposes only and are not to be removed. Match hose size with outlet size and equip with suitable nozzles.

6. Hazard Control: Take necessary precautions to prevent fire during construction. Do not store flammable or combustible liquids in existing buildings. Provide adequate ventilation during use of volatile or noxious substances. Ensure cleanup procedures and storage requirements are followed at close of every work session.
7. Spark Arresters: Equip gasoline or diesel-powered equipment used during periods of potential fire hazards or in potential forest and grass fire locations with spark arresters approved by United States Department of Agriculture (USDA) Forest Service.

    a. Written determinations of periods and areas of potential fire hazard will be issued by Contracting Officer.

8. Buildings: Furnish a minimum of one extinguisher for each 1,500 square feet of area or major fraction thereof.

    a. Travel distance from any workstation to nearest extinguisher shall not exceed 75 feet.

9. Vehicles and Equipment: Provide one extinguisher on each vehicle or piece of equipment.

Service and Refueling Areas: Refueling of equipment is limited to the use of pre-loaded fuel trucks. Long term (1 hr or more) parking or storage of fuel trucks is not allowed in the Park. The handling and design of fueling devices shall comply with O.S.H.A. Comply with the following:

1. Minimize onsite fueling and maintenance.
2. Refuel equipment at least 50 feet from buildings.
3. Refuel equipment at least 95 meters from water bodies when feasible.
4. Two persons shall be utilized during fueling operations, one to operate the fuel nozzle and one on standby at the fuel truck ready to take any necessary actions to prevent any spills.
5. Utilize absorbent spill diapers at all times during fueling, ensuring no droplets hit the ground. When not in use the fuel truck shall be staged outside the Park. Ensure all necessary precautions are taken to ensure no spills will occur.
6. Report and dispose of contaminated soils according to Federal, State, and local regulations. Follow the Hazardous Spill Plan as described in this specification.

S. Nighttime Operations: Provide overhead lighting at road closure points for NPS informational officers during closures.

3.5 OPERATION, TERMINATION, AND REMOVAL

A. Supervision: Enforce strict discipline in use of temporary facilities. Minimize waste and abuse, limit availability of temporary facilities to essential and intended uses.

B. Potable Water Usage: Under no circumstances shall existing system water storage be depleted beyond 50% of existing maximum capacity. Contractor shall monitor storage capacity regularly during high usage periods including filling new water lines.

C. Maintenance: Maintain facilities in good operating condition until removal.

1. Maintain operation of temporary enclosures, heating, cooling, humidity control, ventilation, and similar facilities on 24-hour basis where required to achieve indicated results and avoid possibility of damage.

D. Temporary Facility Changeover: Do not change over from using temporary security and protection facilities to permanent facilities until Substantial Completion.

E. Termination and Removal: Remove each temporary facility when need for its service has ended, when it has been replaced by authorized use of a permanent facility, or no later than Substantial Completion. Complete or, if necessary, restore permanent construction that may have been delayed because of interference with temporary facility. Repair damaged Work, clean exposed surfaces, and replace construction that cannot be satisfactorily repaired.

1. Materials and facilities that constitute temporary facilities are property of Contractor. NPS reserves right to take possession of Project identification signs.
2. At Substantial Completion, clean and renovate permanent facilities used during construction period.
3. Prior to final completion, close-out construction permits, and transfer monitoring permits including the WYPDES SWPPP to GRTE.

END OF SECTION 01 50 00

GRTE 237343 Rehabilitate “4 Lazy F Ranch” Cabins for Housing

SECTION 01 55 26 – TRAFFIC CONTROL

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes traffic control requirements for construction of new water facilities, including, but not limited to following:
   1. Traffic control planning to accommodate construction operations.
   2. Installation and maintenance of traffic control signs, barricades, and warning devices.
   3. Flag persons to direct traffic through or around the project.
   4. Development of a “TRAFFIC CONTROL PLAN” for both vehicular and foot traffic.

B. Related Sections:
   1. Section 01 11 00 "Summary of Work"
   2. Section 01 33 23 " Submittal Procedures"
   3. Section 01 50 00 "Temporary Facilities and Controls"

C. SUBMITTALS
   1. The Contractor shall submit a “TRAFFIC CONTROL PLAN” or modifications to the plan provided in the construction documents prior to commencing construction. Prepare temporary traffic control plan conforming to the current edition of the Manual of Uniform Traffic Control Devices (MUTCD), published by the U.S. Department of Transportation, Federal Highway Administration. No plan may be implemented until approved by the Contracting Officer (CO).
   2. The traffic control plan shall include roadways, pathways and trails affected by construction including, but not limited to, the Teton Park Road, 4 Lazy F Road, and Headquarters Road. Comply with Section 01 33 23, “Submittal Procedures.”

PART 2 - PRODUCTS

2.1 MATERIALS

A. Temporary traffic control equipment and signs shall conform to the current edition of the Manual of Uniform Traffic Control Devices (MUTCD), published by the U.S. Department of Transportation, Federal Highway Administration.

PART 3 - EXECUTION

3.1 REQUIREMENTS

A. Construction shall be performed and phased to minimize disruption to traffic.

B. Develop traffic control plan to ensure safety of the public, Park employees and residents.

C. Provide access to Water System Operators and Maintenance personnel responsible for maintaining existing Government facilities.

D. Maintain access for firefighting equipment and access to fire hydrants.

E. The Contractor shall be responsible for providing 48-hour advance notice in writing to affected residents in cases of periodic local access closures.

F. Contractor shall provide the CO with a minimum of 7 days' advance notice for any work scheduled that will block access to fire facilities.

G. Emergency and special needs traffic shall be accommodated immediately at any location at any time.

H. Modify the traffic control plan as needed if current operations are obviously inadequate to protect the public and/or accommodate traffic volumes or the construction activities.

I. Provide 24-hour emergency telephone number of person responsible for maintaining the traffic control plan.

J. Temporary one lane closures on roads other than the Teton Park Road will be permitted if required, upon specific approval of the CO. Traffic delays should be kept to 10 minutes or less. Flaggers will be required to assist with directing traffic around temporary closures using the adjacent single lane.

K. Provide traffic control devices and flag persons as needed considering the site condition and construction activity.

L. Flag persons shall be adequately trained and shall be certified by successful completion of a traffic control-training course conducted by the Wyoming Department of Transportation or Wyoming Contractors Association.

3.2 EXECUTION

A. Traffic Controls: Comply with requirements of authorities having jurisdiction.

B. Furnish and install necessary traffic control signs and devices. Maintain and touchup signs so they are legible at all times.

C. Provide other signs as required to inform public and individuals seeking entrance to Project.

D. Provide temporary directional signs for construction personnel, Government Officials, and visitors.

E. Provide flag persons when necessary to direct traffic through or around the project.

F. Maintain and modify temporary traffic control equipment as needed to accommodate traffic level and construction operations.

G. Repair or replace temporary traffic control equipment damaged by the public, or the construction operations.

H. Remove all traffic control equipment upon completion of construction operations.

END OF SECTION 01 55 26

GRTE 237343 Rehabilitate "4 Lazy F Ranch" Cabins for Housing

SECTION 01 57 19.11 – INDOOR AIR QUALITY MANAGEMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Special requirements for Indoor Air Quality (IAQ) management during construction operations.

a. Control of emissions during construction.
b. Moisture control during construction.

2. Procedures for testing baseline IAQ. Baseline IAQ requirements, specify maximum indoor pollutant concentrations for acceptance of the facility.

1.2 DEFINITIONS

A. Definitions pertaining to sustainable development: As defined in ASTM E2114.

B. Adequate ventilation: Ventilation, including air circulation and air changes, required to cure materials, dissipate humidity, and prevent accumulation of particulates, dust, fumes, vapors, or gases.

C. Hazardous Materials: Any material regulated as a hazardous material in accordance with 49 CFR 173 (Code of Federal Regulations), requires a Material Safety Data Sheet (MSDS) in accordance with 29 CFR 1910.1200, or which during end use, treatment, handling, storage, transportation or disposal meets or has components which meet or have the potential to meet the definition of a Hazardous Waste in accordance with 40 CFR 261. Throughout this specification, hazardous material includes hazardous chemicals.

1. Hazardous materials include pesticides, biocides, and carcinogens as listed by recognized authorities, such as the Environmental Protection Agency (EPA) and the International Agency for Research on Cancer (IARC).

D. Indoor Air Quality (IAQ): Composition and characteristics of air in an enclosed space that affect occupants of that space. Indoor air quality of a space refers to relative quality of air in a building with respect to contaminants and hazards and is determined by the level of indoor air pollution and other characteristics of the air, including impact on thermal comfort such as air temperature, relative humidity and air speed.

E. Interior final finishes: Materials and products exposed to interior occupied spaces; including flooring, wall covering, finish carpentry, and ceilings.

F. Packaged dry products: Materials and products installed in dry form delivered in manufacturer's packaging; including carpets, resilient flooring, ceiling tiles, and insulation.

G. Wet products: Materials and products installed in wet form, including paints, sealants, adhesives, special coatings, and materials which require curing.

1.3 QUALITY ASSURANCE

A. Inspection and Testing Lab Qualifications: Minimum of 5 years of experience in performing types of testing specified herein.

1.4 SUBMITTALS

Indoor Air Quality (IAQ) Management Plan: After award and before Pre-construction conference, prepare and submit IAQ Management Plan, including:

1. Procedures for control of emissions during construction.

    a. Identify schedule for application of interior finishes: Identify each interior finish that generates odors, moisture, or vapors or is susceptible to adsorption of odors and vapors. Indicate air handling zone, sequence of application, and curing times.
    b. Identify potential sources of odor and dust.
    c. Identify construction activities likely to produce odor or dust.
    d. Identify areas of project potentially affected, especially occupied areas.
    e. Evaluate potential problems by severity and describe methods of control.
    f. Describe construction ventilation to be provided, including type and duration of ventilation, use of permanent Heating HVAC systems, types of filters and schedule for replacement of filters.
    g. Describe cleaning and dust control procedures.
    h. Describe coordination with commissioning procedures.

2. Procedures for moisture control during construction.

    a. Identify porous materials and absorptive materials.
    b. Identify schedule for inspection of stored and installed porous and absorptive materials.

3. Revise and resubmit Plan as required by Contracting Officer (CO).

    a. Approval of Contractor’s Plan will not relieve the Contractor of responsibility for compliance with applicable environmental regulations.

B. Product Data:

1. Submit product data for filtration media used during construction and during operation. Include Minimum Efficiency Reporting Value (MERV).

Material Safety Data Sheets (MSDS): Submit MSDSs for inclusion in Operation and Maintenance Manual for:

a. Adhesives

b. Floor and wall patching/leveling materials
c. Caulking and sealants
d. Insulating materials
e. Fireproofing and firestopping
f. Carpet
g. Paint
h. Clear finish for wood surfaces
i. Lubricants
j. Cleaning products

C. Inspection and Test Reports:

1. Moisture control inspections
2. Moisture content testing
3. Moisture penetration testing
4. Microbial Growth testing

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 IAQ MANAGEMENT - EMISSIONS CONTROL

During construction operations, follow the recommendations in SMACNA IAQ Guidelines for Occupied Buildings under Construction.

B. HVAC Protection:

1. Seal return registers during construction operations.
2. Provide temporary exhaust during construction operations.
3. To greatest extent possible, isolate and/or shut down return side of HVAC system during construction. When ventilation system must be operational during construction activities, provide temporary filters at air inlets (returns) and at locations for filters prescribed in the design.
4. Contractor shall bear cost of cleaning required due to failure to protect ducts and equipment from construction dust.

Source Control: Provide low and zero VOC materials as specified.

D. Pathway Interruption: Isolate areas of work to prevent contamination of clean or occupied spaces. Provide pressure differentials and/or physical barriers to protect clean or occupied spaces.

E. Housekeeping: During construction, maintain project and building products and systems to prevent contamination of building spaces.

Temporary Ventilation: For materials/products that generally require ventilation for off gassing, provide an ACH (air changes per hour) of **1.5**or more and as follows:

1. Provide minimum 48-hour pre-ventilation of packaged dry products prior to installation. Remove from packaging and ventilate in a secure, dry, well-ventilated space free from strong contaminant sources and residues. Provide a temperature range of 60 degrees Fahrenheit minimum to 90-degree Fahrenheit maximum continuously during ventilation period. Do not ventilate within limits of Work unless otherwise approved by Contracting Officer.
2. Provide adequate ventilation during and after installation of interior wet products and interior final finishes.
3. Provide filtration media with a Minimum Efficiency Reporting Value (MERV) of 8 as determined by ASHRAE 52.2 during construction Coordinate with work of Division 23 (15), Heating Ventilating and Air Conditioning (HVAC).

G. Scheduling: Schedule construction operations involving wet products prior to packaged dry products to the greatest extent possible.

H. Flush-Out: After construction ends, prior to occupancy and with all interior finishes installed, perform a building flush-out by supplying a total air volume of 14,000 cubic feet. of outdoor air per square feet of floor area while maintaining an internal temperature of at least 60 degrees Fahrenheit and relative humidity no higher than 60%.

1. Obtain Contracting Officer's concurrence that construction is complete enough before beginning flush-out.
2. If additional construction involving materials that produce particulates or any of specified contaminants is conducted during or after flush-out, then flush-out process must be restarted.
3. Install new HVAC filtration media in locations identified to have permanent filtration in contract documents after completion of flush-out and before occupancy or further testing.

3.2 IAQ MANAGEMENT - MOISTURE CONTROL

Housekeeping:

1. Keep materials dry. Protect stored on-site and installed absorptive materials from moisture damage.
2. Verify installed materials and products are dry prior to sealing and weatherproofing building envelope.
3. Store interior absorptive materials only after building envelope is sealed and weatherproofed.

B. Inspections: Document and report results of inspections; state whether or not inspections indicate satisfactory conditions.

1. Examine materials for dampness as they arrive. If acceptable to Contracting Officer, dry damp materials completely prior to installation; otherwise, reject materials that arrive damp.
2. Examine materials for mold as they arrive and reject materials that arrive contaminated with mold.
3. Inspect stored and installed absorptive materials regularly for dampness and mold growth. Inspect **after each rain event**.

a. If stored or installed absorptive materials become wet, notify Contracting Officer. Inspect for damage. If acceptable to the Contracting Officer, dry completely prior to closing in assemblies; otherwise, remove (in accordance with the Waste Management Plan) and replace with new materials.

4. Basement: Monitor basement and crawlspace humidity and dehumidify when relative humidity is greater than 70 percent for more than 2 weeks or at first sign of mold growth.
5. Site drainage: Verify final grades of site work and landscaping drain surface water and ground water away from building.
6. Weatherproofing: Inspect moisture control materials as they are being installed. Include:

   a. Air barrier: Verify air barrier is installed without punctures and/or other damage. Verify air barrier is sealed completely.
   b. Flashing: Verify correct shingling of flashing for roof, walls, windows, doors, and other penetrations.
   c. Vapor Barrier: Verify vapor barrier is installed in accordance with Contract documents.
   d. Insulation layer: Verify insulation is installed without voids.
   e. Roofing: In accordance with ASTM D7186 Standard Practice for Quality Assurance Observation of Roof Construction and Repair.

7. Plumbing: Verify satisfactory pressure test of pipes and drains is performed before closing in and insulating lines.
8. HVAC: Inspect HVAC system as specified in Section on Commissioning. And, inspect HVAC to verify:

   a. condensate pans are sloped and plumbed correctly;
   b. chilled water line and refrigerant line insulation are installed and sealed.

C. Schedule:

1. Schedule work such that absorptive materials, such as porous insulations, paper-faced gypsum board, ceiling tile, and finish flooring, are not installed until they can be protected from rain and construction-related water.
2. Weather-proof as quickly as possible. Schedule installation of moisture-control materials, including but not limited to air barriers, flashing, exterior sealants and roofing, at earliest possible time.

Testing for Moisture Content: Test moisture content of porous materials and absorptive materials to ensure they are dry before sealing them into an assembly. Document and report results of testing. Where tests are not satisfactory, dry materials and retest. If satisfactory results cannot be obtained with retest, remove and replace with new materials.

1. Wood: Moisture test as per ASTM D4444 - Standard Test Methods for Use and Calibration of Hand-Held Moisture Meters; unless otherwise indicated acceptable upper limits for wood products are less than 20% at center of piece; less than 15% at surface.

E. Testing for Moisture Penetration:
1. Exterior Walls:

a. Air tightness of the enclosure test: ASTM E779 Standard Test Method for Determining Air Leakage Rate by Fan Pressurization or ASTM E1827 Standard Test Methods for Determining Air tightness of Buildings Using an Orifice Blower Door; acceptable upper limits are **0.25 CFM/sf or less at 50 Pascal's**.
b. Water Leakage: Review as per ASTM E2128 Standard Guide for Evaluating Water Leakage of Building Walls.

END OF SECTION 01 57 19.11

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SECTION 01 57 19.12 – NOISE AND ACCOUSTICS MANAGEMENT

GENERAL

1.1 SUMMARY

A. Section includes:

1. Special requirements for noise and acoustics management during **renovation and construction** operations.
2. Additional requirements related to Noise and Acoustic management are included in Section 01 10 00 Summary of Work and in the GRTE Park Specific requirements in the Mitigations and Best Management Practices report. Where conflicting requirements exist between documents, the more restrictive requirement shall take precedence.

1.2 DEFINITIONS

A. Ambient noise level: The total noise associated with a given environment, being usually a composite of normal or existing sounds from all sources near and far, excluding the noise source at issue.

B. Daytime: The hours from 7 A.M. to 9 P.M. on weekdays and 9 A.M. to 9 P.M. on weekends and holidays.

C. Nighttime: All non-daytime hours.

D. Property line: The real or imaginary line along the ground surface and its vertical extension, which separates real property owned or controlled by one person from contiguous real property owned or controlled by another person or from any public right-of-way or from any public space.

E. Receiving noise area: Any real property where people live or work and where noise is heard, excluding the project or source area.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 NOISE MANGEMENT

A. Noise Control: Perform **renovation and construction** operations to minimize noise. Perform noise-producing work in less sensitive hours of the day or week as directed by the Contracting Officer CO).

B. Repetitive and/or intermittent, high-level noise: Permitted only during Daytime.

1. Do not exceed the following dB(A) limitations at 50 feet:

Sound Level in dB(A)	Time Duration of Impact Noise
70	More than 12 minutes in any hour
80	More than 3 minutes in any hour

2. Maximum permissible construction equipment noise levels at 50 feet:

EARTHMOVING	dB(A)	MATERIALS HANDLING	dB(A)
Front Loaders	75	Concrete Mixers	75
Backhoes	75	Concrete Pumps	75
Dozers	75	Cranes	75
Tractors	75	Derricks Impact	75
Scrapers	80	Pile Drivers	95
Graders	75	Jack Hammers	75
Trucks	75	Rock Drills	80
Pavers, Stationary	80	Pneumatic Tools	80
Pumps	75	Saws	75
Generators	75	Vibrators	75
Compressors	75		

C. Ambient Noise:

1. Maximum noise levels (dB (decibel)) for receiving noise area at property line shall be as follows:

a. Residential receiving area
Daytime: 65 dB
Nighttime: 45 dB

b. Commercial/Industrial receiving area
Daytime: 67 dB
Nighttime: 65 dB

c. In the event the existing local ambient noise level exceeds the maximum allowable receiving noise level (dB), the receiving noise level maximum for construction operations shall be adjusted as follows:

d. Residential receiving area: Maximum 3 additional dB above the local ambient as measured at property line.

e. Commercial/Industrial receiving area: Maximum 5 additional dB above the local ambient as measured at the property line.

3.2 FIELD QUALITY CONTROL

A. Assess potential effects of construction noise on **adjacent neighbors** in accordance with ASTM E1686 and as follows:

1. Ambient noise measurement: Measure at property line at a height of at least four (4) feet above the immediate surrounding surface. Average the ambient noise level over a period of at least 15 minutes.

B. Monitor noise produced from construction operations in accordance with ASTM E1780.

END OF SECTION 01 57 19.12

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SECTION 01 57 23 - TEMPORARY STORM WATER POLLUTION PREVENTION

GENERAL

1.1 SUMMARY

A. Federal Regulations for controlling discharges of pollutants (including chemicals, erodible material, and trash) from municipal separate storm sewer systems, construction sites, and industrial activities, were brought under the National Pollution Discharge Elimination System (NPDES) permit process by amendments to the Clean Water Act (CWA), and promulgation of federal stormwater regulations issued by the United States Environmental Protection Agency (USEPA). The USEPA uses amount of ground disturbance as a measure of a project potential to generate pollution from erosion. NPDES Phase I regulates discharges from construction sites that disturb 5 acres or more. NPDES Phase II regulations expand existing General Permit requirements under Phase I to include/regulated discharges from construction sites that disturb land equal to or greater than one (1) acre and less than 5 acres, known as Small Construction Activity. Construction disturbances 1 acre and above typically require a formal NPDES permit and a formal Stormwater Pollution Prevention Plan (SWPPP) must be submitted to Agency(ies) with Jurisdiction for review and approval.

1. Disturbance under this project is over five acres and the Contractor will be required to obtain a formal NPDES permit and develop a formal Temporary Storm Water Pollution Prevention Plan (SWPPP).

B.

C. The work of this section consists of implementing measures to prevent discharges of pollutants, including temporary storm water pollution during construction activities, through compliance with NPDES permit program,

D. Work of this section consists of implementing measures to Temporary Storm Water Pollution during construction activities, through compliance with NPDES permit program;

1.2 Provide necessary equipment, labor and supplies for erosion and sediment control project wide. Work consists of site grading, providing silt fence, wattles, erosion control blankets (ECB's), ground protection mats, tackifier and other necessary measures required for effective control. Sediment and erosion control Work shall comply with requirements provided on the drawings and these specifications, including any additional requirements of the WYPDES permit regulated by the Wyoming Department of Environmental Quality (WDEQ).DEFINITIONS

A. Definitions pertaining to sustainable development: As defined in ASTM E2114.

B. Environmental Pollution and Damage: The presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; or degrade utility of the environment for aesthetic, cultural, or historical purposes.

C. National Pollution Discharge Elimination System (NPDES) Phase I: Regulates discharges from construction sites that disturb 5 acres or more.

D. NPDES Phase II: Regulations expand existing General Permit requirements under Phase I to include and regulate discharges from construction sites that disturb land equal to or greater than one (1) acre and less than 5 acres, known as Small Construction Activity.

E. Storm Water Pollution Prevention Plan (SWPPP): Developed and implemented stormwater management measures to protect surface water from pollutants during construction activities disturbing an acre or more in compliance with federal, state, and local requirements for permit approval under NPDES program.

1.3 SUBMITTALS

A. After contract award and before pre-construction conference, prepare and submit:
   1. A SWPPP showing SWPPP satisfies Federal and State NPDES permit requirements.

B. Inspection Schedule: Submit schedule for inspection and monitoring of pollution prevention measures.

C. Inspection Schedule: Submit schedule for inspection and monitoring of storm water pollution prevention measures.

D. Erosion Control Products: Submit manufacturer's product information and installation recommendations for silt fence, filter fabric, erosion control blanket, straw bales, and other materials proposed for use on this project.

E. Samples: Upon request of the Contracting Officer.

F. Track-out Mitigation Plan: Submit a detailed plan for track-out prevention to the CO for approval. Comply with provisions of PART 3.6, below

1.4 QUALITY ASSURANCE

A. Contractor shall prepare and submit a plan to Contracting Officer (CO) for review and concurrence.

B. Orientation Meeting: Contractor shall arrange and conduct an Erosion and Sediment Control meeting/briefing to inform parties, scheduled to be on-site during project, of measures to be implemented for proper erosion and sediment control (may be included as part of Pre-Construction Meeting).

   1. Installation of silt fences, storm drain protection, and other forms of erosion and sediment control shall not begin until after this meeting has occurred.

C. Pollution Prevention and Erosion Control Manager: Contractor shall designate Pollution Prevention and Erosion Control Manager responsible for implementation, inspection, maintenance, and amendments to approved plan.

1. Pollution Prevention and Erosion Control Manager shall be familiar with temporary storm water pollution prevention procedures and Best Management Practices and ensure emergency procedures and plan are updated as needed and available for inspection.

2. When changes in approved plan are required, Pollution Prevention and Erosion Control Manager shall prepare and certify an amendment and submit to Contracting Officer for review and concurrence.

PART 2 - PRODUCTS

2.1 TEMPORARY STORM WATER POLLUTION PREVENTION PLAN

A. Provide SWPPP which satisfies Federal and State NPDES permit requirements and includes:

1. Site description.
2. Identification and contract information for Pollution Prevention and Erosion Control Manager.
3. Expected sequencing of operations and construction schedule.
4. Weather monitoring procedure.
5. Descriptions and details Best Management Practices for of pollution prevention and erosion controls, including dust control.
6. Pollution prevention and erosion control plans.
7. Controls for other potential onsite storm water pollutants.
8. Applicable specifications.
9. Maintenance and inspection procedures and forms.
10. Description of potential non-storm water discharges at site.
11. Notice of Intent (NOI) form.
12. Notice of Termination (NOT) form.
13. Contractor and Sub-contractor Certification forms.
14. Other record keeping forms and procedures.
15. Housekeeping Best Management Practices, including vehicle wash-down areas, protection of equipment storage and maintenance areas, and sweeping of roadways related to hauling activities.
16.

2.2 SILT FENCE

A. Posts: Use either wooden stakes or steel posts for fence construction. Wooden stakes utilized for silt fence construction, shall have a minimum cross section of 2 by 2 inches when pine is used, and have a minimum length of 5 feet. Steel posts (standard "U" or "T" section) utilized for silt

fence construction, shall have a minimum mass of 1.33 pounds/linear foot and a minimum length of 5 feet.

B. Filter Fabric: Provide geotextile that complies with the requirements of ASTM D4439 and consists of polymeric filaments which are formed into a stable network such that filaments retain their relative positions. The filament shall consist of a long-chain synthetic polymer composed of at least 85 percent by weight of ester, propylene, or amide, and contains stabilizers and/or inhibitors added to the base plastic to make the filaments resistant to deterioration due to ultraviolet and heat exposure. Provide synthetic filter fabric that contains ultraviolet ray inhibitors and stabilizers to assure a minimum of six months of expected usable construction life at a temperature range of 0 to 120 degrees F. The filter fabric shall meet the following requirements:
   1. Minimum Grab Tensile Elongation: 100 lbs. min. – 30 percent max (ASTM D4632)
   2. Minimum Trapezoid Tear: 55 lbs. min. (ASTM D4533)
   3. Permittivity: 0.13 sec-1 (ASTM D4491)
   4. AOS (U.S. Std Sieve): 20-100 (ASTM D4751)
   5. Flow Rate: 10 gal/min/sq. ft (ASTM D4491)

2.3 WATTLES

A. Manufacturer shall meet the following requirements:
   1. Fiber Material: Aspen Wood Fiber or approved equal
   2. Fiber Dimensions: 80% greater than 6 inch
   3. Netting: ½” x ½” UV stable, synthetic
   4. Diameter: 9-inch
   5. Density: 5.7 lbs/cu. Ft
   6. Configuration: Cylindrical with closed ends
   7. Anchoring: 24-inch 2” x 2” sq. (or larger) wooden stakes

B. Requirement non-toxic, environmentally friendly.
   1. Comply with manufacturers’ requirements for mixing, application rates and methods and safety precautions.

2.4 EROSION CONTROL BLANKETS (ECBs)

A. Manufacturer/Make: shall meet the following requirements:
   1. Fiber Material: 100% Rocky Mountain Aspen Wood Fiber or approved equal
   2. Fiber Dimensions: 80% greater than 6 inch
   3. Top Net Type: Biodegradable Weave
   4. Net Material: Cotton thread in natural color
   5. Net Openings: ½-inch
   6. Tensile Strength: 16 lb/in minimum
   7. Water Absorption: 275%
   8. Mass per Unit Area: 11.5 oz/sq. Yd
   9. Fasteners: 12” x 1” 11-gauge staples

B. ECBs shall be free of defects and voids that would interfere with proper installation or impair performance, in conformance with the product manufacturer’s specifications

2.5 TEMPORARY GROUND PROTECTION MATS

A. Furnished by NPS for Contractor use.

PART 3 - EXECUTION

3.1 ENVIRONMENTAL PROTECTION

A. Protection of Natural Resources: Comply with applicable regulations and these specifications. Preserve natural resources within project boundaries and outside limits of work performed under this Contract in their existing condition or restore to an equivalent or improved condition as approved by Contracting Officer.

Construction Zone: Arrange construction activities to minimize pollution (i.e., erosion, trash, etc.) to maximum practical extent.

1. Clearing, excavation, and grading shall be limited to those areas of project site necessary for construction. Minimize area exposed and unprotected.
2. Clearly mark and delineate limits of work activities.
3. Equipment shall not be allowed to operate outside limits of work or to disturb existing vegetation.
4. Excavation and grading shall be completed during dry season to maximum extent possible.
5. Material shall be stored away from locations where water is present to greatest extent practicable.

3.2 REGULATORY REQUIREMENTS

Permits: Contractor shall obtain required NPDES permits resulting in no impacts to scheduled work. Contractor shall account for possibility of significant lead time in scheduling and executing work.

1. Implement requirements of NPDES permit for erosion control due to storm water runoff during construction.
2. Implement good housekeeping practices, inspections and record keeping.
3. Prior to construction, Contractor and Subcontractors shall sign certifications (included in the plan) that they understand requirements of NPDES permit.
4. Subcontractors shall comply with requirements of NPDES under supervision of Contractor.
5. Accepted plan shall comply with terms and conditions of EPA permit.

B. Notice of Intent (NOI): Contractor shall file a Notice of Intent and formal SWPPP as required to the Agency(ies) with Jurisdiction.

C. Permit Transfer: The NPDES permit will be transferred from the Contractor to the GRTE NPS prior or subsequent to final completion of this project
   1. Permit transfer shall take place after the Contracting Officer has approved site conditions and temporary erosion control measures. Contractor shall remove any temporary erosion control that is no longer required as requested by the CO.

2. Contractor shall immediately address any inadequacies in order to expedite permit transfer and subsequent seeding and revegetation.
3. Subsequent to permit transfer, maintenance and removal of all temporary control measures shall become the responsibility of GRTE NPS.
4. Coordinate the completion and submission of the "Notice of Transfer and Acceptance" form attached subsequent to this specification.

D. Notice of Termination (NOT): After Substantial Completion of Work, file a Notice of Termination (NOT) with the Agency(ies) with Jurisdiction.

E. Contracting Officer Notification: Contractor shall notify Contracting Officer in writing and by telephone of these events:

1. Erosion and sediment control meeting/briefing.
2. Following installation of required sediment control structures.
3. Prior to removal of or modification to sediment control structures.
4. Prior to removal of sediment control structures.

3.3 TEMPORARY STORM WATER POLLUTION PREVENTION PLAN

A. Review and Acceptance: Contractor and Contracting Officer will jointly review draft Plan and agree to needed revisions. Contractor shall incorporate revisions, sign, and submit final Plan to Contracting Officer. Final Plan will be the document enforced on the project.

1. Accepted Plan will describe and ensure implementation of practices to be used to reduce pollutants in storm water discharges.
2. Contractor shall maintain current copy of Plan and associated records and forms at jobsite throughout duration of project.
3. Plan shall be available for public inspection and inspection and use of Contracting Officer.
4. Approval of Contractor's Plan will not relieve Contractor of responsibility for compliance with applicable environmental regulations.

B. Implementation: Implement Plan as required throughout construction period and maintain erosion control elements in proper working order.

1. Do not perform clearing and grubbing or earthwork until Plan has been implemented.

C. SWPPP (including inspection forms) and data used to complete the NOI shall be provided to Contracting Officer after Substantial Completion of project.

3.4 SITE INSPECTIONS AND PLAN REVISIONS

A. Inspections: Contractor and Contracting Officer will perform a weekly inspection onsite.

1. Inspection shall include disturbed areas not completely stabilized, areas used for storage of materials, locations where vehicles enter or exit site, and other erosion and sediment controls included in the Plan.
2. Inspections shall be documented.
3. Inspection forms shall be retained onsite in Plan notebook throughout construction period.

B. Plan Revisions: It may be necessary to revise Plan during construction to make necessary improvements, revisions, or to respond to unforeseen conditions noted during construction or site inspections.

1. Plan shall specify mechanism whereby revisions may be proposed by Contractor or Contracting Officer.
2. Contractor and Contracting Officer will jointly review each revision to Plan before changes incorporated and implemented. Contractor will then provide revised copy of Plan to Contracting Officer.
3. Accepted modifications will be implemented within 7 calendar days following date of inspection when deficiencies or necessary corrections are first noted.

C. Negligence: Provide additional temporary erosion and pollution controls made necessary by Contractor's errors or negligence at no additional cost to Government.

3.5 HOUSEKEEPING AND SITE MANAGEMENT

A. Store materials onsite in conformance to Federal, state, local, and manufacturer's regulations and specifications. Use Best Management Practices to minimize risk of materials coming into contact with environmental conditions (i.e. water and wind) that could disperse them.

B. Manage solid waste in conformance to Federal, state, and local regulations. Best Management Practices should be used to minimize risk of materials coming into contact with environmental conditions (i.e. water and wind) that could disperse them.

C. Include a spill prevention and control plan with provisions placed in SWPPP.

D. Manage hazardous waste (including contaminated soil) in conformance to Federal, state, local and NPS regulations and guidelines.

3.6 EROSION CONTROL MEASURES

A. Erosion control measures shall consist of Best Management Practices for storm water discharges, including silt fencing, barrier protectors, straw bales, temporary soil retention blankets, excelsior drainage filters, sediment traps and berms.

B. Berms and excelsior drainage filters shall be used to form sediment traps and control run-on and run-off into other areas, including creeks, streams, marshes, access roads, well areas, and staging areas.

C. Erosion control measures shall be used to contain only direct precipitation in construction zone. Contained water shall be allowed to percolate into ground or drain slowly through drainage filter sediment traps.

D. Contractor's erosion and sediment control measures shall effectively prevent erosion (by wind, water, traffic, and other sources) throughout the Project Site where surface vegetation and soil may be disturbed by the Work.

E. Take action that has only negligible to minor, adverse effects on site hydrology and fluvial processes, including flow, circulation, velocities, hydroperiods, water level fluctuations, sediment transport, channel, morphology, and so on. Take care to avoid any rutting caused by vehicles or equipment.

F. For soil stabilization and erosion control, use only certified weed-free materials to avoid introduction of exotic plant species. Provide product submittals in accordance with Section 01 33 23.

G. Plan and maintain vegetated buffers between areas of soil disturbance and waterways.

H. Do not allow water emanating from the site to enter wetlands, or any natural water body, including those defined by the Wyoming State Engineer's Office (WSEO) as "surface waters of the State."
   1. Throughout the duration of the project, control erosion and prevent siltation to rivers, streams and impoundments. Control storm water by using BMPs applied to erodible materials exposed by any project activity.

I. Control flowing and standing water to prevent erosion and excessive ponding. Do not discharge turbid water from the Site and prevent contamination of water on site by construction materials and debris including fuels and lubricants.

J. Erosion control measures shall be used to contain direct precipitation in the construction zone. The contained water shall be allowed to percolate into the ground or drain slowly through the drainage filter sediment traps.

K. Earthen sediment traps or holding ponds shall not be used unless accepted by Contracting Officer.

L. Reduce runoff velocity and direct surface runoff around and away from fuel containment, storage, and borrow areas.

M. Divert surface runoff around and away from cut and fill slopes.

N. Dress, prepare and install erosion controls on slopes exceeding 3:1 in locations shown in the plans and in accordance with the SWPPP immediately following suspension of grading operations.

O. Fill slopes shall be dressed and prepared as the embankment proceeds to the extent needed to stop erosion and sedimentation.

P. Coordinate grading with site slopes to permit free drainage and prevent impounding of water.

Q. Place drainage filters around catch basins to create sediment traps to control run-off from construction area.

R. Excess water used for dust control shall be contained within demolition areas by erosion control measures.

S. Contractor shall prevent deposition of materials onto paved areas. Contractor shall inspect paved areas for deposited materials weekly and remove materials immediately.

T. Furnish, install, maintain, and operate necessary control measures and other equipment necessary to prevent erosion as described in approved SWPPP.

U. Before work begins, sufficient equipment shall be available on site to assure operation and adequacy of erosion control system can be maintained.

V. Track-out Mitigation: The Contractor shall prevent the deposition of materials onto paved areas. Implement necessary measures and devices to prevent track-out of sediment from tires and heavy equipment onto paved surfaces. Take measures described below to mitigate track-out:
   1. The Contractor shall inspect paved areas for deposited materials weekly and remove the materials immediately. Use cleaning methods approved by the CO.
   2. Install track-out controls and devices including a 30 to 50 feet long track-out pad, or grizzly with rails.
   3. Submit a detailed plan for track-out prevention to the CO for approval.
   4. Remove controls and devises when they are no longer required.

W. Coordinate erosion and sediment control measures with the construction of the permanent drainage facilities which shall be constructed prior to the grading operation to ensure economical, effective, and continuous erosion and siltation control.

3.7 PRODUCT STORAGE

A. Store products including ECB's, wattles and silt fence in a clean and dry locations protected from weather and construction activities.

3.8 SILT FENCE

A. Install silt fence in accordance with the details provided in the drawings and the manufacturer's recommendations.

3.9 WATTLES

A. Install wattles in accordance with details in the drawings, the approved SWPPP and the manufacturer's recommendations. Install wattles on all slopes exceeding 3:1.
   1. Placement: Arrange wattles as shown in the plans, or as required by the SWPPP.
   2. Trench: Installation trench depth shall be 1/3 the diameter of the wattle.
   3. Anchors: Install wooden stakes every 3 feet. Drive stakes until they are flush with the top of the wattle. Do not pinch wattles during stake installation. Replace damaged stakes when required by the CO or representative thereof.
   4. Maximum vertical spacing of wattles for slope installations:
      a. 1:1 slopes – 10 feet
      b. 1:2 slopes – 20 feet
      c. 1:3 slopes – 30 feet

3.10 EROSION CONTROL BLANKETS

A. Install in accordance with manufacturer's recommendations.

B. Install a combination of blankets and wattles on slopes exceeding 2:1 slopes.

C. In areas where both blankets and wattles are used, install blanket under wattle and in the wattle trench.

3.11 TEMPORARY GROUND PROTECTION MATS

A. Place mats in locations notes in the drawings and in accordance with Section 01 12 00 – Natural Features Protection.

3.12 MAINTENANCE OF TEMPORARY FACILITIES

A. Ensure erosion and sediment control structures remain effective throughout excavation and grading operations. Relocate structures as necessary.

B. Contractor shall designate an Emergency Contact Person and provide the CO contact information. When required, the Emergency Contact Person shall be responsible for mobilizing a crew in order to control conditions at the site.

C. Inspect control structures after each significant rainfall. Promptly repair breaches which occur.

D. Contractor shall remove entrapped sediment from behind excelsior drainage filter after each storm.

E. Maintain erosion and sediment control until Substantial Completion; comply with inspection and repair requirements throughout this time period. subsequent to Substantial completion, CO will inspect and approve of erosion control devices prior to transfer of the WYPDES permit. Once the transfer of the permit is complete, removal and disposal of controls shall be the responsibility of GRTE NPS.

3.13 REPORTING

A. If a discharge occurs or if project receives written notice or order from regulatory agency, Contractor shall immediately notify Contracting Officer and shall file written report to Agency(ies) with Jurisdiction within 7 days of discharge event, notice, or order. Corrective measures shall be implemented immediately following discharge, notice, or order. The report to the Agency(ies) with Jurisdiction shall contain:

1. Date, time, location, nature of operation, and type of discharge, including cause or nature of notice or order.
2. Best Management Practices deployed before discharge event, or prior to receiving notice or order.
3. Date of deployment and type of Best Management Practices deployed after discharge event, or after receiving notice or order, including additional Best Management Practices installed or planned to reduce or prevent re-occurrence.
4. An implementation and maintenance schedule for affected Best Management Practices.

3.14 SEDIMENT DISPOSAL

A. Sediment excavated from temporary sediment control structures shall be disposed on site with general fill, or with topsoil. Sediment shall be allowed to dry out as required before reuse.

B. Contractor shall place sediment removed from traps and other structures where it will not enter a storm drain or watercourse and where it will not immediately reenter the basin.

3.15 REMOVAL OF TEMPORARY STORM WATER POLLUTION CONTROL MEASURES

A. All temporary control measures that are no longer required shall be removed by the Contractor prior to transfer of the WYPDES permit.

3.16 PERMIT TRANSFER AND ACCEPTANCE

A. Transfer the WYPDES permit in accordance with 3.12/E above.

END OF SECTION 01 57 23

GRTE 237343 Rehabilitate “4 Lazy F Ranch” Cabins for Housing

SECTION 01 67 00 - PRODUCT REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for selection of products for use in Project; product delivery, storage, and handling; manufacturers' standard warranties on products; special warranties; and environmental requirements.

1.2 DEFINITIONS

Products: Items purchased for incorporating into Work, whether purchased for Project or taken from previously purchased stock. The term "product" includes "material," "equipment," "system," and terms of similar intent.

1. Named Products: Items identified by manufacturer's product name, including make or model number and other designation shown or listed in manufacturer's published product literature, current as of date of Contract Documents.
2. New Products: Items that have not previously been incorporated into another project or facility**,** except that products consisting of recycled-content materials are allowed, unless explicitly stated otherwise. Products salvaged or recycled from other projects are not considered new products.
3. Approved Equals: Product demonstrated and approved through submittal process, or where indicated as a product substitution, to have indicated qualities related to type, function, dimension, in-service performance, physical properties, appearance, and other characteristics that equal or exceed those of specified product.

B. Definitions pertaining to sustainable development: As defined in ASTM E2114.

C. Biobased Materials: As defined in Farm Security and Rural Investment Act, for purposes of Federal procurement of biobased products, "biobased" means a "commercial or industrial product (other than food or feed) composed, in whole or in significant part, of biological products or renewable domestic agricultural materials (including plant, animal, and marine materials) or forestry materials." Biobased materials also include fuels, chemicals, building materials, or electric power or heat produced from biomass as defined by The Biomass Research and Development Act of 2000.

Biobased content: Amount of biobased carbon in material or product as a percentage of weight (mass) of total organic carbon in material or product.

D. Chain-of-Custody: Process whereby a product or material is maintained under physical possession or control during its entire life cycle.

Environmentally preferable products: Products and services with lesser or reduced effect on the environment in comparison to conventional products and services. Refer to Environmental Protection Agency's (EPA) Environmentally Preferable Purchasing Program for more information.

F. Stewardship: Responsible use and management of resources.

G. Maintenance of ecosystem components and functions for future generations.

1. Recycled Content Materials: Products containing pre-consumer or post-consumer materials as all or part of their feedstock. Recycled content claim shall be consistent with International Organization for Standardization (ISO) 140001 Standard for the Use of Environmental Marketing Claims.
2. Rapidly Renewable Material: Material made from plants typically harvested within a ten-year cycle.
3. Regional Materials: Materials manufactured and extracted, harvested, or recovered within a radius of 500 miles from Project location.

1.3 SUBMITTALS

A. Record Submittals and submit:

1. Affirmative Procurement Reporting Form. Submit on form in Appendix A of this Section, or similar form as approved by Contracting Officer (CO).

Submit environmental data in accordance with Table 1 of ASTM E2129 for these products:

a. Finish Carpentry
b. Plastic Fabrications
c. Building Insulation
d. Roofing
e. Joint Sealers
f. Wood & Plastic Doors
g. Gypsum Board
h. Tile
i. Resilient Flooring
j. Paints & Coatings
k. Plumbing fixtures and equipment.
l. HVAC equipment
m. Lighting equipment

.Material Safety Data Sheets (MSDS): For each product required by OSHA to have a MSDS, submit an MSDS. MSDS shall be prepared **no earlier than June 1998**. Include information for MSDS Sections 1 to 16 in accordance with ANSI Z400.1 and as follows:

a. Section 1: Chemical Product and Company Identification
b. Section 2: Composition/Information on Ingredients
c. Section 3: Hazards Identification
d. Section 4: First Aid Measures
e. Section 5: Fire Fighting Measures
f. Section 6: Accidental Release Measures
g. Section 7: Handling and Storage
h. Section 8: Exposure Controls/Person Protection

i. Section 9: Physical and Chemical Properties
j. Section 10: Stability and Reactivity Data
k. Section 11: Toxicological Information. Include data used to determine the hazards cited in Section 3. Identify acute data, carcinogenicity, reproductive effects, and target organ effects.
l. Section 12: Ecological Information. Include data regarding environmental impacts during raw materials acquisition, manufacture, and use. Include data regarding environmental impacts in event of accidental release.
m. Section 13: Disposal Considerations. Include data regarding proper disposal of the chemical. Include information regarding recycling and reuse. Indicate whether or not product is considered to be "hazardous waste" according to United States EPA Hazardous Waste Regulations 40 CFR 261 (Code of Federal Regulations).
n. Section 14: Transportation Information. Identify hazard class for shipping.
o. Section 15: Regulatory Information. Identify federal, state, and local regulations applicable to the material.
p. Section 16: Other Information. Include additional information relative to recycled content, biobased content, and other information regarding environmental and health impacts. Identify the date MSDS was prepared.

QUALITY ASSURANCE

A. Compatibility of Options: If Contractor is given option of selecting between two or more products for use on Project, product selected shall be compatible with products previously selected, even if previously selected products were also options.

1.5 PRODUCT DELIVERY, STORAGE, AND HANDLING

Deliver, store, and handle products using means and methods that prevent damage, deterioration, and loss, including theft. Comply with manufacturer's written instructions.

B. Delivery and Handling:

1. Schedule delivery to minimize long-term storage at Project site and prevent overcrowding of construction spaces.
2. Coordinate delivery with installation time to ensure minimum holding time for items that are flammable, hazardous, easily damaged, or sensitive to deterioration, theft, and other losses.
3. Deliver products to Project site in undamaged condition; in manufacturer's original sealed container or other packaging system; complete with labels and instructions for handling, storing, unpacking, protecting, and installing.
4. Inspect products on delivery to ensure compliance with Contract Documents. Ensure products are undamaged and properly protected.
5. Obtain materials in biodegradable or recyclable/reusable packaging which uses minimum amount of packaging possible.

C. Storage:

1. Allow for inspection and measurement of quantity or counting of units.
2. Store materials in manner to not endanger Project structure.

3. Store products subject to damage by the elements, under cover in weather tight enclosure above ground, with ventilation adequate to prevent condensation.
4. Store cementitious products and materials on elevated platforms.
5. Store foam plastic from exposure to sunlight, except to extent necessary for period of installation and concealment.
6. Comply with product manufacturer's written instructions for temperature, humidity, ventilation, and weather-protection requirements for storage.
7. Protect stored products from damage and liquids from freezing.
8. Store loose granular materials in well drained area on solid surfaces to prevent mixing with foreign matter.

1.6 PACKAGING

A. Where Contractor has option to provide one of listed products or equal, preference shall be given to products with minimal packaging and easily recyclable packaging as defined in ASTM D5834.

B. Maximize use of source reduction and recycling procedures outlined in ASTM D5834.

Provide minimum [45] fourty five percent post-consumer recycled content and minimum 100 percent recovered fiber content of industrial paperboard in accordance with EPA's Comprehensive Procurement Guidelines and ASTM D5663.

D. Provide minimum [10]ten percent post-consumer recycled content and minimum [10] ten percent recovered fiber content of carrier board in accordance with EPA's Comprehensive Procurement Guidelines and ASTM D5663.

E. Provide minimum [5] five percent post-consumer recycled content and minimum [5] five percent recovered fiber content of brown papers (e.g., wrapping papers and bags) in accordance with EPA's Comprehensive Procurement Guidelines and ASTM D5663.

1.7 ENVIRONMENTALLY PREFERABLE PRODUCTS

Provide environmentally preferable products to greatest extent possible.

1. To greatest extent possible, provide products and materials with a lesser or reduced effect on the environment considering raw materials acquisition, production, manufacturing, packaging, distribution, reuse, operation, maintenance, and/or disposal of the product.
2. Eliminate use of ozone depleting compounds during and after construction where alternative environmentally preferable products are available, consistent with either Montreal Protocol and Title VI or Clean Air Act Amendments of 1990, or equivalent overall air quality benefits that take into account life cycle impacts.
3. Use products meeting or exceeding EPA's recycled content recommendations for EPA-designated products. Use materials with recycled content such that the sum of post-consumer recycled content plus one-half of the pre-consumer content constitutes at least 10% (based on cost) of total value of the materials in project.

1.8 PRODUCT WARRANTIES

A. Warranties specified in other Sections shall be in addition to, and run concurrent with, other warranties required by Contract Documents. Manufacturer's disclaimers and limitations on product warranties do not relieve Contractor of obligations under requirements of Contract Documents.

Manufacturer's Warranty: Preprinted written warranty published by individual manufacturer for product specifically endorsed by manufacturer to Owner.

2. Special Warranty: Written warranty required by or incorporated into Contract Documents, either to extend time limit provided by manufacturer's warranty or to provide more rights for Owner.

B. Special Warranties: Prepare written document containing appropriate terms and identification, ready for execution. Submit draft for approval before final execution.

1. Manufacturer's Standard Form: Modified to include Project-specific information and properly executed.
2. Specified Form: When specified forms are included with Specifications, prepare written document using appropriate form properly executed.
3. Refer to Divisions 2 through 49 Sections for specific content requirements and particular requirements for submitting special warranties.

C. Submittal Time: Comply with requirements in Section 01 77 00 "Closeout Procedures."

PART 2 - PRODUCTS

2.1 PRODUCT SELECTION PROCEDURES

General Product Requirements: Provide products to comply with Contract Documents, undamaged and, unless otherwise indicated, new at time of installation.

Provide products complete with accessories, trim, finish, fasteners, and other items needed for a complete installation and indicated use and effect.

2. Standard Products: If available, and unless custom products or nonstandard options are specified, provide standard products of types produced and used successfully in similar situations on other projects.
3. Government reserves right to limit selection to products with warranties not in conflict with requirements of Contract Documents.
4. Where products are accompanied by term "as selected," Contracting Officer will make selection.
5. Where products are accompanied by term "match sample," sample to be matched is Governments.
6. Descriptive, performance, and reference standard requirements in Specifications establish "salient characteristics" of products.

B. Product Selection Procedures:

1. Product: Where Specifications name single product and manufacturer, provide named product that complies with requirements or approved equal.
2. Manufacturer/Source: Where Specifications name single manufacturer or source, provide product by named manufacturer or source that complies with requirements or approved equal.

Available Products: Where Specifications include list of names of both products and manufacturers, provide one of the products listed, or an unnamed product, that complies with requirements. Comply with provisions in Part 2 "Products" Articles for consideration of an unnamed product.

4. Available Manufacturers: Where Specifications include list of manufacturers, provide a product by one of the manufacturers listed, or an unnamed manufacturer, that complies with requirements. Comply with provisions in Part 2 "Products" Articles for consideration of an unnamed product.

Product Options: Where Specifications indicate sizes, profiles, and dimensional requirements on Drawings are based on a specific product or system, provide specified product, system, or approved equal.

6. Visual Matching Specification: Where Specifications require matching an established Sample, select product that complies with requirements and matches Architect's sample. Contracting Officers decision will be final on whether a proposed product matches.

    a. If no product available within specified category matches and complies with other specified requirements, comply with provisions in Part 2 "Product Substitutions" Article for proposal of product.

7. Visual Selection Specification: Where Specifications include phrase "as selected from manufacturer's colors, patterns, textures" or a similar phrase, select a product that complies with other specified requirements.

    a. Standard Range: Where Specifications include phrase "standard range of colors, patterns, textures" or similar phrase, Contracting Officer will select color, pattern, density, or texture from manufacturer's product line that does not include premium items.
    b. Full Range: Where Specifications include phrase "full range of colors, patterns, textures" or similar phrase, Contracting Officer will select color, pattern, density, or texture from manufacturer's product line that includes both standard and premium items.

2.2 APPROVED EQUAL PRODUCTS

A. Conditions: Contracting Officer will consider Contractor's request when the following conditions are satisfied. If following conditions are not satisfied, Contracting Officer will return requests without action, except to record noncompliance with these requirements:

1. Evidence proposed product does not require revisions to Contract Documents, that it is consistent with Contract Documents and will produce indicated results and is compatible with other portions of Work.
2. Detailed comparison of significant qualities of proposed product with those named in Specifications. Significant qualities include attributes such as performance, weight, size, durability, visual effect, and specific features and requirements indicated.
3. Evidence proposed product provides specified warranty.
4. List of similar installations for completed projects with project names and addresses and names and addresses of architects and owners, if requested.

5. Samples, if requested.

PART 3 - EXECUTION

3.1 PROTECTION AFTER INSTALLATION

A. Provide adequate coverings as necessary to protect installed materials from damage resulting from natural elements, traffic, and subsequent construction. Remove when no longer needed.

END OF SECTION 01 67 00

AFFIRMATIVE PROCUREMENT REPORTING FORM
Recycled Content Materials & Biobased Content Materials

Project Name: _____ Project Number: _____

Contractor Name: _____ License Number: _____

Contractor Address: _____

Product	Total $ value provided	Total $ value with recycled content Pre-consumer	Total $ value with recycled content Post-consumer	Total $ value with biobased content	Exempted indicate 1,2,3,4	Comments
Hydraulic Mulch (paper based)						
Hydraulic Mulch (wood based)						
Compost						
Parking Stops (Concrete w/ fly ash, slag cement or low cement content)						
Parking Stops (Plastic/Rubber)						
Patio Blocks/Rubber						
Patio Blocks/Plastic						
Playground Surfaces						
Concrete with fly ash						
Concrete with slag cement						
Concrete with low cement content						
Plastic lumber						
Building Insulation						
Rock Wool						
Fiber glass						
Cellulose						
Perlite Comp Board						
Plastic Rigid Foam						
Glass Fiber Reinforced Foam						
Phenolic Rigid Foam						
Ceramic tile						
Resilient flooring						
Floor Tiles/Rubber						
Floor Tiles/Plastic						
Running Tracks						
Carpet (PET)						
Paint						

Reprocessed Latex Paint White & Light Colors						
Reprocessed Latex Dark Colors						
Consolidated Latex Paint						
Toilet/Shower partitions (plastic or steel)						
Other						

CERTIFICATION

I hereby certify the information provided herein is accurate and that the requisition/procurement of all materials listed on this form comply with current EPA standards for recycled/recovered materials content. The following exemptions may apply to the non-procurement of recycled/recovered content materials:

1. The product does not meet appropriate performance standards.
2. The product is not available within a reasonable time frame.
3. The product is not available competitively (from two or more sources).
4. The product is only available at an unreasonable price (compared with a comparable non-recycled content product.)

Signature: _____ Date: _____

END OF
AFFIRMATIVE PROCUREMENT REPORTING FORM
Recycled Content Materials & Biobased Content Materials

GRTE 237343 Rehabilitate "4 Lazy F Ranch" Cabins for Housing

SECTION 01 73 40 - EXECUTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes general procedural requirements governing execution of Work including:

Coordination with utility service providers

2. Construction layout
3. Field engineering and surveying
4. General installation of products
5. Progress cleaning
6. Starting and adjusting
7. Protection of installed construction
8. Correction of the Work

1.2 SUBMITTALS

A. Certificates: Submit certificate signed by land survey or certifying location and elevation of improvements comply with requirements.

B. Landfill Receipts: Submit copy of receipts issued by landfill facility, licensed to accept hazardous materials, for hazardous waste disposal.

C. Certified Surveys: Submit two> copies signed by land surveyor

1.3 QUALITY ASSURANCE

A. Land Surveyor Qualifications: Professional land surveyor legally qualified to practice in jurisdiction where Project is located and-is experienced in providing land-surveying services of kind indicated.

PART 2 - PRODUCTS (Not Used)

PART 3 - EXECUTION

3.1 EXAMINATION

Existing Conditions: Existence and location of site improvements and other construction indicated as existing are not guaranteed.

1. Before beginning work, investigate and verify the existence and location of mechanical and electrical systems and other construction affecting the Work.

2. Before construction, verify the location and points of connection of utility services.

Existing Utilities: Existence and location of underground and other utilities and construction indicated as existing are not guaranteed. Before beginning sitework, investigate and verify existence and location of underground utilities and other construction affecting Work.

1. Underground utility locations indicated on the POR drawings are provided from information supplied by the various utility companies and a utility location contract performed by an independent company. These locations are intended to reflect the general location of underground facilities. No warranty of accuracy of the provided information is implied.

2. By Wyoming Law, Contractor is required to call One Call of Wyoming at least two business days prior to digging.
   a. Phone: 1(800)849-2476
3. Call Local Utility Line Information service and Park Service Utility Coordinator at least five working days before performing Work.
   a. Request underground utilities to be located and marked within and surrounding construction areas.
   b. Verify utility locations indicated on the POR drawings within the construction limits.
   c. Verify safe working distances from the existing utilities considering the construction operations to be utilized.
   d. Verify depth and location of any existing utilities that have potential to conflict with the work including points of connection or crossings of water distribution mains and water service piping, gravity sanitary sewer, pressure sanitary sewer, storm sewer, culverts, telephone, fiber communication and underground electrical services.
   e. Notify CO of any utility conflicts that are not adequately addressed in the project construction plans and specifications.
   f. Coordinate with each utility as needed to shut down the service as required to construct the work and to facilitate any upgrade or relocation work to be performed by the utility.
   g. Record existing utility locations that are incorrect or not shown on the drawings.
   h. Furnish location data for work related to Project that must be performed by public utilities serving Project site.
   i. Protect existing utilities to prevent damage during construction.
4. Contractor shall be responsible for locating and preventing damage to known utilities. If damage occurs, repair utility at no additional expense to the Government. Provide repair materials and fittings onsite during construction for immediate repair of damaged water mains.
5. If damage occurs to an unknown utility, repair utility. An equitable adjustment will be made in accordance with the Changes clause of the contract. Coordinate with all utility providers to locate existing buried utility facilities within the project.
6. Before construction, verify location and invert elevation at points of connection of sanitary sewer, storm sewer, and water-service piping; and underground electrical services.

C. Acceptance of Conditions: Examine substrates, areas, and conditions, with Installer or Applicator present where indicated, for compliance with requirements for installation tolerances and other conditions affecting performance. Record observations.

1. Verify compatibility with and suitability of substrates, including compatibility with existing finishes or primers.
2. Examine roughing-in for mechanical and electrical systems to verify actual locations of connections before equipment and fixture installation.
3. Examine walls, floors, and roofs for suitable conditions where products and systems are to be installed.
4. Proceed with installation only after unsatisfactory conditions have been corrected. Proceeding with the Work indicates acceptance of surfaces and conditions.

3.2 COORDINATION WITH UTILITY SERVICE PROVIDERS

Coordination with Utility Service Providers: Contact following Utility Service providers, sufficiently in advance to avoid delaying the work, to coordinate Contractor's portion of Work, testing requirements, inspections,

1. Electrical: Service Contact: Contact Lower Valley Energy (LVE), 4000 S. Hwy 89, Jackson, WY 83001, 307-739-6038, **t**to coordinate Electrical service requirements.

2. GRTE Water and Wastewater Service Contact: Request contact information from Contracting Officer.

    a. Construction Contractor Responsibilities: Contractor is responsible for all construction in accordance with the Contract Documents including connection to existing water mains.

3. Fiber Optic & Communications Provider Contact: Contact Diamond Communications, https://www.diamondcomm.com/national-and-state-parks to coordinate fiber optic service requirements.

    a. Construction Contractor Responsibilities: Contractor is responsible for all construction in accordance with the Contract Documents. Coordinate installation of communications conduit and pathway components to meet utility provider requirements.

3.3 PREPARATION

A. Field Measurements: Take field measurements as required to fit Work properly. Recheck measurements before installing each product. Where portions of Work are indicated fit to other construction, verify dimensions of other construction by field measurements before fabrication. Coordinate fabrication schedule with construction progress to avoid delaying Work.

B. Space Requirements: Verify space requirements and dimensions of items shown diagrammatically on Drawings.

C. Review of Contract Documents and Field Conditions: Immediately on discovery of need for clarification of the Contract Documents caused by differing field conditions outside control of

Contractor, submit request for information to Contracting Officer in accordance with Section 01 31 00 "Project Management and Coordination."

3.4 CONSTRUCTION LAYOUT

A. Verification: Verify layout information shown on Drawings, in relation to the existing benchmarks before proceeding to lay out Work. Notify Contracting Officer promptly if discrepancies are discovered.
   1. At the Cabin Cluster work area prepare a certified survey showing of existing conditions. Survey information shall include existing grades, existing building corners, existing building floor elevations. Provide a copy of the existing conditions survey information to the CO for approval.
   2. For work at the Cabin Cluster, provide a coordination drawing based on the existing conditions survey. Indicate proposed finish grade elevations at building corners, pathways, and site elements located above grade. Indicate proposed floor, top of footing, and top of foundation wall elevations. See section 013100 Project Management and Coordination for requirements for coordination drawings.

B. General: Engage a land surveyor to lay out Work using accepted surveying practices.

   1. Establish benchmarks and control points to set lines of construction and elsewhere as needed to locate each element of Project.
   2. Establish dimensions within tolerances indicated. Do not scale Drawings to obtain required dimensions.
   3. Inform installers of lines and levels to which they must comply.
   4. Check location, level and plumb, of every major element as Work progresses.
   5. Notify Contracting Officer when deviations from required lines and levels exceed allowable tolerances.

C. Limits of Disturbance. Identify and define construction zones as described in Section 01 50 00 Temporary Facilities and Controls.

D. Site Improvements: Locate and lay out site improvements, including pavements, grading, fill and topsoil placement, utility slopes, and invert elevations.

E. Building Lines and Levels: Locate and lay out control lines and levels for structures, building foundations, column grids, and floor levels, including those required for mechanical and electrical work. Transfer survey markings and elevations for use with control lines and levels. Level foundations and piers from two or more locations.

F. Record Log: Maintain log of layout control work. Record deviations from required lines and levels. Include beginning and ending dates and times of surveys, weather conditions, name and duty of each survey party member, and types of instruments and tapes used. Make log available for review by National Park Service (NPS).

3.5 FIELD ENGINEERING

A. Reference Points: Locate existing permanent benchmarks, control points, and similar reference points before beginning Work. Preserve and protect permanent benchmarks and control points

during construction operations. Controls destroyed by Contractor will be replaced by Contractor at their expense.

1. Existing Monuments: All benchmarks, land corners, and triangulation points, established by other surveys, existing within construction area shall be preserved. If existing monuments interfere with Work, secure written permission before removing them.

B. Benchmarks: Establish and maintain a minimum of two permanent benchmarks on Project site, referenced to data established by survey control points. Comply with NPS requirements for type and size of benchmark.

1. Record benchmark locations, with horizontal and vertical data, on Project Record Documents.

Certified Survey: On completion of foundation walls, major site improvements, and other work requiring field-engineering services, prepare a certified survey showing dimensions, locations, angles, and elevations of construction and sitework.

3.6 INSTALLATION

A. General: Locate Work and components of Work accurately in correct alignment and elevation, as indicated.

1. Make vertical work plumb and make horizontal work level.
2. Where space is limited, install components to maximize space available for maintenance and ease of removal for replacement.
3. Conceal pipes, ducts, and wiring in finished areas, unless otherwise indicated.

B. Comply with manufacturer's written instructions and recommendations for installing products in applications indicated.

C. Install products at the time and under conditions for best possible results. Maintain conditions required for product performance until Substantial Completion.

D. Conduct construction operations so no part of the Work is subjected to damaging operations or loading in excess of that expected during normal conditions of occupancy.

E. Tools and Equipment: Do not use tools or equipment that produce harmful noise levels.

F. Templates: Obtain and distribute to parties involved templates for work specified to be factory prepared and field installed. Check Shop Drawings of other work to confirm adequate provisions are made for locating and installing products to comply with indicated requirements.

G. Anchors and Fasteners: Provide anchors and fasteners as required to anchor each component securely in place, accurately located and aligned with other portions of Work.

1. Mounting Heights: Where mounting heights are not indicated, mount components at heights directed by Contracting Officer.
2. Allow for building movement, thermal expansion, and contraction.

3. Coordinate installation of anchorages. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors to be embedded in concrete or masonry. Deliver to Project site in time for installation.

H. Joints: Make joints of uniform width. Where joint locations in exposed work are not indicated, arrange joints for best visual effect. Fit exposed connections together to form hairline joints.

I. Hazardous Materials: Use products, cleaners, and installation materials not considered hazardous.

3.7 PROGRESS CLEANING

General: Clean Project site, work areas, and common areas daily. Coordinate progress cleaning for joint-use areas where more than one Installer has worked. Enforce requirements strictly. Dispose of materials lawfully.

1. Comply with requirements in National Fire Protection Association (NFPA) 241 for removal of combustible waste materials and debris at the end of every work session.
2. Do not hold materials more than 7 days during normal weather or 3 days if the temperature is expected to rise above 80 degrees Fahrenheit (27 degrees Celsius).
3. Containerize hazardous and unsanitary waste materials separately from other waste. Mark containers appropriately and dispose of legally, according to regulations.

B. Site: Maintain Project site free of waste materials and debris.

C. Work Areas: Clean areas where work is in progress to level of cleanliness necessary for proper execution of Work.

1. Remove liquid spills promptly.
2. Where dust would impair proper execution of Work, broom-clean or vacuum entire work area, as appropriate.
3. Contractor shall provide progress cleaning that minimizes sources of food, water, and harborage available to pests.

D. Installed Work: Keep installed work clean. Clean installed surfaces according to written instructions of manufacturer or fabricator of product installed, using only cleaning materials specifically recommended. If specific cleaning materials are not recommended, use cleaning materials not hazardous to health or property and will not damage exposed surfaces.

Utilize non-toxic cleaning materials and methods.

a. Comply with Green Seal Standard (GS) 37 for general purpose cleaning and bathroom cleaning.
b. Use natural cleaning materials where feasible. Natural cleaning materials include:

1) Abrasive cleaners: substitute 1/2 lemon dipped in borax.
2) Ammonia: substitute vinegar, salt and water mixture, or baking soda and water.
3) Disinfectants: substitute 1/2 cup borax in gallon water.

4) Drain cleaners: substitute 1/4 cup baking soda and 1/4 cup vinegar in boiling water.
5) Upholstery cleaners: substitute dry cornstarch.

E. Concealed Spaces: Remove debris from concealed spaces before enclosing the space.

F. Exposed Surfaces in Finished Areas: Clean exposed surfaces and protect from damage and deterioration at time of Substantial Completion.

G. Waste Disposal: Do not bury or burn waste materials on-site. Do not wash waste materials down sewers or into waterways.

H. Clean and protect construction in progress and adjoining materials already in place during handling and installation. Apply protective covering where required to ensure protection from damage or deterioration at Substantial Completion.

I. Clean and provide maintenance on completed construction as frequently as necessary through remainder of the construction period. Adjust and lubricate operable components to ensure operability without damaging effects.

J. Limiting Exposures: Supervise construction operations so that no part of construction completed or in progress, is subject to harmful, dangerous, damaging, or deleterious exposure during construction period.

K. Final Cleaning: At completion of Work, remove remaining waste materials, rubbish, tools, equipment, machinery and surplus materials. Clean exposed surfaces and leave Project clean and ready for occupancy.

1. Provide final cleaning in accordance with ASTM E1971.

3.8 STARTING AND ADJUSTING

A. Start equipment and operating components to confirm proper operation. Remove malfunctioning units, replace with new units, and retest.

B. Notify CO 7 days prior to startup of each item.

C. Adjust operating components for proper operation without binding. Adjust equipment for proper operation.

D. Test each piece of equipment to verify proper operation. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

E. Manufacturer's Field Service: If factory-authorized service representative is required to inspect field-assembled components and equipment installation, comply with qualification requirements in Section 01 40 00 "Quality Requirements."

F. Submit a written report according to Section 01 33 23 - Submittal Procedures, that equipment or system has been properly installed and is functioning correctly.

3.9 PROTECTION OF INSTALLED CONSTRUCTION

A. Provide final protection and maintain conditions that ensure installed Work is without damage or deterioration at time of Substantial Completion.

B. Comply with manufacturer's written instructions for temperature and relative humidity.

C. Following construction, furnish the GRTE GIS Office (307-739-3493) with utility location information, including a shapefile using coordinates NAD83 UTM 12N, company name, and type and number of lines.

3.10 CORRECTION OF THE WORK

A. Repair or remove and replace defective construction. Restore damaged substrates and finishes. Comply with requirements in Section 01 73 29 "Cutting and Patching."

1. Repairing includes replacing defective parts, refinishing damaged surfaces, touching up with matching materials, and properly adjusting operating equipment.

B. Restore permanent facilities used during construction to specified condition.

C. Remove and replace damaged surfaces exposed to view if surfaces cannot be repaired without visible evidence of repair.

D. Repair components that do not operate properly. Remove and replace operating components that cannot be repaired.

E. Remove and replace chipped, scratched, and broken glass or reflective surfaces.

END OF SECTION 01 73 40

GRTE 237343 Rehabilitate "4 Lazy F Ranch" Cabins for Housing SECTION 01 74 19 - CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for:

Salvaging nonhazardous demolition and construction waste.

2. Recycling nonhazardous demolition and construction waste.
3. Disposing of nonhazardous demolition and construction waste.

1.2 DEFINITIONS

A. Construction Waste: Building and site improvement materials and other solid waste resulting from construction, remodeling, renovation, or repair operations. Construction waste includes packaging.

B. Demolition Waste: Building and site improvement materials resulting from demolition or selective demolition operations.

C. Solid Waste: Garbage, debris, sludge, or other discharged material (except hazardous waste) including solid, liquid, semisolid, or contained gaseous materials resulting from domestic, industrial, commercial, mining, or agricultural operations.

D. Debris: Non-hazardous solid waste generated during construction, demolition, or renovation of a structure which exceeds 2.5 inch (60 millimeter) particle size that is: a manufactured object; plant or animal matter; or natural geologic material (e.g. cobbles and boulders). A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if mixture is comprised primarily of debris by volume, based on visual inspection.

E. Disposal: Removal off-site of demolition and construction waste and subsequent sale, recycling, reuse, or deposit in landfill or incinerator acceptable to authorities having jurisdiction.

F. Environmental Pollution and Damage: Presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances; or degrade utility of environment for aesthetic, cultural, or historical purposes.

G. Garbage: Refuse and scraps resulting from preparation, cooking, dispensing, and consumption of food.

H. Hazardous Materials: Material regulated as a hazardous material in accordance with 49 CFR 173 (Code of Federal Regulations), requires a Material Safety Data Sheet (MSDS) in accordance with 29 CFR 1910.1200, or which during end use, treatment, handling, storage, transportation or disposal meets or has components which meet or have potential to meet the definition of Hazardous Waste in accordance with 40 CFR 261.

I. Recycle: Recovery of demolition or construction waste for subsequent processing in preparation for reuse.

J. Salvage: Recovery of demolition or construction waste and subsequent sale or reuse in another facility.

K. Single-use plastic products: Plastic items intended to be disposed of immediately after use, including plastic and polystyrene food and beverage containers, bottles, straws, cups, cutlery, and disposable plastic bags.

1.3 PERFORMANCE REQUIREMENTS

A. If waste materials encountered during deconstruction/demolition or construction phase are found to contain lead, asbestos, polychlorinated biphenyls (PCBs), (such as fluorescent lamp ballasts), or other harmful substances, they are to be handled and removed in accordance with local, state, and federal laws and requirements concerning hazardous waste.

B. General: This project shall minimize the creation of construction, deconstruction, and demolition waste to protect and restore natural habitat and resources. Factors that contribute to waste such as over packaging, improper storage, ordering error, poor planning, breakage, mishandling, and contamination shall be minimized. A Waste Management Plan shall be developed to ensure that existing site and building materials are reused, salvaged, or recycled. Waste disposal in landfills shall be minimized.
1. All project zones will be kept free of trash at all times.

Salvage/Recycle Requirements: Government goal is to salvage and recycle as much nonhazardous demolition and construction waste as possible including the following materials:

D.

Salvage for re-use: wood plank flooring as described on the drawings; lavatories as shown on the drawings, roof decking as shown on the drawings, other materials as shown on the drawings.

2. Salvage and re-use top soil to the greatest extend possible.

3. Salvage and donate: To the greatest extent possible salvage and donate serviceable demolition materials, equipment, and fixtures.

4. Recycle: to the greatest extent possible, recycle construction and demolition materials.

E. Wildlife Protection: Project requirements related to wildlife protection are described in Section 01 35 13.05 – Environmental Protection. Required protective measures related to the storage and disposal of bear attractants and materials that may be toxic to bears are included. Please refer to Section 01 35 13.05 – Environmental Protection for specific requirements.
1. Construction debris must be separated from human food garbage and disposed of in dumpsters that can be closed at night. No open dumpsters are allowed. A request for an exception to the open dumpster stipulation, however, can be made to the project manager who will consult with the park's Bear Management Office to determine if such use will be authorized.

1.4 SUBMITTALS

A. Waste Management Plan: After award of contract and prior to scheduled Pre-Construction Conference, Contractor shall submit a draft Waste Management Plan to Contracting Officer for approval. Submit [**3**] copies of plan. Revise and resubmit Plan as required by Contracting Officer. Approval of Contractor's Plan will not relieve Contractor of responsibility for compliance with applicable environmental regulations.

Progress Documentation: Supplemental to Waste Management Plan, document solid waste disposal, diversion, and cost/revenue analysis and submit completed worksheet on a monthly basis. See Project Waste Management Plan Worksheet Sample, attached to the end of the Division 1 Specifications, and report totals to date for column headings.

C. Landfill and Incinerator Disposal Records: Indicate receipt and acceptance of waste by landfills and incinerator facilities licensed to accept them. Include manifests, weight tickets, receipts, and invoices.

Qualification Data: For Waste Management Coordinator

E. Progress payment requirements:

1. With each Application for payment, submit an updated Project Waste Management Plan worksheet for solid waste disposal and diversion.
2. With each Application for Payment, submit manifests, weight tickets, receipts, and invoices specifically identifying the Project and waste material.

F. Closeout Submittals

1. With Closeout Submittals, submit a summary of a Project Waste Management Plan worksheet for solid waste disposal and diversion.

1.5 QUALITY ASSURANCE

A. Regulatory Requirements: Comply with hauling and disposal regulations of authorities having jurisdiction.

Waste Management Meeting: Conduct separate meeting or cover in Pre-Construction Conference and comply with requirements in Section 01 31 00 "Project Management and Coordination." Review methods and procedures related to waste management including:

Review and discuss waste management plan including responsibilities of Waste Management Coordinator.
2. Review requirements for documenting quantities of each type of waste and its disposition.
3. Review and finalize procedures for materials separation and verify availability of containers and bins needed to avoid delays.
4. Review procedures for periodic waste collection and transportation to recycling and disposal facilities.
5. Review procedures for reduction of single-use plastic products on site.
6. Review waste management requirements for each trade.

PART 2 - PRODUCTS

2.1 WASTE MANAGEMENT PLAN

General: Develop plan consisting of waste identification and waste reduction work plan. Include separate sections in plan for demolition and construction waste. Indicate quantities by weight or volume but use same units of measure throughout waste management plan.

B. Waste Identification: Indicate anticipated types and quantities of demolition, site-clearing and construction waste generated by the Work. Include estimated quantities and assumptions for estimates.

C. Waste Reduction Work Plan: List each type of waste and whether it will be salvaged, recycled, or disposed of in landfill or incinerator. Include points of waste generation, total quantity of each type of waste, quantity for each means of recovery, and handling and transportation procedures.

1. Salvaged Materials for Reuse: For materials salvaged and reused in Project, describe methods for preparing salvaged materials before incorporation into the Work.
2. Salvaged Materials for Sale: For materials sold to individuals and organizations, include list of names, addresses, and telephone numbers.

Salvaged Materials for Donation: For materials donated to individuals and organizations, include list of names, addresses, and telephone numbers.

4. Recycled Materials: Include list of local receivers and processors and type of recycled materials each will accept. Include names, addresses, and telephone numbers.
5. Disposed Materials: Indicate how and where materials will be disposed of. Include name, address, and telephone number of each landfill and incinerator facility.
6. Handling and Transportation Procedures: Include method used for separating recyclable waste including sizes of containers, container labeling, and designated location on Project site where materials separation will be located.
7. Reduction of single-use plastic products: Include methods for water distribution to include bottle refilling stations if available for personnel.

D. Cost/Revenue Analysis: Indicate total cost of waste disposal as if there was no waste management plan and net additional cost or net savings resulting from implementing waste management plan. Include:

1. Landfill tip fees per ton.
2. If diverted, tip fee savings from landfill diversion.
3. Costs of recycling, salvage, or reuse.
4. Revenue from recycling, salvage, or reuse.
5. Total cost or savings from diversion. (Calculate by using tip fee savings and subtracting costs of recycling or adding revenue from recycling.)

PART 3 - EXECUTION

3.1 PLAN IMPLEMENTATION

A. General: Implement waste management plan as approved by Contracting Officer. Provide handling, containers, storage, signage, transportation, and other items as required to implement waste management plan during entire duration of Contract.

B. Waste Management Coordinator: Engage waste management coordinator responsible for implementing, monitoring, and reporting status of waste management work plan. Coordinator shall be present at Project site full time for duration of Project.

C. Contractor shall establish contacts with local recycling and reuse companies to set up lines of responsibility. Contractor shall be responsible for coordination in terms of identifying materials, pickup schedules, and standard quality for recycled materials.

D. Training: Train workers, subcontractors, and suppliers on proper waste management procedures, as appropriate for the Work occurring at Project site.

1. Distribute waste management plan to everyone concerned within three days of submittal return.
2. Distribute waste management plan to entities when they begin work on-site. Review plan procedures and locations established for salvage, recycling, and disposal.

E. Site Access and Temporary Controls: Conduct waste management operations to ensure minimum interference with roads, streets, walks, walkways, and other adjacent occupied and used facilities.

F. Separation facilities:

1. Contractor shall designate and Contracting Officer shall approve specific area or areas to facilitate separation of materials for potential reuse, salvage, recycling, and return.
2. Place waste and recycling bins near each other, and close to point of waste generation but out of traffic pattern.
3. Keep recycling and waste bin areas neat, clean, and clearly marked in order to avoid co-mingling of materials.
4. Protect bins during non-working hours from off-site contamination.
5. Check garbage dumpsters periodically for recyclables being thrown away and undocumented materials that could be recycled.

G. Materials handling procedures: Material to be recycled shall be protected from contamination and shall be handled, stored, and transported in a manner that meets requirements set by designated facilities for acceptance. Establish defined area for operations of each trade, especially woodcutting so off-cuts are kept in one area and can be sorted by dimension for future reuse.

3.2 SALVAGING DEMOLITION WASTE

A. Salvaged Items for Reuse in the Work:

1. Clean salvaged items.

2. Pack or crate items after cleaning. Identify contents of containers.
3. Store items in secure area until installation.
4. Protect items from damage during transport and storage.

B. Install salvaged items to comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make items functional for use indicated.Salvaged Items for Donation: Permitted on Project site.

C. Salvaged Items for Governments Use:

1. Clean salvaged items.
2. Pack or crate items after cleaning. Identify contents of containers.
3. Allow for inspection if necessary.
4. Store items in secure area until delivery to Government.
5. Transport items to storage area designated by Government.
6. Protect items from damage during transport and storage.

3.3 RECYCLING DEMOLITION AND CONSTRUCTION WASTE, GENERAL

A. General: Recycle paper and beverage containers used by on-site workers.

Recycling Receivers and Processors: List below is provided for information only; available recycling receivers and processors include:

1. Teton County Recycling Center, 307-733-7678

Recycling Incentives: Revenues, savings, rebates, tax credits, and other incentives received for recycling waste materials shall accrue to Contractor.

Procedures: Separate recyclable waste from other waste materials, trash, and debris. Separate recyclable waste by type at Project site to maximum extent practical.

1. Provide appropriately marked containers or bins for controlling recyclable waste until removed from Project site. Include list of acceptable and unacceptable materials at each container and bin.

    a. Inspect containers and bins for contamination and remove contaminated materials if found.

2. Stockpile processed materials on-site without intermixing with other materials. Place, grade, and shape stockpiles to drain surface water. Cover to prevent windblown dust.
3. Stockpile materials away from construction area. Do not store within drip line of remaining trees.
4. Store components off ground and protect from weather.
5. Remove recyclable waste off Government property and transport to recycling receiver or processor.

3.4 RECYCLING DEMOLITION WASTE

Asphaltic Concrete Paving: Grind asphalt to maximum 1-1/2 inch size.

Concrete: Remove reinforcement and other metals from concrete and sort with other metals.

Pulverize concrete to maximum 8 inch size.

C. Masonry: Remove metal reinforcement, anchors, and ties from masonry and sort with other metals.

Pulverize masonry to maximum 4 inch size.

2. Clean and stack undamaged, whole masonry units on wood pallets.

D. Wood Materials: Sort and stack members according to size, type, and length. Separate lumber, engineered wood products, panel products, and treated wood materials.

E. Metals: Separate metals by type.

1. Structural Steel: Stack members according to size, type of member, and length.
2. Remove and dispose of bolts, nuts, washers, and other rough hardware.

F. Asphalt Shingle Roofing: Separate organic and glass-fiber asphalt shingles and felts. Remove and dispose of nails, staples, and accessories.

G. Gypsum Board: Stack large clean pieces on wood pallets and store in a dry location. Remove edge trim and sort with other metals. Remove and dispose of fasteners.

H. Equipment: Drain tanks, piping, and fixtures. Seal openings with caps or plugs. Protect equipment from exposure to weather.

I. Plumbing Fixtures: Separate by type and size.

J. Piping: Reduce piping to straight lengths and store by type and size. Separate supports, hangers, valves, sprinklers, and other components by type and size.

K. Conduit: Reduce conduit to straight lengths and store by type and size.

L. Electronic Products: Ensure non-usable electronic products are reused, donated, sold, or recycled using environmentally sound management practices at end of life.

3.5 RECYCLING CONSTRUCTION WASTE

Packaging:

1. Cardboard and Boxes: Break down packaging into flat sheets. Bundle and store in dry location.
2. Polystyrene Packaging: Separate and bag materials.
3. Pallets: As much as possible, require deliveries using pallets to remove pallets from Project site. For pallets that remain on-site, break down pallets into component wood pieces and comply with requirements for recycling wood.
4. Crates: Break down crates into component wood pieces and comply with requirements for recycling wood.

B. Site-Clearing Wastes: Chip brush, branches, and trees at landfill facility.

C. Wood Materials:

1. Clean Cut-Offs of Lumber: Grind or chip into small pieces.
2. Clean Sawdust: Bag sawdust not containing painted or treated wood.

D. Gypsum Board: Stack large clean pieces on wood pallets and store in dry location.

1. Clean Gypsum Board: Grind scraps of clean gypsum board using small mobile chipper or hammer mill. Screen out paper after grinding.

3.6 DISPOSAL OF WASTE

General: Except for items or materials to be salvaged, recycled, or otherwise reused, remove waste materials from Project site and legally dispose in landfill or incinerator acceptable to authorities having jurisdiction.

1. Except as otherwise specified, do not allow waste materials to accumulate on-site.
2. Remove and transport debris in manner preventing spillage on adjacent surfaces and areas.

B. Burning: Do not burn waste materials.

Disposal: Transport waste materials off Government property and legally dispose of them.

D. Soil: Dispose of excess soil resulting from excavations at the Kelly Pit.

END OF SECTION 01 74 19

Project Waste Management Plan Worksheet Sample

National Park Service (NPS) - Denver Service Center (DSC) | 10-13-21

	A	B	C	D	E	F	G	H	I	J
Material	**Quantity Recycled (in tons)**	**Quantity Salvaged or Reused (in tons)**	**A + B = Total Quantity Diverted from Landfill**	**Quantity to Landfill (in tons)**	**C + D = Total Quantity Generated (in tons)**	**Tip Fee/Ton at Landfill**	**C x F = Tip Fee Savings resulting from Landfill Diversion**	**Cost of Recycling (R), Salvage (S), or Reuse (Re)** (Specify R, S, or Re)	**Revenue from Recycling (R), Salvage (S), or Reuse (Re)** (Specify R, S, or Re)	**G - H + I = Total Cost (-) or Savings (+) from Diversion**
Asphalt/Concrete										
Brick/Masonry/Tile										
Building Materials (doors, windows, fixtures, shingles, lumber, insulation, sheet goods, etc.)										
Carpet										
Carpet Padding, Foam Only										
Cardboard										
Ceiling Tile										
Drywall										
Glass										
Scrap Metal Aluminum										
Copper										
Steel										
Unpainted Wood & Pallets										
Yard Trimmings, Brush, Trees, Stumps, etc.										
Garbage/Trash										
Other										
Column Totals										
	Total Quantity Recycled	Total Quantity Reused or Salvaged	Total Quantity Diverted from Landfill	Total Quantity to Landfill	Total Quantity Generated		Tip Fee Savings from Diversion	Total Cost of Recycling, Salvage, or Reuse	Revenue from Recycling, Salvage, or Reuse	Total Cost (-) or Savings (+) from Diversion

Percentage Diverted = _____ (C divided by E from Column Totals). Should meet 60% diversion goal.

GRTE 237343 Rehabilitate "4 Lazy F Ranch" Cabins for Housing

SECTION 01 77 00 - CLOSEOUT PROCEDURES

GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for contract closeout, including:

Project Record Drawings
2. Closeout Submittals
3. Substantial Completion and Final Inspection
4. Permit Closure and Transfer
5. Final Acceptance of the Work
6. Warranties

1.2 PROJECT RECORD DRAWINGS

A. Following construction, furnish the GRTE GIS Office (307-739-3493) with utility location information, including a shapefile using coordinates NAD83 UTM 12N, company name, and type and number of lines.

B. Maintain one complete full-size set of contract drawings and one full-size set of vendor-supplied drawings. Clearly mark changes, deletions, and additions using National Park Service (NPS) drafting standards to show actual construction conditions. Show additions in red, deletions in green and special instructions in blue.

C. Keep record drawings current. Make record drawings available to Contracting Officer (CO) for inspection at the time of monthly progress payment requests. If project record drawings are not current, Contracting Officer may retain an appropriate amount of progress payment.

D. Submit complete record drawings on completion of total project. Include shop drawings, sketches, and additional drawings to be included in final set, with clear instructions showing the location of these drawings.

1.3 CLOSEOUT SUBMITTALS

A list of closeout requirements has been attached at the end of the Division 1 Specifications for your convenience. The intent is to provide an overall summary of requirements and not a comprehensive list. Terms and conditions of the contract require satisfaction of requirements of individual specification sections regardless of what is shown on the list. Submit the following before requesting final inspection:

Specific warranties, guarantees, workmanship bonds, final certifications, and similar documents.
2. NPS required forms for occupancy, Fire Sprinkler/Alarm acceptance, and other similar forms or certificates.

3. Project Record Documents, operation and maintenance manuals, final completion construction digital images recorded on CD-R (compact disc-recordable) or DVD-R (digital video disc-recordable) with index and descriptions, and similar final record information.
4. Environmental Record Documents: As specified and as follows:

    a. IAQ Management Plan: As specified in Section 01 57 19 - 11 Indoor Air Quality (IAQ) Management.
    b. Moisture Control inspections and reports: As specified in Section 01 57 19 - 11 Indoor Air Quality (IAQ) Management.
    c. Material Safety Data Sheet (MSDS) Data: As specified
    d. Final Summary of Solid Waste Disposal and Diversion: As specified in Section 01 74 19 Construction Waste Management.

5. Posted Operating Instructions: As specified in individual sections. Furnish operating instructions attached to or posted adjacent to equipment. Include wiring diagrams, control diagrams, control sequence, start-up, adjustment, operation, lubrication, shut-down, safety precautions, procedures in the event of equipment failure, and other items of instruction recommended by manufacturer.
6. Deliver tools, spare parts, extra materials, and similar items to location designated by Contracting Officer. Label with manufacturer's name and model number where applicable.

    a. Special Tools: One set of special tools required to operate, adjust, dismantle, or repair equipment. Special tools are those not normally found in possession of mechanics or maintenance personnel.

Keys and Keying Schedule: Submit keys including duplicates. Wire keys for each lock securely together. Tag and plainly mark with lock number, equipment identification, or panel or switch number, and indicate location, building, and room name or number.

8. Make final changeover of permanent locks and deliver keys to Contracting Officer. Advise Park personnel of changeover in security provisions.

9. Test and balance report.
10. Terminate and remove temporary facilities, mockups, construction tools, and similar elements from Project site, complete final cleaning requirements, including touchup painting.
11. Touch up and repair and restore marred exposed finishes to eliminate visual defects.
12. Instruct NPS personnel in operation, adjustment, and maintenance of products, equipment, and systems.
13. Submit “Notice of Transfer and Acceptance” for SWPPP permit to CO.
14. Close-out construction permits in accordance with AHJ’s

FINAL INSPECTION, SUBSTANTIAL COMPLETION AND ACCEPTANCE PROCEDURES

A. Request final inspection in writing when project or designated portion of project is substantially complete. Contracting Officer will proceed with inspection within 10 days of receipt of written request or will advise Contractor of items that prevent project from being substantially complete.

B. If work is determined substantially complete, following final inspection, Contracting Officer will prepare Punch List and issue a Letter of Substantial Completion.

C. If work is not determined substantially complete following final inspection, Contracting Officer will notify Contractor in writing. Contractor shall request new final inspection after completing work. Re-inspection costs may be charged against Contractor in accordance with Inspection of Construction contract clause.

D. Contractor shall complete Punch List within 30 calendar days, documented weather permitting.

E. If Contractor completes items of work on Punch List and contractually required items, Contracting Officer will issue Letter of final acceptance of work.

F. If Contractor fails to complete work within the time frame, Contracting Officer may correct work with an appropriate reduction in contract price or charge for re-inspection costs in accordance with Inspection of Construction contract clause.

1.5 PERMIT CLOSURE AND TRANSFER

A. When work covered by the permits is complete, create list of tasks required to close or transfer permits to Park. Submit to Contracting Officer for approval.

B. After substantial completion and Punch List completion, permits shall be closed and documented by Agency(ies) with Jurisdiction for the permit.

C. If responsibility for permits is to be transferred to Park, Park shall be informed of permit provisions completed and responsibilities transferring to Park staff.

1.6 WARRANTIES

A. Submittal Time: Submit written warranties on request of Contracting Officer for designated portions of Work where commencement of warranties other than date of Substantial Completion is indicated.

B. Organize warranty documents into an orderly sequence based on table of contents of Project Manual.

1. Bind warranties and bonds in heavy-duty, 3-ring, vinyl-covered, loose-leaf binders, thickness as necessary to accommodate contents, and sized to receive 8-1/2 by 11 inch (215 by 280 millimeters) paper.
2. Provide heavy paper dividers with plastic-covered tabs for each separate warranty. Mark tab to identify product or installation. Provide typed description of product or installation, including name of product and name, address, and telephone number of Installer.
3. Identify each binder on front and spine with typed or printed title "WARRANTIES," Project name, and name of Contractor.

Warranty Electronic File: Scan warranties and bonds and assemble complete warranty and bond submittal package into a single indexed electronic PDF (portable document format) file with links enabling navigation to each item. Provide bookmarked table of contents at beginning of document.

C. Provide additional copies of each warranty in operation and maintenance manuals.

PART 2 - PRODUCTS

2.1 MATERIALS

A. See Division 1 Specification Section "Execution" for information on cleaning agents.

PART 3 - EXECUTION

3.1 FINAL CLEANING

General: Conduct final cleaning and waste-removal operations to comply with local laws and ordinances and Federal and local environmental and antipollution regulations.

B. Cleaning: Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit to condition expected in an average commercial building cleaning and maintenance program. Comply with manufacturer's written instructions.

1. Complete following cleaning operations before requesting inspection for certification of Substantial Completion for entire Project or for a portion of Project:

Clean Project site, yard, and grounds, in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and other foreign substances.

b. Sweep paved areas broom clean. Remove petrochemical spills, stains, and other foreign deposits.
c. Rake grounds that are neither planted nor paved to smooth, even-textured surface.
d. Remove tools, construction equipment, machinery, and surplus material from Project site.
e. Remove snow and ice to provide safe access to building.
f. Clean exposed exterior and interior hard-surfaced finishes to dirt-free condition, free of stains, films, and foreign substances. Avoid disturbing natural weathering of exterior surfaces. Restore reflective surfaces to original condition.
g. Remove debris and surface dust from limited access spaces, including roofs, plenums, shafts, trenches, equipment vaults, manholes, attics, and similar spaces.
h. Sweep concrete floors broom clean in unoccupied spaces.
i. Vacuum carpet and similar soft surfaces, removing debris and excess nap; shampoo, soil or stains remain.
j. Clean transparent materials, including mirrors and glass in doors and windows. Remove glazing compounds and vision-obscuring materials. Replace chipped or broken glass and transparent materials. Polish mirrors and glass.
k. Remove labels that are not permanent.
l. Touch up, repair, and restore marred, exposed finishes and surfaces. Replace finishes and surfaces that cannot be satisfactorily repaired or restored or that already show evidence of repair or restoration.

1) Do not paint over "UL" (Underwriters Laboratories) and similar labels, including mechanical and electrical nameplates.

Wipe surfaces of mechanical and electrical equipment and similar equipment. Remove excess lubrication, paint and mortar droppings, and other foreign substances.

n. Replace parts subject to unusual operating conditions.
o. Clean plumbing fixtures to sanitary condition, free of stains, including stains resulting from water exposure.
p. Replace disposable air filters and clean permanent air filters. Clean exposed surfaces of diffusers, registers, and grills.
q. Clean light fixtures, lamps, globes, and reflectors to function with full efficiency. Replace burned-out and noticeably dimmed bulbs, and defective or noisy starters in fluorescent and mercury vapor fixtures to comply with requirements for new fixtures.
r. Leave Project clean and ready for occupancy.

Pest Control: Engage experienced, licensed exterminator to make a final inspection and rid project of rodents, insects, and other pests. Provide Government with report.

D. Waste Disposal: Comply with requirements of Section 01 74 19 "Construction Waste Management and Disposal."

END OF SECTION 01 77 00

GRTE 237343 Rehabilitate “4 Lazy F Ranch” Cabins for Housing

SECTION 01 78 23 - OPERATION AND MAINTENANCE DATA

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for preparing operation and maintenance manuals, including:

Manuals, general
2. Emergency manuals
3. Operation manuals for systems, subsystems, and equipment
4. Maintenance manuals for care and maintenance of products, materials, and finishes and systems and equipment

B. See Divisions 2 through 49 Sections for additional operation and maintenance manual requirements for Work in those Sections.

SUBMITTALS

A. Manual: Submit one electronic copy at least 15 days before final inspection. Contracting Officer (CO) will return copy or edit version with comments within 15 days of receipt.
1. Include GIS Utility Location information per Section 01 77 00 Closeout Procedures.

B. Format: Submit operations and maintenance manuals in following format:

1. PDF (portable document format) electronic file. Assemble each manual into composite electronically indexed file. Submit on digital media acceptable to Contracting Officer.

a. Name each indexed document file in composite electronic index with applicable item name. Include a complete electronically linked operation and maintenance directory.
b. Enable inserted reviewer comments on draft submittals.

2. Hard copy manual: In accordance with Part 2 of this Section.

3. Correct or modify each manual to comply with Contracting Officers comments. Submit [4] copies of each corrected manual within [**15**] days of receipt of Contracting Officers comments.

PART 2 - PRODUCTS

2.1 MANUALS, GENERAL

A. Organization: Unless otherwise indicated, organize manual into separate sections for each system and subsystem, and separate sections for each piece of equipment not part of a system. Manual shall contain title page, table of contents, and manual contents.

B. Title Page: Enclose title page in transparent plastic sleeve. Include:

Project Title

2. Location
3. Park
4. Contract Number
5. Prime Contractors Name and Address
6. Date of Substantial Completion
7. Binder Volume Number

C. Table of Contents: List each product included in manual, identified by product name, indexed to content of the volume, and cross-referenced to Specification Section number in Project Manual.

D. Manual Contents: Organize into sets of manageable size. Arrange contents alphabetically by system, subsystem, and equipment. Assemble instructions for subsystems, equipment, and components of one system into a single binder if needed.

1. Binders: White, commercial quality, hard back, 3-ring, vinyl-covered, loose-leaf binders, in thickness necessary to accommodate contents, sized to hold 8-1/2 by 11 inch (215 by 280 millimeter) paper; with clear plastic window sleeve on front and spine to hold label describing contents and pockets inside covers to hold folded oversize sheets.

    a. Cover Sheet: Identify binders on front and spine, with project title, location, park, contract number, prime contractor's name and address, date of substantial completion, and binder volume number. Insert cover sheet into clear plastic view pocket on front of binder. Insert sheet into clear plastic view pocket on spine with title "OPERATION AND MAINTENANCE MANUAL," Project title or name, and subject matter of contents. Indicate volume number for multiple-volume sets.

2. Data: Fill binders to no more than 75 percent of capacity. Punch holes shall not obscure any data. When contents of a single tabbed section cover more than one item, provide colored paper sheets to separate the data for each item.

    a. Manufacturers' Data: Provide originals for color or copyrighted data. Black and white data may be originals or clean, good quality reproductions. No copies produced by facsimile transmission and sheets with stamps, such as submittal approval stamps. Include only sheets that apply to items installed; cross out inapplicable data.
    b. Vendor Furnished As-Built Drawings: Maximum 24 inch by 36 inch sheets with minimum character or lettering size of 1/8 inch. Reduced-size reproductions may be provided instead of full-size drawings if reproductions are clear and legible. If

reduced-size drawings are used, identify as "REDUCED SIZE" and provide graphic scales, if applicable.

c. Custom Data: Data supplemented by drawings and schematics necessary to describe systems adequately.
d. Equipment Data Sheet: Data, using form at end of this section.
e. Schedules: Schedules reflecting final, as-installed conditions.
f. Poorly reproduced or illegible data will be rejected.

3. Dividers: Divider sheets with Mylar reinforced edges and pre-printed numbered tabs aligned with numbers and title lines on index sheet. Include typed list of products and major components of equipment included in section on each divider, cross-referenced to Specification Section number and title of Project Manual.
4. Protective Plastic Sleeves: Transparent plastic sleeves designed to enclose diagnostic software diskettes for computerized electronic equipment.
5. Drawings: Attach reinforced, punched binder tabs on drawings and bind with text.

a. If oversize drawings are necessary, fold drawings to same size as text pages and use as foldouts.
b. If drawings are too large to be used as foldouts, fold and place drawings in labeled envelopes and bind envelopes in rear of manual. At appropriate locations in manual, insert typewritten pages indicating drawing titles, descriptions of contents, and drawing locations.

2.2 EMERGENCY MANUALS

Content: Organize manual into separate section for type of emergency, emergency instructions, and emergency procedures.

B. Type of Emergency: Where applicable for each type of emergency indicated below, include instructions and procedures for each system, subsystem, piece of equipment, and component for fire, flood, water leak, power failure, water outage, equipment failure, and chemical release or spill.

C. Emergency Instructions: Describe and explain warnings, trouble indications, error messages, and similar codes and signals. Include responsibilities of National Park Service (NPS) operating personnel for notification of Installer, supplier, and manufacturer to maintain warranties.

D. Emergency Procedures: Include instructions on stopping, shutdown instructions for each type of emergency, operating instructions for conditions outside normal operating limits, and required sequences for electric or electronic systems.

2.3 OPERATION AND MAINTENANCE MANUALS

Operation Requirements

1. Content: In addition to requirements in Section, include operation data required in individual Specification Sections and equipment descriptions, operating standards, operating procedures, operating logs, wiring and control diagrams, and license requirements.

2. Descriptions: Include:

Product name and model number

b. Manufacturer's name
c. Equipment identification with serial number of each component
d. Equipment function
e. Operating characteristics
f. Limiting conditions
g. Performance curves
h. Engineering data and tests
i. Complete nomenclature and number of replacement parts

3. Operating Procedures: Include start-up, break-in, and control procedures; stopping and normal shutdown instructions; routine, normal, seasonal, and weekend operating instructions; and required sequences for electric or electronic systems.
4. Systems and Equipment Controls: Describe sequence of operation, and diagram controls as installed.
5. Piped Systems: Diagram piping as installed and identify color-coding where required for identification.

B. Maintenance Requirements for Systems and Equipment

1. Content: For each system, subsystem, and piece of equipment not part of a system, include source information, manufacturers' maintenance documentation, maintenance procedures, maintenance and service schedules, spare parts list and source information, maintenance service contracts, and warranty and bond information, and equipment data sheets as described below.
2. Source Information: List each system, subsystem, and piece of equipment included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual.
3. Manufacturers' Maintenance Documentation: Manufacturers' maintenance documentation including maintenance instructions, drawings and diagrams for maintenance, nomenclature of parts and components, and recommended spare parts for each component part or piece of equipment:
4. Maintenance Procedures: Test and inspection instructions, troubleshooting guide, disassembly instructions, and adjusting instructions, detailing essential maintenance and environmental procedures.
5. Maintenance and Service Schedules: Service and lubrication requirements, list of required lubricants for equipment, and separate schedules for preventive and routine maintenance and service with standard time allotment.
6. Spare Parts List and Source Information: Lists of replacement and repair parts, with parts identified and cross-referenced to manufacturers' maintenance documentation and local sources of maintenance materials and related services.
7. Warranties and Bonds: Copies of warranties and bonds and lists of circumstances and conditions that affect validity of warranties or bonds.

2.4 PRODUCT MAINTENANCE MANUAL

A. Content: Organize manual into separate section for each product, material, and finish. Include source information, product information, maintenance procedures, repair materials and sources, and warranties and bonds, as described below.

B. Source Information: List each product included in manual identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual.

C. Product Information: Include:

Product name and model number

2. Manufacturer's name
3. Color, pattern, and texture
4. Material and chemical composition
5. Reordering information for specially manufactured products

D. Environmental Requirements

1. Identify environmentally preferable products incorporated into Project. Include: product model; manufacturer’s name, address, phone, and website; and local technical representative.

Verify plastic products to be incorporated into Project are labeled in accordance with ASTM D1972. Where products are not labeled, provide product data indicating polymeric information in Operation and Maintenance Manual.

1) Type 1: Polyethylene Terephthalate (PET, PETE)
2) Type 2: High Density Polyethylene (HDPE)
3) Type 3: Vinyl (Polyvinyl Chloride or PVC)
4) Type 4: Low Density Polyethylene (LDPE)
5) Type 5: Polypropylene (PP)
6) Type 6: Polystyrene (PS)
7) Type 7: Other. Use of this code indicates that package in question is made with a resin other than the six listed above or is made of more than one resin listed above and used in a multi-layer combination.

b. Describe maintenance procedures associated with environmentally preferable materials and systems. Provide cleaning recommendations in accordance with ASTM E1971 and approved Integrated Pest Management (IPM) plan.

1) Include potential environmental impacts of recommended maintenance procedures and materials.
2) Include potential indoor air quality impacts of recommended maintenance procedures and materials.

Where proposed maintenance procedures incorporate composting of plastics, assess potential effect of each type of plastic to be included in composting process in accordance with ASTM D5509 or ASTM D6002

Material Safety Data Sheets (MSDS): Include MSDSs as specified.

E. Maintenance Procedures: Include manufacturer's written recommendations and inspection procedures, types of cleaning agents, methods of cleaning, schedule for cleaning and maintenance, and repair instructions.

F. Repair Materials and Sources: Include lists of materials and local sources of materials and related services.

G. Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that affect validity of warranties or bonds.

2.5 SYSTEMS AND EQUIPMENT MAINTENANCE MANUALS

A. Content: For each system, subsystem, and piece of equipment not part of a system, include source information, manufacturers' maintenance documentation, maintenance procedures, maintenance and service schedules, spare parts list and source information, maintenance service contracts, and warranty and bond information, as described below.

Source Information: List each system, subsystem, and piece of equipment included in manual, identified by product name and arranged to match manual's table of contents. For each product, list name, address, and telephone number of Installer or supplier and maintenance service agent, and cross-reference Specification Section number and title in Project Manual and drawing or schedule designation or identifier where applicable.

C. Manufacturers' Maintenance Documentation: Manufacturers' maintenance documentation including the following information for each component part or piece of equipment:

1. Standard maintenance instructions and bulletins.
2. Drawings, diagrams, and instructions required for maintenance, including disassembly and component removal, replacement, and assembly.
3. Identification and nomenclature of parts and components.
4. List of items recommended to be stocked as spare parts.

D. Seasonal shut down and start-up requirements: All equipment and systems will be shut down for winter months and will be exposed to extended freezing temperatures. Describe specific requirements for shutting down, winterizing, and restarted equipment and systems including the following:
1. Mechanical systems and equipment.
2. Plumbing systems and equipment.
3. Fire sprinkler systems and equipment.
4. Fire alarm systems and equipment.

E. Maintenance Procedures: Include the following and items detailing essential maintenance procedures:

1. Test and inspection instructions
2. Troubleshooting guide
3. Precautions against improper maintenance
4. Disassembly; component removal, repair, and replacement; and reassembly instructions
5. Aligning, adjusting, and checking instructions

F. Maintenance and Service Schedules: Include service and lubrication requirements, list of required lubricants for equipment, and separate schedules for preventive and routine maintenance and service with standard time allotment.

1. Scheduled Maintenance and Service: Tabulate actions for daily, weekly, monthly, quarterly, semiannual, and annual frequencies.
2. Maintenance and Service Record: Include manufacturers' forms for recording maintenance.

G. Spare Parts List and Source Information: Include lists of replacement and repair parts, with parts identified and cross-referenced to manufacturers' maintenance documentation and local sources of maintenance materials and related services.

Warranties and Bonds: Include copies of warranties and bonds and lists of circumstances and conditions that affect validity of warranties or bonds.

1. Include procedures and required notifications for warranty claims.

PART 3 - EXECUTION

3.1 GENERAL

A. At start of project, begin accumulating operation and maintenance data and initiate index. Install and index data in binders within 30 days after delivery of items. As custom written data and test results are produced, add to operation and maintenance data file.

List of Operation and Maintenance requirements has been attached at end of the Division 1 Specifications for your convenience. Intent is to provide an overall summary of requirements and not a comprehensive list. Terms and conditions of the contract require satisfaction of requirements of individual specification sections regardless of what is shown on the list.

C. Keep operation and maintenance data current. Make operation and maintenance binders available to Contracting Officer for inspection at time of monthly progress payment requests. If operation and maintenance binders are not current, Contracting Officer may retain an appropriate amount of the progress payment.

3.2 MANUAL PREPARATION

A. Manual Types

1. Emergency Manual: Assemble complete set of emergency information indicating procedures for use by emergency personnel and by NPS operating personnel for types of emergencies indicated.
2. Product Maintenance Manual: Assemble complete set of maintenance data indicating care and maintenance of each product, material, and finish incorporated into Work.
3. Operation and Maintenance Manuals: Assemble complete set of operation and maintenance data indicating operation and maintenance of each system, subsystem, and piece of equipment not part of a system.

Manual Contents: Including:

1. Manufacturers' Data: Where manuals contain manufacturers' standard printed data, include only sheets pertinent to product or component installed. Mark sheet to identify product or component incorporated into Work. If data include more than one item in a tabular format, identify each item using appropriate references from Contract Documents. Identify data applicable to Work and delete references to information not applicable.

Custom Written Data: For data not in manufacturer's standard literature, provide text, drawings, and schematics specifically applicable to installed systems. Include step-by-step descriptions of operating procedures; identification of individual components and their functions; descriptions of how system components relate to one another and operate together to accomplish a common process or function; and sequence of operation for system control circuits. For seasonally operated systems, provide start-up and shutdown instructions.

3. Equipment Data Sheets: For each item of equipment included in operation and maintenance data, provide Equipment Data Sheet using form at the end of this section. For equipment consisting of a driven machine and a driver (for example, a pump and a motor), equipment data shall cover both the driven machine and the driver. For similar type equipment (for example, multiple exhaust fans of the same model and type), provide a single equipment data sheet with an attached schedule listing individual equipment items.

Vendor Furnished As-Built Drawings: Provide for each electrical and each mechanical control system.

a. For each control system, provide control circuit schematic drawings. Identify each wire and terminal block number. Show terminal numbers on control devices. Show control wires and devices remote from control panel.
b. For each control panel, provide general arrangement drawing showing location of each control component and terminal block on the panel front and interior. Include materials list of panel-mounted control components as well as field-installed control components remote from the panel, identifying components, manufacturer, model number, and initial set points or sensing ranges of devices where applicable.
c. For packaged equipment systems, provide general arrangement drawings showing interrelationships of the various items of equipment and components.
d. In addition to control wiring schematic, provide power wiring schematic drawing showing power flow to each motor. Identify each power conductor. Show over-current protection and motor starting devices.

C. Comply with Section 01 77 00 "Closeout Procedures" for schedule for submitting operation and maintenance documentation.

END OF SECTION 01 78 23

EQUIPMENT DATA SHEET	
Equipment Item: _____ Designation: _____ Function: _____ Location: _____ Project: _____ Model Number: _____ Serial Number: _____	
Manufacturer Address and Phone:	Supplier Address and Phone:
Preventive Maintenance Tasks:	
Nameplate Data:	
Spare Parts Furnished and Other Information:	

GRTE 237343 Rehabilitate “4 Lazy F Ranch” Cabins for Housing

SECTION 01 79 00 - DEMONSTRATION AND TRAINING

GENERAL

1.1 SUMMARY

A. Section includes administrative and procedural requirements for instructing National Park Service (NPS) personnel, including:

Demonstration of operation of systems, subsystems, and equipment.

2. Training in operation and maintenance of systems, subsystems, and equipment, including environmental considerations.
3. Demonstration and training video.

B. See Divisions 2 through 49 Sections for specific requirements for demonstration and training for products in those Sections.

1.2 GENERAL REQUIRMENTS

List of System Demonstration and Training requirements has been attached at end of Division 1 Specifications for your convenience. Intent is to provide overall summary of requirements and not a comprehensive list. Terms and conditions of contract still require satisfaction of requirements of individual specification sections regardless of what is shown on list.

1.3 SUBMITTALS

A. Instruction Program: Submit **two** copies of outline of instructional program for demonstration and training, including schedule of proposed dates, times, length of instruction time, and instructors' names for each training module. Include learning objective and outline for each training module.

1. For each training session, Contractor shall submit for approval a proposed outline of subjects to be covered. Training shall not be conducted until outline is approved.

B. Demonstration and Training Video: Submit [**two**] copies of each DVD (digital video disc) for training sessions within [**seven**] days of end of each training module.

1. Label each DVD with date of demonstration or training, instructor's name, and provide an index of contents. Index shall list start and end time of each subject covered during training session. Sequence of training subjects shall follow sequence listed in approved training outline or as actually conducted

1.4 QUALITY ASSURANCE

A. Facilitator Qualifications: Firm or individual experienced in training or educating maintenance personnel in training program similar in content and extent for this Project, and whose work has resulted in training or education with a record of successful learning performance.

B. Instructor Qualifications: Factory-authorized service representative, complying with requirements in Section 01 40 00 "Quality Requirements," experienced in operation and maintenance procedures and training.

C. Coordinate content of training modules with content of approved emergency, operation, and maintenance manuals. Do not submit instruction program until operation and maintenance data has been reviewed and approved by Contracting Officer (CO).

PART 2 - PRODUCTS

2.1 INSTRUCTION PROGRAM

A. Program Structure: Develop instruction program that includes individual training modules for each system and equipment not part of a system, as required by individual Specification Sections, and:
   1. Environmental Topics

      a. Overview of environmental issues related to building industry.
      b. Overview of environmental issues related to Project.
      c. Review of site-specific procedures and management plans implemented during construction:

         1) Regulatory Requirements
         2) Indoor Air Quality (IAQ) Management
         3) Noise and Acoustics Management
         4) Environmental Management
         5) Construction Waste Management

      d. Review of site-specific procedures and management plans to be implemented during operation and maintenance.

         1) Include review of environmentally related aspects of Operations and Maintenance Manual.
         2) Integrated Pest Management (IPM)

B. Training Modules: Develop a learning objective and teaching outline for each module. Include description of specific skills and knowledge that participant is expected to master. For each module, include instruction for:

Basis of System Design, Operational Requirements, and Criteria: Include system and equipment descriptions, operating standards, regulatory requirements, equipment function, operating characteristics, limiting conditions, and performance curves.

2. Documentation: Review emergency, operations, and maintenance manuals; Project Record Documents; identification systems; warranties and bonds; and maintenance service agreements.
3. Emergencies: Instructions on stopping; shutdown instructions; operating instructions for conditions outside normal operating limits; instructions on meaning of warnings, trouble indications, and error messages; and required sequences for electric or electronic systems.
4. Operations: Startup, break-in, control, and safety procedures; stopping and normal shutdown instructions; routine, normal, seasonal, and weekend operating instructions; operating procedures for emergencies and equipment failure; and required sequences for electric or electronic systems.
5. Adjustments: Alignments and checking, noise, vibration, economy, and efficiency adjustments.
6. Troubleshooting: Diagnostic instructions and test and inspection procedures.
7. Maintenance: Inspection procedures, types of cleaning agents, methods of cleaning, procedures for preventive and routine maintenance, and instruction on use of special tools.
8. Repairs: Diagnosis, repair, and disassembly instructions; instructions for identifying parts; and review of spare parts needed for operation and maintenance.

PART 3 - EXECUTION

3.1 INSTRUCTION

A. Facilitator: Engage qualified facilitator to prepare instruction program and training modules, to coordinate instructors, and to coordinate between Contractor and Contracting Officer for number of participants, instruction times, and location.

B. Engage qualified instructors to instruct NPS personnel to adjust, operate, and maintain systems, subsystems, and equipment not part of a system.

C. Scheduling: Provide instruction at mutually agreed on times. For equipment that requires seasonal operation, provide similar instruction at start of each season.

1. Schedule training with NPS through Contracting Officer with at least **seven** days' advance notice.
2. Conduct training sessions after equipment or system has been accepted and turned over to Government. Coordinate with commissioning requirements.
3. Coordinate with Integrated pest management requirements. Refer to specifications section and approved IPM plan.
4. Individual sections specify duration of training required. If no duration is listed, provide training of sufficient duration to adequately cover subjects.

3.2 DEMONSTRATION AND TRAINING VIDEO

A. General: Engage qualified commercial photographer to record demonstration and training video. Record each training module separately. Include classroom instructions and demonstrations, board diagrams, and other visual aids, but not student practice.

1. At beginning of training module, record each chart containing learning objective and lesson outline.

B. Video Format: Digital Video Disc (DVD).

C. Video Recording: Record sessions with high resolution equipment. Instructor's voice shall be clearly audible and understandable on DVD. Utilize supplemental microphone worn by instructor.

D. Narration: Describe scenes on video by **audio narration by microphone while** video is recorded. Include description of items being viewed. Describe vantage point, indicating location, direction (by compass point), and elevation or story of construction.

1. DVDs with poor video or audio quality will be rejected and training recorded again.

END OF SECTION 01 79 00

SECTION 02 21 19 – UTILITIES SURVEYS

PART 1 GENERAL

1.1 SECTION INCLUDES

A. The work covered by this Section of the Specifications includes the locating, uncovering, crossing, protection of, repair to, relocation of, or any other work associated with crossing under any underground utilities encountered during the excavation and subsequent installation of pipe or other appurtenances.

B. Contractor shall provide insurance adequate to cover the cost of all claims against the Owner by private or public utility owners and for damage to utilities. All claims shall be paid and lien releases provided prior to the release of retainage (lien releases required only if claims are filed). If not settled by the Contractor, the Owner will pay the utility companies out of any progress payment or remaining retainage.

C. The Contractor shall be aware of time requirements for utility companies to complete their work. The Contractor cannot file a "damage for delay" claim due to any activities by a utility company. The Contractor is responsible for all coordination with the utility company.

D. Contractor to obtain control point information from the surveyors that collected the original Lidar information

1.2 RELATED REQUIREMENTS

A. Section 31 10 00 – Site Clearing
B. Section 31 20 00 – Site Preparation
C. Section 31 23 10 – Pipe Trench Excavation and Backfill
D. Section 31 30 00 – Site Earthwork
E. Section 31 23 20 – Street and Site Excavation, Grading, and Backfill.
F. Section 33 40 00 – Special Construction

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION

3.1 GENERAL

A. The Contractor shall have full responsibility for reviewing and checking all information and data regarding any underground facility, for locating all underground facilities for coordination of the work with the owners of such Underground Facilities (utilities) during construction, for the safety and protection thereof and repairing any damage thereto resulting from the Work, the cost of all of which will be included in the contract unit bid price.

B. If a utility is encountered that is not shown in the Plans and is not located in the field by the utility company despite requests for locates, the cost of repair materials only, will be paid in addition to bid prices, if the utility is damaged (and only if it is required to be repaired). If the utility can be abandoned in place, it shall be capped at both ends and plugged with concrete with no separate payment. If existing utility lines are in within 4 feet of new utility lines, they shall be removed with no

separate payment made for the removal and disposal of these utility lines. The ends of the removed lines shall be capped and plugged with concrete as well.

C. After the utility companies have completed their field locates, the Contractor is responsible for recording and preserving this information. The utility companies may charge the Contractor for "excessive locates" based on the respective utility company's policies. The Contractor is responsible for contacting each utility company to verify any policies and/or charges. Any charges incurred for "excessive locates" shall be at the expense of the Contractor.

3.2 UNDERGROUND UTILITY LOCATIONS

A. The Contractor shall locate (uncover) all buried utilities prior to crossing them or connecting to them. This requirement pertains to both utilities that are shown in the plans and those that are marked in the field by the utility owners. The Contractor is responsible for contacting the utility owners to provide these markings; see Section 011100. Location procedures and information gathered by the Contractor shall comply with all state laws regarding utility locations. Wyoming One Call “811” number (1-800-849-2476) shall be utilized and the ticket number shall be provided to the Contracting Officer as soon as one is provided.

B. The Contractor shall provide access to the buried utility that he has located for the Contracting Officer to survey and inspect.

C. Utility locations shall be completed prior to excavation necessary to install the improvements. This requirement will be waived and utility locations may be performed in conjunction with the water, sanitary sewer, or storm drain line installation provided the Contractor furnishes a written statement indicating that he understands the requirements of the Contract Documents relating to utility locations, acknowledges that the utility locations performed in conjunction with the water, sanitary sewer, and storm drain line installation constitute elevation information that is made part of the plan by which the Contractor operates, waives all claims against the Owner and Contracting Officer relating to buried utilities and accepts all risks associated with the locating of utilities while excavating for water, sanitary sewer, storm drain, or general earthwork as opposed to locating them prior to commencing installation.

D. If the Contractor elects to locate buried utilities as part of the construction process, they must still excavate ahead of the installation when a utility or other obstacle is expected to be encountered at the approximate depth of the pipeline, to allow for adjustments in the grade of the pipeline; Utility locations cover underground utilities that are close to the facilities being installed.

3.3 EXISTING UNDERGROUND UTILITIES:

A. Existing underground utility lines and service connections are generally shown in the Plans, but the Contractor shall be solely responsible for locating all existing underground utilities, by calling in for locates.

B. After locates (marking on the ground) have been performed by the utility owner, Contractor shall locate (uncover) the utility, by careful prospecting. Contractor is solely responsible for supporting the existing utilities if support is needed or as directed by the Contracting Officer. Contractor is solely responsible for protection of and repair of any damage thereto as a result of his excavation. Upon discovery of the utility, the Contractor shall notify the Contracting Of-

ficer of its location. The Contractor shall record the size, location and depth of the utility crossing for the compilation of Record Drawings and allow the Contracting Officer the opportunity to do the same.

END OF SECTION

SECTION 02 41 00 - DEMOLITION

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Selective demolition of built site elements.

B. Selective demolition of building elements for alteration purposes.

1.2 RELATED REQUIREMENTS

A. Section 01 10 00 - Summary: Limitations on Contractor's use of site and premises.

B. Section 01 50 00 - Temporary Facilities and Controls: Site fences, security, protective barriers, and waste removal.

C. Section 01 70 00 - Execution and Closeout Requirements: Project conditions; protection of bench marks, survey control points, and existing construction to remain; reinstallation of removed products; temporary bracing and shoring.

1.3 DEFINITIONS

A. Remove: Detach items from existing construction and legally dispose of them off-site, unless indicated to be removed and salvaged or removed and reinstalled.

B. Remove and Salvage: Detach items from existing construction and deliver them to The Park ready for reuse.

C. Remove and Reinstall: Detach items from existing construction, prepare them for reuse, and reinstall them where indicated.

1.4 MATERIALS OWNERSHIP

A. Unless otherwise indicated, demolition waste becomes property of Contractor.

1.5 SUBMITTALS

A. Site Plan: Showing:

1. Vegetation to be protected.

2. Areas for temporary construction and field offices.

B. Schedule of Selective Demolition Activities: Indicate the following:

1. Detailed sequence of selective demolition and removal work, with starting and ending dates for each activity. Ensure Government on-site operations are uninterrupted.

2. Interruption of utility services. Indicate how long utility services will be interrupted.

3. Coordination for shutoff, capping, and continuation of utility services.

4. Use of elevator and stairs.

5. Means of protection for items to remain and items in path of waste removal from building.

C. Project Record Documents: Accurately record actual locations of capped and active utilities and subsurface construction.

D. Predemolition Photographs and Video: Photographs and Videotape: In accordance with Section 01 32 33 "Photographic Documentation," show existing conditions of adjoining construction and site improvements including finish surfaces that might be misconstrued as damage caused by selective demolition operations. Submit before work begins.

E. Statement of Refrigerant Recovery: Signed by refrigerant recovery technician responsible for recovering refrigerant, stating that all refrigerant that was present was recovered and that recovery was performed according to EPA regulations. Include name and address of technician and date refrigerant was recovered.

F. Submittals due at time of closeout:

1. Inventory: Submit a list of items that have been removed and salvaged.

2. Landfill Records: Indicate receipt and acceptance of hazardous wastes by a landfill facility licensed to accept hazardous wastes.

1.6 QUALITY ASSURANCE

A. Predemolition Conference: Conduct conference at Project site to comply with requirements in Division 01 Section 01 31 00 - Project Management and Coordination. Review methods and procedures related to selective demolition including, but not limited to, the following:

1. Inspect and discuss condition of construction to be selectively demolished.

2. Review structural load limitations of existing structure.

3. Review and finalize selective demolition schedule and verify availability of materials, demolition personnel, equipment, and facilities needed to make progress and avoid delays.

4. Review requirements of work performed by other trades that rely on substrates exposed by selective demolition operations.

5. Review areas where existing construction is to remain and requires protection.

B. Landfill Records: Indicate receipt and acceptance of hazardous wastes by a landfill facility licensed to accept hazardous wastes.

1.7 FIELD CONDITIONS

A. Hazardous Materials: Hazardous materials are present in construction to be selectively demolished. A report on the presence of hazardous materials is on file for review and use. Examine report to become aware of locations where hazardous materials are present.

1. Hazardous material remediation is specified in Sections 02 82 13 - Asbestos Abatement; 02 83 13 - Lead Handling Procedures; 02 87 16 - Bird Excrement Removal.

2. Do not disturb hazardous materials or items suspected of containing hazardous materials except under procedures specified elsewhere in the Contract Documents.

B. Utility Service: Maintain existing utilities indicated to remain in service and protect them against damage during selective demolition operations.

1. Maintain fire-protection facilities in service during selective demolition operations.

C. Notify Contracting Officer of discrepancies between existing conditions and Drawings before proceeding with selective demolition.

D. Buildings immediately adjacent to demolition area will be occupied. Conduct building demolition so operations of occupied buildings will not be disrupted.

1. Provide not less than 72 hours' notice of activities that will affect operations of adjacent occupied buildings.

2. Maintain access to existing walkways, exits, and other facilities used by occupants of adjacent buildings.

a. Do not close or obstruct walkways, exits, or other facilities used by occupants of adjacent buildings without written permission from authorities having jurisdiction.

PART 2 PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Regulatory Requirements: Comply with governing EPA notification regulations before beginning selective demolition. Comply with hauling and disposal regulations of authorities having jurisdiction.

B. Standards: Comply with ANSI/ASSE A10.6 and NFPA 241.

2.2 MATERIALS

A. Fill Material: As specified in Section 31 23 23 - Fill.

PART 3 EXECUTION

3.1 GENERAL PROCEDURES AND PROJECT CONDITIONS

A. Comply with applicable codes and regulations for demolition operations and safety of adjacent structures and the public.

1. Obtain required permits.

2. Take precautions to prevent catastrophic or uncontrolled collapse of structures to be removed; do not allow worker or public access within range of potential collapse of unstable structures.

3. Provide, erect, and maintain temporary barriers and security devices.

4. Conduct operations to minimize effects on and interference with adjacent structures and occupants.

5. Do not close or obstruct roadways or sidewalks without permit.

6. Conduct operations to minimize obstruction of public and private entrances and exits; do not obstruct required exits at any time; protect persons using entrances and exits from removal operations.

B. Do not begin removal until receipt of notification to proceed from Contracting Officer.

C. Protect existing structures and other elements that are not to be removed.

3.2 SELECTIVE DEMOLITION FOR ALTERATIONS

A. Drawings showing existing construction and utilities are based on casual field observation and existing record documents only.

1. Verify that construction and utility arrangements are as indicated.

2. Report discrepancies to Contracting Officer before disturbing existing installation.

3. Beginning of demolition work constitutes acceptance of existing conditions that would be apparent upon examination prior to starting demolition.

B. Maintain weatherproof exterior building enclosure, except for interruptions required for replacement or modifications; prevent water and humidity damage.

C. Remove existing work as indicated and as required to accomplish new work.

1. Remove rotted wood, corroded metals, and deteriorated masonry and concrete; replace with new construction indicated.

2. Remove items indicated on drawings.

D. Services (Including but not limited to HVAC, Plumbing, Fire Protection, Electrical, and Telecommunications): Remove existing systems and equipment as indicated.

1. Maintain existing active systems that are to remain in operation; maintain access to equipment and operational components.

2. Remove abandoned pipe, ducts, conduits, and equipment, including those above accessible ceilings; remove back to source of supply where possible, otherwise cap stub and tag with identification.

E. Protect existing work to remain.

1. Refer to section 01 35 91 - Historic Preservation Treatment Procedures.

F. Removed and Salvaged Items:

1. Refer to section 01 35 91 - Historic Preservation Treatment Procedures.

G. Removed and Reinstalled Items:

1. Refer to section 01 35 91 - Historic Preservation Treatment Procedures.

3.3 DEBRIS AND WASTE REMOVAL

A. Remove debris, junk, and trash from site.

B. Leave site in clean condition, ready for subsequent work.

C. Clean up spillage and wind-blown debris from public and private lands.

3.4 SELECTIVE DEMOLITION SCHEDULE (AS NOTED BUT NOT LIMITED TO)

A. Existing items to be removed and salvaged:

1. Door Hardware.

B. Existing items to be removed and reinstalled at similar locations. See Drawings for locations:

1. Wood flooring.

2. Light fixtures.

3. Plumbing fixtures.

4. Wood siding.

5. Wood paneling.

6. Wood roof decking.

7. Wood casework.

END OF SECTION

SECTION 02 41 13 – SELECTIVE SITE DEMOLITION

PART 1 GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section of the Work includes clearing, removal and disposal of trees; including stumps and roots.

1.3 PRE-DEMOLITION MEETING

A. Prior to Commencement of Work, a conference shall be held at the Project site.

1. Inspect and discuss condition of trees to be demolished.
2. Review and finalize tree demolition schedule and verify availability of materials, demolition personnel, equipment, and facilities needed to make progress and avoid delays.

1.4 SUBMITTALS

A. See Section 01 33 23 “Submittal Procedures”.

B. Proposed Protection Measures: Submit report, including drawings, that indicates the measures proposed for protecting individuals and property, for environmental protection, for dust control and for noise control. Indicate proposed locations and construction of barriers.

C. Inventory: Submit a list of trees to be removed and deliver to Contracting Officer prior to start of demolition.

D. Photographs and Videotape: In accordance with Section 01 32 33 "Photographic Documentation," show existing conditions of adjoining construction and site improvements including finish surfaces that might be misconstrued as damage caused by selective demolition operations. Submit before work begins..

1.5 QUALITY ASSURANCE

A. Work Scheduling: Proceed with and complete demolition work as rapidly as portions of the site become available, working within the specified seasonal and inclement weather restrictions and approved schedule.
1. The Contractor shall minimize the impacts to visitors to the extent possible.

B. Do not perform work when climate and existing site conditions will not provide satisfactory results (e.g. wet, freezing). The Contracting Officer reserves the right to deny demolition procedures on days that are deemed to be unfavorable. Proceed with demolition only when existing and forecasted weather conditions permit demolition to be performed when beneficial and optimum results may be obtained.

C. Contractor shall have full use of the approved designated fenced or flagged in project area for specific construction operations during the construction period. Contractor's use of the site is limited to minimizing site disturbance to the extent possible and by the Government's right to perform work or to retain other contractors on portions of Project. There will be no visitor access within the designated fenced in approved project areas.

D. All work is being completed within a historic site and must adhere to the Secretary of Interior's Standards, as per Section 01 35 91 "Historic Preservation Treatment Procedures". Work must be completed without damage to fencing, trees to remain, trails, roads, or other property. The Contractor will be financially liable to the park for any costs incurred as a result of repairing and/or replacing any of the aforementioned items or for the value of the item as assessed by an archeologist, biologist, government and the Contracting Officer.

E. All vehicles and other heavy equipment involved in the completion of this contract will be restricted to roads, parking areas, and road shoulders no more than 10' from the edge of the road surface unless given case-specific permission by the Contracting Officer. Off road equipment use is solely at the discretion of the Contracting Officer and subject to change based on current or anticipated conditions. Rutting or other resource damage will not be tolerated. All work from trails will be considered off road and must adhere to the same guidelines and restrictions.

F. All vehicles should be equipped with absorptive pads while on park property in order to quickly contain any spills of oil or hydraulic fluid. Any such spills must be reported to the Contracting Officer as soon as is safely possible.

G. Damage to natural areas, existing grades, vegetation, pavements, underground utilities, and other improvements and areas of protection, caused during work under this section, shall be repaired to original condition or better at no additional cost to the Government.

H. Vehicular accessibility on site shall be as directed by Contracting Officer. Repair damage to existing surfaces, caused by vehicular access and movement during work under this section, to original condition at no additional cost to the Government.

I. Existing conditions:
   1. Utilities: Determine location of underground utilities. Contractor shall be responsible for contacting One-Call of Wyoming to locate all utilities prior to construction, 811 or 1-800-849-2476. Perform work in a manner to avoid possible damage. Hand excavate, as required.
   2. Any damage to underground piping or wiring arising out of work of this section must be corrected and repaired by the Contractor to the satisfaction of the Contracting Officer at no additional cost to the Government.
   3. Damage to Other Improvements: All costs for repair or replacement of any damage to other work done on-site or adjacent properties by demolition procedures shall be borne by the Contractor completing demolition procedures.

4. When conditions detrimental to demolition are encountered, such as rubble fill, adverse drainage conditions, noxious materials or obstructions, notify Contracting Officer before placement.
   a. Obstructions Below Ground: When obstructions below finished grades prevent demolition procedures, the Contractor shall submit a plan showing proposed adjustments to locations to the Contracting Officer.

J. Keep dust to a minimum at removal areas. Use water trucks as necessary.

K. Noise Control: Perform tree removal operations to minimize noise. Perform noise-producing work in less sensitive hours of the day or week as directed by the Contracting Officer. Reference Section 01 57 19.12 Noise & Acoustics Management.

L. Ensure safety of persons in demolition area. Provide temporary barricades as required per Division 01 Section "Temporary Facilities and Controls".

M. Do not direct vehicle or equipment exhaust towards protection zones.

N. Prohibit heat sources, flames, ignition sources, and smoking within or near protection zones.

O. Construction Equipment Loading and Parking: Load and park only in designated areas.

P. Do not commence tree removal operations until protection measures, temporary fencing/flagging, temporary erosion and sedimentation control measures are in place.

Q. Restore damaged improvements to their original condition, as acceptable to parties having jurisdiction.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 PREPARATION

A. All work shall take place within the confines of the limits of constructions as shown on the drawings.

B. Protect structures, trails, utilities, and vegetation to remain.

C. Should any damage result, it shall be repaired to the satisfaction of the Contracting Officer at no additional cost.

D. Protect and maintain benchmarks, survey control points, and other reference point from disturbance during construction. If disturbed or destroyed, restore or replace at no additional cost.

E. Provide erosion control measures to prevent soil erosion and discharge of soil-bearing water runoff or airborne dust from leaving project site. Reference Section 31 25 00 "Erosion and Sediment Control".

F. Locate and clearly flag trees and vegetation to remain.

G. Set up all barriers, including those for tree protection, in accordance with Division 01 Section "Temporary Facilities and Controls" and Division 01 Section "Tree Retention and Protection", prior to proceeding with any demolition.

3.2 TREE REMOVAL

A. Do not remove trees, shrubs, and other vegetation indicated to remain or to be relocated.

B. Stumps resulting from tree removal shall be flush cut level with the ground and not higher than 2 inches (51 mm) as measured from the uphill side unless specified otherwise.

C. Use only hand methods or air spade for grubbing within protection zones.

D. Chip felled trees and branches and stockpile in areas approved by Contracting Officer.

3.3 RESTORATION

A. Backfilling: Ensure that areas to be filled are free of standing water, frost, frozen material, vegetation, including roots and debris.

B. Grading:

1. Restored Areas: Grade surface to blend with original contours and provide free drainage flow. All ruts and depressions where any amount of standing water collects shall be re-graded to a smooth natural appearance to ensure positive drainage.

END OF SECTION 02 41 13

SECTION 02 42 96 - HISTORIC REMOVAL AND DISMANTLING

PART 1 GENERAL

1.1 SUMMARY

A. Section Includes:

1. Historic treatment procedures for removal and dismantling work for designated historic spaces, areas, rooms, and surfaces and the following specific work:

a. Removal and dismantling of indicated portions of building or structure and debris hauling.

b. Removal and dismantling of indicated site elements and debris hauling.

c. Salvage of existing items to be reused or returned to The Park.

B. Related Requirements:

1. Section 01 3591 "Historic Treatment Procedures" for general historic treatment requirements.

2. Section 02 87 16.13 "Bird Excrement Removal" for removing bird excrement from historic surfaces.

1.2 PRECONSTRUCTION MEETINGS

A. Preconstruction Conference(s): Conduct conference(s) at Project site.

1. Review minutes of Preliminary Historic Treatment Conference that pertain to removal and dismantling procedures and protection of historic areas and surfaces.

2. Review list of items indicated to be salvaged.

3. Verify qualifications of personnel assigned to perform removal and dismantling.

4. Inspect and discuss condition of each construction type to be removed or dismantled.

5. Review requirements of other work that depends on condition of substrates exposed by removal and dismantling work.

6. Review methods and procedures related to removal and dismantling work, including, but not limited to, the following:

a. Historic removal and dismantling specialist's personnel, equipment, and facilities needed to make progress and avoid delays.

b. Materials, material application, sequencing, tolerances, and required clearances.

c. Fire prevention.

d. Coordination with building occupants.

1.3 INFORMATIONAL SUBMITTALS

A. Qualification Statements: For historic removal and dismantling specialist.

B. Preconstruction Documentation: Show preexisting conditions of adjoining construction and site improvements, including finish surfaces, that might be misconstrued as damage caused by Contractor's removal and dismantling operations.

C. Removal and Dismantling Historic Treatment Program: Submit 30 days before work begins.

D. List of Items Indicated To Be Salvaged: Prepare a list of items indicated on Drawings to be salvaged for The Park's use or for reinstallation. Submit 15 days before preconstruction conference.

E. Inventory of Salvaged Items: After removal or dismantling Work is complete, submit a list of items that have been salvaged.

1. Include item description, item condition, number of items if more than one of a type, and tag number. Include photo of item in original location.

2. As work proceeds, include on the inventory items that were indicated to be salvaged and items of historic importance discovered during the work. Document reasons, if any, why an item indicated to be salvaged was not salvaged.

1.4 QUALITY ASSURANCE

A. Removal and Dismantling Historic Treatment Program: Prepare a written, detailed description of materials, methods, equipment, and sequence of operations to be used for each phase of removal and dismantling work, including protection of surrounding and substrate materials and Project site.

1. Dust and Noise Control: Include locations of proposed temporary dust- and noise-control partitions and means of egress from occupied areas coordinated with continuing on-site operations and other known work in progress.

2. Debris Hauling: Include plans clearly marked to show debris-hauling routes, turning radii, and locations and details of temporary protective barriers.

1.5 MOCKUPS

A. Prepare mockups of specific historic removal and dismantling procedures specified in this Section to demonstrate aesthetic effects and to set quality standards for materials and execution.

1. Exploratory Dismantling of Wood Flooring : Perform exploratory dismantling in the presence of Contracting Officer.

2. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations in writing.

B. Regulatory Requirements: Comply with notification regulations of authorities having jurisdiction before beginning removal and dismantling work. Comply with hauling and disposal regulations of authorities having jurisdiction.

1.6 FIELD CONDITIONS

A. Conditions existing at time of inspection for bidding purposes will be maintained by Government as long as practicable.

1. Before removal and dismantling, The Park will remove the following items:

a. Window sashes to be removed by The Park.

B. Notify Contracting Officer of discrepancies between existing conditions and Drawings before proceeding with removal and dismantling work.

C. Hazardous Materials:

1. Hazardous materials are present in construction affected by removal, see Section 02 41 00 - Demolition.

D. Storage or sale of removed or dismantled items on-site is not permitted unless otherwise indicated.

PART 2 PRODUCTS - (NOT USED)

PART 3 EXECUTION

3.1 HISTORIC REMOVAL AND DISMANTLING EQUIPMENT

A. Removal Equipment: Use manual, handheld tools. Handheld power tools may be permitted on a case-by-case basis with approval by Contracting Officer.

B. Dismantling Equipment: Use manual, handheld tools, except as follows or otherwise approved by Contracting Officer on a case-by-case basis:

1. Handheld power tools are permitted only as submitted in the historic treatment program. They must be adjustable so as to penetrate or cut only the thickness of material being removed.

2. Pry bars more than 18 inches (450 mm) long and hammers weighing more than 2 lb (0.9 kg) are not permitted for dismantling work.

3.2 EXAMINATION

A. Preparation for Removal and Dismantling: Examine construction to be removed or dismantled to determine best methods to safely and effectively perform removal and dismantling work. Examine adjacent work to determine what protective measures are necessary. Make explorations, probes, and inquiries as necessary to determine condition of construction to be removed or dismantled and location of utilities and services to remain that may be hidden by construction that is to be removed or dismantled.

1. Verify that affected utilities are disconnected and capped.

2. Inventory and record the condition of items to be removed and dismantled for reinstallation or salvage. Enter this information on the inventory of salvaged items.

3. Before removal or dismantling of existing building elements that will be reproduced or duplicated in final Work, make permanent record of measurements, materials, and construction details required to make exact reproduction.

B. Survey of Existing Conditions: Record existing conditions by use of preconstruction photographs .

1. Comply with requirements specified in Section 01 32 33 "Photographic Documentation."

C. Perform surveys as the Work progresses to detect hazards resulting from historic removal and dismantling procedures.

3.3 HISTORIC REMOVAL AND DISMANTLING

A. Comply with requirements in Section 01 35 91 "Historic Treatment Procedures" for identifying and storing historic items.

B. Perform work according to the historic treatment program and approved mockup(s).

1. Perform removal and dismantling to the limits indicated.

2. Provide supports or reinforcement for existing construction that becomes temporarily weakened by removal and dismantling work, until the Project Work is completed unless otherwise indicated.

3. Perform cutting by hand or with small power tools as permitted by Contracting Officer wherever possible. Cut holes and slots neatly to size required, with minimum disturbance of adjacent work.

4. Do not operate air compressors inside building unless approved by Contracting Officer in each case.

5. Do not drill or cut columns, beams, joints, girders, structural slabs, or other structural supporting elements, without having Contractor's professional engineer's written approval for each location before such work is begun.

6. Dispose of removed and dismantled items off-site unless indicated to be salvaged or reinstalled.

C. Water-Mist Sprinkling: Use water-mist sprinkling and other wet methods to control dust only with adequate, approved procedures and equipment according to the historic treatment program to ensure that such water does not create a hazard or adversely affect other building areas or materials.

D. Unacceptable Equipment: Keep equipment that is not permitted for historic removal or dismantling work away from the vicinity where such work is being performed.

E. Removing and Dismantling Items on or Near Historic Surfaces:

1. Use only dismantling equipment and procedures within 12 inches (300 mm) of historic surface. Protect historic surface from contact with or damage by tools.

2. Unfasten items in the opposite order from which they were installed.

3. Support each item as it becomes loosened to prevent stress and damage to the historic surface.

4. Dismantle anchorages.

F. Masonry Walls:

1. Remove masonry carefully, and erect temporary bracing and supports as needed to prevent collapse of materials being removed.

2. Stop removal work and immediately inform Contracting Officer and if any structural elements above or adjacent to the work show signs of distress or dislocation during any phase of removal work.

3. Remove wall in easily managed pieces.

4. During removal, maintain the stability of the partially remaining wall.

G. Concrete Floor Surface:

1. Remove floor surfaces, fill, and topping to the indicated lower elevations or cleavage planes as indicated on Drawings.

2. Use dismantling methods when removing floor surfaces 12 inches (300 mm) or less away from historic walls.

3. Take away material to a uniform surface at the indicated level.

H. Anchorages:

1. Remove anchorages associated with removed items.

2. Dismantle anchorages associated with dismantled items.

3. In nonhistoric surfaces, patch holes created by anchorage removal or dismantling according to the requirements for new work.

4. In historic surfaces, patch or repair holes created by anchorage removal or dismantling according to Section that is specific to the historic surface being patched.

END OF SECTION

SECTION 02 43 00 - STRUCTURE MOVING

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Preparing structure for moving and lifting.

B. Moving or lifting structure temporarily off the existing foundations.

C. Setting structure on new foundation at exact original location.

1.2 RELATED REQUIREMENTS

A. Section 02 41 00 - Demolition: Removal of existing foundation after structure removal.

1.3 ADMINISTRATIVE REQUIREMENTS

A. Stabilization of structural components such as chimneys, wood logs, and/or framing to be coordinated with and approved by Contracting Officer prior to starting work on this section.

B. Pre-Move Meeting: Convene one week before starting work of this section. Discuss the following:

1. Contractor to determine and justify which structures will be lifted only and which structures need to be moved.

2. Method of determining damage to existing structure and finishes before and after the move.

3. Identify existing damage to sidewalks, roads, and curbs.

4. Method and responsibility for repairs after lift/move.

5. Review the intended route for moving and dimensional clearances of obstructions.

6. Coordination with affected utility companies.

7. Coordination with newly installed utilities from Phase 1 of project

1.4 SUBMITTALS

A. Photographs and Videotape: In accordance with Section 01 32 33 "Photographic Documentation," show existing conditions of adjoining construction and site improvements including finish surfaces that might be misconstrued as damage caused by selective demolition operations. Submit before work begins.

B. Project Record Documents: Record actual locations of new utilities, services, and foundations .\

C. Contractor to submit a structures moving plan for approved by CO and COR before work.

1.5 QUALITY ASSURANCE

A. Mover Qualifications: Company specializing in lifting and relocating building structures with minimum of ten years of documented experience.

PART 2 PRODUCTS

2.1 EQUIPMENT AND MATERIALS

A. Transport, Equipment, and Supports: As required to achieve a successful structure lift/move.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify availability and accessibility of transport routes. Verify route load limits to ensure conditions are adequate to support moving loads of structure.

B. Identify utility services and obstructions to be removed, relocated, or abandoned during progress of the Work.

C. Damage Determination:

1. Before the lift/move, inspect existing structure thoroughly and notify Contracting Officer in writing of visible defects and factors that could affect safe movement of structure to final location.

2. Compile list with photographic documentation and with locations noted on plans/elevations of existing visible defects to building structure, finishes, accessories . This list will form the basis for determining required repair work after the lift/move.

3. Photograph interior and exterior surfaces prior to first and second move for record purposes.

3.2 PREPARATION

A. Prepare site, route of transport, and destination site.

B. Cut, fill, and compact access bed, grade, and surface area as necessary to safely lift/move the structure and to prevent damage.

C. Coordinate the work of municipal utility disconnection and reconnection with the work of this section.

D. Existing site utility services will have been disconnected by Phase 1 work.

E. Remove overhead or exposed utility services to provide clear working and moving space around and below structure.

F. Remove building protrusions prior to lift/move.

G. Stabilize and/or repair structural components prior to lift/move.

H. Secure supplementary framing and bracing to structure.

I. Secure operating, moving, or suspended items such as doors, windows, and light fixtures in a manner to prevent damage to items or to the structure during move.

J. Protect elements surrounding the structure from damage.

3.3 RAISE STRUCTURE

A. Cut structure free of foundation and portions of structure not being moved.

B. Reinforce, brace, and raise structure clear of foundation in manner to prevent damage to historic fabric.

C. Provide necessary framing, bracing, closures, supports, and blocking.

D. Secure structure to temporary supporting structural members to prevent shifting of structure during move.

3.4 MOVE STRUCTURE (AS DETERMINED NEEDED BY CONTRACTOR)

A. Provide transport vehicles for moving structure to temporary site.

B. Move structure, control speed, and provide anchor and restraining devices to maintain the integrity of the structure.

C. During move, protect adjacent structures, and landscape from damage.

1. Coordinate with Natural Resources and Contracting Officer, if they want to be present during the lift/move.

3.5 REINSTALL STRUCTURE

A. Position and anchor structure over prepared foundation and lower onto new foundation.

B. Remove moving equipment.

C. Leave reinforcing, framing, and bracing intact until structure is fully attached and structure loads are supported by new foundation.

D. Reinstall building protrusions removed prior to move.

3.6 TOLERANCES

A. Maximum Variation from Level and Plumb After Reinstallation: 1/4 inch.

B. Adjust structure on foundation:

1. To permit doors to swing freely.

2. So that floor surfaces are level, walls are plumb.

3.7 DAMAGE REPAIR

A. Repair damage to structure not identified in writing prior to move.

1. Follow requirements of this project when repairing damage to structure or fabric of moved building.

B. Refinish repaired surfaces to match adjacent work.

C. Pay all third party claims for incidental or other damage.

3.8 CLEANING

A. Remove moving equipment and materials from original site, final site, and route of travel.

B. Remove road base constructed by this section, fill and return grades and contours to original condition and dimension.

END OF SECTION

SECTION 02 82 00 - ABATEMENT OF ASBESTOS-CONTAINING MATERIALS

PART 1 GENERAL

1.1 SUMMARY OF WORK

A. Contract Documents and Related Requirements

1 Drawings, general provisions of the contract, including General and Supplementary Conditions and other Specification Sections, apply to the work of this Section. The contract documents show the work to be done under the contract and related requirements and conditions impacting the project. Related requirements and conditions include applicable codes and regulations, notices and permits, existing site conditions and restrictions on the use of the site, requirements for partial owner occupancy during the work, coordination with other work, and the phasing of the work. In the event the Asbestos Abatement Contractor discovers a conflict in the contract documents and/or requirements or codes, the conflict must be brought to the immediate attention of the General Contractor for Rehabilitation and the Contracting Officer for resolution. Whenever there is a conflict or overlap in the requirements, the most stringent shall apply. Any actions taken by the Asbestos Abatement Contractor without obtaining guidance from the Contracting Officer shall become the sole risk and responsibility of the Asbestos Abatement Contractor. All costs incurred due to such action are also the responsibility of the Asbestos Abatement Contractor.

B. Extent of Work

1. The Asbestos Abatement Contractor will coordinate all work, phasing, and scheduling with the General Contractor for Rehabilitation, the Contracting Officer, and their field representative. Asbestos Abatement Contractor will have a predetermined number of calendar days to complete asbestos removal, including final clearance monitoring and final teardown per regulated area. The Asbestos Abatement Contractor will coordinate the start date and the number of mobilizations required with the General Contractor for Rehabilitation, the Contracting Officer, and their field representative. Strict adherence to the schedule will be required to allow the work to be completed promptly to accommodate other activities on the site.

2. The Contracting Officer will supply the Asbestos Survey Report, dated July 8, 2025, completed by Northern Industrial Hygiene, Inc. of Billings, Montana, to the Asbestos Abatement Contractor. The asbestos survey report will contain a brief description of the asbestos-containing materials and locations. The Asbestos Abatement Contractor shall satisfy himself as to the actual quantities to be abated and disposed of. Nothing in this section may be interpreted as limiting the extent of work otherwise required by this contract and related documents.

3. Removal, packaging, clean-up, and disposal of asbestos-containing materials (ACM) and asbestos-contaminated elements in an appropriate regulated area as necessary to accommodate the disposal of ACM and the rehabilitation of the buildings.

4. See Section 01 35 91 Historic Preservation Treatment Procedures for additional requirements for the salvage of historic building materials.

5. Any damage to components not scheduled for demolition, resulting from the Asbestos Abatement Contractor's work, shall be repaired or replaced at the sole cost of the Asbestos Abatement Contractor, utilizing appropriately qualified tradespersons.

6. The Asbestos Abatement Contractor will be responsible for moving all property and refuse out of the Asbestos Abatement Contractor's regulated areas.

C. Tasks

The work tasks are summarized as follows:

1. The Asbestos Abatement Contractor will coordinate with the General Contractor for Rehabilitation, the Contracting Officer, and the NPS representative for scheduling access to the structures. The Asbestos Abatement Contractor shall assume that the structures will be partially occupied during this project.

2. Pre-abatement activities, including pre-abatement meeting(s), inspection(s), notifications, permits, submittal approvals, work-site preparations, emergency procedures arrangements, and standard operating procedures (SOPs) for asbestos abatement work.

3. Abatement activities, including removal, packaging, encapsulation, clean-up, storing, and disposal of ACM waste, recordkeeping, security, monitoring, and inspections.

4. Cleaning and decontamination activities, including final visual inspection, air monitoring, and certification of decontamination.

D. Asbestos Abatement Contractors Use of Premises

1. The Asbestos Abatement Contractor and Asbestos Abatement Contractor's personnel shall cooperate fully with the General Contractor for Rehabilitation and Contracting Officer to facilitate efficient use of the site. The Asbestos Abatement Contractor shall perform the work in accordance with the specifications and in compliance with any/all applicable Federal, State, and Local regulations and requirements.

2. The Asbestos Abatement Contractor shall specify the facilities proposed to be utilized in the pre-abatement work plan. The Asbestos Abatement Contractor shall use only the existing facilities in the structures strictly within the limits indicated in the approved pre-abatement work plan. Any variation from the approved work plan shall be secured in writing by the Contracting Officer.

1.2 STOP ASBESTOS REMOVAL

A. If the Contracting Officer, their field representative, the facility Safety Officer/Manager, or their designee presents a verbal **Stop Asbestos Removal Order**, the Asbestos Abatement Contractor/Personnel shall immediately stop all asbestos removal and maintain HEPA filtered negative pressure air flow in the containment and adequately wet any exposed ACM. If a verbal Stop Asbestos Removal Order is issued, the Contracting Officer shall follow up with a written order to the Asbestos Abatement Contractor as soon as it is practicable. The Asbestos Abatement Contractor shall not resume any asbestos removal activity until authorized to do so in writing by the Contracting Officer. A stop asbestos removal order may be issued at any time the Contracting Officer determines abatement conditions or activities are not within specification requirements, regulatory requirements, or that an imminent hazard exists to human health or the environment. The work stoppage will continue until conditions have been corrected to the satisfaction of the Contracting Officer. The occurrence of any of the following events shall be reported immediately by the Asbestos Abatement Contractor's competent person to the Contracting Office and/or their field representative using the most expeditious means (e.g., verbal or telephonic), followed up with a written notification to the Contracting Officer as soon as it is practical. The Asbestos Abatement Contractor shall immediately stop asbestos removal/disturbance activities and initiate fiber reduction activities if:

1. Airborne PCM analysis results equal to or greater than 0.01 f/cc above background levels outside the regulated area.

2. Breach or break in regulated area containment barrier(s).

3. less than –0.02 inch WCG pressure in the regulated area.

4. Serious injury/death at the site.

5. Fire/safety emergency at the site.

6. Respiratory protection system failure.

7. Power failure or loss of wetting agent.

8. Any visible emissions observed outside the regulated area.

9. Failure to follow project specification requirements.

1.3 DEFINITIONS

A. Definitions and explanations here are neither complete nor exclusive of all terms used in the contract documents, but are general for the work to the extent they are not stated more explicitly in another element of the contract documents. Drawings must be recognized as diagrammatic and not completely descriptive of the requirements indicated therein.

B. Glossary

Definitions relative to Asbestos Abatement.

1. Abatement - Procedures to control fiber release from asbestos-containing materials, typically during removal. Includes removal, encapsulation, enclosure, demolition, and renovation activities related to asbestos.

2. ACBM - Asbestos-containing building materials.

3. ACE – Asbestos-contaminated elements.

4. ACM - Asbestos-containing material.

5. Aerosol - Solid or liquid particulate suspended in the air.

6. Adequately wet - Sufficiently mixed or penetrated with liquid to prevent the release of particulates. If visible emissions are observed coming from the ACM, then that material has not been adequately wetted.

7. Aggressive method - Removal or disturbance of building material by sanding, abrading, grinding, or other methods that break, crumble, or disintegrate intact ACM.

8. Aggressive sampling - EPA AHERA defined a clearance sampling method using air-moving equipment such as fans and leaf blowers to aggressively disturb and maintain the air residual fibers after abatement.

9. AHERA - Asbestos Hazard Emergency Response Act. Asbestos regulations for schools were issued in 1987.

10. AIHA - American Industrial Hygiene Association.

11. Aircell - Pipe or duct insulation made of corrugated cardboard that contains asbestos.

12. Air monitoring - The process of measuring the fiber content of a known volume of air collected over a specified period of time. The NIOSH 7400 Method, Issue 2, is used to determine the fiber levels in air.

13. Air monitoring firm - The firm retained by the NPS to conduct baseline, area, and clearance air monitoring before, during, and following the asbestos abatement.

14. Air sample filter - The filter used to collect fibers, which are then counted. The filter is made of mixed cellulose ester membrane for PCM (Phase Contrast Microscopy) and polycarbonate for TEM (Transmission Electron Microscopy).

15. Amended water - Water to which a surfactant (wetting agent) has been added to increase the penetrating ability of the liquid.

16. Asbestos - Includes chrysotile, amosite, crocidolite, tremolite asbestos, anthophyllite asbestos, actinolite asbestos, and any of these minerals that have been chemically treated or altered. Asbestos also includes PACM, as defined below.

17. Asbestos-containing building material (ACBM) - Any building material containing more than one percent by weight of asbestos of any type or mixture.

18. Asbestos-containing material (ACM) - Any material containing more than one percent by weight of asbestos of any type or mixture.

19. Asbestos-contaminated elements (ACE) - Building elements such as ceilings, walls, lights, or ductwork that are contaminated with asbestos.

20. Asbestos-contaminated soil (ACS) - Soil found in the regulated area or in adjacent areas, such as crawlspaces or pipe tunnels, which is contaminated with asbestos-containing material debris and cannot be easily separated from the material.

21. Asbestos-containing waste (ACW) material - Asbestos-containing material or asbestos-contaminated objects requiring disposal.

22. Asbestos waste decontamination facility - A system consisting of drum/bag washing facilities and a temporary storage area for cleaned containers of asbestos waste. Used as the exit for waste and equipment leaving the regulated area. In an emergency, it may be used to evacuate personnel.

23. ASHARA - Asbestos School Hazard Abatement Re-Authorization Act. This act on the regulations for implementation requires individuals conducting asbestos inspections to be AHERA trained with current certification.

24. Authorized person - Any person authorized by the NPS, the Asbestos Abatement Contractor, or government agency and required by work duties to be present in regulated areas.

25. Authorized visitor - Any person approved by the NPS, the Asbestos Abatement Contractor, or any government agency having jurisdiction over the regulated area.

26. Barrier - Any surface that isolates the regulated area and inhibits fiber migration from the regulated area.

27. Containment Barrier - An airtight barrier consisting of walls, floors, and/or ceilings of sealed plastic sheeting that surrounds and seals the outer perimeter of the regulated area.

28. Critical Barrier - The barrier responsible for isolating the regulated area from adjacent spaces, typically constructed of plastic sheeting secured in place at openings such as doors, windows, or any other opening into the regulated area.

29. Primary Barrier - Barriers placed over critical barriers and exposed directly to abatement work.

30. Secondary Barrier - Any additional plastic barriers used to isolate and provide protection from debris during abatement work.

31. Breathing zone - The hemisphere, forward of the shoulders, with a radius of about 150–225 mm (6–9 inches), from the worker's nose.

32. Bridging encapsulant - An encapsulant that forms a layer on the surface of the ACM.

33. Building/facility owner - The legal entity, including a lessee, which exercises control over management and recordkeeping functions relating to a building and/or facility in which asbestos activities take place.

34. Bulk testing - The collection and analysis of suspect asbestos-containing materials.

35. Certified Industrial Hygienist (CIH) - One certified in the practice of industrial hygiene by the American Board of Industrial Hygiene (AIHA). An industrial hygienist certified in Comprehensive Practice by AIHA.

36. Class I asbestos work - Activities involving the removal of Thermal System Insulation (TSI), surfacing ACM, and Presumed Asbestos-containing Material (PACM).

37. Class II asbestos work - Activities involving the removal of ACM, which is not thermal system insulation or surfacing material. This includes but is not limited to the removal of asbestos-containing wallboard, floor tile and sheeting, roofing, siding shingles, and construction mastic.

38. Class III asbestos work - Repair and maintenance operations where ACM, including TSI and surfacing ACM and PACM, may be disturbed.

39. Class IV asbestos work - Maintenance and custodial activities during which employees contact but do not disturb ACM or PACM, and activities to clean up dust, waste, and debris resulting from Class I, II, and III activities.

40. Clean room/Changing room - An uncontaminated room having facilities for the storage of employees' street clothing, uncontaminated materials, and equipment.

41. Clearance sample - The final air sample taken after all asbestos work has been done and visually inspected. Performed by the NPS industrial hygiene consultant (IHC).

42. Closely resemble - The major workplace conditions, which have contributed to the levels of historic asbestos exposure, are no more protective than conditions of the current workplace.

43. Competent person - In addition to the definition in 29 CFR 1926.32(f), one who is capable of identifying existing asbestos hazards in the workplace and selecting the appropriate control strategy for asbestos exposure, who has the authority to take prompt corrective measures to eliminate them, as specified in 29 CFR 1926.32(f); in addition, for Class I and II work, who is specially trained in a training course which meets the criteria of EPA's Model Accreditation Plan (40 CFR 763) for supervisor.

44. Contingency plan - A document setting out an organized, planned, and coordinated course of action to be followed in case of a fire or explosion, or a release of hazardous waste or hazardous waste constituents from a treatment, storage, or disposal facility that could threaten human health or the environment.

45. Contracting Officer - The NPS official responsible for ongoing project work.

46. Count - Refers to the fiber count or the average number of fibers greater than five microns in length with a length-to-width (aspect) ratio of at least 3 to 1, per cubic centimeter of air.

47. Crawlspace - An area that can be found either in or adjacent to the regulated area. This area has limited access and egress and may contain asbestos materials and/or asbestos-contaminated soil.

48. Decontamination area/unit - An enclosed area adjacent to and connected to the regulated area consisting of an equipment room, shower room, and clean room, which is used for the decontamination of workers, materials, and equipment that are contaminated with asbestos.

49. Demolition - The wrecking or taking out of any load-supporting structural member and any related razing, removing, or stripping of asbestos products.

50. Discharge or hazardous waste discharge - The accidental or intentional spilling, hazardous waste leaking, pumping, pouring, emitting, discharging, emptying, or dumping of hazardous wastes onto any land or water.

51. Disposal - The discharge, deposit, injection, dumping, spilling, leaking, or placing of any solid waste or hazardous waste into or on any land or water so that any constituent thereof may enter the environment or be emitted into the air or discharged in any water, including groundwater.

52. Disposal bag - Typically a 6-millimeter (mil) thick sift-proof, dustproof, leak-tight container, used to package and transport asbestos waste from regulated areas to the approved landfill. Each bag or container must be labeled and marked in accordance with EPA, OSHA, and the United States Department of Transportation (USDOT) requirements.

53. Disturbance - Activities that disrupt the matrix of ACM or PACM, crumble or pulverize ACM or PACM, or generate visible debris from ACM or PACM. Disturbance includes cutting away small amounts of ACM or PACM, no greater than the amount that can be contained in one standard-sized glove bag or waste bag, to access a building component. In no event shall the amount of ACM or PACM so disturbed exceed that which can be contained in one glove bag or disposal bag, which shall not exceed 60” in length or width.

54. Drum - A rigid, impermeable container made of cardboard fiber, plastic, or metal that can be sealed to be sift-proof, dustproof, and leak-tight.

55. Employee exposure - The exposure to airborne asbestos that would occur if the employee were not wearing respiratory protection equipment.

56. Encapsulant - A material that surrounds or embeds asbestos fibers in an adhesive matrix and prevents the release of fibers.

57. Enclosure - The construction of an air-tight, impermeable, permanent barrier around ACM to control the release of asbestos fibers from the material and eliminate access to the material.

58. Equipment room - A contaminated room located within the decontamination area that is supplied with impermeable bags or containers for the disposal of contaminated protective clothing and equipment.

59. Fiber - A particulate form of asbestos, 5 microns or longer, with a length-to-width ratio of at least 3 to 1.

60. Fibers per cubic centimeter (f/cc) - Abbreviation for fibers per cubic centimeter, used to describe the level of asbestos fibers in the air.

61. Filter - Media used in respirators, vacuums, or other machines to remove particulates from the air.

62. Firestopping - Material used to close the open parts of a structure to prevent a fire from spreading.

63. Friable asbestos-containing material - Any material containing more than 1 percent asbestos as determined using the method specified in Appendix A, Subpart F, 40 CFR 763, Section 1, Polarized Light Microscopy, that, when dry, can be crumbled, pulverized, or reduced to powder by hand pressure.

64. Glove bag - Not more than a 60 x 60-inch impervious plastic bag-like enclosure affixed around an asbestos-containing material, with glove-like appendages through which materials and tools may be handled.

65. Government industrial hygienist (IHC) – The government retained an Industrial Hygiene Consultant.

66. High-efficiency particulate air (HEPA) filter - A filter capable of trapping and retaining at least 99.97 percent of all mono-dispersed particles of 0.3 microns or greater in diameter.

67. HEPA vacuum - vacuum collection equipment equipped with a HEPA filter system capable of collecting and retaining asbestos fibers.

68. Homogeneous area - An area of surfacing, thermal system insulation, or miscellaneous ACM that is uniform in color, texture, and date of application.

69. HVAC - Heating, Ventilation, and Air Conditioning

70. Industrial hygienist (IH) - A professional qualified by education, training, and experience to anticipate, recognize, evaluate, and develop controls for occupational health hazards and meets the definition requirements of AIHA.

71. Industrial hygienist technician (IH Technician) - A person working under the direction of an IH or CIH with special training, experience, certifications, and licenses required for the assigned industrial hygiene work.

72. Intact - The ACM has not crumbled, been pulverized, or otherwise deteriorated, so the asbestos is no longer likely to be bound with its matrix.

73. Lockdown - Applying encapsulant, after a final visual inspection, on all abated surfaces after ACM removal but before the removal of critical barriers. 70. National Emission Standards for Hazardous Air Pollutants (NESHAP) - EPA's rule to control emissions of asbestos to the environment.

74. NPS industrial hygienist (IHC) – The NPS retained Industrial Hygiene Consultant.

75. Negative initial exposure assessment - A demonstration by the employer that complies with the criteria in 29 CFR 1926.1101 (f) (2) (iii), that employee exposure during an operation is expected to be consistently below the PEL.

76. Negative pressure - Air pressure, which is lower than the surrounding area, is created by exhausting air from a sealed, regulated area through HEPA-equipped filtration units. OSHA requires maintaining a 5.0 Pa (-0.02") water gauge inside the negative pressure enclosure.

77. Negative pressure respirator - A respirator in which the air pressure inside the facepiece is negative during inhalation, relative to the air outside the respirator.

78. Non-friable ACM - Material that contains more than 1 percent asbestos but cannot be crumbled, pulverized, or reduced to powder by hand pressure.

79. Organic vapor cartridge - The type of cartridge used on air purifying respirators for organic vapor exposures.

80. OSHA – Occupational Safety and Health Administration

81. Outside air - The air outside buildings and structures, including, but not limited to, the air under a bridge or in an open ferry dock.

82. Owner/operator - Any person who owns, leases, operates, controls, or supervises the facility being demolished or renovated, or any person who owns, leases, operates, controls, or supervises the demolition or renovation operation, or both.

83. Penetrating encapsulant - Encapsulant that is absorbed into the ACM matrix without leaving a surface layer.

84. Permissible exposure limit (PEL) - The level of exposure OSHA allows for as an eight (8) hour time-weighted average (TWA). For asbestos fibers, the eight (8) hour time-weighted average PEL is 0.1 fibers per cubic centimeter (0.1 f/cc) of air, and the 30-minute Excursion Limit (EL) is 1.0 fibers per cubic centimeter (1 f/cc).

85. Personal protective equipment (PPE) - equipment designed to protect users from injury and/or specific job hazards. Such equipment may include protective clothing, hard hats, safety glasses, and respirators.

86. Personal sampling/monitoring - Representative air samples obtained in the breathing zone for one or more workers within the regulated area using a filter cassette and a calibrated air sampling pump to determine asbestos exposure.

87. Polarized light microscopy (PLM) - Light microscopy uses dispersion staining techniques and refractive indices to identify and quantify the type(s) of asbestos present in a bulk sample.

88. Polyethylene sheeting - Durable plastic barrier material 4 to 6-mil thick, semitransparent, sometimes flame-retardant, which follows NFPA 241.

89. Positive/negative fit check - A method of verifying the fit of a respirator by closing off the filters and breathing in or closing off the exhalation valve and breathing out while detecting leakage of the respirator.

90. Presumed ACM (PACM) - Thermal system insulation, surfacing, and flooring material installed in buildings before 1981. If the building owner has actual knowledge or should have known through the exercise of due diligence that other materials are ACM, they too must be treated as PACM. The designation of PACM may be rebutted pursuant to 29 CFR 1926.1101 (k) (5).

91. Professional IH - An IH who meets the definition requirements of AIHA; meets the definition requirements of OSHA as a "Competent Person" at 29 CFR 1926.1101 (b); Must have AHERA type training for supervisor; has completed two specialized EPA-approved courses on management and supervision of asbestos abatement projects; has formal training in respiratory protection and waste disposal; and has a minimum of four projects of similar complexity with this Project of which at least three projects serving as the supervisory IH.

92. Project designer - A person who has completed the training requirements for an asbestos abatement project designer as required by 40 CFR 763 Appendix C, Part I: (B)(5).

93. Protection factor - A value assigned by OSHA/NIOSH to indicate the assigned protection a respirator should provide if worn properly. The number indicates the reduction of exposure level from outside to inside the respirator.

94. Qualitative fit test (QLFT) - A fit test using a challenging material that can be sensed by the wearer if a leakage in the respirator occurs.

95. Quantitative fit test (QNFT) - A fit test using a challenging material that is quantified outside and inside the respirator, thus allowing the determination of the actual fit factor.

96. Regulated area - An area established by the employer to demarcate where Class I, II, III asbestos work is conducted, and any adjoining area where debris and waste from such asbestos work may accumulate; and a work area within which airborne concentrations of asbestos exceed, or there is a reasonable possibility they may exceed, the PEL.

97. Regulated ACM (RACM) - Friable ACM; Category I nonfriable ACM that has become friable; Category I nonfriable ACM that will be or has been subjected to sanding, grinding, cutting, or abrading or; Category II nonfriable ACM that has a high probability of becoming or has become crumbled, pulverized, or reduced to powder by the forces expected to act on the material in the course of the demolition or renovation operation.

98. Removal - All operations where ACM, PACM, and/or RACM are taken out or stripped from structures or substrates, including demolition operations.

99. Renovation - Altering a facility or one or more facility components in any way, including the stripping or removal of asbestos from a facility component that does not involve demolition activity.

100. Repair - Overhauling, rebuilding, reconstructing, or reconditioning of structures or substrates, including encapsulation or other repairs of ACM or PACM attached to structures or substrates.

101. Shower room - The portion of the personal decontamination facility (PDF) where personnel shower before leaving the regulated area. Also used for bag/drum decontamination in the Waste/Equipment Decontamination Facility (W/EDF).

102. Standard operating procedures (SOPs) - Asbestos work procedures are required to be submitted by the Asbestos Abatement Contractor before work begins.

103. Supplied air respirator (SAR) - A respirator that utilizes an air supply separate from the air in the regulated area.

104. Surfacing ACM - A material containing more than 1 percent asbestos that is sprayed, troweled on, or otherwise applied to surfaces for acoustical, fireproofing, and other purposes.

105. Surfactant - A chemical added to water to decrease water's surface tension.

106. Thermal system ACM - A material containing more than 1 percent asbestos applied to pipes, fittings, boilers, breeching, tanks, ducts, or other structural components to prevent heat loss or gain.

107. Transmission electron microscopy (TEM) - A microscopy method that can identify and count asbestos fibers.

108. USDOT - United States Department of Transportation.

109. Visible emissions - Any emissions that are visually detectable without the aid of instruments come from ACM/PACM/RACM or ACM waste material.

110. Waste/Equipment decontamination facility (W/EDF) - The area in which equipment is decontaminated before removal from the regulated area.

111. Waste generator - Any owner or operator whose act or process produces asbestos-containing waste material.

112. Waste shipment record - The shipping document required to be originated and signed by the waste generator, is used to track and substantiate the disposition of asbestos-containing waste material.

113. Wet cleaning - The process of thoroughly eliminating, by wet methods, any asbestos contamination from surfaces or objects.

C. Referenced Standards Organizations

1. The following acronyms or abbreviations, as referenced in contract/specification documents, are defined to mean the associated names. Names and addresses may be subject to change.

a. ABIH American Board of Industrial Hygiene
6015 West St. Joseph, Suite 102
Lansing, MI 48917-3980
Telephone: 517-321-2638

b. AIHA American Industrial Hygiene Association
2700 Prosperity Avenue, Suite 250
Fairfax, VA 22031
Telephone: 703-849-8888

c. ANSI American National Standards Institute
1430 Broadway Ave.
New York, NY 10018
Telephone: 212-354-3300

d. ASTM American Society for Testing and Materials
1916 Race St.
Philadelphia, PA 19103
Telephone: 215-299-5400

e. CFR Code of Federal Regulations
Government Printing Office
Washington, DC 20420

f. CGA Compressed Gas Association
1235 Jefferson Davis Highway
Arlington, VA 22202
Telephone: 703-979-0900

g. CS Commercial Standard of the National Institute of Standards and Technology (NIST)
U. S. Department of Commerce
Government Printing Office
Washington, DC 20420

h. EPA Environmental Protection Agency
401 M St., SW
Washington, DC 20460
Telephone: 202-382-3949

i. MSHA Mine Safety and Health Administration
Respiratory Protection Division
Ballston Tower #3
Department of Labor
Arlington, VA 22203
Telephone: 703-235-1452

j. NIST National Institute for Standards and Technology
U. S. Department of Commerce
Gaithersburg, MD 20234
Telephone: 301-921-1000

k. NEC National Electrical Code (by NFPA)

l. NEMA National Electrical Manufacturer's Association
2101 L Street, N.W.
Washington, DC 20037

m. NFPA National Fire Protection Association
1 Batterymarch Park
P.O. Box 9101
Quincy, MA 02269-9101
Telephone: 1-800-344-3555

n. NIOSH National Institute for Occupational Safety and Health
4676 Columbia Parkway
Cincinnati, OH 45226
Telephone: 513-533-8236

o. OSHA Occupational Safety and Health Administration
U.S. Department of Labor
Government Printing Office
Washington, DC 20402

p. UL Underwriters Laboratory
333 Pfingsten Rd.
Northbrook, IL 60062
Telephone: 312-272-8800

1.4 APPLICABLE CODES AND REGULATIONS

A. General Applicability of Codes, Regulations, and Standards

1. All work under this contract shall be done in strict accordance with all applicable Federal, State, and local regulations, standards, and codes governing asbestos abatement, and any other trade work done in conjunction with the abatement. All applicable codes, regulations, and standards are adopted into this specification and will have the same force and effect as this specification.

2. The most recent edition of any relevant regulation, standard, document, or code shall be in effect. Where conflict among the requirements or with these specifications exists, the most stringent requirement(s) shall be utilized.

3. Copies of all standards, regulations, codes, and other applicable documents, including this specification and those listed in Section 1.4, shall be available at the worksite. These standards, regulations, codes, and other applicable documents, including this specification and those listed in Section 1.4, may be made available electronically.

B. Asbestos Abatement Contractor Responsibility

1. The Asbestos Abatement Contractor shall assume full responsibility and liability for compliance with all applicable Federal, State, and Local regulations related to all aspects of the abatement project. The Asbestos Abatement Contractor is responsible for providing and maintaining training, accreditations, medical exams, medical records, personal protective equipment (PPE), including respiratory protection, including respirator fit testing, as required by applicable Federal, State, and Local regulations. The Asbestos Abatement Contractor shall hold the Government, NPS, Architect, and IHC harmless for any failure to comply with any applicable work, packaging, transporting, disposal, safety, health, or environmental requirement on the part of himself, his employees, or his subcontractors. The Asbestos Abatement Contractor will incur all IHC costs, including sampling/analytical costs, to ensure compliance with OSHA/EPA/State requirements related to failure to comply with the regulations applicable to the work.

C. Federal Requirements

1. Federal requirements that govern some aspects of asbestos abatement include, but are not limited to, the following regulations.

a. OSHA

i. Title 29 CFR 1910.134 - Respiratory Protection

ii. 29 CFR 1926 - Construction Industry

iii. Title 29 CFR 1926.33 - Access to Employee Exposure and Medical Records

iv. Title 29 CFR 1926.59 same as 1910.1200 - Hazard Communication

v. Title 29 CFR 1926.500 - Guardrails, Handrails, and Covers

vi. Title 29 CFR 1926.1101 - Construction Standard for Asbestos

vii. Title 29 CFR 1926 Subpart C – General Safety and Health Provisions and Subpart D – Occupational Health and Environmental Controls

viii. Title 29 CFR 1926 Subpart E - Personal Protective Equipment and Life Saving Equipment

b. EPA

i. 40 CFR 61 Subpart A and M (Revised Subpart B) - National Emission Standard for Hazardous Air Pollutants - Asbestos

ii. 40 CFR 763.80 - Asbestos Hazard Emergency Response Act (AHERA)

c. USDOT

i. Title 49 CFR 100 - 185 - Transportation

D. State Requirements

1. State requirements that apply to the asbestos abatement work, disposal, clearance, etc., include, but are not limited to, the following:

a. Wyoming Department of Environmental Quality-Air Quality Division
200 West 17th Street
Cheyenne, Wyoming 82002
Phone (307) 777-7394

E. Standards

1. Standards that govern asbestos abatement activities include, but are not limited to, the following:

   a. American National Standards Institute (ANSI)/ASSP Z9.2-2018 - Fundamentals Governing the Design and Operation of Local Exhaust Systems and ANSI/ASSE Z88.2-2015 - Practices for Respiratory Protection.

   b. ANSI Z41.1 - Safety Toe Footwear.

   c. ANSI Z87.1 - Practice for Occupational and Educational Eye and Face Protection

   d. ANSI X88.6 - Respiratory Protection - Respiratory Use Physical Qualifications for Personnel

   e. ANSI Z89.1 - Requirements for Industrial Head Protection

   f. Underwriters Laboratories (UL) 586-2009 - UL Standard for Safety of HEPA Filter Units, 9th Edition; ANSI Approval 2017-12-19.

2. Standards that govern encapsulation work include, but are not limited to, the following:

   a. American Society for Testing and Materials International (ASTM)

3. Standards governing testing laboratories:

   a. American Industrial Hygiene Association (AIHA)

   b. National Institute of Standards and Technology (NIST)

4. Standards that govern the fire and safety concerns in abatement work include, but are not limited to, the following:

   a. National Fire Protection Association (NFPA) 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations.

   b. National Fire Protection Association (NFPA) 10 - Standard for Fire Extinguishers.

   c. NFPA 70 - Standard for National Electric Code.

   d. NFPA 101 - Life Safety Code

   e. NFPA 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations.

   f. NFPA 701 - Standard Methods for Fire Tests for Flame-Resistant Textiles and Film.

   g. Uniform Building Code (UBC) - 2006 Edition

F. EPA Guidance Documents

1. EPA guidance documents that discuss asbestos abatement work activities are listed below. These documents are made part of this section by reference. EPA publications can be ordered from (800) 424-9065.

2. Guidance for Controlling ACM in Buildings (Purple Book) EPA 560/5-85-024

3. Asbestos Waste Management Guidance EPA 530-SW-85-007

4. A Guide to Respiratory Protection for the Asbestos Abatement Industry EPA-560-OPTS-86-001

5. Guide to Managing Asbestos in Place (Green Book) TS 799 20T July 1990

G. Notices

1. State and Local agencies: Send written notification as required by State and Local regulations, including the local fire department, before beginning any work on ACM.

2. Copies of notifications shall be submitted to the NPS for the facility's records in the same time frame as notification is given to the Wyoming Department of Environmental Quality.

H. Permits/Licenses/Fees

1. The Asbestos Abatement Contractor shall apply for and have all required permits and licenses to perform asbestos abatement work, packaging, and storing, and provide timely notification of such actions as may be required by Federal, State, and Local regulations. Written notification must be submitted of the planned asbestos abatement operation as required by US EPA National Emission Standards for Hazardous Air Pollutants (NESHAP) Asbestos Regulations (40 CFR 61, Subpart M) to the regional Asbestos NESHAP Contact. Written notification shall be submitted to:

   a. Wyoming Department of Environmental Quality-Air Quality Division
   200 West 17th Street
   Cheyenne, Wyoming 82002
   Phone (307) 777-7394

2. The Asbestos Abatement Contractor shall be responsible for all applicable fees associated with permits and licenses.

3. The Asbestos Abatement Contractor shall be responsible for all applicable fees associated with patent(s).

I. Posting and Filing of Regulations

1. Maintain two (2) copies of applicable Federal, State, and Local regulations. Post one copy of each at the regulated area where workers will have daily access to the regulations, and keep another copy in the Asbestos Abatement Contractor's office.

J. NPS Responsibilities

1. Before the commencement of work:

a. Contracting Officer will notify others of project dates and requirements for relocation if needed. **Note: Notification of adjacent personnel is required by OSHA in 29 CFR 1926.1101 (k) to prevent unnecessary or unauthorized access to the regulated area.**

b. Submit to the Asbestos Abatement Contractor the results of background air sampling, including the location of samples, the person who collected the samples, the equipment utilized, and the method of analysis. During abatement, submit to the Asbestos Abatement Contractor the results of bulk material analysis and air sampling data collected during the abatement. This information shall not release the Asbestos Abatement Contractor from any responsibility for OSHA compliance.

1.5 EMERGENCY ACTION PLAN AND ARRANGEMENTS

A. An Emergency Action Plan shall be developed before commencing abatement activities and shall be agreed to by the Asbestos Abatement Contractor and the NPS. The Plan shall meet the requirements of 29 CFR 1910.38 Emergency Action Plans.

B. Emergency procedures shall be in written form and prominently posted in the clean room and equipment room of the decontamination unit. Everyone must read and sign these procedures before entering the regulated area to acknowledge understanding of the regulated area layout, the location of emergency exits, and emergency procedures.

C. Emergency planning shall include written notification of police, fire, and emergency medical personnel of planned abatement activities; work schedule, and layout of regulated area, particularly barriers that may affect response capabilities.

D. Emergency planning shall include consideration of fire, explosion, hazardous atmospheres, electrical hazards, slips/trips and falls, confined spaces, and heat stress illness. Written procedures for response to emergencies shall be developed, and employee training in procedures shall be provided.

E. Employees shall be trained in regulated area/site evacuation procedures in the event of workplace emergencies.

1. For non-life-threatening situations - employees injured or otherwise incapacitated shall decontaminate following normal procedures with assistance from fellow workers, if necessary, before exiting the regulated area to obtain proper medical treatment.

2. For life-threatening injury or illness, worker decontamination shall take the least priority after measures to stabilize the injured workers, remove them from the regulated area, and secure proper medical treatment.

F. Telephone numbers of all emergency response personnel shall be kept at the job site along with the location of the nearest telephone.

G. The Asbestos Abatement Contractor shall verify first aid/CPR training for personnel responsible for providing the training. OSHA requires medical assistance within 3-4 minutes of a life-threatening injury/illness. Bloodborne pathogen training shall also be verified for those personnel required to provide first aid/CPR.

H. The Emergency Action Plan shall provide a Contingency Plan if an incident occurs that may require the modification of the SOPs during abatement. Such incidents include, but are not limited to, fire, accident, power failure, and negative pressure failure. The Asbestos Abatement Contractor shall detail procedures to be followed in the event of an incident, assuring that work is stopped, and wetting is continued until correction of the problem.

1.6 PRE-CONSTRUCTION MEETING

A. Before commencing the work, the Asbestos Abatement Contractor shall meet with the Contracting Officer to present and review, as appropriate, the items following this paragraph. The Asbestos Abatement Contractor's Competent Person(s) who will be on-site shall participate in the pre-construction meeting. The pre-construction meeting is to discuss and determine procedures to be used during the project. At this meeting, the Asbestos Abatement Contractor shall provide four copies of:

1. Proof of Asbestos Abatement Contractor licensing.

2. Proof that the Competent Person is trained and accredited for working in this State. Verification of the experience of the Competent Person and proof of certification shall also be presented.

3. A list of all workers who will participate in the project, including experience and verification of training and accreditation.

4. A list of and verification of training for all personnel who have current First Aid/ CPR training. A minimum of one person per shift must have adequate training.

5. Current medical written opinions for all personnel working on-site meeting the requirements of 29 CFR 1926.1101 (m).

6. Current fit tests for all personnel wearing respirators on-site meet the requirements of 29 CFR 1926.1101 (h) and Appendix C.

7. A copy of the Asbestos Abatement Contractor's SOPs for Asbestos Abatement. In these procedures, the following information must be detailed, specifically for this project.

a. Regulated area preparation procedures.

b. Notification requirements procedure of Asbestos Abatement Contractor as required in 29 CFR 1926.1101 (d).

c. Abatement methods/procedures and equipment to be used.

d. PPE to be used.

e. Decontamination area set-up/layout and decontamination procedures for employees; extent of 6-mil polyethylene sheeting placed 25 feet out from the outer perimeter of the work area to aid in cleaning roofing and widow material debris that may fall off building; method of demarcating the regulated work area with Asbestos Danger Tape and Asbestos Danger Signs; method for fall protection; method for accessing roof and windows; and method for bringing material to lined roll-off or approved equivalent.

8. At this meeting, the Asbestos Abatement Contractor shall provide all submittals as required.

9. Procedures for handling, packaging, storage, and disposal of asbestos waste.

10. Emergency Action Plan and Contingency Procedures.

1.7 PROJECT COORDINATION

The following are the minimum administrative and supervisory personnel necessary for the coordination of the work.

A. Personnel

1. Administrative and supervisory personnel shall consist of a qualified Competent Person as defined by OSHA in the Construction Standards and the Asbestos Construction Standard; Asbestos Abatement Contractor, AHERA-trained, and certified personnel. These employees are the Asbestos Abatement Contractor's Representatives responsible for compliance with these specifications and all other applicable requirements.

2. Non-supervisory personnel shall consist of an adequate number of qualified personnel to meet the schedule requirements of the project. Personnel shall meet the required qualifications. Personnel utilized on-site shall be pre-approved by the Contracting Officer. A request for approval shall be submitted for any person to be employed during the project, giving the person's name, qualifications, Certificate of Worker's Acknowledgment, Affidavit of Medical Surveillance and Respiratory Protection, and current Respirator Fit Test.

3. Minimum qualifications for the Asbestos Abatement Contractor and assigned personnel are:

   a. The Asbestos Abatement Contractor has conducted within the last three (3) years, three (3) projects of similar complexity and dollar value as this project; has adequate liability/occurrence insurance for asbestos work as required by the State; is licensed in applicable State; has adequate and qualified personnel available to complete the work; has comprehensive SOPs for asbestos work; and has adequate materials, equipment, and supplies to perform the work.

   b. The Competent Person has four (4) years of abatement experience, of which two (2) years were as the Competent Person on the project; meets the OSHA definition of a Competent Person; has been the Competent Person on two projects of similar size and complexity as this project; has completed EPA AHERA/OSHA/State/Local training requirements/accreditation(s) and refreshers; and has all required OSHA documentation related to medical and respiratory protection.

   c. The Abatement Personnel shall have completed the EPA AHERA/OSHA/State abatement worker course; have training on the Asbestos Abatement Contractor SOPs; have one (1) year of asbestos abatement experience within the past three (3) years of similar size and complexity; has applicable medical and respiratory protection documentation; has a certificate of training and a State accreditation/license.

4. All personnel are to comply with OSHA construction safety training as applicable and submit certification documents.

1.8 RESPIRATORY PROTECTION

A. General – Respiratory Protection Program

1. The Asbestos Abatement Contractor shall develop and implement a Respiratory Protection Program (RPP) that complies with OSHA requirements found at 29 CFR 1926.1101 and 29 CFR 1910.132;134. ANSI Standard Z88.2-2015 provides excellent guidance for developing a respiratory protection program. All respirators used must be NIOSH-approved for asbestos abatement activities. The written respiratory protection shall, at a minimum, contain the basic requirements found at 29 CFR 1910.134 (c) - Respiratory Protection Program.

B. Respiratory Protection Program Coordinator

1. The Respiratory Protection Program Coordinator (RPPC) must be identified and shall have two (2) years of experience coordinating RPP of a similar size and complexity. The RPPC must submit a signed statement attesting to the fact that the program meets the above requirements.

C. Selection and Use of Respirators

1. The procedure for the selection and use of respirators must be submitted to the NPS as part of the Asbestos Abatement Contractor's qualification. The procedure must be written clearly enough for workers to understand. A copy of the Respiratory Protection Program must be available in the clean room of the decontamination unit/area or in the onsite Asbestos Abatement Contractor’s office, for reference by employees or authorized visitors.

D. Minimum Respiratory Protection

1. Minimum Respiratory Protection: Shall be a ½-mask negative pressure air purifying respirator equipped with P100 filters, provided personal air samples in the workplace remain at or below 0.1 f/cc, determined as an 8-hour TWA. Full face powered air purifying respirator equipped with P100 filters shall be required until the Asbestos Abatement Contractor demonstrates that personal air samples are at or below 0.1 f/cc, determined as an 8-hour TWA. A higher level of respiratory protection shall be required if fiber levels exceed 1 f/cc as an 8-hour TWA, inside the regulated work area. Respirator selection shall meet the requirements of 29 CFR 1926.1101 (h) and 29 CFR 1910.134 (d)(3)(i)(A) Table 1, except as indicated in this paragraph. Abatement personnel must have a respirator for their exclusive use.

E. Medical Written Opinion

1. No employee shall be allowed to wear a respirator unless a physician or other licensed health care professional has provided a written determination stating they are medically qualified to wear the class of respirator to be used on the project while wearing whole body impermeable garments and subjected to heat or cold stress.

F. Respirator Fit Test

1. All personnel wearing respirators shall have a current qualitative/quantitative fit test conducted in accordance with 29 CFR 1910.134 (f) and Appendix A. Quantitative fit tests shall be done for PAPRs that have been put into a motor/blower failure mode.

G. Respirator Fit Check

1. The Competent Person shall ensure that the positive/negative pressure user seal check is done each time the respirator is donned by an employee. Head coverings must cover respirator head straps. Any situation that prevents an effective facepiece-to-face seal, as evidenced by the failure of a user seal check, shall preclude that person from entering the regulated area until the resolution of the problem.

H. Maintenance and Care of Respirators

1. The Respiratory Protection Program Coordinator shall submit evidence and documentation showing compliance with 29 CFR 1910.134 (h) Maintenance and Care of Respirators.

I. Supplied Air Systems: If a supplied air system is used, the system shall meet all requirements of 29 CFR 1910.134 and the ANSI/Compressed Gas Association (CGA) Commodity Specification for Air current requirements for Type 1 - Grade D breathing air. Low pressure systems are not allowed to be used on asbestos abatement projects. Supplied Air respirator use shall be in accordance with EPA/NIOSH publication EPA-560-OPTS-86-001 "A Guide to Respiratory Protection for the Asbestos Abatement Industry". The competent person on site will be responsible for the supplied air system to ensure the safety of the worker.

1.9 WORKER PROTECTION

A. Training of Removal Personnel

1. Before beginning any abatement activity, all personnel shall be trained in accordance with OSHA 29 CFR 1926.1101 (k)(9) and any additional State of Wyoming requirements. Training must include, at a minimum, the elements listed in 29 CFR 1926.1101 (k)(9)(viii). Training shall have been conducted by a third-party, EPA/State-approved trainer meeting the requirements of EPA 40 CFR 763 Appendix C (AHERA MAP). Initial training certificates, current refresher, and accreditation proof must be submitted for each person working at the site.

B. Medical Examinations

1. Medical examinations meeting the requirements of 29 CFR 1926.1101 (m) shall be provided for all personnel working in the regulated area, regardless of exposure levels. The physician's written opinion as required by 29 CFR 1926.1101 (m)(4) shall be provided for each person and shall include a statement indicating that the person has been evaluated for working in a heat and cold stress environment while wearing PPE and can perform the work without risk of material health impairment.

C. Personal Protective Clothing

1. Provide whole body clothing, head coverings, foot coverings, and any other personal protective equipment as determined by conducting the hazard assessment required by OSHA at 29 CFR 1910.132 (d). Provide all personnel entering the regulated area with disposable full-body coveralls, disposable head coverings, and 18-inch boot coverings. The Competent Person shall ensure the integrity of personal protective equipment worn for the duration of the project. Duct tape shall be used to secure all suit sleeves to wrists and to secure foot coverings at the ankle.

D. Regulated Area Entry Procedure

1. Worker protection shall meet the most stringent requirement. The Competent Person shall ensure that each time workers enter the regulated area, they remove ALL street clothes in the clean room of the decontamination unit and put on new disposable coveralls, head coverings, a clean respirator, and then proceed through the shower room to the equipment room where they will put on non-disposable required PPE.

E. Decontamination Procedure – Powered Air Purifying Respirator (PAPR)

1. The Competent Person shall require all personnel to adhere to the following decontamination procedures whenever they leave the regulated area.

    a. When exiting the regulated area, remove disposable coveralls and ALL other clothes, including disposable head coverings, and foot coverings or boots in the equipment room.

    b. Still wearing the respirator and without street clothing, proceed to the shower. Showering is MANDATORY. Care must be taken to follow reasonable procedures in removing the respirator to avoid asbestos fibers while showering. The following procedure is required as a minimum:

        i. Thoroughly wet body, including hair and face. If using a PAPR, hold the blower above your head to keep the filters dry.

        ii. With respirator still in place, thoroughly decontaminate body, hair, respirator face piece, and all other parts of the respirator except the blower and battery pack on a PAPR. Pay particular attention to cleaning the seal between the face and the respirator face piece and under the respirator straps.

        iii. Take a deep breath, hold it, and/or exhale slowly, completely wetting hair, face, and respirator. While still holding breath, remove the respirator and hold it away from the face before starting to breathe.

        iv. Carefully decontaminate the face piece of the respirator inside and out. If using a PAPR, shut down using the following sequence: a) first cap inlets to filters; b) turn blower off to keep debris collected on the inlet side of the filter from dislodging and contaminating the outside of the unit; c) thoroughly decontaminate blower and hoses; d) carefully decontaminate battery pack with a wet rag being cautious of getting water in the battery pack thus preventing destruction. (Note: THIS PROCEDURE IS NOT A SUBSTITUTE FOR RESPIRATOR CLEANING!).

        v. Shower and wash body completely with soap and water. Rinse thoroughly.

        vi. Rinse shower room walls and floor to drain before exiting.

        vii. Proceed from the shower to the clean room; dry off and change into street clothes or into new disposable work clothing.

    c. Where containment is not required for ACM removal, an adequate washing station will be provided for the employees for the removal of disposable PPE and to clean the respirator.

F. Decontamination Procedure - Air Purifying, Negative Pressure Respirator

1. The Competent Person shall require all personnel to use the following decontamination procedures, as a minimum, whenever leaving the regulated area with a full face, HEPA filtered respirator:

    a. When exiting the regulated area, remove disposable coveralls and ALL other clothes, disposable head coverings, and disposable foot coverings or boots in the equipment room.

    b. While still wearing the respirator and free of street clothing, proceed to the shower, which is mandatory. Care must be taken to follow reasonable procedures in removing the respirator and filters to avoid asbestos fibers while showering. The following procedure is required, as a minimum:

        i. Thoroughly wet body from neck down. Wet hair as thoroughly as possible without wetting the respirator filters.

        ii. Take a deep breath, hold it, and/or exhale slowly, completely wetting hair, thoroughly wetting face, respirator, and filter(s). While still holding your breath, remove the respirator and hold it away from your face before starting to breathe.

        iii. Dispose of wetted or overloaded filters from the respirator.

        iv. Carefully decontaminate the respirator face piece and respirator inside and out. (NOTE: THIS IS NOT A SUBSTITUTE FOR RESPIRATOR CLEANING!).

        v. Shower and wash body completely with soap and water. Rinse thoroughly.

        vi. Rinse shower room walls and floor to drain before exiting.

        vii. Proceed from the shower to the clean room; dry off and change into street clothes or into new disposable work clothing.

G. Regulated Area Requirements

1. The Competent Person shall meet all requirements of 29 CFR 1926.1101 (o) and ensure that all requirements for OSHA Class I and II regulated areas at 29 CFR 1926.1101 (e) and this specification are met, applicable to OSHA Class I and II work. All personnel in the regulated area shall not be allowed to eat, drink, smoke, chew tobacco or gum, apply cosmetics, or in any way interfere with the fit of their respirator.

1.10 DECONTAMINATION FACILITIES

A. Description

1. Provide each regulated area with a separate PDF and W/EDF. Ensure that the PDF is the only means of ingress and egress to the regulated area and that all equipment, bagged waste, and other material exit the regulated area only through the W/EDF. See OSHA 29 CFR 1926.1101, Appendix F for minimum requirements of each and OSHA 29 CFR 1926.1101, Appendix F.

B. General Requirements

1. All personnel entering or existing in a regulated area shall follow the requirements of 29 CFR 1926.1101 (j) (1) and these specifications. All equipment and materials must exit the regulated area through the W/EDF and be decontaminated in accordance with these specifications. Walls and ceilings of the PDF and W/EDF must be constructed of a minimum of two layers of 6-mil clear, opaque, black or white fire-retardant polyethylene sheeting and be securely attached to existing building components and/or an adequate temporary framework. A minimum of two layers of 6-mil polyethylene shall also be used to cover the floor under the PDF and W/EDF units. Construct doors so that they overlap and secure to adjacent surfaces. Weigh sheets with layers of duct tape so that they close quickly after release. Put arrows on sheets so they show the direction of travel and overlap. If the building adjacent area is occupied, construct a solid barrier on the occupied side(s) to protect the sheeting and reduce potential for non-authorized personnel entering the regulated area.

C. Temporary Facilities to the PDF and W/EDF

1. The Asbestos Abatement Contractor shall provide temporary water service connections to the PDF and W/EDF. Backflow prevention must be provided at the point of connection to the system. Water supply must be under adequate pressure and meet the requirements of 29 CFR 1910.141(d)(3). Provide adequate temporary electric power with ground fault protection and overhead wiring in the PDF and W/EDF. Provide a sub-panel for all temporary power in the clean room. Provide adequate temporary lighting to provide a minimum of 50-foot candles in the PDF and W/EDF. Provide temporary heat, if needed, to maintain 70°F throughout the PDF and W/EDF.

D. Personnel Decontamination Facility (PDF)

1. The Competent Person shall provide a PDF consisting of a shower room which is contiguous to a clean room and equipment room. The PDF must be sized to accommodate the number of personnel scheduled for the Project. The shower room, located in the center of the PDF, shall be fitted with as many portable showers as necessary to ensure all employees can complete the entire decontamination procedure within 15 minutes. The PDF shall be constructed of opaque polyethylene for privacy. The PDF shall be constructed to eliminate any parallel routes of egress without showering.

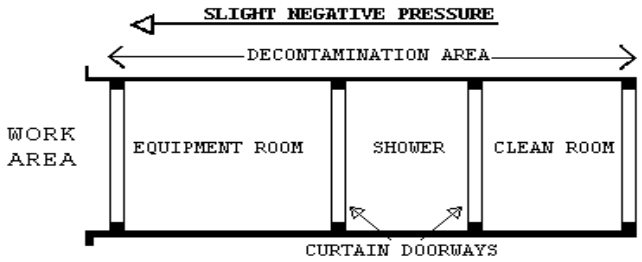
a. Clean Room: The clean room must be physically and visually separated from the rest of the building to protect the privacy of personnel changing clothes. The clean room shall be constructed of at least two layers of 6-mil fire-retardant polyethylene to provide an airtight room. Provide a minimum of 2 - 900 mm (3-foot) wide-flapped doorways. One doorway shall be the entry from outside the PDF, and the second doorway shall be to the shower room of the PDF. The floor of the clean room shall be maintained in a clean, dry condition. An adequate supply of disposable towels and disposable protective clothing must be provided. Provide 1 storage locker or equivalent per person. A portable fire extinguisher, Type ABC, shall be provided in accordance with OSHA and NFPA Standard 10. Any person entering the regulated area shall remove all street clothing in the clean room and dress in disposable protective clothing and respiratory protection. Any person entering the clean room does so either from the outside with street clothing on or is coming from the shower room completely naked and thoroughly washed. Male/Females required to enter the regulated area shall be ensured of their privacy throughout the entry/exit process by posting guards at both entry points to the PDF so that no male/female can enter or exit the PDF during his/her stay in the PDF.

b. Shower Room: The Competent Person shall ensure that the shower room is a completely watertight compartment to be used for the movement of all personnel from the clean room to the equipment room and for the showering of all personnel going from the regulated area to the clean room. Each shower shall be constructed so that water runs down the walls of the shower and into a drip pan. Install a freely draining smooth floor on top of the shower pan. The shower room shall be separated from the rest of the building and from the clean room and equipment room using airtight walls made from at least two layers of 6-mil fire-retardant polyethylene. The shower shall be equipped with a shower head and controls, hot and cold water, drainage, soap dish, and a continuous supply of soap, and shall be maintained in a sanitary condition throughout its use. The controls shall be arranged so that an individual can shower without assistance. Provide a flexible hose shower head, hose bibs, and all other required items. Wastewater will be pumped to a drain after being filtered through a minimum of a 100-micron sock in the shower drain, a 20-micron filter, and a final 5-micron filter. Filter changes must be made in the shower to prevent loss of contaminated water. Hose down all shower surfaces after each shift and clean any debris from the shower pan. Residue is to be disposed of as asbestos waste.

c. Equipment Room: The Competent Person shall provide an equipment room which shall be an airtight compartment for the storage of work equipment, reusable footwear, and for use as a change station for personnel exiting the regulated area. The equipment room shall be separated from the regulated area by a minimum 3-foot-wide door made of three layers of 6-mil fire-retardant polyethylene. The equipment room shall be separated from the regulated area, the shower room, and the rest of the building by airtight walls and a ceiling constructed of a minimum of two layers of 6-mil fire-retardant polyethylene. If the airborne level of asbestos in

the regulated area is expected to exceed 0.5 f/cc, add an intermediate cleaning space between the equipment room and the regulated area. Damp wipe all surfaces of the equipment room after each shift change. Provide an additional loose layer of 6-mil fire-retardant polyethylene per shift change and remove this layer after each shift. Provide a temporary electrical sub-panel in this room to accommodate any power tools and equipment used in the regulated area.

d. An example of the PDF is below: The clean room is at the entrance, followed by a shower room, followed by an equipment room leading to the regulated area. Each doorway in the PDF is at a minimum of 2 layers of 6-mil fire-retardant polyethylene.



E. Waste/Equipment Decontamination Area (W/EDA)

1. The Competent Person shall provide a W/EDF consisting of a washroom, holding room, and clean room for the removal of equipment and material from the regulated area. Personnel shall not enter or exit the W/EDF except in the event of an emergency. Clean debris and residue in the W/EDF daily. All surfaces in the W/EDF shall be wiped/hosed down after each shift, and all debris shall be cleaned from the shower pan. The W/EDF shall consist of the following:

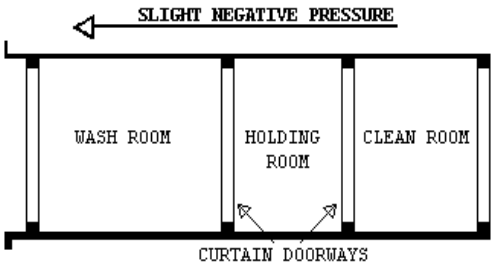
a. Wash Down Station: Provide an enclosed shower unit in the regulated area just outside the washroom as an equipment bag and container cleaning station.

b. Washroom: Provide a washroom for cleaning of bagged or containerized asbestos-containing waste materials passed from the regulated area. Construct the washroom using 50 mm x 100 mm (2" x 4") wood framing and two layers of 6-mil fire-retardant polyethylene. Locate the washroom so that packaged materials, after being wiped clean, can be passed to the holding room. Doorways in the washroom shall be constructed of two layers of 6-mil fire-retardant polyethylene.

c. Holding Room: Provide a holding room as a drop location for bagged materials passed from the washroom. Construct the holding room using 50 x 100 mm (2" x 4") wood framing and three layers of 6-mil fire-retardant polyethylene. The holding room shall be located so that bagged material cannot be passed from the washroom to the clean room unless it goes through the holding room. Doorways in the holding room shall be constructed of two layers of 6-mil fire-retardant polyethylene.

d. Clean Room: Provide a clean room to isolate the holding room from the exterior of the regulated area. Construct a clean room using 50 x 100 mm (2" x 4") wood framing and two layers of 6-mil fire-retardant polyethylene. The clean room shall be located to provide access to the holding room from the building exterior. Doorways to the clean room shall be constructed of two layers of 6-mil fire-retardant polyethylene. When a negative pressure differential system is used, a rigid enclosure separation between the W/EDF clean room and the adjacent areas shall be provided.

e. The W/EDF shall be provided as follows: washroom leading to a holding room, followed by a clean room leading to the outside. See diagram.



F. Waste/Equipment Decontamination Procedures

1. At the washdown station, in the regulated area, thoroughly wet clean contaminated equipment and/or sealed polyethylene bags and pass into the washroom after visual inspection. When passing anything into the washroom, close all doorways of the W/EDF, other than the doorway between the washdown station and the washroom. Keep all outside personnel clear of the W/EDF. Once inside the washroom, wet clean the equipment and/or bags. After cleaning and inspection, pass items into the holding room. Close all doorways except the doorway between the holding room and the clean room. Workers from the clean room/exterior shall enter the holding room and remove the decontaminated/cleaned equipment/bags for removal and disposal by the Asbestos Abatement Contractor. These personnel shall wear full protective clothing and appropriate respirators. At no time shall personnel from the clean side be allowed to enter the washroom.

PART 2 PRODUCTS, MATERIALS, AND EQUIPMENT

2.1 MATERIALS AND EQUIPMENT

A. General Requirements (all asbestos abatement projects)

1. Before the start of work, the Asbestos Abatement Contractor shall provide and maintain enough materials and equipment to ensure continuous and efficient work throughout the duration of the project. Work shall not start unless the following items have been delivered to the site and the Competent Person has submitted verification to the NPS's representative.

2. All materials shall be delivered in their original package, container, or bundle bearing the name of the manufacturer and the brand name (where applicable).

3. Store all materials subject to damage off the ground, away from wet surfaces, and under a cover sufficient to prevent damage or contamination. Flammable materials cannot be stored inside buildings. Replacement materials shall be stored outside of the regulated area until abatement is completed.

4. The Asbestos Abatement Contractor shall not block or hinder the use of the site by staff and visitors in partially occupied buildings by placing materials/equipment in any unauthorized place.

5. The Competent Person shall inspect for damaged, deteriorating, or previously used materials. Such materials shall not be used and shall be removed from the worksite and disposed of properly.

6. Polyethylene sheeting for walls in the regulated area shall be a minimum of 4-mil thickness. For floors and all other uses, sheeting shall be a minimum of 6-mil thickness and shall be used in selected widths to minimize the frequency of joints.

7. Critical penetration points into the building shall be sealed with two (2) layers of 6-mil independently installed plastic sheeting (Polyethylene) secured in place at openings such as ducts, windows, louvers, penetrations, or any other openings into the regulated area. A perimeter drop cloth of one (1) layer of 6-mil plastic sheeting will be placed around the outside perimeter of the roof to capture falling debris and to facilitate clean-up at ground level, if necessary. Fire-retardant poly shall be used throughout.

8. The method of attaching polyethylene sheeting shall be agreed upon in advance by the Asbestos Abatement Contractor and NPS and selected to minimize damage to equipment and surfaces. Method of attachment may include any combination of moisture-resistant duct tape or another waterproof tape, furring strips, spray glue, staples, nails, screws, lumber, and plywood for enclosures or other effective procedures capable of sealing polyethylene to dissimilar finished or unfinished surfaces under both wet and dry conditions. The Asbestos Abatement Contractor shall repair all damage (i.e., fasteners, duct tape damages, etc.) to finishes not scheduled for removal and/or restoration by others.

9. Polyethylene sheeting utilized for the PDF, if used, shall be opaque white or black in color, 6-mil fire-retardant polyethylene.

10. Installation and plumbing hardware, showers, hoses, drain pans, sump pumps, and wastewater filtration systems shall be provided by the Asbestos Abatement Contractor.

11. A minimum of 4 air changes per hour exhausting via HEPA Air Filtration Devices (AFDs) from the regulated area directly to the exterior while maintaining 5.0 Pa (-0.02") water column. Additional AFDs and temporary ductwork may be needed to control air flow. AFDs shall be used as an engineering control to ensure no areas of stagnant air are present in the regulated area and airborne fiber levels in the regulated area are minimized.

12. An adequate number of HEPA vacuums, scrapers, sprayers, nylon brushes, brooms, disposable mops, rags, sponges, staple guns, shovels, ladders, and scaffolding of suitable height and length as well as meeting OSHA requirements, fall protection devices, water hose to reach all areas in the regulated area, airless spray equipment, and any other tools, materials or equipment required to conduct the abatement project. All electrically operated hand tools, equipment, and electric cords shall be connected to GFCI protection.

13. Special protection for objects in the regulated area shall be detailed to prevent damage from scaffolds, water, and falling material.

14. Disposal bags – two (2) layers of 6-mil poly or approved equivalent for asbestos waste shall be pre-printed with labels, markings, and addresses as required by OSHA, EPA, and DOT regulations.

15. Impermeable asbestos disposal drums shall be metal or fiberboard with locking ring tops with required OSHA, EPA, and DOT labels and markings.

16. The NPS shall be provided with a copy of the SDS as required for all hazardous chemicals under OSHA 29 CFR 1910.1200 - Hazard Communication. Methylene chloride shall not be used with any spray adhesive or other product. Appropriate encapsulant(s) shall be provided.

17. OSHA DANGER demarcation signs, as many and as required by OSHA 29 CFR 1926.1101(k)(7), shall be provided and placed by the Competent Person. All other posters and notices required by Federal, State, and Local regulations shall be posted at the outer boundary of the regulated area or approved equivalent.

18. Adequate and appropriate PPE for the project and the number of personnel/shifts shall be provided. All personal protective equipment issued must be based on a written hazard assessment conducted under 29 CFR 1910.132(d).

19. Any damage to building components not scheduled for general renovation, resulting from the Asbestos Abatement Contractor’s work and/or work practices, shall be repaired or replaced at the sole cost of the Asbestos Abatement Contractor utilizing appropriately qualified and insured tradespersons equal to or greater than the original condition.

2.2 MONITORING, INSPECTION, AND TESTING

A. General

1. The Asbestos Abatement Contractor shall perform, throughout abatement work, monitoring of their personnel's exposure, inside the regulated area, in accordance with OSHA requirements and this abatement project specification. The Asbestos Abatement Contractor’s Industrial Hygiene Technician (IHT) or accredited supervisor ("Competent Person") shall personally review conditions inside the regulated area to ensure compliance with EPA and this asbestos abatement project specification. In addition, the IHT shall personally manage air sample collection, analysis, and evaluation for personnel samples and area samples to satisfy OSHA requirements. Additional inspection and testing requirements are specified in other parts of this section.

2. The IHT shall be responsible for managing all personnel monitoring, the inspection and testing required by this asbestos abatement project specification, the OSHA Regulation 29 CFR 1926.1101, and for continuous monitoring of all sub-systems and procedures affecting the safety of the Asbestos Abatement Contractor's employees. The safety of the Asbestos Abatement Contractor's employees and providing safe conditions inside the regulated area for all people entering is the exclusive responsibility of the Asbestos Abatement Contractor. The analytical laboratory that shall be used by the Asbestos Abatement Contractor to analyze the samples shall be AIHA-accredited. The IHT shall keep a daily log of personnel and area samples taken and analyzed, and make such a log available to the Contracting Officer. The log shall contain information on the person's breathing zone sampled, activities being performed, the date of the sample collection, the time of the sample start to finish, flow rate, volume, and fibers/cc. Collect and analyze personnel samples for twenty-five percent of the workforce each day throughout the duration of the project.

3. NPS will employ an independent industrial hygiene consultant (IHC) to perform assorted services on behalf of the Asbestos Abatement Contractor. The IHC will perform the necessary monitoring, inspection, testing, and other support services to ensure that the NPS employees and visitors will not be adversely affected by the abatement work and that the abatement work proceeds in accordance with these specifications, and that the abated areas or abated buildings have been successfully decontaminated. The work of the IHC in no way relieves the Asbestos Abatement Contractor from their responsibility to perform the work in accordance with contract/specification requirements, to perform continuous inspection, monitoring, and testing for the safety of their employees, and to perform other such services as specified. The cost of the IHC and its services will be borne by NPS, excluding any repeat of final inspection and testing that may be required due to unsatisfactory initial results.

4. If fibers counted by the IHC during abatement work, either inside or outside the regulated area, utilizing the NIOSH 7400 air monitoring method, exceed the specified respective limits, the Asbestos Abatement Contractor shall stop work. The Asbestos Abatement Contractor may request confirmation of the results by analysis of the samples by Transmission Electron Microscopy (TEM). The request must be in writing and submitted to the Contracting Officer. The cost for the confirmation of results will be borne by the Asbestos Abatement Contractor for both the collection and analysis of samples and for the time delay that may/does result from this confirmation.

B. Scope of Services of the IHC

1. The purpose of the work of the IHC is to: Assure quality; resolve problems; and prevent the spread of contamination beyond the regulated areas. In addition, their work includes performing the final inspection and testing to determine whether the regulated areas or buildings have been adequately decontaminated. All air monitoring is to be done utilizing PCM/TEM. The IHC may perform the following tasks:

a. Task 1: Establish background levels before abatement begins by collecting background samples. Retain samples for possible PCM/TEM analysis.

b. Task 2: Perform periodic air monitoring, inspection, and testing outside the regulated area during actual abatement work to detect any faults in the regulated area isolation and any adverse impact on the surroundings from regulated area activities.

c. Task 3: Perform unannounced visits to spot-check overall compliance of work with contract/specifications. These visits may include any inspection, monitoring, and testing inside and outside the regulated area, and all aspects of the operation except personnel monitoring.

d. Task 4: Provide support to the Contracting Officer, such as evaluation of submittals from the Asbestos Abatement Contractor, scheduling, resolution of unforeseen developments, etc.

e. Task 5: Perform, in the presence of the Contracting Officer, the final inspection and testing of decontaminated regulated areas at the conclusion of the abatement to certify compliance with the NPS requirements/specifications.

f. Task 6: Issue a Certificate of Decontamination for each regulated area and project report.

2. All documentation, inspection results, and testing results generated by the IHC will be available to the Asbestos Abatement Contractor for information and consideration. The Asbestos Abatement Contractor shall cooperate with and support the IHC for the efficient and smooth performance of their work.

3. The monitoring and inspection results of the IHC will be used by the NPS to issue any stop removal orders to the Asbestos Abatement Contractor during abatement work and to accept or reject a regulated area or building as decontaminated.

2.3 STANDARD OPERATING PROCEDURES

A. The Asbestos Abatement Contractor shall have established SOPs in printed form and a loose-leaf folder consisting of simplified text, diagrams, sketches, and pictures that establish and clearly explain the procedures to be followed during all phases of the work by the Asbestos Abatement Contractor's personnel. The SOPs must be modified as needed to address specific requirements of the project. The SOPs shall be submitted for review and approval before the start of any abatement work. The minimum topics and areas to be covered by the SOPs are:

1. Minimum Personnel Qualifications

2. Contingency Plans and Arrangements

3. Security and Safety Procedures

4. Respiratory Protection/PPE Program and Training

5. Medical Surveillance Program and Recordkeeping

6. Regulated Area Requirements - Establishment and Management of the Regulated Area

7. Decontamination Facilities and Entry/Exit Procedures

8. Negative Pressure Systems Requirements

9. Monitoring, Inspections, and Testing

10. Removal Procedures for ACM and ACE

11. Removal of Contaminated Soil (if applicable)

12. Encapsulation Procedures for ACM

13. Disposal of ACM waste

14. Regulated Area Decontamination/Clean-up

15. Regulated Area Visual and Clearance Standards

16. Project Completion/Closeout

2.4 SUBMITTALS

A. Pre-Construction Meeting Submittals

1. Submit to the NPS a minimum of 14 days before the pre-start meeting the following for review and approval. Meeting this requirement is a prerequisite for the pre-start meeting for this project:

a. Submit an Emergency Action Plan. The Plan shall meet the requirements of 29 CFR 1926, Subpart C, Standard 1926.35 Employee Emergency Action Plans.

b. Submit a detailed work schedule for the entire project reflecting contract documents and the phasing/schedule requirements.

c. Submit a staff organization chart showing all personnel who will be working on the project and their capacity/function. Provide their qualifications, training, accreditations, and licenses, as appropriate. Provide a copy of the "Certificate of Worker's Acknowledgment" and the "Affidavit of Medical Surveillance and Respiratory Protection" for each person.

d. SOPs developed specifically for this project, incorporating the requirements of the specifications, prepared, signed, and dated.

e. Submit the specifics of the materials and equipment to be used for this project, with manufacturer names, model numbers, performance characteristics, pictures/diagrams, and numbers available for the following:

   i. HEPA vacuums, air monitoring pumps, calibration devices, and emergency power-generating systems.

   ii. Wastewater filtration system and containment barriers.

   iii. Encapsulants, surfactants, handheld sprayers, airless sprayers, glove bags, and fire extinguishers.

   iv. Respirators, protective clothing, fall protection, and other required PPE.

   v. Fire safety equipment to be used in the regulated area.

f. Submit the name, location, and phone number of the approved landfill; proof/verification that the landfill is approved for ACM disposal; the landfill's requirements for ACM waste; the type of vehicle to be used for transportation; and the name, address, and phone number of the subcontractor, if used. Proof of asbestos training for transportation personnel shall be provided.

g. Submit required notifications and arrangements made with regulatory agencies having regulatory jurisdiction and the specific contingency/emergency arrangements made with local health, fire, ambulance, and hospital authorities, and any other notifications/arrangements.

h. Submit the name, location, and verification of the laboratory and/or personnel to be used for analysis of air and/or bulk samples. Air monitoring must be done in accordance with OSHA 29 CFR 1926.1101(f) and Appendix A.

i. Submit qualifications verification: Submit the following evidence of qualifications. Make sure that all references are current and verifiable by providing current phone numbers and documentation.

   i. Abatement Company: Project experience within the past three (3) years; listing projects first, most similar to this project: Project Name; Type of Abatement; Duration; Cost; Reference Name/Phone Number; Final Clearance; Completion Date.

   ii. List of project(s) halted by the NPS, A/E, IH, or regulatory agency in the last three (3) years: Project Name; Reason; Date; Reference Name/Number; Resolution.

   iii. List asbestos regulatory citations (e.g., OSHA), notices of violations (e.g., Federal, State, and EPA), penalties, and legal actions taken against the company, including those of the company’s officers (including damages paid) in the last three (3) years. Provide copies and all information needed for verification.

j. Submit personnel information: Provide a resume; address each item completely; provide references; phone numbers; copies of certificates, accreditations, and licenses. Submit an affidavit signed by the Competent Person stating that all personnel submitted below have medical records in accordance with OSHA 29 CFR 1926.1101(m), 29 CFR 1910.134 and that the company has implemented a medical surveillance program and maintains recordkeeping in accordance with the above regulations. Submit the phone number and doctor/clinic/hospital used for medical evaluations.

i. Competent Person(s)/Supervisor(s): Number; names; years of abatement experience as Competent Person/Supervisor; list of similar projects as Competent Person/Supervisor; as a worker; certificates, licenses, accreditations; proof of AHERA/OSHA specialized asbestos training; the maximum number of personnel supervised on a project; medical opinion (asbestos surveillance and respirator use); current respirator fit test.

ii. Workers: Numbers; names; years of abatement experience; certificates, licenses, accreditations; training courses in asbestos abatement and respiratory protection; medical opinion; current respirator fit test.

k. Submit copies of the State license for asbestos abatement; copy of the insurance policy, including exclusions with a letter from the agent stating in English the coverage provided and the fact that asbestos abatement activities are covered by the policy; copy of SOPs incorporating the requirements of this specification; information on who provides your training, how often; who provides medical surveillance, how often; who does and how is air monitoring conducted; a list of references of independent laboratories/IH's familiar with your air monitoring and SOPs; and copies of monitoring results of the five (5) referenced projects listed and analytical method(s) used.

l. Rented equipment must be decontaminated before returning to the rental agency.

m. Submit, before the start of work, the manufacturer's technical data for all types of encapsulants, all SDSs, and application instructions.

B. Submittals During Abatement

1. The Competent Person shall maintain and submit a daily log at the regulated area documenting the dates and times of the following: purpose, attendees, and summary of meetings; all personnel entering/exiting the regulated area; document and discuss the resolution of unusual events such as barrier breaching, equipment failures, emergencies, and any cause for stopping work; representative air monitoring and results/TWA's/EL's. Submit this information weekly to the Contracting Officer.

2. The Competent Person shall document and maintain the inspection and approval of the regulated area preparation before the start of work and daily during work.

a. Removal of any poly barriers

b. Visual inspection/testing by the Competent Person before application of lockdown encapsulant, if used.

c. Packaging and removal of ACM waste from regulated areas.

d. Disposal of ACM waste materials; copies of Waste Shipment Records/landfill receipts to the Contracting Officer weekly.

C. Submittals at Completion of Abatement

1. The Asbestos Abatement Contractor shall submit a project report consisting of the daily logbook requirements and documentation of events during the abatement project, including Waste Shipment Records signed by the landfill's agent. It will also include information on the containment and transportation of waste from the containment with applicable Chain of Custody forms. The report shall include a certificate of completion, signed and dated by the Competent Person, in accordance with Attachment #1. All clearance and perimeter area samples must be submitted. The Contracting Officer will retain the report after the completion of the project and provide copies of the report to the NPS Office of Engineering and the Safety Office.

2.5 ENCAPSULANTS

A. Types of Encapsulants

1. The following four types of encapsulants, if used, must comply with performance requirements as stated in paragraph 2.5.2:

a. Removal encapsulant - used as a wetting agent to remove ACM.

b. Bridging encapsulant - provides a tough, durable coating on ACM.

c. Penetrating encapsulant - penetrates/encapsulates ACM at least 13 mm (1/2 inches).

d. Lockdown encapsulant - seals microscopic fibers on surfaces after ACM removal.

B. Performance Requirements

1. Encapsulants shall meet the latest requirements of EPA; shall not contain toxic or hazardous substances or solvents; and shall comply with the following performance requirements:

a. General Requirements for all Encapsulants:

i. ASTM E84: Flame spread of 25; smoke emission of 50.

ii. University of Pittsburgh Protocol: Combustion Toxicity; zero mortality.

iii. ASTM C732: Accelerated Aging Test; Life Expectancy - 20 years.

iv. ASTM E96: Permeability - minimum of 0.4 perms.

b. Bridging/Penetrating Encapsulants:

i. ASTM E736: Cohesion/Adhesion Test - 24 kPa (50 pounds/square foot).

ii. ASTM E119: Fire Resistance - 3 hours (Classified by UL for use on fibrous/cementitious fireproofing).

iii. ASTM D2794: Gardner Impact Test; Impact Resistance - minimum 11.5 kg-mm (43 in/lb).

iv. ASTM D522: Mandrel Bend Test; Flexibility - no rupture or cracking.

c. Lockdown Encapsulants:

i. ASTM E119: Fire resistance - 3 hours (tested with fireproofing over encapsulant applied directly to steel member).

ii. ASTM E736: Bond Strength - 48 kPa (100 pounds/square foot) (test compatibility with cementitious and fibrous fireproofing).

iii. In certain situations, encapsulants may have to be applied to hot pipes/equipment. The encapsulant must be able to withstand high temperatures without cracking or creating any noxious gases or vapors during or after application.

C. Certificates of Compliance: The Contractor shall submit to NPS representative certification from the manufacturer indicating compliance with performance requirements for encapsulants when applied according to manufacturer recommendations.

PART 3 EXECUTION

3.1 PRE-ABATEMENT ACTIVITIES

A. Pre-Abatement Meeting

1. The Contracting Officer, upon receipt, review, and substantial approval of all pre-abatement submittals and verification that all materials and equipment required for the project are on the site, will arrange for a pre-abatement meeting between the Asbestos Abatement Contractor, Competent Person(s), the Contracting Officer, and the IHC. The purpose of the meeting is to discuss any aspect of the submittals needing clarification or amplification, and to discuss any aspect of the project execution and the sequence of the operation. The Asbestos Abatement Contractor shall be prepared to provide any supplemental information/documentation to the Contracting Officer regarding any submittals, documentation, materials, or equipment. Upon satisfactory resolution of any outstanding issues, the Contracting Officer will issue a written order to proceed to the Asbestos Abatement Contractor. No abatement work of any kind described in the following provisions shall be initiated before the NPS's written order to proceed.

B. Pre-Abatement Inspections and Preparations

1. Before any work begins on the construction of the regulated area, the Asbestos Abatement Contractor will:

a. Conduct a space-by-space inspection with the Contracting Officer and prepare a written inventory of all existing damage in those spaces where asbestos abatement will occur. Still or video photography may be used to supplement the written damage inventory. Documentation will be signed and certified as accurate by both parties.

b. For exterior removal, the Asbestos Abatement Contractor shall cover the ground and adjacent building surfaces with a minimum of one (1) layer of 6-mil fire-retardant poly extending a minimum of 10 feet from the building component being removed.

c. Shut down and lockout in accordance with 29 CFR 1910.147 all electrical circuits that pose a potential hazard. Electrical arrangements will be tailored to the regulated area and the systems involved. All electrical circuits affected will be turned off at the circuit box outside the regulated area, not just the wall switch. The goal is to eliminate the potential for electrical shock, which is a major threat to life in the regulated area due to possible energized circuits. Electrical lines used to power equipment in the regulated area shall conform to all electrical safety standards and shall be isolated using a ground fault circuit interrupter (GFCI). All GFCIs shall be tested before use.

d. Inspect existing firestopping in the regulated area. Correct as needed.

C. Pre-Abatement Construction and Operations

1. Perform all preparatory work for the first regulated area in accordance with the approved work schedule and with this specification.

2. Upon completion of all preparatory work, the IHC will inspect the work and systems and will notify the Contracting Officer when the work is completed in accordance with this specification. The Contracting Officer may inspect the regulated area and the systems with the IHC and may require that, upon satisfactory inspection, the Asbestos Abatement Contractor's employees perform all major aspects of the approved SOPs, especially worker protection, fall protection, respiratory systems, contingency plans, decontamination procedures, and monitoring to demonstrate satisfactory operation.

3. The Competent Person shall document the pre-abatement activities described above and deliver a copy to the Contracting Officer.

4. Upon satisfactory inspection of the installation and operation of systems, the Contracting Officer will notify the Asbestos Abatement Contractor in writing to proceed with the asbestos abatement work in accordance with this specification.

3.2 REGULATED AREA PREPARATIONS

A. Site Security

1. Regulated area access is to be restricted only to authorized, trained/accredited, and protected personnel. These may include the Asbestos Abatement Contractor's employees, employees of subcontractors, NPS employees and representatives, State and local inspectors, and any other designated individuals. A list of authorized personnel shall be established before commencing the project and shall be located immediately outside the boundary of the regulated work area.

2. Entry into a regulated area by unauthorized individuals shall be reported immediately to the Competent Person by anyone observing the entry. The Competent person shall immediately notify the Contracting Officer.

3. A logbook shall be maintained outside the boundary of the regulated work area. Anyone who enters the regulated area must record their name, affiliation, time in, and time out for each entry.

4. Access to a regulated area shall be through a single access point. All other access (doors, windows, hallways, etc.) shall be sealed or locked to prevent entry to or exit from the regulated area. The only exceptions to this requirement are the waste load out area, which shall be sealed or locked except during the removal of containerized asbestos waste from the regulated area, and emergency exits. Emergency exits shall not be locked from the inside or outside.

5. The Asbestos Abatement Contractor Competent Person shall control site security during abatement operations to isolate work in progress and protect adjacent personnel. Security shall be provided at the entrance to the regulated area to ensure that all entrants are logged in/out and that only authorized personnel are allowed entrance.

6. The Asbestos Abatement Contractor will have the NPS's assistance in notifying adjacent personnel of the presence, location, and quantity of ACM in the regulated area and enforcement of restricted access by NPS employees.

7. The regulated area shall be locked during non-working hours.

B. Signage and Power Management

1. Post OSHA DANGER signs meeting the specifications of OSHA 29 CFR 1926.1101 at any location approaching the regulated area where airborne concentrations of asbestos may exceed ambient background levels. Signs shall be posted at a distance sufficiently far away from the regulated area to permit any person to read the sign and take the necessary measures to avoid exposure. Additional signs will be posted following the construction of the regulated area.

2. Shut down and lock out electric power to the regulated area. Provide temporary power and lighting (if working at night). Ensure safe installation, including GFCI of temporary power sources and equipment, in compliance with all applicable electrical code requirements and OSHA requirements for temporary electrical systems. Electricity shall be provided by the NPS at no expense to the Asbestos Abatement Contractor.

3. Shut down and lock out heating, cooling, and air conditioning system (HVAC) components that are in, supply, or pass through the regulated area. Interiors of existing ductwork may require decontamination. This may be done during the pre-cleaning phase of operations before the ductwork is sealed off or during the final cleaning phase before re-engagement of the system. Appropriate equipment and control measures shall be utilized to prevent contamination of building spaces during this operation. Adequate cleaning of ductwork may sometimes be accomplished by drawing high volumes of air through the system using the HEPA-filtered negative-pressure ventilation units. Investigate and document the regulated area and agree on the pre-abatement condition with the Contracting Officer. Seal all intake and exhaust vents in the regulated area with duct tape and two (2) layers of 6-mil plastic. Also, seal any seams in system components that pass through the regulated area.

C. Negative Pressure Filtration System

1. The Contractor shall provide enough HEPA negative air machines to effect $\geq$ - 0.02-inch WCG pressure differential inside the regulated work area. The Competent Person shall determine the number of units needed for the regulated area by dividing the cubic feet in the regulated area by 15 and then dividing that result by the cubic feet per minute (CFM) for each unit to determine the number of units needed to effect $\geq$ - 0.02 inch WCG pressure inside the regulated work area. Provide standby units in the event of machine failure and/or emergency in an adjacent area. HEPA-equipped negative air machines shall be discharged outside of the building a minimum of 30 feet from building make-up air, doors, open windows, patients, visitors, and staff.

2. NIOSH has done extensive studies and has determined that negative air machines typically operate at ~50 percent efficiency. The contractor shall consider this in their determination of the number of units needed to provide $\geq$ - 0.02-inch WCG pressure differential inside the regulated work area. The contractor shall use double the number of machines, based on their calculations, or submit proof that their machines operate at stated capacities, at a 2-inch pressure drop across the filters.

3. Design and Layout:

   a. Before the start of work, submit the design and layout of the regulated area and the negative air machines. The submittal shall indicate the number of, location of, and size of negative air machines. The point(s) of exhaust, air flow within the regulated area, anticipated negative pressure differential, and supporting calculations for sizing shall be provided. In addition, submit the following:

i. Method of supplying power to the units and designation/location of the panels.

ii. Description of testing method(s) for correct air volume and pressure differential.

iii. If auxiliary power supply is to be provided for the negative air machines, provide a schematic diagram of the power supply and manufacturer's data on the generator and switch.

4. Negative Air Machines (Hepa Units):

a. Negative Air Machine Cabinet: The cabinet shall be constructed of steel or other durable material capable of withstanding potential damage from rough handling and transportation. The width of the cabinet shall be less than 30 inches in order to fit in standard doorways. The cabinet must be factory sealed to prevent asbestos fibers from being released during use, transport, or maintenance. Any access to and replacement of filters shall be from the inlet end. The unit must be on casters or wheels.

b. Negative Air Machine Fan: The rating capacity of the fan must indicate the CFM under actual operating conditions. Manufacturers typically use "free-air" (no resistance) conditions when rating fans. The fan must be a centrifugal type fan.

c. Negative Air Machine Final Filter: The final filter shall be a HEPA filter. The filter media must be completely sealed on all edges within a structurally rigid frame. The filter shall align with a continuous flexible gasket material in the negative air machine housing to form an air-tight seal. Each HEPA filter shall be certified by the manufacturer to have an efficiency of not less than 99.97 percent. Testing shall have been done in accordance with Military Standard MIL-STD-282 and Army Instruction Manual 136-300-175A. Each filter must bear a UL586 label to indicate the ability to perform under specified conditions. Each filter shall be marked with the name of the manufacturer, serial number, air flow rating, efficiency, resistance, and the direction of test air flow.

d. Negative Air Machine Pre-filters: The pre-filters, which protect the final HEPA filter by removing larger particles, are required to prolong the operating life of the HEPA filter. Two stages of pre-filtration are required. A first-stage pre-filter shall be a low-efficiency type for particles 10 µm or larger. A second-stage pre-filter shall have a medium efficiency effective for particles down to 5 µm or larger. Pre-filters shall be installed either on or in the intake opening of the NAM, and the second-stage filter must be held in place with a special housing or clamps.

e. Negative Air Machine Instrumentation: Each unit must be equipped with a gauge to measure the pressure drop across the filters and to indicate when filters have become loaded and need to be changed. A table indicating the cfm for various pressure readings on the gauge shall be affixed near the gauge for reference or the reading shall indicate at what point the filters shall be changed, noting cfm delivery. The unit must have an elapsed time meter to show total hours of operation.

f. Negative Air Machine Safety and Warning Devices: An electrical/ mechanical lockout must be provided to prevent the fan from being operated without a HEPA filter. Units must be equipped with an automatic shutdown device to stop the fan in the event of a rupture in the HEPA filter or blockage in the discharge of the fan. Warning lights are required to indicate normal operation; too high a pressure drop across filters, or too low a pressure drop across filters.

g. Negative Air Machine Electrical: All electrical components shall be approved by the National Electrical Manufacturers' Association (NEMA) and Underwriters Laboratories (UL). Each unit must be provided with overload protection, and the motor, fan, fan housing, and cabinet must be grounded.

h. It is essential that replacement HEPA filters be tested using an “in-line” testing method, to ensure the seal around the periphery was not damaged during replacement. Damage to the outer HEPA filter seal could allow contaminated air to bypass the HEPA filter and be discharged to an inappropriate location. Contractor will provide written documentation of test results for negative air machine units with HEPA filters.

5. Pressure Differential: The fully operational negative air system within the regulated area shall continuously maintain a pressure differential $\geq$ - 0.02 inch WCG inside the regulated work area. Before any disturbance of any asbestos material, this shall be demonstrated to the NPS or their representative by use of a pressure differential meter/manometer as required by OSHA 29 CFR 1926.1101(g)(5)(i). The Competent Person shall be responsible for providing, maintaining, and documenting the negative pressure and air changes as required by OSHA and this specification.

6. Monitoring: The pressure differential shall be continuously monitored and recorded between the regulated area and the area outside the regulated area with a monitoring device that incorporates a strip chart recorder. The strip chart recorder shall become part of the project log and shall indicate at least $\geq$ - 0.02 inch WCG pressure differential inside the regulated work area for the duration of the project.

7. Auxiliary Generator: If the building is occupied during abatement, provide an auxiliary gasoline/diesel generator located outside the building in an area protected from the weather. The generator must automatically start and supply power to a minimum of 50 percent of the negative air machines in operation.

8. Supplemental Make-Up Air Inlets: Provide, as needed for proper air flow in the regulated area, in a location approved by NPS, openings in the plastic sheeting to allow outside air to flow into the regulated area. Auxiliary makeup air inlets must be located as far from the negative air machines as possible, off the floor near the ceiling, and away from the barriers that separate the regulated area from the occupied clean areas. Cover the inlets with weighted flaps, which will seal in the event of failure of the negative pressure system.

9. Testing the System: The negative pressure system must be tested before any ACM is disturbed in any way. After the regulated area has been completely prepared, the decontamination units set up, and the negative air machines installed, start the units up one at a time. Demonstrate and document the operation and testing of the negative pressure system to NPS using smoke tubes and a negative pressure gauge. Verification and documentation of adequate negative pressure differential across each barrier must be done at the start of each work shift.

10. Demonstration of the Negative Pressure Filtration System:

    a. The demonstration of the operation of the negative pressure system to NPS or their representative shall include, but not be limited to, the following:

        i. Plastic barriers and sheeting move lightly in toward the regulated area.

        ii. Curtains of the decontamination units move in toward the regulated area.

        iii. There is a noticeable movement of air through the decontamination units. Use the smoke tube to demonstrate air movement from the clean room to the shower room to the equipment room to the regulated area.

        iv. Use smoke tubes to demonstrate that air is moving across all areas in which work is to be done. Use a differential pressure gauge to indicate a negative pressure of at > - 0.02 inches across every barrier separating the regulated area from the rest of the building. Modify the system as necessary to meet the above requirements.

11. Use of the Negative Pressure Filtration System During Abatement Operations:

    a. Start units before beginning any disturbance of ACM occurs. After work begins, the units shall run continuously, maintaining 4 actual air changes per hour at a negative pressure differential of > - 0.02-inch water column gauge, for the duration of the work until a final visual clearance and final air clearance have been completed.

    b. No negative air units shall be shut down at any time unless authorized by the Contracting Officer, verbally and in writing.

b. Pre-cleaning of ACM contaminated items shall be performed after the enclosure has been erected and negative pressure has been established in the work area. After items have been pre-cleaned and decontaminated, they may be removed from the work area for storage until the completion of abatement in the work area.

c. Abatement work shall begin at a location farthest from the units and proceed towards them. If an electric failure occurs, the Competent Person shall stop all abatement work and immediately begin wetting all exposed asbestos materials for the duration of the power outage. Abatement work shall not resume until power is restored and all units are operating properly again.

d. The negative air machines shall continue to run after all work is completed and until a final visual clearance and a final air clearance have been completed for that regulated area.

12. Dismantling The System: After completion of the final visual and final air clearance has been obtained by the IHC, the units may be shut down. The unit exterior surfaces shall have been completely decontaminated; pre-filters are not to be removed, and the unit's inlet/outlet sealed with 2 layers of 6-mil poly immediately after shutdown. No filter removal shall occur at the site following successful completion of site clearance. OSHA/EPA/DOT asbestos labels shall be attached to the units.

3.3 CONTAINMENT BARRIERS AND COVERINGS FOR THE REGULATED AREA

A. General

1. Using critical barriers, seal off the perimeter of the regulated area to completely isolate the regulated area from adjacent spaces. All penetrations in the regulated area must be covered with two (2) layers of independently installed 6-mil fire-retardant poly to prevent contamination and facilitate clean-up. Should adjacent areas become contaminated, immediately stop work and clean up the contamination at no additional cost to the Contracting Officer or the IHC.

B. Preparation Before Sealing Off

1. Place all tools, scaffolding, materials, and equipment needed for working in the regulated area before erecting any plastic sheeting. All uncontaminated non-fixed furniture, equipment, and/or supplies shall be removed by NPS from the regulated area before commencing work. Any objects not scheduled for removal remaining in the regulated area shall be completely covered with 2 layers of 6-mil fire-retardant poly sheeting and secured with duct tape. Lock out and tag out any HVAC/electrical systems in the regulated area.

C. Controlling Access to the Regulated Area

1. Access to the regulated area shall be permitted only through the PDF. All other means of access shall be closed off by proper sealing and by posting DANGER signs on the clean side of the regulated area where it is adjacent to or within view of any occupiable area. An opaque visual barrier of at least 4-mil polyethylene shall be provided so that the abatement work is not visible to any site occupants. If the area adjacent to the regulated area is accessible to the public, construct a solid barrier on the public side of the sheeting for protection and isolation of the Project. The barrier shall be constructed with nominal 2" x 4" (50mm x 100mm) wood or metal studs 16 inches (400mm) on centers, securely anchored to prevent movement, and covered with a minimum of 1/2" (12.5mm) plywood. Provide an appropriate number of OSHA DANGER signs for each visual and physical barrier.

D. Critical Barriers

1. The regulated area must be completely separated from the adjacent areas and the outside by at least two (2) layers of 6-mil polyethylene and duct tape/spray adhesive. Individually seal all supply and exhaust ventilation openings, lighting fixtures, clocks, doorways, windows, convectors, speakers, and other openings into the regulated area with two (2) layers of 6-mil polyethylene and taped securely in place with duct tape/spray adhesive. Critical barriers must remain in place until all work and clearances have been completed. Light fixtures shall not be operational during abatement. Auxiliary lighting shall be provided. If needed, provide plywood squares 6" x 6" x 3/8" (150mm x 150mm x 18mm) held in place with one 6d smooth masonry/galvanized nail driven through the center of the plywood square and duct tape on the polyethylene so as to clamp the polyethylene to the wall/surface. Locate plywood squares at each end, corner, and 4' (1200mm) maximum on centers.

E. Primary Barriers

1. Clean all furniture, equipment, etc., before being moved as specified above. Clean all surfaces in the regulated area, using the HEPA vacuum and/or wet wiping before installing any polyethylene.

2. Floors: Cover the floor of the regulated area with at least two (2) layers of 6-mil polyethylene, turning up the walls at least 12" (300mm). The polyethylene must form a right angle at the floor-wall juncture so no radius can be stepped on, possibly causing detachment of the polyethylene. Spray glue and duct tape must both be used for floor seams. Floor seams must overlap a minimum of 6 feet (1800mm) or be at right angles to each other. The top sheet of polyethylene must be removed independently of the bottom layer. A third loose layer of 6-mil polyethylene shall be used in the area of removal and periodically picked up to reduce contamination of the initial layers.

3. Walls: All walls in the regulated area, including critical barriers, shall be covered with two (2) layers of 4-mil polyethylene, mechanically supported and sealed with duct tape and/or spray glue. Tape all joints, including the floor-wall joint, with duct tape/spray glue. All wall joints must overlap at least 6 feet (1800mm).

4. If stairs and ramps are covered with 6-mil plastic, two layers must be used. Provide 19 mm (3/4 inch) exterior-grade plywood treads held in place with duct tape/glue on the plastic. Do not cover rungs or rails with any isolation materials.

F. Secondary Barriers

1. A loose layer of 6-mil shall be used as a drop cloth to protect the primary layers from debris generated during the abatement. This layer shall be replaced as needed during the work.

G. Extension of the Regulated Area

1. If the regulated area barrier is breached in any manner that could allow the passage of asbestos fibers or debris, the Competent Person shall immediately stop work, continue wetting, and proceed to extend the regulated area to enclose the affected area as per procedures described in this specification. If the affected area cannot be enclosed, decontamination measures and cleanup shall start immediately. All personnel shall be isolated from the affected area until decontamination/cleanup is completed as verified by visual inspection and air monitoring. Air monitoring at completion must indicate background levels.

H. Firestopping

1. Firestop penetrations through walls and ceilings with a fire-rated penetration firestop system offering a tight seal.

2. Products that violate the fire codes by not providing an adequate fire rating seal shall be brought to the attention of the NPS's representative. The Asbestos Abatement Contractor shall list all areas of penetration, the type of sealant used, and whether the location is fire-rated or not. If a penetration is caused or discovered due to abatement work, the Asbestos Abatement Contractor shall notify the NPS's representative immediately for a determination.

3. All visible openings, whether or not caused by a penetration, shall be reported by the Asbestos Abatement Contractor to the NPS's representative for a sealant system determination. Firestops shall be ASTM E814 and UL 1479 tested for that size opening, penetrant, and fire rating.

I. Sanitary Facilities

1. The Asbestos Abatement Contractor shall provide sanitary facilities for abatement personnel and maintain them in a clean and sanitary condition throughout the abatement project.

J. Personal Protective Equipment

1. Provide whole body clothing, head coverings, gloves, and foot coverings, and any other personal protective equipment as determined by conducting the hazard assessment required by OSHA at 29 CFR 1910.132 (d). The Competent Person shall ensure the integrity of personal protective equipment worn for the duration of the project. Duct tape shall be used to secure all suit sleeves to wrists and to secure foot coverings at the ankle.

K. Pre-Cleaning

1. The NPS will provide water for abatement purposes. The Asbestos Abatement Contractor shall connect to the existing system. The service to the shower(s) shall be supplied with backflow prevention.

2. Pre-cleaning of ACM-contaminated items shall be performed after the enclosure has been erected and negative pressure has been established in the work area. All workers performing pre-cleaning activities must don appropriate personal protective equipment (PPE), as specified throughout this document and as approved in the Contractor's work plan. After items have been pre-cleaned and decontaminated, they may be removed from the work area for storage until the completion of abatement in the work area.

3. The Asbestos Abatement Contractor shall pre-clean all movable objects and refuse within the regulated area using a HEPA-filtered vacuum and/or wet cleaning methods as appropriate. Do not use any methods that would raise dust, such as dry sweeping or vacuuming with equipment not equipped with HEPA filters. Do not disturb asbestos-containing materials during this pre-cleaning phase.

4. Pre-clean all fixed objects in the regulated area using HEPA-filtered vacuums and/or wet cleaning techniques as appropriate. Do not use any methods that would raise dust, such as dry sweeping or vacuuming with equipment not equipped with HEPA filters. Do not disturb asbestos-containing materials during this pre-cleaning phase.

5. Pre-clean all surfaces in the regulated area using HEPA-filtered vacuums and/or wet cleaning methods as appropriate. Do not use any methods that would raise dust, such as dry sweeping or vacuuming with equipment not equipped with HEPA filters. Do not disturb asbestos-containing materials during this pre-cleaning phase.

3.4 REMOVAL OF ACM

A. General

1. The NPS must be notified at least 24 hours in advance of any waste removed from the containment. All applicable requirements of OSHA, EPA, State, Local, and DOT shall be followed during Class II work. Keep materials intact; do not disturb; wet while working with it; wrap as soon as possible with two (2) layers of 6-mil plastic for disposal or approved equivalent.

B. Wetting Materials

1. Use amended water for the wetting of ACM before removal. The Competent Person shall ensure the wetting of ACM meets the definition of "adequately wet" in the EPA NESHAP's regulation for the duration of the Project. A removal encapsulant may be used instead of amended water with the written approval of the NPS's representative.

2. Amended Water: Provide water to which a surfactant has been added to wet the ACM and reduce the potential for fiber release during disturbance of the ACM. The mixture must be equal to or greater than the wetting provided by water amended by a surfactant consisting of one ounce of 50% polyoxyethylene ester and 50% polyoxyethylene ether, mixed with 5 gallons (19L) of water.

3. Removal Encapsulant: Provide a penetrating encapsulant designed specifically for the removal of ACM. The material must, when used, result in adequate wetting of the ACM and retard fiber release during disturbance equal to or greater than the amended water described above in 3.4-2.

C. Secondary Barriers and Walkways

1. Install as a drop cloth a 6-mil polyethylene sheet at the beginning of each work shift, where removal is to be done during that shift. Completely cover floors and any walls within 10 feet (3M) of the area where work is to be done. Secure the secondary barrier with duct tape to prevent debris from getting behind it. Remove the secondary barrier at the end of the shift or as work in the area is completed. Keep residue on the secondary barrier wetted. When removing, fold inward to prevent spillage and place in a disposal bag.

2. Install walkways using 6-mil black polyethylene between the regulated area and the decontamination facilities (PDF and W/EDF) to protect the primary layers from contamination and damage. Install the walkways at the beginning of each shift and remove them at the end of each shift.

D. Wet Removal of ACM Other than Amosite

1. Adequately and thoroughly wet the ACM to be removed before removal to reduce/prevent fiber release to the air. Adequate time must be allowed for the amended water to saturate the ACM. Abatement personnel must not disturb dry ACM. Use a fine spray of amended water or removal encapsulant. Saturate the material sufficiently to wet the substrate without causing excessive dripping. The material must be sprayed repeatedly/continuously during the removal process in order to maintain adequately wet conditions. Removal encapsulants must be applied in accordance with the manufacturer's written instructions. (Note: In no event shall dry removal occur except in the case of electrical hazards or a greater safety issue is possible.

2. If ACM does not wet well with amended water due to composition, coating, or jacketing, remove as follows:

   a. Mist work area continuously with amended water whenever necessary to reduce airborne fiber levels.

   b. Remove saturated ACM in small sections. Do not allow material to dry out. As material is removed, the bag material is placed into disposal bags while still wet. Twist the bag neck tightly, bend over (gooseneck), and seal with a minimum of three tight wraps of duct tape. Clean /decontaminate the outside of the bag of any residue and move to the washdown station adjacent to W/EDF.

   c. Fireproofing or Architectural Finish on Scratch Coat: Spray with a fine mist of amended water or removal encapsulant. Allow time for saturation of the substrate. Do not over-saturate, causing excess dripping. Scrape material from the substrate. Remove material in manageable quantities and control falling to staging or the floor. If the falling distance is over 20 feet (6 meters), use a drop chute to contain material through descent. Remove residue remaining on the scratch coat after scraping is done using a stiff-bristle hand brush. If a removal encapsulant is used, remove residue completely before the encapsulant dries. Periodically re-wet the substrate with amended water as needed to prevent drying of the material before the residue is removed from the substrate.

   d. Fireproofing or Architectural Finish on Wire Lath: Spray with a fine mist of amended water or removal encapsulant. Allow time to completely saturate the material. Do not over-saturate, causing excess dripping. If the surface has been painted or otherwise coated, cut small holes as needed and apply amended water or removal encapsulant from above. Cut saturated wire lath into 2 feet x 6 feet (50mm x 150mm) sections and cut hanger wires. Roll up complete with ACM, cover in burlap, and hand place in a disposal bag. Do not drop to the floor. After removal of lath/ACM, remove any overspray on decking and structure using stiff-bristle nylon brushes. Depending on the hardness of overspray, scrapers may be needed for removal.

   e. Pipe/Tank/Vessel/Boiler Insulation: Remove the outer layer of wrap while spraying with amended water to saturate the ACM. Spray ACM with a fine mist of amended water or removal encapsulant. Allow time to saturate the material in the substrate. Cut bands holding pre-formed pipe insulation sections. Slit jacketing at the seams, remove and hand place in a disposal bag. Do not allow dropping to the floor. Remove molded fitting insulation/mud in large pieces and hand place in a disposal bag. Remove any residue on the pipe or fitting with a stiff-bristle nylon brush. In locations where pipe fitting insulation is removed from fibrous glass or other non-asbestos insulated straight runs of pipe, remove fibrous material at least 6 inches from the point it contacts the ACM.

E. Removal of flooring materials

1. All requirements of the OSHA flooring agreement provisions shall be followed:

   a. Flooring shall be removed intact, as much as possible. Do not rip or tear the flooring.

   b. Mechanical chipping or sanding is not allowed.

   c. Wet clean and HEPA vacuum the floor before and after removal of flooring.

F. Removal of Mastic

1. All chemical mastic removers must be low in volatile organic compound (VOC) content, have a flash point greater than 200° Fahrenheit, and contain no chlorinated solvents.

2. Follow all manufacturers' instructions in the use of the mastic removal material.

3. Before application of any liquid material, check the floor for penetrations and seal before removing mastic.

G. Outdoor Work Areas

1. On some projects, work must be performed on exterior areas of the building. If outdoor work is to be performed, all applicable OSHA, State, and Local regulations must be followed to ensure that outdoor work areas comply so that workers, the general public, and the environment are protected.

H. Scaffold Fall Protection

1. Each employee who is more than 10 feet above a lower level shall be protected from falls by guardrails or a fall arrest system. The fall arrest system includes harnesses and components of the harness/belt, such as Dee-rings, snap hooks, lifelines, and anchorage points. Lifelines must be independent of support lines and suspension ropes and not attached to the same anchorage point as the support or suspension rope. OSHA's scaffolding standard defines a competent person as "one who is capable of identifying existing and predictable hazards in the surroundings or working conditions, which are unsanitary, hazardous to employees, and who has the authorization to take prompt corrective measures to eliminate them." The competent person will determine if it is safe for employees to work on or from a scaffold or roof during storms or high winds and ensure that a personal fall arrest system will protect the employees. The competent person will also inspect the scaffold and scaffold components for visible defects before each work shift and after any occurrence that could affect the structural integrity and authorize prompt corrective measures.

I. Roof Fall Protection

1. The competent person shall determine if the walking/working surfaces on which the employees are to work have the strength and structural integrity to support the employees safely. Each employee on a walking/working surface (horizontal and vertical surface) with an unprotected side or edge that is 6 feet or more above a lower level shall be protected from falling by the use of guardrail systems, safety net systems, or personal fall arrest systems.

J. Removal of Roofing and Windows

1. Material shall be removed in an intact state to the extent that it is feasible.

2. Wet methods shall be used to remove materials that are not intact, or that will be rendered not intact during removal, unless such wet methods are not feasible or will create safety hazards.

3. Cutting machines shall be continuously misted during use unless a competent person determines that misting substantially decreases worker safety.

4. When removing materials using a power cutter, all dust resulting from the cutting operation shall be collected by a HEPA dust collector or shall be HEPA vacuumed by vacuuming along the cut line.

5. Asbestos-containing material that has been removed shall not be dropped or thrown to the ground. Unless the material is carried or passed to the ground by hand, it shall be lowered to the ground via a covered, dust-tight chute, crane, or hoist.

6. Any ACM that is not intact shall be lowered to the ground as soon as is practicable, but in any event no later than the end of the work shift. While the material remains, it shall either be kept wet, placed in an impermeable waste bag, or wrapped in plastic sheeting.

7. Intact ACM shall be lowered to the ground as soon as is practicable, but in any event no later than the end of the work shift.

8. Upon being lowered, unwrapped material shall be transferred to a closed receptacle in such a manner as to preclude the dispersion of dust.

9. Notwithstanding any other provision of this section, removal or repair of sections of intact roofing less than 25 square feet in area does not require the use of wet methods or HEPA vacuuming as long as manual methods that do not render the material non-intact are used to remove the material and no visible dust is created by the removal method used. In determining whether a job involves less than 25 square feet, the employer shall include all removal and repair work performed on the same roof on the same day.

10. All waste must be wrapped in two (2) layers of 6-mil poly or approved equivalent and lowered carefully to the ground. Roofing may be lowered by way of a dust-tight chute.

K. Alternate Procedures

1. Procedures described in this specification are to be utilized at all times.

2. If specified procedures cannot be utilized, a request must be made in writing to the Contracting Officer, providing details of the problem encountered and recommended alternatives.

3. Any alternative procedure must be approved in writing by the Contracting Officer before implementation.

L. Monitoring of Regulated Area

1. The Competent Person shall monitor and closely supervise the work carried out in the regulated area, to ensure the protection of workers and the environment. The area shall be secured as detailed in this specification. After the work is finished, the Competent Person will verify and test to determine satisfactory completion and notify the Contracting Officer when the results are satisfactory. The NPS may use its IHC to determine compliance. When the area is found acceptable, the area will be released as decontaminated.

M. Worker Protection

1. Before beginning work with any material for which an SDS has been submitted, provide workers with the required PPE. The required PPE shall be used whenever exposure to the material might occur. In addition to OSHA/specification requirements for respiratory protection, an organic vapor cartridge, at a minimum, shall be used in addition to the HEPA filter when a solvent-based encapsulant is used.

3.5 LOCKDOWN ENCAPSULATION

A. General: Lockdown encapsulation is an integral part of the ACM removal. At the conclusion of ACM removal and before removal of the primary barriers, the contractor shall encapsulate all surfaces with a bridging encapsulant.

B. Delivery and Storage: Deliver materials to the job site in original, new, and unopened containers bearing the manufacturer's name and label, as well as the following information: name of material, manufacturer's stock number, date of manufacture, thinning instructions, application instructions, and the SDS for the material.

C. Worker Protection: Before beginning work with any material for which an SDS has been submitted, provide workers with any required personal protective equipment. The required personal protective equipment shall be used whenever exposure to the material might occur. In addition to OSHA/specification requirements for respiratory protection, a paint pre-filter and an organic vapor cartridge, at a minimum, shall be used in addition to the HEPA filter when an organic solvent-based encapsulant or other solvent is used. Note: Flammable and combustible encapsulants or solvents shall not be used unless authorized in writing by NPS.

3.6 DISPOSAL OF ACM WASTE MATERIALS

A. General

1. Dispose of waste ACM and debris that is packaged in accordance with these specifications, OSHA, EPA, DOT, and the State of Wyoming. The landfill requirements for packaging must also be met. Transport will follow 49 CFR 171–180 regulations. Disposal shall be done at an approved landfill in accordance with State and Local requirements, which are generally more restrictive than Federal EPA requirements. Disposal of non-friable ACM shall be done in accordance with applicable State and Local regulations.

B. Procedures

1. Asbestos waste shall be packaged and moved through the W/EDF into a covered transport container in accordance with NESHAP's packaging requirements. Waste shall be double-bagged before disposal. Wetted waste can be very heavy. Bags shall not be overfilled. Bags shall be securely sealed to prevent accidental opening and/or leakage. The top shall be tightly twisted and goosenecked prior to sealing with at least three wraps of duct tape. Ensure that unauthorized persons do not have access to the waste material once it is outside the regulated area. Transport containers must be covered at all times when not in use. NESHAP's signs must be adhered to containers during loading and unloading. Material shall not be transported in open vehicles. If drums are used for packaging, the drums shall be labeled properly and shall not be reused.

2. Waste Load Out: Waste load out shall be done in accordance with the procedures in the W/EDF Decontamination Procedures. Bags shall be decontaminated on exterior surfaces by wet cleaning and/or HEPA vacuuming before being placed in the second bag.

3. Asbestos waste with sharp-edged components, i.e., nails, screws, lath, strapping, tin sheeting, jacketing, metal mesh, etc., which might tear polyethylene bags, shall be wrapped securely in burlap before packaging and, if needed, use a polyethylene-lined fiber drum as the second container, before disposal.

3.7 PROJECT DECONTAMINATION

A. General

1. The entire work related to project decontamination shall be performed under the close supervision and monitoring of the Competent Person.

2. If the asbestos abatement work is in an area that was contaminated before the start of abatement, the decontamination will be done by cleaning the primary barrier poly before its removal and cleaning of the surfaces of the regulated area after the primary barrier removal.

3. If the asbestos abatement work is in an area that was uncontaminated before the start of abatement, the decontamination will be done by cleaning the primary barrier poly before its removal, thus preventing contamination of the building when the regulated area's critical barriers are removed.

B. Regulated Area Clearance

1. Air testing and other requirements that must be met before the release of the Asbestos Abatement Contractor and re-occupancy of the regulated area space are specified in Final Testing Procedures.

C. Work Description

1. Decontamination includes the cleaning and clearance of the air in the regulated area and the decontamination and removal of the decontamination areas installed before the abatement work, including primary/critical barriers, PDF, and W/EDF facilities, and negative pressure systems.

D. Pre-Decontamination Conditions

1. Before decontamination starts, all ACM waste from the regulated area shall be removed, all waste collected and removed, and the secondary barrier of poly removal and disposed of along with any gross debris generated by the work.

2. At the start of decontamination, the following shall be in place:

a. Primary barriers consisting of two layers of 6-mil polyethylene on the floor and on the walls.

b. Critical barriers consist of two (2) layers of 6-mil polyethylene, which is the sole barrier between the regulated area and the rest of the building or outside.

c. Critical barrier polyethylene over lighting fixtures, clocks, HVAC openings, doorways, windows, convectors, speakers, and other openings in the regulated area.

d. Decontamination facilities for personnel and equipment in operating condition, and the negative pressure system in operation.

E. Cleaning

1. Carry out a first cleaning of all surfaces of the regulated area, including items of remaining polyethylene sheeting, tools, scaffolding, ladders/staging by wet methods and/or HEPA vacuuming. Do not use dry dusting/sweeping methods. Use each surface of a cleaning cloth one time only and then dispose of it as contaminated waste. Continue this cleaning until there is no visible residue from abated surfaces, polyethylene, or other surfaces. Remove all filters in the air handling system and dispose of them as ACM waste in accordance with these specifications. The negative pressure system shall remain in operation during this time. If determined by the IHC, additional cleaning(s) may be needed.

3.8 FINAL VISUAL INSPECTION AND AIR CLEARANCE TESTING

A. General

1. Notify the Contracting Officer 48 hours in advance of the performance of the final visual inspection and testing. The final visual inspection and testing will be performed by the IHC after the cleaning.

B. Final Visual Inspection

1. Final visual inspection will include the entire regulated area, all poly sheeting, seals over ducts, windows, louvers, penetrations, or any other opening into the regulated area. If any debris, residue, dust, or any other suspect material is detected, the cleaning shall be repeated at no additional cost to the Contracting Officer or the IHC. Dust/ material samples may be collected and analyzed at no additional cost to the NPS at the discretion of the IHC to confirm visual findings. When the regulated area is visually clean, perform lockdown encapsulation of all surfaces from which asbestos was abated in accordance with the procedures in this specification. Negative pressure shall be maintained in the regulated area during the lockdown application, and then the final testing can be done.

C. Final Air Clearance Testing

1. After an acceptable final visual inspection by the IHC and Contracting Officer, the IHC will perform aggressive and/or passive final clearance testing. Air samples will be collected and analyzed in accordance with procedures for AHERA. For this project, five (5) PCM samples shall be collected for clearance, and one (1) field blank per 10 samples collected.

2. If release criteria are not met, the Asbestos Abatement Contractor shall repeat the final cleaning and continue the decontamination procedure from that point. Additional review and testing shall be at the expense of the Asbestos Abatement Contractor until the release criteria are met

3. If the results of the PCM are acceptable, remove the critical barriers. Any small quantities of residue material found upon removal of the poly shall be removed with a HEPA vacuum and localized isolation. If significant quantities are found as determined by the IHC, then the entire area affected shall be cleaned as specified in the final cleaning.

4. If release criteria are met, proceed to perform the abatement closeout and issue the certificate of completion in accordance with these specifications.

D. Final Air Clearance Procedures:

1. Asbestos Abatement Contractor's Release Criteria: Work in a regulated area is complete when the regulated area is visually clean and airborne fiber levels have been reduced to or below 0.01 f/cc, as measured by PCM methods or < 70 asbestos structures per square millimeter (s/mm2) by AHERA TEM.

2. Air Monitoring and Final Clearance Sampling: To determine if the elevated airborne fiber counts encountered during abatement operations have been reduced to the specified level, the IHC will secure samples and analyze them according to the following procedures:

   a. Fibers Counted: "Fibers" referred to in this section shall be all fibers regardless of composition, as counted in the NIOSH 7400 PCM method or asbestos fibers counted using the AHERA TEM method.

   b. All clearance air testing samples shall be collected on 0.8µ MCE filters for PCM analysis and 0.45µ MCE for TEM. Air samples will be collected in areas subject to normal air circulation. A minimum of 1200 Liters of air shall be collected for all clearance air samples.

3.9 ABATEMENT CLOSEOUT AND CERTIFICATE OF COMPLIANCE

A. After thorough decontamination, complete asbestos abatement work upon meeting the regulated area clearance criteria and fulfilling the following:

   1. Remove all equipment, materials, and debris from the project area.

   2. Package and dispose of all asbestos waste as required.

   3. Repair or replace all finishes damaged during the abatement work not scheduled for restoration by others.

   4. Fulfill other Project closeout requirements as specified elsewhere in this specification.

B. Certificate of Completion by Asbestos Abatement Contractor

   1. The Asbestos Abatement Contractor shall complete and sign the "Certificate of Completion" in accordance with Attachment #11 after the abatement and decontamination of the regulated area.

C. Work Shifts

   1. All work shall be done during administrative hours, excluding Federal Holidays. Any change in the work schedule must be approved in writing by the Contracting Officer.

ATTACHMENT #1

CERTIFICATE OF COMPLETION

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

ASBESTOS ABATEMENT CONTRACTOR'S NAME AND ADDRESS:_____
_____
_____
_____

1. I certify that I have personally inspected, monitored and supervised the abatement work of (specify regulated areas or buildings):

Which took place from_____to _____

2. That throughout the work, all applicable requirements/regulations and the specifications were met.

3. That any person who entered the regulated area was protected with the appropriate PPE and respirator and that they followed the proper entry and exit procedures and the proper operating procedures for the duration of the work.

4. That all employees of the Asbestos Abatement Contractor engaged in this work were trained in respiratory protection, were experienced with abatement work, had proper medical surveillance documentation, were fit-tested for their respirator, and were not exposed at any time during the work to asbestos without the benefit of appropriate respiratory protection.

5. I performed and supervised all inspection and testing specified and required by applicable regulations and specifications.

6. That the negative pressure system was installed, operated and maintained in order to provide a minimum of 4 air changes per hour with a continuous 5.0 Pa (-0.02") of water column pressure.

Signature:_____ Date:_____
(Printed Name/ Signature and Accreditation No./Exp.)

Signature:_____ Date:_____
(Printed Name/ Signature and Accreditation No./Exp.)

ATTACHMENT #2

CERTIFICATE OF WORKER'S ACKNOWLEDGMENT

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

ASBESTOS ABATEMENT CONTRACTOR'S NAME AND ADDRESS:_____

_____

WORKING WITH ASBESTOS CAN BE HAZARDOUS TO YOUR HEALTH. INHALING ASBESTOS HAS BEEN LINKED WITH VARIOUS TYPES OF CANCERS. IF YOU SMOKE AND INHALE ASBESTOS FIBERS YOUR CHANCES OF DEVELOPING LUNG CANCER ARE 50% TO 90% GREATER THAN THAT OF THE NON-SMOKING PUBLIC.

Your employer's contract with the OWNER for the above project requires that: You must be supplied with the proper PPE including an adequate respirator and be trained in its use. You must be trained in safe and healthy work practices and in the use of the equipment found at an asbestos abatement project. You must receive/have a current medical examination for working with asbestos. These things shall be provided at no cost to you. By signing this certificate, you are indicating to the OWNER that your employer has met these obligations.

RESPIRATORY PROTECTION: I have been trained in the proper use of respirators and have been informed of the type of respirator to be used on the above indicated project. I have a copy of the written Respiratory Protection Program issued by my employer. I have been provided for my exclusive use, at no cost, with a respirator to be used on the above indicated project.

TRAINING COURSE: I have been trained by a third party, State/EPA accredited trainer in the requirements for an AHERA/OSHA Asbestos Abatement Worker training course, 32 hours minimum duration. I currently have a valid State accreditation certificate. The topics covered in the course include, as a minimum, the following:

- Potential Health Effects Related to Exposure to Chemical Hazards
- Potential Health Effects Related to Exposure to Asbestos
- Employee Personal Protective Equipment
- Establishment of a Respiratory Protection Program
- State of the Art Work Practices
- Personal Hygiene
- Additional Safety Hazards
- Medical Monitoring
- Air Monitoring, if required
- Relevant Federal, State, and Local Regulatory Requirements, Procedures, and Standards
- Asbestos Waste Disposal

MEDICAL EXAMINATION: I have had a medical examination within the past 12 months which was paid for by my employer. This examination included: health history, occupational history, pulmonary function test, and may have included a chest x-ray evaluation. The physician issued a positive written opinion after the examination.

Signature:_____ Printed Name:_____

ATTACHMENT #3

AFFIDAVIT OF MEDICAL SURVEILLANCE, RESPIRATORY PROTECTION AND TRAINING/ACCREDITATION

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

_____

ASBESTOS ABATEMENT CONTRACTOR'S NAME AND ADDRESS:_____

_____

_____

1. I verify that the following individual

Name:_____

who is proposed to be employed in asbestos abatement work associated with the above project by the named Asbestos Abatement Contractor, is included in a medical surveillance program in accordance with 29 CFR 1926.1101(m), and that complete records of the medical surveillance program as required by 29 CFR 1926.1101(m)(n), 29 CFR 1910.134, and 29 CFR 1910.20 are kept at the offices of the Asbestos Abatement Contractor at the following address.

Address:_____

_____

2. I verify that this individual has been trained, fit-tested and instructed in the use of all appropriate respiratory protection systems and that the person is capable of working in safe and healthy manner as expected and required in the expected work environment of this project.

3. I verify that this individual has been trained as required by 29 CFR 1926.1101(k). This individual has also obtained a valid accreditation certificate. Documentation will be kept onsite.

Signature of Asbestos Abatement Contractor:_____Date:_____

Printed Name of Asbestos Abatement Contractor:_____

ATTACHMENT #4

DAILY SIGN-IN AND SIGN-OUT SHEET

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

ASBESTOS ABATEMENT CONTRACTOR'S NAME AND ADDRESS:_____

_____

PRINT NAME	CERT. NUMBER EXP. DATE	TIME-IN	TIMEOUT

ATTACHMENT #5

DAILY ACTIVITY REPORT

Project Name:		Project No.	
Building:		Date	
Area:		Project Day No.	of
Asbestos Abatement Contractor:		Personnel	
IHC:		Sheet	of
ACM Removed:			
Estimated Percent of Phase Completed:		On Schedule	Yes No

GENERAL REMARKS/MEETINGS/SITE VISITS/ASBESTOS ABATEMENT CONTRACTOR PROGRESS:

Asbestos Abatement Contractor Supervisor:	Reviewed by:

ATTACHMENT #6

VISITOR RELEASE FORM

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

ASBESTOS ABATEMENT CONTRACTOR'S NAME AND ADDRESS:_____

_____

VISITOR NAME:_____

VISITOR'S COMPANY:_____

PURPOSE OF VISIT:_____

Time of Entry to the Project Site:_____ to the Exclusion Zone:_____

Time of Departure from the Site:_____ from the Exclusion Zone:_____

Personal Protective Equipment Utilized:_____

I acknowledge and understand that I am visiting a hazardous waste control work area. I understand the dangers of exposure to hazardous waste.

I have read and understand the Site Safety and Health Plan for this project and will abide by the directions, stipulations, and terms specified therein.

I knowingly assume all risks in connection with potential exposure to hazardous waste and I do hereby, for myself and my heirs at law, release and forever discharge the property Owner, Owner's Representative, Project Administrator, independent testing laboratory, architects, engineers, or Contracting firms employed by the Owner or Project Administrator, their owners, employees, nominees, personal representative, affiliates, successors, and assigns from and against all liability whatsoever at common law or otherwise. I hereby waive and relinquish all claims of every nature that I now have or may have or claim to have that are in any way, directly or indirectly, related to exposure to hazardous waste and hazardous waste-containing materials.

Furthermore, I know that I am entering a construction area where workplace conditions, such as water on floors, scaffolding, electrical equipment, etc., CAN CREATE HAZARDOUS SITUATIONS FOR VISITORS. I assume all risks of accidental injury or illness, regardless of cause, while visiting this construction site. I hereby waive and relinquish all claims of every nature that I now have or may have or claim to have that are in any way, directly or indirectly, related to such injury.

THIS RELEASES ALL PARTIES ON THE ABOVE NAMED PROJECT:

Visitor's Signature:_____

Date:_____

Witness:_____

ATTACHMENT #7

FINAL CLEARANCE VISUAL INSPECTION

Project:	Project No.:
Building\Address:	Date:
Area:	ACM Removed:
Asbestos Abatement Contractor:	Project Permit No.:

Residual Dust On:	**Yes**	**No**	**Not Applicable**
Floors			
Walls			
Ledges			
Roof Decking			
Pipes			
Hangars			
Conduits			
Cables			
Light Fixtures			
Ductwork			
Equipment			
HEPA machine(s) running			
Containment dry			
Only critical barriers present			
Decontamination unit is attached and functional			

Visual Inspection: ___Pass ___Fail

Remarks:_____

_____

Technician:_____ _____
(Printed Name/ Accreditation No./Exp. Date) Signature

Asbestos C/S:_____ _____
(Printed Name/ Accreditation No./Exp. Date) Signature

END OF SECTION 02 82 00

SECTION 02 83 00 – LEAD-BASED PAINT STABILIZATION, COMPONENT REMOVAL, AND SOIL REMOVAL

PART 1 GENERAL

1.1 SUMMARY OF WORK

A. Contract Documents and Related Requirements

1 Drawings, general provisions of the contract, including General and Supplementary Conditions and other Specification Sections, apply to the work of this Section. The contract documents show the work to be done under the contract and related requirements and conditions impacting the project. Related requirements and conditions include applicable codes and regulations, notices and permits, existing site conditions and restrictions on the use of the site, requirements for partial owner occupancy during the work, coordination with other work, and the phasing of the work. In the event the Contractor discovers a conflict in the contract documents and/or requirements or codes, the conflict must be brought to the immediate attention of the General Contractor for Rehabilitation and the Contracting Officer for resolution. Whenever there is a conflict or overlap in the requirements, the most stringent shall apply. Any actions taken by the Contractor without obtaining guidance from the Contracting Officer shall become the sole risk and responsibility of the Contractor. All costs incurred due to such action are also the responsibility of the Contractor.

B. Extent of Work

1. Contractor will coordinate all work, phasing, and scheduling with the General Contractor for Rehabilitation, the Contracting Officer, and their field representative. The contractor will have a predetermined number of calendar days to complete lead-based paint (LBP) stabilization, component removal, soil removal, cleaning, and backfilling. The contractor will coordinate the start date and the number of mobilizations required with the General Contractor for Rehabilitation and the Contracting Officer. Strict adherence to the schedule will be required to allow the work to be completed promptly to accommodate other activities on the site.

2. Contracting Officer will supply the Lead-Based Paint Survey Report, dated July 8, 2025, completed by Northern Industrial Hygiene, Inc. of Billings, Montana, to the Contractor. The lead-based paint survey report will contain a brief description of the LBP components and locations. The Contractor shall satisfy themselves as to the actual number of building components and quantities to be stabilized, components removed, soil removed, cleaned, disposed of, and soil backfilled by the Contractor in preparation for re-painting and component replacement. NOTE: The Contractor shall also conduct a thorough cleaning of all lateral and horizontal interior and exterior surfaces on the building and exterior drip line around the building where LBP stabilization and component removal are scheduled. Nothing in this section may be interpreted as limiting the extent of work otherwise required by this contract and related documents.

3. Various structures meet the criteria established by the U.S. Environmental Protection Agency (EPA) that would require compliance with *The Residential Lead-Based Paint Hazard Reduction Act* (Title X), as the buildings could be continuously occupied by children. The EPA's regulations on residential property renovation (40 CFR 745 Subpart E) Lead RRP Rule requires that Contractors performing renovation, repair, and painting projects that disturb lead-based paint in homes, childcare facilities, and pre-schools built before 1978 be certified by the EPA (or an EPA-authorized state), use certified renovators who are trained by EPA-approved training providers and follow lead-safe work practices, and firms also must be certified. This is for general contractors who disturb lead-based paint as a side effect of their primary work. To perform maintenance, repair, or aesthetic upgrades safely without necessarily removing all lead.

4. Stabilization, component removal, soil removal, packaging, and clean-up of LBP components and lead-contaminated soils and elements in an appropriate regulated area as necessary to accommodate the disposal of LBP and the renovation of the structures.

5. Any damage to components not scheduled for stabilization, removal, cleaning, and backfill, resulting from the Contractor's work, shall be repaired or replaced at the sole cost of the Contractor utilizing appropriately qualified tradespersons. The damaged building components shall be assessed by the Contracting Officer, with a determination made for either repair or replacement.

6. Contractor will be responsible for moving all property and refuse out of the Contractor's regulated areas.

C. Tasks

The work tasks are summarized as follows:

1. The Contractor will coordinate with the General Contractor for Rehabilitation and the Contracting Officer for scheduling access to the structures. The Contractor shall assume that the structures will be partially occupied during this project.

2. Pre-work activities including pre-work meeting(s), inspection(s), notifications, permits, submittal approvals, work-site preparations, emergency procedures arrangements, and standard operating procedures (SOPs) for LBP stabilization, component removal, soil removal, and cleaning work.

3. LBP-related work activities include stabilization, component removal, soil removal, packaging, clean-up, storing, and disposal of LBP waste, recordkeeping, security, monitoring, and inspections.

4. Access to the site will be restricted to the Contractor, Contractor's Subcontractors, Contracting Officer, Government Representative, and Architect. The work areas shall be demarcated in accordance with the OSHA requirements as a lead work area. Appropriate signage is discussed elsewhere in this specification.

5. The Contractor will prepare the work areas for stabilization, component removal, soil removal, cleaning, and backfill. The work area preparation measures are to be approved by the Contracting Officer before the commencement of work, but may include localized isolation of the area immediately adjacent to the stabilization, component removal, soil removal, cleaning, and backfill utilizing poly sheeting.

6. In the absence of a Negative Exposure Assessment as described in OSHA 1926.62, the Contractor shall utilize respiratory protection while in the regulated work areas or until sufficient air monitoring data has been collected for representative exposures resulting from stabilization, component removal, soil removal, cleaning, and backfill activities. In addition, the Contractor shall maintain the use of disposable coveralls, hand, boot, and head coverings. The Contractor shall also provide for worker/equipment decontamination areas that are within or contiguous to the regulated area.

7. The Contractor shall remove 6 inches of soil contaminated with loose lead-based paint chips and/or HEPA vacuum loose lead-based paint chips from soil around buildings. The extensive use of HEPA-equipped vacuums and wet methods is anticipated to reduce the generation of lead dust and migration of dust or paint chips beyond the localized isolation area.

8. At the conclusion of stabilization, component removal, soil removal, cleaning, and backfill, the adjacent surfaces are to be thoroughly cleaned with a water and TSP solution to remove any remaining loose dust or dirt.

9. Following the completion of the procedures outlined above, the Contractor will thoroughly clean the localized isolation areas and remove visible paint chips that may have been missed by HEPA vacuuming or wet wiping, and remove any poly sheeting.

10. Contractor will not backfill until all lead debris, unsatisfactory soil materials, obstructions, and deleterious materials have been removed from the designated areas. Use excavated materials or borrow for backfill, as applicable. Do not use unsuitable excavated materials. Do not backfill until final soil samples of excavated areas and new backfill have been analyzed for lead and inspected and approved by the Contracting Officer.

11. Following the soil lead abatement, the Contractor shall then proceed to backfill areas with non-contaminated topsoil and compact lightly. Topsoil shall be fertile, friable, natural topsoil of loamy character and characteristic of the locality. Topsoil shall be capable of growing healthy horticultural crops of grasses. Stockpile topsoil and protect as directed by the Contracting Officer. Eliminate foreign material, such as weeds, roots, stones, subsoil, frozen clods, and similar foreign materials, larger than 0.014 m3 (1/2 cubic foot) in volume, from soil as it is stockpiled. Remove foreign materials larger than 50 mm (2”) in any dimension from the topsoil used in the final grading. Topsoil work, such as stripping, stockpiling, and similar topsoil work, shall not, under any circumstances, be carried out when the soil is wet so that the tilth of the soil will be destroyed.

12. Contractor will use approved equipment (hand or mechanical) well-suited to the type of material being compacted. Do not operate mechanized vibratory compaction equipment within 3000 mm (10') of existing building walls without the prior approval of the Contracting Officer. Moisten or aerate material as necessary to provide the moisture content that will readily facilitate obtaining the specified compaction with the equipment used.

13. Contractor will slope backfill outside the buildings away from the building's walls at a minimum five percent (5%) slope.

14. Disposal of all materials removed from the site buildings shall be disposed of at a legally approved site, for the specific materials, and all removals shall be in accordance with all applicable Federal, State, and local regulations.

15. Contractor will install new commercial-grade landscape matting and no larger than 2-inch washed round landscape rock in designated areas.

16. Following the backfill of soil, new commercial-grade matting, and landscape rock, all porous and non-porous horizontal and vertical surfaces near the work areas will require a thorough re-cleaning.

17. Cleaning and decontamination activities are to include final visual inspections, wipe and soil sample collection and analysis, and certification of decontamination.

18. Backfill work activities are to include final visual inspections and certification of completion.

19. The Contractor shall be responsible for characterizing all waste during the project by Toxicity Characteristic Leachate Procedure (TCLP) analysis and proper disposal of the waste and associated materials generated by project activities.

D. Contractors Use of Premises

1. The Contractor and Contractor's personnel shall cooperate fully with the General Contractor for Rehabilitation and the Contracting Officer to facilitate efficient use of the site. The Contractor shall perform the lead work activities in accordance with the specifications and phasing plan and in compliance with any/all applicable Federal, State, and Local regulations and requirements.

2. The Contractor shall specify the facilities proposed to be utilized in the LBP stabilization, component removal, soil removal, cleaning, and backfill work plan. The Contractor shall use only the existing facilities in the structures strictly within the limits indicated in the approved LBP work plan. Any variation from the approved work plan shall be secured in writing by the Contracting Officer.

1.2 STOP LBP STABILIZATION, COMPONENT REMOVAL, AND SOIL REMOVAL

A. If the Contracting Officer or their designee presents a verbal **Stop Work Order**, the Contractor/Personnel shall immediately stop all LBP-related work activities within the regulated areas. If a verbal stop work order is issued, the Contracting Officer shall follow up with a written order to the Contractor as soon as it is practicable. The Contractor shall not resume any LBP-related work activities until authorized to do so in writing by the Contracting Officer. A stop work order may be issued at any time the Contracting Officer determines work conditions or activities are not within specification requirements, regulatory requirements, or that an imminent hazard exists to human health or the environment. The work stoppage will continue until conditions have been corrected to the satisfaction of the Contracting Officer. The occurrence of any of the following events shall be reported immediately by the Contractor and/or their field representative using the most expeditious means (e.g., verbal or telephonic), followed up with written notification to the Contracting Officer as soon as it is practical. The Contractor shall immediately stop LBP-related work activities and initiate lead hazard reduction activities if:

1. Visible evidence of airborne fugitive dust.

2. Contractor employees performing lead work activities without proper Personal Protective Equipment (PPE), in the absence of a Negative Exposure Assessment, which may be performed by the Contractor to eliminate or minimize PPE requirements.

3. Serious injury/death at the site.

4. Fire/safety emergency at the site.

5. Respiratory protection system failure.

6. Visual or analytical evidence of tracking of lead or lead-based paint from beyond the regulated areas.

7. Failure to follow project specification requirements.

1.3 DEFINITIONS

A. Definitions and explanations here are neither complete nor exclusive of all terms used in the contract documents, but are general for the work to the extent they are not stated more explicitly in another element of the contract documents. Drawings must be recognized as diagrammatic in nature and not completely descriptive of the requirements indicated therein.

B. Glossary

Definitions relative to LBP stabilization and component removal.

1. ACGIH - American Conference of Government Industrial Hygienists

2. Accessible surface - Any interior or exterior surface accessible for a young child to mouth or chew, such as sills and protruding surfaces.

3. Action Level - The point at which something needs to be done to correct or eliminate the presence of lead. The Action Level for airborne concentrations of lead is 30 $\mu g/cm^3$.

4. Aerosol - Solid or liquid particulate suspended in the air.

5. Aggressive method - Removal or disturbance of material by sanding, abrading, grinding, or other methods that break, crumble, or disintegrate intact lead components and elements.

6. Adequately wet - Sufficiently mixed or penetrated with liquid to prevent the release of particulates. If visible emissions are observed coming from the LBP building component, then that material has not been adequately wetted.

7. AIHA - American Industrial Hygiene Association

8. Air monitoring - The process of measuring the lead content of a known volume of air collected over a specified period of time.

9. Air monitoring firm - The firm retained by the Government to conduct air monitoring.

10. Air sample media - The filter is used to collect airborne dust, which is then analyzed for total lead concentration.

11. ANSI - American National Standards Institute

12. Amended Water - Water to which a surfactant (wetting agent) has been added to increase the penetrating ability of the liquid.

13. Authorized person - Any person authorized by the NPS, the Removal Contractor, or government agency and required by work duties to be present in regulated areas.

14. Authorized visitor - Any person approved by the NPS, the Removal Contractor, or any government agency having jurisdiction over the regulated area.

15. Barrier - Any surface that isolates the regulated area and inhibits dust migration from the regulated area.

16. Containment Barrier - An airtight barrier consisting of walls, floors, and/or ceilings of sealed plastic sheeting that surrounds and seals the outer perimeter of the regulated area.

17. Critical Barrier - The barrier responsible for isolating the regulated area from adjacent spaces, typically constructed of plastic sheeting secured in place at openings such as doors, windows, or any other opening into the regulated area.

18. Primary Barrier - Barriers placed over critical barriers and exposed directly to LBP stabilization and component removal work.

19. Secondary Barrier - Any additional sheeting used to isolate and provide protection from debris during LBP stabilization and component removal work.

20. Bias - The lead reading provided by any XRF on a lead-free surface, also called a Substrate Equivalent Lead (SEL).

21. Biennial report - A report (EPA Form 8700-13A) submitted to generators of hazardous waste to the Regional Administrator due March 1 of each even-numbered year. The report includes information on the generator's activities during the previous calendar year. The owner or operator of a treatment, storage, and disposal facility must also prepare and submit a biennial report using EPA Form 8700-1313.

22. Biological monitoring - The analysis of a person's blood and/or urine to determine the level of lead contamination in the body.

23. Blank - A non-exposed sample of the medium used for testing, such as a wipe or filter, which is analyzed like other samples to determine whether (1) samples are contaminated with lead before samples are collected (e.g., at the factory, or at the testing site, (2) the samples are contaminated after sample collection (e.g., during transportation to the laboratory or in the laboratory).

24. Breathing zone - The hemisphere forward of the shoulders with a radius of about 150–225 mm (6–9 inches) from the worker's nose.

25. Building/facility owner - The legal entity, including a lessee, which exercises control over management and recordkeeping functions relating to a building and/or facility in which lead activities take place.

26. Certified Industrial Hygienist (CIH) - One certified in the practice of industrial hygiene by the American Board of Industrial Hygiene (AIHA). An industrial hygienist Certified in Comprehensive Practice by the AIHA.

27. Characteristics - EPA has identified four characteristics of hazardous waste: Ignitability, Corrosivity, Reactivity, and Toxicity. Any solid waste that exhibits one or more of these characteristics is classified as hazardous waste under RCRA.

28. Clearance sample - The final sample is taken after all lead work has been done and visually inspected. Performed by the NPS industrial hygiene consultant (IHC).

29. Clearance standards - Lead levels at permissible levels at the conclusion of LBP stabilization and component removal projects and interim control interventions.

30. Common area - A room or area that is accessible to all tenants in a project (e.g., hallway). Generally, any area that is not kept locked.

31.

32. Competent person - One who can identify existing lead hazards in the workplace and select the appropriate control strategy for lead exposure, who has the authority to take prompt corrective measures to eliminate them. In addition, the individual is specially trained in a training course that meets the requirements of (40 CFR part 745) for a supervisor.

33. Contingency plan - A document setting out an organized, planned, and coordinated course of action to be followed in case of a fire or explosion, or a release of hazardous waste or hazardous waste constituents from a treatment, storage, or disposal facility that could threaten human health or the environment.

34. Contracting Officer - The NP official responsible for ongoing project work.

35. Crawlspace - An area that can be found either in or adjacent to the regulated area. This area has limited access and egress.

36. Demolition - The wrecking or taking out of any load-supporting structural member and any related razing.

37. Direct reading XRF - An analyzer that provides the operator with a display of a lead concentration calculated from the lead "K" x-ray intensity.

38. Discharge or hazardous waste discharge - The accidental or intentional spilling, hazardous waste leaking, pumping, pouring, emitting, discharging, emptying, or dumping of hazardous wastes onto any land or water.

39. Disposal - The discharge, deposit, injection, dumping, spilling, leaking, or placing of any solid waste or hazardous waste into or on any land or water so that any constituent thereof may enter the environment or be emitted into the air or discharged in any water, including groundwater.

40. Disposal bag - Typically a 6-mil thick sift-proof, dustproof, leak-tight container used to package and transport lead waste from regulated areas to the approved landfill. Each bag or container must be labeled and marked in accordance with EPA, OSHA, and DOT requirements.

41. Disposal facility - A facility or part of a facility at which hazardous waste is intentionally placed into or on any land or water, and at which waste will remain after closure.

42. Drum - A rigid, impermeable container made of cardboard fiber, plastic, or metal that can be sealed in order to be sift-proof, dustproof, and leak-tight.

43. Dust standards (EPA) - The present LBP standards are window sills, 40 $\mu g/ft^2$; window troughs, 100 $\mu g/ft^2$; bare floors, 5 $\mu g/ft^2$ based on wipe tests.

44. Elevated blood lead level - In children, any blood lead level greater than 10 micrograms per deciliter (µg/dl). In adults, any blood lead level greater than 25 µg/dl.

45. Employee exposure - The exposure to lead that would occur if the employee were not wearing respiratory protection equipment.

46. Encapsulation - Involves resurfacing or covering surfaces, and sealing or caulking with durable materials, to prevent or control chalking, flaking lead-containing substances from becoming part of house dust or accessible to children. There are currently no HUD or ASTM-approved encapsulants.

47. Enclosure - The use of rigid, durable construction materials that are mechanically fastened to the substrate to act as a barrier between LBP and the living space. Drywall, paneling, and metal siding are just a few examples of enclosure materials.

48. Engineering controls - Measures implemented at the work site to contain, control, and/or otherwise reduce exposure to lead dust and debris.

49. EPA - Environmental Protection Agency

50. Exposure monitoring - The personal air monitoring of an employee's breathing zone to determine the amount of contaminant (e.g., lead) to which they are exposed.

51. Filter - Media used in respirators, vacuums, or other machines to remove particulate from the air.

52. Final inspection - Inspection by a qualified inspector, industrial hygienist, or local public health official to determine whether LBP stabilization, component removal, and cleaning activities are complete.

53. Firestopping - Material used to close the open parts of a structure to prevent a fire from spreading.

54. Friction Surface - Any interior or exterior surface subject to abrasion or friction, such as windows or stair treads.

55. Generator - Any person who first creates a hazardous waste or any person who first makes the waste subject to the Subtitle C regulation (e.g., imports a hazardous waste, initiates a shipment of hazardous waste from a TSD, or mixes hazardous wastes of different DOT shipping descriptions by placing them into a single container).

56. Government industrial hygienist (IHC) – The government retained an Industrial Hygiene Consultant.

57. Hazardous Materials Control - Incidental work procedures for control of releases of project-related hazardous materials, including containment, enclosure, wetting, controlled renovations and demolition procedures, removal, storage, and disposal.

58. Hazardous Waste -

    a. Waste material, including asbestos, loose and peeling LBP, PCB ballasts, and any other material which requires management, handling, transport, treatment, storage, or disposal according to the requirements of the Federal Resource, Conservation and Recovery Act (RCRA) and associated regulation 42 U.S.C. 6901 et seq. and 40 CFR Part 260 et seq.

    b. References to hazardous material or contaminated material incorporate definitions of hazardous pollutants, hazardous contaminants, hazardous material, hazardous substance, hazardous waste, toxic pollutants, and toxic substance applicable in accordance with federal, state, regional, and local statutes, laws, regulations, and policies.

59. High efficiency particulate air (HEPA) filter - A filter capable of filtering out particles of 0.3 microns or greater from a body of air at 99.97% efficiency or greater microns or greater from a body of air at 99.97% efficiency or greater.

60. HEPA vacuum - Vacuum collection equipment equipped with a HEPA filter system capable of collecting and retaining lead in dust or air.

61. High phosphate detergent - Detergent that contains at least 5% Tri-Sodium Phosphate (TSP).

62. Homogeneous area - A grouping of painted building components that are uniform in color, shape, and date of application.

63. Heating, Ventilation, and Air Conditioning (HVAC)

64. Impact surface - Any interior or exterior surface subject to damage by repeated impacts, such as the surface on doors and jambs.

65. Industrial hygienist (IH) - A professional qualified by education, training, and experience to anticipate, recognize, evaluate, and develop controls for occupational health hazards and meets the definition requirements of AIHA.

66. Industrial hygienist technician (IH Technician) - A person working under the direction of an IH or CIH who has special training, experience, certifications, and licenses required for the industrial hygiene work assigned.

67. Initial survey - A systematic inspection of a building by a qualified inspector, using a portable XRF analyzer, atomic absorption spectroscopy, or other approved testing techniques, to determine whether an LBP hazard is present.

68. In-place management - See interim controls.

69. Inspection - A surface-by-surface investigation to determine the presence of LBP (and in some cases, sampling for lead in dust and soil) and a report of the results.

70. Interim controls - Any set of measures designed to reduce temporary human exposure or likely exposure to LBP hazards, including specialized cleaning, repairs, maintenance, painting, temporary containment, management, and resident education programs. Monitoring by owners and reevaluation by professionals of lead exposures is an integral element of interim controls. Interim controls include dust removal, paint film stabilization, friction, and impact surface treatments, and installation of soil coverings (such as grass or sod) or land use controls.

71. Lead-Based Paint (LBP)- Any paint, varnish, shellac, or other coating surfaces with lead content that exceeds 1.0 mg/cm^2 as measured by x-ray fluorescence (XRF) detector or laboratory analysis or 0.5 percent weight (5,000 ppm, 5,000 mg/kg, 5,000 µg/g) by laboratory analysis.

72. LBP Hazard - Any condition that causes exposure to lead from contaminated soil, LBP that is deteriorated or present on accessible surfaces, friction surfaces, or impact surfaces that would result in adverse human health effects as established by the EPA administrator under Title IV of the Toxic Substance Control Act.

73. Lead-contaminated soil - Bare soil on residential property that contains lead in excess of the standard established by the EPA Administrator pursuant to Title IV of TSCA. The interim HUD recommendation is 400 ppm.

74. Lead safe - Refers to buildings that meet all the following criteria:

    a. Leaded dust levels on floors, interior windowsills, and exterior windowsills are below HUD clearance standards or EPA health-based standards.

    b. Bare soil lead levels in exterior play areas are below EPA health-based standards.

    c. No deteriorated LBP or no deteriorated paint that is suspected to contain lead is present on any surface inside or outside of the building.

    d. A plan to ensure that intact LBP or suspected LBP does not become a lead hazard during maintenance and renovation activities; and

    e. Periodic surveillance to ensure that these criteria are met for a specific time period.

75. Lead-free building - A lead-free building contains no LBP and has interior dust and exterior soil levels below applicable HUD and EPA standards.

76. Manifest - The shipping document, EPA Form 8700-22, used for identifying the quantity, composition, origin, routing, and destination of hazardous waste during its transportation from the point of generation to the point of treatment, storage, or disposal.

77. Microgram – One-millionth of a gram - 453 grams in a pound, 28,310,000 micrograms (µ) in one ounce.

78. NPS industrial hygienist (IHC) – The NPS retained an Industrial Hygiene Consultant.

79. Negative initial exposure assessment - A demonstration by the employer that complies with the criteria in 29 CFR 1926.1101 (f) (2) (iii), that employee exposure during an operation is expected to be consistently below the PELs.

80. Negative pressure - Air pressure, which is lower than the surrounding area, is created by exhausting air from a sealed, regulated area through HEPA-equipped filtration units. OSHA requires maintaining a 5.0 Pa (-0.02") water gauge inside the negative pressure enclosure.

81. Negative pressure respirator - A respirator in which the air pressure inside the facepiece is negative during inhalation relative to the air outside the respirator.

82. Organic vapor cartridge - The type of cartridge used on air purifying respirators for organic vapor exposures.

83. OSHA – Occupational Safety and Health Administration.

84. Outside air - The air outside buildings and structures, including, but not limited to, the air under a bridge or in an open ferry dock.

85. Owner/operator - Any person who owns, leases, operates, controls, or supervises the facility being demolished or renovated, or any person who owns, leases, operates, controls, or supervises the demolition or renovation operation, or both.

86. Paint inspections - Reveal which specific surfaces in a building are coated with LBP, regardless of their condition.

87. Paint removal - A strategy of stabilizing loose or damaged paint or abatement which entails removing LBP from surfaces of components using chemicals and certain contained abrasive methods, but not open flame burning, open abrasive blasting, sandblasting, water blasting, or extensive dry scraping (methylene chloride paint removers are also not recommended).

88. Pattern - The process of identifying specific components containing LBP at a hazardous level.

89. Personal sampling/monitoring - Representative air samples were obtained in the breathing zone of the person using a cassette and battery-operated pump to determine LBP exposure.

90. Permissible exposure limit (PEL) - The level of exposure OSHA allows for an 8-hour time-weighted average. For LBP, the PEL is 50 $\mu g/cm^3$.

91. Permit - An authorization, license, or equivalent control document issued by the EPA or an authorized State to implement the regulatory requirements of Subtitle C Parts 264 and 265 for TSDs.

92. Personal protective equipment (PPE) - Equipment designed to protect users from injury and/or specific job hazards. Such equipment may include protective clothing, hard hats, safety glasses, and respirators.

93. Personnel samples (for sampling lead dust) - Air samples collected from within the breathing zone of a worker, but outside the respirator. The samples are collected with a personal sampling pump, pulling 1 to 4 liters/minute of air.

94. Polyethylene sheeting - Strong plastic barrier material 4–6 millimeters (mil) thick, semitransparent, sometimes flame-retardant, complies with NFPA 241.

95. Positive/negative fit check - A method of verifying the fit of a respirator by closing off the filters and breathing in or closing off the exhalation valve and breathing out while detecting leakage of the respirator.

96. Protection factor - A Value assigned by OSHA/NIOSH to indicate the assigned protection a respirator should provide if worn properly. The number indicates the reduction of exposure level from outside to inside the respirator.

97. Precision - The degree of repeatability of a series of successive measurements.

98. Qualitative fit test (QLFT) - A fit test using a challenging material that can be sensed by the wearer if a leakage in the respirator occurs.

99. Quantitative fit test (QNFT) - A fit test using a challenging material that is quantified outside and inside the respirator, thus allowing the determination of the actual fit factor.

100. Reading cycle - Direct reading XRF analyzers calculate the lead "K" x-ray intensity in a specific time interval (10 to 30 seconds), which is fixed by the manufacturer and related to the age of the source. The calculated result in this time interval is a reading cycle. To determine a concentration of lead (ALC or SEL), the displayed results of a minimum of three single reading cycles must be averaged. The difference between the lowest and highest reading from at least three reading cycles must be less than 1.7 mg/cm^2.

101. Reevaluation - A visual assessment and collection of environmental samples by a certified risk assessor or certified inspector to determine if a lead hazard control measure previously implemented is still effective and if the building is still lead-safe.

102. Regulated area - An area established by the employer to demarcate where lead-related work activities are conducted, any adjoining area where debris and waste from such lead work may accumulate, and a work area within which airborne or dust concentrations of lead exceed, or there is a reasonable possibility they may exceed, the PEL.

103. Removal - All operations where LBP is taken out from structures or substrates, including demolition operations.

104. Renovation - Altering a facility, or one or more facility components in any way, including the stripping or removal of LBP from a facility component that does not involve demolition activity.

105. Repair - Overhauling, rebuilding, reconstructing, or reconditioning of structures or substrates, including repair of LBP applied to building components or substrates.

106. Replacement - A strategy of abatement that entails the removal of components such as windows, doors, and trim that have LBP surfaces and installing new components free of LBP.

107. Representative sample - A sample of a universe or whole (e.g., waste sample pile, lagoon, groundwater, or waste stream) which can be expected to exhibit the average properties of the universe or whole.

108. Risk Assessor - A certified individual who has completed training from an accredited training program and has been licensed to 1) perform risk assessments, 2) identify acceptable removal and interim control strategies for reducing any LBP hazards identified, 3) perform clearance testing and reevaluations, and 4) document successful completion of lead hazard control activities.

109. Sample site - A specific spot on a surface being tested for lead concentration through portable XRF or laboratory analysis.

110. SEL or Substrate Equivalent Lead Concentration - The average of at least three XRF single-cycle readings on an unpainted surface.

111. Small quantity - A generator that produces less than 100 kg of hazardous waste per month (or accumulates less than 100 kg at any one time) or one that produces less than 1 kg of acutely hazardous waste per month (or accumulates less than 1 kg of acutely hazardous waste at any one time).

112. Sodium sulfide - A chemical used to test a paint sample qualitatively for lead; typical concentrations are from 6% to 10%. A positive test is characterized by a gray, black, or other dark discoloration of the paint film cross-section.

113. Solid waste - Defined in RCRA the term "solid waste" means any garbage, refuse, sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility and other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities, but does not include solid or dissolved material in domestic sewage, or solid or dissolved materials in irrigation return flows or industrial discharges which are point sources subject to permits under the Clean Water Act, or special nuclear or byproduct material as defined by the Atomic Energy Act of 1954.

114. Standard - Used in two ways in this specification - (a) levels established by law or regulation, such as 1.0 mg/cm^2, (b) materials to which known quantities of lead have been applied; used to evaluate the accuracy and performance of the XRF analyzer, usually called Standard Reference Materials.

115. Standard operating procedures (SOPs) - LBP work procedures are required to be submitted by the Contractor before work begins.

116. Storage - The holding of hazardous waste for a temporary period, at the end of which the hazardous waste is treated, disposed of, or stored elsewhere.

117. Substrate - A surface upon which paint or varnish has been or may be applied. Examples of substrates include wood, plaster, metal, and drywall.

118. Substrate effect - The return of backscattered radiation from the paint, substrate, or underlying material to the XRF analyzer. This radiation, when counted as lead X-rays by an XRF, contributes to SEL or bias. The inspector may have to compensate for this effect when using direct reading XRF analyzers.

119. Surfactant - A chemical added to water to decrease water's surface tension, thus making it more penetrating.

120. Supplied air respirator (SAR) - A respirator that utilizes an air supply separate from the air in the regulated area.

121. TCLP - Toxic Characteristic Leaching Procedures, see EPA: Toxicity Test.

122. Transporter - Any person engaged in the off-site transportation of hazardous waste within the United States, by air, rail, highway, or water, if such transportation requires a manifest under 40 CFR Part 262.

123. µg - Micrograms: the prefix "micro" means "1/1,000,000 of" (one-millionth of). A microgram is 1/1,000,000 of a gram and 1/1000 of a milligram. A microgram is equal to about 35/1,000,000,000 (thirty-five billionths) of an ounce. 28,400,000 µg is equal to 1 ounce.

124. USDOT - United States Department of Transportation.

125. Visible emissions - Any emissions that are visually detectable without the aid of instruments come from LBP building components or cleaning operations.

126. Waste generator - Any owner or operator whose act or process produces LBP waste material.

127. Waste shipment record - The shipping document, required to be originated and signed by the waste generator, is used to track and substantiate the disposition of LBP waste material.

128. Work area – An OSHA-defined term applying specifically to potential occupational exposure to lead on this project. The work area for this project includes the interior and exterior portions of the building during the LBP interim stabilization, component removal, and cleaning portion of the project. It also includes any areas adjacent to the building which the Contractor is expected to have under their control.

129. XRF analyzer - An instrument that determines lead concentration in milligrams per square centimeter (mg/cm^2) using the principle of X-ray fluorescence. Two types of XRF analyzers are used: direct readers and spectrum analyzers.

C. Referenced Standards Organizations

1. The following acronyms or abbreviations, as referenced in contract/specification documents, are defined to mean the associated names. Names and addresses may be subject to change.

a. ABIH American Board of Industrial Hygiene
6015 West St. Joseph, Suite 102
Lansing, MI 48917-3980
Telephone: 517-321-2638

b. AIHA American Industrial Hygiene Association
2700 Prosperity Avenue, Suite 250
Fairfax, VA 22031
Telephone: 703-849-8888

c. ANSI American National Standards Institute
1430 Broadway
New York, NY 10018
Telephone: 212-354-3300

d. ASTM American Society for Testing and Materials
1916 Race St.
Philadelphia, PA 19103
Telephone: 215-299-5400

e. CFR Code of Federal Regulations
Government Printing Office
Washington, DC 20420

f. CGA Compressed Gas Association
1235 Jefferson Davis Highway
Arlington, VA 22202
Telephone: 703-979-0900

g. CS Commercial Standard of the National Institute of Standards and Technology (NIST)
U. S. Department of Commerce
Government Printing Office
Washington, DC 20420

h. Department of Housing and Urban Development
451 Seventh Street Southwest
Washington, DC 20410

i. EPA Environmental Protection Agency
401 M St., SW
Washington, DC 20460
Telephone: 202-382-3949

j. MSHA Mine Safety and Health Administration
Respiratory Protection Division
Ballston Tower #3
Department of Labor
Arlington, VA 22203
Telephone: 703-235-1452

k. NIST National Institute for Standards and Technology
U. S. Department of Commerce
Gaithersburg, MD 20234
Telephone: 301-921-1000

l. NEC National Electrical Code (by NFPA)

m. NEMA National Electrical Manufacturer's Association
2101 L Street, N.W.
Washington, DC 20037

n. NFPA National Fire Protection Association
1 Batterymarch Park
P.O. Box 9101
Quincy, MA 02269-9101
Telephone: 1-800-344-3555

o. NIOSH National Institutes for Occupational Safety and Health
4676 Columbia Parkway
Cincinnati, OH 45226
Telephone: 513-533-8236

p. OSHA Occupational Safety and Health Administration
U.S. Department of Labor
Government Printing Office
Washington, DC 20402

q. UL Underwriters Laboratory
333 Pfingsten Rd.
Northbrook, IL 60062

1.4 APPLICABLE CODES AND REGULATIONS

A. General Applicability of Codes, Regulations, and Standards

1. All work under this contract shall be done in strict accordance with all applicable Federal, State, and local regulations, standards, and codes governing LBP-related work activities and any other trade work done in conjunction with the LBP-related work activities. All applicable codes, regulations, and standards are adopted into this specification and will have the same force and effect as this specification.

2. The most recent edition of any relevant regulation, standard, document, or code shall be in effect. Where conflict among the requirements or with these specifications exists, the most stringent requirement(s) shall be utilized.

3. Copies of all standards, regulations, codes, and other applicable documents, including this specification and those listed in Section 1.4, shall be available at the worksite. These standards, regulations, codes, and other applicable documents, including this specification and those listed in Section 1.4, may be made available electronically.

B. Contractor Responsibility

1. The Contractor shall assume full responsibility and liability for compliance with all applicable Federal, State, and Local regulations related to all aspects of the LBP-related work activities. The Contractor is responsible for providing and maintaining training, accreditations, medical exams, medical records, personal protective equipment (PPE), including respiratory protection, including respirator fit testing, as required by applicable Federal, State, and Local regulations. The Contractor shall hold the Government, NPS, Architect, and IHC harmless for any failure to comply with any applicable work, packaging, transporting, disposal, safety, health, or environmental requirement on the part of himself, his employees, or his subcontractors. The Contractor will incur all costs of the IHC, including all sampling/analytical costs, to ensure compliance with OSHA/EPA/State requirements related to failure to comply with the regulations applicable to the work.

C. Federal Requirements

1. Federal requirements that govern some aspects of stabilization and component removal of LBP include, but are not limited to, the following regulations.

a. OSHA

i. Title 29 CFR 1910.134 - Respiratory Protection

ii. Title 29 CFR 1910.1025 – General Industry Standard for Lead

iii. Title 29 CFR 1910.1200 - Hazard Communication

iv. 29 CFR 1926 - Construction Industry

v. Title 29 CFR 1926.62 – Construction Standard for Lead

vi. Title 29 CFR 1926.500 - Guardrails, Handrails, and Covers

vii. Title 29 CFR 1926 Subpart E - Personal Protective Equipment and Life Saving Equipment

b. EPA

i. 42 U.S.C S6901 et. Seq. – Resource Conservation and Recovery Act (RCRA) of 1976

ii. 40 CFR 745 – Lead, Requirements for Lead-Based Paint Activities in Target Housing & Child-Occupied Facilities

c. USDOT

i. Title 49 CFR 100 - 185 - Transportation

D. State Requirements

1. Adhere to LBP interim stabilization and component removal, transportation, and disposal requirements as mandated by the Environmental Protection Agency (EPA). These may include, but are not limited to, licensing, regulations, training, handling, and disposal requirements. Notify the proper authorities of this project, including the EPA, and follow any specific requirements imposed. Ensure that labeling containers are acceptable to the regulatory authorities of the EPA. At this time, the State of Wyoming has not adopted a formal program for licensing, permitting, or certification of workers, supervisors, or firms for LBP-related work activities. Until such a program is adopted, the enforcement of the regulations is the burden of the EPA. EPA requirements that apply to LBP-related work activities can be obtained from:

Environmental Protection Agency
Mail Stop 8AT-AP
999 18th Street, Suite 500
Denver, Colorado 80202

E. Standards

1. Standards that govern stabilization and component removal of LBP activities include, but are not limited to, the following:

a. American National Standards Institute (ANSI)/ASSP Z9.2-2018 - Fundamentals Governing the Design and Operation of Local Exhaust Systems and ANSI/ASSE Z88.2-2015 - Practices for Respiratory Protection.

b. ANSI Z41.1 - Safety Toe Footwear

c. ANSI Z87.1 - Practice for Occupational and Educational Eye and Face Protection

d. ANSI X88.6 - Respiratory Protection - Respiratory Use Physical Qualifications for Personnel

e. ANSI Z89.1 - Requirements for Industrial Head Protection

f. Underwriters Laboratories (UL) 586-2009 - UL Standard for Safety of HEPA Filter Units, 9th Edition; ANSI Approval 2017-12-19.

2. Standards governing testing laboratories:

a. American Industrial Hygiene Association (AIHA)

b. Air, paint chip, wipe, and soil sample analysis – NLLAP Accredited

3. Standards that govern the fire and safety concerns in the stabilization, component removal, and cleaning work areas include, but are not limited to, the following:

a. National Fire Protection Association (NFPA) 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations.

b. National Fire Protection Association (NFPA) 10 - Standard for Fire Extinguishers

c. NFPA 70 - Standard for National Electric Code

d. NFPA 101 - Life Safety Code

e. NFPA 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations

f. NFPA 701 - Standard Methods for Fire Tests for Flame-Resistant Textiles and Film.

g. Uniform Building Code (UBC) - 2006 Edition

F. EPA Guidance Documents

1. Lead, Requirements for Lead-Based Paint Activities in Target Housing and Child-Occupied Facilities; Final Rule EPA-745-OPPTS-62128B; FRL-5389-9

2. Protect Your Family from Lead in Your Home 055-000-0507-9

3. Lead Poisoning and Your Children 055-000-00414-5

G. Notices

1. State and Local agencies: Send written notification as required by state and local regulations, including the local fire department, before beginning any work on LBP.

2. Copies of notifications shall be submitted to the Contracting Officer for the facility's records in the same time frame as notification is given to EPA, State, and Local authorities.

H. Permits/Licenses/Fees

1. The Contractor shall apply for and have all required permits and licenses to perform stabilization and component removal of LBP work, packaging, storing, transportation, disposal, and provide timely notification of such actions as may be required by Federal, State, and Local regulations.

2. The Contractor shall be responsible for all applicable fees associated with permits and licenses.

3. The Contractor shall be responsible for all applicable fees associated with patent(s).

I. Posting and Filing of Regulations

1. Maintain two (2) copies of applicable federal, state, and local regulations. Post one copy of each at the regulated area where workers will have daily access to the regulations, and keep another copy in the Contractor's office.

J. NPS Responsibilities

1. Before the commencement of work:

a. Contracting Officer will notify others of project dates and requirements for relocation if needed.

b. Submit to the Contractor results of the previous lead inspection report of findings, including baseline wipe and soil sampling information; location of samples, the person who collected the samples, equipment utilized, and method of analysis. During the LBP stabilization, component removal, and cleaning operation, submit to the Contractor the results of the clearance wipe and soil sample results collected at the conclusion of the LBP stabilization, component removal, soil removal, and cleaning operation. This information shall not release the Contractor from any responsibility for OSHA compliance, including personnel sample collection and analysis for lead exposure.

1.5 EMERGENCY ACTION PLAN AND ARRANGEMENTS

A. An Emergency Action Plan shall be developed before commencing lead activities and shall be agreed to by the Contractor and the Contracting Officer. The Plan shall meet the requirements of 29 CFR 1926, Subpart C, Standard 1926.35 Employee Emergency Action Plans.

B. Emergency procedures shall be in written form and prominently posted near the regulated area. Everyone, prior to entering the regulated area, must read and sign these procedures to acknowledge understanding of the regulated area layout, location of emergency exits, and emergency procedures.

C. Emergency planning shall include written notification of police, fire, and emergency medical personnel of planned stabilization, component removal, and cleaning activities; work schedule and layout of the regulated area, particularly barriers that may affect response capabilities.

D. Emergency planning shall include consideration of fire, explosion, hazardous atmospheres, electrical hazards, slips/trips and falls, confined spaces, and heat stress illness. Written procedures for response to emergency situations shall be developed, and employee training in procedures shall be provided.

E. Employees shall be trained in regulated area/site evacuation procedures in the event of workplace emergencies.

1. For non-life-threatening situations - employees injured or otherwise incapacitated shall decontaminate following normal procedures with assistance from fellow workers, if necessary, before exiting the regulated area to obtain proper medical treatment.

2. For life-threatening injury or illness, worker decontamination shall take the least priority after measures to stabilize the injured worker, remove them from the regulated area, and secure proper medical treatment.

F. Telephone numbers of all emergency response personnel shall be kept at the job site along with the location of the nearest telephone.

G. The Contractor shall verify first aid/CPR training for personnel responsible for providing the training. OSHA requires medical assistance within 3-4 minutes of a life-threatening injury/illness. Bloodborne pathogen training shall also be verified for those personnel required to provide first aid/CPR.

H. The Emergency Action Plan shall provide a Contingency Plan if an incident occurs that may require the modification of the SOP during stabilization, component removal, soil removal, and cleaning activities. Such incidents include, but are not limited to, fire, accident, power failure, and negative pressure failure. The Contractor shall detail procedures to be followed in the event of an incident, assuring that work is stopped, and wetting is continued until correction of the problem.

1.6 PRE-CONSTRUCTION MEETING

A. Before commencing the LBP-related work activities, the Contractor shall meet with the Contracting Officer to present and review, as appropriate, the items following this paragraph. The Contractor's Competent Person(s) who will be on-site shall participate in the pre-construction meeting. The pre-construction meeting is to discuss and determine procedures to be used during the project. At this meeting, the Contractor shall provide four copies of:

1. Proof of Contractor licensing.

2. Proof the Competent Person is trained, accredited, and certified for working in this State. Verification of the experience of the Competent Person and proof of certification shall also be presented.

3. A list of all workers who will participate in the project, including experience and verification of training and accreditation.

4. A list of and verification of training for all personnel who have current First Aid/ CPR training. A minimum of one person per shift must have adequate training.

5. Current medical written opinions for all personnel working on-site meeting the requirements of 29 CFR 1926.1101 (m).

6. Current fit tests for all personnel wearing respirators on-site.

7. A copy of the Contractor's SOPs for LBP stabilization and component removal. In these procedures, the following information must be detailed, specifically for this project.

    a. Regulated area preparation procedures.

    b. Notification requirements procedure of the Contractor.

    c. Stabilization and component removal methods/procedures and equipment to be used; and

    d. PPE to be used.

    e. If required, decontamination area set-up/layout and decontamination procedures for employees; extent of 6-mil polyethylene sheeting placed 25 feet out from the outer perimeter of the work area to aid in cleaning lead material debris that may fall off building; method of demarcating the regulated work area with Lead Danger Tape and Lead Danger Signs; method for fall protection; method for accessing lead LBP components and method for bringing material to lined roll-off or approved equivalent.

8. At this meeting, the Contractor shall provide all submittals as required.

9. Procedures for handling, packaging, storage, and disposal of lead waste.

10. Emergency Action Plan and Contingency Procedures.

1.7 PROJECT COORDINATION

The following are the minimum administrative and supervisory personnel necessary for the coordination of the work.

A. Personnel

1. Administrative and supervisory personnel shall consist of a qualified Competent Person as defined by OSHA in the Construction Standards and the Lead in Construction Standard and EPA Lead-Based Paint Poisoning Prevention in Certain Residential Structures. These employees are the Contractor's representatives responsible for compliance with these specifications and all other applicable requirements.

2. Non-supervisory personnel shall consist of an adequate number of qualified personnel to meet the schedule requirements of the project. Personnel shall meet the required qualifications. Personnel utilized on-site shall be pre-approved by the Contracting Officer. A request for approval shall be submitted for any person to be employed during the project, giving the person's name, qualifications, Certificate of Worker's Acknowledgment, Affidavit of Medical Surveillance and Respiratory Protection and current Respirator Fit Test.

3. Minimum qualifications for the Contractor and assigned personnel are:

   a. The Contractor has conducted within the last three (3) years, three (3) projects of similar complexity and dollar value as this project; has adequate liability/occurrence insurance for LBP work as required by the State; is licensed in applicable State; has adequate and qualified personnel available to complete the work; has comprehensive standard operating procedures for LBP work; and has adequate materials, equipment and, supplies to perform the work.

   b. The Competent Person has four (4) years of LBP experience, of which two (2) years were as the Competent Person on the project; has been the Competent Person on two projects of similar size and complexity as this project; has completed the EPA RRP training and has all required certification and OSHA lead in construction training; meets the OSHA definition of a Competent Person; and has all required OSHA documentation related to medical and respiratory protection.

   c. Contractor Personnel shall have completed the OSHA lead in construction training; have training on the SOPs of the Contractor; have one (1) year of LBP-related work experience within the past three (3) years of similar size and complexity; have applicable medical and respiratory protection documentation; has a certificate of training.

   All personnel should be in compliance with EPA RRP training and OSHA construction safety training as applicable and submit certification documents.

1.8 RESPIRATORY PROTECTION

A. General – Respiratory Protection Program

1. The Contractor shall develop and implement a Respiratory Protection Program (RPP) which complies with OSHA requirements found at 29 CFR 1926.1101 and 29 CFR 1910.132;134. ANSI Standard Z88.2-2015 provides excellent guidance for developing a respiratory protection program. All respirators used must be NIOSH-approved for asbestos abatement activities. The written respiratory protection shall, at a minimum, contain the basic requirements found at 29 CFR 1910.134 (c) - Respiratory Protection Program.

B. Respiratory Protection Program Coordinator

1. The Respiratory Protection Program Coordinator (RPPC) must be identified and shall have two (2) years of experience coordinating RPP of a similar size and complexity. The RPPC must submit a signed statement attesting to the fact that the program meets the above requirements.

C. Selection and Use of Respirators

1. The procedure for the selection and use of respirators must be submitted to the NPS as part of the Contractor's qualification. The procedure must be written clearly enough for workers to understand. A copy of the Respiratory Protection Program must be available on site, for reference by employees or authorized visitors.

D. Minimum Respiratory Protection

1. Minimum respiratory protection shall be a half-face air purifying respirator equipped with P100 filters, and an Organic Vapor Cartridge shall be required when lead dust levels are maintained consistently at or below 30 $\mu g/cm^3$ determined as an 8-hour TWA. A higher level of respiratory protection shall be required if lead dust levels exceed 30 $\mu g/cm^3$, determined as an 8-hour TWA. Respirator selection shall meet the requirements of 29 CFR 1926.1101 (h) and 29 CFR 1910.134 (d)(3)(i)(A) Table 1, except as indicated in this paragraph. The Contractor personnel must have a respirator for their exclusive use.

E. Medical Written Opinion

1. No employee shall be allowed to wear a respirator unless a physician or other licensed health care professional has provided a written determination stating they are medically qualified to wear the class of respirator to be used on the project while wearing whole body impermeable garments and subjected to heat or cold stress.

F. Respirator Fit Test

1. All personnel wearing respirators shall have a current qualitative/quantitative fit test conducted in accordance with 29 CFR 1910.134 (f) and Appendix A. Quantitative fit tests shall be done for PAPRs that have been put into a motor/blower failure mode.

G. Respirator Fit Check

1. The Competent Person shall ensure that the positive/negative pressure user seal check is done each time the respirator is donned by an employee. Head coverings must cover respirator head straps. Any situation that prevents an effective facepiece-to-face seal, as evidenced by the failure of a user seal check, shall preclude that person from entering the regulated area until resolution of the problem.

H. Maintenance and Care of Respirators

1. The Respiratory Protection Program Coordinator shall submit evidence and documentation showing compliance with 29 CFR 1910.134 (h) Maintenance and Care of Respirators.

1.9 WORKER PROTECTION

A. Training of Removal Personnel

1. Before beginning any LBP-related work activity, all personnel shall be trained in accordance with OSHA 29 CFR 1926.21 subpart C and any additional Federal/State/Local requirements. Training shall have been conducted by a third-party, EPA/State-approved trainer meeting the requirements of EPA 40 CFR 745. Initial training certificates, current refresher, and accreditation proof must be submitted for each person working at the site.

B. Medical Examinations

1. Medical examinations shall be provided for all personnel working in the regulated area, regardless of exposure levels. The physician's written opinion shall be provided for each person and shall include a statement indicating that the person has been evaluated for working in a heat and cold stress environment while wearing PPE and can perform the work without risk of material health impairment.

C. Personal Protective Clothing

1. Provide whole body clothing, head coverings, foot coverings, and any other personal protective equipment as determined by conducting the hazard assessment required by OSHA at 29 CFR 1910.132 (d). The Competent Person shall ensure the integrity of personal protective equipment worn for the duration of the project. Duct tape shall be used to secure all suit sleeves to wrists and to secure foot coverings at the ankle.

D. Regulated Area Entry Procedure

1. The Competent Person shall ensure that each time workers enter the regulated area, they place protective disposable coveralls over ALL street clothes in the area established by the competent person and put on head coverings, fall protection, PPE, a clean respirator, and then proceed to the regulated work area.

E. Decontamination Procedure

1. The Competent Person shall require all personnel to adhere to the following decontamination procedures whenever they leave the regulated area.

   a. When exiting the regulated area, HEPA vacuum contaminated work clothing while it is still being worn. At no time may lead be removed from protective clothing by any means that would result in uncontrolled dispersal of lead dust into the air. Remove shoe coverings and leave them in the decontamination area. Remove protective coveralls by carefully rolling down the garment to reduce exposure to lead dust.

   b. Still wearing the respirator, care must be taken to follow reasonable procedures in decontamination and removing the respirator to avoid the inhalation or ingestion of lead dust. The following procedure is required, as a minimum:

      i. Thoroughly HEPA vacuum body from neck down. Vacuum hair as thoroughly as possible.

      ii. With the respirator still in place, wash hands, exposed areas of the face, and hair with soap and water. Carefully remove the respirator while proceeding out of the decontamination area.

      iii. Dispose of wetted or overloaded filters from the respirator.

      iv. Carefully decontaminate the face piece of the respirator inside and out. If using a PAPR, shut down using the following sequence: a) first cap inlets to filters; b) turn blower off to keep debris collected on the inlet side of the filter from dislodging and contaminating the outside of the unit; c) thoroughly decontaminate blower and hoses; d) carefully decontaminate battery pack with a wet rag being cautious of getting water in the battery pack thus preventing destruction. (Note: THIS PROCEDURE IS NOT A SUBSTITUTE FOR RESPIRATOR CLEANING!).

      v. Shower and wash body completely with soap and water. Rinse thoroughly.

      vi. Carefully decontaminate the respirator face piece and respirator inside and out. (NOTE: THIS IS NOT A SUBSTITUTE FOR RESPIRATOR CLEANING!).

   c. Where containment is not required for LBP removal, an adequate washing station will be provided for the employees for the removal of disposable PPE and to clean the respirator.

F. Regulated Area Requirements

1. The Competent Person shall ensure that all requirements for regulated areas are met. All personnel in the regulated area shall not be allowed to eat, drink, smoke, chew tobacco or gum, apply cosmetics, or in any way interfere with the fit of their respirator.

1.10 DECONTAMINATION FACILITIES

A. Description

1. Provide each regulated area with a fiber drum with a disposal bag in it for personnel waste materials. The Competent Person shall provide a specific area at the outer boundary of the regulated area and designate it specifically for the decontamination of personnel and respirators.

B. Waste/Equipment Decontamination Procedures

1. All personnel entering or exiting a regulated area shall follow the requirements at 29 CFR 1926.62 and in this scope of work. All equipment and materials must exit the regulated areas through the W/EDF and be decontaminated in accordance with this Scope of Work. PDF and W/EDF must consist of 6-mil fire-retardant polyethylene sheeting placed on the ground adjacent to the regulated area. This area must be properly demarcated with barrier tape or appropriate signage similar to the regulated area. The area must be equipped with a HEPA vacuum, soap and water, and disposable towels adequate for the decontamination of workers and equipment.

PART 2 PRODUCTS, MATERIALS, AND EQUIPMENT

2.1 MATERIALS AND EQUIPMENT

A. General Requirements (all LBP stabilization, component removal, and soil removal projects)

1. Before the start of work, the Contractor shall provide and maintain enough materials and equipment to ensure continuous and efficient work throughout the duration of the project. Work shall not start unless the following items have been delivered to the site and the Competent Person has submitted verification to the Contracting Officer.

2. All materials shall be delivered in their original package, container, or bundle bearing the name of the manufacturer and the brand name (where applicable).

3. Store all materials subject to damage off the ground, away from wet or damp surfaces, and under cover sufficient to prevent damage or contamination. Flammable materials cannot be stored inside structures. Replacement materials shall be stored outside of the regulated area until lead work is completed.

4. The Contractor shall not block or hinder the use of the site by staff and visitors in partially occupied buildings by placing materials/equipment in any unauthorized place.

5. The Competent Person shall inspect for damaged, deteriorating, or previously used materials. Such materials shall not be used and shall be removed from the worksite and disposed of properly.

6. If polyethylene sheeting is used for walls in the regulated area, then polyethylene shall be a minimum of 4-mil thickness. For floors and all other uses, sheeting shall be a minimum of 6-mil thickness and shall be used in widths selected to minimize the frequency of joints.

7. The method of attaching polyethylene sheeting shall be agreed upon in advance by the Contractor and Contracting Officer and selected to minimize damage to equipment and surfaces. Method of attachment may include any combination of moisture-resistant duct tape or another waterproof tape, furring strips, spray glue, staples, nails, screws, lumber, and plywood for enclosures or other effective procedures capable of sealing polyethylene to dissimilar finished or unfinished surfaces under both wet and dry conditions. The Contractor shall repair all damage (i.e., fasteners, duct tape damages, etc.) to finishes not scheduled for removal and/or restoration by others.

8. Polyethylene sheeting utilized for the PDF, if used, shall be opaque white or black in color, 6-mil fire-retardant polyethylene.

9. Installation and plumbing hardware, showers, hoses, drain pans, sump pumps, and wastewater filtration system shall be provided by the Contractor.

10. Providing HEPA Air Filtration Devices (AFDs) from the regulated area directly to the exterior. Additional AFDs and temporary ductwork may be needed in the regulated area to control air flow. AFDs shall be used as an engineering control in case of lead contamination.

11. An adequate number of HEPA vacuums, scrapers, sprayers, nylon brushes, brooms, disposable mops, rags, sponges, staple guns, shovels, ladders, and scaffolding of suitable height and length as well as meeting OSHA requirements, fall protection devices, water hose to reach all areas in the regulated area, airless spray equipment, and any other tools, materials or equipment required to conduct the stabilization, component removal, and soil removal project. All electrically operated hand tools, equipment, and electric cords shall be connected to GFCI protection.

12. An adequate number of scrapers, sprayers, nylon brushes, brooms, disposable mops, rags, sponges, staple guns, shovels, ladders and scaffolding of suitable height and length as well as meeting OSHA requirements, fall protection devices, water hose to reach all areas in the regulated area, airless spray equipment, and any other tools, materials or equipment required to conduct the stabilization and component removal project. All electrically operated hand tools, equipment, and electric cords shall be equipped with ground-fault circuit protection.

13. Special protection for objects in the regulated area shall be detailed to prevent damage from scaffolds, water, and falling material.

14. Impermeable containers and 6-mil disposal bags for lead waste shall be preprinted with labels and markings as required by OSHA, EPA, and DOT regulations.

15. Impermeable disposal drums shall be metal or fiberboard with locking ring tops with required OSHA, EPA, and DOT labels and markings.

16. The Contracting Officer shall be provided a copy of the SDS as required for all hazardous chemicals under OSHA 29 CFR 1910.1200 - Hazard Communication. Methylene chloride shall not be used with any spray adhesive or other product.

17. OSHA DANGER demarcation signs, as many and as required by OSHA 29 CFR 1926.62(m) (2) shall be provided and placed by the Competent Person. All other posters and notices required by Federal, State, and Local regulations shall be posted at the outer boundary of the regulated area or approved equivalent.

18. Adequate and appropriate PPE for the project and the number of personnel/shifts shall be provided. All personal protective equipment issued must be based on a written hazard assessment conducted under 29 CFR 1910.132(d).

19. Any damage to building components not scheduled for general renovation, resulting from the Contractor's work and/or work practices, shall be repaired or replaced at the sole cost of the Contractor, utilizing appropriately qualified and insured tradespersons equal to or greater than the original condition.

2.2 MONITORING, INSPECTION, AND TESTING

A. General

1. The Contractor shall perform, throughout the LBP stabilization, component removal, and cleaning work, monitoring of Contractor personnel's exposure, review, and testing inside the work area in accordance with the OSHA requirements and these specifications. The Contractor's Industrial Hygiene Technician (IHT) or accredited supervisor ("competent person") shall personally review conditions inside the work area to ensure compliance with these specifications. In addition, the Contractor's IHT or Competent Person shall personally manage air sample collection, analysis, and evaluation for personnel samples and regulated area samples to satisfy OSHA requirements. Additional inspection and testing requirements are also indicated in other parts of this specification.

2. The IHT shall be responsible for managing all personnel monitoring, the inspection and testing required by this LBP stabilization and component removal project specification, and for continuous monitoring of all sub-systems and procedures affecting the safety of the Contractor's employees. The safety of the Contractor's employees and providing safe conditions inside the regulated area for all persons entering is the exclusive responsibility of the Contractor. The analytical laboratory that shall be used by the Contractor to analyze the samples shall be AIHA-accredited. The IHT shall keep a daily log of personnel and area samples taken and analyzed, and make such a log available to the Contracting Officer. The log shall contain information on the person's breathing zone sampled, activities being performed, the date of the sample collection, the time of the sample start to finish, flow rate, volume, and analytical results. Collect and analyze personnel samples for twenty-five percent of the workforce each day throughout the duration of the project.

3. NPS could employ an independent industrial hygiene consultant (IHC) to perform various services on behalf of NPS. The IHC may perform the necessary monitoring, inspection, testing, and other support services to ensure that the NPS employees and visitors will not be adversely affected by the stabilization and component removal work and that the stabilization and component removal work proceeds in accordance with these specifications, and that the areas have been successfully decontaminated. The work of the IHC in no way relieves the Contractor from their responsibility to perform the work in accordance with contract/specification requirements, to perform continuous inspection, monitoring, and testing for the safety of their employees, and to perform other such services as specified. The cost of the IHC and its services will be borne by NPS, excluding any repeat of final inspection and testing that may be required due to unsatisfactory initial results.

4. If the lead in air or settled dust measured by the IHC during LBP interim stabilization, component removal, and cleaning activities, either inside or outside the regulated area, utilizing total lead analytical methods, exceeds the specified respective limits, the Contractor shall stop work. The Contractor may request confirmation of the results by additional sample collection and analysis. Requests must be in writing and submitted to the Contracting Officer. The cost for the confirmation sample analysis will be borne by the Contractor for both the collection and analysis of samples and for the time delay that may/does result from this confirmation.

B. Scope of Services of the IHC

1. The purpose of the work of the IHC is to: Ensure quality; resolve problems; and prevent the spread of contamination beyond the regulated areas. In addition, their work includes performing the final inspection and testing to determine whether the regulated areas or buildings have been adequately decontaminated. All clearance monitoring is to be done utilizing visual inspection, settled dust wipes, and soil sample collection methods, with an option to discontinue clearance wipes and soil sampling. The IHC may perform the following tasks:

    1. Task 1: Establish background lead in settled dust and soil levels before LBP stabilization, component removal, and cleaning begin by collecting background samples.

    2. Task 2: Perform periodic air monitoring, inspection, and testing outside the regulated area during the actual removal of LBP components to detect any faults in the regulated area isolation and any adverse impact on the surroundings from Contractor activities within the regulated area.

    3. Task 3: Perform unannounced visits to spot-check overall compliance of work with contract/specifications. These visits may include any inspection, monitoring, and testing inside and outside the regulated areas and all aspects of the operation, except personnel monitoring.

4. Task 4: Provide support to the Contracting Officer, such as evaluation of submittals from the Contractor, scheduling, resolution of unforeseen developments, etc.

5. Task 5: Perform, in the presence of the Contracting Officer, the final inspection and testing of decontaminated regulated areas at the conclusion of the LBP stabilization, component removal, soil removal, and clean-up work to certify compliance with the NPS requirements/specifications.

6. Task 6: Issue a Certificate of Decontamination for each regulated area and project report.

2. All documentation, inspection results, and testing results generated by the IHC will be available to the Contractor for information and consideration. The Contractor shall cooperate with and support the IHC for efficient and smooth performance of their work.

3. The monitoring and inspection results of the IHC will be used by the Contracting Officer to issue any stop-work orders to the Contractor during stabilization, component removal, and soil removal work, and to accept or reject a regulated area or building as decontaminated.

2.3 STANDARD OPERATING PROCEDURES

A. The Contractor shall have established SOPs in printed form and loose-leaf folders consisting of simplified text, diagrams, sketches, and pictures that establish and clearly explain the procedures to be followed during all phases of the work by the Contractor's personnel. The SOPs must be modified as needed to address the specific requirements of the project. The SOPs shall be submitted for review and approval before the start of any stabilization and component removal work. The minimum topics and areas to be covered by the SOPs are:

1. Minimum Personnel Qualifications

2. Contingency Plans and Arrangements

3. Security and Safety Procedures

4. Respiratory Protection/PPE Program and Training

5. Medical Surveillance Program and Recordkeeping

6. Regulated Area Requirements - Establishment and Management of the Regulated Area

7. Decontamination Facilities and Entry/Exit Procedures

8. Monitoring, Inspections, and Testing

9. LBP Stabilization, Component Removal, and Cleaning Procedures for LBP and Lead-Containing Dust

10. Removal of Contaminated Soil

11. On-Site Storage of Lead Pending Disposal

12. Regulated Area Decontamination/Clean-up

13. Disposal

14. Regulated Area Visual and Clearance Standards

15. Project Completion/Closeout

2.4 SUBMITTALS

A. Pre-Construction Meeting Submittals

1. Submit to the NPS a minimum of 14 days before the pre-start meeting, the following for review and approval. Meeting this requirement is a prerequisite for the pre-start meeting for this project:

a. Submit an Emergency Action Plan. The Plan shall meet the requirements of 29 CFR 1926, Subpart C, Standard 1926.35 Employee Emergency Action Plans.

b. Submit a detailed work schedule for the entire project reflecting contract documents and the phasing/schedule requirements.

c. Submit a staff organization chart showing all personnel who will be working on the project and their capacity/function. Provide their qualifications, training, accreditations, and licenses, as appropriate. Provide a copy of the "Certificate of Worker's Acknowledgment" and the "Affidavit of Medical Surveillance and Respiratory Protection" for each person.

d. SOPs developed specifically for this project, incorporating the requirements of the specifications, prepared, signed, and dated.

e. Submit the specifics of the materials and equipment to be used for this project, with manufacturer names, model numbers, performance characteristics, pictures/diagrams, and numbers available for the following:

i. HEPA vacuums, air monitoring pumps, calibration devices, and emergency power-generating systems.

ii. Wastewater filtration system, if needed, and barriers.

iii. Respirators, protective clothing, fall protection, and other required PPE.

iv. Fire safety equipment to be used in the regulated area.

f. Submit the name, location, and phone number of the approved landfill; proof/verification that the landfill is approved for lead disposal; the landfill's requirements for lead waste; the type of vehicle to be used for transportation; and the name, address, and phone number of the subcontractor, if used. Proof of training for transportation personnel shall be provided.

g. Submit required notifications and arrangements made with regulatory agencies having regulatory jurisdiction, and the specific contingency/emergency arrangements made with local health, fire, ambulance, and hospital authorities, and any other notifications/arrangements.

h. Submit the name, location, and verification of the laboratory and/or personnel to be used for the analysis of air, wipe, and/or bulk samples. Air monitoring must be done in accordance with OSHA 29 CFR 1926.62.

i. Submit qualifications verification: Submit the following evidence of qualifications. Make sure that all references are current and verifiable by providing current phone numbers and documentation.

    i. Company: Project experience within the past three (3) years; listing projects first, most like this project: Project Name; Type of Stabilization and Component Removal; Duration; Cost; Reference Name/Phone Number; Final Clearance; Completion Date.

    ii. List of the project(s) halted by the NPS, A/E, IH, or regulatory agency in the last three (3) years: Project Name; Reason; Date; Reference Name/Number; Resolution.

    iii. List regulatory citations (e.g., OSHA), notices of violations (e.g., Federal and state EPA), penalties, and legal actions taken against the company, including those of the company’s officers (including damages paid) in the last three (3) years. Provide copies and all information needed for verification.

j. Submit personnel information: Provide a resume; address each item completely; provide references; phone numbers; copies of certificates, accreditations, and licenses. Submit a declaration of facts stating that all personnel submitted below have medical records in accordance with OSHA 29 CFR 1926.62, 29 CFR 1910.134, and that the company has implemented a medical surveillance program and maintains recordkeeping in accordance with the above regulations. Submit the phone number and doctor/clinic/hospital used for medical evaluations.

    i. Competent Person(s)/Supervisor(s): Number; names; years of LBP-related experience as Competent Person/Supervisor; list of similar projects as Competent Person/Supervisor; as a worker; certificates, licenses, accreditations; proof of OSHA specialized lead training; the maximum number of personnel supervised on a project; medical opinion; current respirator fit test.

ii. Workers: Numbers; names; years of LBP-related work experience; certificates, licenses, accreditations; training courses in LBP work activities and respiratory protection; medical opinion; current respirator fit test.

k. Submit copies of the State license; copy of the insurance policy, including exclusions with a letter from the agent stating in English the coverage provided and the fact that LBP stabilization, component removal, and soil removal activities are covered by the policy; copy of SOPs incorporating the requirements of this specification; information on who provides your training, how often; who provides medical surveillance, how often; who does and how is air monitoring conducted.

l. Rented equipment must be decontaminated before returning to the rental agency.

m. Cleaning Agent Requirements:

i. Submit, before the start of work, the manufacturer's technical data for all types of cleaning agents that will be used on the project. Provide application instructions also.

ii. Submit SDSs and applicable instructions

B. Submittals During Lead Work Activities

1. The Competent Person shall maintain and submit a daily log of the regulated area documenting the dates and times of the following: purpose, attendees, and summary of meetings; all personnel entering/exiting the regulated area; document and discuss the resolution of unusual events such as equipment failures, emergencies, and any cause for stopping work; representative air monitoring and results/TWA's/EL's. Submit this information weekly to the Contracting Officer.

2. The Competent Person shall document and maintain the inspection and approval of the regulated area preparation before the start of work and daily during work.

a. Removal of any poly barriers.

b. Visual inspection/testing by the Competent Person.

c. Packaging and removal of lead waste from the regulated area.

d. Disposal of lead waste materials; copies of Waste Shipment Records/landfill receipts to the Contracting Officer weekly.

C. Submittals at Completion of Lead Work Activities

1. The Contractor shall submit a project report consisting of the daily logbook requirements and documentation of events during the LBP stabilization, component removal, and soil removal project, including Waste Shipment Records signed by the landfill's agent. It will also include information on the containment and transportation of waste from the containment with applicable Chain of Custody forms. The report shall include a certificate of completion, signed and dated by the Competent Person, in accordance with Attachment #1. All clearance and perimeter area samples must be submitted. The Contracting Officer will retain the report after completion of the project and provide copies of the report to the NPS Office of Engineering and the Safety Office.

PART 3 - EXECUTION

3.1 PRE-WORK ACTIVITIES

A. Pre-Work Meeting

1. The Contracting Officer, upon receipt, review, and substantial approval of all pre-work submittals and verification that all materials and equipment required for the project are on the site, will arrange for an LBP pre-work meeting between the Contractor, Competent Person(s), the Contracting Officer, and the IHC. The purpose of the meeting is to discuss any aspect of the submittals needing clarification or amplification, and to discuss any aspect of the project execution and the sequence of the operation. The Contractor shall be prepared to provide supplemental information/documentation to the Contracting Officer regarding any submittals, documentation, materials, or equipment. Upon satisfactory resolution of outstanding issues, the Contracting Officer will issue a written order to proceed to the Contractor. No LBP stabilization, component removal, soil removal, and cleaning work of any kind described in the following provisions shall be initiated before the Contracting Officer's written order to proceed.

B. Pre-Work Inspections and Preparations

1. Before any work begins on the construction of the regulated area, the Contractor will:

a. Conduct a space-by-space inspection with the Contracting Officer and prepare a written inventory of all existing damage in those spaces where LBP stabilization, component removal, and soil removal will occur. Still or video photography may be used to supplement the written damage inventory. Documentation will be signed and certified as accurate by both parties.

b. All furniture, curtains, drapes, blinds, and other furnishings are required to be cleaned, or cleaned and relocated, as necessary by the Contractor, if present. Should moving of any items be required to accomplish cleaning, the items are to be replaced by the Contractor.

c. For interior LBP stabilization, the Contractor shall shut down and seal with a minimum of two layers of 6-mil plastic all HVAC systems serving the regulated area. The regulated area environment shall be completely isolated from any other air in the building.

d. For exterior LBP stabilization and component removal, the Contractor shall cover the ground and adjacent building surfaces with a minimum of 1 layer of 6-mil fire-retardant poly extending a minimum of 10 feet from the building component being stabilized, removed, or cleaned.

e. Shut down and lockout in accordance with 29 CFR 1910.147 all electrical circuits that pose a potential hazard. Electrical arrangements will be tailored to the particular regulated area and the systems involved. All electrical circuits affected will be turned off at the circuit box outside the regulated area, not just the wall switch. The goal is to eliminate the potential for electrical shock, which is a major threat to life in the regulated area due to possible energized circuits. Electrical lines used to power equipment in the regulated area shall conform to all electrical safety standards and shall be isolated using a ground fault circuit interrupter (GFCI). All GFCIs shall be tested before use.

f. Ensure that all floors in the regulated area have been cleaned and decontaminated and then properly protected from contamination.

g. Inspect existing firestopping in the regulated area. Correct as needed.

C. Lead Pre-Work Construction and Operations

1. Perform all preparatory work for the first regulated area in accordance with the approved work schedule and with this specification.

2. Upon completion of all preparatory work, the IHC will inspect the work and systems and will notify the Contracting Officer when the work is completed in accordance with this specification. The Contracting Officer may inspect the regulated area and the systems with the IHC and may require that, upon satisfactory inspection, the Contractor's employees perform all major aspects of the approved SOPs, especially worker protection, fall protection, respiratory systems, contingency plans, decontamination procedures, and monitoring to demonstrate satisfactory operation.

3. The Competent Person shall document the LBP pre-work activities described above and deliver a copy to the Contracting Officer.

4. Upon satisfactory inspection of the installation and operation of systems, the Contracting Officer will notify the Contractor in writing to proceed with the LBP stabilization, component removal, soil removal, and cleaning work in accordance with this specification.

3.2 REGULATED AREA PREPARATIONS

A. Site Security

1. Regulated area access is to be restricted only to authorized, trained/accredited, and protected personnel. These may include the Contractor's employees, employees of subcontractors, NPS employees and representatives, State and local inspectors, and any other designated individuals. A list of authorized personnel shall be established before commencing the project and shall be located immediately outside the boundary of the regulated work area.

2. Entry into a regulated area by unauthorized individuals shall be reported immediately to the Competent Person by anyone observing the entry. The Competent person shall immediately notify the Contracting Officer.

3. A logbook shall be maintained outside the boundary of the regulated work area. Anyone who enters the regulated area must record their name, affiliation, time in, and time out for each entry.

4. Access to a regulated area shall be through a single access point. All other access (doors, windows, hallways, etc.) shall be sealed or locked to prevent entry to or exit from the regulated area. The only exceptions to this requirement are the waste load out area, which shall be sealed or locked except during the removal of containerized lead waste from the regulated area and emergency exits. Emergency exits shall not be locked from the inside or outside.

5. The Contractor's Competent Person shall control site security during removal activities to isolate work in progress and protect adjacent personnel. Security shall be provided at the entrance to the regulated area to ensure that all entrants are logged in/out and that only authorized personnel are allowed entrance.

6. The Contractor will have the NPS's assistance in notifying adjacent personnel of the presence, location, and quantity of LBP in the regulated area and enforcement of restricted access by NPS employees.

7. The regulated area shall be locked during non-working hours.

B. Signage and Power Management

1. Post OSHA WARNING signs meeting the specifications of OSHA 29 CFR 1926.62 at any location approaching the regulated area where airborne concentrations of lead may exceed ambient background levels. Signs shall be posted at a distance sufficiently far away from the regulated area to permit any person to read the sign and take the necessary measures to avoid exposure. Additional signs will be posted following the construction of the regulated area enclosure.

2. Shut down and lock out electric power to the regulated area. Provide temporary power and lighting (if working at night). Ensure safe installation, including GFCI of temporary power sources and equipment, in compliance with all applicable electrical code requirements and OSHA requirements for temporary electrical systems. Electricity shall be provided by the NPS at no expense to the Contractor.

3. Shut down and lock out heating, cooling, and air conditioning system (HVAC) components that are in, supply, or pass through the regulated area. Interiors of existing ductwork may require decontamination. This may be done during the pre-cleaning phase of operations before the ductwork is sealed off or during the final cleaning phase before re-engagement of the system. Appropriate equipment and control measures shall be utilized to prevent contamination of building spaces during this operation. Investigate and document the regulated area and agree on the pre-removal condition with the Contracting Officer. Seal all intake and exhaust vents in the regulated area with duct tape and two (2) layers of 6-mil plastic. Also, seal any seams in system components that pass through the regulated area.

3.3 CONTAINMENT BARRIERS AND COVERINGS FOR THE REGULATED AREA

A. General

1. Using critical barriers, seal off the regulated area to completely isolate the regulated area from adjacent spaces. All penetrations to the regulated area must be covered with 2 layers of 6-mil fire-retardant poly to prevent contamination and facilitate clean-up. Should adjacent areas become contaminated, immediately stop work, and clean up the contamination at no additional cost to the Government. Provide firestopping and identify all fire barrier penetrations due to abatement work as specified in Section 3.3 F: Firestopping.

B. Preparation Before Sealing the Regulated Area

1. Place all tools, scaffolding, materials, and equipment needed for working in the regulated area before erecting any plastic sheeting. Remove all uncontaminated removable furniture, equipment, and/or supplies from the regulated area before commencing work or completely cover with two layers of 6-mil polyethylene sheeting and secure with duct tape. Lock out and tag out any HVAC systems in the regulated area, if required.

C. Controlling Access to the Regulated Area

1. Access to the regulated area is allowed only at the Competent Person’s designated location and shall serve as a personnel decontamination facility (PDF)/Area if required. All other means of access shall be eliminated, and OSHA Danger demarcation signs shall be posted as required by OSHA. Any alternative method must be submitted for the Contracting Officer's written approval.

D. Preparation of Regulated Area

1. Lead warning signs shall be posted at all approaches to the regulated area. Post all emergency exits as emergency exits only on the regulated area side, post with lead warning signs on the non-regulated area side. Provide all non-regulated area stairs and corridors accessible to the regulated area with warning tapes at the base of stairs and beginning of corridors. Warning tapes shall be in addition to caution signs.

2. Regulated area preparation for exterior removal:

   a. All ground surfaces exterior to the regulated area shall have a layer of 6-mil polyethylene sheeting attached to the building face, and laid down on the surface below the exterior removal area, at least 10 feet wide or to the furthest point of gravity fall for dislodged debris by methods used, whichever is further. For work at the second story and above, extend 6-mil polyethylene sheeting as necessary. For work above the third story, by sidewalk, street, or property boundary, scaffolding sides shall be covered in 6-mil polyethylene sheeting.

   b. All operable windows within the regulated area and 25 feet from all sides of the regulated area shall be closed.

   c. In the regulated area, isolate all HVAC equipment intakes by temporarily shutting down units during removals and installing polyethylene sheeting over the opening.

3. Regulated area preparation for interior removal:

   a. Isolate all HVAC equipment, including installing polyethylene sheeting on all air returns and exhausts. Turn off all HVAC systems serving regulated areas when feasible.

   b. All floor areas adjacent to the regulated area shall have a layer of 6-mil polyethylene sheeting attached to the interior wall, and laid down on the surfaces below the work area, at least 5 feet wide or to the furthest point of gravity fall for dislodged debris by methods used, whichever is further.

   c. All movable objects shall be removed from the immediate removal area. All non-movable objects shall be covered with one layer of 6-mil polyethylene sheeting and sealed at the edges.

   d. All operable windows within the regulated area shall be closed.

   e. Temporary dust barriers consisting of a minimum of 6-mil polyethylene sheeting shall be installed in hallways, corridors, doorways, and other openings to the regulated area and not used for passage during removals to establish work area containment enclosure.

   f. A 6-mil polyethylene sheeting overlapping the curtained doorway shall be installed at the entrance to the regulated area.

g. For all regulated areas with the use of electromechanical tools for LBP stabilization and component removal, HEPA filtered negative air ventilation units must be installed in regulated areas and operate continuously during removal operations to establish negative pressure.

E. Extension of the Regulated Area

1. If the regulated area is breached in any way that could allow contamination to occur, the affected area shall be included in the regulated area and constructed as per this section. If the affected area cannot be added to the regulated area, decontamination measures must be started immediately and continue until dust wipe and soil sampling indicate clearance levels are met.

F. Firestopping

1. Through penetrations caused by cables, cable trays, and pipes, sleeves must be firestopped with a fire-rated firestop system providing an air-tight seal.

2. Firestop materials that are not equal to the wall or ceiling penetrated shall be brought to the attention of the Contracting Officer. The Contractor shall list all areas of penetration, the type of sealant used, and whether or not the location is fire-rated. Any discovery of penetrations during abatement shall be brought to the attention of the Contracting Officer immediately. All walls, floors, and ceilings are considered fire-rated unless otherwise determined by the Contracting Officer or Fire Marshall.

3. Any visible openings, whether caused by a penetration, shall be reported by the Contractor to the Contracting Officer for a sealant system determination. Firestops shall meet ASTM E814 and UL 1479 requirements for the opening size, penetrant, and fire rating needed.

G. Sanitary Facilities

1. The Contractor shall provide sanitary facilities for abatement personnel and maintain them in a clean and sanitary condition throughout the abatement project.

H. Personal Protective Equipment

I. Provide whole body clothing, head coverings, gloves, and foot coverings, and any other personal protective equipment as determined by conducting the hazard assessment required by OSHA at 29 CFR 1910.132 (d). The Competent Person shall ensure the integrity of personal protective equipment worn for the duration of the project. Duct tape Pre-Cleaning

1. The NPS will provide water for abatement purposes. The Contractor shall connect to the existing system. The service to the shower(s) shall be supplied with backflow prevention.

2. The Contractor shall pre-clean all movable objects and refuse within the regulated area using a HEPA-filtered vacuum and/or wet cleaning methods as appropriate.

3. Clean all fixed objects in the regulated area using HEPA-filtered vacuums and/or wet cleaning techniques as appropriate. Do not use any methods that would raise dust, such as dry sweeping or vacuuming with equipment not equipped with HEPA filters. Surrounding building components are to be thoroughly cleaned following LBP stabilization, component removal, and soil removal operations.

4. Pre-clean all surfaces in the regulated area using HEPA-filtered vacuums and/or wet cleaning methods as appropriate. Do not use any methods that would raise dust, such as dry sweeping or vacuuming with equipment not equipped with HEPA filters. Surrounding building components are to be thoroughly cleaned following LBP stabilization, component removal, and soil removal operations.

3.4 LBP STABILIZATION, COMPONENT REMOVAL, AND SOIL REMOVAL

A. Wetting Materials

1. Use amended water for the wetting of loose or peeling LBP before stabilization and component removal, and soil contaminated with LBP. The Competent Person shall ensure the wetting of LBP is sufficient to minimize the generation of potentially lead-containing dust for the duration of the project.

B. Drop Cloths and Walkways

1. Install as a drop cloth, a 6-mil poly sheet at the beginning of each work shift where LBP stabilization and component removal are to be performed during that shift. Completely cover floors or ground within 10 feet (3M) of the area where work is performed. Secure the drop cloths with duct tape to prevent debris from getting beneath them. Remove the drop cloth at the end of the shift or as work in the area is completed. Keep paint chips and debris on the drop cloth wet. When removing, fold inward to prevent spillage and place in a disposal bag.

2. Install walkways using 6-mil black poly between the regulated area and the decontamination areas to protect the building surfaces and grounds from contamination resulting from the tracking of potentially lead-containing dust. Install the walkways at the beginning of each shift and remove them at the end of each shift.

C. Wet Removal of LBP

1. Adequately wet the building components and soil contaminated with LBP to be removed, before LBP stabilization, component removal, and soil removal to reduce/prevent lead dust release to the air. Adequate time must be allowed for the amended water to saturate the LBP. Contractor personnel must not disturb dry LBP. Use a fine spray of amended water. Saturate the material sufficiently to wet the substrate without causing excessive dripping. The material must be sprayed repeatedly/continuously during the LBP stabilization and component removal process to maintain adequately wet conditions. (Note: In no event shall dry LBP stabilization and component removal occur except in the case of electrical hazards or a greater safety issue.

D. Alternate Procedures

1. Procedures described in this specification are to be utilized at all times.

2. If specified procedures cannot be utilized, a request must be made in writing to the Contracting Officer, providing details of the problem encountered and recommended alternatives.

3. Any alternative procedure must be approved in writing by the Contracting Officer before implementation.

E. Monitoring of Regulated Area

1. The Competent Person shall monitor and closely supervise the work carried out in the regulated area, to ensure the protection of workers and the environment. The area shall be secured as detailed in this specification. After the work is finished, the Competent Person will verify and test to determine satisfactory completion and notify the Contracting Officer when the results are satisfactory. The NPS may use its IHC to determine compliance. When the area is found acceptable, the area will be released as decontaminated.

F. Worker Protection

1. Before beginning work with any material for which an SDS has been submitted, provide workers with the required PPE. The required PPE shall be used whenever exposure to the material might occur. In addition to OSHA/specification requirements for respiratory protection, a paint pre-filter and an organic vapor cartridge, at a minimum, shall be used in addition to the HEPA filter when a solvent-based material is used.

3.5 DISPOSAL OF LBP WASTE MATERIALS FOR DISPOSAL

A. General

1. Dispose of waste, LBP, and debris that is packaged in accordance with these specifications, OSHA, WYDEQ, EPA, and DOT. The landfill requirements for packaging must also be met. Disposal shall be made at an approved landfill. Disposal of soil contaminated with lead-based paint and incidental cleaning materials shall be disposed of by an appropriate licensed transporter in accordance with applicable regulations, including the Resource Conservation and Recovery Act (RCRA) and the Toxic Substances Control Act (TSCA). The Contractor is responsible for all required testing by the Toxicity Characteristic Leaching Procedures (TCLP) to determine proper disposal methods. The cost for testing and disposal costs associated with, or as a result of, the testing shall be at the expense of the Contractor.

B. Transportation and Disposal Site

1. The Owner shall designate the waste Hauler and Disposal Site. For any permitted out-of-state landfill not specifically authorized for disposal of LBP, written notice must be provided 15 days before the first shipment of the same waste stream that contains lead. The letter shall be acknowledged via a disposal facility representative's signature, printed name, and title. If the facility is permitted to accept lead waste, no letter is required.

2. The Removal Contractor shall give 24-hour notification before removing any waste from the site. Waste shall be removed from the site only during normal working hours unless otherwise specified. No waste may be taken from the site unless the Removal Contractor and Contracting Officer are present, and the Owner's Representative authorizes the release of the waste as described herein.

3. All waste generated as part of the LBP project shall be removed from the site within 10 calendar days after the successful completion of all LBP removal work.

4. Upon arrival at the project Site, the hauler must possess and present to the Contracting Officer a valid Waste Hauler's Permit. The Contracting Officer may verify the authenticity of the hauler's permit with the proper authority.

5. The hauler, with the Contractor and the Contracting Officer, shall inspect all material in the transport container before taking possession and signing the Hazardous Waste Manifests.

C. Storage Containers

1. All waste containers shall be fully enclosed and lockable (i.e., enclosed dumpster, trailer, etc.).

2. The Contracting Officer shall verify the waste storage container.

3. The container shall be plasticized and sealed with one layer of 6-mil polyethylene. Once on-site, it shall always be kept locked, except during load out. The waste container shall not be used for storage of equipment or Contractor supplies.

4. While on-site, the container(s) shall be labeled with Lead Warning Labels and Hazardous Waste Labels.

5. The container is not permitted to be loaded unless it is properly plasticized, has the appropriate danger signage affixed, and has the permit number appropriately stenciled on the container.

6. The Contracting Officer may initiate random checks at the Storage Site to ensure that the procedures outlined herein are in compliance.

3.6 HAZARDOUS WASTE MANIFESTS

A. All waste containers shall be fully enclosed and lockable (i.e., enclosed dumpster, trailer, etc.).

B. A Hazardous Waste Manifest shall be utilized for transportation. A Hauler Billing Form may be used if the hauler needs an independent record, but shall not be used as a shipping document.

C. The Manifest shall be completed by the Contractor and verified by the Contracting Officer that all the information and amounts are accurate, and the proper signatures are in place.

D. The Manifest shall have the appropriate signatures of the Contracting Officer (the Generator) and the hauler representative prior to any waste being removed from the site.

E. Copies of the completed Manifest shall be retained by the Contracting Officer and shall remain on-site for inspection.

F. Upon arrival at the Disposal Site, the Manifest shall be signed by the Disposal Facility operator to certify receipt of lead materials covered by the manifest.

G. The Disposal Facility operator shall return the original Manifest to the Contracting Officer within 30 days.

H. The Contractor shall utilize the Waste Disposal Log provided by the Contracting Officer. This log shall be maintained by the project supervisor and shall be kept on-site at all times. (See Attachment #7).

I. Originals of all waste disposal manifests and disposal logs shall be submitted by the Contractor to the Contracting Officer with the final close-out documentation.

3.7 PROJECT DECONTAMINATION

A. General

1. The entire work related to project decontamination shall be performed under close supervision.

2. The LBP stabilization, component removal, and soil removal work areas will be evaluated by the IHC and the Contractor’s representative for visual evidence of cleaning deficiencies, paint chips, or debris.

B. Regulated Area Clearance

1. Settled dust wipe and soil sampling, and other requirements that must be met before the release of the Contractor and re-occupancy of the regulated area space are specified in Final Testing Procedures.

C. Work Description

1. Decontamination includes the cleaning and clearance of the regulated area and the decontamination and removal of the decontamination areas installed before the LBP stabilization and component removal work.

3.8 FINAL VISUAL INSPECTION AND AIR CLEARANCE TESTING

A. General

1. Notify the Contracting Officer 48 hours in advance of the performance of the final visual inspection and testing. The final visual inspection and testing will be performed by the IHC after the cleaning.

B. Final Visual Inspection

1. Final visual inspection will include the entire regulated area, the PDF, and interior and exterior LBP components scheduled for stabilization, component removal, and cleaning. If debris, residue, dust, or any other suspect material is detected, the final cleaning shall be repeated at no cost to the Contracting Officer or the IHC. Dust/material samples may be collected and analyzed at no cost to the NPS at the discretion of the IHC to confirm visual findings. When the regulated area is visually clean, the final testing can be done.

C. Final Clearance Testing

1. After an acceptable final visual inspection by the IHC and the Contracting Officer, the IHC will perform the final testing. Wipe and soil samples will be taken and analyzed in accordance with the procedures for total lead analysis of clearance wipe and soil samples specified elsewhere in this section. At the discretion of the Contracting Officer, visual inspection criteria only may be employed.

2. If release criteria are not met, the Contractor shall repeat the final cleaning and continue the decontamination procedure from that point. Additional review and testing shall be at the expense of the Contractor until the release criteria are met.

3. If release criteria are met, proceed to perform the stabilization and component removal closeout and issue the certificate of completion in accordance with these specifications.

D. Final Clearance Procedures

1. Contractor's Release Criteria: Work in a regulated area is complete when the regulated area is visually clean, and lead concentrations in settled dust and soil have been reduced to or below the level stated below, as measured with total lead concentrations:

a. Windowsills - 100 micrograms per square foot

b. Floor - 10 micrograms per square foot

c. Exterior Bare Soil – 400 parts per million (ppm) where children play and 1,200 ppm in other parts of the yards used less by children

3.9 STABILIZATION, COMPONENT, AND SOIL CLOSEOUT AND CERTIFICATE OF COMPLIANCE

A. Completion of Stabilization, Component Removal, and Soil Removal Work

1. After thorough decontamination, seal negative air machines with two (2) layers of 6-mil polyethylene and duct tape to form a tight seal at the intake/outlet ends before removal from the regulated area. Complete stabilization, component removal, and soil removal of LBP work upon meeting the regulated area clearance criteria and fulfilling the following:

2. Remove all equipment, materials, and debris from the project area.

3. Package and dispose of all lead waste as required.

4. Repair or replace all interior finishes damaged during the stabilization and component removal work not scheduled for restoration by others.

5. Fulfill other project closeout requirements as specified elsewhere in this specification.

B. Certificate of Completion by Contractor

1. The Contractor shall complete and sign the "Certificate of Completion" in accordance with Attachment #1 after the LBP stabilization, component removal, cleaning, and decontamination of the regulated area.

C. Work Shifts

1. All work shall be done during working hours, excluding federal holidays. Any change in the work schedule must be approved in writing by the Contracting Officer.

ATTACHMENT #1

CERTIFICATE OF COMPLETION

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

CONTRACTOR'S NAME AND ADDRESS:_____

_____

_____

1. I certify that I have personally inspected, monitored, and supervised the LBP stabilization and component removal work of (specify regulated areas or structures):

Which took place from_____to _____

2. That throughout the work, all applicable requirements/regulations and the specifications were met.

3. That any person who entered the regulated area was protected with the appropriate PPE and respirator, and that they followed the proper entry and exit procedures and the proper operating procedures for the duration of the work.

4. That all employees of the Contractor engaged in this work were trained in respiratory protection, were experienced with lead-related activities and cleaning work, had proper medical surveillance documentation, were fit-tested for their respirator, and were not exposed at any time during the work to lead without the benefit of appropriate respiratory protection.

5. That I performed and supervised all inspections and testing specified and required by applicable regulations and specifications.

Signature:_____ Date:_____
(Printed Name/ Signature)

Signature:_____ Date:_____
(Printed Name/ Signature)

ATTACHMENT #2

CERTIFICATE OF WORKER'S ACKNOWLEDGMENT

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

CONTRACTOR'S NAME AND ADDRESS:_____

_____

WORKING WITH LEAD CAN BE HAZARDOUS TO YOUR HEALTH. INHALING OR INGESTING LEAD HAS BEEN LINKED WITH VARIOUS TYPES OF PHYSICAL DAMAGE.

Your employer's contract with the OWNER for the above project requires that you be supplied with the proper PPE, including an adequate respirator, and be trained in its use. You must be trained in safe and healthy work practices and in the use of the equipment found at a stabilization and component removal of an LBP project. You must receive/have a current medical examination for working with lead. These things shall be provided at no cost to you. By signing this certificate, you are indicating to the OWNER that your employer has met these obligations.

RESPIRATORY PROTECTION: I have been trained in the proper use of respirators and have been informed of the type of respirator to be used on the above indicated project. I have a copy of the written Respiratory Protection Program issued by my employer. I have been provided for my exclusive use, at no cost, with a respirator to be used on the above indicated project.

TRAINING COURSE: I have been trained by a third-party, State/EPA-accredited trainer in the requirements for an AHERA/OSHA Stabilization and component removal of LBP Worker training course, 32 hours minimum duration. I currently have a valid State accreditation certificate. The topics covered in the course include, as a minimum, the following:

- Potential Health Effects Related to Exposure to Chemical Hazards
- Potential Health Effects Related to Exposure to Lead
- Employee Personal Protective Equipment
- Establishment of a Respiratory Protection Program
- State of the Artwork Practices
- Personal Hygiene
- Additional Safety Hazards
- Medical Monitoring
- Air Monitoring, if required
- Relevant Federal, State, and Local Regulatory Requirements, Procedures, and Standards
- Lead Waste Disposal

MEDICAL EXAMINATION: I have had a medical examination within the past 12 months, which was paid for by my employer. This examination included: health history, occupational history, pulmonary function test, and may have included a chest x-ray evaluation. The physician issued a positive written opinion after the examination.

Signature:_____ Printed Name:_____

ATTACHMENT #3

AFFIDAVIT OF MEDICAL SURVEILLANCE, RESPIRATORY PROTECTION AND TRAINING/ACCREDITATION

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

_____

CONTRACTOR'S NAME AND ADDRESS:_____

_____

1. I verify that the following individual

Name:_____

who is proposed to be employed in stabilization and component removal of LBP related work associated with the above project by the named Contractor, is included in a medical surveillance program in accordance with 29 CFR 1926.62, and that complete records of the medical surveillance program as required by 29 CFR 1926.62, 29 CFR 1910.134, and 29 CFR 1910.20 are kept at the offices of the Contractor at the following address.

Address:_____

_____

2. I verify that this individual has been trained, fit-tested, and instructed in the use of all appropriate respiratory protection systems and that the person is capable of working safely and healthily as expected and required in the expected work environment of this project.

3. I verify that this individual has been trained. This individual has also obtained a valid accreditation certificate. Documentation will be kept on-site.

Signature of Contractor:_____ Date:_____

Printed Name of Contractor:_____

ATTACHMENT #4

DAILY SIGN-IN AND SIGN-OUT SHEET

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

CONTRACTOR'S NAME AND ADDRESS:_____

_____

PRINT NAME	CERT. NUMBER EXP. DATE	TIME-IN	TIMEOUT

ATTACHMENT #5

DAILY ACTIVITY REPORT

Project Name:		Project No.	
Building:		Date	
Area:		Project Day No.	of
Contractor:		Personnel	
IHC:		Sheet	of
LBP Removed:			
Estimated Percent of Phase Completed:		On Schedule	Yes No

GENERAL REMARKS/MEETINGS/SITE VISITS/CONTRACTOR PROGRESS:

Contractor Supervisor:	Reviewed by:

ATTACHMENT #6

VISITOR RELEASE FORM

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

CONTRACTOR'S NAME AND ADDRESS:_____

_____

VISITOR NAME:_____

VISITOR'S COMPANY:_____

PURPOSE OF VISIT:_____

Time of Entry to the Project Site:_____ to the Exclusion Zone:_____

Time of Departure from the Site:_____ from the Exclusion Zone:_____

Personal Protective Equipment Utilized:_____

I acknowledge and understand that I am visiting a hazardous waste control work area. I understand the dangers of exposure to hazardous waste.

I have read and understand the Site Safety and Health Plan for this project and will abide by the directions, stipulations, and terms specified therein.

I knowingly assume all risks in connection with potential exposure to hazardous waste and I do hereby, for myself and my heirs at law, release and forever discharge the property Owner, Owner's Representative, Project Administrator, independent testing laboratory, architects, engineers, or Contracting firms employed by the Owner or Project Administrator, their owners, employees, nominees, personal representative, affiliates, successors, and assigns from and against all liability whatsoever at common law or otherwise. I hereby waive and relinquish all claims of every nature which I now have or may have or claim to have which are in any way, directly or indirectly, related to exposure to hazardous waste and hazardous waste-containing materials.

Furthermore, I know that I am entering a construction area where workplace conditions, such as water on floors, scaffolding, electrical equipment, etc., CAN CREATE HAZARDOUS SITUATIONS FOR VISITORS. I assume all risk of accidental injury or illness, regardless of cause, while visiting this construction site. I hereby waive and relinquish all claims of every nature which I now have or may have or claim to have which are in any way, directly or indirectly, related to such injury.

THIS RELEASES ALL PARTIES ON THE ABOVE-NAMED PROJECT:

Visitor's Signature:_____

Date:_____

Witness:_____

ATTACHMENT #7

WASTE MANIFEST LOG

Facility:	Building:
Project:	Project Number:
Lead Removal Contractor:	Owner's Representative:

Load No.	Hauler	Driver	License Plate No.	Size of Container	Date	DATES (Chain of Events) Departed from Site	Rec'd at Disposal Site	Manifest Returned

COMMENTS:

END OF SECTION 02 84 00

SECTION 02 85 00 – REMEDIATION OF BIRD, BAT, AND RODENT DROPPINGS

PART 1 - GENERAL

1.1 SUMMARY OF WORK

A. Contract Documents and Related Requirements

1 Drawings, general provisions of the contract, including General and Supplementary Conditions and other Specification Sections, apply to the work of this Section. The contract documents show the work to be done under the contract and related requirements and conditions impacting the project. Related requirements and conditions include applicable codes and regulations, notices and permits, existing site conditions and restrictions on the use of the site, requirements for partial owner occupancy during the work, coordination with other work, and the phasing of the work. In the event the Remediation Contractor discovers a conflict in the contract documents and/or requirements or codes, the conflict must be brought to the immediate attention of the General Contractor for Rehabilitation and the Contracting Officer for resolution. Whenever there is a conflict or overlap in the requirements, the most stringent shall apply. Any actions taken by the Remediation Contractor without obtaining guidance from the Contracting Officer shall become the sole risk and responsibility of the Remediation Contractor. All costs incurred due to such action are also the responsibility of the Remediation Contractor.

B. Extent of Work

1. The Remediation Contractor will coordinate all work, phasing, and scheduling with the General Contractor for Rehabilitation, the Contracting Officer, and their field representative. The Remediation Contractor will have a predetermined number of calendar days to complete bird, bat, and rodent remediation and final teardown per regulated area. The Remediation Contractor will coordinate the start date and the number of mobilizations required with the General Contractor for Rehabilitation, the Contracting Officer, and their field representative. Strict adherence to the schedule will be required to allow the work to be completed promptly to accommodate other activities on the site.

2. The Remediation Contractor shall satisfy himself as to the actual quantities to be remediated and disposed of. Nothing in this section may be interpreted as limiting the extent of work otherwise required by this contract and related documents.

3. Removal, packaging, clean-up, decontamination, and disposal of existing bird, bat, and rodent droppings in an appropriate regulated area as necessary to accommodate the rehabilitation of the buildings.

a. The high nutrient content of accumulated bird, bat, and rodent excrement provides an excellent growth medium for organisms of potential human health exposure concern.

b. Humans may become infected by inhaling dust containing dried feces, urine, or respiratory secretions of infected birds, bats, and rodents.

c. Other sources of potential exposures include a bite from an infected bird, bat, or rodent, and handling the plumage, fur, and tissues of infected animals.

d. The main disease organisms found in bird, bat, and rodent droppings are Cryptococcus, Histoplasmosis, and Psittacosis.

4. See Section 01 35 91 Historic Preservation Treatment Procedures for additional requirements for the salvage of historic building materials.

5. Any damage to components not scheduled for demolition, resulting from the Remediation Contractor's work, shall be repaired or replaced at the sole cost of the Remediation Contractor, utilizing appropriately qualified tradespersons.

6. The Remediation Contractor will be responsible for moving all property and refuse out of the Remediation Contractor's regulated areas.

C. Tasks

The work tasks are summarized as follows:

1. The Remediation Contractor will coordinate with the General Contractor for Rehabilitation, the Contracting Officer, and the NPS representative for scheduling access to the structures. The Remediation Contractor shall assume that the structures will be partially occupied during this project.

2. Pre-remediation activities, including pre-remediation meeting(s), inspection(s), notifications, permits, submittal approvals, work-site preparations, emergency procedures arrangements, and standard operating procedures (SOPs) for bird, bat, and rodent dropping remediation work.

3. Remediation activities, including removal, packaging, clean-up, decontamination, storing, and disposal of bird, bat, and rodent droppings waste, recordkeeping, security, monitoring, and inspections.

4. Remediation and decontamination activities, including final visual inspection and certification of decontamination.

D. Remediation Contractors' Use of Premises

1. The Remediation Contractor and the Remediation Contractor's personnel shall cooperate fully with the General Contractor for Rehabilitation and the Contracting Officer to facilitate efficient use of the site. The Remediation Contractor shall perform the work in accordance with the specifications and in compliance with any/all applicable Federal, State, and Local regulations and requirements.

2. The Remediation Contractor shall specify the facilities proposed to be utilized in the pre-remediation work plan. The Remediation Contractor shall use only the existing facilities in the structures strictly within the limits indicated in the approved pre-remediation work plan. Any variation from the approved work plan shall be secured in writing by the Contracting Officer.

1.2 STOP BIRD, BAT, AND RODENT REMEDIATION

A. If the Contracting Officer presents a verbal **Stop Bird, Bat, and Rodent Remediation Order**, the Remediation Contractor/Personnel shall immediately stop all bird, bat, and rodent remediation and adequately wet any exposed bird, bat, and rodent droppings. If a verbal Stop Bird, Bat, and Rodent Remediation Order is issued, the Contracting Officer shall follow up with a written order to the Remediation Contractor as soon as it is practicable. The Remediation Contractor shall not resume any bird, bat, and rodent remediation activity until authorized to do so in writing by the Contracting Officer. A stop bird, bat, and rodent remediation order may be issued at any time the Contracting Officer determines remediation conditions or activities are not within specification requirements, regulatory requirements, or that an imminent hazard exists to human health or the environment. The work stoppage will continue until conditions have been corrected to the satisfaction of the Contracting Officer. The occurrence of any of the following events shall be reported immediately by the Contractor's competent person to the Contracting Office and/or their field representative using the most expeditious means (e.g., verbal or telephonic), followed up with a written notification to the Contracting Officer as soon as it is practical. The Remediation Contractor shall immediately stop bird, bat, and rodent remediation/disturbance activities and initiate bird, bat, and rodent dropping reduction activities if:

1. Visible evidence of airborne fugitive dust.

2. Contractor employees performing remediation activities without proper Personal Protective Equipment (PPE).

3. Serious injury/death at the site.

4. Fire/safety emergency at the site.

5. Respiratory protection system failure.

6. Power failure or loss of wetting agent.

7. Any visible emissions or tracking of bird, bat, and rodent droppings from beyond the regulated area.

8. Failure to follow project specification requirements.

1.3 DEFINITIONS

A. Definitions and explanations here are neither complete nor exclusive of all terms used in the contract documents, but are general for the work to the extent they are not stated more explicitly in another element of the contract documents. Drawings must be recognized as diagrammatic and not completely descriptive of the requirements indicated therein.

B. Glossary

Definitions relative to Bird, Bat, and Rodent Dropping Remediation.

1. Aerosol - Solid or liquid particulate suspended in the air.

2. Adequately wet - Sufficiently mixed or penetrated with liquid to prevent the release of particulates. If visible emissions are observed coming from the bird, bat, and rodent droppings, then that material has not been adequately wetted.

3. Aggressive method - Removal or disturbance of bird, bat, and rodent droppings that crumble or disintegrate intact bird, bat, and rodent droppings.

4. Air monitoring firm - The firm retained by the Government to conduct air monitoring.

5. Air sample filter - The filter is used to collect airborne dust, which is then analyzed for biological contamination.

6. Amended water - Water to which a surfactant (wetting agent) has been added to increase the penetrating ability of the liquid.

7. Authorized person - Any person authorized by the NPS, the Remediation Contractor, or government agency and required by work duties to be present in regulated areas.

8. Authorized visitor - Any person approved by the NPS, the Remediation Contractor, or any government agency having jurisdiction over the regulated area.

9. Barrier - Any surface that isolates the regulated area and inhibits dust migration from the regulated area.

10. Containment Barrier - An airtight barrier consisting of walls, floors, and/or ceilings of sealed plastic sheeting that surrounds and seals the outer perimeter of the regulated area.

11. Critical Barrier - The barrier responsible for isolating the regulated area from adjacent spaces, typically constructed of plastic sheeting secured in place at openings such as doors, windows, or any other opening into the regulated area.

12. Primary Barrier - Barriers placed over critical barriers and exposed directly to remediation work.

13. Secondary Barrier - Any additional plastic barriers used to isolate and provide protection from debris during remediation work.

14. Breathing zone - The hemisphere, forward of the shoulders, with a radius of about 150–225 mm (6–9 inches), from the worker's nose.

15. Building/facility owner - The legal entity, including a lessee, which exercises control over management and recordkeeping functions relating to a building and/or facility in which bird, bat, and rodent droppings activities take place.

16. Certified Industrial Hygienist (CIH) - One certified in the practice of industrial hygiene by the American Board of Industrial Hygiene (AIHA). An industrial hygienist certified in Comprehensive Practice by AIHA.

17. Competent person - One who can identify existing bird, bat, and rodent dropping hazards in the workplace and select the appropriate control strategy for biological exposure, who has the authority to take prompt corrective measures to eliminate them.

18. Contingency plan - A document setting out an organized, planned, and coordinated course of action to be followed in case of a fire or explosion, or a release of hazardous waste or hazardous waste constituents from a treatment, storage, or disposal facility that could threaten human health or the environment.

19. Contracting Officer - The NPS official responsible for ongoing project work.

20. Crawlspace - An area that can be found either in or adjacent to the regulated area. This area has limited access and egress.

21. Demolition - The wrecking or taking out of any load-supporting structural member and any related razing, removing, or stripping.

22. Discharge or hazardous waste discharge - The accidental or intentional spilling, hazardous waste leaking, pumping, pouring, emitting, discharging, emptying, or dumping of hazardous wastes onto any land or water.

23. Disposal - The discharge, deposit, injection, dumping, spilling, leaking, or placing of any solid waste or hazardous waste into or on any land or water so that any constituent thereof may enter the environment or be emitted into the air or discharged in any water, including groundwater.

24. Disposal bag - Typically a 6-millimeter (mil) thick sift-proof, dustproof, leak-tight container, used to package and transport bird, bat, and rodent droppings waste from regulated areas to the approved landfill.

25. Drum - A rigid, impermeable container made of cardboard fiber, plastic, or metal that can be sealed to be sift-proof, dustproof, and leak-tight.

26. Employee exposure - The exposure to bird, bat, and rodent droppings that would occur if the employee were not wearing respiratory protection equipment.

27. Encapsulant - A material that surrounds or embeds bird, bat, and rodent droppings in an adhesive matrix and prevents the release of bird, bat, and rodent droppings.

28. Enclosure - The use of rigid, durable construction materials that are mechanically fastened to the substrate to act as a barrier between bird, bat, and rodent droppings and other spaces.

29. Filter - Media used in respirators, vacuums, or other machines to remove particulates from the air.

30. Final inspection - Inspection by a qualified inspector, industrial hygienist, or local public health official to determine whether bird, bat, and rodent dropping removal and cleaning activities are complete.

31. Firestopping - Material used to close the open parts of a structure to prevent a fire from spreading.

32. Government industrial hygienist (IHC) – The government retained Industrial Hygiene Consultant.

33. High-efficiency particulate air (HEPA) filter - A filter capable of trapping and retaining at least 99.97 percent of all mono-dispersed particles of 0.3 microns or greater in diameter.

34. HEPA vacuum - vacuum collection equipment equipped with a HEPA filter system capable of collecting and retaining bird, bat, and rodent droppings.

35. HVAC - Heating, Ventilation, and Air Conditioning

36. Industrial hygienist (IH) - A professional qualified by education, training, and experience to anticipate, recognize, evaluate, and develop controls for occupational health hazards and meets the definition requirements of AIHA.

37. Industrial hygienist technician (IH Technician) - A person working under the direction of an IH or CIH with special training, experience, certifications, and licenses required for the assigned industrial hygiene work.

38. NPS industrial hygienist (IHC) – The NPS retained Industrial Hygiene Consultant.

39. Negative pressure - Air pressure, which is lower than the surrounding area, created by exhausting air from a sealed regulated area through HEPA-equipped filtration units. OSHA requires maintaining a 5.0 Pa (-0.02") water gauge inside the negative pressure enclosure.

40. Negative pressure respirator - A respirator in which the air pressure inside the facepiece is negative during inhalation, relative to the air outside the respirator.

41. Organic vapor cartridge - The type of cartridge used on air purifying respirators for organic vapor exposures.

42. OSHA – Occupational Safety and Health Administration.

43. Outside air - The air outside buildings and structures, including, but not limited to, the air under a bridge or in an open ferry dock.

44. Owner/operator - Any person who owns, leases, operates, controls, or supervises the facility being demolished or renovated, or any person who owns, leases, operates, controls, or supervises the demolition or renovation operation, or both.

45. Personal protective equipment (PPE) - equipment designed to protect users from injury and/or specific job hazards. Such equipment may include protective clothing, hard hats, safety glasses, and respirators.

46. Personal sampling/monitoring - Representative air samples were obtained in the breathing zone of the person, using a cassette and battery-operated pump to determine biological exposure.

47. Polyethylene sheeting - Durable plastic barrier material 4 to 6-mil thick, semitransparent, sometimes flame-retardant, which follows NFPA 241.

48. Positive/negative fit check - A method of verifying the fit of a respirator by closing off the filters and breathing in or closing off the exhalation valve and breathing out while detecting leakage of the respirator.

49. Protection factor - A value assigned by OSHA/NIOSH to indicate the assigned protection a respirator should provide if worn properly. The number indicates the reduction of exposure level from outside to inside the respirator.

50. Qualitative fit test (QLFT) - A fit test using a challenging material that can be sensed by the wearer if a leakage in the respirator occurs.

51. Quantitative fit test (QNFT) - A fit test using a challenging material that is quantified outside and inside the respirator, thus allowing the determination of the actual fit factor.

52. Regulated area - An area established by the employer to demarcate where bird, bat, and rodent droppings related work activities are conducted, and any adjoining area where debris and waste from such bird, bat, and rodent dropping work may accumulate.

53. Remediation - The process or procedure for removing and controlling the biological release and/or dispersion of microbial agents.

54. Removal - All operations where bird, bat, and rodent droppings are taken out from structures or substrates, including demolition operations.

55. Renovation - Altering a facility or one or more facility components in any way, including the stripping or removal of bird, bat, and rodent droppings from a facility component that does not involve demolition activity.

56. Repair - Overhauling, rebuilding, reconstructing, or reconditioning of structures or substrates, including removal of bird, bat, and rodent droppings.

57. Standard operating procedures (SOPs) - bird, bat, and rodent dropping work procedures are required to be submitted by the Remediation Contractor before work begins.

58. Surfactant - A chemical added to water to decrease water's surface tension.

59. Visible emissions - Any emissions that are visually detectable without the aid of instruments come from bird, bat, and rodent dropping removal or cleaning operations.

60. Wet cleaning - The process of thoroughly eliminating, by wet methods, any bird, bat, and rodent droppings contamination from surfaces or objects.

C. Referenced Standards Organizations

1. The following acronyms or abbreviations, as referenced in contract/specification documents, are defined to mean the associated names. Names and addresses may be subject to change.

a. ABIH American Board of Industrial Hygiene
6015 West St. Joseph, Suite 102
Lansing, MI 48917-3980
Telephone: 517-321-2638

b. AIHA American Industrial Hygiene Association
2700 Prosperity Avenue, Suite 250
Fairfax, VA 22031
Telephone: 703-849-8888

c. ANSI American National Standards Institute
1430 Broadway Ave.
New York, NY 10018
Telephone: 212-354-3300

d. ASTM American Society for Testing and Materials
1916 Race St.
Philadelphia, PA 19103
Telephone: 215-299-5400

e. CFR Code of Federal Regulations
Government Printing Office
Washington, DC 20420

f. CGA Compressed Gas Association
1235 Jefferson Davis Highway
Arlington, VA 22202
Telephone: 703-979-0900

g. CS Commercial Standard of the National Institute of Standards and Technology (NIST)
U. S. Department of Commerce
Government Printing Office
Washington, DC 20420

h. EPA Environmental Protection Agency
401 M St., SW
Washington, DC 20460
Telephone: 202-382-3949

i. MSHA Mine Safety and Health Administration
Respiratory Protection Division
Ballston Tower #3
Department of Labor
Arlington, VA 22203
Telephone: 703-235-1452

j. NIST National Institute for Standards and Technology
U. S. Department of Commerce
Gaithersburg, MD 20234
Telephone: 301-921-1000

k. NEC National Electrical Code (by NFPA)

l. NEMA National Electrical Manufacturer's Association
2101 L Street, N.W.
Washington, DC 20037

m. NFPA National Fire Protection Association
1 Batterymarch Park
P.O. Box 9101
Quincy, MA 02269-9101
Telephone: 1-800-344-3555

n. NIOSH National Institute for Occupational Safety and Health
4676 Columbia Parkway
Cincinnati, OH 45226
Telephone: 513-533-8236

o. OSHA Occupational Safety and Health Administration
U.S. Department of Labor
Government Printing Office
Washington, DC 20402

p. UL Underwriters Laboratory
333 Pfingsten Rd.
Northbrook, IL 60062
Telephone: 312-272-8800

1.4 APPLICABLE CODES AND REGULATIONS

A. General Applicability of Codes, Regulations, and Standards

1. All work under this contract shall be done in strict agreement with all applicable Federal, State, and local regulations, standards, and codes governing biological remediation, and any other trade work done in conjunction with the remediation. All applicable codes, regulations, and standards are adopted into this specification and will have the same force and effect as this specification.

2. The most recent edition of any relevant regulation, standard, document, or code shall be in effect. Where conflict among the requirements or with these specifications exists, the most stringent requirement(s) shall be utilized.

3. Copies of all standards, regulations, codes, and other applicable documents, including this specification and those listed in Section 1.4, shall be available at the worksite. These standards, regulations, codes, and other applicable documents, including this specification and those listed in Section 1.4, may be made available electronically.

B. Remediation Contractor Responsibility

1. The Remediation Contractor shall assume full responsibility and liability for compliance with all applicable Federal, State, and Local regulations related to all aspects of the remediation project. The Remediation Contractor is responsible for providing and maintaining training, accreditations, medical exams, medical records, personal protective equipment (PPE), including respiratory protection, including respirator fit testing, as required by applicable Federal, State, and Local regulations. The Remediation Contractor shall hold the Government, NPS, Architect, and IHC harmless for any failure to comply with any applicable work, packaging, transporting, disposal, safety, health, or environmental requirement on the part of himself, his employees, or his subcontractors. The Contractor will incur all IHC costs, including sampling/analytical costs, to ensure compliance with OSHA/EPA/State requirements related to failure to comply with the regulations applicable to the work.

C. Federal Requirements

1. Federal requirements that govern some aspects of biological remediation include, but are not limited to, the following regulations.

a. OSHA

i. Title 29 CFR 1910.134 - Respiratory Protection

ii. 29 CFR 1926 - Construction Industry

iii. Title 29 CFR 1926.33 - Access to Employee Exposure and Medical Records

iv. Title 29 CFR 1926.59 same as 1910.1200 - Hazard Communication

v. Title 29 CFR 1926.500 - Guardrails, Handrails, and Covers

vi. Title 29 CFR 1926 Subpart C – General Safety and Health Provisions and Subpart D – Occupational Health and Environmental Controls

vii. Title 29 CFR 1926 Subpart E - Personal Protective Equipment and Life Saving Equipment

D. Standards

1. Standards that govern biological remediation activities include, but are not limited to, the following:

a. American National Standards Institute (ANSI)/ASSP Z9.2-2018 - Fundamentals Governing the Design and Operation of Local Exhaust Systems and ANSI/ASSE Z88.2-2015 - Practices for Respiratory Protection.

b. ANSI Z41.1 - Safety Toe Footwear.

c. ANSI Z87.1 - Practice for Occupational and Educational Eye and Face Protection

d. ANSI X88.6 - Respiratory Protection - Respiratory Use Physical Qualifications for Personnel

e. ANSI Z89.1 - Requirements for Industrial Head Protection

f. Underwriters Laboratories (UL) 586-2009 - UL Standard for Safety of HEPA Filter Units, 9th Edition; ANSI Approval 2017-12-19.

2. Standards governing testing laboratories:

a. American Industrial Hygiene Association (AIHA)

b. National Institute of Standards and Technology (NIST)

3. Standards that govern the fire and safety concerns in remediation work include, but are not limited to, the following:

a. National Fire Protection Association (NFPA) 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations.

b. National Fire Protection Association (NFPA) 10 - Standard for Fire Extinguishers.

c. NFPA 70 - Standard for National Electric Code.

d. NFPA 101 - Life Safety Code

e. NFPA 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations.

f. NFPA 701 - Standard Methods for Fire Tests for Flame-Resistant Textiles and Film.

g. Uniform Building Code (UBC) - 2006 Edition

E. Guidance Documents

1. Centers for Disease Control (CDC): Air Pollution and Respiratory Health -Prevention and remediation strategies for the Control and Prevention of Fungal Growth.

2. CDC: Histoplasmosis – Protecting Workers at Risk.

3. National Institute for Occupational Safety and Health (NIOSH): publication 2005-109: Histoplasmosis-Protecting Workers at Risk.

4. The Internet Center for Wildlife Damage Management: Bat Guano & Bat Feces cleanup.

F. NPS Responsibilities

1. Before the commencement of work:

a. Contracting Officer will notify others of project dates and requirements for relocation if needed.

b. Submit to the Remediation Contractor the results of previous reports of findings, including background air sampling, including the location of samples, the person who collected the samples, the equipment utilized, and the method of analysis. During remediation, submit to the Remediation Contractor the results of any sampling data collected during the remediation. This information shall not release the Remediation Contractor from any responsibility for OSHA compliance.

1.5 EMERGENCY ACTION PLAN AND ARRANGEMENTS

A. An Emergency Action Plan shall be developed before commencing remediation activities and shall be agreed to by the Contractor and the NPS. The Plan shall meet the requirements of 29 CFR 1926, Subpart C, Standard 1926.35 Employee Emergency Action Plans.

B. Emergency procedures shall be in written form and prominently posted near the regulated area. Everyone, prior to entering the regulated area, must read and sign these procedures to acknowledge understanding of the regulated area layout, location of emergency exits, and emergency procedures.

C. Emergency planning shall include written notification of police, fire, and emergency medical personnel of planned remediation activities; work schedule, and layout of regulated area, particularly barriers that may affect response capabilities.

D. Emergency planning shall include consideration of fire, explosion, hazardous atmospheres, electrical hazards, slips/trips and falls, confined spaces, and heat stress illness. Written procedures for response to emergencies shall be developed, and employee training in procedures shall be provided.

E. Employees shall be trained in regulated area/site evacuation procedures in the event of workplace emergencies.

1. For non-life-threatening situations - employees injured or otherwise incapacitated shall decontaminate following normal procedures with assistance from fellow workers, if necessary, before exiting the regulated area to obtain proper medical treatment.

2. For life-threatening injury or illness, worker decontamination shall take the least priority after measures to stabilize the injured workers, remove them from the regulated area, and secure proper medical treatment.

F. Telephone numbers of all emergency response personnel shall be kept at the job site along with the location of the nearest telephone.

G. The Remediation Contractor shall verify first aid/CPR training for personnel responsible for providing the training. OSHA requires medical assistance within 3-4 minutes of a life-threatening injury/illness. Bloodborne pathogen training shall also be verified for those personnel required to provide first aid/CPR.

H. The Emergency Action Plan shall provide a Contingency Plan if an incident occurs that may require the modification of the SOPs during remediation. Such incidents include, but are not limited to, fire, accident, power failure, and negative pressure failure. The Remediation Contractor shall detail procedures to be followed in the event of an incident, assuring that work is stopped, and wetting is continued until correction of the problem.

1.7 PRE-CONSTRUCTION MEETING

A. Before commencing the work, the Remediation Contractor shall meet with the Contracting Officer to present and review, as appropriate, the items following this paragraph. The Remediation Contractor's Competent Person(s) who will be on-site shall participate in the pre-construction meeting. The pre-construction meeting is to discuss and determine procedures to be used during the project. At this meeting, the Remediation Contractor shall provide four copies of:

1. A list of all workers who will participate in the project, including their experience.

2. A list of and verification of training for all personnel who have current First Aid/ CPR training. A minimum of one person per shift must have adequate training.

3. Current medical written opinions for all personnel working on-site meeting the requirements of 29 CFR 1926.1101 (m).

4. Current fit tests for all personnel wearing respirators on-site meet the requirements of 29 CFR 1926.1101 (h) and Appendix C.

5. A copy of the Remediation Contractor's SOPs for Bird, Bat, and Rodent Dropping Remediation. In these procedures, the following information must be detailed, specifically for this project.

    a. Regulated area preparation procedures.

    b. Remediation methods/procedures and equipment to be used.

    c. PPE to be used.

    d. Decontamination area set-up/layout and decontamination procedures for employees; method of demarcating the regulated work area with Biological Warning Signs; and method for fall protection, if required.

6. At this meeting, the Remediation Contractor shall provide all submittals as required.

7. Procedures for handling, packaging, storage, and disposal of bird, bat, and rodent dropping waste.

8. Emergency Action Plan and Contingency Procedures.

1.8 PROJECT COORDINATION

The following are the minimum administrative and supervisory personnel necessary for the coordination of the work.

A. Personnel

1. Administrative and supervisory personnel shall consist of a qualified Competent Person. These employees are the Remediation Contractor's representatives responsible for compliance with these specifications and all other applicable requirements.

2. Non-supervisory personnel shall consist of an adequate number of qualified personnel to meet the schedule requirements of the project. A request for approval shall be submitted for any person to be employed during the project, giving the person's name, qualifications, Certificate of Worker's Acknowledgment, Affidavit of Medical Surveillance and Respiratory Protection, and current Respirator Fit Test.

3. Minimum qualifications for the Remediation Contractor and assigned personnel are:

   a. The Remediation Contractor has conducted within the last three (3) years, three (3) projects of similar complexity and dollar value as this project; has adequate liability/occurrence insurance for the work as required by the State; is licensed in applicable State; has adequate and qualified personnel available to complete the work; has comprehensive SOPs for bird, bat, and rodent dropping remediation work; and has adequate materials, equipment, and supplies to perform the work.

   b. The Competent Person has four (4) years of remediation experience, of which two (2) years were as the Competent Person on the project; meets the OSHA definition of a Competent Person; and has all required OSHA documentation related to medical and respiratory protection.

   c. The Remediation Personnel shall have completed training on the Remediation Contractors SOPs; have one (1) year of bird, bat, and rodent dropping remediation experience within the past three (3) years of similar size and complexity; have applicable medical and respiratory protection documentation; have a certificate of training and a State accreditation/license.

4. All personnel are to comply with OSHA construction safety training as applicable and submit certification documents.

1.9 RESPIRATORY PROTECTION

A. General – Respiratory Protection Program

1. The Remediation Contractor shall develop and implement a Respiratory Protection Program (RPP) that complies with OSHA requirements found at 29 CFR 1926.1101 and 29 CFR 1910.132;134. ANSI Standard Z88.2-2015 provides excellent guidance for developing a respiratory protection program. All respirators used must be NIOSH-approved for bird, bat, and rodent dropping remediation activities. The written respiratory protection shall, at a minimum, contain the basic requirements found at 29 CFR 1910.134 (c) - Respiratory Protection Program.

B. Respiratory Protection Program Coordinator

1. The Respiratory Protection Program Coordinator (RPPC) must be identified and shall have two (2) years of experience coordinating RPP of a similar size and complexity. The RPPC must submit a signed statement attesting to the fact that the program meets the above requirements.

C. Selection and Use of Respirators

1. The procedure for the selection and use of respirators must be submitted to the NPS as part of the Remediation Contractor's qualification. The procedure must be written clearly enough for workers to understand. A copy of the Respiratory Protection Program must be available onsite, for reference by employees or authorized visitors.

D. Minimum Respiratory Protection

1. Minimum respiratory protection shall be a half-face air-purifying respirator equipped with P100 filters. Respirator selection shall meet the requirements of 29 CFR 1926.1101 (h) and 29 CFR 1910.134 (d)(3)(i)(A) Table 1, except as indicated in this paragraph. Remediation personnel must have a respirator for their exclusive use.

E. Medical Written Opinion

1. No employee shall be allowed to wear a respirator unless a physician or other licensed health care professional has provided a written determination stating they are medically qualified to wear the class of respirator to be used on the project while wearing whole body impermeable garments and subjected to heat or cold stress.

F. Respirator Fit Test

1. All personnel wearing respirators shall have a current qualitative/quantitative fit test conducted in accordance with 29 CFR 1910.134 (f) and Appendix A.

G. Respirator Fit Check

1. The Competent Person shall ensure that the positive/negative pressure user seal check is done each time the respirator is donned by an employee. Head coverings must cover respirator head straps. Any situation that prevents an effective facepiece-to-face seal, as evidenced by the failure of a user seal check, shall preclude that person from entering the regulated area until the resolution of the problem.

H. Maintenance and Care of Respirators

1. The Respiratory Protection Program Coordinator shall submit documentation showing compliance with 29 CFR 1910.134 (h) Maintenance and Care of Respirators.

I. Supplied Air Systems: If a supplied air system is used, the system shall meet all requirements of 29 CFR 1910.134 and the ANSI/Compressed Gas Association (CGA) Commodity Specification for Air current requirements for Type 1 - Grade D breathing air. Low-pressure systems are not allowed to be used on this remediation. Supplied Air respirator use shall be in accordance with EPA/NIOSH publication EPA-560-OPTS-86-001. The competent person on site will be responsible for the supplied air system to ensure the safety of the worker.

1.10 WORKER PROTECTION

A. Medical Examinations

1. Medical examinations shall be provided for all personnel working in the regulated area, regardless of exposure levels. The physician's written opinion shall be provided for each person and shall include a statement indicating that the person has been evaluated for working in a heat and cold stress environment while wearing PPE and can perform the work without risk of material health impairment.

B. Personal Protective Clothing

1. Provide whole body clothing, head coverings, foot coverings, and any other personal protective equipment as determined by conducting the hazard assessment required by OSHA at 29 CFR 1910.132 (d). The Competent Person shall ensure the integrity of personal protective equipment worn for the duration of the project. Duct tape shall be used to secure all suit sleeves to wrists and to secure foot coverings at the ankle.

C. Regulated Area Entry Procedure

1. The Competent Person shall ensure that each time workers enter the regulated area, they place protective disposable coveralls over ALL street clothes in the area established by the competent person and put on head coverings, fall protection, PPE, a clean respirator, and then proceed to the regulated work area.

D. Decontamination Procedure

1. The Competent Person shall require all personnel to adhere to the following decontamination procedures whenever they leave the regulated area.

a. When exiting the regulated area, HEPA vacuum contaminated work clothing while it is still being worn. At no time may birds, bats, and rodent droppings be removed from protective clothing by any means that would result in uncontrolled dispersal of birds, bats, and rodent droppings/dust into the air. Remove shoe coverings and leave them in the decontamination area. Remove protective coveralls by carefully rolling down the garment to reduce exposure to bird, bat, and rodent dropping dust.

b. Still wearing the respirator, care must be taken to follow reasonable procedures in decontamination and removing the respirator to avoid the inhalation or ingestion of bird, bat, and rodent dropping dust. The following procedure is required, as a minimum:

i. Thoroughly HEPA vacuum body from neck down. Vacuum hair as thoroughly as possible.

ii. With the respirator still in place, wash hands, exposed areas of the face, and hair with soap and water. Carefully remove the respirator while proceeding out of the decontamination area.

iii. Dispose of wetted or overloaded filters from the respirator.

iv. Carefully decontaminate the face piece of the respirator inside and out. If using a PAPR, shut down using the following sequence: a) first cap inlets to filters; b) turn blower off to keep debris collected on the inlet side of the filter from dislodging and contaminating the outside of the unit; c) thoroughly decontaminate blower and hoses; d) carefully decontaminate battery pack with a wet rag being cautious of getting water in the battery pack thus preventing destruction. (Note: THIS PROCEDURE IS NOT A SUBSTITUTE FOR RESPIRATOR CLEANING!).

v. Shower and wash body completely with soap and water. Rinse thoroughly.

vi. Carefully decontaminate the respirator face piece and respirator inside and out. (NOTE: THIS IS NOT A SUBSTITUTE FOR RESPIRATOR CLEANING!).

c. Where containment is not required for bird, bat, and rodent dropping removal, an adequate washing station will be provided for the employees for the removal of disposable PPE and to clean the respirator.

E. Regulated Area Requirements

1. The Competent Person shall ensure that all requirements for regulated areas are met. All personnel in the regulated area shall not be allowed to eat, drink, smoke, chew tobacco or gum, apply cosmetics, or in any way interfere with the fit of their respirator.

1.11 DECONTAMINATION FACILITIES

A. Description

1. Provide each regulated area with a fiber drum with a disposal bag in it for personnel waste materials. The Competent Person shall provide a specific area at the outer boundary of the regulated area and designate it specifically for the decontamination of personnel and respirators.

B. Waste/Equipment Decontamination Area (W/EDA)

1. The Competent Person shall provide a specific area in the regulated area designated for the removal of all waste, equipment, and contaminated material from the regulated area.

C. Waste/Equipment Decontamination Procedures

1. Contain all waste in 6-mil poly bags or approved equivalent. Clean/decontaminate bags and pass them through into another bag or approved equivalent. Remove to the disposal dumpster or enclosed vehicle.

PART 2 - PRODUCTS, MATERIALS, AND EQUIPMENT

2.1 MATERIALS AND EQUIPMENT

A. General Requirements

1. Before the start of work, the Remediation Contractor shall provide and maintain enough materials and equipment to ensure continuous and efficient work throughout the duration of the project. Work shall not start unless the following items have been delivered to the site and the Competent Person has submitted verification to the NPS's representative.

2. All materials shall be delivered in their original package, container, or bundle bearing the name of the manufacturer and the brand name (where applicable).

3. Store all materials subject to damage off the ground, away from wet surfaces, and under a cover sufficient to prevent damage or contamination. Flammable materials cannot be stored inside buildings. Replacement materials shall be stored outside of the regulated area until remediation is completed.

4. The Remediation Contractor shall not block or hinder the use of the site by staff and visitors in partially occupied buildings by placing materials/equipment in any unauthorized place.

5. The Competent Person shall inspect for damaged, deteriorating, or previously used materials. Such materials shall not be used and shall be removed from the worksite and disposed of properly.

6. An adequate number of HEPA vacuums, scrapers, sprayers, nylon brushes, brooms, disposable mops, rags, sponges, staple guns, shovels, ladders, and scaffolding of suitable height and length as well as meeting OSHA requirements, fall protection devices, water hose to reach all areas in the regulated area, airless spray equipment, and any other tools, materials or equipment required to conduct the remediation project. All electrically operated hand tools, equipment, and electric cords shall be connected to GFCI protection.

7. Special protection for objects in the regulated area shall be detailed to prevent damage from scaffolds, water, and falling material.

8. Provide unlabeled, 6-mil thick, leak-tight clear polyethylene bags. Asbestos or lead warning labels are not permitted on the bags.

9. The NPS shall be provided a copy of the SDS as required for all hazardous chemicals under OSHA 29 CFR 1910.1200 - Hazard Communication. Methylene chloride shall not be used with any spray adhesive or other product.

10. Warning demarcation signs, as many as required by shall be provided and placed by the Competent Person. All other posters and notices required by Federal, State, and Local regulations shall be posted at the outer boundary of the regulated area or approved equivalent.

11. Adequate and appropriate PPE for the project and the number of personnel/shifts shall be provided. All personal protective equipment issued must be based on a written hazard assessment conducted under 29 CFR 1910.132(d).

12. Provide standard detergents to be used for cleaning, which shall be diluted in water and used in a manner as directed by the manufacturer's labeling.

13. If approved for use by NPS in writing, any disinfecting agents shall be utilized as directed by the manufacturer's labeling.

14. Cleaners, Disinfectants, and Solutions: Use of any of the following is acceptable:

    a. Disinfectants/Biocides and Cleaners: Any NPS-approved EPA-registered disinfectants appropriate for cleaning bird/animal droppings.

b. Formalin Solution: Dilute commercially available formaldehyde (containing 37-40 percent by weight of formaldehyde gas in water, stabilized with 10-15 percent methanol) with water to create a 5-percent, by volume, formalin solution.

c. Chlorine Solution: 10% Chlorine Bleach and 90% warm water mixture.

15. Any damage to building components not scheduled for general renovation, resulting from the Remediation Contractor's work and/or work practices, shall be repaired or replaced at the sole cost of the Remediation Contractor utilizing appropriately qualified and insured tradespersons equal to or greater than the original condition.

2.3 MONITORING, INSPECTION, AND TESTING

A. General

1. NPS may employ an independent industrial hygiene consultant (IHC) to perform assorted services on behalf of NPS. The IHC will perform the necessary monitoring, inspection, testing, and other support services to ensure that the NPS employees and visitors will not be adversely affected by the remediation work and that the remediation work proceeds in accordance with these specifications, and that the abated areas or abated buildings have been successfully decontaminated. The work of the IHC in no way relieves the Remediation Contractor from their responsibility to perform the work in accordance with contract/specification requirements, to perform continuous inspection, monitoring, and testing for the safety of their employees, and to perform other such services as specified. The cost of the IHC and their services will be borne by the NPS, excluding any repeat of final inspection and testing that may be required due to unsatisfactory initial results.

B. Scope of Services of the IHC

1. The purpose of the work of the IHC is to: Assure quality; resolve problems; and prevent the spread of contamination beyond the regulated areas. In addition, their work includes performing the final inspection and testing to determine whether the regulated areas or buildings have been adequately decontaminated. The IHC may perform the following tasks:

a. Task 2: Perform periodic air monitoring, inspection, and testing outside the regulated area during actual remediation work to detect any faults in the regulated area isolation and any adverse impact on the surroundings from regulated area activities.

b. Task 3: Perform unannounced visits to spot-check overall compliance of work with contract/specifications. These visits may include any inspection, monitoring, and testing inside and outside the regulated area.

c. Task 4: Provide support to the Contracting Officer, such as evaluation of submittals from the Remediation Contractor, scheduling, resolution of unforeseen developments, etc.

d. Task 5: Perform, in the presence of the Contracting Officer, the final inspection and testing of decontaminated regulated areas at the conclusion of the remediation to certify compliance with the NPS requirements/specifications.

e. Task 6: Issue a Certificate of Decontamination for each regulated area and project report.

2. All documentation, inspection results, and testing results generated by the IHC will be available to the Remediation Contractor for information and consideration. The Remediation Contractor shall cooperate with and support the IHC for the efficient and smooth performance of their work.

3. The monitoring and inspection results of the IHC will be used by the NPS to issue any stop removal orders to the Remediation Contractor during remediation work and to accept or reject a regulated area or building as decontaminated.

2.4 STANDARD OPERATING PROCEDURES

A. The Remediation Contractor shall have established SOPs in printed form and a loose-leaf folder consisting of simplified text, diagrams, sketches, and pictures that establish and clearly explain the procedures to be followed during all phases of the work by the Remediation Contractor's personnel. The SOPs must be modified as needed to address specific requirements of the project. The SOPs shall be submitted for review and approval before the start of any remediation work. The minimum topics and areas to be covered by the SOPs are:

1. Minimum Personnel Qualifications

2. Contingency Plans and Arrangements

3. Security and Safety Procedures

4. Respiratory Protection/PPE Program and Training

5. Medical Surveillance Program and Recordkeeping

6. Regulated Area Requirements - Establishment and Management of the Regulated Area

7. Decontamination Facilities and Entry/Exit Procedures

8. Monitoring, Inspections, and Testing

9. Removal Procedures for bird, bat, and rodent droppings

10. Disposal of bird, bat, and rodent dropping waste

11. Regulated Area Decontamination/Clean-up

12. Regulated Area Visual and Clearance Standards

13. Project Completion/Closeout

2.5 SUBMITTALS

A. Pre-Construction Meeting Submittals

1. Submit to the NPS a minimum of 14 days before the pre-start meeting, the following for review and approval. Meeting this requirement is a prerequisite for the pre-start meeting for this project:

    a. Submit an Emergency Action Plan. The Plan shall meet the requirements of 29 CFR 1926, Subpart C, Standard 1926.35 Employee Emergency Action Plans.

    b. Submit a detailed work schedule for the entire project reflecting contract documents and the phasing/schedule requirements.

    c. Submit a staff organization chart showing all personnel who will be working on the project and their capacity/function. Provide their qualifications, training, accreditations, and licenses, as appropriate. Provide a copy of the "Certificate of Worker's Acknowledgment" and the "Affidavit of Medical Surveillance and Respiratory Protection" for each person.

    d. SOPs developed specifically for this project, incorporating the requirements of the specifications, prepared, signed, and dated.

    e. Submit the specifics of the materials and equipment to be used for this project, with manufacturer names, model numbers, performance characteristics, pictures/diagrams, and numbers available for the following:

        i. HEPA vacuums and emergency power-generating systems.

        ii. Respirators, protective clothing, fall protection, and other required PPE.

        iii. Fire safety equipment to be used in the regulated area.

    f. Submit the specific contingency/emergency arrangements made with local health, fire, ambulance, and hospital authorities and any other notifications/arrangements.

    g. Submit the name, location, and verification of the laboratory and/or personnel to be used for analysis of any samples.

    h. Submit qualifications verification: Submit the following evidence of qualifications. Make sure that all references are current and verifiable by providing current phone numbers and documentation.

        i. Remediation Company: Project experience within the past three (3) years; listing projects first, most like this project: Project Name; Type of Remediation; Duration; Cost; Reference Name/Phone Number; Final Clearance; Completion Date.

ii. List of project(s) halted by the NPS, A/E, IH, or regulatory agency in the last three (3) years: Project Name; Reason; Date; Reference Name/Number; Resolution.

iii. List any regulatory citations (e.g., OSHA), notices of violations (e.g., Federal, State, and EPA), penalties, and legal actions taken against the company, including those of the company's officers (including damages paid) in the last three (3) years. Provide copies and all information needed for verification.

i. Submit personnel information: Provide a resume; address each item completely; provide references; phone numbers; copies of certificates, accreditations, and licenses. Submit an affidavit signed by the Competent Person stating that all personnel submitted below have medical records in accordance with OSHA 29 CFR 1926.1101(m), 29 CFR 1910.134, and that the company has implemented a medical surveillance program and maintains recordkeeping in accordance with the above regulations. Submit the phone number and doctor/clinic/hospital used for medical evaluations.

i. Competent Person(s)/Supervisor(s): Number; names; years of remediation experience as Competent Person/Supervisor; list of similar projects as Competent Person/Supervisor; as a worker; the maximum number of personnel supervised on a project; medical opinion; current respirator fit test.

ii. Workers: Numbers; names; years of remediation experience; medical opinion; current respirator fit test.

j. Submit copies of the State license; copy of the insurance policy; copy of SOPs incorporating the requirements of this specification; information on who provides your training, how often, and who provides medical surveillance.

k. Rented equipment must be decontaminated before returning to the rental agency.

l. Procedures to be utilized for any cleaning and disinfecting solutions, and the proposed list of EPA-registered biocides, disinfectants, and microbial coatings to be utilized on the project.

m. Waste transport routes through the building, or exterior to the waste storage container for each regulated area/containment.

n. Submit, before the start of work, the manufacturer's technical data for all types of disinfectants/cleaners, all SDSs, and application instructions.

B. Submittals During Remediation

1. The Competent Person shall maintain and submit a daily log at the regulated area documenting the dates and times of the following: purpose, attendees, and summary of meetings; all personnel entering/exiting the regulated area; document and discuss the resolution of unusual events such as barrier breaching, equipment failures, emergencies, and any cause for stopping work; representative air monitoring and results. Submit this information weekly to the Contracting Officer.

2. The Competent Person shall document and maintain the inspection and approval of the regulated area preparation before the start of work and daily during work.

   a. Removal of any poly barriers

   b. Visual inspection/testing by the Competent Person before application of lockdown encapsulant, if used.

   c. Packaging and removal of bird, bat, and rodent dropping waste from regulated areas.

   d. Disposal of bird, bat, and rodent dropping waste materials.

C. Submittals at Completion of Remediation

1. The Remediation Contractor shall submit a project report consisting of the daily logbook requirements and documentation of events during the remediation project. It will also include information on the containment. The report shall include a certificate of completion, signed and dated by the Competent Person, in accordance with Attachment #1. All sampling documentation must be submitted. The Contracting Officer will retain the report after the completion of the project and provide copies of the report to the NPS Office of Engineering and the Safety Office.

PART 3 - EXECUTION

3.1 PRE-REMEDIATION ACTIVITIES

A. Pre-Remediation Meeting

1. The Contracting Officer, upon receipt, review, and substantial approval of all pre-remediation submittals and verification that all materials and equipment required for the project are on the site, will arrange for a pre-remediation meeting between the Remediation Contractor, Competent Person(s), the Contracting Officer, and the IHC. The purpose of the meeting is to discuss any aspect of the submittals needing clarification or amplification, and to discuss any aspect of the project execution and the sequence of the operation. The Remediation Contractor shall be prepared to provide any supplemental information/documentation to the Contracting Officer regarding any submittals, documentation, materials, or equipment. Upon satisfactory resolution of any outstanding issues, the Contracting Officer will issue a written order to proceed to the Contractor. No remediation work of any kind described in the following provisions shall be initiated before the NPS's written order to proceed.

B. Pre-Remediation Inspections and Preparations

1. Before any work begins on the construction of the regulated area, the Remediation Contractor will:

   a. Conduct a space-by-space inspection with the Contracting Officer and prepare a written inventory of all existing damage in those spaces where bird, bat, and rodent remediation will occur. Still or video photography may be used to supplement the written damage inventory. Documentation will be signed and certified as accurate by both parties.

   b. Shut down and lockout in accordance with 29 CFR 1910.147 all electrical circuits that pose a potential hazard. Electrical arrangements will be tailored to the regulated area and the systems involved. All electrical circuits affected will be turned off at the circuit box outside the regulated area, not just the wall switch. The goal is to eliminate the potential for electrical shock, which is a major threat to life in the regulated area due to possible energized circuits. Electrical lines used to power equipment in the regulated area shall conform to all electrical safety standards and shall be isolated using a ground fault circuit interrupter (GFCI). All GFCIs shall be tested before use.

   c. Inspect existing firestopping in the regulated area. Correct as needed.

C. Pre-Remediation Construction and Operations

1. Perform all preparatory work for the first regulated area in accordance with the approved work schedule and with this specification.

2. Upon completion of all preparatory work, the IHC will inspect the work and systems and will notify the Contracting Officer when the work is completed in accordance with this specification. The Contracting Officer may inspect the regulated area and the systems with the IHC and may require that, upon satisfactory inspection, the Remediation Contractor's employees perform all major aspects of the approved SOPs, especially worker protection, fall protection, respiratory systems, contingency plans, decontamination procedures, and monitoring to demonstrate satisfactory operation.

3. The Competent Person shall document the pre-remediation activities described above and deliver a copy to the Contracting Officer.

4. Upon satisfactory inspection of the installation and operation of systems, the Contracting Officer will notify the Remediation Contractor in writing to proceed with the bird, bat, and rodent droppings remediation work in accordance with this specification.

3.2 REGULATED AREA PREPARATIONS

A. Site Security

1. Regulated area access is to be restricted only to authorized, trained, and protected personnel. These may include the Remediation Contractor's employees, employees of subcontractors, NPS employees and representatives, State and local inspectors, and any other designated individuals. A list of authorized personnel shall be established before commencing the project and shall be located immediately outside the boundary of the regulated work area.

2. Entry into a regulated area by unauthorized individuals shall be reported immediately to the Competent Person by anyone observing the entry. The Competent person shall immediately notify the Contracting Officer.

3. A logbook shall be maintained outside the boundary of the regulated work area. Anyone who enters the regulated area must record their name, affiliation, time in, and time out for each entry.

4. Access to a regulated area shall be through a single access point. All other access (doors, windows, hallways, etc.) shall be sealed or locked to prevent entry to or exit from the regulated area. The only exceptions to this requirement are the waste loadout area, which shall be sealed or locked except during the removal of containerized bird, bat, and rodent dropping waste from the regulated area, and emergency exits. Emergency exits shall not be locked from the inside or outside.

5. The Remediation Contractor Competent Person shall control site security during remediation operations to isolate work in progress and protect adjacent personnel. Security shall be provided at the entrance to the regulated area to ensure that all entrants are logged in/out and that only authorized personnel are allowed entrance.

6. The Remediation Contractor will have the NPS's assistance in notifying adjacent personnel of the presence, location, and quantity of bird, bat, and rodent droppings in the regulated area and enforcement of restricted access by NPS employees.

7. The regulated area shall be locked during non-working hours.

B. Signage and Power Management

1. Post WARNING signs or tape at any location approaching the regulated area. Signs shall be posted at a distance sufficiently far away from the regulated area to permit any person to read the sign and take the necessary measures to avoid exposure. Additional signs will be posted following the construction of the regulated area.

a. Provide signage with a minimum size of 8-1/2 inches by 11 inches with a white background on which is printed in large type with wording in the format shown below:

BIOLOGICAL WASTE REMEDIATION
DO NOT ENTER
AUTHORIZED PERSONNEL ONLY
RESPIRATORS AND PROTECTIVE CLOTHING
ARE REQUIRED IN THIS AREA

2. Shut down and lock out electric power to the regulated area. Provide temporary power and lighting (if working at night). Ensure safe installation, including GFCI of temporary power sources and equipment, in compliance with all applicable electrical code requirements and OSHA requirements for temporary electrical systems. Electricity shall be provided by the NPS at no expense to the Remediation Contractor.

3. Shut down and lock out heating, cooling, and air conditioning system (HVAC) components that are in, supply, or pass through the regulated area. Investigate and document the regulated area and agree on the pre-remediation condition with the Contracting Officer. Seal all intake and exhaust vents in the regulated area with duct tape and two (2) layers of 6-mil plastic. Also, seal any seams in system components that pass through the regulated area.

C. Negative Pressure Filtration System

1. The Remediation Contractor shall provide a HEPA AFD from the regulated area directly to the exterior. The Competent Person shall determine the number of units needed for each regulated area. Provide a standby unit in the event of machine failure and/or emergency in an adjacent area. All AFDs shall be exhausted from the regulated area to the exterior. NIOSH has done extensive studies and has determined that negative air machines typically operate at ~50% efficiency. The Remediation Contractor shall consider this in their determination of the number of units needed in each regulated area.

D. Design and Layout

1. Before the start of work, submit the design and layout of the regulated area and the negative air machines. The submittal shall indicate the number of, location of, and size of negative air machines. The point(s) of exhaust, airflow within the regulated area, anticipated negative pressure differential, and supporting calculations for sizing shall be provided. In addition, submit the following:

a. Manufacturer's information on the negative air machine(s).

b. Method of supplying power to the units and designation/location of the panels.

c. Description of testing method(s) for correct air volume and pressure differential. Provide the manufacturer's product data on the pressure differential measuring device used.

d. If an auxiliary power supply is to be provided for the negative air machines, provide a schematic diagram of the power supply and manufacturer's data on the generator and switch.

E. Negative Air Machines (HEPA Units)

1. Negative Air Machine Cabinet: The cabinet shall be constructed of steel or other durable material capable of withstanding potential damage from rough handling and transportation. The width of the cabinet shall be less than 30" in order to fit in standard doorways. The cabinet must be factory sealed to prevent dust from being released during use, transport, or maintenance. Any access to and replacement of filters shall be from the inlet end. The unit must be on casters or wheels.

2. Negative Air Machine Fan: The rating capacity of the fan must be the air moving capacity under actual operating conditions. Manufacturers typically use "free-air" (no resistance) conditions when rating fans. The fan must be a centrifugal-type fan.

3. Negative Air Machine Final Filter: The final filter shall be a HEPA filter. The filter media must be completely sealed on all edges within a structurally rigid frame. The filter shall align with a continuous flexible gasket material in the negative air machine housing to form an air-tight seal. Each HEPA filter shall be individually tested and certified by the manufacturer to have an efficiency of not less than 99.97% when challenged with 0.3μm dioctylphthalate (DOP) particles. Testing shall have been done in accordance with Military Standard MIL-STD-282 and Army Instruction Manual 136-300-175A. Each filter must bear a UL586 label to indicate the ability to perform under specified conditions. Each filter shall be marked with the name of the manufacturer, serial number, air flow rating, efficiency, resistance, and the direction of test airflow.

4. Negative Air Machine Pre-filters: The pre-filters, which protect the final HEPA filter by removing larger particles, are required to prolong the operating life of the HEPA filter. Two stages of pre-filtration are required. A first-stage pre-filter shall be a low-efficiency type for particles 10 μm or larger. A second-stage pre-filter shall have a medium efficiency effective for particles down to 5 μm or larger. Pre-filters shall be installed either on or in the intake grid of the unit and held in place with special housing or clamps.

5. Negative Air Machine Instrumentation: Each unit must be equipped with a gauge to measure the pressure drop across the filters and to indicate when filters have become loaded and need to be changed. A table indicating the cfm for various pressure readings on the gauge shall be affixed near the gauge for reference, or the reading shall indicate at what point the filters shall be changed, noting cfm delivery at that point. The unit must have an elapsed time meter to show the total hours of operation.

6. Negative Air Machine Safety and Warning Devices: An electrical/ mechanical lockout must be provided to prevent the fan from being operated without a HEPA filter. Units must be equipped with an automatic shutdown device to stop the fan in the event of a rupture in the HEPA filter or blockage in the discharge of the fan. Warning lights are required to indicate normal operation, too high a pressure drop across filters, or too low a pressure drop across filters.

7. Negative Air Machine Electrical: All electrical components shall be approved by the National Electrical Manufacturers' Association (NEMA) and Underwriters' Laboratories (UL). Each unit must be provided with overload protection, and the motor, fan, fan housing, and cabinet must be grounded.

F. Supplemental Make-Up Air Inlets

1. Provide, as needed, proper airflow in the regulated area, in locations approved by the NPS's representative, by making openings in the plastic sheeting to allow outside air to flow into the regulated area. Auxiliary makeup air inlets must be located as far from the negative air machines as possible, off the floor near the ceiling, and away from the barriers that separate the regulated area from the occupied clean areas. Cover the inlets with weighted flaps, which will seal in the event of failure of the negative pressure system. The flap must be sprayed with adhesive to ensure sealing if it closes.

G. Testing the System

1. The negative pressure system must be tested before bird, bat, and rodent droppings are disturbed in any way. After the regulated area has been completely prepared, the decontamination units set up, and the negative air machines installed, start the units up one at a time. Demonstrate and document the operation of the system to the NPS's representative and/or IHC using smoke tubes. Testing must also be done at the start of each work shift.

H. Use of System During Remediation Operations

1. Start units before the disturbance of bird, bat, and rodent droppings occurs. After work begins, the units shall run continuously until final visual clearance is complete.

2. The negative air machines shall not be shut down for the duration of the Project unless authorized by the NPS's representative, in writing.

3. Remediation work shall begin at a location farthest from the units and proceed towards them. If an electric failure occurs, the Competent Person shall stop all remediation work and immediately begin wetting all exposed bird, bat, and rodent droppings for the duration of the power outage. Remediation work shall not resume until power is restored and all units necessary are operating properly again.

4. The negative air machines shall continue to run after all work is completed and until a final visual clearance has been completed for that regulated area.

I. Dismantling the System

1. After completion of the final visual clearance has been obtained, the units may be shut down. The units shall have been completely decontaminated, all pre-filters removed and disposed of, and the unit inlet and outlet sealed with two layers of 6-mil polyethylene.

3.3 CONTAINMENT BARRIERS AND COVERINGS IN THE REGULATED AREA

A. General

1. Using critical barriers, seal off the regulated area to completely isolate it from adjacent spaces. The contamination is at no additional cost to the NPS. Provide firestopping and identify all fire barrier penetrations due to remediation work as specified in Section 2.2 E: Firestopping.

B. Preparation Before Sealing the Regulated Area

1. Place all tools, scaffolding, materials, and equipment needed for working in the regulated area before erecting any plastic sheeting. Lock out and tag out any HVAC systems in the regulated area, if required.

C. Controlling Access to the Regulated Area

1. Access to the regulated area is allowed only at the Competent Person's designated location and shall serve as a personnel decontamination area. All other means of access shall be eliminated, and warning demarcation signs shall be posted as required.

D. Extension of the Regulated Area

1. If the enclosure of the regulated area is breached in any way that could allow contamination to occur, the affected area shall be included in the regulated area and constructed as per this section. If the affected area cannot be added to the regulated area, decontamination measures must be started immediately.

E. Firestopping

1. All walls, floors, and ceilings are considered fire-rated unless otherwise determined by the Contracting Officer or Fire Marshall.

2. Any visible openings, whether caused by a penetration, shall be reported by the Remediation Contractor to the Contracting Officer for a sealant system determination. Firestops shall meet ASTM E814 and UL 1479 requirements for the opening size, penetrant, and fire rating needed.

F. The Remediation Contractor shall provide sanitary facilities for remediation personnel and maintain them in a clean and sanitary condition throughout the remediation project.

G. The NPS will provide water for remediation purposes. The Remediation Contractor shall connect to the existing system. The service to the shower(s), if used, shall be supplied with backflow prevention.

H. The Remediation Contractor shall pre-clean all movable objects and refuse within the regulated area using a HEPA-filtered vacuum and/or wet cleaning methods as appropriate. Do not use any methods that would raise dust, such as dry sweeping or vacuuming with equipment not equipped with HEPA filters. Do not disturb bird, bat, and rodent droppings during this pre-cleaning phase.

3.4 REMOVAL AND DISINFECTION OF BIRD, BAT, AND RODENT DROPPINGS

A. General

1. The NPS must be notified at least 24 hours in advance of any waste removed from the containment. Keep materials wet while working with them; wrap as soon as possible with two (2) layers of 6-mil plastic for disposal or approved equivalent.

B. Wetting Materials

1. Use amended water for the wetting of bird, bat, and rodent droppings. The Competent Person shall ensure the wetting of bird, bat, and rodent droppings is completed for the duration of the Project. A disinfection/biocide and cleaning solution may be used instead of amended water with the written approval of the NPS's representative.

2. Amended Water: Provide water to which a surfactant has been added to wet the bird, bat, and rodent droppings and reduce the potential for dust release during disturbance of the bird, bat, and rodent droppings. The mixture must be equal to or greater than the wetting provided by water amended by a surfactant consisting of one ounce of 50% polyoxyethylene ester and 50% polyoxyethylene ether, mixed with 5 gallons (19L) of water.

3. Cleaners, Disinfectants, and Solutions: Use of any of the following is acceptable:

a. Disinfectants/Biocides and Cleaners: Any NPS-approved EPA-registered disinfectants appropriate for cleaning bird/animal droppings.

b. Formalin Solution: Dilute commercially available formaldehyde (containing 37-40 percent by weight of formaldehyde gas in water, stabilized with 10-15 percent methanol) with water to create a 5-percent, by volume, formalin solution.

c. Chlorine Solution: 10% Chlorine Bleach and 90% warm water mixture.

C. Removal and Disinfection of bird, bat, and rodent droppings

1. All waste shall be decontaminated and/or removed within containment. As waste is removed, it shall be placed into a disposal container promptly. Disposal procedures, at a minimum, shall consist of double bagging using 6-mil clear polyethylene bags or double wrapping with 6-mil polyethylene sheeting. Bags shall be taped to form an airtight seal. Waste from HEPA-filtered vacuums shall be double-bagged in 6-mil clear polyethylene bags.

2. The Remediation Contractor shall keep the site and work areas free from accumulations of bagged material or rubbish caused by its operations and free from any flammable materials or other sources of fire hazard.

3. Work from top to bottom.

    a. Remove gross residue first to prevent spreading material to non-contaminated areas.

    b. Gross removal of areas of accumulated waste can be performed with scrapers, plastic shovels, etc., directly into bags. Keep the material wet.

4. Use of the Disinfectants/Biocides and Cleaner:

    a. Follow the manufacturer's written instructions for product use and applications.

    b. Do not perform final application (if required by manufacturer) until final inspection and approval by the IHC

    c. Upon written notification of completion of initial cleaning, apply approved biocide/disinfectant to all cleaned surfaces.

5. Use of Formalin Solution

    a. Conduct decontamination when the temperature of the material being decontaminated is between 62F and 90F. Formalin solutions are less effective at temperatures outside this range.

    b. Completely saturated infected areas and the immediately adjacent areas that may have little or no droppings visible.

    c. Follow manufacturer's printed instructions (if available), or the following CDC procedures. Quantities may vary; the proper amount of formalin to use is that which completely saturates the contaminated material.

    d. Vertical Surfaces: One gallon per 150 square feet.

    e. Horizontal Surfaces (except on soil): One gallon per six (6) square feet.

    f. Soil Surface: One gallon per one (1) square foot.

    g. Exercise care to avoid formalin runoff.

    h. Where bird, bat, and rodent excrement are deep, apply formalin solution on alternate days to enhance the probability of contact with all of the infected material.

    i. Disinfect contaminated equipment by soaking it in 5 percent formalin solution for 15 minutes.

6. Use of the Chlorine Solution:

    a. Use a light mist of the solution applied with pump action hand sprayer.

    b. Do not use high pressure or a direct stream, as this will scatter waste.

7. Hosing or pressure washing shall not be permitted for removing interior deposited droppings.

8. All visually impacted materials and adjacent visually unimpacted material shall be cleaned and/or removed under full containment unless specified otherwise.

9. Cleaning and disinfecting agents shall be utilized to clean all remaining surfaces within the Work Area(s).

10. If areas adjoining the enclosed project area become or are suspected of becoming impacted because of the Remediation Contractor's work, the Remediation Contractor shall thoroughly clean the affected areas. These areas shall be subject to detailed visual inspection and potentially post-remediation clearance sampling by the Environmental Consultant.

3.5 DISPOSAL OF BIRD, BAT, AND RODENT DROPPING WASTE MATERIALS

A. General

1. Dispose of waste bird, bat, and rodent droppings and debris that is packaged in accordance with these specifications. Bird, bat, and rodent droppings can be disposed of as normal household waste.

3.6 FINAL VISUAL INSPECTION AND AIR CLEARANCE TESTING

B. General

1. Notify the Contracting Officer 48 hours in advance of performance of the final visual inspection. The final visual inspection will be performed by the IHC after the cleaning.

C. Final Visual Inspection

1. Final visual inspection will include the entire regulated area, all poly sheeting, seals over ducts, windows, louvers, penetrations, or any other opening into the regulated area. If any debris, residue, dust, or any other suspect material is detected, the cleaning shall be repeated at no additional cost to the Contracting Officer or the IHC. Dust/ material samples may be collected and analyzed at no additional cost to the NPS at the discretion of the IHC to confirm visual findings. When the regulated area is visually clean, the final testing can be done.

D. Final Air Clearance Testing

1. After an acceptable final visual inspection by the IHC and Contracting Officer, the IHC may perform air or dust sampling at the direction of NPS.

2. If the visual and or air or dust sampling is acceptable, remove the critical barriers. Any small quantities of residue material found upon removal of the poly shall be removed with a HEPA vacuum and localized isolation. If significant quantities are found as determined by the IHC, then the entire area affected shall be cleaned as specified in the final cleaning.

3. If release criteria are met, proceed to perform the remediation closeout and issue the certificate of completion in accordance with these specifications.

3.7 REMEDIATION CLOSEOUT AND CERTIFICATE OF COMPLIANCE

A. After thorough decontamination, complete bird, bat, and rodent dropping remediation work upon meeting the regulated area clearance criteria and fulfilling the following:

1. Remove all equipment, materials, and debris from the project area.

2. Package and dispose of all bird, bat, and rodent dropping waste as required.

3. Repair or replace all finishes damaged during the remediation work, not scheduled for restoration by others.

4. Fulfill other Project closeout requirements as specified elsewhere in this specification.

B. Certificate of Completion by Remediation Contractor

1. The Remediation Contractor shall complete and sign the "Certificate of Completion" in accordance with Attachment #1 after the remediation and decontamination of the regulated area.

C. Work Shifts

1. All work shall be done during administrative hours, excluding Federal Holidays. Any change in the work schedule must be approved in writing by the Contracting Officer.

ATTACHMENT #1

CERTIFICATE OF COMPLETION

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

REMEDIATION CONTRACTOR'S NAME AND ADDRESS:_____
_____
_____

1. I certify that I have personally inspected, monitored, and supervised the remediation work of (specify regulated areas or buildings):

Which took place from_____to _____

2. That throughout the work, all applicable requirements/regulations and the specifications were met.

3. That any person who entered the regulated area was protected with the appropriate PPE and respirator, and that they followed the proper entry and exit procedures and the proper operating procedures for the duration of the work.

4. That all employees of the Remediation Contractor engaged in this work were trained in respiratory protection, were experienced with remediation work, had proper medical surveillance documentation, were fit-tested for their respirator, and were not exposed at any time during the work to bird, bat, and rodent droppings without the benefit of appropriate respiratory protection.

5. I performed and supervised all inspections and testing specified and required by this specification.

Signature:_____ Date:_____
(Printed Name/ Signature)

Signature:_____ Date:_____
(Printed Name/ Signature)

ATTACHMENT #2

CERTIFICATE OF WORKER'S ACKNOWLEDGMENT

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

REMEDIATION CONTRACTOR'S NAME AND ADDRESS:_____

_____

WORKING WITH BIRD, BAT, AND RODENT DROPPINGS CAN BE HAZARDOUS TO YOUR HEALTH. INHALING BIRD, BAT, AND RODENT DROPPINGS HAS BEEN LINKED WITH THE FUNGUS HISTOPLASMA CAPSULATUM THAT CAUSES HISTOPLASMOSIS, A SEVERE RESPIRATORY DISEASE CONTRACTED BY BREATHING IN AIRBORNE SPORES FROM DISTURBED, DRIED DROPPINGS. IT CAN ALSO CAUSE CRYPTOCOCCOSIS AND CAUSE ALLERGIC REACTIONS.

Your employer's contract with the OWNER for the above project requires that You must be supplied with the proper PPE, including an adequate respirator, and be trained in its use. You must be trained in safe and healthy work practices and in the use of the equipment found at a remediation project. You must receive/have a current medical examination. These things shall be provided at no cost to you. By signing this certificate, you are indicating to the OWNER that your employer has met these obligations.

RESPIRATORY PROTECTION: I have been trained in the proper use of respirators and have been informed of the type of respirator to be used on the above indicated project. I have a copy of the written Respiratory Protection Program issued by my employer. I have been provided for my exclusive use, at no cost, with a respirator to be used on the above indicated project.

MEDICAL EXAMINATION: I have had a medical examination within the past 12 months, which was paid for by my employer. This examination included: health history, occupational history, pulmonary function test, and may have included a chest x-ray evaluation. The physician issued a positive written opinion after the examination.

Signature:_____ Printed Name:_____

ATTACHMENT #3

AFFIDAVIT OF MEDICAL SURVEILLANCE, RESPIRATORY PROTECTION AND TRAINING/ACCREDITATION

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

_____

REMEDIATION CONTRACTOR'S NAME AND ADDRESS:_____

_____

_____

1. I verify that the following individual

Name:_____

who is proposed to be employed in remediation work associated with the above project by the named Remediation Contractor, is included in a medical surveillance program in accordance with 29 CFR 1926.1101(m), and that complete records of the medical surveillance program as required by 29 CFR 1926.1101(m)(n), 29 CFR 1910.134, and 29 CFR 1910.20 are kept at the offices of the Remediation Contractor at the following address.

Address:_____

_____

2. I verify that this individual has been trained, fit-tested, and instructed in the use of all appropriate respiratory protection systems and that the person is capable of working safely and healthily as expected and required in the expected work environment of this project.

3. I verify that this individual has been trained as required by 29 CFR 1926.1101(k). This individual has also obtained a valid accreditation certificate. Documentation will be kept on-site.

Signature of Remediation Contractor:_____Date:_____

Printed Name of Remediation Contractor:_____

ATTACHMENT #4

DAILY SIGN-IN AND SIGN-OUT SHEET

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

REMEDIATION CONTRACTOR'S NAME AND ADDRESS:_____

_____

PRINT NAME	TIME-IN	TIMEOUT

ATTACHMENT #5

DAILY ACTIVITY REPORT

Project Name:		Project No.	
Building:		Date	
Area:		Project Day No.	of
Remediation Contractor:		Personnel	
IHC:		Sheet	of
Material Removed:			
Estimated Percent of Phase Completed:		On Schedule	Yes No

GENERAL REMARKS/MEETINGS/SITE VISITS/REMEDIATION CONTRACTOR PROGRESS:

Remediation Contractor Supervisor:	Reviewed by:

ATTACHMENT #6

VISITOR RELEASE FORM

PROJECT NAME AND NUMBER:_____

PROJECT ADDRESS:_____

REMEDIATION CONTRACTOR'S NAME AND ADDRESS:_____

_____

VISITOR NAME:_____

VISITOR'S COMPANY:_____

PURPOSE OF VISIT:_____

Time of Entry to the Project Site:_____ to the Exclusion Zone:_____

Time of Departure from the Site:_____ from the Exclusion Zone:_____

Personal Protective Equipment Utilized:_____

I acknowledge and understand that I am visiting a hazardous waste control work area. I understand the dangers of exposure to hazardous waste.

I have read and understand the Site Safety and Health Plan for this project and will abide by the directions, stipulations, and terms specified therein.

I knowingly assume all risks in connection with potential exposure to hazardous waste and I do hereby, for myself and my heirs at law, release and forever discharge the property Owner, Owner's Representative, Project Administrator, independent testing laboratory, architects, engineers, or Contracting firms employed by the Owner or Project Administrator, their owners, employees, nominees, personal representative, affiliates, successors, and assigns from and against any and all liability whatsoever at common law or otherwise. I hereby waive and relinquish any and all claims of every nature that I now have or may have or claim to have that are in any way, directly or indirectly, related to exposure to hazardous waste and hazardous waste-containing materials.

Furthermore, I know that I am entering a construction area where workplace conditions, such as water on floors, scaffolding, electrical equipment, etc., CAN CREATE HAZARDOUS SITUATIONS FOR VISITORS. I assume all risks of accidental injury or illness regardless of cause while visiting this construction site. I hereby waive and relinquish any and all claims of every nature that I now have or may have or claim to have that are in any way, directly or indirectly, related to such injury.

THIS RELEASES ALL PARTIES ON THE ABOVE NAMED PROJECT:

Visitor's Signature:_____

Date:_____

Witness:_____

ATTACHMENT #7

FINAL CLEARANCE VISUAL INSPECTION

Project:	Project No.:
Building\Address:	Date:
Area:	Material Removed:
Contractor:	Project Permit No.:

Residual Dust On:	**Yes**	**No**	**Not Applicable**
Floors			
Walls			
Ledges			
Roof Decking			
Pipes			
Hangars			
Conduits			
Cables			
Light Fixtures			
Ductwork			
Equipment			
HEPA machine(s) running			
Containment dry			
Only critical barriers present			
Decontamination unit is attached and functional			

Visual Inspection: ___Pass ___Fail

Remarks:_____

_____

Technician:_____ _____

Printed Name Signature

C/S:_____ _____

Printed Name Signature

END OF SECTION 02 85 00

SECTION 03 01 00 - MAINTENANCE OF CONCRETE

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Cleaning of existing concrete surfaces.

B. Repair of exposed structural, shrinkage, and settlement cracks.

C. Repair of deteriorated concrete and parge coats.

D. Crack repair at cast-concrete elements.

1.2 REFERENCE STANDARDS

A. ASTM C33/C33M - Standard Specification for Concrete Aggregates; 2013.

B. ASTM C404 - Standard Specification for Aggregates for Masonry Grout; 2011.

C. ASTM C928/C928M - Standard Specification for Packaged, Dry, Rapid-Hardening Cementitious Materials for Concrete Repairs; 2025.

1.3 SUBMITTALS

A. Product Data: Indicate product standards, physical and chemical characteristics, technical specifications, limitations, maintenance instructions, and general recommendations regarding each material.

B. Manufacturer's instructions.

1.4 QUALITY ASSURANCE

A. Cleaner Qualifications: Company specializing in, and with minimum of five years of commercial experience in, the type of cleaning specified.

B. Consult with Contracting Officer to identify which cracks in concrete can be surface repaired and which cracks are the result from oxidation and expansion of reinforcing steel and if the concrete needs to be removed to expose the reinforcement and arrest the corrosion process.

1.5 SITE CONDITIONS

A. Clean concrete surfaces only when air temperatures are above 40 degrees F (4 deg. C) and will remain so until concrete has dried out.

1. Do not perform any patching unless air temperatures are between 40 degrees Fahrenheit (10 deg. C) and 86 degrees Fahrenheit (30 deg. C) and will remain so for at least 24 hours after completion of work.

2. Do not perform any patching work if precipitation is expected. In case of unexpected precipitation, work shall cease and all uncured material shall be adequately protected with an impermeable polyethylene sheet.

3. If either the ambient or subsurface temperature is expected to fall between 35F (2C) and 40F (4C) during curing and ultimate drying of the patching compound, then the cold weather precautions shall be followed.

    a. Curing times shall be extended to compensate for lower temperature cure.

    b. Do not proceed if temperatures will drop below freezing before patching compound has reached final set. Any material disrupted by early freezing must be removed and replaced under appropriate controls or conditions.

    c. If auxiliary heating will be used to protect freshly placed materials from freezing, equipment must not directly vent exhaust gases onto the repair materials or into repair enclosure air. This may cause carbonation and low strength. Use moderate temperatures and heated air or radiant heat.

4. If ambient or surface temperature is expected to rise above 86F (30C) during application and curing of the patching compound, then the hot weather precautions shall be followed.

    a. All materials shall be kept cool, stored out of direct sun.

    b. Pre-cooling of subsurface shall be carried out by continuous wetting at least one hour before placing patching compounds. This shall be done by covering or draping the entire repair area with burlap and by continuously keeping the burlap wet. The burlap shall be removed just prior to the patching of each area, as work proceeds.

    c. In hot, dry, windy weather, repair areas shall be covered or draped with burlap for a minimum of 3 hours after patch placement, to avoid rapid drying of the patches. Maximum wet cure time shall not exceed 24 hours.

1.6 SEQUENCING OF WORK

A. Perform concrete restoration work in the following sequence:

1. Remove paint, stains and plant material from all surfaces.

2. Remove existing unsound materials from areas indicated to be restored.

3. Clean building and repair surfaces as indicated.

4. Patch and repair existing concrete structures as indicated.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Comply with manufacturers' instructions for storage, shelf life limitations, and handling of products.

PART 2 PRODUCTS

2.1 CLEANING MATERIALS

A. Detergent: Non-ionic detergent.

2.2 CEMENTITIOUS PATCHING AND REPAIR MATERIALS

A. Cementitious Repair Mortar, Trowel Grade: One- or two-component, factory-mixed, polymer modified cementitious mortar.

1. In-place material resistant to freeze/thaw conditions.

2. Mixed with water in proportions as recommended by manufacturer.

3. Integral corrosion inhibitor.

4. Basis of Design:

a. Euclid Chemical Company; Product DuralPrep AC. Product characteristics noted in paragraph 2.2.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2.3 EPOXY PATCHING AND REPAIR MATERIALS

A. Epoxy Injection Adhesive: Two-component, 100 percent solids, moisture sensitive.

B. Basis of Design:

1. Euclid Chemical Company: Product Dural Injection Gel; www.euclidchemical.com.. Product characteristics noted in paragraph 2.3.

2. Substitutions: See Section 01 67 00 – Product Requirements.

2.4 ACCESSORIES

A. Bonding Agent and Anti-Corrosive Coating:

1. Basis of Design:

a. Euclid Chemical Company; Product DURALPREP AC; www.euclidchemicals.com.. Product characteristics noted in paragraph 2.4.

b. Substitutions: See Section 01 67 00 – Product Requirements.

B. Portland Cement: ASTM C150/C150M, Type I, grey.

C. Sand: ASTM C33/C33M or ASTM C404; uniformly graded, clean.

D. Water: Clean and potable.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that surfaces are ready to receive work.

B. Beginning of installation means acceptance of substrate.

3.2 CONCRETE SURFACE AND CRACK REPAIR

A. General: Follow repair and patching compounds manufacturer's written instructions.

3.3 CONCRETE SURFACE REPAIR USING CEMENTITIOUS MATERIALS

A. Clean concrete surfaces, cracks, and joints of dirt, laitance, corrosion, and other contamination using method(s) specified above and allow to dry.

B. Remove broken and soft concrete at least 1/4 inch deep.

C. Remove corrosion from steel and clean mechanically.

D. Cover exposed steel reinforcement with cementitious mortar.

E. Work mortar into broken surface and build up patch to match original.

F. Feather edges of repairs flush to sound surface and trowel surface to match surrounding area.

G. Apply coating of bonding agent to entire concrete surface to be repaired.

H. Fill voids with cementitious mortar flush with surface.

I. Apply repair mortar by steel trowel to a minimum thickness of 1/4 inch over entire surface, terminating at a vertical change in plane on all sides.

J. Trowel finish to match adjacent concrete surfaces.

END OF SECTION

SECTION 03 30 00 - CAST-IN-PLACE CONCRETE

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes cast-in-place concrete, including formwork, reinforcement, concrete materials, mixture design, placement procedures, and finishes.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Design Mixtures: For each concrete mixture.

C. Steel Reinforcement Shop Drawings: Placing Drawings that detail fabrication, bending, and placement.

1.3 INFORMATIONAL SUBMITTALS

A. Material certificates.

B. Material test reports.

1.4 QUALITY ASSURANCE

A. Manufacturer Qualifications: A firm experienced in manufacturing ready-mixed concrete products and that complies with ASTM C 94/C 94M requirements for production facilities and equipment.

1. Manufacturer certified according to NRMCA's "Certification of Ready Mixed Concrete Production Facilities."

B. Testing Agency Qualifications: An independent agency, acceptable to authorities having jurisdiction, qualified according to ASTM C 1077 and ASTM E 329 for testing indicated.

1.5 PRECONSTRUCTION TESTING

A. Preconstruction Testing Service: Engage a qualified testing agency to perform preconstruction testing on concrete mixtures.

1.6 FIELD CONDITIONS

A. Cold-Weather Placement: Comply with ACI 306.1.

1. Do not use calcium chloride, salt, or other materials containing antifreeze agents or chemical accelerators unless otherwise specified and approved in mixture designs.

B. Hot-Weather Placement: Comply with ACI 301 and ACI 305.1.

PART 2 - PRODUCTS

2.1 CONCRETE, GENERAL

A. ACI Publications: Comply with the following unless modified by requirements in the Contract Documents:

1. ACI 301.

2. ACI 117.

2.2 FORM-FACING MATERIALS

A. Smooth-Formed Finished Concrete: Form-facing panels that provide continuous, true, and smooth concrete surfaces. Furnish in largest practicable sizes to minimize number of joints.

B. Rough-Formed Finished Concrete: Plywood, lumber, metal, or another approved material. Provide lumber dressed on at least two edges and one side for tight fit.

2.3 STEEL REINFORCEMENT

A. Reinforcing Bars: ASTM A 615/A 615M, Grade 60, deformed.

B. Plain-Steel Welded-Wire Reinforcement: ASTM A 1064/A 1064M, plain, fabricated from as-drawn steel wire into flat sheets.

C. Deformed-Steel Welded-Wire Reinforcement: ASTM A 1064/A 1064M, flat sheet.

D. Galvanized-Steel Welded-Wire Reinforcement: ASTM A 1064/A 1064M, plain, fabricated from galvanized-steel wire into flat sheets.

E. Bar Supports: Bolsters, chairs, spacers, and other devices for spacing, supporting, and fastening reinforcing bars and welded-wire reinforcement in place. Manufacture bar supports from steel wire, plastic, or precast concrete according to CRSI's "Manual of Standard Practice."

2.4 CONCRETE MATERIALS

A. Cementitious Materials:

1. Portland Cement: ASTM C 150/C 150M, Type I/II.

2. Fly Ash: ASTM C 618, Class C.

B. Normal-Weight Aggregates: ASTM C 33/C 33M, graded.

1. Maximum Coarse-Aggregate Size: 3/4 inch nominal.

2. Fine Aggregate: Free of materials with deleterious reactivity to alkali in cement.

C. Air-Entraining Admixture: ASTM C 260/C 260M.

D. Chemical Admixtures: Certified by manufacturer to be compatible with other admixtures and that do not contribute water-soluble chloride ions exceeding those permitted in hardened concrete. Do not use calcium chloride or admixtures containing calcium chloride.

1. Water-Reducing Admixture: ASTM C 494/C 494M, Type A.

2. Retarding Admixture: ASTM C 494/C 494M, Type B.

3. Water-Reducing and Retarding Admixture: ASTM C 494/C 494M, Type D.

4. High-Range, Water-Reducing Admixture: ASTM C 494/C 494M, Type F.

5. High-Range, Water-Reducing and Retarding Admixture: ASTM C 494/C 494M, Type G.

6. Plasticizing and Retarding Admixture: ASTM C 1017/C 1017M, Type II.

E. Water: ASTM C 94/C 94M.

2.5 VAPOR RETARDERS

A. Sheet Vapor Retarder: Polyethylene sheet, ASTM D 4397, not less than 10 mils thick.

2.6 CURING MATERIALS

A. Evaporation Retarder: Waterborne, monomolecular film forming, manufactured for application to fresh concrete.

B. Absorptive Cover: AASHTO M 182, Class 2, burlap cloth made from jute or kenaf, weighing approximately 9 oz./sq. yd. when dry.

C. Moisture-Retaining Cover: ASTM C 171, polyethylene film or white burlap-polyethylene sheet.

D. Water: Potable.

E. Clear, Waterborne, Membrane-Forming Curing Compound: ASTM C 309, Type 1, Class B, dissipating.

2.7 RELATED MATERIALS

A. Expansion- and Isolation-Joint-Filler Strips: ASTM D 1751, asphalt-saturated cellulosic fiber.

2.8 CONCRETE MIXTURES, GENERAL

A. Prepare design mixtures for each type and strength of concrete, proportioned on the basis of laboratory trial mixture or field test data, or both, according to ACI 301.

B. Cementitious Materials: Use fly ash, pozzolan, slag cement, and silica fume as needed to reduce the total amount of portland cement, which would otherwise be used, by not less than 40 percent.

C. Admixtures: Use admixtures according to manufacturer's written instructions.

1. Use water-reducing or plasticizing admixture in concrete, as required, for placement and workability.

2. Use water-reducing and -retarding admixture when required by high temperatures, low humidity, or other adverse placement conditions.

3. Use water-reducing admixture in pumped concrete, concrete for heavy-use industrial slabs and parking structure slabs, concrete required to be watertight, and concrete with a w/c ratio below 0.50.

2.9 CONCRETE MIXTURES FOR BUILDING ELEMENTS

A. Normal-Weight Concrete used for footings and foundation walls:

1. Minimum Compressive Strength: 4500 psi at 28 days.

2. Maximum W/C Ratio: 0.45.

3. Exposure Class: F2

4. Slump Limit: 3" +/- 1” for footings, 4" +/- 1” for walls

5. Air Content: 6% plus or minus 1.5 percent at point of delivery for 3/4-inch nominal maximum aggregate size.

6. Air Content: Do not allow air content of trowel-finished floors to exceed 3 percent.

B. Normal-Weight Concrete used for slabs on grade:

1. Minimum Compressive Strength: 3,000 psi at 28 days.

2. Maximum W/C Ratio: 0.45.

3. Exposure Class: F0

4. Slump Limit: 3" +/- 1”

5. Air Content: Do not allow air content of trowel-finished floors to exceed 3 percent.

2.10 FABRICATING REINFORCEMENT

A. Fabricate steel reinforcement according to CRSI's "Manual of Standard Practice."

2.11 CONCRETE MIXING

A. Ready-Mixed Concrete: Measure, batch, mix, and deliver concrete according to ASTM C 94/C 94M, and furnish batch ticket information.

1. When air temperature is between 85 and 90 deg F, reduce mixing and delivery time from 1-1/2 hours to 75 minutes; when air temperature is above 90 deg F, reduce mixing and delivery time to 60 minutes.

PART 3 - EXECUTION

3.1 FORMWORK INSTALLATION

A. Design, erect, shore, brace, and maintain formwork, according to ACI 301, to support vertical, lateral, static, and dynamic loads, and construction loads that might be applied, until structure can support such loads.

B. Construct formwork so concrete members and structures are of size, shape, alignment, elevation, and position indicated, within tolerance limits of ACI 117.

C. Chamfer exterior corners and edges of permanently exposed concrete.

3.2 EMBEDDED ITEM INSTALLATION

A. Place and secure anchorage devices and other embedded items required for adjoining work that is attached to or supported by cast-in-place concrete. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.

3.3 VAPOR-RETARDER INSTALLATION

A. Sheet Vapor Retarders: Place, protect, and repair sheet vapor retarder according to ASTM E 1643 and manufacturer's written instructions.

1. Lap joints 6 inches and seal with manufacturer's recommended tape.

3.4 STEEL REINFORCEMENT INSTALLATION

A. General: Comply with CRSI's "Manual of Standard Practice" for fabricating, placing, and supporting reinforcement.

1. Do not cut or puncture vapor retarder. Repair damage and reseal vapor retarder before placing concrete.

3.5 JOINTS

A. General: Construct joints true to line with faces perpendicular to surface plane of concrete.

B. Construction Joints: Install so strength and appearance of concrete are not impaired, at locations indicated or as approved by Architect.

C. Contraction Joints in Slabs-on-Grade: Form weakened-plane contraction joints, sectioning concrete into areas as indicated. Construct contraction joints for a depth equal to at least one-fourth of concrete thickness as follows:

1. Grooved Joints: Form contraction joints after initial floating by grooving and finishing each edge of joint to a radius of 1/8 inch. Repeat grooving of contraction joints after applying surface finishes. Eliminate groover tool marks on concrete surfaces.

2. Sawed Joints: Form contraction joints with power saws equipped with shatterproof abrasive or diamond-rimmed blades. Cut 1/8-inch- wide joints into concrete when cutting action does not tear, abrade, or otherwise damage surface and before concrete develops random contraction cracks.

D. Isolation Joints in Slabs-on-Grade: After removing formwork, install joint-filler strips at slab junctions with vertical surfaces, such as column pedestals, foundation walls, grade beams, and other locations, as indicated.

3.6 WATERSTOP INSTALLATION

A. Waterstops: Install in construction joints and at other locations indicated, according to manufacturer's written instructions.

3.7 CONCRETE PLACEMENT

A. Before placing concrete, verify that installation of formwork, reinforcement, and embedded items is complete and that required inspections are completed.

B. Deposit concrete continuously in one layer or in horizontal layers of such thickness that no new concrete is placed on concrete that has hardened enough to cause seams or planes of weakness. If a section cannot be placed continuously, provide construction joints as indicated. Deposit concrete to avoid segregation.

1. Consolidate placed concrete with mechanical vibrating equipment according to ACI 301.

3.8 FINISHING FORMED SURFACES

A. Rough-Formed Finish: As-cast concrete texture imparted by form-facing material with tie holes and defects repaired and patched. Remove fins and other projections that exceed specified limits on formed-surface irregularities.

1. Apply to concrete surfaces not exposed to public view.

B. Smooth-Formed Finish: As-cast concrete texture imparted by form-facing material, arranged in an orderly and symmetrical manner with a minimum of seams. Repair and patch tie holes and defects. Remove fins and other projections that exceed specified limits on formed-surface irregularities.

1. Apply to concrete surfaces exposed to public view.

C. Rubbed Finish: Apply the following to smooth-formed-finished as-cast concrete where indicated:

1. Smooth-Rubbed Finish: Not later than one day after form removal, moisten concrete surfaces and rub with carborundum brick or another abrasive until producing a uniform color and texture. Do not apply cement grout other than that created by the rubbing process.

2. Grout-Cleaned Finish: Wet concrete surfaces and apply grout of a consistency of thick paint to coat surfaces and fill small holes. Mix 1 part portland cement to 1-1/2 parts fine sand with a 1:1 mixture of bonding admixture and water. Add white portland cement in amounts determined by trial patches, so color of dry grout matches adjacent surfaces. Scrub grout into voids and remove excess grout. When grout whitens, rub surface with clean burlap and keep surface damp by fog spray for at least 36 hours.

3. Cork-Floated Finish: Wet concrete surfaces and apply a stiff grout. Mix 1 part portland cement and 1 part fine sand with a 1:1 mixture of bonding agent and water. Add white portland cement in amounts determined by trial patches, so color of dry grout matches adjacent surfaces. Compress grout into voids by grinding surface. In a swirling motion, finish surface with a cork float.

D. Related Unformed Surfaces: At tops of walls, horizontal offsets, and similar unformed surfaces adjacent to formed surfaces, strike off smooth and finish with a texture matching adjacent formed surfaces. Continue final surface treatment of formed surfaces uniformly across adjacent unformed surfaces unless otherwise indicated.

3.9 CONCRETE PROTECTING AND CURING

A. General: Protect freshly placed concrete from premature drying and excessive cold or hot temperatures. Comply with ACI 306.1 for cold-weather protection and ACI 305.1 for hot-weather protection during curing.

B. Evaporation Retarder: Apply evaporation retarder to unformed concrete surfaces if hot, dry, or windy conditions cause moisture loss approaching 0.2 lb/sq. ft. x h before and during finishing operations. Apply according to manufacturer's written instructions after placing, screeding, and bull floating or darbying concrete, but before float finishing.

C. Formed Surfaces: Cure formed concrete surfaces, including underside of beams, supported slabs, and other similar surfaces. If forms remain during curing period, moist cure after loosening forms. If removing forms before end of curing period, continue curing for remainder of curing period.

D. Cure concrete according to ACI 308.1, by one or a combination of the following methods:

1. Moisture Curing: Keep surfaces continuously moist for not less than seven days.

2. Moisture-Retaining-Cover Curing: Cover concrete surfaces with moisture-retaining cover for curing concrete, placed in widest practicable width, with sides and ends lapped at least 12 inches, and sealed by waterproof tape or adhesive. Cure for not less than seven days. Immediately repair any holes or tears during curing period, using cover material and waterproof tape.

3. Curing Compound: Apply uniformly in continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Maintain continuity of coating and repair damage during curing period.

    a. Removal: After curing period has elapsed, remove curing compound without damaging concrete surfaces by method recommended by curing compound manufacturer unless manufacturer certifies curing compound does not interfere with bonding of floor covering used on Project.

4. Curing and Sealing Compound: Apply uniformly to floors and slabs indicated in a continuous operation by power spray or roller according to manufacturer's written instructions. Recoat areas subjected to heavy rainfall within three hours after initial application. Repeat process 24 hours later and apply a second coat. Maintain continuity of coating and repair damage during curing period.

3.10 CONCRETE SURFACE REPAIRS

A. Defective Concrete: Repair and patch defective areas when approved by Architect. Remove and replace concrete that cannot be repaired and patched to Architect's approval.

3.11 FIELD QUALITY CONTROL

A. Special Inspections: The Contractor is responsible for all testing and inspections. Contractor will engage a qualified testing and inspecting agency to perform field tests and inspections and prepare test reports.

B. Concrete Tests: Testing of composite samples of fresh concrete obtained in accordance with ASTM C 172/C 172M shall be performed in accordance with the following requirements:
   1. Testing Frequency: Obtain one composite sample for each day's pour of each concrete mixture exceeding 5 cu. yd., but less than 25 cu. yd., plus one set for each additional 50 cu. yd. or fraction thereof.
      a. When frequency of testing provides fewer than five compressive-strength tests for each concrete mixture, testing shall be conducted from at least five randomly selected batches or from each batch if fewer than five are used.
   2. Slump: ASTM C143/C143M:
      a. One test at point of placement for each composite sample, but not less than one test for each day's pour of each concrete mixture.
      b. Perform additional tests when concrete consistency appears to change.
   3. Slump Flow: ASTM C1611/C1611M:
      a. One test at point of placement for each composite sample, but not less than one test for each day's pour of each concrete mixture.
      b. Perform additional tests when concrete consistency appears to change.
   4. Air Content: ASTM C231/C231M pressure method, for normal-weight concrete;
      a. One test for each composite sample, but not less than one test for each day's pour of each concrete mixture.
   5. Concrete Temperature: ASTM C1064/C1064M:
      a. One test hourly when air temperature is 40 deg F and below or 80 deg F and above, and one test for each composite sample.
   6. Unit Weight: ASTM C567/C567M fresh unit weight of structural lightweight concrete.
      a. One test for each composite sample, but not less than one test for each day's pour of each concrete mixture.
   7. Compression Test Specimens: ASTM C31/C31M:
      a. Cast and laboratory cure two sets of three 6-inch by 12-inch or 4-inch by 8-inch cylinder specimens for each composite sample.
      b. Cast, initial cure, and field cure two sets of three standard cylinder specimens for each composite sample.
   8. Compressive-Strength Tests: ASTM C39/C39M.
      a. Test one set of three laboratory-cured specimens at seven days and one set of two specimens at 28 days.
      b. Test one set of three field-cured specimens at seven days and one set of two specimens at 28 days.
      c. A compressive-strength test shall be the average compressive strength from a set of two specimens obtained from same composite sample and tested at age indicated.
   9. When strength of field-cured cylinders is less than 85 percent of companion laboratory-cured cylinders, Contractor shall evaluate operations and provide corrective procedures for protecting and curing in-place concrete.
   10. Strength of each concrete mixture will be satisfactory if every average of any three consecutive compressive-strength tests equals or exceeds specified compressive strength, and no compressive-strength test value falls below specified compressive strength by more than 500 psi if specified compressive strength is 5000 psi, or no compressive strength test value is less than 10 percent of specified compressive strength if specified compressive strength is greater than 5000 psi.
   11. Nondestructive Testing: Impact hammer, sonoscope, or other nondestructive device may be permitted by Architect but will not be used as sole basis for approval or rejection of concrete.

12. Additional Tests:
    a. Testing and inspecting agency shall make additional tests of concrete when test results indicate that slump, air entrainment, compressive strengths, or other requirements have not been met, as directed by Architect.
    b. Testing and inspecting agency may conduct tests to determine adequacy of concrete by cored cylinders complying with ASTM C42/C42M or by other methods as directed by Architect.
        1) Acceptance criteria for concrete strength shall be in accordance with ACI 301 section 1.6.6.3.
13. Additional testing and inspecting, at Contractor's expense, will be performed to determine compliance of replaced or additional work with specified requirements.
14. Correct deficiencies in the Work that test reports and inspections indicate do not comply with the Contract Documents.

3.12 POST INSTALLED ANCHORS INTO CONCRETE

A. Anchor type, diameter and embedment shall be as indicated on drawings.

B. Install only where specifically shown in the details or allowed by the SER. All post-installed anchor types and locations shall be approved by the SER and shall have a current ICC-ESR that provides relevant design values necessary to validate the available strength exceeds the required strength. Submit current manufacturer’s data and ICC-ESR to the SER for approval regardless of whether or not it is a pre-approved anchor.

C. Anchors shall be installed in strict accordance with ICC-ESR and the manufacturer’s printed installation instructions (MPII) in conjunction with edge distance, spacing and embedment depth as indicated on the drawings. The contractor shall arrange for a manufacturer’s field representative to provide installation training for all products to be used prior to the commencement of work. Only trained installers shall perform post-installed anchor installation. A record of training shall be kept on site and made available to the SER as requested. Adhesive anchors installed horizontally or upwardly inclined orientations shall be performed by a certified adhesive installer (AAI) as certified through ACI/CRSI or an approved equivalent.

D. No reinforcement bars shall be damaged during installation of post installed anchors.

E. Special inspection shall be per the Statement of Special Inspections.

F. ADHESIVE ANCHORS: The following adhesive type anchors have been used in design and shall be used for anchorage to concrete as applicable and in accordance with the corresponding ICC-ESR report. Reference the corresponding ICC_ESR for required minimum age of concrete, concrete temperature range, moisture condition, lightweight concrete and hole drilling and preparation requirements. Dilled in anchor embedment lengths shall he as shown in drawings or not less than seven times the anchor nominal diameter (7D). Adhesive anchors are to be installed in concrete ages a minimum of 21 days unless otherwise specified in the ICC-ESR.
    1. Hilti “HIT-HY 200-R V3” - ICC-ESR 4868 for anchorage to concrete
    2. Hilti “HIT-HY 200-R V3” - ICC-ESR 4868
    3. Simpson “SET-3G” - ICC-ESR 4057
    4. Simpson STRONG-TIE “AT-3G” ICC-ESR 5026

G. SCREW ANCHORS: The following screw type anchors have been used in design and shall be used for anchorage to concrete as applicable and in accordance with the corresponding ICC-ESR report.
   1. Simpson “TITEN HD” – for interior condition only. ICC-ESR 2713
   2. Simpson “TITEN HD MECHANICALLY GALVANIZED” – For interior and exterior conditions. ICC-ESR 2713
   3. Simpson “STAINLESS STEEL TITEN HD” - For interior and exterior conditions IAPMO 493

END OF SECTION 033000

SECTION 04 03 42 - HISTORIC MASONRY REPAIR

PART 1 GENERAL

1.1 SUMMARY

A. Section includes historic treatment work consisting of repairing historic masonry assemblies as follows:

1. Cleaning historic stone masonry surfaces.

2. Repairing historic stone masonry.

3. Removing abandoned anchors and other miscellaneous attachments.

4. Chimney flashing at historic stone masonry.

1.2 RELATED REQUIREMENTS:

A. Section 01 35 91 - Historic Treatment Procedures" for general historic treatment requirements.

B. Section 02 41 00 - Demolition

C. Section 02 42 96 - Historic Removal and Dismantling" for historic removal and dismantling work.

D. Section 07 62 00 - Sheet Metal Flashing and Trim" for metal flashing installed in or on repaired stonework.

1.3 DEFINITIONS

A. Refer to Section 01 42 16 - Definitions.

1.4 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference on historic masonry cleaning, repair and repointing at Project site with Contracting Officer.

1. Review minutes of Preliminary Historic Treatment Conference that pertain to historic masonry treatment.

2. Review methods and procedures related to repairing historic masonry, including, but not limited to, the following:

a. Verify historic treatment specialist's personnel, equipment, and facilities needed to make progress and avoid delays.

b. Materials, material application, sequencing, tolerances, and required clearances.

c. Quality-control program.

d. Fire-protection plan.

e. Historic masonry treatment program.

1.5 SEQUENCING AND SCHEDULING

A. Order sand and gray portland cement for colored mortar immediately after approval of samples and mockups. Take delivery of and store at Project site a sufficient quantity to complete Project.

B. Work Sequence: Perform historic masonry treatment work in the following sequence, which includes work specified in this and other Sections:

1. Remove plant growth.

2. Inspect masonry for open mortar joints and permanently or temporarily point them before cleaning to prevent the intrusion of water and other cleaning materials into the wall.

3. Remove paint.

4. Clean masonry.

5. Repair masonry, including crack repair, patching, and unit replacement.

6. Point mortar and sealant joints.

7. After repairs and repointing have been completed and cured, perform a final cleaning to remove residues from this work.

1.6 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include construction details, material descriptions, dimensions of individual components and profiles, and finishes.

2. Include recommendations for product application and use.

3. Include test data substantiating that products comply with requirements.

B. Shop Drawings:

1. Include plans, elevations, sections, and locations of masonry repair work on the structure.

2. Indicate complete dimensions for new masonry units and their jointing, showing relation of existing to new units.

3. Indicate setting number of each new masonry unit and its location on the structure in annotated plans and elevations.

4. Show provisions for expansion joints or other sealant joints.

5. Show provisions for flashing, lighting fixtures, conduits, and weep holes as required.

6. Show replacement and repair anchors, including drilled-in pins. Coordinate reinforcement with Structural including direction and angle of holes for pins.

7. Show locations of scaffolding and points of scaffolding in contact with masonry. Include details of each point of contact or anchorage.

C. Samples for Initial Selection: For the following:

1. Pointing Mortar: Submit sets of mortar that will be left exposed in the form of sample mortar strips, 6 inches (150 mm) long by 1/2 inch (13 mm) wide, set in aluminum or plastic channels.

a. Have each set contain a close color range of at least six Samples of different mixes of colored sands and cements that produce a mortar matching existing, cleaned mortar when cured and dry.

b. Submit with precise measurements on ingredients, proportions, gradations, and sources of colored sands from which each Sample was made.

2. Each type of sand used for mortar; minimum 8 oz. (240 mL) of each in plastic screw-top jars.

a. For blended sands, provide Samples of each component and blend. Identify blend ratio.

b. Identify sources, both supplier and quarry, of each type of sand.

3. Include similar Samples of accessories involving color selection.

D. Samples for Verification: For the following:

1. Each type of replacement masonry. Include sets of Samples to show full range of color, texture, grain, veining, and finish to be expected.

2. Sealant Materials.

3. Accessories: Each type of anchor, accessory, and miscellaneous support.

1.7 INFORMATIONAL SUBMITTALS

A. Qualification Data: For historic treatment specialist including field supervisors and workers and testing service.

B. Preconstruction Test Reports: For existing masonry types and mortar, replacement masonry types , and cleaning materials and methods.

C. Quality-control program.

D. Historic masonry treatment program.

1.8 QUALITY ASSURANCE

A. Historic Treatment Specialist Qualifications: A qualified historic masonry repair specialist. Experience installing standard unit masonry or new stone masonry is insufficient experience for historic masonry treatment work.

1. Historic Treatment Worker Qualifications: When stone units are being patched, assign at least one worker per crew who is trained and certified by manufacturer of patching compound to apply its products.

B. Quality-Control Program: Prepare a written quality-control program for this Project to systematically demonstrate the ability of personnel to properly follow methods and use materials and tools without damaging masonry. Include provisions for supervising worker performance and preventing damage.

C. Historic Masonry Treatment Program: Prepare a written, detailed description of materials, products methods, equipment, and sequence of operations to be used for each phase of the historic treatment work, including protection of surrounding materials and Project site. 1. Include cleaning, repairs, and repointing to historic masonry and backup masonry uncovered during Work.

1. Include methods for keeping exposed mortar damp during curing period.

2. If materials and methods other than those indicated are proposed for any phase of historic treatment work, add to the quality-control program a written description of such materials and methods, including evidence of successful use on comparable projects, and demonstrations to show their effectiveness for this Project.

D. No chemical cleaning solutions are to be used for this Project. Cleaning is limited to steam or high temperature low pressure water washing. No water repellent or graffiti coatings are to be applied to brick or stone masonry for this project.

E. Mockups: Prepare mockups of historic treatment to demonstrate aesthetic effects and to set quality standards for materials and execution and for fabrication and installation.

1. Cleaning: Clean area approximately 25 sq. ft. (2.3 sq. m) for each type of masonry and surface condition.

a. Test cleaners and methods on samples of adjacent materials for possible adverse reactions. Do not test cleaners known to have deleterious effects.

b. Allow a waiting period of not less than seven days after completion of sample cleaning to permit a study of sample panels for negative reactions.

2. Masonry Repair: Prepare sample areas for each type of masonry indicated to have repair work performed. If not otherwise indicated, size each mockup not smaller than two

adjacent whole units or approximately 48 inches (1200 mm) in least dimension. Construct sample areas in locations in existing walls where directed by Contracting Officer unless otherwise indicated. Demonstrate quality of materials, workmanship, and blending with existing work. Include the following as a minimum:

a. Replacement: Two masonry units replaced.

b. Stone replacement: Two stone replacements (dutchman repairs).

c. Stone plug and masonry repair: Not allowed. Replace damaged stones and masonry as full size units.

3. Repointing: Rake out joints in two separate areas, each approximately 36 inches (900mm) high by 48 inches (1200 mm) wide, for each type of repointing required, and repoint one of the areas.

4. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations in writing.

5. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.9 PRECONSTRUCTION TESTING

A. Preconstruction Testing Service: Engage a qualified testing agency to perform preconstruction testing on stone masonry as follows:

1. Provide test specimens as indicated and representative of proposed materials and existing construction.

2. Existing Mortar: Test according to ASTM C1324, modified as agreed by testing service and Contracting Officer for Project requirements, to determine proportional composition of original ingredients, sizes and colors of aggregates, and approximate strength. Use x-ray diffraction, infrared spectroscopy, and differential thermal analysis to supplement microscopical methods. Carefully remove existing mortar for testing from within joints at three locations designated by Contracting Officer.

3. Temporary Patch: As directed by Contracting Officer, provide temporary materials followed by permanent repairs at locations from which existing samples were taken.

1.10 DELIVERY, STORAGE, AND HANDLING

A. Deliver packaged materials to Project site in manufacturer's original and unopened containers, labeled with manufacturer's name and type of products.

B. Store cementitious materials on elevated platforms, under cover, and in a dry location. Do not use cementitious materials that have become damp.

C. Store hydrated lime in manufacturer's original and unopened containers. Discard lime if containers have been damaged or have been opened for more than two days.

D. Store lime putty covered with water in sealed containers.

E. Store sand where grading and other required characteristics can be maintained and contamination avoided.

1.11 FIELD CONDITIONS

A. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit repair work to be performed according to product manufacturers' written instructions and specified requirements.

B. Temperature Limits: Repair masonry only when air temperature is between 40 and 90 deg F (4 and 32 deg C) and is predicted to remain so for at least seven days after completion of the Work unless otherwise indicated.

C. Cold-Weather Requirements: Comply with the following procedures for masonry repair unless otherwise indicated:

1. When air temperature is below 40 deg F (4 deg C), heat mortar ingredients, repair materials, and existing masonry to produce temperatures between 40 and 120 deg F (4 and 49 deg C).

2. When mean daily air temperature is below 40 deg F (4 deg C), provide enclosure and heat to maintain temperatures above 32 deg F (zero deg C) within the enclosure for seven days after repair.

D. Hot-Weather Requirements: Protect masonry repairs when temperature and humidity conditions produce excessive evaporation of water from mortar and patching materials. Provide artificial shade and wind breaks, and use cooled materials as required to minimize evaporation. Do not apply mortar to substrates with temperatures of 90 deg F (32 deg C) and above unless otherwise indicated.

E. For manufactured repair materials, perform work within the environmental limits set by each manufacturer.

F. Clean masonry surfaces only when air temperature is 40 deg F (4 deg C) and above and is predicted to remain so for at least seven days after completion of cleaning.

PART 2 PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Source Limitations: Obtain each type of material for repairing historic masonry (stone, cement, sand, etc.) from single source with resources to provide materials of consistent quality in appearance and physical properties.

2.2 CLEANING MATERIALS

A. Water: Potable.

B. Hot Water: Water heated to a temperature of 140 to 160 deg F (60 to 71 deg C).C

C. Detergent:

1. Basis of Design:

a. D/2 Biological Solution: www.d2bio.com. . Product characteristics noted in paragraph 2.2.C.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2. Characteristics:

a. Biodegradable

b. pH neutral.

c. Contains no chlorine or acids.

D. Mold, Mildew, and Algae Remover, Job Mixed: Solution prepared by mixing 2 cups (0.5 L) of tetrasodium pyrophosphate (TSPP), 5 quarts (5 L) of 5 percent sodium hypochlorite (bleach),

2.3 UNIT REPAIR MATERIALS

A. Acquire units from adjacent river.

B. Salient Characteristics:

1. Match existing masonry in size and shape and color

2.4 MORTAR MATERIALS

A. Portland Cement: ASTM C150/C150M, Type I or Type II; white, gray, or both, where required for color matching of mortar.

1. Provide cement containing not more than 0.60 percent total alkali when tested according to ASTM C114.

B. Hydrated Lime: ASTM C207, Type S.

C. Factory-Prepared Lime Putty: ASTM C1489.

D. Quicklime: ASTM C5, pulverized lime.

E. Mortar Sand: ASTM C144 unless otherwise indicated.

1. Match size, texture, and gradation of existing mortar sand as closely as possible. Blend several sands if necessary to achieve suitable match.

2. Colored Mortar: Natural sand or ground marble, granite, or other sound stone of color necessary to produce required mortar color.

3. For exposed mortar, provide sand with rounded edges.

F. Water: ASTM C270, potable.

2.5 MANUFACTURED REPAIR MATERIALS

A. Cementitious Crack Filler: Ultrafine superplasticized grout that can be injected into cracks, is suitable for application to wet or dry cracks, exhibits low shrinkage, and develops high bond strength to all stone types.

2.6 BONDING ADHESIVES

A. General: Products designed specifically for repairing and bonding masonry.

2.7 PAINT REMOVERS

A. Basis of Design:

1. PROSOCO, Inc; Enviro Klean Safety Peel 1: www.prosoco.com. Product characteristics noted in paragraph 2.7.

2. Substitutions: See Section 01 67 00 – Product Requirements.

B. Product Characteristics

1. Water or Solvent-Based: Formulated without methylene chloride or methanol and used for removal of multiple coats of oil-based, water-based, acrylic-based, epoxy-based, urethane-based, elastomeric, and lead-based paints.

a. Formulation: Thixotropic paste or gel.

b. pH neutral and non-flammable.

c. Surface Neutralization After Use: Not required.

d. Restrictions: Not intended by the manufacturer for use on furniture, polished surfaces, or for removal of cementitious coatings.

2.8 ACCESSORY MATERIALS

A. Liquid Strippable Masking Agent: Manufacturer's standard liquid, film-forming, strippable masking material for protecting glass, metal, glazed masonry, and polished stone surfaces from damaging effects of acidic and alkaline masonry cleaners.

B. Masking Tape: Nonstaining, nonabsorbent material; compatible with mortar, joint primers, sealants, and surfaces adjacent to joints; and that easily comes off entirely, including adhesive.

C. Sealant Materials

1. Sealant manufacturer's standard elastomeric sealant(s) of base polymer and characteristics indicated below and according to applicable requirements in Section 07 92 05 "Joint Sealers."

a. Type: Single-component, non-sag urethane sealant.

2. Colors: Provide colors of exposed sealants to match colors of mortar adjoining installed sealant unless otherwise indicated.

3. Ground-Mortar Aggregate: Custom crushed and ground pointing mortar sand or existing mortar retrieved from joints. Grind to a particle size that matches the adjacent mortar aggregate and color. Remove all fines passing the 100 sieve.

D. Joint-Sealant Backing:

1. Cylindrical Sealant Backings: ASTM C1330, Type C (closed-cell material with a surface skin) or Type B (bicellular material with a surface skin), and of size and density to control sealant depth and otherwise contribute to producing optimum sealant performance.

2. Bond-Breaker Tape: Polyethylene tape or other plastic tape recommended in writing by sealant manufacturer for preventing sealant from adhering to rigid, inflexible joint-filler materials or joint surfaces at back of joint where such adhesion would result in sealant failure. Provide self-adhesive tape where applicable. (15 L) of hot water for every 5 gal. (20 L) of solution required.

E. Historic Stone Masonry Chimney Flashing System:

1. Flashing repair system forming a flexible, waterproof membrane. Formulated for permanent adhesion to asphalt shingles, metal flashings, masonry, and roofing materials. Solves leaks around the base of chimneys.

a. Remains flexible over time.

b. Easy brush-on application.

c. Soap and water clean up.

2. Color: Brown.

3. Manufacturer:

a. Saver Systems; Flash Seal Chimney Saver; www.saversystems.com. Product characteristics noted in paragraph 2.8.E.

b. Substitutions: See Section 01 67 00 – Product Requirements.

F. Other Products: Select materials and methods of use based on the following, subject to approval of a mockup:

1. Previous effectiveness in performing work involved.

2. Minimal possibility of damaging exposed surfaces.

3. Consistency of each application.

4. Uniformity of the resulting overall appearance.

5. Do not use products or tools that could do the following:

a. Remove, alter, or harm the present condition or future preservation of existing surfaces, including surrounding surfaces not in contract.

b. Leave residue on surfaces.

2.9 MORTAR MIXES

A. Preparing Lime Putty: Slake quicklime and prepare lime putty according to appendix in ASTM C5 and to manufacturer's written instructions.

B. Measurement and Mixing: Measure cementitious materials and sand in a dry condition by volume or equivalent weight. Do not measure by shovel; use known measure. Mix materials in a clean, mechanical batch mixer.

C. Do not use admixtures in mortar unless otherwise indicated.

D. Mixes: Mix mortar materials in the following proportions:

1. General: Match mortar analysis result.

PART 3 EXECUTION

3.1 HISTORIC TREATMENT SPECIALIST

A. See Section 01 35 91 - Historic Treatment Requirements for requirements of historic treatment specialist.

3.2 PROTECTION

A. Prevent mortar from staining face of surrounding stone and other surfaces.

1. Cover sills, ledges, and other projecting items to protect them from mortar droppings.

2. Keep wall area wet below rebuilding and repair work to discourage mortar from adhering.

3. Immediately remove mortar splatters in contact with exposed masonry and other surfaces.

B. Remove gutters and downspouts and associated hardware adjacent to immediate work area and store during stone repair work. Reinstall when repairs are complete.

1. Provide temporary rain drainage during work to direct water away from building.

3.3 STONE REPAIR, GENERAL

A. Have repair work performed only by qualified historic masonry specialist.

B. Repair Appearance Standard: Repaired surfaces are to have a uniform appearance as viewed from 20 feet (6) (m) away by Contracting Officer.

3.4 PRELIMINARY CLEANING

A. Removing Plant Growth: Completely remove visible plant, moss, and shrub growth from masonry surfaces. Carefully remove plants, creepers, and vegetation by cutting at roots and allowing remaining growth to dry as long as possible before removal. Remove loose soil and plant debris from open masonry joints to whatever depth they occur.

B. Preliminary Cleaning: Before beginning general cleaning, remove extraneous substances that are resistant to planned cleaning methods. Extraneous substances include paint, caulking, asphalt, and tar.

1. Carefully remove heavy accumulations of rigid materials from masonry surface with sharp chisel. Do not scratch or chip masonry surface.

3.5 CLEANING UNPOLISHED STONEWORK

A. Cold-Water Wash: Use cold water applied by low-pressure spray.

B. Hot-Water Wash: Use hot water applied by low-pressure spray.

C. Detergent Cleaning:

1. Wet surface with hot water applied by low-pressure spray.

2. Scrub surface with detergent solution using medium-soft brushes until soil is thoroughly dislodged and can be removed by rinsing. Use small brushes to remove soil from mortar joints and crevices. Dip brush in solution often to ensure that adequate fresh detergent is used and that surface remains wet.

3. Rinse with hot water applied by low-pressure spray to remove detergent solution and soil.

4. Repeat cleaning procedure, where needed to produce cleaning effect established by mockup.

D. Mold, Mildew, and Algae Removal:

1. Wet surface with hot water applied by low-pressure spray.

2. Apply mold, mildew, and algae remover by brush or low-pressure spray.

3. Scrub surface with medium-soft brushes until mold, mildew, and algae are thoroughly dislodged and can be removed by rinsing. Use small brushes for mortar joints and crevices. Dip brush in mold, mildew, and algae remover often to ensure that adequate fresh cleaner is used and that surface remains wet.

4. Rinse with hot water applied by low-pressure spray to remove mold, mildew, and algae remover and soil.

5. Repeat cleaning procedure, where needed to produce cleaning effect established by mockup.

3.6 CLEANING MASONRY, GENERAL

A. Proceed with cleaning in an orderly manner; work from top to bottom of each scaffold width and from one end of each elevation to the other. Ensure that dirty residues and rinse water do not wash over dry, cleaned surfaces.

B. Use only those cleaning methods indicated for each masonry material and location.

1. Brushes: Do not use wire brushes or brushes that are not resistant to chemical cleaner being used.

2. Spray Equipment: Use spray equipment that provides controlled application at volume and pressure indicated, measured at nozzle. Adjust pressure and volume to ensure that cleaning methods do not damage masonry.

a. Equip units with pressure gauges.

b. For water-spray application, use fan-shaped spray that disperses water at an angle of 25 to 50 degrees.

c. For high-pressure water-spray application, use fan-shaped spray that disperses water at an angle of at least 40 degrees.

d. For heated water-spray application, use equipment capable of maintaining temperature between 140 and 160 deg F (60 and 71 deg C) at flow rates indicated.

3. For steam application, use steam generator capable of delivering live steam at nozzle.

C. Perform each cleaning method in a manner that results in uniform coverage of all surfaces, including corners, moldings, and interstices, and that produces an even effect without streaking or damaging masonry surfaces.

1. Keep wall wet below area being cleaned to prevent streaking from runoff.

D. Perform additional general cleaning, paint and stain removal, and spot cleaning of small areas that are noticeably different when viewed according to "Cleaning Appearance Standard" Paragraph, so that cleaned surfaces blend smoothly into surrounding areas.

E. Water-Spray Application Methods:

1. Water-Soak Application: Soak masonry surfaces by applying water continuously and uniformly to limited area for time indicated. Apply water at low pressures and low volumes in multiple fine sprays using perforated hoses or multiple spray nozzles. Erect a protective enclosure constructed of polyethylene sheeting to cover area being sprayed.

2. Water-Spray Applications: Unless otherwise indicated, hold spray nozzle at least 6 inches (150 mm) from masonry surface, and apply water in horizontal back-and-forth sweeping motion, overlapping previous strokes to produce uniform coverage.

F. Steam Cleaning: Apply steam to masonry surfaces at very low pressures indicated for each type of masonry. Hold nozzle at least 6 inches (150 mm) from masonry surface, and apply steam in horizontal back-and-forth sweeping motion, overlapping previous strokes to produce uniform coverage.

3.7 ABANDONED ANCHOR REMOVAL

A. Remove abandoned anchors, brackets, wood nailers, and other extraneous items no longer in use unless indicated to remain.

1. Remove items carefully to avoid spalling or cracking stone.

2. Notify Contracting Officer before proceeding if an item cannot be removed without damaging surrounding stone; do the following where directed:

a. Cut or grind off item approximately 3/4 (20) inch (mm) beneath surface, and core drill a recess of same depth in surrounding stone as close around item as practical.

b. Immediately paint exposed end of item with two coats of antirust coating, following coating manufacturer's written instructions and without exceeding manufacturer's recommended dry film thickness per coat. Keep paint off sides of recess.

3. Patch the hole where each item was removed unless directed to remove and replace the stone unit.

3.8 BACKUP MASONRY REMOVAL AND REPLACEMENT

A. Where backup masonry is fractured or unstable and at locations indicated, remove mortar and masonry units that are broken or deteriorated, and rebuild with whole, new brick or whole, salvaged backup masonry units. Carefully remove entire units from joint to joint, without damaging surrounding masonry, in a manner that permits replacement with full-size units.

B. Support and protect remaining masonry that surrounds removal area.

C. Maintain flashing, reinforcement, anchors, lintels, and adjoining construction in an undamaged condition. Coordinate with new flashing, reinforcement, and lintels, which are specified in other Sections.

D. Notify Contracting Officer of unforeseen detrimental conditions, including voids, cracks, bulges, loose masonry units beyond the removal area, rotted wood, rusted metal, and other deteriorated items.

E. Clean masonry surrounding removal areas by removing mortar, dust, and loose particles in preparation for brick replacement.

F. Replace removed damaged brick with salvaged backup brick in good condition, where possible, or with new building brick matching existing backup brick. Do not use broken units unless they can be cut to usable size.

G. Install replacement brick into bonding and coursing pattern of existing brick. If cutting is required, use a motor-driven saw designed to cut masonry with clean, sharp, unchipped edges.

H. Lay replacement brick with rebuilding (setting) mortar and with completely filled bed, head, and collar joints. Butter ends with sufficient mortar to fill head joints and shove into place. Wet both replacement and surrounding bricks that have ASTM C67 initial rates of absorption (suction) of more than 30 g/30 sq. in. per min. (30 g/194 sq. cm per min.). Use wetting methods that ensure that units are nearly saturated but surface is dry when laid.

I. Curing: Cure mortar by maintaining in thoroughly damp condition for at least 72 consecutive hours, including weekends and holidays.

1. Hairline cracking within the mortar or mortar separation at edge of a joint is unacceptable. Completely remove such mortar and repoint.

3.9 PAINTING STEEL UNCOVERED DURING THE WORK

A. Notify Contracting Officer if steel is exposed during masonry removal. Where Contracting Officer determines that it is structural, or for other reasons cannot be totally removed, prepare and paint steel as follows:

1. Surface Preparation: Remove paint, rust, and other contaminants according to SSPC-SP 2, "Hand Tool Cleaning, as applicable to comply with paint manufacturer's recommended preparation.

2. Antirust Coating: Immediately paint exposed steel with two coats of antirust coating, following coating manufacturer's written instructions and without exceeding manufacturer's recommended rate of application (dry film thickness per coat).

B. If on inspection and rust removal, the thickness of a steel member is found to be reduced from rust by more than 1/16 (1.6) inch (mm), notify Contracting Officer before proceeding.

3.10 UNIT REPAIR MATERIAL INSTALLATION

A. Place in same orientation as existing units.

3.11 REPOINTING

A. Repoint cracks to the following extent, coordinate with Structural the need for reinforcing:

1. All cracks in areas indicated.

2. Cracks at locations of the following defects:

a. Chimney at Main Lodge building:

1) Holes and missing mortar.

2) Cracks that can be penetrated 1/4 inch (6 mm) or more by a knife blade 0.027 inch (0.7 mm) thick.

3) Cracks 1/8 (3) inch (mm) or more in width and of any depth.

4) Hollow-sounding joints when tapped by metal object.

5) Eroded surfaces 1/4 inch (6 mm) or more deep.

6) Deterioration to point that mortar can be easily removed by hand, without tools.

7) Joints filled with substances other than mortar.

B. Do not rake out and repoint joints where not required.

C. Notify Contracting Officer of unforeseen detrimental conditions, including voids in mortar joints, cracks, loose stone, rotted wood, rusted metal, and other deteriorated items.

D. Pointing with Mortar:

1. Rinse joint surfaces with water to remove dust and mortar particles. Time rinsing application so, at time of pointing, joint surfaces are damp but free of standing water. If rinse water dries, dampen joint surfaces before pointing.

2. Apply pointing mortar first to areas where existing mortar was removed to depths greater than surrounding areas. Apply in layers not greater than 3/8 (9) inch (mm)until a uniform depth is formed. Fully compact each layer thoroughly, and allow it to become thumbprint hard before applying next layer.

3. After deep areas have been filled to same depth as remaining joints, point joints by placing mortar in layers not greater than 3/8 (9) inch (mm). Fully compact each layer, and allow it to become thumbprint hard before applying next layer. Where existing stone has worn or rounded edges, slightly recess finished mortar surface below face of stone to

avoid widened joint faces. Take care not to spread mortar beyond joint edges onto exposed stone surfaces or to featheredge the mortar.

4. When mortar is thumbprint hard, tool joints to match original appearance of joints as demonstrated in approved mockup. Remove excess mortar from edge of joint by brushing.

5. Cure mortar by maintaining in thoroughly damp condition for at least 72 consecutive hours, including weekends and holidays.

    a. Acceptable curing methods include covering with wet burlap and plastic sheeting, periodic hand misting, and periodic mist spraying using system of pipes, mist heads, and timers.

    b. Adjust curing methods to ensure that pointing mortar is damp throughout its depth without eroding surface mortar.

6. Hairline cracking within the mortar or mortar separation at edge of a joint is unacceptable. Remove mortar and repoint.

E. Pointing with Sealant: Comply with Section 07 90 05 "Joint Sealers" and as follows:

1. After raking out, keep joints dry and free of mortar and debris.

2. Clean and prepare joint surfaces. Prime joint surfaces unless sealant manufacturer recommends against priming. Do not allow primer to spill or migrate onto adjoining surfaces.

3. Fill sealant joints with specified joint sealant.

    a. Install cylindrical sealant backing beneath the sealant. Where space is insufficient for cylindrical sealant backing, install bond-breaker tape.

    b. Install sealant using only proven installation techniques that ensure that sealant is deposited in a uniform, continuous ribbon, without gaps or air pockets, and with complete wetting of the joint bond surfaces equally on both sides. Fill joint flush with surrounding stonework and matching the contour of adjoining mortar joints.

    c. Install sealant as recommended in writing by sealant manufacturer but within the following general limitations, measured at the center (thin) section of the bead:

    d. Fill joints to a depth equal to joint width, but not more than 1/2 inch (13 mm) deep or less than 1/4 inch (6 mm) deep.

    e. Tool sealant to form smooth, uniform beads, slightly concave. Remove excess sealant from surfaces adjacent to joint.

    f. Sanded Joints: Immediately after first tooling, apply ground-mortar aggregate to sealant, gently pushing aggregate into the surface of sealant. Lightly retool sealant

to form smooth, uniform beads, slightly concave. Remove excess sealant and aggregate from surfaces adjacent to joint.

g. Do not allow sealant to overflow or spill onto adjoining surfaces, or to migrate into the voids of adjoining surfaces, particularly rough textures. Remove excess and spillage of sealant promptly as the work progresses. Clean adjoining surfaces by the means necessary to eliminate evidence of spillage, without damage to adjoining surfaces or finishes, as demonstrated in an approved mockup.

F. Where repointing work precedes cleaning of existing masonry, allow mortar to harden at least 30 days before beginning cleaning work.

3.12 CHIMNEY FLASHING

A. Install chimney flashing product following manufacturer's written installation instructions.

3.13 FINAL CLEANING

A. After mortar has fully hardened, thoroughly clean exposed stone surfaces of excess mortar and foreign matter; use wood scrapers, stiff-nylon or -fiber brushes, and clean water applied by low-pressure spray.

1. Do not use metal scrapers or brushes.

2. Do not use acidic or alkaline cleaners.

B. Clean adjacent non-stone surfaces. Use detergent and soft brushes or cloths.

C. Clean mortar and debris from roof; remove debris from gutters and downspouts. Rinse off roof and flush gutters and downspouts.

D. Remove masking materials, leaving no residues that could trap dirt.

E. Sweep and rake adjacent pavement and grounds to remove mortar and debris. Where necessary, pressure-wash pavement surfaces to remove mortar, dust, dirt, and stains.

3.14 FIELD QUALITY CONTROL

A. Contracting Officer's Project Representatives: Contracting Officer will assign Project representatives to help carry out Contracting Officer's responsibilities at the site, including observing progress and quality of portion of the Work completed. Allow Contracting Officer's Project representatives use of lift devices and scaffolding, as needed, to observe progress and quality of portion of the Work completed.

B. Notify Contracting Officer's Project representatives in advance of times when lift devices and scaffolding will be relocated. Do not relocate lift devices and scaffolding until inspectors and Contracting Officer's Project representatives have had reasonable opportunity to make inspections and observations of work areas at lift device or scaffold location.

END OF SECTION

SECTION 051200 - STRUCTURAL STEEL FRAMING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Structural steel.
2. Grout.

1.2 DEFINITIONS

A. Structural Steel: Elements of the structural frame indicated on Drawings and as described in AISC 303, "Code of Standard Practice for Steel Buildings and Bridges."

1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Shop Drawings: Show fabrication of structural-steel components.

1.5 INFORMATIONAL SUBMITTALS

A. Qualification Data: For fabricator.

B. Welding certificates.

C. Mill test reports for structural steel, including chemical and physical properties.

D. Source quality-control reports.

E. Field quality-control and special inspection reports.

1.6 QUALITY ASSURANCE

A. Fabricator Qualifications: A qualified fabricator that participates in the AISC Quality Certification Program and is designated an AISC-Certified Plant, Category STD.

B. Installer Qualifications: A qualified installer who participates in the AISC Quality Certification Program and is designated an AISC-Certified Erector, Category CSE.

C. Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."

D. Comply with applicable provisions of the following specifications and documents:

1. AISC 303.
2. AISC 360.
3. RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

PART 2 - PRODUCTS

2.1 STRUCTURAL-STEEL MATERIALS

A. W-Shapes: ASTM A 992/A 992M.

B. Channels, Angles: ASTM A 36/A 36M.

C. Plate and Bar: ASTM A 36/A 36M.

D. Cold-Formed Hollow Structural Sections: ASTM A 500/A 500M, Grade B, structural tubing.

E. Steel Pipe: ASTM A 53/A 53M, Type E or Type S, Grade B.

F. Welding Electrodes: Comply with AWS requirements.

2.2 BOLTS, CONNECTORS, AND ANCHORS

A. High-Strength Bolts, Nuts, and Washers: ASTM F 3125/F 3125M, Grade A325, Type 1, heavy-hex steel structural bolts; ASTM A 563, Grade DH, heavy-hex carbon-steel nuts; and ASTM F 436/F 436M, Type 1, hardened carbon-steel washers; all with plain finish.

1. Direct-Tension Indicators: ASTM F 959/F 959M, Type 325-1, compressible-washer type with plain finish.

B. Zinc-Coated High-Strength Bolts, Nuts, and Washers: ASTM F 3125/F 3125M, Grade A325, Type 1, heavy-hex steel structural bolts; ASTM A 563, Grade DH, heavy-hex carbon-steel nuts; and ASTM F 436/F 436M, Type 1, hardened carbon-steel washers.

1. Finish: Hot-dip zinc coating.
2. Direct-Tension Indicators: ASTM F 959/F 959M, Type 325-1, compressible-washer type with mechanically deposited zinc coating finish.

C. Tension-Control, High-Strength Bolt-Nut-Washer Assemblies: ASTM F 3125/F 3125M, Grade F1852, Type 1, heavy-hex head assemblies consisting of steel structural bolts with

splined ends; ASTM A 563, Grade DH, heavy-hex carbon-steel nuts; and ASTM F 436/F 436M, Type 1 hardened carbon-steel washers.

1. Finish: Plain.

D. Shear Connectors: ASTM A 108, Grades 1015 through 1020, headed-stud type, cold-finished carbon steel; AWS D1.1/D1.1M, Type B.

E. Unheaded Anchor Rods: ASTM F 1554, Grade 36.

1. Configuration: Hooked.
2. Finish: Hot-dip zinc coating, ASTM A 153/A 153M, Class C.

F. Headed Anchor Rods: ASTM F 1554, Grade 36, straight.

1. Finish: Hot-dip zinc coating, ASTM A 153/A 153M, Class C.

G. Threaded Rods: ASTM A 36/A 36M.

1. Finish: Plain.

H. Clevises and Turnbuckles: Made from cold-finished carbon steel bars, ASTM A 108, Grade 1035.

2.3 PRIMER

A. Primer: Comply with Section 099113 "Exterior Painting" and Section 099123 "Interior Painting."

B. Primer: SSPC-Paint 25, Type I, zinc oxide, alkyd, linseed oil primer.

C. Primer: Fabricator's standard lead- and chromate-free, non-asphaltic, rust-inhibiting primer complying with MPI#79 and compatible with topcoat.

2.4 GROUT

A. Metallic, Shrinkage-Resistant Grout: ASTM C 1107/C 1107M, factory-packaged, metallic aggregate grout, mixed with water to consistency suitable for application and a 30-minute working time.

B. Nonmetallic, Shrinkage-Resistant Grout: ASTM C 1107/C 1107M, factory-packaged, nonmetallic aggregate grout, noncorrosive and nonstaining, mixed with water to consistency suitable for application and a 30-minute working time.

2.5 FABRICATION

A. Structural Steel: Fabricate and assemble in shop to greatest extent possible. Fabricate according to AISC 303, "Code of Standard Practice for Steel Buildings and Bridges," and to AISC 360.

B. Shear Connectors: Prepare steel surfaces as recommended by manufacturer of shear connectors. Use automatic end welding of headed-stud shear connectors according to AWS D1.1/D1.1M and manufacturer's written instructions.

2.6 SHOP CONNECTIONS

A. High-Strength Bolts: Shop install high-strength bolts according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts" for type of bolt and type of joint specified.

1. Joint Type: Snug tightened.

B. Weld Connections: Comply with AWS D1.1/D1.1M and AWS D1.8/D1.8M for tolerances, appearances, welding procedure specifications, weld quality, and methods used in correcting welding work.

2.7 SHOP PRIMING

A. Shop prime steel surfaces except the following:

1. Surfaces embedded in concrete or mortar. Extend priming of partially embedded members to a depth of 2 inches.
2. Surfaces to be field welded.
3. Surfaces of high-strength bolted, slip-critical connections.
4. Surfaces to receive sprayed fire-resistive materials (applied fireproofing).
5. Galvanized surfaces.
6. Surfaces enclosed in interior construction.

B. Surface Preparation: Clean surfaces to be painted. Remove loose rust and mill scale and spatter, slag, or flux deposits. Prepare surfaces according to the following specifications and standards:

1. SSPC-SP 2, "Hand Tool Cleaning."
2. SSPC-SP 3, "Power Tool Cleaning."
3. SSPC-SP 7/NACE No. 4, "Brush-off Blast Cleaning."

C. Priming: Immediately after surface preparation, apply primer according to manufacturer's written instructions and at rate recommended by SSPC to provide a minimum dry film thickness of 1.5 mils. Use priming methods that result in full coverage of joints, corners, edges, and exposed surfaces.

2.8 SOURCE QUALITY CONTROL

A. Testing Agency: The Contractor is responsible for all testing and inspections. Contrctor will engage a qualified testing agency to perform shop tests and inspections.

1. Provide testing agency with access to places where structural-steel work is being fabricated or produced to perform tests and inspections.

B. Bolted Connections: Inspect shop-bolted connections according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

C. Welded Connections: Visually inspect shop-welded connections according to AWS D1.1/D1.1M and the following inspection procedures, at testing agency's option:

1. Liquid Penetrant Inspection: ASTM E 165.
2. Magnetic Particle Inspection: ASTM E 709; performed on root pass and on finished weld. Cracks or zones of incomplete fusion or penetration are not accepted.
3. Ultrasonic Inspection: ASTM E 164.
4. Radiographic Inspection: ASTM E 94.

D. Prepare test and inspection reports.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify, with certified steel erector present, elevations of concrete- and masonry-bearing surfaces and locations of anchor rods, bearing plates, and other embedments for compliance with requirements.

B. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 ERECTION

A. Set structural steel accurately in locations and to elevations indicated and according to AISC 303 and AISC 360.

B. Baseplates: Clean concrete- and masonry-bearing surfaces of bond-reducing materials, and roughen surfaces prior to setting plates. Clean bottom surface of plates.

1. Set plates for structural members on wedges, shims, or setting nuts as required.
2. Weld plate washers to top of baseplate.
3. Snug-tighten anchor rods after supported members have been positioned and plumbed. Do not remove wedges or shims but, if protruding, cut off flush with edge of plate before packing with grout.
4. Promptly pack grout solidly between bearing surfaces and plates so no voids remain. Neatly finish exposed surfaces; protect grout and allow to cure. Comply with manufacturer's written installation instructions for shrinkage-resistant grouts.

C. Maintain erection tolerances of structural steel within AISC 303, "Code of Standard Practice for Steel Buildings and Bridges."

3.3 FIELD CONNECTIONS

A. High-Strength Bolts: Install high-strength bolts according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts" for type of bolt and type of joint specified.

1. Joint Type: Snug tightened.

B. Weld Connections: Comply with AWS D1.1/D1.1M and AWS D1.8/D1.8M for tolerances, appearances, welding procedure specifications, weld quality, and methods used in correcting welding work.

1. Comply with AISC 303 and AISC 360 for bearing, alignment, adequacy of temporary connections, and removal of paint on surfaces adjacent to field welds.
2. Remove backing bars or runoff tabs, back gouge, and grind steel smooth.
3. Assemble and weld built-up sections by methods that maintain true alignment of axes without exceeding tolerances in AISC 303, "Code of Standard Practice for Steel Buildings and Bridges," for mill material.

3.4 FIELD QUALITY CONTROL

A. Special Inspections: The Contractor is responsible for all testing and inspections. Contractor will engage a qualified special inspector to perform the following special inspections:

1. Verify structural-steel materials and inspect steel frame joint details.
2. Verify weld materials and inspect welds.
3. Verify connection materials and inspect high-strength bolted connections.

B. Testing Agency: Contractor will engage a qualified testing agency to perform tests and inspections.

C. Bolted Connections: Inspect bolted connections according to RCSC's "Specification for Structural Joints Using ASTM A 325 or A 490 Bolts."

D. Welded Connections: Visually inspect field welds according to AWS D1.1/D1.1M.

1. In addition to visual inspection, test and inspect field welds according to AWS D1.1/D1.1M and the following inspection procedures, at testing agency's option:

a. Liquid Penetrant Inspection: ASTM E 165.
b. Magnetic Particle Inspection: ASTM E 709; performed on root pass and on finished weld. Cracks or zones of incomplete fusion or penetration are not accepted.
c. Ultrasonic Inspection: ASTM E 164.
d. Radiographic Inspection: ASTM E 94.

END OF SECTION 051200

SECTION 05 50 00 - METAL FABRICATIONS

PART 1 GENERAL

1.1 SECTION INCLUDES

A. This section describes items made from iron and steel shapes, plates, bars, strips, tubes, pipes, and castings which are not a part of structural steel or other metal systems specified elsewhere.

B. This section describes items made from extruded aluminum shapes, plates, tubes, and pipes, which are not a part of other metal systems specified elsewhere.

C. Shop fabricated steel and aluminum items. as noted but not limited to:

1. Steel Railing

2. Slotted Channel Framing

1.2 REFERENCE STANDARDS

A. ASTM A36/A36M - Standard Specification for Carbon Structural Steel; 2019.

B. ASTM A501/A501M - Standard Specification for Hot-Formed Welded and Seamless Carbon Steel Structural Tubing; 2021.

C. ASTM A1011/A1011M - Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength; 2023.

D. AWS B2.1/B2.1M - Specification for Welding Procedure and Performance Qualification; 2021, with Errata (2023).

E. AWS C3.4M/C3.4 - Specification for Torch Brazing; 2026.

F. AWS C3.5M/C3.5 - Specification for Induction Brazing; 2026.

G. AWS C3.9M/C3.9 - Specification for Resistance Brazing; 2020.

H. AWS D1.1/D1.1M - Structural Welding Code - Steel; 2025.

I. AWS D1.6/D1.6M - Structural Welding Code - Stainless Steel; 2017, with Amendment (2021).

1.3 PERFORMANCE REQUIREMENTS

A. Delegated-Design: Design handrails and railings, including comprehensive engineering analysis by a qualified professional engineer, using performance requirements and design criteria indicated.

1. Delegated-Design Requirements: Contractor is responsible for the design and installation of handrails and railings that comply with all codes in effect at project location.

Contractor is responsible for attaining special permits as required by the building official having jurisdiction over construction at project site. Provide calculations, layouts and other information required for permitting. Coordinate required inspections and attain all required approvals

B. Structural performance of handrails and railings: Provide handrails and railings capable of withstanding structural loads required by ASCE 7, latest edition, without exceeding the allowable design working stress of materials for handrails, railings, anchors and connections.

C. NAAM Stair Standard: Where any condition is not specifically shown comply with "Recommended Voluntary Minimum Standards for fixed Metal Stairs" in NAAM AMP 510, "Metal Stairs Manual" for class of stair designated.

1. Architectural Class.

1.4 SUBMITTALS

A. Shop Drawings: Indicate profiles, sizes, connection attachments, reinforcing, anchorage, size and type of fasteners, and accessories. Include erection drawings, elevations, and details where applicable.

B. Welders' Qualification Statement: Welders' certificates in accordance with AWS B2.1/B2.1M and dated no more than 12 months before start of scheduled welding work.

1.5 COORDINATION

A. Coordinate installation of anchorages for metal fabrications. Furnish setting drawings, templates, and directions for installing anchorages, including sleeves, concrete inserts, anchor bolts, and items with integral anchors, that are to be embedded in concrete or masonry. Deliver such items to Project site in time for installation.

B. Coordinate installation of steel weld plates and angles for casting into concrete that are specified in this Section but required for work of another Section. Deliver such items to Project site in time for installation.

PART 2 PRODUCTS

2.1 MATERIALS - STEEL

A. Steel Sections: ASTM A36/A36M.

B. Steel Tubing: ASTM A501/A501M hot-formed structural tubing.

C. Slotted Channel Fittings: ASTM A1011/A1011M.

2.2 FASTENERS

A. General: Unless otherwise indicated, provide Type 304 stainless-steel fasteners for exterior use and zinc-plated fasteners with coating complying with ASTM B 633, Class Fe/Zn 5, at exterior

walls. Provide stainless-steel fasteners for fastening aluminum. Select fasteners for type, grade, and class required.

B. Steel Bolts and Nuts: Regular hexagon-head bolts, ASTM A 307, Grade A (ASTM F 568M, Property Class 4.6); with hex nuts, ASTM A 563 (ASTM A 563M); and, where indicated, flat washers.

C. Stainless-Steel Bolts and Nuts: Regular hexagon-head annealed stainless-steel bolts, nuts and, where indicated, flat washers; ASTM F 593 (ASTM F 738M) for bolts and ASTM F 594 (ASTM F 836M) for nuts, Alloy Group 1 (A1).

D. Anchor Bolts: ASTM F 1554, Grade 36.

1. Provide hot-dip or mechanically deposited, zinc-coated anchor bolts where item being fastened is indicated to be galvanized.

E. Eyebolts: ASTM A 489.

F. Machine Screws: ASME B18.6.3 (ASME B18.6.7M).

G. Lag Bolts: ASME B18.2.1 (ASME B18.2.3.8M).

H. Wood Screws: Flat head, ASME B18.6.1.

I. Plain Washers: Round, ASME B18.22.1 (ASME B18.22M).

J. Lock Washers: Helical, spring type, ASME B18.21.1 (ASME B18.21.2M).

K. Anchors, General: Anchors capable of sustaining, without failure, a load equal to six times the load imposed when installed in unit masonry and four times the load imposed when installed in concrete, as determined by testing according to ASTM E 488, conducted by a qualified independent testing agency.

L. Cast-in-Place Anchors in Concrete: Anchors capable of sustaining, without failure, a load equal to four times the load imposed, as determined by testing according to ASTM E 488, conducted by a qualified independent testing agency.

1. Threaded or wedge type; galvanized ferrous castings, either ASTM A 47/A 47M malleable iron or ASTM A 27/A 27M cast steel. Provide bolts, washers, and shims as needed, hot-dip galvanized per ASTM A 153/A 153M.

M. Post installed expansion Anchors: Anchor bolt and sleeve assembly with capability to sustain, without failure, a load equal to six times the load imposed when installed in unit masonry and four times the load imposed when installed in concrete, as determined by testing according to ASTM E 488, conducted by a qualified independent testing agency. Post Installed anchors in concrete used to resist seismic forces shall be prequalified in accordance with ACI 355.2

1. Material for Anchors in Interior Locations: Carbon-steel components zinc-plated to comply with ASTM B 633, Class Fe/Zn 5..

2. Material for Anchors in Exterior Locations: Alloy Group 1 (A1) stainless-steel bolts complying with ASTM F 593 (ASTM F 738M) and nuts complying with ASTM F 594 (ASTM F 836M).

2.3 FABRICATED ITEMS

A. Railings:

1. Dimensions: See drawings for configurations and heights.

a. Top Rails and Wall Rails: 1-1/2 inches diameter, round.

2. Provide anchors and other components as required to attach to structure, made of same materials as railing components unless otherwise indicated; where exposed fasteners are unavoidable provide flush countersunk fasteners.

a. For anchorage to concrete, provide inserts to be cast into concrete, for bolting anchors.

3. Provide welding fittings to join lengths, seal open ends, and conceal exposed mounting bolts and nuts, including but not limited to elbows, T-shapes, splice connectors, flanges, escutcheons, and wall brackets.

4. Welded and Brazed Joints: Make visible joints butt tight, flush, and hairline; use methods that avoid discoloration and damage of finish; grind smooth, polish, and restore to required finish.

a. Ease exposed edges to a small uniform radius.

b. Welded Joints:

1) Carbon Steel: Perform welding in accordance with AWS D1.1/D1.1M.

5. Paint railings in field.

2.4 FABRICATION TOLERANCES

A. Squareness: 1/8 inch maximum difference in diagonal measurements.

B. Maximum Offset Between Faces: 1/16 inch.

C. Maximum Misalignment of Adjacent Members: 1/16 inch.

D. Maximum Bow: 1/8 inch in 48 inches.

E. Maximum Deviation From Plane: 1/16 inch in 48 inches.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that field conditions are acceptable and are ready to receive work.

3.2 PREPARATION

A. Clean and strip primed steel items to bare metal where site welding is required.

B. Furnish setting templates to the appropriate entities for steel items required to be cast into concrete or embedded in masonry.

3.3 INSTALLATION

A. General: Install framing and supports to comply with requirements of items being supported, including manufacturers' written instructions and requirements indicated on Shop Drawings. Install items plumb and level, accurately fitted, free from distortion or defects.

B. Fit exposed connections accurately together to form hairline joints. Weld connections that are not to be left as exposed joints but cannot be shop welded because of shipping size limitations. Do not weld, cut, or abrade surfaces of exterior units that have been hot-dip galvanized after fabrication and are for bolted or screwed field connections. Provide for erection loads, and for sufficient temporary bracing to maintain true alignment until completion of erection and installation of permanent attachments.

C. Install items plumb and level, accurately fitted, free from distortion or defects.

D. Provide for erection loads, and for sufficient temporary bracing to maintain true alignment until completion of erection and installation of permanent attachments.

E. Fastening to In-Place Construction: Provide anchorage devices and fasteners where metal fabrications are required to be fastened to in-place construction. Provide threaded fasteners for use with concrete and masonry inserts, toggle bolts, through bolts, lag bolts, wood screws, and other connectors.

F. Provide temporary bracing or anchors in formwork for items that are to be built into concrete, masonry, or similar construction.

G. Obtain approval prior to site cutting or making adjustments not scheduled.

H. After erection, prime welds, abrasions and surfaces not shop primed or galvanized , except surfaces to be in contact with concrete.

I. Follow manufacturer's written instructions when installing Primer.

J. All corners exposed to view shall be mitered unless otherwise shown in drawings

END OF SECTION

SECTION 06 03 12 - HISTORIC WOOD REPAIR

PART 1 GENERAL

1.1 SUMMARY

A. Section includes historic treatment of wood in the form of repairing wood features as follows:

1. Repair of damaged wood components.

2. Replacement of damaged wood components.

3. Removal of existing wood finishes and refinishing.

4. Repairing, refinishing, and replacing hardware.

5. Repair of damaged wood window frames and trim.

6. Cleaning of soiled wood following animal guano remediation

B. RELATED REQUIREMENTS:

1. Section 01 35 91 - Historic Treatment Procedures for general historic treatment requirements.

2. Section 02 42 96 - Historic Removal and Dismantling for historic removal and dismantling work.

3. Section 08 03 14 - Historic Treatment of Wood Doors for historic wood door repairs, including related trim.

4. Section 06 13 13 - Historic Log Construction Repair

1.2 REFERENCE STANDARDS

A. BHMA A156.18 - Standard for Materials and Finishes; 2020.

1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site .

1. Review minutes of Preliminary Historic Treatment Conference that pertain to historic wood repair.

2. Review methods and procedures related to historic wood repair, including, but not limited to, the following:

a. Historic treatment specialist's personnel, equipment, and facilities needed to make progress and avoid delays.

b. Materials, material application, sequencing, tolerances, and required clearances.

c. Fire-protection plan.

d. Wood historic treatment program.

e. Coordination with building occupants.

1.4 SEQUENCING AND SCHEDULING

A. Perform historic wood repair in the following sequence, which includes work specified in this and other Sections:

1. Before removing wood components for on-site or off-site repair, tag each component with location-identification numbers. Indicate on tags and building plans the locations of each component, such as "Baseboard on North Side of Room 101."

2. Dismantle hardware and tag with location-identification numbers.

3. In the shop, label each repaired component and whole or partial replacement with permanent location-identification number in inconspicuous location and remove site-applied tags.

4. Sort units by condition, separating those that need extensive repair.

5. Clean surfaces.

6. General Wood-Repair Sequence:

a. Remove paint to bare wood.

b. Repair wood by consolidation, replacement, and partial replacement.

1) Patching of wood is not allowed.

c. Match existing work in profile, size, arrangement and surface character. All exposed surfaces shall show hand-sanded, hand-planed, or hand-hewn finish. Machine marks are not acceptable

d. Prepare surface for finish, see Section 09 91 13 Exterior Painting and 09 91 23 - Interior Painting, if applicable

7. Repair, refinish, and replace hardware if required. Reinstall operating hardware.

8. Reinstall components.

9. Apply finish coats.

10. Install remaining hardware.

1.5 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include recommendations for product application and use. Include test data substantiating that products comply with requirements.

B. Shop Drawings:

1. Include plans, elevations, and sections showing locations and extent of repair and replacement work, with enlarged details of replacement parts indicating materials, profiles, joinery, reinforcing, method of splicing or attaching wood members to other surfaces, accessory items, and finishes.

2. Include field-verified dimensions and the following:

a. Full-size shapes and profiles with complete dimensions for replacement components and their jointing, showing relationship of existing components to new components.

b. Templates and directions for installing hardware and anchorages.

c. Identification of each new unit and its corresponding location in the building on annotated plans and elevations.

d. Provisions for sealant joints as required for location.

C. Samples for Initial Selection: For each type of exposed wood and finish.

1. Identify wood species, cut, and other features.

2. Include Samples of hardware and accessories involving color or finish selection.

D. Samples for Verification: Actual sample of finished products for the following products in manufacturer's standard sizes unless otherwise indicated:

1. Replacement Wood: 12-inch- (300-mm-) long, full-size molding sections with applied finish.

a. Additional Samples of replacement members that show fabrication techniques, materials, and finishes as requested by Contracting Officer.

2. Repaired Wood: Prepare Samples using existing wood removed from site, repaired, and prepared for refinishing.

3. Refinished Wood: Prepare Samples using existing wood removed from site, repaired, and refinished.

4. Hardware: Full-size units with each factory-applied or restored finish.

1.6 INFORMATIONAL SUBMITTALS

A. Preconstruction Test Reports: For historic wood repair.

B. Wood Species analysis.

C. Qualification Statements: For historic treatment specialist .

D. Wood Historic Treatment Program: Submit before work begins.

1.7 QUALITY ASSURANCE

A. Conduct wood species analysis prior to repair of wood.

B. Historic Treatment Specialist Qualifications: A qualified historic wood-repair specialist, experienced in repairing, refinishing, and replacing wood in whole and in part. Experience only in fabricating and installing new woodwork is insufficient experience for wood historic treatment work.

C. Wood-Repair-Material Manufacturer Qualifications: A firm regularly engaged in producing wood consolidant that have been used for similar historic wood-treatment applications with successful results, and with factory-authorized service representatives who are available for consultation, Project-site inspection, and on-site assistance.

D. Wood Historic Treatment Program: Prepare a written, detailed description of materials, methods, equipment, and sequence of operations to be used for historic treatment work, including protection of surrounding materials and Project site.

1. If materials and methods other than those indicated are proposed for any phase of historic treatment work, add a written description of such materials and methods, including evidence of successful use on comparable projects, and demonstrations to show their effectiveness for this Project.

1.8 MOCKUPS

A. Prepare mockups of historic treatment repair processes to demonstrate aesthetic effects, to set quality standards for materials and execution, and to set quality standards for fabrication and installation. Prepare mockups so they are inconspicuous.

1. Locate mockups on existing surfaces where directed by Contracting Officer.

2. Wood Baseboard Component Repair: Prepare (2) areas of each type of wood repair, including patching and Dutchman repairs, to serve as mockup to demonstrate quality and appearance expectations. Prepare an approximately 72-inch (2000-mm) length of baseboard to serve as mockup to demonstrate samples of each type of wood repair.

3. Wood Refinishing: Prepare (2) areas for each type of finish, where finish is removed from both locations and new finish is applied to one location, to serve as mockup to demonstrate quality and appearance expectations for removal of existing finish and application of new finish.

4. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations by Change Order.

5. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.9 DELIVERY, STORAGE, AND HANDLING

A. Pack, deliver, and store products in suitable packs, heavy-duty cartons, or wooden crates; surround with sufficient packing material to ensure that products will not be deformed, broken, or otherwise damaged.

B. Until installed, store products inside a well-ventilated area and protect from weather, moisture, soiling, abrasion, extreme temperatures, and humidity, and where environmental conditions comply with manufacturer's requirements.

1.10 FIELD CONDITIONS

A. Weather Limitations: Proceed with historic wood repair only when existing and forecasted weather conditions are within environmental limits set by each manufacturer's written instructions and specified requirements.

PART 2 PRODUCTS

2.1 HISTORIC WOOD REPAIR QUALITY STANDARD

A. Quality Standard: Comply with applicable requirements in Section 12, "Historic Restoration Work," and related requirements in AWMAC/WI's "North American Architectural Woodwork Standards" for construction, finishes, grade rules, and other requirements unless otherwise indicated.

1. Exception: Industry practices cited in Section 12, Paragraph 6, "Industry Practices," under Article 12.1, "Basic Considerations," of AWMAC/WI's "North American Architectural Woodwork Standards" do not apply to the Work of this Section.

2.2 WOOD-REPLACEMENT MATERIALS

A. Wood or Wood-Based Materials:

1. Wood fabricated from old growth timber is not permitted.

B. Lumber Materials:

1. Softwood Lumber: match existing, plain sawn, maximum moisture content of 6 percent; B & BTR SELECT grade.

C. Quality Standard: Items fabricated in accordance with AWI/AWMAC/WI (AWS), Section 12, unless noted otherwise.

D. Surface Burning Characteristics: Provide materials having fire and smoke properties as required by applicable code.

E. Wood Paneling: Matching existing elements.

1. Species: Match existing species

2. Joint Construction: Match existing joints

3. Wood Clarity and Cut: Match existing clarity and cut

4. Wood Members and Trim Profiles: Match profiles and detail of existing

5. Date Identification: Emboss Month, Day, and Year on a concealed surface of reach replaced item, in easily read characters following the format ##/##/####

F. Exterior Finish Carpentry Items: Matching existing elements.

1. Species: Match existing

2. Wood Clarity and Cut: Match existing clarity and cut

3. Wood Members and Trim Profiles: Match profiles and detail of existing

4. Window Casings: Prepare for linseed oil-base paint or finish per drawings.

5. Exterior Trim: Prepare for opaque, linseed oil-based finish.

G. Interior Finish Carpentry Items: Matching existing elements.

1. Species: Match existing.

2. Moldings, Bases, Casings, and Miscellaneous Trim: Prepare for linseed-oil based finish.

3. Window Sills: Prepare for linseed-oil based finish as indicated in drawings.

H. Window Frame: Match existing elements

1. Species: Match existing

2. Wood Clarity and Cut: Match existing clarity and cut

3. Window Frames: prepare for linseed oil-base paint or finish as indicated in drawings.

2.3 WOOD-REPAIR MATERIALS

A. Source Limitations: Obtain compatible consolidants and patching and filling compounds from single manufacturer.

B. Consolidants:

1. Epoxy Based: Two-component clear epoxy liquid system to penetrate and restore deteriorated wood.

    a. Formulation: Two-part; resin and hardener.

    b. Mix Ratio: Recommended by manufacturer.

    c. Additional Uses: Primer for patching and filling compounds.

    d. Tensile Strength: 1,460 psi (10.1 MPa), minimum.

    e. Compressive Strength: 5,210 psi (36 MPa), minimum.

    f. Flexural Strength: 900 psi (6.2 MPa), minimum.

    g. Elongation: 84 percent, minimum.

    h. Basis of Design:

        1) Conserve Epoxy: www.conserveepoxy.com. Product characteristics noted in paragraph 2.3.

        2) Substitutions: See Section 01 67 00 – Product Requirements.

2.4 HARDWARE

A. Reuse existing unless otherwise indicated.

B. See Section 08 71 00 - Hardware and provide historic hardware or providing historically appropriate replacements.

C. Hardware, General: Provide hardware required for each type of replicated or repaired wood, including, but not limited to, hinges, pulls, latches, fasteners, and accessories indicated or required for proper operation. Hardware must smoothly operate, tightly close, and secure units appropriately for frequency of use, unit weight, and dimensions.

D. Replacement Hardware: Replace existing damaged or missing hardware with new hardware manufactured by one of the following:

E. Material and Design:

1. Material: Solid bronze of alloy indicated unless otherwise indicated.

2. Design: Custom hardware to replicate existing hardware.

3. Replacement Hardware: Regardless of mechanisms within, match existing, exposed hardware of the following types:

    a. Knobs, levers, and escutcheons.

    b. Latches.

c. Surface-mounted bolts.

d. Handles.

e. Pulls.

f. Handrail brackets.

4. Date Identification: Emboss on a concealed surface of the metal body of each new hardware item, in easily read characters, "MADE [Insert year]" Manufacturer's name may also be embossed. For cast iron or other brittle metals, add the identification to the mold pattern before casting.

F. Hardware Finishes: Comply with BHMA A156.18 for base material and finish requirements indicated by the following:

1. BHMA 609: Antique Brass, clear coated; brass base metal.

2.5 PAINT REMOVERS

A. Basis of Design:

1. PROSOCO, Inc; Enviro Klean Safety Peel 1: www.prosoco.com. Product characteristics noted in paragraph 2.5.

2. Substitutions: See Section 01 67 00 – Product Requirements.

B. Water or Solvent-Based: Formulated without methylene chloride or methanol and used for removal of multiple coats of oil-based, water-based, acrylic-based, epoxy-based, urethane-based, elastomeric, and lead-based paints.

1. Formulation: Thixotropic paste or gel.

2. PH neutral and non-flammable.

3. Surface Neutralization After Use: Not required.

4. Restrictions: Not intended by the manufacturer for use on furniture, polished surfaces, or for removal of cementitious coatings

2.6 MISCELLANEOUS MATERIALS

A. Borate Preservative Treatment: Inorganic, borate-based solution, with disodium octaborate tetrahydrate as the primary ingredient; manufactured for preserving weathered and decayed wood from further damage caused by fungi and wood-boring insects; containing no boric acid.

1. Basis of Design:

a. Tim-Bor Professional: https://nisuscorp.com/product/tim-bor-professional. Product characteristics noted in paragraph 2.6.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2. Product Characteristics:

a. Can be applied as a solution, dust, or foam to bare wood. \

b. Low toxicity.

c. Non-staining.

d. Non-corrosive.

e. Highly soluble.

f. Odorless.

B. Cleaning Materials:

1. Detergent Solution: Solution prepared by mixing 2 cups (0.5 L) of tetrasodium pyrophosphate (TSPP), 1/2 cup (125 mL) of laundry detergent that contains no ammonia, 5 quarts (5 L) of 5 percent sodium hypochlorite bleach, and 15 quarts (15 L) of warm water for each 5 gal. (20 L) of solution required.

2. Mildewcide: Commercial, proprietary mildewcide or a solution prepared by mixing 1/3 cup (80 mL) of household detergent that contains no ammonia, 1 quart (1 L) of 5 percent sodium hypochlorite bleach, and 3 quarts (3 L) of warm water.

C. Adhesives: Wood adhesives with minimum 15- to 45-minute cure at 70 deg F (21 deg C), in gunnable and liquid formulations as recommended in writing by adhesive manufacturer for each type of repair and exposure condition.

1. Wood Glue:

a. Basis of Design:

1) Titebond III Ultimate Wood Glue http://www.titebond.com/titebond_wood_glues/Ultimate_Wood_Glue.asp x. Product characteristics noted in paragraph 2.6.C.

2) Substitutions: See Section 01 67 00 – Product Requirements.

2. Epoxy Glue:

a. Basis of Design:

1) ConServ Epoxy; Product 552 Multipurpose Structural Adhesive www.conservepoxy.com. Product characteristics noted in paragraph 2.6.C.2.

2) Substitutions: See Section 01 67 00 – Product Requirements.

b. Product Characteristics:

1) 2:1 mix ratio with room temperature cure.

2) Medium viscosity with a working time of 60-90 mintues.

3) Must be sand-able, drilled-able, and paint-able when cured.

D. Fasteners: Use fastener metals that are noncorrosive and compatible with each material joined.

1. Match existing fasteners in material and type of fastener unless otherwise indicated.

2. Use concealed fasteners for interconnecting wood components.

3. Use concealed fasteners for attaching items to other work unless exposed fasteners are unavoidable.

4. For fastening metals, use fasteners of same basic metal as fastened metal unless otherwise indicated.

5. For exposed fasteners, use Phillips-type machine screws of head profile flush with metal surface unless otherwise indicated.

6. Finish exposed fasteners to match finish of metal fastened unless otherwise indicated.

2.7 WOOD FINISHES

A. See Section 09 91 23 – Exterior Painting

B. Unfinished Replacement Units: Provide exposed exterior wood surfaces of replacement units unfinished; smooth, filled, and suitably prepared for on-site priming and finishing.

PART 3 EXECUTION

3.1 HISTORIC TREATMENT SPECIALIST

A. Historic Treatment Specialist Firms: Subject to compliance with requirements, provide historic wood repair by one of the following:

1. Firm meeting Part 1 Quality Assurance requirements..

3.2 PREPARATION

A. Protect adjacent materials from damage by historic wood repair.

B. Clean wood of mildew, algae, moss, plant material, loose paint, grease, dirt, and other debris by scrubbing with bristle brush or sponge and detergent solution. Scrub mildewed areas with mildewcide. After cleaning, rinse thoroughly with fresh water. Allow to dry before repairing or painting.

C. Clean wood stains on logs, trim, doors, and flooring following remediation.

D. Condition replacement wood members and replacement units to prevailing conditions at installation areas before installing.

3.3 HISTORIC WOOD REPAIR, GENERAL

A. Historic Treatment Appearance Standard: Completed work is to have a uniform appearance as viewed by Contracting Officer from 5 ft. (1.5 m) away for interior work and from 20 ft. (6 m) away for exterior work.

B. General: In treating historic items, disturb them as minimally as possible and as follows:

1. Stabilize and repair wood to reestablish structural integrity and weather resistance while maintaining the existing form of each item.

2. Remove coatings and apply borate preservative treatment before repair. Remove coatings in accordance with Section 090391 "Historic Treatment of Plain Painting" unless otherwise indicated.

3. Repair items in place where possible.

4. Install temporary protective measures to protect wood-treatment work that is indicated to be completed later.

C. Mechanical Abrasion: Where mechanical abrasion is needed for the Work, use only the gentlest mechanical methods, such as scraping and natural-fiber bristle brushing, that will not abrade wood substrate, reducing clarity of detail. Do not use abrasive methods, such as sanding, wire brushing, or power tools, except as indicated as part of the historic treatment program and as approved by Contracting Officer.

D. Repair and Refinish Existing Hardware (unless otherwise noted): Dismantle hardware; strip paint, repair, and refinish it to match finish samples; and lubricate moving parts just enough to function smoothly.

E. Repair Wood: Match existing materials and features, retaining as much original material as possible to perform repairs.

1. Unless otherwise indicated, repair wood by consolidating, patching, splicing, or otherwise reinforcing wood with new wood matching existing wood or with salvaged, sound, original wood.

2. Where indicated, repair wood by limited replacement matching existing material.

F. Replace Wood: Where indicated, duplicate and replace units with units made from salvaged, sound, original wood or with new wood matching existing wood. Use surviving prototypes to create patterns for duplicate replacements.

1. Do not use substitute materials unless otherwise indicated.

2. Compatible substitute materials may be used.

G. Identify removed items with numbering system corresponding to item locations, to ensure reinstallation in same location. Key items to Drawings showing location of each removed unit. Permanently label units in a location that will be concealed after reinstallation.

3.4 WOOD PATCH-TYPE REPAIR

A. General: Patch wood that exhibits depressions, holes, or similar voids, and that has limited amounts of rotted or decayed wood.

1. Verify that surfaces are sufficiently clean and free of paint residue prior to patching.

2. Treat wood with wood consolidant prior to application of patching compound. Coat wood surfaces by brushing, applying multiple coats until wood is saturated and refuses to absorb more. Allow treatment to harden before filling void with patching compound.

3. Remove rotted or decayed wood down to sound wood.

B. Apply borate preservative treatment to accessible surfaces either before applying wood consolidant or after removing rotted or decayed wood. Apply treatment liberally by brush to joints, edges, and ends; top, sides, and bottom. Allow treatment to dry.

C. Apply wood-patching compound to fill depressions, nicks, cracks, and other voids created by removed or missing wood.

1. Prime patch area with application of wood consolidant or manufacturer's recommended primer.

2. Mix only as much patching compound as can be applied according to manufacturer's written instructions.

3. Apply patching compound in layers as recommended in writing by manufacturer until the void is completely filled.

4. Sand patch surface smooth and flush with adjacent wood, without voids in patch material, and matching contour of wood member.

5. Clean spilled compound from adjacent materials immediately.

3.5 WOOD-REPLACEMENT REPAIR

A. General: Replace parts of or entire wood items at locations indicated on Drawings.

1. Remove surface-attached items from wood surface before performing wood-replacement repairs unless otherwise indicated.

2. Verify that surfaces are sufficiently clean and free of paint residue prior to repair.

3. Remove broken, rotted, and decayed wood down to sound wood.

4. Custom fabricate new wood to replace missing wood; either replace entire wood member or splice new wood part into existing member.

5. Secure new wood using finger joints, multiple dowels, or splines with adhesive and nailing to ensure maximum structural integrity at each splice. Use only concealed fasteners. Fill nail holes and patch surface to match surrounding sound wood.

B. Apply borate preservative treatment to accessible surfaces after replacements are made. Apply treatment liberally by brush to joints, edges, and ends; top, sides, and bottom.

C. Repair remaining depressions, holes, or similar voids with patch-type repairs.

D. Clean spilled materials from adjacent surfaces immediately.

E. Reinstall items removed for repair into original locations.

3.6 FIELD QUALITY CONTROL

A. Manufacturer's Field Service: Engage wood-repair-material manufacturers' factory-authorized service representatives for consultation and Project-site inspection, and provide on-site assistance when requested by Contracting Officer.

3.7 ADJUSTMENT

A. Adjust existing and replacement operating items, hardware, and accessories for a tight fit at contact points and for smooth operation and tight closure. Lubricate hardware and moving parts.

3.8 CLEANING AND PROTECTION

A. Protect wood surfaces from contact with contaminating substances resulting from construction operations. Monitor wood surfaces adjacent to and below exterior concrete and masonry during construction for presence of dirt, scum, alkaline deposits, stains, or other contaminants. If contaminating substances contact wood surfaces, remove contaminants immediately.

B. Clean exposed surfaces immediately after historic wood repair. Avoid damage to coatings and finishes. Remove excess sealants, patching materials, dirt, and other substances.

END OF SECTION

SECTION 06 10 00 – ROUGH CARPENTRY

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Framing with dimension lumber.
2. Framing with engineered wood products.
3. Shear wall panels.
4. Rooftop equipment bases and support curbs.
5. Wood blocking and nailers.
6. Wood furring.
7. Wood sleepers.
8. Plywood backing panels.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of process and factory-fabricated product.

1.3 INFORMATIONAL SUBMITTALS

A. Material Certificates: For dimension lumber specified to comply with minimum allowable unit stresses. Indicate species and grade selected for each use and design values approved by the ALSC Board of Review.

B. Evaluation Reports: For the following, from ICC-ES:

1. Wood-preservative-treated wood.
2. Fire-retardant-treated wood.
3. Engineered wood products.
4. Shear panels.
5. Power-driven fasteners.
6. Post-installed anchors.
7. Metal framing anchors.

PART 2 - PRODUCTS

2.1 WOOD PRODUCTS, GENERAL

A. Lumber: DOC PS 20 and applicable rules of grading agencies indicated. If no grading agency is indicated, comply with the applicable rules of any rules-writing agency certified by the ALSC Board of Review. Grade lumber by an agency certified by the ALSC Board of Review to inspect and grade lumber under the rules indicated.

1. Factory mark each piece of lumber with grade stamp of grading agency.
2. For exposed lumber indicated to receive a stained or natural finish, omit grade stamp and provide certificates of grade compliance issued by grading agency.
3. Dress lumber, S4S, unless otherwise indicated.

B. Maximum Moisture Content of Lumber: 19 percent unless otherwise indicated.

C. Engineered Wood Products: Acceptable to authorities having jurisdiction and for which current model code research or evaluation reports exist that show compliance with building code in effect for Project.

1. Allowable design stresses, as published by manufacturer, shall meet or exceed those indicated. Manufacturer's published values shall be determined from empirical data or by rational engineering analysis and demonstrated by comprehensive testing performed by a qualified independent testing agency.

2.2 WOOD-PRESERVATIVE-TREATED LUMBER

A. Preservative Treatment by Pressure Process: AWPA U1; Use Category UC2 for interior construction not in contact with ground, Use Category UC3b for exterior construction not in contact with ground, and Use Category UC4a for items in contact with ground.

1. Preservative Chemicals: Acceptable to authorities having jurisdiction and containing no arsenic or chromium. Do not use inorganic boron (SBX) for sill plates.

B. Kiln-dry lumber after treatment to a maximum moisture content of 19 percent. Do not use material that is warped or that does not comply with requirements for untreated material.

C. Mark lumber with treatment quality mark of an inspection agency approved by the ALSC Board of Review.

D. Application: Treat items indicated on Drawings, and the following:

1. Wood cants, nailers, curbs, equipment support bases, blocking, stripping, and similar members in connection with roofing, flashing, vapor barriers, and waterproofing.
2. Wood sills, sleepers, blocking, furring, stripping, and similar concealed members in contact with masonry or concrete.
3. Wood framing and furring attached directly to the interior of below-grade exterior masonry or concrete walls.

4. Wood framing members that are less than 18 inches above the ground in crawlspaces or unexcavated areas.
5. Wood floor plates that are installed over concrete slabs-on-grade.

2.3 FIRE-RETARDANT-TREATED MATERIALS

A. General: Where fire-retardant-treated materials are indicated, materials shall comply with requirements in this article, that are acceptable to authorities having jurisdiction, and with fire-test-response characteristics specified as determined by testing identical products per test method indicated by a qualified testing agency.

B. Fire-Retardant-Treated Lumber and Plywood by Pressure Process: Products with a flame-spread index of 25 or less when tested according to ASTM E 84, and with no evidence of significant progressive combustion when the test is extended an additional 20 minutes, and with the flame front not extending more than 10.5 feet beyond the centerline of the burners at any time during the test.

1. Exterior Type: Treated materials shall comply with requirements specified above for fire-retardant-treated lumber and plywood by pressure process after being subjected to accelerated weathering according to ASTM D 2898. Use for exterior locations and where indicated.
2. Interior Type A: Treated materials shall have a moisture content of 28 percent or less when tested according to ASTM D 3201 at 92 percent relative humidity. Use where exterior type is not indicated.

C. Kiln-dry lumber after treatment to maximum moisture content of 19 percent.

D. Identify fire-retardant-treated wood with appropriate classification marking of qualified testing agency.

E. Application: Treat items indicated on Drawings.

2.4 DIMENSION LUMBER FRAMING

A. Non-Load-Bearing Interior Partitions: Standard, Stud, or No. 3 grade.

1. Application: Interior partitions not indicated as load bearing.
2. Species:

    a. Southern pine or mixed southern pine; SPIB.
    b. Northern species; NLGA.
    c. Eastern softwoods; NeLMA.
    d. Western woods; WCLIB or WWPA.

B. Framing Other Than Non-Load-Bearing Partitions: No. 2 grade.

1. Application: Framing other than interior partitions not indicated as load bearing.
2. Species:

a. Douglas fir-larch; WCLIB or WWPA.

C. Framing Other Than Non-Load-Bearing Partitions: Any species and grade with a modulus of elasticity of at least 1,300,000 psi and an extreme fiber stress in bending of at least 850 psi for 2-inch nominal thickness and 12-inch nominal width for single-member use.

1. Application: Framing other than interior partitions not indicated as load-bearing.

D. Exposed Framing: Hand-select material for uniformity of appearance and freedom from characteristics, on exposed surfaces and edges, that would impair finish appearance, including decay, honeycomb, knot-holes, shake, splits, torn grain, and wane.

1. Species and Grade: As indicated above for load-bearing construction of same type.

2.5 ENGINEERED WOOD PRODUCTS

A. Laminated-Veneer Lumber: Structural composite lumber made from wood veneers with grain primarily parallel to member lengths, evaluated and monitored according to ASTM D 5456 and manufactured with an exterior-type adhesive complying with ASTM D 2559.
1. Extreme Fiber Stress in Bending, Edgewise: 2600 psi for 12-inch nominal- depth members.
2. Modulus of Elasticity, Edgewise: 2,000,000 psi.

B. Wood I-Joists: Prefabricated units, I-shaped in cross section, made with solid or structural composite lumber flanges and wood-based structural panel webs, let into and bonded to flanges. Comply with material requirements of and with structural capacities established and monitored according to ASTM D 5055.

1. Web Material: Either OSB or plywood, complying with DOC PS 1 or DOC PS 2, Exposure 1.
2. Structural Properties: Depths and design values not less than those indicated.
3. Comply with APA PRI-400. Factory mark I-joists with APA-EWS trademark indicating nominal joist depth, joist class, span ratings, mill identification, and compliance with APA-EWS standard.

C. Rim Boards: Product designed to be used as a load-bearing member and to brace wood I-joists at bearing ends, complying with research or evaluation report for I-joists.

1. Manufacturer: Provide products by same manufacturer as I-joists.
2. Material: All-veneer product glued-laminated wood or product made from any combination solid lumber, wood strands, and veneers.
3. Thickness: 1-1/4 inches.
4. Comply with APA PRR-401, rim board plus grade. Factory mark rim boards with APA-EWS trademark indicating thickness, grade, and compliance with APA-EWS standard.

D. Wood-Framed Shear Wall Panels: Prefabricated assembly consisting of wood perimeter framing, tie downs, and Exposure I, Structural I plywood or OSB sheathing.

E. Allowable design loads, as published by manufacturer, shall meet or exceed those indicated of basis-of-design products. Manufacturer's published values shall be determined from empirical data or by rational engineering analysis and demonstrated by comprehensive testing performed by a qualified independent testing agency.

2.6 MISCELLANEOUS LUMBER

A. General: Provide miscellaneous lumber indicated and lumber for support or attachment of other construction, including the following:

1. Blocking.
2. Nailers.
3. Rooftop equipment bases and support curbs.
4. Cants.
5. Furring.
6. Grounds.

B. Dimension Lumber Items: Construction or No. 2 grade lumber of any species.

C. Concealed Boards: 19 percent maximum moisture content and the following species and grades:

1. Western woods; Construction or No. 2 Common grade; WCLIB or WWPA.

2.7 PLYWOOD BACKING PANELS

A. Equipment Backing Panels: Plywood, DOC PS 1, Exterior, A-C, in thickness indicated or, if not indicated, not less than 1/2-inch nominal thickness.

2.8 FASTENERS

A. General: Fasteners shall be of size and type indicated and shall comply with requirements specified in this article for material and manufacture.

1. Where rough carpentry is exposed to weather, in ground contact, pressure-preservative treated, or in area of high relative humidity, provide fasteners with hot-dip zinc coating complying with ASTM A 153/A 153M.

B. Power-Driven Fasteners: Fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC70.

C. Post-Installed Anchors: Fastener systems with an evaluation report acceptable to authorities having jurisdiction, based on ICC-ES AC01 ICC-ES AC58 ICC-ES AC193 or ICC-ES AC308 as appropriate for the substrate.

2.9 METAL FRAMING ANCHORS

A. Allowable design loads, as published by manufacturer, shall meet or exceed those indicated. Manufacturer's published values shall be determined from empirical data or by rational engineering analysis and demonstrated by comprehensive testing performed by a qualified independent testing agency. Framing anchors shall be punched for fasteners adequate to withstand same loads as framing anchors.

B. Galvanized-Steel Sheet: Hot-dip, zinc-coated steel sheet complying with ASTM A 653/A 653M, G60 coating designation.

1. Use for interior locations unless otherwise indicated.

C. Hot-Dip, Heavy-Galvanized Steel Sheet: ASTM A 653/A 653M; structural steel (SS), high-strength low-alloy steel Type A (HSLAS Type A), or high-strength low-alloy steel Type B (HSLAS Type B); G185 coating designation; and not less than 0.036 inch thick.

1. Use for wood-preservative-treated lumber and where indicated.

2.10 MISCELLANEOUS MATERIALS

A. Sill-Sealer Gaskets: Closed-cell neoprene foam, 1/4 inch thick, selected from manufacturer's standard widths to suit width of sill members indicated.

B. Flexible Flashing: Composite, self-adhesive, flashing product consisting of a pliable, butyl rubber or rubberized-asphalt compound, bonded to a high-density polyethylene film, aluminum foil, or spunbonded polyolefin to produce an overall thickness of not less than 0.025 inch.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Framing Standard: Comply with AF&PA's WCD 1, "Details for Conventional Wood Frame Construction," unless otherwise indicated.

B. Framing with Engineered Wood Products: Install engineered wood products to comply with manufacturer's written instructions.

C. Set rough carpentry to required levels and lines, with members plumb, true to line, cut, and fitted. Fit rough carpentry accurately to other construction. Locate furring, nailers, blocking and similar supports to comply with requirements for attaching other construction.

D. Install shear wall panels to comply with manufacturer's written instructions.

E. Install metal framing anchors to comply with manufacturer's written instructions. Install fasteners through each fastener hole.

F. Do not splice structural members between supports unless otherwise indicated.

G. Comply with AWPA M4 for applying field treatment to cut surfaces of preservative-treated lumber.

H. Securely attach rough carpentry work to substrate by anchoring and fastening as indicated, complying with the following:

1. Table 2304.9.1, "Fastening Schedule," in ICC's International Building Code (IBC).
2. Table R602.3(1), "Fastener Schedule for Structural Members," and Table R602.3(2), "Alternate Attachments," in ICC's International Residential Code for One- and Two-Family Dwellings.
3. ICC-ES evaluation report for fastener.

3.2 PROTECTION

A. Protect wood that has been treated with inorganic boron (SBX) from weather. If, despite protection, inorganic boron-treated wood becomes wet, apply EPA-registered borate treatment. Apply borate solution by spraying to comply with EPA-registered label.

B. Protect rough carpentry from weather. If, despite protection, rough carpentry becomes wet enough that moisture content exceeds that specified, apply EPA-registered borate treatment. Apply borate solution by spraying to comply with EPA-registered label.

END OF SECTION 061000

SECTION 06 13 13 - HISTORIC LOG TREATMENT/LOG CONSTRUCTION

PART 1 GENERAL

1.1 SUMMARY

A. This Section includes the following:

1. Cleaning existing log construction.

2. Repairing logs.

3. Replacing logs in historically correct style.

4. Finishing of logs.

5. Accessories

1.2 RELATED REQUIREMENTS

A. Section 01 35 91 "Historic Treatment Procedures" for general historic treatment requirements.

B. Section 06 03 12 "Historic Wood Repair" for architectural woodwork.

C. Appendix 1: NPS Standard Operating Procedure (SOP) - Chinking and Daubing

1.3 REFERENCE STANDARDS

A. The Secretary of the Interior's Standards for Preservation, Restoration, Rehabilitation, Reconstruction.

B. Preservation Brief #26: The Preservation and Repair of Historic Log Buildings.

1.4 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at project site.

1. Review minutes of Preliminary Historic Treatment Conference that pertain to historic wood repair.

2. Review methods and procedures related to historic wood repair, including, but not limited to, the following:

a. Historic treatment specialist's personnel, equipment, and facilities needed to make progress and avoid delays.

b. Materials, material application, sequencing, tolerances, and required clearances.

c. Fire-protection plan.

d. Log historic treatment program.

e. Coordination with building occupants.

f. Review of the current log stock pile at INSERT LOCATION for log quality and selection.

g. Discuss Log Replacement Schedule.

1.5 SEQUENCING AND SCHEDULING

A. Perform historic log repair in the following sequence, which includes work specified in this and other Sections:

1. Before removing log components for on-site or off-site repair, tag each component with location-identification numbers. Indicate on tags and building plans and elevations the locations of each component, such as "Wall Log 4, West Elevation." Number all logs from the base of the building up.

2. Dismantle any hardware and tag with location-identification numbers.

3. Clean surfaces.

4. General Wood-Repair Sequence:

a. Remove finish to bare wood as necessary.

b. Repair wood by consolidation, partial replacement, or full in-kind replacement.

5. See Preservation Brief #26: The Preservation and Repair of Historic Log Buildings.

6. Repair, refinish, and replace hardware if required.

7. Reinstall components.

8. Repair chinking and daubing where necessary.

9. Apply finish.

1.6 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include recommendations for product application and use. Include test data substantiating that products comply with requirements.

B. Shop Drawings.

1. Include field-verified dimensions and the following

a. Log Replacement Schedule: Identification of each new unit and its corresponding location in the building on annotated plans and elevations.

b. Provisions for log daubing sealant, and flashing as required for location.

C. Samples:

1. Chinking materials.

2. Daubing - for composition and color.

3. Sealants.

4. Finishes.

5. Others where color selection is required.

1.7 INFORMATIONAL SUBMITTALS

A. Qualification Data: For historic treatment specialist.

B. Pre-construction test reports: For existing chinking materials and daubing composition.

C. Quality Control Program.

D. Log Historic Treatment Program: Submit before work begins.

1.8 QUALITY ASSURANCE

A. Historic Treatment Specialist Qualifications: A qualified historic log repair specialist, experienced in repairing, replacing, and refinishing log construction. Experience only in constructing new log structures is insufficient experience for historic log repair work. Contractor Qualifications: See Supplemental Responsibility Criteria.

B. Wood-Repair-Material Manufacturer Qualifications: A firm regularly engaged in producing wood consolidant and wood-patching compound that have been used for similar historic wood-treatment applications with successful results, and with factory-authorized service representatives who are available for consultation, Project-site inspection, and on-site assistance.

C. Log Historic Treatment Program: Prepare a written, detailed description of materials, methods, equipment, and sequence of operations to be used for historic treatment work, including protection of surrounding materials and Project site.

1. If materials and methods other than those indicated are proposed for any phase of historic treatment work, add a written description of such materials and methods, including evidence of successful use on comparable projects, and demonstrations to show their effectiveness for this Project.

D. Quality Control Program Language: Included as Informational Submittal above.

1. Mortar analysis of Chinking.

E. See NPS Standard Operating Procedure (SOP) - Chinking and Daubing for additional requirements.

1.9 MOCKUPS

A. Prepare mockups of historic treatment repair processes to demonstrate aesthetic effects and to set quality standards for materials and execution, and for fabrication and installation. Prepare mockups so they are as inconspicuous as practicable.

1. Locate mockups on existing surfaces where directed by Project Representative.

2. Log Splice Repair: Prepare (1) of each type of splice repair as a mockup to demonstrate this type of wood repair.

3. Log Unit Replacement: Replace full log.

4. Protective Treatment: Prepare (1) area, 36" high by 36" wide, for each type of protective treatment application.

5. Chinking & Daubing: Prepare (2) areas, 36" high by 36" wide, of chinking repair. Install daubing at one of these areas.

6. Chinking & Daubing: Prepare (2) areas, 36" high by 36" wide, of chinking replacement. Install daubing at one of these areas.

7. Finish: Prepare (1) area, 36" high by 36" wide, for each type of log finish.

8. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Project Representative specifically approves such deviations in writing.

9. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.10 PRE-CONSTRUCTION TESTING:

A. Preconstruction Testing Service: Engage a qualified testing agency to perform preconstruction testing as follows:

1. Provide test specimens as indicated and representative of proposed materials and existing construction.

2. Existing Daubing: Test according to ASTM C1324, modified as agreed by testing service and Project Representative for Project requirements, to determine proportional composition of original ingredients, sizes and colors of aggregates, and approximate strength. Use x-ray diffraction, infrared spectroscopy, and differential thermal analysis to supplement microscopical methods. Carefully remove existing daubing for testing from within joints at two locations designated by Project Representative.

3. Existing Wood Type Identification: Test existing historic wood log sample to identify wood specimen for new material matching.

1.11 DELIVERY, STORAGE, AND HANDLING

A. Pack, deliver, and store products in suitable packs, heavy-duty cartons, or wooden crates; surround with sufficient packing material to ensure that products will not be deformed, broken, or otherwise damaged.

B. Until installed, store products inside a well-ventilated area and protect from weather, moisture, soiling, abrasion, extreme temperatures, and humidity, and where environmental conditions comply with manufacturer's requirements.

C. Protect new and existing log material from rain and related moisture. Logs to be utilized as building logs shall not be in direct ground contact. Store on blocks or construct crib. Logs shall be supported at sufficient intervals not exceeding four (4) feet. Closer support intervals if deemed necessary due to evidence of warping, twisting or other deformation. The Contractor may order additional supports based on their judgement of probable future harm.

D. Maintain full air circulation around raw and finished logs to facilitate further drying and to prevent mildew. Embedded mildew growth shall be cause for log rejection.

1.12 FIELD CONDITIONS

A. Weather Limitations: Proceed with historic log repair only when existing and forecasted weather conditions are within the environmental limits set by each manufacturer's written instructions and specified requirements.

PART 2 PRODUCTS

2.1 HISTORIC LOG REPAIR, GENERAL

A. Quality Standard: Comply with applicable requirements in Section 12, "Historic Restoration Work," and related requirements in AWI/AWMAC/WI's "Project Woodwork Standards" for construction, finishes, grade rules, and other requirements unless otherwise indicated.

1. The historic and significance of the 4 Lazy F Ranch warrants a thoughtful/careful approach to the restoration effort. The Contractor, all subcontractors, the Project Representative, the Project Representative, and others whom may have involvement in the overall restoration effort, shall endeavor to minimize errors and avoid unnecessary destruction of historic fabric.

2. To facilitate this effort, the Project Representative, and Contractor shall closely coordinate in the decision making process as such relates to the removal, replacement, re-use, destruction, alteration or relocation of historic log fabric.

3. When unforeseen conditions are encountered which may require deviation from plans or specifications, coordinate with the Project Representative prior to proceeding.

4. Date Identification: Emboss on a concealed surface of each replaced item, in easily read characters, MADE DATE.

2.2 CLEANING

A. Exterior Cleaning:

1. CL-1: Overall Exterior Cleaning Treatment

2. CL-2: Bio-growth Exterior Cleaning Treatment:

a. D/2 Biological Solution.

B. Interior Cleaning:

1. ICL-1: Overall Interior Cleaning Treatment, consists of two steps

a. Step 1:Bac-A-Zap Odor Eliminator.

1) Basis of Design:

a) Bac-A-Zap Odor Eliminator, https://nisuscorp.com/product/bac-a-zap?srsltid=AfmBOoqSSCXrD2hPBt6LF-QV_8WpPuCtx0nUFuOzN_CUdCVduP0qGC-H,. Product characteristics noted in paragraph 2.2.B.1.a.

b) Substitutions: See Section 01 67 00 – Product Requirements.

2) Salient Characteristics for Bac-AZap:

a) Biodegradable solution that eliminates odor and sanitizes. The solution should contain microbes that attack and consume organic matter such as proteins, starches, oils, cellulose, and urine. Applicable in foam, liquid, or airborne states for maximum flexibility.

b. Step 2: CPR Log Cleaner.

1) Basis of Design:

a) ICL-2: CPR log cleaner, https://www.sashco.com/products/cpr-log-cleaner. Product characteristics noted in paragraph 2.2.B.1.b.

b) Substitutions: See Section 01 67 00 – Product Requirements.

2) Salient Characteristics for CPR Log Cleaner and Brightener:

a) Oxygen bleach and water soluble. Does not contain chlorine bleach or oxalic acid.

2. ICL-3: Interior Wood Stain Cleaning Treatment:

a. Bleaching solution, test least toxic solution and advance in order.

1) Sodium per carbonate based cleaner

2) Household chlorine bleach and/or bleach and Trisodium Phospahte solution (no stronger than a 10% solution)

3) Oxalic acid or 35% hydrogen peroxide (medical grade)

2.3 LOG PROTECTIVE TREATMENT MATERIALS

A. EPDM Sill Gasket:

1. Basis of Design:

a. Conservation Technologies, www.conservationtechnology.com.. Product characteristics noted in paragraph 2.3.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2.4 CHINKING & DAUBING MATERIALS

A. See NPS Standard Operating Procedure (SOP) - Chinking and Daubing.

B. Match existing Chinking and Daubing based on mortar analysis:

1. Match existing chinking materials as provided by chinking analysis.

2. Match mortar analysis results for new daubing in composition and color.

C. Wood strip ledger below daubing joints to be retained, secured with nails, or replaced in kind.

2.5 FINISH MATERIALS

A. See Section 09 91 13 - Exterior Painting

2.6 FULL LOG-REPLACEMENT MATERIALS

A. Logs.

1. Species: Match existing. All logs utilized for this Project shall be new unless otherwise noted in drawings.

2. Quality:

a. All new replacement logs shall be free from the following General Defects:

1) Insect damage.

2) Fire damage.

3) Bleeding pitch pockets.

4) Decay.

5) Fungi growth.

6) Varmints damage.

7) Variations in color/staining.

8) Other deterioration or damaged wood fibers, including handling damage.

b. Logs shall meet the following criteria:

1) Straight with minimum taper. (Two inches in 50 feet maximum.)

2) Minimum of six (6) growth rings per inch.

3) Reverse bends or crooks will not be allowed.

4) Spiral grain twists will not be allowed.

5) Minimal seasoning checks, although not desirable, shall NOT be considered a defect if located on the non-weather side in the finished log, expect that checks shall not exceed 1/8 inch in width at the time of log acceptance (prior to assembly.)

6) Excessive seasoning checks may be cause for log rejection.

7) Logs shall be uniform in color, shape and general appearance.

8) Knots shall be held to a minimum and all knots shall be trimmed flush to the body of the log.

9) Hook butts or ground swell will not be allowed.

10) Logs showing evidence of mechanical abuse, such as produced from rough handling shall be rejected. Handle all logs with soft slings, or chains. Lifting devices which penetrate the skin of the log will not be permitted. Splits or bruises will not be allowed.

11) Logs shall be 100 percent free of all rough outer bark and 98 percent free of inner bark. No strips of inner bark (cambium layer) remaining shall be over 1/4 inch wide or over 1 inch long.

12) The Project Representative shall be the final authority as to the suitability of all logs. The Contractor shall make arrangements to have all logs inspected by the Project Representative prior to construction.

3. Moisture Content

a. Replacement log moisture content is considered to be most critical element in the selection and acceptance of new logs.

b. Replacement log moisture content, as measured at a point approximately 1/5 the log diameter below the log surface, shall be less than 20 percent.

c. The Contract shall verify that moisture content meets the above specifications, by the use of industry standard instruments such as Delmhorst Instrument Company or similar moisture meters. Readings shall be adjusted for species and temperature.

d. Final moisture measurements, which shall constitute acceptance or rejection based on moisture content, shall be performed at the construction site, no earlier than 10 days after delivery.

4. Log Source

a. Contractor will supply all logs for the project. The logs will meet the requirements outlined in this section. The Contractor will be responsible for transport, storage, and installation of the logs.

2.7 LOG-REPAIR MATERIALS

A. Wood Consolidant: As noted in Section 04 03 42 - Historic Wood Repair

B. Dutchman and Lap joint:

1. Provide structural epoxy and fiberglass rebar to secure Dutchman and lap repairs per drawings.

2. Use in-kind wood to match species and characteristics to match existing. See Section 06 13 13 - Historic Log Construction Repair part 2.11 for adhesives and reinforcement.

2.8 PRESERVATIVE TREATMENT MATERIALS

A. Borate Preservative Treatment: Inorganic, borate-based solution, with disodium octaborate tetrahydrate as the primary ingredient; manufactured for preserving weathered and decayed wood from further damage caused by fungi and wood-boring insects; complying with AWPA P5; containing no boric acid.

1. Basis of Design:

a. Nisus Coproration - Bora-Care Termiticide Insecticide and Fungicide Concentrate. Product characteristics noted in paragraph 2.8.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2.9 CLEANING MATERIALS

A. General: Do not use wet cleaning methods.

B. Compressed Air: 30 psi

C. Vacuumed Air with brush attachment.

2.10 MISCELLANEOUS MATERIALS

A. Adhesives: Wood adhesives with minimum 15- to 45-minute cure at 70 deg F (21 deg C), in gunnable and liquid formulations as recommended in writing by adhesive manufacturer for each type of repair and exposure condition.

1. Wood Glue:

a. Basis of Design:

1) Titebond III Ultimate Wood Glue http://www.titebond.com/titebond_wood_glues/Ultimate_Wood_Glue.asp x. Product characteristics noted in paragraph 2.10.1.

2) Substitutions: See Section 01 67 00 – Product Requirements.

2. Epoxy Glue:

a. Basis of Design:

1) ConServ Epoxy system; www.conservepoxy.com. Product characteristics noted in Section 06 13 12 Historic Wood Repair 2.6 C.2

2) Substitutions: See Section 01 67 00 – Product Requirements.

B. Fasteners: Use fastener metals that are noncorrosive and compatible with each material joined.

1. Match existing fasteners in material and type of fastener unless otherwise indicated.

2. Use concealed fasteners for interconnecting wood components.

3. Use concealed fasteners for attaching items to other work unless exposed fasteners are unavoidable or the existing fastening method.

4. Provide sufficient galvanized fasteners and hardware for proper and stable installation.

5. Fiberglass Rebar: Use 3/4 fiberglass rebar set in epoxy at log splice repairs

6. Lag bolts shall be 1/2 inch minimum diameter, complete with washers.

7. All log fasteners shall be of sufficient length, style and diameter to meet code, the design and site requirements as encountered and/or set forth by the Project Representative.

8. Provide stainless steel fasteners and items when indicated on the plans or other sections of these specifications. Fasteners which are fully or partially exposed on the weather side of the building shall be stainless steel.

2.11 WOOD FINISHES

A. See Section 09 91 13 - Exterior Painting and Section 09 91 23 - Interior Painting.

2.12 OTHER MATERIALS

A. Provide other materials, not specifically described but required for a complete and proper installation, as selected by the Contractor and approved by the Project Representative.

PART 3 EXECUTION

3.1 PREPARATION

A. Protect adjacent materials from damage by historic log repair.

B. Confirm all log conditions require action as identified on the plans prior to full execution of repairs and/or replacement.

C. Clean wood of mildew, algae, moss, plant material, loose paint, grease, dirt, and other debris by dry methods.

D. Condition logs to site environment prior to repair and installation.

3.2 TOOLS AND EQUIPMENT

A. Hand tools used during construction produced log tooling marks which are distinctive and give logs character and a connection to the past.

B. Final observable tool marks on all replacement log fabric shall be authentic reproductions of the original tool marks found on the historic fabric in the buildings. The following shall apply:

1. Bark Removal:

a. Bark shall be removed in a manner as to leave no inappropriate mechanical marks, such as those produced by the lathe peeling. Bark shall be removed by peeling spud, drawknife or related appropriate hand tool. The Project Representative will consider alternate methods. Contractor is to submit the request for alternate methods in writing to the Project Representative for approval. High pressure water methods will not be considered.

b. Outer bark and inner bark is responsibility of the Contractor.

2. Roughing Out: The use of modern power tools such as chain saws, power mills, Alaskan chainsaw mills, power planers, power drills and any related hand power tool available for roughing out and preparing the log for final shaping and fitting is permitted.

3. Final Log Tooling: Final fitting, notching, shaping, log surface preparation and related operations shall be performed entirely by the use of non-powered hand tools. The use of such tools as axe, broad axe, framing hatchet, chisel, framing chisel, slick, peeling spud,

froe, adze, beetle, maul, draw knife, in-shave, large and small spoke shave, along with related period tools is strongly encouraged. The style of hand tool, whether they be period tools or modern hardware store tools, shall be the Contractors choice. The use of power hand tools however, excepting drills and boring devices, shall not be allowed during the final log tooling stage.

4. Under no circumstances shall the use of hand held power planers be authorized for use upon the weather or interior surface of any log. Such power tool may be used upon log surfaces in contact with masonry or surfaces which are not in general view.

5. All logs shall run continuous. Splices in logs are not allowed other than as shown on the plans.

6. No peeves, hooks, clamps, or any tool or method which will leave a permanent puncture, mark or scar shall be used during log handling. Use soft slings or other approved methods for handling and turning logs.

7. Fitting clearances between logs shall be as small as possible and in no case exceed 3/32 inch.

8. The Project Representative shall reject logs exhibiting poor or improper fit.

3.3 HISTORIC WOOD REPAIR, GENERAL

A. Historic Treatment Appearance Standard: Completed work is to have a uniform appearance as viewed by Project Representative from 20 feet away for exterior work.

B. General: In treating historic items, disturb them as minimally as possible and as follows:

1. Stabilize and repair wood to reestablish structural integrity and weather resistance while maintaining the existing form of each item.

2. Remove coatings and apply borate preservative treatment before repair. Remove coatings according to Section 09 03 91 "Historic Treatment of Plain Painting" unless otherwise indicated.

3. Repair logs in place where possible.

4. Install temporary protective measures to protect wood-treatment work that is indicated to be completed later.

C. Mechanical Abrasion: Where mechanical abrasion is needed for the work, use only the gentlest mechanical methods.

D. Repair Wood: Match existing materials and features, retaining as much original material as possible to perform repairs.

1. Unless otherwise indicated, repair wood by consolidating, splicing, or otherwise reinforcing wood with new wood matching existing wood or with salvaged, sound, original wood.

2. Where indicated, repair wood by limited replacement matching existing material.

E. Replace Wood: Where indicated, duplicate and replace units with units made from salvaged, sound, original wood or with new wood matching existing wood. Use surviving prototypes to create patterns for duplicate replacements.

1. Do not use substitute materials unless otherwise indicated.

F. Identify removed items with numbering system corresponding to item locations, to ensure reinstallation in same location. Key items to Drawings showing location of each removed unit. Permanently label units in a location that will be concealed after reinstallation.

3.4 LOG CLEANING

A. General:

1. Test cleaning method in inconspicuous area

B. Exterior Cleaning:

1. General Exterior Cleaning Treatment (CL-1): Vacuum with brush attachment and or provide 30psi compressed air to all exterior surfaces to remove loose debris from cracks and crevasses in logs.

2. Bio-growth Exterior Cleaning Treatment (CL-2): Use D/2 detergent using manufacture instructions only where biogrowth is present. Allow to completely dry then use CL-1 cleaning treatment.

C. Interior Cleaning:

1. General Interior Cleaning Treatment (ICL-1): -Apply enzyme cleaner to all surfaces, such as Bac-A-Zap to all surfaces. -Scrub all surfaces with CPR log cleaner and brightener, or equivalent product. Use a nylon bristle brush and follow manufacturer instructions.

2. Interior Wood Stain Cleaning Treatment (ICL-2): -At stained wood areas, apply solution to bleach the wood to matched cleaned log (ICL-2) surfaces.

a. Start with the least aggressive solution as noted in section 2.2.B.2

3.5 LOG REPLACEMENT AND REPAIR

A. Log expert is required for this work.

3.6 PROTECTIVE TREATMENT

A. Follow Tim-Bor Professional Insecticide & Fungicide manufacturer's instructions.

3.7 CHINKING & DAUBING

A. See Appendix 1.2 - SOP - Chinking and Daubing.

3.8 LOG FINISHING

A. See Section 09 91 13 - Exterior Painting and Section 09 91 23 - Interior Painting.

3.9 WOOD-REPLACEMENT REPAIR

A. Prior to full log production, coordinate with the Project Representative for required log moisture content inspection/evaluation. Comply with article "Log Replacement Materials."

B. Replace or repair logs as indicated in drawings

C. Protect and safeguard patina of existing logs scheduled to be retained. Coordinate this effort with the Project Representative.

D. Whenever possible, use existing logs as pattern.

E. Finish log-Work, including such items as final log surface tooling, final notch fitting and final surface to surface interfaces shall be performed entirely with hand tools to the extent feasible, practicable and reasonable, complete Work utilizing historic methods and tools. See article "Tools and Equipment."

F. The applied standard for fitting errors shall be 0 to 3/32 inch. All notches and log surface interfaces will be held to this fitting error standard.

G. Except as noted on the plans or these specifications, duplicate all fastening methods found in the original construction. Use galvanized fasteners unless stainless steel is called for.

H. Unless directed to the contrary by the Project Representative, duplicate all aspects and character defining features of any log to be replaced with new material. Such details as nail holes, screw holes boreholes or existing general damage shall NOT be duplicated.

I. Consider the intermixing of the historic/existing log fabric with new materials. Allow for the appropriate contraction and expansion and the additive effect of log shrinkage. At doors and windows, allow 3/8 inch for each foot of opening height for shrinkage. See appropriate plan details.

J. Reinstall items removed for repair into original locations.

K. Install replacement log.

L. Apply borate preservative treatment to accessible exterior surfaces after replacements are made. Apply treatment liberally by brush to joints, edges, and ends; top, sides, and bottom

3.10 ADJUSTMENT

A. Adjust existing and replacement operating items, hardware, and accessories for a tight fit at contact points and for smooth operation and tight closure. Lubricate hardware and moving parts.

3.11 CLEANING AND PROTECTION

A. Protect wood surfaces from contact with contaminating substances resulting from construction operations. Monitor wood surfaces adjacent to and below exterior concrete and masonry during construction for presence of dirt, scum, alkaline deposits, stains, or other contaminants. If contaminating substances contact wood surfaces, remove contaminants immediately.

B. Clean exposed surfaces immediately after historic log repair. Avoid damage to coatings and finishes. Remove excess sealants, patching materials, dirt, and other substances.

END OF SECTION

SECTION 06 16 00 – SHEATHING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Wall sheathing.
2. Roof sheathing.
3. Subflooring.
4. Underlayment.
5. Sheathing joint and penetration treatment.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of process and factory-fabricated product.

1.3 INFORMATIONAL SUBMITTALS

A. Evaluation Reports: For the following, from ICC-ES:

1. Wood-preservative-treated plywood.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Fire-Resistance Ratings: As tested according to ASTM E 119; testing by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

1. Fire-Resistance Ratings: Indicated by design designations from UL's "Fire Resistance Directory" or from the listings of another qualified testing agency.

2.2 WOOD PANEL PRODUCTS

A. Emissions: Products shall meet the testing and product requirements of the California Department of Public Health's "Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources Using Environmental Chambers."

2.3 PRESERVATIVE-TREATED PLYWOOD

A. Preservative Treatment by Pressure Process: AWPA U1; Use Category UC2 for interior construction not in contact with ground, Use Category UC3b for exterior construction not in contact with ground, and Use Category UC4a for items in contact with ground.

B. Mark plywood with appropriate classification marking of an inspection agency acceptable to authorities having jurisdiction.

C. Application: Treat items indicated on Drawings.

2.4 WALL SHEATHING

A. Plywood Sheathing: Exposure 1, Structural I sheathing.

1. Thickness: As indicated on plan

B. Oriented-Strand-Board Sheathing: DOC PS 2, Exposure 1, Structural I sheathing.

1. Thickness: As indicated on plan

2.5 ROOF SHEATHING

A. Plywood Sheathing: Exposure 1, Structural I sheathing.

1. Thickness: As indicated on plan

B. Oriented-Strand-Board Sheathing: DOC PS 2, Exposure 1, Structural I sheathing.

1. Thickness: As indicated on plan

2.6 SUBFLOORING AND UNDERLAYMENT

A. Plywood Combination Subfloor-Underlayment: DOC PS 1, Exposure 1, Structural I, Underlayment single-floor panels.

B. Oriented-Strand-Board Combination Subfloor-Underlayment: DOC PS 2, Exposure 1 single-floor panels.

C. Plywood Subflooring: Exposure 1, Structural I single-floor panels or sheathing.

D. Oriented-Strand-Board Subflooring: DOC PS 2, Exposure 1, Structural I sheathing.

2.7 FASTENERS

A. General: Provide fasteners of size and type indicated that comply with requirements specified in this article for material and manufacture.

1. For roof and wall sheathing, provide fasteners with hot-dip zinc coating complying with ASTM A 153/A 153M.
2. For roof and wall sheathing, provide fasteners with organic-polymer or other corrosion-protective coating having a salt-spray resistance of more than 800 hours according to ASTM B 117.

PART 3 - EXECUTION

3.1 INSTALLATION, GENERAL

A. Do not use materials with defects that impair quality of sheathing or pieces that are too small to use with minimum number of joints or optimum joint arrangement. Arrange joints so that pieces do not span between fewer than three support members.

B. Cut panels at penetrations, edges, and other obstructions of work; fit tightly against abutting construction unless otherwise indicated.

C. Securely attach to substrate by fastening as indicated, complying with the following:

1. Table 2304.9.1, "Fastening Schedule," in the ICC's International Building Code.
2. Table R602.3(1), "Fastener Schedule for Structural Members," and Table R602.3(2), "Alternate Attachments," in the ICC's International Residential Code for One- and Two-Family Dwellings.
3. ICC-ES evaluation report for fastener.

D. Coordinate wall and roof sheathing installation with flashing and joint-sealant installation so these materials are installed in sequence and manner that prevent exterior moisture from passing through completed assembly.

E. Do not bridge building expansion joints; cut and space edges of panels to match spacing of structural support elements.

3.2 WOOD STRUCTURAL PANEL INSTALLATION

A. General: Comply with applicable recommendations in APA Form No. E30, "Engineered Wood Construction Guide," for types of structural-use panels and applications indicated.

B. Fastening Methods: Fasten panels as indicated below:

1. Combination Subfloor-Underlayment:

a. Glue and nail to wood framing.
b. Screw to cold-formed metal framing.
c. Space panels 1/8 inch apart at edges and ends.

2. Subflooring:

a. Glue and nail to wood framing.
b. Screw to cold-formed metal framing.
c. Space panels 1/8 inch apart at edges and ends.

3. Wall and Roof Sheathing:

a. Nail or staple to wood framing.
b. Screw to cold-formed metal framing.
c. Space panels 1/8 inch apart at edges and ends.

4. Underlayment:

a. Nail or staple to subflooring.
b. Space panels 1/32 inch apart at edges and ends.
c. Fill and sand edge joints of underlayment receiving resilient flooring immediately before installing flooring.

END OF SECTION 061600

SECTION 06 18 00 - GLUED-LAMINATED CONSTRUCTION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes framing using structural glued-laminated timber.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.3 INFORMATIONAL SUBMITTALS

A. Certificates of Conformance: Issued by a qualified testing and inspecting agency indicating that structural glued-laminated timber complies with requirements in AITC A190.1.

1.4 QUALITY ASSURANCE

A. Manufacturer Qualifications: An AITC- or APA-EWS-licensed firm. DELIVERY, STORAGE, AND HANDLING

B. General: Comply with provisions in AITC 111.

C. Individually wrap members using plastic-coated paper covering with water-resistant seams.

PART 2 - PRODUCTS

2.1 STRUCTURAL GLUED-LAMINATED TIMBER

A. General: Provide structural glued-laminated timber that complies with AITC A190.1 and AITC 117 or research/evaluation reports acceptable to authorities having jurisdiction.

1. Factory mark each piece of structural glued-laminated timber with AITC Quality Mark or APA-EWS trademark. Place mark on surfaces that are not exposed in the completed Work.
2. Provide structural glued-laminated timber made with wet-use adhesive complying with AITC A190.1.

B. Species and Grades for Structural Glued-Laminated Timber: Douglas fir-larch that complies with structural properties indicated.

C. Species and Grades for Beams:

1. Species and Beam Stress Classification: Douglas fir-larch, 24F-1.8E.
2. Lay-up: Balanced.

D. Species and Grades for Columns:

1. Species and Beam Stress Classification: Douglas fir-larch, 1.

E. Appearance Grade: Framing, complying with AITC 110.

2.2 TIMBER CONNECTORS

A. Materials: Unless otherwise indicated, fabricate from the following materials:

1. Structural-steel shapes, plates, and flat bars complying with ASTM A 36/A 36M.
2. Round steel bars complying with ASTM A 575, Grade M 1020.
3. Hot-rolled steel sheet complying with ASTM A 1011/A 1011M, Structural Steel, Type SS, Grade 33.

B. Finish steel assemblies and fasteners with rust-inhibitive primer, 2-mil dry film thickness.

C. Hot-dip galvanize steel assemblies and fasteners after fabrication to comply with ASTM A 123/A 123M or ASTM A 153/A 153M.

2.3 MISCELLANEOUS MATERIALS

A. End Sealer: Manufacturer's standard, transparent, colorless wood sealer that is effective in retarding the transmission of moisture at cross-grain cuts and is compatible with indicated finish.

B. Penetrating Sealer: Manufacturer's standard, transparent, penetrating wood sealer that is compatible with indicated finish.

2.4 FABRICATION

A. Shop fabricate for connections to greatest extent possible, including cutting to length and drilling bolt holes.

B. Camber: Fabricate horizontal and inclined members of less than 1:1 slope with either circular or parabolic camber equal to 1/500 of span.

C. End-Cut Sealing: Immediately after end cutting each member to final length, apply a saturation coat of end sealer to ends and other cross-cut surfaces, keeping surfaces flood coated for not less than 10 minutes.

D. Seal Coat: After fabricating, sanding, and end-coat sealing, apply a heavy saturation coat of penetrating sealer on surfaces of each unit.

PART 3 - EXECUTION

3.1 INSTALLATION

A. General: Erect structural glued-laminated timber true and plumb and with uniform, close-fitting joints. Provide temporary bracing to maintain lines and levels until permanent supporting members are in place.

1. Handle and temporarily support glued-laminated timber to prevent surface damage, compression, and other effects that might interfere with indicated finish.

B. Cutting: Avoid extra cutting after fabrication. Where field fitting is unavoidable, comply with requirements for shop fabrication.

C. Fit structural glued-laminated timber by cutting and restoring exposed surfaces to match specified surfacing.

1. Predrill for fasteners using timber connectors as templates.
2. Finish exposed surfaces to remove planing or surfacing marks.
3. Coat cross cuts with end sealer.

3.2 ADJUSTING

A. Repair damaged surfaces after completing erection. Replace damaged structural glued-laminated timber if repairs are not approved by Architect.

3.3 PROTECTION

A. Do not remove wrappings on individually wrapped members until they no longer serve a useful purpose, including protection from weather, sunlight, soiling, and damage from work of other trades.

1. Slit underside of wrapping to prevent accumulation of moisture inside the wrapping.

END OF SECTION 061800

SECTION 06 20 00 - FINISH CARPENTRY

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Finish carpentry items.

B. Crawl space access doors.

C. Hardware and attachment accessories.

1.2 RELATED REQUIREMENTS

A. Section 09 91 13 - Exterior Painting: Painting of finish carpentry items.

B. Section 09 91 23 - Interior Painting: Painting of finish carpentry items.

1.3 REFERENCE STANDARDS

A. AWI (QCP) - Quality Certification Program; Current Edition.

B. AWI/AWMAC/WI (AWS) - Architectural Woodwork Standards, 2nd Edition; 2014, with Errata (2016).

C. BHMA A156.9 - Cabinet Hardware; 2020.

D. PS 1 - Structural Plywood; 2023.

1.4 SUBMITTALS

A. Product Data:

1. Provide data on fire retardant treatment materials and application instructions.

a. Include copies of warranties from chemical treatment manufacturers for each type of treatment.

B. Shop Drawings: Indicate materials, component profiles, fastening methods, jointing details, and accessories.

1. Scale of Drawings: 1-1/2 inch to 1 foot, minimum.

2. Provide the information required by AWI/AWMAC/WI (AWS) Architectural Woodwork Standards.

3. Include certification program label.

C. Samples: Submit two samples of finished wood items 12 inch long illustrating wood grain and specified finish for each type of wood and each type of finish specified.

D. Certificate: Submit labels and certificates required by quality assurance and quality control programs.

1.5 QUALITY ASSURANCE

A. Fabricator Qualifications: Company specializing in fabricating the products specified in this section with minimum five years of documented experience.

1. Accredited participant in the specified certification program prior to the commencement of fabrication and throughout the duration of the project.

B. Quality Certification: Comply with AWI (QCP) woodwork association quality certification service/program in accordance with requirements for work specified in this section.

1. Provide labels or certificates indicating that the work complies with AWI/AWMAC/WI (AWS) requirements for grade or grades specified.

2. Provide designated labels on shop drawings as required by certification program.

3. Provide designated labels on installed products as required by certification program.

4. Submit certifications upon completion of installation that verifies this work is in compliance with specified requirements.

1.6 REGULATORY REQUIREMENTS

A. Conform to applicable code for fire retardant requirements.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Protect from moisture damage.

PART 2 PRODUCTS

2.1 FINISH CARPENTRY ITEMS

A. Quality Grade: Unless otherwise indicated provide products of quality specified by AWI/AWMAC/WI (AWS) Architectural Woodwork Standards for Premium Grade.

B. Exterior Woodwork Items:

1. Window Sills, Casings, and Moldings: Softwood Lumber 2; prepare for paint finish.

2. Soffits and Fascias: Softwood Lumber 1, Prepare for clear finish unless noted otherwise.

3. Wood Railings: Softwood Lumber 1, prepare for paint finish.

C. Interior Woodwork Items:

1. Wood Base (WB-1): Clear finish, profile as indicated in drawings, Softwood Lumber 2

2. Wood Trim: Clear Finish, Softwood Lumber 2

3. Wood Window Sill: Profile as indicated in drawings, Clear Finish, Softwood Lumber 2

4. Wood Apron: Profile as indicated in drawings, Clear Finish, Softwood Lumber 2

5. Wood Casing: Profile as indicated in drawings, Clear Finish, Softwood Lumber 2

6. Door Frames: Profile as indicated in drawings, Clear finish, Softwood Lumber 2

7. Crawl Space Access Doors: Profile as indicated in drawings, Clear finish, Softwood Plywood.

2.2 LUMBER MATERIALS

A. Softwood Lumber 1: Cedar species, plain sawn, maximum moisture content of 6 percent; with vertical grain, #2 or better grade.

B. Softwood Lumber 2: Doug Fir or Pine species, match species at existing interior trims in the individual structure, plain sawn, maximum moisture content of 6 percent, #3 or better with clear mixed grain (CMG) grade.

2.3 SHEET MATERIALS

A. Softwood Plywood, Not Exposed to View: Any face species, medium density fiberboard core; PS 1 Grade A-B, glue type as recommended for application.

B. Softwood Plywood, Exposed to View: Face species as indicated, plain sawn, medium density fiberboard core; PS 1 Grade A-B, glue type as recommended for application.

C. Medium Density Fiberboard (MDF): ANSI A208.2; type as specified in AWI/AWMAC/WI (AWS) Architectural Woodwork Quality Standards Illustrated; composed of wood fibers pressure bonded with moisture resistant adhesive to suit application; sanded faces; thickness as required.

2.4 FASTENINGS

A. Adhesive for Purposes Other Than Laminate Installation: Suitable for the purpose; not containing formaldehyde or other volatile organic compounds.

1. Use wood glue that has a VOC content of 30 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

B. Fasteners for Exterior Applications: Stainless steel; length required to penetrate wood substrate 1-1/2 inch minimum.

2.5 ACCESSORIES

A. Wood Filler: Solvent base, tinted to match surface finish color.

2.6 HARDWARE

A. Hardware: Comply with BHMA A156.9.

B. Crawl Space Access Door:

1. Recessed Pull for access.

2. Two Hinges.

3. Lock.

2.7 WOOD TREATMENT

A. Factory-Treated Lumber: Comply with requirements of AWPA U1 - Use Category System for pressure impregnated wood treatments determined by use categories, expected service conditions, and specific applications.

B. Wood Preservative by Pressure Treatment (PT Type): Provide AWPA U1 treatment using waterborne preservative with 0.25 percent retainage.

1. Basis of Design:

a. Acetylated Wood FSC Certified; Accoya Wood. Product characteristics noted in paragraph 2.7.

b. Substitutions: See Section 01 67 00 – Product Requirements.

C. Water Repellent Preservative Treatment by Dipping Method: WDMA I.S. 4, with 0.25 percent retainage.

2.8 FABRICATION

A. Shop assemble work for delivery to site, permitting passage through building openings.

B. When necessary to cut and fit on site, provide materials with ample allowance for cutting. Provide trim for scribing and site cutting.

C. Back out or kerf backs of the following members except those with ends exposed in finished work:

1. Interior standing and running trim, except shoe and crown molds.

2.9 SHOP FINISHING

A. Sand work smooth and set exposed nails and screws.

B. Apply wood filler in exposed nail and screw indentations.

C. On items to receive transparent finishes, use wood filler that matches surrounding surfaces and is of type recommended for the applicable finish.

D. Finish work in accordance with AWI/AWMAC/WI (AWS) Architectural Woodwork Standards, Section 5 - Finishing for Grade specified and as follows:

1. Transparent:

a. System - 6, Oil, Synthetic Penetrating (transparent only).

2. Opaque:

a. System - 6, Oil, Synthetic Penetrating (transparent only).

b. Color: As selected by Contracting Officer.

E. Back prime woodwork items to be field finished, prior to installation.

F. See Section 09 91 13 - Exterior and Painting and Section 09 91 23 - Interior Painting

PART 3 EXECUTION

3.1 PREPARATION

A. Clean substrates of projections and substances detrimental to application.

B. Prime lumber to be painted, including both faces and edges. Cut to required lengths and prime ends. Comply with requirements in Section 09 91 23 - Interior Painting.

3.2 EXAMINATION

A. Verify adequacy of backing and support framing.

B. Examine finish carpentry materials before installation. Reject materials that are wet, moisture damaged, and mold damaged.

C. Verify mechanical, electrical, and building items affecting work of this section are placed and ready to receive this work.

3.3 INSTALLATION

A. Install work in accordance with AWI/AWMAC/WI (AWS) Architectural Woodwork Standards requirements for grade indicated.

B. Do not use materials that are unsound, warped, improperly treated or finished, inadequately seasoned, or too small to fabricate with proper jointing arrangements.

C. Set and secure materials and components in place, true, plumb and level and aligned with adjacent materials. Use concealed shims where necessary for alignment.

1. Scribe and cut exterior finish carpentry to fit adjoining work. Refinish and seal cuts as recommended by manufacturer.

D. Carefully scribe work abutting other components, with maximum gaps of 1/32 inch. Do not use additional overlay trim to conceal larger gaps.

E. Calk space between wood base and wall with sealant specified in Section 07 90 05 - Joint Sealers.

3.4 TOLERANCES

A. Maximum Variation from True Position: 1/16 inch.

B. Maximum Offset from True Alignment with Abutting Materials: 1/32 inch.

3.5 ADJUSTING

A. Replace interior finish carpentry that is damaged or does not comply with requirements. Interior finish carpentry may be repaired or refinished if work complies with requirements and shows no evidence of repair or refinishing. Adjust joinery for uniform appearance.

3.6 PROTECTION

A. Protect installed products from damage from weather and other causes during remainder of the construction period.

B. Remove and replace finish carpentry materials that are wet, moisture damaged, and mold damaged.

1. Indications that materials are wet or moisture damaged include, but are not limited to, discoloration, sagging, irregular shape, fuzzy or splotchy surface contamination and discoloration.

2. Moldings, Bases, Casings, and Miscellaneous Trim: Clear white pine; prepare for clear finish.

END OF SECTION

SECTION 06 41 00 - ARCHITECTURAL WOOD CASEWORK

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Specially fabricated cabinet units, as noted but not limited to:

1. Wood Veneer Faced Cabinet

2. All Plywood Cabinet

B. Countertops.

C. Hardware.

1.2 REFERENCE STANDARDS

A. ANSI A208.1 - American National Standard for Particleboard; 2022.

B. AWI (QCP) - Quality Certification Program; Current Edition.

C. AWI/AWMAC/WI (AWS) - Architectural Woodwork Standards, 2nd Edition; 2014, with Errata (2016).

1.3 SUBMITTALS

A. Shop Drawings: Indicate materials, component profiles, fastening methods, jointing details, and accessories.

1. Scale of Drawings: 1-1/2 inch to 1 foot, minimum.

2. Provide the information required by AWI/AWMAC/WI (AWS).

3. Include certification program label.

B. Product Data: Provide data for hardware accessories.

C. Samples: Submit actual samples of architectural cabinet construction, minimum 12 inches square, illustrating proposed cabinet, countertop, and shelf unit substrate and finish.

1. See Section 12 36 00 - Countertops for submittal requirements for countertops.

2. Submit countertop color options together with casework material submittals.

D. Samples: Submit actual sample items of proposed pulls, hinges, shelf standards, and locksets, demonstrating hardware design, quality, and finish.

E. Provide centerline of blocking diagrams as required by casework installation in time to be submitted with main interior wall framing shop drawings.

1.4 QUALITY ASSURANCE

A. Fabricator Qualifications: Company specializing in fabricating the products specified in this section with minimum five years of documented experience.

1. Accredited participant in the specified certification program prior to the commencement of fabrication and throughout the duration of the project.

B. Quality Certification: Comply with AWI (QCP) woodwork association quality certification service/program in accordance with requirements for work specified in this section.

1. Provide labels or certificates indicating that the installed work complies with AWI/AWMAC/WI (AWS) requirements for grade or grades specified.

2. Provide designated labels on shop drawings as required by certification program.

3. Provide designated labels on installed products as required by certification program.

4. Submit certifications upon completion of installation that verifies this work is in compliance with specified requirements.

5. Replace, repair, or rework all work for which certification is refused.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Protect units from moisture damage.

B. Do not deliver woodwork until painting and similar operations that could damage woodwork have been completed in installation areas. If woodwork must be stored in other than installation areas, store only in areas where environmental conditions comply with requirements specified in "Environmental requirements" Article below.

1.6 FIELD CONDITIONS

A. During and after installation of custom cabinets, maintain temperature and humidity conditions in building spaces at same levels planned for occupancy.

B. Environmental Limitations: Do not deliver or install woodwork until building is enclosed, wet work is complete, and HVAC system is operating and will maintain temperature between 60 and 90 deg F and relative humidity between 43 and 70 percent.

C. During and after installation of custom cabinets, maintain temperature and humidity conditions in building spaces at same levels planned for occupancy.

D. Field Measurements: Where woodwork is indicated to fit to other construction, verify dimensions of other construction by field measurements before fabrication and indicate measurements on Shop Drawings. Coordinate fabrication schedule with construction progress to avoid delaying the Work.

1. Locate concealed framing, blocking, and reinforcements that support woodwork by field measurements before being enclosed and indicate measurements on Shop Drawings.

2. Established Dimensions: Where field measurements cannot be made without delaying the Work, establish dimensions and proceed with fabricating woodwork without field measurements. Provide allowance for trimming at site, and coordinate construction to ensure that actual dimensions correspond to established dimensions.

PART 2 PRODUCTS

2.1 CABINETS

A. Quality Grade: Unless otherwise indicated provide products of quality specified by AWI/AWMAC/WI (AWS) Architectural Woodwork Standards for Premium Grade.

B. Cabinets:

1. Wood Veneer Faced Cabinet:

a. Locations: Cabinets at Main Lodge Kitchen and Laundry Room.

b. Veneer faced Panel Products (Hardwood Plywood): HPVA HP-1, made with adhesive containing no urea formaldehyde.

c. Cabinet door and drawer front products: Baltic Birch Plywood

d. Finish - Exposed Exterior Surfaces: Baltic Birch Plywood.

e. Finish - Exposed Interior Surfaces: Baltic Birch Plywood.

f. Finish - Semi-Exposed Surfaces: Wood

g. Finish - Concealed Surfaces: Manufacturer's option unless indicated differently in drawings.

h. Door and Drawer Front Edge Profiles: Cabinet door and drawer front products: Baltic Birch Plywood

i. Drawer material: 1/2 veneer core or solid stock match grade of casework.

j. Casework Construction Type: Type A - Frameless.

1) Interface Style for Cabinet and Door: Style 1 - Overlay; flush overlay.

k. Cabinet and Door Fronts: Flush panel.

l. Adjustable Shelf Loading: 40 psf.

1) Deflection: L/144.

m. Drawer Construction Technique: Dovetail joints.

2. All Plywood Cabinet: Fir AC plywood, sanded.

a. Location: Cabinets unless noted otherwise, including wardobe casework..

b. Veneer faced Panel Products (Hardwood Plywood): HPVA HP-1, made with adhesive containing no urea formaldehyde.

c. Finish - Exposed Exterior Surfaces: Solid wood Fir and Finished Plywood.

d. Finish - Exposed Interior Surfaces: Finished Plywood

e. Finish - Semi-Exposed Surfaces: Finished Plywood

f. Finish - Concealed Surfaces: Manufacturer's option unless indicated differently in drawings.

g. Door and Drawer Front Edge Profiles: Square edge, Plywood edge to be filled, sanded, finished, and exposed.

h. Drawer material: 1/2 veneer core or solid stock (for premium MATCH GRADE OF CASEWORK).

i. Casework Construction Type: Type B - Face-frame.

1) Interface Style for Cabinet and Door: Style 1 - Overlay; flush overlay.

j. Cabinet and Door Fronts: Flush panel.

1) Custom Grade: Doors, drawer fronts and false fronts wood grain to run and match vertically within each cabinet unit.

k. Adjustable Shelf Loading: 40 psf.

1) Deflection: L/144.

l. Drawer Construction Technique: Dovetail joints.

C. Complete fabrication, including assembly, finishing, and hardware application, to maximum extent possible before shipment to Project site. Disassemble components only as necessary for shipment and installation. Where necessary for fitting at site, provide ample allowance for scribing, trimming, and fitting.

1. Notify Contracting Officer seven days in advance of the dates and times woodwork fabrication will be complete.

2.2 WOOD-BASED COMPONENTS

A. Wood fabricated from old growth timber is not permitted.

B. Wood Moisture Content: Comply with requirements of referenced quality standard for wood moisture content in relation to ambient relative humidity during fabrication and in installation areas.

C. Particleboard: ANSI A208.1; 45 pound medium density industrial type as specified in AWI/AWMAC/WI (AWS) Architectural Woodwork Quality Standards Illustrated, composed of wood chips bonded with interior grade adhesive under heat and pressure; sanded faces; thickness as required; use for components indicated on drawings.

D. Medium Density Fiberboard (MDF): ANSI A208.2; type as specified in AWI/AWMAC/WI (AWS) Architectural Woodwork Quality Standards Illustrated; composed of wood fibers pressure bonded with moisture resistant adhesive to suit application; sanded faces; thickness as required.

2.3 HARDWOOD PLYWOOD PANELS

A. Hardwood Plywood: Plywood manufactured for nonstructural decorative applications; consisting of faces and backs applied to a variety of core types; comply with HPVA HP-1.

1. Woodwork Quality Standard: Panels complying with specified woodwork quality standard.

2. Face: Wood Veneer Faced Cabinets (Baltic Birch Plywood):

a. Hardwood Plywood: HPVA HP-1, Face species Birch, , Front Face Grade B/BB, Core: Birch Glue Phenolic, plain sliced, book matched;, Thickness 3/4 inch; glue type as recommended for application.

3. Face: Plywood Cabinets

a. Softwood Faced Plywood: HPVA HP-1; Fir, AC sanded, graded in accordance with AWI/AWMAC/WI (AWS) Architectural Woodwork Quality Standards Illustrated, plywood core; type of glue recommended for specific application; thickness as required; face veneer as follows

b. Face: Match existing face species; plain-sliced; grade AA; slip match.

2.4 HARDWARE

A. Exposed Hardware Finishes: For exposed hardware, provide finish that complies with BHMA A156.18 for BHMA finish number indicated.

1. Polished chrome: BHMA 625/US26.

B. Adjustable Shelf Supports: Standard side-mounted system using multiple holes for pin supports and coordinated self rests, polished chrome finish, for nominal 1 1/4 inch spacing adjustments.

C. Drawer and Door Pulls:

1. Door pulls:

    a. Basis of Design:

        1) Emtek, Cadet. Product characteristics noted in paragraph 2.4.C.1.

        2) Substitutions: See Section 01 67 00 – Product Requirements.

    b. Product Characteristics:

        1) ABA compliant.

        2) 1inch diameter, cylindrical shape.

        3) 1 1/2 inch inch projection.

        4) Constructed of solid brass.

        5) Polished chrome

2. Recessed bypass door pull:

    a. Basis of Design:

        1) Emtek, Modern Rectangular Flush Pull. Product characteristics noted in paragraph 2.4.C.2.

        2) Substitutions: See Section 01 67 00 – Product Requirements.

    b. Product Characteristics:

        1) ABA compliant

        2) 4 inch diameter,

        3) Width:1 3/4 inch width

        4) Depth: 1/2 inch depth

        5) Constructed of solid brass.

        6) Polished chrome.

D. Steel rod and shelf bracket:

    1. Finish: black

    2. Capable of supporting 500 lb. each

E. Steel closet rod:

    1. Finish: black

2. Diameter: 1-5/16 inch

3. End Caps: Heavy duty metal with matching finish.

F. Bi-pass door hardware:

1. Material: Steel construction

2. Fabrication: Specifically made for 3/4" thick doors

3. Include all components for a complete system including top rail, bottom guide, roller type door hangers.

G. Drawer Slides:

1. Type: Full extension.

2. Static Load Capacity: Commercial grade.

3. Mounting: Side mounted.

4. Stops: Integral type.

5. Features: Provide self closing/stay closed type.

6. Basis of Design:

a. Knape & Vogt Manufacturing Company; Product 8400 Series: www.kv.com. . Product characteristics noted in paragraph 2.4.G.

b. Substitutions: See Section 01 67 00 – Product Requirements.

H. Hinges: European style concealed self-closing type, BHMA No. 156.9, B01602, 125 degrees of opening, self closing, steel with polished finish, 3 point mounting plate.

I. Adjustable shelf support: Wall mounted

1. Basis of Design:

a. Knape & Vogt Manufacturing Company. Product characteristics noted in paragraph 2.4.I.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2. Provide all hardware for a complete installation, including, but not limited to the following items:

a. Shelving Standards: #85 Standards, Anochrome finish.

1) 16 GA. for 12 inch deep shelving at locations shown on drawings.

2) 14 GA. for 24 inch deep shelving at locations shown on drawings.

b. Shelving Brackets: #185 Brackets, Anochrome finish.

1) 12 inch deep at locations shown on drawings.

2) 24 inch deep at locations shown on drawings.

J. Provide clear rubber bumper pads 5/16 inch diameter and 3/31 inch high at all doors.

2.5 FABRICATION

A. Assembly: Shop assemble cabinets for delivery to site in units easily handled and to permit passage through building openings.

B. Fitting: When necessary to cut and fit on site, provide materials with ample allowance for cutting. Provide matching trim for scribing and site cutting.

C. Matching Wood Grain: Comply with requirements of quality standard for specified Grade and as follows:

1. Provide center matched panels at each elevation.

D. Mechanically fasten back splash to countertops as recommended by laminate manufacturer at 16 inches on center.

E. Provide cutouts for plumbing fixtures. Verify locations of cutouts from on-site dimensions. Prime paint cut edges.

F. Wood Moisture Content: Comply with requirements of referenced quality standard for wood moisture content in relation to ambient relative humidity during fabrication and in installation areas.

G. Fabricate woodwork to dimensions, profiles, and details indicated. Ease edges to radius indicated for the following:

1. Corners of Cabinets and Edges of Solid-Wood (Lumber) Members and Rails: 1/16 inch.

2. Edges of Rails and Similar Members More Than 3/4 Inch Thick: 1/8 inch .

H. Complete fabrication, including assembly, finishing, and hardware application, to maximum extent possible before shipment to Project site. Disassemble components only as necessary for shipment and installation. Where necessary for fitting at site, provide ample allowance for scribing, trimming, and fitting..

1. Notify Contracting Officer seven days in advance of the dates and times woodwork fabrication will be complete.

2. Trial fit assemblies at fabrication shop that cannot be shipped completely assembled. Install dowels, screws, bolted connectors, and other fastening devices that can be

removed after trial fitting. Verify that various parts fit as intended and check measurements of assemblies against field measurements indicated on Shop Drawings before disassembling for shipment.

I. Fitting: When necessary to cut and fit on site, provide materials with ample allowance for cutting. Provide matching trim for scribing and site cutting.

J. Shop-cut openings to maximum extent possible to receive hardware, appliances, plumbing fixtures, electrical work, and similar items. Locate openings accurately and use templates or roughing-in diagrams to produce accurately sized and shaped openings. Sand edges of cutouts to remove splinters and burrs.

1. Seal edges of openings in countertops with a coat of varnish.

2.6 WOOD CABINETS FOR TRANSPARENT FINISH

A. Quality Standard: Comply with AWI/AWMAC/WI (AWS) Section 4 requirements for wood cabinets.

B. Wood Species and Cut for Exposed Surfaces: Baltic Birch Plywood, Quarter sawn

1. Grain Direction: As indicated in drawings.

2. Vertical Matching of Veneer Leaves: End match.

3. Veneer Matching within Panel Face: Running match.

4. Comply with veneer and other matching requirements indicated for blueprint-matched paneling.

C. Semiexposed Surfaces: Provide surface materials indicated below:

1. Surfaces Other Than Drawer Bodies: Thermoset decorative overlay.

2. Drawer Sides and Backs: Thermoset decorative overlay. (If premium match face veneer)

3. Drawer Bottoms: Thermoset decorative overlay.

D. Substrate: Plywood or medium-density fiberboard

E. Provide dust panels of 1/4-inch plywood or tempered hardboard above compartments and drawers, unless located directly under tops.

2.7 SHOP FINISHING

A. Wood Casework in accordance with AWI/AWMAC/WI (AWS), Section 5 - Finishing for grade specified and as follows:

1. Transparent:

a. System - 5, Varnish, Conversion.

b. Stain: As selected Contracting Officer.

c. Sheen: Flat.

1) Premium Quality, satin sheen at 30-50 degrees

B. Plywood Casework in accordance with AWI/AWMAC/WI (AWS), Section 5 - Finishing for grade specified and as follows:

1. Transparent:

a. System - 6, Oil, Synthetic Penetrating (transparent only).

b. Stain: As selected Contracting Officer.

c. Sheen: Flat.

1) Premium Quality, satin sheen at 30-50 degrees

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify adequacy of backing and support framing.

B. Verify location and sizes of utility rough-in associated with work of this section.

3.2 INSTALLATION

A. Set and secure custom cabinets in place, assuring that they are rigid, plumb, and level.

B. Use fixture attachments in concealed locations for wall mounted components.

C. Use concealed joint fasteners to align and secure adjoining cabinet units.

D. Carefully scribe casework abutting other components, with maximum gaps of 1/32 inch. Do not use additional overlay trim for this purpose.

E. Secure desks to floor using appropriate angles and anchorages.

F. Countersink anchorage devices at exposed locations. Conceal with solid wood plugs of species to match surrounding wood; finish flush with surrounding surfaces.

1. Install countertops with no more than 1/8 inch in 96-inch sag, bow, or other variation from a straight line.

2. Secure backsplashes to tops with concealed metal brackets at 16 inches o.c. and to walls with adhesive.

3. Calk space between backsplash and wall with sealant specified in Section 07 90 05 - Joint Sealers.

3.3 ADJUSTING

A. Adjust installed work.

B. Adjust moving or operating parts to function smoothly and correctly.

3.4 CLEANING

A. Clean casework, counters, shelves, hardware, fittings, and fixtures.

END OF SECTION

SECTION 06 42 00 - WOOD PANELING

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Custom solid wood tongue and groove paneling.

B. Custom softwood plywood paneling.

C. Shop finishing.

1.2 REFERENCE STANDARDS

A. AWI (QCP) - Quality Certification Program; Current Edition.

B. AWI/AWMAC/WI (AWS) - Architectural Woodwork Standards, 2nd Edition; 2014, with Errata (2016).

1.3 SUBMITTALS

A. Shop Drawings: Indicate materials, component profiles, fastening methods, jointing details, and accessories.

1. Scale of Drawings: 1-1/2 inch to 1 foot, minimum.

B. Samples: Submit two samples of finished plywood, 12 x 12 inch in size, illustrating wood grain and specified finish.

C. Submit two samples of solid wood tongue and groove material, 12" length, illustrating wood grain and specified finish.

1.4 QUALITY ASSURANCE

A. Fabricator Qualifications: Company specializing in fabricating the products specified in this section with minimum five years of documented experience.

1.5 MOCK-UP

A. Construct mock-up, 8 feet long by 8 feet wide, illustrating full panel sheet, edge trim, joint trim, applied finish.

1. Provide up to three iterations of panel finish

B. Locate where directed.

C. Mock-up may remain as part of the Work.

PART 2 PRODUCTS

2.1 PANELING

A. Quality Standard: Premium Grade, in accordance AWI/AWMAC/WI (AWS), unless noted otherwise.

B. Tongue and groove wood paneling (WP-1):

1. Species: Match species and level of quality of existing wood paneling in the individual cabin, match profile of existing T&G paneling.

2. Cut: Quarter sawn.

3. Visible Edges and Reveals: Filled and painted.

4. Outside Corners: Mitered and splined.

C. Softwood plywood paneling (WP-2).

1. Species: Match species and level of quality of existing plywood paneling in individual cabin..

2. Visible Edges and Reveals: Match existing.

3. Outside Corners: Match existing.

2.2 ADHESIVES AND FASTENERS

A. Fasteners: Of size and type to suit application; stainless steel finish in concealed locations and brushed stainless steel finish in exposed locations.

B. Multipurpose Construction Adhesive: Formulation complying with ASTM D 3498 that is recommended for indicated use by adhesive manufacturer.

1. Use adhesive that has a VOC content of 70 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24) and containing no Urea Formaldehyde.

2.3 FABRICATION

A. Prepare panels for delivery to site, permitting passage through building openings.

B. Finish exposed edges of panels as specified by grade requirements.

C. Trim exposed edges of panels with solid wood trim at locations indicated.

D. When necessary to cut and fit on site, provide materials with ample allowance for cutting and scribing.

2.4 FINISHING

A. Sand work smooth and set exposed nails and screws.

B. Apply wood filler in exposed nail and screw indentations.

C. Finish work in accordance with AWI/AWMAC/WI (AWS), Section 5 - Finishing for grade specified and as follows:

1. Transparent: The intent is to match existing, however existing finish is unknown. Please see this as placeholder:

a. System - 5, Varnish, Conversion.

b. Stain: As selected by Contracting Officer.

c. Sheen: Flat.

2.5 ACCESSORIES

A. Lumber for Shimming, Blocking: Softwood lumber of Doug Fir species.

B. Wood Filler: Tinted to match surface finish color.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that field measurements are as indicated on shop drawings.

B. Verify adequacy of backing and support framing.

C. Verify mechanical, electrical, and building items affecting work of this section are placed and ready to receive this work.

3.2 INSTALLATION

A. Install work in accordance with AWI/AWMAC/WI (AWS) requirements for grade indicated.

B. Do not begin installation until wood materials have been fully acclimated to interior conditions.

C. Set and secure materials and components in place, plumb and level, using concealed fasteners wherever possible.

D. Where necessary to cut and fit on site, scribe work abutting other components. Do not use additional overlay trim to conceal gaps.

E. Plywood paneling: Select and arrange panels on each wall to minimize noticeable variations in grain character and color between adjacent panels.

1. Conceal fasteners to greatest practical extent.

2. Miter corners at edge trim.

3.3 TOLERANCES

A. Maximum Variation from True Position: 1/16 inch.

B. Maximum Offset from True Alignment with Abutting Materials: 1/32 inch.

END OF SECTION

SECTION 07 01 50.19 - PREPARATION FOR RE-ROOFING

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Partial replacement of existing roofing system in preparation for replacement roofing system in designated areas as indicated on drawings.

B. Reroofing work to occur prior to bat roosting season per 01 10 00 - Summary of Work.

C. Temporary roofing protection.

D. Recycle existing roofing system and related materials

1.2 RELATED REQUIREMENTS

A. Section 07 31 13 - Asphalt Shingles: Roof system.

1.3 REFERENCE STANDARDS

A. PS 1 - Structural Plywood; 2023.

1.4 ADMINISTRATIVE REQUIREMENTS

A. Coordinate with affected mechanical and electrical work associated with roof penetrations.

B. Preinstallation Meeting: Convene one week before starting work of this section.

1. Attendees:

a. Contracting Officer.

b. Contractor.

c. Installer.

2. Meeting Agenda: Provide agenda to participants prior to meeting in preparation for discussions on the following:

a. Removal and installation schedule.

b. Necessary preparatory work.

c. Protection before, during, and after roofing system installation.

d. Removal of existing roofing system.

e. Installation of new roofing system.

f. Temporary roofing and daily terminations.

g. Transitions and connection to and with other work.

h. Inspections and testing of installed systems.

1.5 SUBMITTALS

A. Product Data: Submit for each type of material.

B. Shop Drawings: Indicate size, configuration, and installation details.

C. Materials Removal Company Qualification Statement.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Ensure storage and staging of materials does not exceed static and dynamic load-bearing capacities of roof decking.

1.7 FIELD CONDITIONS

A. Existing Roofing System: Asphalt Sheet roofing.

B. Do not remove existing roofing membrane when weather conditions threaten the integrity of building contents or intended continued occupancy.

C. Maintain continuous temporary protection prior to and during installation of new roofing system.

D. Provide notice at least three days before starting activities that will affect normal building operations.

PART 2 PRODUCTS

2.1 COMPONENTS

A. See the following sections for additional information on components relating to this work:

1. Partial removal of existing roofing system in preparation for new roofing system in designated areas as indicated on drawings, see Section 07 31 13.

2.2 MATERIALS

A. Patching Materials: Provide necessary materials in accordance with requirements of existing roofing system.

B. Temporary Roofing Protection Materials:

1. Contractor's responsibility to select appropriate materials for temporary protection of roofing areas as determined necessary for this work.

2. Plywood Sheathing: PS 1, Grade C-D, Exposure I.

2.3 ACCESSORIES

A. Fasteners: Type and size as required and compatible with existing and new roofing system to resist local wind uplift.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that existing roof surface has been cleared of materials being removed from existing roofing system and ready for next phase of work as required.

3.2 PREPARATION

A. Sweep roof surface clean of loose matter.

B. Remove loose refuse and dispose of properly off-site.

3.3 MATERIAL REMOVAL

A. Remove only existing roofing materials that can be replaced with new materials with-in three days.

3.4 INSTALLATION

A. Coordinate scope of this work with requirements for installation of new roofing system, see Section 07 31 13 for additional requirements.

3.5 PROTECTION

A. Provide protection of existing roofing system that is not having work performed on it.

B. Provide temporary protective sheeting over uncovered deck surfaces.

C. Turn sheeting up and over parapets and curbing. Retain sheeting in position with weights.

D. Provide for surface drainage from sheeting to existing drainage facilities.

E. Do not permit traffic over unprotected or repaired deck surface.

END OF SECTION

SECTION 07 11 13 - BITUMINOUS DAMPPROOFING

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Bituminous dampproofing (DAMPPROOFING MEMBRANE).

1. Apply to the following surfaces:

a. Exterior, below-grade surfaces of concrete and masonry walls, except those surfaces indicated to be protected with Waterproofing.

1.2 REFERENCE STANDARDS

A. ASTM D41/D41M - Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing; 2011 (Reapproved 2023).

B. ASTM D3747 - Standard Specification for Emulsified Asphalt Adhesive for Adhering Roof Insulation; 1979 (Reapproved 2007).

C. NRCA (WM) - The NRCA Waterproofing Manual; 2021.

1.3 SUBMITTALS

A. Product Data: Provide properties of primer, bitumen, and mastics.

1.4 FIELD CONDITIONS

A. Maintain ambient temperatures above 40 degrees F for 24 hours before and during application until dampproofing has cured.

PART 2 PRODUCTS

2.1 MANUFACTURERS (DAMPPROOFING MEMBRANE)

A. Basis of Design:

1. BASF; Product MasterSeal 610, 614, 615. Product characteristics noted in paragraph 2.2.

2. Substitutions: See Section 01 67 00 – Product Requirements.

2.2 COLD ASPHALTIC MATERIALS

A. Bitumen: Emulsified asphalt, ASTM D1227; with fiber reinforcement other than asbestos (Type II).

B. Asphalt Primer: ASTM D41, compatible with substrate.

C. Sealing Mastic: Asphalt roof cement, ASTM D2822, Type I.

D. Asphalt-Coated Glass Fabric: ASTM D 1668, Type I.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify existing conditions are acceptable prior to starting this work.

B. Verify substrate surfaces are durable, free of matter detrimental to adhesion or application of dampproofing system.

C. Verify that items penetrating surfaces to receive dampproofing are securely installed.

3.2 PREPARATION

A. Protect adjacent surfaces not designated to receive dampproofing.

B. Clean and prepare surfaces to receive dampproofing in accordance with manufacturer's instructions.

C. Do not apply dampproofing to surfaces unacceptable to manufacturer.

D. Apply mastic to seal penetrations, small cracks, or minor honeycombs in substrate.

3.3 APPLICATION, GENERAL

A. Comply with manufacturer's written recommendations unless more stringent requirements are indicated or required by Project conditions to ensure satisfactory performance of dampproofing.

1. Apply additional coats if recommended by manufacturer or if required to achieve coverages indicated.

2. Allow each coat of dampproofing to cure 24 hours before applying subsequent coats.

3. Allow 36 hours drying time prior to backfilling.

B. Apply dampproofing to footings and foundation walls where opposite side of wall faces building interior or occupied space.

1. Apply from finished-grade line to top of footing, extend over top of footing, and down a minimum of 6 inches (150 mm) over outside face of footing.

2. Extend 12 inches (300 mm) onto intersecting walls and footings, but do not extend onto surfaces exposed to view when Project is completed.

3. Install flashings and corner protection stripping at internal and external corners, changes in plane, construction joints, cracks, and where shown as "reinforced," by embedding an 8-inch- (200-mm-) wide strip of asphalt-coated glass fabric in a heavy coat of dampproofing. Dampproofing coat for embedding fabric is in addition to other coats required.

3.4 APPLICATION

A. Foundation Walls: Apply two coats of asphalt dampproofing.

B. Foundation Walls: Patch disturbed areas of existing dampproofing with two additional coats of dampproofing of the same generic type.

C. Perform this work in accordance with manufacturer's instructions and NRCA (WM) applicable requirements.

D. Prime surfaces in accordance with manufacturer's instructions and NRCA (WM) applicable requirements.

E. Apply bitumen with mop.

F. Apply bitumen in one coat, continuous and uniform, at a rate of 25 sq ft/gal per coat.

G. Seal items watertight with mastic, that project through dampproofing surface.

3.5 FIELD QUALITY CONTROL

A. Notify Contracting Officer when application of Dampproofing is complete. Contracting Officer will observe work prior to covering.

3.6 CLEANING

A. Remove dampproofing materials from surfaces not intended to receive dampproofing.

END OF SECTION

SECTION 07 21 00 - THERMAL INSULATION

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Batt insulation in exterior wall construction.

B. Board insulation in crawlspace wall and roof construction

1.2 RELATED REQUIREMENTS

A. Section 01 81 13 - Sustainable Design Requirements.

B. Section 06 10 00 - Rough Carpentry: Installation requirements for board insulation over steep slope roof sheathing or roof structure.

1.3 REFERENCE STANDARDS

A. ASTM C518 - Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus; 2021.

B. ASTM C612 - Standard Specification for Mineral Fiber Block and Board Thermal Insulation; 2014 (Reapproved 2019).

C. ASTM C665 - Standard Specification for Mineral-Fiber Blanket Thermal Insulation for Light Frame Construction and Manufactured Housing; 2024.

D. ASTM C726 - Standard Specification for Mineral Wool Roof Insulation Board; 2024.

E. ASTM E84 - Standard Test Method for Surface Burning Characteristics of Building Materials; 2024.

F. ASTM E96/E96M - Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials; 2024a.

G. ASTM E136 - Standard Test Method for Assessing Combustibility of Materials Using a Vertical Tube Furnace at 750 Degrees C; 2024c.

H. NAIMA - Recommendations for Installation in Residential and Other Light-Frame Construction - Fiber Glass Building Insulation; North American Insulation Manufacturers Association; 1999.

1.4 SUBMITTALS

A. Product Data: Provide data on product characteristics, performance criteria, and product limitations.

B. Manufacturer's Certificate: Certify that products meet or exceed specified requirements.

1.5 QUALITY ASSURANCE

A. Surface-Burning Characteristics: As determined by testing identical products according to ASTM E 84 by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

1.6 FIELD CONDITIONS

A. Do not install insulation adhesives when temperature or weather conditions are detrimental to successful installation.

PART 2 PRODUCTS

2.1 APPLICATIONS

A. INSULATION R1: Insulation at Roof Construction: Mineral Wool Roof Insulation.

B. INSULATION B1: Insulation at Perimeter of Foundation: Mineral Fiber board.

C. INSULATION A1: Acoustical insulation at interior walls: Mineral Fiber Batt Insulation

D. INSULATION W1: Insulation in Wood Framed Walls: Mineral Fiber Batt insulation with separate vapor retarder.

2.2 MINERAL FIBER BOARD INSULATION MATERIALS

A. INSULATION R1: Mineral Wool Roof Insulation Boards: Comply with ASTM C726, with high-density top layer. Mold and microbial growth resistant.

1. Face Coating: None, unfaced.

2. Board Size: 48 by 48 inches.

3. Board Thickness: (2) layers required for 3 inch total thickness, at River Cabin, provide 4 inch total thickness.

4. Thermal Conductivity (k-factor): Btu inch/hr sq ft degrees F of 3.8 per inch, minimum, at 75 degrees F when tested in accordance with ASTM C518.

5. Top Layer Density: 13.75 pcf, nominal.

6. Bottom Layer Density: 9.36 pcf for thicknesses greater than 2-1/2 inches.

7. Basis of Design:

a. ROCKWOOL: ComfortBoard 110; www.rockwool.com. Product characteristics noted in paragraph 2.2.A.

b. Substitutions: See Section 01 67 00 – Product Requirements.

B. INSULATION B1: Stone Wool Insulation Board: Rigid or semi-rigid mineral fiber, ASTM C612 or C553; unfaced flame spread index of 0 (zero) when tested in accordance with ASTM E84.

1. Smoke Developed Index: 450 or less, when tested in accordance with ASTM E84.

2. Board Size: 48 by 96 inches.

3. Board Thickness: 3 inches.

4. Density: 8 pcf

5. Thermal Resistance: R-value of 4.2 per inch at 75 degrees F, minimum, when tested in accordance with ASTM C518.

6. Basis of Design:

a. ROCKWOOL; Comfortboard 80: www.rockwool.com. . Product characteristics noted in paragraph 2.2.B.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2.3 MINERAL FIBER BLANKET INSULATION MATERIALS

A. INSULATION A1: Mineral Wool Insulation: Flexible preformed batt or blanket, complying with ASTM C665; friction fit.

1. Flame Spread Index: 25 or less, when tested in accordance with ASTM E84.

2. Smoke Developed Index: 450 or less, when tested in accordance with ASTM E84.

3. Combustibility: Non-combustible, when tested in accordance with ASTM E136, except for facing, if any.

4. Facing: Unfaced.

5. Basis of Design:

a. RockWool; Safe n' Sound: www.rockwool.com. Product characteristics noted in paragraph 2.3.A.

b. Substitutions: See Section 01 67 00 – Product Requirements.

B. INSULATION W1: Mineral Fiber Batt Insulation: Flexible or semi-rigid preformed batt or blanket, complying with ASTM C665; friction fit; unfaced flame spread index of 0 (zero) when tested in accordance with ASTM E84.

1. Flame Spread Index: 25 or less, when tested in accordance with ASTM E84.

2. Smoke Developed Index: 450 or less, when tested in accordance with ASTM E84.

3. Thermal Resistance: R-value of R21.

4. Basis of Design:

a. ROCKWOOL, Inc; ComfortBatt: www.rspec.com. Product characteristics noted in paragraph 2.3.B.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2.4 ACCESSORIES

A. Sheet Vapor Retarder ASTM D 4397: Clear polyamide film for above grade application, 2 mil thick. Class A Fire rated

1. Water Vapor Permeance: ASTM C665 Section 7.4, Water-Vapor Permeance • and ASTM E96.

2. Basis of Design:

a. CertainTeed Corporation; MemBrain: www.certainteed.com. Product characteristics noted in paragraph 2.4.

b. Substitutions: See Section 01 67 00 – Product Requirements.

3. Vapor-Retarder Tape: Pressure-sensitive tape of type recommended by vapor-retarder manufacturer for sealing joints and penetrations in vapor retarder.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that substrate, adjacent materials, and insulation materials are dry and that substrates are ready to receive insulation.

B. Verify substrate surfaces are flat, free of honeycomb, fins, irregularities, or materials or substances that may impede adhesive bond.

3.2 BOARD INSTALLATION AT FOUNDATION PERIMETER

A. Install boards horizontally on foundation perimeter.

B. Cut and fit insulation tightly to protrusions or interruptions to the insulation plane.

3.3 BOARD INSTALLATION OVER STEEP SLOPE ROOF SHEATHING OR ROOF STRUCTURE

A. Installation of board insulation over steep slope roof structure or roof sheathing, see Section 06 10 00.

3.4 BATT INSTALLATION

A. Install insulation and vapor retarder in accordance with manufacturer's instructions.

B. Install in exterior wall spaces without gaps or voids. Do not compress insulation.

C. Trim insulation neatly to fit spaces. Insulate miscellaneous gaps and voids.

D. Fit insulation tightly in cavities and tightly to exterior side of mechanical and electrical services within the plane of the insulation.

E. Install with factory-applied vapor retarder membrane facing warm side of building spaces. Lap ends and side flanges of membrane over framing members.

F. Tape insulation batts in place.

G. Tape seal butt ends, lapped flanges, and tears or cuts in membrane.

H. Install Insulation with facing contacting ceiling system. Butt tightly together leaving no gap for thermal or acoustical leakage.

3.5 PROTECTION

A. Do not permit installed insulation to be damaged prior to its concealment.

END OF SECTION

SECTION 07 25 00 - WEATHER BARRIERS

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Water-resistive barriers.

1.2 RELATED REQUIREMENTS

A. Section 09 21 16 - Gypsum Board Assemblies

1.3 DEFINITIONS

A. Weather Barriers: Assemblies that form either water-resistive barriers, air barriers, or vapor retarders.

B. Water-Resistive Barrier: A material behind an exterior wall covering that is intended to resist liquid water that has penetrated behind the exterior covering from further intruding into the exterior wall assembly.

1.4 REFERENCE STANDARDS

A. ASTM D1970/D1970M - Standard Specification for Self-Adhering Polymer Modified Bituminous Sheet Materials Used as Steep Roofing Underlayment for Ice Dam Protection; 2021.

B. ASTM E84 - Standard Test Method for Surface Burning Characteristics of Building Materials; 2024.

C. ASTM E96/E96M - Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials; 2024a.

D. ASTM E2178 - Standard Test Method for Determining Air Leakage Rate and Calculation of Air Permeance of Building Materials; 2021a.

1.5 SUBMITTALS

A. Product Data: Provide data on material characteristics.

B. Shop Drawings: Provide drawings of special joint conditions.

1.6 FIELD CONDITIONS

A. Maintain temperature and humidity recommended by materials manufacturers before, during, and after installation.

PART 2 PRODUCTS

2.1 AIR AND WATER BARRIER MATERIALS

A. FIELD MEMBRANE: Air Barrier Sheet, Mechanically Fastened; Sheet

1. Air Permeance: 0.004 cubic feet per minute per square foot, maximum, when tested in accordance with ASTM E2178.

a. Type 1 when tested in accordance with ASTM E 1677 .

2. Water Vapor Permeance: 5 perms, minimum, when tested in accordance with ASTM E96/E96M Procedure A (desiccant procedure).

3. Ultraviolet (UV) and Weathering Resistance: Approved in writing by manufacturer for up to 180 days of weather exposure.

4. Surface Burning Characteristics: Flame spread index of 25 or less, and smoke developed index of 50 or less, when tested in accordance with ASTM E84.

5. Basis of Design:

a. DuPont Building Innovations; Tyvek Commercial Wrap D with Tyvek Fluid Applied Flashing - Brush Formulation, Tyvek Fluid Applied Flashing and Joint Compound, FlexWrap NF, StraightFlash, StraightFlash VF, Tyvek Wrap Caps, and Tyvek Tape: www.dupont.com.. Product characteristics noted in paragraph 2.1.

b. Substitutions: See Section 01 67 00 – Product Requirements.

6. Install 1 layer directly onto sheathing where indicated in drawings.

B. ACCESSORIES

1. Sealants, Tapes, and Accessories Used to construct a complete primary air and water resistive barrier system at new construction walls: As indicated or complying with field membrane manufacturer's installation instructions.

2. Flexible Flashing: Self-adhesive sheet flashing complying with ASTM D1970/D1970M, except slip resistance requirement is waived if not installed on a roof.

a. Width: 4 inches.

b. Ultraviolet (UV) and Weathering Resistance: Approved by manufacturer for up to 30 days of weather exposure.

c. Products: Product characteristics indicated in 2.1B

3. Thinners and Cleaners: As recommended by field membrane manufacturer.

4. Fasteners and Anchors:

a. Metal and Finish: Hot-dipped galvanized steel per ASTM A 153/A 153M at all locations.

b. Nails, Brads, and Staples: ASTM F 1667.

c. Staple Caps:

1) Basis of Design:

a) DuPont; Product Tyvek Wrap Caps Staple Pack; www.dupont.com . Product characteristics noted in paragraph 2.1.B.4.c.

b) Substitutions: See Section 01 67 00 – Product Requirements.

2) Product Characteristics:

a) Caps prevent tears.

b) Caps improve holing power.

c) Caps provide better air and wind resistance.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that surfaces and conditions comply with requirements of this section.

3.2 PREPARATION

A. Remove projections, protruding fasteners, and loose or foreign matter that might interfere with proper installation.

3.3 INSTALLATION

A. Install materials in accordance with manufacturer's installation instructions.

B. Water-Resistive Barriers: Install continuous air and water-resistive barrier over surfaces indicated, with sheets lapped to shed water but with seams not sealed.

C. Mechanically Fastened Exterior Sheets:

1. Install sheets shingle-fashion to shed water, with seams aligned horizontal.

2. Overlap seams as recommended by manufacturer, 6 inches, minimum.

3. Overlap at outside and inside corners as recommended by manufacturer, 12 inches, minimum.

4. For applications indicated to be airtight, seal seams, laps, penetrations, tears, and cuts with self-adhesive tape; use only large-headed, gasketed fasteners as recommended by manufacturer.

5. Install air and water-resistive barrier over jamb flashings.

6. Install head flashings under air and water-resistive barrier.

7. At framed openings with frames having nailing flanges, extend sheet into opening and over flanges; at head of opening, seal sheet over flange and flashing.

D. Openings and Penetrations in Exterior Water-Resistive Barriers:

1. Install flashing over sills, covering entire sill framing member, and extend at least 5 inches onto air and water-resistive barrier and at least 6 inches up jambs; mechanically fasten stretched edges.

2. At openings filled with frames having nailing flanges, seal head and jamb flanges using a continuous bead of sealant compressed by flange and cover flanges with sealing tape at least 4 inches wide; do not seal sill flange.

3. At openings filled with nonflanged frames, seal air and water-resistive barrier to each side of framing at opening using flashing at least 9 inches wide, and covering entire depth of framing.

4. At head of openings, install flashing under air and water-resistive barrier extending at least 2 inches beyond face of jambs; seal air and water-resistive barrier to flashing.

5. At interior face of openings, seal gaps between window and door frames and rough framing using appropriate joint sealant over backer rod.

6. Service and Other Penetrations: Form flashing around penetrating items and seal to surface of air and water-resistive barrier.

3.4 FIELD QUALITY CONTROL

A. See Section 01 40 00 - Quality Requirements for additional requirements.

B. Third Party Inspection and Testing: Engage third part testing agency.

1. Allow access to work areas and staging.

2. Coordinate with testing agency and notify Contracting Officer in writing of schedule for work of this section to allow sufficient time for testing and inspection.

3. Do not cover work of this section until testing and inspection is accepted.

C. Obtain approval of installation procedures from air and water-resistive barrier manufacturer based on a mock-up installed in place, prior to proceeding with remainder of installation.

3.5 PROTECTION

A. Do not leave materials exposed to weather longer than recommended by manufacturer.

END OF SECTION

SECTION 07 31 13 - ASPHALT SHINGLES

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Asphalt shingle roofing.

B. Sheet membranes for underlayment and valley protection.

C. Associated metal flashings and accessories.

D. Patching of existing sheathing.

1.2 RELATED REQUIREMENTS

A. Section 02 41 00 - Demolition

1.3 REFERENCE STANDARDS

A. ASTM D1970/D1970M - Standard Specification for Self-Adhering Polymer Modified Bituminous Sheet Materials Used as Steep Roofing Underlayment for Ice Dam Protection; 2021.

B. ASTM D3161/D3161M - Standard Test Method for Wind Resistance of Steep Slope Roofing Products (Fan-Induced Method); 2020 (Reapproved 2025).

C. ASTM D3462/D3462M - Standard Specification for Asphalt Shingles Made from Glass Felt and Surfaced with Mineral Granules; 2023.

D. ASTM D4586/D4586M - Standard Specification for Asphalt Roof Cement, Asbestos-Free; 2007 (Reapproved 2024).

E. ASTM F1667 - Standard Specification for Driven Fasteners: Nails, Spikes, and Staples; 2021.

F. UL (DIR) - Online Certifications Directory; Current Edition.

1.4 SUBMITTALS

A. Product Data: Provide data indicating material characteristics.

B. Shop Drawings: For metal flashings, indicate specially configured metal flashings jointing methods and locations, fastening methods and locations, and installation details.

C. Samples: Submit two samples of each shingle color indicating color range and finish texture/pattern ; for color selection.

1. Asphalt Shingle: Full size.

2. Ridge and Hip Cap Shingles: Full size.

3. Style D Roof Edge" 12 inch section.

4. Exposed Valley Lining: 12 inch section..

5. Self-Adhering Underlayment: 12 inches square.

D. Manufacturer's Installation Instructions: Indicate installation criteria and procedures.

E. Manufacturer's Certificate: Certify that products meet or exceed specified requirements.

F. Product test reports.

G. Report of condition of deck following demolition of existing roofing and related substrates.

H. Maintenance Materials: Furnish the following for Contracting Officer's use in maintenance of project.

1. Provide 5% of total amount of shingles as extra stock.

2. Provide 5% of total amount of ridge shingles as extra stock.

I. Warranties: Sample of special warranties.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: Manufacturer's authorized representative who is trained and approved for installation of units required for this Project.

B. Source Limitations: Obtain ridge and hip cap shingles ridge vents and self-adhering sheet underlayment from single source from single manufacturer.

C. Perform Work in accordance with the recommendations of NRCA Steep Roofing Manual.

1. Maintain one copy of document on site.

D. Products are Required to Comply with Fire Resistance Criteria: UL (DIR) listed and labeled.

E. Preinstallation Conference: Conduct conference at Project site.

1.6 MOCK-UPS

A. Provide mock-up of 100 sq ft, including all types of underlayment, drip edge, and valley flashing.

B. Locate where directed.

C. Mock-up may remain as part of work.

1.7 WARRANTY

A. When warranties are required, verify with Contracting Officer's counsel that special warranties stated in this article are not less than remedies available to Owner under prevailing local laws.

B. Special Warranty: Standard form in which manufacturer agrees to repair or replace asphalt shingles that fail in materials or workmanship within specified warranty period.

1. Material Warranty Period: 40 years from date of Substantial Completion, prorated.

2. Algae-Discoloration Warranty Period: Asphalt shingles will not discolor 10 years from date of Substantial Completion.

C. Limited Wind-Speed Warranty Period: Asphalt shingles will resist blow-off or damage caused by wind speeds of up to 80 mph for 15 years from date of Substantial Completion.

D. Roofing Installer's Warranty: On warranty form at end of this Section, signed by Installer, in which Installer agrees to repair or replace components of asphalt-shingle roofing that fail in materials or workmanship within specified warranty period.

1. Warranty Period: Two years from date of Substantial Completion.

PART 2 PRODUCTS

2.1 GLASS FIBER REINFORCED ASPHALT SHINGLES

A. Basis of Design:

1. GAF; Timberline HDZ RS Shingles: www.gaf.com. Product characteristics noted in paragraph 2.2.

2. Substitutions: See Section 01 67 00 – Product Requirements.

2.2 ASPHALT SHINGLES

A. Asphalt Shingles: Asphalt-coated glass felt, laminated, multi ply overlay construction, mineral granule surfaced, complying with ASTM D3462; Class A fire resistance.

1. Wind Resistance: Class A, when tested in accordance with ASTM D3161/D3161M.

2. Warranted Wind Speed: Not greater than 60 mph.

3. Algae Resistant.

4. Weight: 210 lb/100 sq ft.

5. Self-sealing type.

6. Style: Square.

7. Color:

a. GAF Timberline "Hunter Green" (Basis of Design)

b. Any proposed substitutions must offer a similar color.

2.3 SHEET MATERIALS

A. Eave Edge Starter Shingles: Glass felt base, with ceramic coated mineral granules tightly embedded in refined, water-resistant asphalt, complying with ASTM D3462/D3462M.

1. Shingle Size: 8-3/16 by 38-1/4 inches, nominal.

B. Self-adhering rubberized asphalt sheet underlayment complying with ASTM D1970; 40 mil total thickness; with strippable release film and Polyethylene faced top surface.

1. Top Surface: Polyethylene film.

2. Self Sealability: Passing nail sealability test specified in ASTM D1970/D1970M.

3. Low Temperature Flexibility: Passing test specified in ASTM D1970/D1970M.

4. Functional Temperature Range: From minus 70 degrees F to 212 degrees F.

5. Basis of Design:

a. Grace; Product Ice and Watershield . Product characteristics noted in paragraph 2.3.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2.4 ACCESSORIES

A. Roofing Nails: Standard round wire shingle type, galvanized steel, stainless steel, aluminum roofing nails, or copper roofing nails, minimum 3/8-inch head diameter, 12-gauge, 0.109-inch nail shank diameter, 1-1/2 inches long and complying with ASTM F1667.

B. Plastic Cement: ASTM D4586, Type II asphalt roof cement.

C. Lap Cement: Fibrated cutback asphalt type, recommended for use in application of underlayment, and free of toxic solvents.

D. Patching materials for existing sheathing:

1. Roof Sheathing: Any PS 2 type, rated Structural I Sheathing.

a. Bond Classification: Exterior.

b. Span Rating: 60.

c. Performance Category: 3/4 PERF CAT.

2. Thickness: Match thickness of existing sheathing

2.5 METAL FLASHINGS

A. Metal Flashings: Provide sheet metal gable edge, ridge, open valley flashing, chimney flashing, and dormer flashing.

1. Form flashings to profiles indicated on drawings.

2. Form sections square and accurate to profile, in maximum possible lengths, free from distortion or defects detrimental to appearance or performance.

3. Hem exposed edges of flashings minimum 1/4 inch on underside.

4. Coat concealed surfaces of flashings with bituminous paint.

B. Sheet Metal: Prefinished galvanized steel, 0.025 inch/24 gage thick, minimum G90/Z275 hot-dipped galvanized; PVC coated, match color of roofing.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify existing conditions prior to beginning work.

B. Verify that roof deck is of sufficient thickness to accept fasteners.

C. Verify that roof penetrations and plumbing stacks are in place and flashed to deck surface.

D. Verify roof openings are correctly framed.

E. Verify deck surfaces are dry, free of ridges, warps, or voids.

F. Notify Contracting Officer if defects listed above are present following demolition of existing roofing material and related substrates.

3.2 PREPARATION

A. Seal roof deck joints wider than 1/16 inch as recommended by shingle manufacturer.

B. At areas where eave protection membrane is to be adhered to substrate, fill knot holes and surface cracks with latex filler.

C. Broom clean deck surfaces before installing underlayment or eave protection.

D. Install eave edge flashings tight with fascia boards. Weather lap joints 2 inches and seal with plastic cement. Secure flange with nails spaced as required by shingle manufacturer.

E. Patching of existing roof:

1. Install construction panels of same thickness as existing sheathing tight at locations of removed roof vents and other penetrations in the existing sheathing.

3.3 UNDERLAYMENT INSTALLATION

A. General: Comply with underlayment manufacturer's written installation instructions applicable to products and applications indicated unless more stringent requirements apply.

B. Self-Adhering rubberized asphalt Sheet Underlayment: Install, wrinkle free, on roof deck. Comply with low-temperature installation restrictions of underlayment manufacturer if applicable. Install lapped in direction that sheds water. Lap sides not less than 3-1/2 inches. Lap ends not less than 6 inches staggered 24 inches between courses. Roll laps with roller. Cover underlayment within seven days.

1. Install self-adhering rubberized asphalt sheet underlayment at the following locations:

a. All roof surfaces.

b. Sidewalls: Extend beyond sidewall 18 inches, and return vertically against sidewall not less than 4 inches.

c. Chimneys, Roof Monitors, and Other Roof-Penetrating Elements: Extend 24 inches around penetrating element, and return vertically against penetrating element not less than 8 inches.

3.4 INSTALLATION - VALLEY PROTECTION

A. Install flexible flashing in accordance with manufacturer's instructions and {\rs\#1} applicable requirements.

B. At Exposed Valleys: Install one layer of sheet metal flashing, minimum 24 inches wide, centered over open valley and crimped to guide water flow; weather lap joints minimum 2-inch wide band of lap cement along each edge of first layer, press roll roofing into cement, and nail in place minimum 18 inches on center and 1 inch from edges.

3.5 INSTALLATION - METAL FLASHING AND ACCESSORIES

A. Install flashings in accordance with manufacturer's instructions and {\rs\#1} applicable requirements.

B. Weather lap metal flashing joints at least 2 inches and seal joints weathertight.

C. Apron Flashings: Extend lower flange over and beyond each side of downslope asphalt shingles and up the vertical surface.

D. Step Flashings: Install with a headlap of 2 inches and extend over the underlying asphalt shingle and up the vertical surface. Fasten to roof deck only.

E. Cricket or Backer Flashings: Install against the roof-penetrating element extending concealed flange beneath upslope asphalt shingles and beyond each side.

F. Open-Valley Flashings: Install centered in valleys, lapping ends at least 8 inches (200 mm) in direction to shed water. Fasten upper end of each length to roof deck beneath overlap.

1. Adhere 9-inch- (225-mm-) wide strip of self-adhering sheet to metal flanges and to self-adhering sheet underlayment.

G. Rake Drip Edges: Install rake drip-edge flashings over underlayment and fasten to roof deck.

H. Eave Drip Edges: Install eave drip-edge flashings below underlayment and fasten to roof sheathing.

I. Pipe Flashings: Form flashing around pipe penetrations and asphalt shingles. Fasten and seal to asphalt shingles as recommended by manufacturer.

J. Install soffit and ridge vents following manufacturer's written installation instructions.

3.6 INSTALLATION - SHINGLES

A. General: Install asphalt shingles according to manufacturer's written instructions, recommendations in ARMA's "Residential Asphalt Roofing Manual," and recommendations in NRCA's "NRCA Guidelines for Asphalt Shingle Roof Systems."

B. Install shingles in accordance with manufacturer's instructions manufacturer's instructions and applicable requirements.

C. Place shingles in straight coursing pattern with 5-inch weather exposure to produce double thickness over full roof area, and provide double course of shingles at eaves.

D. Project first course of shingles 3/4 inch beyond fascia boards.

E. Extend shingles 1/2 inch beyond face of gable edge fascia boards.

F. Install first and remaining courses of asphalt shingles stair-stepping diagonally across roof deck with manufacturer's recommended offset pattern at succeeding courses, maintaining uniform exposure.

G. Fasten asphalt-shingle strips with a minimum of six roofing nails located according to manufacturer's written instructions.

H. Hip and Ridge Shingles: Maintain same exposure of cap shingles as roofing shingle exposure. Lap cap shingles at ridges to shed water away from direction of prevailing winds. Fasten with roofing nails of sufficient length to penetrate sheathing.

1. Fasten ridge cap asphalt shingles to cover ridge vent without obstructing airflow.

I. Cap hips with individual shingles, maintaining 5-inch weather exposure, and place to avoid exposed nails.

J. After installation, place one daub of roof cement, 1-inch diameter, under each individual shingle tab exposed to weather to prevent lifting.

K. Coordinate installation of roof-mounted components or work projecting through roof with weathertight placement of counterflashing.

L. Complete installation to provide weathertight benefits.

3.7 CLEANING

A. Clean exposed work upon completion of installation; remove grease and oil films, excess joint sealer, handling marks, and debris from installation, leaving work clean and unmarked, free from dents, creases, waves, scratch marks, or other damage to finish.

3.8 PROTECTION

A. Do not permit traffic over finished roof surface.

END OF SECTION

SECTION 07 46 23 - WOOD SIDING

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Board siding for exterior Walls .

B. Trim, flashings, accessories, and fastenings.

1.2 RELATED REQUIREMENTS

A. Section 05 40 00 - Cold-Formed Metal Framing: Water-resistive barrier under siding.

1.3 REFERENCE STANDARDS

1.4 SUBMITTALS

A. Product Data: Provide manufacturer's data on materials, component profiles, fastening methods, jointing details, sizes, surface texture, finishes, and accessories; showing compliance with requirements, including:

1. Physical characteristics of components shown on shop drawings.

2. Storage and handling requirements and recommendations.

3. Installation instructions and recommendations.

B. Samples: Submit two samples 12 inches by 12 inches in size to applicator of finish paint for use in preparation of finish samples.

C. Installer's qualification statement.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: Company specializing in installing products of the type specified in this section with minimum three years of documented experience.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Store in ventilated areas with constant minimum temperature of 60 degrees F and maximum relative humidity of 55 percent.

PART 2 PRODUCTS

2.1 SIDING

A. Board and Batten Siding: Grade A Clear or better with a maximum of 5% B clear) finger jointed, clear, maximum moisture content of 10 percent. Species match existing.

1. Size: Match existing

2. Profile: Match existing.

3. Surface Texture: Match existing.

B. Log Siding: Ship lap type siding with log shaped profile. Match existing profile and species.

C. Trim: Match existing.

2.2 ACCESSORIES

A. Fasteners for Exterior Finish Carpentry: Provide nails, in sufficient length to penetrate not less than 1-1/2 inches (38 mm) into wood furring.

1. Face-fasten siding, provide stainless steel waffle head ringed-shank siding nails with small heads.

a. Product: Stainless Steel "Slim-Jim Wood siding Nails or approved equal

1) Type: 1 3/4 inch length: SS-5-WS

2) Type: 2 inch length: SS-6-WS

B. Drainage mat: Vertically-channeled, three dimensional matrix designed to form a continuous space for drainage and air movement.

1. Material: Vinyl

2. Thickness ASTM D 5199: ¼ inch

3. Weight ASTM D 3776: 256 GSM

4. Tensile Strength ASTM D 5034: Machine 26 lbf, Crosswise 32 lbf

5. Surface Burning Characteristic ASTM E 84: Class A

C. Flashing: As specified in Section 07 62 00 - Sheet Metal Flashing and Trim.

D. Screws: Hot dipped galvanized type; non-staining, of size and strength to securely and rigidly retain the work ; prefinished to match siding finish.

E. Insect Screen: Flexible Fiberglass Screening

1. Basis of Design:

a. Adfors Saint Gobain; Product Fiberglass Screen; www.newyorkwire.com. Product characteristics noted in paragraph 2.2.H.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2. Product Characteristics:

a. Color: Charcoal

b. Mesh: 18 x 16

c. Diameter: 0.011

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that substrates are ready to receive work.

B. Verify that water-resistive barrier has been correctly and completely installed over substrate; see Section 05 40 00.

C. Do not begin until unacceptable conditions have been corrected.

D. Examine finish carpentry materials before installation. Reject materials that are wet, moisture damaged, and mold damaged.

E. If substrate preparation is the responsibility of another installer, notify Contracting Officer of unsatisfactory preparation before proceeding.

3.2 SHOP PREPARATION

A. In shop prestain all sides and edges.

B. Apply dip- or brush-type finish to site-sawn ends of pressure preservative treated materials. Allow finish to cure prior to erecting materials.

3.3 PREPARATION

A. Protect surrounding areas and adjacent surfaces during execution of this work.

B. Apply preservative treatment in accordance with manufacturer's instructions.

1. Verify materials do not exceed specified percent moisture content before applying wood preservative treatment.

2. Brush apply one coat of preservative treatment.

3.4 INSTALLATION

A. Install siding, soffits, batten strips, and trim in accordance with manufacturer's instructions.

B. Install siding in accordance with manufacturer's instructions and following the Technical Guide for Natural Wood Siding of the WWPA .

C. Install drainage mat in accordance with manufacturer's instruction.

D. Install board and batten siding:

1. Mix of new and salvaged material.

2. Finished siding should match existing.

E. Log Siding: Install siding to align with and match existing adjacent siding.

1. Nail at each stud. Do not allow nails to penetrate more than one thickness of siding.

a. Install without vertical end joints.

2. Install all nails in-line, vertically plumb and evenly spaced relative to the other nail lines. Drive nails so that heads are set slightly below the surface of the siding boards.

3. Miter external corners.

4. Butt joints only over framing or blocking, nailing top and bottom on each side and staggering joints in subsequent courses.

5. Flashing: Install metal flashing as indicated on Drawings and as recommended by siding manufacturer.

6. Allow a 1/8 inch expansion gap between pieces if siding is air or kiln-dried. Do not nail through overlaps. For siding up to 1 6 inches wide use 1 nail 1 inch up from the lap.

7. Install with minimum number of joints practical using full-length pieces for maximum length of lumber available.

8. Finish and Color:

a. Stain and seal all edges and sides

F. Fasten siding in place, level and plumb.

1. Arrange for orderly nailing pattern, blind nail except over trim.

2. Install siding for natural shed of water.

3. Position cut ends over bearing surfaces, and sand cut edges smooth and clean.

4. Scribe and cut exterior finish carpentry to fit adjoining work. Refinish and seal cuts as recommended by manufacturer.

G. Seal exposed wood substrates exposed to weather to prevent water accumulation and moisture intrusion.

1. Seal penetrations.

2. Seal exposed cuts of siding and trim; use of field-applied coatings is not permitted.

H. Set exposed nails and screws.

3.5 STANDING AND RUNNING TRIM INSTALLATION

A. Install flat grain lumber with bark side exposed to weather.

B. Install trim with minimum number of joints practical, using full-length pieces from maximum lengths of lumber available. Do not use pieces less than 24 inches long except where necessary.

1. Use scarf joints for end-to-end joints.

2. Stagger end joints in adjacent and related members.

C. Fit exterior joints to exclude water. Cope at returns and miter at corners to produce tight-fitting joints with full-surface contact throughout length of joint. Plane backs of casings to provide uniform thickness across joints, where necessary for alignment.

D. Unless otherwise indicated, countersink fasteners, fill surface flush, and sand where face fastening is unavoidable.

3.6 TOLERANCES

A. Maximum Variation From Plumb and Level: 1/4 inch per 10 feet.

B. Maximum Offset From Joint Alignment: 1/16 inch.

3.7 ADJUSTING

A. Replace siding that is damaged or does not comply with requirements. Exterior finish carpentry may be repaired or refinished if work complies with requirements and shows no evidence of repair or refinishing. Adjust joinery for uniform appearance.

3.8 CLEANING

A. Clean siding on exposed and semiexposed surfaces. Touch up factory-applied finishes to restore damaged or soiled areas.

3.9 PROTECTION

A. Protect installed products from damage from weather and other causes during construction.

B. Remove and replace finish carpentry materials that are wet, moisture damaged, and mold damaged.

1. Indications that materials are wet or moisture damaged include, but are not limited to, discoloration, sagging, or irregular shape.

2. Indications that materials are mold damaged include, but are not limited to, fuzzy or splotchy surface contamination and discoloration.

END OF SECTION

SECTION 07 62 00 - SHEET METAL FLASHING AND TRIM

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Fabricated sheet metal items, including flashings, counterflashings, gutters, downspouts, sheet metal roofing, exterior penetrations, and other items indicated in Schedule.

B. Sealants for joints within sheet metal fabrications.

1.2 REFERENCE STANDARDS

A. ASTM C920 - Standard Specification for Elastomeric Joint Sealants; 2018 (Reapproved 2024).

B. ASTM D4586/D4586M - Standard Specification for Asphalt Roof Cement, Asbestos-Free; 2007 (Reapproved 2024).

1.3 CONVERSION TABLES

A. Galvanized and Stainless Steel (per Table A-1, SMACNA)

Gauge	Steel (galv) - inch (min/max)	Steel(galv)- mm (min/max)	Stainless - inch (nom)	Stainless - mm (nom)
10	0.129 - 0.147	3.28 - 3.74	0.141	3.57
12	0.099 - 0.117	2.52 - 2.98	0.109	2.78
14	0.071 - 0.087	1.78 - 2.20	0.078	1.98
16	0.058 - 0.070	1.46 - 1.76	0.063	1.59
18	0.047 - 0.057	1.18 - 1.44	0.050	1.27
20	0.036 - 0.044	0.91 - 1.11	0.038	0.95
22	0.030 - 0.038	0.75 - 0.95	0.031	0.80
24	0.024 - 0.032	0.60 - 0.80	0.025	0.64
26	0.019 - 0.025	0.47 - 0.63	0.019	0.48
28	0.016 - 0.022	0.40 - 0.56	0.016	0.40

1.4 SUBMITTALS

A. Shop Drawings: Indicate material profile, jointing pattern, jointing details, fastening methods, flashings, terminations, and installation details.

B. Selection samples: Provide color choices for selection of finish by Contracting Officer

C. Samples: Submit two samples, 6 by 6 inches in size, illustrating metal finish color.

1.5 PROJECT CONDITIONS

A. Coordinate with the work of the following sections:

1. Section 07 31 13 - Asphalt Shingles for base flashing

1.6 WARRANTY

A. Installers warranty: The contractor shall warrant that the work will remain free from defects of labor and materials used in conjunction with his work in accordance with the general conditions for this project or a maximum of three (3) years.

1. Warranty includes installation, leaks, displacement, failed joints, and damages to other building components impacted by failure.

B. Special Finish Warranty: Submit a written warranty, signed by manufacturer, covering failure of the factory-applied exterior finish on metal wall panels within the specified warranty period and agreeing to repair finish or replace wall panels that show evidence of finish deterioration. Deterioration of finish includes, but is not limited to, color fade, chalking, cracking, peeling, and loss of film integrity.

1. Finish Warranty Period: 20 years from date of Substantial Completion.

PART 2 PRODUCTS

2.1 SHEET MATERIALS

A. General: Protect mechanical and other finishes on exposed surfaces from damage by applying a strippable, temporary protective film before shipping.

B. Unless noted otherwise provide sheet metal components from the following materials.

1. Non-exposed (non replaceable) sheet metal: Stainless-Steel Sheet: ASTM A 240/A 240M or ASTM A 666, Type 304.

a. Finish: 2D (dull, cold rolled) soft annealed, except where harder temper is required for forming or performance; minimum 0.0187 inch thick, unless otherwise indicated.

b. Surface: Smooth, flat.

2. Exposed sheet metal: Aluminum-Zinc Alloy-Coated Steel Sheet: ASTM A 792/A 792M, Class AZ50 coating designation, Grade 40 (Class AZM150 coating designation, Grade 275); structural quality or to suit project conditions, with 55 percent aluminum, not less than 0.0396 inch (1.00 mm) thick, unless otherwise indicated.

a. Exposed Finishes: Apply the following coil coating:

1) High-Performance Organic Finish: Prepare, pretreat, and apply coating to exposed metal surfaces to comply with coating and resin manufacturers' written instructions.

a) Fluoropolymer 2-Coat System: Manufacturer's standard 2-coat, thermocured system consisting of specially formulated inhibitive primer and fluoropolymer color topcoat containing not less than 70

percent polyvinylidene fluoride resin by weight; complying with physical properties and coating performance requirements of AAMA 2605, except as modified below:

1 Humidity Resistance: 2000 hours.

2 Salt-Spray Resistance: 2000 hours.

2) Color: To be selected from full range of manufacturer color options.

b. Concealed Finish: Pretreat with manufacturer's standard white or light-colored acrylic or polyester backer finish, consisting of prime coat and wash coat with a minimum total dry film thickness of 0.5 mil (0.013 mm).

C. Nails and Staples: ASTM F 1667.

2.2 FLEXIBLE FLASHING

A. General:

1. Install aluminum foil faced flashing at locations where flashing will be in contact with sealants listed in Section 07 90 05 - Joint Sealers.

2. Primer for Flexible Flashing: Product recommended by manufacturer of flexible flashing for substrate.

B. Sheet applied metal clad air barrier (metal clad air barrier): rubberized asphalt sheet membrane flashing - aluminum foil faced.(FLEXIBLE FLASHING - METAL CLAD)

1. Basis of Design:

a. Henry; Product HE-200AM. Product characteristics noted in paragraph 2.2.B.

b. Substitutions: See Section 01 67 00 – Product Requirements.

C. Air Barrier Sheet Termination Flashing: As specified in section 07 25 00 - Weather Barriers.

D. Self-Adhering, High-Temperature Sheet (HIGH TEMP MEMBRANE): Minimum 30 to 40 mils (0.76 to 1.0 mm) thick, consisting of slip-resisting polyethylene-film top surface laminated to layer of butyl or SBS-modified asphalt adhesive, with release-paper backing; cold applied. Provide primer when recommended by underlayment manufacturer.

1. Thermal Stability: ASTM D 1970; stable after testing at 240 deg F (116 deg C).

2. Low-Temperature Flexibility: ASTM D 1970; passes after testing at minus 20 deg F (29 deg C).

3. Basis of Design:

a. Henry Company; Blueskin PE200 HT.. Product characteristics noted in paragraph 2.2.D.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2.3 FABRICATED SHEET METAL COMPONENTS

A. Sheet Metal Reglet: Units of type, material, and profile indicated, formed to provide secure interlocking of separate reglet and counterflashing pieces, and compatible with flashing indicated with factory-mitered and -welded corners and junctions.

1. Basis of Design:

a. Fry Reglet Corporation. Product characteristics noted in paragraph 2.3.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2. Material: Stainless steel, 0.019 inch (0.48 mm) thick.

3. Surface-Mounted Type: Provide with slotted holes for fastening to substrate, with neoprene or other suitable weatherproofing washers, and with channel for sealant at top edge.

4. Concrete Type: Provide temporary closure tape to keep reglet free of concrete materials, special fasteners for attaching reglet to concrete forms, and guides to ensure alignment of reglet section ends.

5. Accessories:

a. Flexible-Flashing Retainer: Provide resilient plastic or rubber accessory to secure flexible flashing in reglet where clearance does not permit use of standard metal counterflashing or where Drawings show reglet without metal counterflashing.

b. Counterflashing Wind-Restraint Clips: Provide clips to be installed before counterflashing to prevent wind uplift of counterflashing lower edge.

6. Finish: Mill.

B. Flue Cover: Embossed steel cover, with integral fastening mechanism.

1. Size: 9 5/8 inch diameter.

2. Thickness: 24 gage steel.

3. Finish: Black.

2.4 FABRICATION

A. Form sections true to shape, accurate in size, square, and free from distortion or defects.

B. Form pieces in longest possible lengths.

C. Hem exposed edges on underside 1/2 inch; miter and seam corners.

D. Form material with flat lock seams, except where otherwise indicated; at moving joints, use sealed lapped, bayonet-type or interlocking hooked seams.

E. Fabricate corners from one piece with minimum 18-inch long legs; seam for rigidity, seal with sealant.

F. Fabricate flashings to allow toe to extend 2 inches over roofing gravel. Return and brake edges.

2.5 ACCESSORIES

A. Fasteners: Galvanized steel, with soft neoprene washers.

B. Primer: Zinc chromate type.

C. Concealed Sealants: Non-curing butyl sealant.

D. Exposed Sealants: ASTM C920; elastomeric sealant, with minimum movement capability as recommended by manufacturer for substrates to be sealed; color to match adjacent material.

1. As specified in Section 07 90 05

E. Asphalt Roof Cement: ASTM D4586/D4586M, Type I, asbestos-free.

F. Termination bar: Stainless steel; compatible with membrane and adhesives approximately 1 inch by 1/8 inch thick, predrilled at 9 inch centers.

1. Manufacturer: MetalEra Inc. or equal

G. Sheet Metal Skirt Flashing: Galvanized Steel; Thickness 0.028 inches

H. Sheet Metal Closure: Galvanized Steel; Thickness 0.028 inches

I. Sheet Metal Flashing Collar: Galvanized Steel; Thickness 0.028 inches

J. Sheet Metal Rain Collar: Galvanized Steel; Thickness 0.028 inches.

K. Drawband: Galvanized Steel; Thickness 0.028 inches

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify roof openings, curbs, pipes, sleeves, ducts, and vents through roof are solidly set, reglets in place, and nailing strips located.

B. Verify roofing termination and base flashings are in place, sealed, and secure.

3.2 PREPARATION

A. Install starter and edge strips, and cleats before starting installation.

B. Back paint concealed metal surfaces with protective backing paint to a minimum dry film thickness of 15 mil, 0.015 inch.

3.3 INSTALLATION

A. Secure flashings in place using concealed fasteners, and use exposed fasteners only where permitted..

B. Apply plastic cement compound between metal flashings and felt flashings.

C. Fit flashings tight in place; make corners square, surfaces true and straight in planes, and lines accurate to profiles.

D. Joint Style at roof edge flashing as noted but not limited to Gravel Stops, Facias, and Copings:

1. Lap, 4 inches wide.

2. Butt, with 12-inch- wide, concealed backup plate.

E. Metal Protection: Where dissimilar metals will contact each other or corrosive substrates, protect against galvanic action by painting contact surfaces with bituminous coating or by other permanent separation as recommended by SMACNA.

1. Coat back side of uncoated aluminum and stainless-steel sheet metal flashing and trim with bituminous coating where flashing and trim will contact wood, ferrous metal, or cementitious construction.

2. Underlayment: Where installing metal flashing directly on cementitious or wood substrates, install a course of felt underlayment and cover with a slip sheet or install a course of polyethylene sheet.

END OF SECTION

SECTION 07 90 05 - JOINT SEALERS

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Sealants and joint backing.

1.2 RELATED REQUIREMENTS

A. Section 09 21 16 - Gypsum Board Assemblies

B. Section 09 30 00 - Tiling

1.3 REFERENCE STANDARDS

A. ASTM C834 - Standard Specification for Latex Sealants; 2017 (Reapproved 2023).

B. ASTM C919 - Standard Practice for Use of Sealants in Acoustical Applications; 2024.

C. ASTM C920 - Standard Specification for Elastomeric Joint Sealants; 2018 (Reapproved 2024).

D. ASTM C1193 - Standard Guide for Use of Joint Sealants; 2025.

E. ASTM D1056 - Standard Specification for Flexible Cellular Materials--Sponge or Expanded Rubber; 2014.

F. SCAQMD 1168 - South Coast Air Quality Management District Rule No.1168; current edition.

1.4 ADMINISTRATIVE REQUIREMENTS

A. Coordinate the work with other sections referencing this section.

1.5 SUBMITTALS

A. Preinstallation Field Test Reports: Indicate which sealants and joint preparation methods resulted in optimum adhesion to joint substrates based on preinstallation testing specified in "Quality Assurance" Article.

B. Compatibility and Adhesion Test Reports: From sealant manufacturer, indicating the following:

1. Materials forming joint substrates and joint-sealant backings have been tested for compatibility and adhesion with joint sealants.

2. Interpretation of test results and written recommendations for primers and substrate preparation needed for adhesion.

C. Product Data: Provide data indicating sealant chemical characteristics.

D. Samples: Submit two samples 4 inch long, illustrating profile, dimension, color, and finish selected.

1.6 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing the Products specified in this section with minimum five years documented commercial experience.

B. Applicator Qualifications: Company specializing in performing the work of this section with minimum 5 years commercial experience.

1. Installer Qualifications: An experienced installer who has specialized in installing joint sealants similar in material, design, and extent to those indicated for this Project and whose work has resulted in joint-sealant installations with a record of successful in-service performance over the past 5 years

C. Preinstallation Compatibility and Adhesion Testing: Submit to joint-sealant manufacturers, for testing indicated below, samples of materials that will contact or affect joint sealants.

1. Use ASTM C 1087 or manufacturer's standard test method, whichever is more stringent, to determine whether priming and other specific joint preparation techniques are required to obtain rapid, optimum adhesion of joint sealants to joint substrates.

2. Schedule sufficient time for testing and analyzing results to prevent delaying the Work.

3. Submit not fewer than eight pieces of each kind of material, including joint substrates, shims, joint-sealant backings, secondary seals, and miscellaneous materials.

4. For materials failing tests, obtain joint-sealant manufacturer's written instructions for corrective measures including use of specially formulated primers.

5. Testing will not be required if joint-sealant manufacturers submit joint preparation data that are based on previous testing of current sealant products for adhesion to, and compatibility with, joint substrates and other materials matching those submitted.

D. Product Testing: Obtain test results for "Product Test Reports" Paragraph in "Submittals" Article from a qualified testing agency based on testing current sealant formulations within a 36-month period preceding the commencement of the Work.

1. Testing Agency Qualifications: An independent testing agency qualified according to ASTM C 1021 to conduct the testing indicated, as documented according to ASTM E 548.

2. Test elastomeric joint sealants for compliance with requirements specified by reference to ASTM C 920, and where applicable, to other standard test methods.

3. Include test results for hardness, stain resistance, adhesion, and cohesion under cyclic movement (per ASTM C719), low-temperature flexibility, modules of elasticity at 100 percent strain, effects of heat aging and effects of accelerated weathering.

4. Test other joint sealants for compliance with requirements indicated by referencing standard specifications and test methods.

E. Preinstallation Field-Adhesion Testing: Before installing elastomeric sealants, field test their adhesion to Project joint substrates as follows:

1. Locate test joints where indicated on Project or, if not indicated, as directed by Contracting Officer.

2. Conduct field tests for each application indicated below:

3. Revise list below to suit Project.

a. Each type of elastomeric sealant and joint substrate indicated.

4. Notify Contracting Officer seven days in advance of dates and times when test joints will be erected.

5. Arrange for tests to take place with joint-sealant manufacturer's technical representative present.

a. Test Method: Test joint sealants according to Method A, Field-Applied Sealant Joint Hand Pull Tab, in Appendix X1 in ASTM C 1193.

1) For joints with dissimilar substrates, verify adhesion to each substrate separately; extend cut along one side, verifying adhesion to opposite side. Repeat procedure for opposite side.

6. Report whether sealant in joint connected to pulled-out portion failed to adhere to joint substrates or tore cohesively. Include data on pull distance used to test each type of product and joint substrate. For sealants that fail adhesively, retest until satisfactory adhesion is obtained.

7. Evaluation of Preinstallation Field-Adhesion-Test Results: Sealants not evidencing adhesive failure from testing, in absence of other indications of noncompliance with requirements, will be considered satisfactory. Do not use sealants that fail to adhere to joint substrates during testing.

1.7 FIELD CONDITIONS

A. Maintain temperature and humidity recommended by the sealant manufacturer during and after installation.

1.8 WARRANTY

A. Correct defective work within a five year period after Date of Substantial Completion.

B. Warranty: Include coverage for installed sealants and accessories which fail to achieve watertight seal, exhibit loss of adhesion or cohesion, or do not cure.

C. Special Manufacturer's Warranty: Contractor to provide manufacturer's standard form in which elastomeric sealant manufacturer agrees to furnish elastomeric joint sealants to repair or replace those that do not comply with performance and other requirements specified in this Section within specified warranty period.

1. Warranty Period: 20 years from date of Substantial Completion.

PART 2 PRODUCTS

2.1 SEALANTS

A. Sealants and Primers - General: Provide only products having lower volatile organic compound (VOC) content than required by South Coast Air Quality Management District Rule No.1168.

B. Type PRIMARY SEAL -General Purpose Exterior Sealant: Polyurethane; ASTM C920, Grade NS, Class 25, Uses M, G, and A; single component.

1. Color: To be selected by Contracting Officer from manufacturer's standard range.

2. Applications: Use at locations indicated and not exposed to ultraviolet rays:

a. Control, expansion, and soft joints in masonry.

b. Joints between concrete and other materials.

c. Joints between metal frames and other materials.

3. Basis of Design:

a. Pecora Corporation; DynaTrol I-XL General Purpose One Part Polyurethane Sealant: www.pecora.com. Product characteristics noted in paragraph 2.1..

b. Substitutions: See Section 01 67 00 – Product Requirements.

C. Concealed Exterior Metal Lap Joint Sealant: Butyl or polyisobutylene, nondrying, nonskinning, noncuring.

1. Applications: Exposed joints in sheet metal flashing and trim or as indicated on the drawings.

D. Type SEALANT B1 -General Purpose Interior Sealant: Acrylic emulsion latex; ASTM C834, Type OP, Grade NF single component, paintable.

1. Color: To be selected by Contracting Officer from manufacturer's standard range.

2. Applications: Use for:

a. Interior wall and ceiling control joints.

b. Joints between door and window frames and wall surfaces.

c. Other interior joints for which no other type of sealant is indicated.

3. Basis of Design:

a. Pecora Corporation; AC-20 + Silicone Acrylic Latex Caulking Compound: www.pecora.com. Product characteristics noted in paragraph 2.1.D.

b. Substitutions: See Section 01 67 00 – Product Requirements.

E. Type SEALANT B3 -Bathtub/Tile Sealant: White silicone; ASTM C920, Uses I, M and A; single component, mildew resistant.

1. Applications: Use for:

a. Joints between plumbing fixtures and floor and wall surfaces.

b. Joints between kitchen and bath countertops and wall surfaces.

2. Basis of Design:

a. Pecora Corporation; 898NST Sanitary Silicone Sealant - Class 50: www.pecora.com. Product characteristics noted in paragraph 2.1.E.

b. Substitutions: See Section 01 67 00 – Product Requirements.

F. Type SEALANT B2 -Acoustical Sealant for Concealed Locations:

1. Composition: Acrylic latex emulsion sealant.

2. Applications: Use for concealed locations only at all acoustic walls:

a. Sealant bead between top stud runner and structure and between bottom stud track and floor.

3. Basis of Design:

a. Pecora Corporation; AC-20 FTR Acoustical and Insulation Sealant: www.pecora.com. Product characteristics noted in paragraph 2.1.F.

b. Substitutions: See Section 01 67 00 – Product Requirements.

4. Products: Sprays

a. Hilti, Inc.; CDS 572: www.us.hilti.com.

b. STI; Smoke "N" Sound 205; www.stifirestop.com

c. Or approved equal

G. Type SEALANT A -Silicone Sealant: ASTM C920, Grade NS, Class 50 or 100/50, Uses NT, A, G, M; single component, solvent curing, non-sagging, non-staining, fungus resistant, non-bleeding.

1. Color: To be selected by Contracting Officer from manufacturer's range.

2. Basis of Design:

a. Pecora Corporation; 890NST Ultra Low Modulus Architectural Silicone Sealant - Class 100: www.pecora.com. Product characteristics noted in paragraph 2.1.G.

b. Substitutions: See Section 01 67 00 – Product Requirements.

3. Application: Use for:

a. All locations unless noted otherwise.

H. Fire rated Sealant:

1. Type: SEALANT B4-1: Products that do not need to be painted:

a. Basis of Design:

1) Tremco Inc; Fyre-Sil. Product characteristics noted in paragraph 2.1.H.

2) Substitutions: See Section 01 67 00 – Product Requirements.

2. Type: SEALANT B4-2: Products that can be painted:

a. Basis of Design:

1) Pecora - AC-20 FTR. Product characteristics noted in paragraph 2.1.H.

2) Substitutions: See Section 01 67 00 – Product Requirements.

I. Air Seal: Where joint sealants of this type are required, provide products complying with the following:

1. Application (use for): Wood framed construction only: Spray applied insulation for filling perimeter window and door shim spaces and crevices in exterior wall and roof.

2. Basis of Design:

a. Pur Fill 1G. Product characteristics noted in paragraph 2.1.I.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2.2 ACCESSORIES

A. Primer: Non-staining type, recommended by sealant manufacturer to suit application.

B. Joint Cleaner: Non-corrosive and non-staining type, recommended by sealant manufacturer; compatible with joint forming materials.

C. Backerrod:

1. Joint Backing: Round foam rod compatible with sealant; ASTM D1056 sponge or expanded rubber; oversized 30 to 50 percent larger than joint width.

a. Basis of Design:

1) ChemLink Sof Rod. Product characteristics noted in paragraph 2.2.C.

2) Substitutions: See Section 01 67 00 – Product Requirements.

D. Bond Breaker: Pressure sensitive tape recommended by sealant manufacturer to suit application.

E. Masking Tape: Nonstaining, nonabsorbent material compatible with joint sealants and surfaces adjacent to joints.

F. Pest Control Mesh: Copper wool with special interlocked structure.

1. Basis of Design:

a. Boegger; Stuff Fit Copper Mesh; www.copper-mesh.com. Product characteristics noted in paragraph 2.2.F.

b. Substitutions: See Section 01 67 00 – Product Requirements.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that substrate surfaces are ready to receive work.

B. Verify that joint backing and release tapes are compatible with sealant.

3.2 PREPARATION

A. Remove loose materials and foreign matter that could impair adhesion of sealant.

B. Clean and prime joints in accordance with manufacturer's instructions.

C. Perform preparation in accordance with manufacturer's instructions and ASTM C1193.

D. Protect elements surrounding the work of this section from damage or disfigurement.

E. Joint Priming: Prime joint substrates where recommended by joint-sealant manufacturer or as indicated by preconstruction joint-sealant-substrate tests or prior experience. Apply primer to comply with joint-sealant manufacturer's written instructions. Confine primers to areas of joint-sealant bond; do not allow spillage or migration onto adjoining surfaces.

F. Provide continuous undisturbed separation between any bituminous materials and installed sealant to avoid discoloration of the installed sealant.

G. Masking Tape: Use masking tape where required to prevent contact of sealant or primer with adjoining surfaces that otherwise would be permanently stained or damaged by such contact or by cleaning methods required to remove sealant smears. Remove tape immediately after tooling without disturbing joint seal.

3.3 INSTALLATION

A. Perform work in accordance with sealant manufacturer's requirements for preparation of surfaces and material installation instructions.

B. Perform installation in accordance with ASTM C1193.

C. Perform acoustical sealant application work in accordance with ASTM C919.

D. Measure joint dimensions and size joint backers to achieve width-to-depth ratio, neck dimension, and surface bond area as recommended by manufacturer.

E. Install sealant backings of kind indicated to support sealants during application and at position required to produce cross-sectional shapes and depths of installed sealants relative to joint widths that allow optimum sealant movement capability.

1. Do not leave gaps between ends of sealant backings.

2. Do not stretch, twist, puncture, or tear sealant backings.

3. Remove absorbent sealant backings that have become wet before sealant application and replace them with dry materials.

F. Install bond breaker where joint backing is not used.

G. Install sealant free of air pockets, foreign embedded matter, ridges, and sags.

H. Apply sealant within recommended application temperature ranges. Consult manufacturer when sealant cannot be applied within these temperature ranges.

1. Produce uniform, cross-sectional shapes and depths relative to joint widths that allow optimum sealant movement capability.

I. Tool joints .

J. At sound-rated assemblies and elsewhere as indicated, seal construction at perimeters, behind control joints, and at openings and penetrations with a continuous bead of acoustical sealant. Install acoustical sealant at both faces of partitions at perimeters and through penetrations. Perform acoustical sealant application work in accordance with ASTM C 919 and with manufacturer's written recommendations.

K. Tooling of Nonsag Sealants: Immediately after sealant application and before skinning or curing begins, tool sealants according to requirements specified in subparagraphs below to form smooth, uniform beads of configuration indicated; to eliminate air pockets; and to ensure contact and adhesion of sealant with sides of joint.

1. Remove excess sealant from surfaces adjacent to joints.

2. Use tooling agents that are approved in writing by sealant manufacturer and that do not discolor sealants or adjacent surfaces.

3. Provide concave joint profile per Figure 8A in ASTM C 1193, unless otherwise indicated.

4. Provide flush joint profile where indicated per Figure 8B in ASTM C 1193.

5. Provide recessed joint configuration of recess depth and at locations indicated per Figure 8C in ASTM C 1193.

a. Use masking tape to protect surfaces adjacent to recessed tooled joints.

6. For traffic joint installations, tool joints to match grout installation. Sand joints after tooling.

L. Provide sanded joints at exposed locations matching grout product used in adjacent masonry, portland cement stucco, Portland cement plastering or ceramic tile.

M. Weep sealants at bottom edge of water shedding assemblies for 1 inch every 2 feet on center.

N. Install pest control mesh following manufacturer's written installation instructions.

3.4 FIELD QUALITY CONTROL

A. Field-Adhesion Testing: Field test joint-sealant adhesion to joint substrates as follows:

1. Extent of Testing: Test completed and cured sealant joints as follows:

a. Perform 5 tests for the first 1000 feet of joint length for each kind of sealant and joint substrate.

b. Perform 1 test for each 1000 feet of joint length thereafter or 1 test per each floor per elevation.

2. Test Method: Test joint sealants according to Method A, Field-Applied Sealant Joint Hand Pull Tab, in Appendix X1 in ASTM C 1193 or Method A, Tail Procedure, in ASTM C 1521.

a. For joints with dissimilar substrates, verify adhesion to each substrate separately; extend cut along one side, verifying adhesion to opposite side. Repeat procedure for opposite side.

3. Inspect tested joints and report on the following:

   a. Whether sealants filled joint cavities and are free of voids.

   b. Whether sealant dimensions and configurations comply with specified requirements.

   c. Whether sealants in joints connected to pulled-out portion failed to adhere to joint substrates or tore cohesively. Include data on pull distance used to test each kind of product and joint substrate. Compare these results to determine if adhesion passes sealant manufacturer's field-adhesion hand-pull test criteria.

4. Record test results in a field-adhesion-test log. Include dates when sealants were installed, names of persons who installed sealants, test dates, test locations, whether joints were primed, adhesion results and percent elongations, sealant fill, sealant configuration, and sealant dimensions.

5. Repair sealants pulled from test area by applying new sealants following same procedures used originally to seal joints. Ensure that original sealant surfaces are clean and that new sealant contacts original sealant.

B. Evaluation of Field-Adhesion Test Results: Sealants not evidencing adhesive failure from testing or noncompliance with other indicated requirements will be considered satisfactory. Remove sealants that fail to adhere to joint substrates during testing or to comply with other requirements. Retest failed applications until test results prove sealants comply with indicated requirements.

3.5 CLEANING

A. Clean adjacent soiled surfaces as the Work progresses by methods and with cleaning materials approved in writing by manufacturers of joint sealants and of products in which joints occur..

3.6 PROTECTION

A. Protect joint sealants during and after curing period from contact with contaminating substances and from damage resulting from construction operations or other causes so sealants are without deterioration or damage at time of Substantial Completion. If, despite such protection, damage or deterioration occurs, cut out and remove damaged or deteriorated joint sealants immediately so installations with repaired areas are indistinguishable from original work.

END OF SECTION

SECTION 08 03 14 - HISTORIC TREATMENT OF WOOD DOORS

PART 1 GENERAL

1.1 SUMMARY

A. Section Includes:

1. Historic treatment of wood doors in the form of the following:

a. Repairing wood doors and trim.

b. Replacing wood door units and trim with custom-fabricated replicated units.

c. Reglazing.

d. Repairing, refinishing, and replacing hardware.

e. Repairing storm doors.

f. Replacing storm-door units with custom-fabricated replicated units.

g. Providing new storm-door units.

B. Related Requirements:

1. Section 01 35 91 "Historic Treatment Procedures" for general historic treatment requirements.

2. Section 02 42 96 "Historic Removal and Dismantling" for historic removal and dismantling work.

3. Section 08 14 33 "Stile and Rail Wood Doors" for replacement wood doors or new replacement leaves not included in this Section.

1.2 DEFINITIONS

A. Door: Generally, this term includes door frame, leaves, hardware, side panels or lights, fan light, transom, storm and screen doors, and storm vestibule unless otherwise indicated by context.

B. Glazing: Includes glass, glazing points, glazing tapes, glazing sealants, and glazing compounds.

C. Storm Vestibule: Removable winter enclosure erected on a covered porch, which includes side panels or lights and door leaf and may include top panels.

D. Wood Door Component Terminology: Wood door components for historic treatment work include the following classifications:

1. Frame Components: Head, jambs, stop, and threshold or sill.

2. Leaf Components: Stiles, rails, and muntins.

3. Exterior Trim: Exterior casing, brick mold, and cornice or drip cap.

4. Interior Trim: Casing.

1.3 REFERENCE STANDARDS

A. ASTM B633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel; 2013.

B. ASTM C509 - Standard Specification for Elastomeric Cellular Preformed Gasket and Sealing Material; 2006 (Reapproved 2021).

C. ASTM C864 - Standard Specification for Dense Elastomeric Compression Seal Gaskets, Setting Blocks, and Spacers; 2005 (Reapproved 2011).

D. ASTM C920 - Standard Specification for Elastomeric Joint Sealants; 2018 (Reapproved 2024).

E. BHMA A156.18 - Standard for Materials and Finishes; 2020.

1.4 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1. Review minutes of Preliminary Historic Treatment Conference that pertain to historic treatment of wood doors.

2. Review methods and procedures related to historic treatment of wood doors including, but not limited to, the following:

a. Historic treatment specialist's personnel, equipment, and facilities needed to make progress and avoid delays.

b. Materials, material application, sequencing, tolerances, and required clearances.

c. Fire-protection plan.

d. Wood door historic treatment program.

e. Coordination with building occupants.

1.5 SEQUENCING AND SCHEDULING

A. Perform historic treatment of wood doors in the following sequence, which includes Work specified in this and other Sections:

1. Label each door frame with permanent opening-identification number in inconspicuous location.

2. Tag existing door leaves, storm doors, storm-vestibule panels, and screen doors with opening-identification numbers and remove for on-site or off-site repair. Indicate on tags

the locations of each component, such as "left-hand door leaf," "right-hand reverse door leaf," "top dutch-door leaf," "bottom dutch-door leaf," "first left-side storm-vestibule panel," and "second left-side storm-vestibule panel."

3. Remove door, dismantle hardware, and tag hardware with door opening-identification numbers.

4. In the shop, label each leaf, storm door, storm-vestibule panel, and screen door with permanent opening-identification number in inconspicuous location and remove site-applied tags.

5. Install temporary protection and security at door openings.

6. Sort units by condition, separating those that need extensive repair.

7. Clean surfaces.

8. General Wood-Repair Sequence:

    a. Remove paint to bare wood.

    b. Rack frames slightly to inject adhesive into mortise and tenon joints; square frames to proper fit before adhesive sets.

    c. If glass thicker than original is required, rout existing muntins to required rebate size.

    d. Repair wood by consolidation, member replacement, partial member replacement, and patching.

    e. Sand, prime, fill, sand again, and prime surfaces again for refinishing.

9. Repair, refinish, and replace hardware if required. Reinstall operating hardware.

10. Install glazing.

11. Remove temporary protection and security at door openings.

12. Reinstall units.

13. Apply finish coats.

14. Install remaining hardware and weather stripping.

1.6 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include recommendations for product application and use. Include test data substantiating that products comply with requirements.

B. Shop Drawings: For locations and extent of wood-door repair and replacement work.

1. Include plans, elevations, sections, and details of replacement parts indicating materials, profiles, joinery, reinforcing, method of splicing into or attaching to existing wood door, accessory items, and finishes.

2. Include field-verified dimensions and the following:

a. Full-size shapes and profiles with complete dimensions for replacement components and their jointing, showing relation of existing to new components.

b. Templates and directions for installing hardware and anchorages.

c. Identification of each new unit and its corresponding door locations in the building on annotated plans and elevations.

d. Provisions for sealant joints as required for location.

C. Samples for Initial Selection: For each type of exposed wood and finish.

1. Identify wood species, cut, and other features.

2. Include Samples of hardware and accessories involving color selection.

D. Samples for Verification: Actual sample of finished products for the following, in manufacturer's standard sizes unless otherwise indicated:

1. Replacement Units: 12-inch- (300-mm-) long, full-size frame sections with applied finish.

2. Replacement Members: 12 inches (300 mm) long for each replacement member, including parts of frame, leaf, exterior trim, and interior trim.

a. Additional Samples of replacement members that show fabrication techniques, materials, and finishes as requested by Contracting Officer.

3. Repaired Wood Door Members: Prepare Samples using existing wood door members removed from site, repaired, and prepared for refinishing.

4. Refinished Wood Door Members: Prepare Samples using existing wood door members removed from site, repaired, and refinished.

5. Hardware: Full-size units with each factory-applied or restored finish.

6. Weather Stripping: 12-inch- (300-mm-) long sections.

7. Glass: Full-size units of each type and appearance.

1.7 INFORMATIONAL SUBMITTALS

A. Test and Evaluation Reports:

1. Preconstruction Test Reports: For historic treatment of wood doors.

B. Qualification Statements: For historic treatment specialist.

C. Wood Door Historic Treatment Program: Submit before work begins.

1.8 QUALITY ASSURANCE

A. Historic Treatment Specialist Qualifications: A qualified historic wood door specialist, experienced in repairing, refinishing, and replacing wood doors in whole and in part. Experience only in fabricating and installing new wood doors is insufficient experience for wood-door historic treatment work.

B. Wood-Repair-Material Manufacturer Qualifications: A firm regularly engaged in producing wood consolidant that have been used for similar historic wood-treatment applications with successful results, and with factory-authorized service representatives who are available for consultation and Project-site inspection and on-site assistance.

C. Wood Door Historic Treatment Program: Prepare a written, detailed description of materials, methods, equipment, and sequence of operations to be used for historic treatment work, including protection of surrounding materials and Project site.

1. If materials and methods other than those indicated are proposed for any phase of historic treatment work, add a written description of such materials and methods, including evidence of successful use on comparable projects, and demonstrations to show their effectiveness for this Project.

1.9 MOCKUPS

A. Prepare mockups of historic treatment repair processes to demonstrate aesthetic effects, to set quality standards for materials and execution, and to set quality standards for fabrication and installation. Prepare mockups so they are inconspicuous.

1. Locate mockups in locations that enable viewing under same conditions as the completed Work.

2. Wood Door Repair: Prepare one entire door unit to serve as mockup to demonstrate Samples of each type of repair of wood door members including frame, leaves, trim, glazing, and hardware.

a. Provide three level repair sample (good, better, best) for Contracting officer to decide which one becomes the basis of the execution of repair.

3. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations by Change Order.

4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

1.10 PRECONSTRUCTION TESTING

A. Preconstruction Testing Service: Engage a qualified historic treatment specialist to perform preconstruction testing on historic wood doors.

1. Provide test specimens representative of proposed materials and existing construction.

2. Test historic treatment products and methods for effectiveness and compliance with specified requirements.

1.11 DELIVERY, STORAGE, AND HANDLING

A. Pack, deliver, and store products in suitable packs, heavy-duty cartons, or wooden crates; surround with sufficient packing material to ensure that products will not be deformed, broken, or otherwise damaged.

B. Store products inside a well-ventilated area, protected from weather, moisture, soiling, abrasion, extreme temperatures, and humidity, and where environmental conditions comply with manufacturer's requirements.

1.12 FIELD CONDITIONS

A. Weather Limitations: Proceed with historic treatment of wood doors only when existing and forecasted weather conditions are within environmental limits set by each manufacturer's written instructions and specified requirements.

PART 2 PRODUCTS

2.1 HISTORIC TREATMENT OF WOOD DOORS QUALITY STANDARD

A. Quality Standard: Comply with applicable requirements in Section 12, "Historic Restoration Work," and related requirements in AWMAC/WI's "North American Architectural Woodwork Standards" for construction, finishes, grades of wood doors, and other requirements unless otherwise indicated.

1. Exception: Industry practices cited in Section 12, Paragraph 6, "Industry Practices," under Article 12.1, "Basic Considerations," of AWMAC/WI's "North American Architectural Woodwork Standards" do not apply to the Work of this Section.

2.2 REPLACEMENT WOOD DOOR UNITS

A. Replacement Wood Door Units: Custom-fabricated, replicated wood door units and trim with operating and latching hardware.

1. Wood Door Components: Replace frames.

2. Joint Construction: Joints matching existing.

3. Wood Species: Match wood species of existing door components.

4. Wood Cut: Match cut of existing wood door components.

5. Wood Member and Trim Profiles: Match profiles and detail of existing door members and trim.

6. Hardware: Reuse existing unless otherwise indicated on drawings and per Section 08 71 00 - Door Hardware.

7. Hardware Set: Door Hardware Set No. in accordance with Section 08 71 00 - Door Hardware.

8. Glazing Stops: Provide replacement glazing stops coordinated with glazing system indicated.

9. Weather Stripping: Full-perimeter weather stripping for each exterior door leaf.

10. Date Identification: Emboss on a concealed surface of each replaced door frame and leaf, in easily read characters, "DOOR FRAME MADE [Insert year]" or "DOOR LEAF MADE [Insert year]." Manufacturer's name may also be embossed.

2.3 SCREEN DOORS

A. General: Custom fabricated, tight fitting, replicating appearance of existing screen door, and with operating and latching hardware.

B. Joint Construction: Joints matching existing .

C. Wood Species: Match wood species of existing screen door.

D. Wood Cut: Match cut of existing screen door .

E. Wood Member Profiles: Match design reference sample.

F. Date Identification: Emboss on a concealed surface of each screen door, in easily read characters, "SCREEN DOOR MADE [Insert year]." Manufacturer's name may also be embossed.

2.4 WOOD-REPLACEMENT MATERIALS

A. Wood, General: Clear fine-grained lumber; kiln dried to a moisture content of 6 to 12 percent at time of fabrication; free of visible finger joints, blue stain, knots, pitch pockets, and surface checks larger than 1/32 inch (0.8 mm) deep by 2 inches (51 mm) wide.

1. Species: Match species of each existing type of wood component or assembly unless otherwise indicated.

B. Frame Heads and Jambs and Exterior Trim: Match existing species.

C. Exterior Trim: Match existing species.

D. Thresholds or Sills: Match existing species.

E. Leaf Components: Match existing species.

F. Interior Trim: Match existing species

2.5 WOOD-REPAIR MATERIALS

A. Source Limitations: Obtain wood consolidant and wood-patching compound from single source from single manufacturer.

B. Wood Consolidant: Ready-to-use product designed to penetrate, consolidate, and strengthen soft fibers of wood materials that have deteriorated because of weathering and decay and designed specifically to enhance the bond of wood-patching compound to existing wood.

1. As noted in Section 04 03 42 - Historic Wood Repair

C. Wood-Patching Compound: Two-part, epoxy-resin, wood-patching compound; knife-grade formulation as recommended in writing by manufacturer for type of wood repair indicated, tooling time required for the detail of work, and site conditions. Compound to be designed for filling voids in damaged wood materials that have deteriorated because of weathering and decay. Compound to be capable of filling deep holes and spreading to feather edge.

2.6 GLAZING MATERIALS

A. Glass:

1. See Section 08 80 00 "Glazing."

B. Glazing Systems:

1. Traditional Glazing Products: Glazing points and oil-based glazing putty. Tint to required color in accordance with manufacturer's written instructions.

2. Basis of Design:

a. Dana Lim: https://www.danalim.com/product/linseed-oil-putty-683. Product characteristics noted in paragraph 2.6.

b. Substitutions: See Section 01 67 00 – Product Requirements.

3. Primers and Cleaners for Glazing: As recommended in writing by glazing material manufacturer.

2.7 WEATHER STRIPPING

A. Metal Weather Stripping: Bronze weather stripping; designed either as one piece to seal door at head and jambs by door sliding against it or as two pieces that interlock; and completely concealed when door is closed.

2.8 MISCELLANEOUS MATERIALS

A. Insect Screening: Replace existing screen "in-kind:"

1. Bronze Wire Fabric: 18-by-14 count per sq. in. (1.1-by-1.5-mm) mesh of 0.009-inch-(0.23-mm-) diameter bronze wire with a clear varnish finish.

B. Borate Preservative Treatment: Inorganic, borate-based solution, with disodium octaborate tetrahydrate as the primary ingredient; manufactured for preserving weathered and decayed wood from further damage by decay fungi and wood-boring insects; containing no boric acid.

C. Cleaning Materials:

1. Detergent Solution: Solution prepared by mixing 2 cups (0.5 L) of tetrasodium pyrophosphate, 1/2 cup (125 mL) of laundry detergent that contains no ammonia, 5 quarts (5 L) of 5 percent sodium hypochlorite bleach, and 15 quarts (15 L) of warm water for each 5 gal. (20 L) of solution required.

2. Mildewcide: Commercial, proprietary mildewcide or a solution prepared by mixing 1/3 cup (80 mL) of household detergent that contains no ammonia, 1 quart (1 L) of 5 percent sodium hypochlorite bleach, and 3 quarts (3 L) of warm water.

D. Adhesives: Wood adhesives with minimum 15- to 45-minute cure at 70 deg F (21 deg C), in gunnable and liquid formulations as recommended in writing by adhesive manufacturer for each type of repair and exposure conditions.

E. Fasteners: Use fastener metals that are noncorrosive and compatible with each material joined.

1. Match existing fasteners in material and type of fastener unless otherwise indicated.

2. Use concealed fasteners for interconnecting wood components.

3. Use concealed fasteners for attaching items to other work unless exposed fasteners are unavoidable.

4. For fastening metals, use fasteners of same basic metal as fastened metal unless otherwise indicated.

5. For exposed fasteners, use Phillips-type machine screws of head profile flush with metal surface unless otherwise indicated.

6. Finish exposed fasteners to match finish of metal fastened unless otherwise indicated.

F. Anchors, Clips, and Accessories: Fabricate anchors, clips, and door accessories of aluminum, nonmagnetic stainless steel, or hot-dip zinc-coated steel complying with requirements in ASTM B633 for SC 3 (Severe) service condition.

2.9 WOOD DOOR FINISHES

A. Unfinished Replacement Units: Provide exposed exterior wood surfaces of replacement units unfinished; smooth, filled, and suitably prepared for on-site priming and finishing.

B. Linseed Oil Paint or transparent wood Finish: See 09 91 13 Exterior Painting and 09 91 23 - Interior Painting.

PART 3 EXECUTION

3.1 HISTORIC TREATMENT SPECIALIST

A. Historic Treatment Specialist Firms: Subject to compliance with requirements, provide historic treatment of wood doors by one of the following:

1. Firm meeting Part 1 Quality Assurance requirements.

3.2 PREPARATION

A. Protect adjacent materials from damage by historic treatment of wood doors.

B. Clean wood doors and trim of mildew, algae, moss, plant material, loose paint, grease, dirt, and other debris by scrubbing with bristle brush or sponge and detergent solution. Scrub mildewed areas with mildewcide. After cleaning, rinse thoroughly with fresh water. Allow to dry before repairing or painting.

C. Condition replacement wood members and replacement units to prevailing conditions at installation areas before installing.

3.3 HISTORIC TREATMENT OF WOOD DOORS, GENERAL

A. Historic Treatment Appearance Standard: Completed work is to have a uniform appearance as viewed by Contracting Officer from the door interior at 5 ft. (1.5 m) away and from the door exterior at 20 ft. (6 m) away.

B. General: In treating historic items, disturb them as minimally as possible and as follows:

1. Stabilize and repair wood doors to reestablish structural integrity and weather resistance while maintaining the existing form of each item.

2. Remove coatings and apply borate preservative treatment before repair. Remove coatings in accordance with Section 09 91 13 - Exterior Painting unless otherwise indicated.

3. Repair items in place where possible.

4. Install temporary protective measures to protect wood door work that is indicated to be completed later.

C. Mechanical Abrasion: Where mechanical abrasion is needed for the Work, use only the gentlest mechanical methods, such as scraping and natural-fiber bristle brushing, that will not abrade wood substrate, reducing clarity of detail. Do not use abrasive methods such as sanding, wire brushing, or power tools except as indicated as part of the historic treatment program and as approved by Contracting Officer.

D. Repair and Refinish Existing Hardware: Dismantle door hardware; strip paint, repair, and refinish it to match finish Samples; and lubricate moving parts just enough to function smoothly.

E. Repair Wood Doors: Match existing materials and features, retaining as much original material as possible to perform repairs.

1. Unless otherwise indicated, repair wood doors by consolidating, patching, splicing, or otherwise reinforcing wood with new wood matching existing wood or with salvaged, sound, original wood.

2. Where indicated, repair wood doors by limited replacement matching existing material.

F. Replace Wood Units: Where indicated, duplicate and replace units with units made from salvaged, sound, original wood or with new wood matching existing wood. Use surviving prototypes to create patterns for duplicate replacements.

1. Do not use substitute materials unless otherwise indicated.

2. Compatible substitute materials may be used.

G. Protection of Openings: Where doors are indicated for removal, cover resultant openings with temporary enclosures so that openings are weathertight during repair period.

H. Identify removed doors, frames, leaves, trim, and members with numbering system corresponding to door locations to ensure reinstallation in same location. Key doors, frames, leaves, trim, and members to Drawings showing location of each removed unit. Permanently label units in a location that will be concealed after reinstallation.

3.4 WOOD DOOR PATCH-TYPE REPAIR

A. General: Patch wood members that exhibit depressions, holes, or similar voids and that have limited amounts of rotted or decayed wood.

1. Remove leaves from door frames before performing patch-type repairs at meeting or sliding surfaces unless otherwise indicated. Reglaze units prior to reinstallation.

2. Verify that surfaces are sufficiently clean and free of paint residue before patching.

3. Treat wood members with wood consolidant before applying patching compound. Coat wood surfaces by brushing, applying multiple coats until wood is saturated and unable to absorb more. Allow treatment to harden before filling void with patching compound.

4. Remove rotted or decayed wood down to sound wood.

B. Apply borate preservative treatment to accessible surfaces either before applying wood consolidant or after removing rotted or decayed wood. Apply treatment liberally by brush to joints, edges, and ends; top, sides, and bottom. Allow treatment to dry.

C. Apply wood-patching compound to fill depressions, nicks, cracks, and other voids created by removed or missing wood.

1. Prime patch area with application of wood consolidant or manufacturer's recommended primer.

2. Mix only as much patching compound as can be applied in accordance with manufacturer's written instructions.

3. Apply patching compound in layers as recommended in writing by manufacturer until the void is completely filled.

4. Sand patch surface smooth and flush with adjacent wood, without voids in patch material, and matching contour of wood member.

5. Clean spilled compound from adjacent materials immediately.

3.5 WOOD DOOR MEMBER-REPLACEMENT REPAIR

A. General: Replace parts of or entire wood door members at locations indicated on Drawings.

1. Remove leaves from door frames before performing member-replacement repairs unless otherwise indicated.

2. Verify that surfaces are sufficiently clean and free of paint residue before repair.

3. Remove broken, rotted, and decayed wood down to sound wood.

4. Custom fabricate new wood to replace missing wood; either replace entire wood member or splice new wood part into existing member.

5. Secure new wood using finger joints, multiple dowels, or splines with adhesive and nailing to ensure maximum structural integrity at each splice. Use only concealed fasteners. Fill nail holes and patch surface to match surrounding sound wood.

B. Apply borate preservative treatment to accessible surfaces after replacements are made. Apply treatment liberally by brush to joints, edges, and ends; top, sides, and bottom.

C. Repair remaining depressions, holes, or similar voids with patch-type repairs.

D. Clean spilled materials from adjacent surfaces immediately.

E. Glazing: Reglaze units before reinstallation.

1. Mill new and rout existing glazed members to accommodate new glass thickness.

2. Provide replacement glazing stops coordinated with glazing system indicated.

3. Provide glazing stops to match contour of door frames.

F. Reinstall units removed for repair into original openings.

G. Weather Stripping: Replace nonfunctioning and install missing weather stripping to ensure full-perimeter weather stripping for each exterior leaf.

3.6 GLAZING

A. Comply with combined written instructions of glass, glazing system, and glazing material manufacturers, unless more stringent requirements are indicated.

B. Remove cracked and damaged glass and glazing materials from openings and prepare surfaces for reglazing.

C. Remove existing glass and glazing where indicated on Drawings, and prepare surfaces for reglazing.

D. Size glass as required by Project conditions to provide necessary bite on glass, minimum edge and face clearances, with reasonable tolerances.

E. Apply primers to joint surfaces where required for adhesion of glazing system, as determined by preconstruction testing (such as shellac at consolidated areas).

F. Apply linseed oil to all bed surfaces

G. Install setting bead, side beads, and back bead against stop in glazing rabbets before setting glass.

H. Install glass with proper orientation so that coatings, if any, face exterior or interior as required.

I. Install glazing points.

J. Reinstall wood trim if present or install a chamfered glazing bead at exterior.

K. Disposal of Removed Glass: Remove from Government property and legally dispose unless otherwise indicated.

3.7 WOOD DOOR UNIT REPLACEMENT

A. General: Replace existing wood door-frame units with new custom-fabricated replicated units at locations indicated on Drawings.

B. Apply borate preservative treatment to accessible surfaces before finishing. Apply treatment liberally by brush to joints, edges, and ends; top, sides, and bottom.

C. Mill glazed members to accommodate glass thickness. Glaze units before installation.

D. Install units, hardware, weather stripping, accessories, and other components as indicated on Drawings.

E. Install units level, plumb, square, true to line, without distortion or impeding movement, anchored securely in place to structural support, and in proper relation to wall flashing, trim, and other adjacent construction.

F. Set threshold or sill members in bed of sealant for weathertight construction unless otherwise indicated.

G. Install door units with new anchors into existing openings.

H. Install full-perimeter weather stripping for each operable exterior leaf.

I. Metal Protection: Separate aluminum and other corrodible surfaces from sources of corrosion or electrolytic action at points of contact with other materials.

J. Disposal of Removed Units: Remove from Government property and legally dispose of them.

3.8 INSTALLATION OF SCREEN DOORS

A. Fit wood screen doors at each door jamb indicated. Verify that screen door and door frames are correctly tagged with opening-identification numbers.

B. Adjust hardware, clips, and removable fasteners for a tight fit with uniform joints. Uninstall and reinstall door at least once to ensure proper fit.

C. Install and leave screen doors in place when directed by Contracting Officer.

D. Install units by mounting to door frames in accordance with manufacturer's written instructions.

E. Replace existing insect screening where indicated; remove from Government's property.

F. Install insect screening to be smooth, flat, and uniformly taut.

3.9 INSTALLATION OF WEATHER STRIPPING

A. Install weather stripping for tight seal of joints as determined by preconstruction testing and demonstrated in mockup.

3.10 ADJUSTING

A. Adjust existing and replacement operating leaves, screens, hardware, weather stripping, and accessories for a tight fit at contact points and weather stripping for smooth operation and weathertight closure. Lubricate hardware and moving parts.

3.11 CLEANING AND PROTECTION

A. Protect door surfaces from contact with contaminating substances resulting from construction operations. Monitor door surfaces adjacent to and below exterior concrete and masonry during construction for presence of dirt, scum, alkaline deposits, stains, or other contaminants. If contaminating substances contact door surfaces, remove contaminants immediately.

B. Clean exposed surfaces immediately after historic treatment of wood doors. Avoid damage to coatings and finishes. Remove excess sealants, glazing and patching materials, dirt, and other substances.

C. Remove and replace glass that has been broken, chipped, cracked, abraded, or damaged during construction period.

END OF SECTION

SECTION 08 11 13 - HOLLOW METAL DOORS AND FRAMES

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Thermally insulated hollow metal doors with frames.

1.2 RELATED REQUIREMENTS

A. Section 08 71 00 - Door Hardware.

1.3 REFERENCE STANDARDS

A. ADA Standards - 2010 ADA Standards for Accessible Design; 2010.

B. ANSI/ICC A117.1 - American National Standard for Accessible and Usable Buildings and Facilities; International Code Council; 2009.

C. ANSI/SDI A250.4 - Test Procedure and Acceptance Criteria for Physical Endurance for Steel Doors, Frames and Frame Anchors; 2026.

D. ANSI/SDI A250.8 - Specifications for Standard Steel Doors and Frames (SDI-100); 2023.

E. ANSI/SDI A250.10 - Test Procedure and Acceptance Criteria for Prime Painted Steel Surfaces for Steel Doors and Frames; 2025.

F. ASTM A653/A653M - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process; 2023.

G. ASTM A1008/A1008M - Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Required Hardness, Solution Hardened, and Bake Hardenable; 2024.

H. ASTM A1011/A1011M - Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength; 2023.

I. ASTM E84 - Standard Test Method for Surface Burning Characteristics of Building Materials; 2024.

J. BHMA A156.115 - Hardware Preparation in Steel Doors and Frames; 2016.

K. ICC A117.1 - Accessible and Usable Buildings and Facilities; 2017.

L. NAAMM HMMA 830 - Hardware Selection for Hollow Metal Doors and Frames; 2002.

M. NAAMM HMMA 831 - Hardware Locations for Hollow Metal Doors and Frames; 2024.

N. NAAMM HMMA 840 - Guide Specifications for Receipt, Storage and Installation of Hollow Metal Doors and Frames; 2024.

1.4 SUBMITTALS

A. Product Data: Materials and details of design and construction, hardware locations, reinforcement type and locations, anchorage and fastening methods, and finishes; and one copy of referenced standards/guidelines.

1.5 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section, with not less than three years documented experience.

B. Maintain at project site copies of reference standards relating to installation of products specified.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Comply with NAAMM HMMA 840 or ANSI/SDI A250.8 (SDI-100) in accordance with specified requirements.

B. Protect with resilient packaging; avoid humidity build-up under coverings; prevent corrosion and adverse effects on factory applied painted finish.

PART 2 PRODUCTS

2.1 MANUFACTURERS

A. Basis of Design:

1. Stiles; www.stilesdoors.com. Product characteristics noted in paragraph 2.2 through 2.6.

2. Substitutions: See Section 01 67 00 – Product Requirements.

2.2 PERFORMANCE REQUIREMENTS

A. Requirements for Hollow Metal Doors and Frames:

1. Steel Sheet: Comply with one or more of the following requirements; galvannealed steel complying with ASTM A653/A653M, cold-rolled steel complying with ASTM A1008/A1008M, or hot-rolled pickled and oiled (HRPO) steel complying with ASTM A1011/A1011M, commercial steel (CS) Type B, for each.

2. General: Provide flush-design doors, 1-3/4 inches thick, of seamless hollow construction, unless otherwise indicated. Construct doors with smooth, flush surfaces without visible joints or seams on exposed faces or stile edges.

a. Visible joints or seams around glazed or louvered panel inserts are permitted.

b. For single-acting swing doors, bevel both vertical edges 1/8 inch in 2 inches.

3. Accessibility: Comply with ICC A117.1 and ADA Standards.

4. Door Edge Profile: Manufacturers standard for application indicated.

5. Typical Door Face Sheets: Flush.

6. Vertical steel stiffeners: 0.026-inch steel vertical stiffeners extending full-door height, spaced not more than 6 inches apart and spot welded to face sheets a maximum of 6 inches o.c. Fill spaces between stiffeners with insulation of minimum 0.6-lb/cu. ft. density or sound deadener applied to inside surfaces of face sheets.

7. Hardware Preparations, Selections and Locations: Comply with NAAMM HMMA 830 and NAAMM HMMA 831 or BHMA A156.115 and ANSI/SDI A250.8 (SDI-100) in accordance with specified requirements.

8. Zinc Coating for Typical Interior and/or Exterior Locations: Provide metal components zinc-coated (galvanized) and/or zinc-iron alloy-coated (galvannealed) by the hot-dip process in accordance with ASTM A653/A653M, with manufacturer's standard coating thickness, unless noted otherwise for specific hollow metal doors and frames.

B. Combined Requirements: If a particular door and frame unit is indicated to comply with more than one type of requirement, comply with the specified requirements for each type; for instance, an exterior door that is also indicated as being sound-rated must comply with the requirements specified for exterior doors and for sound-rated doors; where two requirements conflict, comply with the most stringent.

2.3 HOLLOW METAL DOORS

A. Exterior Doors: Thermally insulated.

1. Based on SDI Standards: ANSI/SDI A250.8 (SDI-100).

a. Level 3 - Extra Heavy-duty.

b. Physical Performance Level A, 1,000,000 cycles; in accordance with ANSI/SDI A250.4.

c. Model 1 - Full Flush.

d. Door Face Metal Thickness: 16 gauge, 0.053 inch, minimum.

e. Zinc Coating: A60/ZF180 galvannealed coating; ASTM A653/A653M.

2. Door Core Material: Polyisocyanurate, 2 lbs/cu ft minimum density.

a. Foam Plastic Insulation: Manufacturer's standard board insulation with maximum flame spread index (FSI) of 75, and maximum smoke developed index (SDI) of

450 in accordance with ASTM E84, and completely enclosed within interior of door.

3. Door Thermal Resistance: R-Value of 9.9, minimum, for installed thickness of polyisocyanurate.

4. Door Thickness: 1-3/4 inches, nominal.

5. Door Finish: Factory primed and field finished.

2.4 HOLLOW METAL FRAMES

A. Comply with standards and/or custom guidelines as indicated for corresponding door in accordance with applicable door frame requirements.

B. General:

1. Comply with the requirements of grade specified for corresponding door.

a. ANSI A250.8 Level 3 Doors:

1) For exterior use, form frames from 0.067-inch- thick, metallic-coated steel sheets.

2) For interior use, form frames from cold- or hot-rolled steel sheet of the following thicknesses:

a) Openings up to 48 Inches Wide: 0.053 inch (16 gage).

2. Spreader Bars: Provide removable spreader bar across bottom of frames, tack welded to jambs and mullions.

3. Hardware Reinforcement: Fabricate according to ANSI/SDI A250.6 with reinforcement plates from same material as frames.

4. Mullions and Transom Bars: Provide closed or tubular mullions and transom bars where indicated. Fasten mullions and transom bars at crossings and to jambs by butt welding. Reinforce joints between frame members with concealed clip angles or sleeves of same metal and thickness as frame.

a. Provide false head member to receive lower ceiling where frames extend to finish ceilings of different heights.

5. Head Reinforcement: Provide minimum 0.093-inch- thick, steel channel or angle stiffener for opening widths more than 48 inches .

C. Exterior Door Frames: Full profile/continuously welded type.

1. Galvanizing: Components hot-dipped zinc-iron alloy-coated (galvannealed) in accordance with ASTM A653/A653M, with A40/ZF120 coating.

2. Frame Metal Thickness: 14 gauge, 0.067 inch, minimum.

3. Frame Finish: Factory primed and field finished.

4. Weatherstripping: Separate, see Section 08 71 00.

2.5 FINISHES

A. Primer: Rust-inhibiting, complying with ANSI/SDI A250.10, door manufacturer's standard.

2.6 FRAME ANCHORS

A. Jamb Anchors:

1. Stud-Wall Type: Designed to engage stud, welded to back of frames; not less than 0.042 inch thick.

B. Floor Anchors: Formed from same material as frames, not less than 0.042 inch thick, and as follows:

1. Monolithic Concrete Slabs: Clip-type anchors, with two holes at each jamb to receive fasteners.

C. Head Anchors: Provide 2 head anchors for frames more than 42 inches wide and mounted in steel-stud walls.

2.7 ACCESSORY MATERIALS

A. Silencers: Resilient rubber, fitted into drilled hole; 3 on strike side of single door, 3 on center mullion of pairs, and 2 on head of pairs without center mullions.

2.8 FABRICATION

A. Fabricate hollow metal work to be rigid and free of defects, warp, or buckle. Accurately form metal to required sizes and profiles, with minimum radius for thickness of metal. Where practical, fit and assemble units in manufacturer's plant. To ensure proper assembly at Project site, clearly identify work that cannot be permanently factory assembled before shipment.

B. Tolerances: Fabricate hollow metal work to tolerances indicated in SDI 117 [ANSI/NAAMM-HMMA 861].

C. Hollow Metal Doors:

D. Hollow Metal Frames: Where frames are fabricated in sections due to shipping or handling limitations, provide alignment plates or angles at each joint, fabricated of same thickness metal as frames.

1. Welded Frames: Weld flush face joints continuously; grind, fill, dress, and make smooth, flush, and invisible.

2. Provide countersunk, flat- or oval-head exposed screws and bolts for exposed fasteners unless otherwise indicated.

3. Floor Anchors: Weld anchors to bottom of jambs and mullions with at least four spot welds per anchor.

4. Jamb Anchors: Provide number and spacing of anchors as follows:

   a. Stud-Wall Type: Locate anchors not more than 18 inches from top and bottom of frame. Space anchors not more than 32 inches o.c. and as follows:

      1) Four anchors per jamb from 60 to 90 inches high.

5. Door Silencers: Except on weather-stripped doors, drill stops to receive door silencers as follows. Keep holes clear during construction.

   a. Single-Door Frames: Drill stop in strike jamb to receive three door silencers.

   b. Double-Door Frames: Drill stop in head jamb to receive two door silencers.

E. Fabricate concealed stiffeners, edge channels, and hardware reinforcement from either cold- or hot-rolled steel sheet.

F. Hardware Preparation: Factory prepare hollow metal work to receive templated mortised hardware; include cutouts, reinforcement, mortising, drilling, and tapping according to the Door Hardware Schedule and templates furnished as specified in Division 08 Section "Door Hardware."

1. Locate hardware as indicated, or if not indicated, according to ANSI/SDI A250.8

2. Comply with applicable requirements in ANSI/SDI A250.6 and ANSI/DHI A115 Series specifications for preparation of hollow metal work for hardware.

G. Stops and Moldings: Provide stops and moldings around glazed lites where indicated. Form corners of stops and moldings with mitered hairline joints.

1. Single Glazed Lites: Provide fixed stops and moldings welded on secure side of hollow metal work.

2. Multiple Glazed Lites: Provide fixed and removable stops and moldings so that each glazed lite is capable of being removed independently.

3. Provide fixed frame moldings on outside of exterior and on secure side of interior doors and frames.

4. Provide loose stops and moldings on inside of hollow metal work.

5. Coordinate rabbet width between fixed and removable stops with type of glazing and type of installation indicated.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify existing conditions before starting work.

B. Verify that opening sizes and tolerances are acceptable.

C. Verify that finished walls are in plane to ensure proper door alignment.

3.2 PREPARATION

A. Coat inside of frames to be installed in masonry or to be grouted, with bituminous coating, prior to installation.

3.3 INSTALLATION

A. Install doors and frames in accordance with manufacturer's instructions and related requirements of specified door and frame standards or custom guidelines indicated.

B. Coordinate frame anchor placement with wall construction.

C. Install door hardware as specified in Section 08 71 00.

D. Hollow-Metal Frames: Install hollow-metal frames of size and profile indicated. Comply with SDI A250.11 or NAAMM-HMMA 840 as required by standards specified.

1. Set frames accurately in position; plumbed, aligned, and braced securely until permanent anchors are set. After wall construction is complete, remove temporary braces, leaving surfaces smooth and undamaged.

    a. Where frames are fabricated in sections because of shipping or handling limitations, field splice at approved locations by welding face joint continuously; grind, fill, dress, and make splice smooth, flush, and invisible on exposed faces.

    b. Install frames with removable stops located on secure side of opening.

    c. Install door silencers in frames before grouting.

    d. Remove temporary braces necessary for installation only after frames have been properly set and secured.

    e. Check plumb, square, and twist of frames as walls are constructed. Shim as necessary to comply with installation tolerances.

    f. Field apply bituminous coating to backs of frames that will be filled with grout containing anti-freezing agents.

2. Floor Anchors: Provide floor anchors for each jamb and mullion that extends to floor, and secure with postinstalled expansion anchors.

a. Floor anchors may be set with power-actuated fasteners instead of postinstalled expansion anchors if so indicated and approved on Shop Drawings.

3. Metal-Stud Partitions: Solidly pack mineral-fiber insulation inside frames.

4. In-Place Metal or Wood-Stud Partitions: Secure slip-on drywall frames in place according to manufacturer's written instructions.

3.4 TOLERANCES

A. Maximum Diagonal Distortion: 1/16 inch measured with straight edge, corner to corner.

B. Installation Tolerances: Adjust hollow-metal door frames for squareness, alignment, twist, and plumb to the following tolerances:

1. Squareness: Plus or minus 1/16 inch (1.6 mm), measured at door rabbet on a line 90 degrees from jamb perpendicular to frame head.

2. Alignment: Plus or minus 1/16 inch (1.6 mm), measured at jambs on a horizontal line parallel to plane of wall.

3. Twist: Plus or minus 1/16 inch (1.6 mm), measured at opposite face corners of jambs on parallel lines, and perpendicular to plane of wall.

4. Plumbness: Plus or minus 1/16 inch (1.6 mm), measured at jambs at floor.

3.5 ADJUSTING

A. Final Adjustments: Check and readjust operating hardware items immediately before final inspection. Leave work in complete and proper operating condition. Remove and replace defective work, including hollow-metal work that is warped, bowed, or otherwise unacceptable.

B. Remove grout and other bonding material from hollow-metal work immediately after installation.

C. Prime-Coat Touchup: Immediately after erection, sand smooth rusted or damaged areas of prime coat and apply touchup of compatible air-drying, rust-inhibitive primer.

D. Metallic-Coated Surface Touchup: Clean abraded areas and repair with galvanizing repair paint according to manufacturer's written instructions.

E. Factory-Finish Touchup: Clean abraded areas and repair with same material used for factory finish according to manufacturer's written instructions.

F. Touchup Painting: Cleaning and touchup painting of abraded areas of paint are specified in painting Sections.

END OF SECTION

SECTION 08 14 33 - STILE AND RAIL WOOD DOORS

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Wood doors, stile and rail design; non-fire rated.

1.2 REFERENCE STANDARDS

A. AWI (QCP) - Quality Certification Program; Current Edition.

B. AWI/AWMAC/WI (AWS) - Architectural Woodwork Standards, 2nd Edition; 2014, with Errata (2016).

1.3 SUBMITTALS

A. Product Data: Indicate stile and rail core materials and construction; type and characteristics.

B. Shop Drawings: Illustrate door opening criteria, elevations, sizes, types, swings, undercuts required, special beveling, special blocking for hardware, factory machining criteria, factory finishing criteria, cutouts for glazing, and cutouts for louvers.

C. Samples: Submit two samples of door construction, 12 by 12 inches in size cut from top corner of door.

D. Warranty, executed in Contracting Officer's name.

1.4 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace doors that fail in materials or workmanship, or have warped (bow, cup, or twist) more than 1/4 inch (6.4 mm) in a 42-by-84-inch (1067-by-2134-mm) section.

1. Warranty shall also include installation and finishing that may be required due to repair or replacement of defective doors.

2. Warranty shall be in effect during the following period of time from date of Substantial Completion:

a. Interior Doors: Life of installation.

PART 2 PRODUCTS

2.1 MANUFACTURERS

A. Basis of Design:

1. Simpson Nantucket Series; www.simpsondoor.com. Product characteristics noted in paragraph 2.2 through 2.6..

B. Substitutions: See Section 01 67 00 – Product Requirements.

2.2 MATERIALS

A. General: Use only materials that comply with referenced standards and other requirements specified.

1. Assemble interior doors including components, with either dry-use or wet-use adhesives complying with ASTM D 5572 for finger joints and with ASTM D 5751 for joints other than finger joints.

2. Veneer core plywood.

B. Low-Emitting Materials: Provide doors made with adhesives and composite wood products that do not contain urea-formaldehyde resins.

2.3 DOORS

A. Quality Standard: Premium Grade, Heavy Duty performance, in accordance with AWI/AWMAC/WI (AWS), unless otherwise indicated.

B. Exterior Doors: 1-3/4 inches thick unless otherwise indicated; solid lumber construction; mortise and tenon joints; water repellent treated. Primed finish.

1. Material: Accoya

2. Door design per drawings

3. No exposed pins.

C. Interior Doors: 1-3/8 inches thick unless otherwise indicated; solid lumber construction; mortise and tenon joints. Transparent finish.

1. Material: Fir

2. Door design per drawings

2.4 DOOR AND PANEL FACINGS

A. Veneer Facing for Transparent Finish: Species as specified above, veneer grade in accordance with quality standard indicated, plain sliced (flat cut), with book match between leaves of veneer, running match of spliced veneer leaves assembled on door or panel face.

B. Adhesive: Type I - Waterproof.

2.5 DOOR CONSTRUCTION

A. Factory fit doors to suit frame-opening sizes indicated, with the following uniform clearances and bevels unless otherwise indicated:

1. Clearances: Provide 1/8 inch at heads, jambs, and between pairs of doors. Provide 1/2 inch from bottom of door to top of decorative floor finish or covering. Where threshold is shown or scheduled, provide not more than 3/8 inch (10 mm) from bottom of door to top of threshold

B. Vertical Exposed Edge of Stiles: Of same species as veneer facing.

C. Fit door edge trim to edge of stiles after applying veneer facing.

D. Bond edge banding to cores.

E. Factory machine doors for finish hardware in accordance with hardware requirements and dimensions. Do not machine for surface hardware.

1. Locate hardware to comply with DHI-WDHS-3. Comply with final hardware schedules, door frame Shop Drawings, DHI A115-W Series standards, and hardware templates.

2.6 FINISHES

A. Finish work in accordance with AWI/AWMAC/WI (AWS), Section 5 - Finishing for grade specified and as follows:

1. Door prepared for finish as indicated in door schedule.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify existing conditions before starting work.

B. Verify that opening sizes and tolerances are acceptable.

C. Do not install doors in frame openings that are not plumb or are out of tolerance for size or alignment.

3.2 INSTALLATION

A. Install doors in accordance with manufacturer's instructions and specified quality standards.

B. Factory-Finished Doors: Do not field cut or trim; if fit or clearance is not correct, replace door.

C. Adjust width of non-rated doors by cutting equally on both jamb edges.

D. Trim door height by cutting bottom edges to a maximum of 3/4 inch.

E. Machine cut for hardware.

F. Coordinate installation of doors with installation of frames and hardware.

3.3 ADJUSTING

A. Adjust doors for smooth and balanced door movement.

B. Finished Doors: Replace doors that are damaged or do not comply with requirements. Doors may be repaired or refinished if Work complies with requirements and shows no evidence of repair or refinishing.

END OF SECTION

SECTION 08 31 00 - ACCESS DOORS AND PANELS

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Ceiling-mounted attic access units.

B. Bulkhead doors.

1.2 SUBMITTALS

A. Product Data: Provide sizes, types, finishes, hardware, scheduled locations, and details of adjoining work.

B. Shop Drawings:

1. Indicate exact position of each access door and exact position of each access panel unit.

2. Provide details of every bulkhead door connection. Include project specific conditions.

C. Manufacturer's Installation Instructions: Indicate installation requirements.

1.3 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing the products specified in this section with minimum ten years documented experience.

B. Installer Qualifications: Company specializing in performing work of the type specified and with at least five years documented experience.

PART 2 PRODUCTS

2.1 CEILING-MOUNTED ACCESS UNITS

A. Basis of Design:

1. Best Access Doors;[BA-ATCR80-22-30]. Product characteristics noted in paragraph 2.1.B.

2. Substitutions: See Section 01 67 00 – Product Requirements.

B. Ceiling-Mounted Units: Factory-fabricated door and frame, fully assembled units with corner joints welded, filled and ground flush; square and without rack or warp; coordinate requirements with type of installation assembly being used for each unit.

1. Material: Steel.

2. Style: Exposed frame with door surface flush with frame surface.

3. Door Style: Single thickness with rolled or turned in edges.

4. Heavy-Duty Frames: 14-gauge, 0.0747-inch minimum thickness.

5. Heavy-Duty Single Steel Sheet Door Panels: 14-gauge, 0.0747-inch minimum thickness.

6. Insulation: Non-combustible mineral wool or glass fiber.

7. Door/Panel Size: 22 by 30 inches.

8. Hardware:

    a. Hinges for Non-Fire-Rated Units:Concealed, constant force closure spring type.

    b. Handle: Fixed.

    c. Latch/Lock: Cylinder lock-operated cam latch, two keys for each unit.

2.2 CURB-MOUNTED BULKHEAD DOORS

A. Basis of Design:

1. LuciGold; www.lucigold.com. Product characteristics noted in paragraph 2.2.B.

2. Substitutions: See Section 01 67 00 – Product Requirements.

B. Floor-Mounted Access Units: Factory fabricated, fully assembled units with corner joints welded, filled, and ground flush; square and without rack or warp; coordinate requirements with type of installation assembly being used for each unit.

1. Size: As indicated on drawings.

2. Hardware: Steel, hot-dipped galvanized.

    a. Hinges: Nonremovable pin.

2.3 FABRICATION

A. General: Provide access door and frame assemblies manufactured as integral units ready for installation.

B. Metal Surfaces: For metal surfaces exposed to view in the completed Work, provide materials with smooth, flat surfaces without blemishes. Do not use materials with exposed pitting, seam marks, roller marks, rolled trade names, or roughness.

C. Doors and Frames: Grind exposed welds smooth and flush with adjacent surfaces. Furnish attachment devices and fasteners of type required to secure access panels to types of supports indicated.

D. Latching Mechanisms: Furnish number required to hold doors in flush, smooth plane when closed.

1. For recessed panel doors, provide access sleeves for each locking device. Furnish plastic grommets and install in holes cut through finish.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that rough openings are correctly sized and located.

B. Verify the structure surrounding opening is framed soundly to withstand the repeated opening of hatch without sustaining visible damage.

C. Begin installation only after substrates have been properly prepared, and if the responsibility of another installer, notify Contracting Officer of unsatisfactory preparation before proceeding.

3.2 PREPARATION

A. Clean surfaces thoroughly prior to proceeding with this work.

3.3 INSTALLATION

A. Install units in accordance with manufacturer's instructions.

B. Install frames plumb and level in openings, and secure units rigidly in place.

C. Install doors flush with adjacent finish surfaces or recessed to receive finish material.

D. Position units to provide convenient access to concealed equipment when necessary.

3.4 ADJUSTING AND CLEANING

A. Adjust doors and hardware after installation for proper operation.

B. Remove and replace doors and frames that are warped, bowed, or otherwise damaged.

END OF SECTION

SECTION 08 36 13 - SECTIONAL DOORS

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Overhead sectional doors, operated.

B. Operating hardware and supports.

1.2 REFERENCE STANDARDS

A. ASTM A653/A653M - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process; 2023.

B. ASTM C1036 - Standard Specification for Flat Glass; 2025.

C. ASTM C1048 - Standard Specification for Heat-Strengthened and Fully Tempered Flat Glass; 2025.

D. ASTM E283/E283M - Standard Test Method for Determining Rate of Air Leakage Through Exterior Windows, Skylights, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen; 2019.

E. ASTM E330/E330M - Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights and Curtain Walls by Uniform Static Air Pressure Difference; 2014 (Reapproved 2021).

F. DASMA 102 - American National Standard Specifications for Sectional Doors; 2018.

1.3 SUBMITTALS

A. Shop Drawings: Indicate opening dimensions and required tolerances, connection details, anchorage spacing, hardware locations, and installation details.

B. Product Data: Show component construction, anchorage method, and hardware.

C. Samples: Two panel finish samples, 6 by 6 inch in size, illustrating color and finish.

D. Manufacturer's Installation Instructions: Include any special procedures required by project conditions.

1.4 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing the products specified in this section with minimum three years of documented experience.

B. Contractor to provide attachment of sectional door tracks and accessories to existing building structure.

1.5 WARRANTY

A. Correct defective Work within a five year period after Date of Substantial Completion.

PART 2 PRODUCTS

2.1 MANUFACTURERS

A. Basis of Design:

1. Wayne-Dalton, a Division of Overhead Door Corporation[<>]: www.wayne-dalton.com. Product characteristics noted in paragraph 2.2 and 2.3.

2. Substitutions: See Section 01 67 00 – Product Requirements.

2.2 PERFORMANCE REQUIREMENTS

A. Performance: Withstand positive and negative wind loads equal to 1.5 times design wind loads specified by local code without damage or permanent set, when tested in accordance with ASTM E330/E330M, using 10 second duration of maximum load.

B. Air Leakage Rate: Less than 0.40 cfm/sq ft when tested in accordance with ASTM E283/E283M at test pressure difference of 1.57 psf.

C. Thermal Transmittance: U-factor of 0.31 Btu/hr sq ft degrees F, maximum, in accordance with DASMA 102.

2.3 STEEL DOORS

A. Doors: Flush steel, insulated; standard lift operating style with track and hardware; complying with DASMA 102, Commercial application.

1. Door Panels: Steel construction; outer steel sheet of 26 gauge, 0.016 inch minimum thickness, flush profile; inner steel sheet of 20 gauge, 0.0359 inch minimum thickness, flat profile; core reinforcement sheet steel roll formed to channel shape, rabbeted weather joints at meeting rails; polyurethane insulation.

2. Door Nominal Thickness: 1 5/8 inches thick.

3. Exterior Finish:

a. Factory finished with acrylic baked enamel; Custom color to match other openings as selected by Contracting Officer.

4. Interior Finish:

a. Factory finished with powder coated finish; color as selected from manufacturer's standard line.

5. Glazed Lites: Full panel width, one row; set in place with resilient glazing channel.

a. Glazing: Fully tempered glass; insulated glass units; clear; 1/2 inch nominal overall thickness.

6. Manual Operation: Chain hoist.

7. Provide sectional door with integral person door where indicated on drawings.

2.4 COMPONENTS

A. Track: Galvanized steel angles, 0.094 inch minimum thickness; 2-5/16 x 4 inch size, continuous one piece per side; galvanized steel mounting brackets 1/4 inch thick.

B. Lift Mechanism: Torsion spring on cross head shaft, with braided galvanized steel lifting cables.

1. For Manual Operation: Requiring maximum exertion of 25 lbs force to open.

C. Sill Weatherstripping: Resilient hollow rubber strip, one piece; fitted to bottom of door panel, full length contact.

D. Jamb Weatherstripping: Roll formed steel section full height of jamb, fitted with resilient weatherstripping, placed in moderate contact with door panels.

E. Head Weatherstripping: EPDM rubber seal, one piece full length.

F. Panel Joint Weatherstripping: Neoprene foam seal, one piece full length.

G. Lock: Inside center mounted, adjustable keeper, spring activated latch bar with feature to retain in locked or retracted position; interior and exterior handle.

H. Lock Cylinders: Master keyed to building keying system.

I. Springs: 25,000 cycles

2.5 MATERIALS

A. Sheet Steel: Hot-dipped galvanized steel sheet, ASTM A653/A653M, with G60/Z180 coating, plain surface.

B. Float Glass: Provide float glass glazing, unless noted otherwise.

1. Annealed Type: ASTM C1036, Type I, transparent flat, Class 1 clear, Quality Q3 (glazing select).

2. Heat-Strengthened and Fully Tempered Types: ASTM C1048.

C. Insulation: Foamed-in-place polyurethane, bonded to facing.

2.6 ACCESSORIES

A. All hardware, structural bracing and structural components to provide complete attachment to existing building structure.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that wall openings are ready to receive work and opening dimensions and tolerances are within specified limits.

3.2 PREPARATION

A. Prepare opening to permit correct installation of door unit to perimeter air and vapor barrier seal.

B. Apply primer to wood frame.

3.3 INSTALLATION

A. Install door unit assembly in accordance with manufacturer's instructions.

B. Anchor assembly to wall construction and building framing without distortion or stress.

C. Securely brace door tracks suspended from structure. Secure tracks to structural members only.

D. Fit and align door assembly including hardware.

3.4 TOLERANCES

A. Maximum Variation from Plumb: 1/16 inch.

B. Maximum Variation from Level: 1/16 inch.

C. Longitudinal or Diagonal Warp: Plus or minus 1/8 inch from 10 ft straight edge.

D. Maintain dimensional tolerances and alignment with adjacent work.

3.5 ADJUSTING

A. Adjust door assembly for smooth operation and full contact with weatherstripping.

B. Have manufacturer's field representative present to confirm proper operation and identify adjustments to door assembly for specified operation.

3.6 CLEANING

A. Clean doors and frames and glazing.

B. Remove temporary labels and visible markings.

3.7 PROTECTION

A. Protect installed products from damage until Date of Substantial Completion.

B. Do not permit construction traffic through overhead door openings after adjustment and cleaning.

END OF SECTION

SECTION 08 52 00 - WOOD WINDOWS

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Factory-fabricated wood windows.

B. Operating hardware.

C. Installation into existing frames.(TEAM TO VERIFY)

1.2 RELATED REQUIREMENTS

A. Section 07 25 00 - Weather Barriers: Sealing frames to water-resistive barrier installed on adjacent construction.

B. Section 07 90 05 - Joint Sealers

C. Section 09 91 13 - Exterior Painting: Site finishing wood surfaces with Linseed Oil Paint.

1.3 REFERENCE STANDARDS

A. AAMA 502 - Voluntary Specification for Field Testing of Newly Installed Fenestration Products; 2021.

B. AAMA/WDMA/CSA 101/I.S.2/A440 - North American Fenestration Standard/Specification for Windows, Doors, and Skylights; 2022, with Errata (2023).

C. ASTM E783 - Standard Test Method for Field Measurement of Air Leakage Through Installed Exterior Windows and Doors; 2002 (Reapproved 2018).

D. ASTM E1105 - Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls, by Uniform or Cyclic Static Air Pressure Difference; 2015 (Reapproved 2023).

1.4 SUBMITTALS

A. Product Data: Show component dimensions, anchorage and fasteners, and glass.

B. Shop Drawings: Indicate opening dimensions, framed opening tolerances, affected related work, and installation requirements.

1. Include details at existing log wall assembly.

C. Samples: Two samples 12 by 12 inch in size illustrating window frame section.

D. Operating Hardware: Two samples of each type of operating hardware.

E. Grade Substantiation: Prior to submitting shop drawings or starting fabrication, submit one of the following showing compliance with specified grade:

1. Evidence of AAMA Certification; label or other documentation.

2. Evidence of WDMA Certification.

3. Evidence of CSA Certification.

4. Test report(s) by independent testing agency itemizing compliance and acceptable to authorities having jurisdiction.

F. Field Quality Control Submittals: Report of field testing for water penetration and air leakage.

1.5 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section with not less than three years of documented experience.

B. Installer Qualifications: Company specializing in performing work of the type specified and with at least three years of documented experience.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Protect factory finished surfaces with wrapping. Do not use adhesive papers or sprayed coatings that bond when exposed to sunlight or weather.

B. Install window in dry weather conditions.

1.7 FIELD CONDITIONS

A. Do not install sealants when ambient temperature is less than 40 degrees F.

B. Maintain this minimum temperature during and after installation of sealants.

1.8 WARRANTY

A. See Section 01 78 00 - Closeout Submittals for additional warranty requirements.

B. Manufacturer Warranty: Provide 5-year manufacturer warranty for insulated glass units against seal failure, interpane dusting or misting, and replacement of same. Complete forms in Contracting Officer's name and register with manufacturer.

PART 2 PRODUCTS

2.1 WOOD WINDOWS

A. Wood Windows: Wood frame inset and sash, factory fabricated and assembled.

1. Exterior Finish: Linseed oil paint to match other sashes. Coordinate with Contracting Officer..

2. Interior Finish: Linseed oil finish to match other sashes. Coordinate with Contracting Officer..

3. Construction: Match existing windows.

4. Configuration: As indicated on drawings.

5. Window Product Types: Triple C - Casement window, in accordance with AAMA/WDMA/CSA 101/I.S.2/A440.

6. Single pane glazing and putty as noted in Section 08 80 00 - Glazing.

7. Wood Species: Wood species to match existing historic windows to be confirmed by material testing by contractor.

8. Frame and Sash Members: Mortise and tenon joints. Glue and steel pin joints to hairline fit, weather tight.

    a. Provide factory fabricated rame that can be placed into existing frame.

9. Weather Stop Flange: Match existing windows

10. Fasteners: Concealed from view.

11. Operable Units: Weather stripping to match other historic windows. Coordinate with Contracting Officer.

2.2 COMPONENTS

A. Single pane glazing and glazing putty to match historic windows.

B. Frames: Match existing.

C. Sills: Wood, with thickness matching existing sills; sloped for positive drainage; fits under sash and projects same distance as existing windows beyond exterior face of wall; single piece full width of opening; ; single full length kerf cut on underside to break water tension.

D. Fasteners: Stainless steel.

E. Sealant and Backing Materials: See Section 07 90 05 of types as indicated.

F. Flashing: Provide related flashings, with necessary anchors and attachment devices.

2.3 HARDWARE

A. Operator: Lever action handle fitted to projecting sash arms with limit stops; baked enamel finish. (TEAM TO VERIFY)

B. Pulls: Manufacturer's standard type. (TEAM TO VERIFY)

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify wall openings and adjoining water-resistive barrier materials are ready to receive wood windows; see Section 07 25 00.

3.2 INSTALLATION

A. Attach infill window frame and shims to existing frame opening to accommodate construction tolerances and other irregularities. (TEAM TO VERIFY)

B. Set sill members and sill flashing in continuous bead of sealant.

C. Install operating hardware.

D. Finish exterior surfaces with opaque materials; see Section 09 91 13 - Exterior Painting.

E. Finish interior surfaces with transparent materials; see Section 09 91 23 - Interior Painting.

3.3 TOLERANCES

A. Maximum Variation from Level or Plumb: 1/16 inch per 3 ft non-cumulative or 1/8 inch per 10 ft, whichever is less.

3.4 FIELD QUALITY CONTROL

A. See Section 01 40 00 - Quality Requirements for independent field testing and inspection requirements, and requirements for monitoring quality of specified product installations.

B. Provide field testing of installed wood windows by independent laboratory in accordance with AAMA 502 and AAMA/WDMA/CSA 101/I.S.2/A440 during construction process and before installation of interior finishes.

1. Field test for water penetration in accordance with ASTM E1105 using Procedure B - cyclic static air pressure difference; test pressure shall not be less than 1.9 psf.

2. Field test for air leakage in accordance with ASTM E783 with uniform static air pressure difference of 6.27 psf.

C. Repair or replace fenestration components that have failed designated field testing, and retest to verify performance complies with specified requirements.

3.5 ADJUSTING

A. Adjust hardware for smooth operation and secure weathertight closure.

3.6 CLEANING

A. Remove protective material from factory finished surfaces.

B. Wash surfaces by method recommended and acceptable to window manufacturer; rinse and wipe surfaces clean.

C. Remove excess glazing sealant by moderate use of mineral spirits or other solvent acceptable to sealant manufacturer.

END OF SECTION

SECTION 08 71 00 - DOOR HARDWARE

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Hardware for wood doors.

1.2 RELATED REQUIREMENTS

A. Section 08 14 33 - Stile and Rail Wood Doors

1.3 REFERENCE STANDARDS

A. 36 CFR 1191 - Americans with Disabilities Act Accessibility Guidelines for Buildings and Facilities; Final Rule; current edition; (ADA Standards for Accessible Design).

B. ANSI/ICC A117.1 - American National Standard for Accessible and Usable Buildings and Facilities; International Code Council; 2017.

C. DHI (LOCS) - Recommended Locations for Architectural Hardware for Standard Steel Doors and Frames; Door and Hardware Institute; 2004.

D. UL (BMD) - Building Materials Directory; Underwriters Laboratories Inc.; current edition.

1.4 ADMINISTRATIVE REQUIREMENTS

A. Coordinate the manufacture, fabrication, and installation of products onto which door hardware will be installed.

B. Furnish templates for door and frame preparation to manufacturers and fabricators of products requiring internal reinforcement for door hardware.

C. Convey Park's keying requirements to manufacturers.

D. Preinstallation Meeting: Convene a preinstallation meeting one week prior to commencing work of this section; require attendance by all affected installers.

E. Sequence installation to ensure utility connections are achieved in an orderly and expeditious manner.

1.5 SUBMITTALS

A. Product Data: Manufacturer's catalog literature for each type of hardware, marked to clearly show products to be furnished for this project.

B. Samples: Prior to preparation of hardware schedule:

1. Submit 1 sample of hinge, latchset, lockset, closer, and door actuator illustrating style, color, and finish.

2. Samples will be returned to supplier.

C. Hardware Schedule: Submit for approval six copies of schedule per Division 1. Organize vertically formatted schedule into “Hardware Sets” with index of doors and headings, indicating complete designations of every item required for each door or opening. Include following information:

1. Type, style, function, size, quantity and finish of hardware items.

   a. Use BHMA Finish codes per ANSI A156.18.

2. Name, part number and manufacturer of each item.

3. Fastenings and other pertinent information.

4. Location of hardware set coordinated with floor plans and door schedule.

5. Explanation of abbreviations, symbols, and codes contained in schedule.

6. Mounting locations for hardware.

7. Door and frame sizes, materials, hand, rating, degrees of swing, and jamb depth.

8. List of manufacturers used and their nearest representative with address and phone number.

9. Catalog cuts.

10. Manufacturer’s technical data and installation instructions for electronic hardware.

11. Date of jobsite visit if existing conditions.

12. Cylinder Schedule less the keying. Include key system description and keys to be provided.

13. Provide point to point wiring and riser diagrams for each electric hardware application. Include electrical items provided by others where specified. Include a system description.

D. Keying Schedule: Submit for approval separate detailed keying schedule for Contracting Officer review after Hardware Schedule has been approved by Contracting Officer and keying requirements established with Contracting Officer .

E. Manufacturer's Installation Instructions: Indicate special procedures, perimeter conditions requiring special attention.

F. Maintenance Data: Include data on operating hardware, lubrication requirements, and inspection procedures related to preventative maintenance.

G. Warranty: Submit manufacturer's warranty and ensure that forms have been completed in Park's name and registered with manufacturer.

H. Closeout Submittals: Furnish 2 copies of the as-built/as-installed hardware schedule with closeout documents including copy of keying schedule, wiring diagrams, manufacturers installation instructions, and suppliers final inspection report.

1.6 QUALITY ASSURANCE

A. Supplier Qualifications:

1. Direct factory contract supplier who has been furnishing hardware in the projects vicinity for a period of not less then 5 years.

2. Employs a certified architectural hardware consultant (AHC), available at reasonable times during course Work for project hardware consultation to Contracting Officer and Contractor. Responsible for detailing, scheduling and ordering of finish hardware.

3. Supplier willing to agree in writing to maintain parts inventory of items supplied for future service.

4. Supplier must meet manufacturers requirements to provide and install power operators where specified. Hardware supplier shall provide an installation cost estimate to the GC at time of bid for GC to include in bid package. Local factory authorized installation companies can be obtained through the automatic operator representatives.

B. Hardware: New, free of defects, blemishes and excessive play. Obtain each kind of hardware (latch and locksets, exit devices, hinges and closers) from one manufacturer.

C. Exit Doors: Operable from inside with single motion without the use of a key or special knowledge or effort.

D. Notify Contracting Officer of any code conflicts before ordering material.

E. Pre-Installation Meetings: Initiate and conduct with supplier, installer, and related trades through Contracting Officer. Coordinate materials and techniques, and sequence complex mechanical and electrical hardware items and systems installation. Convene at least two weeks prior to commencement of related work.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Package hardware items individually; label and identify each package with door opening code to match hardware schedule.

1.8 WARRANTY

A.	Closers:	35 years mechanical, 2 years electrical.
B.	Hinges:	5 years.
C.	Locksets	5 years
D.	Other Hardware:	2 years.

1.9 MAINTENANCE:

A. Tools: After final adjustment of door hardware turn over to Owner tools furnished during construction used for installation and adjustment.

PART 2 PRODUCTS

2.1 MANUFACTURERS - BASIS OF DESIGN

A. Listed below are specified manufacturers and acceptable alternates:

	ITEM:	MANUFACTURER:	ACCEPTABLE SUB:
B.	Hinges	(BES) Best	Bommer, Hager, Ives
C.	Locks	(BES) Best	Park Standard
D.	Cylinders	(BES) Best	Park Standard
E.	Closers	(BES) Best	LCN 4040XP
F.	Kick Plates	(BRN) Burns	Ives, Trimco
G.	Threshold	(NGP) National G	Zero, Pemko
H.	Curtain Sash Rod	Tyron	As approved by Contracting Officer

I. Provide hardware items required to complete the work in accordance with these specifications and manufacturers' instructions.

1. Include items inadvertently omitted from this specification. Note these items in submittal for review.

2.2 DOOR HARDWARE - GENERAL

A. Provide all hardware specified or required to make doors fully functional, compliant with

applicable codes, and secure to the extent indicated.

B. Provide all items of a single type of the same model by the same manufacturer.

C. Provide products that comply with the following:

1. Applicable provisions of federal, state, and local codes.

2. Architectural Barriers Act (ABA)

3. ANSI/ICC A117.1, American National Standard for Accessible and Usable Buildings and Facilities.

4. Products Requiring Electrical Connection: Listed and classified by UL as suitable for the purpose specified and indicated.

D. Function: Lock and latch function numbers and descriptions of manufactures series as listed in hardware schedule.

E. Electrically Operated and/or Controlled Hardware: Provide all power supplies, power transfer hinges, relays, and interfaces required for proper operation; provide wiring between hardware and control components and to building power connection.

F. Finishes: Identified in schedule.

2.3 HANGING MEANS:

A. Conventional Hinges: Hinge open widths minimum, but, of sufficient throw to permit maximum door swing. Steel or stainless steel pins and concealed bearings.

1. Three hinges per leaf to 7 foot, 6 inch height. Add one for each additional 30 inches in height, or any fraction thereof.

a. All existing hinges to be restored and reused. Where hinge is damaged beyond repair, provide new set of hinges to match existing.

b. Heavy weight hinges on all exterior doors. 4 ½ x 4 ½ standard. 5 x 4 ½ on doors over 3 foot 5 inches in width.

c. Out swinging exterior doors: non-ferrous with non-removable (NRP) pins.

d. Non-ferrous material exteriors and at doors subject to corrosive atmospheric conditions.

e. Provide shims and shimming instructions for proper door adjustment.

f. Provide decorative ball tips for aesthetics.

B. Strap Hinges: Provide as or equal to specified.

1. Verify with the owner and architect for additional applications other than specified.

2.4 LOCKSETS, LATCHSETS, DEADBOLTS:

A. Best locks, mortised or cylindrical as specified (Basis of Design)

B. Cylinders: Cylinders per the Park's standard

1. Case: heavy gauge steel, handing field-changeable without disassembly.

a. Mortise Latchbolts: 3/4 inch throw with stainless steel inserts.

b. Cylindrical Latchbolts: 1/2 inch minimum throw

c. Lever Trim: through-bolted, accessible. BOD: Schlage Broadway Lever Set with Decorative Greyson Trim.

d. Thumbturns: accessible design not requiring pinching or twisting motions to operate.

e. Deadbolts: stainless steel 1-inch throw.

f. Strikes: 16 gauge curved brass with 1 inch deep box construction, lips of sufficient length to clear trim and protect clothing.

2.5 CLOSERS

A. Surface Closers:

1. Full rack-and-pinion type cylinder with removable non-ferrous cover and cast aluminum body.

a. Thru-bolts at all doors. Non-sized and adjustable. Place closers inside building, stairs and rooms.

b. Plates, brackets and special templating when needed for interface with particular header, door and wall conditions and hardware.

c. Opening pressure: Exterior doors 8.5 lb., interior doors 5 lb., labeled fire doors 15 lb.

d. Separate adjusting valves for closing speed, latching speed and backcheck, fourth valve for delayed action where scheduled.

e. Heavy duty arms (PR) for doors scheduled with parallel arm units.

2.6 OTHER HARDWARE

A. Flush Bolts: Low operating force design.

B. Kick Plates: Four beveled edges,.050 inches minimum thickness, height and width as scheduled.

1. Sheet-metal screws of bronze or stainless steel to match other hardware.

C. Door Stops: Provide stops to protect walls, casework or other hardware.

1. Unless otherwise noted in Hardware Sets, provide wall type with appropriate fasteners.

Where wall type cannot be used, provide floor type. If neither can be used, provide overhead type.

D. Thresholds: As scheduled and per details. Substitute products: certify that the products equal or exceed specified material's thickness. Proposed substitutions: submit for approval.

1. Exteriors: Set in full bed of butyl-rubber or polyisobutylene mastic sealant. Non-ferrous 1/4-inch fasteners and lead expansion shield anchors, or Red-Head #SFS-1420 (or approved equivalent) Flat Head Sleeve Anchors (SS/FHSL).

a. Sound control openings: Set in bed of mastic sealant.

2. Provide solid brass products, finished as specified.

3. Provide thresholds at new doors only.

E. Fasteners: Generally, exposed screws to be Phillips or Robertson drive. Pinned TORX drive at high security areas. Flat head sleeve anchors (FHSL) may be slotted drive. Sheet metal and wood screws: full-thread. Sleeve nuts: full length to prevent door compression.

F. Silencers: Interior hollow metal frames, 3 for single doors, 4 for pairs of doors. Omit where adhesive mounted seal occurs. Leave no unfilled/uncovered pre-punched silencer holes.

G. Knox Box: (if required)

1. Assume (1) per building. Coordinate locations with Contracting Officer.

2. High-security lock box with hinged door and weather-resistant door gasket.

3. Mounting: Surface

4. Color: Black

5. Product:

a. KNOX; Knox box 3200: www.knoxbox.com

b. Ensure that the product is compatible with Park requirements.

H. Curtain Sash Rod: Adjustable steel metal rod. 3/8in diameter with adjustable range of the rod (18-30 inches). Powder-coated black finish. Exposed screws.

2.7 KEYING REQUIREMENTS:

A. Key System: Existing Park's Best Patented Key System. Initiate and conduct meeting(s) with Contracting Officer to determine system structure, furnish Park's written approval of the system.

1. Construction keying: brass keyed-alike temporary cores plus 10 operating keys. Temporary cores and keys remain property of hardware supplier.

2. Provide the following permanent keys:

a. 6 Grandmaster keys

b. 6 Masterkeys

c. 6 Construction keys

d. 1 Control Key

e. 1 Construction Control Key

f. 3 Keys per lock

B. Interchangeable Core cylinders at all cylinders.

C. Permanent cores: factory-keyed.

D. Permanent keys and cores: secured shipment direct from point of origination to the Park .

E. Bitting List: Secured shipment direct from point of origination to the Park upon completion.

F. Pinning for lock cylinders: provided by the Park.

PART 3 EXECUTION

3.1 ACCEPTABLE INSTALLER

A. Factory trained, certified, and carries a factory-issued card certifying that person as a “Certified installer”. Alternative: can demonstrate suitably equivalent competence and experience.

3.2 PREPARATION

A. Ensure that walls and frames are square and plumb before hardware installation.

B. Clarify placement if new hardware is to be installed near existing doors/hardware scheduled to remain.

C. Mounting Heights: Locate hardware units at heights indicated in DHI’s recommended locations unless otherwise indicated or required to comply with governing regulations.

D. Existing frames and doors: catalogue and carefully remove existing hardware. Restore and reinstall hardware to be reused. Tag and bag hardware to be replaced, and turn over to the Park.

1. Patch and fill wood frames and doors with solid in-kind wood stock or dowel material before cutting for new hardware. Do not reuse existing screw holes - - fill and re-pilot.

3.3 EXAMINATION

A. Verify that doors and frames are ready to receive work prior to harwdware install.

B. Verify that existing hardware to be reused has been properly restored prior to reinstallation.

C. Assure proper install and operability of all hardware.

D. Verify that electric power is available to power operated devices and of the correct characteristics.

3.4 INSTALLATION

A. Install hardware per manufacturer’s instructions and recommendations. do not install surface-mounted items until finishes have been completed on substrate. Set units level, plumb and true to line and location. Adjust and reinforce attachment substrate for proper installation and operation.

1. Gaskets: install jamb-applied gaskets before closers, overhead stops, rim strikes, etc. Install sweeps across bottoms of doors before astragals, cope sweeps around bottom pivots, trim astragals to tops of sweeps.

2. When hardware is to be attached to existing metal surface and insufficient reinforcement exists, use RivNuts, Nutserts or similar anchoring device for screws.

B. Use templates provided by hardware item manufacturer.

C. Install hardware on fire-rated doors and frames in accordance with code and NFPA 80.

D. Mounting heights for hardware from finished floor to center line of hardware item: As listed in Schedule, unless otherwise noted:

1. For steel doors and frames: Comply with DHI "Recommended Locations for Architectural Hardware for Steel Doors and Frames."

E. Locate floor stops not more than 4 inches from the wall.

F. Drill pilot holes for fasteners in wood doors and/or frames.

G. Lubricate and adjust existing hardware scheduled to remain. Carefully remove and give to the Park items not scheduled for reuse.

3.5 FIELD QUALITY CONTROL

A. Provide an Architectural Hardware Consultant to inspect installation and certify that hardware and installation has been furnished and installed in accordance with manufacturer's instructions and as specified.

B. Intent is to provide an aesthetic finish consistent with the existing structures.

1. Reuse existing hardware where feasible – consult with the architect and owner.

2. Comply with applicable codes where required.

3.6 ADJUSTING

A. Adjust and check for proper operation and function. Replace units, which cannot be adjusted to

operate freely and smoothly.

1. Hardware damaged by improper installation or adjustment methods to be repaired or replaced to Contracting Officer's satisfaction

B. Inspection: Use hardware supplier. Include suppliers with closeout documents.

C. Follow-up inspection: Installer to provide letter of agreement to Contracting Officer's that approximately 6 months after substantial completion, installer will visit Project with representatives of the manufacturers of the locking devices and door closers to accomplish following:

1. Re-adjust hardware.

2. Evaluate maintenance procedures and recommend changes or additions, and instruct the Park's personnel.

3. Identify items that have deteriorated or failed.

4. Submit written report identifying problems and likely future problems.

3.7 CLEANING

A. Clean adjacent surfaces soiled by hardware installation. Clean finished hardware per manufacturer's instructions after final adjustments has been made. Replace items that cannot be cleaned to manufacturer's level of finish quality at no additional cost.

3.8 PROTECTION

A. Do not permit adjacent work to damage hardware or finish.

3.9 SCHEDULE OF FINISH HARDWARE.

MANUFACTURERS

Code	**Name**	
ABH	Architectural Builders	Overhead Stops
BES	Best Access Systems	Locks, Cylinders, Hinges, Closers
R/W	Richards Wilcox	Strap Hinges. Bolts, Pulls
NGP	National Guard	Gaskets, Thresholds
BRN	Burns	Door Stops, Flat Goods

FINISH LIST

Code	Description
606	Satin Brass
613/641	Oil Rubbed Bronze, Oxidized
622	Flat Black Painted
693	Black Painted
BLK	Black
Gray	Gray

OPTIONS

Code	Description
B4E	Beveled 4 Edges – Kick, Mop & Armor Plates (Burns)
CSK	Counter Sinking Of Kick, Mop & Armor Plates (Burns)
MS/EA	Stainless Machine Screws/Expansion Anchors (NGP)

Set #1 - Basement - New Door

Doors: 399.00A

2.0	Hinge	BT FBB199 4.5x4.5	613	BES
1.0	Mortise Lock	45H 7 AB 15 M PATD	613	BES
1.0	Door Closer	EHD9016 JT90	693	BES
1.0	Kick Plate	KP50 10" x 2" LDW B4E CSK	622	BRN
1.0	Wall Stop	575	622	BRN
1.0	Gasketing	5075 C Head & Jambs		NGP
1.0	Door Bottom	36EV BLK		NGP
1.0	Threshold	426 B HD (1/4-20 MS/EA)	613	NGP

Set #2 - Existing Dutch Door

Doors: 399.01A, 399.05B, 399.09A, 399.11A

4.0	Hinge	BT FBB199 4.5x4.5 as required	613	BES
1.0	Mortise Lock	45H 7 AB 15 M PATD	613	BES
1.0	Wall Stop	575 or 521 as required	622	BRN
1.0	Astragal	139 BLK @ Exterior Top Half to Bottom		NGP
1.0	Surface Bolt	594	613	BRN
1.0	Gasketing	5075 C Head & Jambs		NGP
1.0	Gasketing	5060 C @ Astragal		NGP
1.0	Door Bottom	36EV BLK		NGP

NOTE: Modify existing door to accept new hardware as specified. Lockset in top half, surface bolt top to bottom. Mount new pull set on bottom half. Notify the Contracting Officer of any conditions that may need to be addressed.

Set #2A - Existing Dutch Door

Doors: 399.02A, 399.05A

4.0	Hinge	BT FBB199 4.5x4.5 as required	613	BES
1.0	Mortise Lock	45H 7 AB 15 M PATD	613	BES
1.0	Overhead Stop	9020 Series	622	ABH
1.0	Astragal	139 BLK @ Exterior Top Half to Bottom		NGP
1.0	Surface Bolt	594	613	BRN
1.0	Gasketing	5075 C Head & Jambs		NGP
1.0	Gasketing	5060 C @ Astragal		NGP
1.0	Door Bottom	36EV BLK		NGP

Set #3 - Existing Bedroom

Doors: 399.02B

2.0	Hinge	BT FBB179 4.5X4.5 as required	641	BES
1.0	Cylindrical Privacy	9K 3 0 L 15 D	613	BES
1.0	Wall Stop	575	622	BRN
3.0	Silencers	501	Gray	BRN

NOTE: Modify existing door to accept new hardware as specified. Notify the Contracting Officer of any conditions that may need to be addressed.

Set #4 - Restroom - New Door

Doors: 399.03A, 399.06A, 401.04A, 401.05A

2.0	Hinge	BT FBB179 4.5X4.5	641	BES
1.0	Cylindrical Privacy	9K 3 0 L 15 D	613	BES
1.0	Wall Stop	575 or 521 as required	622	BRN
1.0	Gasketing	5050 C Head & Jambs		NGP

Set #5 - Closet - New Door

Doors: 399.04C

2.0	Hinge	BT FBB179 4.5X4.5	641	BES
1.0	Cylindrical Latch	9K 3 0 N 15 D	613	BES
1.0	Wall Stop	575	622	BRN
3.0	Silencers	501	Gray	BRN

Set #6 - Laundry/Food Storage – Existing

Doors: 399.09B

2.0	Hinge	BT FBB179 4.5X4.5 as required	641	BES
1.0	Cylindrical Latch	9K 3 0 N 15 D	613	BES

1.0	Wall Stop	575	622	BRN
3.0	Silencers	501	Gray	BRN

NOTE: Modify existing door to accept new hardware as specified. Notify the Contracting Officer of any conditions that may need to be addressed.

Set #7 - Electrical/Mechanical Closet - New Door

Doors: 399.09C, 400.05A

2.0	Hinge	BT FBB179 4.5X4.5 NRP	641	BES
1.0	Cylindrical Lock	9K 3 7 D 15 D PATD	613	BES
1.0	Wall Stop	575	622	BRN
1.0	Gasketing	5050 C Head & Jambs		NGP

Set #7A - Mechanical - New Door

Doors: 404.03A

2.0	Hinge	BT FBB179 4.5X4.5	641	BES
1.0	Cylindrical Lock	9K 3 7 D 15 D PATD	613	BES
1.0	Wall Stop	575	622	BRN
1.0	Gasketing	5050 C Head & Jambs		NGP

Set #8 - Restroom - New Door

Doors: 399.12A

2.0	Hinge	BT FBB199 4.5x4.5 NRP	613	BES
1.0	Mortise Lock	45H 7 T 15 M PATD VIB	613	BES
1.0	Gasketing	5075 C Head & Jambs		NGP
1.0	Sweep	200N BLK		NGP
1.0	Threshold	426 B HD (1/4-20 MS/EA)	613	NGP

Set #9 - Restroom - New Door

Doors: 399.12B

2.0	Hinge	BT FBB179 4.5X4.5	641	BES
1.0	Cylindrical Privacy	9K 3 0 L 15 D	613	BES
1.0	Gasketing	5050 C Head & Jambs		NGP

Set #10 - Kitchen/Bedroom - Existing/New Door

Doors: 399.12C, 407.1B

2.0	Hinge	BT FBB179 4.5X4.5 as required	641	BES
1.0	Cylindrical Latch	9K 3 0 N 15 D	613	BES
1.0	Wall Stop	575 or 521 as required	622	BRN
3.0	Silencers	501	Gray	BRN

NOTE: Modify existing door and frame to accept new hardware as specified. Notify the Contracting Officer of any conditions that may need to be addressed.

Set #11 - Cabin - Existing/New Door

Doors: 400.02A, 401.01A, 401.06A, 404.04A, 405.01A, 407.1A, 407.2A, 410.1A, 411.01A

2.0	Hinge	BT FBB199 4.5x4.5 as required	613	BES
OR				
2.0	Strap Hinge	1035/1036.01038SC as required		R/W
1.0	Mortise Lock	45H 7 AB 15 M PATD	613	BES
1.0	Locking Bolt	0128.00010 x 2/0126.00014	BLK	R/W
1.0	Padlock	41B 7 22 PATD	606	BES
1.0	Wall Stop	575 or 521 as required	622	BRN
1.0	Gasketing	5075 C Head & Jambs		NGP
1.0	Door Bottom	36EV BLK		NGP
1.0	Threshold	426 B HD (1/4-20 MS/EA)	613	NGP

NOTE: Restore existing hinges; if beyond repair, provide new to match existing in-kind. Modify existing door and frame to accept new hardware as specified. Hasp bolt for off season securing. Install staples for locked/unlocked positions. Notify the Contracting Officer of any conditions that may need to be addressed.

Set #12 - Restroom - Existing Door

Doors: 400.04A, 407.3A, 407.3B, 410.2A, 411.02A

2.0	Hinge	BT FBB179 4.5X4.5 as required	641	BES
1.0	Cylindrical Lock	9K 3 0 L 15 D	613	BES
1.0	Wall Stop	575 or 521 as required	622	BRN
1.0	Gasketing	5050 C Head & Jambs		NGP

NOTE: Modify existing door to accept new hardware as specified. Notify the Contracting Officer of any conditions that may need to be addressed.

Set #13 - Overhead Door

Doors: 403.1A, 403.1C, 403.1B

1.0	Padlock	41B 7 22 PATD	606	BES

NOTE: Padlock specified for securing. Revise to cylinder if the manufacturer requires. Confirm and coordinate with the manufacturer. Balance by door manufacturer.

Set #14 - Bedroom/Restroom - New Door

Doors: 404.01A, 404.02A, 404.05A

2.0	Hinge	BT FBB179 4.5X4.5	641	BES
1.0	Cylindrical Privacy	9K 3 0 L 15 D	613	BES
1.0	Wall Stop	575 or 521 as required	622	BRN
3.0	Silencers	501	Gray	BRN

Set #15 - Ice House - Existing Door

Doors: 406.01A

NOTE: Restore per drawings.

Set #16 - 4-Door Bifold – Existing

Doors: 411.01B

1.0	Bi-Fold Door Set	SLAL-75-BF4DR LAR	NGP

END OF SECTION

SECTION 08 80 00 - GLAZING

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Glass.

B. Glazing compounds and accessories.

1.2 REFERENCE STANDARDS

A. ASTM C1048 - Standard Specification for Heat-Strengthened and Fully Tempered Flat Glass; 2025.

B. GANA (GM) - GANA Glazing Manual; 2022.

1.3 PERFORMANCE REQUIREMENTS

A. Select type and thickness of exterior glass to withstand dead loads and wind loads acting normal to plane of glass at design pressures calculated in accordance with IBC code.

1. Use the procedure specified in ASTM E 1300 to determine glass type and thickness.

2. Limit glass deflection to 1/200 or flexure limit of glass, whichever is less, with full recovery of glazing materials.

3. Thicknesses listed are minimum.

B. Comply with Requirements of Local Jurisdictions for glass and glazing requirements.

1. At exterior and interior locations, where subject to human impact, provide Tempered Glass as required by IBC, unless laminated safety glass are specifically noted.

1.4 SUBMITTALS

A. Product Data on Glass Types: Provide structural, physical and environmental characteristics, size limitations, special handling or installation requirements.

B. Product Data on Glazing Compounds: Provide chemical, functional, and environmental characteristics, limitations, special application requirements. Identify available colors.

C. Samples: Submit two samples 10 x 10 inch in size of glass units.

1.5 QUALITY ASSURANCE

A. Perform Work in accordance with GANA Glazing Manual for glazing installation methods. Maintain one copy on site.

B. Installer Qualifications: Company specializing in performing the work of this section with minimum 5 years documented experience and who employs glass installers for this Project who are certified under the National Glass Association's Certified Glass Installer Program.

C. Source Limitations for Glass: Obtain coated float glass laminated glass and insulating glass from single source from single manufacturer for each glass type.

D. Source Limitations for Glazing Accessories: Obtain from single source from single manufacturer for each product and installation method.

E. Glazing Publications: Comply with published recommendations of glass product manufacturers and organizations below, unless more stringent requirements are indicated. Refer to these publications for glazing terms not otherwise defined in this Section or in referenced standards.

1. GANA Publications: GANA's "Laminated Glazing Reference Manual" and GANA's "Glazing Manual."

2. AAMA Publications: AAMA GDSG-1, "Glass Design for Sloped Glazing," and AAMA TIR-A7, "Sloped Glazing Guidelines."

3. IGMA Publication for Sloped Glazing: IGMA TB-3001, "Guidelines for Sloped Glazing."

4. IGMA Publication for Insulating Glass: SIGMA TM-3000, "North American Glazing Guidelines for Sealed Insulating Glass Units for Commercial and Residential Use."

1.6 FIELD CONDITIONS

A. Do not install glazing when ambient temperature is less than 50 degrees F.

B. Maintain minimum ambient temperature before, during and 24 hours after installation of glazing compounds.

PART 2 PRODUCTS

2.1 GLAZING UNITS

A. Type G-1 -Single Vision Glazing:

1. Application: All exterior glazing unless otherwise noted.

2. Type: Fully tempered float glass.

3. Tint: Clear.

4. Thickness: 1/4 inch.

5. Glazing Method: Interior wet method, glazing compound.

a. See Section 08 03 14 Historic Treatment of Wood Doors.

B. Type G-2 -Single Vision Glazing:

1. Applications: All interior glazing unless otherwise indicated.

2. Type: Laminated float glass.

3. Tint: Clear.

4. Thickness: 1/4 inch.

5. Glazing Method: Interior wet method, glazing compound.

a. See Section 08 03 14 Historic Treatment of Wood Doors

2.2 GLASS MATERIALS

A. Float Glass: All glazing is to be float glass unless otherwise indicated.

1. Heat-Strengthened and Fully Tempered Types: ASTM C1048.

2. Thicknesses: As indicated; for exterior glazing comply with specified requirements for wind load design regardless of specified thickness.

2.3 GLAZING COMPOUNDS

A. Basis of Design:

1. Dana Lim: https://www.danalim.com/product/linseed-oil-putty-683. Product characteristics noted in paragraph 2.3.

2. Substitutions: See Section 01 67 00 – Product Requirements.

B. Glazing Putty: Linseed oil putty recommended by manufacturer for outdoor use, knife grade consistency; Match existing color.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that openings for glazing are correctly sized and within tolerance.

B. Verify that surfaces of glazing channels or recesses are clean, free of obstructions that may impede moisture movement, weeps are clear, and ready to receive glazing.

3.2 PREPARATION

A. Clean contact surfaces with solvent and wipe dry.

B. Apply linseed oil to rabet prior to installing linseed oil-based glazing putty.

3.3 INSTALLATION - INTERIOR WET METHOD (COMPOUND AND COMPOUND)

A. Install glazing resting on setting blocks. Install applied stop and center pane by use of spacer shims at 24 inch centers, kept 1/4 inch below sight line.

B. Locate and secure glazing pane using spring wire clips.

C. Fill gaps between glazing and stops with glazing compound until flush with sight line. Tool surface to straight line.

3.4 CLEANING

A. Remove glazing materials from finish surfaces.

B. Remove labels after work is complete.

C. Clean glass and adjacent surfaces.

3.5 PROTECTION

A. After installation, mark pane with an 'X' by using removable plastic tape or paste; do not mark heat absorbing or reflective glass units.

B. Protect exterior glass from damage immediately after installation by attaching crossed streamers to framing held away from glass. Do not apply markers to glass surface. Remove non-permanent labels, and clean surfaces.

C. Protect glass from contact with contaminating substances resulting from construction operations, including weld splatter. If, despite such protection, contaminating substances do come into contact with glass, remove substances immediately as recommended by glass manufacturer.

D. Examine glass surfaces adjacent to or below exterior concrete and other masonry surfaces at frequent intervals during construction, but not less than once a month, for buildup of dirt, scum, alkaline deposits, or stains; remove as recommended in writing by glass manufacturer.

END OF SECTION

SECTION 08 95 16 - WALL VENTS

PART 1 GENERAL

1.1 SUMMARY

A. Section Includes:

1. Wall vents.

1.2 REFERENCE STANDARDS

A. AAMA 611 - Specification for Anodized Architectural Aluminum; 2026.

B. ASTM D4214 - Standard Test Methods for Evaluating the Degree of Chalking of Exterior Paint Films; 2023.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Samples: For each type of metal finish required.

1.4 INFORMATIONAL SUBMITTALS

A. Evaluation Reports: For flood vents, from ICC-ES.

B. Sample Warranties: For manufacturer's special warranties.

1.5 WARRANTY

A. Special Finish Warranty, Anodized Finishes: Standard form in which manufacturer agrees to repair finishes or replace aluminum that shows evidence of deterioration of anodized finishes within specified warranty period.

1. Deterioration includes, but is not limited to, the following:

2. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D4214.

3. Cracking, peeling, or chipping.

4. Warranty Period: 10 years from date of Substantial Completion.

PART 2 PRODUCTS

2.1 MANUFACTURERS

A. Source Limitations: Obtain vents from single source from single manufacturer.

2.2 WALL VENTS

A. Extruded-Aluminum Wall Vents:

1. Extruded-aluminum louvers and frames, not less than 0.125-inch (3.18-mm) nominal thickness, assembled by welding; with 18-by-14- (1.4-by-1.8-mm-) mesh, aluminum insect screening on inside face; incorporating weep holes, continuous drip at sill, and integral waterstop on inside edge of sill; of load-bearing design and construction.

2. Dampers: Aluminum blades and frames mounted on inside of wall vents; operated from exterior with Allen wrench in socket-head cap screw. Fabricate operating mechanism from Type 304 stainless steel components.

3. Finish: Anodized.

4. Basis of Design:

   a. Sunvent Industries; Product extruded Aluminum Vent with operable damper; www.sunventindustries.com. Product characteristics noted in paragraph 2.2.

   b. Substitutions: See Section 01 67 00 – Product Requirements.

2.3 MATERIALS

A. Aluminum Extrusions: ASTM B221 (ASTM B221M), Alloy 6063-T5, T-52, or T6.

B. Bituminous Paint: Cold-applied asphalt emulsion complying with ASTM D1187.

2.4 ALUMINUM FINISHES

A. Clear Anodic Finish: AAMA 611, AA-M12C22A41, Class I, 0.018 mm or thicker.

PART 3 EXECUTION

3.1 INSTALLATION

A. Install according to manufacturer's written instructions.

B. Locate and place vents level, plumb, and at indicated alignment with adjacent work.

C. Attach vents securely in place using fasteners supplied or approved by manufacturer.

D. Protect unpainted surfaces that are in contact with concrete, masonry, or dissimilar metals from corrosion and galvanic action by applying a heavy coating of bituminous paint or by separating surfaces with waterproof gaskets or nonmetallic flashing.

E. Build vents into concrete work as construction progresses; comply with requirements in Section 03 30 00 "Cast-in-Place Concrete."

F. Provide perimeter reveals of uniform width for sealants and joint fillers, where indicated.

G. Use concealed anchorages.

3.2 ADJUSTING AND CLEANING

A. Before final inspection, clean exposed surfaces with water and a mild soap or detergent not harmful to finishes. Thoroughly rinse surfaces and dry.

B. Restore vents damaged during installation and construction, so no evidence remains of corrective work. If results of restoration are unsuccessful, as determined by Contracting Officer , remove damaged units and replace with new units.

END OF SECTION

SECTION 09 21 16 - GYPSUM BOARD ASSEMBLIES

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Gypsum wallboard.

B. Joint treatment and accessories.

C. Water-resistive barrier over exterior wall sheathing.

1.2 RELATED REQUIREMENTS

A. Section 07 21 00 - Thermal Insulation: Acoustic insulation.

B. Section 07 90 05 - Joint Sealers: Acoustic sealant.

1.3 REFERENCE STANDARDS

A. ANSI A108.11 - American National Standard Specifications for Interior Installation of Cementitious Backer Units; 2023.

B. ASTM C840 - Standard Specification for Application and Finishing of Gypsum Board; 2024.

C. ASTM C1002 - Standard Specification for Steel Self-Piercing Tapping Screws for Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs; 2022.

D. ASTM C1047 - Standard Specification for Accessories for Gypsum Wallboard and Gypsum Veneer Base; 2019.

E. ASTM C1325 - Standard Specification for Fiber-Mat Reinforced Cementitious Backer Units; 2025.

F. ASTM C1396/C1396M - Standard Specification for Gypsum Board; 2024.

G. ASTM D3273 - Standard Test Method for Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber; 2021.

H. GA-216 - Application and Finishing of Gypsum Panel Products; 2024.

I. ESR 1308 - Fire and non-fire resistant rated suspended ceiling framing systems; 2006

J. ER 4071 - Seismic connection details for gypsum wallboard nonbearing partitions and suspended ceilings - 2007

K. USG Gypsum Handbook; latest edition

L. CA 0135 - Green Guard Certification.

1.4 SUBMITTALS

A. Product Data: Provide data on gypsum board, accessories, and joint finishing system.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: Company specializing in performing gypsum board installation and finishing, with minimum five years of experience.

B. STC-Rated Assemblies: For STC-rated assemblies, provide materials and construction identical to those tested in assembly indicated according to ASTM E 90 and classified according to ASTM E 413 by an independent testing agency.

C. ANSI/ASTM C 840 Application and Finishing of Gypsum Board and recommendations and specifications of The Gypsum Association, and The Western Lath/Plaster/ Drywall Industries Association.

D. Single Source Requirements: Work of Gypsum Board, Gypsum Board Shaft Wall Assembly, and Non Structural Metal Framing, shall be performed by or under direct supervision of one installer.

E. Do not mix different brands of Gypsum Board at rated assemblies.

F. Comply with details indicated and with gypsum board manufacturer's written recommendations or, if none available, with United States Gypsum's "Gypsum Construction Handbook."

PART 2 PRODUCTS

2.1 BOARD MATERIALS

A. Gypsum Wallboard: Paper-faced gypsum panels as defined in ASTM C1396/C1396M; sizes to minimize joints in place; ends square cut.

1. Application: Use for vertical surfaces, unless otherwise indicated.

2. Mold Resistance: Score of 10, when tested in accordance with ASTM D3273.

3. At Assemblies Indicated with Fire-Resistance Rating: Use type required by indicated tested assembly; if no tested assembly is indicated, use Type X board, UL or WH listed.

4. Thickness: 5/8 inches

a. Multi-Layer Assemblies: Thicknesses as indicated on drawings.

B. Backing Board For Wet Areas: One of the following products:

1. Application: Surfaces behind tile in wet areas including restrooms.

2. Mold Resistance: Score of 10, when tested in accordance with ASTM D3273.

3. ANSI Cement-Based Board (CEMENT BACKER BOARD): Non-gypsum-based; aggregated Portland cement panels with glass fiber mesh embedded in front and back surfaces complying with ANSI A118.9 or ASTM C1325.

   a. Thickness: 1/2 inch.

C. Backing Board For Non-Wet Areas: Water-resistant gypsum backing board as defined in ASTM C1396/C1396M; sizes to minimum joints in place; ends square cut.

1. Application: Vertical surfaces behind thinset tile, except in wet areas.

2. Mold Resistance: Score of 10, when tested in accordance with ASTM D3273.

3. At Assemblies Indicated with Fire-Resistance Rating: Use type required by indicated tested assembly; if no tested assembly is indicated, use Type X board, UL or WH listed.

4. Type: Regular and Type X, in locations indicated.

5. Type X Thickness: 5/8 inch.

6. Regular Board Thickness: 1/2 inch.

7. Edges: Tapered.

D. Gypsum Ceiling Board: Special sag-resistant gypsum ceiling board as defined in ASTM C1396/C1396M; sizes to minimize joints in place; ends square cut.

1. Application: Ceilings, unless otherwise indicated.

2. Thickness: 5/8 inch.

3. Edges: Tapered.

2.2 GYPSUM WALLBOARD ACCESSORIES

A. Acoustic Insulation: See Section 07 21 00.

B. Acoustic Sealant: As specified in Section 07 90 05.

C. Beads, Joint Accessories, and Other Trim: ASTM C1047, rolled zinc, unless noted otherwise.

D. Joint Materials: ASTM C475 and as recommended by gypsum board manufacturer for project conditions.

1. Joint Tape:

   a. Interior Gypsum Wallboard: Paper.

   b. Tile Backing Panels: As recommended by panel manufacturer.

2. Joint Compound for Interior Gypsum Wallboard: For each coat use formulation that is compatible with other compounds applied on previous or for successive coats.

    a. Prefilling: At open joints, rounded or beveled panel edges, and damaged surface areas, use setting-type taping compound.

    b. Embedding and First Coat: For embedding tape and first coat on joints, fasteners, and trim flanges, use setting-type taping or drying-type, all-purpose compound.

        1) Use setting-type compound for installing paper-faced metal trim accessories.

    c. Fill Coat: For second coat, use setting-type, sandable topping or drying-type, all-purpose compound.

    d. Finish Coat: For third coat, use setting-type, sandable topping or drying-type, all-purpose compound.

3. Acoustical Sealant: As specified in Section 07 90 05 - Joint Sealants.

4. Joint Compound for Exterior Applications:

    a. Glass-Mat Gypsum Sheathing Board: As recommended by sheathing board manufacturer.

5. Joint Compound for Tile Backing Panels:

    a. Water-Resistant Gypsum Backing Board: Use setting-type taping compound and setting-type, sandable topping compound.

E. Screws for Attachment to Steel Members Less Than 0.03 inch In Thickness, to Wood Members, and to Gypsum Board: ASTM C1002; self-piercing tapping type. fasteners at exterior applications shall be corrosion resistive.

1. For fastening cementitious backer units, use screws of type and size recommended by panel manufacturer.

F. Mechanical Fasteners: ASTM C 1513, corrosion-resistant-coated, self-drilling, self-tapping, steel drill screws.

1. Head Type: Low-profile head beneath sheathing, manufacturer's standard elsewhere.

G. Laminating Adhesive: Adhesive or joint compound recommended for directly adhering gypsum panels to continuous substrate.

H. Use adhesives that have a VOC content of 50 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

I. Adhesive for bonding insulation: Product with demonstrated capability to bond insulation securely to substrates indicated, where mechanical fasteners cannot be used, without damaging insulation and substrates

2.3 TRIM ACCESSORIES

A. Accessories for Interior Installation: Cornerbead, edge trim, and control joints complying with ASTM C 1047 and requirements indicated below.

B. Aluminum Trim: Extruded accessories of profiles and dimensions indicated.

1. Basis of Design:

a. Fry Reglet Corporation.. Product characteristics noted in paragraph 2.3.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2. Shapes as indicated by reference to Fig. 1 designations in ASTM C 1047.

a. Material: Galvanized or aluminum-coated steel sheet or rolled zinc Cornerbead on outside corners, unless otherwise indicated.

1) LC-bead with both face and back flanges; face flange formed to receive joint compound. Use LC beads for edge trim.

2) PRM Molding.

3) (CONTROL JOINT) One-piece control joint formed with V-shaped slot and removable strip covering slot opening.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that project conditions are appropriate for work of this section to commence.

3.2 ACOUSTIC ACCESSORIES INSTALLATION

A. Erect insulation (specified in Section 07 21 00 - Thermal Insulation) vertically and hold in place with Z-furring members spaced 24 inches (610 mm) o.c.

B. Acoustic Insulation: Place tightly within spaces, around cut openings, behind and around electrical and mechanical items within partitions, and tight to items passing through partitions.

C. Acoustic Sealant: Install in accordance with manufacturer's instructions.

3.3 BOARD INSTALLATION

A. Comply with ASTM C840, GA-216, and manufacturer's instructions. Install to minimize butt end joints, especially in highly visible locations.

B. Single-Layer Nonrated: Install gypsum board in most economical direction, with ends and edges occurring over firm bearing.

1. Exception: Tapered edges to receive joint treatment at right angles to framing.

2. On ceilings, apply gypsum panels before wall/partition board application to greatest extent possible and at right angles to framing, unless otherwise indicated.

3. On partitions/walls, apply gypsum panels horizontally (perpendicular to framing), unless otherwise indicated or required by fire-resistance-rated assembly, and minimize end joints.

a. Stagger abutting end joints not less than one framing member in alternate courses of panels.

b. At stairwells and other high walls, install panels horizontally, unless otherwise indicated or required by fire-resistance-rated assembly.

4. On Z-furring members, apply gypsum panels vertically (parallel to framing) with no end joints. Locate edge joints over furring members.

5. Fastening Methods: Apply gypsum panels to supports with steel drill screws.

C. Cementitious Backing Board: Install over steel framing members and plywood substrate where indicated, in accordance with ANSI A108.11 and manufacturer's instructions.

1. Areas Not Subject to Wetting: Install regular-type gypsum wallboard panels to produce a flat surface except at showers, tubs, and other locations indicated to receive water-resistant panels.

2. Where tile backing panels abut other types of panels in same plane, shim surfaces to produce a uniform plane across panel surfaces.

D. Install sound attenuation blankets before installing gypsum panels, unless blankets are readily installed after panels have been installed on one side.

E. Extend insulation full thickness as indicated to envelop entire area to be insulated. Cut and fit tightly around Obstructions, and fill voids with insulation. At indoor face of exterior wall, bond units to substrate with adhesive to provide permanent placement and support of units.

F. Use blanket widths and lengths that fill cavities formed by framing members. Where more than one length is required to fill cavity, provide lengths that will produce a snug fit between ends.

G. Place blankets in cavities formed by framing members to produce a friction fit between edges of insulation and adjoining framing members.

H. Install thermal insulation blanket at locations indicated. Extend insulation full thickness as indicated to envelop entire area to be insulated. Cut and fit tightly around obstructions, and fill voids with insulation.

1. Use blanket widths and lengths that fill cavities formed by framing members. Where more than one length is required to fill cavity, provide lengths that will produce a snug fit between ends.

2. Place blankets in cavities formed by framing members to produce a friction fit between edges of insulation and adjoining framing members.

I. Install ceiling panels across framing to minimize the number of abutting end joints and to avoid abutting end joints in central area of each ceiling. Stagger abutting end joints of adjacent panels not less than one framing member.

J. Install panels with face side out. Butt panels together for a light contact at edges and ends with not more than 1/16 inch (1.5 mm) of open space between panels. Do not force into place.

K. Locate edge and end joints over supports, except in ceiling applications where intermediate supports or gypsum board back-blocking is provided behind end joints. Do not place tapered edges against cut edges or ends. Stagger vertical joints on opposite sides of partitions. Do not make joints other than control joints at corners of framed openings.

L. Form control and expansion joints with space between edges of adjoining gypsum panels.

M. Cover both faces of support framing with gypsum panels in concealed spaces (above ceilings, etc.), except in chases braced internally.

1. Unless concealed application is indicated or required for sound, fire, air, or smoke ratings, coverage may be accomplished with scraps of not less than 8 sq. ft. (0.7 sq. m) in area.

2. Fit gypsum panels around ducts, pipes, and conduits.

3. Where partitions intersect structural members projecting below underside of floor/roof slabs and decks, cut gypsum panels to fit profile formed by structural members; allow 1/4- to 3/8-inch- (6.4- to 9.5-mm-) wide joints to install sealant.

N. Isolate perimeter of gypsum board applied to non-load-bearing partitions at structural abutments, except floors. Provide 1/4- to 1/2-inch- (6.4- to 12.7-mm-) wide spaces at these locations, and trim edges with edge trim where edges of panels are exposed. Seal joints between edges and abutting structural surfaces with acoustical sealant.

O. Attach gypsum panels to framing provided at openings and cutouts.

P. STC-Rated Assemblies: Seal construction at perimeters, behind control joints, and at openings and penetrations with a continuous bead of acoustical sealant. Install acoustical sealant at both faces of partitions at perimeters and through penetrations. Comply with ASTM C 919 and with manufacturer's written recommendations for locating edge trim and closing off sound-flanking paths around or through assemblies, including sealing partitions above acoustical ceilings.

Q. Water resistant sealant: Provide sealant per manufacturer's instructions at edges, around cutouts, and at all penetrations in water resistant gypsum board. Where partition is fire rated, provide fire rated water resistant sealant as required to maintain the fire rating.

R. Space fasteners in gypsum panels according to referenced gypsum board application and finishing standard and manufacturer's written recommendations.

S. Line or otherwise seal penetrations: or openings in fire resistive rated partitions, walls or ceilings for pipe sleeves, telephone specialties, toilet accessories, electrical devices or similar items, to maintain the required ratings.

T. Space fasteners in gypsum panels according to referenced standards and manufacturer's recommendations

1. Space screws a maximum of 12 inches o.c. for vertical applications.

U. Space fasteners in panels that are tile substrates a maximum of 8 inches o.c.

V. Install Gypsum Board panels ½ inch distance of off floor at all wall locations.

W. Any Gypsum Board that gets wet shall be removed and replaced.

3.4 INSTALLATION OF TRIM AND ACCESSORIES

A. Control Joints: Place control joints consistent with lines of building spaces and as indicated.

B. Corner Beads: Install at external corners, using longest practical lengths.

C. Edge Trim: Install at locations where gypsum board abuts dissimilar materials.

3.5 JOINT TREATMENT

A. Finish gypsum board in accordance with levels defined in GA-214 ASTM C840, as follows:

1. Level 4: Walls and ceilings to receive flat paint finish or wall coverings, unless otherwise indicated. Gloss and semi-gloss enamel paints are not usually recommended over this level of finish. ASTM C 840 requires application of "drywall primer" on surfaces before final decoration. Priming of gypsum board surfaces specified in Section 09 90 00 - Painting and Coating.

a. Embed tape in joint compound and apply first, fill (second), and finish (third) coats of joint compound over joints, angles, fastener heads, and accessories. Touch up and sand between coats and after last coat as needed to produce a surface free of visual defects and ready for decoration.

2. Level 3: Walls to receive textured wall finish before painting or heavy wallcoverings where lighting conditions are not critical and in closets.

a. Embed tape and apply separate first and fill coats of joint compound to tape, fasteners, and trim flanges

3. Level 2: In utility areas, behind cabinetry, and on backing board to receive tile finish and panels that are substrate for acoustical tile.

a. Embed tape and apply separate first and fill coat of joint compound to tape, fasteners, and trim flanges

4. Level 1: Fire-resistance-rated wall areas above finished ceilings, whether or not accessible in the completed construction.

a. Embed tape at joints.

B. Use the following joint compound combinations as applicable to finish levels:

1. Level 3:

a. Embedding and first coat: Ready mixed, drying type, all purpose or topping compound.

b. Fill (second) coat: Ready mixed, drying type, all purpose or topping compound.

2. Level 4:

a. Embedding and first coat: Setting type joint compound.

b. Fill (second) coat: Setting type joint compound.

c. Finish (third) coat: Ready mixed, drying type, all purpose or topping compound.

3. Level 5:

a. Embedding and first coat: Setting type joint compound.

b. Fill (second) coat: Setting type joint compound.

c. Fill (third) coat: Setting type joint compound.

d. Finish (fourth) coat: Ready mixed, drying type, all purpose or topping compound.

C. Primer and its application to surfaces are specified in other Division 9 Sections.

D. Tape, fill, and sand exposed joints, edges, and corners to produce smooth surface ready to receive finishes.

1. Feather coats of joint compound so that camber is maximum 1/32 inch.

3.6 TOLERANCES

A. Maximum Variation of Finished Gypsum Board Surface from True Flatness: 1/8 inch in 10 feet in any direction.

B. Suspended Ceilings and Soffits: Space framing and furring members as indicated.

1. Level ceiling system to a tolerance of 1/1200.

3.7 FIELD QUALITY CONTROL

A. Above-Ceiling Observation: Before Contractor installs final gypsum board ceilings, Contracting Officer will conduct an above-ceiling observation with consultants and report deficiencies in the work observed. Do not proceed with installation of final gypsum board to ceiling support framing until the deficiencies of the above ceiling work have been corrected.

1. Notify Contracting Officer seven days in advance of date and time when project, or part of project, will be ready for above-ceiling observation.

2. Before notifying Contracting Officer, complete gypsum board ceilings only to the extent that the following installation work has been completed:

a. 80 percent of lighting fixtures, powered for operation.

b. Insulation, and leak and pressure testing of water piping systems.

c. Air-duct systems, and air devices.

d. Mechanical system control-air tubing.

e. Support framing for items suspended from ceiling.

f. Acoustic isolation, sound control, and audio-visual components.

3.8 CLEANING, PROTECTION, FINAL CLEAN-UP

A. Remove any residual joint compound from adjacent surfaces as soon as possible.

B. Provide final protection and maintain conditions in a manner acceptable to installer that will make sure that gypsum board assemblies are undamaged or deteriorated, and ready for finishing by other trades prior to the time of Substantial Completion.

C. Screw Pop: Repair by installing new screw approximately 1-1/2 inches away from projecting screw and re-set projecting screw.

1. Where face is fractured, install new fastener approximately 1-1/2 inches away from projecting screw and remove projecting screw.

2. Fill damaged surface with joint compound and finish flush and to match adjacent surfaces.

D. Clean, repair, touch-up, replace, or make good the restoration of any work of other trades which have been soiled, or damaged by the work of this section.

E. Do not permit debris to accumulate. Remove debris from the site at the end of each day's work and legally dispose of debris.

END OF SECTION

SECTION 09 30 00 - TILING

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Tile for wall applications.

B. Cementitious backer board as tile substrate.

C. Ceramic trim.

D. Non-ceramic trim.

1.2 REFERENCE STANDARDS

A. ANSI A108.1a - American National Standard Specifications for Installation of Ceramic Tile in the Wet-Set Method, with Portland Cement Mortar; 2023.

B. ANSI A108.1b - American National Standard Specifications for Installation of Ceramic Tile on a Cured Portland Cement Mortar Setting Bed with Dry-Set or Latex-Portland Cement Mortar; 1999 (Reaffirmed 2010).

C. ANSI A108.1c - Specifications for Contractors Option: Installation of Ceramic Tile in the Wet-Set Method with Portland Cement Mortar or Installation of Ceramic Tile on a Cured Portland Cement Mortar Bed with Dry-Set or Latex-Portland Cement; 1999 (Reaffirmed 2010).

D. ANSI A108.2 - American National Standard General Requirements: Materials, Environmental and Workmanship; 2019.

E. ANSI A108.4 - American National Standard Specifications for Installation of Ceramic Tile with Organic Adhesives or Water Cleanable Tile-Setting Epoxy Adhesive; 2009 (Revised).

F. ANSI A108.5 - American National Standard Specifications for Installation of Ceramic Tile with Dry-Set Portland Cement Mortar or Latex-Portland Cement Mortar; 1999 (Reaffirmed 2010).

G. ANSI A108.6 - American National Standard Specifications for Installation of Ceramic Tile with Chemical Resistant, Water Cleanable Tile-Setting and -Grouting Epoxy; 1999 (Reaffirmed 2010).

H. ANSI A108.8 - American National Standard Specifications for Installation of Ceramic Tile with Chemical Resistant Furan Resin Mortar and Grout; 1999 (Reaffirmed 2010).

I. ANSI A108.9 - American National Standard Specifications for Installation of Ceramic Tile with Modified Epoxy Emulsion Mortar/Grout; 1999 (Reaffirmed 2010).

J. ANSI A108.10 - American National Standard Specifications for Installation of Grout in Tilework; 1999 (Reaffirmed 2010).

K. ANSI A108.11 - American National Standard Specifications for Interior Installation of Cementitious Backer Units; 2023.

L. ANSI A108.12 - American National Standard for Installation of Ceramic Tile with EGP (Exterior glue plywood) Latex-Portland Cement Mortar; 1999 (Reaffirmed 2010).

M. ANSI A108.13 - American National Standard for Installation of Load Bearing, Bonded, Waterproof Membranes for Thin-Set Ceramic Tile and Dimension Stone; 2005 (Reaffirmed 2021).

N. ANSI A108.19 - American National Standard Specifications for Interior Installation of Gauged Porcelain Tiles and Gauged Porcelain Tile Panels/Slabs by the Thin-Bed Method Bonded with Modified Dry-Set Cement Mortar or Improved Modified Dry-Set Cement Mortar; 2020.

O. ANSI A118.1 - American National Standard Specifications for Dry-Set Cement Mortar; 2023.

P. ANSI A118.4 - American National Standard Specifications for Modified Dry-Set Cement Mortar; 2023.

Q. ANSI A118.6 - American National Standard Specifications for Standard Cement Grouts for Tile Installation; 2019.

R. ANSI A118.9 - American National Standard Specifications for Test Methods and Specifications for Cementitious Backer Units; 2023.

S. ANSI A137.1 - American National Standard Specifications for Ceramic Tile; 2013.1.

T. ASTM C373 - Standard Test Methods for Determination of Water Absorption and Associated Properties by Vacuum Method for Pressed Ceramic Tiles and Glass Tiles and Boil Method for Extruded Ceramic Tiles and Non-tile Fired Ceramic Whiteware Products; 2018 (Reapproved 2023).

U. TCNA (HB) - Handbook for Ceramic, Glass, and Stone Tile Installation; 2025.

1.3 SUBMITTALS

A. Product Data: Provide manufacturers' data sheets on tile, mortar, grout, and accessories. Include instructions for using grouts and adhesives.

B. Shop Drawings: Indicate tile layout, patterns, color arrangement, perimeter conditions, junctions with dissimilar materials, control and expansion joints, thresholds, ceramic accessories, and setting details.

C. Samples: Mount tile and apply grout on two plywood panels, minimum 18 by 18 inches in size illustrating pattern, color variations, and grout joint size variations.

D. Installer's Qualification Statement:

1. Submit documentation of National Tile Contractors Association (NTCA) or Tile Contractors' Association of America (TCAA) accreditation.

1.4 QUALITY ASSURANCE

A. Single source limitation: Provide mortar and grout from same manufacturer.

B. Source limitation for tile: Obtain tile of each type and color or finish from one source or producer.

1. Obtain tile of each type and color or finish from same production run and of consistent quality in appearance and physical properties for each contiguous area.

C. Source limitations for setting and grouting materials: Obtain ingredients of a uniform quality for each mortar, adhesive, and grout component from one manufacturer and each aggregate from one source or producer.

D. Source limitations for other products: Obtain each of the following products specified in this section from a single manufacturer for each product:

1. Waterproof membrane.

2. Crack isolation membrane.

3. Joint sealants.

4. Cementitious backer units.

5. Metal edge strips.

1.5 MOCK-UPS

A. Construct tile mock-up where indicated on drawings, incorporating all components specified for the location.

1. Minimum size of mock-up is indicated on drawings.

2. Approved mock-up may remain as part of work.

1.6 DELIVERY, STORAGE, AND HANDLING

A. See Section 01 74 19 - Construction Waste Management and Disposal for packaging waste requirements.

B. Protect adhesives from freezing or overheating in accordance with manufacturer's instructions.

1.7 FIELD CONDITIONS

A. Do not install solvent-based products in an unventilated environment.

B. Maintain ambient and substrate temperature above 50 degrees F and below 100 degrees F during installation and curing of setting materials.

PART 2 PRODUCTS

2.1 PERFORANCE REQUIREMENTS:

A. Dynamic Coefficient of Friction: Not less than 0.42.

2.2 TILE

A. Glazed Wall Tile (WT-1): ANSI A137.1 standard grade.

1. Moisture Absorption: 7.0 to 20.0 percent as tested in accordance with ASTM C373.

2. Size: 4-1/4 by 4-1/4 inch, nominal.

3. Thickness: 5/16 inches.

4. Edges: Cushioned.

5. Surface Finish: High gloss.

6. Color: White 0100.

7. Basis of Design:

a. Dal-Tile Corporation; Color Wheel Collection - Glazed Ceramic; www.daltile.com. Product characteristics noted in paragraph 2.2A

b. Substitutions: See Section 01 67 00 – Product Requirements.

B. Glazed Wall Tile (WT-2): ANSI A137.1 standard grade.

1. Moisture Absorption: 3.0 to 6.0 percent as tested in accordance with ASTM C373.

2. Size: 4 by 4 inch (108 by 108 mm), nominal.

3. Thickness: 5/16 inch

4. Surface Finish: Gloss.

5. Color: Bone

6. Basis of Design:

a. Dal-Tile Corporation; Nu Bare - NB03. Product characteristics noted in paragraph 2.2B

b. Substitutions: See Section 01 67 00 – Product Requirements.

2.3 TRIM AND ACCESSORIES

A. Ceramic Trim Matching jolly, bullnose and quarter round ceramic shapes in sizes coordinated with field tile.

B. Non-Ceramic Trim (FLOOR TRANSITION): Satin brass anodized extruded aluminum, style and dimensions to suit application, for setting using tile mortar or adhesive.

1. Applications:

a. Open edges of floor tile.

b. Thresholds at door openings.

C. POST INSTALLED TILE FLANGE

1. Application a. Existing bathtubs with new tile surround.

2. Vinyl tiling-in bead designed allow alcove installation of a drop-in bath. Provide means of sealing to tub surface.

2.4 SETTING MATERIALS

A. Trowelable underlayments and patching compounds: Latex modified, portland cement based formulation provided or approved by manufacturer of tile setting materials for installations indicated.

B. Latex-Portland Cement Mortar Bond Coat: ANSI A118.4.

1. Applications: Use this type of bond coat where indicated, and where no other type of bond coat is indicated.

C. Thin-Set Portland Cement Mortar Bond Coat: ANSI A118.1.

1. For wall applications, provide non sagging mortar that complies with Paragraph C-4.6.1 in addition to the other requirements in ANSI A118.1.

D. Trowelable underlayments and patching compounds: Latex modified, portland cement based formulation provided or approved by manufacturer of tile setting materials for installations indicated.

E. Provide prepackaged, dry-mortar mix containing dry, redispersable, vinyl acetate or acrylic additive to which only water must be added at work site.

F. For wall applications, provide mortar that complies with requirements for nonsagging mortar in addition to the other requirements in ANSI A118.4.

2.5 GROUTS

A. Standard Grout: ANSI A118.6 standard cement grout.

1. Applications: Use where indicated on drawings.

2.6 ACCESSORY MATERIALS

A. Waterproofing Membrane (WATERPROOFING MEMBRANE SHEET): Polyethylene sheet membrane, faced on both sides with fleece webbing 0.008 inch thick, minimum; specifically designed for bonding to solid backing under thin-set tile; complying with ANSI A118.10.

1. Width:

a. 39 inches wide for singular application.

B. Provide accessories as required for a complete waterproof shower system including seals at plumbing elements.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that wall surfaces are smooth and flat within the tolerances specified for that type of work, are dust-free, and are ready to receive tile.

3.2 PREPARATION

A. Protect surrounding work from damage.

B. Vacuum clean surfaces and damp clean.

C. Seal substrate surface cracks with filler. Level existing substrate surfaces to acceptable flatness tolerances.

D. Fill cracks, holes, and depressions in concrete substrates for tile floors installed with thin-set mortar with trowelable leveling and patching compound specifically recommended by tile-setting material manufacturer.

E. Install backer board in accordance with ANSI A108.11 and board manufacturer's instructions. Tape joints and corners, cover with skim coat of setting material to a feather edge.

3.3 INSTALLATION - GENERAL

A. Install tile, thresholds, and stair treads and grout in accordance with applicable requirements of ANSI A108.1a through ANSI A108.19 , manufacturer's instructions, and TCNA (HB) recommendations.

B. Lay tile to pattern indicated. Do not interrupt tile pattern through openings.

C. Cut and fit tile to penetrations through tile, leaving sealant joint space. Form corners and bases neatly. Align floor joints.

D. Place tile joints uniform in width, subject to variance in tolerance allowed in tile size. Make grout joints without voids, cracks, excess mortar or excess grout, or too little grout.

E. Lay tile to pattern indicated. Do not interrupt tile pattern through openings.

F. Extend tile work into recesses and under or behind equipment and fixtures to conform complete covering without interruptions unless otherwise indicated. Terminate work neatly at obstructions, edges, and corners without disrupting patterns or joint alignments.

G. Accurately form intersections and returns. Perform cutting and drilling of tile without marring visible surfaces. carefully grind cut edges of tile abutting trim, finish or built in items for straight aligned joints. Fit tile closely to electrical outlets, piping, fixtures and other penetrations so plates, collars, or covers overlap tile.

H. Provide manufacturer's standard trim shapes where necessary to eliminate exposed tile edges.

I. Jointing pattern: lay tile in grid pattern unless otherwise indicated. Lay out tile work and center tile fields in both directions in each space or at each wall area. Lay out tile work to minimize the use of pieces that are less than half of a tile. Provide uniform joint widths unless otherwise indicated.

J. Cut and fit tile to penetrations through tile, leaving sealant joint space. Form corners and bases neatly. Align joints with existing joints.

K. For tile mounted in sheets, make joints between tile sheets same width as joints within tile sheets so joints between sheets are not apparent in finished work.

L. Where adjoining tiles on floor, base, walls, or trim are specified or indicated to be same size, align joints.

M. Where tiles specified or indicated to be whole integer multiples of adjoining tiles on floor, base, walls, or trim, align joints unless otherwise indicated.

N. Trim and Metal Edge Strips: Install where exposed edge of tile flooring meets carpet, wood, or other flooring that finishes flush with or below top of tile, and where no threshold is indicated.

O. Form internal angles square and external angles bullnosed.

P. Install non-ceramic trim in accordance with manufacturer's instructions.

Q. Sound tile after setting. Replace hollow sounding units.

R. Keep control and expansion joints free of mortar, grout, and adhesive.

S. Prior to grouting, allow installation to completely cure; minimum of 48 hours.

T. Grout tile joints unless otherwise indicated. Use standard grout unless otherwise indicated.

U. At changes in plane and tile-to-tile control joints, use tile sealant instead of grout, with either bond breaker tape or backer rod as appropriate to prevent three-sided bonding.

3.4 INSTALLATION - WALL TILE

A. Over cementitious backer units on studs, install in accordance with TCNA (HB) Method W422 at shower walls and W245 at dry location walls.

1. Joint Widths: Install tile on walls with the following joint widths:

a. Glazed Wall Tile: 1/8 to 3/8 inch (1.6 mm) as recommended by tile manufacturer.

3.5 CLEANING

A. Clean tile and grout surfaces.

3.6 PROTECTION

A. Do not permit traffic over finished floor surface for 4 days after installation.

END OF SECTION

SECTION 09 64 29 - WOOD STRIP AND PLANK FLOORING

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Wood strip and plank flooring, nailed.

B. Sleepers.

1.2 REFERENCE STANDARDS

A. NWFA (IG) - Installation Guidelines; current edition located at www.nwfa.org.

1.3 SUBMITTALS

A. Product Data: Provide data for flooring, floor finish materials, and accessories.

B. Shop Drawings: Indicate floor joint pattern and termination details.

C. Samples: Submit two samples 12 by 12 inch in size illustrating floor finish, color, and sheen.

D. Installation Instructions: Indicate standard and special installation procedures.

1.4 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section with minimum three years documented experience.

B. Installer Qualifications: Company specializing in performing work of the type specified and with at least three years of documented experience.

1.5 FIELD CONDITIONS

A. Do not install wood flooring until wet construction work is complete and ambient air at installation space has moisture content stabilized at maximum moisture content of 40 percent.

B. Provide heat, light, and ventilation prior to installation.

C. Store materials in area of installation for minimum period of 24 hours prior to installation.

D. Maintain minimum room temperature of 65 degrees F for a period of two days prior to delivery of materials to installation space, during installation, and after installation.

PART 2 PRODUCTS

2.1 MATERIALS

A. Wood Plank Flooring:

1. Species: Grade C and better, mixed grain Fir .

2. Actual Thickness: 25/32 inch.

3. Actual Width: Match existing

4. Edge: Tongue and groove.

5. End: End matched.

6. Finish: Linseed oil based varnish - see Section 09 91 23 Interior Painting.

PART 3 EXECUTION

3.1 INSTALLATION

A. Coordinate installation of new wood flooring material with salvaged material.

B. Consolidate salvaged material and new material in individual buildings.

C. Sheathing Paper: Place over wood subfloor; lap edges and ends 2 inches, staple in place.

D. Vapor Retarder: Comply with NOFMA's "Installing Hardwood Flooring" for vapor retarder installation and the following:

1. Wood Flooring Nailed to Wood Subfloor: Install flooring over a layer of asphalt-saturated felt.

E. Wood Flooring:

1. Install in accordance with manufacturer's and NWFA instructions; predrill and blind nail to subfloor.

2. Lay flooring parallel to length of room areas. Verify alignment as work progresses.

3. Arrange flooring with end matched grain set flush and tight.

4. Terminate flooring at centerline of door openings where adjacent floor finish is dissimilar; provide divider strips and transition strips in accordance with flooring manufacturer's recommendations and as indicated.

5. Install edge strips at unprotected or exposed edges, and where flooring terminates.

6. Secure edge strips before installation of flooring with stainless steel screws.

7. Install flooring tight to floor access covers.

8. Provide no less than 3/4 inch expansion space at fixed walls and other interruptions.

END OF SECTION

SECTION 09 65 00 - RESILIENT FLOORING

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Resilient sheet flooring.

B. Resilient tile flooring.

C. Installation accessories.

1.2 REFERENCE STANDARDS

A. ASTM E648 - Standard Test Method for Critical Radiant Flux of Floor-Covering Systems Using a Radiant Heat Energy Source; 2014c.

B. ASTM F2034 - Standard Specification for Sheet Linoleum Floor Covering; 2018 (Reapproved 2025).

C. ASTM F2195 - Standard Specification for Linoleum Floor Tile; 2018 (Reapproved 2023).

D. BAAQMD 8-51 - Bay Area Air Quality Management District Regulation 8, Rule 51, Adhesive and Sealant Products; www.baaqmd.gov; 2002.

E. GEI (SCH) - GREENGUARD "Children and Schools" Certified Products; GREENGUARD Environmental Institute; current listings at www.greenguard.org.

F. NFPA 253 - Standard Method of Test for Critical Radiant Flux of Floor Covering Systems Using a Radiant Heat Energy Source; 2015.

G. SCAQMD 1168 - South Coast Air Quality Management District Rule No.1168; current edition; www.aqmd.gov.

1.3 SUBMITTALS

A. Product Data: Provide data on specified products, describing physical and performance characteristics; including sizes, patterns and colors available; and installation instructions.

B. Shop Drawings: Indicate seaming plans and floor patterns.

C. Samples: Submit two samples, 12 by 12 inch in size illustrating color and pattern for each resilient flooring product specified.

1.4 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing specified flooring with minimum ten years documented experience.

B. Installer Qualifications: Company specializing in installing specified flooring with minimum five years documented experience.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: Engage an experienced installer trained by flooring manufacturer, equal to Forbo's Master Mechanic Program, to perform work of this Section who has specialized in installing linoleum floor coverings similar to those required for this Project and with a record of successful in-service performance.

B. Source Limitations: Obtain each type, color, and pattern of floor covering specified from one source with resources to provide products of consistent quality in appearance and physical properties without delaying the Work.

C. Fire-Test-Response Characteristics: Provide products with the following fire-test-response characteristics as determined by testing identical products according to test method indicated below by a testing and inspecting agency acceptable to authorities having jurisdiction.

1. Critical Radiant Flux: 0.45 W/sq. cm or greater when tested according to ASTM E 648.

2. Smoke Density: Maximum specific optical density of 450 or less when tested according to ASTM E 662.

D. Standard of Quality: For the purpose of evaluating the quality of workmanship, a mock up installation of the specified floor covering shall be provided by the Floor covering Contractor in an area designated by the Contracting Officer. Upon approval, this test installation shall then be considered the standard of quality and basis of comparison for the balance of the project. Areas found to be deficient by specification standards or application procedures shall be repaired/replaced at contractor's expense.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Upon receipt, immediately remove any shrink-wrap and check materials for damage and the correct style, color, quantity and run numbers.

B. Store all materials off of the floor in an acclimatized, weather-tight space.

C. Maintain temperature in storage area between 55 degrees F and 90 degrees F.

D. Protect roll materials from damage by storing on end.

E. Do not double stack pallets.

1.7 FIELD CONDITIONS

A. Maintain temperature in storage area between 55 degrees F and 90 degrees F.

B. Store materials for not less than 48 hours prior to installation in area of installation at a temperature of 70 degrees F to achieve temperature stability. Thereafter, maintain conditions above 55 degrees F.

1.8 EXTRA MATERIALS

A. Provide 10 lineal feet of base, of each type and color specified.

B. Provide 10 linear feet, full width of each type of sheet good flooring.

C. Provide 100 square feet of each type of resilient tile flooring.

PART 2 PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Coefficient of Friction: For flooring surfaces, provide products with the following values as determined by testing identical products per ASTM D 2047:

1. Level Surfaces: Minimum 0.6

2.2 SHEET FLOORING

A. Linoleum Sheet Flooring - (LIN-1) and Wall Covering: Homogeneous wear layer bonded to backing, with color and pattern through wear layer thickness.

1. Manufacturers: Product characteristics indicated in 2.2A

2. Basis of Design:

a. Forbo Flooring, Inc; Marmoleum: www.forboflooringna.com. Product characteristics noted in paragraph 2.2.A.

b. Substitutions: See Section 01 67 00 – Product Requirements.

3. Minimum Requirements: Comply with ASTM F2034, Type corresponding to type specified.

4. Critical Radiant Flux (CRF): Minimum 0.45 watt per square centimeter, when tested in accordance with ASTM E 648, NFPA 253, ASTM E 648, or NFPA 253.

5. VOC Content: Certified as Low Emission by one of the following :

a. GreenGuard Children and Schools; www.greenguard.org.

6. Backing: Jute fabric.

7. Thickness: 0.100 inch, minimum, excluding backing.

8. Sheet Width: 79 inch, minimum.

9. Seams: Heat welded.

10. Color: T3233 SHitake.

B. Welding Rod: Solid bead in material compatible with flooring, produced by flooring manufacturer for heat welding seams, and in color matching field color.

2.3 TILE FLOORING

A. Linoleum Tile: - (LIN-2) Homogeneous wear layer bonded to backing, with color and pattern through wear layer thickness.

1. Basis of Design:

a. Forbo Flooring, Inc; Marmoleum Modular: www.forboflooringna.com. Product characteristics noted in paragraph 2.3.A.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2. Minimum Requirements: Comply with ASTM F2195, Type corresponding to type specified.

3. Critical Radiant Flux (CRF): Minimum 0.45 watt per square centimeter, when tested in accordance with ASTM E 648, NFPA 253, ASTM E 648, or NFPA 253.

4. Backing: Synthetic fabric.

5. Thickness: 0.100 inch, minimum, excluding backing.

6. Color: MCT-629 Eiger.

2.4 ACCESSORIES

A. 1/2 inch underlayment, compatible with linoleum sheet flooring, for floor installations.

B. Subfloor Filler: White premix latex; type recommended by adhesive material manufacturer.

C. Primers, Adhesives, and Seam Sealer: Waterproof; types recommended by flooring manufacturer.

1. Provide only products having lower volatile organic compound (VOC) content than required by the more stringent of the South Coast Air Quality Management District Rule No.1168 and the Bay Area Air Quality Management District Regulation 8, Rule 51.

a. Cove Base Adhesives: Not more than 50 g/L.

b. Rubber Floor Adhesives: Not more than 60 g/L.

D. Reducer Strips for Resilient Floor Covering - (TS-#):

1. Material: Rubber Molding

2. Color: Burnt Umber

E. Seamless-Installation Accessories:

1. Heat-Welding Bead: Manufacturer's solid-strand product for heat welding seams.

a. Color: As selected by Contracting Officer from manufacturer's full range to match flooring.

F. J mold and corner mold for wall installations:

1. Material: Extruded aluminum alloy 6063 T5.

2. Wall thickness: - 0.050” nominal or greater.

3. Length: 10-foot lengths.

4. Finish: Clear Anodized.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that surfaces are flat to tolerances acceptable to flooring manufacturer, free of cracks that might telegraph through flooring, clean, dry, and free of curing compounds, surface hardeners, and other chemicals that might interfere with bonding of flooring to substrate.

B. Verify that wall surfaces are smooth and flat within the tolerances specified for that type of work, are dust-free, and are ready to receive resilient base.

C. Verify that required floor-mounted utilities are in correct location.

3.2 PREPARATION

A. Remove subfloor ridges and bumps. Fill minor low spots, cracks, joints, holes, and other defects with subfloor filler to achieve smooth, flat, hard surface.

B. Prohibit traffic until filler is fully cured.

C. Clean substrate.

3.3 INSTALLATION - GENERAL

A. Starting installation constitutes acceptance of subfloor conditions.

B. Install in accordance with manufacturer's written instructions.

C. Adhesive-Applied Installation:

1. Fit joints and butt seams tightly.

2. Set flooring in place, press with heavy roller to attain full adhesion.

D. Where type of floor finish, pattern, or color are different on opposite sides of door, terminate flooring under centerline of door.

E. Install edge strips at unprotected or exposed edges, where flooring terminates, and where indicated.

F. Scribe flooring to walls, columns, cabinets, floor outlets, and other appurtenances to produce tight joints.

G. Contractor shall provide Testing information to Floor covering installer

H. Install concrete surface treatment where required per manufacturer's written recommendations

3.4 INSTALLATION - SHEET FLOORING

A. Lay flooring with joints and seams parallel to longer room dimensions, to produce minimum number of seams. Lay out seams to avoid widths less than 1/3 of roll width; match patterns at seams.

B. Seal seams by heat welding where indicated.

C. Linoleum Sheet: Apply adhesive using manufacturer's recommended square notch trowel and lay flooring into wet adhesive and roll with a 100 pound roller to ensure proper bonding,

1. For wet areas and/or heavy rolling traffic refer to manufacturer recommendations for adhesive type and requirements

D. Do not install resilient flooring over expansion joints. Use expansion joint covers manufactured for use with resilient flooring.

E. Ramp up to match carpet height at transition between Linoleum Flooring and Carpet. Install Transition strip with leg embedded into Portland cement surfacing.

3.5 INSTALLATION - TILE FLOORING

A. Mix tile from container to ensure shade variations are consistent when tile is placed, unless otherwise indicated in manufacturer's installation instructions.

3.6 RESILIENT ACCESSORY INSTALLATION

A. Comply with manufacturer's written instructions for installing resilient accessories.

B. Resilient Molding Accessories: Butt to adjacent materials and tightly adhere to substrates throughout length of each piece. Install reducer strips at edges of carpet and resilient floor covering that would otherwise be exposed.

C. Seamless Installation:

1. Provide heat welded seams from Linoleum Flooring Manufacturer at all Linoleum to Vinyl transitions.

    a. Heat-Welded Seams: Comply with ASTM F 1516. Rout joints and heat weld with welding bead to permanently fuse sections into a seamless flooring. Prepare, weld, and finish seams to produce surfaces flush with adjoining flooring surfaces.

3.7 CLEANING

A. Remove excess adhesive from floor, base, and wall surfaces without damage.

B. Clean in accordance with manufacturer's written instructions.

3.8 PROTECTION

A. Prohibit traffic on resilient flooring for 48 hours after installation.

B. Cover resilient products until Substantial Completion.

END OF SECTION

SECTION 09 91 13 - EXTERIOR PAINTING

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Surface preparation.

B. Field application of paints.

1.2 RELATED REQUIREMENTS

A. Appendix 2: NPS Standard Operating Procedure (SOP) - Linseed Oil Painting

B. Appendix 3:NPS Standard Operating Procedure (SOP) - Oil and Pine Tar

1.3 DEFINITIONS

A. Conform to ASTM D16 for interpretation of terms used in this section.

B. MPI Gloss Level 1 - FLAT: Not more than five units at 60 degrees and 10 units at 85 degrees, according to ASTM D 523.

C. MPI Gloss Level 2 - VELVET: Not more than 10 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D 523.

D. MPI Gloss Level 3 - EGGSHELL: 10 to 25 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D 523.

E. MPI Gloss Level 4 - SATIN: 20 to 35 units at 60 degrees and not less than 35 units at 85 degrees, according to ASTM D 523.

F. MPI Gloss Level 5 - SEMI GLOSS: 35 to 70 units at 60 degrees, according to ASTM D 523.

G. MPI Gloss Level 6 - GLOSS: 70 to 85 units at 60 degrees, according to ASTM D 523.

H. MPI Gloss Level 7 - HIGH GLOSS: More than 85 units at 60 degrees, according to ASTM D 523.

1.4 REFERENCE STANDARDS

A. ASTM D16 - Standard Terminology for Paint, Related Coatings, Materials, and Applications; 2024.

B. MPI (APSM) - Master Painters Institute Architectural Painting Specification Manual; Current Edition.

1.5 SUBMITTALS

A. Product Data: Provide complete list of products to be used, with the following information for each:

1. Manufacturer's name, product name and/or catalog number, and general product category (e.g. "alkyd enamel").

2. MPI product number (e.g. MPI #47).

3. Cross-reference to specified paint system(s) product is to be used in; include description of each system.

B. Samples: Submit three paper "draw down" samples, 8-1/2 by 11 inches in size, illustrating range of colors available for each finishing product specified.

1. Where sheen is specified, submit samples in only that sheen.

2. Allow 30 days for approval process, after receipt of complete samples by Contracting Officer.

C. Manufacturer's Instructions: Indicate special surface preparation procedures.

D. Maintenance Data: Submit data including product technical data sheets, material safety data sheets (MSDS), care and cleaning instructions, and touch-up procedures.

1.6 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing the products specified, with minimum three years documented experience.

B. Applicator Qualifications: Company specializing in performing the type of work specified with minimum five years experience and approved by manufacturer.

C. See appendices for additional requirements:

1. Appendix 2: NPS Standard Operating Procedure (SOP) - Linseed Oil Painting.

2. Appendix 3:NPS Standard Operating Procedure (SOP) - Oil and Pine Tar.

1.7 MOCK-UP

A. Mockups: Apply mockups of each paint system indicated and each color and finish selected to verify preliminary selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

1. Contracting Officer will select one surface to represent surfaces and conditions for application of each paint system.

a. Vertical and Horizontal Surfaces: Provide samples of at least 100 sq. ft. (9 sq. m).

b. Other Items: Contracting Officer will designate items or areas required.

2. Final approval of color selections will be based on mockups.

a. If preliminary color selections are not approved, apply additional mockups of additional colors selected by Contracting Officer at no added cost.

3. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations in writing.

4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

B. Provide door and frame assembly mockup illustrating paint color, texture, and finish.

C. Provide full window frame and exterior casing mockup illustrating paint color, texture, and finish

D. Locate where directed by Contracting Officer .

E. Mock-up may remain as part of the work.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Deliver products to site in sealed and labeled containers; inspect to verify acceptability.

B. Container Label: Include manufacturer's name, type of paint, brand name, lot number, brand code, coverage, surface preparation, drying time, cleanup requirements, color designation, and instructions for mixing and reducing.

C. Paint Materials: Store at minimum ambient temperature of 45 degrees F and a maximum of 90 degrees F, in ventilated area, and as required by manufacturer's instructions.

1.9 FIELD CONDITIONS

A. Do not apply materials when surface and ambient temperatures are outside the paint product manufacturer's temperature ranges.

B. Follow manufacturer's recommended procedures for producing best results, including testing of substrates, moisture in substrates, and humidity and temperature limitations.

C. Do not apply exterior paint and finishes during rain or snow, or when relative humidity is outside the humidity ranges required by the paint product manufacturer.

D. Minimum Application Temperatures for Latex Paints: 50 degrees F for exterior; unless required otherwise by manufacturer's instructions.

E. Provide lighting level of 80 ft candles measured mid-height at substrate surface.

PART 2 PRODUCTS

2.1 PAINTS AND FINISHES - GENERAL

A. Paints and Finishes: Linseed Oil/Pine Tar finish is mixed on site, unless required to be a field-catalyzed paint.

1. Provide paints and finishes of a soft paste consistency, capable of being readily and uniformly dispersed to a homogeneous coating, with good flow and brushing properties, and capable of drying or curing free of streaks or sags.

2. Supply each paint material in quantity required to complete entire project's work from a single production run.

3. Do not reduce, thin, or dilute paint or finishes or add materials unless such procedure is described explicitly in manufacturer's product instructions.

2.2 PAINT SYSTEMS - EXTERIOR

A. Exterior surfaces to be painted or coated as noted on drawings and/or listed on Schedule.

B. Substrate Preparation:

1. Remove loose or failing coatings per 06 03 12 - Historic Wood Repair and 06 13 13 Historic Log Construction Repair

2. Repair as needed per 06 03 12 - Historic Wood Repair and 06 13 13 Historic Log Construction Repair

3. Provide sound and dry wood for paint finish application. Moisture content to be under 15%.

4. Provide lightly sanded surface to promote adhesion.

C. Linseed Oil and Blond Pine Tar Clear Coat (ST-2):

1. Provide a finish mixture of 20 percent (one part) blond pine tar and 80 percent (4 four parts) organic purified boiled linseed oil. Heat both pine tar and linseed oil to minimum temperature of 70°F and mix per manufacturer’s written instructions.

2. NPS Standard Operating Procedure (SOP) - Oil and Pine Tar

D. Linseed Oil Paint (PT-1):

1. Linseed Oil Priming Coat: Apply 1 thin coat of linseed oil applied as a primer. Allow for appropriate drying time (24 - 48 hours).

2. Linseed Oil Paint Coats: Apply 3 thin coats of linseed oil. Allow for appropriate drying time between coats (24 - 48 hours).

3. NPS Standard Operating Procedure (SOP) - Oil and Pine Tar

E. Exterior Surfaces to be Painted, Unless Otherwise Indicated: Including as indicated in Schedule at end of section.

1. Two top coats and one coat primer.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that surfaces are ready to receive work as instructed by the product manufacturer.

B. Examine surfaces scheduled to be finished prior to commencement of work. Report any condition that may potentially effect proper application.

C. Test shop-applied primer for compatibility with subsequent cover materials.

3.2 PREPARATION

A. Remove loose or failing coatings per 06 03 12 - Historic Wood Repair and 06 13 13 Historic Log Construction Repair.

B. Repair as needed per 06 03 12 - Historic Wood Repair and 06 13 13 Historic Log Construction Repair.

C. Provide sound and dry wood for paint finish application. Moisture content to be under 15%.

D. Provide lightly sanded surface to promote adhesion.

E. Clean surfaces thoroughly and correct defects prior to application.

F. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

G. Remove or mask surface appurtenances, including electrical plates, hardware, light fixture trim, escutcheons, and fittings, prior to preparing surfaces for finishing.

3.3 APPLICATION

A. See Standard Operating Procedures from NPS for Linseed oil application:

1. Appendix 2: NPS Standard Operating Procedure (SOP) - Linseed Oil Painting

2. Appendix 3:NPS Standard Operating Procedure (SOP) - Oil and Pine Tar

B. Apply products in accordance with manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual".

C. Do not apply finishes to surfaces that are not dry. Allow applied coats to dry before next coat is applied.

D. Apply paint such that it does not impede window operation.

E. Apply each coat to uniform appearance.

F. Vacuum clean surfaces of loose particles. Use tack cloth to remove dust and particles just prior to applying next coat.

G. Reinstall electrical cover plates, hardware, light fixture trim, escutcheons, and fittings removed prior to finishing.

3.4 CLEANING

A. Collect waste material that could constitute a fire hazard, place in closed metal containers, and remove daily from site.

3.5 PROTECTION

A. Protect finishes until completion of project.

B. Touch-up damaged finishes after Substantial Completion.

3.6 SCHEDULE - LOCATIONS (AS NOTED BUT NOT LIMITED TO):

A. As indicated on drawings.

END OF SECTION

SECTION 09 91 23 - INTERIOR PAINTING

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Surface preparation.

B. Field application of paints.

C. Do Not Paint or Finish the Following Items:

1. Items factory-finished unless otherwise indicated; materials and products having factory-applied primers are not considered factory finished.

2. Items indicated to receive other finishes.

3. Items indicated to remain unfinished.

4. Fire rating labels, equipment serial number and capacity labels, bar code labels, and operating parts of equipment.

5. Floors, unless specifically indicated.

6. Glass.

7. Concealed pipes, ducts, and conduits.

1.2 DEFINITIONS

A. Conform to ASTM D16 for interpretation of terms used in this section.

B. MPI Gloss Level 1 - FLAT: Not more than five units at 60 degrees and 10 units at 85 degrees, according to ASTM D 523.

C. MPI Gloss Level 2 - VELVET: Not more than 10 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D 523.

D. MPI Gloss Level 3 - EGGSHELL: 10 to 25 units at 60 degrees and 10 to 35 units at 85 degrees, according to ASTM D 523.

E. MPI Gloss Level 4 - SATIN: 20 to 35 units at 60 degrees and not less than 35 units at 85 degrees, according to ASTM D 523.

F. MPI Gloss Level 5 - SEMI GLOSS: 35 to 70 units at 60 degrees, according to ASTM D 523.

G. MPI Gloss Level 6 - GLOSS: 70 to 85 units at 60 degrees, according to ASTM D 523.

H. MPI Gloss Level 7 - HIGH GLOSS: More than 85 units at 60 degrees, according to ASTM D 523.

1.3 REFERENCE STANDARDS

A. 40 CFR 59, Subpart D - National Volatile Organic Compound Emission Standards for Architectural Coatings; U.S. Environmental Protection Agency; Current Edition.

B. ASTM D16 - Standard Terminology for Paint, Related Coatings, Materials, and Applications; 2024.

C. MPI (APL) - Master Painters Institute Approved Products List; Master Painters and Decorators Association; current edition, www.paintinfo.com.

D. MPI (APSM) - Master Painters Institute Architectural Painting Specification Manual; Current Edition.

E. SSPC-SP 1 - Solvent Cleaning; 2015.

1.4 SUBMITTALS

A. Product Data: Provide complete list of products to be used, with the following information for each:

1. Manufacturer's name, product name and/or catalog number, and general product category (e.g., "alkyd enamel").

2. MPI product number (e.g., MPI #47).

3. Cross-reference to specified paint system(s) product is to be used in; include description of each system.

B. Samples: Submit three paper "draw down" samples, 8-1/2 by 11 inches in size, illustrating range of colors available for each finishing product specified.

1. Where sheen is specified, submit samples in only that sheen.

2. Allow 30 days for approval process, after receipt of complete samples by Contracting Officer.

C. Manufacturer's Instructions: Indicate special surface preparation procedures.

D. Maintenance Data: Submit data including product technical data sheets, material safety data sheets (MSDS), care and cleaning instructions, and touch-up procedures.

1.5 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing the products specified, with minimum five years documented experience.

B. Applicator Qualifications: Company specializing in performing the type of work specified with minimum five years experience and approved by manufacturer.

1.6 MOCK-UP

A. Mockups: Apply mockups of each paint system indicated and each color and finish selected to verify preliminary selections made under Sample submittals and to demonstrate aesthetic effects and set quality standards for materials and execution.

1. Contracting Officer will select one surface to represent surfaces and conditions for application of each paint system.

a. Vertical and Horizontal Surfaces: Provide samples of at least 100 sq. ft. (9 sq. m).

b. Other Items: Contracting Officer will designate items or areas required.

2. Final approval of color selections will be based on mockups.

a. If preliminary color selections are not approved, apply additional mockups of additional colors selected by Contracting Officer at no added cost.

3. Approval of mockups does not constitute approval of deviations from the Contract Documents contained in mockups unless Contracting Officer specifically approves such deviations in writing.

4. Subject to compliance with requirements, approved mockups may become part of the completed Work if undisturbed at time of Substantial Completion.

B. Provide door and frame assembly illustrating paint color, texture, and finish.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Deliver products to site in sealed and labeled containers; inspect to verify acceptability.

B. Container Label: Include manufacturer's name, type of paint, brand name, lot number, brand code, coverage, surface preparation, drying time, cleanup requirements, color designation, and instructions for mixing and reducing.

C. Paint Materials: Store at minimum ambient temperature of 45 degrees F and a maximum of 90 degrees F, in ventilated area, and as required by manufacturer's instructions.

1.8 FIELD CONDITIONS

A. Do not apply materials when surface and ambient temperatures are outside the temperature ranges required by the paint product manufacturer.

B. Follow manufacturer's recommended procedures for producing best results, including testing of substrates, moisture in substrates, and humidity and temperature limitations.

C. Provide lighting level of 80 ft candles measured mid-height at substrate surface.

PART 2 PRODUCTS

2.1 MANUFACTURERS

A. Provide paints and finishes from the same manufacturer to the greatest extent possible.

B. Basis of Design:

1. Sherwin-Williams Company: www.sherwin-williams.com. Product characteristics noted in paragraph 2.3.B

2. Substitutions: See Section 01 67 00 – Product Requirements.

2.2 PAINTS AND FINISHES - GENERAL

A. Sheens: Provide the sheens specified; where sheen is not specified, sheen will be selected later by Contracting Officer from the manufacturer's full line.

2.3 PAINT SYSTEMS - INTERIOR

A. Interior Surfaces to be Painted or Coated as noted on drawings and/or listed on Schedules.

B. Moisture Resistant Paint (PT-2):

1. Top Coat(s): High Performance Architectural Interior Latex; MPI #138, 139, 140, or 141.

2. Top Coat Sheen:

a. Eggshell: MPI gloss level 3; use this sheen at all locations.

C. Linseed Oil & Turpentine Coat (ST-1):

1. Provide an equal part finish mixture of (one part) boiled Linseed Oil and (one part) Turpentine.

2. Apply (3) thin coats. With each coat applied, let absorb for 30min and then wipe off excess. Allow for appropriate drying time between coats (24-48 hours).

D. Linseed Oil Varnish (ST-3):

1. Basis of Design:

a. Le Tonkinois No 1. Product characteristics noted in paragraph 2.3.D.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2. Linseed oil, tung oil, natural resins, isoaliphates.

3. UV Resistant.

4. Apply per manufacturer instructions.

E. Linseed Oil Wood Floor Varnish Coat (ST-4)

1. Basis of Design:

a. Le Tonkinois Parquets. Product characteristics noted in paragraph 2.3.E.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2. Open pore varnish.

3. Linseed oil, tung oil, natural resins, isoaliphates.

4. UV Resistant.

5. Apply per manufacturer instructions.

2.4 PRIMERS

A. Primers: Provide the following unless other primer is required or recommended by manufacturer of top coats.

1. Alkali Resistant Water Based Primer; MPI #3.

a. Location: Where latex paint is being used.

2. Linseed Oil Base Finish

a. Location: Where a linseed oil based product is being used.

2.5 ACCESSORY MATERIALS

A. Accessory Materials: Provide primers, sealers, cleaning agents, cleaning cloths, sanding materials, and clean-up materials as required for final completion of painted surfaces.

B. Patching Material for gypsum board walls Latex filler.

C. Fastener Head Cover Material for gypsum board walls: Latex filler.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify that surfaces are ready to receive work as instructed by the product manufacturer.

B. Examine surfaces scheduled to be finished prior to commencement of work. Report any condition that may potentially affect proper application.

C. Test shop-applied primer for compatibility with subsequent cover materials.

D. Measure moisture content of surfaces using an electronic moisture meter. Do not apply finishes unless moisture content of surfaces is below the following maximums:

3.2 PREPARATION

A. Clean surfaces thoroughly and correct defects prior to application.

B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

C. Remove or mask surface appurtenances, including electrical plates, hardware, light fixture trim, escutcheons, and fittings, prior to preparing surfaces or finishing.

D. Seal surfaces that might cause bleed through or staining of topcoat.

E. Gypsum Board: Fill minor defects with filler compound. Spot prime defects after repair.

F. Plaster: Fill hairline cracks, small holes, and imperfections with latex patching plaster. Make smooth and flush with adjacent surfaces. Wash and neutralize high alkali surfaces.

3.3 APPLICATION

A. Apply products in accordance with manufacturer's written instructions and recommendations in "MPI Architectural Painting Specification Manual".

B. Do not apply finishes to surfaces that are not dry. Allow applied coats to dry before next coat is applied.

C. Apply each coat to uniform appearance in thicknesses specified by manufacturer.

D. Vacuum clean surfaces of loose particles. Use tack cloth to remove dust and particles just prior to applying next coat.

E. Reinstall electrical cover plates, hardware, light fixture trim, escutcheons, and fittings removed prior to finishing.

3.4 CLEANING

A. Collect waste material that could constitute a fire hazard, place in closed metal containers, and remove daily from site.

3.5 PROTECTION

A. Protect finishes until completion of project.

B. Touch-up damaged finishes after Substantial Completion.

3.6 SCHEDULE - LOCATIONS (AS NOTED BUT NOT LIMITED TO):

A. As indicated on drawings, unless otherwise noted.

B. Ferrous Metals:

1. Exposed electrical piping.

2. Exposed conduit and J-boxes.

3.7 COLOR SCHEDULE (P-#)

A. PT-3:

1. Color and Sheen: Match colors used by Park provided by Contracting Officer.

END OF SECTION

SECTION 10 14 00 - SIGNAGE

PART 1 - GENERAL

1.1 SUMMARY:

A. Work of this Section includes:

1. Code-required interior signs

2. Room ID

3. Fabrication and installation for all signs in package.

1.2 SUBMITTALS:

A. Schedule:

1. Sign Contractor is responsible for providing all submittals in a timely manner, allowing sufficient time for review by Contracting Officer, revisions and re-submittals without jeopardizing the sign installation schedule set by the Contracting Officer. Any delays in the submittal process will have to be made up by Sign Contractor during the fabrication process.

B. Shop Drawings

1. Must be approved before any sign samples or signs are fabricated, or materials are ordered.

2. Submit a digital set of shop drawings (PDF format) for review by Contracting Officer .

3. Sign Contractor’s drawings must include necessary product data and mounting details for each sign in this package. Indicate sign layout, sign messages, letter forms, spacing, dimensions, elevations and connection details for each sign type. Indicate sign locations and connection to mounting surface.

C. Material and Finish Samples:

1. Submit for approval (3) three sets of samples of each sign component material showing finishes, colors, and surface textures. Component material samples must be approved before completed product samples are fabricated.

a. Sample Size: 6" x 6"

D. Sign Samples:

1. Production to commence after approval of shop drawings & material and finish samples.

2. Submission prior to approval of shop drawings & material and finish samples will be grounds for rejection without review.

3. Sign Contractor must submit (1) one sample of each sign types to Contracting Officer for review and approval prior to producing signs for this package (note: submit complete, full-size sign unless otherwise indicated). The Contracting Officer reserves the right to disassemble any item and to subject items to any test necessary to determine its strength or character without being responsible for any damage to the item caused thereby. Samples will be retained by the Contracting Officer and cannot be used as part of the project.

E. Written Warranty

1. Provide a written warranty indicating all work of this Bid Lot will be free from defects in material and workmanship for a period of (1) one year after Contracting Officer acceptance, and that for a period of (1) one year after Contracting Officer acceptance repairs and replacements of said defects shall be performed in a timely manner at no expense to the Government.

1.3 QUALITY ASSURANCE:

A. Fabricator Qualifications: For each sign form and graphic image process indicated, furnished products from manufacturers regularly engaged in work of this magnitude and scope for minimum of five years.

B. Uniformity of Manufacturers: For each sign form and graphic image process indicated, furnish products of a single manufacturer.

C. Fabrication Observation: In addition to review for submittals, fabricator may be required to provide photo documentation and allow intermittent shop visits to review progress and quality. Photos of reviewed sign types shall be part of meeting record. 14-day notification shall be provided.

D. Where specified materials or methods exceed minimum standards allowed by applicable codes, the more stringent requirement shall apply.

1.4 PROJECT CONDITIONS

A. Field Measurements: Take field measurements prior to preparation of shop drawings and fabrication, where possible, to insure proper fitting of work.

1.5 DELIVERY AND STORAGE

A. Delivery of Materials: Deliver work to project site when adjacent finishes are complete and ready for immediate installation.

B. Handling Materials: Handle finished product in a careful manner in order not to damage or mar surfaces of finished product or adjacent surfaces.

1.6 WARRANTY

A. Special Warranty: Manufacturer's standard form in which the manufacturer agrees to repair or replace components of signs that fail in materials or workmanship within a specified warranty period.

1. Warranty periods from substantial completion:

a. Vinyl graphics: 6 years

b. Direct print graphics: Minimum of 5 years against fading, cracking, chipping or interior use.

PART 2 - PRODUCTS

2.1 MATERIALS AND FABRICATION:

A. General:

1. All materials shall be new stock, free from defects impairing strength, durability and appearance. All fabrication and installation shall be in accordance with the highest standards of the trade. All signs and components shall be complete and free from visual and structural/mechanical flaws.

2. All graphics, including text, symbols and arrows shall be executed in such a manner that all edges and corners are true and clean.

3. Type Sizes: As indicated on drawings for particular units.

4. Typefaces: All work to precisely replicate the typefaces as indicated on drawings.

5. Typographic Spacing: Match letter, word and line spacing as indicated on drawings for all text configurations.

6. Symbols and Arrows: Match artwork as indicated on the drawings.

B. Tactile Text:

1. Tactile text is produced by using a laser or rotary engraver to cut each character. Weed the scrap from around each character. Use a cutter that leaves an edge that is perfectly perpendicular (no bevel). Cut edges shall be smooth. Divots or start/stop cutting marks shall not be visible on finished product.

2. Each sign panel to receive tactile text shall be routed to accept each individual character. Each character is adhered to routed panel with thin, clear liquid adhesive (no adhesive shall be visible on finished product). Tactile text shall be raised 1/32” above face of panel.

3. Align character baselines to maintain a baseline parallel to the sign format. Placement of tactile text shall be as shown on drawings.

4. If any recent governmental promulgations have altered the use of tactile text, so as to impact the accompanying drawings, those new uses shall be utilized and indicated in fabricator's shop drawings.

C. Braille

1. Braille is produced as clear acrylic rastor beads embedded into surface of the sign panel. Use an engraver to drill Braille holes and a pen applicator to insert Braille beads.

2. All Braille shall be Grade 2 Braille. Placement of the Braille shall be located as shown on drawings. The fabricator is responsible for proper use of Grade 2 Braille in terms of shape, size, placement, and spelling, including proper contractions.

3. Sign Contractor must verify that the Braille translations included on the signs fabricated and installed for this project are correct and compliant with the ABAAS.

4. If any recent governmental promulgations have altered the use of Braille, so as to impact the accompanying drawings, those new uses shall be utilized and indicated in fabricator's shop drawings.

5. Braille Dots shall be 1/10 inch on center in each cell with 2/10 inch space between cells, measured from the second column of dots in the first cell to the first column of dots in the second cell. Dots shall be raised a minimum of 1/40 inch above the background. Braille dots shall be domed or rounded.

D. Ink & Silkscreen:

1. Ink shall be compatible with the surface on which it is applied and recommended by the manufacturer of the ink.. All ink used shall be of a type containing ultraviolet inhibitor to protect against fading and discoloration. All inks shall be evenly applied and without roughness, pinholes, scratches, orange peeling, dust, application marks, etc.

2. Disposition of silkscreened color shall be even and opaque.

3. There shall be 100% fidelity between film positives and screen. Provide 100% film positive, emulsion side up and clean. Screen artwork shall not noticeably deviate from the film positives in line weight.

4. Double strikes of screened art which show as double images, faint, or fuzzy line/edge quality will not be accepted if visible when viewed from 5 feet or normal viewing distance, whichever is closer.

5. The fabricator shall determine the optimum mesh size for screenprinting.

6. Ink colors specified:

a. Gray - PMS 424

E. Paint & Painting:

1. All paint shall be compatible with the surface on which it is applied and recommended by the manufacturer of the paint. All paint used shall be of a type containing ultraviolet inhibitor to protect against fading and discoloration. All paint shall be evenly applied and without roughness, pinholes, scratches, orange peeling, dust, application marks, etc.

2. Unless otherwise indicated, painted surfaces shall have a semi-matte or “Eggshell” finish.

3. Paint: Provide Matthews Brand paint finishes, color matched to the following:

a. White Matthews Paint 32071 White Wonder

F. Colors: **Contracting Officer to confirm color paint spec**

1. Gray - Matthews Paint 04761 Steel Wool

2. Red - Matthews Paint 10252 Campbells Soup Red

G. Hardware:

1. Provide hardware in sizes, types and finishes shown and specified. Where hardware is indicated but not explicitly specified, submit hardware sizes, types and finishes for review.

2. Fasteners: Of same basic metal and alloy as fastened metal, unless otherwise indicated. Do not use metals which are corrosive or otherwise incompatible with metals joined.

a. Fastening devices between dissimilar materials shall be 300 Series non-magnetic stainless steel.

b. Material: Galvanically compatible with adjacent materials.

c. Finish: Where exposed to view match adjacent material.

d. Provide concealed fasteners for interconnection of metal work components and for their attachment to other work except where exposed fasteners are indicated on the drawings.

e. Provide socket flat-head machine screws or bolts for exposed fasteners, unless otherwise indicated.

H. Adhesive:

1. General: Provide low or no VOC adhesives.

2. High Bond Tape: Product as recommended by manufacturer for type of use, materials and fabrication.

a. 3M Company VHB tape

b. Duraco high bond tape

c. Or approved equal.

3. Foam Tape: Double faced pressure sensitive foam tape, product in width and thickness as recommended by manufacturer for type of use, materials and fabrication.

a. 3M Company

b. Avery Dennison Acrylic Foam Bonding tapes

c. Duraco

d. Or approved equal.

4. Silicone: FS TT-S-001543B, Class A, silicone sealant.

a. Dowsil; 795 silicone building sealant.

b. Or approved equal.

5. Epoxy: Epoxy shall be two-component thermosetting epoxy adhesive with 100% solids content.

a. Miracle Adhesives Corporation; #NP-428

b. Hughson Chemical Division of Lord Corporation; ChemLok #304

c. Or approved equal.

6. Magnetic Tape: Scotch Brand, or approved equal, self-adhesive magnetic tape in size and quantity to assure adhesion.

7. Adhesive products must never be visible on finished, installed signs.

I. Acrylic:

1. Unless otherwise indicated, acrylic surfaces shall be “non-glare.” Where non-glare acrylic is specified, use ACRYCAST cell cast acrylic manufactured by Calsak Plastics, or comparable.

2. All sign surfaces (including edges) must be smooth and free of scratches, gouges, cutting marks or other imperfections. Incorporate ultraviolet inhibitor into sign materials to produce maximum color stability.

J. Rowmark ABA Alternative:

1. Single-ply modified acrylic with a matte non-glare finish. All edges shall be smooth-finished.

2. Color for tactile text is 595 “Cinder”

K. Typography

1. There are two families of typefaces used in this sign program; alternate fonts will not be accepted. Letter spacing shall be as shown on drawings.

2. The fabricator is responsible for purchasing all specified typefaces and for setting appropriate Grade 2 Braille.

a. Font: Futura Book

b. Grade 2 Braille

2.2 SIGN TYPE SPECIFICATIONS

A. The following specifications are meant to be accompanied with sign drawings. See Sign Drawings for additional fabrication and installation details.

1. Sign Type D3

a. Materials:

1) 1/8 inch thick aluminum sign panel

2) 2.5 inch x 2.5 inch extruded aluminum tube post

3) 1/8 inch thick aluminum post top

4) exterior-grade mounting adhesive.

5) Concrete footing, as necessary

b. Color:

1) Acrylic sign panel face is painted White. Returns are painted White

2) Graphics are surface-silkscreened Gray to sign panel face.

3) Aluminum post and cap are raw.

c. Size:

1) Sign panel is 8 inches wide x 8 inches high

2) Post is 2.5 inch square x 48 high (above ground). Sign Contractor to engineer size of concrete footing and overall height of post.

d. Graphics:

1) Typeface used is Futura Book.

e. Construction:

1) See sections 2.1.D / E / H and sign drawing.

f. Installation:

1) Install sign post directly in ground or into a concrete footing.

2) Adhere panel directly to post with exterior-grade adhesive.

2. Sign Type E1

a. Materials:

1) 1/4 inch thick clear non-glare acrylic sign panel

2) 1/8 inch thick clear acrylic offset panel

3) 3M VHB mounting tape adhesive

b. Color:

1) Acrylic sign panel face is clear non-glare; backpainted White. Returns are painted White

2) Graphics are surface-silkscreened Gray.

3) Acrylic offset panel is clear.

c. Size:

1) Sign panel is 4 inches wide x 6 inches high

2) Offset panel is 3.5 inches wide x 5.5 inches high

d. Graphics:

1) Typeface used is Futura Book.

e. Construction:

1) See sections 2.1.D / E / H / I and sign drawing.

f. Installation:

1) Adhere sign flush to wall with bottom edge 1/2 inch above call button panel; vertical edges to align exactly with call button panel edges.

3. Sign Type F1

a. Materials:

1) 1/4 inch thick clear non-glare acrylic sign panel

2) Rowmark® tactile text

3) Clear Braille beads

4) 1/8 inch thick clear acrylic offset panel

5) 3M VHB mounting tape adhesive

b. Color:

1) Acrylic sign panel face is clear non-glare; backpainted White. Returns are painted White

2) Rowmark text color is Cinder.

3) Braille beads are clear

4) Acrylic offset panel is clear.

c. Size:

1) Sign panel is 8.5 inches wide x 3 inches high

2) Offset panel is 8 inches wide x 2.5 inches high

d. Graphics:

1) Typefaces used are Futura Book and Grade 2 Braille.

e. Construction:

1) See sections 2.1.E / H / I / J and sign drawing.

f. Installation:

1) Adhere sign flush to wall with top edge 60 inches AFF. Locate sign on latch-side of door, 4 inches from door frame. If less than 24 inches is available at latchside of door, center sign horizontally in available space.

2) See section 2.3.B for wall-mounted signs

3) See section 2.3.C for glass-mounted signs.

4. Sign Type H2

a. Materials:

1) 1/4 inch thick clear non-glare acrylic sign panel

2) Rowmark® tactile text

3) Clear Braille beads

4) 1/8 inch thick clear acrylic offset panel

5) 3M VHB mounting tape adhesive

b. Color:

1) Acrylic sign panel face is clear non-glare; backpainted White. Returns are painted White

2) Rowmark text color is Cinder.

3) Braille beads are clear

4) Graphics are surface-silkscreened Gray.

5) Acrylic offset panel is clear.

c. Size:

1) Sign panel is 7 inches wide x 8.5 inches high

2) Offset panel is 6.5 inches wide x 8 inches high

d. Graphics:

1) Typefaces used are Futura Book and Grade 2 Braille.

e. Construction:

1) See sections 2.1.D / E / H / I / J and sign drawing.

f. Installation:

1) Adhere sign flush to wall with top edge 60 inches AFF. Locate sign 4 inches from door frame. If less than 24 inches is available at latchside of door, center sign horizontally in available space.

PART 3 - EXECUTION

3.1 INSTALLATION

A. General:

1. All sign installations shall meet the established guidelines of the Architectural Barriers Act as well as relevant American National Standards Institute (ANSI) guidelines adopted into local building codes.

2. The supplier is responsible for staking all sign locations and field verifying with Contracting Officer and Contracting Officer before installing signs.

3. The supplier may be asked to install specific sign mock-ups on site for approval prior to beginning final installation.

4. The supplier is responsible for securely installing all signs. When the type of mounting is not other-wise specified, all signs shall be permanently mounted.

5. Install all items as described in specifications and shown on sign drawings. If site conditions do not allow for sign placement as described herein, or for the sign to function as described herein, seek advice from Contracting Officer. Check all items for correct placement

6. Make all provisions necessary and take special precautions to protect and prevent damage to Contracting Officer's property. Any items damaged shall be restored to the original condition and the supplier charged with the expense thereof.

7. All signs are to be installed in such a way that wall surfaces are not damaged.

8. Installers shall maintain a clean work area and keep walkways clear of debris. All crating materials must be removed from project site at the end of each workday and when installation is complete.

B. Wall-mounted Signs

1. Foam Tape Mounting: Use 3M product as specified for mounting signs to smooth, non-porous surfaces as indicated on the drawings. Do not use this method for vinyl-covered or rough surfaces.Position tape at back of sign for optimum performance and hidden from view.

2. Silicone Adhesive Mounting: Use liquid silicone adhesive as specified to attach sign units to irregular, porous or vinyl-covered surfaces as indicated on the drawings. Use foam tape where recommended by the sign manufacturer to hold the sign in place until the adhesive has fully cured. Remove visible adhesive.

3. Mechanical Fastener Mounting: Install signs securely to wall with fasteners and anchoring devices as indicated on drawings and as specified.

4. VHB Adhesive Mounting: Install signs with full sheet VHB adhesive where indicated on the drawings. Trim adhesive flush to exposed edges of material being installed.

3.2 CLEANING AND PROTECTION:

A. General:

1. Protect finishes from damage during construction period, field handling and installation by use of temporary protective coverings. Protect adjacent surfaces from damage during field fabrication and installation. Remove protective covering at time of substantial completion.

2. Restore finishes damaged during installation and construction period so that no evidence remains of corrective work. Touch up any exposed fasteners and connecting hardware to match color and finish of surrounding surface. Touch up damaged surfaces carefully, using airbrush technique where necessary. Return items which cannot be refinished in the field to the shop; make required alterations and refinish entire unit, or provide new units as required.

3. Clean all exposed surfaces just prior to substantial completion in accordance with manufacturer's written cleaning instructions. Protect signs from damage until acceptance.

END OF SECTION

SECTION 10 28 00 - TOILET, BATH, AND LAUNDRY ACCESSORIES

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Commercial toilet accessories.

B. Utility room accessories.

C. Piping protectors

1.2 RELATED REQUIREMENTS

A. Section 22 40 00 - Plumbing Fixtures: Under-lavatory pipe and supply covers.

1.3 REFERENCE STANDARDS

A. ASTM A666/A666M - Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate, and Flat Bar; 2024.

B. ASTM C1036 - Standard Specification for Flat Glass; 2025.

C. ASTM C1503 - Standard Specification for Silvered Flat Glass Mirror; 2024.

D. ICC A117.1 - Accessible and Usable Buildings and Facilities; 2017.

1.4 SUBMITTALS

A. Product Data: Submit data on accessories describing size, finish, details of function, and attachment methods.

PART 2 PRODUCTS

2.1 MATERIALS (COMMERCIAL ACCESSORIES)

A. Accessories - General: Shop assembled, free of dents and scratches and packaged complete with anchors and fittings, steel anchor plates, adapters, and anchor components for installation.

B. Stainless Steel Sheet: ASTM A666/A666M, Type 304.

C. Mirror Glass: Annealed float glass, ASTM C1036 Type I, Class 1, Quality Q2, with silvering, protective and physical characteristics complying with ASTM C1503.

D. Adhesive: Two component epoxy type, waterproof.

E. Fasteners, Screws, and Bolts: Hot dip galvanized; tamper-proof; security type.

2.2 FINISHES

A. Stainless Steel: Satin finish, unless otherwise noted.

2.3 COMMERCIAL TOILET ACCESSORIES

A. Combination Towel Dispenser/Waste Receptacle: (PTDISP) Recessed flush with wall, stainless steel; seamless wall flanges, continuous piano hinges.

B. Mirrors: Stainless steel framed, 1/4 inch thick annealed float glass; ASTM C1036.

1. Annealed Float Glass: Silvering, protective and physical characteristics in compliance with ASTM C1503.

2. Size:

a. MR-1= 24" X 36"

b. MR-2 = 18" X 36"

C. Grab Bars (GB-#): Stainless steel, smooth surface.

1. Standard Duty Grab Bars:

a. Push/Pull Point Load: 250 pound-force, minimum.

b. Dimensions: 1-1/4 inch outside diameter, minimum 0.05 inch wall thickness, exposed flange mounting, 1-1/2 inch clearance between wall and inside of grab bar.

c. Length and Configuration:

1) (GB36) - 36 ich length bar

2) (GB42) - 42 inch length bar

3) (GB2448) - 24" X 48" L-shaped bar

D. (PIPING PROTECTORS) Provide Underlavatory Guards for P-traps and Valves, allowing access to valves if required:

1. Basis of Design:

a. Truebro, Inc.; Product Lav Guard 2 with E-Z grip fasteners . Product characteristics noted in paragraph 2.4.D.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2.4 RESIDENTIAL SHOWER, TOILET, AND BATH ACCESSORIES

A. Materials (Residential Accessories)

1. Accessories - General: Shop assembled, free of dents and scratches and packaged complete with anchors and fittings, steel anchor plates, adapters, and anchor components for installation.

2. Stainless Steel Sheet: ASTM A666/A666M, Type 304.

3. Adhesive: Two component epoxy type, waterproof.

4. Fasteners, Screws, and Bolts: Hot dip galvanized; tamper-proof; security type.

B. Toilet Paper Dispenser (TPD): Single roll, surface mounted bracket type, stainless steel, spindleless type for tension spring delivery designed to prevent theft of tissue roll.

C. Shower Curtain Rod (SC-#):

1. Basis of Design:

a. Kohler; Line: Elate; Model K-33810-BS. Product characteristics noted in paragraph 2.5.A&C.

b. Substitutions: See Section 01 67 00 – Product Requirements.

2. Custom: Configuration as indicated in the drawings.

a. Material: Brushed Stainless Steel

b. Dimensions: 1-1/4 inch outside diameter, minimum 0.05 inch wall thickness, exposed flange mounting.

D. Towel Bar (TB):Stainless steel:

1. Length: 24 inches.

2. Finish: Polished nickel

3. Basis of Design:

a. Kohler Occasion K-27062 stainless steel, 24" double tower bar. Product characteristics noted in paragraph 2.5.A&D.

b. Substitutions: See Section 01 67 00 – Product Requirements.

E. Towel Ring (TR): Stainless steel, 2-1/2 inch extension from wall, with 1/4 inch diameter trapezoidal shaped ring, rectangular-shaped bracket and backplate for concealed attachment, satin finish.

1. Finish: Polished nickel.

2. Basis of Design:

a. Kohler Occasion K-27063-CP stainless steel, towel ring. Product characteristics noted in paragraph 2.5.A&E.

b. Substitutions: See Section 01 67 00 – Product Requirements.

F. Robe Hook (RH): Heavy-duty stainless steel, double-prong, rectangular-shaped bracket and backplate for concealed attachment, satin finish.

1. Provide (1) robe hook at interior side of each bathroom door.

G. Medicine Cabinet: Metal cabinet, shelves, and door; surface mounted.

1. Overall Size: 22 inches wide by 34 inches high.

2. Cabinet Construction: Heavy-gauge steel, factory-applied, gloss white, baked-enamel finish. Form body in one piece, without seams, and with rounded inside corners.

3. Shelves: Adjustable, glass; provide not less than three shelves.

4. Door Type: Mirror with satin finish stainless steel frame.

5. Size:

a. 22 inches by 34 inches rectangular framed medicine cabinet.

6. Features:

a. Front mirror, inside door mirror, and back panel mirror provide superior visibility

b. Three adjustable tempered-glass shelves

c. Finish-matched cabinet interior

d. Half-shelf adjustable accessory for additional storage included

e. Hinges have 110° opening capability

7. Material:

a. Anodized aluminum construction with a durable rustproof and chip-resistant finish

8. Installation:

a. Surface-mount or recessed. As indicated on Drawings.

b. Can be installed with left- or right-hand swing

9. Color: Brushed Nickel

10. Basis of Design:

a. Kohler; Line: Verdera; Model: K-35570. Product characteristics noted in paragraph 2.5.A&G.

b. Substitutions: See Section 01 67 00 – Product Requirements.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify existing conditions before starting work.

B. Verify exact location of accessories for installation.

C. For electrically-operated accessories, verify that electrical power connections are ready and in the correct locations.

3.2 PREPARATION

A. Deliver inserts and rough-in frames to site for timely installation.

B. Provide templates and rough-in measurements as required.

3.3 INSTALLATION

A. Install accessories in accordance with manufacturers' instructions in locations indicated on drawings.

B. Install plumb and level, securely and rigidly anchored to substrate.

C. Mounting Heights: As required by accessibility regulations, unless otherwise indicated.

3.4 PROTECTION

A. Protect installed accessories from damage due to subsequent construction operations.

END OF SECTION

SECTION 10 44 00 - FIRE PROTECTION SPECIALTIES

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Fire extinguisher cabinets.

1.2 REFERENCE STANDARDS

A. NFPA 10 - Standard for Portable Fire Extinguishers; 2026.

1.3 SUBMITTALS

A. Product Data: Provide extinguisher operational features.

B. Shop Drawings: Indicate locations of cabinets and cabinet physical dimensions.

PART 2 PRODUCTS

2.1 FIRE EXTINGUISHER CABINETS

A. Cabinet Construction: Non-fire rated.

1. Formed primed steel sheet; 0.036 inch thick base metal.

B. Cabinet Configuration: Semi-recessed type.

1. Size to accommodate accessories.

2. Projected Trim: Returned to wall surface, with 3 1/2 inch projection.

C. Door: 0.036 inch metal thickness, reinforced for flatness and rigidity with nylon catch. Hinge doors for 180 degree opening with two butt hinges.

D. Door Glazing: Float glass, clear, 1/8 inch thick, and set in resilient channel glazing gasket.

E. Cabinet Mounting Hardware: Appropriate to cabinet, with pre-drilled holes for placement of anchors.

F. Fabrication: Weld, fill, and grind components smooth.

G. Finish of Cabinet Exterior Trim and Door: No.4 - Brushed stainless steel.

H. Finish of Cabinet Interior: White colored enamel.

I. Die cute lettering style & color: Vertical, White.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify existing conditions before starting work.

B. Verify rough openings for cabinet are correctly sized and located.

3.2 INSTALLATION

A. Install in accordance with manufacturer's instructions.

END OF SECTION

SECTION 11 30 13 - RESIDENTIAL APPLIANCES

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Kitchen appliances.

1.2 REFERENCE STANDARDS

A. UL (DIR) - Online Certifications Directory; Current Edition.

1.3 SUBMITTALS

A. Product Data: Manufacturer's data indicating dimensions, capacity, and operating features of each piece of residential equipment specified.

1.4 QUALITY ASSURANCE

A. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section, with not less than three years of documented experience.

B. Electric Appliances: Listed and labeled by UL (DIR) and complying with NEMA Standards (National Electrical Manufacturers Association).

PART 2 PRODUCTS

2.1 KITCHEN APPLIANCES

A. Cooking Exhaust: Range hood.

1. Size: 30 inches wide.

2. Fan: Two-speed, 300 cfm

3. Exhaust: Rectangular, vented to exterior.

4. Features: Include cooktop light, night light, backdraft damper, removable grease filter, and retractable visor.

5. Exterior Finish: Stainless steel.

6. Provide low voltage wiring kit and any associated accessories required for ABA compliant hood controls. Electronic control hood to be wired to a momentary wall switch for full speed and lighting control.

7. Basis of Design:

a. Broan-NuTone, LLC[<>]: Product Under-Cabinet Glacier GLA2303SS; www.broan-nutone.com. Product characteristics noted in paragraph 2.1.

b. Substitutions: See Section 01 67 00 – Product Requirements.

PART 3 EXECUTION

3.1 EXAMINATION

A. Verify utility rough-ins are provided and correctly located.

3.2 INSTALLATION

A. Install in accordance with manufacturer's instructions.

3.3 ADJUSTING

A. Adjust equipment to provide efficient operation.

3.4 CLEANING

A. Remove packing materials from equipment and properly discard.

B. Wash and clean equipment.

END OF SECTION

SECTION 12 36 00 - COUNTERTOPS

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Countertops for architectural cabinet work.

B. Sinks molded into countertops.

1.2 RELATED REQUIREMENTS

A. Section 06 41 00 - Architectural Wood Casework.

1.3 REFERENCE STANDARDS

A. ASTM E84 - Standard Test Method for Surface Burning Characteristics of Building Materials; 2024.

B. AWI (QCP) - Quality Certification Program; Current Edition.

C. AWI/AWMAC/WI (AWS) - Architectural Woodwork Standards, 2nd Edition; 2014, with Errata (2016).

D. IAPMO Z124 - Plastic Plumbing Fixtures; 2012.

E. ISFA 2-01 - Classification and Standards for Solid Surfacing Material; 2013.

F. NEMA LD 3 - High-Pressure Decorative Laminates; 2005.

1.4 SUBMITTALS

A. Product Data: Manufacturer's data sheets on each product to be used, including:

1. Preparation instructions and recommendations.

2. Storage and handling requirements and recommendations.

3. Specimen warranty.

B. Shop Drawings: Complete details of materials and installation ; combine with shop drawings of cabinets and casework specified in other sections.

C. Samples: For each finish product specified, minimum size 6 inches square, representing actual product, color, and patterns.

D. Test Reports: Chemical resistance testing, showing compliance with specified requirements.

E. Certificate: Submit labels and certificates required by quality assurance and quality control programs.

F. Installer's qualification statement.

1.5 QUALITY ASSURANCE

A. Installer Qualifications: Company specializing in performing work of the type specified in this section, with not less than three years of documented experience.

B. Quality Certification:

1. Comply with AWI (QCP) woodwork association quality certification service/program in accordance with requirements for work specified in this section: www.awiqcp.org.

2. Provide labels or certificates indicating that the installed work complies with AWI/AWMAC/WI (AWS) requirements for grade or grades specified.

3. Provide designated labels on shop drawings as required by certification program.

4. Provide designated labels on installed products as required by certification program.

5. Submit certifications upon completion of installation that verifies this work is in compliance with specified requirements.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Store products in manufacturer's unopened packaging until ready for installation.

B. Store and dispose of solvent-based materials, and materials used with solvent-based materials, in accordance with requirements of local authorities having jurisdiction.

1.7 FIELD CONDITIONS

A. Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's absolute limits.

PART 2 PRODUCTS

2.1 COUNTERTOPS

A. Quality Standard: See Section 06 41 00 - Architectural Wood Casework.

B. Solid Surfacing Countertops: Solid surfacing sheet or plastic resin casting over continuous substrate.

1. Flat Sheet Thickness: 1/2 inch, minimum.

2. Solid Surfacing Sheet and Plastic Resin Castings: Complying with ISFA 2-01 and NEMA LD 3; acrylic or polyester resin, mineral filler, and pigments; homogenous, non-porous and capable of being worked and repaired using standard woodworking tools; no surface coating; color and pattern consistent throughout thickness.

a. Basis of Design:

1) Wilsonart; Solid Surface. Product characteristics noted in paragraph 2.1.B.

2) Substitutions: See Section 01 67 00 – Product Requirements.

b. Surface Burning Characteristics: Flame spread index of 25, maximum; smoke developed index of 450, maximum; when tested in accordance with ASTM E84.

c. NSF approved for food contact.

d. Sinks and Bowls: Integral castings; minimum 3/4 inch wall thickness; comply with IAPMO Z124.

e. Finish on Exposed Surfaces: Matte, gloss rating of 5 to 20.

f. Color and Pattern: As selected by Contracting Officer from Manufacturer's full line of standard colors.

3. Other Components Thickness: 1/2 inch, minimum.

4. Back and End Splashes: Same sheet material, square top; minimum 4 inches high.

2.2 MATERIALS

A. Adhesives: Chemical resistant waterproof adhesive as recommended by manufacturer of materials being joined.

B. Counter Support Brackets

1. Definitions:

a. Flush Mounted Installation: Counter support brackets are attached to stud prior to application of the wall board.

2. Basis of Design:

a. A&M Hardware Inc.; Concealed brackets; www.aandmhardware.com. Product characteristics noted in paragraph 2.2.B.

b. Substitutions: See Section 01 67 00 – Product Requirements.

3. Products:

a. C-9 Bracket

b. C-12 Bracket

c. C-24 Bracket

4. Color: White.

5. Width 2 inches.

6. Load limit range of 1,000 lbs. to 7,960 lbs. per pair.

7. Material: Extruded aluminum

8. Provide installation material as required by counter bracket manufacturer for specific wall surfaces.

2.3 FABRICATION

A. Fabricate tops and splashes in the largest sections practicable, with top surface of joints flush.

1. Join lengths of tops using best method recommended by manufacturer.

2. Fabricate to overhang fronts and ends of cabinets 1 inch except where top butts against cabinet or wall.

3. Prepare all cutouts accurately to size; replace tops having improperly dimensioned or unnecessary cutouts or fixture holes.

B. Provide back/end splash wherever counter edge abuts vertical surface unless otherwise indicated.

1. Secure to countertop with concealed fasteners and with contact surfaces set in waterproof glue.

2. Height: 4 inches, unless otherwise indicated.

PART 3 EXECUTION

3.1 EXAMINATION

A. Do not begin installation until substrates have been properly prepared.

B. If substrate preparation is the responsibility of another installer, notify Contracting Officer of unsatisfactory preparation before proceeding.

C. Verify that wall surfaces have been finished and mechanical and electrical services and outlets are installed in proper locations.

3.2 PREPARATION

A. Clean surfaces thoroughly prior to installation.

B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

3.3 INSTALLATION

A. Securely attach countertops to cabinets using concealed fasteners. Make flat surfaces level; shim where required.

B. Seal joint between back/end splashes and vertical surfaces.

3.4 TOLERANCES

A. Variation From Horizontal: 1/8 inch in 10 feet, maximum.

B. Offset From Wall, Countertops: 1/8 inch maximum; 1/16 inch minimum.

C. Field Joints: 1/8 inch wide, maximum.

3.5 CLEANING

A. Clean countertops surfaces thoroughly.

3.6 PROTECTION

A. Protect installed products until completion of project.

B. Touch-up, repair or replace damaged products before Date of Substantial Completion.

END OF SECTION

SECTION 21 00 00 - GENERAL REQUIREMENTS OF FIRE-SUPPRESSION

PART 1 GENERAL

1.1 SUMMARY

A. The requirements listed in this section are supplemental to the Division 01 General Requirements.

B. It shall be the responsibility of the Fire Suppression Contractor to examine and refer to Architectural, Civil, Structural, Mechanical, Plumbing, Electrical, and Landscape and specifications for construction conditions which may affect the scope of Fire Suppression work. Inspect the building site and existing facilities for verification of present conditions. Make proper provisions for these conditions in performance of the work and cost thereof.

C. Fire Suppression work for this project shall include items, articles, materials, and the associated labor mentioned, scheduled or shown in these specifications and in the accompanying drawings.

D. Furnish and install equipment, materials and any required incidental items required by good practice to complete the systems described herein. Retain subparagraphs below to cross-reference requirements Contractor might expect to find in this Section but are specified in other Sections.

1.2 CODES AND STANDARDS

A. Work shall meet the requirements of the plans and specifications and shall not be less than the minimum requirements of applicable sections of the latest Codes and Standards of the following Organizations:

1. American Water Works Association (AWWA)
2. National Electrical Code (NEC)
3. National Electrical Manufacturers Association (NEMA)
4. National Fire Protection Association (NFPA)
5. Uniform Plumbing Code (UPC)
6. Occupational Safety & Health Act (OSHA)
7. International Building Code (IBC)
8. International Existing Building Code (IEBC)
9. International Fire Code (IFC)
10. Requirements of the Serving Utility Company
11. Local and State Codes and Ordinances

1.3 FEES AND PERMITS

A. The Fire Suppression Contractor shall pay fees and arrange permits required for work done under their contract and under their supervision by subcontract.

B. All usage contracts between the Owner and the serving utilities company, such as membership and usage charges or fees, etc., for the purpose of obtaining the services for the utility company shall be applied for and paid for by the Owner.

1.4 MATERIALS AND EQUIPMENT

A. Manufacturer's trade names and catalog numbers listed are intended to indicate the quality of equipment or materials desired. Manufacturers not listed in the specification will be considered substitutions and must have prior approval.

B. See Division 01 for Substitutions Procedures. Requests for substitution are to be submitted sufficiently ahead of the deadline, to give ample time for examination. Prior approval request for substitution must indicate the specific item or items to be furnished in lieu of those scheduled, together with complete technical and comparative data on scheduled items and items proposed for substitution.

C. If the engineer approves any proposed substitution, the approved product will be listed in an addendum. Bidders shall not rely on approval made in any other manner.

D. Fire Suppression equipment shall be installed with manufacturer's standard finish and color except where specific color, finish or choice is indicated.

E. High altitude operation: Capacity of equipment is to be sized and manufactured to perform at the elevation of the project site. If not specifically indicated in the equipment schedule or in the specifications provide required accessories and equipment for proper operation at elevation of the project site.

F. The Contractor shall be responsible for materials and equipment installed under this contract. Contractor shall also be responsible for the protection of materials and equipment of others from damage as a result of their work.

G. Manufactured material and equipment shall be applied, installed, connected, erected, used, cleaned, and conditioned as directed by manufacturer unless herein specified to the contrary.

H. The Contractor shall make the required arrangement with General Contractor or Construction Manager for the introduction into the building of equipment too large to pass through finished openings.

I. Store materials and equipment indoors at the job site or, if this is not possible, store on raised platforms and protect from the weather by means of waterproof covers. Coverings shall permit

circulation of air around the materials to prevent condensation of moisture. Screen or cap openings in equipment to prevent the entry of vermin.

1.5 INTENT OF DRAWINGS

A. The drawings show intended locations of piping and sprinklers unless specifically noted otherwise. Riser and other diagrams show the physical location and arrangement of the equipment. The drawings may be used for obtaining lineal runs of piping, piping fabrication, and ordering. Discrepancies between plans and actual field conditions shall be brought to the attention of the contract officer for resolution.

1.6 RESPONSIBILITY

A. Fire Suppression work shall conform to requirements of division 21 specifications.

B. The Fire Suppression Contractor shall be responsible for the installation of a satisfactory and complete system in accordance with the intent of the drawing and specifications. Provide, at no extra cost, incidental items, materials, accessories, and labor required for completion of the work even though they are not specifically mentioned or indicated on the drawings or in the specifications.

C. The drawings do not attempt to show complete details of the building construction which affect Fire Suppression; and reference is therefore required to the Architectural, Civil, Structural, Landscape, Plumbing, Mechanical, and Electrical drawings, and specifications and to shop drawings of trades for additional details which affect the installation of the work covered under this Division of the Contract.

D. Location of Fire Suppression system components shall be checked for conflicts with openings, structural members and components of other systems having fixed locations. In the event of any conflicts, the Contracting officer shall be consulted, and their decision shall govern. Necessary changes shall be made at the Contractor’s expense.

E. Determine, and be responsible for, the proper location and character of inserts for hangers, sleeves, and other openings in the construction required for the work and obtain this information well in advance of the construction progress so work will not be delayed.

F. Final location of inserts, hangers, etc., required for each installation, must be coordinated with facilities required for other installations to prevent interference.

G. Do not to install work that connects to equipment until such time as complete Shop Drawings of such equipment have been approved by the Contracting officer. Any work installed by the Contractor, prior to approval of Shop Drawings, will be at the Contractor's risk.

H. All modifications and changes required due to installation of substituted equipment shall be made at the contractor’s expense.

I. It shall be the responsibility of the installing contractor to coordinate changes to work by other trades that result from the installation of equipment other than the scheduled equipment.

J. If the provided equipment is heavier or larger than the scheduled or specified equipment, it shall be the responsibility of the installing contractor to coordinate the required structural changes and pay for any associated cost.

K. If the provided equipment has different motor characteristics or electrical requirements than the scheduled or specified equipment, it shall be the responsibility of the installing contractor to coordinate the required changes and pay for any associated cost.

L. If larger or additional electrical conduits are required due to the installation of equipment other than the scheduled or specified equipment it shall be the responsibility of the installing contractor to coordinate the required changes and pay for any associated cost.

M. If the provided equipment requires different fluid flow rates than the scheduled or specified equipment, it shall be the responsibility of the installing contractor to coordinate required changes including but not limited to pumps, piping, valves, etc. and pay for any associated cost.

N. During the performance of this Contract, properly protect work from damage and protect the Owner's property from injury of loss. Make good any damage, injury or loss, except such as may be directly due to errors in the Contract Documents or caused by Agents or Employees of the Owner. Adequately protect adjacent property as provided by law and the Contract Documents. Provide and maintain passageways, guard fences, lights and other facilities for protection required by Public Authority or Local conditions.

O. The Contractor shall be responsible for damages due to the work of their contractors, to the building or its contents, people, etc.

1.7 REVIEW

A. All work and material are subject to review at any time by the Contracting officer or their representative. If the Contracting officer or their representative finds material that does not conform to these specifications or that is not properly installed or finished, correct the deficiencies in a manner satisfactory to the Contracting officer at the Contractor’s expense.

1.8 WORKMANSHIP

A. Work under this contract shall be performed by workmen skilled in the particular trade, including work necessary to properly complete the installation in a manner to present a neat and finished appearance.

B. Obtain Architect's/Engineer's approval before performing any cutting on structural members or patching of building surfaces. Any damage to the building or equipment by the Fire Suppression Contractor shall be the responsibility of the Fire Suppression Contractor and shall be repaired by skilled craftsmen of the trades involved at the Contractor’s expense.

C. Chases, openings, sleeves, hangers, anchors, recesses, equipment pads, framing for equipment, provided by others only if so noted on the drawings. Otherwise, they will be provided by the Fire Suppression Contractor for their work.

1.9 COORDINATION

A. The Fire Suppression Contractor shall plan their work to proceed with a minimum interference with other trades and it shall be their responsibility to inform the General Contractor of openings required in the building structure for installation of work, and to provide sleeves as required. Dimensions of equipment installed and/or provided by others shall be checked in order that correct clearances and connections may be made.

B. In general, pipes requiring gravity drainage shall be installed first, followed by ductwork, large piping mains and electrical conduit. The location fire Suppression piping, and heads shall be coordinated with other trades to ensure that installations by other trades do not block heads.

C. Leave sufficient space for the installation of insulation on piping as specified. It is not acceptable to compress pipe or duct insulation for any reason.

1.10 CLEANING

A. Keep the job site clean. The Fire Suppression Contractor shall remove waste and rubbish associated with their work.

B. Upon completion of work, remove materials, scraps and debris related to fire suppression work and leave spaces including tunnels, crawlspaces, pipe or duct chases and ceiling plenums clean and orderly.

C. The Fire Suppression Contractor shall be responsible for cleaning the exterior and interior of equipment prior to start-up. Once equipment has been cleaned it shall be inspected by the Contracting officer prior to start-up.

D. The Fire Suppression Contractor shall provide dust protection of existing materials and equipment as well as new materials and equipment for the duration of the project. Protect existing materials and equipment from damage for the duration of the project. Clean the exterior and interior of existing equipment at the completion of the project.

1.11 TEMPORARY FACILITIES

A. Offices

1. The Fire Suppression Contractor must have the permission of the Owner and General Contractor or Construction Manager to install a temporary office/job trailer on the project site.

2. Contractor shall completely remove their temporary installations when no longer needed and the premises shall be completely clean, disinfected, patched, and refinished to match adjacent areas.

B. Ladders and Scaffolds

1. The Fire Suppression Contractor shall provide their own ladders, scaffolds, etc. of for access to their work in various portions of the building as may be required. When no longer needed, they shall be removed by the Contractor.

C. Protection Devices

1. The Fire Suppression Contractor shall provide and maintain their own necessary barricades, fences, signal lights, etc., required by governing authorities or shown on the drawings. When no longer needed, they shall be removed by the Contractor.

D. TEMPORARY FIRE PROTECTION

1. The Fire Suppression Contractor shall provide necessary first aid and hand fire extinguishers for Class A, B, C, and special hazards as may exist in their own work area only in accordance with good and safe practice and as required by jurisdictional safety authority.

1.12 SUBMITTALS

A. Submittals shall be required for each piece of equipment, material or product that is referenced within the Division 21 specification sections. Equipment, material or product not listed in the Division 21 specification sections that effects the performance, aesthetic or integrity of the Fire Suppression System shall be included.

B. Submittal Definitions

1. Product Data: Provide manufacturers cut sheets that include general product information including but not limited to: Model Number, physical data, nominal capacities, rough-in requirements.

2. Performance Data: Provide detailed performance and capacities based on project specific requirements including but not limited to: flow rates, capacities, pressure loss, temperatures, fan curves, pump curves, part load performance, sound data, and electrical characteristics.

3. Delegated Design: Provide detailed drawings prepared and stamped by a registered Professional Engineer that detail pertinent design criteria, the materials and products to be installed and the required installation locations.

4. Wiring Diagram: Provide diagrams that identify and detail required field wiring.

5. Color Chart: Provide a physical color chart of material samples required for selection of equipment colors.

C. Submittal Formats:

1. Product Data

a. Product data submittals required in division 21 specification sections shall be combined into a single file and submitted under this section as “210000 – Fire Suppression Product Data”. The product data submittal shall contain a table of contents that clearly identifies each specification section and the equipment, material, and products that are contained within. Each specification section shall begin with a title page and contain a hyperlink that corresponds with the table of contents. Each submittal not following this layout will be rejected and returned without review.

2. Include the following information with each submittal:

a. Project Name

b. Submittal Date

c. Name of Architect

d. Name of Engineer

e. Name of General Contractor or Construction Manager

f. Name of Sub-Contractor

g. Name of firm or entity that prepared the submittal.

h. Unique Submittal Number

i. Type of Submittal

j. Specification Section

k. Name or Mark of equipment or material and detail or drawings reference.

3. Submittals, with the exception of color charts or material samples, shall be electronically transmitted PDFs.

D. Submittal Requirements

1. Submittals shall be complete, clearly show item used, size, dimensions, capacity, rough in, etc., as required for complete review and installation. Manufacturer’s literature

showing more than one item shall be clearly marked as to which item is being furnished or it will be rejected and returned without review.

2. Each submittal shall be thoroughly checked by the Contractor for compliance with the Contract Document requirements, accuracy of dimensions, relationship to the work of other trades, and conformance with sound, safe practices as to erection and installation. Each submittal shall then bear a stamp evidencing such checking and shall show corrections made, if any. Submittals requiring extensive corrections shall be revised before submission. Each submittal not stamped and signed by the Contractor evidencing such checking will be rejected and returned without review.

3. On each submittal, clearly indicate deviations from requirements in the Contract Documents, including minor variations and limitations; include relevant additional information and revisions, other than those requested on previous submittals. Indicate by highlighting on each submittal or noting on attached separate sheet.

4. Review of the shop drawings and literature by the engineer shall not relieve the contractor of responsibility for deviations of the drawings or specifications, nor shall it relieve the contractor of responsibility for errors in the shop drawings or literature. It is the responsibility of the contractor to provide materials and equipment which meet the specifications and job requirements.

1.13 OPERATION AND MAINTENANCE

A. Operation and Maintenance Manuals (O&M Manuals) shall contain:

1. Names and contact information for the Project Architect and Project Engineer.

2. Names and contact information for the General Contractor or Construction Manager.

3. Names and contact information for sub-contractors.

4. Bound copy of NFPA 25 (Standard for Inspection, Testing, and Maintenance of Water-Based Fire Protection Systems) of applicable year.

5. Installation, maintenance, and operating instructions for each piece of equipment.

6. Parts lists

7. Wiring Diagrams

8. Equipment Start-up and inspection certificates

9. Winterization Procedure

10. Commissioning Reports

11. Copies of Equipment Warranties

12. Copies of Installation Warranties

13. Copies of Submittals

14. Approved Record Drawings

B. Prior to substantial completion submit an electronic copy of the O&M manual in PDF format to the Architect, Engineer and Owner for Review for approval. The PDF shall be one file with an index and hyperlinks to each section. Individual bound PDFs without automated navigation will be rejected.

1.14 AS-BUILT RECORD DRAWINGS

A. The Contractor shall furnish to the Owner and Contracting officer a marked print showing the location of fire suppression equipment, pipe routing, valves, drains, sprinklers, etc. installed other than as shown on the drawings. Accurate dimensions and elevations shall be indicated from walls, obstructions, and finished floors where there are deviations from the bid documents.

B. The location of maintenance related items such as access doors, floor drain, isolation valves, filters, etc., shall be highlighted on as-built drawing.

1.15 PLACING SYSTEM INTO OPERATION

A. Prior to starting of equipment, the Fire Suppression Contractor shall thoroughly inspect the installation and work completed to verify compliance with the contract documents.

B. Start-up of Fire Suppression equipment shall be completed by factory trained representatives. At the completion of start-up, the factory representative shall submit to the architect and engineer, a start-up report that indicates any problems encountered, potential problems including installation issues, adjustments made or required to be made to ensure proper operation. Any installation deficiencies identified shall be corrected at no additional cost to the owner.

1.16 OWNER TRAINING

A. The Contractor shall coordinate with owner on fire suppression equipment training.

1. Training shall include but not be limited to winterization procedures, seasonal system commissioning, maintenance, and emergency shutoff.

1.17 WARRANTY

A. The Contractor shall guarantee that materials and labor installed are new and of first quality and that any material or labor found defective shall be replaced without cost to the Owner within one (1) year after substantial completion of the Contract. The guarantee shall list the date of the beginning of the one (1) year period, which shall be the date that the Substantial Completion Certificate is issued.

B. See other specification sections for specific warranty information.

C. Any damage to the building, caused by defective work or material of the Contractor within the above-mentioned period, shall be satisfactorily repaired without cost to the Owner.

D. The guarantee does not include maintenance of equipment. The Owner shall accept full responsibility for proper operation and maintenance of equipment immediately upon substantial completion and occupancy of the building.

E. Final acceptance by the Owner shall not occur until operating instructions are mounted in Equipment Rooms and Operating Personnel thoroughly indoctrinated in the operation of fire suppression equipment by the Contractor.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION (NOT USED)

END OF SECTION

SECTION 21 05 23 - GENERAL-DUTY VALVES FOR FIRE-SUPPRESSION PIPING

PART 1 GENERAL

1.1 SUMMARY

A. Section Includes:

1. Iron butterfly valves with indicators.

2. Check valves.

3. Trim and drain valves.

4. Backflow Preventers.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of valve.

1. Division 21 Product Data submittals shall be combined into a single file and submitted in accordance with the requirements specified in Section 21 00 00 "General Requirements of Fire-Suppression".

PART 2 PRODUCTS

1.3 GENERAL REQUIREMENTS FOR VALVES

A. Source Limitations for Valves: Obtain valves for each valve type from single manufacturer.

B. ASME Compliance:

1. ASME B16.1 for flanges on iron valves.

2. ASME B1.20.1 for threads for threaded-end valves.

3. ASME B31.9 for building services piping valves.

C. AWWA Compliance: Comply with AWWA C606 for grooved-end connections.

D. NFPA Compliance: Comply with NFPA for valves.

E. Valve Pressure Ratings: Not less than the minimum pressure rating indicated or higher as required by system pressures.

F. Valve Sizes: Same as upstream piping unless otherwise indicated.

G. Valve Actuator Types:

1. Worm-gear actuator with handwheel for quarter-turn valves, except for trim and drain valves.

2. Handwheel: For other than quarter-turn trim and drain valves.

3. Handlever: For quarter-turn trim and drain valves NPS 2 and smaller.

1.4 IRON BUTTERFLY VALVES WITH INDICATORS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Anvil International.

2. Globe Fire Sprinkler Corporation.

3. Tyco Fire Products LP.

4. Victaulic Company.

5. Zurn Industries, LLC.

B. Description:

1. Standard: UL 1091 and FM Global standard for indicating valves, (butterfly or ball type), Class Number 112.

2. Minimum Pressure Rating: 175 psig.

3. Body Material: Cast or ductile iron with nylon, EPDM, epoxy, or polyamide coating.

4. Seat Material: EPDM.

5. Stem: Stainless steel.

6. Disc: Ductile iron, nickel plated and EPDM or SBR coated.

7. Actuator: Worm gear or traveling nut.

8. Supervisory Switch: Internal or external.

1.5 CHECK VALVES

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Anvil International.

2. Globe Fire Sprinkler Corporation.

3. Mueller Co.

4. Tyco Fire Products LP.

5. Victaulic Company.

6. Zurn Industries, LLC.

B. Description:

1. Standard: UL 312 and FM Global standard for swing check valves, Class Number 1210.

2. Minimum Pressure Rating: 175 psig.

3. Type: Single swing check.

4. Body Material: Cast iron, ductile iron, or bronze.

5. Clapper: Bronze, ductile iron, or stainless steel with elastomeric seal.

6. Clapper Seat: Brass, bronze, or stainless steel.

7. Hinge Shaft: Bronze or stainless steel.

8. Hinge Spring: Stainless steel.

9. End Connections: Flanged, grooved, or threaded.

1.6 TRIM AND DRAIN VALVES

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Fire Protection Products, Inc.

2. NIBCO INC.

3. United Brass Works, Inc.

B. Description:

1. Minimum Pressure Rating: 250 psig.

2. Body Material: Brass or bronze.

3. Ends: Threaded.

4. Stem: Bronze.

5. Disc: Bronze.

6. Packing: Asbestos free.

7. Handwheel: Malleable iron, bronze, or aluminum.

1.7 BACKFLOW PREVENTERS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Ames

2. Deringer

3. Watts

4. Zurn Wilkins

5. Febco

B. Description

1. Standard: UL 1469.

2. Pressure Rating: 175 psig.

3. Ends: Threaded, Grooved, or Flanged.

4. Body Material: Stainless Steel.

PART 3 EXECUTION

1.8 GENERAL REQUIREMENTS FOR VALVE INSTALLATION

A. Install listed fire suppression shutoff valves supervised-open, located to control sources of water supply except from fire-department connections. Install permanent identification signs indicating portion of system controlled by each valve.

B. Install check valve in each water-supply connection. Install backflow preventers instead of check valves in potable-water-supply sources.

C. Install valves having threaded connections with unions at each piece of equipment arranged to allow easy access, service, maintenance, and equipment removal without system shutdown. Provide separate support where necessary.

D. Install valves in horizontal piping with stem at or above the pipe center.

E. Install valves in position to allow full stem movement.

F. Install valve tags. Comply with requirements in Section 21 05 53 "Identification for Fire-Suppression Piping and Equipment" for valve tags and schedules and signs on surfaces concealing valves; and the NFPA standard applying to the piping system in which valves are installed. Install permanent identification signs indicating the portion of system controlled by each valve.

G. Install listed fire suppression shutoff valves supervised-open, located to control sources of water supply except from fire-department connections.

H. Install check valve in each water-supply connection. Install backflow preventers instead of check valves in potable-water-supply sources.

END OF SECTION

SECTION 21 05 29 - HANGERS AND SUPPORTS FOR FIRE-SUPPRESSION PIPING AND EQUIPMENT

PART 1 GENERAL

1.1 SUMMARY

A. Section Includes:

1. Metal pipe hangers, supports, and structural attachments.

2. Trapeze pipe hangers.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Division 21 Product Data submittals shall be combined into a single file and submitted in accordance with the requirements specified in Section 21 00 00 “General Requirements of Fire-Suppression”.

PART 2 PRODUCTS

1.3 PERFORMANCE REQUIREMENTS

A. Structural Performance: Hangers and supports for fire-suppression piping and equipment shall withstand the effects of gravity loads and stresses within limits and under conditions indicated according to NFPA 13.

B. NFPA Compliance: Comply with NFPA 13D.

C. UL Compliance: Comply with UL 203.

1.4 STEEL PIPE HANGERS, SUPPORTS, AND STRUCTURAL ATTACHMENTS

A. Description: Factory-fabricated components, NFPA approved and UL listed or FM approved for fire-suppression piping support.

B. Galvanized Metallic Coatings: Pre-galvanized or hot-dip galvanized.

C. Hanger Rods: Continuous-thread rod, nuts, and washer made of carbon steel or stainless steel.

D. Surge Restraints: Extended continuous-thread rod or factor-fabricated components, NFPA approved, UL listed, or FM approved for fire-suppression piping support.

1.5 TRAPEZE PIPE HANGERS

A. Description: MSS SP-58, Type 59, shop- or field-fabricated pipe-support assembly, made from structural-carbon-steel shapes, with NFPA-approved, UL-listed, or FM-approved carbon-steel hanger rods, nuts, saddles, and U-bolts.

B. Section modulus for trapeze members shall be according to NFPA 13.

PART 3 EXECUTION

1.6 APPLICATION

A. Comply with NFPA and IBC requirements for firestopping penetrations through fire-rated walls and ceilings.

1.7 INSTALLATION OF HANGERS AND SUPPORTS

A. Metal Pipe-Hanger Installation: Comply with installation requirements of approvals and listings. Install hangers, supports, clamps, and attachments as required to properly support piping from building structure.

B. Metal Trapeze Pipe-Hanger Installation: Comply with installation requirements of approvals and listings. Install trapeze hangers, supports, clamps, and attachments are required to properly support piping from building structure.

1.8 ADJUSTING

A. Hanger Adjustments: Adjust hangers to distribute loads equally on attachments and to achieve indicated slope of pipe as required.

B. Trim excess length of continuous-thread hanger and support rods to 1-1/2 inches.

1.9 HANGER AND SUPPORT SCHEDULE

A. See drawings for hanger types.

END OF SECTION

SECTION 21 05 53 - IDENTIFICATION FOR FIRE-SUPPRESSION PIPING AND EQUIPMENT

PART 1 GENERAL

1.1 SUMMARY

A. Section Includes:

1. Equipment labels.

2. Warning signs and labels.

3. Pipe labels.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Division 21 Product Data submittals shall be combined into a single file and submitted in accordance with the requirements specified in Section 21 00 00 "General Requirements of Fire-Suppression".

PART 2 PRODUCTS

2.1 EQUIPMENT LABELS

A. Plastic Labels for Equipment:

1. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, with predrilled holes for attachment hardware.

2. Maximum Temperature: Able to withstand temperatures up to 160 deg F.

3. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.

4. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.

5. Fasteners: Stainless-steel rivets or self-tapping screws.

6. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

B. Label Content: Include equipment's Drawing designation or unique equipment number, Drawing numbers where equipment is indicated (plans, details, and schedules), and the Specification Section number and title where equipment is specified.

C. Equipment-Label Schedule: For each item of equipment to be labeled, on 8-1/2-by-11-inch bond paper. Tabulate equipment identification number and identify Drawing numbers where equipment is indicated (plans, details, and schedules) and the Specification Section number and title where equipment is specified. Equipment schedule shall be included in operation and maintenance data.

2.2 WARNING SIGNS AND LABELS

A. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, with predrilled holes for attachment hardware.

B. Letter Color: White.

C. Background Color: Red.

D. Maximum Temperature: Able to withstand temperatures up to 160 deg F.

E. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.

F. Minimum Letter Size: 1/4 inch for name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-fourths the size of principal lettering.

G. Fasteners: Stainless-steel rivets or self-tapping screws.

H. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

I. Label Content: Include caution and warning information, plus emergency notification instructions.

2.3 PIPE LABELS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1. Seton Identification Products.

B. General Requirements for Manufactured Pipe Labels: Preprinted, color-coded, with lettering indicating service and showing flow direction.

C. Pretensioned Pipe Labels: Precoiled, semirigid plastic formed to cover full circumference of pipe and to attach to pipe without fasteners or adhesive.

D. Pipe-Label Contents: Include identification of piping service using same designations or abbreviations as used on Drawings, pipe size, and an arrow indicating flow direction.

1. Flow-Direction Arrows: Integral with piping-system service lettering to accommodate both directions or as separate unit on each pipe label to indicate flow direction.

2. Lettering Size: At least 1/2 inch for viewing distances up to 72 inches and proportionately larger lettering for greater viewing distances.

PART 3 EXECUTION

3.1 PREPARATION

A. Clean piping and equipment surfaces of incompatible primers, paints, and encapsulants, as well as dirt, oil, grease, release agents, and other substances that could impair bond of identification devices.

3.2 LABEL INSTALLATION REQUIREMENTS

A. Coordinate installation of identifying devices with completion of covering and painting of surfaces where devices are to be installed.

B. Coordinate installation of identifying devices with locations of access panels and doors.

C. Install or permanently fasten labels on each major item of mechanical equipment.

D. Locate equipment labels where accessible and visible.

E. Pipe-Label Locations: Locate pipe labels where piping is exposed or above accessible ceilings in finished spaces; machine rooms; accessible maintenance spaces such as shafts, tunnels, and plenums; and exterior exposed locations as follows:

1. Near each valve and control device.

2. Near each branch connection excluding short takeoffs. Where flow pattern is not obvious, mark each pipe at branch.

3. Near penetrations and on both sides of through walls, floors, ceilings, and inaccessible enclosures.

4. At access doors, manholes, and similar access points that permit a view of concealed piping.

5. Near major equipment items and other points of origination and termination.

6. Spaced at maximum intervals of 50 feet along each run. Reduce intervals to 25 feet in areas of congested piping and equipment.

7. On piping above removable acoustical ceilings. Omit intermediately spaced labels.

END OF SECTION

SECTION 21 10 00 - FIRE-SUPPRESSION ACCESSORIES

PART 1 GENERAL

1.1 SUMMARY

A. Section Includes:

1. Wall plates.

2. Floor plates.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

1. Division 21 Product Data submittals shall be combined into a single file and submitted in accordance with the requirements specified in Section 21 00 00 "General Requirements of Fire-Suppression".

PART 2 PRODUCTS

1.3 WALL PLATES

A. One-piece, Chrome-plated steel

1.4 FLOOR PLATES

A. One-piece, Cast-iron flange.

PART 3 EXECUTION

1.5 WALL PLATE AND FLOOR PLATE INSTALLATION

A. Install wall plates for piping penetrations of walls and floors in finished rooms.

B. Install wall plates with the inside diameter to closely fit around pipe and the outside diameter that completely covers opening.

C. Install floor plates for piping penetrations of equipment-room floors.

D. Install floor plates with the inside diameter to closely fit around pipe and the outside diameter that completely covers opening.

1.6 FIRE DEPARTMENT CONNECTION INSTALLATION

A. Coordinate fire department connection location with the authority having jurisdiction.

B. Provide a minimum distance of 4 feet between the check valve and the exterior wall.

C. Install drain elbow and automatic (ball-drip) drain valve at the check valve for the fire-department

connection.

1.7 FIELD QUALITY CONTROL

A. Replace broken and damaged wall plates and floor plates using new materials.

END OF SECTION

SECTION 21 13 13 - WET-PIPE SPRINKLER SYSTEMS

PART 1 GENERAL

1.1 SUMMARY

A. Section Includes:

1. Pipes, fittings, and specialties.

2. Specialty valves.

3. Sprinklers.

4. Pressure gauges.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Division 21 Product Data submittals shall be combined into a single file and submitted in accordance with the requirements specified in Section 21 00 00 "General Requirements of Fire-Suppression".

1.3 INFORMATIONAL SUBMITTALS

A. Approved Sprinkler Piping Drawings: Working plans, prepared according to NFPA 13, that have been approved by authorities having jurisdiction, including hydraulic calculations if applicable.

B. Field Test Reports and Certificates: Indicate and interpret test results for compliance with performance requirements and as described in NFPA 13. Include "Contractor's Material and Test Certificate for Aboveground Piping."

1.4 CLOSEOUT SUBMITTALS

A. Operation and maintenance data.

1.5 QUALITY ASSURANCE

A. Installer Qualifications:

1. State or municipal certified/licensed sprinkler contractor

2. NICET Level II

3. As approved by the FCO

B. Designer Qualifications:

1. A fire protection engineer

2. When a delegated design is required, preparation of working plans and calculations shall be completed by a qualified NICET Level III designer or professional engineer using performance requirements and design criteria indicated.

3. As approved by the FCO.

PART 2 PRODUCTS

1.6 PERFORMANCE REQUIREMENTS

A. Sprinkler system equipment, specialties, accessories, installation, and testing shall comply with the following:

1. NFPA 13D.

2. Local Authority Having Jurisdiction.

B. Standard-Pressure Piping System Component: Listed for 175-psig minimum working pressure.

C. Fire Sprinkler System design shall be approved by the authority having jurisdiction. Provide approval letter from the authority having jurisdiction to the engineer prior to installation.

D. Margin of Safety for Available Water Flow and Pressure: 10 percent, including losses through water-service piping, valves, and backflow preventers.

E. Sprinkler Occupancy Hazard Classifications: According to NFPA 13D unless otherwise indicated.

F. Maximum Protection Area per Sprinkler: According to NFPA and UL listing.

G. Seismic Performance: Sprinkler piping shall withstand the effects of earthquake motions determined according NFPA 13D.

1.7 STEEL PIPE AND FITTINGS

A. Schedule 40 - Galvanized and Black-Steel Pipe: ASTM A 135/795, Type E, Grade A. Pipe ends may be factory or field formed to match joining method.

B. Schedule 10, Galvanized and Black-Steel Pipe: ASTM A 135/795, Type E, Grade A. Pipe ends may be factory or field formed to match joining method.

C. Cast-Iron Threaded Fittings: ANSI B16.4, Class 125, Standard Pattern. Threads shall conform to ANSI B1.20.1

D. Malleable-Iron Threaded Fittings: ANSI B16.3, Class 150, Standard Pattern. Threads shall conform to ANSI B1.20.1.

E. Ductile-Iron Threaded Fittings: ANSI B16.42, Class 300, Standard Pattern. Threads shall conform to ANSI B1.20.1.

F. Cast-Iron Flanges: ASME 16.1, Class 125.

G. Steel Flanges and Flanged Fittings: ASME B16.5, Class 150.

1. Pipe-Flange Gasket Materials: AWWA C110, rubber, flat face, 1/8 inch thick ASME B16.21, nonmetallic and asbestos free or EPDM rubber gasket.

2. Class 125 and Class 250, Cast-Iron, Flat-Face Flanges: Full-face gaskets.

3. Class 150 and Class 300, Ductile-Iron or -Steel, Raised-Face Flanges: Ring-type gaskets.

H. Grooved-Joint, Steel-Pipe Appurtenances:

1. Galvanized or Painted Grooved-End Fittings for Steel Piping: ASTM A 47/A 47M, malleable-iron casting or ASTM A 536, ductile-iron casting, with dimensions matching steel pipe.

2. Grooved-End-Pipe Couplings for Steel Piping: AWWA C606 and UL 213 rigid pattern, unless otherwise indicated, for steel-pipe dimensions. Include ferrous housing sections, EPDM-rubber gasket, and bolts and nuts.

1.8 SPECIALTY VALVES

A. Listed in UL's "Fire Protection Equipment Directory" or FM Global's "Approval Guide."

B. Specialty Valves Pressure Rating: 175-psig minimum.

C. Body Material: Cast or ductile iron.

D. Size: Same as connected piping.

E. End Connections: Flanged, threaded, or grooved.

F. Riser Manifolds:

1. Standard: UL 193.

2. Design: For horizontal or vertical installation.

3. Include trims sets for test and drain valve, gauge, pressure relief valve, flow switch, and test orifice in accordance with NFPA 13.

G. Automatic (Ball Drip) Drain Valves:

1. Standard: UL 1726.

2. Pressure Rating: 175-psig minimum.

3. Type: Automatic draining, ball check.

1.9 SPRINKLER PIPING SPECIALTIES

A. Branch Outlet Fittings:

1. Standard: UL 213.

2. Pressure Rating: 300 psig .

3. Body Material: Ductile-iron housing with EPDM seals and bolts and nuts.

4. Type: Mechanical-tee and -cross fittings.

5. Configurations: Snap-on and strapless, ductile-iron housing with branch outlets.

6. Size: Of dimension to fit onto sprinkler main and with outlet connections as required to match connected branch piping.

7. Branch Outlets: Grooved, plain-end pipe, or threaded.

B. Flow Detection and Test Assemblies:

1. Standard: UL's "Fire Protection Equipment Directory" or FM Global's "Approval Guide."

2. Pressure Rating: 300 psig.

3. Body Material: Cast- or ductile-iron housing with orifice, sight glass, and integral test valve.

4. Size: Same as connected piping.

5. Inlet and Outlet: Threaded or grooved.

C. Branch Line Testers:

1. Standard: UL 199.

2. Pressure Rating: 300 psig.

3. Body Material: Brass.

4. Size: Same as connected piping.

5. Inlet: Threaded.

6. Drain Outlet: Threaded and capped.

7. Branch Outlet: Threaded, for sprinkler.

D. Sprinkler Inspector's Test Fittings:

1. Standard: UL's "Fire Protection Equipment Directory" or FM Global's "Approval Guide."

2. Pressure Rating: 300 psig.

3. Body Material: Cast- or ductile-iron housing with sight glass.

4. Size: Same as connected piping.

5. Inlet and Outlet: Threaded.

E. Flexible Sprinkler Hose Fittings:

1. Standard: UL 1474.

2. Type: Flexible hose for connection to sprinkler, and with bracket for connection to ceiling grid.

3. Pressure Rating: 300 psig.

4. Size: Same as connected piping, for sprinkler.

5. Inlet and Outlet: Threaded or grooved.

1.10 SPRINKLERS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Globe Fire Sprinkler Corporation.

2. Reliable Automatic Sprinkler Co., Inc.

3. Tyco Fire & Building Products LP.

4. Victaulic Company.

5. Viking Corporation.

B. Listed in UL's "Fire Protection Equipment Directory" or FM Global's "Approval Guide."

C. Pressure Rating for Residential Sprinklers: 175-psig minimum.

D. Pressure Rating for Automatic Sprinklers: 175-psig minimum.

E. Automatic Sprinklers with Heat-Responsive Element:

1. Early-Suppression, Fast-Response Applications: UL 1767.

2. Nonresidential Applications: UL 199.

3. Residential Applications: UL 1626.

4. Characteristics: Nominal 1/2-inch orifice with Discharge Coefficient K of 5.6, and for "Or-

dinary" or “Intermediate” temperature classification rating unless otherwise indicated or required by application.

F. Sprinkler Finishes: See drawings.

G. Sprinkler Escutcheons: See drawings.

H. Sprinkler Guards:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

2. Reliable Automatic Sprinkler Co., Inc.

3. Tyco Fire & Building Products LP.

4. Victaulic Company.

5. Viking Corporation.

6. Standard: UL 199.

7. Type: Wire cage with fastening device for attaching to sprinkler.

1.11 PRESSURE GAUGES

A. Standard: UL 393.

B. Dial Size: 3-1/2- to 4-1/2-inch diameter.

C. Pressure Gauge Range: 0- to 250-psig minimum.

D. Label: Include "WATER" label on dial face.

PART 3 EXECUTION

1.12 SERVICE-ENTRANCE PIPING

A. Install shutoff valve, backflow preventer, pressure gauge, drain, and other accessories indicated at connection to water-service piping.

B. Install shutoff valve, check valve, pressure gauge, and drain at connection to water service.

1.13 PIPING INSTALLATION

A. Locations and Arrangements: Drawing plans, schematics, and diagrams indicate general location and arrangement of piping. Install piping as indicated on approved working plans.

1. Deviations from approved working plans for piping require written approval from authorities having jurisdiction. File written approval with contracting officer or Engineer before de-

viating from approved working plans.

2. Coordinate layout and installation of sprinklers with other construction that penetrates ceilings, including light fixtures, HVAC equipment, and partition assemblies.

B. Piping Standard: Comply with NFPA 13 requirements for installation of sprinkler piping.

C. Install seismic bracing and restraint on piping. Comply with NFPA 13 and ASCE 7 requirements.

D. Use listed fittings to make changes in direction, branch takeoffs from mains, and reductions in pipe sizes.

E. Install unions adjacent to each valve with threaded-end connections.

F. Install flanges, flange adapters, or couplings for grooved-end piping on valves, apparatus, and equipment with grooved end connections.

G. Install "Inspector's Test Connections" in sprinkler system piping, complete with shutoff valve, and sized and located according to NFPA 13.

H. Install sprinkler piping with drains for complete system drainage.

I. Install sprinkler control valves, check valves, test assemblies, and drain risers adjacent to standpipes when sprinkler piping is connected to standpipes.

J. Install alarm devices in piping systems.

K. Install hangers and supports for sprinkler system piping according to NFPA 13. Comply with requirements for hanger materials in NFPA 13.

L. Install pressure gauges on riser or feed main, at each sprinkler test connection, and at top of each standpipe. Include pressure gauges with connection not less than NPS 1/4 and with soft-metal seated globe valve, arranged for draining pipe between gauge and valve. Install gauges to permit removal, and install where they are not subject to freezing.

M. Fill sprinkler system piping with water.

N. Install sleeves for piping penetrations of walls, ceilings, and floors as required.

O. Install sleeve seals for piping penetrations of concrete walls and slabs as required.

1.14 JOINT CONSTRUCTION

A. Install couplings, flanges, flanged fittings, unions, nipples, and transition and special fittings that have finish and pressure ratings same as or higher than system's pressure rating for aboveground applications unless otherwise indicated.

B. Install unions adjacent to each valve in pipes with threaded end connections.

C. Install flanges, flange adapters, or couplings for grooved-end piping on valves, apparatus, and

equipment with grooved end connections.

D. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.

E. Remove scale, slag, dirt, and debris from inside and outside of pipes, tubes, and fittings before assembly.

F. Flanged Joints: Select appropriate gasket material in size, type, and thickness suitable for water service. Join flanges with gasket and bolts according to ASME B31.9.

G. Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:

1. Apply appropriate tape or thread compound to external pipe threads.

2. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged.

H. Steel-Piping, Pressure-Sealed Joints: Join lightwall steel pipe and steel pressure-seal fittings with tools recommended by fitting manufacturer.

I. Steel-Piping, Cut-Grooved Joints: Cut square-edge groove in end of pipe according to AWWA C606. Assemble coupling with housing, gasket, lubricant, and bolts. Join steel pipe and grooved-end fittings according to AWWA C606 for steel-pipe joints.

J. Steel-Piping, Roll-Grooved Joints: Roll groove in end of pipe according to AWWA C606. Assemble coupling with housing, gasket, lubricant, and bolts. Join steel pipe and grooved-end fittings according to AWWA C606 for steel-pipe joints.

K. Dissimilar-Material Piping Joints: Make joints using adapters compatible with materials of both piping systems.

1.15 VALVE AND SPECIALTIES INSTALLATION

A. Install listed fire suppression valves, trim and drain valves, specialty valves and trim, controls, and specialties according to NFPA 13 and authorities having jurisdiction.

B. Install listed fire suppression shutoff valves supervised open, located to control sources of water supply except from fire-department connections. Install permanent identification signs indicating portion of system controlled by each valve.

C. Install check valve in each water-supply connection. Install backflow preventers instead of check valves in potable-water-supply sources.

D. Specialty Valves:

1. Install valves in approved position for proper direction of flow, in main supply to system.

2. Install alarm valves with bypass check valve and retarding chamber drain-line connection.

1.16 SPRINKLER INSTALLATION

A. Install sprinklers in suspended ceilings in center of narrow dimension of acoustical ceiling panels.

B. Install dry-type sprinklers with water supply from heated space. Do not install pendent or side-wall, wet-type sprinklers in areas subject to freezing.

C. Install sprinklers into flexible, sprinkler hose fittings, and install hose into bracket on ceiling grid.

1.17 IDENTIFICATION

A. Install labeling and pipe markers on equipment and piping according to requirements in NFPA 13.

B. Identify system components, wiring, cabling, and terminals.

1.18 FIELD QUALITY CONTROL

A. Perform the following tests and inspections with the assistance of a factory-authorized service representative:

1. Leak Test: After installation, charge systems and test for leaks. Repair leaks and retest until no leaks exist.

2. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

3. Flush, test, and inspect sprinkler systems according to NFPA 13, "Systems Acceptance" Chapter.

4. Energize circuits to electrical equipment and devices.

5. Coordinate with fire-alarm tests. Operate as required.

6. Coordinate with fire-pump tests. Operate as required.

7. Verify that equipment hose threads are same as local fire department equipment.

B. Sprinkler piping system will be considered defective if it does not pass tests and inspections.

C. Prepare test and inspection reports.

1.19 CLEANING

A. Clean dirt and debris from sprinklers.

B. Only sprinklers with their original factory finish are acceptable. Remove and replace any sprinklers that are painted or have any other finish than their original factory finish.

END OF SECTION

SECTION 22 00 00 - GENERAL REQUIREMENTS OF PLUMBING

PART 1 - GENERAL

1.1 SUMMARY

A. The requirements listed in this section are supplemental to the Division 01 General Requirements.

B. It shall be the responsibility of the Plumbing Contractor to examine and refer to all Architectural, Civil, Structural, Electrical, and Landscape and specifications for construction conditions which may affect the scope of Plumbing work. Inspect the building site and existing facilities for verification of present conditions. Make proper provisions for these conditions in performance of the work and cost thereof.

C. Plumbing work for this project shall include all items, articles, materials and the associated labor mentioned, scheduled or shown in these specifications and in the accompanying drawings.

D. Furnish and install all equipment, materials and any required incidental items required by good practice to complete the systems described herein.

1.2 CODES AND STANDARDS

A. Work shall meet the requirements of the plans and specifications and shall not be less than the minimum requirements of applicable sections of the latest Codes and Standards of the following Organizations:

1. American Gas Association (AGA)

2. American Society of Heating, Refrigeration and Air Conditioning Engineers (ASHRAE)

3. American Society of Mechanical Engineers (ASME)

4. American Water Works Association (AWWA)

5. National Electrical Code (NEC)

6. National Electrical Manufacturers Association (NEMA)

7. National Fire Protection Association (NFPA)

8. International Plumbing Code (IPC)

9. Occupational Safety & Health Act (OSHA)

10. Plastic Pipe Institute (PPI)

11. International Mechanical Code (IMC)

12. International Fuel Gas Code (IFGC)

13. International Building Code (IBC)

14. International Energy Conservation Code (IECC)

15. Requirements of the Serving Utility Company

16. Local and State Codes and Ordinances

1.3 FEES AND PERMITS

A. The Plumbing Contractor shall pay all fees and arrange all permits required for work done under their contract and under their supervision by subcontract.

B. All usage contracts between the Owner and the serving utilities company, such as membership and usage charges or fees, etc., for the purpose of obtaining the services for the utility company shall be applied for and paid for by the Owner.

1.4 MATERIALS AND EQUIPMENT

A. Manufacturer's trade names and catalog numbers listed are intended to indicate the quality of equipment or materials desired. Manufacturers not listed in the specification will be considered substitutions and must have prior approval.

B. See Division 01 for Substitutions Procedures. Requests for substitution are to be submitted sufficiently ahead of the deadline, to give ample time for examination. Prior approval request for substitution must indicate the specific item or items to be furnished in lieu of those scheduled, together with complete technical and comparative data on scheduled items and items proposed for substitution.

C. If the engineer approves any proposed substitution, the approved product will be listed in an addendum. Bidders shall not rely on approval made in any other manner.

D. Plumbing equipment may be installed with manufacturer's standard finish and color except where specific color, finish or choice is indicated. If the manufacturer has no standard finish, equipment shall have a prime coat and two finish coats of gray enamel.

E. High altitude operation: Capacity of all equipment is to be sized and manufactured to perform at the elevation of the project site. If not specifically indicated in the equipment schedule or in the specifications provide all required accessories and equipment for proper operation at elevation of the project site.

F. This Contractor shall be responsible for materials and equipment installed under this contract. Contractor shall also be responsible for the protection of materials and equipment of others from damage as a result of his work.

G. Manufactured material and equipment shall be applied, installed, connected, erected, used, cleaned and conditioned as directed by manufacturer unless herein specified to the contrary.

H. This Contractor shall make the required arrangement with the General Contractor or Construction Manager for the introduction into the building of equipment too large to pass through finished openings.

I. Store materials and equipment indoors at the job site. If this is not possible, store on raised platforms and protect from the weather by means of waterproof covers. Coverings shall permit circulation of air around the materials to prevent condensation of moisture. Screen or cap openings in equipment to prevent the entry of vermin.

1.5 INTENT OF DRAWINGS

A. The drawings are diagrammatic and do not necessarily show exact location of piping and ductwork unless specifically dimensioned. Riser and other diagrams are schematic and do not necessarily show the physical arrangement of the equipment. They shall not be used for obtaining lineal runs of piping or ductwork, nor shall they be used for shop drawings for piping and ductwork fabrication or ordering. Discrepancies shown on different plans, or between plans and actual field conditions shall be brought to the attention of the Architect/Engineer for resolution.

1.6 COMMISSIONING OF SYSTEMS

A. Mechanical systems with a total cooling capacity exceeding 480,000 Btu/hr or a combined heating and service water-heating capacity exceeding 600,000 Btu/hr shall be commissioned by a registered professional in accordance with Section C408 of the International Energy Conservation Code. The mechanical contractor shall correct all installation deficiencies identified by the commissioning agent at no cost to the owner.

B. See Sections 019113 “Commissioning Requirements of Contractor”, 220800 “Commissioning of Plumbing”, 230800 “Commissioning of HVAC”, 260800 “Commissioning of Electrical Systems”.

1.7 RESPONSIBILITY

A. Plumbing work shall conform to requirements of all division 22 specifications.

B. The Plumbing Contractor shall be responsible for the installation of a satisfactory and complete system in accordance with the intent of the drawing and specifications. Provide, at no extra cost, all incidental items, materials, accessories and labor required for completion of the work even though they are not specifically mentioned or indicated on the drawings or in the specifications.

C. The drawings do not attempt to show complete details of the building construction which affect the mechanical and plumbing installation; and reference is therefore required to the Architectural, Civil, Structural, Landscape and Electrical drawings and specifications and to shop drawings of all trades for additional details which affect the installation of the work covered under this Division of the Contract.

D. Location of mechanical and plumbing system components shall be checked for conflicts with openings, structural members and components of other systems having fixed locations. In the event of any conflicts, the Architect/Engineer shall be consulted, and their decision shall govern. Necessary changes shall be made at the Contractor’s expense.

E. Determine, and be responsible for, the proper location and character of inserts for hangers, chases, sleeves, and other openings in the construction required for the work and obtain this information well in advance of the construction progress so work will not be delayed.

F. Final location of inserts, hangers, etc., required for each installation, must be coordinated with facilities required for other installations to prevent interference.

G. Take extreme caution not to install work that connects to equipment until such time as complete Shop Drawings of such equipment have been approved by the Architect/Engineer. Any work installed by the Contractor, prior to approval of Shop Drawings, will be at the Contractor's risk.

H. All modifications and changes required due to installation of equipment other than the scheduled equipment shall be made at the contractor’s expense.

I. It shall be the responsibility of the installing contractor to coordinate changes to work by other trades that result from the installation of equipment other than the scheduled equipment.

J. If the provided equipment is heavier or larger than the scheduled or specified equipment, it shall be the responsibility of the installing contractor to coordinate the required structural changes and pay for any and all associated cost.

K. If the provided equipment has different motor characteristics or electrical requirements than the scheduled or specified equipment, it shall be the responsibility of the installing contractor to coordinate the required changes and pay for any and all associated cost.

L. If larger or additional electrical conduits, wire or breakers are required due to the installation of equipment other than the scheduled or specified equipment it shall be the responsibility of the installing contractor to coordinate the required changes and pay for any and all associated cost.

M. If the provided equipment requires different fluid flow rates than the scheduled or specified equipment, it shall be the responsibility of the installing contractor to coordinate all required changes including but not limited to pumps, piping, valves, etc and pay for any and all associated cost.

N. At all times during the performance of this Contract, properly protect work from damage and protect the Owner's property from injury of loss. Make good any damage, injury or loss, except such as may be directly due to errors in the Bidding Documents or caused by Agents or Employees of the Owner. Adequately protect adjacent property as provided by law and the Bidding Documents. Provide and maintain passageways, guard fences, lights and other facilities for protection required by Public Authority or Local conditions.

O. The Contractor shall be responsible for damages incurred due to the work of their contractors, to the building or its contents, people, etc.

1.8 REVIEW

A. All work and material is subject to review at any time by the Architect/Engineer or his representative. If the Architect/Engineer or his representative finds material that does not conform to these specifications or that is not properly installed or finished, correct the deficiencies in a manner satisfactory to the Architect/Engineer at the Contractor’s expense.

1.9 WORKMANSHIP

A. Work under this contract shall be performed by workmen skilled in the particular trade, including work necessary to properly complete the installation in a workmanlike manner to present a neat and finished appearance.

B. Obtain Architect's/Engineer's approval before performing any cutting on structural members or patching of building surfaces. Any damage to the building or equipment by the Plumbing Contractor shall be the responsibility of the Plumbing Contractor and shall be repaired by skilled craftsmen of the trades involved at the Contractor’s expense.

C. Chases, openings, sleeves, hangers, anchors, recesses, equipment pads, and framing for equipment; shall be provided by others only if so noted on the drawings. Otherwise, they will be provided by the Mechanical or Plumbing Contractor for their work.

1.10 COORDINATION

A. The Plumbing Contractor shall plan their work to proceed with a minimum interference with other trades and it shall be their responsibility to inform the General Contractor of all openings required in the building structure for installation of work, and to provide sleeves as required. Dimensions of equipment installed and/or provided by others shall be checked so that correct clearances and connections may be made.

B. In general, pipelines requiring gravity drainage shall be installed first, followed by ductwork, large piping mains and electrical conduit. The location of fire protection piping and heads shall be coordinated with other trades to ensure that installations by other trades do not block heads.

C. Leave sufficient space for the installation of insulation on piping and ductwork as specified. It is not acceptable to compress pipe or duct insulation for any reason.

1.11 CLEANING

A. Keep the job site clean. The Plumbing Contractor shall remove all waste and rubbish associated with their work.

B. Upon completion of work, remove materials, scraps and debris related to plumbing and mechanical work and leave all spaces including tunnels, crawlspaces, pipe or duct chases and ceiling plenums clean and orderly.

C. The Plumbing Contractor will be responsible for cleaning the exterior and interior of all equipment prior to star-up. Once all equipment has been cleaned it shall be inspected by the Architect/Engineer prior to start-up.

D. The Plumbing Contractor shall provide dust protection of existing materials and equipment as well as new materials and equipment for the duration of the project. Protect existing materials and equipment from damage for the duration of the project. Clean the exterior and interior of all existing equipment at the completion of the project.

1.12 TEMPORARY FACILITIES

A. Offices

1. The Plumbing Contractor must have the permission of the Owner and General Contractor or Construction Manager to install a temporary office/job trailer on the project site.

2. The Contractor shall completely remove his temporary installations when no longer needed and the premises shall be completely clean, disinfected, patched, and refinished to match adjacent areas.

B. Ladders and Scaffolds

1. The Plumbing Contractor shall provide their own ladders, scaffolds, etc. of substantial construction for access to their work in various portions of the building as may be required. When no longer needed, they shall be removed by the Contractor.

C. Protection Devices

1. The Plumbing Contractor shall provide and maintain his own necessary barricades, fences, signal lights, etc., required by all governing authorities or shown on the drawings. When no longer needed, they shall be removed by the Contractor.

D. TEMPORARY FIRE PROTECTION

1. The Plumbing Contractor shall provide all necessary first aid hand fire extinguishers for Class A, B, C and special hazards as may exist in his own work area only in accordance with good and safe practice and as required by jurisdictional safety authority.

1.13 SUBMITTALS

A. Submittals will be required for each piece of equipment, material or product as noted in the table below. All submittals shall be submitted, reviewed and all discrepancies addressed prior to ordering equipment or starting work. Any equipment ordered without having first completed the submittal process is done at the risk of the contractor. Any work performed prior to completing the submittal process is done at the risk of the contractor.

Specification Section	Product Data	Performance Data	Shop Drawing	Delegated Design	Wiring Diagram	Color Chart	Notes
220500	X			X			
220519	X						
220523	X						
220529	X			X			Provide Delegated Design per the requirements of this section
220548	X			X			Provide Delegated Design per the requirements of this section
220553	X						
220716	X						
221116	X						
221119	X						
221316	X						
221319	X						
221429	X	X			X		
223300	X	X			X		
224100	X	X	X			X	

B. Submittal Definitions

1. Product Data: Provide manufacturers' cut sheets that include general product information including but not limited to, model number, physical data, nominal capacities, and rough-in requirements.

2. Performance Data: Provide detailed performance and capacities based on project specific requirements including but not limited to: flow rates, capacities, pressure loss, temperatures, fan curves, pump curves, part load performance, sound data, and electrical characteristics.

3. Shop Drawings: Provide detailed drawings of the equipment showing overall dimensions, location of electrical and piping connection, location of anchorage points, location of electrical and control panels, and all operating, service and maintenance clearances.

4. Delegated Design: Provide detailed drawings prepared and stamped by a registered Professional Engineer, that detail pertinent design criteria, the materials and products to be installed and the required installation locations.

5. Wiring Diagram: Provide diagrams that identify, and detail required field wiring.

6. Color Chart: Provide a physical color chart of material samples required for selection of equipment colors.

C. Submittal Formats:

1. Include the following information with each submittal:

a. Project Name

b. Submittal Date

c. Name of Architect

d. Name of Engineer

e. Name of General Contractor or Construction Manager

f. Name of Sub-Contractor

g. Name of firm or entity that prepared the submittal

h. Unique Submittal Number

i. Type of Submittal

j. Specification Section

k. Name or Mark of equipment or material and detail or drawings reference.

2. All Submittals with the exception of color charts or material samples shall be electronically transmitted PDFs.

D. Submittal Requirements

1. Submittals shall be submitted as a complete specification section. The submittal must include all materials and equipment for that specification section. Submittals for individual materials of equipment will be rejected without review.

2. Submittals shall be complete, clearly show item used, size, dimensions, capacity, rough in, etc., as required for complete check and installation. Manufacturer's literature showing more than one item shall be clearly marked as to which item is being furnished or it will be rejected and returned without review.

3. Each submittal shall be thoroughly checked by the Contractor for compliance with the Contract Document requirements, accuracy of dimensions, relationship to the work of

other trades, and conformance with sound, safe practices as to erection and installation. Each submittal shall then bear a stamp evidencing such checking and shall show corrections made, if any. Submittals requiring extensive corrections shall be revised before submission. Each submittal not stamped and signed by the Contractor evidencing such checking will be rejected and returned without review.

4. On each submittal, clearly indicate deviations from requirements in the Contract Documents, including minor variations and limitations. Include relevant additional information and revisions, other than those requested on previous submittals. Indicate by highlighting on each submittal or noting on attached separate sheet.

5. Review of the shop drawings and literature by the engineer shall not relieve the contractor for responsibility for deviations from the drawings or specifications, nor shall it relieve the contractor from responsibility for errors in the shop drawings or literature. It is the responsibility of the contractor to provide materials and equipment which meet the specifications and job requirements.

1.14 OPERATION AND MAINTENANCE MANUALS

A. Operation and Maintenance Manuals (O&M Manuals) shall contain:

1. Names and contact information for the Project Architect, Project Engineer.

2. Names and contact information for the General Contractor or Construction Manager.

3. Names and contact information for sub-contractors.

4. Installation, maintenance and operating instructions for each piece of equipment.

5. Parts lists

6. Wiring Diagrams

7. Equipment Start-up and inspection certificates

8. Test and Balance Reports

9. Commissioning Reports

10. Copies of Equipment Warranties

11. Copies of Submittals

12. Record Drawings.

B. Prior to substantial completion, submit an electronic copy of the O&M manual in PDF format to the Architect, Engineer and Owner for Review and approval. The PDF shall be one file with an index and hyperlinks to each section. Individual bound PDFs without automated navigation will be rejected. All O&M data shall be grouped by the equipment type and ordered by the specification numbering.

C. Prior to final payment a final electronic copy of the O&M manual on an archival quality DVD as well as two printed copies, shall be furnished to the owner. Printed copies shall have commercial quality 8-1/2" x 11" 3-ring binders with tabbed dividers for each section.

1.15 AS-BUILT RECORD DRAWINGS

A. The Contractor shall furnish to the Owner and Architect/Engineer a marked print showing the location of all concealed or underground pipe or conduit runs and other equipment installed other than as shown on the drawings. Dimension underground lines from established building lines. Indicate all installed pull boxes in conduit runs.

B. The Contractor shall furnish to the Architect/Engineer a marked print showing the location of all mechanical equipment, plumbing fixtures, piping, ductwork, diffusers, grilles, etc. The location of any item which deviates from the bid documents shall be accurately drawn and dimensioned.

C. All underground piping and ductwork shall be dimensioned from nearest column and/or exterior walls. The location of all maintenance related items, such as duct access doors, fire dampers, isolation valves, filters, etc., shall be highlighted on the as built drawing.

1.16 PLACING SYSTEM INTO OPERATION

A. Prior to the starting of equipment, the Plumbing Contractor shall thoroughly inspect the installation and any work completed by other trades and subcontractors to verify compliance with the contract documents.

1.17 OWNER TRAINING

A. General

1. The system training is intended to familiarize the Owner's operating and maintenance staff with all systems requiring maintenance. Training is to be provided after the systems are in place and operational, after issues noted during commissioning have been resolved, and before final acceptance.

2. Provide second set of training sessions for automatic control systems about 6-9 months after the first sessions.

B. Systems Requiring Training

1. All mechanical, electrical, safety, standby, and automatic control systems in the project, and other systems specified elsewhere to have training.

C. Attendance:

1. Training is to be provided by contractor's representatives that are familiar with the system's operation and maintenance requirements. Individual training sessions (modules) shall be provided for each type or group of systems, separated roughly by trade group that will be performing maintenance on the system. The trades groups and systems typically requiring training are:

a. Plumbers (Domestic and Sanitary Plumbing, gas-fired heating, packaged rooftop units, miscellaneous process piping systems)

D. Schedule:

1. Duplicate training sessions are to be provided for each training module, so that the Owner's operating personnel can be split into two groups during training. Duplicate training sessions shall be scheduled on different days. Length of training sessions will be determined by scope of training indicated below, and as coordinated with Owner after draft copy of training documents have been reviewed.

E. Training Documentation:

1. Contractor to submit draft copy of agenda and training documents to Owner for review at least two weeks prior to training date.

2. Provide a copy of the following items for each person that will be attending the training sessions. Coordinate required number with the Owner.

a. Training agenda.

b. Summary of new systems and existing systems affected by this project.

c. Summary of work performed under this project.

d. Control system drawings and sequences of operation.

e. List of important maintenance and trouble-shooting operations for all systems.

3. Provide minimum of 2 copies of following items:

a. Contract documents including all drawings, specifications, addendums, and change orders.

F. Training Sessions:

1. Assemble at location to be determined by the Owner.

2. Distribute training documentation as indicated above.

3. Provide classroom style training if required for orientation and discussion of new systems and existing systems affected by this project, and other issues appropriate for a classroom format.

4. Visit site and review locations; and perform detailed review of operation and maintenance requirements for current systems.

1.18 WARRANTY

A. The Contractor shall guarantee that all materials and labor installed are new and of first quality and that any material or labor found defective shall be replaced without cost to the Owner within one (1) year after substantial completion of the Contract or one (1) full season of heating and cooling operation, whichever is the greater. The guarantee shall list the date of the beginning of the one (1) year period, which shall be the date that the Substantial Completion Certificate is issued.

B. Any damage to the building, caused by defective work or material of the Contractor within the above-mentioned period, shall be satisfactorily repaired without cost to the Owner.

C. The guarantee does not include maintenance of equipment. The Owner shall accept full responsibility for proper operation and maintenance of equipment immediately upon substantial completion and occupancy of the building.

D. Final acceptance by the Owner will not occur until all operating instructions are mounted in Equipment Rooms and Operating Personnel are thoroughly indoctrinated in the operation of all mechanical equipment by the Contractor.

E. No equipment installed as part of this project shall be used for temporary heat during construction.

END OF SECTION

SECTION 22 05 00 - GENERAL PROVISIONS OF PLUMBING

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes the following:

1. Expansion Fittings and Loops for Piping Systems
2. Alignment Guides and Anchors
3. Dielectric Fittings
4. Pipe Sleeves
5. Sleeve Seals Systems for Piping
6. Silicone Sealant
7. Escutcheons for Piping
8. Floor Plates

1.2 SUBMITTALS

A. See Section 22 00 00 “General Requirements of Plumbing” for Submittal requirements.

1.3 QUALITY ASSURANCE

A. Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."

B. Pipe and Pressure-Vessel Welding Qualifications: Qualify procedures and operators according to ASME Boiler and Pressure Vessel Code.

1.4 PERFORMANCE REQUIREMENTS

A. Compatibility: Products shall be suitable for piping service fluids, materials, working pressures, and temperatures.

B. Capability: Products to absorb 200 percent of maximum axial movement between anchors.

PART 2 - PRODUCTS

2.1 DIELECTRIC FITTINGS

A. General Requirements: Assembly of copper alloy and ferrous materials with separating nonconductive insulating material. Include end connections compatible with pipes to be joined.

B. Dielectric Unions:

1. Dielectric Unions are not allowed.

C. Dielectric Flanges:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

    a. Capitol Manufacturing Company; member of the Phoenix Forge Group.

    b. Central Plastics Company.

    c. Matco-Norca.

    d. Watts; a division of Watts Water Technologies, Inc.

    e. Wilkins; a Zurn company.

2. Standard: ASSE 1079.

3. Factory-fabricated, bolted, companion-flange assembly.

4. Pressure Rating: 175 psig.

5. End Connections: Solder-joint copper alloy and threaded ferrous; threaded solder-joint copper alloy and threaded ferrous.

D. Dielectric-Flange Insulating Kits:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

    a. Advance Products & Systems, Inc.

    b. Calpico, Inc.

    c. Central Plastics Company.

    d. Pipeline Seal and Insulator, Inc.

2. Nonconducting materials for field assembly of companion flanges.

3. Pressure Rating: 150 psig.

4. Gasket: Neoprene or phenolic.

5. Bolt Sleeves: Phenolic or polyethylene.

6. Washers: Phenolic with steel backing washers.

E. PEX Dielectric Separator:

1. Description: 6” long section of pex piping shall be installed between dis-similar piping materials.

2. Pipe Material: PEX plastic according to ASTM F 876.

3. Oxygen Barrier: O2 permeability <= 0.32 mg/m2/day in accordance with DIN 4726.

4. Fittings: ASTM F 1960, cold expansion fittings and reinforcing rings.

5. Pressure/Temperature Rating: Minimum 100 psig and 180 deg F.

2.2 SLEEVES

A. Galvanized-Steel Sheet Pipe Sleeves: 0.0239-inch minimum thickness; round tube closed with welded longitudinal joint.

2.3 SLEEVE-SEAL SYSTEMS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Advance Products & Systems, Inc.

2. CALPICO, Inc.

3. GPT; an EnPro Industries company.

4. Metraflex Company (The).

B. Description:

1. Modular sealing-element unit, designed for field assembly, for filling annular space between piping and sleeve.

2. Designed to form a hydrostatic seal of 20-psig.

3. Sealing Elements: EPDM-rubber interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size.

4. Pressure Plates: Composite plastic.

5. Connecting Bolts and Nuts: Stainless steel of length required to secure pressure plates to sealing elements.

2.4 ASSEMBLY PENETRATIONS

A. All penetrations through a fire rated assembly shall be protected with an approved fire stop system in compliance with the rated assemblies as outlined in the Underwriters Laboratory Listing.

1. Manufacturers: Subject to compliance with requirements, provide products by the following:

a. 3M Company

b. Holdrite

c. Hilti

2.5 SILICONE SEALANTS

A. Silicone, S, P, 25, T, NT: Single-component, pourable, plus 25 percent and minus 25 percent movement capability, traffic- and nontraffic-use, neutral-curing silicone joint sealant; ASTM C

920, Type S, Grade P, Class 25, Uses T and NT. Grade P Pourable (self-leveling) formulation is for opening in floors and other horizontal surfaces that are not fire rated.

2.6 ESCUTCHEONS

A. One-Piece, Cast-Brass Type: With polished, chrome-plated finish and setscrew fastener.

B. One-Piece, Deep-Pattern Type: Deep-drawn, box-shaped brass with chrome-plated finish and spring-clip fasteners.

C. One-Piece, Stamped-Steel Type: With chrome-plated finish and spring-clip fasteners.

2.7 FLOOR PLATES

A. One-Piece Floor Plates: Cast-iron flange with holes for fasteners.

PART 3 - EXECUTION

3.1 EXPANSION JOINT INSTALLATION

A. Install expansion joints of sizes matching sizes of piping in which they are installed.

B. Install expansion joint per the manufacturer’s written instructions.

3.2 ALIGNMENT-GUIDE AND ANCHOR INSTALLATION

A. Install alignment guides to guide expansion and to avoid end-loading and torsional stress.

B. Install two guide(s) on each side of pipe expansion fittings and loops. Install guides nearest to expansion joint not more than four (4) pipe diameters from expansion joint.

C. Attach guides to pipe, and secure guides to building structure.

D. Install anchors at locations to prevent stresses from exceeding those permitted by ASME B31.9 and to prevent transfer of loading and stresses to connected equipment.

E. Anchor Attachments:

1. Anchor Attachment to Steel Pipe: Attach by welding. Comply with ASME B31.9 and ASME Boiler and Pressure Vessel Code: Section IX, "Welding and Brazing Qualifications."

2. Anchor Attachment to Copper Tubing: Attach with pipe hangers. Use MSS SP-69, Type 24; U bolts bolted to anchor.

F. Fabricate and install steel anchors by welding steel shapes, plates, and bars. Comply with ASME B31.9 and AWS D1.1/D1.1M.

1. Anchor Attachment to Steel Structural Members: Attach by welding.

2. Anchor Attachment to Concrete Structural Members: Attach by fasteners. Follow fastener manufacturer's written instructions.

G. Use grout to form flat bearing surfaces for guides and anchors attached to concrete.

3.3 DIELECTRIC FITTING INSTALLATION

A. Install dielectric fittings in piping at connections of dissimilar metal piping and tubing.

B. Install Dielectric fittings per the manufacturers written instructions.

C. Install pipe hangers immediately upstream and downstream of dielectric fittings.

D. Install isolation valves immediately upstream and downstream of dielectric fittings.

E. Dielectric Fittings for NPS 2 and Smaller: PEX Dielectric Separator.

F. Dielectric Fittings for NPS 2-1/2 and Larger: Dielectric Flange.

3.4 SLEEVE INTALLATION

A. Install sleeves for piping passing through penetrations in floors, partitions, roofs, and walls.

B. For sleeves that will have sleeve-seal system installed, select sleeves of size large enough to provide 1-inch annular clear space between piping and concrete slabs and walls.

C. Install sleeves in concrete floors, concrete roof slabs, and concrete walls as new slabs and walls are constructed.

1. Cut sleeves to length for mounting flush with both surfaces.

a. Exception: Extend sleeves installed in floors of mechanical equipment areas or other wet areas 2 inches above finished floor level.

2. Using silicone sealant, seal space outside of sleeves in slabs and walls without sleeve-seal system.

D. Fire-Resistance-Rated Penetrations, Horizontal Assembly Penetrations, and Smoke-Barrier Penetrations: Maintain indicated fire or smoke rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with fire- and smoke-stop materials. Comply with requirements for firestopping and fill materials specified in Section 07 84 13 "Penetration Firestopping."

3.5 SLEEVE-SEALS SYSTEM INSTALLATION

A. Install sleeve-seal systems in sleeves in exterior concrete walls at piping entries into building.

B. Select type, size, and number of sealing elements required for piping material and size and for sleeve ID or hole size. Position piping in center of sleeve. Center piping in penetration, assemble sleeve-seal-system components, and install in annular space between piping and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make a watertight seal.

3.6 SLEEVE-SEAL SCHEDULE

A. Use sleeve and sleeve-seals for the following piping-penetration applications:

1. Exterior Concrete Walls Above Grade: Galvanized-Steel Sheet Pipe Sleeves with Sleeve-seal system

2. Exterior Concrete Walls Below Grade: Galvanized-Steel Sheet Pipe Sleeves with Sleeve-seal system

3. Interior or Exterior Concrete Slabs-on-Grade: Sleeve not required.

4. Interior Concrete Slabs Above Grade: Galvanized-Steel Sheet Pipe Sleeves with Silicone Sealant or Fire calk

5. Interior Partitions: Sleeve not required – fire calk penetrations of rated assemblies.

3.7 ESCUTCHEON INSTALLATION

A. Install escutcheons for piping penetrations of walls, ceilings, and finished floors.

B. Install escutcheons with ID to closely fit around pipe, tube, and insulation of piping and with OD that completely covers opening.

3.8 FLOOR PLATE INSTALLATION

A. Install floor plates for piping penetrations of equipment-room floors.

B. Install floor plates with ID to closely fit around pipe, tube, and insulation of piping and with OD that completely covers opening.

END OF SECTION

SECTION 22 05 19 - METERS AND GAGES FOR PLUMBING PIPING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Bimetallic-actuated thermometers.

2. Thermowells.

3. Dial-type pressure gages.

4. Gage attachments.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.4 INFORMATIONAL SUBMITTALS

A. Product Certificates: For each type of meter and gage.

1.5 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For meters and gages to include in operation and maintenance manuals.

PART 2 - PRODUCTS

2.1 BIMETALLIC-ACTUATED THERMOMETERS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Ashcroft Inc.

2. Miljoco Corporation.

3. Tel-Tru Manufacturing Company.

4. Trerice, H. O. Co.

5. Prior Approved Equal

B. Standard: ASME B40.200.

C. Case: Liquid-filled and sealed type(s); stainless steel with 3-inch nominal diameter.

D. Dial: Nonreflective aluminum with permanently etched scale markings and scales in deg F.

E. Connector Type(s): Union joint, adjustable angle, with unified-inch screw threads.

F. Connector Size: 1/2 inch, with ASME B1.1 screw threads.

G. Stem: 0.25 or 0.375 inch in diameter; stainless steel.

H. Window: Plain glass.

I. Ring: Stainless steel.

J. Element: Bimetal coil.

K. Pointer: Dark-colored metal.

L. Accuracy: Plus or minus 1 percent of scale range.

2.2 THERMOWELLS

A. Thermowells:

1. Standard: ASME B40.200.

2. Description: Pressure-tight, socket-type fitting made for insertion into piping tee fitting.

3. Material for Use with Copper Tubing: CNR or CUNI.

4. Material for Use with Steel Piping: CRES or CSA.

5. Type: Stepped shank unless straight or tapered shank is indicated.

6. External Threads: NPS 1/2, NPS 3/4, or NPS 1, ASME B1.20.1 pipe threads.

7. Internal Threads: 1/2, 3/4, and 1 inch, with ASME B1.1 screw threads.

8. Bore: Diameter required to match thermometer bulb or stem.

9. Insertion Length: Length required to match thermometer bulb or stem.

10. Lagging Extension: Include on thermowells for insulated piping and tubing.

11. Bushings: For converting size of thermowell's internal screw thread to size of thermometer connection.

B. Heat-Transfer Medium: Mixture of graphite and glycerin.

2.3 PRESSURE GAGES

A. Direct-Mounted and Remote, Metal-Case, Dial-Type Pressure Gages:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Ashcroft Inc.

b. Miljoco Corporation.

c. Tel-Tru Manufacturing Company.

d. Trerice, H. O. Co.

e. Prior Approved Equal

2. Standard: ASME B40.100.

3. Case: Liquid-filled, Sealed type(s); 4-1/2-inch nominal diameter.

4. Pressure-Element Assembly: Bourdon tube unless otherwise indicated.

5. Pressure Connection: Brass, with NPS 1/4 or NPS 1/2, ASME B1.20.1 pipe threads and bottom-outlet type unless back-outlet type is indicated.

6. Movement: Mechanical, with link to pressure element and connection to pointer.

7. Dial: Nonreflective aluminum with permanently etched scale markings graduated in psi.

8. Pointer: Dark-colored metal.

9. Window: Glass.

10. Ring: Metal.

11. Accuracy: Grade B, plus or minus 2 percent of middle half of scale range.

2.4 GAGE ATTACHMENTS

A. Snubbers: ASME B40.100, brass; with NPS 1/4 or NPS 1/2, ASME B1.20.1 pipe threads and piston or porous-metal-type surge-dampening device. Include extension for use on insulated piping.

B. Valves: Brass or stainless-steel needle, with NPS 1/4 or NPS 1/2, ASME B1.20.1 pipe threads.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install thermowells with socket extending a minimum of 2 inches into fluid and in vertical position in piping tees.

B. Install thermowells of sizes required to match thermometer connectors. Include bushings if required to match sizes.

C. Install thermowells with extension on insulated piping.

D. Fill thermowells with heat-transfer medium.

E. Install direct-mounted thermometers in thermowells and adjust vertical and tilted positions.

F. Install remote-mounted thermometer bulbs in thermowells and install cases on panels; connect cases with tubing and support tubing to prevent kinks. Use minimum tubing length.

G. Install direct-mounted pressure gages in piping tees with pressure gage located on pipe at the most readable position.

H. Install remote-mounted pressure gages on panel.

I. Install valve and snubber in piping for each pressure gage for fluids.

J. Install thermometers in the following locations:

1. Inlet and outlet of each domestic water heater.

2. Inlet and outlet of each domestic hot-water storage tank.

3. Two inlets and two outlets of each heat pump.

4. Inlet and outlet of each hydronic coil in air-handling units.

5. Inlet and outlet of each thermal-storage tank.

6. Inlet and outlet of each refrigeration condenser.

K. Install pressure gages in the following locations:

1. Building water service entrance into building.

2. Inlet and outlet of each pressure-reducing valve.

3. Suction and discharge of each pump.

L. Install meters and gages adjacent to machines and equipment to allow service and maintenance of meters, gages, machines, and equipment.

M. Adjust faces of meters and gages to proper angle for best visibility.

3.2 THERMOMETER SCALE-RANGE SCHEDULE

A. Edit section or insert additional paragraphs for pressure-gage applications.

B. Domestic Cold-Water Piping: 0 to 150 deg F.

C. Domestic Hot-Water Piping: 50 to 200 deg F.

3.3 PRESSURE-GAGE SCALE-RANGE SCHEDULE

A. Water Service Piping: 0 to 150 psi.

B. Domestic Water Piping: 0 to 100 psi.

END OF SECTION

SECTION 22 05 23 - GENERAL DUTY VALVES FOR PLUMBING

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Ball Valves

2. Bronze Check Valves

1.2 SUBMITTALS

A. See Section 22 00 00 “General Requirements for Pluming” for submittal requirements.

PART 2 - PRODUCTS

2.1 GENERAL REQUIREMENTS FOR VALVES

A. Source Limitations for Valves: Obtain each type of valve from single source from single manufacturer.

B. ASME Compliance:

1. ASME B1.20.1 for threads for threaded-end valves.

2. ASME B16.1 for flanges on iron valves.

3. ASME B16.10 and ASME B16.34 for ferrous valve dimensions and design criteria.

4. ASME B16.18 for solder-joint connections.

5. ASME B31.1 for power piping valves.

6. ASME B31.9 for building services piping valves.

C. Bronze valves shall be made with dezincification-resistant materials. Bronze valves made with copper alloy (brass) containing more than 15 percent zinc are not permitted.

D. Refer to valve schedule articles for applications of valves.

E. Valve Pressure-Temperature Ratings: Not less than indicated and as required for system pressures and temperatures.

F. Valve Sizes: Same as upstream piping unless otherwise indicated.

G. Valves in Insulated Piping:

1. Include stem extensions.

2. Extended operating handle of nonthermal-conductive material, and protective sleeves that allow operation of valves without breaking the vapor seals or disturbing insulation.

3. Memory stops that are fully adjustable after insulation is applied.

2.2 BRONZE BALL VALVES, TWO-PIECE WITH FULL PORT AND STAINLESS-STEEL TRIM:

A. Manufacturers: Provide products from one of the following:

1. Apollo
2. Nibco
3. Milwaukee
4. Red-White Valves
5. Watts

B. Description:

1. Standard: MSS SP-110.
2. SWP Rating: 150 psig.
3. CWP Rating: 600 psig.
4. Body Design: Two piece.
5. Body Material: Bronze.
6. Ends: Solder or Threaded.
7. Seats: PTFE.
8. Stem: Stainless steel.
9. Ball: Stainless steel, vented.
10. Port: Full.

2.3 BRONZE SWING CHECK VALVES WITH BRONZE DISC, CLASS 125:

A. Manufacturer: Provide products from one of the following:

1. Apollo
2. Crane
3. Milwaukee
4. Nibco
5. Watts

B. Description:

1. Standard: MSS SP-80, Type 3.
2. CWP Rating: 1380 kPa.

3. Body Design: Horizontal flow.

4. Body Material: ASTM B 62, bronze.

5. Ends: Threaded or soldered. See valve schedule articles.

6. Disc: Bronze.

PART 3 - EXECUTION

3.1 VALVE INSTALLATION

A. Install valves with unions or flanges at each piece of equipment arranged to allow service, maintenance, and equipment removal without system shutdown.

B. Locate valves for easy access and provide separate support where necessary.

C. Install valves in horizontal piping with stem at or above center of pipe.

D. Install valves in position to allow full stem movement.

3.2 GENERAL REQUIREMENTS FOR VALVE APPLICATIONS

A. If valves with specified SWP classes or CWP ratings are unavailable, the same types of valves with higher SWP classes or CWP ratings may be substituted.

B. Select valves with the following end connections:

1. For Copper Tubing, and Smaller: solder ends.

2. For Steel Piping, and Smaller: Threaded ends.

3. For Steel Piping NPS 2-1/2” and larger: Flanged or Grooved ends.

3.3 VALVE SCHEDULE

A. Domestic Water ½” – 2” NPS: Ball Valve, Solder or Threaded Ends

3.4 CHECK VALVE SCHEDULE

A. Pump Discharge ½” – 2” NPS: Bronze Swing Check, Threaded or Solder Ends

END OF SECTION

SECTION 22 05 29 - HANGERS AND SUPPORTS FOR PLUMBING PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Metal pipe hangers and supports.

2. Trapeze pipe hangers.

3. Thermal-hanger shield inserts.

4. Fastener systems.

5. Pipe positioning systems.

6. Equipment supports.

1.2 PERFORMANCE REQUIREMENTS

A. Delegated Design: Design trapeze pipe hangers and equipment supports, including comprehensive engineering analysis by a qualified professional engineer, using performance requirements and design criteria indicated.

B. Structural Performance: Hangers and supports for plumbing piping and equipment shall withstand the effects of gravity loads and stresses within limits and under conditions indicated according to ASCE/SEI 7.

1. Design supports for multiple pipes capable of supporting combined weight of supported systems, system contents, and test water.

2. Design equipment supports capable of supporting combined operating weight of supported equipment and connected systems and components.

3. Design seismic-restraint hangers and supports for piping and equipment.

1.3 SUBMITTALS

A. See Section 22 00 00 “General Requirements of Plumbing” for submittal requirements.

B. Shop Drawings: Signed and sealed by a qualified professional engineer. Show fabrication and installation details and include calculations for the following; include Product Data for components:

1. Trapeze pipe hangers.

2. Equipment supports.

C. Delegated-Design Submittal: For trapeze hangers indicated to comply with performance requirements and design criteria, including analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

1.4 QUALITY ASSURANCE

A. Structural Steel Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."

B. Pipe Welding Qualifications: Qualify procedures and operators according to ASME Boiler and Pressure Vessel Code.

PART 2 - PRODUCTS

2.1 METAL PIPE HANGERS AND SUPPORTS

A. Carbon-Steel Pipe Hangers and Supports:

1. Description: MSS SP-58, Types 1 through 58, factory-fabricated components.

2. Galvanized Metallic Coatings: Pre-galvanized or hot dipped.

3. Nonmetallic Coatings: Plastic coating, jacket, or liner.

4. Padded Hangers: Hanger with fiberglass or other pipe insulation pad or cushion to support bearing surface of piping.

5. Hanger Rods: Continuous-thread rod, nuts, and washer made of carbon steel.

B. Copper Pipe Hangers:

1. Description: MSS SP-58, Types 1 through 58, copper-coated-steel, factory-fabricated components.

2. Hanger Rods: Continuous-thread rod, nuts, and washer made of carbon steel.

2.2 TRAPEZE PIPE HANGERS

A. Description: MSS SP-69, Type 59, shop- or field-fabricated pipe-support assembly made from structural carbon-steel shapes with MSS SP-58 carbon-steel hanger rods, nuts, saddles, and U-bolts.

2.3 THERMAL-HANGER SHIELD INSERTS

A. Insulation-Insert Material for Cold Piping: ASTM C 591, Type VI, Grade 1 polyisocyanurate with 125-psig minimum compressive strength and vapor barrier.

B. Insulation-Insert Material for Hot Piping: ASTM C 591, Type VI, Grade 1 polyisocyanurate with 125-psig minimum compressive strength.

C. For Trapeze or Clamped Systems: Insert and shield shall cover entire circumference of pipe.

D. For Clevis or Band Hangers: Insert and shield shall cover lower 180 degrees of pipe.

E. Insert Length: Extend 2 inches beyond sheet metal shield for piping operating below ambient air temperature.

2.4 FASTENER SYSTEMS

A. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

B. Mechanical-Expansion Anchors: Insert-wedge-type, zinc-coated steel anchors, for use in hardened portland cement concrete; with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

2.5 PIPE POSITIONING SYSTEMS

A. Description: IAPMO PS 42, positioning system of metal brackets, clips, and straps for positioning piping in pipe spaces; for plumbing fixtures in commercial applications.

2.6 EQUIPMENT SUPPORTS

A. Description: Welded, shop- or field-fabricated equipment support made from structural carbon-steel shapes.

2.7 MISCELLANEOUS MATERIALS

A. Structural Steel: ASTM A 36/A 36M, carbon-steel plates, shapes, and bars; black and galvanized.

B. Grout: ASTM C 1107, factory-mixed and -packaged, dry, hydraulic-cement, nonshrink and nonmetallic grout; suitable for interior and exterior applications.

1. Properties: Nonstaining, noncorrosive, and nongaseous.

2. Design Mix: 5000-psi, 28-day compressive strength.

PART 3 - EXECUTION

3.1 HANGER AND SUPPORT INSTALLATION

A. Metal Pipe-Hanger Installation: Comply with MSS SP-69 and MSS SP-89. Install hangers, supports, clamps, and attachments as required to properly support piping from the building structure.

B. Metal Trapeze Pipe-Hanger Installation: Comply with MSS SP-69 and MSS SP-89. Arrange for grouping of parallel runs of horizontal piping, and support together on field-fabricated trapeze pipe hangers.

1. Pipes of Various Sizes: Support together and space trapezes for smallest pipe size or install intermediate supports for smaller diameter pipes as specified for individual pipe hangers.

2. Field fabricate from ASTM A 36/A 36M, carbon-steel shapes selected for loads being supported. Weld steel according to AWS D1.1/D1.1M.

C. Thermal-Hanger Shield Installation: Install in pipe hanger or shield for insulated piping.

D. Fastener System Installation:

1. Install powder-actuated fasteners for use in lightweight concrete or concrete slabs less than 4 inches thick in concrete after concrete is placed and completely cured. Use

operators that are licensed by powder-actuated tool manufacturer. Install fasteners according to powder-actuated tool manufacturer's operating manual.

2. Install mechanical-expansion anchors in concrete after concrete is placed and completely cured. Install fasteners according to manufacturer's written instructions.

E. Pipe Positioning-System Installation: Install support devices to make rigid supply and waste piping connections to each plumbing fixture.

F. Install hangers and supports complete with necessary attachments, inserts, bolts, rods, nuts, washers, and other accessories.

G. Equipment Support Installation: Fabricate from welded-structural-steel shapes.

H. Install hangers and supports to allow controlled thermal and seismic movement of piping systems, to permit freedom of movement between pipe anchors, and to facilitate action of expansion joints, expansion loops, expansion bends, and similar units.

I. Install lateral bracing with pipe hangers and supports to prevent swaying.

J. Install building attachments within concrete slabs or attach to structural steel. Install additional attachments at concentrated loads, including valves, flanges, and strainers, NPS 2-1/2 and larger and at changes in direction of piping. Install concrete inserts before concrete is placed; fasten inserts to forms and install reinforcing bars through openings at top of inserts.

K. Load Distribution: Install hangers and supports so that piping live and dead loads and stresses from movement will not be transmitted to connected equipment.

L. Pipe Slopes: Install hangers and supports to provide indicated pipe slopes and to not exceed maximum pipe deflections allowed by ASME B31.9 for building services piping.

M. Insulated Piping:

1. Attach clamps and spacers to piping.

    a. Piping Operating above Ambient Air Temperature: Clamp may project through insulation.

    b. Piping Operating below Ambient Air Temperature: Use thermal-hanger shield insert with clamp sized to match OD of insert.

    c. Do not exceed pipe stress limits allowed by ASME B31.9 for building services piping.

2. Install MSS SP-58, Type 39, protection saddles if insulation without vapor barrier is indicated. Fill interior voids with insulation that matches adjoining insulation.

    a. Option: Thermal-hanger shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.

3. Install MSS SP-58, Type 40, protective shields on cold piping with vapor barrier. Shields shall span an arc of 180 degrees.

a. Option: Thermal-hanger shield inserts may be used. Include steel weight-distribution plate for pipe NPS 4 and larger if pipe is installed on rollers.

4. Shield Dimensions for Pipe: Not less than the following:

a. NPS 1/4 to NPS 3-1/2: 12 inches long and 0.048 inch thick.

b. NPS 4: 12 inches long and 0.06 inch thick.

c. NPS 5 and NPS 6: 18 inches long and 0.06 inch thick.

d. NPS 8 to NPS 14: 24 inches long and 0.075 inch thick.

e. NPS 16 to NPS 24: 24 inches long and 0.105 inch thick.

5. Pipes NPS 8 and Larger: Include wood or reinforced calcium-silicate-insulation inserts of length at least as long as protective shield.

6. Thermal-Hanger Shields: Install with insulation same thickness as piping insulation.

3.2 EQUIPMENT SUPPORTS

A. Fabricate structural-steel stands to suspend equipment from structure overhead or to support equipment above floor.

B. Grouting: Place grout under supports for equipment and make bearing surface smooth.

C. Provide lateral bracing, to prevent swaying, for equipment supports.

3.3 METAL FABRICATIONS

A. Cut, drill, and fit miscellaneous metal fabrications for trapeze pipe hangers and equipment supports.

B. Fit exposed connections together to form hairline joints. Field weld connections that cannot be shop welded because of shipping size limitations.

C. Field Welding: Comply with AWS D1.1/D1.1M procedures for shielded, metal arc welding; appearance and quality of welds; and methods used in correcting welding work; and with the following:

1. Use materials and methods that minimize distortion and develop strength and corrosion resistance of base metals.

2. Obtain fusion without undercut or overlap.

3. Remove welding flux immediately.

4. Finish welds at exposed connections so no roughness shows after finishing and so contours of welded surfaces match adjacent contours.

3.4 ADJUSTING

A. Hanger Adjustments: Adjust hangers to distribute loads equally on attachments and to achieve indicated slope of pipe.

B. Trim excess length of continuous-thread hanger and support rods to 1-1/2 inches.

3.5 PAINTING

A. Touchup: Clean field welds and abraded areas of shop paint. Paint exposed areas immediately after erecting hangers and supports. Use same materials as used for shop painting. Comply with SSPC-PA 1 requirements for touching up field-painted surfaces.

1. Apply paint by brush or spray to provide a minimum dry film thickness of 2.0 mils.

B. Touchup: Cleaning and touchup painting of field welds, bolted connections, and abraded areas of shop paint on miscellaneous metal are specified in Section 09 91 13 "Exterior Painting.", Section 09 91 23 "Interior Painting.", Section 09 96 00 "High-Performance Coatings."

C. Galvanized Surfaces: Clean welds, bolted connections, and abraded areas and apply galvanizing-repair paint to comply with ASTM A 780.

3.6 HANGER AND SUPPORT SCHEDULE

A. Specific hanger and support requirements are in Sections specifying piping systems and equipment.

B. Comply with MSS SP-69 for pipe-hanger selections and applications that are not specified in piping system Sections.

C. Use hangers and supports with galvanized metallic coatings for piping and equipment that will not have field-applied finish.

D. Use nonmetallic coatings on attachments for electrolytic protection where attachments are in direct contact with copper tubing.

E. Use carbon-steel pipe hangers and supports and metal trapeze pipe hangers and attachments for general service applications.

F. Use copper-plated pipe hangers and copper attachments for copper piping and tubing.

G. Use padded hangers for piping that is subject to scratching.

H. Use thermal-hanger shield inserts for insulated piping and tubing.

I. Horizontal-Piping Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Adjustable, Steel Clevis Hangers (MSS Type 1): For suspension of non-insulated or insulated, stationary pipes NPS 1/2 to NPS 30.

2. Yoke-Type Pipe Clamps (MSS Type 2): For suspension of up to 1050 deg F, pipes NPS 4 to NPS 24, requiring up to 4 inches of insulation.

3. Carbon- or Alloy-Steel, Double-Bolt Pipe Clamps (MSS Type 3): For suspension of pipes NPS 3/4 to NPS 36, requiring clamp flexibility and up to 4 inches of insulation.

4. Adjustable, Steel Band Hangers (MSS Type 7): For suspension of non-insulated, stationary pipes NPS 1/2 to NPS 8.

5. U-Bolts (MSS Type 24): For support of heavy pipes NPS 1/2 to NPS 30.

6. Pipe Saddle Supports (MSS Type 36): For support of pipes NPS 4 to NPS 36, with steel-pipe base stanchion support and cast-iron floor flange or carbon-steel plate.

7. Pipe Stanchion Saddles (MSS Type 37): For support of pipes NPS 4 to NPS 36, with steel-pipe base stanchion support and cast-iron floor flange or carbon-steel plate, and with U-bolt to retain pipe.

8. Single-Pipe Rolls (MSS Type 41): For suspension of pipes NPS 1 to NPS 30, from two rods if longitudinal movement caused by expansion and contraction might occur.

9. Complete Pipe Rolls (MSS Type 44): For support of pipes NPS 2 to NPS 42 if longitudinal movement caused by expansion and contraction might occur but vertical adjustment is not necessary.

J. Vertical-Piping Clamps: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Extension Pipe or Riser Clamps (MSS Type 8): For support of pipe risers NPS 3/4 to NPS 24.

2. Carbon- or Alloy-Steel Riser Clamps (MSS Type 42): For support of pipe risers NPS 3/4 to NPS 24 if longer ends are required for riser clamps.

K. Hanger-Rod Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel Turnbuckles (MSS Type 13): For adjustment up to 6 inches for heavy loads.

2. Steel Clevises (MSS Type 14): For 120 to 450 deg F piping installations.

L. Building Attachments: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel or Malleable Concrete Inserts (MSS Type 18): For upper attachment to suspend pipe hangers from concrete ceiling.

2. Top-Beam C-Clamps (MSS Type 19): For use under roof installations with bar-joist construction, to attach to top flange of structural shape.

3. Side-Beam or Channel Clamps (MSS Type 20): For attaching to bottom flange of beams, channels, or angles.

4. Center-Beam Clamps (MSS Type 21): For attaching to center of bottom flange of beams.

5. Welded Beam Attachments (MSS Type 22): For attaching to bottom of beams if loads are considerable and rod sizes are large.

6. C-Clamps (MSS Type 23): For structural shapes.

7. Welded-Steel Brackets: For support of pipes from below, or for suspending from above by using clip and rod. Use one of the following for indicated loads:

a. Light (MSS Type 31): 750 lb.

b. Medium (MSS Type 32): 1500 lb.

c. Heavy (MSS Type 33): 3000 lb.

8. Side-Beam Brackets (MSS Type 34): For sides of steel or wooden beams.

9. Plate Lugs (MSS Type 57): For attaching to steel beams if flexibility at beam is required.

M. Saddles and Shields: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Steel-Pipe-Covering Protection Saddles (MSS Type 39): To fill interior voids with insulation that matches adjoining insulation.

2. Protection Shields (MSS Type 40): Of length recommended in writing by manufacturer to prevent crushing insulation.

3. Thermal-Hanger Shield Inserts: For supporting insulated pipe.

N. Spring Hangers and Supports: Unless otherwise indicated and except as specified in piping system Sections, install the following types:

1. Spring Cushions (MSS Type 48): For light loads if vertical movement does not exceed 1-1/4 inches.

2. Spring-Cushion Roll Hangers (MSS Type 49): For equipping Type 41, roll hanger with springs.

3. Variable-Spring Base Supports (MSS Type 52): Preset to indicated load and limit variability factor to 25 percent to allow expansion and contraction of piping system from base support.

O. Comply with MSS SP-69 for trapeze pipe-hanger selections and applications that are not specified in piping system Sections.

P. Use powder-actuated fasteners or mechanical-expansion anchors instead of building attachments where required in concrete construction.

Q. Use pipe positioning systems in pipe spaces behind plumbing fixtures to support supply and waste piping for plumbing fixtures.

END OF SECTION

SECTION 22 05 48 - VIBRATION AND SEISMIC CONTROLS FOR PLUMBING PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Elastomeric isolation pads.
2. Elastomeric isolation mounts.
3. Restrained elastomeric isolation mounts.
4. Open-spring isolators.
5. Housed-spring isolators.
6. Restrained-spring isolators.
7. Housed-restrained-spring isolators.
8. Pipe-riser resilient supports.
9. Resilient pipe guides.
10. Elastomeric hangers.
11. Spring hangers.
12. Snubbers.
13. Restraint channel bracings.
14. Restraint cables.
15. Seismic-restraint accessories.
16. Mechanical anchor bolts.

1.2 ACTION SUBMITTALS

A. See Section 22 00 00 “General Requirements for Plumbing” for submittal requirements.

B. Delegated-Design Submittal: For each vibration isolation and seismic-restraint device.

1. Include design calculations and details for selecting vibration isolators and seismic restraints complying with performance requirements, design criteria, and analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

1.3 QUALITY ASSURANCE

A. Comply with seismic-restraint requirements in the IBC unless requirements in this Section are more stringent.

B. Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."

C. Seismic-restraint devices shall have horizontal and vertical load testing and analysis and shall bear anchorage preapproval OPA number from OSHPD, preapproval by ICC-ES, or preapproval by another agency acceptable to authorities having jurisdiction, showing maximum seismic-restraint ratings. Ratings based on independent testing are preferred to ratings based on calculations. If preapproved ratings are unavailable, submittals based on independent testing are preferred. Calculations (including combining shear and tensile loads) to support seismic-restraint designs must be signed and sealed by a qualified professional engineer.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Seismic-Restraint Loading:

1. Design seismic restraints for components for seismic design forces defined in Chapter 13 of ASCE 7-16.

a. Design Spectral Response Acceleration at Short Periods, See Structural Drawings for Site Specific S_{DS} Value

b. Component Importance Factor, I_P = 1.0; except for components conveying, supporting, or otherwise containing natural gas or other flammable and/or explosive contents, I_P = 1.5.

c. Component Response Modification Factor, R_P: See Table 13.6-1 of ASCE 7-16

d. Component Amplification Factor, a_P: See Table 13.6-1 of ASCE 7-16

2.2 ELASTOMERIC ISOLATION PADS

A. Elastomeric Isolation Pads:

1. Fabrication: Single or multiple layers of sufficient durometer stiffness for uniform loading over pad area.

2. Size: Factory or field cut to match requirements of supported equipment.

3. Pad Material: Oil and water resistant with elastomeric properties.

4. Surface Pattern: Smooth, Ribbed or Waffle pattern.

5. Infused nonwoven cotton or synthetic fibers.

6. Load-bearing metal plates adhered to pads.

2.3 ELASTOMERIC ISOLATION MOUNTS

A. Double-Deflection, Elastomeric Isolation Mounts:

1. Mounting Plates:

a. Top Plate: Encapsulated steel load transfer top plates, factory drilled and threaded with threaded studs or bolts.

b. Baseplate: Encapsulated steel bottom plates with holes provided for anchoring to support structure.

2. Elastomeric Material: Molded, oil-resistant rubber, neoprene, or other elastomeric material.

2.4 RESTRAINED ELASTOMERIC ISOLATION MOUNTS

A. Restrained Elastomeric Isolation Mounts:

1. Description: All-directional isolator with seismic restraints containing two separate and opposing elastomeric elements that prevent central threaded element and attachment hardware from contacting the housing during normal operation.

a. Housing: Cast-ductile iron or welded steel.

b. Elastomeric Material: Molded, oil-resistant rubber, neoprene, or other elastomeric material.

2.5 OPEN-SPRING ISOLATORS

A. Freestanding, Laterally Stable, Open-Spring Isolators:

1. Outside Spring Diameter: Not less than 80 percent of the compressed height of the spring at rated load.

2. Minimum Additional Travel: 50 percent of the required deflection at rated load.

3. Lateral Stiffness: More than 80 percent of rated vertical stiffness.

4. Overload Capacity: Support 200 percent of rated load, fully compressed, without deformation or failure.

5. Baseplates: Factory-drilled steel plate for bolting to structure with an elastomeric isolator pad attached to the underside. Baseplates shall limit floor load to 500 psig (3447 kPa).

6. Top Plate and Adjustment Bolt: Threaded top plate with adjustment bolt and cap screw to fasten and level equipment.

2.6 HOUSED-SPRING ISOLATORS

A. Freestanding, Laterally Stable, Open-Spring Isolators in Two-Part Telescoping Housing:

1. Outside Spring Diameter: Not less than 80 percent of the compressed height of the spring at rated load.

2. Minimum Additional Travel: 50 percent of the required deflection at rated load.

3. Lateral Stiffness: More than 80 percent of rated vertical stiffness.

4. Overload Capacity: Support 200 percent of rated load, fully compressed, without deformation or failure.

5. Two-Part Telescoping Housing: A steel top and bottom frame separated by an elastomeric material and enclosing the spring isolators.

    a. Drilled base housing for bolting to structure with an elastomeric isolator pad attached to the underside. Bases shall limit floor load to 500 psig.

    b. Top housing with attachment and leveling bolt.

2.7 RESTRAINED-SPRING ISOLATORS

A. Freestanding, Laterally Stable, Open-Spring Isolators with Vertical-Limit Stop Restraint:

1. Housing: Steel housing with vertical-limit stops to prevent spring extension due to weight being removed.

    a. Base with holes for bolting to structure with an elastomeric isolator pad attached to the underside. Bases shall limit floor load to 500 psig.

    b. Top plate with threaded mounting holes.

    c. Internal leveling bolt that acts as blocking during installation.

2. Restraint: Limit stop as required for equipment and authorities having jurisdiction.

3. Outside Spring Diameter: Not less than 80 percent of the compressed height of the spring at rated load.

4. Minimum Additional Travel: 50 percent of the required deflection at rated load.

5. Lateral Stiffness: More than 80 percent of rated vertical stiffness.

6. Overload Capacity: Support 200 percent of rated load, fully compressed, without deformation or failure.

2.8 HOUSED-RESTRAINED-SPRING ISOLATORS

A. Freestanding, Steel, Open-Spring Isolators with Vertical-Limit Stop Restraint in Two-Part Telescoping Housing:

1. Two-Part Telescoping Housing: A steel top and bottom frame separated by an elastomeric material and enclosing the spring isolators. Housings are equipped with adjustable snubbers to limit vertical movement.

    a. Drilled base housing for bolting to structure with an elastomeric isolator pad attached to the underside. Bases shall limit floor load to 500 psig.

    b. Threaded top housing with adjustment bolt and cap screw to fasten and level equipment.

2. Outside Spring Diameter: Not less than 80 percent of the compressed height of the spring at rated load.

3. Minimum Additional Travel: 50 percent of the required deflection at rated load.

4. Lateral Stiffness: More than 80 percent of rated vertical stiffness.

5. Overload Capacity: Support 200 percent of rated load, fully compressed, without deformation or failure.

2.9 PIPE-RISER RESILIENT SUPPORT

A. Description: All-directional, acoustical pipe anchor consisting of two steel tubes separated by a minimum 1/2-inch- thick neoprene.

1. Vertical-Limit Stops: Steel and neoprene vertical-limit stops arranged to prevent vertical travel in both directions.

2. Maximum Load Per Support: 500 psig on isolation material providing equal isolation in all directions.

2.10 RESILIENT PIPE GUIDES

A. Description: Telescopic arrangement of two steel tubes or post and sleeve arrangement separated by a minimum 1/2-inch- thick neoprene.

1. Factory-Set Height Guide with Shear Pin: Shear pin shall be removable and reinsertable to allow for selection of pipe movement. Guides shall be capable of motion to meet location requirements.

2.11 ELASTOMERIC HANGERS

A. Elastomeric Mount in a Steel Frame with Upper and Lower Steel Hanger Rods:

1. Frame: Steel, fabricated with a connection for an upper threaded hanger rod and an opening on the underside to allow for a maximum of 30 degrees of angular lower hanger-rod misalignment without binding or reducing isolation efficiency.

2. Dampening Element: Molded, oil-resistant rubber, neoprene, or other elastomeric material with a projecting bushing for the underside opening preventing steel to steel contact.

2.12 SPRING HANGERS

A. Combination Coil-Spring and Elastomeric-Insert Hanger with Spring and Insert in Compression:

1. Frame: Steel, fabricated for connection to threaded hanger rods and to allow for a maximum of 30 degrees of angular hanger-rod misalignment without binding or reducing isolation efficiency.

2. Outside Spring Diameter: Not less than 80 percent of the compressed height of the spring at rated load.

3. Minimum Additional Travel: 50 percent of the required deflection at rated load.

4. Lateral Stiffness: More than 80 percent of rated vertical stiffness.

5. Overload Capacity: Support 200 percent of rated load, fully compressed, without deformation or failure.

6. Elastomeric Element: Molded, oil-resistant rubber or neoprene. Steel-washer-reinforced cup to support spring and bushing projecting through bottom of frame.

7. Adjustable Vertical Stop: Steel washer with neoprene washer "up-stop" on lower threaded rod.

8. Self-centering hanger-rod cap to ensure concentricity between hanger rod and support spring coil.

2.13 SNUBBERS

A. Description: Factory fabricated using welded structural-steel shapes and plates, anchor bolts, and replaceable resilient isolation washers and bushings.

1. Anchor bolts for attaching to concrete shall be seismic-rated, drill-in, and stud-wedge or female-wedge type.

2. Resilient Isolation Washers and Bushings: Oil- and water-resistant neoprene.

3. Maximum 1/4-inch air gap, and minimum 1/4-inch- thick resilient cushion.

2.14 RESTRAINT CHANNEL BRACINGS

A. Description: MFMA-4, shop- or field-fabricated bracing assembly made of slotted steel channels with accessories for attachment to braced component at one end and to building structure at the other end and other matching components and with corrosion-resistant coating; rated in tension, compression, and torsion forces.

2.15 RESTRAINT CABLES

A. Restraint Cables: ASTM A 603 galvanized-steel cables. End connections made of steel assemblies with thimbles, brackets, swivel, and bolts designed for restraining cable service; with a minimum of two clamping bolts for cable engagement.

2.16 SEISMIC-RESTRAINT ACCESSORIES

A. Hanger-Rod Stiffener: Reinforcing steel angle clamped to hanger rod.

B. Bushings for Floor-Mounted Equipment Anchor Bolts: Neoprene bushings designed for rigid equipment mountings, and matched to type and size of anchor bolts and studs.

C. Bushing Assemblies for Wall-Mounted Equipment Anchorage: Assemblies of neoprene elements and steel sleeves designed for rigid equipment mountings, and matched to type and size of attachment devices used.

D. Resilient Isolation Washers and Bushings: One-piece, molded, oil- and water-resistant neoprene, with a flat washer face.

E. Mechanical Anchor Bolts: Drilled-in and stud-wedge or female-wedge type in zinc-coated steel for interior applications and stainless steel for exterior applications. Select anchor bolts with strength required for anchor and as tested according to ASTM E 488.

2.17 EXECUTION

A. components so strength is adequate to carry present and future static and seismic loads within specified loading limits.

2.18 VIBRATION CONTROL AND SEISMIC-RESTRAINT DEVICE INSTALLATION

A. Coordinate the location of embedded connection hardware with supported equipment attachment and mounting points and with requirements for concrete reinforcement and formwork specified in Section 03 30 00 "Cast-in-Place Concrete." or Section 03 30 53 "Miscellaneous Cast-in-Place Concrete."

B. Installation of vibration isolators must not cause any change of position of equipment, piping, or ductwork resulting in stresses or misalignment.

C. Comply with requirements in Section 07 72 00 "Roof Accessories" for installation of roof curbs, equipment supports, and roof penetrations.

D. Equipment Restraints:

1. Install seismic snubbers on plumbing equipment mounted on vibration isolators. Locate snubbers as close as possible to vibration isolators and bolt to equipment base and supporting structure.

2. Install resilient bolt isolation washers on equipment anchor bolts where clearance between anchor and adjacent surface exceeds 0.125 inch.

3. Install seismic-restraint devices using methods approved by an evaluation service member of ICC-ES that provides required submittals for component.

E. Piping Restraints:

1. Comply with requirements in MSS SP-127.

2. Space lateral supports a maximum of 40 feet o.c., and longitudinal supports a maximum of 80 feet o.c.

3. Brace a change of direction longer than 12 feet.

F. Install cables so they do not bend across edges of adjacent equipment or building structure.

G. Install seismic-restraint devices using methods approved by an evaluation service member of ICC-ES that provides required submittals for component.

H. Install bushing assemblies for anchor bolts for floor-mounted equipment, arranged to provide resilient media between anchor bolt and mounting hole in concrete base.

I. Install bushing assemblies for mounting bolts for wall-mounted equipment, arranged to provide resilient media where equipment or equipment-mounting channels are attached to wall.

J. Attachment to Structure: If specific attachment is not indicated, anchor bracing to structure at flanges of beams, at upper truss chords of bar joists, or at concrete members.

K. Drilled-in Anchors:

1. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Do not damage existing reinforcing or embedded items during coring or drilling.

Notify the structural engineer if reinforcing steel or other embedded items are encountered during drilling. Locate and avoid prestressed tendons, electrical and telecommunications conduit, and gas lines.

2. Do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength.

3. Wedge Anchors: Protect threads from damage during anchor installation. Heavy-duty sleeve anchors shall be installed with sleeve fully engaged in the structural element to which anchor is to be fastened.

4. Set anchors to manufacturer's recommended torque, using a torque wrench.

5. Install zinc-coated steel anchors for interior and stainless-steel anchors for exterior applications.

2.19 ACCOMMODATION OF DIFFERENTIAL SEISMIC MOTION

A. Install flexible connections in piping where they cross seismic joints, where adjacent sections or branches are supported by different structural elements, and where the connections terminate with connection to equipment that is anchored to a different structural element from the one supporting the connections as they approach equipment. Comply with requirements in Section 22 11 16 "Domestic Water Piping" for piping flexible connections.

2.20 ADJUSTING

A. Adjust isolators after piping system is at operating weight.

B. Adjust limit stops on restrained-spring isolators to mount equipment at normal operating height. After equipment installation is complete, adjust limit stops so they are out of contact during normal operation.

END OF SECTION

SECTION 22 05 53 - IDENTIFICATION FOR PLUMBING PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Equipment labels.

2. Warning signs and labels.

3. Pipe labels.

4. Valve Tags.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

PART 2 - PRODUCTS

2.1 EQUIPMENT LABELS

A. Plastic Labels for Equipment:

1. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, and having predrilled holes for attachment hardware.

2. Letter Color: White

3. Background Color: Black

4. Maximum Temperature: Able to withstand temperatures up to 160 deg F.

5. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.

6. Minimum Letter Size: 1/4 inch For name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-quarters the size of principal lettering.

7. Fasteners: Stainless-steel rivets or self-tapping screws.

8. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

B. Label Content: Include equipment's Drawing designation or unique equipment number, Drawing numbers where equipment is indicated (plans, details, and schedules), and the Specification Section number and title where equipment is specified.

C. All labels on equipment visible to the public/occupant shall be coordinated with the architect or engineer prior to installation.

2.2 WARNING SIGNS AND LABELS

A. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, and having predrilled holes for attachment hardware.

B. Letter Color: White

C. Background Color: Red

D. Maximum Temperature: Able to withstand temperatures up to 160 deg F.

E. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.

F. Minimum Letter Size: 1/4 inch (6.4 mm) for name of units if viewing distance is less than 24 inches (600 mm), 1/2 inch (13 mm) for viewing distances up to 72 inches (1830 mm), and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-quarters the size of principal lettering.

G. Fasteners: Stainless-steel rivets or self-tapping screws.

H. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

I. Label Content: Include caution and warning information plus emergency notification instructions.

2.3 PIPE LABELS

A. General Requirements for Manufactured Pipe Labels: Preprinted, color-coded, with lettering indicating service, and showing flow direction.

B. Self-Adhesive Pipe Labels: Printed plastic with contact-type, permanent-adhesive backing.

C. Pipe Label Contents: Include identification of piping service using same designations or abbreviations as used on Drawings; also include pipe size and an arrow indicating flow direction.

1. Flow-Direction Arrows: Integral with piping-system service lettering to accommodate both directions or as separate unit on each pipe label to indicate flow direction.

2. Lettering Size: At least 1/2 inch for viewing distances up to 72 inches and proportionately larger lettering for greater viewing distances.

2.4 VALVE TAGS

A. Description: Stamped or engraved with 1/4-inch letters for piping system abbreviation and 1/2-inch numbers, with numbering scheme approved by Engineer. Provide 5/32-inch hole for fastener.

1. Material: 0.032-inch thick brass or aluminum.

2. Valve-Tag Fasteners: Brass wire-link or beaded chain; or S-hook.

B. Valve Schedules: For each piping system, on 8-1/2-by-11-inch bond paper. Tabulate valve number, piping system, system abbreviation (as shown on valve tag), location of valve (room or space), normal-operating position (open, closed, or modulating), and variations for identification. Mark valves for emergency shutoff and similar special uses.

1. Valve-tag schedule shall be included in operation and maintenance data.

PART 3 - EXECUTION

3.1 EQUIPMENT LABEL INSTALLATION

A. Install or permanently fasten labels on each major item of mechanical equipment.

B. Locate equipment labels where accessible and visible.

3.2 PIPE LABEL INSTALLATION

A. Pipe Label Locations: Locate pipe labels where piping is exposed or above accessible ceilings in finished spaces; machine rooms; accessible maintenance spaces such as shafts, tunnels, and plenums; and exterior exposed locations as follows:

1. Near each valve and control device.

2. Near each branch connection, excluding short takeoffs for fixtures and terminal units. Where flow pattern is not obvious, mark each pipe at branch.

3. Near penetrations and on both sides of through walls, floors, ceilings, and inaccessible enclosures.

4. At access doors, manholes, and similar access points that permit view of concealed piping.

5. Near major equipment items and other points of origination and termination.

6. Spaced at maximum intervals of 50 feet along each run. Reduce intervals to 25 feet in areas of congested piping and equipment.

7. On piping above removable acoustical ceilings. Omit intermediately spaced labels.

B. Pipe Label Color Schedule:

1. Compressed Air Piping:

a. Background: Blue

b. Letter Colors: White

2. Domestic Water Piping:

a. Background: Green

b. Letter Colors: White

3. Sanitary Waste and Storm Drainage Piping:

a. Background Color: Black

b. Letter Color: White

3.3 VALVE-TAG INSTALLATION

A. Install tags on valves and control devices in piping systems, except check valves; valves within factory-fabricated equipment units; plumbing fixture supply stops; shutoff valves; faucets; convenience and lawn-watering hose connections; and similar roughing-in connections of end-use fixtures and units. List tagged valves in a valve schedule.

1. Valve-Tag Size and Shape: 1-1/2 inches.

2. Valve-Tag Color: Natural Brass or Aluminum.

3.4 VALVE-SCHEDULE INSTALLATION

A. Mount valve schedule on wall in accessible location in each major equipment room.

3.5 CLEANING

A. Clean faces of mechanical identification devices and glass frames of valve schedules.

END OF SECTION

SECTION 22 07 16 - PLUMBING EQUIPMENT AND PIPING INSULATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes insulating requirements for equipment, piping:

1.2 SUBMITTALS

A. See section 22 00 00 “General Requirements of Plumbing” for submittal requirements.

1.3 QUALITY ASSURANCE

A. Installer Qualifications: Skilled mechanics who have successfully completed an apprenticeship program or another craft training program certified by the Department of Labor, Bureau of Apprenticeship and Training.

B. Surface-Burning Characteristics: For insulation and related materials, as determined by testing identical products according to ASTM E 84 by a testing agency acceptable to authorities having jurisdiction. Factory label insulation and jacket materials and adhesive, mastic, tapes, and cement material containers, with appropriate markings of applicable testing agency.

1. Insulation Installed Indoors: Flame-spread index of 25 or less, and smoke-developed index of 50 or less.

2. Insulation Installed Outdoors: Flame-spread index of 75 or less, and smoke-developed index of 150 or less.

PART 2 - PRODUCTS

2.1 INSULATION MATERIALS

A. Comply with requirements in "Equipment Insulation Schedule" "Piping Insulation Schedule," and “Duct Insulation Schedule" articles for where insulating materials shall be applied.

B. Products shall not contain asbestos, lead, mercury, or mercury compounds.

C. Products that come in contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested according to ASTM C 871.

D. Insulation materials for use on austenitic stainless steel shall be qualified as acceptable according to ASTM C 795.

E. Foam insulation materials shall not use CFC or HCFC blowing agents in the manufacturing process.

F. Flexible Elastomeric Insulation: Closed-cell, sponge- or expanded-rubber materials. Comply with ASTM C 534, Type I for tubular materials and Type II for sheet materials.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Aeroflex USA, Inc.

b. Armacell LLC.

c. K-Flex USA.

G. Mineral-Fiber, Pipe and Tank Insulation: Mineral or glass fibers bonded with a thermosetting resin. Semirigid board material with factory-applied ASJ complying with ASTM C 1393, Type II or Type IIIA Category 2, or with properties similar to ASTM C 612, Type IB. Nominal density is 2.5 lb/cu. ft. or more. Thermal conductivity (k-value) at 100 deg F is 0.29 Btu x in./h x sq. ft. x deg F or less. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. CertainTeed Corporation.

b. Johns Manville; a Berkshire Hathaway company.

c. Knauf Insulation.

d. Owens Corning.

H. Mineral-Fiber, Preformed Pipe Insulation:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Johns Manville; a Berkshire Hathaway company.

b. Knauf Insulation.

c. Owens Corning.

2. Type I, 850 Deg F Materials: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C 547, Type I, Grade A, with factory-applied ASJ-SSL. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.

I. Thermal Insulating Wool:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Owens Corning

b. Prior Approved Equal

2. Type I, 1000 Deg F Materials: Inorganic glass fibers bonded with thermosetting resin. Comply with ASTM C553, TIW Type I.

2.2 INSULATING CEMENTS

A. Mineral-Fiber, Hydraulic-Setting Insulating and Finishing Cement: Comply with ASTM C 449.

2.3 ADHESIVES

A. Materials shall be compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated unless otherwise indicated.

B. Flexible Elastomeric and Polyolefin Adhesive: Comply with MIL-A-24179A, Type II, Class I.

C. Mineral-Fiber Adhesive: Comply with MIL-A-3316C, Class 2, Grade A.

D. ASJ Adhesive, and FSK and PVDC Jacket Adhesive: Comply with MIL-A-3316C, Class 2, Grade A for bonding insulation jacket lap seams and joints.

E. PVC Jacket Adhesive: Compatible with PVC jacket.

2.4 MASTICS

A. Materials shall be compatible with insulation materials, jackets, and substrates; comply with MIL-PRF-19565C, Type II.

B. Vapor-Barrier Mastic: Water based; suitable for indoor use on below ambient services.

1. Water-Vapor Permeance: ASTM E 96/E 96M, Procedure B, 0.013 perm at 43-mil dry film thickness.

2. Service Temperature Range: Minus 20 to plus 180 deg F.

3. Solids Content: ASTM D 1644, 58 percent by volume and 70 percent by weight.

4. Color: White.

C. Breather Mastic: Water based; suitable for indoor and outdoor use on above ambient services.

1. Water-Vapor Permeance: ASTM F 1249, 1.8 perms at 0.0625-inch dry film thickness.

2. Service Temperature Range: Minus 20 to plus 180 deg F.

3. Solids Content: 60 percent by volume and 66 percent by weight.

4. Color: White.

2.5 SEALANTS

A. Metal Jacket Flashing Sealants:

1. Materials shall be compatible with insulation materials, jackets, and substrates.

2. Fire- and water-resistant, flexible, elastomeric sealant.

3. Service Temperature Range: Minus 40 to plus 250 deg F.

4. Color: Aluminum.

B. ASJ Flashing Sealants, and PVC Jacket Flashing Sealants:

1. Materials shall be compatible with insulation materials, jackets, and substrates.

2. Fire- and water-resistant, flexible, elastomeric sealant.

3. Service Temperature Range: Minus 40 to plus 250 deg F.

4. Color: White.

2.6 FACTORY-APPLIED JACKETS

A. Insulation system schedules indicate factory-applied jackets on various applications. When factory-applied jackets are indicated, comply with the following:

1. ASJ: White, kraft-paper, fiberglass-reinforced scrim with aluminum-foil backing; complying with ASTM C 1136, Type I.

2. ASJ-SSL: ASJ with self-sealing, pressure-sensitive, acrylic-based adhesive covered by a removable protective strip; complying with ASTM C 1136, Type I.

2.7 FIELD-APPLIED JACKETS

A. Field-applied jackets shall comply with ASTM C 921, Type I, unless otherwise indicated.

B. PVC Jacket: High-impact-resistant, UV-resistant PVC complying with ASTM D 1784, Class 16354-C; thickness as scheduled; roll stock ready for shop or field cutting and forming. Thickness is indicated in field-applied jacket schedules.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Johns Manville; a Berkshire Hathaway company.

b. P.I.C. Plastics, Inc.

c. Proto Corporation.

d. Speedline Corporation.

2. Adhesive: As recommended by jacket material manufacturer.

3. Color: White.

4. Factory-fabricated fitting covers to match jacket if available; otherwise, field fabricate.

a. Shapes: 45- and 90-degree, short- and long-radius elbows, tees, valves, flanges, unions, reducers, end caps, soil-pipe hubs, traps, mechanical joints, and P-trap and supply covers for lavatories.

C. Aluminum Jacket: Comply with ASTM B 209, Alloy 3003, 3005, 3105, or 5005, Temper H-14.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Childers Brand; H. B. Fuller Construction Products.

b. ITW Insulation Systems; Illinois Tool Works, Inc.

c. RPR Products, Inc.

2. Sheet and roll stock ready for shop or field sizing or factory cut and rolled to size.

3. Finish and thickness are indicated in field-applied jacket schedules.

4. Moisture Barrier for Outdoor Applications: 3-mil-thick, heat-bonded polyethylene and kraft paper or 2.5-mil-thick polysurlyn.

5. Factory-Fabricated Fitting Covers:

   a. Same material, finish, and thickness as jacket.

   b. Preformed 2-piece or gore, 45- and 90-degree, short- and long-radius elbows.

   c. Tee covers.

   d. Flange and union covers.

   e. End caps.

   f. Beveled collars.

   g. Valve covers.

   h. Field fabricate fitting covers only if factory-fabricated fitting covers are not available.

D. Underground Direct-Buried Jacket: 125-mil-thick vapor barrier and waterproofing membrane consisting of a rubberized bituminous resin reinforced with a woven-glass fiber or polyester scrim and laminated aluminum foil.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

   a. Pittsburgh Corning Corporation.

   b. Polyguard Products, Inc.

2.8 TAPES

A. ASJ Tape: White vapor-retarder tape matching factory-applied jacket with acrylic adhesive, complying with ASTM C 1136.

1. Width: 3 inches.

2. Thickness: 11.5 mils.

3. Adhesion: 90 ounces force/inch in width.

4. Elongation: 2 percent.

5. Tensile Strength: 40 lbf/inch in width.

6. ASJ Tape Disks and Squares: Precut disks or squares of ASJ tape.

B. PVC Tape: White vapor-retarder tape matching field-applied PVC jacket with acrylic adhesive; suitable for indoor and outdoor applications.

1. Width: 2 inches.

2. Thickness: 6 mils.

3. Adhesion: 64 ounces force/inch in width.

4. Elongation: 500 percent.

5. Tensile Strength: 18 lbf/inch in width.

C. Aluminum-Foil Tape: Vapor-retarder tape with acrylic adhesive.

1. Width: 2 inches.

2. Thickness: 3.7 mils.

3. Adhesion: 100 ounces force/inch in width.

4. Elongation: 5 percent.

5. Tensile Strength: 34 lbf/inch in width.

2.9 SECUREMENTS

A. Aluminum Bands: ASTM B 209, Alloy 3003, 3005, 3105, or 5005; Temper H-14, 0.020 inch thick, 1/2 inch wide with wing seal or closed seal.

B. Insulation Pins and Hangers:

1. Metal, Adhesively Attached, Perforated-Base Insulation Hangers: Baseplate welded to projecting spindle that is capable of holding insulation, of thickness indicated, securely in position indicated when self-locking washer is in place.

a. Baseplate: Perforated, galvanized carbon-steel sheet, 0.030 inch thick by 2 inches square.

b. Spindle: Copper- or zinc-coated, low-carbon steel, fully annealed, 0.106-inch-diameter shank, length to suit depth of insulation indicated.

c. Adhesive: Recommended by hanger manufacturer. Product with demonstrated capability to bond insulation hanger securely to substrates indicated without damaging insulation, hangers, and substrates.

2. Insulation-Retaining Washers: Self-locking washers formed from 0.016-inch-thick, galvanized-steel sheet, with beveled edge sized as required to hold insulation securely in place but not less than 1-1/2 inches in diameter.

a. Protect ends with capped self-locking washers incorporating a spring steel insert to ensure permanent retention of cap in exposed locations.

C. Staples: Outward-clinching insulation staples, nominal 3/4-inch-wide, stainless steel or Monel.

D. Wire: 0.062-inch soft-annealed, stainless steel.

2.10 CORNER ANGLES

A. PVC Corner Angles: 30 mils thick, minimum 1 by 1 inch, PVC according to ASTM D 1784, Class 16354-C. White or color-coded to match adjacent surface.

B. Aluminum Corner Angles: 0.040 inch thick, minimum 1 by 1 inch, aluminum according to ASTM B 209, Alloy 3003, 3005, 3105, or 5005; Temper H-14.

2.11 PROTECTIVE SHIELDING GUARDS

A. Protective Shielding Pipe Covers:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Plumberex Specialty Products, Inc.

b. Truebro.

c. Zurn Industries, LLC.

2. Description: Manufactured plastic wraps for covering plumbing fixture hot- and cold-water supplies and trap and drain piping. Comply with Americans with Disabilities Act (ADA) requirements.

B. Protective Shielding Piping Enclosures:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Truebro.

b. Zurn Industries, LLC.

2. Description: Manufactured plastic enclosure for covering plumbing fixture hot- and cold-water supplies and trap and drain piping. Comply with ADA requirements.

PART 3 - EXECUTION

3.1 PREPARATION

A. Surface Preparation: Clean and dry surfaces to receive insulation. Remove materials that will adversely affect insulation application.

B. Coordinate insulation installation with the trade installing heat tracing. Comply with requirements for heat tracing that apply to insulation.

C. Mix insulating cements with clean potable water; if insulating cements are to be in contact with stainless-steel surfaces, use demineralized water.

3.2 GENERAL INSTALLATION REQUIREMENTS

A. Install insulation materials, accessories, and finishes with smooth, straight, and even surfaces; free of voids throughout the length of equipment.

B. Install insulation materials, forms, vapor barriers or retarders, jackets, and thicknesses required for each item as specified in insulation schedules.

C. Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, soften, or otherwise attack insulation or jacket in either wet or dry state.

D. Install insulation with longitudinal seams at top and bottom of horizontal runs.

E. Install multiple layers of insulation with longitudinal and end seams staggered.

F. Do not weld brackets, clips, or other attachment devices to piping, fittings, and specialties.

G. Keep insulation materials dry during application and finishing.

H. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.

I. Install insulation with least number of joints practical.

J. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic.

1. Install insulation continuously through hangers and around anchor attachments.

2. For insulation application where vapor barriers are indicated, extend insulation on anchor legs from point of attachment to supported item to point of attachment to structure. Taper and seal ends at attachment to structure with vapor-barrier mastic.

3. Install insert materials and install insulation to tightly join the insert. Seal insulation to insulation inserts with adhesive or sealing compound recommended by insulation material manufacturer.

4. Cover inserts with jacket material matching adjacent pipe insulation. Install shields over jacket, arranged to protect jacket from tear or puncture by hanger, support, and shield.

K. Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.

L. Install insulation with factory-applied jackets as follows:

1. Draw jacket tight and smooth.

2. Cover circumferential joints with 3-inch-wide strips, of same material as insulation jacket. Secure strips with adhesive and outward clinching staples along both edges of strip, spaced 4 inches o.c.

3. Overlap jacket longitudinal seams at least 1-1/2 inches. Install insulation with longitudinal seams at bottom of pipe. Clean and dry surface to receive self-sealing lap. Staple laps with outward clinching staples along edge at 2 inches o.c.

a. For below ambient services, apply vapor-barrier mastic over staples.

4. Cover joints and seams with tape, according to insulation material manufacturer's written instructions, to maintain vapor seal.

5. Where vapor barriers are indicated, apply vapor-barrier mastic on seams and joints.

M. Cut insulation in a manner to avoid compressing insulation more than 75 percent of its nominal thickness.

N. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.

O. Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches beyond damaged areas. Adhere, staple, and seal patches similar to butt joints.

P. For above ambient services, do not install insulation to the following:

1. Vibration-control devices.

2. Testing agency labels and stamps.

3. Nameplates and data plates.

4. Manholes.

5. Handholes.

6. Cleanouts.

Q. Insulate instrument connections for thermometers, pressure gages, pressure temperature taps, test connections, flow meters, sensors, switches, and transmitters on insulated pipes. Shape insulation at these connections by tapering it to and around the connection with insulating cement and finish with finishing cement, mastic, and flashing sealant.

3.3 PENETRATIONS

A. Insulation Installation at Roof Penetrations: Install insulation continuously through roof penetrations.

1. Seal penetrations with flashing sealant.

2. For applications requiring only indoor insulation, terminate insulation above roof surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.

3. Extend jacket of outdoor insulation outside roof flashing at least 2 inches (50 mm) below top of roof flashing.

4. Seal jacket to roof flashing with flashing sealant.

B. Insulation Installation at Underground Exterior Wall Penetrations: Terminate insulation flush with sleeve seal. Seal terminations with flashing sealant.

C. Insulation Installation at Aboveground Exterior Wall Penetrations: Install insulation continuously through wall penetrations.

1. Seal penetrations with flashing sealant.

2. For applications requiring only indoor insulation, terminate insulation inside wall surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.

3. Extend jacket of outdoor insulation outside wall flashing and overlap wall flashing at least 2 inches.

4. Seal jacket to wall flashing with flashing sealant.

D. Insulation Installation at Interior Wall and Partition Penetrations (That Are Not Fire Rated): Install insulation continuously through walls and partitions.

E. Insulation Installation at Fire-Rated Wall and Partition Penetrations: Install insulation continuously through penetrations of fire-rated walls and partitions.

1. Comply with requirements in Section 07 84 13 "Penetration Firestopping" for firestopping and fire-resistive joint sealers.

F. Insulation Installation at Floor Penetrations:

1. Pipe: Install insulation continuously through floor penetrations.

2. Seal penetrations through fire-rated assemblies. Comply with requirements in Section 07 84 13 "Penetration Firestopping."

3.4 INSTALLATION OF EQUIPMENT, TANK, AND VESSEL INSULATION

A. Mineral-Fiber, Pipe, and Tank Insulation Installation for Tanks and Vessels: Secure insulation with adhesive and anchor pins and speed washers.

1. Apply adhesives according to manufacturer's recommended coverage rates per unit area, for 100 percent coverage of tank and vessel surfaces.

2. Groove and score insulation materials to fit as closely as possible to equipment, including contours. Bevel insulation edges for cylindrical surfaces for tight joints. Stagger end joints.

3. Protect exposed corners with secured corner angles.

4. Install adhesively attached or self-sticking insulation hangers and speed washers on sides of tanks and vessels as follows:

a. Do not weld anchor pins to ASME-labeled pressure vessels.

b. Select insulation hangers and adhesive that are compatible with service temperature and with substrate.

c. On tanks and vessels, maximum anchor-pin spacing is 3 inches from insulation end joints, and 16 inches o.c. in both directions.

d. Do not over compress insulation during installation.

e. Cut and miter insulation segments to fit curved sides and domed heads of tanks and vessels.

f. Impale insulation over anchor pins and attach speed washers.

g. Cut excess portion of pins extending beyond speed washers or bend parallel with insulation surface. Cover exposed pins and washers with tape matching insulation facing.

5. Secure each layer of insulation with stainless-steel or aluminum bands. Select band material compatible with insulation materials.

6. Where insulation hangers on equipment and vessels are not permitted or practical and where insulation support rings are not provided, install a girdle network for securing insulation. Stretch pre-stressed aircraft cable around the diameter of vessel and make taut with clamps, turnbuckles, or breather springs. Place one circumferential girdle around equipment approximately 6 inches from each end. Install wire or cable between two circumferential girdles 12 inches o.c. Install a wire ring around each end and around outer periphery of center openings, and stretch pre-stressed aircraft cable radially from the wire ring to nearest circumferential girdle. Install additional circumferential girdles along the body of equipment or tank at a minimum spacing of 48 inches o.c. Use this network for securing insulation with tie wire or bands.

7. Stagger joints between insulation layers at least 3 inches.

8. Install insulation in removable segments on equipment access doors, manholes, handholes, and other elements that require frequent removal for service and inspection.

9. Bevel and seal insulation ends around manholes, handholes, ASME stamps, and nameplates.

10. For equipment with surface temperatures below ambient, apply mastic to open ends, joints, seams, breaks, and punctures in insulation.

B. Flexible Elastomeric Thermal Insulation Installation for Tanks and Vessels: Install insulation over entire surface of tanks and vessels.

1. Apply 100 percent coverage of adhesive to surface with manufacturer's recommended adhesive.

2. Seal longitudinal seams and end joints.

C. Insulation Installation on Pumps:

1. Fabricate metal boxes lined with insulation. Fit boxes around pumps and coincide box joints with splits in pump casings. Fabricate joints with outward bolted flanges. Bolt flanges on 6-inchcenters, starting at corners. Install 3/8-inch-diameter fasteners with wing nuts. Alternatively, secure the box sections together using a latching mechanism.

2. Fabricate boxes from galvanized steel, at least 0.050 inch thick.

3. For below ambient services, install a vapor barrier at seams, joints, and penetrations. Seal between flanges with replaceable gasket material to form a vapor barrier.

3.5 INSTALLATION OF FLEXIBLE ELASTOMERIC INSULATION

A. Seal longitudinal seams and end joints with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

B. Insulation Installation on Pipe Flanges:

1. Install pipe insulation to outer diameter of pipe flange.

2. Make width of insulation section same as overall width of flange and bolts, plus twice the thickness of pipe insulation.

3. Fill voids between inner circumference of flange insulation and outer circumference of adjacent straight pipe segments with cut sections of sheet insulation of same thickness as pipe insulation.

4. Secure insulation to flanges and seal seams with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

C. Insulation Installation on Pipe Fittings and Elbows:

1. Install mitered sections of pipe insulation.

2. Secure insulation materials and seal seams with manufacturer's recommended adhesive to eliminate openings in insulation that allow passage of air to surface being insulated.

3.6 INSTALLATION OF MINERAL-FIBER PREFORMED PIPE INSULATION

A. Insulation Installation on Straight Pipes and Tubes:

1. Secure each layer of preformed pipe insulation to pipe with wire or bands and tighten bands without deforming insulation materials.

2. Where vapor barriers are indicated, seal longitudinal seams, end joints, and protrusions with vapor-barrier mastic and joint sealant.

3. For insulation with factory-applied jackets on above-ambient surfaces, secure laps with outward clinched staples at 6 inches o.c.

4. For insulation with factory-applied jackets on below-ambient surfaces, do not staple longitudinal tabs. Instead, secure tabs with additional adhesive as recommended by insulation material manufacturer and seal with vapor-barrier mastic and flashing sealant.

B. Insulation Installation on Pipe Flanges:

1. Install preformed pipe insulation to outer diameter of pipe flange.

2. Make width of insulation section same as overall width of flange and bolts, plus twice the thickness of pipe insulation.

3. Fill voids between inner circumference of flange insulation and outer circumference of adjacent straight pipe segments with mineral-fiber blanket insulation.

4. Install jacket material with manufacturer's recommended adhesive, overlap seams at least 1 inch, and seal joints with flashing sealant.

C. Insulation Installation on Fittings, Joints and Couplings:

1. All piping fittings shall be insulated by filling the total void over all fittings between straight runs of pipe insulation with thermal insulating wool, forming a uniform insulation thickness equal to, or exceeding, the adjacent pipe insulation.

2. Finish all insulated pipe fittings by applying PVC fitting covers overlapping the adjacent pipe insulation outer covering.

3. For hot service piping (105F and above), secure the PVC fitting covers stainless steel tack fasteners.

4. For cold service piping (60F and below), seal the ends of the adjacent pipe insulation with vapor barrier mastic, ensure that the PVC fitting cover overlaps the adjacent pipe insulation jacket by 2” minimum and secure PVC fitting covers to adjacent pipe insulation with 2” wide PVC Tape.

5. Fitting covers for grooved piping systems shall be the type specifically manufactured for grooved piping systems.

3.7 INSULATION INSTALLATION ON VALVES AND PIPE SPECIALTIES

A. Install removable insulation covers on all valves and specialties 1-1/2” and larger.

1. Valves, Strainers, and Unions 1-1/2 – 2 NPS: “No Sweat” re-usable valve covers or approved equal product.

2. Valves, Strainers and Unions 2-1/2” and larger use removable insulation jackets from Thermaxx or prior approved manufacturer.

3.8 FIELD-APPLIED JACKET INSTALLATION

A. Where PVC jackets are indicated, install with 1-inch overlap at longitudinal seams and end joints. Seal with manufacturer's recommended adhesive.

1. Apply two continuous beads of adhesive to seams and joints, one bead under lap and the finish bead along seam and joint edge.

B. Where metal jackets are indicated, install with 2-inch overlap at longitudinal seams and end joints. Overlap longitudinal seams arranged to shed water. Seal end joints with weatherproof sealant recommended by insulation manufacturer. Secure jacket with stainless-steel bands 12 inches o.c. and at end joints.

C. Where underground direct-buried jacket are indicated, install per the manufacturers instructions.

3.9 FINISHES

A. Insulation with ASJ or Other Paintable Jacket Material and where Required: Paint jacket with paint system identified below and as specified in Section 09 91 13 "Exterior Painting" and Section 09 91 23 "Interior Painting."

1. Flat Acrylic Finish: Two finish coats over a primer that is compatible with jacket material and finish coat paint. Add fungicidal agent to render fabric mildew proof.

a. Finish Coat Material: Interior, flat, latex-emulsion size.

B. Flexible Elastomeric Thermal Insulation: After adhesive has fully cured, apply two coats of insulation manufacturer's recommended protective coating.

C. Do not field paint aluminum or PVC jacketing.

3.10 FIELD QUALITY CONTROL

A. Perform tests and inspections.

B. Tests and Inspections:

1. Inspect field-insulated equipment, randomly selected by Architect, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to one location(s) for each type of equipment defined in the "Equipment Insulation Schedule" Article. For large equipment, remove only a portion adequate to determine compliance.

C. All insulation applications will be considered defective Work if sample inspection reveals noncompliance with requirements.

3.11 EQUIPMENT INSULATION SCHEDULE

A. Insulation materials and thicknesses for Plumbing equipment are identified in the table below. If more than one material is listed for a type of equipment, selection from materials listed is Contractor's option.

Equipment Type	System Temp	Insulation Type	Insulation Conductivity (Btu x in) / (hr x ft2 x F)	Insulation Thickness (in)	Vapor Barrier	Factory Installed Jacket Type
Tanks	Below 60ºF and above 105ºF	Glass Fiber or Flexible Elastomeric	0.27	2	Cold Service <60ºF only	ASJ
Pumps	Below 60ºF and above 105ºF	Glass Fiber or Flexible Elastomeric	0.27	2	Cold Service <60ºF only	ASJ
Roof Drain Sumps	NA	Glass Fiber or Flexible Elastomeric	0.27	1	No	ASJ

3.12 PIPING INSULATION SCHEDULE

A. Insulation materials and thicknesses for Plumbing piping are identified in the table below. If more than one material is listed for an application, selection from materials listed is at the Contractor's option.

Application	Nominal Pipe Size	Insulation Type	Insulation Conductivity (Btu x in) / (hr x ft2 x F)	Insulation Thickness (in)	Vapor Barrier	Factory Installed Jacket Type
Domestic Cold Water Piping	All	Glass Fiber or Flexible Elastomeric	0.27	1	Yes	ASJ
Plumbing Vents within 6ft of roof termination	All	Glass Fiber or Flexible Elastomeric	0.27	1	Yes	ASJ
Roof Drain Piping	All	Glass Fiber or Flexible Elastomeric	0.27	1	Yes	ASJ

3.13 FIELD APPLIED JACKETING SCHEDULE

A. Field applied jackets for Plumbing piping are identified in the table below. If more than one material is listed for an application, selection from materials listed is at the Contractor's option.

Application	Installation Location	Field Applied Jacketing
Domestic Water Piping	Indoors	PVC when piping is exposed and within 7ft of the floor.
Domestic Water Piping	Outdoors	Aluminum Jacket

END OF SECTION

SECTION 22 11 16 - DOMESTIC WATER PIPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Copper tube and fittings.

2. Piping joining materials.

3. Transition fittings.

B. Related Requirements:

1. Section 22 05 00 “General Provisions of Plumbing”

1.2 ACTION SUBMITTALS

A. See Section 22 00 00 “General Requirements of Plumbing” for submittal requirements.

PART 2 - PRODUCTS

2.1 PIPING MATERIALS

A. Comply with requirements in "Piping Schedule" Article for applications of pipe, tube, fitting materials, and joining methods for specific services, service locations, and pipe sizes.

B. Potable-water piping and components shall comply with NSF 14 and NSF 61 Annex G. and Plastic piping components shall be marked with "NSF-pw."

C. Comply with NSF Standard 372 for low lead.

2.2 COPPER TUBE AND FITTINGS

A. Hard Copper Tube: ASTM B 88, Type L water tube, drawn temper.

B. Soft Copper Tube: ASTM B 88, Type K water tube, annealed temper.

C. Cast-Copper, Solder-Joint Fittings: ASME B16.18, pressure fittings.

D. Wrought-Copper, Solder-Joint Fittings: ASME B16.22, wrought-copper pressure fittings.

E. Copper Unions:

1. MSS SP-123.

2. Cast-copper-alloy, hexagonal-stock body.

3. Ball-and-socket, metal-to-metal seating surfaces.

4. Solder-joint or threaded ends.

F. Copper Pressure-Seal-Joint Fittings:

1. Fittings for NPS 2 and Smaller: Wrought-copper fitting with EPDM-rubber, O-ring seal in each end.

2. Fittings for NPS 2-1/2 to NPS 4: Cast-bronze or wrought-copper fitting with EPDM-rubber, O-ring seal in each end.

2.3 PIPING JOINING MATERIALS

A. Pipe-Flange Gasket Materials:

1. AWWA C110/A21.10, rubber, flat face, 1/8 inch thick or ASME B16.21, nonmetallic and asbestos free unless otherwise indicated.

2. Full-face or ring type unless otherwise indicated.

B. Metal, Pipe-Flange Bolts and Nuts: ASME B18.2.1, carbon steel unless otherwise indicated.

C. Solder Filler Metals: ASTM B 32, lead-free alloys.

D. Flux: ASTM B 813, water flushable.

E. Brazing Filler Metals: AWS A5.8M/A5.8, BCuP Series, copper-phosphorus alloys for general-duty brazing unless otherwise indicated.

2.4 TRANSITION FITTINGS

A. General Requirements:

1. Same size as pipes to be joined.

2. Pressure rating at least equal to pipes to be joined.

3. End connections compatible with pipes to be joined.

B. Fitting-Type Transition Couplings: Manufactured piping coupling or specified piping system fitting.

PART 3 - EXECUTION

3.1 EARTHWORK

A. Comply with requirements in Section 31 20 00 "Earth Moving" for excavating, trenching, and backfilling.

3.2 PIPING INSTALLATION

A. Drawing plans, schematics, and diagrams indicate general location and arrangement of domestic water piping. Indicated locations and arrangements are used to size pipe and calculate friction loss, expansion, and other design considerations. Install piping as indicated unless deviations to layout are approved on coordination drawings.

B. Install copper tubing under building slab according to CDA's "Copper Tube Handbook."

C. Install shutoff valve, hose-end drain valve, strainer, pressure gage, and test tee with valve inside the building at each domestic water-service entrance. Comply with requirements for pressure

gages in Section 22 05 19 "Meters and Gages for Plumbing Piping" and with requirements for drain valves and strainers in Section 22 11 19 "Domestic Water Piping Specialties."

D. Install water-pressure-reducing valves downstream from shutoff valves. Comply with requirements for pressure-reducing valves in Section 22 11 19 "Domestic Water Piping Specialties."

E. Install domestic water piping level and plumb.

F. Rough-in domestic water piping for water-meter installation according to utility company's requirements.

G. Install seismic restraints on piping. Comply with requirements for seismic-restraint devices in Section 22 05 48 "Vibration and Seismic Controls for Plumbing Piping and Equipment."

H. Install piping concealed from view and protected from physical contact by building occupants unless otherwise indicated and except in equipment rooms and service areas.

I. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.

J. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal, and coordinate with other services occupying that space.

K. Install piping to permit valve servicing.

L. Install nipples, unions, special fittings, and valves with pressure ratings the same as or higher than the system pressure rating used in applications below unless otherwise indicated.

M. Install piping free of sags and bends.

N. Install fittings for changes in direction and branch connections.

O. Install unions in copper tubing at final connection to each piece of equipment, machine, and specialty.

P. Install pressure gages on suction and discharge piping for each plumbing pump and packaged booster pump. Comply with requirements for pressure gages in Section 22 05 19 "Meters and Gages for Plumbing Piping."

Q. Install thermometers on inlet and outlet piping from each water heater. Comply with requirements for thermometers in Section 22 05 19 "Meters and Gages for Plumbing Piping."

R. Install sleeves for piping penetrations of walls, ceilings, and floors. Comply with requirements for sleeves specified in Section 22 05 00 "General Provisions of Plumbing."

S. Install sleeve seals for piping penetrations of concrete walls and slabs. Comply with requirements for sleeve seals specified in Section 22 05 00 "General Provisions of Plumbing."

T. Install escutcheons for piping penetrations of walls, ceilings, and floors. Comply with requirements for escutcheons specified in Section 22 05 00 "General Provisions of Plumbing."

3.3 JOINT CONSTRUCTION

A. Ream ends of pipes and tubes and remove burrs. Bevel plain ends of steel pipe.

B. Remove scale, slag, dirt, and debris from inside and outside of pipes, tubes, and fittings before assembly.

C. Threaded Joints: Thread pipe with tapered pipe threads according to ASME B1.20.1. Cut threads full and clean using sharp dies. Ream threaded pipe ends to remove burrs and restore full ID. Join pipe fittings and valves as follows:

1. Apply appropriate tape or thread compound to external pipe threads.

2. Damaged Threads: Do not use pipe or pipe fittings with threads that are corroded or damaged.

D. Brazed Joints for Copper Tubing: Comply with CDA's "Copper Tube Handbook," "Brazed Joints" chapter.

E. Soldered Joints for Copper Tubing: Apply ASTM B 813, water-flushable flux to end of tube. Join copper tube and fittings according to ASTM B 828 or CDA's "Copper Tube Handbook."

F. Flanged Joints: Select appropriate asbestos-free, nonmetallic gasket material in size, type, and thickness suitable for domestic water service. Join flanges with gasket and bolts according to ASME B31.9.

G. Joints for Dissimilar-Material Piping: Make joints using adapters compatible with materials of both piping systems.

3.4 TRANSITION FITTING INSTALLATION

A. Install transition couplings at joints of dissimilar piping.

B. Transition Fittings in Underground Domestic Water Piping:

1. Fittings for NPS 1-1/2 and Smaller: Fitting-type coupling.

2. Fittings for NPS 2 and Larger: Sleeve-type coupling.

C. Transition Fittings in Aboveground Domestic Water Piping NPS 2 and Smaller: Plastic-to-metal transition fittings or unions.

3.5 HANGER AND SUPPORT INSTALLATION

A. Comply with requirements for seismic-restraint devices in Section 22 05 48 "Vibration and Seismic Controls for Plumbing Piping and Equipment."

B. Comply with requirements for pipe hanger, support products, and installation in Section 22 05 29 "Hangers and Supports for Plumbing Piping and Equipment."

1. Vertical Piping: MSS Type 8 or 42, clamps.

2. Individual, Straight, Horizontal Piping Runs:

a. 100 Feet and Less: MSS Type 1, adjustable, steel clevis hangers.

b. Longer Than 100 Feet: MSS Type 43, adjustable roller hangers.

c. Longer Than 100 Feet if Indicated: MSS Type 49, spring cushion rolls.

3. Multiple, Straight, Horizontal Piping Runs 100 Feet or Longer: MSS Type 44, pipe rolls. Support pipe rolls on trapeze.

4. Base of Vertical Piping: MSS Type 52, spring hangers.

C. Support vertical piping and tubing at base and at each floor.

D. Rod diameter may be reduced one size for double-rod hangers, to a minimum of 3/8 inch.

E. Install hangers for copper tubing with the following maximum horizontal spacing and minimum rod diameters:

1. NPS 3/4 and Smaller: 60 inches with 3/8-inch rod.

2. NPS 1 and NPS 1-1/4: 72 inches with 3/8-inch rod.

3. NPS 1-1/2 and NPS 2: 96 inches with 3/8-inch rod.

4. NPS 2-1/2: 108 inches with 1/2-inch rod.

5. NPS 3 to NPS 5: 10 feet with 1/2-inch rod.

6. NPS 6: 10 feet with 5/8-inch rod.

7. NPS 8: 10 feet with 3/4-inch rod.

F. Install supports for vertical copper tubing every 10 feet.

G. Support piping and tubing not listed in this article according to MSS SP-58 and manufacturer's written instructions.

3.6 CONNECTIONS

A. Drawings indicate general arrangement of piping, fittings, and specialties.

B. When installing piping adjacent to equipment and machines, allow space for service and maintenance.

C. Connect domestic water piping to exterior water-service piping. Use transition fitting to join dissimilar piping materials.

D. Connect domestic water piping to water-service piping with shutoff valve; extend and connect to the following:

1. Domestic Water Booster Pumps: Cold-water suction and discharge piping.

2. Water Heaters: Cold-water inlet and hot-water outlet piping in sizes indicated, but not smaller than sizes of water heater connections.

3. Plumbing Fixtures: Cold- and hot-water-supply piping in sizes indicated, but not smaller than that required by plumbing code.

4. Equipment: Cold- and hot-water-supply piping as indicated, but not smaller than equipment connections. Provide shutoff valve and union for each connection. Use flanges instead of unions for NPS 2-1/2 and larger.

3.7 IDENTIFICATION

A. Identify system components. Comply with requirements for identification materials and installation in Section 22 05 53 "Identification for Plumbing Piping and Equipment."

B. Label pressure piping with system operating pressure.

3.8 FIELD QUALITY CONTROL

A. Perform the following tests and inspections:

1. Piping Inspections:

a. Do not enclose, cover, or put piping into operation until it has been inspected and approved by authorities having jurisdiction.

b. During installation, notify authorities having jurisdiction at least one day before inspection must be made. Perform tests specified below in presence of authorities having jurisdiction:

1) Roughing-in Inspection: Arrange for inspection of piping before concealing or closing in after roughing in and before setting fixtures.

2) Final Inspection: Arrange for authorities having jurisdiction to observe tests specified in "Piping Tests" Subparagraph below and to ensure compliance with requirements.

c. Reinspection: If authorities having jurisdiction find that piping will not pass tests or inspections, make required corrections and arrange for reinspection.

d. Reports: Prepare inspection reports and have them signed by authorities having jurisdiction.

2. Piping Tests:

a. Fill domestic water piping. Check components to determine that they are not air bound and that piping is full of water.

b. Test for leaks and defects in new piping and parts of existing piping that have been altered, extended, or repaired. If testing is performed in segments, submit a separate report for each test, complete with diagram of portion of piping tested.

c. Leave new, altered, extended, or replaced domestic water piping uncovered and unconcealed until it has been tested and approved. Expose work that was covered or concealed before it was tested.

d. Cap and subject piping to static water pressure of 50 psig above operating pressure, without exceeding pressure rating of piping system materials. Isolate test source and allow it to stand for four hours. Leaks and loss in test pressure constitute defects that must be repaired.

e. Repair leaks and defects with new materials, and retest piping or portion thereof until satisfactory results are obtained.

f. Prepare reports for tests and for corrective action required.

B. Domestic water piping will be considered defective if it does not pass tests and inspections.

C. Prepare test and inspection reports.

3.9 ADJUSTING

A. Perform the following adjustments before operation:

1. Close drain valves, hydrants, and hose bibbs.

2. Open shutoff valves to fully open position.

3. Open throttling valves to proper setting.

4. Adjust balancing valves in hot-water-circulation return piping to provide adequate flow.

a. Manually adjust ball-type balancing valves in hot-water-circulation return piping to provide hot-water flow in each branch.

b. Adjust calibrated balancing valves to flows indicated.

5. Remove plugs used during testing of piping and for temporary sealing of piping during installation.

6. Remove and clean strainer screens. Close drain valves and replace drain plugs.

7. Remove filter cartridges from housings and verify that cartridges are as specified for application where used and are clean and ready for use.

8. Check plumbing specialties and verify proper settings, adjustments, and operation.

3.10 CLEANING

A. Clean and disinfect potable domestic water piping as follows:

1. Purge new piping and parts of existing piping that have been altered, extended, or repaired before using.

2. Use purging and disinfecting procedures prescribed by authorities having jurisdiction; if methods are not prescribed, use procedures described in either AWWA C651 or AWWA C652 or follow procedures described below:

a. Flush piping system with clean, potable water until dirty water does not appear at outlets.

b. Fill and isolate system according to either of the following:

1) Fill system or part thereof with water/chlorine solution with at least 50 ppm of chlorine. Isolate with valves and allow to stand for 24 hours.

2) Fill system or part thereof with water/chlorine solution with at least 200 ppm of chlorine. Isolate and allow to stand for three hours.

c. Flush system with clean, potable water until no chlorine is in water coming from system after the standing time.

d. Repeat procedures if biological examination shows contamination.

e. Submit water samples in sterile bottles to authorities having jurisdiction.

B. Prepare and submit reports of purging and disinfecting activities. Include copies of water-sample approvals from authorities having jurisdiction.

C. Clean interior of domestic water piping system. Remove dirt and debris as work progresses.

3.11 PIPING SCHEDULE

A. Transition and special fittings with pressure ratings at least equal to piping rating may be used in applications below unless otherwise indicated.

B. Flanges and unions may be used for aboveground piping joints unless otherwise indicated.

C. Fitting Option: Extruded-tee connections and brazed joints may be used on aboveground copper tubing.

Application	**Location**	**Size**	**Material**	**Fittings**
Domestic Water Piping	Below Grade	All	Type K Soft Copper	None – below grade piping shall be continuous without fitting or joints
Domestic Water Piping	Indoor Above Grade	All	Type L Copper	Copper: Sweat or Pressure Seal

END OF SECTION

SECTION 22 11 19 - DOMESTIC WATER PIPING SPECIALTIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Vacuum breakers.

2. Backflow preventers.

3. Water pressure-reducing valves.

4. Temperature-actuated, water mixing valves.

5. Strainers.

6. Drain valves.

7. Water-hammer arresters.

8. Trap-seal primer valves.

B. Related Requirements:

1. Section 22 05 00 “General Provisions of Plumbing” for Expansion Loops, Alignment Guides, Dielectric Fittings, Sleeves and Sleeve Seals, Sealants, Escutcheons and floor plates.

2. Section 22 05 19 "Meters and Gages for Plumbing Piping" for thermometers, pressure gages.

3. Section 22 11 16 "Domestic Water Piping" for piping and fittings.

1.2 ACTION SUBMITTALS

A. See Section 22 00 00 “General Requirements of Plumbing” for submittal requirements.

PART 2 - PRODUCTS

2.1 GENERAL REQUIREMENTS FOR PIPING SPECIALTIES

A. Potable-water piping and components shall comply with NSF 61 Annex G and NSF 14. Mark "NSF-pw" on plastic piping components.

2.2 PERFORMANCE REQUIREMENTS

A. Minimum Working Pressure for Domestic Water Piping Specialties: 125 psig unless otherwise indicated.

2.3 VACUUM BREAKERS

A. Pipe-Applied, Atmospheric-Type Vacuum Breakers:

1. Standard: ASSE 1001.

2. Size: NPS 1/4 to NPS 3, as required to match connected piping.

3. Body: Bronze.

4. Inlet and Outlet Connections: Threaded.

5. Finish: Chrome plated.

B. Hose-Connection Vacuum Breakers:

1. Standard: ASSE 1011.

2. Body: Bronze, nonremovable, with manual drain.

3. Outlet Connection: Garden-hose threaded complying with ASME B1.20.7.

4. Finish: Chrome or nickel plated.

2.4 BACKFLOW PREVENTERS

A. Reduced-Pressure-Principle Backflow Preventers:

1. Manufacturers:

a. Watts

b. Zurn Wilkins

c. Prior approved Equal

2. Description:

a. Standard: ASSE 1013.

b. Operation: Continuous-pressure applications.

c. Pressure Loss: 12 psig maximum, through middle third of flow range.

d. Size: See Drawings – Line size of size not specifically listed.

e. Body: Bronze for NPS 2 and smaller; cast iron with interior lining that complies with AWWA C550 or that is FDA approved for NPS 2-1/2 and larger.

f. End Connections: Threaded for NPS 2 and smaller; flanged for NPS 2-1/2 and larger.

g. Accessories:

1) Valves NPS 2 and Smaller: Ball type with threaded ends on inlet and outlet.

2) Valves NPS 2-1/2 and Larger: Outside-screw and yoke-gate type with flanged ends on inlet and outlet.

3) Air-Gap Fitting: ASME A112.1.2, matching backflow-preventer connection.

2.5 WATER PRESSURE-REDUCING VALVES

A. Water Regulators:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

    a. Apollo Valves; Conbraco Industries, Inc.

    b. Watts; a Watts Water Technologies company.

    c. Zurn Industries, LLC.

2. Standard: ASSE 1003.

3. Pressure Rating: Initial working pressure of 150 psig.

4. Size: See Drawings – Line size of size not specifically listed.

5. Body: Bronze for NPS 2 and smaller; cast iron with interior lining that complies with AWWA C550 or that is FDA approved for NPS 2-1/2 and NPS 3.

6. Valves for Booster Heater Water Supply: Include integral bypass.

7. End Connections: Threaded for NPS 2 and smaller; flanged for NPS 2-1/2 and NPS 3.

2.6 TEMPERATURE-ACTUATED, WATER MIXING VALVES

A. Water-Temperature Limiting Devices:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

    a. Leonard Valve Company.

    b. Symmons Industries, Inc.

    c. Watts; a Watts Water Technologies company.

    d. Zurn Industries, LLC.

2. Standard: ASSE 1017.

3. Pressure Rating: 125 psig.

4. Type: Thermostatically controlled, water mixing valve.

5. Material: Bronze body with corrosion-resistant interior components.

6. Connections: Threaded union inlets and outlet.

7. Accessories: Check stops on hot- and cold-water supplies, and adjustable, temperature-control handle.

8. Valve Finish: Chrome plated.

B. Primary, Thermostatic, Water Mixing Valves:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Armstrong International, Inc.

b. Leonard Valve Company.

c. Symmons Industries, Inc.

d. Zurn Industries, LLC.

2. Standard: ASSE 1017.

3. Pressure Rating: 125 psig minimum unless otherwise indicated.

4. Type: Cabinet-type, thermostatically controlled, water mixing valve.

5. Material: Bronze body with corrosion-resistant interior components.

6. Connections: Threaded union inlets and outlet.

7. Accessories: Manual temperature control, check stops on hot- and cold-water supplies, and adjustable, temperature-control handle.

8. Valve Finish: Chrome plated.

9. Piping Finish: Chrome plated.

10. Cabinet: Factory fabricated, stainless steel, for surface mounting and with hinged, stainless-steel door.

2.7 STRAINERS FOR DOMESTIC WATER PIPING

A. Y-Pattern Strainers:

1. Pressure Rating: 125 psig minimum unless otherwise indicated.

2. Body: Bronze for NPS 2 and smaller; cast iron with interior lining that complies with AWWA C550 or that is FDA approved, epoxy coated and for NPS 2-1/2 and larger.

3. End Connections: Threaded for NPS 2 and smaller; flanged for NPS 2-1/2 and larger.

4. Screen: Stainless steel with round perforations unless otherwise indicated.

5. Perforation Size:

a. Strainers NPS 2 and Smaller: 0.020 inch.

b. Strainers NPS 2-1/2 to NPS 4: 0.045 inch.

6. Drain: Factory-installed, hose-end drain valve with cap.

2.8 DRAIN VALVES

A. Ball-Valve-Type, Hose-End Drain Valves:

1. Standard: MSS SP-110 for standard-port, two-piece ball valves.

2. Pressure Rating: 400-psig minimum CWP.

3. Size: NPS 3/4.

4. Body: Copper alloy.

5. Ball: Chrome-plated brass.

6. Seats and Seals: Replaceable.

7. Handle: Vinyl-covered steel.

8. Inlet: Threaded or solder joint.

9. Outlet: Threaded, short nipple with garden-hose thread complying with ASME B1.20.7 and cap with brass chain.

2.9 WATER-HAMMER ARRESTERS

A. Water-Hammer Arresters:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Jay R. Smith Mfg. Co.

b. Sioux Chief Manufacturing Company, Inc.

c. Watts; a Watts Water Technologies company.

d. Zurn Industries, LLC.

2. Standard: ASSE 1010 or PDI-WH 201.

3. Type: Metal bellows or Copper tube with piston.

4. Size: ASSE 1010, Sizes AA and A through F, or PDI-WH 201, Sizes A through F.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install double check backflow preventers in each water supply to mechanical equipment and systems and to other equipment and water systems that may be sources of contamination. Comply with authorities having jurisdiction.

1. Locate backflow preventers in same room as connected equipment or system.

2. Install drain for backflow preventers with atmospheric-vent drain connection with air-gap fitting, fixed air-gap fitting, or equivalent positive pipe separation of at least two pipe

diameters in drain piping and pipe-to-floor drain. Locate air-gap device attached to or under backflow preventer. Simple air breaks are unacceptable for this application.

3. Do not install bypass piping around backflow preventers.

B. Install water regulators with inlet and outlet shutoff valves. Install pressure gages on inlet and outlet.

C. Install temperature-actuated, water mixing valves with check stops or shutoff valves on inlets and with shutoff valve on outlet.

1. Install cabinet-type units recessed in or surface mounted on wall as specified.

D. Install Y-pattern strainers for water on supply side of each water pressure-reducing valve, solenoid valve and pump.

E. Install water-hammer arresters in water piping according to PDI-WH 201.

3.2 CONNECTIONS

A. Comply with requirements for ground equipment in Section 26 05 26 "Grounding and Bonding for Electrical Systems."

B. Fire-retardant-treated-wood blocking is specified in Section 26 05 19 "Low-Voltage Electrical Power Conductors and Cables" for electrical connections.

3.3 FIELD QUALITY CONTROL

A. Perform the following tests and inspections:

1. Test each reduced-pressure-principle backflow preventer and double-check, detector-assembly backflow preventer according to authorities having jurisdiction and the device's reference standard.

B. Domestic water piping specialties will be considered defective if they do not pass tests and inspections.

C. Prepare test and inspection reports.

3.4 ADJUSTING

A. Set field-adjustable pressure set points of water pressure-reducing valves. Verify pressure set points with Engineer prior to setting

B. Set field-adjustable flow set points of balancing valves. Verify flow rates with Engineer prior to setting.

C. Set field-adjustable temperature set points of temperature-actuated, water mixing valves. Verify temperature setting with engineer prior to setting.

END OF SECTION

SECTION 22 13 16 - SANITARY WASTE AND VENT PIPING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Pipe, tube, and fittings.

2. Specialty pipe fittings.

1.2 ACTION SUBMITTALS

A. See section 22 00 00 “General Requirements of Plumbing” for submittal requirements.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Components and installation shall be capable of withstanding the following minimum working pressure unless otherwise indicated:

1. Soil, Waste, and Vent Piping: 10-foot head of water.

B. Seismic Performance: Soil, waste, and vent piping and support and installation shall withstand the effects of earthquake motions determined according to ASCE/SEI 7. See section 22 05 48 “Vibration and Seismic Controls for Plumbing Piping and Equipment”

2.2 PIPING MATERIALS

A. Piping materials shall bear label, stamp, or other markings of specified testing agency.

B. Comply with requirements in "Piping Schedule" Article for applications of pipe, tube, fitting materials, and joining methods for specific services, service locations, and pipe sizes.

2.3 HUBLESS, CAST-IRON SOIL PIPE AND FITTINGS

A. Pipe and Fittings: ASTM A 888 or CISPI 301.

B. CISPI, Hubless-Piping Couplings:

1. Standards: ASTM C 1277 and CISPI 310.

2. Description: Stainless-steel corrugated shield with stainless-steel bands and tightening devices; and ASTM C 564, rubber sleeve with integral, center pipe stop.

2.4 COPPER TUBE AND FITTINGS

A. Copper Type DWV Tube: ASTM B 306, drainage tube, drawn temper.

B. Copper Drainage Fittings: ASME B16.23, cast copper or ASME B16.29, wrought copper, solder-joint fittings.

C. Copper Pressure Fittings:

1. Copper Fittings: ASME B16.18, cast-copper-alloy or ASME B16.22, wrought-copper, solder-joint fittings. Furnish wrought-copper fittings if indicated.

2. Copper Unions: MSS SP-123, copper-alloy, hexagonal-stock body with ball-and-socket, metal-to-metal seating surfaces, and solder-joint or threaded ends.

D. Copper Flanges: ASME B16.24, Class 150, cast copper with solder-joint end.

1. Flange Gasket Materials: ASME B16.21, full-face, flat, nonmetallic, asbestos-free, 1/8-inch maximum thickness unless thickness or specific material is indicated.

2. Flange Bolts and Nuts: ASME B18.2.1, carbon steel unless otherwise indicated.

E. Solder: ASTM B 32, lead free with ASTM B 813, water-flushable flux.

2.5 PVC PIPE AND FITTINGS

A. Comply with NSF 14, "Plastics Piping Systems Components and Related Materials," for plastic piping components. Include marking with "NSF-dwv" for plastic drain, waste, and vent piping and "NSF-sewer" for plastic sewer piping.

B. Solid-Wall PVC Pipe: ASTM D 2665, drain, waste, and vent.

C. PVC Socket Fittings: ASTM D 2665, made to ASTM D 3311, drain, waste, and vent patterns and to fit Schedule 40 pipe.

D. Adhesive Primer: ASTM F 656.

E. Solvent Cement: ASTM D 2564.

2.6 SPECIALTY PIPE FITTINGS

A. Transition Couplings:

1. Fitting-Type Transition Couplings: Manufactured piping coupling or specified piping system fitting.

2. Unshielded, Non-pressure Transition Couplings:

a. Standard: ASTM C 1173.

b. Description: Elastomeric, sleeve-type, reducing or transition pattern. Include shear ring and corrosion-resistant-metal tension band and tightening mechanism on each end.

c. End Connections: Same size and compatible with pipes to be joined.

d. Sleeve Materials:

1) For Cast-Iron Soil Pipes: ASTM C 564, rubber.

2) For Plastic Pipes: ASTM F 477, elastomeric seal or ASTM D 5926, PVC.

3) For Dissimilar Pipes: ASTM D 5926, PVC or other material compatible with pipe materials being joined.

3. Shielded, Non-pressure Transition Couplings:

a. Standard: ASTM C 1460.

b. Description: Elastomeric or rubber sleeve with full-length, corrosion-resistant outer shield and corrosion-resistant-metal tension band and tightening mechanism on each end.

c. End Connections: Same size as and compatible with pipes to be joined.

PART 3 - EXECUTION

3.1 EARTH MOVING

A. Comply with requirements for excavating, trenching, and backfilling specified in Section 31 20 00 "Earth Moving."

3.2 PIPING INSTALLATION

A. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems.

1. Indicated locations and arrangements were used to size pipe and calculate friction loss, expansion, pump sizing, and other design considerations.

2. Install piping as indicated unless deviations to layout are approved on coordination drawings.

B. Install piping in concealed locations unless otherwise indicated and except in equipment rooms and service areas.

C. Install piping indicated to be exposed and piping in equipment rooms and service areas at right angles or parallel to building walls. Diagonal runs are prohibited unless specifically indicated otherwise.

D. Install piping above accessible ceilings to allow sufficient space for ceiling panel removal.

E. Install piping to permit valve servicing.

F. Install piping at indicated slopes.

G. Install piping free of sags and bends.

H. Install fittings for changes in direction and branch connections.

I. Install piping to allow application of insulation.

J. Install seismic restraints on piping. Comply with requirements for seismic-restraint devices specified in Section 22 05 48 "Vibration and Seismic Controls for Plumbing Piping and Equipment."

K. Make changes in direction for soil and waste drainage and vent piping using appropriate branches, bends, and long-sweep bends.

1. Sanitary tees and short-sweep 1/4 bends may be used on vertical stacks if change in direction of flow is from horizontal to vertical.

2. Use long-turn, double Y-branch and 1/8-bend fittings if two fixtures are installed back to back or side by side with common drain pipe.

    a. Straight tees, elbows, and crosses may be used on vent lines.

3. Do not change direction of flow more than 90 degrees.

4. Use proper size of standard increasers and reducers if pipes of different sizes are connected.

    a. Reducing size of waste piping in direction of flow is prohibited.

L. Lay buried building waste piping beginning at low point of each system.

1. Install true to grades and alignment indicated, with unbroken continuity of invert. Place hub ends of piping upstream.

2. Install required gaskets according to manufacturer's written instructions for use of lubricants, cements, and other installation requirements.

3. Maintain swab in piping and pull past each joint as completed.

M. Install soil and waste and vent piping at the following minimum slopes unless otherwise indicated:

1. Horizontal Sanitary Waste: 1/4” per foot downward in direction of flow. 1/8” per foot is allowable if necessitated by site conditions.

2. Vent Piping: 1/8” per foot down toward vertical fixture vent or toward vent stack.

N. Install cast-iron soil piping according to CISPI's "Cast Iron Soil Pipe and Fittings Handbook," Chapter IV, "Installation of Cast Iron Soil Pipe and Fittings."

O. Install aboveground copper tubing according to CDA's "Copper Tube Handbook."

P. Install aboveground PVC piping according to ASTM D 2665.

Q. Install underground PVC piping according to ASTM D 2321.

R. Plumbing Specialties:

1. Install cleanouts at grade and extend to where building sanitary drains connect to building sanitary sewers in sanitary waste gravity-flow piping.

    a. Comply with requirements for cleanouts specified in Section 22 13 19 "Sanitary Waste Piping Specialties."

S. Do not enclose, cover, or put piping into operation until it is inspected and approved by authorities having jurisdiction.

T. Install sleeves for piping penetrations of walls, ceilings, and floors.

1. Comply with requirements for sleeves specified in Section 22 05 00 "General Provisions of Plumbing."

U. Install sleeve seals for piping penetrations of concrete walls and slabs.

1. Comply with requirements for sleeves specified in Section 22 05 00 "General Provisions of Plumbing."

V. Install escutcheons for piping penetrations of walls, ceilings, and floors.

1. Comply with requirements for escutcheons specified in Section 22 05 00 "General Provisions of Plumbing."

3.3 JOINT CONSTRUCTION

A. Join hub-and-spigot, cast-iron soil piping with gasket joints according to CISPI's "Cast Iron Soil Pipe and Fittings Handbook" for compression joints.

B. Join copper tube and fittings with soldered joints according to ASTM B 828. Use ASTM B 813, water-flushable, lead-free flux and ASTM B 32, lead-free-alloy solder.

C. Grooved Joints: Cut groove ends of pipe according to AWWA C606. Lubricate and install gasket over ends of pipes or pipe and fitting. Install coupling housing sections, over gasket, with keys seated in piping grooves. Install and tighten housing bolts.

D. Plastic, Non-pressure-Piping, Solvent-Cement Joints: Clean and dry joining surfaces. Join pipe and fittings according to the following:

1. Comply with ASTM F 402 for safe-handling practice of cleaners, primers, and solvent cements.

2. ABS Piping: Join according to ASTM D 2235 and ASTM D 2661 appendixes.

3. PVC Piping: Join according to ASTM D 2855 and ASTM D 2665 appendixes.

3.4 SPECIALTY PIPE FITTING INSTALLATION

A. Transition Couplings:

1. Install transition couplings at joints of piping with small differences in ODs.

2. In Waste Drainage Piping: [**Unshielded**] [**Shielded**], nonpressure transition couplings.

3.5 HANGER AND SUPPORT INSTALLATION

A. Comply with requirements for seismic-restraint devices specified in Section 22 05 48 "Vibration and Seismic Controls for Plumbing Piping and Equipment."

B. Comply with requirements for pipe hanger and support devices and installation specified in Section 22 05 29 "Hangers and Supports for Plumbing Piping and Equipment."

1. Vertical Piping: MSS Type 8 or Type 42, clamps.

2. Install individual, straight, horizontal piping runs:

a. 100 Feet and Less: MSS Type 1, adjustable, steel clevis hangers.

b. Longer Than 100 Feet: MSS Type 43, adjustable roller hangers.

c. Longer Than 100 Feet if Indicated: MSS Type 49, spring cushion rolls.

3. Multiple, Straight, Horizontal Piping Runs 100 Feet or Longer: MSS Type 44, pipe rolls. Support pipe rolls on trapeze.

4. Base of Vertical Piping: MSS Type 52, spring hangers.

C. Support horizontal piping and tubing within 12 inches of each fitting, valve, and coupling.

D. Support vertical piping and tubing at base and at each floor.

E. Rod diameter may be reduced one size for double-rod hangers, with 3/8-inch minimum rods.

F. Install hangers for cast-iron soil piping with the following maximum horizontal spacing and minimum rod diameters:

1. NPS 1-1/2 and NPS 2: 60 inches with 3/8-inch rod.

2. NPS 3: 60 inches with 1/2-inch rod.

3. NPS 4 and NPS 5: 60 inches with 5/8-inch rod.

4. NPS 6 and NPS 8: 60 inches with 3/4-inch rod.

5. NPS 10 and NPS 12: 60 inches with 7/8-inch rod.

6. Spacing for 10-foot lengths may be increased to 10 feet. Spacing for fittings is limited to 60 inches.

G. Install supports for vertical cast-iron soil piping every 15 feet.

H. Install hangers for copper tubing with the following maximum horizontal spacing and minimum rod diameters:

1. NPS 1-1/4: 72 inches with 3/8-inch rod.

2. NPS 1-1/2 and NPS 2: 96 inches with 3/8-inch rod.

3. NPS 2-1/2: 108 inches with 1/2-inch rod.

4. NPS 3 and NPS 5: 10 feet with 1/2-inch rod.

5. NPS 6: 10 feet with 5/8-inch rod.

6. NPS 8: 10 feet with 3/4-inch rod.

I. Install supports for vertical copper tubing every 10 feet.

J. Install hangers for PVC piping with the following maximum horizontal spacing and minimum rod diameters:

1. NPS 1-1/2 and NPS 2: 48 inches with 3/8-inch rod.

2. NPS 3: 48 inches with 1/2-inch rod.

3. NPS 4 and NPS 5: 48 inches with 5/8-inch rod.

4. NPS 6 and NPS 8: 48 inches with 3/4-inch rod.

5. NPS 10 and NPS 12: 48 inches with 7/8-inch rod.

K. Install supports for vertical PVC piping every 48 inches.

L. Support piping and tubing not listed above according to MSS SP-58 and manufacturer's written instructions.

3.6 CONNECTIONS

A. Drawings indicate general arrangement of piping, fittings, and specialties.

B. Connect soil and waste piping to exterior sanitary sewerage piping. Use transition fitting to join dissimilar piping materials.

C. Connect waste and vent piping to the following:

1. Plumbing Fixtures: Connect waste piping in sizes indicated, but not smaller than required by plumbing code.

2. Plumbing Fixtures and Equipment: Connect atmospheric vent piping in sizes indicated, but not smaller than required by authorities having jurisdiction.

3. Plumbing Specialties: Connect waste and vent piping in sizes indicated, but not smaller than required by plumbing code.

4. Install test tees (wall cleanouts) in conductors near floor and floor cleanouts with cover flush with floor.

5. Equipment: Connect waste piping as indicated.

a. Provide shutoff valve if indicated and union for each connection.

b. Use flanges instead of unions for connections NPS 2-1/2 and larger.

D. Where installing piping adjacent to equipment, allow space for service and maintenance of equipment.

E. Make connections according to the following unless otherwise indicated:

1. Install unions, in piping NPS 2 and smaller, adjacent to each valve and at final connection to each piece of equipment.

2. Install flanges, in piping NPS 2-1/2 and larger, adjacent to flanged valves and at final connection to each piece of equipment.

3.7 IDENTIFICATION

A. Identify exposed sanitary waste and vent piping.

B. Comply with requirements for identification specified in Section 22 05 53 "Identification for Plumbing Piping and Equipment."

3.8 FIELD QUALITY CONTROL

A. During installation, notify authorities having jurisdiction at least 24 hours before inspection must be made. Perform tests specified below in presence of authorities having jurisdiction.

1. Roughing-in Inspection: Arrange for inspection of piping before concealing or closing-in after roughing-in and before setting fixtures.

2. Final Inspection: Arrange for final inspection by authorities having jurisdiction to observe tests specified below and to ensure compliance with requirements.

B. Re-inspection: If authorities having jurisdiction find that piping will not pass test or inspection, make required corrections and arrange for re-inspection.

C. Reports: Prepare inspection reports and have them signed by authorities having jurisdiction.

D. Test sanitary waste and vent piping according to procedures of authorities having jurisdiction or, in absence of published procedures, as follows:

1. Test for leaks and defects in new piping and parts of existing piping that have been altered, extended, or repaired.

a. If testing is performed in segments, submit separate report for each test, complete with diagram of portion of piping tested.

2. Leave uncovered and unconcealed new, altered, extended, or replaced waste and vent piping until it has been tested and approved.

a. Expose work that was covered or concealed before it was tested.

3. Roughing-in Plumbing Test Procedure: Test waste and vent piping except outside leaders on completion of roughing-in.

a. Close openings in piping system and fill with water to point of overflow, but not less than 10-foot head of water.

b. From 15 minutes before inspection starts to completion of inspection, water level must not drop.

c. Inspect joints for leaks.

4. Finished Plumbing Test Procedure: After plumbing fixtures have been set and traps filled with water, test connections and prove they are gastight and watertight.

a. Plug vent-stack openings on roof and building drains where they leave building. Introduce air into piping system equal to pressure of 1-inch wg.

b. Use U-tube or manometer inserted in trap of water closet to measure this pressure.

c. Air pressure must remain constant without introducing additional air throughout period of inspection.

d. Inspect plumbing fixture connections for gas and water leaks.

5. Repair leaks and defects with new materials and retest piping, or portion thereof, until satisfactory results are obtained.

6. Prepare reports for tests and required corrective action.

3.9 CLEANING AND PROTECTION

A. Clean interior of piping. Remove dirt and debris as work progresses.

B. Protect sanitary waste and vent piping during remainder of construction period to avoid clogging with dirt and debris and to prevent damage from traffic and construction work.

C. Place plugs in ends of uncompleted piping at end of day and when work stops.

D. Exposed PVC Piping: Protect plumbing vents exposed to sunlight with two coats of water-based latex paint.

E. Repair damage to adjacent materials caused by waste and vent piping installation.

3.10 PIPING SCHEDULE

A. Piping system materials are identified in the table below. If more than one material is listed, selection from the materials listed is at the Contractor's option.

Application	**Location**	**Size**	**Material**	**Fittings**
Sanitary Waste and Vent	Below Grade	All	PVC	Solvent Joint
Sanitary Waste and Vent	Above Grade	All	PVC or Copper	Solvent Joint Soldered
Condensate Drain	Above Grade	All	PVC or Copper	Solvent Joint

END OF SECTION

SECTION 22 13 19 - SANITARY WASTE PIPING SPECIALTIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Cleanouts.

2. Miscellaneous sanitary drainage piping specialties.

B. Related Requirements:

1. Section 22 13 16 “Sanitary Waste and Vent Piping”

1.2 SUBMITTALS

A. See Section 22 00 00 “General Requirements of Plumbing” for submittal requirements.

PART 2 - PRODUCTS

2.1 ASSEMBLY DESCRIPTIONS

A. Sanitary waste piping specialties shall bear label, stamp, or other markings of specified testing agency.

B. Comply with NSF 14 for plastic sanitary waste piping specialty components.

2.2 CLEANOUTS

A. Above Grade Wall Cleanout

1. Provide JR Smith 4422 or approved equal

2. Description: Cast iron caulked spigot ferrule with cast bronze taper thread plug and stainless-steel round cover and screw.

B. Finished Floor Cleanout

1. Provide JR Smith 4100 or approved equal

2. Description: Cast iron cleanout with extra heavy-duty round, adjustable, scoriated, secured nickel bronze top, and no-hub outlet, gasket seal bronze plug and flashing clamp for.

C. Outdoor Cleanout

1. Provide JR Smith 4241S or approved equal

2. Description: Cast iron floor level cleanout assembly with heavy duty, round, adjustable, scoriated cast iron top, non-tilt tractor cover, gasket seal bronze plug.

2.3 MISCELLANEOUS SANITARY DRAINAGE PIPING SPECIALTIES

A. Open Drains or Hub Drains:

1. Description: Shop or field fabricate from ASTM A 74, Service class, hub-less, cast-iron soil-pipe fittings. Include P-trap, riser section; and where required, increaser fitting joined with ASTM C 564 rubber gaskets.

2. Size: See drawings. If not shown drain shall 2” minimum or one size larger than piping discharging to the drain.

B. Floor-Drain, Trap-Seal Primer Fittings:

1. Description: Cast iron, with threaded inlet and threaded or spigot outlet, and trap-seal primer valve connection.

2. Size: Same as floor drain outlet with NPS 1/2 side inlet.

C. Floor-Drain, Trap Seal:

1. Barrier type floor drain or sink trap seal device.

2. IAPMO 7479 and ASSE std. 1072 listed.

D. Air-Gap Fittings:

1. Standard: ASME A112.1.2, for fitting designed to ensure fixed, positive air gap between installed inlet and outlet piping.

2. Body: Bronze or cast iron.

3. Inlet: Opening in top of body.

4. Outlet: Larger than inlet.

5. Size: Same as connected waste piping and with inlet large enough for associated indirect waste piping.

E. Expansion Joints:

1. Standard: ASME A112.6.4.

2. Body: Cast iron with bronze sleeve, packing, and gland.

3. End Connections: Matching connected piping.

4. Size: Same as connected soil, waste, or vent piping.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install cleanouts in aboveground piping and building drain piping according to the following, unless otherwise indicated:

1. Size same as drainage piping up to NPS 4. Use NPS 4 for larger drainage piping unless larger cleanout is indicated.

2. Locate at each change in direction of piping greater than 45 degrees.

3. Locate at minimum intervals of 50 feet for piping NPS 4 and smaller and 100 feet for larger piping.

4. Locate at base of each vertical soil and waste stack.

B. For floor cleanouts for piping below floors, install cleanout with top flush with finished floor. It shall be the responsibility of the plumbing contractor to coordinate the installation of cleanouts with the general contractor and floor contractor to ensure that floor cleanouts are properly adjusted so that the top is flush and level with finished flooring material. Cleanout covers that are not flush and level with the finished floor will be rejected and the plumbing contractor will be required to sawcut or core drill the floor, provide and install and new cleanout, coordination installation of new concrete and new finished flooring material.

C. For cleanouts located in concealed piping, install cleanout wall access covers, of types indicated, with frame and cover flush with finished wall.

D. Coordinate installation of roof flashing assemblies on sanitary stack vents and vent stacks that extend through roof with the general contractor or construction manager.

E. Assemble open drain fittings and install with top of hub 2 inches above floor.

F. Install deep-seal traps on floor drains and other waste outlets, if indicated.

G. Install floor-drain, trap-seal primer fittings on inlet to floor drains that require trap-seal primer connection.

1. Exception: Fitting may be omitted if trap has trap-seal primer connection.

2. Size: Same as floor drain inlet.

H. Install air-gap fittings on draining-type backflow preventers and on indirect-waste piping discharge into sanitary drainage system.

I. Install expansion joints on vertical stacks and conductors. Position expansion joints for easy access and maintenance.

J. Install wood-blocking reinforcement for wall-mounting-type specialties.

K. Install traps on plumbing specialty drain outlets. Omit traps on indirect wastes unless trap is indicated.

3.2 CONNECTIONS

A. Comply with requirements in Section 22 13 16 "Sanitary Waste and Vent Piping" for piping installation requirements. Drawings indicate general arrangement of piping, fittings, and specialties.

B. Install piping adjacent to equipment to allow service and maintenance.

C. Ground equipment according to Section 26 05 26 "Grounding and Bonding for Electrical Systems."

D. Connect wiring according to Section 26 05 19 "Low-Voltage Electrical Power Conductors and Cables."

3.3 PROTECTION

A. Protect drains during remainder of construction period to avoid clogging with dirt or debris and to prevent damage from traffic or construction work.

B. Place plugs in ends of uncompleted piping at end of each day or when work stops.

END OF SECTION

SECTION 22 14 29 - SUMP PUMPS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Submersible sump/effluent pumps.

1.2 SUBMITTALS

A. See section 22 00 00 “General Requirements of Plumbing” for submittal requirements.

1.3 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. UL Compliance: Comply with UL 778 for motor-operated water pumps.

PART 2 - PRODUCTS

2.1 SUBMERSIBLE SUMP/EFFLUENT PUMPS

A. Submersible, Fixed-Position, Single-Seal Sump Pumps:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Liberty Pumps.

b. Weil Pump Company, Inc.

c. Zoeller Company.

2. Description: Factory-assembled and -tested sump-pump unit.

3. Pump Type: Submersible, end-suction, single-stage, close-coupled, overhung-impeller, centrifugal sump pump as defined in HI 1.1-1.2 and HI 1.3.

4. Pump Casing: Cast iron, with strainer inlet, legs that elevate pump to permit flow into impeller, and vertical discharge for piping connection.

5. Impeller: Statically and dynamically balanced, ASTM A 48/A 48M, Class No. 25 A cast iron or engineered polymer, designed for clear wastewater handling, and keyed and secured or screw mounted to the shaft.

6. Pump and Motor Shaft: Stainless steel, with factory-sealed, grease-lubricated ball bearings.

7. Seal: Mechanical.

8. Motor: Hermetically sealed, capacitor-start type; with built-in overload protection; lifting eye or lug; and three-conductor, waterproof power cable of length required and with grounding plug and cable-sealing assembly for connection at pump.

a. Motor Housing Fluid: Oil.

9. Controls:

a. Enclosure: NEMA 250, [**Type 1**] [**Type 4X**]; [**pedestal**] [**wall**]-mounted.

b. Switch Type: Mechanical-float Pressure Insert type type, in NEMA 250, Type 6 enclosures with mounting rod and electric cables.

c. Magnetic Motor Contactor

d. Hand/Off/Auto switches for duplex units

e. Automatic Alternator: Start pumps on successive cycles and start multiple pumps if one cannot handle load.

f. High-Water Alarm: Rod-mounted, NEMA 250, Type 6 enclosure with mechanical-float matching control and electric bell; 120-V ac, with transformer and contacts for remote alarm bell.

10. Control-Interface Features:

a. Remote Alarm Contacts: For remote alarm interface.

b. Building Automation System Interface: Auxiliary contacts in pump controls for interface to building automation system and capable of providing the following:

1) On-off status of pump.

2) Alarm status.

11. Capacities and Characteristics:

a. See Drawings.

2.2 MOTORS

A. Comply with NEMA designation, temperature rating, service factor, enclosure type, and efficiency requirements for motors specified in Section 22 05 13 "Common Motor Requirements for Plumbing Equipment."

1. Motor Sizes: Minimum size as indicated. If not indicated, large enough so driven load will not require motor to operate in service factor range above 1.0.

B. Motors for submersible pumps shall be hermetically sealed.

PART 3 - EXECUTION

3.1 EARTHWORK

A. Excavation and filling are specified in Section 31 20 00 "Earth Moving."

3.2 INSTALLATION

A. Pump Installation Standard: Comply with HI 1.4 for installation of sump pumps.

END OF SECTION

SECTION 22 33 00 - ELECTRIC, DOMESTIC-WATER HEATERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Residential, storage, electric, domestic-water heaters.

2. Domestic-water heater accessories.

1.2 SUBMITTALS

A. See Section 22 00 00 “General Requirements of Plumbing” for submittal requirements.

1.3 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. ASME Compliance: Where ASME-code construction is indicated, fabricate and label residential, domestic-water heater storage tanks to comply with ASME Boiler and Pressure Vessel Code: Section VIII, Division 1.

C. NSF Compliance: Fabricate and label equipment components that will be in contact with potable water to comply with NSF 61 Annex, "Drinking Water System Components - Health Effects."

1.4 WARRANTY

A. Special Warranty: Manufacturer's standard form in which manufacturer agrees to repair or replace components of electric, domestic-water heaters that fail in materials or workmanship within specified warranty period.

a. Residential, Electric, Storage, Domestic-Water Heaters:

1) Storage Tank: Six years.

2) Controls and Other Components: Six years.

b. Compression Tanks: Five years.

PART 2 - PRODUCTS

2.1 RESIDENTIAL, ELECTRIC, DOMESTIC-WATER HEATERS

A. Residential, Storage, Electric, Domestic-Water Heaters:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Bradford White Corporation.

b. Lochinvar, LLC.

c. Rheem Manufacturing Company.

d. Smith, A. O. Corporation.

2. Standard: UL 174.

3. Storage-Tank Construction: Steel, vertical arrangement.

a. Tappings: ASME B1.20.1 pipe thread.

b. Pressure Rating: 150 psig.

c. Interior Finish: Comply with NSF 61 Annex barrier materials for potable-water tank linings, including extending lining material into tappings.

4. Factory-Installed Storage-Tank Appurtenances:

a. Anode Rod: Replaceable magnesium.

b. Dip Tube: Required unless cold-water inlet is near bottom of tank.

c. Drain Valve: ASSE 1005.

d. Insulation: Comply with ASHRAE/IESNA 90.1.

e. Jacket: Steel with enameled finish.

f. Heat-Trap Fittings: Inlet type in cold-water inlet and outlet type in hot-water outlet.

g. Heating Elements: Two; electric, screw-in immersion type; wired for simultaneous operation unless otherwise indicated. Limited to 12 kW total.

h. Temperature Control: Adjustable thermostat.

i. Safety Control: High-temperature-limit cutoff device or system.

j. Relief Valve: ASME rated and stamped for combination temperature-and-pressure relief valves. Include relieving capacity at least as great as heat input, and include pressure setting less than domestic-water heater working-pressure rating. Select relief valve with sensing element that extends into storage tank.

B. Capacity and Characteristics:

1. See Drawings.

2.2 DOMESTIC-WATER HEATER ACCESSORIES

A. Domestic-Water Expansion Tanks:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. AMTROL, Inc.

b. Taco

c. Armstrong

d. John Wood

2. Description: AMSE rated, steel pressure-rated tank constructed with welded joints and factory-installed butyl-rubber diaphragm. Include air precharge to minimum system-operating pressure at tank.

3. Construction:

a. Tappings: Factory-fabricated steel, welded to tank before testing and labeling. Include ASME B1.20.1 pipe thread.

b. Interior Finish: Comply with NSF 61 Annex barrier materials for potable-water tank linings, including extending finish into and through tank fittings and outlets.

c. Air-Charging Valve: Factory installed.

4. Capacity and Characteristics:

a. See Drawings.

B. Drain Pans: Corrosion-resistant metal with raised edge. Comply with ANSI/CSA LC 3. Include dimensions not less than base of domestic-water heater, and include drain outlet not less than NPS 3/4 with ASME B1.20.1 pipe threads or with ASME B1.20.7 garden-hose threads.

C. Piping-Type Heat Traps: Field-fabricated piping arrangement according to ASHRAE/IESNA 90.1.

D. Heat-Trap Fittings: ASHRAE 90.2.

E. Pressure-Reducing Valves: ASSE 1003 for water. Set at 25-psig-maximum outlet pressure unless otherwise indicated.

F. Combination Temperature-and-Pressure Relief Valves: ASME rated and stamped. Include relieving capacity at least as great as heat input, and include pressure setting less than domestic-water heater working-pressure rating. Select relief valves with sensing element that extends into storage tank.

G. Pressure Relief Valves: ASME rated and stamped. Include pressure setting less than domestic-water heater working-pressure rating.

H. Vacuum Relief Valves: ANSI Z21.22/CSA 4.4.

I. Shock Absorbers: ASSE 1010 or PDI-WH 201, Size A water hammer arrester.

J. Domestic-Water Heater Stands: Manufacturer's factory-fabricated steel stand for floor mounting, capable of supporting domestic-water heater and water. Include dimension that will support bottom of domestic-water heater a minimum of 18 inches above the floor.

K. Domestic-Water Heater Mounting Brackets: Manufacturer's factory-fabricated steel bracket for wall mounting, capable of supporting domestic-water heater and water.

2.3 SOURCE QUALITY CONTROL

A. Factory Tests: Test and inspect domestic-water heaters specified to be ASME-code construction, according to ASME Boiler and Pressure Vessel Code.

B. Electric, domestic-water heaters will be considered defective if they do not pass tests and inspections. Comply with requirements in Section 01 40 00 "Quality Requirements" for retesting and reinspecting requirements and Section 01 73 00 "Execution" for requirements for correcting the Work.

C. Prepare test and inspection reports.

PART 3 - EXECUTION

3.1 DOMESTIC-WATER HEATER INSTALLATION

A. Residential, Electric, Domestic-Water Heater Mounting: Install residential, electric, domestic-water heaters on concrete base. Comply with requirements for concrete bases specified in Section 03 30 00 "Cast-in-Place Concrete."

1. Exception: Omit concrete bases for residential, electric, domestic-water heaters if installation on stand, bracket, suspended platform is indicated.

2. Maintain manufacturer's recommended clearances.

3. Arrange units so controls and devices that require servicing are accessible.

4. Install dowel rods to connect concrete base to concrete floor. Unless otherwise indicated, install dowel rods on 18-inch centers around the full perimeter of concrete base.

5. For supported equipment, install epoxy-coated anchor bolts that extend through concrete base and anchor into structural concrete floor.

6. Place and secure anchorage devices. Use setting drawings, templates, diagrams, instructions, and directions furnished with items to be embedded.

7. Install anchor bolts to elevations required for proper attachment to supported equipment.

8. Anchor domestic-water heaters to substrate.

B. Install electric, domestic-water heaters level and plumb, according to layout drawings, original design, and referenced standards. Maintain manufacturer's recommended clearances. Arrange units so controls and devices needing service are accessible.

1. Install shutoff valves on domestic-water-supply piping to domestic-water heaters and on domestic-hot-water outlet piping. Comply with requirements for shutoff valves specified in Section 22 05 23 "General Duty Valves for Plumbing".

C. Install residential, electric, domestic-water heaters with seismic-restraint devices. Comply with requirements for seismic-restraint devices specified in Section 22 05 48 "Vibration and Seismic Controls for Plumbing Piping and Equipment."

D. Install combination temperature-and-pressure relief valves in top portion of storage tanks. Use relief valves with sensing elements that extend into tanks. Extend residential-water-heater relief-

valve outlet, with drain piping same as domestic-water piping in continuous downward pitch, and discharge by positive air gap onto closest floor drain.

E. Install combination temperature-and-pressure relief valves in water piping for electric, domestic-water heaters without storage. Extend residential-water-heater relief-valve outlet, with drain piping same as domestic-water piping in continuous downward pitch, and discharge by positive air gap onto closest floor drain.

F. Install water-heater drain piping as indirect waste to spill by positive air gap into open drains or over floor drains. Install hose-end drain valves at low points in water piping for electric, domestic-water heaters that do not have tank drains. Comply with requirements for hose-end drain valves specified in Section 22 11 19 "Domestic Water Piping Specialties."

G. Install thermometers on outlet piping of electric, domestic-water heaters. Comply with requirements for thermometers specified in Section 22 05 19 "Meters and Gages for Plumbing Piping."

H. Install piping-type heat traps on inlet and outlet piping of electric, domestic-water heater storage tanks without integral or fitting-type heat traps.

I. Fill electric, domestic-water heaters with water.

J. Charge domestic-water compression tanks with air.

3.2 CONNECTIONS

A. Comply with requirements for piping specified in Section 22 11 16 "Domestic Water Piping." Drawings indicate general arrangement of piping, fittings, and specialties.

B. Where installing piping adjacent to electric, domestic-water heaters, allow space for service and maintenance of water heaters. Arrange piping for easy removal of domestic-water heaters.

3.3 IDENTIFICATION

A. Identify system components. Comply with requirements for identification specified in Section 22 05 53 "Identification for Plumbing Piping and Equipment."

3.4 FIELD QUALITY CONTROL

A. Perform tests and inspections.

1. Leak Test: After installation, charge system and test for leaks. Repair leaks and retest until no leaks exist.

2. Operational Test: After electrical circuitry has been energized, start units to confirm proper operation.

3. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

B. Electric, domestic-water heaters will be considered defective if they do not pass tests and inspections. Comply with requirements in Section 01 40 00 "Quality Requirements" for retesting and reinspecting requirements and Section 01 73 00 "Execution" for requirements for correcting the Work.

C. Prepare test and inspection reports.

END OF SECTION

SECTION 22 41 00 - PLUMBING FIXTURES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Plumbing fixtures shown and scheduled on the drawings.

1.2 SUBMITTALS

A. See section 22 00 00 “General Requirements of Plumbing” for submittal requirements.

PART 2 - PRODUCTS

2.1 PLUMBING FIXTURE MANUFACTURERS

A. FLOOR DRAINS & SINKS

1. Fixtures

a. Jr Smith

b. Zurn

c. Josam

2. Trap Seal

a. Trap Guard

b. Jr Smith

3. Trap Primers

a. Jr Smith

b. Watts

c. Zurn

d. Sioux Chief

B. HOSE BIBBS / WALL HYDRANTS

1. Woodford

2. Watts

3. Zurn

C. LAVATORIES

1. Fixtures

a. Kohler

b. American Standard

c. Toto

2. Carriers And Supports

a. Jr Smith

b. Zurn

c. Josam

3. Faucets

a. Moen Commercial

b. Sloan

c. Chicago Faucets

4. Piping Covers

a. Trubro

b. Plummerex

D. ROOF DRAINS

1. Jr Smith

2. Zurn

3. Sioux Chief

E. STAINLESS STEEL SINKS

1. Fixtures

a. Elkay

b. Just

c. Kohler

2. Faucets

a. Moen Commercial

b. T&S Brass

c. Chicago Faucet

F. STOP VALVES

1. Brasscraft

2. Watts

3. Kingston Brass

G. THERMOSTATIC MIXING VALVES

1. Symmons
2. Watts
3. Leonard

H. UTILITY SINKS & MOP SINKS

1. Fixtures
    a. Fiat
    b. Mustee
    c. Kohler
2. Faucets
    a. Moen Commercial
    b. T&S Brass
    c. Chicago Faucet
3. Accessories
    a. Fiat
    b. Mustee

I. WATER CLOSETS

1. Fixtures
    a. Kohler
    b. American Standard
    c. Toto
    d. Sloan
2. Seats
    a. Kohler
    b. Church
    c. Olsonite
3. Carriers and Supports

a. Jr Smith
b. Zurn
c. Josam

2.2 GROUT

A. Standard: ASTM C 1107/C 1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.

B. Characteristics: Nonshrink; recommended for interior and exterior applications.

C. Design Mix: 5000-psi, 28-day compressive strength.

D. Packaging: Premixed and factory packaged.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install plumbing fixtures level and plumb according to roughing-in drawings.

B. Install floor-mounted water closets on closet flange attachments to drainage piping.

C. Install counter-mounting fixtures in and attached to casework.

D. Install pedestal lavatories on pedestals and secured to wood blocking in wall.

E. Install water-supply piping with stop on each supply to each fixture to be connected to water distribution piping. Attach supplies to supports or substrate within pipe spaces behind fixtures. Install stops in locations where they can be easily reached for operation.

1. Exception: Use ball, gate, or globe valves if supply stops are not specified with fixture. Comply with valve requirements specified in Section 22 05 23 "General-Duty Valves for Plumbing Piping."

F. Install tanks for accessible, tank-type water closets with lever handle mounted on wide side of compartment.

G. Install toilet seats on water closets.

H. Install faucet flow-control fittings with specified flow rates and patterns in faucet spouts if faucets are not available with required rates and patterns. Include adapters if required.

I. Install shower flow-control fittings with specified maximum flow rates in shower arms.

J. Install traps on fixture outlets.

1. Exception: Omit trap on fixtures with integral traps.

2. Exception: Omit trap on indirect wastes unless otherwise indicated.

K. Install disposer in outlet of each sink indicated to have disposer. Install switch where indicated or in wall adjacent to sink if location is not indicated.

L. Install dishwasher air-gap fitting at each sink indicated to have air-gap fitting. Install in sink deck. Connect inlet hose to dishwasher and outlet hose to disposer.

M. Set bathtubs and shower receptors in leveling bed of cement grout.

N. Install protective shielding pipe covers and enclosures on exposed supplies and waste piping of accessible lavatories and sinks.

O. Install wall flanges or escutcheons at piping wall penetrations in exposed, finished locations. Use deep-pattern escutcheons if required to conceal protruding fittings.

P. Seal joints between plumbing fixtures, counters, floors, and walls using sanitary-type, one-part, mildew-resistant silicone sealant. Match sealant color to fixture color.

Q. The Plumbing contractor shall furnish a 24V control transformer to all hard wired optical/handsfree fixtures. The Plumbing contractor shall coordinate with the electrical contractor to install all line and low voltage wiring in compliance with section 26 05 19 “Low-voltage Electrical Power Conductors and Cables”, and section 26 05 23 “Control-Voltage Electrical Power Cables”.

3.2 CONNECTIONS

A. Connect fixtures with water supplies, stops, and risers, and with traps, soil, waste, and vent piping. Use size fittings required to match fixtures.

B. Comply with water piping requirements specified in Section 22 11 16 "Domestic Water Piping."

C. Comply with soil and waste piping requirements specified in Section 22 13 16 "Sanitary Waste and Vent Piping."

D. Install protective shielding pipe covers and enclosures on exposed supplies and waste piping of accessible lavatories and sinks.

E. All electrical connections shall be coordinated by the plumbing contractor with the electrical contractor.

3.3 ADJUSTING

A. Operate and adjust plumbing fixtures and controls. Replace damaged and malfunctioning fixtures, fittings, and controls.

B. Adjust water pressure at faucets to produce proper flow.

3.4 CLEANING AND PROTECTION

A. After completing installation of plumbing fixtures, inspect and repair damaged finishes.

B. Clean plumbing fixtures, faucets, and other fittings with manufacturers' recommended cleaning methods and materials.

C. Provide protective covering for installed plumbing fixtures and fittings.

D. Do not allow use of plumbing fixtures for temporary facilities unless approved in writing by Owner.

END OF SECTION

SECTION 23 00 00 - GENERAL REQUIREMENTS OF HVAC

PART 1 - GENERAL

1.1 SUMMARY

A. The requirements listed in this section are supplemental to the Division 01 General Requirements.

B. It shall be the responsibility of the Mechanical Contractor to examine and refer to all Architectural, Civil, Structural, Plumbing, Electrical, and Landscape plans and specifications for construction conditions which may affect the scope of HVAC work. Inspect the building site and existing facilities for verification of present conditions. Make proper provisions for these conditions in performance of the work and cost thereof.

C. Mechanical work for this project shall include all items, articles, materials and the associated labor mentioned, scheduled or shown in these specifications and in the accompanying drawings.

D. Furnish and install all equipment, materials and any required incidental items required by good practice to complete the systems described herein.

1.2 CODES AND STANDARDS

A. Work shall meet the requirements of the plans and specifications and shall not be less than the minimum requirements of applicable sections of the latest Codes and Standards of the following Organizations:

1. American Gas Association (AGA)

2. American Society of Heating, Refrigeration and Air Conditioning Engineers (ASHRAE)

3. American Society of Mechanical Engineers (ASME)

4. Sheet Metal and Air Conditioning Contractors' National Association Inc. (SMACNA)

5. American Water Works Association (AWWA)

6. National Electrical Code (NEC)

7. National Electrical Manufacturers Association (NEMA)

8. National Fire Protection Association (NFPA)

9. International Plumbing Code (IPC)

10. Occupational Safety & Health Act (OSHA)

11. Plastic Pipe Institute (PPI)

12. International Mechanical Code (IMC)

13. International Fuel Gas Code (IFGC)

14. International Building Code (IBC)

15. International Energy Conservation Code (IECC)

16. Requirements of the Serving Utility Company

17. Local and State Codes and Ordinances

1.3 FEES AND PERMITS

A. The Mechanical Contractor shall pay all fees and arrange all permits required for work done under their contract and under their supervision by subcontract.

B. All usage contracts between the Owner and the serving utilities company, such as membership and usage charges or fees, etc., for the purpose of obtaining the services for the utility company shall be applied for and paid for by the Owner.

1.4 MATERIALS AND EQUIPMENT

A. Manufacturer's trade names and catalog numbers listed are intended to indicate the quality of equipment or materials desired. Manufacturers not listed in the specification will be considered substitutions and must have prior approval.

B. See Division 01 for Substitutions Procedures. Requests for substitution are to be submitted sufficiently ahead of the deadline, to give ample time for examination. Prior approval request for substitution must indicate the specific item or items to be furnished in lieu of those scheduled, together with complete technical and comparative data on scheduled items and items proposed for substitution.

C. If the engineer approves any proposed substitution, the approved product will be listed in an addendum. Bidders shall not rely on approval made in any other manner.

D. Mechanical equipment may be installed with manufacturer's standard finish and color except where specific color, finish or choice is indicated. If the manufacturer has no standard finish, equipment shall have a prime coat and two finish coats of gray enamel.

E. High altitude operation: Capacity of all equipment is to be sized and manufactured to perform at the elevation of the project site. If not specifically indicated in the equipment schedule or in the specifications provide all required accessories and equipment for proper operation at elevation of the project site.

F. This Contractor shall be responsible for materials and equipment installed under this contract. Contractor shall also be responsible for the protection of materials and equipment of others from damage as a result of his work.

G. Manufactured material and equipment shall be applied, installed, connected, erected, used, cleaned and conditioned as directed by manufacturer unless herein specified to the contrary.

H. This Contractor shall make the required arrangement with the General Contractor or Construction Manager for the introduction into the building of equipment too large to pass through finished openings.

I. Store materials and equipment indoors at the job site. If this is not possible, store on raised platforms and protect from the weather by means of waterproof covers. Coverings shall permit circulation of air around the materials to prevent condensation of moisture. Screen or cap openings in equipment to prevent the entry of vermin.

1.5 INTENT OF DRAWINGS

A. The drawings are diagrammatic and do not necessarily show exact location of piping and ductwork unless specifically dimensioned. Riser and other diagrams are schematic and do not necessarily show the physical arrangement of the equipment. They shall not be used for obtaining lineal runs of piping or ductwork, nor shall they be used for shop drawings for piping and ductwork fabrication or ordering. Discrepancies shown on different plans, or between plans and actual field conditions shall be brought to the attention of the Architect/Engineer for resolution.

1.6 COMMISSIONING OF SYSTEMS

A. Mechanical systems with a total cooling capacity exceeding 480,000 Btu/hr or a combined heating and service water-heating capacity exceeding 600,000 Btu/hr shall be commissioned by a registered professional in accordance with Section C408 of the International Energy Conservation Code. The mechanical contractor shall correct all installation deficiencies identified by the commissioning agent at no cost to the owner.

B. See Sections 019113 “Commissioning Requirements of Contractor”, 220800 “Commissioning of Plumbing”, 230800 “Commissioning of HVAC”, 260800 “Commissioning of Electrical Systems”.

1.7 RESPONSIBILITY

A. HVAC work shall conform to requirements of all divisions 22 and 23 specifications.

B. The Mechanical Contractor shall be responsible for the installation of a satisfactory and complete system in accordance with the intent of the drawing and specifications. Provide, at no extra cost, all incidental items, materials, accessories and labor required for completion of the work even though they are not specifically mentioned or indicated on the drawings or in the specifications.

C. The drawings do not attempt to show complete details of the building construction which affect the mechanical installation; and reference is therefore required to the Architectural, Civil, Structural, Landscape and Electrical drawings and specifications and to shop drawings of all trades for additional details which affect the installation of the work covered under this Division of the Contract.

D. Location of mechanical system components shall be checked for conflicts with openings, structural members and components of other systems having fixed locations. In the event of any conflicts, the Architect/Engineer shall be consulted, and their decision shall govern. Necessary changes shall be made at the Contractor’s expense.

E. Determine, and be responsible for, the proper location and character of inserts for hangers, chases, sleeves, and other openings in the construction required for the work and obtain this information well in advance of the construction progress so work will not be delayed.

F. Final location of inserts, hangers, etc., required for each installation, must be coordinated with facilities required for other installations to prevent interference.

G. Take extreme caution not to install work that connects to equipment until such time as complete Shop Drawings of such equipment have been approved by the Architect/Engineer. Any work installed by the Contractor, prior to approval of Shop Drawings, will be at the Contractor's risk.

H. All modifications and changes required due to installation of equipment other than the scheduled equipment shall be made at the contractor’s expense.

I. It shall be the responsibility of the installing contractor to coordinate changes to work by other trades that result from the installation of equipment other than the scheduled equipment.

J. If the provided equipment is heavier or larger than the scheduled or specified equipment, it shall be the responsibility of the installing contractor to coordinate the required structural changes and pay for any and all associated cost.

K. If the provided equipment has different motor characteristics or electrical requirements than the scheduled or specified equipment, it shall be the responsibility of the installing contractor to coordinate the required changes and pay for any and all associated cost.

L. If larger or additional electrical conduits, wire or breakers are required due to the installation of equipment other than the scheduled or specified equipment it shall be the responsibility of the installing contractor to coordinate the required changes and pay for any and all associated cost.

M. If the provided equipment requires different fluid flow rates than the scheduled or specified equipment, it shall be the responsibility of the installing contractor to coordinate all required changes including but not limited to pumps, piping, valves, etc and pay for any and all associated cost.

N. At all times during the performance of this Contract, properly protect work from damage and protect the Owner's property from injury of loss. Make good any damage, injury or loss, except such as may be directly due to errors in the Bidding Documents or caused by Agents or Employees of the Owner. Adequately protect adjacent property as provided by law and the Bidding Documents. Provide and maintain passageways, guard fences, lights and other facilities for protection required by Public Authority or Local conditions.

O. The Contractor shall be responsible for damages incurred due to the work of their contractors, to the building or its contents, people, etc.

1.8 REVIEW

A. All work and material is subject to review at any time by the Architect/Engineer or his representative. If the Architect/Engineer or his representative finds material that does not conform to these specifications or that is not properly installed or finished, correct the deficiencies in a manner satisfactory to the Architect/Engineer at the Contractor’s expense.

1.9 WORKMANSHIP

A. Work under this contract shall be performed by workmen skilled in the particular trade, including work necessary to properly complete the installation in a workmanlike manner to present a neat and finished appearance.

B. Obtain Architect's/Engineer's approval before performing any cutting on structural members or patching of building surfaces. Any damage to the building or equipment by the Mechanical Contractor shall be the responsibility of the Mechanical Contractor and shall be repaired by skilled craftsmen of the trades involved at the Contractor’s expense.

C. Chases, openings, sleeves, hangers, anchors, recesses, equipment pads, and framing for equipment; shall be provided by others only if so noted on the drawings. Otherwise, they will be provided by the Mechanical Contractor for their work.

1.10 COORDINATION

A. The Mechanical Contractor shall plan their work to proceed with a minimum interference with other trades and it shall be their responsibility to inform the General Contractor of all openings required in the building structure for installation of work, and to provide sleeves as required. Dimensions of equipment installed and/or provided by others shall be checked so that correct clearances and connections may be made.

B. In general, pipelines requiring gravity drainage shall be installed first, followed by ductwork, large piping mains and electrical conduit. The location of fire protection piping and heads shall be coordinated with other trades to ensure that installations by other trades do not block heads.

C. Leave sufficient space for the installation of insulation on piping and ductwork as specified. It is not acceptable to compress pipe or duct insulation for any reason.

1.11 CLEANING

A. Keep the job site clean. The Mechanical Contractor shall remove all waste and rubbish associated with their work.

B. Upon completion of work, remove materials, scraps and debris related to mechanical work and leave all spaces including tunnels, crawlspaces, pipe or duct chases and ceiling plenums clean and orderly.

C. The Mechanical contractor will be responsible for cleaning the exterior and interior of all equipment prior to star-up. Once all equipment has been cleaned it shall be inspected by the Architect/Engineer prior to start-up.

D. The Mechanical Contractor shall provide dust protection of existing materials and equipment as well a new materials and equipment for the duration of the project. Protect existing materials and equipment from damage for the duration of the project. Clean the exterior and interior of all existing equipment at the completion of the project.

1.12 TEMPORARY FACILITIES

A. Offices

1. The Mechanical Contractor must have the permission of the Owner and General Contractor or Construction Manager to install a temporary office/job trailer on the project site.

2. The Contractor shall completely remove his temporary installations when no longer needed and the premises shall be completely clean, disinfected, patched, and refinished to match adjacent areas.

B. Ladders and Scaffolds

1. The Mechanical Contractor shall provide their own ladders, scaffolds, etc. of substantial construction for access to their work in various portions of the building as may be required. When no longer needed, they shall be removed by the Contractor.

C. Protection Devices

1. The Mechanical Contractor shall provide and maintain his own necessary barricades, fences, signal lights, etc., required by all governing authorities or shown on the drawings. When no longer needed, they shall be removed by the Contractor.

D. TEMPORARY FIRE PROTECTION

1. The Mechanical Contractor shall provide all necessary first aid hand fire extinguishers for Class A, B, C and special hazards as may exist in his own work area only in accordance with good and safe practice and as required by jurisdictional safety authority.

1.13 SUBMITTALS

A. Submittals will be required for each piece of equipment, material or product as noted in the table below. All submittals shall be submitted, reviewed and all discrepancies addressed prior to ordering equipment or starting work. Any equipment ordered without having first completed the submittal process is done at the risk of the contractor. Any work performed prior to completing the submittal process is done at the risk of the contractor.

Specification Section	Product Data	Performance Data	Shop Drawing	Delegated Design	Wiring Diagram	Color Chart	Notes
230500	X			X			
230548	X			X			Provide Delegated Design per the requirements of this section
230553	X						
230593							Provide T&B Certifications
230713	X	X	X				Provide Shop Drawings with Ductwork Plans
231123	X						
233300	X	X					
233423	X	X			X		
238231	X	X			X		

B. Submittal Definitions

1. Product Data: Provide manufacturers' cut sheets that include general product information including but not limited to, model number, physical data, nominal capacities, and rough-in requirements.

2. Performance Data: Provide detailed performance and capacities based on project specific requirements including but not limited to: flow rates, capacities, pressure loss, temperatures, fan curves, pump curves, part load performance, sound data, and electrical characteristics.

3. Shop Drawings: Provide detailed drawings of the equipment showing overall dimensions, location of electrical and piping connection, location of anchorage points, location of electrical and control panels, and all operating, service and maintenance clearances.

4. Delegated Design: Provide detailed drawings prepared and stamped by a registered Professional Engineer, that detail pertinent design criteria, the materials and products to be installed and the required installation locations.

5. Wiring Diagram: Provide diagrams that identify, and detail required field wiring.

6. Color Chart: Provide a physical color chart of material samples required for selection of equipment colors.

C. Submittal Formats:

1. Include the following information with each submittal:

a. Project Name

b. Submittal Date

c. Name of Architect

d. Name of Engineer

e. Name of General Contractor or Construction Manager

f. Name of Sub-Contractor

g. Name of firm or entity that prepared the submittal

h. Unique Submittal Number

i. Type of Submittal

j. Specification Section

k. Name or Mark of equipment or material and detail or drawings reference.

2. All Submittals with the exception of color charts or material samples shall be electronically transmitted PDFs.

D. Submittal Requirements

1. Submittals shall be submitted as a complete specification section. The submittal must include all materials and equipment for that specification section. Submittals for individual materials of equipment will be rejected without review.

2. Submittals shall be complete, clearly show item used, size, dimensions, capacity, rough in, etc., as required for complete check and installation. Manufacturer’s literature showing more than one item shall be clearly marked as to which item is being furnished or it will be rejected and returned without review.

3. Each submittal shall be thoroughly checked by the Contractor for compliance with the Contract Document requirements, accuracy of dimensions, relationship to the work of

other trades, and conformance with sound, safe practices as to erection and installation. Each submittal shall then bear a stamp evidencing such checking and shall show corrections made, if any. Submittals requiring extensive corrections shall be revised before submission. Each submittal not stamped and signed by the Contractor evidencing such checking will be rejected and returned without review.

4. On each submittal, clearly indicate deviations from requirements in the Contract Documents, including minor variations and limitations. Include relevant additional information and revisions, other than those requested on previous submittals. Indicate by highlighting on each submittal or noting on attached separate sheet.

5. Review of the shop drawings and literature by the engineer shall not relieve the contractor for responsibility for deviations from the drawings or specifications, nor shall it relieve the contractor from responsibility for errors in the shop drawings or literature. It is the responsibility of the contractor to provide materials and equipment which meet the specifications and job requirements.

1.14 OPERATION AND MAINTENANCE MANUALS

A. Operation and Maintenance Manuals (O&M Manuals) shall contain:

1. Names and contact information for the Project Architect, Project Engineer.

2. Names and contact information for the General Contractor or Construction Manager.

3. Names and contact information for sub-contractors.

4. Installation, maintenance and operating instructions for each piece of equipment.

5. Parts lists

6. Wiring Diagrams

7. Equipment Start-up and inspection certificates

8. Test and Balance Reports

9. Commissioning Reports

10. Copies of Equipment Warranties

11. Copies of Submittals

12. Record Drawings.

B. Prior to substantial completion, submit an electronic copy of the O&M manual in PDF format to the Architect, Engineer and Owner for Review and approval. The PDF shall be one file with an index and hyperlinks to each section. Individual bound PDFs without automated navigation will be rejected. All O&M data shall be grouped by the equipment type and ordered by the specification numbering.

C. Prior to final payment a final electronic copy of the O&M manual on an archival quality DVD as well as two printed copies, shall be furnished to the owner. Printed copies shall have commercial quality 8-1/2" x 11" 3-ring binders with tabbed dividers for each section.

1.15 AS-BUILT RECORD DRAWINGS

A. The Contractor shall furnish to the Owner and Architect/Engineer a marked print showing the location of all concealed or underground pipe or conduit runs and other equipment installed other than as shown on the drawings. Dimension underground lines from established building lines. Indicate all installed pull boxes in conduit runs.

B. The Contractor shall furnish to the Architect/Engineer a marked print showing the location of all mechanical equipment, piping, ductwork, diffusers, grilles, etc. The location of any item which deviates from the bid documents shall be accurately drawn and dimensioned.

C. All underground piping and ductwork shall be dimensioned from nearest column and/or exterior walls. The location of all maintenance related items, such as duct access doors, fire dampers, isolation valves, filters, etc., shall be highlighted on the as built drawing.

1.16 PLACING SYSTEM INTO OPERATION

A. Prior to the starting of equipment, the Mechanical Contractor shall thoroughly inspect the installation and any work completed by other trades and subcontractors to verify compliance with the contract documents.

B. Start-up of all HVAC equipment shall be completed by factory trained representatives. At the completion of start-up, the factory representative shall submit to the architect and engineer, a start-up report that indicates any problems encountered, potential problems including installation issues, adjustments made or required to be made to ensure proper operation of the equipment. Any installation deficiencies identified shall be corrected at no additional cost to the owner.

1.17 OWNER TRAINING

A. General

1. The system training is intended to familiarize the Owner's operating and maintenance staff with all systems requiring maintenance. Training is to be provided after the systems are in place and operational, after issues noted during commissioning have been resolved, and before final acceptance.

2. Provide second set of training sessions for automatic control systems about 6-9 months after the first sessions.

B. Systems Requiring Training

1. All mechanical, electrical, safety, standby, and automatic control systems in the project, and other systems specified elsewhere to have training.

C. Attendance:

1. Training is to be provided by contractor's representatives that are familiar with the system's operation and maintenance requirements. Individual training sessions (modules) shall be provided for each type or group of systems, separated roughly by trade

group that will be performing maintenance on the system. The trades groups and systems typically requiring training are:

D. Schedule:

1. Duplicate training sessions are to be provided for each training module, so that the Owner's operating personnel can be split into two groups during training. Duplicate training sessions shall be scheduled on different days. Length of training sessions will be determined by scope of training indicated below, and as coordinated with Owner after draft copy of training documents have been reviewed.

E. Training Documentation:

1. Contractor to submit draft copy of agenda and training documents to Owner for review at least two weeks prior to training date.

2. Provide a copy of the following items for each person that will be attending the training sessions. Coordinate required number with the Owner.

a. Training agenda.

b. Summary of new systems and existing systems affected by this project.

c. Summary of work performed under this project.

d. Control system drawings and sequences of operation.

e. List of important maintenance and trouble-shooting operations for all systems.

3. Provide minimum of 2 copies of following items:

a. Contract documents including all drawings, specifications, addendums, and change orders.

F. Training Sessions:

1. Assemble at location to be determined by the Owner.

2. Distribute training documentation as indicated above.

3. Provide classroom style training if required for orientation and discussion of new systems and existing systems affected by this project, and other issues appropriate for a classroom format.

4. Visit site and review locations; and perform detailed review of operation and maintenance requirements for current systems.

1.18 WARRANTY

A. The Contractor shall guarantee that all materials and labor installed are new and of first quality and that any material or labor found defective shall be replaced without cost to the Owner within one (1) year after substantial completion of the Contract or one (1) full season of heating and cooling operation, whichever is the greater. The guarantee shall list the date of the beginning of the one (1) year period, which shall be the date that the Substantial Completion Certificate is issued.

B. Any damage to the building, caused by defective work or material of the Contractor within the above-mentioned period, shall be satisfactorily repaired without cost to the Owner.

C. The guarantee does not include maintenance of equipment. The Owner shall accept full responsibility for proper operation and maintenance of equipment immediately upon substantial completion and occupancy of the building.

D. Final acceptance by the Owner will not occur until all operating instructions are mounted in Equipment Rooms and Operating Personnel are thoroughly indoctrinated in the operation of all mechanical equipment by the Contractor.

E. No equipment installed as part of this project shall be used for temporary heat during construction.

END OF SECTION

SECTION 23 05 00 - GENERAL PROVISIONS OF HVAC

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes the following:

1. Expansion Fittings and Loops for Piping Systems
2. Alignment Guides and Anchors
3. Dielectric Fittings
4. Pipe Sleeves
5. Sleeve Seals Systems for Piping
6. Silicone Sealant
7. Escutcheons for Piping
8. Floor Plates

1.2 SUBMITTALS

A. See Section 23 00 00 “General Requirements of HVAC” for Submittal requirements.

1.3 QUALITY ASSURANCE

A. Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."

B. Pipe and Pressure-Vessel Welding Qualifications: Qualify procedures and operators according to ASME Boiler and Pressure Vessel Code.

1.4 PERFORMANCE REQUIREMENTS

A. Compatibility: Products shall be suitable for piping service fluids, materials, working pressures, and temperatures.

B. Capability: Products to absorb 200 percent of maximum axial movement between anchors.

PART 2 - PRODUCTS

2.1 EXPANSION FITTINGS AND LOOPS FOR PIPING SYSTEMS

A. Rubber Union Connector Expansion Joints

1. Manufacturers: Subject to compliance with requirements, provide products by the following:

a. Engineered Flexible Products EFP

b. Mason Industries, Inc.

c. MetraFlex.

d. Twin City Hose.

e. Vibro-Acoustics

2. Material: Twin reinforced-rubber spheres with external restraining cables.

3. Minimum Pressure Rating: 150 psig at 170 deg F, unless otherwise indicated.

4. End Connections for NPS 2 and Smaller: Threaded.

B. Flexible-Hose Packless Expansion Joints:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Engineered Flexible Products EFP

b. Mason Industries, Inc.

c. Metraflex Company (The).

d. Twin City Hose Inc.

e. Vibro-Acoustics

2. Description: Manufactured assembly with inlet and outlet elbow fittings and two flexible-metal-hose legs joined by long-radius, 180-degree return bend or center section of flexible hose.

3. Flexible Hose: Corrugated-metal inner hoses and braided outer sheaths.

4. Expansion Joints for Copper Tubing NPS 2 and Smaller: Copper-alloy fittings with solder-joint end connections.

a. Bronze hoses and single-braid bronze sheaths with 450 psig at 70 deg F and 340 psig at 450 deg F ratings.

5. Expansion Joints for Copper Tubing NPS 2-1/2 to NPS 4: Copper-alloy fittings with threaded end connections.

a. Stainless-steel hoses and single-braid, stainless-steel sheaths with 300 psig at 70 deg F and 225 psig at 450 deg F ratings.

6. Expansion Joints for Steel Piping NPS 2 and Smaller: Carbon-steel fittings with threaded end connections.

a. Stainless-steel hoses and single-braid, stainless-steel sheaths with 450 psig at 70 deg F and 325 psig at 600 deg F ratings.

7. Expansion Joints for Steel Piping NPS 2-1/2 to NPS 6: Carbon-steel fittings with flanged end connections.

a. Stainless-steel hoses and single-braid, stainless-steel sheaths with 200 psig at 70 deg F and 145 psig at 600 deg F ratings.

8. Expansion Joints for DWV and Storm Drainage PVC or Cast Iron Piping NPS 2 and Smaller: Copper-alloy fittings with no hub or flanged end connections.

a. Bronze hoses and single-braid bronze sheaths with 450 psig at 70 deg F and 340 psig at 450 deg F ratings.

b. UPC listed with cleanout plugs located in the U-Bend

9. Expansion Joints for DWV and Storm Drainage or Cast Iron Piping NPS 4 to NPS 8: Stainless steel fittings with no hub or flanged end connections.

a. Stainless-steel hoses and single-braid, stainless-steel sheaths with 300 psig at 70 deg F and 225 psig at 450 deg F ratings.

b. UPC listed with cleanout plugs located in the U-Bend

2.2 ALIGNMENT GUIDES AND ANCHORS

A. Alignment Guides

1. Description: Steel, factory-fabricated alignment guide, with bolted two-section outer cylinder and base for attaching to structure; with two-section guiding slider for bolting to pipe.

B. Anchor Materials:

1. Steel Shapes and Plates: ASTM A 36/A 36M.

2. Bolts and Nuts: ASME B18.10 or ASTM A 183, steel hex head.

3. Washers: ASTM F 844, steel, plain, flat washers.

4. Mechanical Fasteners: Insert-wedge-type stud with expansion plug anchor for use in hardened portland cement concrete, with tension and shear capacities appropriate for application.

a. Stud: Threaded, zinc-coated carbon steel.

b. Expansion Plug: Zinc-coated steel.

c. Washer and Nut: Zinc-coated steel.

5. Chemical Fasteners: Insert-type stud, bonding-system anchor for use with hardened portland cement concrete, with tension and shear capacities appropriate for application.

a. Bonding Material: ASTM C 881/C 881M, Type IV, Grade 3, two-component epoxy resin suitable for surface temperature of hardened concrete where fastener is to be installed.

b. Stud: ASTM A 307, zinc-coated carbon steel with continuous thread on stud, unless otherwise indicated.

c. Washer and Nut: Zinc-coated steel.

2.3 DIELECTRIC FITTINGS

A. General Requirements: Assembly of copper alloy and ferrous materials with separating nonconductive insulating material. Include end connections compatible with pipes to be joined.

B. Dielectric Unions:

1. Dielectric Unions are not allowed.

C. Dielectric Flanges:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Capitol Manufacturing Company; member of the Phoenix Forge Group.

b. Central Plastics Company.

c. Matco-Norca.

d. Watts; a division of Watts Water Technologies, Inc.

e. Wilkins; a Zurn company.

2. Standard: ASSE 1079.

3. Factory-fabricated, bolted, companion-flange assembly.

4. Pressure Rating: 175 psig.

5. End Connections: Solder-joint copper alloy and threaded ferrous; threaded solder-joint copper alloy and threaded ferrous.

D. Dielectric-Flange Insulating Kits:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Advance Products & Systems, Inc.

b. Calpico, Inc.

c. Central Plastics Company.

d. Pipeline Seal and Insulator, Inc.

2. Nonconducting materials for field assembly of companion flanges.

3. Pressure Rating: 150 psig.

4. Gasket: Neoprene or phenolic.

5. Bolt Sleeves: Phenolic or polyethylene.

6. Washers: Phenolic with steel backing washers.

E. PEX Dielectric Separator:

1. Description: 6” long section of pex piping shall be installed between dis-similar piping materials.

2. Pipe Material: PEX plastic according to ASTM F 876.

3. Oxygen Barrier: O2 permeability <= 0.32 mg/m2/day in accordance with DIN 4726.

4. Fittings: ASTM F 1960, cold expansion fittings and reinforcing rings.

5. Pressure/Temperature Rating: Minimum 100 psig and 180 deg F.

2.4 SLEEVES

A. Galvanized-Steel Sheet Pipe Sleeves: 0.0239-inch minimum thickness; round tube closed with welded longitudinal joint.

2.5 SLEEVE-SEAL SYSTEMS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Advance Products & Systems, Inc.

2. CALPICO, Inc.

3. GPT; an EnPro Industries company.

4. Metraflex Company (The).

B. Description:

1. Modular sealing-element unit, designed for field assembly, for filling annular space between piping and sleeve.

2. Designed to form a hydrostatic seal of 20-psig.

3. Sealing Elements: EPDM-rubber interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size.

4. Pressure Plates: Composite plastic.

5. Connecting Bolts and Nuts: Stainless steel of length required to secure pressure plates to sealing elements.

2.6 ASSEMBLY PENETRATIONS

A. All penetrations through a fire rated assembly shall be protected with an approved fire stop system in compliance with the rated assemblies as outlined in the Underwriters Laboratory Listing.

1. Manufacturers: Subject to compliance with requirements, provide products by the following:

    a. 3M Company

    b. Holdrite

    c. Hilti

2.7 SILICONE SEALANTS

A. Silicone, S, P, 25, T, NT: Single-component, pourable, plus 25 percent and minus 25 percent movement capability, traffic- and nontraffic-use, neutral-curing silicone joint sealant; ASTM C 920, Type S, Grade P, Class 25, Uses T and NT. Grade P Pourable (self-leveling) formulation is for opening in floors and other horizontal surfaces that are not fire rated.

2.8 ESCUTCHEONS

A. One-Piece, Cast-Brass Type: With polished, chrome-plated finish and setscrew fastener.

B. One-Piece, Deep-Pattern Type: Deep-drawn, box-shaped brass with chrome-plated finish and spring-clip fasteners.

C. One-Piece, Stamped-Steel Type: With chrome-plated finish and spring-clip fasteners.

2.9 FLOOR PLATES

A. One-Piece Floor Plates: Cast-iron flange with holes for fasteners.

PART 3 - EXECUTION

3.1 EXPANSION JOINT INSTALLATION

A. Install expansion joints of sizes matching sizes of piping in which they are installed.

B. Install expansion joint per the manufacture’s written instructions.

3.2 ALIGNMENT-GUIDE AND ANCHOR INSTALLATION

A. Install alignment guides to guide expansion and to avoid end-loading and torsional stress.

B. Install two guide(s) on each side of pipe expansion fittings and loops. Install guides nearest to expansion joint not more than four (4) pipe diameters from expansion joint.

C. Attach guides to pipe, and secure guides to building structure.

D. Install anchors at locations to prevent stresses from exceeding those permitted by ASME B31.9 and to prevent transfer of loading and stresses to connected equipment.

E. Anchor Attachments:

1. Anchor Attachment to Steel Pipe: Attach by welding. Comply with ASME B31.9 and ASME Boiler and Pressure Vessel Code: Section IX, "Welding and Brazing Qualifications."

2. Anchor Attachment to Copper Tubing: Attach with pipe hangers. Use MSS SP-69, Type 24; U bolts bolted to anchor.

F. Fabricate and install steel anchors by welding steel shapes, plates, and bars. Comply with ASME B31.9 and AWS D1.1/D1.1M.

1. Anchor Attachment to Steel Structural Members: Attach by welding.

2. Anchor Attachment to Concrete Structural Members: Attach by fasteners. Follow fastener manufacturer's written instructions.

G. Use grout to form flat bearing surfaces for guides and anchors attached to concrete.

3.3 DIELECTRIC FITTING INSTALLATION

A. Install dielectric fittings in piping at connections of dissimilar metal piping and tubing.

B. Install Dielectric fittings per the manufacturers written instructions.

C. Install pipe hangers immediately upstream and downstream of dielectric fittings.

D. Install isolation valves immediately upsteam and downstream of dielectric fittings.

E. Dielectric Fittings for NPS 2 and Smaller: PEX Dielectric Separator.

F. Dielectric Fittings for NPS 2-1/2 and Larger: Dielectric Flange.

3.4 SLEEVE INTALLATION

A. Install sleeves for piping passing through penetrations in floors, partitions, roofs, and walls.

B. For sleeves that will have sleeve-seal system installed, select sleeves of size large enough to provide 1-inch annular clear space between piping and concrete slabs and walls.

C. Install sleeves in concrete floors, concrete roof slabs, and concrete walls as new slabs and walls are constructed.

1. Cut sleeves to length for mounting flush with both surfaces.

a. Exception: Extend sleeves installed in floors of mechanical equipment areas or other wet areas 2 inches above finished floor level.

2. Using silicone sealant, seal space outside of sleeves in slabs and walls without sleeve-seal system.

D. Fire-Resistance-Rated Penetrations, Horizontal Assembly Penetrations, and Smoke-Barrier Penetrations: Maintain indicated fire or smoke rating of walls, partitions, ceilings, and floors at pipe penetrations. Seal pipe penetrations with fire- and smoke-stop materials. Comply with requirements for firestopping and fill materials specified in Section 07 84 13 "Penetration Firestopping."

3.5 SLEEVE-SEALS SYSTEM INSTALLATION

A. Install sleeve-seal systems in sleeves in exterior concrete walls at piping entries into building.

B. Select type, size, and number of sealing elements required for piping material and size and for sleeve ID or hole size. Position piping in center of sleeve. Center piping in penetration, assemble sleeve-seal-system components, and install in annular space between piping and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make a watertight seal.

3.6 SLEEVE-SEAL SCHEDULE

A. Use sleeve and sleeve-seals for the following piping-penetration applications:

1. Exterior Concrete Walls Above Grade: Galvanized-Steel Sheet Pipe Sleeves with Sleeve-seal system

2. Exterior Concrete Walls Below Grade: Galvanized-Steel Sheet Pipe Sleeves with Sleeve-seal system

3. Interior or Exterior Concrete Slabs-on-Grade: Sleeve not required.

4. Interior Concrete Slabs Above Grade: Galvanized-Steel Sheet Pipe Sleeves with Silicone Sealant or Fire calk

5. Interior Partitions: Sleeve not require – fire calk penetrations of rated assemblies.

3.7 ESCUTCHEON INSTALLATION

A. Install escutcheons for piping penetrations of walls, ceilings, and finished floors.

B. Install escutcheons with ID to closely fit around pipe, tube, and insulation of piping and with OD that completely covers opening.

3.8 FLOOR PLATE INSTALLATION

A. Install floor plates for piping penetrations of equipment-room floors.

B. Install floor plates with ID to closely fit around pipe, tube, and insulation of piping and with OD that completely covers opening.

END OF SECTION

SECTION 23 05 48 - VIBRATION AND SEISMIC CONTROLS FOR HVAC PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Elastomeric isolation pads.
2. Elastomeric isolation mounts.
3. Restrained elastomeric isolation mounts.
4. Open-spring isolators.
5. Housed-spring isolators.
6. Restrained-spring isolators.
7. Housed-restrained-spring isolators.
8. Pipe-riser resilient supports.
9. Resilient pipe guides.
10. Elastomeric hangers.
11. Spring hangers.
12. Snubbers.
13. Restraint channel bracings.
14. Restraint cables.
15. Seismic-restraint accessories.
16. Mechanical anchor bolts.

1.2 ACTION SUBMITTALS

A. See Section 23 00 00 “General Requirements for HVAC” for submittal requirements.

B. Delegated-Design Submittal: For each vibration isolation and seismic-restraint device.

1. Include design calculations and details for selecting vibration isolators and seismic restraints complying with performance requirements, design criteria, and analysis data signed and sealed by the qualified professional engineer responsible for their preparation.

1.3 QUALITY ASSURANCE

A. Comply with seismic-restraint requirements in the IBC unless requirements in this Section are more stringent.

B. Welding Qualifications: Qualify procedures and personnel according to AWS D1.1/D1.1M, "Structural Welding Code - Steel."

C. Seismic-restraint devices shall have horizontal and vertical load testing and analysis and shall bear anchorage preapproval OPA number from OSHPD, preapproval by ICC-ES, or preapproval by another agency acceptable to authorities having jurisdiction, showing maximum seismic-restraint ratings. Ratings based on independent testing are preferred to ratings based on calculations. If preapproved ratings are unavailable, submittals based on independent testing are preferred. Calculations (including combining shear and tensile loads) to support seismic-restraint designs must be signed and sealed by a qualified professional engineer.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Seismic-Restraint Loading:

1. Design seismic restraints for components for seismic design forces defined in Chapter 13 of ASCE 7-16.

a. Design Spectral Response Acceleration at Short Periods, See Structural Drawings for Site Specific S_{DS} Value

b. Component Importance Factor, I_P = 1.0; except for components conveying, supporting, or otherwise containing natural gas or other flammable and/or explosive contents, I_P = 1.5.

c. Component Response Modification Factor, R_P: See Table 13.6-1 of ASCE 7-16

d. Component Amplification Factor, a_P: See Table 13.6-1 of ASCE 7-16

2.2 ELASTOMERIC ISOLATION PADS

A. Elastomeric Isolation Pads:

1. Fabrication: Single or multiple layers of sufficient durometer stiffness for uniform loading over pad area.

2. Size: Factory or field cut to match requirements of supported equipment.

3. Pad Material: Oil and water resistant with elastomeric properties.

4. Surface Pattern: Smooth, Ribbed or Waffle pattern.

5. Infused nonwoven cotton or synthetic fibers.

6. Load-bearing metal plates adhered to pads.

2.3 ELASTOMERIC ISOLATION MOUNTS

A. Double-Deflection, Elastomeric Isolation Mounts:

1. Mounting Plates:

a. Top Plate: Encapsulated steel load transfer top plates, factory drilled and threaded with threaded studs or bolts.

b. Baseplate: Encapsulated steel bottom plates with holes provided for anchoring to support structure.

2. Elastomeric Material: Molded, oil-resistant rubber, neoprene, or other elastomeric material.

2.4 RESTRAINED ELASTOMERIC ISOLATION MOUNTS

A. Restrained Elastomeric Isolation Mounts:

1. Description: All-directional isolator with seismic restraints containing two separate and opposing elastomeric elements that prevent central threaded element and attachment hardware from contacting the housing during normal operation.

a. Housing: Cast-ductile iron or welded steel.

b. Elastomeric Material: Molded, oil-resistant rubber, neoprene, or other elastomeric material.

2.5 OPEN-SPRING ISOLATORS

A. Freestanding, Laterally Stable, Open-Spring Isolators:

1. Outside Spring Diameter: Not less than 80 percent of the compressed height of the spring at rated load.

2. Minimum Additional Travel: 50 percent of the required deflection at rated load.

3. Lateral Stiffness: More than 80 percent of rated vertical stiffness.

4. Overload Capacity: Support 200 percent of rated load, fully compressed, without deformation or failure.

5. Baseplates: Factory-drilled steel plate for bolting to structure with an elastomeric isolator pad attached to the underside. Baseplates shall limit floor load to 500 psig (3447 kPa).

6. Top Plate and Adjustment Bolt: Threaded top plate with adjustment bolt and cap screw to fasten and level equipment.

2.6 HOUSED-SPRING ISOLATORS

A. Freestanding, Laterally Stable, Open-Spring Isolators in Two-Part Telescoping Housing:

1. Outside Spring Diameter: Not less than 80 percent of the compressed height of the spring at rated load.

2. Minimum Additional Travel: 50 percent of the required deflection at rated load.

3. Lateral Stiffness: More than 80 percent of rated vertical stiffness.

4. Overload Capacity: Support 200 percent of rated load, fully compressed, without deformation or failure.

5. Two-Part Telescoping Housing: A steel top and bottom frame separated by an elastomeric material and enclosing the spring isolators.

    a. Drilled base housing for bolting to structure with an elastomeric isolator pad attached to the underside. Bases shall limit floor load to 500 psig.

    b. Top housing with attachment and leveling bolt.

2.7 RESTRAINED-SPRING ISOLATORS

A. Freestanding, Laterally Stable, Open-Spring Isolators with Vertical-Limit Stop Restraint:

1. Housing: Steel housing with vertical-limit stops to prevent spring extension due to weight being removed.

    a. Base with holes for bolting to structure with an elastomeric isolator pad attached to the underside. Bases shall limit floor load to 500 psig.

    b. Top plate with threaded mounting holes.

    c. Internal leveling bolt that acts as blocking during installation.

2. Restraint: Limit stop as required for equipment and authorities having jurisdiction.

3. Outside Spring Diameter: Not less than 80 percent of the compressed height of the spring at rated load.

4. Minimum Additional Travel: 50 percent of the required deflection at rated load.

5. Lateral Stiffness: More than 80 percent of rated vertical stiffness.

6. Overload Capacity: Support 200 percent of rated load, fully compressed, without deformation or failure.

2.8 HOUSED-RESTRAINED-SPRING ISOLATORS

A. Freestanding, Steel, Open-Spring Isolators with Vertical-Limit Stop Restraint in Two-Part Telescoping Housing:

1. Two-Part Telescoping Housing: A steel top and bottom frame separated by an elastomeric material and enclosing the spring isolators. Housings are equipped with adjustable snubbers to limit vertical movement.

    a. Drilled base housing for bolting to structure with an elastomeric isolator pad attached to the underside. Bases shall limit floor load to 500 psig.

    b. Threaded top housing with adjustment bolt and cap screw to fasten and level equipment.

2. Outside Spring Diameter: Not less than 80 percent of the compressed height of the spring at rated load.

3. Minimum Additional Travel: 50 percent of the required deflection at rated load.

4. Lateral Stiffness: More than 80 percent of rated vertical stiffness.

5. Overload Capacity: Support 200 percent of rated load, fully compressed, without deformation or failure.

2.9 PIPE-RISER RESILIENT SUPPORT

A. Description: All-directional, acoustical pipe anchor consisting of two steel tubes separated by a minimum 1/2-inch- thick neoprene.

1. Vertical-Limit Stops: Steel and neoprene vertical-limit stops arranged to prevent vertical travel in both directions.

2. Maximum Load Per Support: 500 psig on isolation material providing equal isolation in all directions.

2.10 RESILIENT PIPE GUIDES

A. Description: Telescopic arrangement of two steel tubes or post and sleeve arrangement separated by a minimum 1/2-inch- thick neoprene.

1. Factory-Set Height Guide with Shear Pin: Shear pin shall be removable and reinsertable to allow for selection of pipe movement. Guides shall be capable of motion to meet location requirements.

2.11 ELASTOMERIC HANGERS

A. Elastomeric Mount in a Steel Frame with Upper and Lower Steel Hanger Rods:

1. Frame: Steel, fabricated with a connection for an upper threaded hanger rod and an opening on the underside to allow for a maximum of 30 degrees of angular lower hanger-rod misalignment without binding or reducing isolation efficiency.

2. Dampening Element: Molded, oil-resistant rubber, neoprene, or other elastomeric material with a projecting bushing for the underside opening preventing steel to steel contact.

2.12 SPRING HANGERS

A. Combination Coil-Spring and Elastomeric-Insert Hanger with Spring and Insert in Compression:

1. Frame: Steel, fabricated for connection to threaded hanger rods and to allow for a maximum of 30 degrees of angular hanger-rod misalignment without binding or reducing isolation efficiency.

2. Outside Spring Diameter: Not less than 80 percent of the compressed height of the spring at rated load.

3. Minimum Additional Travel: 50 percent of the required deflection at rated load.

4. Lateral Stiffness: More than 80 percent of rated vertical stiffness.

5. Overload Capacity: Support 200 percent of rated load, fully compressed, without deformation or failure.

6. Elastomeric Element: Molded, oil-resistant rubber or neoprene. Steel-washer-reinforced cup to support spring and bushing projecting through bottom of frame.

7. Adjustable Vertical Stop: Steel washer with neoprene washer "up-stop" on lower threaded rod.

8. Self-centering hanger-rod cap to ensure concentricity between hanger rod and support spring coil.

2.13 SNUBBERS

A. Description: Factory fabricated using welded structural-steel shapes and plates, anchor bolts, and replaceable resilient isolation washers and bushings.

1. Anchor bolts for attaching to concrete shall be seismic-rated, drill-in, and stud-wedge or female-wedge type.

2. Resilient Isolation Washers and Bushings: Oil- and water-resistant neoprene.

3. Maximum 1/4-inch air gap, and minimum 1/4-inch- thick resilient cushion.

2.14 RESTRAINT CHANNEL BRACINGS

A. Description: MFMA-4, shop- or field-fabricated bracing assembly made of slotted steel channels with accessories for attachment to braced component at one end and to building structure at the other end and other matching components and with corrosion-resistant coating; rated in tension, compression, and torsion forces.

2.15 RESTRAINT CABLES

A. Restraint Cables: ASTM A 603 galvanized-steel cables. End connections made of steel assemblies with thimbles, brackets, swivel, and bolts designed for restraining cable service; with a minimum of two clamping bolts for cable engagement.

2.16 SEISMIC-RESTRAINT ACCESSORIES

A. Hanger-Rod Stiffener: Reinforcing steel angle clamped to hanger rod.

B. Bushings for Floor-Mounted Equipment Anchor Bolts: Neoprene bushings designed for rigid equipment mountings, and matched to type and size of anchor bolts and studs.

C. Bushing Assemblies for Wall-Mounted Equipment Anchorage: Assemblies of neoprene elements and steel sleeves designed for rigid equipment mountings, and matched to type and size of attachment devices used.

D. Resilient Isolation Washers and Bushings: One-piece, molded, oil- and water-resistant neoprene, with a flat washer face.

E. Mechanical Anchor Bolts: Drilled-in and stud-wedge or female-wedge type in zinc-coated steel for interior applications and stainless steel for exterior applications. Select anchor bolts with strength required for anchor and as tested according to ASTM E 488.

2.17 EXECUTION

A. components so strength is adequate to carry present and future static and seismic loads within specified loading limits.

2.18 VIBRATION CONTROL AND SEISMIC-RESTRAINT DEVICE INSTALLATION

A. Coordinate the location of embedded connection hardware with supported equipment attachment and mounting points and with requirements for concrete reinforcement and formwork specified in Section 03 30 00 "Cast-in-Place Concrete." or Section 03 30 53 "Miscellaneous Cast-in-Place Concrete."

B. Installation of vibration isolators must not cause any change of position of equipment, piping, or ductwork resulting in stresses or misalignment.

C. Comply with requirements in Section 07 72 00 "Roof Accessories" for installation of roof curbs, equipment supports, and roof penetrations.

D. Equipment Restraints:

1. Install seismic snubbers on HVAC equipment mounted on vibration isolators. Locate snubbers as close as possible to vibration isolators and bolt to equipment base and supporting structure.

2. Install resilient bolt isolation washers on equipment anchor bolts where clearance between anchor and adjacent surface exceeds 0.125 inch.

3. Install seismic-restraint devices using methods approved by an evaluation service member of ICC-ES that provides required submittals for component.

E. Piping Restraints:

1. Comply with requirements in MSS SP-127.

2. Space lateral supports a maximum of 40 feet o.c., and longitudinal supports a maximum of 80 feet o.c.

3. Brace a change of direction longer than 12 feet.

F. Install cables so they do not bend across edges of adjacent equipment or building structure.

G. Install seismic-restraint devices using methods approved by an evaluation service member of ICC-ES that provides required submittals for component.

H. Install bushing assemblies for anchor bolts for floor-mounted equipment, arranged to provide resilient media between anchor bolt and mounting hole in concrete base.

I. Install bushing assemblies for mounting bolts for wall-mounted equipment, arranged to provide resilient media where equipment or equipment-mounting channels are attached to wall.

J. Attachment to Structure: If specific attachment is not indicated, anchor bracing to structure at flanges of beams, at upper truss chords of bar joists, or at concrete members.

K. Drilled-in Anchors:

1. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Do not damage existing reinforcing or embedded items during coring or drilling.

Notify the structural engineer if reinforcing steel or other embedded items are encountered during drilling. Locate and avoid prestressed tendons, electrical and telecommunications conduit, and gas lines.

2. Do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength.

3. Wedge Anchors: Protect threads from damage during anchor installation. Heavy-duty sleeve anchors shall be installed with sleeve fully engaged in the structural element to which anchor is to be fastened.

4. Set anchors to manufacturer's recommended torque, using a torque wrench.

5. Install zinc-coated steel anchors for interior and stainless-steel anchors for exterior applications.

2.19 ACCOMMODATION OF DIFFERENTIAL SEISMIC MOTION

A. Install flexible connections in piping where they cross seismic joints, where adjacent sections or branches are supported by different structural elements, and where the connections terminate with connection to equipment that is anchored to a different structural element from the one supporting the connections as they approach equipment. Comply with requirements in Section 22 11 16 "Domestic Water Piping" for piping flexible connections.

2.20 ADJUSTING

A. Adjust isolators after piping system is at operating weight.

B. Adjust limit stops on restrained-spring isolators to mount equipment at normal operating height. After equipment installation is complete, adjust limit stops so they are out of contact during normal operation.

END OF SECTION

SECTION 23 05 53 - IDENTIFICATION FOR HVAC PIPING AND EQUIPMENT

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Equipment labels.

2. Warning signs and labels.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product indicated.

PART 2 - PRODUCTS

2.1 EQUIPMENT LABELS

A. Plastic Labels for Equipment:

1. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, and having predrilled holes for attachment hardware.

2. Letter Color: White

3. Background Color: Black

4. Maximum Temperature: Able to withstand temperatures up to 160 deg F.

5. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.

6. Minimum Letter Size: 1/4 inch For name of units if viewing distance is less than 24 inches, 1/2 inch for viewing distances up to 72 inches, and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-quarters the size of principal lettering.

7. Fasteners: Stainless-steel rivets or self-tapping screws.

8. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

B. Label Content: Include equipment's Drawing designation or unique equipment number, Drawing numbers where equipment is indicated (plans, details, and schedules), and the Specification Section number and title where equipment is specified.

C. All labels on equipment visible to the public/occupant shall be coordinated with the architect or engineer prior to installation.

2.2 WARNING SIGNS AND LABELS

A. Material and Thickness: Multilayer, multicolor, plastic labels for mechanical engraving, 1/8 inch thick, and having predrilled holes for attachment hardware.

B. Letter Color: White

C. Background Color: Red

D. Maximum Temperature: Able to withstand temperatures up to 160 deg F.

E. Minimum Label Size: Length and width vary for required label content, but not less than 2-1/2 by 3/4 inch.

F. Minimum Letter Size: 1/4 inch (6.4 mm) for name of units if viewing distance is less than 24 inches (600 mm), 1/2 inch (13 mm) for viewing distances up to 72 inches (1830 mm), and proportionately larger lettering for greater viewing distances. Include secondary lettering two-thirds to three-quarters the size of principal lettering.

G. Fasteners: Stainless-steel rivets or self-tapping screws.

H. Adhesive: Contact-type permanent adhesive, compatible with label and with substrate.

I. Label Content: Include caution and warning information plus emergency notification instructions.

PART 3 - EXECUTION

3.1 EQUIPMENT LABEL INSTALLATION

A. Install or permanently fasten labels on each major item of mechanical equipment.

B. Locate equipment labels where accessible and visible.

3.2 PIPE LABEL INSTALLATION

A. Pipe Label Locations: Locate pipe labels where piping is exposed or above accessible ceilings in finished spaces; machine rooms; accessible maintenance spaces such as shafts, tunnels, and plenums; and exterior exposed locations as follows:

1. Near each valve and control device.

2. Near each branch connection, excluding short takeoffs for fixtures and terminal units. Where flow pattern is not obvious, mark each pipe at branch.

3. Near penetrations and on both sides of through walls, floors, ceilings, and inaccessible enclosures.

4. At access doors, manholes, and similar access points that permit view of concealed piping.

5. Near major equipment items and other points of origination and termination.

6. Spaced at maximum intervals of 50 feet along each run. Reduce intervals to 25 feet in areas of congested piping and equipment.

7. On piping above removable acoustical ceilings. Omit intermediately spaced labels.

3.3 VALVE-TAG INSTALLATION

A. Install tags on valves and control devices in piping systems, except check valves; valves within factory-fabricated equipment units; plumbing fixture supply stops; shutoff valves; faucets;

convenience and lawn-watering hose connections; and HVAC terminal devices and similar roughing-in connections of end-use fixtures and units. List tagged valves in a valve schedule.

1. Valve-Tag Size and Shape: 1-1/2 inches.

2. Valve-Tag Color: Natural Brass or Aluminum.

3.4 VALVE-SCHEDULE INSTALLATION

A. Mount valve schedule on wall in accessible location in each major equipment room.

3.5 CLEANING

A. Clean faces of mechanical identification devices and glass frames of valve schedules.

END OF SECTION

SECTION 23 07 13 - DUCT INSULATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes insulating the following duct services:

1. Indoor, Duct Insulation.

B. Related Sections:

1. Section 23 07 16 "HVAC Equipment and Piping Insulation."

2. Section 23 31 13 "Metal Ducts" for duct liners.

1.2 ACTION SUBMITTALS

A. See Section 23 00 00 “General Requirements of HVAC” for submittal requirements.

B. Provide floor plan shop drawings showing intended locations of insulation. Exposed ductwork that is to not be insulated shall be specifically noted.

1.3 INFORMATIONAL SUBMITTALS

A. Field quality-control reports.

1.4 QUALITY ASSURANCE

A. Surface-Burning Characteristics: For insulation and related materials, as determined by testing identical products according to ASTM E 84, by a testing agency acceptable to authorities having jurisdiction. Factory label insulation and jacket materials and adhesive, mastic, tapes, and cement material containers, with appropriate markings of applicable testing agency.

1. Insulation Installed Indoors: Flame-spread index of 25 or less, and smoke-developed index of 50 or less.

2. Insulation Installed Outdoors: Flame-spread index of 75 or less, and smoke-developed index of 150 or less.

PART 2 - PRODUCTS

2.1 INSULATION MATERIALS

A. Comply with requirements in "Duct Insulation Schedule, General," "Indoor Duct and Plenum Insulation Schedule," and "Aboveground, Outdoor Duct and Plenum Insulation Schedule" articles for where insulating materials shall be applied.

B. Products shall not contain asbestos, lead, mercury, or mercury compounds.

C. Products that come in contact with stainless steel shall have a leachable chloride content of less than 50 ppm when tested according to ASTM C 871.

D. Insulation materials for use on austenitic stainless steel shall be qualified as acceptable according to ASTM C 795.

E. Foam insulation materials shall not use CFC or HCFC blowing agents in the manufacturing process.

F. Mineral-Fiber Blanket Insulation: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C 553, Type II and ASTM C 1290, Type III with factory-applied FSK jacket. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. CertainTeed Corporation.

b. Johns Manville; a Berkshire Hathaway company.

c. Knauf Insulation.

d. Owens Corning.

G. Jacketed Mineral-Fiber Board Insulation: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C 612, Type IA or Type IB. For duct and plenum applications, provide insulation with factory-applied ASJ. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. CertainTeed Corporation.

b. Johns Manville; a Berkshire Hathaway company.

c. Knauf Insulation.

d. Owens Corning.

H. Mineral-Fiber Board Insulation: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C 612, Type IA or Type IB. For duct and plenum applications, provide insulation without factory-applied jacket. Factory-applied jacket requirements are specified in "Factory-Applied Jackets" Article.

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. CertainTeed Corporation.

b. Johns Manville; a Berkshire Hathaway company.

c. Knauf Insulation.

d. Owens Corning.

2.2 ADHESIVES

A. Materials shall be compatible with insulation materials, jackets, and substrates and for bonding insulation to itself and to surfaces to be insulated unless otherwise indicated.

B. Mineral-Fiber Adhesive: Comply with MIL-A-3316C, Class 2, Grade A.

C. ASJ Adhesive, and FSK Jacket Adhesive: Comply with MIL-A-3316C, Class 2, Grade A for bonding insulation jacket lap seams and joints.

2.3 MASTICS

A. Materials shall be compatible with insulation materials, jackets, and substrates; comply with MIL-PRF-19565C, Type II.

B. Vapor-Barrier Mastic: Water based; suitable for indoor use on below ambient services.

1. Water-Vapor Permeance: ASTM E 96/E 96M, Procedure B, 0.013 perm at 43-mil dry film thickness.

2. Service Temperature Range: Minus 20 to plus 180 deg F.

3. Solids Content: ASTM D 1644, 58 percent by volume and 70 percent by weight.

4. Color: White.

C. Breather Mastic: Water based; suitable for indoor and outdoor use on above ambient services.

1. Water-Vapor Permeance: ASTM F 1249, 1.8 perms at 0.0625-inch dry film thickness.

2. Service Temperature Range: Minus 20 to plus 180 deg F.

3. Solids Content: 60 percent by volume and 66 percent by weight.

4. Color: White.

2.4 SEALANTS

A. FSK and Metal Jacket Flashing Sealants:

1. Materials shall be compatible with insulation materials, jackets, and substrates.

2. Fire- and water-resistant, flexible, elastomeric sealant.

3. Service Temperature Range: Minus 40 to plus 250 deg F.

4. Color: Aluminum.

B. ASJ Flashing Sealants, and Vinyl and PVC Jacket Flashing Sealants:

1. Materials shall be compatible with insulation materials, jackets, and substrates.

2. Fire- and water-resistant, flexible, elastomeric sealant.

3. Service Temperature Range: Minus 40 to plus 250 deg F.

4. Color: White.

2.5 FACTORY-APPLIED JACKETS

A. Insulation system schedules indicate factory-applied jackets on various applications. When factory-applied jackets are indicated, comply with the following:

1. ASJ: White, kraft-paper, fiberglass-reinforced scrim with aluminum-foil backing; complying with ASTM C 1136, Type I.

2. ASJ-SSL: ASJ with self-sealing, pressure-sensitive, acrylic-based adhesive covered by a removable protective strip; complying with ASTM C 1136, Type I.

3. FSK Jacket: Aluminum-foil, fiberglass-reinforced scrim with kraft-paper backing; complying with ASTM C 1136, Type II.

2.6 FIELD-APPLIED JACKETS

A. Self-Adhesive Outdoor Jacket: 60-mil-thick, laminated vapor barrier and waterproofing membrane for installation over insulation located aboveground outdoors; consisting a rubberized bitumen compound; heat applied to a multi-ply embossed UV-resistant aluminum foil/polymer laminate, and polyester/foil multiple layer laminate with acrylic adhesive.

1. Manufacturers: Subject to compliance with requirements, provide products by the following:

a. Polyguard Products, Inc.

b. Prior Approved Equal

2.7 TAPES

A. ASJ Tape: White vapor-retarder tape matching factory-applied jacket with acrylic adhesive, complying with ASTM C 1136.

1. Width: 3 inches.

2. Thickness: 11.5 mils.

3. Adhesion: 90 ounces force/inch in width.

4. Elongation: 2 percent.

5. Tensile Strength: 40 lbf/inch in width.

6. ASJ Tape Disks and Squares: Precut disks or squares of ASJ tape.

B. FSK Tape: Foil-face, vapor-retarder tape matching factory-applied jacket with acrylic adhesive; complying with ASTM C 1136.

1. Width: 3 inches.

2. Thickness: 6.5 mils.

3. Adhesion: 90 ounces force/inch in width.

4. Elongation: 2 percent.

5. Tensile Strength: 40 lbf/inch in width.

6. FSK Tape Disks and Squares: Precut disks or squares of FSK tape.

C. Aluminum-Foil Tape: Vapor-retarder tape with acrylic adhesive.

1. Width: 2 inches.

2. Thickness: 3.7 mils.

3. Adhesion: 100 ounces force/inch in width.

4. Elongation: 5 percent.

5. Tensile Strength: 34 lbf/inch in width.

2.8 SECUREMENTS

A. Cupped Head Weld Pins:

1. Material: Low carbon steel.

2. Finish: Copper coated pins with galvanized washer

3. Pin gauge: 12 Ga.

B. Staples: Outward-clinching insulation staples, nominal 3/4-inch-wide, stainless steel or Monel.

C. Wire: 0.080-inch nickel-copper alloy.

PART 3 - EXECUTION

3.1 PREPARATION

A. Surface Preparation: Clean and dry surfaces to receive insulation. Remove materials that will adversely affect insulation application.

3.2 GENERAL INSTALLATION REQUIREMENTS

A. Install insulation materials, accessories, and finishes with smooth, straight, and even surfaces; free of voids throughout the length of ducts and fittings.

B. Install insulation materials, vapor barriers or retarders, jackets, and thicknesses required for each item of duct system as specified in insulation system schedules.

C. Install accessories compatible with insulation materials and suitable for the service. Install accessories that do not corrode, soften, or otherwise attack insulation or jacket in either wet or dry state.

D. Install insulation with longitudinal seams at top and bottom of horizontal runs.

E. Install multiple layers of insulation with longitudinal and end seams staggered.

F. Keep insulation materials dry during application and finishing.

G. Install insulation with tight longitudinal seams and end joints. Bond seams and joints with adhesive recommended by insulation material manufacturer.

H. Install insulation with least number of joints practical.

I. Where vapor barrier is indicated, seal joints, seams, and penetrations in insulation at hangers, supports, anchors, and other projections with vapor-barrier mastic.

1. Install insulation continuously through hangers and around anchor attachments.

2. For insulation application where vapor barriers are indicated, extend insulation on anchor legs from point of attachment to supported item to point of attachment to structure. Taper and seal ends at attachment to structure with vapor-barrier mastic.

3. Install insert materials and install insulation to tightly join the insert. Seal insulation to insulation inserts with adhesive or sealing compound recommended by insulation material manufacturer.

J. Apply adhesives, mastics, and sealants at manufacturer's recommended coverage rate and wet and dry film thicknesses.

K. Install insulation with factory-applied jackets as follows:

1. Draw jacket tight and smooth.

2. Cover circumferential joints with 3-inch-wide strips, of same material as insulation jacket. Secure strips with adhesive and outward clinching staples along both edges of strip, spaced 4 inches o.c.

3. Overlap jacket longitudinal seams at least 1-1/2 inches. Clean and dry surface to receive self-sealing lap. Staple laps with outward clinching staples along edge at 4 inches o.c.

a. For below ambient services, apply vapor-barrier mastic over staples.

4. Cover joints and seams with tape, according to insulation material manufacturer's written instructions, to maintain vapor seal.

5. Where vapor barriers are indicated, apply vapor-barrier mastic on seams and joints and at ends adjacent to duct flanges and fittings.

L. Cut insulation in a manner to avoid compressing insulation more than 75 percent of its nominal thickness.

M. Finish installation with systems at operating conditions. Repair joint separations and cracking due to thermal movement.

N. Repair damaged insulation facings by applying same facing material over damaged areas. Extend patches at least 4 inches beyond damaged areas. Adhere, staple, and seal patches similar to butt joints.

3.3 PENETRATIONS

A. Insulation Installation at Roof Penetrations: Install insulation continuously through roof penetrations.

1. Seal penetrations with flashing sealant.

2. For applications requiring only indoor insulation, terminate insulation above roof surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.

3. Extend jacket of outdoor insulation outside roof flashing at least 2 inches below top of roof flashing.

4. Seal jacket to roof flashing with flashing sealant.

B. Insulation Installation at Aboveground Exterior Wall Penetrations: Install insulation continuously through wall penetrations.

1. Seal penetrations with flashing sealant.

2. For applications requiring only indoor insulation, terminate insulation inside wall surface and seal with joint sealant. For applications requiring indoor and outdoor insulation, install insulation for outdoor applications tightly joined to indoor insulation ends. Seal joint with joint sealant.

3. Extend jacket of outdoor insulation outside wall flashing and overlap wall flashing at least 2 inches.

4. Seal jacket to wall flashing with flashing sealant.

C. Insulation Installation at Interior Wall and Partition Penetrations (That Are Not Fire Rated): Install insulation continuously through walls and partitions.

D. Insulation Installation at Fire-Rated Wall and Partition Penetrations: Terminate insulation at fire damper sleeves for fire-rated wall and partition penetrations. Externally insulate damper sleeves to match adjacent insulation and overlap duct insulation at least 2 inches.

1. Comply with requirements in Section 07 84 13 "Penetration Firestopping" for firestopping and fire-resistive joint sealers.

E. Insulation Installation at Floor Penetrations:

1. Duct: For penetrations through fire-rated assemblies, terminate insulation at fire damper sleeves and externally insulate damper sleeve beyond floor to match adjacent duct insulation. Overlap damper sleeve and duct insulation at least 2 inches.

2. Seal penetrations through fire-rated assemblies. Comply with requirements in Section 07 84 13 "Penetration Firestopping."

3.4 INSTALLATION OF MINERAL-FIBER INSULATION

A. Blanket Insulation Installation on Ducts and Plenums: Secure with adhesive and insulation pins.

1. Apply adhesives according to manufacturer's recommended coverage rates per unit area, for 100 percent coverage of duct and plenum surfaces.

2. Apply adhesive to entire circumference of ducts and to all surfaces of fittings and transitions.

3. Install cupped-head, capacitor-discharge-weld pins on sides and bottom of horizontal ducts and sides of vertical ducts as follows:

   a. On duct sides with dimensions 18 inches and smaller, place pins along longitudinal centerline of duct. Space 3 inches maximum from insulation end joints, and 16 inches o.c.

   b. On duct sides with dimensions larger than 18 inches, place pins 16 inches o.c. each way, and 3 inches maximum from insulation joints. Install additional pins to hold insulation tightly against surface at cross bracing.

   c. Pins may be omitted from top surface of horizontal, rectangular ducts and plenums.

   d. Do not overcompress insulation during installation.

   e. Impale insulation over pins and attach speed washers.

   f. Cut excess portion of pins extending beyond speed washers or bend parallel with insulation surface. Cover exposed pins and washers with tape matching insulation facing.

4. For ducts and plenums with surface temperatures below ambient, install a continuous unbroken vapor barrier. Create a facing lap for longitudinal seams and end joints with insulation by removing 2 inches from one edge and one end of insulation segment. Secure laps to adjacent insulation section with 1/2-inch outward-clinching staples, 1 inch o.c. Install vapor barrier consisting of factory- or field-applied jacket, adhesive, vapor-barrier mastic, and sealant at joints, seams, and protrusions.

   a. Repair punctures, tears, and penetrations with tape or mastic to maintain vapor-barrier seal.

   b. Install vapor stops for ductwork and plenums operating below 50 deg F at 18-foot intervals. Vapor stops shall consist of vapor-barrier mastic applied in a Z-shaped pattern over insulation face, along butt end of insulation, and over the surface. Cover insulation face and surface to be insulated a width equal to two times the insulation thickness, but not less than 3 inches.

5. Overlap unfaced blankets a minimum of 2 inches on longitudinal seams and end joints. At end joints, secure with steel bands spaced a maximum of 18 inches o.c.

6. Install insulation on rectangular duct elbows and transitions with a full insulation section for each surface. Install insulation on round and flat-oval duct elbows with individually mitered gores cut to fit the elbow.

7. Insulate duct stiffeners, hangers, and flanges that protrude beyond insulation surface with 6-inch-wide strips of same material used to insulate duct. Secure on alternating sides of stiffener, hanger, and flange with pins spaced 6 inches o.c.

B. Board Insulation Installation on Ducts and Plenums: Secure with adhesive and insulation pins.

1. Apply adhesives according to manufacturer's recommended coverage rates per unit area, for 100 percent coverage of duct and plenum surfaces.

2. Apply adhesive to entire circumference of ducts and to all surfaces of fittings and transitions.

3. Install either capacitor-discharge-weld pins and speed washers or cupped-head, capacitor-discharge-weld pins on sides and bottom of horizontal ducts and sides of vertical ducts as follows:

    a. On duct sides with dimensions 18 inches and smaller, place pins along longitudinal centerline of duct. Space 3 inches maximum from insulation end joints, and 16 inches o.c.

    b. On duct sides with dimensions larger than 18 inches, space pins 16 inches o.c. each way, and 3 inches maximum from insulation joints. Install additional pins to hold insulation tightly against surface at cross bracing.

    c. Pins may be omitted from top surface of horizontal, rectangular ducts and plenums.

    d. Do not overcompress insulation during installation.

    e. Cut excess portion of pins extending beyond speed washers or bend parallel with insulation surface. Cover exposed pins and washers with tape matching insulation facing.

4. For ducts and plenums with surface temperatures below ambient, install a continuous unbroken vapor barrier. Create a facing lap for longitudinal seams and end joints with insulation by removing 2 inches from one edge and one end of insulation segment. Secure laps to adjacent insulation section with 1/2-inch outward-clinching staples, 1 inch o.c. Install vapor barrier consisting of factory- or field-applied jacket, adhesive, vapor-barrier mastic, and sealant at joints, seams, and protrusions.

    a. Repair punctures, tears, and penetrations with tape or mastic to maintain vapor-barrier seal.

    b. Install vapor stops for ductwork and plenums operating below 50 deg F at 18-foot intervals. Vapor stops shall consist of vapor-barrier mastic applied in a Z-shaped pattern over insulation face, along butt end of insulation, and over the surface. Cover insulation face and surface to be insulated a width equal to two times the insulation thickness, but not less than 3 inches.

5. Install insulation on rectangular duct elbows and transitions with a full insulation section for each surface. Groove and score insulation to fit as closely as possible to outside and inside radius of elbows. Install insulation on round and flat-oval duct elbows with individually mitered gores cut to fit the elbow.

6. Insulate duct stiffeners, hangers, and flanges that protrude beyond insulation surface with 6-inch-wide strips of same material used to insulate duct. Secure on alternating sides of stiffener, hanger, and flange with pins spaced 6 inches o.c.

3.5 FINISHES

A. Insulation with ASJ or Other Paintable Jacket Material: Paint jacket with paint system identified below and as specified in Section 09 91 13 "Exterior Painting" and Section 09 91 23 "Interior Painting."

B. Color: Final color as selected by Architect. Vary first and second coats to allow visual inspection of the completed Work.

C. Do not field paint outdoor ductwork.

3.6 FIELD QUALITY CONTROL

A. Perform tests and inspections.

B. Tests and Inspections:

1. Inspect ductwork, randomly selected by engineer, by removing field-applied jacket and insulation in layers in reverse order of their installation. Extent of inspection shall be limited to one location(s) for each duct system defined in the "Duct Insulation Schedule, General" Article.

C. All insulation applications will be considered defective Work if sample inspection reveals noncompliance with requirements.

3.7 DUCT INSULATION SCHEDULE, GENERAL

A. Duct insulation shall not be installed on indoor supply or return ductwork that is exposed to view in a normally occupied and conditioned space unless otherwise indicated on the drawings.

B. Insulation materials and thicknesses for ductwork are identified in the table below. If more than one material is listed for an application, selection from materials listed is at the Contractor's option. Ductwork that is not listed below or is exposed to view shall not be insulated.

Location	Application	Insulation Type	Installed R-Value **	Vapor Barrier	Factory Installed Jacket Type
Indoor	Supply	Mineral-Fiber Blanket	6	YES	FSK
Indoor	Exhaust*	Mineral-Fiber Blanket	12	YES	FSK
Indoor	Type I Kitchen Exhaust	Fire-rated Insulation Systems	NA	NO	FSK
Indoor	Outdoor Air	Mineral-Fiber Board	12	YES	FSK

Outdoor	Supply & Return	Mineral-Fiber Board	12	YES	FSK
Unconditioned Space	Supply, Return & Exhaust	Mineral-Fiber Board	12	YES	FSK

*Indoor Exhaust Ductwork shall be insulated from the penetration of the building envelope to 10ft upstream of a backdraft of shutoff damper.

** All above values are a minimum and shall be superseded by more stringent currently adopted Energy or Mechanical Code.

3.8 FIELD APPLIED JACKETING SCHEDULE

A. Field applied jackets for HVAC piping are identified in the table below. If more than one material is listed for an application, selection from materials listed is at the Contractor's option.

Application	Location	Field Applied Jacketing
Supply	Outdoor	Self-Adhesive Outdoor Jacket
Return	Outdoor	Self-Adhesive Outdoor Jacket

END OF SECTION

SECTION 23 31 13 - METAL DUCTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Rectangular ducts and fittings.

2. Round ducts and fittings.

3. Sheet metal materials.

4. Sealants and gaskets.

5. Acoustic Liner.

6. Hangers and supports.

7. Seismic-restraint devices.

B. Related Sections:

1. Section 23 05 29 “Hangars and Supports for HVAC Piping and Equipment.”

2. Section 23 05 48 “Vibration and Seismic Controls for HVAC Piping and Equipment”

3. Section 23 05 93 "Testing, Adjusting, and Balancing for HVAC" for testing, adjusting, and balancing requirements for metal ducts.

4. Section 23 33 00 "Air Duct Accessories" for dampers, sound-control devices, duct-mounting access doors and panels, turning vanes, and flexible ducts.

C. Airstream Surfaces: Surfaces in contact with the airstream shall comply with requirements in ANSI/ASHRAE 62.1.

1.2 SUBMITTALS

A. See Section 23 00 00 “General Requirements of HVAC” for submittal requirements.

1.3 QUALITY ASSURANCE

A. ASHRAE Compliance: Applicable requirements in ASHRAE 62.1, Section 5 - "Systems and Equipment" and Section 7 - "Construction and System Start-up."

B. ASHRAE/IES Compliance: Applicable requirements in ASHRAE/IES 90.1, Section 6.4.4 - "HVAC System Construction and Insulation."

PART 2 - PRODUCTS

2.1 RECTANGULAR DUCTS AND FITTINGS

A. General Fabrication Requirements: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible" based on indicated static-pressure class unless otherwise indicated.

B. Transverse Joints: Select joint types and fabricate according to SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 2-1, "Rectangular Duct/Transverse Joints," for static-pressure class, applicable sealing requirements, materials involved, duct-support intervals, and other provisions in SMACNA's "HVAC Duct Construction Standards - Metal and Flexible."

C. Longitudinal Seams: Select seam types and fabricate according to SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 2-2, "Rectangular Duct/Longitudinal Seams," for static-pressure class, applicable sealing requirements, materials involved, duct-support intervals, and other provisions in SMACNA's "HVAC Duct Construction Standards - Metal and Flexible."

D. Elbows, Transitions, Offsets, Branch Connections, and Other Duct Construction: Select types and fabricate according to SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Chapter 4, "Fittings and Other Construction," for static-pressure class, applicable sealing requirements, materials involved, duct-support intervals, and other provisions in SMACNA's "HVAC Duct Construction Standards - Metal and Flexible."

2.2 ROUND DUCTS AND FITTINGS

A. General Fabrication Requirements: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Chapter 3, "Round, Oval, and Flexible Duct," based on indicated static-pressure class unless otherwise indicated.

B. Transverse Joints: Select joint types and fabricate according to SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 3-1, "Round Duct Transverse Joints," for static-pressure class, applicable sealing requirements, materials involved, duct-support intervals, and other provisions in SMACNA's "HVAC Duct Construction Standards - Metal and Flexible."

1. Transverse Joints in Ducts Larger Than 24 in Diameter: Flanged.

C. Tees and Laterals: Select types and fabricate according to SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 3-5, "90 Degree Tees and Laterals," and Figure 3-6, "Conical Tees," for static-pressure class, applicable sealing requirements, materials involved, duct-support intervals, and other provisions in SMACNA's "HVAC Duct Construction Standards - Metal and Flexible."

2.3 SHEET METAL MATERIALS

A. General Material Requirements: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible" for acceptable materials, material thicknesses, and duct construction methods unless otherwise indicated. Sheet metal materials shall be free of pitting, seam marks, roller marks, stains, discolorations, and other imperfections.

B. Galvanized Sheet Steel: Comply with ASTM A 653/A 653M.

1. Galvanized Coating Designation: G90.

2. Finishes for Surfaces Exposed to View: Mill phosphatized.

C. Carbon-Steel Sheets: Comply with ASTM A 1008/A 1008M, with oiled, matte finish for exposed ducts.

D. Stainless-Steel Sheets: Comply with ASTM A 480/A 480M, Type 304 or 316, as indicated in the "Duct Schedule" Article; cold rolled, annealed, sheet. Exposed surface finish shall be No. 2B, No. 2D, No. 3, or No. 4 as indicated in the "Duct Schedule" Article.

E. Aluminum Sheets: Comply with ASTM B 209 Alloy 3003, H14 temper; with mill finish for concealed ducts, and standard, one-side bright finish for duct surfaces exposed to view.

F. Reinforcement Shapes and Plates: ASTM A 36/A 36M, steel plates, shapes, and bars; black and galvanized.

1. Where black- and galvanized-steel shapes and plates are used to reinforce aluminum ducts, isolate the different metals with butyl rubber, neoprene, or EPDM gasket materials.

G. Tie Rods: Galvanized steel, 1/4-inch minimum diameter for lengths 36 inches or less; 3/8-inch minimum diameter for lengths longer than 36 inches.

2.4 SEALANT AND GASKETS

A. General Sealant and Gasket Requirements: Surface-burning characteristics for sealants and gaskets shall be a maximum flame-spread index of 25 and a maximum smoke-developed index of 50 when tested according to UL 723; certified by an NRTL.

B. Water-Based Joint and Seam Sealant:

1. Application Method: Brush on.

2. Solids Content: Minimum 65 percent.

3. Shore A Hardness: Minimum 20.

4. Water resistant.

5. Mold and mildew resistant.

6. VOC: Maximum 75 g/L (less water).

7. Maximum Static-Pressure Class: 10-inch wg, positive and negative.

8. Service: Indoor or outdoor.

9. Substrate: Compatible with galvanized sheet steel (both PVC coated and bare), stainless steel, or aluminum sheets.

C. Flanged Joint Sealant: Comply with ASTM C 920.

1. General: Single-component, acid-curing, silicone, elastomeric.

2. Type: S.

3. Grade: NS.

4. Class: 25.

5. Use: O.

6. For indoor applications, sealant shall have a VOC content of 250 g/L or less when calculated according to 40 CFR 59, Subpart D (EPA Method 24).

7. Sealant shall comply with the testing and product requirements of the California Department of Health Services' "Standard Practice for the Testing of Volatile Organic Emissions from Various Sources Using Small-Scale Environmental Chambers."

D. Flange Gaskets: Butyl rubber, neoprene, or EPDM polymer with polyisobutylene plasticizer.

E. Round Duct Joint O-Ring Seals:

1. Seal shall provide maximum leakage class of 3 cfm/100 sq. ft. at 1-inch wg and shall be rated for10-inch wg static-pressure class, positive or negative.

2. EPDM O-ring to seal in concave bead in coupling or fitting spigot.

3. Double-lipped, EPDM O-ring seal, mechanically fastened to factory-fabricated couplings and fitting spigots.

2.5 ACOUSTIC LINER

A. General: The liner shall meet the Life Safety Standards as established by NFPA 90A and 90B, FHC 25/50 and Limited Combustibility and the airstream surface coating should contain an immobilized, EPA-registered, antimicrobial agent so it will not support microbial growth as tested in accordance with ASTM G 21 and G 22.

B. Performance: The duct liner shall conform to the requirements of ASTM C 1071, with an NRC not less than 0.75 as tested per ASTM C 423 using a Type “A” mounting, and a thermal conductivity no higher than .25 Btu•in/(hr•ft2•°F) at 75° mean temperature.

C. Fibrous-Glass Duct Liner: Comply with ASTM C1071 and with NAIMA AH124, "Fibrous Glass Duct Liner Standard."

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. CertainTeed LLC; Saint-Gobain North America.

b. Johns Manville; a Berkshire Hathaway company.

c. Knauf Insulation.

d. Owens Corning.

2. Maximum Thermal Conductivity:

a. Type I, Flexible: 0.27 Btu x in./h x sq. ft. x deg F at 75 deg F mean temperature.

b. Type II, Rigid: 0.23 Btu x in./h x sq. ft. x deg F at 75 deg F mean temperature.

3. Liner Adhesive: Comply with NFPA 90A or NFPA 90B and with ASTM C916.

a. Adhesive shall have a VOC content of 80 g/L or less.

2.6 HANGERS AND SUPPORTS

A. Hanger Rods for Noncorrosive Environments: Cadmium-plated steel rods and nuts.

B. Hanger Rods for Corrosive Environments: Electrogalvanized, all-thread rods or galvanized rods with threads painted with zinc-chromate primer after installation.

C. Strap and Rod Sizes: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Table 5-1, "Rectangular Duct Hangers Minimum Size," and Table 5-2, "Minimum Hanger Sizes for Round Duct."

D. Steel Cables for Galvanized-Steel Ducts: Galvanized steel complying with ASTM A 603.

E. Steel Cables for Stainless-Steel Ducts: Stainless steel complying with ASTM A 492.

F. Steel Cable End Connections: Cadmium-plated steel assemblies with brackets, swivel, and bolts designed for duct hanger service; with an automatic-locking and clamping device.

G. Duct Attachments: Sheet metal screws, blind rivets, or self-tapping metal screws; compatible with duct materials.

H. Trapeze and Riser Supports:

1. Supports for Galvanized-Steel Ducts: Galvanized-steel shapes and plates.

2. Supports for Stainless-Steel Ducts: Stainless-steel shapes and plates.

3. Supports for Aluminum Ducts: Aluminum or galvanized steel coated with zinc chromate.

PART 3 - EXECUTION

3.1 DUCT INSTALLATION

A. Drawing plans, schematics, and diagrams indicate general location and arrangement of duct system. Indicated duct locations, configurations, and arrangements were used to size ducts and calculate friction loss for air-handling equipment sizing and for other design considerations. Install duct systems as indicated unless deviations to layout are approved on Shop Drawings and Coordination Drawings.

B. Install ducts according to SMACNA's "HVAC Duct Construction Standards - Metal and Flexible" unless otherwise indicated.

C. Install ducts in maximum practical lengths.

D. Install ducts with fewest possible joints.

E. Install factory- or shop-fabricated fittings for changes in direction, size, and shape and for branch connections.

F. Unless otherwise indicated, install ducts vertically and horizontally, and parallel and perpendicular to building lines.

G. Install ducts close to walls, overhead construction, columns, and other structural and permanent enclosure elements of building.

H. Install ducts with a clearance of 1 inch, plus allowance for insulation thickness.

I. Route ducts to avoid passing through transformer vaults and electrical equipment rooms and enclosures.

J. Where ducts pass through non-fire-rated interior partitions and exterior walls and are exposed to view, cover the opening between the partition and duct or duct insulation with sheet metal flanges of same metal thickness as the duct. Overlap openings on four sides by at least 1-1/2 inches.

K. Where ducts pass through fire-rated interior partitions and exterior walls, install fire dampers. Comply with requirements in Section 23 33 00 "Air Duct Accessories" for fire and smoke dampers.

L. Protect duct interiors from moisture, construction debris and dust, and other foreign materials. Comply with SMACNA's "IAQ Guidelines for Occupied Buildings Under Construction," Appendix G, "Duct Cleanliness for New Construction Guidelines."

3.2 INSTALLATION OF EXPOSED DUCTWORK

A. Protect ducts exposed in finished spaces from being dented, scratched, or damaged.

B. Trim duct sealants flush with metal. Create a smooth and uniform exposed bead. Do not use two-part tape sealing system.

C. Grind welds to provide smooth surface free of burrs, sharp edges, and weld splatter. When welding stainless steel with a No. 3 or 4 finish, grind the welds flush, polish the exposed welds, and treat the welds to remove discoloration caused by welding.

D. Maintain consistency, symmetry, and uniformity in the arrangement and fabrication of fittings, hangers and supports, duct accessories, and air outlets.

E. Repair or replace damaged sections and finished work that does not comply with these requirements.

3.3 DUCT SEALING

A. Seal ducts for duct static-pressure, seal classes, and leakage classes specified in "Duct Schedule" Article according to SMACNA's "HVAC Duct Construction Standards - Metal and Flexible."

B. Seal ducts at a minimum to the following seal classes according to SMACNA's "HVAC Duct Construction Standards - Metal and Flexible":

1. Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible."

2. Outdoor, Supply-Air Ducts: Seal Class A.

3. Outdoor, Exhaust Ducts: Seal Class C.

4. Outdoor, Return-Air Ducts: Seal Class C.

5. Unconditioned Space, Supply-Air Ducts in Pressure Classes 2-Inch wg and Lower: Seal Class B.

6. Unconditioned Space, Supply-Air Ducts in Pressure Classes Higher Than 2-Inch wg: Seal Class A.

7. Unconditioned Space, Exhaust Ducts: Seal Class C.

8. Unconditioned Space, Return-Air Ducts: Seal Class B.

9. Conditioned Space, Supply-Air Ducts in Pressure Classes 2-Inch wg and Lower: Seal Class C.

10. Conditioned Space, Supply-Air Ducts in Pressure Classes Higher Than 2-Inch wg: Seal Class B.

11. Conditioned Space, Exhaust Ducts: Seal Class B.

12. Conditioned Space, Return-Air Ducts: Seal Class C.

3.4 INSTALLATION OF ACOUTIC LINER

A. Liner shall be adhered to the sheet metal with full coverage of an approved adhesive that conforms to ASTM C 916, and all exposed leading edges and transverse joints shall be coated with an approved adhesive and shall be neatly butted without gaps. Shop or field cuts shall be liberally coated with an approved adhesive.

B. Metal nosings shall be securely installed over transversely oriented liner edges facing the airstream at forward discharge and at any point where lined duct is preceded by unlined duct.

C. Acoustic liner shall be additionally secured with mechanical fasteners spaced per the manufacturer’s recommendations. The pin length should be such as to hold the material firmly in place with minimum compression of the material.

D. Acoustic liner shall be installed in ductwork as indicated on the plans. Additional acoustic liner shall be installed as indicated below:

1. Acoustic liner shall be installed in rectangular supply, return and exhaust ductwork within 10’ of any fan or fan powered equipment.

E. The dimensions of ductwork listed on the plans indicates the clear width and height and has not been increased to accommodate indicated liner. Duct sizes shall be increased to accommodate the thickness of the liner.

3.5 HANGER AND SUPPORT INSTALLATION

A. Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Chapter 5, "Hangers and Supports."

B. Building Attachments: Concrete inserts, powder-actuated fasteners, or structural-steel fasteners appropriate for construction materials to which hangers are being attached.

1. Where practical, install concrete inserts before placing concrete.

2. Install powder-actuated concrete fasteners after concrete is placed and completely cured.

3. Use powder-actuated concrete fasteners for standard-weight aggregate concretes or for slabs more than 4 inches thick.

4. Do not use powder-actuated concrete fasteners for lightweight-aggregate concretes or for slabs less than 4 inches thick.

5. Do not use powder-actuated concrete fasteners for seismic restraints.

C. Hanger Spacing: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Table 5-1, "Rectangular Duct Hangers Minimum Size," and Table 5-2, "Minimum Hanger Sizes for Round Duct," for maximum hanger spacing; install hangers and supports within 24 inches of each elbow and within 48 inches of each branch intersection.

D. Hangers Exposed to View: Threaded rod and angle or channel supports.

E. Support vertical ducts with steel angles or channel secured to the sides of the duct with welds, bolts, sheet metal screws, or blind rivets; support at each floor and at a maximum intervals of 16 feet.

F. Install upper attachments to structures. Select and size upper attachments with pull-out, tension, and shear capacities appropriate for supported loads and building materials where used.

3.6 SEISMIC-RESTRAINT-DEVICE INSTALLATION

A. Install ducts with hangers and braces designed to support the duct and to restrain against seismic forces required by applicable building codes. Comply with SMACNA's "Seismic Restraint Manual: Guidelines for Mechanical Systems." And ASCE/SEI 7.

3.7 CONNECTIONS

A. Make connections to equipment with flexible connectors complying with Section 23 33 00 "Air Duct Accessories."

B. Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible" for branch, outlet and inlet, and terminal unit connections.

3.8 DUCT SCHEDULE

A. Fabricate ducts with galvanized sheet steel except as otherwise indicated and as follows:

B. Exhaust Ducts:

1. Ducts Connected to Fans Exhausting (ASHRAE 62.1, Class 1 and 2) Air:

a. Pressure Class: Negative 2-inch wg.

b. Minimum SMACNA Seal Class: C if negative pressure, and A if positive pressure.

c. SMACNA Leakage Class for Rectangular: 16.

d. SMACNA Leakage Class for Round and Flat Oval: 8.

C. Intermediate Reinforcement:

1. Galvanized-Steel Ducts: Galvanized steel.

2. PVC-Coated Ducts:

    a. Exposed to Airstream: Match duct material.

    b. Not Exposed to Airstream: Galvanized.

3. Stainless-Steel Ducts:

    a. Exposed to Airstream: Match duct material.

    b. Not Exposed to Airstream: Match duct material.

4. Aluminum Ducts: Aluminum.

D. Elbow Configuration:

1. Rectangular Duct: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 4-2, "Rectangular Elbows."

    a. Velocity 1000 fpm or Lower:

        1) Radius Type RE 1 with minimum 0.5 radius-to-diameter ratio.

        2) Mitered Type RE 4 without vanes.

    b. Velocity 1000 to 1500 fpm:

        1) Radius Type RE 1 with minimum 1.0 radius-to-diameter ratio.

        2) Radius Type RE 3 with minimum 0.5 radius-to-diameter ratio and two vanes.

        3) Mitered Type RE 2 with vanes complying with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 4-3, "Vanes and Vane Runners," and Figure 4-4, "Vane Support in Elbows."

    c. Velocity 1500 fpm or Higher:

        1) Radius Type RE 1 with minimum 1.5 radius-to-diameter ratio.

        2) Radius Type RE 3 with minimum 1.0 radius-to-diameter ratio and two vanes.

        3) Mitered Type RE 2 with vanes complying with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 4-3, "Vanes and Vane Runners," and Figure 4-4, "Vane Support in Elbows."

2. Rectangular Duct: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 4-2, "Rectangular Elbows."

    a. Radius Type RE 1 with minimum 1.5 radius-to-diameter ratio.

    b. Radius Type RE 3 with minimum 1.0 radius-to-diameter ratio and two vanes.

c. Mitered Type RE 2 with vanes complying with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 4-3, "Vanes and Vane Runners," and Figure 4-4, "Vane Support in Elbows."

3. Round Duct: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 3-4, "Round Duct Elbows."

a. Minimum Radius-to-Diameter Ratio and Elbow Segments: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Table 3-1, "Mitered Elbows." Elbows with less than 90-degree change of direction have proportionately fewer segments.

1) Velocity 1000 fpm or Lower: 0.5 radius-to-diameter ratio and three segments for 90-degree elbow.

2) Velocity 1000 to 1500 fpm: 1.0 radius-to-diameter ratio and four segments for 90-degree elbow.

3) Velocity 1500 fpm or Higher: 1.5 radius-to-diameter ratio and five segments for 90-degree elbow.

4) Radius-to Diameter Ratio: 1.5.

b. Round Elbows, 12 Inches and Smaller in Diameter: Stamped or pleated.

c. Round Elbows, 14 Inches and Larger in Diameter: Standing seam.

E. Branch Configuration:

1. Rectangular Duct: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 4-6, "Branch Connection."

a. Rectangular Main to Rectangular Branch: 45-degree entry.

b. Rectangular Main to Round Branch: Spin in.

2. Round: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible," Figure 3-5, "90 Degree Tees and Laterals," and Figure 3-6, "Conical Tees." Saddle taps are permitted in existing duct.

a. Velocity 1000 fpm or Lower: 90-degree tap.

b. Velocity 1000 to 1500 fpm: Conical tap.

c. Velocity 1500 fpm or Higher: 45-degree lateral.

F. Liner:

1. Supply-Air Ducts: [**Fibrous glass, Type I**] [**Flexible elastomeric**], [**1**] [**1-1/2**] [**2**] Insert dimension inches thick.

2. Return-Air Ducts: [**Fibrous glass, Type I**] [**Flexible elastomeric**], [**1**] [**1-1/2**] [**2**] inches thick.

3. Exhaust-Air Ducts: [**Fibrous glass, Type I**] [**Flexible elastomeric**], [**1**] inches thick.

4. Supply Fan Plenums: [**Fibrous glass, Type II**] [**Flexible elastomeric**], [**1**] [**1-1/2**] [**2**] inches thick.

5. Return- and Exhaust-Fan Plenums: [**Fibrous glass, Type II**] [**Flexible elastomeric**] [**Natural fiber**], [**2**] inches thick.

6. Transfer Ducts: [**Fibrous glass, Type I**] [**Flexible elastomeric**] [**Natural fiber**], [**1**] [**1-1/2**] [**2**] inches thick.

END OF SECTION

SECTION 23 33 00 - AIR DUCT ACCESSORIES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Backdraft and pressure relief dampers.

2. Manual volume dampers.

3. Flange connectors.

4. Turning vanes.

5. Duct-mounted access doors.

6. Flexible connectors.

7. Duct accessory hardware.

B. Related Requirements:

1. Section 23 37 23 "HVAC Gravity Ventilators" for roof-mounted ventilator caps.

2. Section 28 31 11 "Digital, Addressable Fire-Alarm System" for duct-mounted fire and smoke detectors.

1.2 SUBMITTALS

A. See section 23 00 00 “General Requirements of HVAC” for submittal requirements.

PART 2 - PRODUCTS

2.1 ASSEMBLY DESCRIPTION

A. Comply with NFPA 90A, "Installation of Air Conditioning and Ventilating Systems," and with NFPA 90B, "Installation of Warm Air Heating and Air Conditioning Systems."

B. Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible" for acceptable materials, material thicknesses, and duct construction methods unless otherwise indicated. Sheet metal materials shall be free of pitting, seam marks, roller marks, stains, discolorations, and other imperfections.

2.2 MATERIALS

A. Galvanized Sheet Steel: Comply with ASTM A 653/A 653M.

1. Galvanized Coating Designation: G90.

2. Exposed-Surface Finish: Mill phosphatized.

B. Stainless-Steel Sheets: Comply with ASTM A 480/A 480M, Type 304, and having a 2D finish for concealed ducts and 2BA finish for exposed ducts.

C. Aluminum Sheets: Comply with ASTM B 209, Alloy 3003, Temper H14; with mill finish for concealed ducts and standard, 1-side bright finish for exposed ducts.

D. Extruded Aluminum: Comply with ASTM B 221, Alloy 6063, Temper T6.

E. Reinforcement Shapes and Plates: Galvanized-steel reinforcement where installed on galvanized sheet metal ducts; compatible materials for aluminum and stainless-steel ducts.

F. Tie Rods: Galvanized steel, 1/4-inch minimum diameter for lengths 36 inches or less; 3/8-inch minimum diameter for lengths longer than 36 inches.

2.3 BACKDRAFT AND PRESSURE RELIEF DAMPERS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Cesco Products; a divsion of MESTEK, Inc.

2. Nailor Industries Inc.

3. Ruskin Company.

B. Description: Gravity balanced.

C. Maximum Air Velocity: 1000 fpm.

D. Maximum System Pressure: 4.5 inch wg.

E. Frame: Hat-shaped, 0.063-inch-thick extruded aluminum, with welded corners or mechanically attached and mounting flange.

F. Blades: Multiple single-piece blades, end pivoted, maximum 6-inch width, 0.025-inch-thick, roll-formed aluminum with sealed edges.

G. Blade Action: Parallel.

H. Blade Seals: Extruded vinyl, mechanically locked.

I. Blade Axles:

1. Material: Nonmetallic.

2. Diameter: 0.20 inch.

J. Bearings: synthetic pivot bushings.

K. Accessories:

1. Adjustment device to permit setting for varying differential static pressure.

2. Counterweights and spring-assist kits for vertical airflow installations.

3. Electric actuators.

4. Chain pulls.

5. Screen Mounting: Front mounted in sleeve.

    a. Sleeve Thickness: 20 gage minimum.

    b. Sleeve Length: 6 inches minimum.

6. Screen Mounting: Rear mounted.

7. Screen Material: Aluminum.

8. Screen Type: Bird.

9. 90-degree stops.

2.4 MANUAL VOLUME DAMPERS

A. Standard, Steel, Manual Volume Dampers:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

    a. Cesco Products; a division of MESTEK, Inc.

    b. Nailor Industries Inc.

    c. Ruskin Company.

2. Standard leakage rating.

3. Suitable for horizontal or vertical applications.

4. Frames:

    a. Frame: 16 gauge galvanized steel, 5 in deep

    b. Mitered and welded corners.

    c. Flanges for attaching to walls and flangeless frames for installing in ducts.

5. Blades:

    a. Multiple or single blade.

    b. Parallel- or opposed-blade design.

    c. Stiffen damper blades for stability.

    d. 16 gauge galvanized steel with V groove for stiffness.

6. Blade Axles: Galvanized steel.

7. Bearings:

    a. Molded synthetic.

b. Dampers in ducts with pressure classes of 3-inch wg or less shall have axles full length of damper blades and bearings at both ends of operating shaft.

8. Tie Bars and Brackets: Galvanized steel.

9. Locking Quadrant handles

2.5 FLANGE CONNECTORS

A. Description: Roll-formed, factory-fabricated, slide-on transverse flange connectors, gaskets, and components.

B. Material: Galvanized steel.

C. Gage and Shape: Match connecting ductwork.

2.6 TURNING VANES

A. Manufactured Turning Vanes for Metal Ducts: Curved blades of galvanized sheet steel; support with bars perpendicular to blades set; set into vane runners suitable for duct mounting.

1. Acoustic Turning Vanes: Fabricate airfoil-shaped aluminum extrusions with perforated faces and fibrous-glass fill.

B. Manufactured Turning Vanes for Nonmetal Ducts: Fabricate curved blades of resin-bonded fiberglass with acrylic polymer coating; support with bars perpendicular to blades set; set into vane runners suitable for duct mounting.

C. General Requirements: Comply with SMACNA's "HVAC Duct Construction Standards - Metal and Flexible"; Figures 4-3, "Vanes and Vane Runners," and 4-4, "Vane Support in Elbows."

D. Vane Construction: Double wall.

2.7 DUCT-MOUNTED ACCESS DOORS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Cesco Products; a divsion of MESTEK, Inc.

2. Ductmate Industries, Inc.

3. Flexmaster U.S.A., Inc.

4. Nailor Industries Inc.

5. Ruskin.

B. Duct-Mounted Access Doors: Fabricate access panels according to SMACNA's "HVAC Duct Construction Standards - Metal and Flexible"; Figures 7-2, "Duct Access Doors and Panels," and 7-3, "Access Doors - Round Duct."

1. Door:

a. Double wall, rectangular.

b. Galvanized sheet metal with insulation fill and thickness as indicated for duct pressure class.

c. Vision panel.

d. Hinges and Latches: 1-by-1-inchbutt or piano hinge and cam latches.

e. Fabricate doors airtight and suitable for duct pressure class.

2. Frame: Galvanized sheet steel, with bend-over tabs and foam gaskets.

3. Number of Hinges and Locks:

a. Access Doors Less Than 12 Inches Square: No hinges and two sash locks.

b. Access Doors up to 18 Inches Square: Continuous and two sash locks.

c. Access Doors up to 24 by 48 Inches: Continuous and two compression latches with outside and inside handles.

d. Access Doors Larger Than 24 by 48 Inches: Continuous and two compression latches with outside and inside handles.

C. Pressure Relief Access Door:

1. Door and Frame Material: Galvanized sheet steel.

2. Door: Double wall with insulation fill with metal thickness applicable for duct pressure class.

3. Operation: Open outward for positive-pressure ducts and inward for negative-pressure ducts.

4. Factory set at 3.0- to 8.0-inch wg.

5. Doors close when pressures are within set-point range.

6. Hinge: Continuous piano.

7. Latches: Cam.

8. Seal: Neoprene or foam rubber.

9. Insulation Fill: 1-inch-thick, fibrous-glass or polystyrene-foam board.

2.8 FLEXIBLE CONNECTORS

A. Materials: Flame-retardant or noncombustible fabrics.

B. Coatings and Adhesives: Comply with UL 181, Class 1.

C. Metal-Edged Connectors: Factory fabricated with a fabric strip 3-1/2 inches wide attached to two strips of 2-3/4-inch-wide, 0.028-inch-thick, galvanized sheet steel or 0.032-inch-thick aluminum sheets. Provide metal compatible with connected ducts.

D. Indoor System, Flexible Connector Fabric: Glass fabric double coated with neoprene.

1. Minimum Weight: 26 oz./sq. yd..

2. Tensile Strength: 480 lbf/inch in the warp and 360 lbf/inch in the filling.

3. Service Temperature: Minus 40 to plus 200 deg F.

E. Outdoor System, Flexible Connector Fabric: Glass fabric double coated with weatherproof, synthetic rubber resistant to UV rays and ozone.

1. Minimum Weight: 24 oz./sq. yd..

2. Tensile Strength: 530 lbf/inch in the warp and 440 lbf/inch in the filling.

3. Service Temperature: Minus 50 to plus 250 deg F.

2.9 DUCT ACCESSORY HARDWARE

A. Instrument Test Holes: Cast iron or cast aluminum to suit duct material, including screw cap and gasket. Size to allow insertion of pitot tube and other testing instruments and of length to suit duct-insulation thickness.

B. Adhesives: High strength, quick setting, neoprene based, waterproof, and resistant to gasoline and grease.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install duct accessories according to applicable details in SMACNA's "HVAC Duct Construction Standards - Metal and Flexible" for metal ducts and in NAIMA AH116, "Fibrous Glass Duct Construction Standards," for fibrous-glass ducts.

B. Install duct accessories of materials suited to duct materials; use galvanized-steel accessories in galvanized-steel and fibrous-glass ducts, stainless-steel accessories in stainless-steel ducts, and aluminum accessories in aluminum ducts.

C. Install volume dampers at points on supply, return, and exhaust systems where branches extend from larger ducts. Where dampers are installed in ducts having duct liner, install dampers with hat channels of same depth as liner, and terminate liner with nosing at hat channel.

D. Set dampers to fully open position before testing, adjusting, and balancing.

E. Install test holes at fan inlets and outlets and elsewhere as indicated.

F. Install duct access doors on sides of ducts to allow for inspecting, adjusting, and maintaining accessories and equipment at the following locations:

1. On both sides of duct coils.

2. Upstream and downstream from duct filters.

3. At outdoor-air intakes and mixed-air plenums.

4. At drain pans and seals.

5. Downstream from control dampers, backdraft dampers, and equipment.

6. Adjacent to and close enough to fire or smoke dampers, to reset or reinstall fusible links. Access doors for access to fire or smoke dampers having fusible links shall be pressure relief access doors and shall be outward operation for access doors installed upstream from dampers and inward operation for access doors installed downstream from dampers.

7. At each change in direction and at maximum 50-foot spacing.

8. Control devices requiring inspection.

9. Elsewhere as indicated.

G. Install access doors with swing against duct static pressure.

H. Access Door Sizes:

1. One-Hand or Inspection Access: 8 by 5 inches.

2. Two-Hand Access: 12 by 6 inches.

3. Head and Hand Access: 18 by 10 inches.

4. Head and Shoulders Access: 21 by 14 inches.

5. Body Access: 25 by 14 inches.

6. Body plus Ladder Access: 25 by 17 inches.

I. Label access doors according to Section 23 05 53 "Identification for HVAC Piping and Equipment" to indicate the purpose of access door.

J. Install flexible connectors to connect ducts to equipment.

K. Connect terminal units to supply ducts directly or with maximum 12-inch lengths of flexible duct. Do not use flexible ducts to change directions.

L. Connect diffusers or light troffer boots to ducts directly or with maximum 60-inch lengths of flexible duct clamped or strapped in place.

M. Connect flexible ducts to metal ducts with draw bands.

N. Install duct test holes where required for testing and balancing purposes.

3.2 FIELD QUALITY CONTROL

A. Tests and Inspections:

1. Operate dampers to verify full range of movement.

2. Inspect locations of access doors and verify that purpose of access door can be performed.

3. Operate fire and smoke dampers to verify full range of movement and verify that proper heat-response device is installed.

4. Inspect turning vanes for proper and secure installation.

END OF SECTION

SECTION 23 34 23 - HVAC POWER VENTILATORS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Ceiling-mounted ventilators.

1.2 SUBMITTALS

A. See Section 23 00 00 “General Requirements of HVAC” for submittal requirements.

1.3 QUALITY ASSURANCE

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. AMCA Compliance: Fans shall have AMCA-Certified performance ratings and shall bear the AMCA-Certified Ratings Seal.

PART 2 - PRODUCTS

2.1 CEILING-MOUNTED VENTILATORS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Acme Engineering & Manufacturing Corp.

2. Loren Cook Company.

3. Panasonic.

B. Housing: Steel, lined with acoustical insulation.

C. Fan Wheel: Centrifugal wheels directly mounted on motor shaft. Fan shrouds, motor, and fan wheel shall be removable for service.

D. Grille: Plastic, louvered grille with spring clip or flange on intake and thumbscrew attachment to fan housing.

E. Electrical Requirements: Junction box for electrical connection on housing and receptacle for motor plug-in.

F. Accessories:

1. Variable-Speed Controller: Solid-state control to reduce speed from 100 to less than 50 percent.

2. Manual Starter Switch: Single-pole rocker switch assembly with cover and pilot light.

3. Time-Delay Switch: Assembly with single-pole rocker switch, timer, and cover plate.

4. Motion Sensor: Motion detector with adjustable shutoff timer.

5. Ceiling Radiation Damper: Fire-rated assembly with ceramic blanket, stainless-steel springs, and fusible link.

6. Isolation: Rubber-in-shear vibration isolators.

7. Manufacturer's standard roof jack or wall cap, and transition fittings.

G. Capacities and Characteristics: See Drawings

2.2 MOTORS

A. Comply with NEMA designation, temperature rating, service factor, enclosure type, and efficiency requirements for motors specified in Section 23 05 13 "Common Motor Requirements for HVAC Equipment."

1. Motor Sizes: Minimum size as indicated. If not indicated, large enough so driven load will not require motor to operate in service factor range above 1.0.

B. Enclosure Type: Totally enclosed, fan cooled.

2.3 SOURCE QUALITY CONTROL

A. Certify sound-power level ratings according to AMCA 301, "Methods for Calculating Fan Sound Ratings from Laboratory Test Data." Factory test fans according to AMCA 300, "Reverberant Room Method for Sound Testing of Fans." Label fans with the AMCA-Certified Ratings Seal.

B. Certify fan performance ratings, including flow rate, pressure, power, air density, speed of rotation, and efficiency by factory tests according to AMCA 210, "Laboratory Methods of Testing Fans for Aerodynamic Performance Rating." Label fans with the AMCA-Certified Ratings Seal.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Equipment Mounting:

1. Install power ventilators on cast-in-place concrete equipment base(s). Comply with requirements for equipment bases and foundations specified in Section 03 30 00 "Cast-in-Place Concrete." or Section 03 30 53 "Miscellaneous Cast-in-Place Concrete."

2. Comply with requirements for vibration isolation and seismic control devices specified in Section 23 05 48 "Vibration and Seismic Controls for HVAC."

B. Secure roof-mounted fans to roof curbs with cadmium-plated hardware. See Section 07 72 00 "Roof Accessories" for installation of roof curbs.

C. Ceiling Units: Suspend units from structure; use steel wire or metal straps.

D. Support suspended units from structure using threaded steel rods and elastomeric hangers or spring hangers having a static deflection of 1 inch. Vibration-control devices are specified in Section 23 05 48 "Vibration and Seismic Controls for HVAC Piping and Equipment."

E. Install units with clearances for service and maintenance.

F. Label units according to requirements specified in Section 23 05 53 "Identification for HVAC Piping and Equipment."

3.2 CONNECTIONS

A. Drawings indicate general arrangement of ducts and duct accessories. Make final duct connections with flexible connectors. Flexible connectors are specified in Section 23 33 00 "Air Duct Accessories."

B. Install ducts adjacent to power ventilators to allow service and maintenance.

C. Ground equipment according to Section 26 05 26 "Grounding and Bonding for Electrical Systems."

D. Connect wiring according to Section 26 05 19 "Low-Voltage Electrical Power Conductors and Cables."

3.3 FIELD QUALITY CONTROL

A. Perform tests and inspections.

1. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect components, assemblies, and equipment installations, including connections, and to assist in testing.

B. Tests and Inspections:

1. Verify that shipping, blocking, and bracing are removed.

2. Verify that unit is secure on mountings and supporting devices and that connections to ducts and electrical components are complete. Verify that proper thermal-overload protection is installed in motors, starters, and disconnect switches.

3. Verify that cleaning and adjusting are complete.

4. Disconnect fan drive from motor, verify proper motor rotation direction, and verify fan wheel free rotation and smooth bearing operation. Reconnect fan drive system, align and adjust belts, and install belt guards.

5. Adjust belt tension.

6. Adjust damper linkages for proper damper operation.

7. Verify lubrication for bearings and other moving parts.

8. Verify that manual and automatic volume control and fire and smoke dampers in connected ductwork systems are in fully open position.

9. Disable automatic temperature-control operators, energize motor and adjust fan to indicated rpm, and measure and record motor voltage and amperage.

10. Shut unit down and reconnect automatic temperature-control operators.

11. Remove and replace malfunctioning units and retest as specified above.

C. Test and adjust controls and safeties. Replace damaged and malfunctioning controls and equipment.

D. Prepare test and inspection reports.

3.4 ADJUSTING

A. Adjust damper linkages for proper damper operation.

B. Adjust belt tension.

C. Comply with requirements in Section 23 05 93 "Testing, Adjusting, and Balancing for HVAC" for testing, adjusting, and balancing procedures.

D. Replace fan and motor pulleys as required to achieve design airflow.

E. Lubricate bearings.

END OF SECTION

SECTION 23 82 31 - HEATERS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes:

1. Electric Wall and Ceiling Heaters

1.2 SUBMITTALS

A. See Section 23 00 00 “General Requirements of HVAC” for submittal requirements.

PART 2 - PRODUCTS

2.1 ELECTRIC WALL AND CEILING HEATERS

A. Manufacturers:

1. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

a. Berko; Marley Engineered Products.

b. Indeeco

c. Markel Products; TPI Corporation.

d. QMark; Marley Engineered Products.

B. Description

1. Assembly including chassis, electric heating coil, fan, motor, and controls. Comply with UL 2021.

2. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

C. Cabinet

1. Front Panel: Extruded-aluminum bar grille, with removable panels fastened with tamperproof fasteners.

2. Finish: Baked enamel over baked-on primer with manufacturer's standard color selected by Architect, applied to factory-assembled and -tested wall and ceiling heaters before shipping.

3. Surface-Mounted Cabinet Enclosure: Steel with finish to match cabinet.

D. Coil

1. Electric-Resistance Heating Coil: Nickel-chromium heating wire, free from expansion noise and 60-Hz hum, embedded in magnesium oxide refractory and sealed in corrosion-resistant metallic sheath. Terminate elements in stainless-steel, machine-staked terminals

secured with stainless-steel hardware, and limit controls for high-temperature protection. Provide integral circuit breaker for overcurrent protection.

E. Fan and Motor

1. Fan: Aluminum propeller directly connected to motor.

2. Motor: Permanently lubricated, multispeed.

F. Controls

1. Controls: Unit-mounted thermostat.

2. Electrical Connection: Factory wire motors and controls for a single field connection with disconnect switch.

3. See Section 23 09 00 “HVAC Controls” for additional information.

G. Capacities and Characteristics: See Drawings

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas to receive hot water cabinet heaters for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

B. Examine roughing-in for piping and electrical connections to verify actual locations before installation.

C. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Install wall and ceiling unit heaters to comply with NFPA 90A.

B. Install wall and ceiling unit heaters level and plumb.

C. Suspend cabinet unit heaters from structure with elastomeric hangers. Vibration isolators shall comply with Section 23 05 48 "Vibration and Seismic Controls for HVAC Piping and Equipment."

D. Install wall-mounted thermostats and switch controls in electrical outlet boxes at heights to match lighting controls. Verify location of thermostats and other exposed control sensors with Drawings and room details before installation.

E. Ground equipment according to Section 26 05 26 "Grounding and Bonding for Electrical Systems."

F. Connect wiring according to Section 26 05 19 "Low-Voltage Electrical Power Conductors and Cables."

G. Install new filters in each fan-coil unit within two weeks of Substantial Completion.

H. Piping installation requirements are specified in Section 23 21 13 "Hydronic Piping," Section 23 21 16 "Hydronic Piping Specialties." Drawings indicate general arrangement of piping, fittings, and specialties.

I. Install piping adjacent to machine to allow service and maintenance.

3.3 FIELD QUALITY CONTROL

A. Perform the following tests and inspections:

1. Operational Test: After electrical circuitry has been energized, start units to confirm proper motor rotation and unit operation.

2. Test and adjust controls and safety devices. Replace damaged and malfunctioning controls and equipment.

B. Units will be considered defective if they do not pass tests and inspections.

END OF SECTION

SECTION 260010 – GENERAL ELECTRICAL REQUIREMENTS

PART 1 - GENERAL

1.1 SUMMARY

A. The requirements listed in this section are supplemental to the Division 01 General Requirements.

B. It shall be the responsibility of the Electrical and Low-voltage Contractors to examine and refer to all Architectural, Civil, Structural, Mechanical, Plumbing and Landscape drawings and specifications for construction conditions which may affect the scope of Electrical, Communications, Electronic Safety and Security work. Inspect the building site and existing facilities for verification of present conditions. Make proper provisions for these conditions in performance of the work and cost thereof.

C. Electrical, Communications, Electronic Safety and Security work for this project shall include all items, articles, materials and the associated labor mentioned, schedules or shown in these specifications and in the accompanying drawings.

D. Furnish and install all equipment, materials and any required incidental items required by good practice to complete the systems described herein.

E. Refer to Division 01 for all listed Alternates and provide separate pricing and work as indicated in Division 01 and Contract Documents.

1.2 DEFINITIONS - Throughout contract documents these words and phrases are used:

A. Contract documents - All drawings, specifications, addenda and change orders that document work to be done.

B. Demolition – Carefully disconnect and remove items. All reasonable caution shall be taken to avoid damaging removed equipment and to retain its operability.

C. Remove back to source - Remove all conduit and wire back to panelboard or last live device.

D. Equivalent or equal - Product of like type and function that complies with all applicable provisions of drawings and specifications and which has been approved as substitute for specified item.

E. Furnish - Purchase material as shown and specified, and place material to approved location on site or elsewhere as noted or agreed upon.

F. Install - Set in place and connect, ready for use and in complete and properly operating finished condition.

G. Provide - Furnish and install with all products, labor, sub-contracts, and appurtenances required for a complete and properly operating, finished condition.

H. Rough-in - Provide conduit raceway system with junction boxes, fittings, straps, BUSHINGS, etc., for future installation of wiring, devices, disconnects and breakers. Provision shall be made in panelboard (hardware, etc.) for future installation of breakers.

I. Serviceable - Arranged so that component or product in question may be properly removed and replaced without disassembly, destruction or damage to surrounding installation.

1.3 CODES, STANDARDS AND REGULATIONS

A. Codes - Perform all work in strict accordance with all applicable national, state and local codes; including, but not limited to latest legally enacted editions of following codes:

1. International Building Code – IBC
2. NFPA 70, National Electric Code – NEC
3. NFPA 72, National Fire Alarm Code
4. International Fire Code – IFC
5. International Energy Conservation Code – IECC
6. ANSI-C2, National Electrical Safety Code – NESC

B. Standards - Reference to standards infers that installation, equipment and material shall be within limits for which it was designed, tested and approved, in conformance with current publications and standards of following organizations:

1. American National Standards Institute – ANSI
2. American Society for Testing and Materials – ASTM
3. American Society of Heating Refrigerating and Air Conditioning Engineers – ASHRAE (Standard 90-75)
4. Institute of Electrical and Electronics Engineers – IEEE
5. Insulated Cable Engineers Association – ICEA
6. National Electrical Contractors Association – NECA
7. National Electrical Manufacturers' Association – NEMA
8. National Fire Protection Association – NFPA
9. Occupational Safety and Health Administration – OSHA
10. Underwriters' Laboratories, Inc. – UL
11. Rules and Regulations of the State/Local Fire Marshal
12. Standards and Requirement of the Serving Utilities
13. State and Local Ordinances

C. Regulations - Design has been performed in accordance with applicable regulations and guidelines noted below. Contractor shall carefully apply these regulations and bring any discrepancies to immediate attention of Architect/Engineer.

1. Americans with Disabilities Act – ADA

2. Architectural Barriers Act – ABA

1.4 FEES AND PERMITS

A. Electrical Contractor shall pay for all permits or fees in connection with electrical work. Fees shall include any or all user fees, government fees, system development fees, connection fees or other fees that are required to be paid before systems can be connected or used.

B. Schedule all required electrical inspections with local electrical inspector. Notify engineer of all items of discrepancy noted by electrical inspector if those items affect cost or function of system, or if they conflict with electrical drawings and specifications.

C. All Utility Cost and fees from the utility work shall be the responsibility of the Owner. Contractor to coordinate all utility requirements, standards and responsibilities with serving utility for a complete scope of work prior to bid.

D. Deliver all inspection certificates to Architect/Engineer prior to final acceptance of work.

1.5 INTENT OF SPECIFICATIONS AND DRAWINGS

A. Plans and specifications are intended to result in complete electrical installation in full compliance with all applicable codes, standards and ordinances.

B. Plans and specifications are to supplement each other and any details contained in one shall be included as if contained in both.

C. Electrical drawings shall serve as working drawings, but Architectural drawings shall take precedence if any dimensional discrepancies exist.

D. Drawings are partly diagrammatic and do not show routing of conduits, exact location of products, or installation features in exact detail. Locations of devices, fixtures and equipment are approximate unless dimensioned.

E. Riser diagrams and control schematics are not to scale and do not show physical arrangement of equipment. Do not use riser diagrams or schematics to obtain lineal conduit and cabling distances.

F. Items are shown on drawings in locations to minimize interference with other equipment, structural members, etc. Exact finish locations are not indicated, however, and all work shall be done to avoid interference, preserve headroom and keep openings and passageways clear.

G. In event that discrepancies of any kind exist or required items/details have been omitted, Contractor shall notify Architect/Engineer in writing of such discrepancy or omission at least ten days prior to bid date. Failure to do so shall be construed as willingness of Contractor to supply all necessary materials and labor required for proper completion of work.

1.6 CONTRACTOR'S RESPONSIBILITY - Contractor shall be responsible for installation of complete and functional piece of work in accordance with true intent of contract documents. Provide all incidental items required for complete installation and satisfactory operation of all equipment, whether or not specifically noted in contract documents.

A. QUALIFICATIONS

1. Contractor shall employ on this project, capable, experienced and reliable foreman and such skilled workmen as may be required for various classes of work to be performed.
2. Where special skills and certification are required, Contractor shall ensure that work is performed by individuals with required experience, skill and certification.
3. If, in Engineer's opinion, Contractor's employees do not possess necessary qualifications to perform specialty work, Contractor will be required to obtain services of workmen who are approved by manufacturer and certified by applicable agency or group. These workmen, if required, shall be provided at no additional expense.
4. Refer to other specification sections for additional required contractor qualifications and certification.

B. LICENSING AND CERTIFICATION - All Division 26 work shall be accomplished by Electricians, licensed by state in which work is being done, certified as required, and skilled in their craft. Electrician may elect to hire subcontractors for portions of work (such as systems described in Divisions 27 and 28) who are not licensed electricians, but have required certificates and are licensed in their discipline by state in which work is being done.

C. COORDINATION

1. Contractor shall consult all contract documents, shop drawings of other trades, and actual building dimensions to predetermine that his work and equipment will fit as planned. Do not scale drawings for fabrication. No extra payment will be issued for materials or items which do not fit because of Contractor's failure to verify as-built building dimensions.
2. Contractor shall check location of fixtures, outlets, equipment, conduit, etc., to determine they clear all openings, structural members, piping, ducts and miscellaneous equipment having fixed locations.
3. Changes in location of electrical work, necessary due to obstacles or installation of other trades shown on contract documents, shall be made by Electrical Contractor at no extra cost.
4. Contractor shall coordinate with Plumbing and Mechanical Contractors to avoid installation of piping and ductwork above or below panelboards in violation of National Electrical Code.
5. Lay out all work in advance and avoid conflict with other work in progress. Physical dimensions shall be determined from architectural and structural plans. Verify locations for junction boxes, disconnect switches, stub-ups, etc., for connection to equipment furnished by others, or in other Divisions of this work.
6. Contractor shall coordinate and plan work to proceed with work of other trades.
7. Contractor shall inform General Contractor of all required openings in building structure for installation of electrical equipment.
8. Contractor shall check dimensions of all electrical equipment installed, provided by himself or by others, so correct clearances and connections can be made.
9. Consulting all contract documents and shop drawings of other trades, contractor shall determine where electrical junction/pull boxes and equipment can be installed to maintain proper accessibility. Where accessibility cannot be maintained by judicious placement of boxes, Electrical Contractor shall coordinate with General Contractor to provide, fabricate, install, adjust, paint, etc. access doors through non-accessible floor, wall, and ceiling finishes to allow access to all electrical junction and pull boxes, electrical devices, electrical equipment, etc. at all required locations whether shown or not shown on plans. Electrical Contractor is responsible for determining size and location of the access doors. Report any conflicts to Architect/Engineer.

1.7 REVIEW

A. All work and material is subject to review at any time by the Architect/Engineer or his representative. If the Architect/Engineer or his representative finds material that does not conform to these specifications or that is not properly installed or finished, correct the deficiencies in a manner satisfactory to the Architect/Engineer at the Contractor's expense.

1.8 TEMPORARY FACILITIES

A. ELECTRICAL UTILITIES

1. The Electrical Contractor shall provide temporary electrical power to the construction site as directed by the General Contractor. No connections to the owner's system shall be allowed without owner's written approval. Provide a separate utility service as required.
2. The Electrical Contractor shall provide temporary electrical power to job trailers as directed by the General Contractor.
3. The Electrical Contractor shall provide temporary communications to job trailers as directed by the General Contractor.
4. All Costs associated with temporary power, communications and utility cost shall be paid by to the General Contractor.
5. The Electrical Contractor shall provide temporary construction lighting as directed by the General Contractor to provide a safe working environment.
6. All temporary services are to be removed in their entirety prior to occupancy as directed by the General Contractor.

B. OFFICES

1. The Electrical Contractor must have the permission of the Owner and General Contractor or Construction Manager to install a temporary office/job trailer on the project site.
2. Contractor shall completely remove his temporary installations when no longer needed and the premises shall be completely clean, disinfected, patched, and refinished to match adjacent areas.

C. LADDERS AND SCAFFOLDS

1. The Electrical and Low-voltage Contractors shall provide their own ladders, scaffolds, etc. of substantial construction for access to their work in various portions of the building as may be required. When no longer needed, they shall be removed by the Contractor.

D. PROTECTION DEVICES

1. The Electrical and Low-voltage Contractors shall provide and maintain their own necessary barricades, fences, signal lights, etc., required by all governing authorities or shown on the drawings. When no longer needed, they shall be removed by the Contractor.

E. TEMPORARY FIRE PROTECTION

1. The Electrical and Low-voltage Contractors shall provide all necessary first aid hand fire extinguishers for Class A, B, C and special hazards as may exist in his own work area only in accordance with good and safe practice and as required by jurisdictional safety authority.

1.9 RECORD DOCUMENTS (AS-BUILT DRAWINGS)

A. See requirements regarding record documents in General Division and Division 1.

B. At beginning of work, Contractor shall set aside one complete set of drawings which shall be maintained as complete "As-Built" set. Drawings shall be updated daily in neat and legible manner and shall not be used for any other purpose. Drawings, specification, addenda, change orders, etc. shall be maintained at job site and available for review at any time.

C. Show dimensioned location and routing of all electrical work that will become permanently concealed, cast in concrete or buried underground.

D. Show complete routing and sizing of any significant revisions to systems shown.

E. Show provisions for future connection, referenced to building lines or approved bench marks.

F. Provide wiring diagrams for all individual communications systems as installed. Identify all components and show all wire and terminal numbers and connections.

G. At completion of project, deliver drawings to Engineer for review.

1.10 WARRANTY

A. The Contractor shall guarantee that all materials and labor installed are new and of first quality and that any material or labor found defective shall be replaced without cost to the Owner within one (1) year after substantial completion of the Contract or one (1) full season of heating and cooling operation, whichever is the greater. The guarantee shall list the date of the beginning of the one (1) year period, which shall be the date that the Substantial Completion Certificate is issued.

B. Any damage to the building, caused by defective work or material of the Contractor within the above-mentioned period, shall be satisfactorily repaired without cost to the Owner.

C. The guarantee does not include maintenance of equipment. The Owner shall accept full responsibility for proper operation and maintenance of equipment immediately upon substantial completion and occupancy of the building.

D. Final acceptance by the Owner will not occur until all operating instructions are mounted in Equipment Rooms and Operating Personnel thoroughly indoctrinated in the operation of all electrical equipment by the Contractor.

E. No equipment installed as part of this project shall be used for temporary heat during construction.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

A. Manufacturer's trade names and catalog numbers listed are intended to indicate the quality of equipment or materials desired. Manufacturers not listed in the specification will be considered substitutions and must have prior approval.

B. See Division 01 for Substitutions Procedures. Requests for substitution are to be submitted sufficiently ahead of the deadline, to give ample time for examination. Prior approval request for substitution must indicate the specific item or items to be furnished in lieu of those scheduled, together with complete technical and comparative data on scheduled items and items proposed for substitution.

C. If the engineer approves any proposed substitution, the approved product will be listed in an addendum. Bidders shall not rely on approval made in any other manner.

D. Electrical equipment may be installed with manufacturer's standard finish and color except where specific color, finish or choice is indicated. If the manufacturer has no standard finish, equipment shall have a prime coat and two finish coats of gray enamel.

E. High altitude operation: Capacity of all equipment is to be sized and manufactured to perform at the elevation of the project site. If not specifically indicated in the equipment schedule or in the specifications provide all required accessories and equipment for proper operation at elevation of the project site.

F. This Contractor shall be responsible for materials and equipment installed under this contract. Contractor shall also be responsible for the protection of materials and equipment of others from damage as a result of his work.

G. Manufactured material and equipment shall be applied, installed, connected, erected, used, cleaned and conditioned as directed by manufacturer unless herein specified to the contrary.

H. This Contractor shall make the required arrangement with General Contractor or Construction Manager for the introduction into the building of equipment too large to pass through finished openings.

I. Store materials and equipment indoors at the job site or, if this is not possible, store on raised platforms and protect from the weather by means of waterproof covers. Coverings shall permit circulation of air around the materials to prevent condensation of moisture. Screen or cap openings in equipment to prevent the entry of vermin.

2.2 SUBSTITUTION OF MATERIALS - Where substituted equipment requires structural, architectural, mechanical, plumbing or electrical work that differs from basic design, cost of all changes, including re-design, shall be responsibility of contractor using substitution.

A. APPROVED MANUFACTURERS

1. In general, one particular manufacturer and part number or series is listed to describe equipment. Equivalent equipment of other manufacturers listed for that item may be substituted without prior approval. It shall be Contractor's responsibility to ensure that item used for bidding purposes is truly equivalent to that specified. If it is not equivalent, it will be rejected at shop drawing review and Contractor shall supply specified item at his own cost.

2. It is understood that manufacturers listed may not actually have equivalent product to that specified. If contractor/distributor has any questions regarding desired product characteristics and suitability of proposed substitution, he is encouraged to submit for prior approval. Also, any manufacturer not listed shall be submitted for prior approval.

B. PRIOR APPROVALS

1. Manufacturers not listed in specification or on schedule for a particular item are open for substitution prior to bid opening only.
2. Manufacturers desiring approval shall submit catalog cuts that define quality of product and ability to perform as specified. It is understood that no two manufactures use identical methods or make identical products. Any and all deviations from that specified shall be clearly noted.
3. Submittals shall arrive at Engineer at least ten (10) days prior to bid opening. All approvals will be listed in last addendum as being approved to bid. Items substituted, but not listed in contract documents, will not be considered if submitted on shop drawings.
4. Approval of substitute equipment is on basis of quality only. Materials supplier shall be responsible for his quotation reflecting proper selection of his particular equipment with regard to proper capacities, physical dimensions, requirements, intended function, finish, color, etc. Engineer will not give approval to specific model numbers or check capacities, dimensions, or requirements. Evaluation will be on basis of quality and equality to specified items.
5. Prior approval shall be obtained from engineer and no other entity (architect, owner, etc.) is authorized to give such approval.

C. SAMPLES

1. Where, in Engineer/Architect's opinion, product sample is required in order to determine appearance, quality, workmanship or operation, Contractor shall submit actual production samples of item in question.
2. Samples will be returned to Contractor. Approved samples may be used.
3. All costs incurred in providing and returning samples will be Contractor's responsibility.

2.3 PRODUCT AND SYSTEM SUBMITTALS

A. Submittals will be required for each piece of equipment, material or product as noted in the table below. All submittal shall be submitted, reviewed and all discrepancies addressed prior to ordering equipment or starting work. Any equipment ordered without having first completed the submittal process is done at the risk of the contractor. Any work performed prior to completing the submittal process is done at the risk of the contractor.

B. SUBMITTAL DEFINITIONS

1. Product Data: Provide manufacturers cut sheets that include general product information including but not limited to: Model Number, physical data, nominal capacities, rough-in requirements.
2. Performance Data: Provide detailed performance and capacities based on project specific requirements including but not limited to: voltage, phase, amperage, overcurrent protection, conductor size, conductor material, conduit size, color temperature, color rendering

index, life expectance, efficacy, efficiency, IP ratings, light distribution types and lighting control.

3. Shop Drawings: Provide detailed drawings of the equipment showing overall dimensions, location of electrical connection, location of anchorage points, location of electrical and control panels, and all operating, service and maintenance clearances.
4. Delegated Design: Provide detailed drawings prepared and stamped by a registered Professional Engineer that detail pertinent design criterial, the materials and products to be installed and the required installation locations.
5. Wiring Diagram: Provide diagrams that identify and detail required field wiring.
6. Color Chart: Provide a physical color chart of material samples required for selection of equipment colors.
7. Sustainability: Provide literature that indicates product compliance. See Division 01 for additional information and requirements.

C. SUBMITTAL FORMATS

1. Include the following information with each submittal:
   a. Project Name
   b. Submittal Date
   c. Name of Architect
   d. Name of Engineer
   e. Name of General Contractor or Construction Manager
   f. Name of Sub-Contractor
   g. Name of firm or entity that prepared the submittal
   h. Unique Submittal Number
   i. Type of Submittal
   j. Specification Section
   k. Name or Mark of equipment or material and detail or drawings reference.

2. All Submittal with the exception of color charts or material samples shall be electronically transmitted PDFs. All submittals over 8 Mb shall be setup on a share file site and access granted through email with folder's link for download.

D. SUBMITTAL REQUIREMENTS

1. Submittals shall be submitted as a complete specification section. The submittal must include all materials and equipment for that specification section. Submittals for individual materials of equipment will be rejected without review.
2. Submittals shall be complete, clearly show item used, size, dimensions, capacity, rough in, etc., as required for complete check and installation. Manufacturer's literature showing more than one item shall be clearly marked as to which item is being furnished or it will be rejected and returned without review.
3. Each submittal shall be thoroughly checked by the Contractor for compliance with the Contract Document requirements, accuracy of dimensions, relationship to the work of other trades, and conformance with sound, safe practices as to erection and installation. Each submittal shall then bear a stamp evidencing such checking and shall show corrections made, if any. Submittals requiring extensive corrections shall be revised before submission. Each submittal not stamped and signed by the General and Electrical Contractors evidencing such checking will be rejected and returned without review.
4. On each submittal, clearly indicate deviations from requirements in the Contract Documents, including minor variations and limitations; include relevant additional information

and revisions, other than those requested on previous submittals. Indicate by highlighting on each submittal or noting on attached separate sheet.

5. Review of the shop drawings and literature by the engineer shall not relieve the contractor for responsibility for deviations for the drawings or specifications, nor shall it relieve the contractor from responsibility for errors in the shop drawings or literature. It is the responsibility of the contractor to provide materials and equipment which meet the specifications and job requirements.
6. Luminaires submittals shall include dimensions, quality, distribution, color rendering index, color temperature, optics, photometrics, all listings (UL, DLC, Energy Star, Made in America, etc.), IP ratings, voltage, wattage, warranty, installation methods, control methods, efficacy, efficiency, diffuser options, emergency operation and any required accessories. Provide IES and Revit files upon request.

E. ENGINEER'S REVIEW - Submittal review is for general design and arrangement only and does not relieve Contractor from any requirements of contract documents. Submittals will not be checked for quantity, dimension, fit or proper technical design of manufactured equipment. Where product or system performance deviations have not been specifically noted in submittal by Contractor, Engineer's review will not relieve Contractor's responsibility to provide complete and satisfactory working installation of equal quality and performance to specified system. Ordering, manufacture, shipment or installation of equipment prior to receipt of Engineer's written review is strictly at Contractor's risk and all costs associated with shipping, changes, replacement or restocking shall be Contractor's responsibility.

2.4 SUB-CONTRACTORS - With shop drawing submittals, Contractor shall submit list of all subcontractors to be used for the project.

2.5 OPERATION AND MAINTENANCE MANUALS

A. Operation and Maintenance Manuals (O&M Manuals) shall contain:

1. Names and contact information for the Project Architect, Project Engineer.
2. Names and contact information for the General Contractor or Construction Manager.
3. Names and contact information for sub-contractors.
4. Installation, maintenance and operating instructions for each piece of equipment.
5. Parts lists
6. Wiring Diagrams
7. Equipment Start-up and inspection certificates
8. Test and Balance Reports
9. Commissioning Reports
10. Copies of Equipment Warranties
11. Copies of Submittals
12. Record Drawings.
13. Training DVD’s.

B. Prior to substantial completion submit an electronic copy of the O&M manual in PDF format to the Architect, Engineer and Owner for Review and approval. The PDF shall be one file with an index and hyperlinks to each section. Individual bound PDFs without automated navigation will be rejected. All O&M data shall be grouped by the equipment type and ordered by the specification numbering.

C. Prior to final payment a final electronic copy of the O&M manual on an archival quality DVD as well as two printed copies shall be furnished to the owner. Printed copies shall have commercial quality 8-1/2" x 11" 3-ring binders with tabbed dividers for each section.

PART 3 - EXECUTION

3.1 SITE EXAMINATION

A. Prior to submitting bid, Contractor shall visit site of proposed work and familiarize himself with conditions affecting work. Allowance shall be made in bid for these conditions and no additional allowance shall be granted because of lack of knowledge of such conditions.

B. Contractor shall verify all measurements at building site.

3.2 CUTTING AND PATCHING

A. Obtain written permission of Architect/Engineer before cutting or piercing structural members.

B. Sleeves through floors and walls shall be black iron pipe, flush with walls, ceilings or finished floors, sized to accommodate raceway. Grout all penetrations through concrete walls or floors. Holes through existing concrete and concrete block (CMU) shall be core drilled.

3.3 CLEAN-UP AND COMMISSIONING

A. DURING CONSTRUCTION - Throughout construction, keep work area reasonably neat and orderly by periodic clean-ups.

B. COMMISSIONING - As independent parts of construction are completed, they may be commissioned and utilized during construction. See various sections for restrictions.

C. AT COMPLETION OF WORK

1. Clean equipment of dirt and debris, including interior of panels, outlet boxes, etc. Remove labels from and clean all fixture lenses.
2. Remove materials, scraps, etc., relative to this work and leave premises in clean and orderly condition. This includes all tunnels, attics, ceiling and crawl spaces.
3. Remove all temporary facilities and restore to conditions present prior to work.

3.4 PROJECT COMPLETION AND DEMONSTRATION

A. TESTING

1. Prior to final test, all switches, panelboards, devices, and fixtures shall be in place.
2. At completion of work, or upon request from Architect/Engineer, place entire electrical installation, and/or any portion thereof, in operation to demonstrate satisfactory operation.
3. All electrical systems shall be free from short circuits and unintentional grounds.

4. Furnish one (1) copy of certified test results to Architect/Engineer prior to final inspection and include one (1) copy in each Brochure of Equipment.

B. ADJUSTMENTS

1. Make all changes necessary to balance connected electrical loads on complete system. Arrange for balanced conditions of circuits under connected load demands, as contemplated by normal working conditions. Final load and balance test shall be demonstrated in presence of Architect/Engineer.
2. Immediately correct all deficiencies which are evidenced during tests and repeat tests until system is approved. Do not cover or conceal electrical installations until satisfactory tests are made and approved.

C. FINAL WALK-THRU

1. Conduct operating tests during final inspection. Demonstrate installation to operate satisfactorily in accordance with requirements of Contract Documents. Should any portion of installation fail to meet requirements of Contract Documents, repair or replace items failing to meet requirements until items can be demonstrated to comply.
2. Have instruments available for measuring light intensities, voltage and current values and for demonstration of continuity, grounds, or open circuit conditions.
3. Furnish personnel to assist in taking measurements and making tests. In event that systems are not complete and fully operational at time of final inspection, all costs of any subsequent inspections shall be borne by Contractor at no additional cost to Owner.

3.5 OWNER ORIENTATION AND TRAINING

A. GENERAL

1. The system training is intended to familiarize the Owner's operating and maintenance staff with all systems requiring maintenance. Training is to be provided after the systems are in place and operational, after issues noted during commissioning have been resolved, and before final acceptance.
2. Provide second set of training sessions for automatic control systems about 6-9 months after the first sessions.
3. All Training shall be videotaped and reproduced on DVD's and given to the owner. Provide a copy for each O&M manual produced.
4. See Individual specification sections for additional training requirements.

B. ATTENDANCE

1. Training is to be provided by contractor's representatives that are familiar with the system's operation and maintenance requirements. Individual training sessions (modules) are to provided for each type or group of systems, separated roughly by trade group that will be performing maintenance on the system.

C. SCHEDULE

1. Duplicate training sessions are to be provided for each training module, so that Owner's operating personnel can be split into two groups during training. Duplicate training sessions

to be scheduled on different days. Length of training sessions will be determined by scope of training indicated below, and as coordinated with Owner after draft copy of training documents have been reviewed.

D. TRAINING DOCUMENTATION

1. Contractor to submit draft copy of agenda and training documents to Owner for review at least two weeks prior to training date.
2. Provide a copy of the following items for each person that will be attending the training sessions. Coordinate required number with the Owner.
   a. Training agenda.
   b. Summary of new systems and existing systems affected by this project.
   c. Summary of work performed under this project.
   d. Control system drawings and sequences of operation.
   e. List of important maintenance and trouble-shooting operations for all systems.

3. Provide minimum of 2 copies of following items:
   a. Contract documents including all drawings, specifications, addendums, and change orders.

E. TRAINING SESSIONS

1. Assemble at location to be determined by the Owner.
2. Distribute training documentation as indicated above.
3. Provide classroom style training if required for orientation, discussion of new systems and existing systems affected by this project, and other issues appropriate for a classroom format.
4. Visit site and review locations, and perform detailed review of operation and maintenance requirements for current systems.
5. All training session shall be video recorded and distributed to the owner upon completion in DVD format, or owner desired format. Include all training videos in the O&M manual.

END OF SECTION 260010

SECTION 260505 - SELECTIVE DEMOLITION OF ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. This section describes general requirements and methods of execution relating to selective demolition of electrical systems.

B. Not all removal and revision work required as part of the demolition work is shown on the plans. The plans are intended to indicate areas where demolition will occur and to establish the intent of the demolition work. It is the Contractor's responsibility to remove all existing electrical raceways, wires, devices and equipment that fall within the area affected by demolition of the structure.

C. The Contractor shall thoroughly familiarize himself with work and local conditions under which the work is to be performed. Using original design drawings and walk-through inspections, a concerted effort was made to place pertinent information on contract drawings. However, due to nature of demo/remodel work, the Contractor must bear in mind that unforeseen conditions may exist, and shall thoroughly inspect work area prior to his bid. The Contractor shall include in his bid any incidental items which may be required to provide complete demolition and rework associated systems in adjacent areas where no demolition is occurring.

PART 2 - PRODUCTS

2.1 MATERIALS AND EQUIPMENT

A. Provide materials in accordance with applicable sections in these specifications where:

1. Additional conduit, fittings, conductors, etc., are required for re-connection of circuits that extend beyond the demolition area.
2. Devices or equipment need to be temporarily or permanently relocated.
3. Portions of the remaining structure need to be patched or resurfaced.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify field measurements and circuiting arrangements as shown on Drawings.

B. Verify that raceways, wiring and equipment being demo'ed only serve facilities in the designated demolition area.

C. Examine existing light fixtures being removed to verify if ballasts contain PCB's.

3.2 PREPARATION

A. Provide temporary wiring and connections to maintain existing systems in service during construction. When work must be performed on energized equipment or circuits, use personnel experienced in such operations and follow the safe working practice requirements of NFPA 70E.

B. PRE-DEMOLITION MEETING - Participate in a pre-demolition meeting at the project site with Owner and all affected stakeholders.

1. Inspect and discuss the condition of construction to be selectively demolished.
2. Review all asbestos reports and plan electrical demo work to comply with report findings.
3. Review and finalize selective demolition schedule and verify availability of materials, demolition personnel, equipment, and facilities needed to make progress and avoid delays.
4. Review and coordinate requirements of work performed by other trades.
5. Review areas where existing construction is to remain and requires protection.
6. Review procedures to be followed when critical systems are inadvertently interrupted. The Contractor shall be responsible for the coordination required with Owner prior to device/system removal to ensure systems that must remain operational are not compromised during the demolition process.

C. SURVEY OF EXISTING CONDITIONS - Record existing conditions by use of preconstruction photographs or video.

1. Inventory and record the condition of items to be removed and salvaged. Provide photographs or video of conditions that might be misconstrued as damage caused by salvage operations.
2. Before selective demolition or removal of existing building elements that will be reproduced or duplicated in final Work, make permanent record of measurements, materials, and construction details required to make exact reproduction.

D. EXISTING ELECTRICAL SERVICE

1. Make provisions to maintain existing electrical services in service at all times.
2. Disable the power system only to make switchovers and connections.
3. Obtain permission from the Owner and the Architect/Engineer at least [48] hours prior to partially or completely disabling the system.
4. Minimize the duration of any outages.
5. If required, make temporary connections to maintain service in areas adjacent to the demolition work area.

3.3 COORDINATION

A. The Contractor is responsible for providing and coordinating phased activities and construction methods that minimize disruption to facility operations. Ensure that any portion of systems or devices to remain continue to be complete and operational. Equipment and devices shall not be removed or reconfigured until coordinated with owner.

B. The Contractor shall coordinate interfaces to existing systems that are being demolished in order to minimize disruption to the existing systems operations. Coordinate all utility service and

system outages with the Owner's Representative, the Architect/Engineer and the local Utility Company as applicable.

C. Demolition and remodel shall be done quickly so as to not hinder other trades.

D. Refer to demolition drawings, new drawings and site drawings to coordinate demolition and remodel efforts. Notify Architect/Engineer of any discrepancies.

3.4 EXISTING SERVICES/SYSTEMS TO REMAIN - Maintain services/systems indicated to remain and protect them against damage.

A. Comply with requirements for existing services/systems interruptions.

B. When temporary bypass systems are installed, test and get approval from Engineer before proceeding with demolition of existing systems.

C. For existing equipment cabinets with active components in them, provide an air tight dust seal around the cabinet and circulate cooling air with a portable air conditioning unit or other means to ensure equipment does not overheat.

3.5 DEMOLITION

A. Revise electrical connections as required to remove all equipment and items listed herein or shown on plans. Demolish and remove existing construction only to the extent required by new construction and as indicated. Use methods required to complete the Work within limitations of governing regulations

B. Remove all electrical devices from walls, floors and ceilings that are to be demolished or moved. This includes but is not limited to:

1. Abandoned panelboards and distribution equipment along with the conduits and wires that constitute their feeders.
2. Starters, disconnects and other devices and equipment serving utilization equipment that is being removed.
3. Light fixtures including brackets, stems, hangers, and other accessories.
4. Switches, outlets, horns, bells, intercom stations, clocks, etc.

C. Remove abandoned outlets if conduit and wiring servicing them is abandoned and removed. Provide blank cover for any abandoned boxes which are noted on the plans as not removed.

D. Remove conduit to point where it no longer interferes with construction and is concealed. For conduit buried in concrete or CMU walls, cut conduit off flush with floor and plug conduit.

E. If certain conduits and boxes are abandoned but not scheduled for removal, they shall be shown on the "As Built Drawings".

F. If the plans specifically call for conduits that are routed through the demolition area, and are to remain, provide supplemental support to meet the requirements in:
1. Section 260529 "Hangers and Supports for Electrical Systems."
2. Section 260533 "Raceways and Boxes for Electrical Systems."

3. Section 260548.16 "Seismic Controls for Electrical Systems."

G. Remove all conductors back to source (panelboard or last live device). Remove all abandoned communications and security systems cable from origin to destination (do not abandon in place UNO).

H. Contractor shall give Owner option to keep demo'ed electrical items of his choice. Contractor is responsible for disposal of all remaining electrical items.

I. Contractor shall be responsible for disposal of all removed lamps and ballasts. Ballasts may contain PCB's and lamps may contain Mercury. These shall be disposed of according to environmental regulations.

J. Provide revised typed circuit directory in panelboards that have circuits removed.

K. Repair adjacent construction and finishes damaged during demolition and extension work.

L. Maintain access to existing electrical installations that remain active. Modify installation or provide access panel as appropriate.

M. Neatly cut openings and holes plumb, square, and true to dimensions required. Use cutting methods least likely to damage construction to remain or adjoining construction. Use hand tools or small power tools designed for sawing or grinding, not hammering and chopping, to minimize disturbance of adjacent surfaces. Temporarily cover any openings to remain.

N. Cut or drill from the exposed or finished side into concealed surfaces to avoid marring existing finished surfaces.

O. Do not use cutting torches until work area is cleared of flammable materials. At concealed spaces, such as duct and pipe interiors, verify condition and contents of hidden space before starting flame-cutting operations. Maintain fire watch and/or portable fire suppression devices during flame-cutting operations.

P. Maintain adequate ventilation when using cutting torches.

Q. Locate selective demolition equipment and remove debris and materials so as not to impose excessive loads on supporting walls, floors, or framing.

R. Dispose of demolished items and materials promptly.

3.6 RELOCATION OF EXISTING EQUIPMENT

A. Equipment to be relocated shall be serviced, modified and repaired as necessary to place it in good working order and to satisfaction of Architect/Engineer.

B. Pack or crate items after cleaning and repairing. Identify contents of containers.

C. Protect items from damage during transport and storage.

D. Reinstall items in locations indicated. Comply with installation requirements for new materials and equipment. Provide connections, supports, and miscellaneous materials necessary to make the item functional for use at its new location.

E. Equipment shall be tested in the new location and proper function demonstrated.

3.7 HANDLING OF BALLASTS WITH PCBs - Generally, all high power factor fluorescent light ballasts manufactured before 1978 and some HID ballasts contain polychlorinated biphenyl (PCB) compounds in their capacitors. The Contractor shall inspect all ballasts in all light fixtures and take the actions described below:

A. The disposal of all ballasts labeled as "NON-PCBs" or "NO PCBs" shall become the responsibility of the Contractor. If the PCB content is not stated on the ballast label, the ballast shall be handled as a PCB ballast.

B. All PCB ballasts shall be removed from the light fixtures and shall have the wires clipped off. However, before removal, all PCB ballasts shall be carefully inspected for leaks. If a ballast appears to be leaking (evidenced by potting compound leaking out or by an oily film on the ballast surface) the ballast must be handled per EPA and DNR PCB regulations. Basically, this means the ballast is to be carefully removed from the fixture and placed in an approved drum as noted below. The person removing the ballast from the fixture shall wear protective gloves, eye protection, and protective clothing as necessary.

C. If the fixture has also been contaminated, it must be cleaned to less than 10 micrograms/100 square centimeters contamination before disposal. Contact Architect/Engineer as this cleaning must be done by an approved PCB contractor and is not considered part of this contract.

D. The PCB ballasts shall then be placed in US DOT approved drums (barrels). The quantity and size of the drums will be determined by the contractor at the time of construction, 30 and 55 gallon drums are typically available.

E. PCB BALLASTS ARE NOT TO BE REMOVED FROM THE WORK SITE BY THE CONTRACTOR. To do so would be a violation of DNR and DOT hazardous waste regulations and may result in a fine to the Contractor.

F. The Contractor shall label and mark the PCB storage drums with EPA approved PCB labels and the storage area with signs, marks and lines.

G. The Contractor shall also provide approved PCB absorbent materials to be stored immediately adjacent to the drum storage area. Do not place loose absorbent material in the drums.

3.8 DISPOSAL OF DEMOLISHED MATERIALS

A. General: Except for items or materials indicated to be recycled, reused, salvaged, reinstalled, or otherwise indicated to remain Owner's property, remove demolished materials from Project site and legally dispose of them in an EPA-approved landfill.

1. Do not allow demolished materials to accumulate on-site.

2. Remove and transport debris in a manner that will prevent spillage on adjacent surfaces and areas.
3. Remove debris from elevated portions of building by chute, hoist, or other device that will convey debris to grade level in a controlled descent.

B. Burning: Do not burn demolished materials.

C. Disposal: Transport demolished materials off Owner's property and legally dispose of them.

3.9 LAMP DISPOSAL

A. All lamps contain mercury and/or lead, as well as other heavy metals and compounds which are regulated by the EPA. As a result, regulations have been issued covering the handling and disposal of all lamps. Lamps which have been removed from service for disposal shall be handled as follows by the Contractor:

1. The Contractor shall very carefully remove all lamps (fluorescent, incandescent, and HID) from light fixtures before removal of the fixture from its mounted position. This is to reduce the likelihood that the lamps will be broken.
2. All fluorescent, neon, mercury vapor, high-pressure sodium, and metal halide lamps shall be recycled in accordance with jurisdictional requirements. The contractor shall be responsible for storing, labeling, shipping and training workers in accordance with recycling requirements. Include recycling receipts in O&M Manuals at the completion of the project.

3.10 CLEANING

A. Clean adjacent structures and improvements of dust, dirt, and debris caused by demolition operations. Return adjacent areas to condition existing before demolition operations began.

B. The contractor shall be required, on a daily basis, to dispose of any demolished material not required to be returned to the Owner. All materials shall be transported off of the Owner’s property at the expense of the Contractor.

C. At the end of each work day or shift, the Contractor shall be required to clean up the work area and remove all construction debris such that the site is clean and usable without hazard to workers.

END OF SECTION 260505

SECTION 260519 - LOW-VOLTAGE ELECTRICAL POWER CONDUCTORS AND CABLES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Copper building wire rated 600 V or less.
2. Aluminum building wire rated 600 V or less.
3. Metal-clad cable, Type MC, rated 600 V or less.
4. Nonmetallic sheathed cable (aka Romex).
5. Connectors, splices, and terminations rated 600 V and less.

B. Related Requirements:

1. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

PART 2 - PRODUCTS

2.1 COPPER BUILDING WIRE

A. Description: Flexible, insulated and uninsulated, drawn copper current-carrying conductor with an overall insulation layer or jacket, or both, rated 600 V or less.

B. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Alcan Products Corporation; Alcan Cable Division.
2. Alpha Wire Company.
3. Belden Inc.
4. Cerro Wire LLC.
5. Encore Wire Corporation.
6. General Cable Technologies Corporation.
7. Okonite Company.
8. Service Wire Co.
9. Southwire Incorporated.
10. WESCO

C. Standards:

1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.
2. RoHS compliant.
3. Conductor and Cable Marking: Comply with wire and cable marking according to UL's "Wire and Cable Marking and Application Guide."

D. Conductors: Copper, complying with ASTM B 3 for bare annealed copper and with ASTM B 8 for stranded conductors.

E. Conductor Insulation:

1. Type USE-2 and Type SE: Comply with UL 854.
2. Type THHN and Type THWN-2: Comply with UL 83.
3. Type THW-2: Comply with NEMA WC-70/ICEA S-95-658 and UL 83.
4. Type XHHW-2: Comply with UL 44.

2.2 ALUMINUM BUILDING WIRE

A. Description: Flexible, insulated and uninsulated, drawn aluminum current-carrying conductor with an overall insulation layer or jacket, or both, rated 600 V or less.

B. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Alcan Products Corporation; Alcan Cable Division.
2. Alpha Wire Company.
3. Belden Inc.
4. Cerro Wire LLC.
5. Encore Wire Corporation.
6. General Cable Technologies Corporation.
7. Okonite Company.
8. Service Wire Co.
9. Southwire Incorporated.
10. WESCO

C. Standards:

1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.
2. RoHS compliant.
3. Conductor and Cable Marking: Comply with wire and cable marking according to UL's "Wire and Cable Marking and Application Guide."

D. Conductors: Aluminum, complying with ASTM B 800 and ASTM B 801.

E. Conductor Insulation:

1. Type USE-2 and Type SE: Comply with UL 854.
2. Type THHN and Type THWN-2: Comply with UL 83.
3. Type THW and Type THW-2: Comply with NEMA WC-70/ICEA S-95-658 and UL 83.

4. Type XHHW-2: Comply with UL 44.

2.3 METAL-CLAD CABLE, TYPE MC

A. Description: A factory assembly of one or more current-carrying insulated conductors in an overall metallic sheath.

B. Approved for:
1. Lighting whips 6’ or less.
2. Installation into existing walls and ceilings.
3. Installation in crawl spaces.

C. Not approved for:
1. Exposed applications.

D. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. AFC Cable Systems.
2. Alpha Wire Company.
3. Belden Inc.
4. Encore Wire Corporation.
5. General Cable Technologies Corporation.
6. Okonite Conpany.
7. Service Wire Co.
8. Southwire Incorporated.
9. WESCO

E. Standards:

1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.
2. Comply with UL 1569.

F. Conductor and Cable Marking: Comply with wire and cable marking according to UL's "Wire and Cable Marking and Application Guide."

G. Conductors:

1. Copper, complying with ASTM B 3 for bare annealed copper and with ASTM B 8 for stranded conductors.
2. Maximum size wire: #10 AWG.

H. Ground Conductor: Insulated.

I. Conductor Insulation:

1. Type TFN/THHN/THWN-2: Comply with UL 83.

J. Armor: Steel, interlocked.

K. Jacket: PVC applied over armor for mechanical connection or wet/damp environments

2.4 NONMETALLIC-SHEATHED CABLE, TYPE NM

A. Description: A factory assembly of one or more current-carrying insulated conductors in an overall metallic sheath.

B. Approved for:
1. Lighting whips 6’ or less.
2. Installation into existing walls and ceilings.
3. Installation in crawl spaces.

C. Not approved for:
1. Exposed applications.

D. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Cable Cabana
2. Encore Wire Corporation.
3. Nassau National Cable.
4. Southwire Incorporated.

E. Standards:

1. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.
2. Comply with UL 719.

F. Conductor and Cable Marking: Comply with wire and cable marking according to UL's "Wire and Cable Marking and Application Guide."

G. Conductors - Copper, complying with ASTM B 3 for bare annealed copper and with ASTM B 8 for stranded conductors.

H. Ground Conductor: Uninsulated, inside outer sheathing.

I. Conductor Insulation:

1. Type THHN: Comply with UL 83.

2.5 CONNECTORS AND SPLICES

A. Description: Factory-fabricated connectors, splices, and lugs of size, ampacity rating, material, type, and class for application and service indicated; listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and use.

B. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. 3M Electrical Products
2. AFC Cable Systems, Inc.
3. Gardner Bender.
4. Hubbell Power Systems, Inc.
5. Ideal Industries, Inc.
6. Ilsco; a branch of Bardes Corporation.
7. NSi Industries LLC.
8. O-Z/Gedney; a brand of the EGS Electrical Group.
9. Service Wire Co.
10. TE Connectivity Ltd.
11. Thomas and Betts Corp

PART 3 - EXECUTION

3.1 CONDUCTOR MATERIAL APPLICATIONS

A. Branch Circuits: Copper; solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.

B. Feeders: Copper for feeds smaller than 100 A; copper or aluminum for feeders 100 A and larger. Conductors shall be solid for No. 10 AWG and smaller; stranded for No. 8 AWG and larger.

3.2 CONDUCTOR INSULATION AND MULTICONDUCTOR CABLE APPLICATIONS AND WIRING METHODS

A. Service Entrance:

1. Type THHN/THWN-2, single conductors in raceway.
2. Type SE or Type USE, multi-conductor cable.

B. Branch Circuits:

1. Type THHN/THWN-2, single conductors in raceway.
2. Metal-clad cable, Type MC.
3. Nonmetallic-sheathed cable, Type NM.

4. Surface mount panels:
   a. Homeruns from panel to j-box or gutter in accessible ceiling or floor space shall be Type THHN/THWN-2, single conductors in raceway.
   b. From box/gutter in concealed accessible space to point of use:
      1) Type THHN/THWN-2, single conductors in raceway.
      2) Metal-clad cable, Type MC.
      3) Nonmetallic-sheathed cable, Type NM.

5. Recessed, flush mount panels:
   a. Type THHN/THWN-2, single conductors in raceway.
   b. Metal-clad cable, Type MC.
   c. Nonmetallic-sheathed cable, Type NM.

C. Cord Drops and Portable Appliance Connections: Type SO, hard service cord with stainless-steel, wire-mesh, strain relief device at terminations to suit application.

3.3 INSTALLATION OF CONDUCTORS AND CABLES

A. Conceal cables in finished walls, ceilings, and floors unless otherwise indicated.

B. Complete raceway installation between conductor and cable termination points according to Section 260533 "Raceways and Boxes for Electrical Systems" prior to pulling conductors and cables.

C. Use manufacturer-approved pulling compound or lubricant where necessary; compound used must not deteriorate conductor or insulation. Do not exceed manufacturer's recommended maximum pulling tensions and sidewall pressure values.

D. Use pulling means, including fish tape, cable, rope, and basket-weave wire/cable grips, that will not damage cables or raceway.

E. Install exposed cables parallel and perpendicular to surfaces of exposed structural members, and follow surface contours where possible.

F. Support cables according to Section 260529 "Hangers and Supports for Electrical Systems."

G. **Provide a dedicated neutral conductor for <u>each</u> 120 V branch circuit**.

3.4 CONNECTIONS

A. Tighten electrical connectors and terminals according to manufacturer's published torque-tightening values. If manufacturer's torque values are not indicated, use those specified in UL 486A-486B.

B. Make splices, terminations, and taps that are compatible with conductor material and that possess equivalent or better mechanical strength and insulation ratings than unspliced conductors.

C. Wiring at Outlets: Install conductor at each outlet, with at least 6 inches (150 mm) of slack.

3.5 NM CABLE APPLICATIONS

A. Installation shall be in strict accordance with NEC Article 334.

B. Wires shall be sized in accordance with the 60 degree C rating column of NEC Table 310.16.

3.6 IDENTIFICATION

A. Identify and color-code conductors and cables according to Section 260553 "Identification for Electrical Systems."

B. Identify each spare conductor at each end with identity number and location of other end of conductor, and identify as spare conductor.

3.7 SLEEVE AND SLEEVE-SEAL INSTALLATION FOR ELECTRICAL PENETRATIONS

A. Install sleeves and sleeve seals at penetrations of exterior floor and wall assemblies. Comply with requirements in Section 260544 "Sleeves and Sleeve Seals for Electrical Raceways and Cabling."

3.8 FIRESTOPPING

A. Apply firestopping to electrical penetrations of fire-rated floor and wall assemblies to restore original fire-resistance rating of assembly.

END OF SECTION 260519

SECTION 260526 - GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. RATIONALE – Grounding provides the foundation to the entire electrical system. This system is designed to:

1. Protect personnel.
2. Minimize damage to equipment and property in the event of high fault current situations,
3. Improve overall electrical system reliability, and
4. Minimize the effects of transient overvoltages.

B. Section includes grounding and bonding systems and equipment.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.4 INFORMATIONAL SUBMITTALS

A. Coordination Drawings: Plans showing dimensioned as-built locations of grounding features specified in "Field Quality Control" Article.

1. Ground rods

B. Field quality-control reports.

C. Photographs of ground rod installation prior to burial.

D. Photographs of Cadweld connections to rebar and building steel.

E. Photographs of connection to water service and all bonding jumpers.

F. Certified test results from ground resistance measurements.

PART 2 - PRODUCTS

2.1 SYSTEM DESCRIPTION

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Comply with UL 467 for grounding and bonding materials and equipment.

2.2 CONDUCTORS

A. Insulated Conductors: Copper wire or cable insulated for 600 V unless otherwise required by applicable Code or authorities having jurisdiction.

B. Equipment and wiring device grounding conductor shall be as follows:

1. Bare copper or have green insulation of same type as circuit conductors (larger wires may be permanently marked with green).
2. Properly sized in accordance with the NEC.

C. Bare Copper Conductors:

1. Solid Conductors: ASTM B 3.
2. Stranded Conductors: ASTM B 8.
3. Tinned Conductors: ASTM B 33.
4. Bonding Cable: 28 kcmil, 14 strands of No. 17 AWG conductor, 1/4 inch (6 mm) in diameter.
5. Bonding Conductor: No. 4 or No. 6 AWG, stranded conductor.
6. Bonding Jumper: Copper tape, braided conductors terminated with copper ferrules; 1-5/8 inches (41 mm) wide and 1/16 inch (1.6 mm) thick.
7. Tinned Bonding Jumper: Tinned-copper tape, braided conductors terminated with copper ferrules; 1-5/8 inches (41 mm) wide and 1/16 inch (1.6 mm) thick.

D. Grounding Bus: Predrilled rectangular bars of annealed copper, 1/4 by 4 inches (6.3 by 100 mm) in cross section, with 9/32-inch (7.14-mm) holes spaced 1-1/8 inches (28 mm) apart. Stand-off insulators for mounting shall comply with UL 891 for use in switchboards, 600 V and shall be Lexan or PVC, impulse tested at 5000 V.

2.3 CONNECTORS

A. Listed and labeled by an NRTL acceptable to authorities having jurisdiction for applications in which used and for specific types, sizes, and combinations of conductors and other items connected.

B. Bolted Connectors for Conductors and Pipes: Copper or copper alloy, pressure type with at least two bolts.

1. Pipe Connectors: Clamp type, sized for pipe.

C. Welded Connectors: Exothermic-welding kits of types recommended by kit manufacturer for materials being joined and installation conditions.

D. Bus-Bar Connectors: Mechanical type, cast silicon bronze, solderless compression-type wire terminals, and long-barrel, two-bolt connection to ground bus bar.

E. Beam Clamps: Mechanical type, terminal, ground wire access from four directions, with dual, tin-plated or silicon bronze bolts.

F. Cable-to-Cable Connectors: Compression type, copper or copper alloy.

G. Conduit Hubs: Mechanical type, terminal with threaded hub.

H. Ground Rod Clamps: Mechanical type, copper or copper alloy, terminal with hex head bolt.

I. Lay-in Lug Connector: Mechanical type, copper rated for direct burial terminal with set screw.

J. Service Post Connectors: Mechanical type, bronze alloy terminal, in short- and long-stud lengths, capable of single and double conductor connections.

K. Signal Reference Grid Clamp: Mechanical type, stamped-steel terminal with hex head screw.

L. Straps: Solid copper, copper lugs. Rated for 600 A.

M. Tower Ground Clamps: Mechanical type, copper or copper alloy, terminal one-piece clamp.

N. U-Bolt Clamps: Mechanical type, copper or copper alloy, terminal listed for direct burial.

O. Water Pipe Clamps:

1. Mechanical type, two pieces with zinc-plated bolts.

a. Material: Die-cast zinc alloy.
b. Listed for direct burial.

2.4 GROUNDING ELECTRODES

A. Ground Rods: Copper-clad steel; 3/4 inch by 10 feet (19 mm by 3 m).

PART 3 - EXECUTION

3.1 APPLICATIONS

A. Conductors: Install solid conductor for No. 10 AWG and smaller, and stranded conductors for No. 8 AWG and larger unless otherwise indicated.

B. Grounding Bus: Install in electrical equipment rooms, in rooms housing service equipment, and elsewhere as indicated.

1. Install bus horizontally, on insulated spacers 2 inches (50 mm) minimum from wall, 6 inches (150 mm) above finished floor unless otherwise indicated.

C. Conductor Terminations and Connections:

1. Pipe and Equipment Grounding Conductor Terminations: Bolted connectors.
2. Underground Connections: Welded connectors except at test wells and as otherwise indicated.
3. Connections to Ground Rods at Test Wells: Bolted connectors.
4. Connections to Structural Steel: Welded connectors.
5. Any threaded bolt connectors shall be torqued in accordance with manufacturer's guidelines.

3.2 GROUNDING AT THE SERVICE

A. Equipment grounding conductors and grounding electrode conductors shall be connected to the ground bus. Install a main bonding jumper between the neutral and ground buses.

3.3 EQUIPMENT GROUNDING

A. Install insulated equipment grounding conductors with all feeders and branch circuits. Do not rely on conduit for the grounding path.

B. Multiple circuits sharing a raceway may share a single grounding conductor if all of the following requirements are met:

1. All circuits originate in the same panel.
2. No more than three single pole circuits may share a ground conductor.
3. Size the ground conductor for the largest circuit.

C. Install insulated equipment grounding conductors with the following items, in addition to those required by NFPA 70:

1. Feeders and branch circuits.
2. Lighting circuits.
3. Receptacle circuits.
4. Single-phase motor and appliance branch circuits.
5. Three-phase motor and appliance branch circuits.
6. Flexible raceway runs.
7. Armored and metal-clad cable runs.

D. Air-Duct Equipment Circuits: Install insulated equipment grounding conductor to duct-mounted electrical devices operating at 120 V and more, including air cleaners, heaters, dampers, humidifiers, and other duct electrical equipment. Bond conductor to each unit and to air duct and connected metallic piping.

E. Water Heater, Heat-Tracing, and Antifrost Heating Cables: Install a separate insulated equipment grounding conductor to each electric water heater and heat-tracing cable. Bond conductor to heater units, piping, connected equipment, and components.

3.4 INSTALLATION

A. Grounding Conductors: Route along shortest and straightest paths possible unless otherwise indicated or required by Code. Avoid obstructing access or placing conductors where they may be subjected to strain, impact, or damage.

B. Ground Rods: Drive rods until tops are 2 inches (50 mm) below finished floor or final grade unless otherwise indicated.

1. Interconnect ground rods with grounding electrode conductor below grade and as otherwise indicated. Make connections without exposing steel or damaging coating if any.
2. For grounding electrode system, install at least 2 rods spaced at least one-rod length from each other and located at least the same distance from other grounding electrodes, and connect to the service grounding electrode conductor.

C. Bonding Straps and Jumpers: Install in locations accessible for inspection and maintenance except where routed through short lengths of conduit.

1. Bonding to Structure: Bond straps directly to basic structure, taking care not to penetrate any adjacent parts.
2. Bonding to Equipment Mounted on Vibration Isolation Hangers and Supports: Install bonding so vibration is not transmitted to rigidly mounted equipment.
3. Use exothermic-welded connectors for outdoor locations; if a disconnect-type connection is required, use a bolted clamp.

D. Grounding and Bonding for Piping:

1. Metal Water Service Pipe: Install insulated copper grounding conductors, in conduit, from building's main service equipment, or grounding bus, to main metal water service entrances to building. Connect grounding conductors to main metal water service pipes; use a bolted clamp connector or bolt a lug-type connector to a pipe flange by using one of the lug bolts of the flange. Where a dielectric main water fitting is installed, connect grounding conductor on street side of fitting. Bond metal grounding conductor conduit or sleeve to conductor at each end.
2. Water Meter Piping: Use braided-type bonding jumpers to electrically bypass water meters. Connect to pipe with a bolted connector.
3. Bond each aboveground portion of gas piping system downstream from equipment shutoff valve.

E. Bonding Interior Metal Ducts: Bond metal air ducts to equipment grounding conductors of associated fans, blowers, electric heaters, and air cleaners. Install bonding jumper to bond across flexible duct connections to achieve continuity. Size bonding conductors and jumpers in accordance with NEC 250.122, using the rating of the circuit that is likely to energize the ducts.

F. Ufer Ground (Concrete-Encased Grounding Electrode): Fabricate according to NFPA 70, and as detailed on drawings. If drawing does not provide Ufer ground detail install the grounding electrode as defined below:

1. Use a minimum of 20 feet (6 m) of bare copper conductor not smaller than No. 4 AWG.
2. If concrete foundation is less than 20 feet (6 m) long, coil excess conductor within base of foundation.

3. Bond grounding conductor to reinforcing steel in at least four locations and to anchor bolts. Extend grounding conductor below grade and connect to building's grounding grid or to grounding electrode external to concrete.

G. Structural Steel: When available, bond structural steel to grounding electrode system, according to NFPA 70.

3.5 FIELD QUALITY CONTROL

A. Perform tests and inspections.

B. Tests and Inspections:

1. After installing grounding system but before permanent electrical circuits have been energized, test for compliance with requirements.
2. Inspect physical and mechanical condition. Verify tightness of accessible, bolted, electrical connections with a calibrated torque wrench according to manufacturer's written instructions.
3. Test completed grounding system at each location where a maximum ground-resistance level is specified, at service disconnect enclosure grounding terminal, and at individual ground rods. Make tests at ground rods before any conductors are connected.

    a. Measure ground resistance no fewer than two full days after last trace of precipitation and without soil being moistened by any means other than natural drainage or seepage and without chemical treatment or other artificial means of reducing natural ground resistance.
    b. Perform tests by fall-of-potential method according to IEEE 81.

4. Prepare dimensioned Drawings locating each test well, ground rod and ground-rod assembly, and other grounding electrodes. Identify each by letter in alphabetical order, and key to the record of tests and observations. Include the number of rods driven and their depth at each location, and include observations of weather and other phenomena that may affect test results. Describe measures taken to improve test results.

C. Grounding system will be considered defective if it does not pass tests and inspections.

D. Prepare test and inspection reports.

E. Report measured ground resistances that exceed 25 ohms to ground.

F. Excessive Ground Resistance: If resistance to ground exceeds specified values, notify Architect promptly and include recommendations to reduce ground resistance.

END OF SECTION 260526

SECTION 260529 - HANGERS AND SUPPORTS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Steel slotted support systems.
2. Conduit and cable support devices.
3. Support for conductors in vertical conduit.
4. Structural steel for fabricated supports and restraints.
5. Mounting, anchoring, and attachment components, including powder-actuated fasteners, mechanical expansion anchors, concrete inserts, clamps, through bolts, toggle bolts, and hanger rods.
6. Fabricated metal equipment support assemblies.

B. Related Requirements:

1. Section 260548.16 "Seismic Controls for Electrical Systems" for products and installation requirements necessary for compliance with seismic criteria.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1.3 INFORMATIONAL SUBMITTALS

A. Seismic Qualification Data: Certificates, for hangers and supports for electrical equipment and systems, accessories, and components, from manufacturer.

1.4 COORDINATION

A. Coordinate size and location of concrete bases. Cast anchor-bolt inserts into bases. Concrete, reinforcement, and formwork requirements are specified together with concrete Specifications.

B. Coordinate installation of roof curbs, equipment supports, and roof penetrations.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Seismic Performance: Hangers and supports shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

1. The term "withstand" means "the supported equipment and systems will remain in place without separation of any parts when subjected to the seismic forces specified and the supported equipment and systems will be fully operational after the seismic event."

2.2 SUPPORT, ANCHORAGE, AND ATTACHMENT COMPONENTS

A. Steel Slotted Support Systems: Preformed steel channels and angles with minimum 13/32-inch-(10-mm-) diameter holes at a maximum of 8 inches (200 mm) o.c. in at least one surface.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

    a. Allied Tube & Conduit; a part of Atkore International.
    b. B-line, an Eaton business.
    c. ERICO International Corporation.
    d. Flex-Strut Inc.
    e. Gripple Inc.
    f. G-Strut.
    g. Thomas & Betts Corporation; A Member of the ABB Group.
    h. Unistrut; Part of Atkore International.
2. Standard: Comply with MFMA-4 factory-fabricated components for field assembly.
3. Material for Channel, Fittings, and Accessories: Galvanized steel.
4. Channel Width: Selected for applicable load criteria.
5. Metallic Coatings: Hot-dip galvanized after fabrication and applied according to MFMA-4.
6. Nonmetallic Coatings: Manufacturer's standard PVC, polyurethane, or polyester coating applied according to MFMA-4.
7. Painted Coatings: Manufacturer's standard painted coating applied according to MFMA-4.
8. Protect finishes on exposed surfaces from damage by applying a strippable, temporary protective covering before shipping.

B. Conduit and Cable Support Devices: Steel hangers, clamps, and associated fittings, designed for types and sizes of raceway or cable to be supported.

C. Support for Conductors in Vertical Conduit: Factory-fabricated assembly consisting of threaded body and insulating wedging plug or plugs for nonarmored electrical conductors or cables in riser conduits. Plugs shall have number, size, and shape of conductor gripping pieces as required to suit individual conductors or cables supported. Body shall be made of malleable iron.

D. Structural Steel for Fabricated Supports and Restraints: ASTM A 36/A 36M steel plates, shapes, and bars; black and galvanized.

E. Mounting, Anchoring, and Attachment Components: Items for fastening electrical items or their supports to building surfaces include the following:

1. Powder-Actuated Fasteners: Threaded-steel stud, for use in hardened portland cement concrete, steel, or wood, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.

2. Mechanical-Expansion Anchors: Insert-wedge-type, zinc-coated steel, for use in hardened portland cement concrete, with tension, shear, and pullout capacities appropriate for supported loads and building materials where used.
3. Concrete Inserts: Steel or malleable-iron, slotted support system units are similar to MSS Type 18 units and comply with MFMA-4 or MSS SP-58.
4. Clamps for Attachment to Steel Structural Elements: MSS SP-58 units are suitable for attached structural element.
5. Through Bolts: Structural type, hex head, and high strength. Comply with ASTM A 325.
6. Toggle Bolts: All-steel springhead type.
7. Hanger Rods: Threaded steel.

2.3 FABRICATED METAL EQUIPMENT SUPPORT ASSEMBLIES

A. Description: Welded or bolted structural-steel shapes, shop or field fabricated to fit dimensions of supported equipment.

PART 3 - EXECUTION

3.1 APPLICATION

A. Comply with the following standards for application and installation requirements of hangers and supports, except where requirements on Drawings or in this Section are stricter:

1. NECA 1.
2. NECA 101
3. NECA 102.
4. NECA 105.
5. NECA 111.

B. Comply with requirements for firestopping materials and installation for penetrations through fire-rated walls, ceilings, and assemblies.

C. Comply with requirements for raceways and boxes specified in Section 260533 "Raceways and Boxes for Electrical Systems."

D. Maximum Support Spacing and Minimum Hanger Rod Size for Raceways: Space supports for EMT, IMC, and RMC as required by NFPA 70. Minimum rod size shall be 1/4 inch (6 mm) in diameter.

E. Multiple Raceways or Cables: Install trapeze-type supports fabricated with steel slotted support system, sized so capacity can be increased by at least 25 percent in future without exceeding specified design load limits.

1. Secure raceways and cables to these supports with two-bolt conduit clamps.

F. Spring-steel clamps designed for supporting single conduits without bolts may be used for 1-1/2-inch (38-mm) and smaller raceways serving branch circuits and communication systems above suspended ceilings, and for fastening raceways to trapeze supports.

3.2 SUPPORT INSTALLATION

A. Comply with NECA 1 and NECA 101 for installation requirements except as specified in this article.

B. Raceway Support Methods: In addition to methods described in NECA 1, EMT, IMC and RMC may be supported by openings through structure members, according to NFPA 70.

C. Strength of Support Assemblies: Where not indicated, select sizes of components so strength will be adequate to carry present and future static loads within specified loading limits. Minimum static design load used for strength determination shall be weight of supported components plus 200 lb (90 kg).

D. Mounting and Anchorage of Surface-Mounted Equipment and Components: Anchor and fasten electrical items and their supports to building structural elements by the following methods unless otherwise indicated by code:

1. To Wood: Fasten with lag screws or through bolts.
2. To New Concrete: Bolt to concrete inserts.
3. To Masonry: Approved toggle-type bolts on hollow masonry units and expansion anchor fasteners on solid masonry units.
4. To Existing Concrete: Expansion anchor fasteners.
5. Instead of expansion anchors, powder-actuated driven threaded studs provided with lock washers and nuts may be used in existing standard-weight concrete 4 inches (100 mm) thick or greater. Do not use for anchorage to lightweight-aggregate concrete or for slabs less than 4 inches (100 mm) thick.
6. To Steel: Beam clamps (MSS SP-58,Type 19, 21, 23, 25, or 27), complying with MSS SP-69.
7. To Light Steel: Sheet metal screws.
8. Items Mounted on Hollow Walls and Nonstructural Building Surfaces: Mount cabinets, panelboards, disconnect switches, control enclosures, pull and junction boxes, transformers, and other devices on slotted-channel racks attached to substrate by means that comply with seismic-restraint strength and anchorage requirements.

E. Drill holes for expansion anchors in concrete at locations and to depths that avoid the need for reinforcing bars.

3.3 INSTALLATION OF FABRICATED METAL SUPPORTS

A. Cut, fit, and place miscellaneous metal supports accurately in location, alignment, and elevation to support and anchor electrical materials and equipment.

B. Field Welding: Comply with AWS D1.1/D1.1M.

END OF SECTION 260529

SECTION 260533 - RACEWAYS AND BOXES FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Metal conduits and fittings.
2. Nonmetallic conduits and fittings.
3. Metal wireways and auxiliary gutters.
4. Surface raceways.
5. Boxes, enclosures, and cabinets.
6. Handholes for exterior underground cabling.

1.2 ACTION SUBMITTALS

A. Product Data: For surface raceways, wireways and fittings, floor boxes, hinged-cover enclosures, and cabinets.

1.3 INFORMATIONAL SUBMITTALS

A. Seismic Qualification Data: Certificates, for enclosures, cabinets, and conduit racks and their mounting provisions, including those for internal components, from manufacturer.

1.4 COORDINATION OF BURIED RACEWAYS ON SITE

A. Coordinate layout and installation of ducts and handholes with final arrangement of other utilities, site grading, and surface features as determined in the field.

B. Coordinate elevations of ducts and duct-bank entrances into handholes with final locations and profiles of ducts and duct banks as determined by coordination with other utilities, underground obstructions, and surface features. Revise locations and elevations from those indicated as required to suit field conditions and to ensure that duct runs drain to manholes and handholes.

PART 2 - PRODUCTS

2.1 METAL CONDUITS AND FITTINGS

A. Metal Conduit:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

a. Allied Tube & Conduit; a part of Atkore International.
b. Electri-Flex Company.
c. O-Z/Gedney; a brand of Emerson Industrial Automation.
d. Patriot Aluminum Products, LLC.
e. Perma-Cote.
f. Picoma Industries, Inc.
g. Plasti-Bond.
h. Republic Conduit.
i. Southwire Company.
j. Thomas & Betts Corporation; A Member of the ABB Group.
k. Western Tube and Conduit Corporation.

2. Listing and Labeling: Metal conduits, tubing, and fittings shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
3. GRC: Comply with ANSI C80.1 and UL 6.
4. IMC: Comply with ANSI C80.6 and UL 1242.
5. PVC-Coated Steel Conduit: PVC-coated rigid steel conduit.

a. Comply with NEMA RN 1.
b. Coating Thickness: 0.040 inch, minimum.

6. EMT: Comply with ANSI C80.3 and UL 797.
7. FMC: Comply with UL 1; zinc-coated steel.
8. LFMC: Flexible steel conduit with PVC jacket and complying with UL 360.

B. Metal Fittings: Comply with NEMA FB 1 and UL 514B.

1. Listing and Labeling: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.
2. Fittings, General: Listed and labeled for type of conduit, location, and use.
3. Conduit Fittings for Hazardous (Classified) Locations: Comply with UL 1203 and NFPA 70.
4. Fittings for EMT:
   a. Material: Steel.
   b. Type: Setscrew.

5. Expansion Fittings: PVC or steel to match conduit type, complying with UL 651, rated for environmental conditions where installed, and including flexible external bonding jumper.
6. Coating for Fittings for PVC-Coated Conduit: Minimum thickness of 0.040 inch, with overlapping sleeves protecting threaded joints.

C. Joint Compound for IMC, GRC: Approved, as defined in NFPA 70, by authorities having jurisdiction for use in conduit assemblies, and compounded for use to lubricate and protect threaded conduit joints from corrosion and to enhance their conductivity.

2.2 NONMETALLIC CONDUITS AND FITTINGS

A. Nonmetallic Conduit:

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

    a. Arnco Corporation.
    b. CANTEX INC.
    c. CertainTeed Corporation.
    d. Champion Fiberglass, Inc.
    e. Condux International, Inc.
    f. Electri-Flex Company.
    g. FRE Composites.
    h. Kraloy.
    i. Lamson & Sessions.
    j. Niedax Inc.
    k. RACO; Hubbell.
    l. Thomas & Betts Corporation; A Member of the ABB Group.

B. Listing and Labeling: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

1. RNC: Type EPC-40-PVC or Type EPC-80-PVC, complying with NEMA TC 2 and UL 651 unless otherwise indicated.
2. LFNC: Comply with UL 1660.
3. Rigid HDPE: Comply with UL 651A.
4. Continuous HDPE: Comply with UL 651B.

C. Nonmetallic Fittings:

1. Fittings, General: Listed and labeled for type of conduit, location, and use.
2. Fittings for ENT and RNC: Comply with NEMA TC 3; match to conduit or tubing type and material.
3. Fittings for LFNC: Comply with UL 514B.
4. Solvents and Adhesives: As recommended by conduit manufacturer.

2.3 STANDARD CONDUIT SEALS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1. American Polywater Corporation
2. Dura-Line, Inc.
3. FS3, Inc.

B. Description: Sealing compound for use in underground conduit to prevent water and gas infiltration in non-classified locations.

1. Semi-permanent, re-enterable seal.
2. Compatible with PVC, rigid steel, EMT, IMC, fiberglass and polyethylene conduits.
3. Keeps water, acids, greases, gases, insects, rodents, etc., out of the conduit.
4. Two-part high-expansion urethane foam with 98% closed cell content.

5. Cured compressive strength of 300 lbs. (ASTM D790), tensile strength of 250 lbs. (ASTM D1623), and flexural strength of 450 lbs. (ASTM D790) and temperature range of -20º to 200ºF.
6. Cured sealant will be capable of holding 10 psi water pressure continuously.
7. Meets NEC requirements for raceway seals per Articles 225.27, 230.8 and 300.5
8. FST™ Sealant or equivalent.

2.4 CLASSIFIED LOCATION CONDUIT SEALS

A. Available Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Appleton Electric
2. Cooper Crouse-Hinds
3. Hubbell Killark

B. Description: Sealing fittings for use between a classified and non-classified location.

1. Explosion proof and dust-ignition proof, raintight construction.
2. Smooth, rounded integral bushing to protect conductor insulation.
3. Tapped and tapered threads for tight, rigid joints and ground continuity.
4. Removable nipple in male sealing fitting that can be used inter-changeably in top or bottom hub.
5. Fiber filler to prevent sealing compound from leaking while being poured in its liquid state.
6. Sealing cement which hardens to contain and restrict the passage of gasses and explosions in classified areas.

2.5 METAL WIREWAYS AND AUXILIARY GUTTERS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1. B-line, an Eaton business.
2. Hoffman; a brand of Pentair Equipment Protection.
3. MonoSystems, Inc.

B. Description: Sheet metal, complying with UL 870 and NEMA 250, Type 1, Type 3R, Type 4 or Type 12 unless otherwise indicated, and sized according to NFPA 70.

1. Metal wireways installed outdoors shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

C. Fittings and Accessories: Include covers, couplings, offsets, elbows, expansion joints, adapters, hold-down straps, end caps, and other fittings to match and mate with wireways as required for complete system.

D. Wireway Covers: Screw-cover type unless otherwise indicated.

E. Finish: Manufacturer's standard enamel finish.

2.6 SURFACE RACEWAYS

A. Listing and Labeling: Surface raceways and tele-power poles shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Surface Metal Raceways: Galvanized steel with snap-on covers complying with UL 5. Manufacturer's standard enamel finish in color selected by Architect.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:
   a. Mono-Systems, Inc.
   b. Panduit Corp.
   c. Wiremold / Legrand.

2.7 J-HOOKS

A. Description: Prefabricated sheet metal cable supports for low-voltage cables (lighting controls).

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Eaton, B-line.
2. Panduit Corp.
3. Wiremold / Legrand.

C. Listed and labeled as defined in NFPA 70, by an NRTL, and marked for intended location and application.

2.8 BOXES, ENCLOSURES, AND CABINETS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1. Crouse-Hinds, an Eaton business.
2. Erickson Electrical Equipment Company.
3. Hoffman; a brand of Pentair Equipment Protection.
4. Hubbell Incorporated.
5. Hubbell Incorporated; Wiring Device-Kellems.
6. Milbank Manufacturing Co.
7. MonoSystems, Inc.
8. Oldcastle Enclosure Solutions.
9. O-Z/Gedney; a brand of Emerson Industrial Automation.
10. RACO; Hubbell.
11. Stahlin Non-Metallic Enclosures.
12. Thomas & Betts Corporation; A Member of the ABB Group.

B. General Requirements for Boxes, Enclosures, and Cabinets: Boxes, enclosures, and cabinets installed in wet locations shall be listed for use in wet locations.

C. Sheet Metal Outlet and Device Boxes: Comply with NEMA OS 1 and UL 514A.

D. Cast-Metal Outlet and Device Boxes: Comply with NEMA FB 1, aluminum, Type FD, with gasketed cover.

E. Nonmetallic Outlet and Device Boxes: Comply with NEMA OS 2 and UL 514C.

F. Metal Floor Boxes. See drawings for differing floor box requirements based on location, floor material and box use.

1. All floor boxes shall be:
   a. Fully adjustable.
   b. Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

2. Specific conditions include:
   a. Concrete floors (3" min. pour depth) - 4-gang floor box with corrosion resistant coating for on-grade use and up to 2" conduit feed.
   b. Raised access floors - 4-gang floor box for up to 2" conduit feed.
   c. Fire rated poke-through floor box for elevated concrete slabs:
      1) Small - 3" diameter core.
      2) Large - 8" diameter for up to 2" conduit feed.
   d. Flush, round single surface floor box for concrete floors with up to 1" conduit feed.
   e. Tombstone pedestal floor box with 1" conduit feed.

3. Include all interior box dividers, flanges, mounting hardware, wiring devices, faceplates, etc. to provide complete floor box outlet in accordance with drawings.

G. Luminaire Outlet Boxes: Nonadjustable, designed for attachment of luminaire weighing 50 lb (23 kg). Outlet boxes designed for attachment of luminaires weighing more than 50 lb (23 kg) shall be listed and marked for the maximum allowable weight.

H. Paddle Fan Outlet Boxes: Nonadjustable, designed for attachment of paddle fan weighing 70 lb (32 kg).

1. Listing and labeling: Paddle fan outlet boxes shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

I. Small Sheet Metal Pull and Junction Boxes: NEMA OS 1.

J. Cast-Metal Access, Pull, and Junction Boxes: Comply with NEMA FB 1 and UL 1773, cast aluminum with gasketed cover.

K. Box extensions used to accommodate new building finishes shall be of same material as recessed box.

L. Device Box Dimensions: 4 inches square by 2-1/8 inches deep with single gang mud ring unless device(s) requires otherwise.

M. Gang-able boxes are allowed for 6-gang or larger.

N. Hinged-Cover Enclosures: Comply with UL 50 and NEMA 250, Type 1, Type 3R, Type 4 or Type 12 with continuous-hinge cover with flush latch unless otherwise indicated.

1. Metal Enclosures: Steel, finished inside and out with manufacturer's standard enamel.
2. Nonmetallic Enclosures: Plastic.
3. Interior Panels: Steel; all sides finished with manufacturer's standard enamel.

O. Cabinets:

1. NEMA 250, Type 1, Type 3R or Type 12 galvanized-steel box with removable interior panel and removable front, finished inside and out with manufacturer's standard enamel.
2. Hinged door in front cover with flush latch and concealed hinge.
3. Key latch to match panelboards.
4. Metal barriers to separate wiring of different systems and voltage.
5. Accessory feet where required for freestanding equipment.
6. Nonmetallic cabinets shall be listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

2.9 DUCT ACCESSORIES FOR BURIED CONDUITS

A. Duct Separators: Factory-fabricated rigid PVC interlocking, intermediate (horizontal) spacers, sized for type and sizes of ducts with which used, and selected to provide minimum duct spacings indicated while supporting ducts during concreting or backfilling.

2.10 HANDHOLES FOR EXTERIOR UNDERGROUND WIRING

A. General Requirements for Handholes:

1. Handholes for use in underground systems shall be designed and identified as defined in NFPA 70, for intended location and application.
2. Handholes shall be listed and labeled for installation in wet areas as defined in NFPA 70. They shall be tested by a qualified testing agency, and marked for intended location and application.

B. Polymer-Concrete Handholes with Polymer-Concrete Cover: Molded of sand and aggregate, bound together with polymer resin, and reinforced with steel, fiberglass, or a combination of the two.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:
   a. Armorcast Products Company.
   b. NewBasis.
   c. Oldcastle Enclosure Solutions.
   d. Oldcastle Precast, Inc.
   e. Quazite (Hubbell).

2. Standard: Comply with ANSI/SCTE 77.
3. Load Ratings:
    a. Tier 5 for non-traffic areas and sidewalk applications with a safety factor for occasional non-deliberate vehicular traffic.
    b. Tier 22 for driveway, parking lot, and off-road applications subject to occasional non-deliberate heavy vehicular traffic.
    c. AASHTO H-20 for roadways and other deliberate vehicular traffic applications.
4. Configuration: Designed for flush burial with open bottom unless otherwise indicated.
5. Sides:
    a. Provide a tapered-side unit when handhole is located in landscaped areas.
    b. Provide a straight-side unit when handhole is located in hardscape areas.
6. Cover: Weatherproof, secured by stainless steel, tamper-resistant, locking devices and having structural load rating consistent with enclosure and handhole location.
7. Cover Finish: Nonskid finish shall have a minimum coefficient of friction of 0.50.
8. Cover Legend: Molded lettering, "ELECTRIC" or “COM” as indicated on drawings.
9. Conduit Entrance Provisions: Conduit-terminating fittings shall mate with entering ducts for secure, fixed installation in enclosure wall.

PART 3 - EXECUTION

3.1 RACEWAY APPLICATION

A. Outdoors: Apply raceway products as specified below unless otherwise indicated:

1. Exposed Conduit: GRC or Schedule 80 PVC where required by utility or required for special environments. Exposed Schedule 40 PVC is not allowed.
2. Concealed Aboveground Conduit: EMT.
3. Underground Conduit: RNC, Type EPC-40-PVC, or HDPE in fine bedded trench.
4. Elbows for underground conduit:
    a. Long sweep
    b. GRS with PVC or bitumastic coating, or fiberglass as required by the utility.
5. Under roadways and paved or concrete walkways: Type EPC-80-PVC or HDPE in fine bedded trench.
6. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): LFMC.
7. Boxes and Enclosures, Aboveground: NEMA 250, Type 3R.
8. Provide expansion-joint fittings where required below.

B. Indoors: Apply raceway products as specified below unless otherwise indicated.

1. Exposed, Not Subject to Physical Damage: EMT.
2. Exposed and Subject to Severe Physical Damage: GRC.
3. Concealed in Ceilings and Interior Walls and Partitions: EMT or as specified in Section 260519, “Low-Voltage Electrical Power Conductors and Cables”.

4. Connection to Vibrating Equipment (Including Transformers and Hydraulic, Pneumatic, Electric Solenoid, or Motor-Driven Equipment): FMC, except use LFMC in damp or wet locations.
5. Damp or Wet Locations: GRC.
6. Boxes and Enclosures: NEMA 250, Type 1, except use NEMA 250, Type 4 nonmetallic in institutional and commercial kitchens and damp or wet locations.
7. Concealed in CMU block wall: Type EPC-40-PVC.

C. Minimum Raceway Size: 1 inch trade size for telecom/data and 3/4 inch trade size for all other applications.

D. Raceway Fittings: Compatible with raceways and suitable for use and location.

1. Rigid and Intermediate Steel Conduit: Use threaded rigid steel conduit fittings unless otherwise indicated. Comply with NEMA FB 2.10.
2. PVC Externally Coated, Rigid Steel Conduits: Use only fittings listed for use with this type of conduit. Patch and seal all joints, nicks, and scrapes in PVC coating after installing conduits and fittings. Use sealant recommended by fitting manufacturer and apply in thickness and number of coats recommended by manufacturer.
3. EMT: Use setscrew, steel fittings. Comply with NEMA FB 2.10.
4. Flexible Conduit: Use only fittings listed for use with flexible conduit. Comply with NEMA FB 2.20.

E. Install surface raceways only where specifically indicated on Drawings.

F. Install nonmetallic conduit or tubing for protecting bare grounding conductors.

G. Do not install nonmetallic conduit where ambient temperature exceeds 120 deg F (49 deg C).

3.2 LOW VOLTAGE CABLE INSTALLATION

A. Any low voltage cables in exposed or finished areas shall be in raceway.

B. In accordance with NEC 300.11 and NEC 800.24, any low voltage cables installed in accessible ceilings without conduit, including lighting control cables, shall be as follows:

1. Cables shall not be draped over air ducts, pipes, or conduits, shall not rest on the ceiling grid or tiles, and shall not use ceiling grid support wires or rods.
2. Cables shall be supported using j-hooks at intervals not to exceed 48”. J-hooks shall be attached to the structure with dedicated support wires, and a j-hook shall be installed at each change in cabling direction.
3. Written approval shall be obtained from the IT designer prior to any use of communications system cable/ladder tray or j-hooks. Wherever cable tray or communication system j-hooks are used, the lighting controls cabling shall be bundled with cable ties. Any non-metallic cable ties used to bundle the cables shall be plenum rated.

3.3 INSTALLATION

A. Comply with requirements in Section 260529 "Hangers and Supports for Electrical Systems" for hangers and supports.

B. Comply with NECA 1 and NECA 101 for installation requirements except where requirements on Drawings or in this article are stricter. Comply with NFPA 70 limitations for types of raceways allowed in specific occupancies and number of floors.

C. Do not install raceways or electrical items on any "explosion-relief" walls or rotating equipment.

D. Do not fasten conduits onto the bottom side of a metal deck roof.

E. Keep raceways at least 6 inches away from parallel runs of flues and steam or hot-water pipes. Install horizontal raceway runs above water and steam piping.

F. Whenever routed in parallel, maintain 12” minimum separation between communications conduits and power conduits. Where these conduits must intersect, cross at 90 degrees.

G. Comply with requirements in Section 260529 "Hangers and Supports for Electrical Systems" for hangers and supports.

H. Install no more than the equivalent of three 90-degree bends in any conduit run except for control wiring conduits, for which fewer bends are allowed. Support within 12 inches of changes in direction.

I. Make bends in raceway using large-radius preformed ells. Field bending shall be according to NFPA 70 minimum radii requirements. Use only equipment specifically designed for material and size involved.

J. Conceal conduit and EMT within finished walls, ceilings, and floors unless otherwise indicated. Install conduits parallel or perpendicular to building lines.

K. Support conduit within 12 inches of enclosures to which attached.

L. Raceways Below Slab:

1. Conduits are permitted under the slab in the base material only (not within the concrete slab).
2. All routing must be approved by the structural engineer prior to rough-in.
3. Arrange stub-ups so that curved portions of bends are not visible above finished slab.
4. Change from RNC, Type EPC-40-PVC to PVC Coated GRC bend and thru slab stub before rising above floor.

M. Stub-ups to Above Recessed Ceilings:

1. Use EMT for raceways.
2. Use a conduit bushing or insulated fitting to terminate stub-ups not terminated in hubs or in an enclosure.

N. Threaded Conduit Joints, Exposed to Wet, Damp, Corrosive, or Outdoor Conditions: Apply listed compound to threads of raceway and fittings before making up joints. Follow compound manufacturer's written instructions.

O. Coat field-cut threads on PVC-coated raceway with a corrosion-preventing conductive compound prior to assembly.

P. Raceway Terminations at Locations Subject to Moisture or Vibration: Use insulating bushings to protect conductors including conductors smaller than No. 4 AWG.

Q. Install raceways square to the enclosure and terminate at enclosures with locknuts. Install locknuts hand tight plus 1/4 turn more.

R. Do not rely on locknuts to penetrate nonconductive coatings on enclosures. Remove coatings in the locknut area prior to assembling conduit to enclosure to assure a continuous ground path.

S. Cut conduit perpendicular to the length. For conduits 2-inch trade size and larger, use roll cutter or a guide to make cut straight and perpendicular to the length.

T. Terminate threaded conduits into threaded hubs or with locknuts on inside and outside of boxes or cabinets. Install bushings on conduits up to 1-1/4-inch trade size and insulated throat metal bushings on 1-1/2-inch trade size and larger conduits terminated with locknuts. Install insulated throat metal grounding bushings on service conduits.

U. Install pull wires in empty raceways. Use polypropylene or monofilament plastic line with not less than 200-lb (90-kg) tensile strength. Leave at least 12 inches of slack at each end of pull wire.

V. Surface Raceways:

1. Install surface raceway with a minimum 2-inch radius control at bend points.
2. Secure surface raceway with screws or other anchor-type devices at intervals not exceeding 48 inches and with no less than two supports per straight raceway section. Support surface raceway according to manufacturer's written instructions. Tape and glue are not acceptable support methods.

W. Install raceway sealing fittings at accessible locations according to NFPA 70 and fill them with listed sealing compound. For concealed raceways, install each fitting in a flush steel box with a blank cover plate having a finish similar to that of adjacent plates or surfaces.

X. Standard Conduit Seals:

1. Install devices to seal raceway interiors at accessible locations. Locate seals so no fittings or boxes are between the seal and the following changes of environments. Seal the interior of all raceways at the following points:

    a. Where conduits pass from warm to cold locations, such as boundaries of refrigerated spaces.
    b. Where an underground service raceway enters a building or structure.
    c. Conduit extending from interior to exterior of building.
    d. Conduit extending into pressurized duct and equipment.

e. Conduit extending into pressurized zones that are automatically controlled to maintain different pressure set points.
f. Where otherwise required by NFPA 70.

Y. Expansion-Joint Fittings:

1. Install where RNC (Schedule 80 PVC) is allowed/required for utility riser at service entrance.
2. Install in each run of aboveground RNC that is located where environmental temperature change may exceed 30 deg F (17 deg C) and that has straight-run length that exceeds 25 feet (7.6 m).
3. Install type and quantity of fittings that accommodate temperature change listed for each of the following locations:
   a. Outdoor Locations Not Exposed to Direct Sunlight: 125 deg F (70 deg C) temperature change.
   b. Outdoor Locations Exposed to Direct Sunlight: 155 deg F (86 deg C) temperature change.
   c. Indoor Spaces Connected with Outdoors without Physical Separation: 125 deg F (70 deg C) temperature change.
   d. Attics: 135 deg F (75 deg C) temperature change.
4. Install fitting(s) that provide expansion and contraction for at least 0.00041 inch per foot of length of straight run per degree F of temperature change for PVC conduits.
5. Install expansion fittings at all locations where conduits cross building or structure expansion joints.
6. Install each expansion-joint fitting with position, mounting, and piston setting selected according to manufacturer's written instructions for conditions at specific location at time of installation. Install conduit supports to allow for expansion movement.

Z. Flexible Conduit Connections: Comply with NEMA RV 3. Use a maximum of 72 inches of flexible conduit for recessed and semi-recessed luminaires, equipment subject to vibration, noise transmission, or movement; and for transformers and motors.

1. Use LFMC in damp or wet locations subject to severe physical damage.
2. Use LFMC or LFNC in damp or wet locations not subject to severe physical damage.

AA. Mount boxes at heights indicated on Drawings. If mounting heights of boxes are not individually indicated, give priority to ADA requirements. Install boxes with height measured to center of box unless otherwise indicated.

BB. Recessed Boxes in Masonry Walls: Saw-cut opening for box in center of cell of masonry block, and install box flush with surface of wall. Prepare block surfaces to provide a flat surface for a raintight connection between the box and cover plate or the supported equipment and box.

CC. Horizontally separate boxes mounted on opposite sides of walls so they are not in the same vertical channel.

DD. Locate boxes so that cover or plate will not span different building finishes.

EE. Support boxes of three gangs or more from more than one side by spanning two framing members or mounting on brackets specifically designed for the purpose.

FF. Fasten junction and pull boxes to or support from building structure. Do not support boxes by conduits.

GG. Set metal floor boxes level and flush with finished floor surface.

HH. Set nonmetallic floor boxes level. Trim after installation to fit flush with finished floor surface.

3.4 INSTALLATION OF UNDERGROUND CONDUIT

A. General

1. Electrical contractor (EC) shall review all specifications (including Civil) for this project and be responsible for all trenching associated with electrical work, whether the actual work is accomplished by the electrical contractor, the general contractor or other contractors.
2. Prior to any trenching, contact 811 'Call-Before-You-Dig' and coordinate with owner and utilities to locate all buried power, communications, gas, water, sewer, irrigation piping, etc. From this information, establish the best routing and plan for areas that will require hand digging.
3. EC shall be responsible for repair of any damage to existing buried power, communications, gas, water, sewer, irrigation piping, etc. that occurs as a result of trenching for the electrical work. EC shall hire trained and certified craftsmen to perform the repairs and bring them back to 'like existing conditions'. Repair work will not be considered complete until all systems are once again functioning properly and owner is satisfied with the repairs.
4. Any time there are two or more conduits in a trench, intermediate (horizontal) spacers shall be used. These allow for the following:
    a. Maintaining conduit separation in accordance with NEC 310.15(C)(2).
    b. Allowing for proper bedding and compaction to a dense and unyielding condition.

B. Excavation

1. Carefully cut and retain sod alive for reinstallation. Saw cut, remove and dispose of concrete and asphalt.
2. Excavate trenches to depth required to install top of duct bank at least 24 inches below finished grade, unless otherwise indicated.
3. Excavate trenches to uniform widths to provide 3 inches on each side of conduits, and sufficiently wide to maintain required separations as noted below.
4. Slope: Pitch ducts a minimum slope of 1:300 down toward manholes and handholes and away from buildings and equipment. Slope ducts from a high point in runs between two manholes to drain in both directions.
5. Trenches in Tree- and Plant-Protection Zones:
    a. Hand-excavate to indicated lines, cross sections, elevations, and subgrades. Use narrow-tine spading forks to comb soil and expose roots. Do not break, tear, or chop exposed roots. Do not use mechanical equipment that rips, tears, or pulls roots.
    b. Do not cut main lateral roots or taproots; cut only smaller roots that interfere with installation of utilities.

C. Trench Bedding

1. Excavate and shape trench bottoms to provide uniform bearing and support of conduits. Shape subgrade to provide continuous support for joints, fittings, and bodies of conduits. Remove projecting stones and sharp objects along trench subgrade. Excavate trenches 6 inches deeper than elevation required to allow for bedding course.
2. Provide 6" of clean, crushed Class I granular bedding material to be installed and compacted in the bottom of the trench prior to installation of conduits. Place backfill on subgrades free of mud, frost, snow, or ice. Shape bedding course to provide continuous support for joints, fittings, and bodies of conduits.

D. Conduit Installation

1. Direct-Buried Duct Banks - Whenever two or more conduits (including spares), or other utilities, are installed in a trench, spacing shall be as follows:
   a. All conduits/pipes shall be at the same horizontal level within the trench.
   b. Install molded plastic horizontal spacers every six feet.
   c. Maintain a minimum 3-inch separation between power conduits.
   d. Maintain a minimum 12-inch separation between power and communications conduits.
   e. Maintain a minimum 24-inch separation from water lines.
   f. Maintain a minimum 24-inch separation from gas or sewer lines.
   g. All conduit separations are measured surface-to-surface and not center-to-center.
2. Support ducts on duct separators coordinated with duct size, duct spacing, and outdoor temperature.
3. Space separators close enough to prevent sagging and deforming of ducts, with not less than 4 spacers per 20 feet of duct. Secure separators to earth and to ducts to prevent displacement during backfill and yet permit linear duct movement due to expansion and contraction as temperature changes. Stagger spacers approximately 6 inches between tiers.
4. After installing first tier of ducts, backfill and compact. Start at tie-in point and work toward end of duct run, leaving ducts at end of run free to move with expansion and contraction as temperature changes during this process. Repeat procedure after placing each tier. After placing last tier, hand-place backfill to 4 inches (100 mm) over ducts and hand tamp. Firmly tamp backfill around ducts to provide maximum supporting strength. Use hand tamper only. After placing controlled backfill over final tier, make final duct connections at end of run and complete backfilling.
5. Where spare buried conduits are installed, they shall have pull string as noted above. Cap each end of conduit to avoid entrance of moisture or vermin. Mark dimensioned termination locations on as-built, red-line drawings. Where noted on plans, spare conduits shall terminate in handholes or above grade alongside raceways in use.
6. Curves and Bends: Use 5-degree angle couplings for small changes in direction. Use manufactured RGS or fiberglass, long sweep (large radius) bends for both horizontal and vertical bends 45-degree or greater.
7. Install manufactured duct elbows for stub-up at poles and equipment and at building entrances through floor unless otherwise indicated. Encase elbows for stub-up ducts throughout length of elbow.
8. Install manufactured rigid steel conduit elbows for stub-ups at poles and equipment and at building entrances through floor.
   a. Couple steel conduits to ducts with adapters designed for this purpose, and encase coupling with 3 inches of concrete for a minimum of 12 inches on each side of the coupling.
   b. For stub-ups at equipment mounted on outdoor concrete bases and where conduits penetrate building foundations, extend steel conduit horizontally a minimum of 60

inches from edge of foundation or equipment base. Install insulated grounding bushings on terminations at equipment.

E. Backfill - After installing conduit, backfill and compact to 95 percent of adjacent undisturbed soil. Start at tie-in point, and work toward end of conduit run, leaving conduit at end of run free to move with expansion and contraction as temperature changes during this process.

1. Trenches under Footings: Refer to and comply with structural details and specifications.
2. Trenches under Roadways: Backfill only after concrete is set.
3. Initial Backfill: Place and compact initial backfill of subbase material, free of particles larger than 1 inch in any dimension, to a height of 12 inches over the conduits. Carefully compact initial backfill under conduit haunches and compact evenly up on both sides and along the full length of conduit to avoid damage or displacement of piping or conduit. Coordinate backfilling with utilities testing.
4. After placing controlled backfill to within 12 inches of finished grade, make final conduit connection at end of run and complete backfilling with normal compaction.

F. Underground Warning Tape: Comply with requirements in Section 260553 "Identification for Electrical Systems."

G. Compaction - Firmly hand tamp backfill around conduit to provide maximum supporting strength. Prior to commencing compaction work, coordinate all requirements with site civil specifications. The most stringent compaction requirements shall take precedence.

1. Compact each layer of initial and final backfill soil material at 95 percent. Refer to geotechnical engineer for relative density of adjacent soils.

H. Finishing

1. Replace sod to match existing. Pour concrete and replace asphalt to match adjacent surfaces.

3.5 INSTALLATION OF UNDERGROUND HANDHOLES

A. Install handholes level and plumb and with orientation and depth coordinated with connecting conduits to minimize bends and deflections required for proper entrances.

B. Unless otherwise indicated, support units on a level bed of crushed stone or gravel, graded from 1/2-inch sieve to No. 4 sieve and compacted to same density as adjacent undisturbed earth.

C. Elevation: In paved areas, set so cover surface will be flush with finished grade. Set covers of other enclosures 1 inch above finished grade.

D. Where required, field-cut openings for conduits according to enclosure manufacturer's written instructions. Cut wall of enclosure with a tool designed for material to be cut. Size holes for terminating fittings to be used, and seal around penetrations after fittings are installed.

3.6 SLEEVE AND SLEEVE-SEAL INSTALLATION FOR ELECTRICAL PENETRATIONS

A. Install sleeves and sleeve seals at penetrations of exterior floor and wall assemblies. Comply with requirements in Section 260544 "Sleeves and Sleeve Seals for Electrical Raceways and Cabling."

3.7 FIRESTOPPING

A. Install firestopping at penetrations of fire-rated floor and wall assemblies.

3.8 PROTECTION

A. Protect coatings, finishes, and cabinets from damage and deterioration.

1. Repair damage to galvanized finishes with zinc-rich paint recommended by manufacturer.
2. Repair damage to PVC coatings or paint finishes with matching touchup coating recommended by manufacturer.

END OF SECTION 260533

SECTION 260544 - SLEEVES AND SLEEVE SEALS FOR ELECTRICAL RACEWAYS AND CABLING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Sleeves for raceway and cable penetration of non-fire-rated construction walls and floors.
2. Sleeve-seal systems.
3. Sleeve-seal fittings.
4. Grout.
5. Silicone sealants.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

PART 2 - PRODUCTS

2.1 SLEEVES

A. Wall Sleeves:

1. Steel Pipe Sleeves: ASTM A 53/A 53M, Type E, Grade B, Schedule 40, zinc coated, plain ends.

B. Sleeves for Conduits Penetrating Non-Fire-Rated Gypsum Board Assemblies: Galvanized-steel sheet; 0.0239-inch (0.6-mm) minimum thickness; round tube closed with welded longitudinal joint, with tabs for screw-fastening the sleeve to the board.

C. Sleeves for Rectangular Openings:

1. Material: Galvanized sheet steel.
2. Minimum Metal Thickness:

a. For sleeve cross-section rectangle perimeter less than 50 inches (1270 mm) and with no side larger than 16 inches (400 mm), thickness shall be 0.052 inch (1.3 mm).
b. For sleeve cross-section rectangle perimeter 50 inches (1270 mm) or more and one or more sides larger than 16 inches (400 mm), thickness shall be 0.138 inch (3.5 mm).

2.2 SLEEVE-SEAL SYSTEMS

A. Description: Modular sealing device, designed for field assembly, to fill annular space between sleeve and raceway or cable.

1. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

a. Advance Products & Systems, Inc.
b. CALPICO, Inc.
c. Metraflex Company (The).
d. Pipeline Seal and Insulator, Inc.
e. Proco Products, Inc.
f. 3M

2. Sealing Elements: EPDM rubber interlocking links shaped to fit surface of pipe. Include type and number required for pipe material and size of pipe.
3. Pressure Plates: Stainless Steel.
4. Connecting Bolts and Nuts: Stainless Steel of length required to secure pressure plates to sealing elements.

2.3 SLEEVE-SEAL FITTINGS

A. Description: Manufactured plastic, sleeve-type, waterstop assembly made for embedding in concrete slab or wall. Unit shall have plastic or rubber waterstop collar with center opening to match piping OD.

2.4 GROUT

A. Description: Non-shrink; recommended for interior and exterior sealing openings in non-fire-rated walls or floors.

B. Standard: ASTM C 1107/C 1107M, Grade B, post-hardening and volume-adjusting, dry, hydraulic-cement grout.

C. Design Mix: 5000-psi (34.5-MPa), 28-day compressive strength.

D. Packaging: Premixed and factory packaged.

2.5 SILICONE SEALANTS

A. Silicone Sealants: Single-component, silicone-based, neutral-curing elastomeric sealants of grade indicated below.

1. Grade: Pourable (self-leveling) formulation for openings in floors and other horizontal surfaces that are not fire rated.

B. Silicone Foams: Multicomponent, silicone-based liquid elastomers that, when mixed, expand and cure in place to produce a flexible, non-shrinking foam.

PART 3 - EXECUTION

3.1 SLEEVE INSTALLATION FOR NON-FIRE-RATED ELECTRICAL PENETRATIONS

A. Comply with NECA 1.

B. Comply with NEMA VE 2 for cable tray and cable penetrations.

C. Sleeves for Conduits Penetrating Above-Grade Non-Fire-Rated Concrete and Masonry-Unit Floors and Walls:

1. Interior Penetrations of Non-Fire-Rated Walls and Floors:

    a. Seal annular space between sleeve and raceway or cable, using joint sealant appropriate for size, depth, and location of joint.
    b. Seal space outside of sleeves with mortar or grout. Pack sealing material solidly between sleeve and wall so no voids remain. Tool exposed surfaces smooth; protect material while curing.

2. Use pipe sleeves unless penetration arrangement requires rectangular sleeved opening.
3. Size pipe sleeves to provide 1/4-inch (6.4-mm) annular clear space between sleeve and raceway or cable unless sleeve seal is to be installed, **unless seismic criteria require different clearance**.
4. Install sleeves for wall penetrations unless core-drilled holes or formed openings are used. Install sleeves during erection of walls. Cut sleeves to length for mounting flush with both surfaces of walls. Deburr after cutting.
5. Install sleeves for floor penetrations. Extend sleeves installed in floors 2 inches (50 mm) above finished floor level. Install sleeves during erection of floors.

D. Sleeves for Conduits Penetrating Non-Fire-Rated Gypsum Board Assemblies:

1. Use circular metal sleeves unless penetration arrangement requires rectangular sleeved opening.
2. Seal space outside of sleeves with approved joint compound for gypsum board assemblies.

E. Roof-Penetration Sleeves: Seal penetration of individual raceways and cables with flexible boot-type flashing units applied in coordination with roofing work, and in accordance with roof system manufacturer's warranty requirements.

F. Aboveground, Exterior-Wall Penetrations: Seal penetrations using steel pipe sleeves and mechanical sleeve seals. Select sleeve size to allow for 1-inch (25-mm) annular clear space between pipe and sleeve for installing mechanical sleeve seals.

G. Underground, Exterior-Wall and Floor Penetrations: Install cast-iron pipe sleeves. Size sleeves to allow for 1-inch (25-mm) annular clear space between raceway or cable and sleeve for installing sleeve-seal system.

3.2 SLEEVE-SEAL-SYSTEM INSTALLATION

A. Install sleeve-seal systems in sleeves in exterior concrete walls and slabs-on-grade at raceway entries into building.

B. Install type and number of sealing elements recommended by manufacturer for raceway or cable material and size. Position raceway or cable in center of sleeve. Assemble mechanical sleeve seals and install in annular space between raceway or cable and sleeve. Tighten bolts against pressure plates that cause sealing elements to expand and make watertight seal.

3.3 SLEEVE-SEAL-FITTING INSTALLATION

A. Install sleeve-seal fittings in new walls and slabs as they are constructed.

B. Assemble fitting components of length to be flush with both surfaces of concrete slabs and walls. Position waterstop flange to be centered in concrete slab or wall.

C. Secure nailing flanges to concrete forms.

D. Using grout, seal the space around outside of sleeve-seal fittings.

END OF SECTION 260544

SECTION 260548.16 - SEISMIC CONTROLS FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Restraint channel bracings.
2. Restraint cables.
3. Seismic-restraint accessories.
4. Mechanical anchor bolts.

1.2 ACTION SUBMITTALS

A. Delegated-Design Submittal: For each seismic-restraint device.

1. Include design calculations and details for selecting seismic restraints complying with performance requirements, design criteria, and analysis data signed and sealed by the qualified professional engineer responsible for their preparation.
2. Design Calculations: Calculate static and dynamic loading caused by equipment weight, operation, and seismic forces required to select seismic restraints and for designing vibration isolation bases.

a. Coordinate design calculations with wind load calculations required for equipment mounted outdoors. Comply with requirements in other Sections for equipment mounted outdoors.

3. Seismic Restraint Details:

a. Design Analysis: To support selection and arrangement of seismic restraints. Include calculations of combined tensile and shear loads.
b. Details: Indicate fabrication and arrangement. Detail attachments of restraints to the restrained items and to the structure. Show attachment locations, methods, and spacings. Identify components, list their strengths, and indicate directions and values of forces transmitted to the structure during seismic events. Indicate association with vibration isolation devices.
c. Coordinate seismic-restraint and vibration isolation details with wind-restraint details required for equipment mounted outdoors. Comply with requirements in other Sections for equipment mounted outdoors.
d. In lieu of design analysis and details, preapproval documentation by OSHPD, ICC-ES or another agency acceptable to authorities having jurisdiction are acceptable.

1.3 QUALITY ASSURANCE

A. Comply with seismic-restraint requirements in the IBC unless requirements in this Section are more stringent.

B. Seismic-restraint devices shall have horizontal and vertical load testing and analysis. They shall bear anchorage preapproval from OSHPD in addition to preapproval, showing maximum seismic-restraint ratings, by ICC-ES or another agency acceptable to authorities having jurisdiction. Ratings based on independent testing are preferred to ratings based on calculations. If preapproved ratings are not available, submittals based on independent testing are preferred. Calculations (including combining shear and tensile loads) that support seismic-restraint designs must be signed and sealed by a qualified professional engineer.

C. Comply with NFPA 70.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Seismic-Restraint Loading:

1. Design seismic restraints for components for seismic design forces defined in Chapter 13 of ASCE 7-10.
   a. Building Risk Category: See Structural/Architectural.
   b. Design Spectral Response Acceleration at Short Periods, S_{DS} = See Structural.
   c. Component Importance Factor, I_P = 1.0 for electrical equipment except for components required for life-safety purposes after an earthquake such as egress lighting and fire alarm control panel where I_P = 1.5.
   d. Component Response Modification Factor, R_P: See Table 13.6-1 of ASCE 7-10
   e. Component Amplification Factor, a_P: See Table 13.6-1 of ASCE 7-10

2.2 RESTRAINT CHANNEL BRACINGS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1. Atkore Unistrut
2. B-line, an Eaton business.
3. Hilti, Inc.
4. Mason Industries, Inc.

B. Description: MFMA-4, shop- or field-fabricated bracing assembly made of slotted steel channels with accessories for attachment to braced component at one end and to building structure at the other end, with other matching components, and with corrosion-resistant coating; rated in tension, compression, and torsion forces.

2.3 RESTRAINT CABLES

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1. Gripple Inc.
2. Kinetics Noise Control, Inc.

3. Vibration & Seismic Technologies, LLC.
4. Vibration Mountings & Controls, Inc.

B. Restraint Cables: ASTM A 603 galvanized steel cables. End connections made of steel assemblies with thimbles, brackets, swivel, and bolts designed for restraining cable service; with a minimum of two clamping bolts for cable engagement.

2.4 SEISMIC-RESTRAINT ACCESSORIES

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1. Atkore Unistrut
2. B-line, an Eaton business.
3. Kinetics Noise Control, Inc.
4. Mason Industries, Inc.

B. Hanger-Rod Stiffener: Steel tube or steel slotted-support-system sleeve with internally bolted connections to hanger rod.

C. Hinged and Swivel Brace Attachments: Multifunctional steel connectors for attaching hangers to rigid channel bracings and restraint cables.

D. Bushings for Floor-Mounted Equipment Anchor Bolts: Neoprene bushings designed for rigid equipment mountings and matched to type and size of anchor bolts and studs.

E. Bushing Assemblies for Wall-Mounted Equipment Anchorage: Assemblies of neoprene elements and steel sleeves designed for rigid equipment mountings and matched to type and size of attachment devices used.

F. Resilient Isolation Washers and Bushings: One-piece, molded, oil- and water-resistant neoprene, with a flat washer face.

2.5 MECHANICAL ANCHOR BOLTS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1. B-line, an Eaton business.
2. Hilti, Inc.
3. Kinetics Noise Control, Inc.
4. Mason Industries, Inc.

B. Mechanical Anchor Bolts: Drilled-in and stud-wedge or female-wedge type in zinc-coated steel for interior applications and stainless steel for exterior applications. Select anchor bolts with strength required for anchor and as tested according to ASTM E 488.

PART 3 - EXECUTION

3.1 APPLICATIONS

A. Multiple Raceways or Cables: Secure raceways and cables to trapeze member with clamps complying with delegated design submittal requirements.

B. Hanger-Rod Stiffeners: Install hanger-rod stiffeners where indicated or scheduled on Drawings to receive them and where required to prevent buckling of hanger rods caused by seismic forces.

C. Strength of Support and Seismic-Restraint Assemblies: Where not indicated, select sizes of components so strength will be adequate to carry present and future static and seismic loads within specified loading limits.

3.2 SEISMIC-RESTRAINT DEVICE INSTALLATION

A. Coordinate the location of embedded connection hardware with supported equipment attachment and mounting points and with requirements for concrete reinforcement and formwork.

B. Equipment and Hanger Restraints:

1. Install resilient, bolt-isolation washers on equipment anchor bolts where clearance between anchor and adjacent surface exceeds 0.125 inch (3.2 mm).

C. Install cables so they do not bend across edges of adjacent equipment or building structure.

D. Install bushing assemblies for mounting bolts for wall-mounted equipment, arranged to provide resilient media where equipment or equipment-mounting channels are attached to wall.

E. Attachment to Structure: If specific attachment is not indicated, anchor bracing to structure at flanges of beams, at upper truss chords of bar joists, or at concrete members.

F. Drilled-in Anchors:

1. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Locate and avoid prestressed tendons, electrical and telecommunications conduit, and gas lines.
2. Do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength.
3. Wedge Anchors: Protect threads from damage during anchor installation. Heavy-duty sleeve anchors shall be installed with sleeve fully engaged in the structural element to which anchor is to be fastened.
4. Adhesive Anchors: Clean holes to remove loose material and drilling dust prior to installation of adhesive. Place adhesive in holes proceeding from the bottom of the hole and progressing toward the surface in such a manner as to avoid introduction of air pockets in the adhesive.
5. Set anchors to manufacturer's recommended torque using a torque wrench.
6. Install zinc-coated steel anchors for interior and stainless-steel anchors for exterior applications.

3.3 ACCOMMODATION OF DIFFERENTIAL SEISMIC MOTION

A. Install flexible connections in runs of raceways, cables, wireways, cable trays, and busways where they cross seismic joints, where adjacent sections or branches are supported by different structural elements, and where connection is terminated to equipment that is anchored to a different structural element from the one supporting them as they approach equipment.

END OF SECTION 260548.16

SECTION 260553 - IDENTIFICATION FOR ELECTRICAL SYSTEMS

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section Includes:

1. Color and legend requirements for raceways, conductors, and warning labels and signs.
2. Tapes and stencils.
3. Signs.
4. Cable ties.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Comply with ASME A13.1.

B. Comply with NFPA 70.

C. Comply with 29 CFR 1910.144 and 29 CFR 1910.145.

D. Comply with ANSI Z535.4 for safety signs and labels.

E. Comply with NFPA 70E requirements for arc-flash warning labels.

F. Adhesive-attached labeling materials, including label stocks, laminating adhesives, and inks used by label printers, shall comply with UL 969.

2.2 COLOR AND LEGEND REQUIREMENTS

A. Raceways and Cables Carrying Circuits within Buildings. Identify the covers of each junction and pull box.
1. Affix label with black letters indicating voltage and system or service type.

B. Conductor Color-Coding for Phase and Voltage-Level Identification, 600 V or Less: Use colors listed below for ungrounded service, feeder and branch-circuit conductors.

1. Utilize factory applied, colored insulation for No. 8 AWG and smaller.

2. If Authority Having Jurisdiction permits, for sizes larger than No. 8 AWG, where conductors with factory colored insulation are not commonly available, colored non-aging, plastic tape may be field applied. Apply in half-lapped turns for a minimum distance of 6 inches (150 mm) from terminal points and in boxes where splices or taps are made. Apply last two turns of tape with no tension to prevent possible unwinding. Locate bands to avoid obscuring factory cable markings.

3. Colors for Single-Phase, 240/120V Circuits:

   a. Phase A: Black.
   b. Phase B: Red.
   c. Neutral: White.

4. Color for Equipment Grounds: Bare copper or Green.
5. Lighting Circuit Switched Legs and 3-way/4-way Traveler: Color unique to those listed above.

C. Warning Label Colors:

1. Identify system voltage with black letters on an orange background.

D. Warning labels and signs shall include, but are not limited to, the following legends:
1. Workspace Clearance Warning: "WARNING - OSHA REGULATION - AREA IN FRONT OF ELECTRICAL EQUIPMENT MUST BE KEPT CLEAR FOR 36 INCHES (915 MM)."
2. Arc Flash Warning: “WARNING – KEEP CLEAR. RISK OF ELECTRIC SHOCK OR ARC FLASH. PPE REQUIRED.”.

E. Equipment Identification Labels:

1. Black letters on a white field, or white letters on a black field.
2. Include equipment designation and circuit.
3. Exterior equipment labels shall have a rivet or screw mounted label on the exterior door.
4. 1” minimum height letters for service disconnect and emergency shut-off switches.
5. 1/2" minimum height letters for panelboards, switchboards, relay enclosures and transformers.
6. 1/4" minimum height letters for disconnect switches and motor starters.

2.3 TAPES AND STENCILS

A. Self-Adhesive Vinyl Tape: Colored, heavy duty, waterproof, fade resistant; not less than 3 mils (0.08 mm) thick by 1 to 2 inches (25 to 50 mm) wide; compounded for outdoor use.

B. Floor Marking Tape: 2-inch- (50-mm-) wide, 5-mil (0.125-mm) pressure-sensitive vinyl tape, with yellow and black stripes and clear vinyl overlay.

C. Underground-Line Warning Tape:

1. Tape:

a. Recommended by manufacturer for the method of installation and suitable to identify and locate underground electrical and communications utility lines.
b. Printing on tape shall be permanent and shall not be damaged by burial operations.
c. Tape material and ink shall be chemically inert and not subject to degradation when exposed to acids, alkalis, and other destructive substances commonly found in soils.

2. Color and Printing:

a. Comply with ANSI Z535.1, ANSI Z535.2, ANSI Z535.3, ANSI Z535.4, and ANSI Z535.5.
b. Inscriptions for Red-Colored Tapes: "ELECTRIC LINE, HIGH VOLTAGE".
c. Inscriptions for Orange-Colored Tapes: "TELEPHONE CABLE, CATV CABLE, COMMUNICATIONS CABLE, OPTICAL FIBER CABLE".

3. Type:

a. Detectable three-layer laminate, consisting of a printed pigmented polyolefin film, a solid aluminum-foil core, and a clear protective film that allows inspection of the continuity of the conductive core; bright colored, continuous-printed on one side with the inscription of the utility, compounded for direct-burial service.
b. Width: 3 inches (75 mm).
c. Overall Thickness: 5 mils (0.125 mm).
d. Foil Core Thickness: 0.35 mil (0.00889 mm).
e. Weight: 28 lb/1000 sq. ft. (13.7 kg/100 sq. m).
f. Tensile according to ASTM D 882: 70 lbf (311.3 N) and 4600 psi (31.7 MPa).

2.4 SIGNS

A. Baked-Enamel Signs:

1. Preprinted aluminum signs, high-intensity reflective, punched or drilled for fasteners, with colors, legend, and size required for application.
2. 1/4-inch (6.4-mm) grommets in corners for mounting.
3. Nominal Size: 7 by 10 inches (180 by 250 mm).

B. Laminated Acrylic or Melamine Plastic Signs:

1. Engraved legend.
2. Thickness:

a. For signs up to 20 sq. in. (129 sq. cm), minimum 1/16 inch (1.6 mm) thick.
b. For signs larger than 20 sq. in. (129 sq. cm), 1/8 inch (3.2 mm) thick.
c. Engraved legend with black letters on white face
d. Punched or drilled for mechanical fasteners with 1/4-inch (6.4-mm) grommets in corners for mounting.
e. Framed with mitered acrylic molding and arranged for attachment at applicable equipment.

2.5 CABLE TIES

A. General-Purpose Cable Ties: Fungus inert, self-extinguishing, one piece, self-locking, and Type 6/6 nylon.

1. Minimum Width: 3/16 inch (5 mm).
2. Tensile Strength at 73 Deg F (23 Deg C) according to ASTM D 638: 12,000 psi (82.7 MPa).
3. Temperature Range: Minus 40 to plus 185 deg F (Minus 40 to plus 85 deg C).
4. Color: Black, except where used for color-coding.

B. UV-Stabilized Cable Ties: Fungus inert, designed for continuous exposure to exterior sunlight, self-extinguishing, one piece, self-locking, and Type 6/6 nylon.

1. Minimum Width: 3/16 inch (5 mm).
2. Tensile Strength at 73 Deg F (23 Deg C) according to ASTM D 638: 12,000 psi (82.7 MPa).
3. Temperature Range: Minus 40 to plus 185 deg F (Minus 40 to plus 85 deg C).
4. Color: Black.

C. Plenum-Rated Cable Ties: Self-extinguishing, UV stabilized, one piece, and self-locking.

1. Minimum Width: 3/16 inch (5 mm).
2. Tensile Strength at 73 Deg F (23 Deg C) according to ASTM D 638: 7000 psi (48.2 MPa).
3. UL 94 Flame Rating: 94V-0.
4. Temperature Range: Minus 50 to plus 284 deg F (Minus 46 to plus 140 deg C).
5. Color: Black.

2.6 MISCELLANEOUS IDENTIFICATION PRODUCTS

A. Paint: Comply with requirements in painting Sections for paint materials and application requirements. Retain paint system applicable for surface material and location (exterior or interior).

B. Fasteners for Labels and Signs: Self-tapping, stainless-steel screws or stainless-steel machine screws with nuts and flat and lock washers.

PART 3 - EXECUTION

3.1 COORDINATION

A. Verify and coordinate identification names, abbreviations, colors, and other features with requirements in other Sections requiring identification applications, Drawings, Shop Drawings, manufacturer's wiring diagrams, and operation and maintenance manual. Use consistent designations throughout Project.

B. Coordinate identification with Project Drawings, manufacturer's wiring diagrams, and operation and maintenance manual.

C. Coordinate installation of identifying devices with location of access panels and doors.

D. Install identifying devices before installing acoustical ceilings and similar concealment.

3.2 INSTALLATION

A. Verify identity of each item before installing identification products.

B. Apply identification devices to surfaces that require finish after completing finish work.

C. Install signs with approved legend to facilitate proper identification, operation, and maintenance of electrical systems and connected items.

D. Self-Adhesive Identification Products: Before applying electrical identification products, clean substrates of substances that could impair bond, using materials and methods recommended by manufacturer of identification product.

E. Self-Adhesive Identification Products used on the exterior of the building: Before applying electrical identification products, clean substrates of substances that could impair bond, using materials and methods recommended by manufacturer of identification product. Labels shall have a rivet or screw mounted on each side of the label, located on the exterior door.

F. Elevated Components: Increase sizes of labels, signs, and letters to those appropriate for viewing from the floor.

G. Floor Marking Tape: Apply stripes to finished surfaces following manufacturer's written instructions.

H. Underground Line Warning Tape:

1. During backfilling of trenches, install continuous underground-line warning tape directly above cable or raceway at 6 to 8 inches (150 to 200 mm) below finished grade. Use multiple tapes where width of multiple lines installed in a common trench or concrete envelope exceeds 16 inches (400 mm) overall.
2. Install underground-line warning tape for direct-buried cables and cables in raceways.

I. Laminated Acrylic or Melamine Plastic Signs:

1. Attach signs and plastic labels that are not self-adhesive type with mechanical fasteners appropriate to the location and substrate.

J. Cable Ties: General purpose, for attaching tags, except as listed below:

1. Outdoors: UV-stabilized nylon.
2. In Spaces Handling Environmental Air: Plenum rated.

3.3 IDENTIFICATION SCHEDULE

A. Install identification materials and devices at locations for most convenient viewing without interference with operation and maintenance of equipment. Install access doors or panels to provide view of identifying devices.

B. Identify conductors, cables, and terminals in enclosures and at junctions, terminals, pull points, and locations of high visibility. Identify by system and circuit designation.

C. Accessible Fittings for Raceways and Cables within Buildings: Identify the covers of each junction and pull box with self-adhesive labels containing the wiring system legend and system voltage.

D. Control-Circuit Conductor Identification: For conductors and cables in pull and junction boxes, manholes, and handholes, use write-on tags with the conductor or cable designation, origin, and destination.

E. Control-Circuit Conductor Termination Identification: For identification at terminations, provide self-adhesive wraparound labels with the conductor designation.

F. Conductors to Be Extended in the Future: Attach write-on tags to conductors and list source.

G. Auxiliary Electrical Systems Conductor Identification: Identify field-installed alarm, control, and signal connections.

1. Identify conductors, cables, and terminals in enclosures and at junctions, terminals, and pull points. Identify by system and circuit designation.
2. Use system of marker tape designations that is uniform and consistent with system used by manufacturer for factory-installed connections.
3. Coordinate identification with Project Drawings, manufacturer's wiring diagrams, and the Operation and Maintenance Manual.

H. Locations of Underground Lines: Underground-line warning tape for power, lighting, communication, and control wiring and optical-fiber cable.

I. Workspace Indication: Apply floor marking tape to finished surfaces. Show working clearances in the direction of access to live parts. Workspace shall comply with NFPA 70 and 29 CFR 1926.403 unless otherwise indicated. Do not install at flush-mounted panelboards and similar equipment in finished spaces.

J. Instructional Signs: Self-adhesive labels, including the color code for grounded and ungrounded conductors.

K. Warning Labels for Indoor Cabinets, Boxes, and Enclosures for Power and Lighting: Self-adhesive equipment labels.

1. Apply to exterior of door, cover, or other access.
2. For equipment with multiple power or control sources, apply to door or cover of equipment, including, but not limited to, the following:

    a. Power-transfer switches.

b. Controls with external control power connections.

L. Arc Flash Warning Labeling: Self-adhesive labels.

M. Operating Instruction Signs: Install instruction signs to facilitate proper operation and maintenance of electrical systems and items to which they connect. Install instruction signs with approved legend where instructions are needed for system or equipment operation.

N. Emergency Operating Instruction Signs: Self-adhesive labels, Laminated acrylic or melamine plastic signs with white legend on a red background with minimum 3/8-inch- (10-mm-) high letters for emergency instructions at equipment used for power transfer and load shedding.

O. Equipment Identification Labels:

1. Indoor Equipment: Engraved, laminated acrylic or melamine plastic label.
2. Outdoor Equipment: Engraved, Laminated acrylic or melamine label.
3. Equipment to Be Labeled:

a. Panelboards/Switchboards:

1) Label shall be self-adhesive, engraved, laminated acrylic or melamine. Label shall include: Panelboard/switchboard name, voltage, amperage, number of phases and wires, source and available fault current with date calculated.
2) Typewritten directory of circuits in the location provided by panelboard manufacturer. Spares shall be filled in by hand with pencil.
3) On main distribution panel door / switchboard front provide a laminated one-line diagram of the electrical system and all panel configurations.

b. Enclosures and electrical cabinets.
c. Access doors and panels for concealed electrical items.
d. Switchgear.
e. Transformers.
f. Emergency system boxes and enclosures.
g. Enclosed switches.
h. Enclosed circuit breakers.
i. Enclosed controllers.
j. Variable-speed controllers.
k. Push-button stations.
l. Power transfer equipment.
m. Contactors.
n. Remote-controlled switches, dimmer modules, and control devices.
o. Battery-inverter units.
p. Power-generating units.
q. Monitoring and control equipment.
r. UPS equipment.
s. Wiring devices: See specification section “Wiring Devices”.

END OF SECTION 260553

SECTION 262416 - PANELBOARDS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Lighting and appliance branch-circuit panelboards.
   2. Load centers.

1.2 RELATED SECTIONS

A. Section 264313 - Surge Protection for Low-Voltage Electrical Power Circuits

1.3 DEFINITIONS

A. OCPD: Overcurrent protective device.

B. MCCB: Molded-case circuit breaker.

C. SPD: Surge protective device.

D. NRTL: Nationally Recognized Testing Laboratory.

1.4 ACTION SUBMITTALS

A. Product Data: For each type of panelboard, overcurrent protective device, surge protection device, ground-fault protector, accessory, and component.

B. Shop Drawings: For each panelboard and related equipment.

   1. Include dimensioned plans, elevations, sections, and details.
   2. Detail enclosure types including mounting and anchorage, environmental protection, knockouts, corner treatments, covers and doors, gaskets, hinges, and locks.
   3. Detail bus configuration, current, and voltage ratings.
   4. Short-circuit current rating of panelboards and all individual overcurrent protective devices.
   5. Schematic and wiring diagrams for power, signal, and control wiring.

1.5 INFORMATIONAL SUBMITTALS

A. Panelboard schedules for installation in panelboards.

B. Seismic Qualification Data: Certificates, for panelboards, overcurrent protective devices, accessories, and components, from manufacturer.

1. Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.
2. Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.
3. Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.

1.6 CLOSEOUT SUBMITTALS

A. Operation and maintenance data.

1. Manufacturer's written instructions for testing and adjusting overcurrent protective devices.

1.7 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Keys: Two (2) spares for each type of panelboard cabinet lock.

1.8 QUALITY ASSURANCE

A. Installer Qualifications: Workers qualified as defined in NEMA PB 1.1 and trained in electrical safety as required by NFPA 70E.

1.9 FIELD CONDITIONS

A. Environmental Limitations:

1. Do not deliver or install panelboards until spaces are enclosed and weathertight, wet work in spaces is complete and dry, work above panelboards is complete, and temporary HVAC system is operating and maintaining ambient temperature and humidity conditions at occupancy levels during the remainder of the construction period.
2. Rate equipment for continuous operation under the following conditions unless otherwise indicated:

    a. Ambient Temperature: Not exceeding 23 deg F (minus 5 deg C) to plus 104 deg F (plus 40 deg C).
    b. Altitude not exceeding 8000 feet.

B. Service Conditions: NEMA PB 1.1, usual service conditions, as noted above.

1.10 COORDINATION

A. Coordinate layout and installation of panelboards and components with other construction that penetrates walls or is supported by them, including electrical and other types of equipment, raceways, piping, ductwork, encumbrances to workspace clearance requirements and adjacent

surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels in accordance with NEC 110.26.

1.11 WARRANTY

A. Manufacturer's Warranty: Manufacturer agrees to repair or replace panelboards enclosures, buswork, overcurrent protective devices, accessories that fail in materials or workmanship within specified warranty period.

1. Panelboard Warranty Period: 12 months from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Seismic Requirements: Fabricate and test panelboards according to IEEE 344 to withstand seismic forces defined in Section 260548.16 "Seismic Controls for Electrical Systems."

1. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified and the unit will be fully operational after the seismic event."

2.2 PANELBOARDS COMMON REQUIREMENTS

A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Square D by Schneider Electric.
2. Eaton Cutler-Hammer.
3. ABB/General Electric Company.

B. Source Limitations: Obtain panelboards, overcurrent protective devices, components, and accessories from single source from single manufacturer.

C. Product Selection for Restricted Space: Drawings indicate maximum dimensions for panelboards including clearances between panelboards and adjacent surfaces and other items. Comply with indicated maximum dimensions.

D. Fabricate and test panelboards according to IEEE 344 to withstand seismic forces defined in Section 260548.16 "Seismic Controls for Electrical Systems."

E. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

F. Comply with NEMA PB 1.

G. Comply with NFPA 70.

H. Enclosures: Flush and Surface-mounted (as noted on plans), dead-front cabinets.

1. Rated for environmental conditions at installed location.
   a. Indoor Dry and Clean Locations: NEMA 250, Type 1.
   b. Outdoor Locations: NEMA 250, Type 3R.
   c. Kitchen or Wash-Down Areas: NEMA 250, Type 4X, stainless steel.
   d. Other Wet or Damp Indoor Locations: NEMA 250, Type 4.
   e. Indoor Locations Subject to Dust, Falling Dirt, and Dripping Noncorrosive Liquids: NEMA 250, Type 12.

2. Mounting Height:
   a. Standard: 84 inches to top of enclosure (so that maximum height of highest breaker is 79 inches maximum).
   b. ADA dwelling units: 53 inches to top of enclosure (so that maximum height of highest breaker is 48 inches maximum and the minimum height of lowest breaker is 16 inches).

3. Front: Secured to box with concealed trim clamps. For surface-mounted fronts, match box dimensions; for flush-mounted fronts, overlap box. Trims shall cover all live parts and shall have no exposed hardware.
4. Hinged Front Cover: Entire front trim hinged to box and with standard door within hinged trim cover. Trims shall cover all live parts and shall have no exposed hardware.

I. Incoming Mains Location: Top or Bottom as determined by Contractor, based on field conditions, UNO.

J. Phase, Neutral, and Ground Buses: Hard-drawn copper (98 percent conductivity).

K. Conductor Connectors: Suitable for use with conductor material, quantity and sizes. Refer to the Feeder Schedule on the contract documents.

1. Material: Hard-drawn copper (98 percent conductivity).
2. Main and Neutral Lugs: Mechanical type, with a lug on the neutral bar for each pole in the panelboard.
3. Ground Lugs and Bus - Configured Terminators: Mechanical type, with a lug on the bar for each pole in the panelboard.

L. NRTL Label: Panelboards shall be labeled by an NRTL acceptable to authority having jurisdiction for use as service equipment with one or more main service disconnecting and overcurrent protective devices.

M. Future Devices: Panelboards shall have mounting brackets, bus connections, filler plates, and necessary appurtenances required for future installation of devices.

N. Panelboard Short-Circuit Current Rating: Fully rated to interrupt symmetrical short-circuit current available at terminals. Assembly listed by an NRTL for 100 percent interrupting capacity.

2.3 DISCONNECTING AND OVERCURRENT PROTECTIVE DEVICES

A. All OCPDs shall be fully rated for available fault current. No series rating will be allowed.

B. Manufacturers – Breakers shall be manufactured by the same manufacturer as the panelboard in which they are installed.

C. Branch Overcurrent Protective Devices - Bolt-on circuit breakers or plug-in circuit breakers where individual positive-locking device requires mechanical release for removal. Replaceable without disturbing adjacent units.

2.4 CIRCUIT BREAKERS

A. General requirements

1. Breakers shall meet current NEMA and UL specifications as applicable to frame size, standard rating and interrupting capability.
2. Breakers shall be one-, two-, or three-pole as scheduled, operate manually for normal ON-OFF switching and automatically under overload and short circuit conditions.
3. The operating handle shall open and close all poles simultaneously on multi-pole breakers. The operating mechanism shall be trip-free so that contacts cannot be held closed against abnormal overcurrent or short circuit conditions. Do not use single-pole circuit breakers with handle ties where multi-pole breakers are indicated on the panel schedule or where required for poly-phase loads.
4. Breakers shall be of the type noted on panel schedule (shunt-trip, GFCI, arc-fault, etc.) or as required by the equipment being provided.
5. Unless specifically noted otherwise, **no** tandem breakers shall be allowed.
6. Breakers noted as GFI protected for equipment shall have a 30mA or greater trip.
7. Breakers noted as GFI protected for personnel shall have a 6mA trip.
8. A control transformer with primary and secondary fusing shall be provided as required for control of shunt-trip breakers.

B. Molded-Case Circuit Breaker (MCCB): Comply with UL 489, with interrupting capacity to meet available fault currents.

1. Molded case circuit breakers shall be bolt-on type only and suitable for individual as well as panelboard mounting. No breakers designated "plug-on" type allowed unless specifically noted on plans.
2. Thermal-Magnetic Circuit Breakers:

    a. Inverse time-current element for low-level overloads.
    b. Instantaneous magnetic trip element for short circuits.
    c. Adjustable magnetic trip setting for circuit-breaker frame sizes 250 A and larger.

3. Adjustable Instantaneous-Trip Circuit Breakers: Magnetic trip element with front-mounted, field-adjustable trip setting.
4. Electronic trip circuit breakers with rms sensing; field-replaceable rating plug or field-replicable electronic trip; and the following field-adjustable settings (LSIG):

    a. Instantaneous trip.
    b. Long- and short-time pickup levels.
    c. Long- and short-time adjustments.
    d. Ground-fault pickup level, time delay, and I squared t response.

5. Current-Limiting Circuit Breakers: Frame sizes 400 A and smaller; let-through ratings less than NEMA FU 1, RK-5.
6. GFCI Circuit Breakers: Single- and double-pole configurations with Class A ground-fault protection (6-mA trip).
7. GFEP Circuit Breakers: Class B ground-fault protection (30-mA trip).
8. Arc-Fault Circuit Interrupter Circuit Breakers: Comply with UL 1699; 120/240-V, single-pole configuration.
9. Sub-feed Circuit Breakers: Vertically mounted.
10. MCCB Features and Accessories:

    a. Standard frame sizes, trip ratings, and number of poles.
    b. Breaker handle indicates tripped status.
    c. UL listed for reverse connection without restrictive line or load ratings.
    d. Lugs: Mechanical style, suitable for number, size, trip ratings, and conductor materials.
    e. Application Listing: Appropriate for application; Type SWD for switching fluorescent lighting loads; Type HID for feeding fluorescent and HID lighting circuits.
    f. Ground-Fault Protection: Integrally mounted relay and trip unit with adjustable pickup and time-delay settings, push-to-test feature, and ground-fault indicator.
    g. Handle Padlocking Device: Fixed attachment, for locking circuit-breaker handle in on or off position.

2.5 SURGE PROTECTION DEVICES

A. Refer to Section 264313; Surge Protection for Low-Voltage Electrical Power Circuits.

2.6 LIGHTING AND APPLIANCE BRANCH-CIRCUIT PANELBOARDS

A. Panelboards: NEMA PB 1, lighting and appliance branch-circuit type.

B. Mains: As noted on drawings.

C. Branch Overcurrent Protective Devices: Bolt-on circuit breakers, replaceable without disturbing adjacent units.

D. Doors: Concealed hinges; secured with flush latch with tumbler lock; keyed alike.

E. Column-Type Panelboards: Single row of overcurrent devices with narrow gutter extension and overhead junction box equipped with ground and neutral terminal buses.

2.7 LOAD CENTERS

A. Load Centers: Comply with UL 67.

B. Mains: As noted on Drawings.

C. Branch Overcurrent Protective Devices: Plug-in circuit breakers, replaceable without disturbing adjacent units.

D. Conductor Connectors: Mechanical type for main, neutral, and ground lugs and buses.

2.8 IDENTIFICATION

A. Service Equipment Label: NRTL labeled for use as service equipment for switchboards (as applicable) with one or more service disconnecting and overcurrent protective devices.

B. Breaker Labels - Faceplate shall list current rating, UL and IEC certification standards, and AIC rating.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Receive, inspect, handle, and store panelboards according to NEMA PB 1.1.

B. Examine panelboards before installation. Reject panelboards that are damaged or rusted or have been subjected to water saturation.

C. Examine elements and surfaces to receive panelboards for compliance with installation tolerances and other conditions affecting performance of the Work.

D. Examine all OCPDs before installation. Reject any that are moisture damaged or physically damaged.

E. Examine utilization equipment nameplates and installation instructions. Install OCPDs of sizes and with characteristics appropriate for each piece of equipment.

F. Proceed with installation only after unsatisfactory conditions have been corrected.

3.2 INSTALLATION

A. Comply with NECA 1.

B. Install panelboards and accessories according to NEMA PB 1.1.

C. Comply with mounting and anchoring requirements specified in Section 260548.16 "Seismic Controls for Electrical Systems."

D. Mount top of enclosure (standard panelboards or ADA dwelling unit panelboards) in accordance with mounting heights noted in paragraph 2.2 above.

E. Mount panelboard cabinet plumb and rigid without distortion of box.

F. Mount recessed panelboards with fronts uniformly flush with wall finish and mating with back box.

G. Install filler plates in unused spaces.

H. Install overcurrent protective devices and controllers not already factory installed.

1. Set field-adjustable, circuit-breaker trip ranges.

I. Make grounding connections and bond neutral for services and separately derived systems to ground. Make connections to grounding electrodes, separate grounds for isolated ground bars, and connections to separate ground bars.

J. Spare conduit stub-outs at recessed panels

1. In the following paragraphs, accessible is defined as being arranged so that an appropriately dressed person, 6'-2" tall, weighing 250 pounds, may approach the area in question with tools and products necessary for the work intended, and may then position himself/herself to properly and safely perform the task to be accomplished, without disassembly or damage to the surrounding installation.
2. All spare conduits shall be terminated in locations where they are accessible from a crawlspace, attic, or by ladder in areas that have t-grid ceilings. They shall be terminated away from equipment, ducts or pipes that would obstruct access.
3. Stub four (4) 1-inch empty conduits from panelboard into accessible ceiling space above the panel, or a space designated to be ceiling space in the future.
4. Where applicable, stub four (4) 1-inch empty conduits into accessible floor space or accessible ceiling space on the level below.

K. Arrange conductors in gutters into groups and bundle and wrap with wire ties.

L. Panelboards shall not be used as pull-boxes for any wiring that does not terminate in that panelboard.

3.3 IDENTIFICATION

A. Identify field-installed conductors, interconnecting wiring, and components; install warning signs complying with requirements in Section 260553 "Identification for Electrical Systems."

B. Circuit Directory: Directory card inside panelboard door, mounted in transparent card holder. Indicate installed circuit loads; incorporate Owner's final room designations. Obtain approval before installing. Handwritten directories are not acceptable. Install directory inside panelboard door.

C. Panelboard Nameplates: Label each panelboard with a nameplate complying with requirements for identification specified in Section 260553 "Identification for Electrical Systems." Include: Panel name, voltage, amperage, number of phases and wires, source and available fault current with date calculated.

D. Device Nameplates: Label each branch circuit device in power panelboards with a nameplate complying with requirements for identification specified in Section 260553 "Identification for Electrical Systems."

E. Install warning signs/labels complying with requirements in Section 260553 "Identification for Electrical Systems" identifying source of remote circuit.

F. On main distribution panel door provide a laminated one-line diagram of the electrical system and all panel configurations.

3.4 FIELD QUALITY CONTROL

A. Perform the following tests and inspections:

1. Acceptance:

a. Test insulation resistance for each panelboard bus, component, connecting supply, feeder, and control circuit. Open control and metering circuits within the panelboard, and remove neutral connection to surge protection and other electronic devices prior to insulation test. Reconnect after test.
b. Test continuity of each circuit.

2. Test ground-fault protection of equipment for service equipment per NFPA 70.
3. Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.
4. Test and adjust controls, remote monitoring, and safeties. Replace any damaged and malfunctioning controls and equipment.
5. Test and demonstrate proper function of all GFCI, AFCI and shunt-trip breakers.

B. Panelboards will be considered defective if they do not pass tests and inspections.
C. Prepare test and inspection reports, including a certified report that identifies panelboards included. Include notation of deficiencies detected, remedial action taken, and observations after remedial action.

3.5 DEMONSTRATION

A. Train Owner's maintenance personnel:

1. To adjust, operate, and maintain panelboards, overcurrent protective devices, instrumentation, and accessories.
2. How to set and reset arc fault reduction switches for maintenance.

END OF SECTION 262416

SECTION 262726 - WIRING DEVICES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Straight-blade convenience receptacles.
2. TV outlets (power & signal).
3. USB charger devices.
4. GFCI receptacles.
5. Toggle switches.
6. Wall-box dimmers.
7. Wall plates.
8. Finishes.

1.2 DEFINITIONS

A. Abbreviations of Manufacturers' Names:

1. Cooper: Copper Wiring Devices; Division of Cooper Industries, Inc.
2. Hubbell: Hubbell Incorporated: Wiring Devices-Kellems.
3. Leviton: Leviton Mfg. Company, Inc.
4. P&S: Pass & Seymour/Legrand.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Samples: One for each type of device and wall plate specified, in each color specified.

1.4 CLOSEOUT SUBMITTALS

A. Operation and maintenance data.

PART 2 - PRODUCTS

2.1 GENERAL WIRING-DEVICE REQUIREMENTS

A. Wiring Devices, Components, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Comply with NFPA 70.

C. Devices that are manufactured for use with modular plug-in connectors may be substituted under the following conditions:

1. Connectors shall comply with UL 2459 and shall be made with stranding building wire.
2. Devices shall comply with the requirements in this Section.

D. Devices for Owner-Furnished Equipment:

1. Receptacles: Match plug configurations including wire count, poles, twistlock, etc.

E. Source Limitations: Obtain each type of wiring device and associated wall plate from single source from single manufacturer.

2.2 RESIDENTIAL DEVICES

A. Residential-Grade, Tamper-Resistant Convenience Receptacles, 125V, 15A: Comply with NEMA WD 1, NEMA WD 6 Configuration 5-15R, and UL 498.

1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
   a. Cooper; TR270-9B (duplex).
   b. Hubbell; RR15STR (duplex).
   c. Leviton; T5320 (duplex).
   d. P&S; 3232TR (duplex).

2. Description: Labeled to comply with NFPA 70, "Receptacles, Cord Connectors, and Attachment Plugs (Caps)" Article, "Tamper-Resistant Receptacles in Dwelling Units" Section.

B. Tamper-Resistant, USB Charger Receptacles: 12V, 2.0A, USB Type A; Comply with NEMA WD 1, NEMA WD 6 Configuration 5-15R, UL 498 Supplement sd, UL 1310, and FS W-C-596.

1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
   a. Cooper; TR7755.
   b. Hubbell; USB15A5.
   c. Leviton; T5632.
   d. P&S; TM826USB.

2. Description: Single-piece, rivetless, nickel-plated, all-brass grounding system. Nickel-plated, brass mounting strap.
3. USB Receptacles: Dual, Type A.
4. Line Voltage Receptacles: Dual, two pole, three wire, and self-grounding.

C. Tamper-Resistant, Duplex GFCI Convenience Receptacles, 125V, 15A:

1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
   a. Cooper; TRSGF15.

b. Hubbell; GFTR15.
c. Leviton; GFTR1.
d. P&S; 1597TR

2. Description: Labeled to comply with NFPA 70, "Receptacles, Cord Connectors, and Attachment Plugs (Caps)" Article, "Tamper-Resistant Receptacles in Dwelling Units" Section.

D. Weather Resistant, Self-Testing, Duplex GFCI Receptacles, 125V, 15A:

1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
   a. Cooper; TWRSGF15.
   b. Hubbell; GFTWRST15.
   c. Leviton; GFWT1.
   d. P&S; 1597TRWR

2. For use only with wet or damp location covers.

E. Telephone Outlet:

1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
   a. Cooper; 5547-5E.
   b. Hubbell; NSJ5E.
   c. Leviton; 40249.

2. Description: Single RJ-45 jack for terminating Category 5e, twisted pair cable complying with Section 260523 "Control-Voltage Electrical Power Conductors."
3. Description: Single RJ-45 jack for terminating Category 5e, twisted pair cable complying with Section 271513 "Communications Copper Horizontal Cabling." Comply with UL 1863.

F. TV Outlet (Power & Signal), recessed, 2-gang, in-wall enclosure with tamper resistant, duplex receptacle, 125V, 15A; comply with NEMA WD 1, NEMA WD 6 Configuration 5-15R, and UL 498; and 1" conduit with pullstring provision for signal cables.

1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
   a. Cooper; 35MRW
   b. Hubbell; RR1512.

2. Description: Single RJ-45 jack for terminating Category 5e, twisted pair cable complying with Section 260523 "Control-Voltage Electrical Power Conductors" and a single BNC connector for terminating coaxial cable."
3. Description: Single RJ-45 jack for terminating Category 5e, twisted pair cable complying with Section 271513 "Communications Copper Horizontal Cabling" and a single BNC connector for terminating coaxial cable complying with Section 271533 "Communications Coaxial Horizontal Cabling."
4. Comply with UL 1863.

G. Toggle Switches, 120/277V, 15A:

1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
   a. Single Pole:
      1) Cooper; 1301-7 (standard), 7501 (decorator), 1301-7LT (lit handle).
      2) Hubbell; RS115 (standard), RSD115 (decorator), RS115IL (lit handle).
      3) Leviton; 1451-2 (standard), 5601-2 (decorator), 1461-LC (lit handle).
      4) P&S; 660G (standard), TM870 (decorator), 660-SLG (lit handle).
   b. Two Pole:
      1) Hubbell; RS215 (standard), RSD215 (decorator), RS215IL (lit handle).
      2) Leviton; CS215 (standard), 5602-2 (decorator).
      3) P&S; CS-B15AC2 (standard); 2622 (decorator).
   c. Three Way:
      1) Cooper; 1303-7 (standard), 7503 (decorator) 1303-7LT (lit handle).
      2) Hubbell; RS315 (standard), RSD315 (decorator), RS315IL (lit handle).
      3) Leviton; 1453-2 (standard), 5603-2 (decorator), 1463-LC (lit handle).
      4) P&S; 663G (standard), TM873 (decorator), 663-SLG (lit handle).
   d. Four Way:
      1) Cooper; 1242-7 (standard), 7504 (decorator).
      2) Hubbell; RS415 (standard), RSD415 (decorator), RS415IL (lit handle).
      3) Leviton; 54504-2 (standard), 5604-2 (decorator).
      4) P&S; 664G (standard), TM874 (decorator), 664-SLG (lit handle).

H. Fan-Speed Controls:

1. Modular, 120-V, full-wave, solid-state units with integral, quiet on-off switches and audible frequency and EMI/RFI filters.

2. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
   a. Cooper; RI061.
   b. Leviton; RNL06-10Z.
   c. P&S; DRSC53PV.

3. Comply with UL 1917.
4. Continuously adjustable rotary knob, 5A.

2.3 WALL-BOX DIMMERS

A. 0-10V Dimming for LED fixtures:

1. Products: Subject to compliance with requirements, available products that may be incorporated into the Work include, but are not limited to, the following:
   a. Philips; Sunrise series.
   b. Watt Stopper; Radiant series.
   c. Synergy; ISD-BC.
   d. Lutron; Diva series.

2. Slide dimmer with separate ON/OFF switch button. Lighting is switched ON or OFF at the level currently set by the slider.
3. Designed for use with standard three-way and four-way switches for applications requiring control from multiple locations.
4. Match dimmer to lamp(s) being dimmed in accordance with manufacturer's guidelines.

B. Dimmer Switches for screw-in LED lamps (LED lamps with integral drivers, designed to replace incandescent or screw-in compact fluorescent lamps):

1. Modular, full-wave, solid-state units with integral, quiet on-off switches, with audible frequency and EMI/RFI suppression filters.
2. Match dimmer to lamp(s) being dimmed in accordance with manufacturer's guidelines.
3. 600 W dimmers shall require no derating when ganged with other devices.

2.4 WALL PLATES

A. Single and combination types shall match corresponding wiring devices.

1. Plate-Securing Screws: Metal with head color to match plate finish.
2. Material for Finished Spaces: Smooth, high-impact thermoplastic.
3. Material for Unfinished Spaces: Galvanized steel.
4. Material for Damp Locations: Cast aluminum with spring-loaded lift cover, and listed and labeled for use in wet and damp locations.

B. Damp/Wet Location Covers

1. General:
   a. All wiring devices installed in damp or wet locations shall have cast metal covers.
   b. Covers shall be UL listed and labeled for use in wet and damp locations.
   c. Distinction between damp and wet locations shall be in accordance with NEC 406.9.
   d. Cover shall be appropriate for the device orientation with the hinge on top.
   e. Gasketing shall be provided to seal the cover to the box. Caulking shall be provided as required to seal any gaps between the cover and wall finish material.

2. Damp Location Covers:
   a. Cast metal with spring-loaded lift cover to seal the device when it is NOT in use.
   b. Leviton Series 6196 or equivalent.

3. Wet Location (Weatherproof-in-Use) Covers:
   a. Heavy Duty, Lockable, cast metal cover to seal the device whether it is in use or not.
   b. Intermatic Series WP1010MXD or equivalent.

2.5 FINISHES

A. Device Color:

1. Wiring Devices Connected to Normal Power System: As selected by Architect unless otherwise indicated or required by NFPA 70 or device listing.

B. Wall Plate Color: For thermoplastic covers, match device color, unless noted otherwise.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Comply with NECA 1, including mounting heights listed in that standard, unless otherwise indicated.

B. Coordination with Other Trades:

1. Protect installed devices and their boxes. Do not place wall finish materials over device boxes and do not cut holes for boxes with routers that are guided by riding against outside of boxes.
2. Keep outlet boxes free of plaster, drywall joint compound, mortar, cement, concrete, dust, paint, and other material that may contaminate the raceway system, conductors, and cables.
3. Install device boxes in brick or block walls so that the cover plate does not cross a joint unless the joint is troweled flush with the face of the wall.
4. Install wiring devices after all wall preparation, including painting, is complete.
5. Coordinate receptacle configuration, location and mounting height with equipment/ function it serves.

C. Conductors:

1. Do not strip insulation from conductors until right before they are spliced or terminated on devices.
2. Strip insulation evenly around the conductor using tools designed for the purpose. Avoid scoring or nicking of solid wire or cutting strands from stranded wire.
3. The length of free conductors at outlets for devices shall meet provisions of NFPA 70, Article 300, without pigtails.
4. Existing Conductors:
    a. Cut back and pigtail, or replace all damaged conductors.
    b. Straighten conductors that remain and remove corrosion and foreign matter.
    c. Pigtailing existing conductors is permitted, provided the outlet box is large enough.

D. Device Installation:

1. Replace devices that have been in temporary use during construction and that were installed before building finishing operations were complete.
2. Keep each wiring device in its package or otherwise protected until it is time to connect conductors.
3. Do not remove surface protection, such as plastic film and smudge covers, until the last possible moment.
4. Connect devices to branch circuits using pigtails that are not less than 6 inches (152 mm) in length.
5. When there is a choice, use side wiring with binding-head screw terminals. Wrap solid conductor tightly clockwise, two-thirds to three-fourths of the way around terminal screw.
6. Use a torque screwdriver when a torque is recommended or required by manufacturer.

7. When conductors larger than No. 12 AWG are installed on 15- or 20-A circuits, splice No. 12 AWG pigtails for device connections.
8. Tighten unused terminal screws on the device.
9. When mounting into metal boxes, remove the fiber or plastic washers used to hold device-mounting screws in yokes, allowing metal-to-metal contact.
10. Damp Location Covers: Not permitted UNO.
11. Wet Location Covers: Install everywhere outside UNO.

E. Device Plates: Do not use oversized or extra-deep plates. Repair wall finishes and remount outlet boxes when standard device plates do not fit flush or do not cover rough wall opening.

F. Dimmers:

1. Install dimmers within terms of their listing.
2. Verify that dimmers used for fan-speed control are listed for that application.
3. Install unshared neutral conductors on line and load side of dimmers according to manufacturers' device listing conditions in the written instructions.
4. Match dimmer to lamp(s) being dimmed in accordance with manufacturer’s guidelines.

G. Arrangement of Devices: Unless otherwise indicated, mount flush, with long dimension vertical and with grounding terminal of receptacles on top. Group adjacent switches under single, multi-gang wall plates.

H. GFCI Receptacles: Install non-feed-through-type GFCI receptacles.

3.2 FIELD QUALITY CONTROL

A. Test Instruments: Use instruments that comply with UL 1436.

B. Test Instrument for Convenience Receptacles: Digital wiring analyzer with digital readout or illuminated digital-display indicators of measurement.

C. Perform the following tests and inspections:

1. Tests for Convenience Receptacles:
   a. Line Voltage: Acceptable range is 105 to 132V.
   b. Percent Voltage Drop under 15-A Load: A value of 6 percent or higher is unacceptable.
   c. Ground Impedance: Values of up to 2 ohms are acceptable.
   d. GFCI Trip: Test for tripping values specified in UL 1436 and UL 943.
   e. Using the test plug, verify that the device and its outlet box are securely mounted.
   f. Tests shall be diagnostic, indicating damaged conductors, high resistance at the circuit breaker, poor connections, inadequate fault current path, defective devices, or similar problems. Correct circuit conditions, remove malfunctioning units and replace with new ones, and retest as specified above.

D. Wiring device will be considered defective if it does not pass tests and inspections.

3.3 IDENTIFICATION

A. Receptacles: Identify panelboard and circuit number from which the device is served.

1. Mark inside of box or coverplate with permanent marker. Test to ensure that marker lines are not visible on outside of cover when it is installed.

3.4 WEATHER STRIPPING

A. Behind exterior wall devices

1. Install a precut foam insulation pad over the fixture and reinstall the cover.

END OF SECTION 262726

SECTION 262816 - ENCLOSED SWITCHES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Fusible switches.
2. Nonfusible switches.
3. Enclosed circuit breaker switches.
4. Fustats
5. Enclosures.

1.2 DEFINITIONS

A. NC: Normally closed.

B. NO: Normally open.

C. SPDT: Single pole, double throw.

1.3 ACTION SUBMITTALS

A. Product Data: For each type of enclosed switch, accessory, and component indicated. Include nameplate ratings, dimensioned elevations, sections, weights, and manufacturers' technical data on features, performance, electrical characteristics, ratings, accessories, and finishes.

B. Shop Drawings: For enclosed switches.

1. Include plans, elevations, sections, details, and attachments to other work.
2. Include wiring diagrams for power, signal, and control wiring.

1.4 INFORMATIONAL SUBMITTALS

A. Seismic Qualification Certificates: For enclosed switches, accessories, and components, from manufacturer.

B. Field quality-control reports.

1.5 CLOSEOUT SUBMITTALS

A. Operation and maintenance data.

1.6 WARRANTY

A. Manufacturer's Warranty: Manufacturer and Installer agree to repair or replace components that fail in materials or workmanship within specified warranty period.

1. Warranty Period: One year(s) from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Seismic Performance: Enclosed switches shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

1. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified and the unit will be fully operational after the seismic event."

2.2 GENERAL REQUIREMENTS

A. Source Limitations: Obtain enclosed switches, overcurrent protective devices, components, and accessories, within same product category, from single manufacturer.

B. Product Selection for Restricted Space: Drawings indicate maximum dimensions for enclosed switches, including clearances between enclosures, and adjacent surfaces and other items. Comply with indicated maximum dimensions.

C. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by an NRTL, and marked for intended location and application.

D. Comply with NFPA 70.

2.3 FUSIBLE SWITCHES

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Square D by Schneider Electric.
2. Eaton Cutler-Hammer.
3. ABB/General Electric Company.

B. Type HD, Heavy Duty:

1. Single throw.
2. Two pole.
3. 240Vac.
4. 1200 A and smaller.

5. UL 98 and NEMA KS 1, horsepower rated, with clips or bolt pads to accommodate fuses indicated on plans, or as required by manufacture of the equipment being protected.
6. Lockable handle (in the OFF position) with capability to accept three padlocks, and interlocked with cover in closed position.
7. Ability to also lock switch handle in the ON position for exterior main disconnect.

C. Accessories:

1. Equipment Ground Kit: Internally mounted and labeled for copper and aluminum ground conductors.
2. Neutral Kit: Internally mounted; insulated, capable of being grounded and bonded; labeled for copper and aluminum neutral conductors.
3. Class R Fuse Kit: Provides rejection of other fuse types when Class R fuses are specified.
4. Lugs: Suitable for number, size, and conductor material.
5. Service-Rated Switches: Labeled for use as service equipment.

2.4 ENCLOSED CIRCUIT BREAKER SWITCHES

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Square D by Schneider Electric.
2. Eaton Cutler-Hammer.
3. ABB/General Electric Company.

B. MOLDED CASE PROTECTIVE DEVICES

1. Protective devices shall be molded case circuit breakers with inverse time and instantaneous tripping characteristics.
2. Circuit breakers shall be operated by a toggle-type handle and shall have a quick-make, quick-break over-center switching mechanism that is mechanically trip-free. Automatic tripping of the breaker shall be clearly indicated by the handle position. Contacts shall be non-welding silver alloy and arc extinction shall be accomplished by means of DE-ION arc chutes. A push-to-trip button on the front of the circuit breaker shall provide a local manual means to exercise the trip mechanism.
3. Circuit breakers shall have a minimum symmetrical interrupting capacity as indicated on the drawings.
4. Circuit breakers shall have thermal-magnetic trip units and inverse time-current characteristics.

C. ACCESSORIES

1. Provide shunt trips, bell alarms and auxiliary switches as shown on the contract drawings.

2.5 NONFUSIBLE SWITCHES

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Square D by Schneider Electric.
2. Eaton Cutler-Hammer.
3. ABB/General Electric Company.

B. Type HD, Heavy Duty:

1. Single throw.
2. Two pole.
3. 240Vac.
4. 1200 A and smaller.
5. UL 98 and NEMA KS 1, horsepower rated.
6. Lockable handle (in the OFF position) with capability to accept three padlocks, and interlocked with cover in closed position.

C. Accessories:

1. Equipment Ground Kit: Internally mounted and labeled for copper and aluminum ground conductors.
2. Neutral Kit: Internally mounted; insulated, capable of being grounded and bonded; labeled for copper and aluminum neutral conductors.
3. Lugs: Suitable for number, size, and conductor material.

2.6 FUSTAT

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. Bussmann.
2. Littelfuse.

B. NEMA 1 rated and designed to mount on square, pressed steel box.
1. 15A and smaller.
2. 120V ac.
3. Type S fuse socket.
4. Toggle switch.
5. Pilot light.

2.7 ENCLOSURES

A. Enclosed Switches: UL 489, NEMA KS 1, NEMA 250, and UL 50, to comply with environmental conditions at installed location.

B. Enclosure Finish: The enclosure shall be finished with:

1. Gray baked enamel paint, electrodeposited on cleaned, phosphatized steel (NEMA 250 Type 1).
2. Gray baked enamel paint, electrodeposited on cleaned, phosphatized galvannealed steel (NEMA 250 Types 3R, 12).
3. Brush finish on Type 304 stainless steel (NEMA 250 Type 4-4X stainless steel).

4. Copper-free cast aluminum alloy (NEMA 250 Types 7, 9).

C. Conduit Entry: NEMA 250 Types 4, 4X, and 12 enclosures shall contain no knockouts. NEMA 250 Types 7 and 9 enclosures shall be provided with threaded conduit openings in both endwalls.

D. Operating Mechanism: The switch operating handle shall be externally operable with the operating mechanism being an integral part of the box, not the cover. The cover interlock mechanism shall have an externally operated override. The override shall not permanently disable the interlock mechanism, which shall return to the locked position once the override is released. The tool used to override the cover interlock mechanism shall not be required to enter the enclosure in order to override the interlock.

E. Enclosures designated as NEMA 250 Type 4, 4X stainless steel, 12, or 12K shall have a dual cover interlock mechanism to prevent unintentional opening of the enclosure cover when the switch is ON and to prevent turning the switch ON when the enclosure cover is open.

F. NEMA 250 Type 7/9 enclosures shall be furnished with a breather and drain kit to allow their use in outdoor and wet location applications.

PART 3 - EXECUTION

3.1 ENCLOSURE ENVIRONMENTAL RATING APPLICATIONS

A. Enclosed Switches: Provide enclosures at installed locations with the following environmental ratings.

1. Indoor, Dry and Clean Locations: NEMA 250, Type 1.
2. Outdoor Locations: NEMA 250, Type 3R.
3. Kitchen/Wash-Down Areas: NEMA 250, Type 4X, stainless steel.
4. Other Wet or Damp, Indoor Locations: NEMA 250, Type 4.
5. Indoor Locations Subject to Dust, Falling Dirt, and Dripping Noncorrosive Liquids: NEMA 250, Type 12.
6. Hazardous Areas Indicated on Drawings: NEMA 250, Type 7/Type 9 with cover attached by Type 316 stainless steel bolts.

3.2 INSTALLATION

A. Coordinate layout and installation of switches and components with equipment served and adjacent surfaces. Maintain required workspace clearances and required clearances for equipment access doors and panels.

B. Install individual wall-mounted switches with tops at uniform height unless otherwise indicated.

C. Comply with mounting and anchoring requirements specified in Section 260548.16 "Seismic Controls for Electrical Systems."

D. Temporary Lifting Provisions: Remove temporary lifting of eyes, channels, and brackets and temporary blocking of moving parts from enclosures and components.

E. Install fuses in fusible devices.

F. Comply with NFPA 70 and NECA 1.

3.3 IDENTIFICATION

A. Comply with requirements in Section 260553 "Identification for Electrical Systems."

1. Identify field-installed conductors, interconnecting wiring, and components; provide warning signs.
2. Label each enclosure with engraved metal or laminated-plastic nameplate.

3.4 FIELD QUALITY CONTROL

A. Perform tests and inspections.

B. Tests and Inspections for Switches:

1. Visual and Mechanical Inspection:

a. Inspect physical and mechanical condition.
b. Inspect anchorage, alignment, grounding, and clearances.
c. Verify that the unit is clean.
d. Verify blade alignment, blade penetration, travel stops, and mechanical operation.
e. Verify that fuse sizes and types match the Specifications and Drawings.
f. Verify that each fuse has adequate mechanical support and contact integrity.
g. Inspect bolted electrical connections for high resistance using one of the two following methods:

1) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method in accordance with manufacturer's published data or NETA ATS Table 100.12.

a) Bolt-torque levels shall be in accordance with manufacturer's published data. In the absence of manufacturer's published data, use NETA ATS Table 100.12.

h. Verify correct phase barrier installation.
i. Verify lubrication of moving current-carrying parts and moving and sliding surfaces.

C. Enclosed switches will be considered defective if they do not pass tests and inspections.

D. Prepare test and inspection reports.

1. Test procedures used.
2. Include identification of each enclosed switch tested and describe test results.
3. List deficiencies detected, remedial action taken, and observations after remedial action.

END OF SECTION 262816

SECTION 264313 - SURGE PROTECTION FOR LOW-VOLTAGE ELECTRICAL POWER CIRCUITS

PART 1 - GENERAL

1.1 SUMMARY

A. Section includes SPDs for low-voltage (120 to 600 V) power distribution and control equipment.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product.

1. Include rated capacities, operating characteristics, electrical characteristics, and furnished specialties and accessories.
2. Copy of UL Category Code VZCA certification, as a minimum, listing the tested values for VPRs, I-nominal ratings, MCOVs, type designations, OCPD requirements, model numbers, system voltages, and modes of protection.

1.3 INFORMATIONAL SUBMITTALS

A. Field quality-control reports.

B. Sample Warranty: For manufacturer's special warranty.

1.4 CLOSEOUT SUBMITTALS

A. Maintenance data.

1.5 WARRANTY

A. Manufacturer's Warranty: Manufacturer agrees to replace or replace SPDs that fail in materials or workmanship within specified warranty period.

1. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 GENERAL SPD REQUIREMENTS

A. SPD with Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Comply with NFPA 70.

C. Comply with UL 1449.

D. MCOV of the SPD shall be the nominal system voltage.

2.2 EXTERIOR MOUNTED 'LIGHTNING ARRESTOR' SPD

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1. Schneider Electric, Square D.
2. ABB, General Electric.
3. Eaton.

B. SPD:

1. Comply with UL 1449, Type 1.
2. Designed to be installed without OCPD.
3. NEMA 4X enclosure with nipple mount for installation on outside of exterior main disconnect switch, panel, or switchboard.
4. 15" #14 AWG connection leads (cut as necessary to maintain shortest lead length possible).

C. Characteristics and protection modes and UL 1449 VPR for 240/120 V, single-phase, three-wire circuits shall be as follows:

1. Maximum Continuous Operating Voltage (MCOV):
   a. Line to Neutral: 150 V.
   b. Line to Line: 300 V.

2. Peak Voltage Let-Through (Vpk):
   a. Line to Neutral: 700 V.
   b. Line to Line: 1200 V.

3. Peak Surge Current Rating per Phase: 36 kA.
4. Nominal Current Rating (I-nom): 10 kA
5. Short Circuit Current Rating (SCCR): 25 kA.

2.3 SERVICE ENTRANCE SUPPRESSOR

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1. Advanced Protection Technologies Inc. (APT).
2. Current Technology Inc.
3. Eaton.
4. ABB, General Electric.
5. Leviton Manufacturing Co., Inc.
6. Liebert; a brand of Vertiv.
7. Schneider Electric, Square D.

B. SPDs: Comply with UL 1449, Type 1.

1. Permanently connected, hard-wired SPD that can be installed in both locations noted:
    a. Between the secondary of the service transformer and the line side of the main service equipment overcurrent protective device.
    b. On the load side of the main service equipment.

2. SPDs with the following features and accessories:
    a. Internal thermal protection that disconnects the SPD before damaging internal suppressor components.
    b. Indicator light display for protection status.

C. Features and Accessories:

1. Internal thermal protection that disconnects the SPD before damaging internal suppressor components.
2. Indicator light display for protection status.
3. Surge counter.

D. Peak Surge Current Rating: The minimum single-pulse surge current withstand rating per phase shall not be less than 200 kA. The peak surge current rating shall be the arithmetic sum of the ratings of the individual MOVs in a given mode.

E. Characteristics and protection modes and UL 1449 VPR for 240/120 V, single-phase, three-wire circuits shall be as follows:

1. Maximum Continuous Operating Voltage (MCOV):
    a. Line to Neutral: 150 V.
    b. Line to Line: 300 V.

2. Peak Voltage Let-Through (Vpk):
    a. Line to Ground: 700 V.
    b. Line to Neutral: 700 V.
    c. Line to Line: 1000 V.

3. Nominal Current Rating (I-nom): 20 kA
4. Short Circuit Current Rating (SCCR): 25 kA.

2.4 PANEL SUPPRESSORS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1. Advanced Protection Technologies Inc. (APT).
2. Current Technology Inc.
3. Eaton.
4. ABB, General Electric.
5. Leviton Manufacturing Co., Inc.
6. Liebert; a brand of Vertiv.
7. Schneider Electric, Square D.

B. SPDs: Comply with UL 1449, Type 2.

1. Permanently connected, hard-wired SPD intended for installation on the load side of the main service equipment overcurrent protective device.
2. Include LED indicator lights for power and protection status.
3. Internal thermal protection that disconnects the SPD before damaging internal suppressor components.

C. Peak Surge Current Rating: The minimum single-pulse surge current withstand rating per phase shall not be less than 100 kA. The peak surge current rating shall be the arithmetic sum of the ratings of the individual MOVs in a given mode.

D. Characteristics and protection modes and UL 1449 VPR for 240/120 V, single-phase, three-wire circuits shall be as follows:

1. Maximum Continuous Operating Voltage (MCOV):
   a. Line to Neutral: 150 V.
   b. Line to Line: 300 V.

2. Peak Voltage Let-Through (Vpk):
   a. Neutral to Ground: 700 V.
   b. Line to Ground: 700 V.
   c. Line to Neutral: 700 V.
   d. Line to Line: 1200 V.

3. Nominal Current Rating (I-nom): 20 kA
4. Short Circuit Current Rating (SCCR): 25 kA.

2.5 ENCLOSURES

A. Service Entrance and Panel Suppressors - Install integral SPD within switchboard/panelboard.

B. ‘Lightning Arrestor’ surge protective devices - NEMA 250, Type 3R

C. Exterior SPDs - NEMA 250, Type 3R as appropriate for the environment.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Comply with NECA 1.

B. Integral Surge Protection - All SPDs shall be provided integral with the switchboard or panelboard, mounted inside the enclosure.

C. Exterior Surge Protection

1. In general, exterior SPDs are not allowed. If proposed for use:
   a. Contractor shall provide documented motivation for installation of exterior mounted device.

b. Device shall be mounted in location to allow for shortest possible wiring to switchboard/panelboard bus or breaker.

D. Install an OCPD or disconnect as required to comply with the UL listing of the SPD.

E. Install SPDs with conductors between suppressor and points of attachment as short and straight as possible, and adjust circuit-breaker positions to achieve shortest and straightest leads. Do not splice and extend SPD leads unless specifically permitted by manufacturer. Do not exceed manufacturer's recommended lead length. Do not bond neutral and ground.

F. Use crimped connectors and splices only. Wire nuts are unacceptable.

G. Only mount ‘Lightning Arrestor’ SPDs outside the building on the exterior of the main disconnect, switchboard or panelboard.

H. Complete startup checks according to manufacturer's written instructions. Energize SPDs after power system has been energized, stabilized, and tested.

3.2 FIELD QUALITY CONTROL

A. Perform the following inspections:

1. Compare equipment nameplate data for compliance with Drawings, Specifications, and protected equipment electrical configuration.
2. Inspect anchorage, alignment, grounding, and clearances.
3. Verify that electrical wiring installation complies with manufacturer's written installation requirements.

3.3 DEMONSTRATION

A. Train Owner's maintenance personnel to inspect and maintain SPDs.

END OF SECTION 264313

SECTION 265100 - LED LIGHTING

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Lighting fixtures that are designed for and exclusively use LED technology.
2. Luminaire supports.

B. Related Sections:
1. Section 262726 "Wiring Devices".

1.2 DEFINITIONS

A. CCT: Correlated color temperature.

B. CRI: Color Rendering Index.

C. Fixture: See "Luminaire."

D. IP: International Protection or Ingress Protection Rating.

E. LED: Light-emitting diode.

F. Lumen: Measured output of lamp and luminaire, or both.

G. Luminaire: Complete lighting unit, including lamp, reflector, and housing.

H. THD: Total Harmonic Distortion.

1.3 PRIOR APPROVAL

A. Prior approvals are not required unless otherwise noted on the Luminaire Schedule.

1. All material supplied to the project must meet or exceed the quality, performance, and have similar features to the product originally specified. It is the contractor's responsibility to ensure that substituted equipment matches the exterior dimensions, weight, and configuration of the specified equipment.

1.4 ACTION SUBMITTALS

A. Product Data: For each type of lighting fixture, arranged in order of fixture designation. Include data on features, accessories, finishes, and the following:

1. Physical description of lighting fixture including dimensions.
2. Ballast/Driver, including THD.
3. Emergency lighting units including battery and charger.
4. Energy-efficiency data.
5. Life, output (lumens, CCT, and CRI), and energy-efficiency data.
6. Fixture UL/ETL rating.
7. Design Lights Consortium (DLC) certification and/or Energy Star rating.
8. Photometric data and adjustment factors based on laboratory tests, complying with IESNA Lighting Measurements Testing & Calculation Guides, of each lighting fixture type. The adjustment factors shall be for lamps, ballasts, and accessories identical to those indicated for the lighting fixture as applied in this Project.
    a. Manufacturer Certified Data: Photometric data shall be certified by a manufacturer's laboratory with a current accreditation under the National Voluntary Laboratory Accreditation Program for Energy Efficient Lighting Products.
9. Color samples (if color is to be chosen by architect/engineer).
10. Foot-candle calculations for spot lights and flood lights.
11. List of all parts necessary for particular installation configuration.

B. Shop Drawings: For nonstandard or custom luminaires.

1. Include plans, elevations, sections, and mounting and attachment details.
2. Include details of luminaire assemblies. Indicate dimensions, weights, loads, required clearances, method of field assembly, components, and location and size of each field connection.
3. Include diagrams for power, signal, and control wiring.

1.5 INFORMATIONAL SUBMITTALS

A. Seismic Qualification Certificates: For luminaires, accessories, and components, from manufacturer.

B. Sample warranty.

1.6 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data.

1.7 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
1. Plastic Diffusers and Lenses: One for every 50 of each type and rating installed. Furnish at least one of each type.
2. Ballasts/Drivers: One for every 50 of each type and rating installed. Furnish at least one of each type.
3. Globes and Guards: One for every 20 of each type and rating installed. Furnish at least one of each type.

1.8 QUALITY ASSURANCE

A. Luminaire Photometric Data Testing Laboratory Qualifications: Provided by manufacturers' laboratories that are accredited under the National Volunteer Laboratory Accreditation Program for Energy Efficient Lighting Products.

B. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

C. Comply with NFPA 70.

1.9 COORDINATION

A. Coordinate layout and installation of lighting fixtures and suspension system with other construction that penetrates ceilings or is supported by them, including HVAC equipment, fire-suppression system, and partition assemblies.

B. Fire rated assemblies: Fixtures installed in fire rated assemblies shall maintain the fire rating of said assembly. Contractor is required to coordinate with Architectural draws to verify assembly ratings.

C. Insulated ceiling space: Fixtures installed in an insulated ceiling be IC rated or manufacturer recommended clearances between fixture and insulation. Contractor is required to coordinate with Architectural draws to verify insulated areas above ceilings.

1.10 WARRANTY

A. Warranty: Manufacturer and Installer agree to repair or replace components of luminaires that fail in materials or workmanship within specified warranty period.

B. Warranty Period: Five years from date of Substantial Completion.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Seismic Performance: Luminaires shall withstand the effects of earthquake motions as noted in 260548.16, “Seismic Controls for Electrical Systems”.

B. Seismic Performance: Luminaires and lamps shall be labeled vibration and shock resistant.

1. The term "withstand" means "the luminaire will remain in place without separation of any parts when subjected to the seismic forces specified and the luminaire will be fully operational during and after the seismic event."

2. Component Importance Factor: 1.5.

2.2 GENERAL LUMINAIRE REQUIREMENTS

A. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

B. Standards - Where noted on plans, comply with the following:

1. ENERGY STAR or Design Lights Consortium (DLC) certified.
2. NRTL Compliance: Luminaires for hazardous locations shall be listed and labeled for indicated class and division of hazard by an NRTL.
3. UL Listing: Listed for damp and/or wet locations as required.
4. Recessed luminaires shall comply with NEMA LE 4.

C. Indoor fixtures shall have a minimum CRI of 80 UNO and a CCT of 3000 K UNO.

D. Outdoor fixtures shall have a minimum CRI of 65 UNO and a CCT of 3000 K UNO.

E. Outdoor fixtures shall have mounting type and distribution as noted on plans.

F. Minimum rated LED lamp life of 50,000 hours to L70.

G. Lamps dimmable from 100 percent to 10 percent of maximum light output.

H. Internal ballast/driver, UNO.

I. Nominal Operating Voltage: As noted on the plans.

J. Lens Thickness: At least 0.125 inch minimum UNO.

K. Reflecting surfaces shall have minimum reflectance as follows unless otherwise indicated:

1. White Surfaces: 85 percent.
2. Specular Surfaces: 83 percent.
3. Diffusing Specular Surfaces: 75 percent.

L. Lens and Refractor Gaskets for Exterior Luminaires: Use heat- and aging-resistant resilient gaskets to seal and cushion lenses and refractors in luminaire doors.

M. Source Limitations: For luminaires, obtain each color, grade, finish, type, and variety of luminaire from single source with resources to provide products of consistent quality in appearance and physical properties.

N. Housings:

1. Rigidly formed, light-tight enclosure that will not warp, sag, or deform in use.
2. Provide weather-tight enclosure with filter/breather for enclosed exterior luminaires.

O. Metal Parts:

1. Free of burrs and sharp corners and edges.
2. Indoor applications: Sheet metal components shall be steel unless otherwise indicated.
3. Exterior applications: Sheet metal components shall be corrosion-resistant aluminum.

4. Form and support to prevent warping and sagging

P. Doors, Frames, and Other Internal Access: Smooth operating, free of light leakage under operating conditions, and designed to permit relamping without use of tools. Designed to prevent doors, frames, lenses, diffusers, and other components from falling accidentally during relamping and when secured in operating position. Doors shall be removable for cleaning or replacing lenses.

Q. Diffusers, and Globes - Tempered glass, acrylic or polycarbonate as noted on plans.

1. Acrylic: One hundred percent virgin acrylic plastic, with high resistance to yellowing and other changes due to aging, exposure to heat, and UV radiation.

a. Lens Thickness: At least 0.125 inch (3.175 mm) minimum unless otherwise indicated.
b. UV stabilized.

2. Glass: Annealed crystal glass unless otherwise indicated.

2.3 METAL FINISHES

A. Variations in finishes are unacceptable in the same piece. Variations in finishes of adjoining components are acceptable if they are within the range of approved Samples and if they can be and are assembled or installed to minimize contrast.

B. Source Limitations: For luminaires, obtain each color, grade, finish, type, and variety of luminaire from single source with resources to provide products of consistent quality in appearance and physical properties.

C. Factory-Applied, powder-coat finish, UNO, with standard color chosen by Architect or as noted on plans.

1. Factory-Applied Finish for Aluminum Luminaires: Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.

a. Finish designations prefixed by AA comply with the system established by the Aluminum Association for designating aluminum finishes.
b. Class I, Color-Anodic Finish: AA-M32C22A42/A44 (Mechanical Finish: Medium satin; Chemical Finish: Etched, medium matte; Anodic Coating: Architectural Class I, integrally colored or electrolytically deposited color coating 0.018 mm or thicker), complying with AAMA 611.

2. Factory-Applied Finish for Steel Luminaires: Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes.

a. Surface Preparation: Clean surfaces to comply with SSPC-SP 1, to remove dirt, oil, grease, and other contaminants that could impair paint bond. Grind welds and polish surfaces to a smooth, even finish. Remove mill scale and rust, if present, from uncoated steel, complying with SSPC-SP 5/NACE No. 1 or SSPC-SP 8.

b. Exterior Surfaces: Manufacturer's standard finish consisting of one or more coats of primer and two finish coats of high-gloss, high-build polyurethane enamel.

2.4 LED ASSEMBLIES

A. Products UL rated for 40 degree C (104 degrees F) ambient environments.

B. Minimum 3000K color temperature unless noted otherwise in the drawings.

C. 50,000 hour fixture life including driver, 5 year warranty.

D. All products compliant with IESNA LM-79 and LM-80 standards.

2.5 LUMINAIRE SUPPORTS

A. Comply with requirements in Section 260529 "Hangers and Supports for Electrical Systems" for channel and angle iron supports and nonmetallic channel and angle supports.

B. Single-Stem Hangers: 1/2-inch (13-mm) steel tubing with swivel ball fittings and ceiling canopy. Finish same as luminaire.

C. Twin-Stem Hangers: Two, 1/2-inch (13-mm) steel tubes with single canopy designed to mount a single fixture. Finish same as fixture.

D. Wires: ASTM A 641/A 641 M, Class 3, soft temper, zinc-coated steel, 12 gauge (2.68 mm).

E. Rod Hangers: 3/16-inch (5-mm) minimum diameter, cadmium-plated, threaded steel rod.

F. Hook Hangers: Integrated assembly matched to luminaire, line voltage, and equipment with threaded attachment, cord, and locking-type plug.

2.6 SCREW-IN BASE FIXTURES

A. Fixture shall be compatible with common type Medium Base E26 bulb.

B. Fixture shall be dimming compatible and compatible with dimmable LED type bulb.

C. Lamp shall be compatible with dimmer switch to allow full, flicker-free dimming throughout the lamp’s full dimming range.

D. Comply with UL1598 and comply with minimum performance requirements for retrofit lamps - NEMA SSL 4.

E. SPARE LAMPS - Spare lamps shall be provided in quantities of 10% of total lamps of each screw-in base type installed. Quantity of spares shall not be less than four (4) spare lamps for any one lamp type.

PART 3 - EXECUTION

3.1 GENERAL INSTALLATION

A. Comply with NECA 1.

B. Install luminaires level, plumb, and square with ceilings and walls unless otherwise indicated.

C. Use fastening methods and materials selected to resist seismic forces defined for the application and approved by manufacturer.

D. Fasten luminaire to structural support.

E. Supports:

1. Sized and rated for luminaire weight, and weight of emergency power unit where applicable.
2. Able to maintain luminaire position after cleaning, while relamping and when testing emergency power unit.
3. Provide support for luminaire and emergency power unit without causing deflection of ceiling or wall.
4. Luminaire-mounting devices shall be capable of supporting a horizontal force of 100 percent of luminaire and emergency power unit weight and vertical force of 400 percent of fixture weight.
5. Use fastening methods and materials selected to resist seismic forces defined for the application and approved by manufacturer.

F. Flush-Mounted Luminaire Support: Secured to outlet box.

G. Wall-Mounted Luminaire Support:

1. Attached to structural members in walls or to a minimum 20 gauge backing plate attached to wall structural members.
2. Do not attach luminaires directly to gypsum board.

H. Ceiling-Mounted Luminaire Support:

1. Secure to any required outlet box and attach to structural member in ceiling or to a minimum 20 gauge backing plate attached to ceiling structural members.
2. Do not attach luminaires directly to gypsum board.
3. Provide offset from ceiling as required by luminaire manufacturer.
4. Secure emergency power unit using approved fasteners in a minimum of four locations, spaced near corners of emergency power unit.

I. Suspended Luminaire Support:

1. Pendants and Rods: Where longer than 48 inches (1200 mm), brace to limit swinging.
2. Stem-Mounted, Single-Unit Luminaires: Suspend with twin-stem hangers. Support with approved outlet box and accessories that hold stem and provide damping of luminaire oscillations. Support outlet box vertically to building structure using approved devices.

3. Continuous Rows of Luminaires: Use tubing or stem for wiring at one point and tubing or rod for suspension for each unit length of luminaire chassis, including one at each end.
4. Do not use ceiling grid as support for pendant luminaires. Connect support wires or rods to building structure.

J. Lay-in Ceiling Lighting Fixtures Supports: Use grid as a support element.

1. Install ceiling support system rods or wires, independent of the ceiling suspension devices, for each fixture. Locate not more than 6 inches (150 mm) from lighting fixture corners.
2. Support Clips: Fasten to lighting fixtures and to ceiling grid members at or near each fixture corner with clips that are UL listed for the application.
3. Fixtures of Sizes Less Than Ceiling Grid: Install as indicated on reflected ceiling plans or center in acoustical panel, and support fixtures independently with at least two 3/4-inch (20-mm) metal channels spanning and secured to ceiling tees.
4. Install at least two independent support rods or wires from structure to a tab on lighting fixture. Wire or rod shall have breaking strength of the weight of fixture at a safety factor of 3.

K. Comply with requirements in Section 260519 "Low-Voltage Electrical Power Conductors and Cables" for wiring connections.

L. Temporary Lighting: If it is necessary, and approved by Architect, to use permanent luminaires for temporary lighting, install and energize the minimum number of luminaires necessary. When construction is sufficiently complete, remove the temporary luminaires, disassemble, clean thoroughly and reinstall.

M. Remote Mounting of Ballasts/Drivers: Distance between the driver and fixture shall not exceed that recommended by luminaire manufacturer.

3.2 IDENTIFICATION

A. Install labels with panel and circuit numbers on concealed junction and outlet boxes. Comply with requirements for identification specified in Section 260553 "Identification for Electrical Systems."

3.3 INSULATED CEILING SPACES

A. Provide IC rated fixture assemblies or manufacturer recommended clearances between fixture and insulation.

3.4 FIRE RATED ASSEMBLIES

A. Provide fire rated fixture assemblies or a third party fire rated cover.
1. Fire rated covers
a. Provide manufacturer recommended clearances for all non IC rated fixtures.

3.5 FIELD QUALITY CONTROL

A. Perform the following tests and inspections:

1. Operational Test: After installing luminaires, switches, and accessories, and after electrical circuitry has been energized, test units to confirm proper operation.
2. Test for Emergency Lighting: Interrupt power supply to demonstrate proper operation. Verify transfer from normal power to battery power and retransfer to normal.

B. Luminaire will be considered defective if it does not pass operation tests and inspections.

C. Prepare test and inspection reports.

END OF SECTION 265100

SECTION 27 0000

COMMUNICATIONS PROJECT OVERVIEW

PART 1 - GENERAL

1.1 SUMMARY OF WORK

A. This project includes the installation of infrastructure-only for telecommunications cabling and systems. Site communications conduit and handholes will be furnished by Diamond Communications as Government Furnished Material (GFM), and will be installed by the Contractor. The Contractor will be responsible for all installation of said materials, including excavation, trenching, and backfill. Site communications conduit shall be provided to each structure indicated on the plans, and stubbed in to the crawl space of each structure. From that point, within the structures, the Contractor will be responsible for furnishing and installing conduit between the entrance from site into the crawl space and extending conduit over and up to the location of the fiber panel in each structure.

1.2 RELATED PROJECTS

A. The Contractor for this project will be required to coordinate with Diamond Communications. Diamond Communications will be providing fiber optic cabling and telecommunications service installation directly to the Owner, utilizing the pathways provided by the Contractor under this project.

1.3 CONTRACTOR QUALIFICATIONS

A. Contractor shall employ, in conjunction with construction of the project, a capable, experienced, and reliable foreperson and such skilled workers as may be required for the various classes of work to be performed.

END OF SECTION 27 0000

SECTION 27 0528

PATHWAYS FOR COMMUNICATIONS SYSTEMS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Metal conduits and fittings.
2. Nonmetallic conduits and fittings.
3. Optical-fiber-cable pathways and fittings.
4. Metallic surface pathways.
5. Nonmetallic surface pathways.
6. J-Hooks.
7. Boxes, enclosures, and cabinets.
8. Polymer-concrete handholes and boxes for exterior underground cabling.

1.2 ACTION SUBMITTALS

A. Product Data: For each type of product used on this project, provide cut sheet with highlighted part number of product.

PART 2 - PRODUCTS

2.1 METAL CONDUITS AND FITTINGS

A. Description: Metal raceway of circular cross section with manufacturer-fabricated fittings.

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. O-Z/Gedney; a brand of Emerson Industrial Automation.
2. Southwire Company.
3. Thomas & Betts Corporation; A Member of the ABB Group.

C. General Requirements for Metal Conduits and Fittings:

1. Listed and labeled as defined in NFPA 70, by a nationally recognized testing laboratory, and marked for intended location and application.
2. Comply with TIA-569-E.

D. GRC: Comply with ANSI C80.1 and UL 6.

E. ARC: Comply with ANSI C80.5 and UL 6A.

F. IMC: Comply with ANSI C80.6 and UL 1242.

G. PVC-Coated Steel Conduit: PVC-coated IMC.

1. Comply with NEMA RN 1.
2. Coating Thickness: 0.040 inch (1 mm), minimum.

H. EMT: Comply with ANSI C80.3 and UL 797.

I. Fittings for Metal Conduit: Comply with NEMA FB 1 and UL 514B.

1. Conduit Fittings for Hazardous (Classified) Locations: Comply with UL 1203 and NFPA 70.
2. Fittings for EMT:

a. Material: Steel
b. Type: Setscrew

3. Expansion Fittings: PVC or steel to match conduit type, complying with UL-467, rated for environmental conditions where installed, and including flexible external bonding jumper.
4. Coating for Fittings for PVC-Coated Conduit: Minimum thickness of 0.040 inch (1 mm), with overlapping sleeves protecting threaded joints.

J. Joint Compound for IMC, GRC, or ARC: Approved, as defined in NFPA 70, by authorities having jurisdiction for use in conduit assemblies, and compounded for use to lubricate and protect threaded conduit joints from corrosion and to enhance their conductivity.

2.2 NONMETALLIC CONDUITS AND FITTINGS

A. Description: Nonmetallic raceway of circular section with manufacturer-fabricated fittings.

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Carlon; a brand of Thomas & Betts Corporation.
2. Dura-Line.
3. RACO; Hubbell.

C. General Requirements for Nonmetallic Conduits and Fittings:

1. Listed and labeled as defined in NFPA 70, by an NRTL, and marked for intended location and application.
2. Comply with TIA-569-E.

D. RNC: Type EPC-40-PVC Type EPC-80-PVC, complying with NEMA TC 2 and UL 651 unless otherwise indicated.

E. Rigid HDPE: Comply with UL 651A.

F. Continuous HDPE: Comply with UL 651A.

G. RTRC: Comply with UL 2515A and NEMA TC 14.

1. Fittings: Comply with NEMA TC 3; match to conduit or tubing type and material.

H. Solvents and Adhesives: As recommended by conduit manufacturer.

2.3 OPTICAL-FIBER-CABLE PATHWAYS AND FITTINGS

A. Description: Comply with UL 2024; flexible-type pathway with a circular cross section, approved for plenum or riser installation unless otherwise indicated.

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Carlon; a brand of Thomas & Betts Corporation.
2. Dura-Line.
3. Hubbell.

2.4 J-HOOKS

A. Description: Prefabricated sheet metal cable supports for telecommunications cable.

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Eaton, B-line.
2. Panduit Corp.
3. Wiremold / Legrand.

C. Listed and labeled as defined in NFPA 70, by an NRTL, and marked for intended location and application.

D. Comply with TIA-569-E.

2.5 BOXES, ENCLOSURES, AND CABINETS

A. Description: Enclosures for communications.

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. O-Z/Gedney; a brand of Emerson Industrial Automation.
2. RACO; Hubbell.
3. Thomas & Betts Corporation; A Member of the ABB Group.

C. General Requirements for Boxes, Enclosures, and Cabinets:

1. Comply with TIA-569-E.

2. Boxes, enclosures, and cabinets installed in wet locations shall be listed and labeled as defined in NFPA 70, by an NRTL, and marked for use in wet locations.
3. Box extensions used to accommodate new building finishes shall be of same material as recessed box.
4. Device Box Dimensions: 4 inches square by 2-1/8 inches deep, with single gang mud ring, U.N.O.

D. Small Sheet Metal Pull and Junction Boxes: NEMA OS 1.

E. Nonmetallic Outlet and Device Boxes: Comply with NEMA OS 2 and UL 514C.

F. Hinged-Cover Enclosures: Comply with UL 50 and NEMA 250, Type 3R, with continuous-hinge cover with flush latch unless otherwise indicated.

1. Metal Enclosures: Steel, finished inside and out with manufacturer's standard enamel.
2. Nonmetallic Enclosures:

a. Material: Fiberglass.
b. Finished inside with radio-frequency-resistant paint.

3. Interior Panels: Steel; all sides finished with manufacturer's standard enamel.

2.6 POLYMER-CONCRETE HANDHOLES

A. Description: Molded of sand and aggregate; bound together with polymer resin; and reinforced with steel, fiberglass, or a combination of the two.

B. Manufacturers: Subject to compliance with requirements, provide products by one of the following:

1. Armorcast Products Company.
2. NewBasis.
3. Quazite: Hubbell Power Systems, Inc.

C. General Requirements for Polymer Concrete Handholes:

1. Boxes and handholes for use in underground systems shall be listed and labeled as defined in NFPA 70, by an NRTL, and marked for intended location and application.
2. Boxes installed in wet areas shall be listed and labeled as defined in NFPA 70, by an NRTL, and marked for intended location and application.
3. Comply with TIA-569-E.

D. Configuration: Designed for flush burial with open bottom unless otherwise indicated.

E. Cover: Weatherproof, secured by tamper-resistant locking devices and having structural load rating consistent with enclosure and handhole location.

1. Cover Finish: Nonskid finish shall have a minimum coefficient of friction of 0.50.
2. Cover Legend: Molded lettering, "COMMUNICATIONS".

PART 3 - EXECUTION

3.1 PATHWAY APPLICATION

A. Minimum Pathway Size: refer to drawings.

B. Pathway Fittings: Compatible with pathways and suitable for use and location.

C. Do not install aluminum conduits, boxes, or fittings in contact with concrete or earth.

D. Install surface pathways only where indicated on Drawings.

E. Do not install nonmetallic conduit where ambient temperature exceeds 120 deg F.

3.2 INSTALLATION

A. Comply with the following standards for installation requirements except where requirements on Drawings or in this Section are stricter:

1. NECA 1.
2. TIA-569-E.
3. NECA 101.
4. NECA 102.
5. NECA 105.
6. NECA 111.

B. Comply with NFPA 70 limitations for types of pathways allowed in specific occupancies and number of floors.

C. Keep pathways at least 12 inches away from parallel runs of flues and steam or hot-water pipes. Install horizontal pathway runs above water and steam piping.

D. Complete pathway installation before starting conductor installation.

E. Install no more than the equivalent of 180-degree bends in any pathway run. Provide junction box at locations exceeding 180-degree bends. Support within 12 inches of changes in direction. Utilize long radius elbows for all optical-fiber cables.

F. Conceal rigid conduit within finished walls, ceilings, and floors unless otherwise indicated. Install conduits parallel or perpendicular to building lines.

G. Support conduit within 12 inches of enclosures to which attached.

H. Pathways Embedded in Slabs:

1. Run conduit larger than 1-inch trade size, parallel or at right angles to main reinforcement. Where at right angles to reinforcement, place conduit close to slab support. Secure pathways to reinforcement at maximum 10-foot intervals.
2. Arrange pathways to cross building expansion joints at right angles with expansion fittings. Comply with requirements for expansion joints specified in this article.

3. Arrange pathways to keep a minimum of 1 inch of concrete cover in all directions.
4. Do not embed threadless fittings in concrete unless specifically approved by Architect for each specific location.

I. Stub-ups to Above Recessed Ceilings:

1. Use EMT, IMC, or RMC for pathways.
2. Use a conduit bushing or insulated fitting to terminate stub-ups not terminated in hubs or in an enclosure.

J. Threaded Conduit Joints, Exposed to Wet, Damp, Corrosive, or Outdoor Conditions: Apply listed compound to threads of pathway and fittings before making up joints. Follow compound manufacturer's written instructions.

K. Coat field-cut threads on PVC-coated pathway with a corrosion-preventing conductive compound prior to assembly.

L. Do not rely on locknuts to penetrate nonconductive coatings on enclosures. Remove coatings in the locknut area prior to assembling conduit to enclosure, to assure a continuous ground path.

M. Cut conduit perpendicular to the length. For conduits of 2-inch trade size and larger, use roll cutter or a guide to ensure cut is straight and perpendicular to the length.

N. Install pull wires in empty pathways. Use polypropylene or monofilament plastic line with not less than 200-lb tensile strength. Leave at least 12 inches of slack at each end of pull wire. Secure pull wire, so it cannot fall into conduit. Cap pathways designated as spare alongside pathways in use.

O. Surface Pathways:

1. Install surface pathway for surface telecommunications outlet boxes only where indicated on Drawings.
2. Install surface pathway with a minimum 2-inch radius control at bend points.
3. Secure surface pathway with screws or other anchor-type devices at intervals not exceeding 48 inches and with no less than two supports per straight pathway section. Support surface pathway according to manufacturer's written instructions. Tape and glue are not acceptable support methods.

P. Pathways for Optical-Fiber and Communications Cable: Install pathways, metal and nonmetallic, rigid and flexible, as follows:

1. 1-Inch Trade Size and Larger: Install pathways in maximum lengths of 75 feet.
2. Install with a maximum of two 90-degree bends or equivalent for each length of pathway unless Drawings show stricter requirements. Separate lengths with pull or junction boxes or terminations at distribution frames or cabinets where necessary to comply with these requirements.

Q. Install pathway-sealing fittings at accessible locations according to NFPA 70 and fill them with listed sealing compound. For concealed pathways, install each fitting in a flush steel box with a blank cover plate having a finish similar to that of adjacent plates or surfaces. Install pathway-sealing fittings according to NFPA 70.

R. Install devices to seal pathway interiors at accessible locations. Locate seals, so no fittings or boxes are between the seal and the following changes of environments. Seal the interior of all pathways at the following points:

1. Where conduits pass from warm to cold locations, such as boundaries of refrigerated spaces.
2. Where an underground service pathway enters a building or structure.
3. Where otherwise required by NFPA 70.

S. Comply with manufacturer's written instructions for solvent welding PVC conduit and fittings.

T. J-Hooks:

1. Size to allow a minimum of 25 percent future capacity without exceeding design capacity limits.
2. Shall be supported by dedicated support wires. Do not use ceiling grid support wire or support rods.
3. Hook spacing shall allow no more than 6 inches of slack. The lowest point of the cables shall be no less than 6 inches adjacent to ceilings, mechanical ductwork and fittings, luminaires, power conduits, power and telecommunications outlets, and other electrical and communications equipment.
4. Space hooks no more than 4 feet o.c.
5. Provide a hook at each change in direction.

U. Mount boxes at heights indicated on Drawings. Install boxes with height measured to center of box unless otherwise indicated.

V. Recessed Boxes in Masonry Walls: Saw-cut opening for box in center of cell of masonry block, and install box flush with surface of wall. Prepare block surface to provide a flat surface for a raintight connection between box and cover plate or supported equipment and box.

W. Horizontally separate boxes mounted on opposite sides of walls, so they are not in the same vertical channel.

X. Fasten junction and pull boxes to or support from building structure. Do not support boxes by conduits.

3.3 INSTALLATION OF UNDERGROUND CONDUIT

A. Refer to specification section 260533 for further details and requirements.

B. Direct-Buried Conduit:

1. Excavate trench bottom to provide firm and uniform support for conduit. Install backfill.
2. After installing conduit, backfill and compact.
3. Install manufactured duct elbows for stub-ups at poles and equipment and at building entrances through floor unless otherwise indicated. Install manufactured rigid steel conduit elbows for stub-ups at poles and equipment and at building entrances through floor.

a. Couple steel conduits to ducts with adapters designed for this purpose, and encase coupling with 3 inches of concrete around conduit for a minimum of 12 inches on each side of the coupling.
b. For stub-ups at equipment mounted on outdoor concrete bases and where conduits penetrate building foundations, extend steel conduit horizontally a minimum of 60 inches from edge of foundation or equipment base. Install insulated grounding bushings on terminations at equipment.

4. Underground Warning Tape: Provide warning tape 12” above trenched conduit.

3.4 INSTALLATION OF UNDERGROUND HANDHOLES AND BOXES

A. Install handholes and boxes level and plumb and with orientation and depth coordinated with connecting conduits to minimize bends and deflections required for proper entrances.

B. Unless otherwise indicated, support units on a level bed of crushed stone or gravel, graded from 1/2-inch sieve to No. 4 sieve and compacted to same density as adjacent undisturbed earth.

C. Elevation: In paved areas, set so cover surface will be flush with finished grade. Set covers of other enclosures 1 inch above finished grade.

D. Field cut openings for conduits according to enclosure manufacturer's written instructions. Cut wall of enclosure with a tool designed for material to be cut. Size holes for terminating fittings to be used, and seal around penetrations after fittings are installed.

3.5 SLEEVE AND SLEEVE-SEAL INSTALLATION FOR COMMUNICATIONS PENETRATIONS

A. Install sleeves and sleeve seals at penetrations of exterior floor and wall assemblies.

3.6 FIRESTOPPING

A. Install firestopping at penetrations of fire-rated floor and wall assemblies.

3.7 PROTECTION

A. Protect coatings, finishes, and cabinets from damage or deterioration.

1. Repair damage to galvanized finishes with zinc-rich paint recommended by manufacturer.
2. Repair damage to PVC coatings or paint finishes with matching touchup coating recommended by manufacturer.

END OF SECTION 27 0528

SECTION 27 1116

COMMUNICATIONS RACKS, FRAMES, AND ENCLOSURES

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:
   1. Fiber panels.

1.2 DEFINITIONS

A. Access Provider: An operator that provides a circuit path or facility between the service provider and user. An access provider can also be a service provider.

B. Service Provider: The operator of a telecommunications transmission service delivered through access provider facilities.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Seismic Performance: Equipment shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

   1. The term "withstand" means "the unit will remain in place without separation of any parts when subjected to the seismic forces specified.

B. UL listed.

C. RoHS compliant.

2.2 WALL MOUNTED FIBER PANELS

A. Description: Wall mounted panel designed for mounting telecommunications equipment.

B. Equipment to be furnished and installed by Diamond Communications.

C. Contractor shall provide 120V power and dedicated space on wall as shown on plans.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Comply with NECA 1.

B. Comply with BICSI TDMM for layout of communications equipment spaces.

C. Comply with BICSI ITSIMM for installation of communications equipment spaces.

D. Coordinate layout and installation of communications infrastructure with Diamond Communications prior to rough-in.

E. Coordinate location of power raceways and receptacles with locations of communications equipment requiring electrical power to operate.

END OF SECTION 27 1116

SECTION 284621 - ADDRESSABLE FIRE ALARM SYSTEM

PART 1 - GENERAL

1.1 SUMMARY

A. System Description:
   1. New fully automated addressable fire alarm system.

B. Related Requirements:

   1. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.
   2. In general, fire alarm system shall be installed in accordance with Section 260010 - General Electrical Requirements.

1.2 DEFINITIONS

A. NICET: National Institute for Certification in Engineering Technologies.

B. NRTL: Nationally Recognized Testing Laboratory.

1.3 REFERENCES AND STANDARDS

A. The fire alarm system shall be designed, manufactured and installed in accordance with the following standards:

   1. NFPA 70, National Electric Code – NEC
   2. NFPA 72, National Fire Alarm Code
   3. International Building Code – IBC
   4. International Fire Code – IFC
   5. FM – Factory Mutual.
   6. UL – Underwriters Laboratories – All applicable standards
      a. Note that all devices shall be cross-listed for use with control unit
   7. ADA – Americans with Disabilities Act
   8. ABA – Architectural Barriers Act

1.4 REGULATORY AGENCIES

A. All work shall be done in accordance with requirements of the following regulatory agencies:

   1. Local Building Department.
   2. Local Fire Department.
   3. State Fire Marshall.
   4. Insurance Services Office or Insuring Authority having jurisdiction.
   5. Owner.

1.5 ACTION SUBMITTALS

A. General Submittal Requirements:

1. Submittals shall be approved by authority having jurisdiction prior to submitting them to Architect.
2. Shop Drawings shall be prepared by persons with the following qualifications:
   a. Trained and certified by manufacturer in fire alarm system design.
   b. NICET-certified, fire alarm technician; Level III minimum.
   c. Licensed or certified by authorities having jurisdiction.

B. Product Data: For each type of product, including furnished options and accessories.

C. Shop Drawings for fire alarm system.

1. Comply with recommendations and requirements in the "Documentation" section of the "Fundamentals" chapter in NFPA 72.
2. Include plans, elevations, sections, details, and attachments to other work.
3. Include details of equipment assemblies. Indicate dimensions, weights, loads, required clearances, method of field assembly, components, and locations. Indicate conductor sizes, indicate termination locations and requirements, and distinguish between factory and field wiring.
4. Detail assembly and support requirements.
5. Include voltage drop calculations for notification-appliance circuits.
6. Include battery-size calculations.
7. Include input/output matrix.
8. Include statement from manufacturer that all equipment and components have been tested as a system and meet all requirements in this Specification and in NFPA 72.
9. Include performance parameters and installation details for each detector, verifying that each duct detector is listed for complete range of air velocity, temperature, and humidity possible when air-handling system is operating.
10. Include plans, sections, and elevations of heating, ventilating, and air-conditioning ducts, drawn to scale; coordinate location of duct smoke detectors and access to them.
    a. Show critical dimensions that relate to placement and support of sampling tubes, detector housing, and remote status and alarm indicators.
    b. Show field wiring required for HVAC unit shutdown on alarm.
    c. Locate detectors according to manufacturer's written recommendations.
11. Include floor plans to indicate final outlet locations showing address of each addressable device. Show size and route of cable and conduits and point-to-point wiring diagrams.
12. Include elevation view of equipment rack or console, grounding schematic and single-line connection diagram.

D. Delegated-Design Submittal: Basic device placement is shown on the electrical plans for system compliance. Supplier shall provide additional devices (including auxiliary power supplies with associated smoke detection, etc.) as required to provide a complete and functional system. Since connections between devices, circuit sizing, voltage drop and device performance are dependent on the system manufacturer’s specific equipment and circuiting, the contractor shall provide system layout drawings that include interconnections and calculations for the specific equipment. These shall be prepared by a Level III, or greater, NICET-certified designer. Drawings shall include:

1. Device Locations:
    a. Location of each initiation device with ratings and installation details as needed to comply with listing conditions the devices and to give proper coverage of the area.

2. Design Calculations:
    a. Voltage drop and loading calculations for each circuit.
    b. Calculate requirements for selecting the spacing and sensitivity of detection devices to comply with NFPA 72.
    c. Calculate spacing and intensities for strobe signals and sound-pressure levels for audible appliances.

1.6 INFORMATIONAL SUBMITTALS

A. Qualification Data: For qualified Installer.

B. Seismic Qualification Certificates: For fire alarm control unit, accessories, and components, from manufacturer.

1. Basis for Certification: Indicate whether withstand certification is based on actual test of assembled components or on calculation.
2. Dimensioned Outline Drawings of Equipment Unit: Identify center of gravity and locate and describe mounting and anchorage provisions.
3. Detailed description of equipment anchorage devices on which the certification is based and their installation requirements.

C. Field quality-control reports.

D. Sample warranty.

1.7 CLOSEOUT SUBMITTALS

A. Operation and Maintenance Data: For fire alarm systems and components to include in emergency, operation, and maintenance manuals. Include the following and deliver copies to authorities having jurisdiction:

1. Comply with the "Records" section of the "Inspection, Testing and Maintenance" chapter in NFPA 72.
2. Provide "Fire Alarm and Emergency Communications System Record of Completion Documents" according to the "Completion Documents" article in the "Documentation" section of the "Fundamentals" chapter in NFPA 72.
3. Complete wiring diagrams showing connections between all devices and equipment.
4. Riser diagram.
5. Record copy of site-specific software.
6. Provide "Inspection and Testing Form" according to the "Inspection, Testing and Maintenance" chapter in NFPA 72, and include the following:
    a. Equipment tested.
    b. Frequency of testing of installed components.
    c. Frequency of inspection of installed components.
    d. Requirements and recommendations related to results of maintenance.

e. Manufacturer's user training manuals.

B. Software and Firmware Operational Documentation:

1. Software operating and upgrade manuals.
2. Program Software Backup: On magnetic media or compact disk, complete with data files.
3. Device address list.
4. Printout of software application and graphic screens.

C. As-built drawings that include:

1. Plan views showing the location of all equipment, outlets and cabling routes.
2. Notation for each outlet and cable to signify the labeling scheme.

1.8 MAINTENANCE MATERIAL SUBMITTALS

A. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.

1. Lamps for Remote Indicating Lamp Units: Quantity equal to 10 percent of amount installed, but no fewer than 1 unit.
2. Lamps for Strobe Units: Quantity equal to 10 percent of amount installed, but no fewer than 1 unit.
3. Smoke Detectors, Fire Detectors : Quantity equal to 10 percent of amount of each type installed, but no fewer than 1 unit of each type.
4. Detector Bases: Quantity equal to 2 percent of amount of each type installed, but no fewer than 1 unit of each type.
5. Keys and Tools: One extra set for access to locked and tamper-proofed components.
6. Audible and Visual Notification Appliances: One of each type installed.
7. Fuses: Two of each type installed in the system.

1.9 QUALITY ASSURANCE

A. Installer Qualifications:

1. Personnel shall be trained and certified by manufacturer for installation of units required for this Project.
2. Installer Qualifications: Installation shall be by personnel certified by NICET as minimum fire alarm Level II technician.

B. Source Limitations for Fire alarm System and Components: Obtain fire alarm system from single source from single manufacturer. Components shall be compatible with, and operate as, an extension of existing system.

C. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

D. NFPA Certification: Obtain certification according to NFPA 72 by a UL-listed alarm company.

1.10 RECORD DRAWINGS

A. At the completion of work, the Contractor shall turn over one CAD file (DXF) of fire alarm system as-built drawings to the Engineer for review. After approval, the Contractor shall reproduce the drawings and submit to the Owner.

B. The drawings shall include:

1. Plan views showing the location of all equipment, outlets and cabling routes.
2. Notation for each outlet and cable to signify the labeling scheme.

1.11 OPERATION AND MAINTENANCE MANUAL: Provide all operation and maintenance information for the system. Additional information shall be provided as follows:

A. Electrical schematics showing all devices.

B. As-built drawings showing device locations, conduits, wires, etc.

C. Operational sequences including A narrative describing elevator recall and shut-down sequences.

D. Final test certificates showing compliance with NFPA 72 testing procedures.

E. Final approval of the authority having jurisdiction.

F. A certificate of equipment manufacturer stating that the system has been installed in accordance with manufacturer's guidelines and applicable codes and regulations.

1.12 WARRANTY

A. Standard Warranty: Fire alarm contractor shall supply complete parts and labor warranty (including travel expenses) for one (1) year from date of substantial completion.

1. Warranty shall cover entire fire alarm system and all associated equipment and devices.
2. Warranty shall include necessary repairs or loaner replacement assuring complete restoration of operation within 48 hours from time service call is requested.
3. Warranty shall include job site visit at completion of warranty period to inspect, clean, adjust, repair and re-certify entire system.
4. Site visit shall also include basic orientation and operator training review with owner's designated personnel.

1.13 SERVICE CONTRACT

A. Fire alarm contractor shall offer Owner and annual service contract at end of warranty period. Acceptance or rejection of service contract will be Owner's option.

1.14 SOFTWARE SERVICE AGREEMENT

A. Comply with UL 864.

B. Technical Support: Beginning with Substantial Completion, provide software support for two years.

C. Upgrade Service: Update software to latest version at Project completion. Install and program software upgrades that become available within two years from date of Substantial Completion. Upgrading software shall include operating system. Upgrade shall include new or revised licenses for use of software.

1. Provide 30 days' notice to Owner to allow scheduling and access to system and to allow Owner to upgrade computer equipment if necessary.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Manufacturers: Subject to compliance with requirements, available manufacturers offering products that may be incorporated into the Work include, but are not limited to, the following:

1. EST Edwards Fire Safety; an Edwards Signal company.
2. Notifier; a Honeywell company.
3. SimplexGrinnell LP; a Tyco International company.
4. Faraday; a Honeywell company.
5. Fire Lite Alarms; a Honeywell company.
6. FireNET; a Hochiki company.
7. Gamewell; a Honeywell company.
8. GE Vigilant; a General Electric company.
9. National Time and Signal
10. Potter; Potter Electric Signal Company.
11. Silent Knight; a Honeywell company.

2.2 SYSTEM DESCRIPTION

A. Noncoded, UL-certified addressable system, with multiplexed signal transmission and horn/strobe evacuation.

B. Automatic sensitivity control of certain smoke detectors.

C. All components provided shall be listed for use with the selected system.

D. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, by a qualified testing agency, and marked for intended location and application.

2.3 SYSTEM OPERATION

A. Alarm Signal

1. Initiation shall be by one or more of the following devices or systems:
   a. Manual pull stations.

b. Smoke detectors (except where installed in separate units of multi-unit dwelling facilities).
c. Automatic sprinkler system water flow.

2. The following actions shall occur when an alarm is initiated:
a. Continuously operate alarm notification appliances including the exterior fire alarm notification appliance.
b. Identify alarm and specific initiating device at fire alarm control unit at and remote annunciators.
c. Record events in the system memory.
d. Transmit an alarm signal to the remote alarm receiving station.

B. System Trouble Signal

1. System trouble signal initiation shall be by one or more of the following devices and actions:
a. Open circuits, shorts, and grounds in designated circuits.
b. Opening, tampering with, or removing alarm-initiating and supervisory signal-initiating devices.
c. Loss of communication with any addressable sensor, input module, relay, control module, or remote annunciator.
d. Loss of primary power at fire alarm control unit.
e. Ground or a single break in internal circuits of fire alarm control unit.
f. Abnormal ac voltage at fire alarm control unit.
g. Break in standby battery circuitry.
h. Failure of battery charging.
i. Abnormal position of any switch at fire alarm control unit or annunciator.

2. The following shall occur when a trouble signal is initiated:
a. Initiate beeping notification at the FACP.
b. Identify specific device/condition initiating the event at fire alarm control unit and remote annunciators.
c. After a maximum time-delay of 200 seconds, transmit a trouble signal to the remote alarm receiving station.

C. System Supervisory Signal

1. Initiation shall be by one or more of the following devices or actions:
a. Loss of communication with any panel on the network.
b. Valve supervisory switch.
c. Automatic sprinkler system Tamper Switch.

2. The following shall occur when a trouble signal is initiated:
a. Initiate beeping notification at the FACP.
b. Identify specific device/condition initiating the event at fire alarm control unit and remote annunciators.
c. After a maximum time-delay of 200 seconds, transmit a supervisory signal to the remote alarm receiving station.

2.4 PERFORMANCE REQUIREMENTS

A. Seismic Performance: Fire alarm control unit and raceways shall withstand the effects of earthquake motions determined according to ASCE/SEI 7.

1. The term "withstand" means "the unit will remain in place without separation of any parts from the device when subjected to the seismic forces specified and the unit will be fully operational after the seismic event."

2.5 FIRE ALARM CONTROL PANEL (FACP)

A. General Requirements for Fire alarm Control Unit:

1. Field-programmable, microprocessor-based, modular, power-limited design with electronic modules, complying with UL 864 and listed and labeled by an NRTL.
   a. System software and programs shall be held in flash electrically erasable programmable read-only memory (EEPROM), retaining the information through failure of primary and secondary power supplies.
   b. Include a real-time clock for time annotation of events on the event recorder.

2. Addressable Initiation Device Circuits: The FACP shall indicate which communication zones have been silenced and shall provide selective silencing of alarm notification appliance by building communication zone.
   a. Smoke sensors shall additionally communicate sensitivity setting and allow for adjustment of sensitivity at fire alarm control unit.
   b. Temperature sensors shall additionally test for and communicate the sensitivity range of the device.

3. Addressable Control Circuits for Operation of Notification Appliances and Mechanical Equipment: The FACP shall be listed for releasing service.

B. Alphanumeric Display and System Controls: Arranged for interface between human operator at fire alarm control unit and addressable system components including annunciation and supervision. Display alarm, supervisory, and component status messages and the programming and control menu.

1. Annunciator and Display: Liquid-crystal type.
2. Keypad: Arranged to permit entry and execution of programming, display, and control commands.

C. Initiating-Device, Notification-Appliance, and Signaling-Line Circuits:

1. Pathway Class Designations: NFPA 72, Class B.
2. Pathway Survivability: Level 1. – There are various Survivability Levels and these are defined in NFPA 72, Section 12.4.

D. Notification-Appliance Circuit:

1. Audible appliances shall sound in a three-pulse temporal pattern, as defined in NFPA 72.

2. Where notification appliances provide signals to sleeping areas, the alarm signal shall be a 520-Hz square wave with an intensity 15 dB above the average ambient sound level or 5 dB above the maximum sound level, or at least 75 dBA, whichever is greater, measured at the pillow.
3. Visual alarm appliances shall flash in synchronization where multiple appliances are in the same field of view, as defined in NFPA 72.

E. Smoke-Alarm Verification:

1. Initiate audible and visible indication of an "alarm-verification" signal at fire alarm control unit.
2. Activate an NRTL-listed and -approved "alarm-verification" sequence at fire alarm control unit and detector.
3. Sound general alarm if the alarm is verified.
4. Cancel fire alarm control unit indication and system reset if the alarm is not verified.

F. Remote Smoke-Detector Sensitivity Adjustment: Controls shall select specific addressable smoke detectors for adjustment, display their current status and sensitivity settings, and change those settings. Allow controls to be used to program repetitive, time-scheduled, and automated changes in sensitivity of specific detector groups. Record sensitivity adjustments and sensitivity-adjustment schedule changes in system memory.

G. Transmission to Remote Alarm Receiving Station: Automatically transmit alarm, supervisory, and trouble signals to a remote alarm station.

H. Primary Power: 24-V dc obtained from 120-V ac service and a power-supply module. Initiating devices, notification appliances, signaling lines, trouble signals, supervisory and digital alarm communicator transmitters shall be powered by 24-V dc source.

1. Alarm current draw of entire fire alarm system shall not exceed 80 percent of the power-supply module rating.

I. Secondary Power: 24-V dc supply system with batteries, automatic battery charger, and automatic transfer switch.

1. System shall be provided with sufficient battery capacity to operate entire system upon loss of primary power.
    a. 5 minutes of operation in Alarm Mode, after 24 hours of continuous operation in Supervisory Mode.

2. System shall automatically transfer to standby batteries upon primary power failure.
3. All battery charging and recharging operations shall be automatic and monitored by the control panel.
4. Batteries: Sealed lead calcium.
5. Auxiliary power supplies, meeting battery backup requirements as listed above, shall be provided as required to power all NAC's.

J. Instructions: Computer printout or typewritten instruction card mounted behind a plastic or glass cover in a stainless-steel or aluminum frame. Include interpretation and describe appropriate response for displays and signals. Briefly describe the functional operation of the system under normal, alarm, and trouble conditions.

2.6 REMOTE ANNUNCIATOR

A. Description: Annunciator functions shall match those of fire alarm control unit for alarm, supervisory, and trouble indications. Manual switching functions shall match those of fire alarm control unit, including acknowledging, silencing, resetting, and testing.

1. Mounting: Flush cabinet, NEMA 250, Type 1.

B. Display Type and Functional Performance: Alphanumeric display and LED indicating lights shall match those of fire alarm control unit. Provide controls to acknowledge, silence, reset, and test functions for alarm, supervisory, and trouble signals.

2.7 ADDRESSABLE INTERFACE DEVICE

A. General:

1. Include address-setting means on the module.
2. Store an internal identifying code for control panel use to identify the module type.
3. Listed for controlling HVAC fan motor controllers.

B. Monitor Module: Microelectronic module providing a system address for alarm-initiating devices for wired applications with normally open contacts.

C. Integral Relay: Capable of providing a direct signal to the elevator controller to initiate elevator recall, and to the circuit-breaker shunt trip for power shutdown.

1. Allow the control panel to switch the relay contacts on command.
2. Have a minimum of two normally open and two normally closed contacts available for field wiring.

D. Control Module:

1. Operate notification devices.
2. Operate solenoids for use in sprinkler service.

2.8 DIGITAL ALARM COMMUNICATOR TRANSMITTER (DACT)

A. DACT shall be acceptable to the remote central station, shall comply with UL 632 and UL 864, shall be listed and labeled by an NRTL, and shall be acceptable to the AHJ.

B. Telephone DACT Functional Performance: Unit shall receive an alarm, supervisory, or trouble signal from fire alarm control unit and automatically capture one of two telephone lines and dial a preset number for a remote central station. When contact is made with central station, signals shall be transmitted. If service on either line is interrupted for longer than 45 seconds, transmitter shall initiate a local trouble signal and transmit the signal indicating loss of telephone line to the remote alarm receiving station over the remaining line. Transmitter shall automatically report telephone service restoration to the central station. If service is lost on both telephone lines, transmitter shall initiate the local trouble signal.

C. IP-DACT Functional Performance: Unit shall receive an alarm, supervisory, or trouble signal from fire alarm control unit and automatically transmit on IP Network line. If service on network is interrupted for longer than 90 seconds, transmitter shall initiate a local trouble signal and transmit the signal indicating loss of network. Transmitter shall automatically report IP network service restoration to the central station. If network service is lost for more than 90 seconds, transmitter shall initiate the local trouble signal.

D. Cellular Communicator: Digital cellular radio alarm transmission device, Telguard No. TG-7FS or approved equivalent.

1. System shall include cellular antenna and other hardware to tie into the FACP.
2. Capable of providing the following three methods of communications for alarm panels via a cellular network:
    a. Sole path (Cellular Transmission Only)
    b. Primary path (Cellular Primary & Telco Backup)
    c. Back-up path (Telco Primary & Cellular Backup)
3. Support multiple communication formats and shall be capable to automatically adjust to the FACP communication format.
4. Continuously supervise communication path(s).
5. Supervise and report status of backup battery and AC power source.
6. Automatically perform self-test.
7. EC shall coordinate with owner and cellular service provider to acquire and set up required cellular service for the unit. Owner will contract with the cellular service and will pay applicable setup fees and monthly expenses.

E. Local functions and display at the digital alarm communicator transmitter shall include the following:

1. Verification that both telephone lines are available.
2. Programming device.
3. LED display.
4. Manual test report function and manual transmission clear indication.
5. Communications failure with the central station or fire alarm control unit.

F. Digital data transmission shall include the following:

1. Address of the alarm-initiating device.
2. Address of the supervisory signal.
3. Address of the trouble-initiating device.
4. Loss of ac supply.
5. Loss of power.
6. Low battery.
7. Abnormal test signal.
8. Communication bus failure.

G. Secondary Power: Integral rechargeable battery and automatic charger.

H. Self-Test: Conducted automatically every 24 hours with report transmitted to central station.

2.9 MANUAL FIRE ALARM BOXES

A. General Requirements for Manual Fire alarm Boxes: Comply with UL 38. Boxes shall be finished in red with molded, raised-letter operating instructions in contrasting color; shall show visible indication of operation; and shall be mounted on recessed outlet box. If indicated as surface mounted, provide manufacturer's surface back box.

1. Double-action mechanism requiring two actions to initiate an alarm, pull-lever type; with integral addressable module arranged to communicate manual-station status (normal, alarm, or trouble) to fire alarm control unit.
2. Station Reset: Key- or wrench-operated switch.

2.10 DETECTOR BASES

A. Two-wire operation. Class B wiring.

2.11 SYSTEM SMOKE DETECTORS

A. Smoke detectors shall be compatible (UL cross-listed) with the FACP.

B. General Requirements for System Smoke Detectors:

1. Comply with UL 268; operating at 24-V dc, nominal.
2. Detectors shall be two-wire type.
3. Integral Addressable Module: Arranged to communicate detector status (normal, alarm, or trouble) to fire alarm control unit.
4. Base Mounting: Detector and associated electronic components shall be mounted in a twist-lock module that connects to a fixed base. Provide terminals in the fixed base for connection to building wiring.
5. Self-Restoring: Detectors do not require resetting or readjustment after actuation to restore them to normal operation.
6. Integral Visual-Indicating Light: LED type indicating detector has operated and power-on status.
7. Remote Control: Unless otherwise indicated, detectors shall be analog-addressable type, individually monitored at fire alarm control unit for calibration, sensitivity, and alarm condition and individually adjustable for sensitivity by fire alarm control unit.
    a. Rate-of-rise temperature characteristic shall be selectable at fire alarm control unit for 15 or 20 deg F (8 or 11 deg C) per minute.
    b. Fixed-temperature sensing shall be independent of rate-of-rise sensing and shall be settable at fire alarm control unit to operate at 135 or 155 deg F (57 or 68 deg C).
    c. Provide multiple levels of detection sensitivity for each sensor.
    d. Sensitivity levels can be based on time of day.

C. Sounder Bases:

1. In sleeping rooms and living rooms of dwelling units, dormitories, hotels, etc. smoke detectors shall be provided with low-frequency sounder bases, unless a horn/strobe is supplied in that room.

D. Photoelectric Smoke Detectors:

1. Detector address shall be accessible from fire alarm control unit and shall be able to identify the detector's location within the system and its sensitivity setting.
2. An operator at fire alarm control unit, having the designated access level, shall be able to manually access the following for each detector:
   a. Primary status.
   b. Device type.
   c. Present average value.
   d. Present sensitivity selected.
   e. Sensor range (normal, dirty, etc.).

2.12 CARBON MONOXIDE DETECTORS

A. General: Carbon monoxide detector shall be compatible (UL cross-listed as required) with the FACP.

1. Mounting: Adapter plate for outlet box mounting.
2. Testable by introducing test carbon monoxide into the sensing cell.
3. Detector shall provide alarm contacts and trouble contacts.
4. Detector shall send trouble alarm when nearing end-of-life, power supply problems, or internal faults.
5. Comply with UL 2075.
6. Locate, mount, and wire according to manufacturer's written instructions.
7. Provide means for addressable connection to fire alarm system.
8. Test button simulates an alarm condition.

2.13 HEAT DETECTORS

A. Heat detectors shall be compatible (UL cross-listed as required) with the FACP.

B. General Requirements for Heat Detectors: Comply with UL 521.

1. Temperature sensors shall test for and communicate the sensitivity range of the device.

C. Heat Detector, Fixed-Temperature Type: Required when installed in elevator hoistways. Actuated by temperature that exceeds a fixed temperature. Provide fixed-temperature devices where installed adjacent to fire sprinkler heads, such as those associated with the elevator equipment.

1. Mounting: Twist-lock base interchangeable with smoke-detector bases.
2. Integral Addressable Module: Arranged to communicate detector status (normal, alarm, or trouble) to fire alarm control unit.

2.14 FIRE SPRINKLER SYSTEM SWITCHES

A. Flow, tamper and pressure switches will be provided by fire sprinkler subcontractor. Fire alarm contractor shall coordinate with fire sprinkler contractor to ensure that switches are compatible with (UL cross-listed as required) the FACP. Fire alarm contractor shall connect sprinkler system

switches via addressable monitoring modules. See fire suppression plans for switch locations and quantities.

2.15 NOTIFICATION APPLIANCES

A. All notification appliances shall be compatible (UL cross-listed as required) with the FACP.

B. General Requirements for Notification Appliances: Individually addressed, connected to a signaling line circuit, equipped for mounting as indicated and with screw terminals for system connections.

1. Combination Devices: Factory-integrated audible and visible devices in a single-mounting assembly, equipped for mounting as indicated and with screw terminals for system connections.

C. Horns: Electric-vibrating-polarized type, 24-V dc; with provision for housing the operating mechanism behind a grille. Comply with UL 464. Horns shall produce a sound-pressure level of 90 dBA, measured 10 feet (3 m) from the horn, using the coded signal prescribed in UL 464 test protocol.

D. Low Frequency Sounder for Sleeping Rooms: 520Hz tone, 24-V dc; with provision for housing the operating mechanism behind a grille. Comply with UL 464. Horns shall produce a sound-pressure level of 90 dBA, measured 10 feet (3 m) from the horn, using the coded signal prescribed in UL 464 test protocol.

E. Visible Notification Appliances: Xenon strobe lights comply with UL 1971, with clear or nominal white polycarbonate lens mounted on an aluminum faceplate. The word “FIRE” is engraved in minimum 1-inch- (25-mm-) high letters on the lens.

1. Rated Light Output: 15/30/75/110 cd, selectable in the field.
2. Mounting: Wall mounted unless otherwise indicated.
3. For units with guards to prevent physical damage, light output ratings shall be determined with guards in place.
4. Flashing shall be in a temporal pattern, synchronized with other units.
5. Strobe Leads: Factory connected to screw terminals.
6. Mounting Faceplate: Factory finished, **red**.

2.16 CABLES AND RACEWAYS

A. Minimum raceway size: 3/4”.

B. Exposed conduit in non-public areas shall be painted with red enamel. Exposed conduit in public areas shall be painted with color as directed by architect.

C. Junction boxes shall have covers painted red with label to identify it as fire alarm.

D. Conductors for all initiating and signal circuits shall be solid copper, with minimum gauge of #18. All conductors shall terminate under screw terminal.

E. Conductors for all notification appliance circuits (NACs) shall be sized for distance and current load, in strict accordance with manufacturer's recommendations. Minimum wire size shall be #14 AWG. Whenever practical, NAC wiring should not be larger than #12 AWG wire. Where larger size is required to serve load, additional circuit module shall be provided to split load current.

F. To minimize voltage drop, conductors serving activation devices shall be #12 AWG minimum. Conductors for external control circuits and annunciator panels should be sized according to applicable code requirements and manufacturer's recommendations.

G. Fire alarm raceways shall not contain any conductors not part of fire alarm system.

PART 3 - EXECUTION

3.1 EQUIPMENT INSTALLATION

A. Comply with NFPA 72, NFPA 101, and requirements of authorities having jurisdiction for installation and testing of fire alarm equipment. Install all electrical wiring to comply with requirements in NFPA 70 including, but not limited to, Article 760, "Fire Alarm Systems."

B. Installation of all smoke detectors shall be in strict accordance with NFPA 72, paragraph 17.7.1.11. - Detectors shall be covered, or not be installed until after construction clean-up of all trades is complete and final.

C. Install wall-mounted equipment, with tops of cabinets not more than 72 inches above the finished floor.

1. Comply with requirements for seismic-restraint devices specified in Section 260548.16 "Seismic Controls for Electrical Systems."

D. Manual Fire alarm Boxes:

1. Install manual fire alarm box in the normal path of egress within 60 inches of the exit doorway.
2. Mount manual fire alarm box on a background of a contrasting color.
3. The operable part of manual fire alarm box shall be between 42 inches and 48 inches above floor level. All devices shall be mounted at the same height unless otherwise indicated.

E. Whether shown on plans or not, a smoke detector shall be installed at FACP and at any/all NAC power extenders.

F. Smoke- or Heat-Detector Spacing:

1. Comply with NFPA 72, "Smoke-Sensing Fire Detectors" Section in the "Initiating Devices" Chapter, for smoke-detector spacing.
2. Comply with NFPA 72, "Heat-Sensing Fire Detectors" Section in the "Initiating Devices" Chapter, for heat-detector spacing.
3. Smooth ceiling spacing shall not exceed 30 feet.
4. Spacing of detectors for irregular areas, for irregular ceiling construction, and for high ceiling areas shall be determined according to Appendix A in NFPA 72.

5. HVAC: Locate detectors not closer than 5 feet from air-supply diffuser or return-air opening.
6. Lighting Fixtures: Locate detectors not closer than 12 inches from any part of a lighting fixture.

G. Single-Station Smoke Detectors: Where more than one smoke alarm is installed within a dwelling or suite, they shall be connected so that the operation of any smoke alarm causes the alarm in all smoke alarms to sound.

H. Remote Status and Alarm Indicators: Install near each smoke detector and each sprinkler water-flow switch and valve-tamper switch that is not readily visible from normal viewing position.

I. Audible Alarm-Indicating Devices: Install not less than 6 inches below the ceiling, unless noted otherwise on the plans. Install bells and horns on flush-mounted back boxes with the device-operating mechanism concealed behind a grille.

1. Provide low frequency sounder where audible notification is intended for sleeping areas.

J. Visible Alarm-Indicating Devices: Install adjacent to each alarm bell or alarm horn and at least 6 inches below the ceiling, unless noted otherwise on the plans. Install all devices at the same height unless otherwise indicated.

K. Device Location-Indicating Lights: Locate in public space near the device they monitor.

L. Fire alarm Control Unit: Surface mounted, with tops of cabinets not more than 72 inches above the finished floor.

M. Remote Annunciator: Install with top of panel not more than 72 inches above the finished floor.

3.2 CONDUCTOR INSTALLATION

A. All cables/conductors associated with the fire alarm system shall be installed in conduit.

3.3 CONNECTIONS

A. For fire-protection systems related to doors in fire-rated walls and partitions and to doors in smoke partitions, comply with requirements in Section 087100 "Door Hardware." Connect hardware and devices to fire alarm system.

1. Verify that hardware and devices are NRTL listed for use with fire alarm system in this Section before making connections.

B. Make addressable connections with a supervised interface device to the following devices and systems. Install the interface device less than 36 inches from the device controlled. Make an addressable confirmation connection when such feedback is available at the device or system being controlled.
1. Supervisory connections at valve supervisory switches.

C. Coordinate with IT contractor to ensure that the following are connected to the Main Ethernet Switch that is backed up by a Line Interactive UPS unit capable of maintaining a minimum 39 minutes of backup power for the 48-port switch (1100W load).

1. Fire Alarm IP-DACT.

3.4 IDENTIFICATION

A. Identify system components, wiring, cabling, and terminals. Comply with requirements for identification specified in Section 260553 "Identification for Electrical Systems."

B. Install framed instructions in a location visible from fire alarm control unit.

C. Test switches shall have plastic laminated labels with 1/2" letters describing the switch function.

3.5 GROUNDING

A. Ground fire alarm control unit and associated circuits; comply with IEEE 1100. Install a ground wire from main service ground to fire alarm control unit.

B. Ground shielded cables at the control panel location only. Insulate shield at device location.

3.6 DACT PROGRAMMING:

A. Program DACT for alarm transmission central station receiver. Provide all programming & integration of the DACT, receiver, and the alarm server necessary for full addressable ID including point description reporting transmission and annunciation for each individual alarm point.

3.7 FIELD QUALITY CONTROL

A. Field tests shall be witnessed by authorities having jurisdiction.

B. Manufacturer's Field Service: Engage a factory-authorized service representative to inspect, test, and adjust components, assemblies, and equipment installations, including connections.

C. Perform tests and inspections with the assistance of a factory-authorized service representative:

1. Visual Inspection: Conduct visual inspection prior to testing.
   a. Inspection shall be based on completed Record Drawings and system documentation that is required by NFPA 72 in its "Completion Documents, Preparation" Table in the "Documentation" Section of the "Fundamentals of Fire Alarm Systems" Chapter.
   b. Comply with "Visual Inspection Frequencies" Table in the "Inspection" Section of the "Inspection, Testing and Maintenance" Chapter in NFPA 72; retain the "Initial/Reacceptance" column and list only the installed components.

2. System Testing: Comply with "Test Methods" Table in the "Testing" Section of the "Inspection, Testing and Maintenance" Chapter in NFPA 72.

3. Test audible appliances for the public operating mode according to manufacturer's written instructions. Perform the test using a portable sound-level meter complying with Type 2 requirements in ANSI S1.4.
    a. The Contractor shall test dB levels in all rooms throughout facility. Assuming that testing will occur before or after business hours, to avoid business interruption, allowances shall be made for expected differential between business operation noise and ambient conditions when building is not in use.
    b. The sound level differential shall be at least 15 dBA, above average ambient, in rooms that also have visual annunciation.
    c. The sound level differential shall be at least 20 dBA, above average ambient, in rooms (such as offices) where visual annunciation is not present. For readings taken in individual offices with no audible or visual device, meter should be pointed toward office's closed entry door.
    d. Sound levels shall not be more than 100 dBA in any location.
    e. The Contractor shall adjust horn volume levels as required to "balance" the sound and ensure that it is at proper dBA levels throughout facility.
4. Test visible appliances for the public operating mode according to manufacturer's written instructions.
5. Demonstrate proper function of fire door release/closure based on fire alarm status.
6. Demonstrate proper signaling via DACT whenever system goes into trouble or alarm condition.
7. Factory-authorized service representative shall prepare the "Fire Alarm System Record of Completion" in the "Documentation" Section of the "Fundamentals of Fire Alarm Systems" Chapter in NFPA 72 and the "Inspection and Testing Form" in the "Records" Section of the "Inspection, Testing and Maintenance" Chapter in NFPA 72.
8. A 24-hour battery test will be required prior to final testing in accordance with local Authority Having Jurisdiction.

D. Reacceptance Testing: Perform reacceptance testing to verify the proper operation of added or replaced devices and appliances.

E. Fire alarm system will be considered defective if it does not pass tests and inspections.

F. Prepare test and inspection reports.

G. Annual Test and Inspection: One year after date of Substantial Completion, test fire alarm system complying with visual and testing inspection requirements in NFPA 72. Use forms developed for initial tests and inspections.

3.8 DEMONSTRATION

A. Train Owner's maintenance personnel to adjust, operate, and maintain fire alarm system. Training shall continue until Owner is satisfied that he understands the operation of his system. Training shall be accomplished using the operation and maintenance manual and shall be video recorded and distributed to the owner upon completion in DVD format.

END OF SECTION 284621

SECTION 31 05 00 – GEOTEXTILES FOR EARTHWORK

PART 1 - GENERAL

1.1 SECTION INCLUDES

A. Reinforcement mats (woven and nonwoven).

B. Furnishing and placing geotextile fabric to the lines and grades shown on the Drawings as a subsurface drainage fabric permeable separator between dissimilar materials (such as between native subgrade and sub-base/base materials), stabilization fabric, temporary and/or permanent erosion control measures fabric, or as a waterproofing/stress releasing membrane fabric within pavement structures.

1.2 RELATED REQUIREMENTS

A. Section 01 31 00 – Project Management and Coordination
B. Section 31 23 10 – Pipe Trench Excavation and Backfill
C. Section 31 23 20 – Site Excavation, Grading, and Backfill
D. Section 31 30 20 – Site Earthwork
E. Section 32 20 00 – Site Improvements

1.3 SUBMITTALS

A. The Contractor shall provide product data and manufacturer information including tensile strength, elongation, thickness, UV resistance and other material specifications for each geotextile fabric used on the Project.

B. Provide manufacture's certificate certifying that products meet or exceed specified requirements. Along with the statement that the material meets or exceeds the specifications, certificate must include name of the manufacturer, the chemical composition of the filaments or yarns, and other information fully describing the geotextile.

C. Submit detailed instructions on installation requirements from the manufacturer, including storage and handling procedures.

1.4 QUALITY ASSURANCE

A. Assure that fibers used in the manufacture of geotextiles, and the threads used in joining geotextiles by sewing, consist of long-chain synthetic polymers, composed of at least 95 percent by weight polyester, polypropylene, and/or polyethylene. They must be formed into a network so the filaments on yarns retain dimensional stability relative to each other, including selvedges.

B. Furnish materials meeting the physical requirements for the indicated application as described in the corresponding table(s) of properties in AASHTO M288-17, Geotextile Specifications for Highway Applications.

PART 2 PRODUCTS

2.1 WOVEN AND NONWOVEN GEOTEXTILE FABRICS

A. Woven and nonwoven geotextile fabrics shall be non-biodegradable, UV resistant fabric made of polyester, polypropylene, or polyethylene. Edges shall be selvedged or finished to prevent separation of outer material. Calendar such that yarns will retain relative positions.

B. Geotextile fabrics shall comply with the requirements of AASHTO M288-17 as summarized below:

1. Separation geotextile fabric shall be AASHTO M288, Separation, Class 1, and have the following properties:
   a. 6 oz per square yard
   b. 315 lbs grab tensile strength
   c. 15% elongation at break
   d. 900 lbs CBR Puncture
   e. 4 gal/min/square foot water flow rate.
   f. Products:
      1) Basis of Design: U.S. Fabrics Inc. US 315. Product characteristics as noted above.

2.2 ACCESSORIES

A. Sewing thread shall be of polypropylene, polyester, or Kevlar material and shall be of equal or greater durability than the associated geotextile.

B. Securing pins shall be steel rods or bars, minimum 3/16-inch diameter, and minimum 12-inch length. Provide washers made of steel, minimum 1-1/2-inch outside diameter and 1/8-inch thick.

C. Wire staples shall be steel, minimum size 10 gauge and minimum length of 6 inches.

PART 3 EXECUTION

3.1 SHIPMENT AND STORAGE

A. Geosynthetics shall be uniformly rolled onto a cardboard core and shall be wrapped in plastic to protect the material from moisture and damage during shipment. Protective wrapping shall be left on the material until installation. The product must not be allowed to get wet prior to installation to prevent weakening of the cardboard core. Rolls shall be externally tagged for easy field identification. External tagging shall include the following: 1) name of manufacturer; 2) product type; 3) product grade; 4) lot number; and, 5) physical dimensions.

3.2 GENERAL INSTALLATION REQUIREMENTS

A. Prepare subgrade prior to placing geotextile. Remove all sharp objects and large rocks more than 6 inches in size within 6 inches of geotextile. Fill depressions or holes with a suitable material to provide a firm foundation with no sharp materials.

B. Lay and maintain geotextile fabric smooth and free of tensile stresses, folds, wrinkles, or creases. Ensure that material is in direct contact with subgrade. Orientate with long dimension of each sheet transverse to direction of slope, and/or as specified hereinafter for specific applications. Provide a minimum un-seamed joint overlap of 12 to 24 inches at the ends and sides of adjoining sheets as noted hereinafter for specific applications or sew the geotextile joints according to the manufacturer's recommendations. For seams across slopes, lay upper panel over lower panel.

C. Replace or repair all geotextile that is torn, punctured, or muddy. Remove the damaged area and place a patch of the same type of geotextile overlapping 3 feet in all directions beyond the damaged area.

D. Install securement pins or staples as recommended by the manufacturer, generally between edges of overlaps and a minimum of 6 inches from free edges. Provide minimum spacing as recommended by manufacturer for the site conditions.

E. Place fill in a manner to prevent tensile stress or wrinkles in the geotextile. Place fill from bottom of side slopes upward. Do not drop fill from a height greater than 4 feet, or less as noted hereinafter, on top of geotextile. Do not operate equipment directly on top of geotextile.

F. Do not leave geotextile material exposed or uncovered for more than 14 days after installation.

G. Install materials and systems in accordance with manufacturer's instructions and approved submittals. Install materials and systems in proper relation with adjacent construction. Coordinate with work of other sections.

3.3 DRAINAGE, SEPARATION, AND STABILIZATION APPLICATIONS

A. Shape the subgrade to a smooth surface and to the cross section required. Shape slopes to gradually transition into slope adjustments without noticeable breaks. At the ends of cuts, the intersection of cuts, and embankments, adjust slopes in the horizontal and vertical planes to blend into each other or into the natural ground.

B. Remove unsuitable material defined hereinbefore from the roadbed or trench and replace with suitable material. Finish the roadbed, ditches, and/or trenches to the required elevation and cross section.

C. Place geotextile smooth and free of tension, stress, or wrinkles. Fold and cut the geotextile to conform to curves. Generally, overlap in the direction of construction. Overlap the geotextile a minimum of 24 inches or sew as noted hereinbefore. Do not place longitudinal overlaps below anticipated wheel loads. Hold the geotextile in place with pins, staples, or piles of cover material.

D. End or side dump cover material onto the geotextile from the edge of the geotextile or from previously placed cover material. Do not operate equipment directly on the geotextile. Spread the end or side dumped pile of cover material maintaining a minimum loose lift thickness of 9 inches and a maximum loose lift thickness of 12 inches prior to compaction. Compact the cover material with rubber tired or non-vibratory smooth drum rollers. Avoid sudden stops, starts, or turns of the construction equipment. Fill all ruts from construction equipment with additional cover material. Do not re-grade ruts with placement equipment.

E. Place and anchor the geotextile on an approved smooth graded surface. For slope and dike protection applications, place the long dimension of the geotextile down the slope. For stream bank, ditch, and other similar protection applications, place the long dimension of the geotextile parallel to the centerline of the channel.

3.4 TEMPORARY AND PERMANENT EROSION CONTROL APPLICATIONS

A. Place and anchor the geotextile on an approved smooth graded surface. For slope and dike protection applications, place the long dimension of the geotextile down the slope. For stream bank, ditch, and other similar protection applications, place the long dimension of the geotextile parallel to the centerline of the channel.

B. Overlap the geotextile a minimum of 12 inches at the ends and sides of adjoining sheets or sew as noted hereinbefore. Overlap the uphill or upstream sheet over the downhill or downstream sheet. Offset end joints of adjacent sheets a minimum of 5 feet. Pins may be used to hold the geotextile sheets in place. Space pins along the overlaps at approximately 3-foot centers.

C. Place aggregate, slope protection, or riprap on the geotextile starting at the toe of the slope and proceed upward. Place riprap onto the geotextile from a height of less than 12 inches. Place slope protection rock or aggregate backfill onto the geotextile from a height of less than 3 feet. In underwater applications, place the geotextile and cover material in the same day.

END OF SECTION

SECTION 31 10 00 - SITE CLEARING

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Removal (clearing and grubbing) of existing vegetation, and topsoil removal and salvage.

B. Removal of trees, stumps and roots, abandoned structures, fences, and other objectionable materials that will interfere with the construction of the project.

C. Removal of natural deleterious materials encountered during vegetation and topsoil removal.

D. Also included is the stripping and stockpiling of existing topsoil onsite.

E. Work also consists of removal of existing gravel roadways.

F. Removed items shall be disposed of out of the park and as required by code.

G. Submittals:

a. Clearing and grubbing plan, including topsoil storage.

1.2 RELATED REQUIREMENTS

A. Section 02 21 19 – Utilities Surveys
B. Section 02 41 00 - Site Demolition
C. Section 31 23 20 – Street and Site Excavation, Grading and Backfill
D. Section 31 25 00 – Erosion and Sediment Control.
E. Section 31 30 00 - Site Earthwork.

1.3 COORDINATION

A. Before starting the project, the Contractor shall notify and coordinate with the Contracting Officer regarding construction traffic and activities, and vehicular and pedestrian traffic controls on or adjacent to public streets.

B. Traffic:
1. Conduct site clearing operations to ensure minimum interference with roads, streets, walks, and other adjacent occupied or used facilities.
2. Do not close or obstruct streets, walks, or other occupied or used facilities without permission from authorities having jurisdiction.
3. Provide temporary alternate vehicular and pedestrian traffic routes as required to complete the work.

PART 2 PRODUCTS

2.1 TEMPORARY FACILITIES

A. The Contractor shall provide all temporary facilities as required for performing the work.

PART 3 DESIGN CRITERIA

3.1 BASIC FUNCTION

A. Clear all trees, brush, and strip topsoil in all the work areas. Including tree stump removal.

B. Conserve onsite existing topsoil.

C. Remove and dispose of existing asphalt surfacing to install proposed facilities.

D. Stripping of existing gravel roadway and parking to install proposed facilities.

E. Abandon and remove existing septic tanks and identified propane tanks.

3.2 PROTECTION

A. Protect existing utilities, structures and other facilities to remain, including supporting soils, from damage at all times during performance of the work.

B. Protect improvements on adjoining properties as well as those on Owner's property.

C. Provide barricades, coverings or other types of protection necessary to prevent damage to existing improvements indicated to remain in place.

D. Protect trees, shrubs and other vegetation beyond the grading limits as shown on the drawings. Where trees are indicated to remain, refrain from disturbing existing ground under the drip line, and install fencing to protect these areas as directed by the Contracting Officer. If stripping, trench excavation, fill or other disturbance within the drip line is required to perform the Work, mark the ground with proposed disturbance area limits, and obtain Contracting Officer approval before excavating or otherwise disturbing existing ground below the drip line.

E. Cut and remove tree branches only where, in the opinion of the Contracting Officer, such cutting is necessary to perform the specified work.

F. Restore improvements damaged by Work to their original condition as acceptable to Contracting Officer or other parties or authorities having jurisdiction.

G. Salvageable Improvements: Carefully remove items indicated to be salvaged, and store on Owner's premises as directed by the Contracting Officer.

H. Monuments and Markers: Preserve and protect survey monuments, benchmarks and markers from damage or displacement throughout construction. If damage occurs or removal becomes necessary, immediately notify Contracting Officer and restore monument or marker to original condition.

3.3 CLEARING AND GRUBBING

A. Definitions:

1. Clearing: Clearing shall consist of cutting, removing, and disposing of all trees, snags, stumps, shrubs, brush, limbs, grass mats, and other vegetative growth with the exception of those trees and shrubs designated to remain, and stockpiling of topsoil for re-use in proposed landscaped areas. Clearing shall be performed in such a manner as to remove

all evidence of their presence from the surface and shall be inclusive of sticks and branches greater than 2 inches in diameter or thickness.

2. Grubbing: Grubbing shall consist of removal and disposal of the following materials to a minimum of 12 inches below the ground surface remaining after clearing: previously buried concrete rubble, wood, tree trunks, root matter and other miscellaneous large objects encountered.

B. Clearing and grubbing shall not extend beyond the grading limits as shown on the Drawings without prior approval by the Contracting Officer.

3.4 TOPSOIL SALVAGE

A. Topsoil is defined as friable clay loam surface soil found in depth of not less than 4 inches. Satisfactory topsoil is reasonably free of subsoil, clay, lumps, stones and other objects over 2 inches in diameter and without weeds, roots, grass mats or other objectionable material.

B. Strip and salvage topsoil to depth encountered. Perform stripping to prevent intermingling with underlying subsoil or other objectionable materials.

C. Stockpile topsoil onsite in designated areas until final placement in landscape areas is complete. Construct storage piles to freely drain surface water. Cover storage piles if required to prevent wind-blown dust.

D. Dispose of any unsuitable and excess topsoil, or at the Owner's option, stockpile suitable topsoil in another location as directed by the Contracting Officer.

3.5 DISPOSAL

A. Legally dispose of all waste materials.

B. The Contractor shall confirm an acceptable method of disposal with the Contracting Officer and state and local agencies before proceeding. No burning of debris is allowed on site.

END OF SECTION

SECTION 31 30 00 – SITE EARTHWORK

PART 1 - GENERAL

1.1 SECTION INCLUDES

A. Site earthwork comprises the following elements:
   1. Site Preparation: All modifications to the site and grades required for construction of the new work, surface reclamation and grading of the existing facilities abandoned, comprised of the following elements:
      a. Earthwork: Changing of grade levels, removal of soil, modifying existing soils in preparation for construction. Excavation for the installation of proposed facilities.
      b. Erosion Control: Temporary erosion control during construction.
   2. Excavation of all below grade facilities, including wet wells, valve vaults, grinder/lift stations/ manholes, and all other site utility services.

B. Submittals:
   1. Excavation and shoring plans

1.2 RELATED REQUIREMENTS
A. Section 02 21 19 – Utilities Survey
B. Section 02 41 00 - Site Demolition
C. Section 31 23 20 – Street and Site Excavation, Grading and Backfill
D. Section 31 25 00 – Erosion and Sediment Control
E. Section 31 30 00 - Site Earthwork.

1.3 REFERENCES

A. ASTM D 698 Test for Laboratory Compaction Characteristics of Soil Using Standard Effort

B. 29 CFR 1910 – OSHA, Occupational Safety and Health Standards

C. 29 CFR 1926 – OSHA, Safety and Health Regulations for Construction

1.4 FIELD CONDITIONS

A. The following existing site elements must be preserved:
   1. All underground utility services that are not abandoned or to be abandoned.
   2. All resources outside the limits of disturbance indicated on the POR Drawings.

B. Salvage all topsoil as noted in other design criteria sections.

PART 2 - PRODUCTS AND METHODS

2.1 TEMPORARY EROSION AND SEDIMENTATION BMPS

A. Temporary Sediment Barriers and Traps:
   1. Use one or more of the following

a. Silt fences
b. Straw wattles

B. Temporary Erosion Control Structures:
1. Check dams in swales and ditches
2. Level spreaders across slopes to convert concentrated flow to sheet flow

2.2 GRADING AND EXCAVATION

A. Excavated materials not utilized in the backfill process must be stored within the limits of disturbance.

B. Excess material that is not topsoil and not used in backfill operations will be hauled from the site and out of the park.

C. Excavation methods are machine or hand excavation.

2.3 EROSION AND SEDIMENTATION CONTROLS

A. Prepare and submit a Storm Water Pollution Prevention Plan (SWPPP) to Wyoming Department of Environmental Quality. Contractor shall obtain all NPDES and Temporary Storm Water permits required.

B. Develop and implement temporary erosion and sediment control measures and other Best Management Practices (BMP's) prior to commencement of earthwork. Remove all non-permanent erosion control measures after vegetation is fully established.

2.4 SITE EARTHWORK

A. Contractor shall preserve natural topographical features to minimize the impact on the existing drainage patterns at the site.

B. Finish Grading shall provide drainage towards drainage features. Finish grading shall not result in low spots that hold water.

C. All earthwork shall be designated "Unclassified". All surplus excavation shall be disposed of outside of the park.

D. Remove unsatisfactory soil materials from the site.

E. Provide compaction in accordance with a geotechnical report.

PART 3 - DESIGN CRITERIA

3.1 BASIC FUNCTION

A. Provide all modifications to the site and site improvements required for proper functioning of the project and as indicated on the plans.

B. Prepare the site for execution of earthwork under G10 – Site Preparation Criteria.

C. Modify the site grades and soils as required for construction of facilities.

D. Excavation for all structures and underground facilities, to be constructed abiding by OSHA safety standards.

E. In addition to the requirements of this section, comply with all applicable requirements of Section DC 00 – Facility Design Criteria.

3.2 HEALTH AND SAFETY CRITERIA

A. Safety:
   1. Workers:
      a. Provide excavations in compliance with code and OSHA requirements.
      b. Excavations shall be sloped adequately or all excavations shall be shored.
   2. Park Visitors
      a. Inhibit access in the areas of the excavations from park visitors.
      b. Prevent access during all times when open excavations are present, both work and non-workhours.

B. Maximum Slopes:
   1. Maximum finish grade slopes at 2:1.

C. Prevent all standing water that could become stagnant.

3.3 STRUCTURAL CRITERIA

A. Earthwork: Provide structural design as required by code.
   1. Bearing Capacity: Under structure, paving, and site structural elements, maintain natural bearing capacity or achieve or correct compaction as required to prevent subsidence.

B. Site Structures:
   1. Provide earth foundations as required to support the completed wet well without subsidence.

3.4 DURABILITY CRITERIA

A. Soil Erosion Resistance: comply with the code and the following:
   1. Maintain the existing site features that contribute to erosion resistance to the greatest extent possible.

2. Maintain temporary and permanent erosion and sediment controls during the work.
3. Design to minimize soil erosion.
4. If erosion occurs during construction and within one year after completion, eroded areas shall be repaired by the Contractor at no cost to the Government.
5. During construction, take measures to minimize the amount of eroded soil that is transported into waterways.
6. Design erosion control measures in accordance with "Best Management Practices" (BMPs) and follow procedures prescribed by code.

B. Flooding:
1. Control storm water runoff as required to prevent damage to project elements, including vegetation.
2. Minimize increase in storm water runoff into streams and lakes.

3.5 OPERATION AND MAINTENANCE CRITERIA

A. Ease of maintenance
1. Design and construct earthwork elements so that they are permanent, not requiring periodic maintenance to maintain stability or appearance.

END OF SECTION

SECTION 31 21 13 – RADON MITIGATION

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Pipe, tube, and fittings.

2. Soil-gas collection materials.

3. Sealants and Accessories

1.2 ACTION SUBMITTALS

A. See section 22 00 00 “General Requirements of Plumbing” for submittal requirements.

PART 2 - PRODUCTS

2.1 PERFORMANCE REQUIREMENTS

A. Components and installation shall be capable of withstanding the following minimum working pressure unless otherwise indicated:

1. Soil, Waste, and Vent Piping: 10-foot head of water.

B. Seismic Performance: Soil, waste, and vent piping and support and installation shall withstand the effects of earthquake motions determined according to ASCE/SEI 7. See section 22 05 48 “Vibration and Seismic Controls for Plumbing Piping and Equipment”

2.2 PIPING MATERIALS

A. Piping materials shall bear label, stamp, or other markings of specified testing agency.

B. Comply with requirements in "Piping Schedule" Article for applications of pipe, tube, fitting materials, and joining methods for specific services, service locations, and pipe sizes.

2.3 PVC PIPE AND FITTINGS

A. Comply with NSF 14, "Plastics Piping Systems Components and Related Materials," for plastic piping components. Include marking with "NSF-dwv" for plastic drain, waste, and vent piping and "NSF-sewer" for plastic sewer piping.

B. Solid-Wall PVC Pipe: ASTM D 2665, drain, waste, and vent.

C. PVC Socket Fittings: ASTM D 2665, made to ASTM D 3311, drain, waste, and vent patterns and to fit Schedule 40 pipe.

D. Adhesive Primer: ASTM F 656.

E. Solvent Cement: ASTM D 2564.

2.4 SOIL-GAS COLLECTION MATERIALS

A. Perforated Piping:

1. Single wall HDPE perforated piping, ASTM F667

B. Aggregate Layer:

1. Clean, coarse aggregate, ¾ inch nominal size, free of fines.

C. Soil-Gas Collection Mat:

1. High-density polyethylene (HDPE) geocomposite mat designed for sub-slab depressurization.

D. Sealants and Accessories:

1. Slab penetration sealant:

a. Polyurethane or silicone sealant compatible with concrete.

PART 3 - EXECUTION

3.1 EARTH MOVING

A. Comply with requirements for excavating, trenching, and backfilling specified in Section 31 20 00 "Earth Moving."

3.2 PIPING INSTALLATION

A. Drawing plans, schematics, and diagrams indicate general location and arrangement of piping systems.

1. Install piping as indicated unless deviations to layout are approved on coordination drawings.

B. Install radon vent piping in accordance with ASTM E165.

C. Provide continuous, unobstructed vertical riser from slab to roof penetration.

D. Slope horizontal piping a minimum of 1/8 in per foot toward the slab penetration.

E. Sub-Slab depressurization system:

1. Install soil-gas collection point(s) at locations indicated on drawings.

2. Seal all slab penetrations, control joints, and utility opening airtight.

3. Connect collection point to vertical riser using solvent-welded PVC fittings.

F. Vent Riser Installation:

1. Route riser through interior spaces; exterior routing is not permitted unless approved.

2. Maintain minimum 10 feet horizontal separation from operable windows at roof termination.

3. Terminate vent a minimum of 12 inches above roof surface and 10 feet from any air intake.

3.3 IDENTIFICATION

A. Identify exposed radon piping.

B. Comply with requirements for identification specified in Section 22 05 53 "Identification for Plumbing Piping and Equipment."

3.4 FIELD QUALITY CONTROL

A. During installation, notify authorities having jurisdiction at least 24 hours before inspection must be made. Perform tests specified below in presence of authorities having jurisdiction.

1. Roughing-in Inspection: Arrange for inspection of piping before concealing or closing-in after roughing-in and before setting fixtures.

2. Final Inspection: Arrange for final inspection by authorities having jurisdiction to observe tests specified below and to ensure compliance with requirements.

B. Re-inspection: If authorities having jurisdiction find that piping will not pass test or inspection, make required corrections and arrange for re-inspection.

C. Reports: Prepare inspection reports and have them signed by authorities having jurisdiction.

3.5 CLEANING AND PROTECTION

A. Clean interior of piping. Remove dirt and debris as work progresses.

B. Protect piping during remainder of construction period to avoid clogging with dirt and debris and to prevent damage from traffic and construction work.

C. Place plugs in ends of uncompleted piping at end of day and when work stops.

D. Exposed PVC Piping: Protect plumbing vents exposed to sunlight with two coats of water-based latex paint.

E. Repair damage to adjacent materials caused by waste and vent piping installation.

END OF SECTION

SECTION 31 23 10 – PIPE TRENCH EXCAVATION AND BACKFILL

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Excavation, trenching and backfilling for pipelines and appurtenances.

B. Necessary handling and storage of materials to be used for fill and backfill; including bracing, shoring, protection, pumping, dewatering, and cleanup.

1.2 RELATED REQUIREMENTS

A. Section 31 23 20 – Street and Site Excavation, Grading, and Backfill.

B. Section 31 25 00 – Erosion and Sediment Control.

C. Section 33 14 15 – Site Utilities

D. Standard Specifications:
   1. The Specifications consist of the Wyoming Public Works Standard Specifications, 2015 edition (WPWSS).
   2. Except as specifically noted otherwise in the contract documents, all work shall be performed in accordance with the Standard Specifications.
   3. The information in these project specifications shall take precedence in the event of any discrepancies. Any discrepancies discovered by the Contractor shall be brought to the attention of the Contracting Officer before performing the associated work.

1.3 TESTING

A. Field Density Testing
   1. Testing will comply with WPWSS.
   2. Contractor shall accommodate testing by Owner.
   3. Test Frequency: Quality assurance testing frequency will average three tests for every 50 lineal feet of trench. Testing will cover both subgrade and backfill compaction.

B. Laboratory Maximum Density and Optimum Moisture Testing
   1. Testing will comply with WPWSS.
   2. Provide material submittals to the Contracting Officer per WPWSS.

PART 2 PRODUCTS - REFER TO SECTION 02210 – TRENCH EXCAVATION, AND SECTION 02225 – TRENCH BACKFILL IN WPWSS.

PART 3 EXECUTION - REFER TO SECTION 02210 – TRENCH EXCAVATION, AND SECTION 02225 – TRENCH BACKFILL IN WPWSS.

END OF SECTION

SECTION 31 23 20 – STREET AND SITE EXCAVATION, GRADING, AND BACKFILL

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Excavation of roadway, embankment, and preparing of subgrade for asphalt and concrete paving in streets, sidewalks, parking access drives and parking areas.

B. Strip, stockpile, and place topsoil.

C. Placement of imported borrow.

D. Finish grading.

E. Backfill materials, placement, and compaction

1.2 RELATED REQUIREMENTS

A. Section 31 1000 – Site Clearing.

B. Section 31 2500 – Erosion and Sediment Control.

C. Section 32 1300 - Exterior Concrete.

1.3 SUBMITTALS

A. Submit shop drawings for approval of all backfill materials with corresponding proctor data for testing.

1.4 TESTING

A. Field Density Testing
   1. Contractor is responsible for all testing.
   2. Laboratory Maximum Density and Optimum Moisture: Quality assurance tests will be made for each on-site natural soil. These tests shall determine, for each material, the laboratory maximum density values and optimum compaction moisture content under AASHTO T99 or ASTM D698.
   3. Test Frequency: Quality assurance testing frequency will average of 1 test for every 2,000 square feet of hard surfacing (concrete or asphalt). Testing will cover subgrade preparation and backfill compaction of sub-base and base course materials.
   4. Provide material test reports to Contracting Officer.

PART 2 PRODUCTS

2.1 EXCAVATION AND ON-SITE EMBANKMENT

A. Excavation is unclassified and shall include removal of all soil and other materials which are encountered in excavating to the required grades, as required to accomplish the construction.

B. Fill and backfill materials obtained on-site shall consist of native soils free from organic matter, frozen material, refuse, construction debris, or other deleterious or man-made items. Obtain approval from the Contracting Officer for all fill before placing and only use fill from designated on-site borrow areas, or use imported borrow as specified hereinafter.

2.2 IMPORTED BORROW MATERIALS FOR EMBANKMENTS IN PLACE

A. If required, obtain borrow soil for embankments from areas off the project site. Furnish imported borrow at Contractor's expense. Obtain Contracting Officer approval of borrow sources and materials. Imported borrow shall meet the requirements of On-site Embankment hereinbefore.

2.3 SUB-EXCAVATION / REPLACEMENT BELOW SUBGRADE

A. Sub-excavation consists of removing and disposing of unstable material from below planned subgrade elevation in cut sections or from below the natural ground line in embankment sections as specified further in Part 3, EXECUTION of this section.

B. Replacement material for sub-excavation shall consist of either:
1. Suitable native materials from within the project limits if suitable material, as determined by the Contracting Officer, is available; or
2. Imported gravel material if suitable material is not available within the project limits. Where imported gravel is used, furnish replacement material meeting the following gradation requirement:

Sieve Opening	Percent Passing
3-Inch	100%
No. 4	25 – 60%
No. 200	12% Max.

C. Alternative sub-excavation replacement may include placement of a geotextile or geo-grid material at the bottom of the sub-excavation and a possible reduction of the sub-excavation depth, as determined by the Contracting Officer.

2.4 EQUIPMENT

A. Furnish all necessary equipment required to accomplish the excavating, shaping, embankment, grading and rolling, moisture control, and compaction specified herein.

PART 3 EXECUTION

3.1 EXCAVATION STABILITY AND SAFETY

A. Contractor shall meet OSHA requirements for excavations and excavated material stockpiles. This may require design of temporary slopes and/or shoring by a licensed professional engineer.

3.2 PROTECTION OF PROPERTY

A. Contractor shall take precautions to protect all adjoining public and private property and facilities, including underground and overhead utilities, curbs, sidewalks, driveways, structures, fences, and vegetation. Any disturbed or damaged facilities shall be suitably restored or replaced consistent with condition(s) which existed prior to construction by the Contractor at Contractor's expense.

3.3 SUBGRADE EXCAVATION

A. Excavate and shape subgrade to line, grade, and cross-section specified on the Drawings. Excavate without causing rutting, pumping, or other disturbances to underlying materials.

B. Restore any sub-excavated areas as directed by the Contracting Officer and correct subgrade disturbances by removing the disturbed soil and replacing with suitable excavated material or imported material and compacting to reach at least 95 percent of the maximum laboratory dry density at optimum moisture content determined by AASHTO T99 or ASTM D698. Correct any subgrade disturbances before placing overlying fill, backfill, base course or other courses.

C. Maintain the subgrade to drain at all times. Construct side ditches or gutters from cuts to embankments to prevent erosion damage to embankments.

D. Construct and maintain temporary drainage facilities where existing surface drainage, sewers, or under-drainage facilities are disturbed during the work until permanent drainage facilities are completed and are operational. Protect and preserve all existing drains, sewers, sub-surface drains, conduits, gas lines, and other underground utilities and structures which may be affected by the work. Any damage to these facilities or structures resulting from the work shall be repaired by the Contractor to the satisfaction of the Contracting Officer.

E. Excavate to minimize foundation soil and/or subgrade soil exposure to erosion, drying, or infiltrating moisture. Perform excavation to provide drainage away from foundation/subgrade soils and minimize the potential for surface runoff to enter the foundation/subgrade soils.

F. Grade all intersecting streets and approaches within the project limits as specified or as directed using suitable materials on the surfaces to produce smooth riding and satisfactory approaches to the intersections.

3.4 DISPOSAL OF EXCAVATED MATERIAL

A. Contractor shall dispose of all debris and unused excavated materials off the project site in accordance with all applicable state and local regulations. Contractor shall be responsible for locating suitable disposal area(s) and all costs for disposal.

B. Contractor to refer to and adhere to hazardous material specifications located elsewhere in these project specifications.

3.5 SUBGRADE PREPARATION AND COMPACTION

A. Assure the subgrade beneath pavements, curb, or sidewalks is natural soil free of topsoil, organic material, or refuse. Place sub-base course, base course, pavement, curb and gutter, and sidewalk over the prepared subgrade as soon as practical. Do not place base course materials on frozen subgrade.

B. If the surface of a previous roadbed or pavement surface matches the surface of the finished subgrade, scarify the top 6 inches of the previous surface the full width of the subgrade to permit uniform reshaping and compaction, unless shown otherwise on the Drawings.

3.6 FINE GRADING

A. The finished subgrade surface shall be within a tolerance of 0.10 feet of the design subgrade and cross-section shown on the Drawings, shall be smooth and free from irregularities, and shall be compacted to the specified density. In addition, the sum of the deviations from true grade of any two points less than 30 feet apart shall not exceed 0.10 feet.

B. The subgrade shall be considered to extend over the full width of the base course and compaction shall extend 1 foot beyond the edge of paving, curb, or forms.

3.7 COMPACTION

A. Compact subgrade with an approved roller until the top 8 inches of the subgrade is compacted to at least 95 percent of the maximum laboratory dry density at optimum moisture content in accordance with AASHTO T99 or ASTM D698. Proof roll the subgrade surface for observation by the Contracting Officer. Compact all soft, yielding or otherwise unstable areas to provide adequate support of construction equipment, and meet the specified density. Remove and replace any unstable or otherwise unsuitable subgrade as specified herein.

3.8 EMBANKMENT PREPARATION AND COMPACTION

A. Place fill materials (embankment) to the specified lines and grades shown on the Drawings. Place fill in uniform layers not exceeding 8 inches loose thickness. Once placed, moisten, or aerate to approximate optimum moisture content, mix, and compact each layer as specified hereinafter. Work clay soils to maximum 2-inch nominal size before compacting. Do not begin fill placement until the subgrade has been approved by the Contracting Officer. Do not place fill on wet or frozen areas. Do not operate heavy equipment for spreading or compacting fill within four feet of structures.

B. If grading operations are suspended due to weather or other circumstances, blade the entire area until it is smooth, free of depressions and ruts, and crowned to drain water.

C. Control the fill moisture content to assist in obtaining the specified field density. Maintain the moisture content of fill soils to within plus or minus 3 percent of optimum moisture. Compact each fill layer (and the top 8 inches of subgrade soil) to at least 95 percent of the maximum laboratory dry density at optimum moisture content determined by AASHTO T99 or ASTM D698. Compact areas within four feet of structures in maximum 8-inch loose lifts using power-driven handheld tampers.

D. Apply water when required at the locations and in the amounts needed to compact the material to the specified density. Maintain an adequate water supply during the work. Assure the equipment used for watering is of the capacity and design to provide uniform water application. Also apply water during the work to control dust and to maintain all embankment and base courses in a damp condition. Water required for compacting subgrade and/or embankments must be obtained by the Contractor from municipal system or other sources, as needed.

E. Do not place fill or embankment when moisture content prevents effective compaction or causes rutting. Dry all embankments having excessive moisture by scarifying and blading the affected areas before compacting or placing succeeding layers.

3.9 SUB-EXCAVATION / REPLACEMENT BELOW SUBGRADE

A. Sub-excavation consists of the following:

1. In excavation (cut) areas, removing and disposing of unsuitable material, as determined by the Contracting Officer, from below planned (design) subgrade elevation.

2. In embankment (fill) areas and following clearing and grubbing, removing and disposing of unsuitable material, as determined by the Engineer, from below planned (design) subgrade elevation.

B. Soil is unsuitable if, in the opinion of the Contracting Officer, it contains excessive organics, refuse, construction or other debris, or other objectionable material; if it is unstable, rutting, or yielding; or if it contains excessive moisture. Generally, soils shall be sub-excavated and replaced if they are unable to adequately support equipment typically used for excavation and soil transport.

C. Contractor shall assure that the Contracting Officer has measured the area where unstable materials have been removed before backfilling. Do not backfill any area where unstable foundation soils have been excavated until authorized by the Contracting Officer. Backfill placed without approval may be ordered removed and replaced at Contractor's expense.

D. Backfill with material meeting with the requirements of Part 2, PRODUCTS of this section. Compact the replacement material to at least 95 percent of the maximum laboratory dry density at optimum moisture content in accordance with AASHTO T99 or ASTM D698.

E. Compacted backfill shall extend to planned (design) subgrade elevations.

3.10 PROTECTION OF SUBGRADE WORK

A. After preparing the subgrade as specified hereinbefore, all unnecessary traffic shall be kept off. Should it be necessary to haul over the prepared subgrade, the Contractor shall drag and roll the travelled way as frequently as may be necessary to remove ruts, cuts, and breaks in the surface. All ruts, cuts, and breaks in the surface of the subgrade that are not removed by the above operations shall be raked and hand tamped. All equipment used for transporting materials over the prepared subgrade shall be equipped with pneumatic tires.

B. Continued use of sections of prepared subgrade for hauling, so as to cut up or deform the true cross-section, will not be permitted. The Contractor shall protect the prepared subgrade from both the Contractor's own and public traffic.

C. If excessive damage is being caused, the Contractor will be required, at the Contractor's sole expense, to restore the subgrade surface as specified hereinbefore and to plank the subgrade before hauling any additional materials or equipment over it.

D. In all cases, the subgrade shall be maintained in the finished condition, meeting the tolerances specified hereinbefore, until the first succeeding course is placed.

END OF SECTION

SECTION 31 25 00 – EROSION AND SEDIMENT CONTROL

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Erosion and sediment control requirements.

1.2 RELATED REQUIREMENTS

A. Section 02 41 00 – Site Demolition.

B. Section 31 10 00 – Site Clearing.

C. Section 31 23 10 – Pipe Trench Excavation and Backfill.

D. Section 31 23 20 – Street and Site Excavation, Grading, and Backfill.

1.3 SUMMARY

A. Contractor shall be fully responsible for the Storm Water Discharge Permit and fully comply with the Environmental Protection Agency (EPA) regulations in regard to Storm Water Discharges associated with Construction Activity including, but not limited to, any and all submittals, inspections, fees, reporting, training, and installation of Best Management Practices (BMP's). CONTRACTOR shall sign all permits and forms and assumes all responsibility of management of the Storm Water Erosion Control Plan and any associated records or fines.

B. The CONTRACTOR is required to develop a Storm Water Pollution Prevention Plan (SWPPP) per EPA standards. The SWPPP must clearly address the effluent limitations and the selected BMP's to be used to manage pollutant sources and ensure appropriate protection of state surface waters as outlined in EPA's General Permit for Storm Water Discharges Associated with Construction Activity (called "General Permit"). In case of conflict between this specification and the General Permit, the General permit shall prevail.

C. The site is required to reach "final stabilization" before permit coverage may be terminated. In Wyoming's semi-arid climate, the time necessary to achieve this "final stabilization" often requires maintenance and permit coverage well beyond the Substantial Completion phase to ensure vegetation or other site stabilization measures are in-place.

1.4 DEFINITIONS

A. Best Management Practices (BMPs)
   1. Schedule of activities, prohibition of practices, maintenance procedures, and other management practices to prevent or reduce the pollution of state surface waters. BMPs also include treatment requirements, operating procedures, and practices to control plant site runoff, spillage or leaks, sludge or waste disposal, or drainage from raw material storage.

B. Final Stabilization
   1. The time at which all soil-disturbing activities at the site have been completed, and a vegetative cover has been established with a density of at least 70% of the pre-disturbance levels, or equivalent permanent, physical erosion reduction methods have been employed.

Final stabilization using vegetation must be accomplished using seeding mixtures or forbs, grasses, and shrubs that are adapted to the conditions of the site. Establishment of a vegetative cover capable of providing erosion control equivalent to pre-existing conditions at the site will be considered final stabilization.

1.5 SUBMITTALS

A. Prior to receiving a Notice to Proceed, contractor shall submit to the Contracting Officer the following documentation:
   1. Evidence that a Notice of Intent (NOI) has been submitted to the EPA via the NPDES eReporting Tool (NeT) and accepted by the EPA;
   2. Qualifications of the inspector to conduct the required inspections to verify compliance with the permit; and
   3. A copy of the SWPPP has submitted to the EPA; and
   4. Inspection schedule for inspection and monitoring of pollution prevention measures; and
   5. Erosion control product submittal including manufacturer's product information and installation instructions; and
   6. Notice of Termination after Final Stabilization.

B. In addition, the Contracting Officer shall be copied on all documentation submitted to or received from the EPA including but not limited to notices of noncompliance, if necessary and the Notice of Termination, when submitted.

1.6 QUALITY ASSURANCE

A. Contractor shall prepare and submit a plan to Contracting Officer (CO) for review and concurrence.

B. Contractor shall be responsible for arranging and conducting Pollution Prevention, Erosion, and Sediment Control meeting/briefing to inform parties scheduled to be onsite during Project of measures to be implemented for proper pollution prevention and control (may be included as part of Pre-Construction Meeting).
   1. Installation of silt fences, storm drain protection, and other forms of pollution prevention, erosion, and sediment controls shall not begin until after this meeting has occurred.

C. Pollution Prevention and Erosion Control Manager: Contractor shall designate a Pollution Prevention and Erosion Control Manager who will be responsible for implementation, inspection, maintenance, and amendments to approved plan.
   1. Pollution Prevention and Erosion Control Manager shall be familiar with EPA procedures, the temporary storm water pollution prevention procedures, and BMPs and shall ensure emergency procedures and plan are updated as needed and available for inspection.
   2. When changes in approved plan are required, Pollution Prevention and Erosion Control Manager shall prepare and certify an amendment and submit to Contracting Officer for review and concurrence.
   3. Pollution Prevention and Erosion Control Manager will conduct documented routine inspections of BMPs and implement any corrective measures required.

PART 2 PRODUCTS – NOT USED

PART 3 EXECUTION

3.1 ENVIRONMENTAL PROTECTION

A. Protection of Natural Resources: Comply with applicable regulations and these specifications. Preserve natural resources within Project boundaries and outside limits of work performed under this Contract in their existing condition or restore to an equivalent or improved condition as approved by Contracting Officer.

B. Construction Zone: Arrange construction activities to minimize pollution (i.e., erosion, trash, etc.) to maximum practical extent.

1. Clearing, excavation, and grading shall be limited to those areas of Project Site necessary for construction. Minimize area exposed and unprotected.
2. Clearly mark and delineate limits of work activities.
3. Equipment shall not be allowed to operate outside limits of work or to disturb existing vegetation.
4. Excavation and grading shall be completed during dry season to maximum extent possible.
5. Material shall be stored away from locations where water is present to greatest extent practicable.

3.2 REGULATORY REQUIREMENTS

A. Permits: Contractor shall follow requirements listed in the Summary and Submittal sections.

B. The Contractor will be responsible for annual renewals of the permit until “final stabilization” has occurred and the Notice of Termination (NOT) can be submitted. The Contractor must submit a NOT form when the construction activity is complete and the site has achieved “final stabilization.” Contracting Officer shall be notified of any inspection made for the purpose of determining if the site has achieved final stabilization.

C. Contracting Officer Notification:

1. Contractor shall notify Contracting Officer in writing and by telephone of these events:
   a. Erosion and sediment control meeting/briefing.
   b. Following installation of required sediment control structures.
   c. Prior to removal of or modification to sediment control structures.
   d. Prior to removal of sediment control structures.

3.3 TEMPORARY STORM WATER POLLUTION PREVENTION PLAN

A. Records and requirements:

1. Contractor shall maintain current copy of SWPPP and associated records and forms at Job Site throughout duration of Project.
2. SWPPP shall be available for public inspection and inspection and use of Contracting Officer.

B. Implementation: Implement the SWPPP as required throughout construction period and maintain erosion control elements in proper working order.

1. Do not perform clearing and grubbing or earthwork until initial site stabilization has been implemented.

C. SWPPP (including inspection forms) and data used to complete the NOI shall be provided to Contracting Officer after Final Stabilization has been achieved and Notice of Termination has been accepted by the EPA.

3.4 SITE INSPECTIONS AND PLAN REVISIONS

A. Inspections: Contractor will perform a weekly inspection onsite. Contractor shall inform the Contracting Officer of schedule of inspections so the Contracting Officer can witness the inspection. Documentation of each inspection is required and made available to the Contracting Officer as requested.

B. Plan Revisions: It may be necessary to revise Plan during construction to make necessary improvements, revisions, or to respond to unforeseen conditions noted during construction or Site inspections.

C. Corrections: Provide additional temporary erosion and pollution controls made necessary by Contractor's errors or changes in plans at no additional cost to Government.

3.5 MAINTENANCE

All BMPs identified in the SWPPP must be maintained in effective operating condition. If site inspections identify BMPs which are not in effective operating condition, maintenance must be performed before the next storm event. If existing BMPs need to be modified, or if additional BMPs are necessary for any reason, implementation of these additional measures must be completed before the next storm event. All changes in the design, implementation, or installation of erosion and sediment control or other BMPs must be documented where applicable in the SWPPP.

3.6 HOUSEKEEPING AND SITE MANAGEMENT

A. Store materials onsite in conformance to federal, state, local, and manufacturer's regulations and specifications. Use Best Management Practices to minimize risk of materials coming into contact with environmental conditions (i.e., water and wind) that could disperse them.

B. Manage solid waste in conformance to federal, state, and local regulations. BMPs should be used to minimize risk of materials coming into contact with environmental conditions (i.e., water and wind) that could disperse them.

C. Include spill prevention and control in the SWPPP.

D. Manage hazardous waste (including contaminated soil) in conformance to federal, state, local, and NPS regulations and guidelines.

3.7 EROSION CONTROL MEASURES

A. Erosion control measures shall consist of Best Management Practices for storm water discharges, including items such as silt fencing, barrier protectors, coconut husk wattles, temporary soil retention blankets, excelsior drainage filters, sediment traps and berms.

B. Berms and excelsior drainage filters shall be used to form sediment traps and control run-on and run-off into other areas, including creeks, streams, marshes, access roads, well areas, and staging areas.

C. Erosion control measures shall be used to contain only direct precipitation in construction zone. Contained water shall be allowed to percolate into ground or drain slowly through drainage filter sediment traps.

D. Earthen sediment traps or holding ponds shall not be used unless accepted by Contracting Officer.

E. Reduce runoff velocity and direct surface runoff around and away from fuel containment, storage, and borrow areas.

F. Divert surface runoff around and away from cut and fill slopes.

G. Place drainage filters around catch basins to create sediment traps to control run-off from construction area.

H. Excess water used for dust control shall be contained within demolition areas by erosion control measures.

I. Contractor shall prevent deposition of materials onto paved areas. Contractor shall inspect paved areas for deposited materials weekly and remove materials immediately.

J. Furnish, install, maintain, and operate necessary control measures and other equipment necessary to prevent erosion as described in approved SWPPP.

K. Before work begins, sufficient equipment shall be available onsite to assure operation and adequacy of erosion control system can be maintained.

3.8 MAINTENANCE OF TEMPORARY FACILITIES

A. Ensure erosion and sediment control structures remain effective throughout excavation and grading operations. Relocate structures as necessary.

B. Inspect all BMPs after each significant rainfall. Promptly repair any problems.

3.9 REPORTING

A. If a discharge occurs or if Project receives written notice or order from regulatory agency, Contractor shall immediately notify Contracting Officer and shall file written report to Agency(ies) with Jurisdiction within 7 days of discharge event, notice, or order or sooner if required by the notice or order. Corrective measures shall be implemented immediately following discharge, notice, or order. The report to the Agency(ies) with Jurisdiction shall contain:

1. Date, time, location, nature of operation, and type of discharge, including cause or nature of notice or order.

2. Best Management Practices deployed before discharge event, or prior to receiving notice or order.
3. Date of deployment and type of Best Management Practices deployed after discharge event, or after receiving notice or order, including additional Best Management Practices installed or planned to reduce or prevent re-occurrence.
4. An implementation and maintenance schedule for affected Best Management Practices.

3.10 SEDIMENT DISPOSAL

A. Sediment excavated from temporary sediment control structures shall be disposed onsite with general fill, with topsoil or hauled off site. Sediment shall be allowed to dry out as required before reuse.

B. Contractor shall place sediment removed from traps and other structures where it will not enter a storm drain or watercourse and where it will not immediately reenter the basin.

3.11 REMOVAL OF TEMPORARY STORM WATER POLLUTION CONTROL MEASURES

A. Temporary control measures shall be removed after Notice of Termination has been accepted by the EPA and after notifying the Contracting Officer and obtaining permission. Temporary control measures shall be removed within 20 calendar days after notification by the Contracting Officer if the site has reached Final Stabilization.

END OF SECTION

SECTION 32 10 00 – GENERAL FACILITY DESIGN

PART 1 - GENERAL

1.1 SECTION INCLUDES

A. Section Includes: The construction generally consists of a sewer collection and conveyance system, a water distribution system, buried power, and fiber optic communications for the 4 Lazy F Ranch withing Grand Teton National Park. All utilities or conduits will be constructed to each structure to be served, with the exception of the lift station. The structures are indicated on the drawings.
   1. Sewer System to consist of sewer services, sewer gravity collection system, a lift station, forcemain, small grinder pump stations, and associated site requirements.
   2. Water Distribution System consists of water mains, water valves, water services, drains, blow offs, water meter and vault, abandonment of an existing water well, and associated site requirements.
   3. Buried Power consists of providing power from existing LVE infrastructure to the new lift station and installing direct bury conductor from an existing H panel and meter to the southernmost cabin as indicated on the POR drawings.
   4. Fiber optic to consist of the extension of existing facilities at the Caretaker's Cabin to the lift station and future use conduit to within 5 feet of each structure to be served in the future.

1.2 RELATED REQUIREMENTS
   1. Section 02 21 19 – Utilities Survey
   2. Section 02 41 00 - Site Demolition
   3. Section 31 23 20 – Street and Site Excavation, Grading and Backfill
   4. Section 31 25 00 – Erosion and Sediment Control
   5. Section 31 30 00 - Site Earthwork.
   6. Section 32 20 00 - Site Improvements.
   7. Section 33 14 15 - Site Utilities.

1.3 DEFINITIONS

A. Code: The code referred to herein consists of all applicable local, state, and federal regulations, and the following:
   1. Non-Regulatory Criteria Documents: In addition to specific regulatory requirements, the following documents are also incorporated into the definition of "the code" for the purposes of this project, except for administrative provisions contained therein; where referenced, the role of the code official described in the document will be performed by Owner. Wyoming Department of Environmental Quality Rules and Regulations
      a. 2024 ICC Codes and Standards
      b. NFPA 70 – National Electrical Code
      c. Chapter 11, Chapter 12, and Chapter 26 Wyoming Department of Environmental Quality
   2. Electrical: Provision and distribution of electrical power to operate all electrically-operated devices, including those included under other services both interior and exterior.
   3. Site Elements and Work; Modifications to the site, site improvements, and site utilities.

1.4 FIELD CONDITIONS

A. Project Site: the project site is a mix of area occupied by existing structures and previously undisturbed ground.

B. Verification of Field Conditions:
   1. Additional investigations shall be conducted by the Contractor for the design of the project.
      a. A geotechnical investigation shall be conducted to determine ground water depth through the project, more specifically in the location of the lift station.
      b. A site topographical survey shall be conducted through the limits of disturbance indicated on the POR drawings to verify surface elevations.
      c. An in depth site investigation shall be performed on the existing septic systems to determine the number and locations of each. These are critical to determine in the design so that they are abandoned properly during construction.

C. Demolition: The abandonment and removal of existing septic tanks are included in the design. An existing water well shall be abandoned per the requirements of WYDEQ Chapter 26 Water Quality Regulations.

PART 2 - PRODUCTS (SEE OTHER SECTIONS)

PART 3 - DESIGN CRITERIA

3.1 BASIC FUNCTION

A. Code: Make all portions of the project comply with the 2024 ICC codes and standards, along with the WYDEQ requirements.

B. Provide built elements and site modifications required to fulfill the needs described in the project program, POR drawings, and as specified.

C. Provide permanently enclosed spaces for all lift station valves and above ground piping shown in the approved POR drawings. This above ground enclosure shall be subject to the requirements of the 4LF historic district.

D. Provide a physical enclosure that keeps out weather, unwelcome people, animals, and insects.

3.2 AMENITY AND COMFORT CRITERIA

A. Conduct construction operations to minimum disruption to public and park service personnel. Including vehicle access to 4 Lazy F Ranch on the existing road during construction of the sewer force main.

B. Minimize the closures to Menor’s Ferry Area to a limited number of temporary closures. Temporary closures must note exceed 2 days for each closure, as approved by the CO no less than 7 days in advance.

C. Environmental Responsibility:

1. In addition to other requirements, provide design and construction that minimizes adverse effects on the exterior environment, minimizes consumption of energy, water, construction materials, and other resources.
2. Provide design and construction that is compatible with the applicable SOI preservation standards for the historic district.

3.3 HEALTH AND SAFETY CRITERIA

A. Fire Resistance: Design and select materials to provide fire resistance in accordance with code

B. Prevention of Accidental Injury: As required by code and as follows:
1. Avoid safety hazards whenever possible where services much involve flammable materials or hazardous operations and comply with code.

3.4 STRUCTURAL CRITERIA

A. Structural Performance: Design and select materials to support all loads without damage due to loads, in accordance with code.
1. Design & detail in accordance with Current building codes & standards.
2. For structures engineered by their manufacture or fabricator, detailed design analysis prepared by and shop drawings stamped by a licensed structural engineer, with approval of engineer of record.

3.5 DURABILITY CRITERIA

A. Expected Service Life Span: Expected functional service life of the built portions of this project is 50 years.
1. All exterior underground utilities: not less than 50 years.
2. Software and Firmware Integral to Operation of Equipment: Minimum 20 years functional life without reprogramming required.
3. Service life spans of individual elements that differ from the overall project life span are defined in other sections.

B. Weather Resistance: Design and select materials to minimize deterioration due to precipitation, sunlight, ozone, and normal temperature changes.
1. Weather resistance requirements apply to all components exposed to the outdoor environment.
2. Deterioration includes corrosion, shrinking, cracking, spalling, delamination, abnormal oxidation, decay, and rot.
3. Surfaces Exposed to View: Deterioration adversely affecting aesthetic life span includes color fading, crazing, and delamination of applied coatings.
4. Joint Components and Penetration Seals: Capable of resisting expected thermal expansion and contraction. Use overlapping joints that shed water wherever possible.
5. Service Temperature: Low temperature equal to historically recorded low. High temperature equal to that expected due to any combination of air temperature and heat gain from solar and other sources.
6. Freeze Thaw Resistance: Adequate for the climate of the project.
7. Buried Water Piping: Minimum 7’ below finish grade, insulate if less depth.
8. Buried Sewer Piping: Minimum depth 6’ below finish grade, insulate if less depth.

C. Temperature and Humidity Endurance: Design equipment to endure temperature and humidity that will be encountered and to resist damage due to thermal expansion and contraction.

D. Impact Resistance: Design and select materials to resist damage due to impact in accordance with code and the following:
   1. Design and select materials to resist damage from hail of size up to ½ inch.
   2. Minimize damage due to potential vandalism.
   3. Natural hazards: Design to resist damage from perching, nesting, and feeding birds.

E. Accidental Damage Resistance:
   1. Minimize potential for damage to build elements due to accidents.
   2. Accidental Water Leakage: Locate components that would be damaged by water leakage from pipes or through foundations.

F. Water Resistance: Design and select materials to provide resistance to normal wear and tear in accordance with code.

G. Resistance to Biological Factors:
   1. Animals: Do not use materials that are attractive to or edible by animals or birds.
   2. Insects: Do not use materials that are edible by insects unless access by insects is prevented.
   3. Wood: When wood is used, provide at least the protection recommended by American Wood Protection Association (AWPA) as contained in AWPA U1.

3.6 OPERATIONAL AND MAINTENANCE CRIERIA

A. Waste (Trash/Rubbish) Removal: As described in the project program and as follows:
   1. See Division 1 Specifications in the appendix and NPS Project Sustainability Checklist for construction waste management requirements.

B. Ease of Operation and Use:
   1. Intended operating personnel are personnel with a reasonable level of training for similar activities.
   2. Provide facility, equipment, and systems that are easily operated by intended personnel.
      a. Space Around Components: Working clearances and access routes as required by code and indicated on the POR drawings.
      b. Access: All mechanical and electrical equipment is located to allow easy access.
   3. Minimize the need for specialized training in operation of specific equipment or systems. Identify all equipment and systems for which the manufacturer recommends or provides training programs.
   4. Preparation for Use: Prepare services for use by testing appropriately for proper operation before start-up, eliminating operation anomalies, adjusting control systems for optimum operation, and demonstrating proper function.

C. Ease of Maintenance
   1. Minimize the amount of maintenance required.
   2. Do not locate any equipment requiring maintenance on the roof, in attics, in crawl spaces, where access must be through attics or crawl spaces, or where access is not possible using removal panels or doors.
   3. Light Levels: Provide adequate lighting for locating and maintaining equipment.
   4. Cleaning: Where not otherwise specified, design equipment mountings to allow easy cleaning around and under equipment, if applicable, without crevices, cracks, and

concealed spaces where dirt and grease can accumulate and with raised, closed bases for equipment mounted on the floor.

5. Equipment Enclosures: Provide removable access panels to allow cleaning.
6. Site Utilities: Record or mark locations of existing, abandoned, and new utility lines in such a manner that they are easily located during and after the completion of construction.
7. Piping Systems:
   a. Piping other than Gravity Drains: Provide means of isolating convenient portions of piping systems so that small portions may be shut down leaving the remainder in operation.
   b. Piping: Entire systems drainable without disassembly of piping.
8. Replaceability of Parts:
   a. Parts Having Service Life Less Than That Specified for Element: Easily replaceable without de-installation or de-mounting of the entire element, component, or equipment item.
   b. Valves: Easily replaceable internal parts, eliminating necessity of removal of entire valve for repair.
   c. Parts Availability: Readily available from stocking distributors within 100 miles of project location.

D. Ease of Replacement
   1. Elements Not Required to have Expected Service Life Span Equal to that Specified for Facility as a Whole: Make provisions for replacement without undue disruption of building operation.

END OF SECTION

SECTION 32 12 43 - RECLAIMED ASPHALT PAVEMENT (RAP) TRAIL SURFACE

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Contract Plans and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes requirements for furnishing, placement, mixing, stabilization, and compaction of reclaimed asphalt pavement (RAP).
   1. Work includes furnishing, placing, and securing geotextile fabric to prevent intermixing of subgrade and trail base, promote drainage, and improve long-term performance of the trail.
   2. Furnish and place RAP, in accordance with these specifications and in conformity with the dimensions, typical cross section, and the lines and grades shown on the Contract Plans. The locations where RAP will be used are shown on the Contract Plans.

1.3 REFERENCES

A. ASTM D1557 – Modified Proctor for Compaction

B. ASTM D2950 – Nuclear Density of Bituminous Materials

C. ABAAS – Architectural Barriers Act Accessibility Standards

1.4 SUBMITTALS

A. Samples: 12x12” sample of geotextile fabric for approval.

B. Product Data: Provide a complete list of products to be used, with the following information for each:
   1. Manufacturer’s name, product name and/or catalog number, and general product category
   2. Manufacturer’s installation instructions.
   3. Material performance information

C. Manufacturer’s Installation Instructions

D. Equipment list confirming compliance with access limitations

E. Field quality control compaction reports (if requested by the Contracting Officer)

1.5 DELIVERY, STORAGE, AND HANDLING

A. Packaged Materials: Deliver packaged materials in original, unopened containers showing weight, certified analysis, name and address of manufacturer, and indication of conformance with state and federal laws if applicable. Store in cool, shaded area away from direct heat or freezing conditions.

1. Geotextile fabric: Do not drag rolls across ground. Handle in a manner to avoid tearing or puncturing fabric.

B. Bulk Materials:
1. Do not dump or store bulk materials near structures, utilities, walkways and pavements, or on existing plant materials or within critical root zones.
2. Provide erosion-control measures to prevent erosion or displacement of bulk materials, discharge of soil-bearing water runoff, and airborne dust reaching adjacent properties, water conveyance systems, or walkways.
3. Accompany each delivery of bulk materials with appropriate certificates.

C. Rejection of Material.
1. Evidence of inadequate protection or improper handling or storage shall be cause for rejection.
2. Any product or material exhibiting signs of damage due to nonconformity to specifications or due to delivery, storage or handling shall be rejected by the Contracting Officer. Contractor shall be responsible for hauling off-site and disposing of according to general conditions and codes of the governing jurisdiction.

1.6 PROJECT CONDITIONS

A. Environmental Requirements: Work shall occur only when weather and soil conditions permit in accordance with locally accepted practice.

B. Field Measurements: Verify actual grade elevations, service and utility locations, and dimensions of revegetation and construction contiguous with proposed RAP areas by field measurements before proceeding with work.

C. Interruption of Existing Services or Utilities: Do not interrupt services or utilities to facilities occupied by the Government or others.

D. Existing Conditions:
1. Utilities: Determine location of existing and proposed underground utilities. Perform work in a manner to avoid damage. Hand excavate as required.
2. Excavation: Maintain grade stakes set by others until removal is mutually agreed upon by parties concerned.

E. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit placement to be performed when beneficial and optimum results may be obtained.

PART 2 - PRODUCTS

2.1 RECLAIMED ASPHALT PAVEMENT (RAP)

Type: Reclaimed asphalt pavement. Shall be free of debris and contain no pieces greater than 1 inch in any dimension. Contractor shall screen or process RAP as needed to meet gradation requirements.

1. Color: Gray
2. Depth: Approximately 6” Depth

2.2 SOIL (not Topsoil)

A. Contractor shall furnish soil from native onsite materials, taken from areas below the soil, within the trail alignment, as approved by the Contracting Officer.

B. Soil shall be screened to ½-inch minus, and free of organic debris, roots, clay lenses or unsuitable material.

2.3 STABILIZATION EMULSION

A. Stabilization Emulsion: Eco Pave by Midwest Industrial Supply, 1-866-662-3878 or approved equal.

B. Emulsion shall be a polymer-enhanced, resin-based organic emulsion designed for use with RAP or aggregate surfaces.

C. Apply at manufacturer-recommended rates and installation procedures.

2.4 GEOTEXTILE

A. Type: Nonwoven needle-punched geotextile suitable for separation and stabilization on trails.

B. Mirafi 140N by TenCate or approved equal.

C. Salient Characteristics:
1. Grab Tensile Strength (ASTM D4632): 120 lb
2. Elongation at Break: ≥ 50%
3. Puncture Resistance (ASTM D6241): 320 lb
4. Apparent Opening Size (AOS, ASTM D4751): No. 70 sieve (0.21 mm)
5. Permittivity (ASTM D4491): 1.3 sec^{-1}
6. Roll size: Minimum 12.5 ft width to minimize seams.
7. Seaming: Seams shall be overlapped, not sewn. No sewn seams permitted in trail construction.

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas where the Work of this Section will be performed for compliance with requirements and conditions affecting installation and performance.
1. Verify that no foreign or deleterious material or liquid such as paint, paint washout, concrete slurry, concrete layers or chunks, cement, plaster, oils, gasoline, diesel fuel, paint thinner, turpentine, tar, roofing compound, or acid has been deposited in soil within the work area. Remove undesirable materials and debris prior to trail placement.
2. Verify that subgrade is uniform, compacted to a minimum of 95%, and graded to drain prior to placement of geotextile fabric and aggregate materials.
3. Remove sharp rocks, roots or debris that may puncture geotextile.
4. Verify that final grades are completed in accordance with the plans.

B. Proceed with installation only after unsatisfactory conditions have been corrected and approved by Contracting Officer.

3.2 QUALITY CONTROL

A. Mock-up: Provide field constructed sample installation of RAP and prepared subgrade.
   1. Mock-up to be 20 feet (20') x width of trail and located where directed by Contracting Officer. Mock-up shall include proposed edge and banding as applicable. Ensure firm and stable surface per accessibility requirements is met.
   2. Contracting Officer shall review mock-up within forty-eight (48) hours of notification by the contractor.
   3. Make necessary adjustments as directed by Contracting Officer.
   4. Obtain approval from Contracting Officer before proceeding with the Work.
   5. Retain and protect mock-up during construction as a standard for judging completed RAP paving work. Do not remove or destroy mock-up until work is completed.
   6. Accepted and properly maintained sample installations may remain in completed work if approved in writing by Contracting Officer.
   7. All work shall match accepted field mock-up.

3.3 PREPARATION

A. Protect structures, utilities, sidewalks, pavements, and other facilities, existing landscape areas, and trees from damage.

B. Install erosion-control measures to prevent erosion or displacement of soils.

C. Obtain acceptance of layout by Contracting Officer before excavating or installing. Make minor adjustments as required.

3.4 TRAIL BED PREPARATION

A. Cut earthwork to width of area to receive RAP paving to approximate depth section as specified on the Contract Plans. Remove, haul and dispose of excess material off site, or use on-site with approval of Contracting Officer.

B. Complete excavation required in sub-grade before fine grading and final compaction of sub-grade is performed. Extend sub-grade compaction one foot (1') beyond proposed edge of RAP or as indicated on plans.
   1. Subgrade shall be shaped to the line and grade required for positive drainage on all surfaces to receive stabilization emulsion.
   2. Where earth moving is required the sub-grade shall be compacted to ninety five percent (95%) standard proctor within two percent (2%) of the optimum moisture.
   3. Keep areas being graded or compacted shaped and drained during construction. Ruts greater than or equal to 1 inch deep in sub-grade shall be graded out and reshaped as required, and re-compacted before RAP placement.
   4. If the area is part of a cross slope it should drain in the direction of the slope no greater than two percent (2%). Ensure that no low spots exist so that ponding does not occur.

C. Trail clearing and geotextile placement shall be completed prior to RAP installation. Contractor shall protect placed geotextile from damage during all phases of RAP work.
   1. Unroll geotextile fabric in the direction of trail alignment. Do not stretch or pull taut.

2. Overlap edges a minimum of 18 inches, or as directed by Contracting Officer.
3. Anchor fabric with landscape staples or sod pins at a minimum rate of 1 per 3 square feet, or per manufacturer's recommendation.
4. Install fabric to cover entire subgrade width, extending a minimum of 12 inches beyond the edge of trail subgrade.
5. Protect fabric from exposure to sunlight for longer than 14 days.
6. Immediately place RAP material over geotextile after installation. Avoid driving equipment directly on geotextile.

3.5 RAP PLACEMENT

A. Place RAP over geotextile mat to a loose depth of 6 inches, maintaining full trail width as shown on drawings. Care shall be taken by the contractor to ensure geotextile is not damaged during spreading and leveling of RAP.
   1. Manual placement of RAP shall be required to spread and adequately place material to the width and depth required. Contractor may dump RAP in up to 3 stockpiles conducive for placement on trail.

B. Use mechanical spreading equipment no wider than the trail width; hand tools shall be used for shaping and finishing edges. Access new trail in approved locations by Contracting Officer.

C. Ensure material is evenly spread and roughly leveled prior to mixing and compaction.

D. Maintain a consistent cross slope across the trail as indicated on the drawings.

3.6 SOIL MIXING

A. Place a loose layer of soil over the placed RAP at a rate sufficient to achieve a 2-inch-deep blended layer of RAP and soil.

B. Mixing ratio shall be between 1/3 to 1/2 parts soil to 2/3 to 1/2 parts RAP by volume.

C. Mix the top 2 inches of trail surface using a rotary tiller or approved method.

D. Restore soil borrow areas to original contours and natural slope upon completion.

E. Moisture content shall be adjusted during mixing to achieve optimum conditions for compaction. Contractor shall apply water as needed to ensure mixed material holds shape when compressed but does not release free water.

3.7 STABILIZATION EMULSION – INTIAL APPLICATION

A. Spray stabilization emulsion uniformly over the loose soil prior to tilling and mixing.

B. Apply at manufacturer-recommended rates. Allow sufficient absorption time before tilling.

3.8 COMPACTION

A. Compact trail surface to a minimum of 90% modified proctor density.

B. A minimum of three overlapping passes with a smooth drum roller or other approved compaction equipment shall be made. If surface appears dry prior to compaction, contractor

shall apply a light mist of water to aid in particle cohesions. Excess moisture resulting in soft unstable conditions shall be avoided.

C. Add additional material and regrade areas showing dips, depressions, or surface irregularities prior to final compaction to remove all surface deformities.

D. Finished surface shall be free of protruding material greater than ¼ inch in any dimensions and meet accessibility guidelines for firmness and stability.

E. Contracting Officer will test with a rotational penetrometer ensuring that the trail has a firmness (penetration resistance) of less than 0.5 inches of penetration under a 50 lb load and that the stability (recovery after deformation) does not remain indented after penetration.

3.9 STABILIZATION EMULSION – FINISH APPLICATION

A. After final compaction, apply a finish coat of stabilization emulsion per manufacturer's instructions.

B. Contractor shall coordinate with manufacturer's representative to be on-site during application.

C. Repair or recompact any areas affected by final application process prior to acceptance.

3.10 MAINTENANCE AND REPAIRS:

A. GEOTEXTILE:
1. Inspect for tears or displacement before and during placement of RAP.
2. Replace damaged geotextile fabric prior to continuing trail work.

B. RAP:
1. Protect RAP from soil or other contaminates during and following installation.
2. Do not allow traffic on RAP for two to three days.
3. Areas that do not compact, become eroded or are degraded in visual quality and/or performance as determined by the Contracting Officer are to be removed and/or repaired. Obtain approval of repair methods from Contracting Officer prior to affecting repairs.
4. To repair, excavate damaged area leaving a minimum one inch depth of existing RAP. Complete installation per preparation, placement and application described above.
5. Do not allow traffic on repaired RAP for two-three days or until paving has fully cured.

3.11 CLEANUP AND PROTECTION

A. All areas shall be clean at the end of each workday.

B. The Contractor shall maintain protection during installation, curing, and maintenance periods.
1. Protect installed geotextile from exposure to UV light, excessive moisture, and construction traffic until covered with RAP. Replace or repair damaged sections at no additional cost to the Government.
2. Erect temporary fencing or barricades and warning signs as required protecting newly installed RAP areas from traffic, other trades, and trespassers. Maintain fencing and barricades throughout initial maintenance period and remove with approval of Contracting Officer.

C. Project completion: All debris, soil, trash, and excavated and/or stripped material resulting from RAP operations and unsuitable for or in excess of requirements for completing work of this Section shall be disposed of off-site.
   1. Remove any excess material from adjacent areas.
   2. Leave trail and surrounding areas in clean condition. Feather surrounding area grades to create a smooth and finished appearance.

D. Maintain protection during installation and maintenance periods. See Division 01. Treat, repair or replace damaged work as required.

3.12 QUALITY ASSURANCE

A. Refer to Division 1 Section Quality Requirements.

END OF SECTION 32 12 43

SECTION 32 13 00 – EXTERIOR CONCRETE

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Construction of concrete sidewalks, slabs, drainage chases, curbs, curb & gutter, and all other miscellaneous new exterior concrete construction.

1.2 RELATED REQUIREMENTS

A. Section 03 30 00 – Cast-in-place concrete

B. Section 31 23 20 - Street Excavation and Backfill.

1.3 SUBMITTALS

A. Per Section 03 30 00 – Cast-in-place concrete

B. Submit photo of finished concrete example as identified by Contracting Officer.

PART 2 PRODUCTS

2.1 MATERIALS

A. Per Section 03 30 00.

PART 3 EXECUTION

3.1 TESTING

A. The Contractor shall furnish Portland cement concrete tests to concrete placed.

B. One set of tests shall be required for every 50 cubic yards of concrete placed with a minimum of one test per project and one test per day. Separate pours during a given day shall have separate tests.

C. Testing will include slump, air content and temperature, collecting samples and performing strength testing per:

1. Sampling fresh concrete ASTM C-172

2. Slump ASTM C-143 or AASHTO T119

3. Air Content ASTM C-231 or C-173 or C-138 or AASHTO T152

4. Compressive Strength ASTM C-39 or AASHTO T22

5. Making and curing test specimens in the field ASTM C-31 or AASHTO T23

3.2 FINISHING

A. Finished surface of concrete shall match sample area of concrete as identified by the Contracting Officer and approved submittal.

END OF SECTION

SECTION 32 15 40 - CRUSHED STONE PAVING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes requirements for demolition, earthwork, grading, furnishing, and placement of crushed stone paving.
   1. Furnish and place crushed stone paving, bonded with fine aggregate, constructed on compacted sub-grade in accordance with these specifications and in conformity with the dimensions, typical cross section, and the lines and grades shown on the Contract Drawings. The locations where crushed stone paving and soil stabilizer will be used are shown on the Contract Drawings.

1.3 REFERENCES

A. ASTM C117 – Test Method for Materials Finer than No. 200 (75-um) Sieve in Mineral Aggregates by Washing.

B. ASTM C136 – Method for Sieve Analysis of Fine and Coarse Aggregates.

C. ASTM D4318 – Test Method for Liquid Limit, Plastic Limit and Plasticity Index of Soils.

1.4 SUBMITTALS

A. See Division 01 Section 01 33 23 “Submittal Procedures”.

B. Material Analysis: Contractor shall provide copies of the following test data required by ASTM:
   1. ASTM C136 - Sieve Analysis.
   2. ASTM C127 - Specific Gravity and Absorption.
   3. ASTM C131 - L.A. Abrasion.

C. Samples: Provide a one (1) quart sample of crushed stone material for approval.

D. Soil stabilizer product data.

E. Compaction Testing: The Contractor shall provide compaction testing results.

F. Installer Qualification Data: Submit Installer qualifications and experience on projects, including reference contact information.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Packaged Materials: Deliver packaged materials in original, unopened containers showing weight, certified analysis, name and address of manufacturer, and indication of conformance with state and federal laws if applicable.

B. Bulk Materials:
   1. Do not dump or store bulk materials near structures, utilities, walkways, and pavements, or on existing turf areas, plant materials or within critical root zones.
   2. Provide erosion-control measures to prevent erosion or displacement of bulk materials, discharge of soil-bearing water runoff, and airborne dust reaching adjacent properties, water conveyance systems, or walkways.
   3. Accompany each delivery of bulk materials with appropriate certificates.

C. Rejection of material.
   1. Evidence of inadequate protection or improper handling or storage shall be cause for rejection.
   2. Any product or material exhibiting signs of damage due to nonconformity to specifications or due to delivery, storage or handling shall be rejected by the Contracting Officer. Contractor shall be responsible for hauling off-site and disposing of according to general conditions and codes of the governing jurisdiction.

1.6 PROJECT CONDITIONS

A. Environmental requirements: Work shall occur only when weather and soil conditions permit in accordance with locally accepted practice.

B. Field Measurements: Verify actual grade elevations, service and utility locations, irrigation system components, and dimensions of plantings and construction contiguous with proposed crushed stone paving areas by field measurements before proceeding with work.

C. Interruption/Damage of Existing Services or Utilities: Do not interrupt or damage services or utilities. The Contractor is responsible for repair to services and utilities if interrupted or damaged.

D. Existing Conditions:
   1. Utilities: Determine location of existing and proposed underground utilities. Perform work in a manner to avoid damage. Hand excavation as required.
   2. Excavation: Maintain grade stakes set by others until removal is mutually agreed upon by parties concerned.

E. Weather Limitations: Proceed with installation only when existing and forecasted weather conditions permit planting to be performed when beneficial and optimum results may be obtained.

PART 2 - PRODUCTS

2.1 CRUSHER FINES

A. Type: Crushed granite stone or gravel. Shall be unused material free of shale, lay, friable materials, organics, and debris.
   1. Size Range: 3/8 inch maximum

Sieve Size	Percent Passing
2 inch	100
3/8 inch	100
No. 4	85
No. 8	63
No. 16	50
No. 30	39
No. 50	29
No. 100	18

   2. Color: Uniform grey or as approved by the Contracting Officer.

2.2 SOIL STABILIZER

A. Soil stabilizer or binder:
   1. StaLok or approved equal.
   2. Salient Characteristics:
      a. Non-toxic
      b. Organic
      c. Colorless
      d. Odorless powder

2.3 GEOTEXTILE

A. Type: Nonwoven needle-punched geotextile suitable for separation and stabilization on trails.

B. Product: Mirafi 140N by TenCate or approved equal.

C. Salient Characteristics:
   1. Grab Tensile Strength (ASTM D4632): 120 lb
   2. Elongation at Break: ≥ 50%
   3. Puncture Resistance (ASTM D6241): 320 lb
   4. Apparent Opening Size (AOS, ASTM D4751): No. 70 sieve (0.21 mm)
   5. Permittivity (ASTM D4491): 1.3 sec^{-1}
   6. Roll size: Minimum 12.5 ft width to minimize seams.
   7. Seaming: Seams shall be overlapped, not sewn. No sewn seams permitted in trail construction.

2.4

PART 3 - EXECUTION

3.1 EXAMINATION

A. Examine areas where the Work of this Section will be performed for compliance with requirements and conditions affecting installation and performance.
   1. Verify that no foreign or deleterious material or liquid such as paint, paint washout, concrete slurry, concrete layers or chunks, cement, plaster, oils, gasoline, diesel fuel, paint thinner, turpentine, tar, roofing compound, or acid has been deposited in soil within the work area.
   2. Verify that final grades are completed in accordance with the drawings.

B. Proceed with installation only after unsatisfactory conditions have been corrected and approved by Contracting Officer..

3.2 QUALITY CONTROL

A. Mock-up: Provide field constructed sample installation of stabilized aggregate and prepared subgrade.
   1. Mock-up to be (20') linear feet by width of trail and located where directed by Contracting Officer. Mock-up shall include proposed edge and surface stabilization specified.
   2. Contracting Officer shall review mock up within forty-eight (48) hours of notification by the contractor.
   3. Make necessary adjustments as directed by Contracting Officer.
   4. Obtain approval from Contracting Officer before proceeding with the Work.
   5. Retain and protect mock-up during construction as a standard for judging completed stabilized soil work. Do not remove or destroy mock-up until work is completed.
   6. Accepted and properly maintained sample installations may remain in completed work if approved in writing by Contracting Officer.
   7. All work shall match accepted field mock-up.

3.3 PREPARATION

A. Protect structures, utilities, sidewalks, pavements, and other facilities, turf areas, existing landscape areas, and trees from damage.

B. Install erosion-control measures to prevent erosion, displacement of soils, or discharge of water.

3.4 BLENDING STABILIZER

A. Stabilizer shall be thoroughly pre-mixed with aggregate at the rate of 15-lbs of stabilizer per 1-ton of aggregate or as directed by manufacturer. Drop spreading of stabilizer over pre-placed aggregate or mixing by rototilling is not acceptable. Stabilizer shall be mechanically pre-mixed per manufacturer's recommendations using an approved mechanical blending unit to adequately blend stabilizer with aggregate (Bucket blending is not an approved blending apparatus). Blend stabilizer and aggregate dry.

3.5 PLACEMENT OF STABLIZED CRUSHED STONE PAVING

A. Cut earthwork to width of trail/area to receive crushed stone paving to approximate depth section as specified on the Drawings. Remove, haul, and dispose of excess material off site, or use on-site with approval of Contracting Officer.

B. Complete excavation required in sub-grade before fine grading and final compaction of sub-grade is performed. Extend sub-grade compaction one foot (1') beyond proposed edge of crushed stone paving or as indicated on drawings.
   1. Where earth moving is required the sub-grade shall be compacted to ninety five percent (95%) standard proctor within two percent (2%) of the optimum moisture.
   2. Keep areas being graded or compacted shaped and drained during construction. Ruts greater than or equal to 1 inch deep in sub-grade shall be graded out and reshaped as required, and re-compacted before crushed stone paving placement.
   3. If the trail is part of a cross slope it should drain in the direction of the slope no greater than two percent (2%). Ensure that no low spots exist so that ponding does not occur.

C. Prior to placement of Crushed Stone Paving material, the sub-grade shall be proof rolled. Where soft spots are detected, scarify subgrade beneath Crushed Stone Paving trail to a minimum of six-inch (6") depth. Moisture treat and compact to a minimum ninety five percent (95%) proctor density as determined by ASTM D698 or AASHTO T-99. Take moisture density tests every two hundred fifty (250) lineal feet of trail or proof roll. Treat and compact sub-grade, leaving it 5-inches below final grade for placement of Crushed Stone Paving. Compact material and retest by proof rolling to achieve approval of Contracting Officer..

D. Install crushed stone paving only after excavation and construction work which might injure it have been completed. Install crushed stone paving, over compacted sub-grade in areas indicated on plan.

E. Compact to ninety-five percent (95%) of maximum density as determined by ASTM D1557.
   1. Maintain surface course moisture content within plus/minus three percent (± 3%) of optimum. Add water to quarry fines as required to achieve optimum moisture content and a uniform, compacted surface conforming to the finish grades indicated.
   2. Compact areas inaccessible to rolling by mechanical tamping.

F. Protect crushed stone paving from soil or other contaminates during and following installation.

G. Spread and compact additional crushed stone paving to achieve the required minimum compacted thickness per compaction requirements in this section

H. Do not allow any traffic on stabilized crushed stone paving for a minimum of two (2) days or until paving has fully cured.

3.6 MAINTENANCE AND REPAIRS:

A. Stabilized Crushed Stone Paving:
   1. Areas that do not compact, become eroded, or are degraded in visual quality and/or performance as determined by the Contracting Officer are to be removed and/or repaired.

The Contractor shall obtain approval of repair methods from the Contracting Officer prior to completing repairs.

3.7 CLEANUP AND PROTECTION

A. All areas shall be clean at the end of each workday.

B. The Contractor shall maintain protection during installation and maintenance periods.
   1. Erect temporary fencing or barricades and warning signs as required protecting newly installed Crushed Stone Paving areas from traffic, other trades, and trespassers. Maintain fencing and barricades throughout initial maintenance period and remove with approval of the Contracting Officer.

C. Project completion: All debris, soil, trash, excavated, and stripped material resulting from Crushed Stone Paving operations used for completing the Work in this Section shall be disposed of off-site.

D. Maintain protection during installation and maintenance periods. See Division 1. Treat, repair or replace damaged work as required.

END OF SECTION 32 15 40

SECTION 32 15 50 –WASHED ROCK

PART 1 GENERAL

1.1 SECTION INCLUDES

A. Washed Rock (for perimeter gravel beds).

B. Landscape Edgings.

C. Geotextile Landscape Fabric (for perimeter gravel beds)

1.2 SUBMITTALS

A. Submit shop drawings for approval all products noted.

B. The Contractor shall provide product data and manufacturer information including tensile strength, elongation, thickness, UV resistance and other material specifications for each geotextile fabric used on the Project.

B. Provide manufacture's certificate certifying that products meet or exceed specified requirements. Along with the statement that the material meets or exceeds the specifications, certificate must include name of the manufacturer, the chemical composition of the filaments or yarns, and other information fully describing the geotextile.

B. Submit detailed instructions on installation requirements from the manufacturer, including storage and handling procedures.

PART 2 PRODUCTS

2.1 WASHED ROCK

A. Water-torn, hard, durable gravel, washed free of loam, sand, clay and other foreign substances, and of the following size range and color:
   1. For bidding purposes
      a. Size Range: ¾" angular mountain granite
      b. Color: Brown tone to match building foundation rock/stone/concrete.
   2. Contracting Officer to select final materials.

B. Free from clay lumps, vegetable matter and deleterious material.

C. Submit one (1) quart sample for approval of rock.

2.2 LANDSCAPE EDGINGS

A. Steel Edging: Standard perforated commercial-steel edging, fabricated in sections of standard lengths, with loops stamped from or welded to face of sections to receive stakes.
   1. Edging Size: Perforated, 14ga thick by 4 inches (100 mm) deep.
   2. Stakes: 11ga, 8" overall.
   3. Finish: Weathering steel.

2.3 GEOTEXTILE FABRIC

A. Filter fabric shall meet the following requirements: Weight oz/s.y. 4.0, Grab tensile strength lbs. 145, thickness 15 mill., Grab tensile elongation 70%, trapezoid tear strength 75 lbs., Mullen Burst strength 175 psi, puncture strength 50 lbs., Apparent Opening size 70-100 U.S. Std. Sieve, permittivity 0.8sec., flow rate 60gpm/ft and resistance after 500 hours 50% strength required.

a. Products:

i. Basis of Design: Mirafi 140N. Product characteristics listed above.

PART 3 EXECUTION

3 .1 Subgrade Preparation: Prior to placing washed rock, shape, grade, and compact subgrade.

3 .2 Placing Washed Rock: Place to avoid segregation, in maximum layers of 3 inches. Compact by wetting. Roll with hand operated power roller of type and weight as approved by the Contracting Officer to obtain a dense, smooth, uniform texture.

3 .3 Steel Edging: Install steel edging where indicated according to manufacturer's written instructions. Provide a 2” gap every 20’-0” maximum for drainage. Anchor with steel stakes spaced approximately 36 inches (760 mm) apart, driven below top elevation of edging. Spot weld stakes to edging.

3 .4 Finishing: Grade each area to ensure positive drainage away from structures. Finish adjacent shoulders and slopes to required grade and cross section.

3 .5 Inspection: Final thickness of each area shall not vary more than ½ inch from indicated. Make measurements by means of test holes taken at random. Correct any variations in the thickness beyond the allowable ½ inch by adding material, shaping, and compacting.

END OF SECTION

SECTION 32 17 23 – PAVEMENT MARKINGS

PART 1 GENERAL

1.1 SECTION INCLUDES

A. The work included in this section consists of painting lines, words, and/or symbols on project roads or parking areas, in conformity with the lines and dimensions shown on the Drawings, complete.

1.2 SUBMITTALS

A. Submit for approval all pavement marking data sheets.

PART 2 PRODUCTS

2.1 PAVEMENT MARKING PAINT

A. Pavement marking paint shall be a waterborne acrylic latex white and/or lead-free yellow paint, ready mixed, conventional, fast dry, specifically made for concrete painting.

B. The exact composition shall be at the manufacturer's discretion except that the paint shall be 100 percent acrylic polymer based and the paint shall not contain any of the following ingredients:

1. Lead or chromate compounds, mercury, chlorinated solvents, hydrolysable chlorine derivatives, ethylene based glycol ethers and their acetates.

C. Pavement marking paint shall also meet the following requirements:

Parameter and Test	**White**	**Yellow**
Pigment, % solids ASTM D3723	68 Max 60 Min	68 Max 60 Min
Total Solids, % by weight ASTM D2369	75 Min	75 Min
Titanium Dioxide, lb/gal ASTM D4563, D1394	1.00 Min	0.15 Min
% Non-volatile Vehicle of Total Vehicle Weight ASTM D2697	41 Min	41 Min
VOC Content, g/L EPA Method 24	150 Max	150 Max
pH ASTM E70	9.6 Min	9.6 Min

Viscosity, Kreb Units ASTM D562 @ 77 degF	80-95	80-95
Grind, Hegeman Gage, minutes ASTM D1210	2 Max	2 Max
Deviation in % Weight/Gallon (From Mfr. Specified Weight)	+-0.30 Max	+-0.30 Max
Daylight Reflectance ASTM D2805	85 Min	59 Min
Contrast Ratio, 15 mils wet ASTM D2805	0.92 Min	0.88 Min
Dry Time, 15 mil wet film, 65 % RH, minutes ASTM D711	10 Max	10 Max
Dry Through, 15 mil wet film, 90 % RH, minutes ASTM D1640	125 Max	125 Max
Freeze-Thaw ASTM D2243	Pass	Pass
Scrub Resistance, cycles ASTM D2486	600 Min	600 Min
Bleeding Ratio ASTM D969	0.95 Min	0.95 Min

D. Pavement marking paint shall use titanium dioxide meeting ASTM D476, Type I or Type II.

E. Performance and characteristics criteria:

1. Flexibility and Adhesion: Paint shall adhere to road surface, forming smooth continuous film one minute after application. Paint shall adhere firmly without showing cracking or flaking.

2. Paint Drying: Tack free by touch and will not transfer to vehicle tires within 10 to 15 minutes after application.

3. Water Resistance: When immersed in distilled water at 77 degrees F for 24 hours, after a 15 mil wet film thickness has been applied to a glass plate and dried at 77 degrees F for 72 hours, paint shall not show blistering or adhesion loss after air drying for 2 hours.

4. Bleeding: When tested under ASTM D969, paint shall not show perceptible bleeding when painted on a bituminous surface.

5. Freeze-Thaw Stability: When tested under ASTM D2243, paint shall not show coagulation or viscosity change exceeding 10 Kreb Units.

6. Static Heat Stability: After heating a pint of paint in an oven at 140 degrees F for seven days, allow paint to cool to 77 degrees F. Paint viscosity shall not vary by more 10 Kreb Units from the original viscosity measured at 77 degrees F per ASTM D562

PART 3 EXECUTION

3.1 GENERAL PAVEMENT MARKING PAINT APPLICATION REQUIREMENTS

A. Follow the manufacturer's requirements for pavement cleaning and paint application or as follows, whichever is more stringent:

1. Only apply paint to a dry surface. Clean the pavement by blowing and/or sweeping of all loose rock, dirt, and debris immediately before applying paint.

2. Do not heat paint to exceed 110 degrees F before and/or during application.

3. Apply paint when the ambient temperature is at least 50 degrees F and rising. Stop application when the temperature is 50 degrees F and falling, and when rain, fog, or other weather adverse to the paint during its drying time is imminent. Remove and replace all defective pavement marking paint damaged by weather at Contractor's sole expense.

4. Agitate paint for 5 to 15 minutes prior to application to ensure even distribution of pigment. Apply the paint at 15 mils wet thickness in a single application, plus or minus 1 mil.

5. Re-paint, at Contractor expense, all paint markings represented by paint samples where any specified property is outside 20 percent of the specified value.

3.2 PAINTING TRAFFIC OR PARKING LINES

A. Apply paint by hand or mechanical means consistent with the scope of the project. Assure the width and layout of the stripes or the area to be painted meets the Drawings and details.

B. Apply paint pneumatically, using guidelines and templates as necessary to control application. Manually paint numbers, letters and symbols. Prevent splattering and overspray when applying markings.

C. Where shown on the Drawings to paint curbs or other concrete, paint the top and traffic sides by uniformly applying one coat of yellow paint. Paint curbs or other concrete after it has cured for at least 30 days after being placed. Apply the paint at a rate to ensure that the curb or other concrete surface is completely covered and hidden.

D. Apply paint with equipment manufactured specifically for that purpose and using workmen experienced in operating such equipment. Assure that the paint applicator is shielded to avoid spraying of paint outside of the designated width of line. Contractor shall also assure that the equipment is capable of painting a strip or stripes of the specified width with a tolerance of plus or minus ¼-inch.

E. Use hand operated equipment such as walk-behind stripers, hand sprayers, or stencil trucks to stripe stop bars, walk areas, or other areas not readily accessible to a pavement striping machine. Do not use hand brushes or rollers, except in areas inaccessible to equipment.

F. For a 4-inch wide pavement stripe, apply paint at a rate of at least 250 feet per gallon, but not more than 275 feet per gallon, unless recommended otherwise by the manufacturer. In other areas, apply

paint at a rate of between 80 and 90 square feet per gallon, unless recommended otherwise by the manufacturer.

G. Protect all markings from vehicular and pedestrian traffic until paint is dry and track free by placing appropriate guarding or warning devices as needed. Follow manufacturer instructions or use a minimum of 45 minutes dry time. Correct any markings smeared or otherwise damaged at no additional expense to the Owner. If a vehicle or other object crosses a marking and tracks it, or if splattering or overspray occurs, eradicate the affected marking and resultant tracking and apply new markings.

H. Repair or replace lines and markings, which after application and curing, do not meet the following criteria:

1. Incorrect location
2. Insufficient thickness, width, coverage, or retention
3. Uncured or discolored material
4. Insufficient bonding

3.3 CLEANING

A. Collect and legally dispose of all residues and debris from painting operations.

END OF SECTION

SECTION 32 20 00 – SITE IMPROVEMENTS

PART 1 - GENERAL

1.1 SECTION INCLUDES

A. Site improvements comprise the following elements:
   1. Tree protection for existing trees and vegetation to remain.
   2. Soil Preparation
   3. Restoration Revegetation:
   4. Invasive species removal
   5. Watering by water truck
   6. Pavements and Gravel Surfacing:
      a. Gravel surfaces where construction activities required the removal of existing gravel roads and parking areas to install utilities.

B. Products: Where specific products are required or allowed, use products complying with the additional requirements specified elsewhere.

1.2 RELATED REQUIREMENTS

A. Section 02 21 19 – Utilities Survey
B. Section 02 41 00 - Site Demolition
C. Section 31 23 20 – Street and Site Excavation, Grading and Backfill
D. Section 31 25 00 – Erosion and Sediment Control
E. Section 31 30 00 - Site Earthwork.
F. Section 32 20 00 - Site Improvements.
G. Section 33 14 15 - Site Utilities.
H. Section 33 20 00 – Sewer Wet Well Design Criteria
I. Section 33 30 00 – Submersible Centrifugal Pump Design
J. Section 33 40 00 – Special Construction

1.3 REFERENCES

A. ASTM C117 – Test Method for Materials Finer than No. 200 (75-um) Sieve in Mineral Aggregates by Washing.

B. ASTM C136 – Method for Sieve Analysis of Fine and Coarse Aggregates.

C. ASTM D4318 – Test Method for Liquid Limit, Plastic Limit and Plasticity Index of Soils.

D. ASTM D698 or AASHTO T-99 – Proctor density of subgrade material.

E. AASHTO M147 – Standard Specification for Materials for Aggregate and Soil Aggregate Subbase, Base and Surface Courses.

1.4 SUBMITTALS

A. Tree Protection
   1. Tree Protection Plan: Submit tree protection plan for approval by the CO.

2. Proposed methods and schedule for effectuating tree and other plant protection shall be submitted for approval. Contractor shall submit construction schedule which includes a time frame for work near existing plants. Approval of such shall be obtained from the CO prior to commencement of construction near tree protection areas.
3. Proposed methods, materials, and schedule for root pruning, branch pruning, and other tree maintenance shall be submitted for approval.
    a. The CO shall mark the location of root pruning lines in the field prior to the operation.
    b. If possible, root pruning should occur between autumnal leaf fall and spring foliation.
    c. Root pruning during the growing season shall require approval of the CO.
4. Maintenance Schedule: Submit maintenance schedule to CO for approval.
5. Quality Control Submittals:
    a. Certificates: State, Federal and other inspection certificates shall accompany invoice for materials showing source or origin. Submit it to Contracting Officer prior to acceptance of material.
    b. Material Analysis: Provide soil conditioner analysis performed no more than three months prior to delivery to site.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Handle and store material to preserve its quality and fitness for the work. Stored material approved before storage may again be inspected before use in the work. Locate stored material to facilitate prompt inspection.

1.6 PROJECT/FIELD CONDITIONS

A. Soil Preparation
    1. General:
        a. Do not perform work when climate and existing site conditions will not provide satisfactory results (e.g. wet, freezing).
        b. Vehicular site access shall be limited to the area(s) indicated on the drawings or as defined by the Contracting Officer.
        c. Damage to natural areas, pavements, underground utilities, and other improvements shall be repaired by the contractor at no additional cost to the NPS.

1.7 ENVIRONMENTAL PROTECTION

A. Protection of Natural Resources: Comply with applicable regulations and these specifications. Preserve the natural resources within the project boundaries and outside the limits of work performed under this Contract in their existing condition or restore to an equivalent or improved condition as approved by NPS.

B. Construction Zone: Arrange construction activities to minimize pollution (i.e., erosion, trash, etc.) to the maximum practical extent.

C. Clearing, excavation, and grading shall be limited to those areas of the project site necessary for construction. Minimize the area exposed and unprotected.

D. Clearly mark and delineate the limits of work activities.

E. Equipment shall not be allowed to operate outside the limits of work or to disturb existing vegetation.

F. Excavation and grading shall be completed during the dry season to the maximum extent possible

G. Material should be stored away from locations where water is present to the greatest extent practicable.

1.8 CONSTRUCTION EQUIPMENT CLEANING

A. Contractor equipment and vehicles will be pressure washed to remove seed-bearing material prior to entering the park. Contractor to follow requirements within FP-14= The 2014 Standard Specifications for the Construction of Roads and Bridges on Federal Highway Projects, WEED WASHING OF EQUIPMENT & USE OF WEED FREE MATERIAL SOURCES- (in FP-14 under 107.10)

B. All vehicles and equipment entering the project area must be clean of plant materials and dirt and free from oil leaks and are subject to inspection. Wash all construction and restoration/ revegetation equipment (including the inside beds of vehicles hauling materials) to thoroughly remove all dirt, plant, and other foreign material prior to entering the project. Particular attention must be shown to the under carriage, mower decks (top and bottom) and any surface where soil containing exotic seeds may exist. Allow the NPS to inspect each piece of equipment before entering the project. Provide the cleaning and inspection records to the CO. Equipment found operating on the project that has not been inspected or has oil leaks shall be shut down and subject to citation.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Before delivery of rock, sand, aggregate, gravel, soil or other natural material to the project from a material source outside the project area, Contractor shall obtain clearance or eradicate exotic plants by:
   1. Providing certification by a county extension agent, agronomist, or other qualified invasive species expert that the source is free from exotic plants; or
   2. Notifying the CO at least 1 month before beginning operations or starting crushing to allow for investigation for exotic plants. If exotics are found, the CO will determine if the upper portion of the source will need to be stripped or sprayed with an herbicide approved by the CO. If spraying is required, provide a licensed operator to spray according to applicable state regulations. Do not spray any herbicides until approved in writing by the CO. Spraying or stripping of material does not, necessarily, constitute approval.
   3. If exotic plant clearance is not obtained, heat all material to 150 degrees Centigrade to ensure sterilization of any exotic plants before delivery of uncertified material. Do not perform hearing operations within park boundaries. Do not stockpile the heated material outside the park boundaries prior to delivery. These requirements are waived if the contractor uses a source that meets the requirements stated in items (1) and (2) above.

2.2 EROSION CONTROL MATERIALS

A. A. On disturbed slopes greater than 3H:1V and less than 2H:1V furnish the following:
   1. Wattles meeting the requirements of specification 01 57 23.
   2. Space wattle at the top of each slope and bottom and every 10’ O.C.

B. On disturbed slopes greater than 2H:1V furnish the following:
   1. Furnish Rolled Erosion Control Product & wattles (see above for type and spacing).
   2. Furnish ECB’s meeting the requirements of specification 01 57 23 and the properties below.
   3. The erosion control blanket shall be used to prevent surface erosion and enhance vegetation establishment.
   4. The ECB shall be mechanically (stitch) bonded on two inch centers between two bio-degradable, jute/scrim nets. Excel excelsior matrix consists of curled, machine produced fibers with greater than eighty percent longer than six inches. Stitching is comprised of cotton thread in natural color.
   5. The ECB shall be identified by the following properties:

Property	Test Method	Value	Unit
Tensile Strength	ASTM D6818	12.5 (MD), 12.5 (TD)	lb/in
Elongation	ASTM D6818	5.0 (MD), 5.0 (TD)	%
Mass per Unit Area	ASTM D6566	11.5	oz/yd^2
Thickness	ASTM D1777	10.0	mm
Light Penetration	ECTC TASC 00197	50	% open

2.3 CONSERVED TOPSOIL

A. Topsoil refers to the uppermost soil horizon which includes organic duff and other materials capable of supporting vegetation. Break down live vegetation as described below and incorporate into the topsoil during removal. Rocks less than 2-inches in diameter and branches less than 2-inchesin any dimension may be left in the topsoil.

B. Excavate material conserved topsoil from the excavation areas that is reasonably free from hard soil, rock, clay, toxic substances, litter, or other deleterious material, and is suitable for growth of grass, cover crops, or native vegetation.

C. Conserve topsoil from excavation and embankment foundation areas. Remove the topsoil in one pass to a minimum depth of 6 inches. Use equipment capable of excavating small isolated pockets

of topsoil. Manually remove topsoil in areas where topsoil cannot be removed mechanically. Do not mix topsoil with subsoil. Do not stockpile topsoil longer than three months. Stockpile topsoil in windrows less than 3 feet high. Do not compact or drive upon topsoil during removal.

D. Identify removal areas and stockpiles with stakes so that topsoil may be replaced in the same locations. If it is impractical to windrow topsoil adjacent to the disturbed area, remove the topsoil and stockpile at designated areas as described above.

E. Provide diagrams of each stockpile storage areas so they can be identified if the stakes are lost or removed.

F. Haul excess topsoil to the nearest area requiring topsoil, but lacking in quantity as directed by NPS.

G. Do not conserve topsoil from the locations with severe concentrations of noxious weeds.

H. Place conserved topsoil on finished slopes according to PLACING TOPSOIL section in the specification.

2.4 CLEARING AND GRUBBING

A. Clearing. Within the clearing limits clear trees, brush, downed timber, and other vegetation as follows:
1. Cut trees so they fall within the clearing limits;
2. In areas of cut slope rounding, cut stumps flush with or below the final ground-line;
3. In areas outside the excavation, embankment, and slope rounding limits, cut stumps to within 6 inches (150 millimeters) of the ground; and
4. Trim tree branches that extend over the construction surface and shoulders to attain a clear height that prevents damage to the tree by equipment. If required, remove other branches to present a balanced appearance. Trim according to accepted agriculture practices.

B. Grubbing. Grub deep enough to remove stumps, roots, buried logs, moss, turf, or other vegetative debris as follows:
1. Grub areas to be excavated, except for cut slope rounding areas
2. Grub embankment areas. Undisturbed stumps less than 24 inches in diameter may be left in place if they protrude less than 6 inches above the original ground and will be covered with more than 48 inches of embankment. Remove all other stumps;
3. Grub pits, channel changes, and ditches only to the depth necessary for the excavation; and
4. Backfill stump holes and other grubbing holes with backfill material to the level of the surrounding ground.

2.5 FINAL SITE PREPARATION

A. Following site compaction:
1. Site compaction shall be defined as any area where equipment accessed, drove on, or parked, including standard trucks. It may also include areas where materials were stockpiled. A National Park Service representative will define all areas designated compacted post construction.
2. Scarify or rip the site prior to topsoil application. Rip all areas compacted to a depth of 12 inches. Ripping shall be a minimum of 12” depth, 12” spacing between teeth. Subgrade to be smooth prior to topsoil placement.

3. All excess spoils and excess cobbles shall be removed by Contractor off site and disposed of outside the park.

B. Following construction (areas not compacted but disturbed):
   1. Roughen all areas requiring topsoil to create a minimum of 2-inch deep grooves parallel with the slope contours during dry weather. Leave the soil surface roughened such that obvious furrows are eliminated and a clear break between topsoil and subgrade layers is not produced.

2.6 PLACING TOPSOIL

A. Place a minimum of 6 inches of topsoil in all areas requiring topsoil. Do not place topsoil when the ground or topsoil is frozen, excessively wet, or otherwise in a condition detrimental to the work. Use all conserved topsoil before furnishing topsoil. NPS to approve final topsoil placement. Keep the roadway surfaces clean during hauling and spreading operations.

B. Spread topsoil to a depth that, after settlement, provides the required depth. Break clods and lumps with harrows, disks, or other appropriate equipment to provide a uniform textured soil.

C. Remove and dispose of all stumps, roots, and branches larger than 2-inches in any direction, and rocks larger than 2-inches upon completion of processing conserved topsoil as described in Grubbing. Clean and process all topsoil conserved from the project.

D. Place topsoil onto ledges, pockets and depressions on slopes by mechanical or hand labor means.

E. Roughen the surface to create 2-inch deep groves perpendicular with the slope in areas steeper than 1:3 slope. Ensure work is performed in dry weather.

F. Place conserved topsoil before October 30th of the year it is salvaged.

G. Place topsoil by hand in areas where machine placement is not allowed or practical.

H. A National Park Representative will approve all work upon completion.

I. Seeding and planting will be completed by the National Park Service.

2.7 GRAVEL SURFACING

A. Aggregate Base Course
   1. Material shall meet the requirements of Wyoming Public Works Specification Section 02231 Type H or Type W.
   2. Placed at minimum depths of 4 inches.
   3. Surface shall be smooth.

B. Existing utility structures shall be adjusted to meet the new finished pavement grades as required.

2.8 BASES, SUB-BASES, AND CRUSHED AGGREGATE

A. Prepare subgrade in accordance with Section G10 Site Preparation.

B. Compact aggregate base course at optimum moisture content to 95 percent maximum dry density per AASHTO T180.

2.9 EXTERIOR FENCING

A. Tree Protection Fencing
   1. Existing trees to be protected during demolition and construction.
   2. Tree protecting fencing shall be installed prior to any site activity and shall remain until its removal is authorized by the Contracting Officer.
   3. A continuous line of fencing may be provided at the limits of disturbance line to protect existing trees that are to remain.
   4. Tree protection fence material, size, and installation to be approved by Contracting Officer prior to installation.
   5. A standard tree protection sign shall be mounted on tree protection fencing at 50-foot intervals.
   6. Contractor to use caution during removal of trees to minimize injury to tree root systems to remain.

PART 3 - DESIGN CRITERIA

3.1 BASIC FUNCTION

A. Code: Make all portions of the project comply with the code.

B. Provide built elements and site modifications as required to fulfill the needs described in the project program and POR drawings.

C. Provide pavements and surfacing that are adequate in extent and sufficiently durable to accommodate without damage the types of traffic that can be reasonably anticipated for the facility type and intended user population.

D. Where site improvements are integral with elements defined within another element group, meet requirements of both element groups.

E. In addition to the requirements of this section, comply with all applicable requirements of Section DC G10 – Site Preparation Criteria.

3.2 HEALTH AND SAFETY CRITERIA

A. Fire Resistance: Design and select materials to provide fire resistance in accordance with code.

B. Prevention of Accidental Injury: As required by code and as follows:
   1. Avoid safety hazards whenever possible where services much involve flammable materials or hazardous operations and comply with code.

3.3 DURABILITY CRITERIA

A. Expected Service Life Span:
   1. Gravel driving surface: 15 years under normal usage.

B. Weather Resistance: Design and select materials to minimize deterioration due to precipitation, sunlight, ozone, and normal temperature changes.
   1. Weather resistance requirements apply to all components exposed to the outdoor environment.
   2. Deterioration includes corrosion, shrinking, cracking, spalling, delamination, abnormal oxidation, decay, and rot.
   3. Surfaces Exposed to View: Deterioration adversely affecting aesthetic life span includes color fading, crazing, and delamination of applied coatings.
   4. Joint Components and Penetration Seals: Capable of resisting expected thermal expansion and contraction. Use overlapping joints that shed water wherever possible.
   5. Service Temperature: Low temperature equal to historically recorded low. High temperature equal to that expected due to any combination of air temperature and heat gain from solar and other sources.
   6. Freeze Thaw Resistance: Adequate for the climate of the project.

C. Temperature and Humidity Endurance: Design equipment to endure temperature and humidity that will be encountered and to resist damage due to thermal expansion and contraction.

D. Impact Resistance: Design and select materials to resist damage due to impact in accordance with code and the following:
   1. Minimize damage from high winds.
   2. Minimize damage due to potential vandalism.
   3. Natural hazards: Design to resist damage from perching, nesting, and feeding birds.

E. Water Resistance: Design and select materials to provide resistance to normal wear and tear in accordance with code.

SECTION 32 91 13 – SOIL PREPARATION

PART 1 GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. The Work of this Section includes preparation of soil for the purpose of soil conditioning for all seeding and planting areas.
   1. Soil preparation consists of decompaction, soil conditioning and fine grading the topsoil. Soil preparation as specified herein MUST precede all seeding and planting.
   2. Seeding shall be performed by Park once soil preparation is accepted by the Contracting Officer.
   3. Dates for seeding and required soil preparation are October 1 – October 31 unless advised otherwise by the Contracting Officer.

1.3 PRE-INSTALLATION MEETING

A. Conduct meeting at Project site with Contracting Officer to coordinate the process with other trades, to coordinate equipment movement within landscape improvement areas and to avoid soil compaction, to review proposed methods of placement and maintenance. Review project limits, existing survey data, including utilities, and any additional supplementary reports and plans. This meeting shall be coordinated by the Contractor, and comply with requirements in Division 01.

1.4 DEFINITIONS

A. Finish Grade: Elevation of finished surface of planting soil.

B. Planting Area: Areas to be planted.

C. Planting Soil: Standardized topsoil; existing, native surface topsoil and existing, in-place surface soil,

D. Subgrade: Surface or elevation of subsoil remaining after excavation is complete, or the top surface of a fill or backfill before planting soil is placed.

E. Subsoil: All soil beneath the topsoil layer of the soil profile and typified by the lack of organic matter and soil organisms.

F. Surface Soil: Soil that is present at the top layer of the existing soil profile at the Project site. In undisturbed areas, the surface soil is typically topsoil; but in disturbed areas such as urban environments, the surface soil can be subsoil.

G. Compacted Soil: Soil where bulk density or resistance exceeds acceptable limits for root penetration. Compaction shall be defined using a handheld, spring loaded soil penetrometer which is capable of reading soil resistance from 0-300 psi, with depth increments to at least 24 inches. Compaction is defined as a reading greater than 250 psi in the upper 12 inches of

soil OR continuous refusal of the penetrometer within the top 12 inches. Testing shall occur every 10 feet in a grid pattern.

1.5 SUBMITTALS

A. See Division 01 Section 01 33 23 “Submittal Procedures”.

B. Field quality-control reports.

C. Product Data: For each type of product.
   1. Include recommendations for application and use.
   2. Submit a delivery schedule at least forty-five (45) calendar days prior to the first day of delivery.
   3. Submit a list of equipment to be used for topsoil placement operations two (2) weeks prior to starting work.
   4. Substitutions of approved products or the desired quantity is not available will review only on the basis of written proof of unavailability.

D. Weekly Inspection and Maintenance Reports; Submit a weekly inspection report once material has been placed for the first four months and then monthly until the maintenance period has been satisfied.

E. Obstructions Below Ground Reporting: When at grade obstructions prevent soil procedures to take place, the Contractor shall submit a plan showing proposed adjustments to locations to the Contracting Officer.

F. Soil Preparation Damage Reporting: Contractor shall document the site conditions (weather, etc. when damage occurred), provide photos of damage, describe the damage, and file this information with the Contracting Officer within forty-eight (48) hours.

G. Materials List: The Contractor shall submit to the Contracting Officer for approval a complete list of all materials to be used during this portion of the work prior to delivery of any materials to the site. Include complete data on source, amount and quality.

H. Construction Equipment Cleaning Records: Submit the cleaning and inspection records to the Contracting Officer.

I. The Contracting Officer reserves the right to reject materials that fail to meet specified requirements.

1.6 CLOSEOUT SUBMITTALS

A. See Section 01 7700 - Closeout Procedures.

1.7 DELIVERY, STORAGE, AND HANDLING

A. Notify Contracting Officer of delivery schedule two weeks in advance so material can be inspected upon arrival at project site. Immediately remove unacceptable material from project site.

B. Construction Equipment Cleaning: All vehicles and equipment entering the project area must be clean of plant materials and dirt and free from oil leaks and are subject to inspection. Wash all construction and restoration/vegetation equipment (including the inside beds of vehicles hauling materials) to thoroughly remove all dirt, plant, and other foreign material

prior to entering the project. Particular attention must be shown to the under carriage, mower decks (top and bottom) and any surface where soil containing exotic seeds may exist. Allow the NPS to inspect each piece of equipment before entering the project. Provide the cleaning and inspection records to the Contracting Officer. Equipment found operating on the project that has not been inspected or has oil leaks shall be shut down and subject to citation.

C. Soil Preparation Schedule – Within forty-five (45) calendar days of Notice to Proceed submit electronic Word documents and PDF of proposed soil preparation schedule for all aspects of soil preparation procedures, including but not limited to; indicating dates for subgrade approval, site soil preparation, decompaction, topsoil and mulch placing, seeding and coordination with plant procurement & delivery, plant material storage, staging areas, list of equipment for soil preparations, erosion and sediment control, storage procedures, and schedule for seeding for approval. Schedule all Work during specified planting seasons, indicate optimal time for planting. Once accepted, revise dates only as approved in writing, after documentation of reasons for delays.

D. The Contractor shall be responsible for all certificates of inspection of materials that may be required by federal, state or other authorities to accompany each shipment of materials and on arrival; the certificates along with the accepted delivery invoice, shall be filed with the Contracting Officer.

E. Material will be inspected upon arrival at project site. Contracting Officer will reject any opened or unacceptable materials as described above.
   1. Immediately remove unacceptable material from job site.

F. Bulk Materials:
   1. Do not dump or store bulk materials near structures, utilities, walkways and pavements, or on existing plants.
   2. Provide erosion-control measures to prevent erosion or displacement of bulk materials; discharge of soil-bearing water runoff; and airborne dust reaching adjacent properties, water conveyance systems, or walkways.
   3. Accompany each delivery of bulk materials with appropriate certificates.

1.8 PROJECT/SITE CONDITIONS

A. Grand Teton National Park – 4 Lazy F Ranch; Elevation: ±6,475 ft.

B. Work Scheduling: Proceed with and complete landscape work as rapidly as portions of the site become available, working within the specified planting season and approved schedule.

C. Soil Preparation Coordination:
   1. Subgrade - verify that existing site conditions are as specified and indicated on drawings before beginning work under this Section.
   2. All invasive species and other vegetation are to be removed prior to soil preparation by the Government.
   3. Coordinate soil preparations with seeding installation schedules.
      1) Seeding shall take place between October 1 and October 31 or when all the following conditions are met:
         a) Average soil temperatures are below 48 degrees, typically between October and November

b) Soil is decompacted and all preparation work is complete
c) Soil seed bed is friable and is soft enough to show a footprint and firm enough to hold seed in the top ¼
d) Soil is not excessively wet or muddy
e) Soil is not frozen
f) Snow is less than ¼" deep
g) Following seeding no vehicle or equipment will be allowed on the seeded area and minimize all other ground disturbance, including excessive walking, in seeded areas.
h) Seed will be applied by drill-seeding to the highest extent feasible.
i) Areas that cannot be drill-seeding will be seeded with hand seeding method.

4. Prior to the application of seed, the area shall be recontoured to final grade, topsoil placed, and soil thoroughly decompacted. Follow all of the specifications in Section 32 91 13 "Soil Preparation".
    1) Until the areas meet these conditions the area shall not be seeded.
    2) Undertake a walk-through investigation with the Contracting Officer to ensure the conditions are met for seeding operations.
    3) Review areas to be seeded with Contracting Officer for determination of where additional soils may be required for establishing seed mix.
5. Seeding shall occur within 48 hours of decompaction. If a precipitation event takes place between decompaction and seeding the site shall be scarified down to 2 inches, as determined by the Contracting Officer.
6. Coordinate soil preparation actions, planting and seeding with Contracting Officer.
7. Coordinate erosion and sediment control procedures and maintenance. Reference Section 31 25 00 "Erosion and Sediment Control".

D. Final work schedule will be coordinated with and approved by Contracting Officer.

E. Coordinate approved hours of soil preparation with Contracting Officer prior to operations.

F. Do not perform work when climate and existing site conditions will not provide satisfactory results (e.g. wet, freezing). The Contracting Officer reserves the right to deny soil preparation on days that are deemed to be unfavorable for installation. Proceed with soil preparation only when existing and forecasted weather conditions permit soil preparation to be performed when beneficial and optimum results may be obtained. Apply products during favorable weather conditions according to manufacturer's written instructions and warranty requirements.

G. Contractor shall have full use of the approved designated fenced or flagged in project area for specific construction operations during the construction period. Contractor's use of the site is limited to minimizing site disturbance to the extent possible and by the Government's right to perform work or to retain other contractors on portions of Project. There will be no visitor access within the designated fenced in approved project areas.

H. All work is being completed within a historic site and must adhere to the Secretary of Interior's Standards, as per Section 01 35 91 "Historic Preservation Treatment Procedures". Work must be completed without damage to without damage to fencing, trees to remain, trails, roads, or other property. The Contractor will be financially liable to the park for any costs incurred as a result of repairing and/or replacing any of the aforementioned items or for

the value of the item as assessed by an archeologist, biologist, government and the Contracting Officer.

I. All work is being completed within a natural resource area and must adhere to Section 01 35 13.05 “Environmental Protection”. Protect natural resources in and around the construction site including but not limited to requirements for wildlife protection, inspection and maintenance of equipment, vegetation/site protection, and pollution prevention.

J. Contractor’s operations shall be conducted reasonably to prevent rutting or other soil damage. No work shall be allowed for at least a 24-hour period following a rain accumulation of one (1) inch or more during the proceeding 24-hour period. After any work stoppage, it will be the responsibility of the Contracting Officer to determine when work may re-commence.

K. All vehicles and other heavy equipment involved in the completion of this contract will be restricted to roads, parking areas, and road shoulders no more than 10’ from the edge of the road surface unless given case-specific permission by the Contracting Officer. Off road equipment use is solely at the discretion of the Contracting Officer and subject to change based on current or anticipated conditions. Rutting or other resource damage will not be tolerated. All work from trails will be considered off road and must adhere to the same guidelines and restrictions.

L. All vehicles should be equipped with absorptive pads while on park property in order to quickly contain any spills of oil or hydraulic fluid. Any such spills must be reported to the Contracting Officer as soon as is safely possible.

M. Damage to natural areas, existing grades, vegetation, pavements, underground utilities, and other improvements and areas of protection, caused during work under this section, shall be repaired to original condition or better at no additional cost to the Government.

N. Vehicular accessibility on site shall be as directed by Contracting Officer. Repair damage to prepared topsoil and existing surfaces, caused by vehicular access and movement during work under this section, to original condition at no additional cost to the Government.

O. Existing conditions:
   1. Utilities: Determine location of underground utilities. Perform work in a manner to avoid possible damage. Hand excavate, as required.
   2. Any damage to underground piping or wiring arising out of work of this section must be corrected and repaired by the Contractor to the satisfaction of the Contracting Officer.
   3. Damage to Other Improvements: All costs for repair or replacement of any damage to other work done on-site or adjacent properties by soil preparation procedures shall be borne by the Contractor completing soil preparation.
   4. Excavation: Maintain grade stakes set by others until removal is mutually agreed upon by parties concerned.
   5. Existing grades and vegetation damaged during soil preparations shall be restored at the Contractor’s expense.
   6. Excavation: When conditions detrimental to soil preparations are encountered, such as rubble fill, adverse drainage conditions, noxious materials or obstructions, notify Contracting Officer before planting.
      a. Obstructions Below Ground: When obstructions below finished grades prevent soil preparation, the Contractor shall submit a plan showing proposed adjustments to

locations to the Contracting Officer.

7. If weeds / invasive species are present on site, Government to assume responsibilities for removal.

P. Field Measurements:

1. Verify actual grade elevations, service and utility locations, and dimensions of plantings and construction contiguous with new plantings by field measurements before proceeding with soil preparation work.

1.9 QUALITY ASSURANCE

A. Perform the following Field Tests, Inspections and Remedial Action:

2. Compaction: Test planting soil compaction after placing each lift and at completion using a densitometer or soil compaction meter calibrated to a reference test value based on laboratory testing according to ASTM D698. Space tests at no less than one for each 1000 sq ft.
3. Soil will be considered defective if it does not pass tests.
4. Prepare test reports.
5. Inspection: Provide notice to the Contracting Officer requesting inspection at least seven (7) days prior to anticipated date of completion.
6. Deficiencies: The Contracting Officer will specify deficiencies to Contractor, who shall make satisfactory adjustments and shall notify Contracting Officer for final inspection.

PART 2 - PRODUCTS

2.1 MATERIALS

A. On-Site Topsoil or Topsoil from another approved source within the Park as required in Section 32 91 19.13 “Topsoil”.

PART 3 - EXECUTION

3.1 EXAMINATION

A. General: Verify that existing site conditions are as specified and indicated on drawings before beginning work under this Section.

1. Grades: Inspect to verify subgrade is within +/- 0.1-foot of grades indicated and specified.
2. Damaged Earth: Inspect to verify that soil rendered unfit to support planting due to concrete, water, mortar, limewater or any other contaminant dumped on it has been removed and replaced with clean soil from a source approved by the Contracting Officer.

B. Unsatisfactory Conditions: Report in writing to General Contractor with copy to Contracting Officer.

C. Acceptance: Beginning of installation means acceptance of existing conditions by installer.

3.2 PREPARATION

A. Weed Control: Treatment of invasive species prior to construction shall be conducted in areas of infestation by the Government.

B. Topsoil:
   1. Strip topsoil as required in Section 02 41 00 “Site Demolition”.
   2. Stockpile and place topsoil as required in Section 32 91 19.13 “Topsoil”.

C. Areas of Newly Placed Topsoil:
   3. Protection:
      a. Locate sewer, water, irrigation, gas, electric, phone and other pipelines or conduits and equipment prior to commencing work.
      b. Contractor shall be responsible for proper repair to landscape, utilities, walls, pavements and other site improvements damaged by operations under this section.

D. Areas of Compacted Topsoil: Areas within the work limits or as defined on Drawings or by the Contracting Officer that are defined as compacted with the soil penetrometer or was used as a construction staging, parking area and/or subjected to heavy use will require decompaction to prepare the soil for revegetation. Extent of areas to be decompacted to be approved by Contracting Officer.

E. Areas of New and Disturbed Topsoil: All new areas of topsoil to be scarified/decompacted to prepare the soil for revegetation. Areas of disturbed topsoil but not severely compacted as determined by the Contracting Officer, shall be scarified to prepare the soil for revegetation, per below specifications.

3.3 PREPARATION OF UNDISTURBED AREAS

A. For preparation of soils in areas not otherwise modified by site and landscape improvements, grading and drainage or construction activities. See Section 32 92 10 “Seeding”.

3.4 INSTALLATION

A. Install topsoil as required in Section 32 91 19.13 “Topsoil”

B. Contractor to prepare the area for optimal germination and establishment of seed and so that the performance standards are met. After understory removals and invasive species suppression has been completed and specifications met, recontouring the site to subgrade may be undertaken. Once sub-grade is achieved, topsoil may be placed and the site decompacted according to the specifications to achieve finish design grade. Topsoil depth as determined by the Contracting Officer and as conditions allow following the specifications in Section 32 91 19.13 “Topsoil”. Where salvaged topsoil is insufficient, another approved source from within the park may be used as approved.

C. The topsoil surface shall be left in a roughened state, but obvious furrows are eliminated and clear break between topsoil and subgrade layers are not produced.

D. Decompaction: Occurs prior to topsoil placement and occurs in the subgrade.

E. Soil Preparation for Areas of Compacted Topsoil:
   1. Scarify or rip the site prior to topsoil application. Scarify and loosen compacted subsoils to a minimum depth of 12 inches prior to replacing the salvaged topsoil.

2. Scarification/de-compaction equipment (e.g. subsoiler, deep spader, ripping tines) will be used in such a manner that scarification furrows are no further than 6-9 inches apart and, where possible, equipment will travel across the site in 2 passes in perpendicular directions (cross-hatched pattern). When decompacting minimize water usage to only the amount needed to abate dust. In addition, the maximum dirt clod size should be 4 inches in order to obtain a roughened surface that will maximum soil/seed contact. Take care to not re-compact the topsoil (e.g. do not repeatedly move heavy equipment over the site).
3. Areas include, but are not limited to; former walking paths or otherwise compacted areas caused by visitor use. It may also include areas where equipment and vehicles traveled or parked and where materials were stockpiled. All areas defined as compacted will be designed by the Contracting Officer.
4. Adjacent trees and their roots and areas of archeological sensitivity shall be avoided during decompaction activities. Review areas with Contracting Officer.

F. Soil Preparation for Areas of New and Disturbed Topsoil:
1. Scarify or rip the site prior to topsoil application. Scarify and loosen compacted subsoils to a minimum depth of 12 inches prior to replacing the salvaged topsoil. Soil decompaction shall be done with equipment that has ripping teeth, placed a maximum of 12 inches apart.Adjacent trees and their roots and areas of archeological sensitivity shall be avoided during decompaction activities.

G. Fine Grading in all Landscape Areas:
1. Complete fine grading for all areas prior to seeding. Allow for natural settlement.
2. For ground surface areas surrounding historical buildings, maintain required positive drainage away from buildings.
3. Establish finish grades to within plus or minus 0.10-foot of grades indicated, in order to prevent “bird-baths” or ponding.
4. Finish grade shall be below edge of pavement prior to seeding or planting.
   a. Seeding Areas: Allow one (1) inch for seed.
5. Noxious weeds or parts thereof shall not be present in the surface grade prior to seeding, coordinate with Contracting Officer.
6. Compaction of surface grade prior to seeding/container plant installation: Firm, but not hard, 85% standard Proctor density within 2% optimum moisture. Extent determined by soils analysis tests and will be provided to contractor.
7. Prior to acceptance, native seed areas shall be hand-raked to a roughened surface, free of debris, clods, rocks and organic matter greater than 1-inch.
8. Restore planting areas to specified condition if eroded or otherwise disturbed after fine grading and prior to planting.

H. Seeding is to take place within 24 hours of decompaction. If a precipitation event takes place between decompaction and seeding the site shall be scarified down to 2 inches, as determined by the Contracting Officer.

3.5 PROTECTION

A. Provide and install barriers as required and as directed by Contracting Officer to protect completed areas against damage from pedestrian and vehicular traffic until acceptance by Contracting Officer.

B. Protect areas of in-place soil from additional compaction, disturbance, and contamination. Prohibit the following practices within these areas except as required to perform planting operations:
   1. Storage of construction materials, debris, or excavated material.
   2. Parking vehicles or equipment.
   3. Vehicle traffic.
   4. Foot traffic.
   5. Erection of sheds or structures.
   6. Impoundment of water.
   7. Excavation or other digging unless otherwise indicated.

C. If planting soil or subgrade is over compacted, disturbed, or contaminated by foreign or deleterious materials or liquids, remove the planting soil and contamination; restore the subgrade as directed by Contracting Officer and replace contaminated planting soil with new planting soil.

D. Erosion Control: Take measures and furnish equipment and labor necessary to control and prevent soil erosion, blowing soil and accumulation of wind-deposited materials on the site throughout the duration of work.

3.6 CLEANING

A. Protect plants from damage due to landscape operations and operations of other contractors and trades. Maintain protection during installation and until final acceptance. Treat, repair, or replace damaged plantings.

B. Protect areas adjacent to planting-soil preparation and placement areas from contamination. Keep adjacent paving and construction clean and work area in an orderly condition. Clean wheels of vehicles before leaving site to avoid tracking soil onto roads, walks, or other paved areas.

C. Any damage to other work done by landscape crew is to be reported and repaired immediately at no additional cost to the Government.

D. When soil preparation and planting operations have been completed, the Contractor shall restore any damaged areas to their original conditions as specified on the construction drawings and in these specifications and approved by the Contracting Officer.

E. Remove debris and excess materials from site. Clean out drainage inlet structures. Clean paved and finished surfaces soiled as a result of work under this Section, in accordance with the General Specifications or as directed by the Contracting Officer.

END OF SECTION 32 91 13

SECTION 32 91 19.13 –TOPSOIL

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. This Section includes requirements for stockpiling, furnishing, and placing topsoil on a previously prepared subgrade and use for seeding and planting areas.

1.3 PRE-INSTALLATION MEETING

A. Conduct meeting at Project site with Contracting Officer to coordinate the process with other trades, to coordinate equipment movement within landscape improvement areas and to avoid soil compaction, to review proposed methods of placement and maintenance. Review project limits, existing survey data, including utilities, and any additional supplementary reports and plans. This meeting shall be coordinated by the Contractor and comply with requirements in Division 01.

1.4 DEFINITIONS

A. Duff Layer: The surface layer of native topsoil that is composed of mostly decayed leaves, twigs, and detritus.

B. Finish Grade: Elevation of finished surface of planting soil.

C. Pesticide: A substance or mixture intended for preventing, destroying, repelling, or mitigating a pest. This includes insecticides, miticides, herbicides, fungicides, rodenticides, and molluscicides. It also includes substances or mixtures intended for use as a plant regulator, defoliant, or desiccant.

D. Planting Area: Areas to be planted.

E. Planting Soil: Standardized topsoil; existing, native surface topsoil and existing, in-place surface soil.

F. Subgrade: Surface or elevation of subsoil remaining after excavation is complete, or the top surface of a fill or backfill before planting soil is placed.

G. Subsoil: All soil beneath the topsoil layer of the soil profile and typified by the lack of organic matter and soil organisms.

H. Surface Soil: Soil that is present at the top layer of the existing soil profile at the Project site. In undisturbed areas, the surface soil is typically topsoil; but in disturbed areas such as urban environments, the surface soil can be subsoil.

1.5 SUBMITTALS

A. See Section 01 33 23 “Submittal Procedures”.

B. Field quality-control reports.

1. Weekly Inspection and Maintenance Reports; Submit a weekly inspection report once material has been placed for the first four months and then monthly until the maintenance period has been satisfied.

C. Topsoil Schedule: Within forty-five (45) calendar days of Notice to Proceed submit electronic Word documents and PDF of proposed topsoil schedule for all aspects of topsoil procedures, including but not limited too; indicating dates for subgrade approval, site soil preparation, decompaction, topsoil and mulch placing, seeding, staging areas, list of equipment for topsoil placement, erosion and sediment control, stockpile procedures and maintenance and schedule for seeding. Schedule all Work during specified seeding seasons, indicate optimal time for seeding. Once accepted, revise dates only as approved in writing, after documentation of reasons for delays.

1. Delivery schedule: Indicate delivery schedule and minimum forty-five (45) calendar day notice of delivery.

2. Equipment summary: Indicate issuance of a list of equipment to be utilized for topsoil activities two (2) weeks prior to work.

D. Topsoil Stockpile and Maintenance Plan: The Contractor shall submit a topsoil stockpile and maintenance plan concurrently with the topsoil schedule for approval by the Contracting Officer. The use of multiple topsoil shipping dates is acceptable. Protections During Delivery – duration and methodologies shall be submitted with the topsoil schedule detailed under the Delivery subparagraph.

1. To the extent possible, windrow stripped topsoil adjacent the perimeter of stripped areas. Minimize handling and reuse topsoil close to its point of origin.

2. Plan sheet indicating the location of the proposed field stockpile sites. The number and location of the field stockpile sites shall be developed in such a manner as to promote efficient topsoil operations. If required, additional stockpile area shall be established at locations selected by the Contracting Officer at no additional cost to the government.

3. Maintenance Instructions for supplementing landscape improvement areas throughout the maintenance period until Final Acceptance, as applicable.

E. Obstructions Below Ground Reporting: When at grade obstructions prevent topsoil placement, the Contractor shall submit a plan showing proposed adjustments to locations to the Contracting Officer.

F. Topsoil Damage Reporting: Contractor shall document the site conditions (weather, etc. when damage occurred), provide photos of damage, describe the damage, and file this information with the Contracting Officer within forty-eight (48) hours.

G. Construction Equipment Cleaning Records: Submit the cleaning and inspection records to the Contracting Officer.

H. The Contracting Officer reserves the right to reject any material that fails to meet specified requirements.

1.6 CLOSEOUT SUBMITTALS

A. See Section 01 7700 – "Closeout Procedures".

1.7 DELIVERY, STORAGE, AND HANDLING

A. Construction Equipment Cleaning: All vehicles and equipment entering the project area must be clean of plant materials and dirt and free from oil leaks and are subject to inspection. Wash all construction and restoration/vegetation equipment (including the inside beds of vehicles hauling materials) to thoroughly remove all dirt, plant, and other foreign material prior to entering the project. Particular attention must be shown to the under carriage, mower decks (top and bottom) and any surface where soil containing exotic seeds may exist. Allow the NPS to inspect each piece of equipment before entering the project. Provide the cleaning and inspection records to the Contracting Officer. Equipment found operating on the project that has not been inspected or has oil leaks shall be shut down and subject to citation.

B. Delivery Schedule – No later than forty-five (45) days prior to the first date of topsoil delivery, the Contractor shall submit to the Contracting Officer a topsoil schedule for approval. This estimated schedule shall include the topsoil shipping dates for each of the shipments, delivery dates at the construction site, placement dates for each of the shipments, quantities and an estimate of the crew size. It shall also include details of the shipping methods, estimated times for placement and protection. The Contractor shall submit a soil material storage and maintenance plan concurrently with the seeding schedule for approval by the Contracting Officer. The use of multiple material shipping dates is acceptable.

C. Protection During Delivery – The Contractor is responsible for the protection of soil during delivery. Soil shall be transported in vehicles free/clean of former soil materials to limit the spread of weeds and pathogens. Cover soil with tarpaulins or other suitable covers securely fastened to the vehicle to prevent loss of topsoil and prevent introduction of weed seed and pathogens during transportation.

D. The Contracting Officer shall inspect all soil material upon delivery and determine if they meet the requirements detailed herein. If the delivery date is delayed or cancelled the Contractor shall notify the Contracting Officer immediately.

E. Bulk Materials:

1. Do not dump or store bulk materials near structures, utilities, walkways and pavements, or on existing plants.

2. Provide erosion-control measures to prevent erosion or displacement of bulk materials; discharge of soil-bearing water runoff; and airborne dust reaching adjacent properties, water conveyance systems, or walkways.
3. Accompany each delivery of bulk materials with appropriate certificates.

F. Do not deliver or place topsoil in a frozen, wet, or muddy condition.

G. Protect stored and placed topsoil from vehicular traffic, equipment storage, material storage, or from contaminants or pollution sources. Topsoil that is tainted during construction is to be removed from site and disposed of at a licensed landfill at no additional cost to the NPS.

1.8 PROJECT/SITE CONDITIONS

A. Grand Teton National Park – 4 Lazy F Ranch; Elevation: ±6,475 ft.

B. Work Scheduling: Proceed with and complete landscape work as rapidly as portions of the site become available, working within the specified planting season and approved schedule.

C. Topsoil Coordination:
1. All invasive species are to be removed by the Government prior to seeding.
2. Coordinate soil preparations with seeding installations schedules.

D. Final work schedule will be coordinated with and approved by Contracting Officer.

E. Coordinate approved hours of topsoil procedures with Contracting Officer prior to operations.

F. Do not perform work when climate and existing site conditions will not provide satisfactory results (e.g. wet, freezing). The Contracting Officer reserves the right to deny topsoil procedures on days that are deemed to be unfavorable for installation. Proceed with topsoil only when existing and forecasted weather conditions permit topsoil to be performed when beneficial and optimum results may be obtained. Apply products during favorable weather conditions according to manufacturer's written instructions and warranty requirements.

G. Contractor shall have full use of the approved designated fenced or flagged in project area for specific construction operations during the construction period. Contractor's use of the site is limited to minimizing site disturbance to the extent possible and by the Government's right to perform work or to retain other contractors on portions of Project. There will be no visitor access within the designated fenced in approved project areas.

H. All work is being completed within a historic site and must adhere to the Secretary of Interior’s Standards, as per Section 01 35 91 “Historic Preservation Treatment Procedures”. Work must be completed without damage to fencing, trees to remain, trails, roads, or other property. The Contractor will be financially liable to the park for any costs incurred as a result of repairing and/or replacing any of the aforementioned items or for the value of the item as assessed by an archeologist, biologist, government and the Contracting Officer.

I. All work is being completed within a natural resource area and must adhere to Section 01 35 13.05 “Environmental Protection”. Protect natural resources in and around the construction site including but not limited to requirements for wildlife protection, inspection and maintenance of equipment, vegetation/site protection, and pollution prevention.

J. Contractor's operations shall be conducted reasonably to prevent rutting or other soil damage. No work shall be allowed for at least a 24-hour period following a rain accumulation of one (1) inch or more during the proceeding 24-hour period. After any work stoppage, it will be the responsibility of the Contracting Officer to determine when work may re-commence.

K. All vehicles and other heavy equipment involved in the completion of this contract will be restricted to roads, parking areas, and road shoulders no more than 10' from the edge of the road surface unless given case-specific permission by the Contracting Officer. Off road equipment use is solely at the discretion of the Contracting Officer and subject to change based on current or anticipated conditions. Rutting or other resource damage will not be tolerated. All work from trails will be considered off road and must adhere to the same guidelines and restrictions.

L. All vehicles should be equipped with absorptive pads while on park property in order to quickly contain any spills of oil or hydraulic fluid. Any such spills must be reported to the Contracting Officer as soon as is safely possible.

M. Damage to natural areas, existing grades, vegetation, pavements, underground utilities, and other improvements and areas of protection, caused during work under this section, shall be repaired to original condition or better at no additional cost to the Government.

N. Vehicular accessibility on site shall be as directed by Contracting Officer. Repair damage to prepared topsoil and existing surfaces, caused by vehicular access and movement during work under this section, to original condition at no additional cost to the Government.

O. Existing conditions:
   1. Utilities: Determine location of underground utilities. Perform work in a manner to avoid possible damage. Hand excavate, as required.
   3. Any damage to underground piping or wiring arising out of work of this section must be corrected and repaired by the Contractor to the satisfaction of the Contracting Officer.
   4. Damage to Other Improvements: All costs for repair or replacement of any damage to other work done on-site or adjacent properties by topsoil procedures shall be borne by the Contractor placing topsoil.
   5. Existing grades and vegetation damaged during topsoil procedures shall be restored at the Contractor's expense.
   6. Excavation: When conditions detrimental to topsoil procedures are encountered, such as rubble fill, adverse drainage conditions, noxious materials or obstructions, notify Contracting Officer before placement.

      a. Obstructions Below Ground: When obstructions below finished grades prevent topsoil procedures, the Contractor shall submit a plan showing proposed adjustments to locations to the Contracting Officer.

   7. If weeds / invasive species are present on site, Government to assume responsibilities for removal.

P. Field Measurements:
   1. Verify actual grade elevations, service and utility locations, and dimensions of construction contiguous with new seeding by field measurements before proceeding with seeding work.

PART 2 - PRODUCTS

2.1 ON-SITE TOPSOIL OR TOPSOIL FROM ANOTHER APPROVED SOURCE WITHIN THE PARK

A. Topsoil shall be stripped and stockpiled prior to earthwork operations. Salvaged topsoil will be obtained from areas to be cleared, grubbed or graded as demarcated in Demolition Plan Sets. Topsoil refers to the uppermost soil horizon (typically top 6 inches) which includes organic duff and other materials capable of supporting vegetation. Break down grasses, shrubs, and other minor brush by chipping, mowing, shredding or removing from the site. Chipped, mowed, or shredded material less than 2" diameter and height may be incorporated with the conserved topsoil. Remove the topsoil in one pass under dry soil conditions, only using water to abate dust. Use equipment capable of excavating small isolated pockets of topsoil. Do not mix topsoil with subsoil.

1. Coordinate at least two (2) weeks prior to topsoil salvage with Contracting Officer to ensure their on-site presence at the initiation of salvage activities.

PART 3 - EXECUTION

3.1 TOPSOIL STOCKPILING

A. Topsoil required for establishment of vegetation shall be stockpiled on site immediately adjacent to salvage location where feasible and at designated locations as approved by Contracting Officer, in the amount necessary to complete the finished grading of all exposed areas that are to be re-vegetated. Topsoil shall be stored in windrow or pile formation with a maximum height of 3 feet. If topsoil piles have to be stockpiled consult with Contracting Officer and turn piles from bottom to top, mixing thoroughly to ensure air circulation, once weekly. Stockpiles shall be stabilized with silt fence if determined necessary by the Contracting Officer, see Section 31 25 00 "Erosion and Sediment Control". Flag or install barriers to keep heavy equipment/vehicles off stockpiled topsoil.

B. Stockpile topsoil material away from edge of excavations without intermixing with subsoil. Shape stockpiles to drain surface water. Water once weekly with enough water to prevent windblown dust.

1. Care will be taken to assure that topsoil and fill material are not mixed and are stockpiled in separate areas (i.e., topsoil to the right of the trench and fill to the left).
2. Equipment capable of working around areas of designated vegetation or tree stumps prior to grubbing operations will be required to perform the work.
3. Limit height of topsoil stockpiles to 36 inches (900 mm) unless authorized by Contracting Officer.
4. Do not stockpile topsoil within drip line of remaining trees.
5. Stockpile surplus topsoil to allow for re-spreading a thicker layer of topsoil.
6. Do not compact or drive upon topsoil during any stage of the process. Prior to construction, remove or strip topsoil from all areas to be impacted by construction, including staging areas, unless otherwise specified.

C. Topsoil Handling: Sections will be rehabilitated as soon as possible. Topsoil may not be stockpiled over the winter or for longer than six (6) months in grass and tree dominated zones. Any deviation must be approved by the Contracting Officer.

1. Topsoil may not be used as bedding material. Separate bedding material must be obtained from sources approved by the Contracting Officer.
2. All topsoil that is to be reused on-site is to be stockpiled on-site with erosion control surrounding the pile in the location established at the pre-demolition meeting. Reference Section 31 25 00 “Erosion and Sediment Control”.
3. Suspended topsoil operations during periods of excessive soil moisture until the moisture content reaches acceptable levels to attain the required results.

3.2 EXAMINATION

A. Any vegetation within preserved areas damaged by the work under this or any other section, as determined by the Contracting Officer, shall be replaced immediately after the completion of the work at no additional cost to the Government.

B. Examine areas where the Work of this Section will be performed for compliance with requirements and conditions affecting installation and performance.

1. Verify that no foreign or deleterious material or liquid such as paint, paint washout, concrete slurry, concrete layers or chunks, cement, plaster, oils, gasoline, diesel fuel, paint thinner, turpentine, tar, roofing compound, or acid has been deposited in soil within the work area.
2. Verify that final grades are completed in accordance with Contract Drawings.
3. Suspend topsoil operations during periods of excessive soil moisture until the moisture content reaches acceptable levels to attain the required results.
4. Uniformly moisten excessively dry soil that is not workable or which is dusty.

C. If contamination by foreign or deleterious material or liquid is present in soil within a planting area, remove the soil and contamination as directed by Contracting Officer and replace with new planting soil.

D. If existing grade or finish subgrade is not complete immediately notify the Contracting Officer.

E. Proceed with installation only after unsatisfactory conditions have been corrected and approved by Contracting Officer.

F. Existing grade to be considered as final or finish subgrade upon commencement of work. Contractor to examine areas prior to installing topsoil and decompaction.

3.3 PLACING TOPSOIL

A. Protect structures, utilities, sidewalks, pavements, and other facilities and landscape improvement areas and existing plants from damage caused by topsoil operations.

B. Install erosion-control measures to prevent erosion or displacement of soils and discharge of soil-bearing water runoff or airborne dust to adjacent properties and walkways.

C. Scarification, decompaction, and fine grading as per requirements in Section 32 91 13 “Soil Preparation”. Soil decompaction to occur prior to topsoil placement.

D. Before topsoil procedures, obtain Contracting Officer’s acceptance of subgrade grading; restore landscape improvement areas if eroded or otherwise disturbed after finish grading.

E. Topsoil shall be free from weeds, sod, and material larger than 1-inch (1”), toxic substances, litter or other deleterious material. Topsoil shall be re-spread to a uniform depth consistent with site conditions for adjacent undisturbed areas (roughly 2-4 inches depending on the amount of available on-site topsoil). Leave the soil surface roughened such that obvious furrows are eliminated and a clear break between topsoil and subgrade layers is not produced. Hand place topsoil in areas inaccessible to machinery. Once topsoil is placed decompaction may occur as per Section 32 91 19 Soil Preparation.

1. For bidding purposes only:

    a. Topsoil (on-site or from a source within the Park): Twelve (12.35) cubic yards per thousand (1000) square feet at 4” depth.
    b. Topsoil settled depth to be 3” minimum.

F. Do not place topsoil when the ground or topsoil is frozen, excessively wet, or otherwise in a condition detrimental to the work. Topsoil should not be spread with a substantial use of water. Take care to not compact the topsoil by minimizing the traffic of heavy equipment over the site once the topsoil has been spread and using rubber tracked tired equipment.

G. Utilize salvaged topsoil as the top layer to the extent available.

3.4 PROTECTION AND REPAIR

A. The Contractor shall carefully protect all preserved areas as shown on the plans and shall also be liable for any and all damages to property caused by the work under this section. Any damages to property or to vegetation within preserved areas shall be restored to the original conditions or to the satisfaction of the Contracting Officer.

B. Protect completed areas where topsoil has been spread from traffic which will compact the soil volume. Any areas that, as determined by the Contracting Officer, become compacted due to Contractor’s construction traffic shall be reconstructed to specified requirements and approved by the Contracting Officer.

3.5 CLEAN UP AND PROTECTION

A. Protect plants from damage due to topsoil procedures and operations of other contractors and trades. Maintain protection during installation and maintenance periods. Treat, repair, or replace damaged plantings.

B. During topsoil procedures work, keep pavements clean and work area in an orderly condition. Clean wheels of vehicles before leaving site to avoid tracking soil onto roads, walks, or other paved areas.

C. Any damage to other work done by landscape crew is to be reported and repaired immediately at no additional cost to the Government.

D. When topsoil procedures have been completed, the Contractor shall restore any damaged areas to their original conditions as specified on the construction drawings and in these specifications and approved by the Contracting Officer.

E. Upon completion of work, remove debris and leave area in clean, acceptable condition. The Contractor shall remove damaged, excess, and waste materials including excess subsoil, unsuitable soil, trash, and debris from the Project Site and legally dispose of them off the Project Site and properly dispose of the materials at an approved disposal facility.

F. Protect landscape work and materials from damage due to topsoil procedures, operations by other Contractors and trades and trespassers by erecting barricades and/or providing signage as directed by the Contracting Officer. Maintain protection during installation periods. Treat, repair, or replace damaged topsoil work as directed until final acceptance.

END OF SECTION 32 91 19.13

SECTION 32 93 00 - PLANTS

PART 1 - GENERAL

1.1 SUMMARY

A. Section Includes:

1. Plants.
2. Mulches.
3. Landscape edgings

1.2 DEFINITIONS

A. Backfill: The earth used to replace or the act of replacing earth in an excavation.

B. Pesticide: A substance or mixture intended for preventing, destroying, repelling, or mitigating a pest. Pesticides include insecticides, miticides, herbicides, fungicides, rodenticides, and molluscicides. They also include substances or mixtures intended for use as a plant regulator, defoliant, or desiccant. Some sources classify herbicides separately from pesticides.

C. Planting Soil: Existing, on-site soil; imported soil; or manufactured soil that has been modified with soil amendments and perhaps fertilizers to produce a soil mixture best for plant growth. See Section 329113 "Soil Preparation" for drawing designations for planting soils.

D. Root Flare: Also called "trunk flare." The area at the base of the plant's stem or trunk where the stem or trunk broadens to form roots; the area of transition between the root system and the stem or trunk.

1.3 PREINSTALLATION MEETINGS

A. Preinstallation Conference: Conduct conference at Project site.

1.4 ACTION SUBMITTALS

A. Product Data: For each type of product.

B. Samples of each type of mulch.

1.5 INFORMATIONAL SUBMITTALS

A. Product certificates.

B. Sample warranty.

1.6 CLOSEOUT SUBMITTALS

A. Maintenance Data: Recommended procedures to be established by Owner for maintenance of plants during a calendar year.

1.7 QUALITY ASSURANCE

A. Provide quality, size, genus, species, and variety of plants indicated, complying with applicable requirements in ANSI Z60.1.

1.8 DELIVERY, STORAGE, AND HANDLING

A. Do not prune shrubs before delivery. Protect bark, branches, and root systems from sun scald, drying, wind burn, sweating, whipping, and other handling and tying damage. Do not bend or bind-tie trees or shrubs in such a manner as to destroy their natural shape. Provide protective covering of plants during shipping and delivery. Do not drop plants during delivery and handling.

B. Handle planting stock by root ball.

C. Deliver plants after preparations for planting have been completed, and install immediately. If planting is delayed more than six hours after delivery, set plants and trees in their appropriate aspect (sun, filtered sun, or shade), protect from weather and mechanical damage, and keep roots moist.

1.9 WARRANTY

A. Special Warranty: Installer agrees to repair or replace plantings and accessories that fail in materials, workmanship, or growth within specified warranty period.

1. Failures include, but are not limited to, the following:

a. Death and unsatisfactory growth, except for defects resulting from abuse, lack of adequate maintenance, or neglect by Owner.
b. Structural failures including plantings falling or blowing over.

2. Warranty Periods: From date of planting completion .

a. Shrubs, Vines, and Ornamental Grasses: 12 months.
b. Ground Covers, Biennials, Perennials, and Other Plants: 12 months.

PART 2 - PRODUCTS

2.1 PLANT MATERIAL

A. General: Furnish nursery-grown plants true to genus, species, variety, cultivar, stem form, shearing, and other features indicated in Plant List, Plant Schedule, or Plant Legend indicated on Drawings and complying with ANSI Z60.1; and with healthy root systems developed by

transplanting or root pruning. Provide well-shaped, fully branched, healthy, vigorous stock, densely foliated when in leaf and free of disease, pests, eggs, larvae, and defects such as knots, sun scald, injuries, abrasions, and disfigurement.

B. Root-Ball Depth: Furnish trees and shrubs with root balls measured from top of root ball, which begins at root flare according to ANSI Z60.1. Root flare shall be visible before planting.

2.2 MULCHES

A. Organic Mulch: Shredded hardwood.

2.3 LANDSCAPE EDGINGS

A. Steel Edging: Standard commercial-steel edging, fabricated in sections of standard lengths, with loops stamped from or welded to face of sections to receive stakes.
   1. Edging Size: 14ga thick by 4 inches (100 mm)
   2. Finish: Manufacturer's standard paint: Brown

PART 3 - EXECUTION

3.1 PLANTING AREA ESTABLISHMENT

A. General: Prepare planting area for soil placement and mix planting soil according to Section 32 91 13 "Soil Preparation."

B. Placing Planting Soil: Place and mix planting soil in-place over exposed subgrade.

C. Before planting, obtain Contracting Officer’s acceptance of finish grading; restore planting areas if eroded or otherwise disturbed after finish grading.

3.2 INVASIVE SPECIES CONTROL

A. Seed mixes used for revegetation in roadside corridors will be composed of native species and approved by SRM Vegetation Branch staff.

B. To reduce the threat of nonnative, invasive vegetation being introduced to the park, all imported material (i.e., sand, gravel, rock, rip-rap, etc.) must be obtained from a park approved or county weed district approved source. Teton County Weed and Pest has currently approved as weed-free the various Evans Construction pits in Teton County. If a new material source is requested, park vegetation staff will seek county-approved material source pits and/or perform a nonnative, invasive plant inspection.

3.3 EXCAVATION FOR TREES AND SHRUBS

A. Planting Pits and Trenches: Excavate circular planting pits.

1. Excavate planting pits with sides sloping inward at a 45-degree angle. Excavations with vertical sides are unacceptable. Trim perimeter of bottom leaving center area of bottom raised slightly to support root ball and assist in drainage away from center. Do not further disturb base. Ensure that root ball will sit on undisturbed base soil to prevent settling. Scarify sides of planting pit smeared or smoothed during excavation.

2. Excavate approximately three times as wide as ball diameter.
3. Excavate at least 12 inches (300 mm) wider than root spread and deep enough to accommodate vertical roots for bare-root stock.
4. Do not excavate deeper than depth of the root ball, measured from the root flare to the bottom of the root ball.

B. Backfill Soil: Subsoil and topsoil removed from excavations may be used as backfill soil unless otherwise indicated.

3.4 SHRUB PLANTING

A. Inspection: At time of planting, verify that root flare is visible at top of root ball according to ANSI Z60.1. If root flare is not visible, remove soil in a level manner from the root ball to where the top-most root emerges from the trunk. After soil removal to expose the root flare, verify that root ball still meets size requirements.

B. Roots: Remove stem girdling roots and kinked roots. Remove injured roots by cutting cleanly; do not break.

C. Set each plant plumb and in center of planting pit or trench with root flare 1 inch (25 mm) above finish grades.

1. Backfill: Planting soil from amended, on-site soil

2. Balled and Burlapped Stock: After placing some backfill around root ball to stabilize plant, carefully cut and remove burlap, rope, and wire baskets from tops of root balls and from sides, but do not remove from under root balls. Remove pallets, if any, before setting. Do not use planting stock if root ball is cracked or broken before or during planting operation.
3. Container-Grown Stock: Carefully remove root ball from container without damaging root ball or plant.
4. Fabric Bag-Grown Stock: Carefully remove root ball from fabric bag without damaging root ball or plant. Do not use planting stock if root ball is cracked or broken before or during planting operation.
5. Bare-Root Stock: Support stem of each plant and spread roots without tangling or turning toward surface. Plumb before backfilling, and maintain plumb while working. Carefully work backfill around roots by hand. Bring roots into close contact with the soil.
6. Backfill around root ball in layers, tamping to settle soil and eliminate voids and air pockets. When planting pit is approximately one-half filled, water thoroughly before placing remainder of backfill. Repeat watering until no more water is absorbed.
7. Continue backfilling process. Water again after placing and tamping final layer of soil.

D. Slopes: When planting on slopes, set the plant so the root flare on the uphill side is flush with the surrounding soil on the slope; the edge of the root ball on the downhill side will be above the surrounding soil. Apply enough soil to cover the downhill side of the root ball.

3.5 SHRUB PRUNING

A. Remove only dead, dying, or broken branches. Do not prune for shape.

B. Prune, thin, and shape trees, shrubs, and vines as directed by Arborist.

C. Prune, thin, and shape trees, shrubs, and vines according to standard professional horticultural and arboricultural practices. Unless otherwise indicated by Arborist, do not cut tree leaders; remove only injured, dying, or dead branches from trees and shrubs; and prune to retain natural character.

D. Do not apply pruning paint to wounds.

3.6 GROUND COVER AND PLANT PLANTING

A. Set out and space ground cover and shrubs, as indicated on Drawings.

B. Use amened, on-site planting soil backfill.

C. Dig holes large enough to allow spreading of roots.

D. Work soil around roots to eliminate air pockets and leave a slight saucer indentation around plants to hold water.

E. Water thoroughly after planting, taking care not to cover plant crowns with wet soil.

F. Protect plants from hot sun and wind; remove protection if plants show evidence of recovery from transplanting shock.

G. Mulch backfilled surfaces of planting areas and other areas indicated.

1. Shrubs: Apply organic mulch ring of 3-inch (75-mm) average thickness, with 12-inch (300-mm radius around trunks or stems. Do not place mulch within 3 inches 75 mm of trunks or stems.

3.7 EDGING INSTALLATION

A. Steel Edging: Install steel edging where indicated according to manufacturer's written instructions. Anchor with steel stakes spaced approximately 30 inches (760 mm) apart, driven below top elevation of edging.

3.8 PLANT MAINTENANCE

A. Maintain plantings by pruning, cultivating, watering, weeding, fertilizing, mulching, restoring planting saucers, resetting to proper grades or vertical position, and performing other operations as required to establish healthy, viable plantings.

B. Fill in, as necessary, soil subsidence that may occur because of settling or other processes. Replace mulch materials damaged or lost in areas of subsidence.

C. Apply treatments as required to keep plant materials, planted areas, and soils free of pests and pathogens or disease. Use integrated pest management practices when possible to minimize use of pesticides and reduce hazards. Treatments include physical controls such as hosing off foliage, mechanical controls such as traps, and biological control agents.

D. Apply biological control agents according to authorities having jurisdiction and manufacturer's written recommendations. Coordinate applications with Owner's operations and others in proximity to the Work. Notify Owner before each application is performed.

E. Protect plants from damage due to landscape operations and operations of other contractors and trades. Maintain protection during installation and maintenance periods. Treat, repair, or replace damaged plantings.

F. At time of Substantial Completion, verify that tree-watering devices are in good working order and leave them in place. Replace improperly functioning devices.

3.9 MAINTENANCE SERVICE

A. Maintenance Service: Provide maintenance by skilled employees of landscape Installer. Maintain as required in "Plant Maintenance" Article. Begin maintenance immediately after plants are installed and continue until plantings are acceptably healthy and well established, but for not less than maintenance period below:

1. Maintenance Period for Shrubs: 12 months from date of planting completion.
2. Maintenance Period for Ground Cover and Other Plants: 12 months from date of planting completion.

END OF SECTION 32 93 00

SECTION 33 14 15 - SITE UTILITIES

1.1 SUMMARY

A. Section Includes:

1. Installation of site utilities to within 5 feet of structures at the Lazy F Ranch in Grand Teton National Park. Utilities include sewer collection and water distribution infrastructure. All water and sewer utility design and construction shall meet Wyoming Public Works Standard Specifications 2023 editions and Wyoming Department of Environmental Quality regulations, as well as relevant 2024 ICC codes and standards.

2. Sanitary Sewer:
   a. Gravity Sewer Collection includes sewer services, cleanouts, gravity sewer mains and sewer manholes
   b. Sewer lift station and force main
   c. Construction technique for the sewer force main along the gravel road alignment from Moose to 4 Lazy F as noted on the drawings shall be constructed via plowing methods to limit disturbance.
   d. Packaged sewer grinder/lift stations and small force main for individual structures as noted on the drawings.

3. Water:
   a. Includes the water connection to an existing HDPE water main at the 4 Lazy F connected to the Moose Water System
   b. Water services and operation valves
   c. Water meter and meter vault
   d. Blowoff hydrants and drains

4. Submittals:
   a. Product Data on the Following:
      1) Gravity sewer pipe and fittings
      2) Force main piping and fittings
      3) Lift station enclosure and piping
      4) Small packaged grinder/lift station
      5) Concrete manholes
      6) Water main and fittings
      7) Water valves
      8) Water meter and vault

1.2 RELATED REQUIREMENTS

1. Section 02 21 19 – Utilities Survey
2. Section 02 41 00 - Site Demolition
3. Section 31 23 20 – Street and Site Excavation, Grading and Backfill
4. Section 31 25 00 – Erosion and Sediment Control
5. Section 31 30 00 - Site Earthwork.
6. Section 32 20 00 - Site Improvements.
7. Section 33 14 15 - Site Utilities.
8. Section 33 20 00 – Sewer Wet Well Design Criteria

9. Section 33 30 00 – Submersible Centrifugal Pump Design
10. Section 33 40 00 – Special Construction

1.3 REFERENCE STANDARDS

A. Wyoming Public Works Standard Specifications 2023 Edition

B. Wyoming Department of Environmental Quality Rules and Regulations

C. 2024 ICC Codes and Standards

D. National Fire Protection Association (NFPA)

E. ASTM D-3034 Standard Specification for Polyvinyl Chloride Sewer Pipe and Fittings

F. AWWA C906 Polyethylene Pressure Pipe and Fittings

G. AWWA C-509; Resilient-Seated Gate Valves for Water and Sewerage Systems

H. ASTM C-478 Precast Concrete

I. ASTM A615 Rebar

1.4 QUALITY ASSURANCE

A. Materials and assemblies installed in the work shall be inspected and found to be in compliance with industry standards and these specifications prior to acceptance of the work. Items found not to be in compliance shall be removed, or corrective measures taken, to assure compliance with the referenced standard. The Contractor shall perform field tests and provide labor, equipment and incidentals required for testing.

B. All materials shall be new, and shall bear the label of standardizing agency whenever standards have been established and label service is normally and regularly furnished by the agency. All equipment provided shall be listed and labled suitable for the specified purposed, environment, and application and installed in accordance with manufacturer's recommendations.

C. Provide such other labor and materials as are required for a complete and usable system in accordance with the requirements of the criteria listed, regardless of whether such material and associated labor are called for elsewhere,

D. Provide testing on water mains and services in accordance with Wyoming Public Works Standard Specifications 2023 Edition

1. Field quality-control reports.

1.5 COORDINATION

A. The contractor shall verify that the locations of the inverts of all site utility lines are coordinated with surrounding grades and existing utility lines. If necessary, the Contractor shall make

adjustments to the locations indicated on the drawings in accordance with applicable codes and standards.

1.6 FIELD CONDITIONS

A. Protect all existing water mains and sanitary sewer mainlines and related appurtenances in the project area during construction and installation of the new facilities

B. Protect existing utilities to remain. Where removal of existing utilities and pavement is required, provide approved barricades, temporary covering of exposed areas, and temporary services or connections. Repair damage to existing utilities to remain at no additional expense to the Government,

C. Terminate utilities in accordance with state and local rules and regulations the nationally recognized code; and the requirements of the utility provide covering the specific utility; NPS Standards; and approved by the Contracting Officer.

D. Record actual locations of piping mains, services, valves, utility connections, cleanouts, thrust restraints, and invert elevations

1.7 DELIVERY, STORAGE, AND HANDLING OF MATERIALS

A. Inspect materials delivered to the site for damage. Unload and store with minimum handling. Store materials on site in enclosures or under protective covering. Store plastic piping, jointing materials and rubber gaskets under cover out of direct sunlight. Do not store materials directly in a manner to ensure delivery to the trench in sound undamaged condition. Take special care to avoid injury to coatings and linings on pipe and fittings; make satisfactory repairs if coatins or linings are damaged. Carry, do not drag pipe to the trench.

PART 2 - PRODUCTS

2.1 WATER SYSTEM

A. Install a water system connected to the existing 8-inch water main at 4 Lazy F meeting WYDEQ Water Quality Chapter 12 regulations. All water system components must be NSF certified.
   1. Water Mains
      a. Material: HDPE PE 4710
   2. Isolation Valves:
      a. Resilient seated meet AWWA C509
      b. Iron body, non rising steams
      c. 2-inch square wrench nut for key operation
      d. Valves shall incorporate 8mil factory applied fusion bonded exterior epoxy coating
      e. All valve metal accessory items shall be Grade 316 Stainless Steel
      f. Valve boxes shall be cast iron, 5 ¼” diameter adjustable valve boxes with “Water” stamped on the lid
   3. Blow Off Hydrants

a. Non-freezing, self draining type
b. Working parts shall be bronze to bronze design and serviceable from above grade with no digging
c. Shall be lockable
d. 2 ½" NST Bronze outlet
e. Provided with non turning operating valve

4. Water Services
   a. HDPE PE 4710 CTS
   b. Saddle; ductile iron with epoxy body and 2 stainless steel bands
   c. Corporation Stop; brass, ground key type
   d. Curb Box; arch pattern, designed to accommodate HS-20 loadings
   e. Curb Stop; brass, H pattern type
5. Utility Line Marking Tape and Tracer Wire
   a. Metallic foil laminated between 2 layers of impervious plastic film
   b. Minimum 2-inches wide and a thickness of 5 mil
   c. Imprint: ¾ inch or larger bold black letters
   d. Legend: Identify buried utility line tape with the actual utility type and imprint such as "Caution Waterline Below" repeat approximately 24-inch interval
   e. Background Color: American Public Works Association (APWA) color code
   f. Tracer wire shall be 12 awg copper direct bury tracer wire, with termination posts
6. Water Meter Vault
   a. Precast concrete meeting ASTM C-478
   b. Cast iron ring and cover
   c. Wate Meter
      1) Compatible with NSP software provided with digital read on a post
      2) Fusion bonded epoxy coated iron body

2.2 SANITARY SEWER SYSTEM

A. Install a sewer collection system to convey sanitary sewer from 4 Lazy F to the Moose Sewer Collection System meeting WYDEQ Water Quality Chapter 11 regulations.
   1. Gravity Sewer Pipe
      a. Gravity ewer main shall be PVC SDR 35, with bell and spigot joints
      b. Fittings shall be same material as pipe
   2. Manholes
      a. Pre cast concrete in accordance with requirements with flowlines cast in the base with a minimum 48-inch diameter
      b. Manhole frames and covers shall be cast iron, designed to accommodate HS-20 traffic loading. The word "Sewer" shall be cast into the cover
   3. Sewer Services
      a. Pipe material shall be PVC SDR 35
      b. Fittings shall be same material as pipe
   4. Utility Line Marking Tape
      a. Metallic foil laminated between 2 layers of impervious plastic film
      b. Minimum 2-inches wide and a thickness of 5 mil.
      c. Imprint: ¾ inch or larger bold black letters
      d. Legend: Identify buried utility line tape with the actual utility type and imprint such as "Caution Sewerline Below" repeat approximately 24-inch interval.
      e. Background Color: American Public Works Association (APWA) code color
   5. Duplex Submersible Sewage Pump Station

a. Pump station shall be an automatically controlled duplex pump station capable of handling raw sewage effluent.
b. Pump control panel, liquid level control, and system valving shall be installed in an above ground enclosure.
c. Station Enclosure
    1) Shall contain all valves and associated controls
    2) Access panels shall be sized and placed to permit routine maintenance operations through the openings of the enclosure above ground
    3) Enclosure base shall function as the wet well top and incorporate a duplex access lid
    4) Ventilation blower shall be provided with the enclosure along with a wet well vent
    5) Enclosure appearances and finishes shall be subject to historic district requirements and will be approved by the CO
d. Pumps
    1) Submersible pumps per Submersible Centrifugal Pumps
    2) Pump casing shall be gray iron with a gray iron or ductile iron slide rail attached to the discharge flange as an integral assembly
    3) Pump shall be sized to pass 3” diameter spheres
    4) Provided with a hoisting bail utilitzing a single lift cable
    5) Provided with a submersible discharge connection system to permit removal and installation without entering the wet well
e. Wet Well access
    1) Shall meet the requirements of technical specification DC A20 Wet Well Design.
f. Valves and Piping
    1) Each pump shall be provided with a full flow type check valve capable of passing a 3” solid with flanged ends
    2) Each discharge line shall be equipped with a 2-way plug valve,
    3) Plug valve to be non-lubricated, tapered type
    4) Pump station piping to be ductile iron class 53 thickness.
g. Force Main and Accessories
    1) Pipe
        a) HDPE PE 4719
        b) Fittings to meet same requirements as the pipe. Connections to flanged fittings shall be made with PE flange adapters and ductile iron backup rings
        c) Tracer wire to be 14 ga. Solid conductor copper direct bury with appropriate terminal spacing.
        d) Utility marking tape same as specified above
h. Cleanouts, Air Reliefs, and Drains
    1) Enclosure shall be precast concrete in accordance with the requirements with a minimum 48” diameter
    2) Frames and covers shall be cast iron, designed to accommodate HS-20 traffic loading. The word “Sewer” shall be cast into the cover.
    3) Internal equipment shall include ductile iron components having interior and exterior epoxy coating meeting AWWA C-110 or C-153. All hardware to be stainless steel.
    4) Plug Valves

a) Worn-gear actuated. Gear enclosed in a cast iron housing, with outboard seals. Actuator shaft and gear quadrant supported on permanently lubricated bronze bearings.
b) Plug to be resilient faced cylindrical plugs eccentrically offset from seat.
c) Geared operator with 3-inch operation nut and had level for operation.

5) Air release valves shall be sized appropriately with a maximum operating pressure of 150 psi. Valve shall be provided with back flush attachments, a method to protect dirt and debris from entering the top, and soft seating material used on valves operating at pressure less than 20 psi.

i. Small Packaged Grinder/Lift Station

1) Grinder Pump Station with wired level sensor shall be provided where indicated on drawings at the New Cabin.
2) Manufacturer shall provide a semi-positive displacement type grinder station within a minimum of 10 years experience and supported by a representative within 300 miles of the project site
3) Grinder Station
   a) Manufacturer shall furnish a complete factory built and tested wetwell/drywell grinder station
   b) Components:
      - NEMA 6P quick disconnect
      - Pump removal system
      - Stainless steel discharge assembly and shut off valve
      - Anti-siphon check valve
      - Electrical alarm panel
      - Necessary Controls
      - Ensure there is an internal check valve
   c) Pump
      - Casting shall be cast iron
      - 8 mil minimum epoxy coating
      - Suitable for domestic wastewater service
   d) Grinder
      - Directly below pumping elements
      - Direct driven single one piece motor shaft
      - Capable of reducing all components in normal domestic sewage to finely divided particles
   e) Tank
      - Wetwell/dry well design made of HDPE
   f) Forcemain
      - Shall be HDPE PE 4710 CTS

PART 3 - DESIGN CRITERIA

3.1 BASIC FUNCTION

A. Provide a year round water and sewer system to all the cabins at 4 Lazy F that can easily be winterized each season

B. Install new gravity sewer facilities to convey wastewater to new lift station wet well. Gravity collection mains, 6-inch minimum diameter, with manholes as required shall be installed within the 4LF area in alignment that keeps all individual gravity sewer services from each cabin less than 100 feet long

C. Install a new sewer lift station and force main to convey wastewater to the Moose Sewer Collection System. Force main will have adequate cleanouts for maintenance, with drains at low point and air reliefs at high points in the alignment.

D. Install new gravity sewer services from each cabin indicated on the drawings to the sewer collection system that maintains a maximum length of 100 feet from cabin to sewer main.

E. Install packaged grinder lift station to the New Cabin

F. Install a water distribution system that provides not only potable water to each of the cabins but also fire suppression with adequate valves for isolation and maintenance.

G. Install individual water services to each cabin that can provide potable and fire flows with separate connections and curb stop valves for each service

3.2 INSTALLATION CRITERIA

A. All site mechanical utilities must be designed and installed per WYDEQ Water Quality Regulations and Wyoming Public Works Standard Specifications 2023 Edition. Including all disinfection and testing requirement for water and testing requirements for sewer.

B. All water and sewer must be installed and approved while maintain existing sewer collection in Moose and water delivery in Moose.

C. Water Distribution System
1. The alignment of the water distribution system and services must be located within the Limits of Disturbance. The water main alignment must keep all water services less than 200 feet in length,
2. All water mains and services will be marked with marking tape and tracer wire. Tracer wire will include termination posts that are flush with the ground surface.
3. All water mains and services shall be buried a minimum of 5 feet deep and require a minimum of 2” blue board insulation over the pipe. Water mains and services buried 7 feet deep to NOT require blue board insulation.
4. The alignment of the water system shall be surveyed following design and prior to construction to verify location in the field.
5. The water main shall be looped leaving o dead-end lines.
6. Water main shall include isolation valves at each branch to control the system, and one mid loop.
7. All water main valve boxes shall include a concrete collar.
8. All dead end water mains shall include a blow off.
9. No high points will be accepted without an air relief valve.
10. All water services shall be constructed to within 5 feet of each structure with a maximum length of 200 feet.
11. Water services shall include a drain out per the detail in the drawing.

12. Water services shall be sloped either back to the main or the service connection for winterization.
13. Water meter and meter vault shall be located at the connection with the existing water main on site. Water vault shall be properly insulated to prevent freezing. Meter readout shall be installed on a post at the vault.

D. Sewer Collection System
1. The alignment of the sewer collection within 4 Lazy F must be located within the limits of disturbance in the drawings.
2. All sewer mains and services will be marked with marking tape and tracer wire. Tracer wire will include termination posts that are flush with the ground surface.
3. All sewer services and sewer mains shall be buried a minimum of 5 feet deep and require a minimum 2" blue board insulation. Sewer mains and services buried a minimum of 6 feet deep do NOT require blue board insulation,
4. Gravity sewer services connected to a manhole will require a single direction cleanout 5 feet from each structure. Sewer services that are connected to the gravity main will require a double direction cleanout.
5. All gravity sewer services connected to the main will be completed by the use of a wye fitting, no service saddles will be accepted.
6. No gravity sewer services shall exceed 100 feet in length.

E. Lift Station
1. Lit station to be located within the area noted on the drawings.
2. Wet well size and depth will be dictated by the WYDEQ Chapter 11 design requirements
3. Lift station enclosure appearance and finishes shall be subject to historic district requirements and will be approved by the CO.

F. Force Main
1. Location of force main rom the lift station to the connection with the Moose Collection System must be located within the Limits of Disturbance noted on the drawings.
2. Construction of the force main along the 4 Lazy F road shall be completed within the limits and keep one 10' wide travel lane at all times unless approved by the CO.
3. To keep within the proposed limits of disturbance, contractors may utilize alternative construction techniques such as plowing the force main.
4. Force main cleanouts, cleanouts with pump out drains, or cleanouts with air reliefs shall be located not more than 800 feet apart.
5. Force main pump out drains shall be located at all low points in the alignment with air relief valves at the high points.
6. Force main pipe shall be marked with tape and tracer wire. Tracer wire shall have termination posts flush with the ground and a concrete collar.
7. Force main shall be buried a minimum 5 feed deep and require blue board insulation. Force main buried 6 feet or deeper does NOT require blue board insulation.
8. Force main shall discharge in the existing collection manhole at an elevation matching the flowline.

G. Site Mechanical Backfill Requirements
1. Compaction tests shall be required per Section 312310 Trench Backfill, Wyoming Public Works Specifications. Intervals shall not exceed 200' for each one foot lift. Densities shall meet at least 96% maximum density as determined by AASHTO T-99 (Standard Proctor.)

3.3 HEALTH AND SAFETY CRITERIA

A. Safety Hazards: Avoid safety hazards wherever possible; where services must involve flammable materials or hazardous operations, comply with code

B. Unauthorized Access: Provide locking devices to stop unauthorized access.

C. Excess Pressure: Design pressurized components to withstand operational pressures without failure and to relieve or reduce excessive pressure to prevent failure.

D. Misuse: Minimize misuse that could result in damage to property, injury, or loss of life.

E. Vermin Resistance: Use components that are resistant to the entry of rodents and insects.

F. Maintain minimum required separation of water piping from sanitary sewer piping in accordance with local code and requirements.

3.4 STRUCTURAL CRITERIA

A. Buried Piping and Components: Design cover of concealment so that they are not subjected to damaging stress due to applied loads. Follow manufacturers' recommendations for bedding of pipelines.

3.5 DURABILITY CRITERIA

A. Expected Service Life Span: 50 years for all except pumps, meters and controls

B. Weather Resistance:
   1. Minimum burial depth of piping as noted above. If that depth is not achievable, blue board insulation shall be installed.

3.6 OPERATION AND MAINTENANCE CRITERIA

A. Provide easy access to and working clearance around system components,

B. Ease of Maintenance of force main, install cleanouts at a minimum 500 feet spacing.

END OF SECTION

SECTION 33 20 00 – PREFABRICATED SEWAGE PUMP STATION WITH 2 SUBMERSIBLE PUMPS

PART 1. GENERAL

1.1. SCOPE OF WORK

A. The work in this section shall include furnishing and placing into operation X prefabricated sewage pump station(s) complete with 2 submersible pumps, internal piping, lifting chains, guide rails, valve vault and access covers as specified herein and as indicated on the drawings. The complete pump station shall be submersible up to 65 feet above the inlet pipe level.

1.2. RELATED SECTIONS

A. Section 20 00 00 MONITORING AND CONTROL PANEL FOR SEWAGE PUMP STATION

1.3. REFERENCES

A. American Society for testing and material (ASTM) International
   1. A 48: Standard Specification for Gray Iron Castings.
   2. A743: Standard Specification Iron-Chromium Nickel, Corrosion Resistant,

B: American National Standards Institute (ANSI):
   1. B16.1: Standard for Cast Iron Pipe Flanges and Flanged Fittings, 125 lb.

C. Hydraulic Institute: Current Standards.
   1. HI 14.6: Hydrodynamic Pumps for Hydraulic Performance Acceptance Tests.
   2. HI 11.6: Submersible Pump Tests

1.4. SUBMITTALS

A. Submittal data shall be provided to show compliance with these specifications, plans or other specifications that will influence the proper operation of the pump(s).

B. Standard submittal data for approval must consist of:

1. Pump Performance Curves.
2. Pump Outline Drawing.
3. Station Drawing for Accessories.
4. Electrical Motor Data.
5. Typical Installation Guides.
6. Technical Manuals and Parts List.
8. Printed Warranty.
9. Management system certificate ISO 9001.
10. Manufacturer's Equipment Storage Recommendations.
11. Manufacturer's Standard Recommended Start-Up Report Form.

C. Lack of the above requested submittal data is cause for rejection.

1.5. QUALIFICATION REQUIREMENTS

A. The manufacturer shall provide data on alternate equipment manufacturer's experience. Only Manufacturers with 20 or more years of experience who have furnished at least 5 similar lift stations shall be considered.

B. After installation, a pump station start-up shall be performed by the installing contractor under the supervision of the manufacture's authorized representative. 8 hours of field service shall be provided by an authorized, factory trained representative of the pump manufacturer. Services shall include, but not be limited to, inspection of the completed pump station installation to ensure that it has been performed in accordance with the manufacturer's instructions and recommendations, supervision of all field-testing and activation of the Pump Manufacturer's Warranty. The test shall demonstrate to the satisfaction of the Owner that the equipment meets all specified performance criteria, is properly installed and anchored, and operates smoothly without exceeding the full load amperage rating of the motor. The Contractor shall be responsible for coordinating the required field services with the Pump Manufacturer.

1.6. DELIVERY, STORAGE AND HANDLING

A. Deliver, store and handle products to site per direction of contracting officer.

1.7. OPERATIONAL REQUIREMENTS AND WARRANTY

A. The contractor shall supply and install (1) prefabricated and preassembled sewage pump station(s).

B. Each sewage pump station shall be completely issued with discharge connections, discharge pipes, guide bars, safety grit and cable holder to assemble 2 submersible waste water pumps.

C. For each pump station the contractor shall supply and install a separate valve vault.

D. The contractor shall supply and install 2 submersible sewage pumps

E. The submersible pumps shall have a semi open multi vane self-cleaning impeller designed to transport wastewater with fibrous materials like wet wipes.

F. The impeller shall be wear resistant and made of high chromium cast iron with at least 24% chrome against sand and grit which is expected to enter the pump station with the sewage or the storm water. Impellers that have surface hardening (by thermal, coating, etc.) will not be allowed.

G. Each pump shall be capable to lift 82.73 US gpm at a total dynamic head of 73.63 **feet**.

H. The pumps shall be provided with prorated 60 months (5 years) warranty against defects in materials and or workmanship. Unless otherwise specified, all other equipment shall be

warrantied for 12 months (1 year). The warranty shall be in printed form and previously published as the manufacturer's standard warranty for all similar units manufactured, latest revision. Upon warranty occurrence, the manufacturer's authorized service center shall remove the pump, repair, reinstall and provide start up on the repaired pump. A detailed failure analysis shall be submitted to the Owner for their records summarizing corrective action taken.

I. The manufacturer shall guarantee clog-free operation for a period of 24 months from the date of start-up of the pumps by the local authorized factory representative. A certificate shall be provided to the Owner on the day of start up with the local contact information and effective date. If the impeller clogs with typical solids or modern trash debris normally found in domestic wastewater during this period, an authorized representative shall travel to the jobsite, remove the pump, clear the obstruction and reinstall the pump at no cost for the Owner. A written report shall be provided to the Owner detailing the service call with pictures for verification purposes.

PART 2. PRODUCTS

2.1. PREFABRICATED SEWAGE PUMP STATION (FLYGT TOP 4 OR EQUIVALENT)

A. The sewage pump station shall be made of Fiberglass reinforced polymer (FRP) equipped with discharge connections, discharge pipes, valves, guide bars and cable holder to assemble 2 submersible wastewater pumps. The material of the complete station shall be applicable for wastewater with a PH value from 5.5 to 8.0. The Inner diameter shall be 60”.

B. All dimensions incl. inlet and outlet positions shall be made according to drawings.

C. The station cylinder shall be affixed to the station bottom such that the assembled components are structurally integrated, resulting in a watertight vessel which is capable of withstanding the full hydrostatic head from the exterior of the station while the station is completely empty.

D. The cylinder shall be made of FRP using the filament winding process. A safety factor of two (2) on the minimum ultimate tensile strength of the laminate bottom shall be used in designing the basin and cylinder wall thicknesses for the station, taking into account all normally imposed loads arising from flotation, soil pressures, normal backfill, handling loads, operating loads and static loads imposed by equipment used in hoisting the pumps in and out of the station. All inside surfaces shall be smooth and free of cracks and crazing. The inside surface shall be pigmented or gel coated to a bright white finish. All surfaces other than those made in contact with the mold surface shall be coated with air-inhibited resin or gel coat; this includes any cut edges of laminate.

E. The bottom part, the inlet and the outlet connection pipes shall be laminated to the station cylinder. The station shall have an anti-flotation flange and be approved to withstand the full hydrostatic head from the exterior of the station and to withstand the forces acting upon the station due to the subsoil water pressure while the station is completely empty.

F. The manufacturer shall provide with the delivery a manual with detailed installation and safety instructions. The manual shall include information about the backfill material to provide adequate ballast against buoyancy under full hydrostatic head conditions.

G. The sloping walls of the pump station bottom shall transport all solids, trash and sludge, normally found in sewage and wastewater, to the suction of the submersible pumps to facilitate removal and effectively clean the bottom. The maximum volume left in the station after the pumps have stopped shall be 58 US gallons or 25 US gallons if the pumps are connected to an electronic sump cleaning device.

H. The discharge connections shall be mounted on a sloped wall so solids can't settle below the discharge pipes.

I. Lifting eyes and anchoring steel bars shall enable a quick and easy installation of the station.

J. The station must be protected against electrical shock according the IEC regulation 611 40.

K. The pump station shall be ready for operation and be furnished with following equipment:
   a. All fasteners shall be made of stainless steel AISI 316.
   b. 2x 3"x4" reducers and 2x 4" discharge connections made of cast iron ASTM A-48, Class 35B with MULTI/JOINT coupling.
   c. 2 x 2" guide bars per discharge connection made of stainless steel AIAI 316.
   d. 2 upper guide bar holder made of stainless steel 316.
   e. 2 x 4" discharge pipe made of PE 100, PVC or stainless steel 316. The length of the pipes shall be sufficient to connect these pipes to the valves in the valve vault.
   f. 1 Cable holder with 4 hooks made in Stainless steel AISI 316.

L. The station cover shall be of ¼-inch thick Type-5086 aluminum diamond plate with an integral Safe-Hatch access cover. All bars, angles and shapes shall be type 6061-T6 aluminum. The access cover frame shall be a minimum of 4-inches deep and shall be adequately sized to allow for easy passage of the submersible pumps. The Safe-Hatch access cover shall be designed to support the weight of the pump unit plus pedestrian traffic. The access door(s) shall be equipped with a hold-open arm, held open in the 90-degree position. Cover door hinges shall be heavy-duty design and be cast 1/4-inch thick Type 316 stainless steel with 3/8-inch diameter stainless steel hinge pins. All fasteners shall be type-316 stainless steel. Each hatch shall be supplied with a type-316 stainless steel slam lock, having a key-way protected by a threaded plug. The plug shall be flush with the diamond plate cover. The hatch shall be equipped with an aluminum lift handle that shall be flush to the top of the diamond plate cover. The station lid shall have an integral four-inch diameter stub-pipe connection for the purpose of venting the pump station. The inverted J-shaped vent pipe shall be schedule 40 PVC pipe and shall end at a point at least

3-foot above the elevation of the station cover. There shall also be an option for a second vent to accommodate positive ventilation of the wet well. The access cover unit shall be equipped with a Safe-Hatch hinged safety grate to provide protection against fall-through and to control access into the confined space. Grate openings shall be sized to allow for routine maintenance inspection without having to open the safety grate. The closed safety grate shall be designed to support the weight of one pump to facilitate site pump wash-down and inspection. The hatch opening will have a 4" elevated toe board to prevent tools from being kicked into the wet well (per OSHA 1926.502)

2.2. VALVE VAULT

A. Each discharge pipe from the sewage pump station shall be connected to a check valve and a gate valve in a separate valve vault. The valve vault shall be fabricated in either concrete or FRP and shall be large enough to allow entry for routine maintenance and inspection. The valve vault shall be equipped with an integral drain and check valve to facilitate drainage from the valve vault back into the pump station.

B. The check valves shall be 4" Ball-type or ValMatic flap-type. The 4" gate valves shall be ¼-turn eccentric plug-type. All valves shall be made of cast iron with flanges according ANSI and be connected to a single 4" discharge pipe which shall last at least 5 feet outside the valve vault. The pipes shall be made of either of PE 100 or Stainless steel AISI 316.

C. The valve vault shall be housed within an above-ground enclosure.

2.3. SUBMERSIBLE SEWAGE PUMPS (FLYGT NP 3085 SH 255 OR EQUIVALENT)

A. Each station shall be equipped with **2** submersible, close-coupled wastewater pumps.

B. Each pump shall be equipped with a **4 HP** submersible electric motor, capable to operate on a **240** volt, 3 phases, 60 hertz voltage supply. Single phase power will be converted to 3 phase at the control panel by panel-mounted VFDs.

C. The hydraulic of the pump shall be capable of handling raw domestic wastewater and storm water with fibrous materials like wet wipes**.**

D. The impeller blades shall be self-cleaning upon each rotation as they pass across a sharp relief groove in the Insert ring and shall keep the impeller blades clear of debris. The insert ring shall have a guide pin which moves fibers from the center of the impeller to the leading edges of the impeller. The impeller shall move axially upwards to allow larger debris to pass through and immediately return to normal operating position. The clearance between the insert ring and the impeller leading edges shall be adjustable.

E. The pump shall be capable to operate without any limitation between 50% and 150% of the Best efficiency point (B.E.P) of the performance curve.

	Flow in USgpm	TDH in feet	Hydraulic efficiency %	NPSHre in feet
Required Duty Point	82.5	73.5	42.5	17.5

F. A performance chart shall be provided upon request showing curves for torque, current, power factor, input/output HP and efficiency. This chart shall also include data on starting and no-load characteristics

G. The impeller shall be mounted on the motor shaft. Couplings shall not be accepted.

H. The pump motor shall be induction type with a squirrel cage rotor, shell type design, housed in an air filled, watertight chamber. It shall be permanently submersible according standard IEC 60034 and protection class IP 68.

I. The stator windings shall be insulated with moisture resistant Class H insulation rated for 356°F.

J. The motor shall be capable of no less than 30 evenly spaced starts per hour and be able to operate throughout the entire pump performance curve from shut-off through run-out even when the motor is not submerged

K. Sealing of the pumping unit to the discharge connection shall be accomplished by a machined metal to metal watertight contact. Sealing of the discharge interface with a diaphragm, O-ring or profile gasket will not be acceptable.

L. It shall be possible to lift and lower the pumps on parallel guide bars and connect them to wet well mounted discharge connection. There shall be no need for personal to enter the wet well when removing or reinstalling the pumps.

M. The pump housing shall be prepared for the assembling of a sump mixing valve. The discharge flange of the pump housing shall be 4”.

N. The junction chamber containing the terminal board shall be hermetically sealed from the motor by an elastomeric compression seal. Connection between the cable conductors and stator leads shall be made with threaded compression type binding posts permanently affixed to a terminal board. The motor and the pump shall be produced by the same manufacturer.

O. The motor shall be protected by 3 thermal switches embedded in the stator set to open at 260°F and one leakage sensor floating type located in the stator chamber. The sensor and the switches shall be connected to the control panel which shall stop the motor and send an alarm when the sensors are activated.

P. The pump shall be Explosion approved according FM CLASS 1. DIV 1 "C" & "D"

Q. The cable entry shall consist of dual cylindrical elastomer sleeves, flanked by washers, all having a close tolerance fit against the cable and the cable entry. Epoxies, silicones, or other secondary sealing systems shall not be considered acceptable.

R. The pump shaft shall rotate on two bearings. Motor bearings shall be permanently grease lubricated and have a nominal L10 lifetime of 50.000 hours. The upper bearing shall be a single deep groove ball bearing. The lower bearing shall be a two row angular contact bearing to compensate for axial thrust and radial forces. Single row lower bearings are not acceptable.

S. The shaft shall be sealed by a tandem mechanical shaft seal system consisting of two seals, each having an independent spring system. The seals shall require neither maintenance nor adjustment and shall be capable of operating in either clockwise or counter clockwise direction of rotation without damage or loss of seal function.

T. Each pump shall be provided with a lubricant chamber for the shaft sealing system. The lubricant chamber shall be designed to prevent overfilling and to provide lubricant expansion capacity. The drain and inspection plug, with positive anti-leak seal shall be easily accessible from the outside. The seal system shall not rely upon the pumped media for lubrication. Seal lubricant shall be non-hazardous.

U. Where a seal cavity is present in the seal chamber, the area about the exterior of the lower mechanical seal in the cast iron housing shall have cast in an integral concentric spiral groove. This groove shall protect the seals by causing abrasive particulate entering the seal cavity to be forced out away from the seal due to centrifugal action

V. The Materials of construction shall be as follows:

   a. Pump housing: ASTM A-48, Class 35B
   b. Impeller and insert ring: A 532 ALLOY III A (25% Chrome)
   c. Stator housing: ASTM A-48, Class 35B
   d. Shaft: ASTM A479 S43100-T.
   e. Shaft seal: Pump side: - Corrosion resistant Tungsten carbide WCCR
   f. Shaft seal Motor side: - Corrosion resistant Tungsten carbide WCCR

W. All castings must be blasted before coating. All wet surfaces are to be coated with two-pack oxyrane ester Duasolid 50. The total layer thickness should be at least 120 microns. Zink dust primer shall not be used.

X. The motor shall be equipped with 30 feet of screened cable suitable for submersible pump applications. The power cable shall be sized according to NEC and ICEA standards. The outer jacket of the cable shall be oil resistant chlorinated polyethylene rubber. The cable shall be capable of continuous submergence underwater without loss of watertight integrity to a depth of 65 feet.

Y. Each completed and assembled pump/motor unit shall undergo the following factory tests at the manufacturer's plant prior to shipment. The Manufacturer shall provide on demand a copy of his quality control plan for these tests and an ISO 9001 factory certificate:

a. Minimum 3-point hydraulic performance test
b. No-Leak seal integrity test
c. Electrical integrity test

2.4. LIFTING EQUIPMENT FOR PUMPS

A. Each pump shall be fitted with 30 feet of stainless steel lifting chain or lifting cable. The working load of the lifting system shall be 50% greater than the pump unit weight.

2.5. SUMP MIXING VALVE (FLYGT 4901 0R EQUIVALENT)

A. One pump unit in each pump station shall be equipped with an automatically operating flush valve mounted directly to the pump volute. During the starting the valve shall redirect a portion of the pumped media into the sump to re-suspend solids and grease by the turbulent action of its discharge.

B. The valve shall be equipped with an adjustable, wear-resistant discharge nozzle that can be used to direct flow within the sump. The valve shall operate by differential pressure across the valve and shall not require any electric or pneumatic power source to operate. The valve shall be suitable for use in Class I, Division 1 hazardous locations.

C. The valve shall open at the beginning of each pumping cycle and shall automatically close during the pump operation after a pre-set time. A method of adjusting the valve operating time shall be provided.

2.6. SUBMERSIBLE CABLE CONNECTION BOX (INTEX, RALSTON OR SIMILAR)

A. The submersible cable of the pump shall be connected to the cable from the Control panel in a floor or wall mounted cable connection box to ease the installation and disassembling of the pumps and keep the submersible cables as short as possible.

B. The cable connection box shall be submersible NEMA 6P (IP 68) to secure that no water can enter the motor via the cables even when the complete area is flooded.

SECTION 33 30 00 PUMP CONTROL SYSTEM FOR DUPLEX SEWAGE PUMP STATION

PART 1. GENERAL

1.1. SCOPE

A. The work in this section shall include furnishing and placing into operation the monitoring and control panel a pump station with **2** submersible wastewater pumps.

B. The controller shall alternate the pumps and operate the pumps always at its best efficiency

1.2. RELATED SECTIONS

A. SECTION 33 20 00 PREFABRICATED SEWAGE PUMP STATION WITH 2 SUBMERSIBLE PUMPS

1.3. REFERENCES

A. Electrical work shall be in accordance with Division 16 - Electrical
B. UL 508A and 698A
C. NFPA 70 - NEC (2017)
D. NFPA 820
E. ISA 5.1 and 5.4

1.4. SUBMITTALS

A. Submittals shall include but not be limited to the following:

1. Shop drawings
2. Bill of Material (BOM)
3. Wiring diagrams
4. Outline and dimension drawings
5. Enclosure mounting details

1.5. DELIVERY, STORAGE, AND HANDLING

A. Deliver, store, protect and handle products to site under provisions of Specifications
B. Store in a clean, dry space.
C. Lift only with lugs provided for the purpose.
D. Handle carefully to avoid damage to internal components, enclosure and finish.

1.6. QUALIFICATION REQUIREMENTS

A. The manufacturer shall provide data on alternate equipment manufacturer's experience. Only Manufacturers with 20 or more years of experience who have furnished at least 20 similar lift stations that have been in regular operation not less than 5 years will be considered. Evidence of experience and operational data may be required from the manufacturer to determine the suitability and efficiency of the equipment offered.

B. The Manufacturer shall be able to offer Alarm Transmission Service via CLOUD (Internet) for a monthly fee. Information such as SMS or email shall be available 24 hours / 7 days with and update interval: 60 seconds

1.7. INSTALLATION

A. After installation, a pump station start-up shall be performed by the installing contractor under the supervision of the manufacture's authorized representative 3 days of field service shall be provided by an authorized, factory trained representative of the pump manufacturer. Services shall include, but not be limited to, inspection of the completed pump station installation to ensure that it has been performed in accordance with the manufacturer's instructions and recommendations, supervision of all field-testing and activation of the Pump Manufacturer's Warranty. The test shall demonstrate to the satisfaction of the Owner that the equipment meets all specified performance criteria, is properly installed and anchored, and operates smoothly without exceeding the full load amperage rating of the motor. The Contractor shall be responsible for coordinating the required field services with the Pump Manufacturer.

B. All electrical work performed shall be in full accordance to the requirements of the Division 16 (Electrical) Specifications.

1.8. WARRANTY

A. When installed in accordance with all the provisions of this section, related sections and the supplier's installation manual, the warranty on the Monitoring and Control equipment shall be 12 months.

B. The warranty period shall start from the date of equipment delivery to the job site.

1.9. LEVEL CONTROL (FLYGT ENM 10)

A. Each station shall be equipped with 2 Float switches for backup level sump control approved according LVD EN61058.

B. CSA approved according Class I Zone 0, Gr. IIC and Div.1 Gr A, B, C&D

C. Material of casing: Polypropylene.

D. Degree of protection: NEMA 6.

E. They shall be applicable for liquids with a density of 0.95 – 1.10 g/cm3. The Level control shall include 40 feet submersible cable.

1.10. LEVEL TRANSMITTER (FLYGT LTU 801)

A. Each station shall be equipped 1 Level transmitter approved acc. EN 61000-6-2, EN 61000-6-3, EN 61326-1**.** It shall be approved for explosive areas according UL Class 1, 2 and 3 Division 1 Group A-D T4/T5/T6

B. Output 4–20 mA direct current, proportional to the measured level. Low supply voltage 8–30 V DC – or battery operation.

C. It shall be suitable for waste water with a diaphragm made of ceramic.

D. Insulated > 100 MΩ at 500 V DC. Material of sensor body: Ryton PPS.

E. Degree of protection: Nema 6.

F. The transmitter shall include 35 feet submersible cable.

1.11. CABINET

A. The Monitoring & Control and the telemetry and 2 pump drives shall be assembled in a cabinet. It shall be made of Reinforced plastic or stainless steel and it shall be prepared for an installation on a concrete floor.

B. Beside the Monitoring & Control unit it shall have enough space for the customer-installed telemetry equipment and include.

- 1 backup battery for the telemetry unit.
- Red-dome style flashing alarm light
- Alarm Horn or Bell with Alarm silence button
- Intrinsically safe barrier for UL 913 requirements
- Anti-condensation heater and thermostat
- Generator receptacle and Emergency circuit breaker (walking-beam interlocked with Main breaker)
- Enclosure legs
- Power switch, which can be locked in the off position by a pad-lock.
- 1 light group with socket

C. The inner dimension of the cabinet shall reserve at least 10% back panel space for future adjustments.

D. All components on the front of the machine should be provided with function plate.

PART 2. EXECUTION

2.1. GENERAL

A. Perform installation in accordance with Contract Documents and manufacturers specifications.

2.2. EXAMINATION

A. A factory trained technician shall examine the work area prior to beginning work and check the following:

1. The environment is safe to begin working in
2. All surfaces are ready to receive work
3. All tools are in the proper location and are in good condition
4. Grounding of the system

2.3. FIELD QUALITY CONTROL

A. The follow field tests shall be performed by a factory trained technician

1. Point to point wiring verification

2. Utility power verification
3. Site acceptance testing
4. System demonstration

B. Point to Point I/O Verification

1. After installation of the pumps and the control panel, a factory trained technician shall prepare the I/O checklist. The checklist shall include the following:
    a. All inputs and outputs connected to the control panel
    b. All alarms that can be generated by the control panel
2. The technician shall follow a test procedure to test all I/O and alarms.
    a. All digital inputs shall be tested from point of origin unless it is unsafe.
    b. All digital outputs shall be tested by running a simulation test from the controller or by simulating the fault condition.
    c. All analog inputs shall be tested from the point of origin where possible and by use of a signal generator otherwise.
    d. All analog outputs shall be tested by running a simulation program or by forcing the output to a value.
3. The technician shall follow a test procedure to ensure the system operation parameters are met.

C. Configuration Verification

1. The factory trained technician shall document the settings using a factory provided configuration checklist. Each parameter shall be verified prior to the beginning of testing and then again after testing is completed.
2. The configuration of the pump station manager as well as the IPS gateways shall be documented.
3. The pump station manager configuration shall be saved to a factory provided SD card after testing is completed.

2.4. FACTORY TRAINED SUPERVISION

A. The contractor shall procure a factory trained technician to check over equipment prior to putting the equipment into operation.

B. Point to point test of all wiring.

C. Functional test of all equipment alarms and controls.

2.5. CERTIFICATION OF TESTING

A. All tests shall be performed in the presence of a duly authorized representative of the Owner. If the presence is waived, certified results shall be provided by the Contractor.

B. Written notice of all tests shall be given two weeks in advance.

2.6. TEST EQUIPMENT

A. All test equipment shall be provided by the Contractor.

2.7. TRAINING

A. Training shall be a minimum of four (4) hours and cover the complete Pumping System and related controls.
B. Instruction material shall be provided for four (4) trainees.

2.8. ANNUAL PREVENTATIVE MAINTENANCE AGREEMENT

A. Training In addition to the pump(s) the supplier shall provide a minimum one-year preventative maintenance agreement that includes a minimum of one annual site visit with a multi-point pump(s) inspection performed by manufacturer certified and manufacturer trained technician(s) within 12 months of installation.
B. The multi-point inspection shall include visual and functional inspections of all wet well components, pumps and control panels; oil inspection and oil change of submersible pumps; pertinent physical and electrical data to reasonably expect prolonged equipment operation for a year of additional service.
C. The end-user shall also be provided a copy of a summarized report at the conclusion of the multi-point inspection.

SECTION 33 40 00 – SPECIAL CONSTRUCTION

PART 1 - GENERAL

1.1 SECTION INCLUDES

A. Abandonment and demolition of the existing propane tanks on site.

B. Abandonment of the existing water well on site.

C. Submittals:
   1. Schedule of demolition activities
   2. Abandonment and rerouting of utility services, if any
   3. Photographs and video of pre existing conditions adjoining the work area

1.2 RELATED REQUIREMENTS
A. Section 02 21 19 – Utilities Survey
B. Section 02 41 00 - Site Demolition
C. Section 33 14 15 - Site Utilities.

1.3 FIELD CONDITIONS

A. On site storage or sale of removed and abandoned items or materials is not permitted

PART 2 - PRODUCTS (NOT USED)

PART 3 - DESIGN CRITERIA

3.1 BASIC FUNCTION

A. Demolition and removal of the existing septic tanks on site as required in WYDEQ Chapter

B. Abandonment of the existing water well per WYDEQ Chapter 26 requirements.

C. Coordinate with Lower Valley Energy for the abandonment of the power to the well if it exists.

3.2 EXECUTION

A. Examination
   1. Verify that utilities have been disconnected and capped before starting demolition operations.

B. Preparation and Protection
   1. Prior to abandonment of the existing water well, the existing historic wellhouse fabric should be documented in place, set aside for future reconstruction.
      a. This will include photos and videos of the wellhouse in the current state prior to construction.

2. Provide and maintain interior and exterior shoring, bracing, or support as needed to provide stability and prevent unexpected movement or collapse of structures.
3. Protect adjacent facilities not slated for removal.
4. Erect a visible fence around the demolition footprint.
5. Provide protection to ensure safe passage of people around the demolition area.
6. Limit and mitigate dust and noise to surrounding areas.
7. Leave barriers in place until open excavations are eliminated.

END OF SECTION

APPENDIX 1

7. CHINKING AND DAUBING SOP

APPLICABLE CONDITION

This SOP is applicable where daubing is deteriorated, unstable, friable, spalling, or missing. Additionally, it is suitable for daubing to be replaced if there are gaps between the daubing and the wood (¼" inch or more), leading to pest intrusion that must be addressed for continued safe use of the log structure. Lastly, daubing may be replaced if current daubing (for example, elastomeric caulking products) is causing prolonged moisture in wood that has led to active decay. This SOP also applies to new log buildings that require daubing for use, comfort, and/or esthetic purposes.

LIMITATIONS AND CONSIDERATIONS

A note on terminology:

In the Mountain West, "chinking" and "daubing" are often used interchangeably. This document uses the term "daubing" exclusively to describe the soft material troweled into the horizontal spaces or joints between logs. For round logs, daubing is held in place by a variety of mechanisms that include but are not limited to: willow branches or wood furring strips acting as a "ledger" nailed into the log, and/or nails hammered shallowly into the joint to act as anchors, and/or metal lath nailed or friction-fit into the joint.

"Chinking" is used to describe the dry, bulky material that is packed into the joint prior to the application of daubing, such as oakum.

"Quarter-round chinking" describes pieces of small diameter wood that have been quartered, inserted into the joint, and secured with nails. There is no daubing when quarter round chinking is used.

Daubing is the least durable part of a log building and was historically understood to be sacrificial; as such, it was patched or replaced as part of regular maintenance cycles. Daubing is only suitable for buildings that were daubed as part of their Period of Significance and is not appropriate for log structures that originally had quarter-round chinking.

The color of daubing should be considered and an effort made to match existing daubing while also taking into consideration a daubing mix that is appropriate for the Period of Significance. For example, if a structure is currently daubed with a portland cement-based daubing (grey in color) but would have been daubed during the Period of Significance with a lime-based daubing (bright white), building stakeholders should discuss what material is most reasonable and appropriate for the building's daubing campaign moving forward if complete replacement is not an option.

In some cases where daubing is considered particularly character defining and original daubing is still present, a mortar analysis via acid digestion method can be done to determine the original daubing recipe. Mortar analysis can also determine sand characteristics that can then be matched at a local construction material supplier, such as color, grain size, and shape.

Safety must be considered. The demolition of daubing as well as mixing new daubing poses a risk of crystalline silica exposure. Workers must be familiar with OSHA standards for Personal Protective Equipment (PPE) when working with the binders most often used in daubing mixes: non-hydraulic lime, Naturally Hydraulic Lime (NHL), natural cement, and/or artificial cement.

If daubing is painted, test for lead before removing or disturbing any material. If lead is found in the paint, consult a certified lead paint removal specialist.

This document does not cover using elastomeric daubing products, such as Permachink. Synthetic daubing products are often not appropriate for historic structures, nor is their record of performance grossly superior to more traditional materials that are property applied. However, there may be instances where synthetic daubing is appropriate. Always consult with a preservationist, building stakeholders, and design team (where applicable) before deciding on the use of a synthetic daubing product.

Always exercise caution when attempting to make a historic building "airtight", especially when paired with heating/cooling and year-round use for a structure that was meant to be used seasonally. Any "tightening" of a structure must come with adequate planned ventilation and consideration of where and how moisture may increase and cause problems.

HOW TO USE THIS SOP

This is the Standard Operating Procedure for applying daubing and/or chinking to log buildings.

Each manufacturer's product varies, so make sure to follow their instructions and learning resources to correctly use their product (for example, knowing the difference between a "Type N" pre-mixed mortar versus mixing your own with a Naturally Hydraulic Lime (NHL) 3.5).

Each building varies, so make sure to follow the character defining features of the building. For example, if daubing was historically held in place with willow branches forming a ledge under the joint, continue to use willow branches when repairing/replacing daubing.

MATERIALS AND EQUIPMENT LIST

Materials:

- Binder appropriate for structure/historically matched:
    - Non-hydraulic lime (aka lime putty): https://atlaspreservation.com/collections/traditional-lime-putty
    - Naturally Hydraulic Lime (NHL) 2.0, 3.5, or 5: https://atlaspreservation.com/collections/restoration-mortars
    - Natural Cement: https://www.preservationworks.us/product-category/materials/ *There are huge variations in natural cement. Consult a preservationist if natural cement is required for your daubing mix
    - Artificial Cement (e.g. portland cement): Quikrete is the most common option
    - OR
    - Appropriate pre-blended mortar: https://amerimix.com/amerimix-supporting-resources-technical-documents/technical-documents/type-n-s-m-o-masonry-mortar/
- Sand

Equipment:

- Hammer

- Chisel
- Stiff bristle brush
- Wheelbarrow
- Buckets
- Hoe
- Water
- Hawk/Mortar Board: https://marshalltown.com/cat-1235-hawks-accessories
- Brick Trowel: https://marshalltown.com/pro-4096-london--wide-brick-trowels
- Margin Trowel: https://marshalltown.com/pro-1899-margin-trowels
- Re-pointing ("tuckpointing") Trowel: https://marshalltown.com/pro-1585-tuck-pointers
- Spray Bottle
- Sponge
- Burlap

Depending on joint historical condition:

- Furring Strips/Willow branches
- Metal Lath
- Nails
- Oakum
- Other chinking/anchoring materials deemed appropriate for the project, such as backer rod

Personal Protective Equipment (PPE)

- For removing daubing:
    - Safety glasses
    - Mouth and nose covering (N95 respirator recommended)
    - Ear protection
    - Work Gloves
- For mixing daubing:
    - Safety glasses
    - Mouth and nose covering (N95 respirator recommended)
    - Nitrile gloves (especially if using a hand trowel to mix daubing)
- For application:
    - Nitrile Gloves
        - Simple daubing recipes based on lime, natural cement, artificial cement, clays, etc. will not hurt skin but will dry it out
        - Gloves are generally not required, but are recommended

WORK METHOD

Joint Preparation:

Use appropriate PPE

- Remove old daubing with hammer and chisel or similar method

- Clean joints with stiff bristle brush to remove any remaining daubing
- Take note of wood condition during daubing removal
    - If any decay is uncovered by the removal of old daubing, find and address the root cause of moisture intrusion and make repairs as necessary before preceding with daubing
- Note any chinking material such as oakum or small rocks
- Note any anchoring material such as metal lath or nails

Mixing Daubing:

Prior to daubing project, the following should be determined for the daubing mixture:

- Type of binder (lime, NHL, natural cement, portland cement, clay)
- Type of aggregate (angular, round, grain-size distribution, color)
- Ratio of binder to aggregate (generally 1:2 - 1:3)
    - OR
- Appropriate Pre-blended mortar type (M, S, N, O)
- Presence of additives: (animal dung, animal hair, clay)
- Use appropriate PPE
- If mixing your own daubing, always mix dry ingredients first in a wheelbarrow or bucket, using a hoe or margin trowel; take the time to mix dry ingredients well
- Add water in small increments, and mix well with hoe or trowel before adding more water
- Add water until a trowel with the daubing mix can be turned upside down and the daubing stays on the trowel and doesn't fall off
    - For the first couple attempts, it is recommended to mix only small batches to get a feel for how much water is needed to achieve the right consistency. This will avoid adding too much water to a large batch of dry ingredients, thus ruining the daubing and wasting material

Application:

- Use appropriate PPE
- Based on the project's specifications, add chinking material and anchoring material to the joint (backer rod, oakum, metal lath, nails, etc.)
- Using a spray bottle or mister, wet the joint and allow a couple minutes for the water to absorb enough into the wood that it is not dripping down the face of logs, this will keep the daubing from excessive shrinking and help with adhesion
    - Avoid overwetting or applying daubing when water is running down logs, which will carry binder out of the daubing mix and stain the face of the logs
- Lightly mist all tools with water; this makes working the daubing easier and keeps the daubing the right consistency
- If preferred, scoop a small pile of daubing onto a hawk, mortar board, or brick trowel
    - If using small buckets, this may not be necessary
- Using the back of a margin or repointing trowel, press a small amount of daubing into the joint and spread with pressure laterally
    - This can be done either:
        - By pushing the daubing off a hawk/mortar board/brick trowel with a margin or repointing trowel directly into the joint; or

        - By flipping the daubing onto the back of a margin or repointing trowel, and then pressing into the joint
- Apply daubing in ¼" lifts, building to the depth required/desired
- If using furring strips or willows as a ledge, build the joint to the level of the ledge so that the ledge "holds" the daubing.
- Finish the surface texture of the joint in a manner consistent with the historic condition: smooth, rough, tool marks etc.
    - Light wiping with a sponge can provide some texture to the surface by exposing the aggregate
    - Light tapping with a soft or coarse bristle brush when the daubing has set up but not dry can also create a roughened texture
- Avoid overworking the daubing (jokingly referred to as "petting the daubing") as this pulls the binder to the surface and can leave a milky film as well as cause excessive shrinking upon drying and/or weaken the daubing
- Take care to keep work clean and tight; avoid getting daubing on log faces
- Use a clean, damp (but not wet!) sponge to lightly wipe areas if daubing gets on log faces/outside of the joint, but avoid scrubbing or using a sponge that already has daubing on it, as this will spread the binder and create milky white stains when dried
    - These stains are cosmetic only - they do not hurt the logs and will generally disappear within the year due to weathering
- Avoid applying daubing in direct sunlight, as this can cause "flash drying" that causes rapid shrinking, cracking, and poor adhesion

Aftercare/Cleanup:

- As the daubing dries, mist periodically to slow the drying process and thus avoid excessive shrinking, cracking, or poor adhesion
- Wetted burlap placed over the face of the wall will also slow the drying process and is recommended in particularly dry or sunny environments
- If small cracks do form during drying, lightly mist the joint and press the cracks closed with fingers
- Use clean water to scrub and rinse tools of any residual daubing and store in a dry place

INSPECTION AND MAINTENANCE

This procedure can be repeated as often as needed by the building owner or manager, but it is likely only necessary on a 15-20-year or longer cycle.

APPENDIX 2

3. LINSEED OIL PAINTING

APPLICABLE CONDITION

This SOP is applicable where waterborne or oil-based paint has failed on wood trim, fascia, soffits, jams, and similar surfaces and is no longer providing protection from UV rays and excess moisture. Wood often appears dry and splintered after years of exposure without proper protection. Pockets of inactive rot or decay are acceptable, but wood that is constantly wet from roof runoff, exposure to wet earth, or other sources should not be repainted until the source of excess moisture is addressed. This SOP also applies to new replacement wood used in repairs or replacements.

LIMITATIONS AND CONSIDERATIONS

Linseed oil paint is not suitable for every situation. The building owner or manager must assess the condition of the wood (i.e., enough exposed wood to ensure proper penetration of an oil-based paint), the environmental conditions (i.e., sufficiently low relative humidity and high enough temperature at the time of application), and the project schedule and budget.

A note on current paint condition: While it's not necessary for the wood surface to be completely bare, there must be enough exposed area for penetration. Test several different spots on the building to see how well the linseed oil paint can penetrate after surface preparation. If it's not possible to prepare the surface in a way that allows proper penetration—or if doing so would damage the wood—then this SOP isn't suitable for your building.

Do not apply to wood with a moisture content above 15% or in environments with an ambient relative humidity above X% or below 70 degrees.

Always test for lead before removing or disturbing painted surfaces. If lead is found in the paint, consult a certified lead paint removal specialist to prepare the windows for repainting.

HOW TO USE THIS SOP

This is the Standard Operating Procedure for applying linseed oil paint to new, unfinished wood or properly prepared existing wood surfaces. Select only the necessary steps based on the initial condition of the wood. New wood requires almost no preparation, while wood with decades of accumulated paint layers needs much more preparation.

Each manufacturer's product varies, so make sure to follow their instructions and learning resources to correctly apply their product (for example, some require adding zinc to the paint before use). This SOP offers general guidance. For instance, there are extra steps not included for managing mildew growth and humid climates.

MATERIALS AND EQUIPMENT LIST

Materials:

- Linseed Oil Soap made from purified organic raw linseed oil and without additives such as Allback Linoljeprodukter of Sweden, distributed by Viking Sales, http://www.solventfreepaint.com.

- Organic Raw Linseed Oil $45/gallon http://solventfreepaint.com/cleaned_linseed_oil.htm
- Linseed Oil Paint $160/3 liters http://solventfreepaint.com/linseed_paint.htm
- 3M LeadCheck Swabs, 8 Pack $25 https://www.amazon.com/3M-717834209102DUPE-LeadCheck-Swabs-8-Pack/dp/B008BK15PU/ref=sr_1_1?ie=UTF8&qid=1473960532&sr=8-1&keywords=lead+test+kit

Equipment (select the most appropriate for your site and building):

- Nylon scrub brush or hog/china bristle brush
- Angle grinder and Osborne brush attachment
- Palm sander with medium grit sandpaper
- Empty mini metal paint can for pouring smaller quantities of paint (do not use plastic—BPAs can affect adhesion)
- Stiff natural bristle brush
- Rags
- Paint mixing stick
- Silent Paint Remover SH1000 $295 http://www.silentpaintremover.com/spr/index.htm

WORK METHOD

Surface Preparation

1. For loose, severely failing paint—use the pull scraper, Osborne brush, or palm sander with medium grit paper to remove loose paint chunks and expose the wood. The goal isn't to reveal perfect, bare wood; it's to create a surface that can allow penetration. If you are removing wood fibers during your preparation, you're using a too abrasive technique.
2. For mostly intact paint, use the silent paint remover infrared heat tool to remove the finish. Do not use chemical solvents, as they are not compatible with linseed oil-based products.
    a. Apply one coat of raw degummed linseed oil and allow it to soak for at least 24 hours.
    b. Position the Silent Paint Remover: Hold the device a few inches from the surface, ensuring the infrared element is directed at the paint.
    c. Heat for a short duration: The heating time will vary depending on the paint's thickness and type, but generally, it's a matter of seconds (e.g., 2-3 seconds for Speedheater Cobra, 20-30 seconds for the original Speedheater).
    d. Watch for bubbling and lifting: The paint will start to bubble and soften as it heats up.
    e. Avoid overheating: Do not let the paint scorch or smoke, as this can damage the wood.
    f. Scraping: Use a pull scraper, as it's less likely to mar the wood. Once the paint is softened, scrape it off quickly while the heat is still effective.
    g. Work in sections: Remove paint in manageable sections, allowing the heat to dissipate between sections to avoid scorching.
    h. Reheating: Avoid reheating sections that have already been heated, as this can lead to scorching.
3. For new wood- sand as you would to prepare for any type of paint.

Painting Procedure

1. Apply purified raw linseed oil to fill wood pores and allow for maximum coverage of linseed oil paint coats.
    a. Heat the linseed oil to a minimum temperature of 140°F (optional) and mix in complete compliance with manufacturer's written instructions.
    b. Apply the finish warm (optional) at ambient air temperature of at least 70°F with a brush in thin coats in complete compliance with the manufacturer's written instructions.
    c. Allow manufacturer's recommended dwell time (usually minimum of two days).
    d. Wipe off excess finish.
2. Apply Linseed Oil Paint
    a. Pigment can settle to the bottom of the can and must be mixed back into the paint. Stir the paint thoroughly. If the paint has skinned over on the top, carefully remove it before mixing your paint.
    b. Apply two or more thin, even coats as needed for coverage. Thicker coats will take much longer to cure with possible imperfections like surface wrinkling. Allow 24 hours between coats in ideal conditions, and longer in humid or cool environments.
    c. Maintain brushes by suspending them in Purified Linseed Oil between coats. Cleaning isn't necessary between coats; remove excess paint with a brush, plywood, cardboard, or rags. To clean, gently massage linseed oil or dish soap into the bristles, then rinse with hot water, repeating if needed. Avoid soaking natural bristle brushes, store in oil up to the ferrule, and use linseed oil or dish soap to clean linseed oil paints from surfaces.

INSPECTION AND MAINTENANCE

The natural aging process of linseed oil paint involves the paint developing a matte finish upon exposure to sunlight. The rate at which this ageing occurs depends on the regional climate, weather conditions, and sunlight exposure. While the substrate remains protected by the paint, the coating will continue to dry out over time. This aging process can be mitigated through surface cleaning, the application of Purified Linseed Oil to the paint, and the subsequent wiping off of excess oil after 30 minutes. Applying oil to the paint enhances the natural binder content, thereby re-establishing the pigment's stability. Such maintenance oiling is generally recommended for darker colors. Conversely, some individuals find that lighter shades tend to appear excessively yellow after oiling; in such cases, recoating is considered the most suitable option.

APPENDIX 3

2. LINSEED OIL AND PINE TAR FINISHING

APPLICABLE CONDITION

This SOP applies to various situations. A building's waterborne or oil-based finish might be past its usable life, or a building or a replacement wood member may be unfinished, and the building owner or manager may want to add UV and moisture protection by applying a finish.

LIMITATIONS AND CONSIDERATIONS

The primary limitation of this procedure is that the oil and pine tar must penetrate the wood to work effectively. While the wood surface doesn't need to be completely bare or raw, enough exposed area must be available for penetration. Test several spots on the building to see how well the oil/tar mixture penetrates after surface preparation. If it isn't possible to prepare the surface for proper penetration—or if doing so would damage the wood—then this SOP isn't suitable for your building. This method can be used on log buildings with mortar chinking and doesn't require taping or protection of the chinking. The craftsperson should use a small enough brush to generally avoid the chinking, mainly for aesthetic reasons; it will not harm the mortar.

HOW TO USE THIS SOP

This is the Standard Operating Procedure for applying an 80/20 mixture of linseed oil and pine tar to log or wood structures. This process should follow the wet and dry cleaning steps outlined in the "Wood and Log Building Cleaning SOP." It is crucial that the wood surface is reasonably clean so the new finish doesn't trap organic material, which will be impossible to remove once the oil and pine tar penetrate. All repairs should be finished before this process, as the oil/tar mixture will create a uniform appearance across all elements.

MATERIALS AND EQUIPMENT LIST

Materials:

- Authentic Light Pine Tar
- Viking Purified Raw Linseed Oil
- Linseed Oil Soap made from purified organic raw linseed oil and without additives such as Allback Linoljeprodukter of Sweden, distributed by Viking Sales, http://www.solventfreepaint.com.

Equipment (select the most appropriate for your site and building):

- Brush- Anza 120mm sturdy, stiff-bristle brush with ergonomic handle or similar.
- Mixing bucket (non-plastic preferred as BPAs can interfere with performance)
- Measuring utensils
- Wood stirring stick
- Nylon scrub brush or hog/china bristle brush
- Angle grinder and Osborne brush attachment

WORK METHOD

Application Instructions for Authentic (Dark, Light, Black, Black-Brown, Brown and Red) Pine Tar:

1. Ensure the wood surface is clean and dry before application. (See Log and Wood Cleaning SOP)
2. If a previous barrier-forming coating remains after cleaning, a more aggressive removal method may be necessary. In areas where some adhesion from the old stain persists, the craftsperson should loosen the finish using either a scrub brush or an Osborne brush attached to an angle grinder (with linseed oil soap). This more intensive surface preparation should be tested in a low-visibility area to ensure it can be done without abrading the log surface. If the previous paint or stain cannot be removed by this method, the building owner or manager may consider corncob blasting or using an infrared heat paint remover tool. Do not use chemical strippers on porous wood surfaces. Avoid power washing.
3. For roughcut, kiln-dried, or other very dry timber, apply an initial "primer" coat of Viking Purified Raw Linseed Oil for extra moisture protection and to extend Pine Tar coverage. (optional)
4. Mix Authentic Pine Tar with Viking Purified Linseed Oil. No additional solvents are needed. Use an 80/20 mixture to start, but the ratio can be adjusted based on personal preference. It is recommended to do a test first to see if you prefer a lighter or darker finish.
5. If pine tar and purified oil are cold, they will be thick or appear syrupy. Ensure materials are at least room temperature before application, so it is easy to brush or wipe on.
6. Apply one coat in an environment that is 60°F or warmer. With this method, a higher ratio of pine tar may be desirable to immediately get more opacity or color intensity (a second coat may be applied the next season to ensure protection while keeping a matte surface).
7. Apply thin, even coats of pine tar mixture with a stiff bristle brush or a clean cloth. Ensure each coat is fully dry before applying the next. Two coats minimum are recommended.
8. Brush out as needed to avoid drip marks. Let dry. Typically, drying takes up to 72 hours in temperatures above 60°F, but it may take longer depending on climate and wood type.
9. The mixture may chalk slightly between coats and requires about one to two weeks to fully cure and reach the final hardened finish.
10. For some woods or when ambient temperatures are cooler, it may help to apply the pine tar mixture, let it soak in, and wipe off any excess sitting on the surface before the end of the day to prevent unevenness.
11. Wipe off any excess finish.

INSPECTION AND MAINTENANCE

Annual inspection should include a thorough evaluation of how the oil/tar mixture performs. Spot reapplication may be necessary in areas exposed to high UV or moisture. Reapplication cycles will depend on the type of wood and climate.